

50TH ANNIVERSARY EDITION



PHYSICS

for JEE Main and
Advanced

Electrostatics and Current Electricity



Mc
Graw
Hill

Shashi Bhushan Tiwari

Electrostatics and Current Electricity

YOUR PERSONAL COACH
PHYSICS
for **JEE** Main and
Advanced

Electrostatics and Current Electricity

YOUR PERSONAL COACH
PHYSICS
for **JEE** Main and
Advanced

Shashi Bhushan Tiwari



McGraw Hill Education (India) Private Limited



Published by McGraw Hill Education (India) Private Limited,
444/1, Sri Ekambara Naicker Industrial Estate, Alapakkam, Porur, Chennai - 600 116, Tamil Nadu, India

Electrostatics and Current Electricity

Copyright © 2020 by McGraw Hill Education (India) Private Limited

No Part of this publication may be reproduced or distributed in any form or by any means, electronic, mechanical, photocopying, recording, or otherwise or stored in a database or retrieval system without the prior written permission of the publishers. The program listings (if any) may be entered, stored and executed in a computer system, but they may not be reproduced for publication.

This edition can be exported from India only by the publishers.
McGraw Hill Education (India) Private Limited

1 2 3 4 5 6 7 8 9 7085462 23 22 21 20 19

ISBN (13) : 978-93-5316-389-1

ISBN (10) : 93-5316-389-7

Information contained in this work has been obtained McGraw Hill Education (India), from sources believed to be reliable. However, neither, McGraw Hill nor its authors guarantee the accuracy or completeness of any information published herein, and neither McGraw Hill Education (India) nor its authors shall be responsible for any errors, omissions, or damages arising out of use of this information. This work is published with the understanding that McGraw Hill Education (India) and its authors are supplying information but are not attempting to render engineering or other professional services. If such services are required, the assistance of an appropriate professional should be sought.

Cover Designer: Creative Designer

Cover Image Source: DAL

Visit us at: www.mheducation.co.in

Write to us at: info.india@mheducation.com

CIN: U80302TN2010PTC111532

Toll Free Number: 1800 103 5875

Dedicated to

*All those who are behind technological innovations
to fight climate change*

Preface

This book forms a part of the series “Your Personal Coach”. Like other books in this series, this book has been written on two core principles:

- (i) a text book must have continuity and flow in what it discusses.
- (ii) nothing contributes more in understanding Physics than a good example.

I have tried to unfold the concepts gradually, one-by-one; illustrating each of them with examples. The main aim is to make the students learn the basic principles of Physics independently.

Many of the experiments that you do in your school laboratory are related to the electric circuits. I found it appropriate to include a chapter on Measurements and Errors in this book. A lot of questions in this chapter have been deliberately framed on measurements related to the electric circuits.

I shall be grateful to everyone who would provide feedback or help me with useful suggestions.

S.B. Tiwari

How to use this Book

To make full use of this book one must go through the topics sequentially while working through the examples and in-chapter problems given under heading “**Your Turn**”. By doing this you will have a fair amount of grasp over all the essentials in a chapter.

Miscellaneous examples given at the end of each chapter have problems which involve multiple concepts or have some mathematical complexity or are tricky. *If you are studying the subject for the first time or are hard pressed for time, you may skip the section on miscellaneous examples.*

Almost every solved example starts with explanation of physical situation and basic principles involved. This feature comes under heading “**Concepts**” at the beginning of each example.

I have highlighted the important points of learning under the heading “**In short**”. Here, I have also taken important learning points from the examples. While going through the chapter it is essential to go through these points.

Physics cannot be mastered without practice. Keeping this in view I have given three Worksheets (exercises) after every chapter. **Worksheet 1** has multiple choice objective type questions with single correct answer. **Worksheet 2** has multiple choice questions having *one or more than one correct answers*. **Worksheet 3** has subjective problems. A good number of problems has been given in the Worksheets to give you a good practice on concepts learnt.

After few chapters, at regular intervals, you will find separate assignments on miscellaneous type problems. These are problems based on latest trend of competitive examinations and contain **Match the Column type questions and problems based on a given paragraph**. Attempt these questions only after you gain enough confidence in the related chapters. I have kept these problems in separate chapters so that you have no bias or hint about the equation/s to use.

In the last chapter, you will find a collection of questions asked in competitive examinations since 2005. This is an ideal collection of problems for revision.

In the end of the book, solutions to all questions has been given. Solutions are quite descriptive and easy to understand.

Those who desire to practice at even higher level, I recommend my book – “*Problems in Physics for JEE Advanced*”.

I hope you will enjoy this book.

S.B. Tiwari



Acknowledgements

I thank to all who helped me in preparation of this book. My special thanks to –

- My students, who have taught me a lot.
- The management at McGraw-Hill which has shown a lot of patience.

S.B. Tiwari

Contents

<i>Preface</i>	vii	8. Electric Flux	1.19
<i>How to use this Book</i>	ix	• <i>Your Turn</i> 1.21	
<i>Acknowledgements</i>	xi	9. Gauss's Law	1.21
		• <i>Your Turn</i> 1.24	
1. Electrostatic Field	1.1–1.70	10. Application of Gauss's Law in	
1. Introduction	1.1	Calculation of Electric Field	1.25
2. Charge	1.1	10.1 Spherical symmetry	1.25
2.1 Charging of material bodies	1.1	• <i>Your Turn</i> 1.27	
2.2 Properties of charge	1.3	10.2 Cylindrical symmetry	1.28
2.3 Electroscope	1.3	• <i>Your Turn</i> 1.29	
• <i>Your Turn</i> 1.5		10.3 Symmetry on a plane	1.30
3. Fundamental Question in Electrostatics	1.5	<i>Your Turn</i> 1.31	
4. Coulomb's Law	1.5	11. Spherical Charge with Cavity	1.32
4.1 Coulomb's law in a medium	1.6	• <i>Your Turn</i> 1.34	
4.2 Coulomb's law in vector notation	1.7	12. Electrostatic Properties of Conductors	1.34
• <i>Your Turn</i> 1.8		• <i>Your Turn</i> 1.39	
5. Principle of Superposition	1.8	13. Motion of a Charged Particle	
• <i>Your Turn</i> 1.10		in an Electric Field	1.40
6. Electric Field	1.10	• <i>Your Turn</i> 1.41	
6.1 Electric field due to a point charge	1.10	14. Conservative Nature of Coulomb Force	1.41
6.2 Electric field obeys superposition		• <i>Your Turn</i> 1.42	
principle	1.11	15. Gauss's Law in Gravitation	1.42
• <i>Your Turn</i> 1.13		• <i>Your Turn</i> 1.43	
6.3 Electric field due to continuous		16. Dielectric Breakdown	1.43
distribution of charge	1.14	• <i>Your Turn</i> 1.43	
• <i>Your Turn</i> 1.16		■ MISCELLANEOUS EXAMPLES	1.43
7. Electric Field Lines	1.17	◇ Worksheet 1	1.55
• <i>Your Turn</i> 1.19		◇ Worksheet 2	1.61

✧ Worksheet 3	1.64		
✧ Answer Sheet	1.68		
2. Electrostatic Potential	2.1–2.59		
1. Introduction	2.1		
2. Electrical Potential Energy	2.1		
2.1 Potential energy of a pair of point charges	2.1		
• <i>Your Turn</i>	2.4		
2.2 Potential energy for a system of more than two charges	2.4		
3. Electric Potential	2.5		
• <i>Your Turn</i>	2.5		
3.1 Electric potential due to a point charge	2.6		
3.2 Potential due to multiple charges	2.6		
• <i>Your Turn</i>	2.8		
3.3 Electric potential on the axis of a charged ring	2.9		
3.4 Electric potential on the axis of a uniformly charged disc	2.9		
• <i>Your Turn</i>	2.11		
4. Conservation of Energy and Work–Energy Theorem	2.11		
• <i>Your Turn</i>	2.14		
5. Relation between Electric Field and Potential	2.15		
5.1 Potential difference in a uniform field	2.15		
5.2 Potential difference in terms of rectangular components of field	2.16		
• <i>Your Turn</i>	2.18		
5.3 Getting electric field from potential function	2.18		
• <i>Your Turn</i>	2.19		
6. Equipotential Surfaces	2.20		
• <i>Your Turn</i>	2.22		
7. Potential due to a Uniformly Charged Shell	2.22		
8. Electric Potential due to a Uniform Spherical Volume Charge	2.22		
• <i>Your Turn</i>	2.24		
9. Potential of a Conductor	2.24		
9.1 Charge sharing	2.24		
9.2 Surface charge density	2.25		
9.3 Grounding a conductor	2.25		
• <i>Your Turn</i>	2.27		
10. Electrostatic Self-Energy	2.28		
10.1 Electrostatic potential energy of a uniformly charged, thin spherical shell	2.28		
10.2 Electrostatic potential energy of a uniformly charged spherical volume	2.28		
• <i>Your Turn</i>	2.30		
11. Accelerating and Retarding Potential Difference	2.30		
12. Electric Dipole	2.31		
12.1 Potential due to a dipole	2.31		
12.2 Electric field due to a dipole	2.31		
• <i>Your Turn</i>	2.31		
12.3 Field and potential on axis and equator	2.32		
• <i>Your Turn</i>	2.33		
13. Electric Dipole in an Electric Field	2.33		
13.1 Potential energy of a dipole in a uniform field	2.34		
13.2 Dipole in a non-uniform field	2.34		
• <i>Your Turn</i>	2.35		
14. Dielectric in Electric Field	2.35		
■ MISCELLANEOUS EXAMPLES	2.36		
✧ Worksheet 1	2.45		
✧ Worksheet 2	2.51		
✧ Worksheet 3	2.55		
✧ Answer Sheet	2.58		
3. Miscellaneous Problems on Chapters 1 & 2	3.1–3.9		
Match the Column	3.1		
Passage based Problems	3.3		
✧ Answer Sheet	3.9		

4. Capacitors	4.1–4.63		
1. Introduction	4.1		
2. Capacitance of a Conductor	4.1		
2.1 Capacitance of a spherical conductor	4.1		
• <i>Your Turn</i>	4.2		
2.2 Energy of a charged conductor	4.2		
3. Charge Sharing between Two Conductors	4.2		
• <i>Your Turn</i>	4.3		
4. Capacitor	4.4		
4.1 Capacitance of a parallel plate capacitor	4.4		
4.2 Parallel plate capacitor with dielectric	4.5		
4.3 Parallel plate capacitor with multiple dielectrics	4.6		
4.4 Capacitance of a spherical capacitor	4.6		
4.5 Capacitance of a cylindrical capacitor	4.7		
• <i>Your Turn</i>	4.8		
5. Energy Stored in a Capacitor	4.9		
6. Energy Density in an Electric Field	4.9		
7. Force between Plates of a Parallel Plate Capacitor	4.10		
• <i>Your Turn</i>	4.11		
8. Cell (or, battery)	4.11		
9. Charging a Capacitor with a Cell	4.12		
• <i>Your Turn</i>	4.15		
10. Grouping of Capacitors	4.15		
10.1 Parallel combination of capacitors	4.16		
10.2 Series combination of capacitors	4.16		
• <i>Your Turn</i>	4.20		
11. Balanced Wheatstone Bridge of Capacitors	4.21		
• <i>Your Turn</i>	4.23		
12. Use of Series–Parallel Grouping to find Capacitance of a Capacitor filled with Multiple Dielectrics	4.23		
12.1 Using series grouping	4.23		
12.2 Using parallel grouping	4.24		
12.3 Mixed grouping	4.24		
• <i>Your Turn</i>	4.25		
13. Short Circuited Capacitor	4.26		
• <i>Your Turn</i>	4.27		
14. Multi Plate Capacitor	4.27		
• <i>Your Turn</i>	4.28		
15. Charge Sharing between Two Capacitors	4.28		
• <i>Your Turn</i>	4.30		
16. Complex Capacitor Circuits	4.30		
• <i>Your Turn</i>	4.35		
17. Symmetry in Circuits	4.35		
• <i>Your Turn</i>	4.39		
■ MISCELLANEOUS EXAMPLES	4.40		
✧ Worksheet 1	4.49		
✧ Worksheet 2	4.56		
✧ Worksheet 3	4.59		
✧ Answer Sheet	4.62		
5. Miscellaneous Problems on Chapter 4	5.1–5.6		
Match the Column	5.1		
Passage based Problems	5.3		
✧ Answer Sheet	5.6		
6. Electric Current and Resistance	6.1–6.55		
1. Introduction	6.1		
2. Electric Current	6.1		
3. Current Density	6.2		
• <i>Your Turn</i>	6.2		
4. Electric Current in Conductors	6.3		
4.1 Drift speed	6.3		
4.2 Relation between current and drift speed	6.3		
5. Ohm's Law	6.4		
6. Resistance	6.5		
6.1 Temperature dependence of resistivity	6.6		
6.2 Different type of material	6.6		
• <i>Your Turn</i>	6.7		
7. Heating Effect of Current	6.8		
7.1 Battery as a source of energy	6.9		
7.2 Incandescent bulb	6.9		
7.3 Electric heater	6.9		
7.4 Fuse	6.10		
• <i>Your Turn</i>	6.11		

8. Kirchhoff's Laws	6.11	9. Compact Wheatstone Bridge (Post Office Box)	7.9
9. Resistors in Series and Parallel	6.12	• <i>Your Turn</i> 7.10	
9.1 Series combination	6.13	10. Potentiometer	7.10
9.2 Parallel combination	6.13	10.1 Comparing emfs of two cells	7.11
• <i>Your Turn</i> 6.17		10.2 Measuring internal resistance of a cell	7.11
10. Wheatstone Bridge of Resistors	6.19	• <i>Your Turn</i> 7.13	
• <i>Your Turn</i> 6.20		■ MISCELLANEOUS EXAMPLES 7.14	
11. Complex Circuits Using Resistances	6.21	✧ Worksheet 1 7.17	
11.1 Another way to use Kirchhoff's laws	6.26	✧ Worksheet 2 7.20	
11.2 Superposition theorem for cells	6.27	✧ Worksheet 3 7.22	
• <i>Your Turn</i> 6.28		✧ Answer Sheet 7.24	
12. Real Cells	6.29		
• <i>Your Turn</i> 6.30		8. Transient RC Circuits	8.1–8.24
12.1 Capacity of a battery (The Ampere Hour)	6.30	1. Introduction	8.1
12.2 Maximum power transfer	6.31	2. Charging of a Capacitor	8.1
• <i>Your Turn</i> 6.32		2.1 Condition immediately after the switch is closed	8.1
12.3 Series grouping of cells	6.32	2.2 Condition after a long time	8.1
12.4 Parallel grouping of cells	6.32	2.3 Dependence of charge and current on time	8.2
• <i>Your Turn</i> 6.34		• <i>Your Turn</i> 8.5	
■ MISCELLANEOUS EXAMPLES 6.35		3. Charging of a Capacitor in a Complex Network of Resistors	8.5
✧ Worksheet 1 6.42		• <i>Your Turn</i> 8.7	
✧ Worksheet 2 6.48		4. Discharging of a Capacitor	8.8
✧ Worksheet 3 6.50		• <i>Your Turn</i> 8.9	
✧ Answer Sheet 6.54		5. Leakage in a Capacitor	8.9
7. Basic Electrical Measurement	7.1–7.24	• <i>Your Turn</i> 8.9	
1. Introduction	7.1	■ MISCELLANEOUS EXAMPLES 8.10	
2. Galvanometer	7.1	✧ Worksheet 1 8.14	
3. Ammeter	7.1	✧ Worksheet 2 8.16	
• <i>Your Turn</i> 7.3		✧ Worksheet 3 8.17	
4. Voltmeter	7.3	✧ Answer Sheet 8.19	
• <i>Your Turn</i> 7.4		9. Miscellaneous Problems on Chapters 6–8	9.1–9.13
5. Resistance Box	7.5	Match the Column	9.1
6. Rheostat	7.5	Passage based Problems	9.3
7. Verifying Ohm's Law and Measuring a Resistance	7.6	✧ Answer Sheet 9.13	
8. Meter Bridge	7.6		
• <i>Your Turn</i> 7.8			

10. Error and Measurement	10.1–10.24	10. Precision and Accuracy	10.17
1. Introduction	10.1	■ MISCELLANEOUS EXAMPLES	10.18
2. Reporting a Single Measurement	10.1	✧ Worksheet 1	10.20
2.1 Uncertainty in a single measurement	10.1	✧ Worksheet 2	10.22
2.2 Rules to identify significant figures	10.2	✧ Worksheet 3	10.23
• <i>Your Turn</i>	10.4	✧ Answer Sheet	10.24
3. Errors in Measurement	10.4	11. Miscellaneous Problems on	
4. Estimating Uncertainty in Multiple Measurements	10.5	Chapter 10	11.1–11.3
• <i>Your Turn</i>	10.6	Match the Column	11.1
4.1 Standard deviation	10.6	Passage based Problems	11.1
4.2 Fractional and percentage uncertainty	10.7	✧ Answer Sheet	11.3
• <i>Your Turn</i>	10.7	12. Previous Years' JEE Questions	12.1–12.22
5. Significant Figures in Algebraic Operations	10.8	Electrostatics	12.1
5.1 Multiplication and division	10.8	AIEEE/JEE Main Questions	12.1
5.2 Addition and subtraction	10.8	IIT–JEE/JEE Advanced Questions	12.3
5.3 Rules for rounding off	10.8	Capacitor	12.9
6. Propagation of Uncertainty (Error)	10.9	AIEEE/JEE Main Questions	12.9
• <i>Your Turn</i>	10.9	IIT–JEE/JEE Advanced Questions	12.10
6.1 Error in sum or difference of quantities	10.9	Current, Electrical Measurements and RC Circuits	12.11
6.2 Error in product of quantities	10.10	AIEEE/JEE Main	12.11
• <i>Your Turn</i>	10.12	IIT–JEE/JEE Advanced Questions	12.14
7. Sensitivity of a Measuring Instrument	10.13	Error and Measurement	12.18
• <i>Your Turn</i>	10.13	AIEEE / JEE Main Questions	12.18
8. Vernier Caliper	10.14	IIT–JEE / JEE Advanced Questions	12.19
8.1 Measurement with a Vernier caliper	10.14	✧ Answer Sheet	12.22
8.2 Zero error and correction	10.15		
• <i>Your Turn</i>	10.15		
9. Screw Gauge or Micrometer	10.16	Solutions	S.1–S.160
• <i>Your Turn</i>	10.17		

Electrostatic Field

“On graduating from school, a studious young man who would withstand the tedium and monotony of his duties has no choice but to lose himself in some branch of science or literature completely irrelevant to his assignment”.

—Charles Augustin Coulomb

1. INTRODUCTION

Mass is a property which causes objects to apply force on one another. Force of gravitation is caused by mass. There is yet another property of matter which makes interaction amongst objects possible. This property is known as *charge*.

Electric forces resulting due to property of charge are far stronger than gravitational force. Apart from electric phenomena, all magnetic phenomena we are aware of results from the property of charge. In our entire course of electricity and magnetism we will be dealing with this property of matter called *charge*.

Force which keeps a building intact in a strong wind is electromagnetic. Friction which helps you in walking also arises due to complex interactions of charged particles. In fact, all forces which you and I experience in daily life are fundamentally electromagnetic or gravitational.

In this chapter and the next one, we will discuss the properties of *charges at rest* and develop the concept of electric field and potential.

2. CHARGE

It was Benjamin Franklin, in 18th century, who introduced the idea that all substances contain what he called “electric fluid”, or “electric fire”. He introduced the convention of positive and negative signs. He argued that if a body gets too much of fire then it gets positively charged and if a body has deficiency of it then it is negatively charged. What Franklin had not figured out is what the fire really consists of.

Today, we know that every piece of matter is composed of atoms and every atom is made of fundamental particles – electron, proton and neutron. Neutrons are neutral. What it means is that they do not apply electric force on one another.

Two electrons placed 1 cm apart *repel* each other with a force equal to 2.3×10^{-24} N. This is same as force of repulsion between two protons placed 1 cm apart. [Note that

forces are repulsive and not attractive and therefore they are not the gravitational force]. The electrons and protons have a god gifted property known as charge which helps them exert force on one another.

An electron and a proton placed 1 cm apart *attract* each other with a force of 2.3×10^{-24} N. We conclude that charges can attract as well as repel (recall that force of gravitation is attractive only). There are two types of charges. *Like charges repel and unlike charges attract*. To differentiate between the two types of charges we call them as *positive* and *negative* charges. [We could have called them as blue and red charges as well; or for that matter dark and bright charges. In fact any name assigned to the two types of charges would not have changed the Physics.

Unit of charge is Coulomb (C). At this stage it is difficult to define how much charge exactly is 1 Coulomb. Assuming that you have studied a little about electric current in junior classes, all I can say is *1 C is the quantity of charge that flows through a wire in 1 second if it carries a current of 1 ampere*.

Charge on one electron is equal to -1.602×10^{-19} C and charge on a proton is positive of this value. We often denote this value by symbol *e*.

2.1 Charging of Material Bodies

All matter is a mixture of positive protons, negative electrons and neutral neutrons. There are numerous attractive and repulsive force pairs inside a piece of matter. But we do not notice any such force. The balance is perfect! When you stand near someone you don’t feel any force though both the bodies have literally infinite amount of charged particles. Can you imagine what will happen if the God just removed 1 % of electrons in your bodies? You will repel each other with a force that would be enough to throw both of you out of the gravitational field of the Earth – somewhere deep into space travelling at terrible speed!

A piece of matter is neutral as it has equal number of evenly distributed protons and electrons when viewed at macroscopic scale. If some of the electrons leave a piece of object and move on to some other object both of them get charged. *The one that loses electrons has excess of protons (or, is electron deficient) and is said to be positively charged. The other object which gains electrons acquires negative charge.* A positively or negatively charged body has ability to exert force on any other charged body.

How can we remove electrons from a body? Once you provide sufficient energy to an electron it may leave the body. The smallest energy that an electron will need to leave a material is known as *work function* of the material. Metals have smaller work functions. It is easier to remove electrons from metals. This energy can be provided in many different ways. Just by heating a piece of metal or, making light incident on it we can cause electrons to leave the metal piece. But there are even simpler ways. Let's discuss some simple practical ways of charging a body.

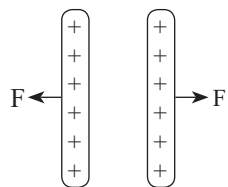
Rubbing

When we rub two objects made of *different substance*, enough energy may be imparted to electrons in one object causing them to leave the object. These electrons will move to the other object. The electron loser becomes positively charged and the one which gains electrons becomes negatively charged. This effect in which certain materials become electrically charged after they come into frictional contact with a different material is known as *triboelectric effect*.

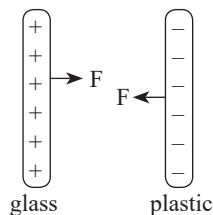
When a glass rod is rubbed with a silk cloth, glass happens to lose electrons easily, and silk grabs them away from the glass atoms. So, after rubbing, the glass becomes positively charged and the silk becomes negatively charged.

When you rub a plastic body with fur, it easily grabs electrons from the fur. It becomes negatively charged while the fur becomes positively charged.

If two glass rods are rubbed to silk cloth and they (the glass rods) are brought close, they repel. When you bring the glass rod (after being rubbed with silk) close to the plastic rod (which has been rubbed to fur), they attract.



Two glass rods repel each other after they have been rubbed with a silk cloth.



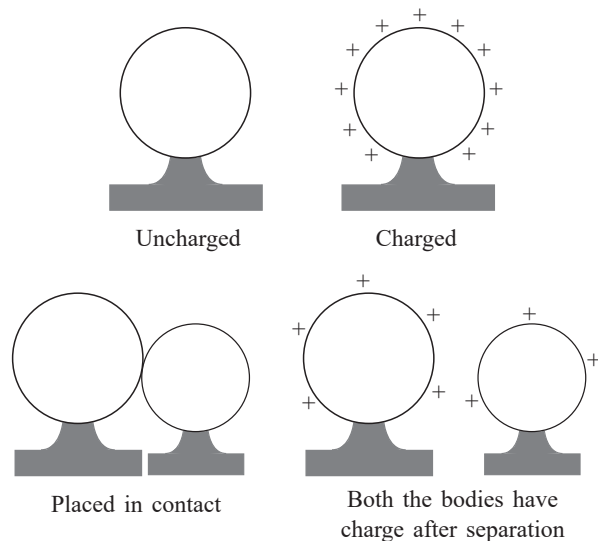
A glass rod rubbed with a silk cloth and a plastic rod rubbed with fur, attract one another.

Contact Charging

Substances which allow easy movement of charge are known as **conductors**. Metals have free electrons in them which can move from one point to another inside the material. For example, when copper atoms come close together to form a solid, the electrons of neighbouring atoms exert force on one another. This causes one or two electrons in an atom to become free. Becoming free means that the electron is not bound to a particular atom. It is free to move inside the material without being tied to any particular nucleus. The free electrons act as *charge carriers* in such material and the material act as conductor. In fact, metals are also good conductors of heat. Also, it is much easier to charge a conductor by rubbing.

Materials which do not have free charge carriers do not conduct. Such materials are *insulators* or *non conductors*.

When an uncharged (neutral) conductor is put into contact with another charged conductor, the uncharged conductor gets charged. Charge on the previously charged conductor reduces. The charge spreads on both the conductors till both of them have same *potential* (We will study about potential later). Remember, charge is free to move inside a conductor. This method of charging a neutral conductor by putting it into contact with a charged conductor is known as *charging by conduction*.



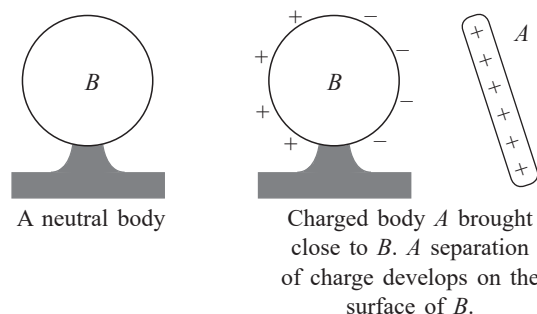
When a neutral non conducting object is placed in contact to a charged body, will it acquire charge? Most likely, yes. This does not happen because of free movement of charge from one object to another. It happens mostly because of jumping of charge from one body to another when there is a loose contact between them. It is similar to *sparking* that we observe when two loose electric wires rub against one another. Do not worry much about this phenomenon at this moment. We will learn about it later.

Induction

Suppose a charged body A (say, having positive charge) is brought near a neutral body B . The face of body B that is close to A becomes negatively charged and the away face acquires positive charge. The mechanism is very simple in case of conductors. The free electrons get attracted towards A and pile up on the face of B that is near to A . The other face becomes positively charged due to deficiency of electrons. This kind of separation of charge in a neutral body in presence of other charged body is called *induction* and the charges appearing on faces of B are known as *induced charges*. Notice that net charge on B is still zero. It is just a rearrangement of charge particles in B which results in separation of charges. If A is moved far away from B , the body B returns back to its original state with no charge separation.

If the positive face of B is touched by a third neutral conductor (C), some positive charge will move to C and now B will have a net negative charge. Now B will be a negatively charged body if A and C are moved far away.

The induced charges appear on the faces of a non conductor also when it is placed near a charged body. We will discuss the exact mechanism of this induction later on.

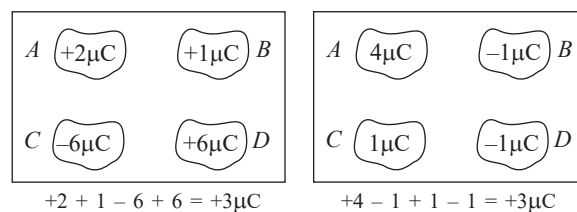


can take only discrete values is said to be quantized. Charge is quantized.

In practice, even a small charge contains a very large number of electronic charges. For example, $1\mu\text{C}$ charge has 6×10^{12} units of basic charge e . In such situations there is nothing wrong in regarding charge as continuous, much in the same way as we regard matter to be continuous even though it is made of discrete grains called atoms.

(iii) Conservation of charge

Total charge of an isolated system remains conserved. Isolated system is one from which no charge can leak and no charge from outside world can enter its boundary. The figure shows four interacting bodies (A , B , C and D) forming an isolated system. Charge on the bodies change to values shown in second figure due to mutual exchange. Note that *algebraic sum of charge remains conserved*. The amount of positive and negative charges are not conserved separately; it is their sum which remains conserved.



Interacting charged bodies in an isolated system

(iv) Charge is relativistically invariant

In his theory of relativity, Einstein said that mass of a body increases with its speed. But so is not the case with charge. Charge on a body does not change with its speed.

2.2 Properties of Charge

- (i) It is matter that carries charge. Electric charge cannot exist without mass. There is no particle is nature which has charge but no mass. Mass of an electron is $9.11 \times 10^{-31}\text{kg}$ and that of a proton is $1.6727 \times 10^{-27}\text{kg}$.

(ii) Charge is quantized

All experiments so far have revealed that electric charge occurs in integral multiple of electronic charge (e). The smallest charge that we can get in nature is $\pm e = \pm 1.6 \times 10^{-19}$. A body can acquire a charge q given by

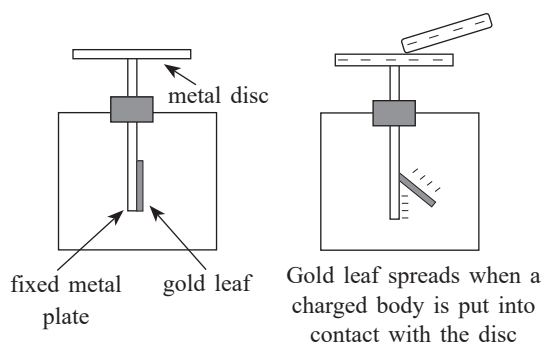
$$q = \pm ne, \text{ where } n = 0, 1, 2, 3, \dots \quad (1)$$

Thus charge can have only discrete values given by the above equation. We cannot have a body having charge equal to $2.5e$ or, $0.5e$ or, $-9.9e$. Charge is available in *packet size* of e only. A quantity that

2.3 Electroscope

This is an instrument used to detect charge. A narrow metal plate is connected to a metal disc. A thin piece of gold leaf is fixed to the plate. The gold leaf and the metal plate are inside a dust proof glass box. The metal disc is outside the box. When a charged body is touched with the disc, the charge conducts to the metal plate and the leaf. Both the plate and the leaf have like charges on them. They repel and the gold leaf spreads out. If the leaf spreads a lot it implies that it acquired a lot of charge. This means that the charged body was having a lot of charge (in fact, lot of potential as we will learn later). To discharge the electroscope, its disc is connected to ground. The entire charge on the disc, metal plate and the gold leaf flows to the Earth. The electroscope is ready to be used again.

1.4 Electrostatics and Current Electricity



Example 1 A body has 1 mC of negative charge on it. How many excess electrons does it have?

Solution

Concepts

Charge is quantized. Adding n electron to a body results in the body acquiring a charge $q = -ne$

$$n = \frac{q}{e} = \frac{1 \times 10^{-3} \text{ C}}{1.6 \times 10^{-19} \text{ C}} = 6.25 \times 10^{15}$$

Example 2 Even a small piece of matter has huge amount of charge

A copper wire has mass 6.35 g. Knowing that molar mass of copper is 63.5 g mol^{-1} and its atomic number is 29, find

- Total charge on all the protons in the wire.
- The charge acquired by the wire if it loses 0.0001% of its total number of electrons.

Solution

Concepts

- One mole Cu atoms have a mass of 63.5 g.
- Each Cu atom has 29 electrons.
- $q = ne$

$$\begin{aligned} \text{Number of moles of Cu in the wire} &= \frac{6.35 \text{ g}}{63.5 \text{ g mol}^{-1}} \\ &= 0.1 \text{ mol.} \end{aligned}$$

$$\begin{aligned} \text{Number of Cu atoms in the wire} &= 0.1 \times 6.02 \times 10^{23} \\ &= 6.02 \times 10^{22} \end{aligned}$$

$$\begin{aligned} \text{Number of protons in the wire} &= 6.02 \times 10^{22} \times 29 \\ &= 1.75 \times 10^{24} \end{aligned}$$

$$\begin{aligned} \text{(a) Charge on all protons, } Q &= +ne \\ &= 1.75 \times 10^{24} \times 1.6 \times 10^{-19} \text{ C} = 2.8 \times 10^5 \text{ C} \end{aligned}$$

This is a huge amount of charge.

- (b) Number of electrons = Number of protons in neutral wire.

Charge on $10^{-4} \%$ of all electrons will be

$$q = -(10^{-4} \% \text{ of } Q) = -\frac{10^{-4}}{100} \times 2.8 \times 10^5 = -0.28 \text{ C}$$

When 0.28 C of negative charge is removed, the wire acquires 0.28 C of positive charge.

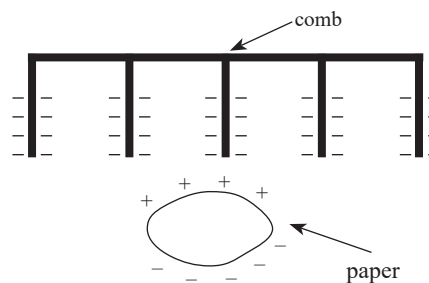
Example 3 A charged body can attract a neutral object
When you comb your hair on a dry day and bring the comb near bits of paper, the comb attracts the small paper pieces. Explain.

Solution

Concepts

Phenomenon of induction

When you comb your hair, it gains electrons from hairs and becomes negatively charged. [In fact, you need not worry much about the sign of the charge]. When the comb is brought close to the pieces of paper, the phenomenon of induction takes place. The edge of the paper closer to the comb becomes positively charged and the further side of the paper piece acquires negative charge. The comb attracts the positive side of the paper and repels the negative side. Overall force on the paper is towards the comb because the positive edge is closer to the comb and experiences a higher force. The force is quite strong. Pieces of paper jump against gravity and cling to the comb.



Charge on edges of the paper piece is due to induction.

Note: If you comb your hair for a long time all your hairs will stand up and can spark. This happens prominently during dry winter days. [Actually moist air takes away charge from charged bodies]. Due to combing your hairs get charged and repel each other to remain separated from one another.

Example 4 Inside a nucleus, a neutron can convert itself into a proton and an electron. A third particle known as antineutrino is also formed. Can you guess the charge present on antineutrino?

Solution

Concepts

Conservation of charge

$$n \rightarrow p + e \text{ (antineutrino)}$$

Charge on a neutron is zero. Sum of charges on the three products must also be zero. Sum of charges on proton and electron is zero. Therefore, third particle must have no charge.

In Short

- (i) Charge is a property of matter which is responsible for all electric and magnetic phenomena.
- (ii) Charge is of two types – positive and negative.
- (iii) Smallest charge in nature is electronic charge $\pm e = 1.6 \times 10^{-19} \text{C}$.
- (iv) Charge of an isolated system is conserved.
- (v) In real life a body acquires charge due to gaining electrons or losing them.
- (vi) A body which loses electrons gets positively charged and one which gains them becomes negatively charged.
- (vii) Charge on a body which loses n electrons is $q = +ne$ and the one which gains n electrons has a charge $q = -ne$.
- (viii) If two bodies, made of different substances, are rubbed, electrons may get transferred from one of them to another. Therefore, the two bodies acquire equal and opposite charge.
- (ix) A conductor is a substance which has plenty of free electrons in it. These electrons help in movement of charge inside a conductor. An insulator does not have free electrons.
- (x) When a neutral body is brought close to a charged body (without touching it), there is separation of charge in the neutral body. Its face closer to the charged body acquires an opposite charge and the far face gets a charge similar to the charged body. This phenomenon is called induction. It happens in conductors as well as insulators.

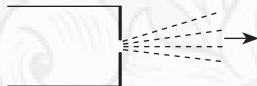


YOUR TURN

Q.1 A neutral body becomes positively charged due to rubbing. Has its mass changed?

Q.2 How many electrons will make 1 coulomb of negative charge?

Q.3 A spray-painting machine has a nozzle that produces droplets of paint in a cone shaped beam. By some mechanism each droplet is given a positive charge as it leaves the nozzle. What effect will it have on the beam?



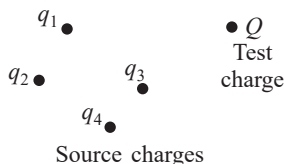
Q.4 A petrol tanker has a metallic chain suspended from its body touching the road. Why?

Q.5 A charged comb attracts bits of paper (refer to example 3). Paper pieces often jump away from the comb after contact. Why?

Q.6 A piece of solid has 10^{22} atoms. If one electron is removed from each of 0.01 % atoms of the solid, find the charge acquired by the solid.

3. FUNDAMENTAL QUESTION IN ELECTROSTATICS

Suppose $q_1, q_2, q_3, \dots$ are a set of fixed charges. We will call these fixed charges as **source charges**. Consider another charge Q and call it **test charge**. The fundamental problem is to find the force experienced by the test charge (Q) due to the source charges. This is our basic aim in this chapter.



Note that our source charges are stationary though the test charge may be moving. [If source charges are moving then they may apply another force – known as magnetic force – on Q . We will study about it later].

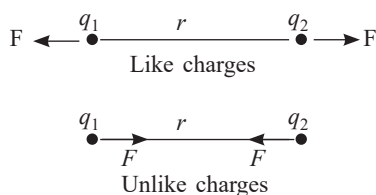
The Coulomb's law and the principle of superposition will help us answer our fundamental question in electrostatics. Whatever else we study in this chapter is mathematics to simplify the application of these fundamental laws.

4. COULOMB'S LAW

Charles Augustin de Coulomb, a French physicist, experimentally discovered the law that governs force between two charges. He gave his law in 1785. Coulomb's law tells us how to find force between two *point charges*. [It cannot be used directly to find force between two extended charges like a charged stick and a charged disk].

The law states that *the electrostatic force between two point charges is directly proportional to the product of the magnitude of the charges and inversely proportional to square of the distance between them. Like charges repel and unlike charges attract. The force always acts along the line joining the two charges.*

Consider two point charges q_1 and q_2 kept at a separation r in vacuum. They repel if both charges have same sign and attract if they are of opposite sign. The magnitude of force that they apply on one another is given by Coulomb's law as:



$$F \propto \frac{|q_1 q_2|}{r^2}$$

$$\Rightarrow F = k \frac{|q_1 q_2|}{r^2} \quad (2)$$

where k is a constant. Its experimentally obtained value is:

$$k = 9 \times 10^9 \frac{\text{Nm}^2}{\text{C}^2}$$

Constant k is often written in terms of another constant ϵ_0 (pronounced as “epsilon nought”) as

$$k = \frac{1}{4\pi\epsilon_0} \quad (3)$$

$$\text{where } \epsilon_0 = 8.85 \times 10^{-12} \frac{\text{C}^2}{\text{Nm}^2} \quad (4)$$

$$[\text{Note that } \frac{1}{4\pi \times 8.85 \times 10^{-12}} = 9 \times 10^9]$$

Constant k is written in terms of ϵ_0 for historical reasons and it is very insightful. This simplifies many results in Physics. However, you should not worry much about this right now.

The constant ϵ_0 is known as *electrical permittivity of free space*. Coulomb’s law in terms of ϵ_0 is

$$F = \frac{1}{4\pi\epsilon_0} \frac{|q_1 q_2|}{r^2} \quad (5)$$

4.1 Coulomb’s Law in a Medium

Consider two point charges in a medium. For example, consider two small charged balls inside a large tank filled with some insulating oil. Apparent force between two point charges in a medium is less than the force between them when they are in vacuum (given by equation 5).

Force between point charges q_1 and q_2 kept inside a medium (which extends to a large distance) at a separation r is given by

$$F = \frac{1}{4\pi\epsilon_0\epsilon_r} \frac{|q_1 q_2|}{r^2} \quad (6)$$

ϵ_r is a *dimensionless* number greater than or equal to 1. ϵ_r is known as *relative permittivity* of the medium. Sometimes it is also known as *dielectric constant* and denoted by (K) .

For air $\epsilon_r = 1.00054$. We will consider ϵ_r to be 1 for air. Force between two charges kept in air is almost same as force between them when they are in vacuum.

For all other media $\epsilon_r > 1$.

$\epsilon = \epsilon_0 \epsilon_r$ is known as electrical permittivity of the medium. The reason for decrease in force will be understood clearly in next chapter when we discuss dielectrics.

Example 5 Two small particles having charge $+1\text{C}$ and -1C are kept at a separation of 1 m . Find electrostatic force between them. What conclusion can you draw from the result?

Solution

Concepts

Force is attractive. When direction of force is known, we will use Coulomb’s law to find magnitude of force only. We should write both q_1 and q_2 with positive sign only in equation (2).

$$F = k \frac{|q_1 q_2|}{r^2} = 9 \times 10^9 \frac{\text{Nm}^2}{\text{C}^2} \cdot \frac{(1\text{C})(1\text{C})}{(1\text{m})^2} = 9 \times 10^9 \text{ N}$$

Force is attractive.

Force is enormous. It is nearly equal to weight of two hundred thousand elephants! We can draw two important conclusions from this simple example:

- Electrostatic force is quite strong and
- 1 C is a huge amount of charge. In real life we do not get objects having few coulomb charge on them. Practical units of charge are mC , μC , nC etc.

Example 6 Force between two electrons

Find electrostatic force between two electrons kept at a separation of 1 cm . Also find gravitational force between them. What do you conclude?

Solution

Concepts

Electrostatic force arises due to charge possessed by the electrons. It is calculated using Coulomb’s law. Gravitational force arises due to mass possessed by the electrons. It is calculated using Newton’s law of gravitation. To compare the two forces (or any two quantities), one way is to get their ratio.

Electrostatic force is

$$F_e = k \frac{e \cdot e}{r^2} = 9 \times 10^9 \times \frac{(1.6 \times 10^{-19})^2}{(0.01)^2}$$

$$= 2.3 \times 10^{-24} \text{ N} \quad [\text{Repulsion}]$$

Gravitational force is

$$F_g = G \frac{m \cdot m}{r^2} = 6.7 \times 10^{-11} \times \frac{(9.1 \times 10^{-31})^2}{(0.01)^2}$$

$$= 5.5 \times 10^{-67} \text{ N} \quad [\text{Attraction}]$$

Ratio of electric and gravitational force is

$$\frac{F_e}{F_g} = \frac{2.3 \times 10^{-24}}{5.5 \times 10^{-67}} = 4.1 \times 10^{42}$$

We conclude that:

- Electrostatic force is far more stronger than gravitational force.
- While studying interactions between light weight charged particles their gravitational interaction can be neglected.

Example 7 Force between two point charges in vacuum is F_0 when they are placed at separation r . Find apparent force between them when they are kept well immersed in a medium of dielectric constant (or relative permittivity) 3, at separation $2r$.

Solution

Concepts

Force between two charges in a medium is given by (6)

$$F_0 = \frac{1}{4\pi\epsilon_0} \frac{q_1 q_2}{r^2} \quad (i)$$

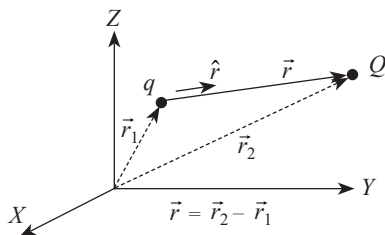
Force in medium, with separation $2r$ is

$$F = \frac{1}{4\pi\epsilon_0\epsilon_r} \frac{q_1 q_2}{(2r)^2} = \frac{1}{3 \times 4} \cdot \frac{1}{4\pi\epsilon_0} \frac{q_1 q_2}{r^2} \quad [\because \epsilon_r = 3]$$

$$= \frac{F_0}{12}$$

4.2 Coulomb's law in vector notation

Consider two point charges q_1 and q_2 kept at positions $\vec{r}_1$ and $\vec{r}_2$ in the co-ordinate system shown. Position vector of charge Q with respect to q is $\vec{r}$. Unit vector from q to Q is $\hat{r}$. Obviously, $\hat{r} = \frac{\vec{r}}{|\vec{r}|}$



Assume that we wish to write force on Q due to q . You can say that q is the source charge and Q is a test charge. Force on Q can be written as

$$\vec{F} = k \frac{qQ}{|\vec{r}|^2} \hat{r} \quad (7)$$

While calculating $\vec{F}$, we must substitute values of q and Q with their respective signs. The above equation can be written as

$$\vec{F} = k \frac{qQ}{|\vec{r}|^2} \frac{\vec{r}}{|\vec{r}|} = k \frac{qQ}{|\vec{r}|^3} \vec{r} \quad (8)$$

$$\text{Since } \vec{r} = \vec{r}_2 - \vec{r}_1$$

$$\therefore \vec{F} = k \frac{qQ}{|\vec{r}_2 - \vec{r}_1|^3} (\vec{r}_2 - \vec{r}_1) \quad (9)$$

Example 8 A point charge $q = -2\mu\text{C}$ is kept at the origin of a co-ordinate system. Another charge $Q = 1\mu\text{C}$ is kept at a point having co-ordinates $(0.6, 0.8)\text{m}$. Find force on q due to Q .

Solution

Concepts

$\vec{F} = k \frac{qQ}{|\vec{r}|^2} \hat{r}$. In this equation $|\vec{r}| = r$ is nothing but distance between the two charges.

You must be careful in writing $\hat{r}$. It is a unit vector from source (Q) towards the test charge (q).

Put q and Q with sign in the above equation.

Distance between charges

$$r = \sqrt{0.6^2 + 0.8^2} = 1.0 \text{ m}$$

Vector joining Q to q is

$$\vec{r} = (0 - 0.6)\hat{i} + (0 - 0.8)\hat{j}$$

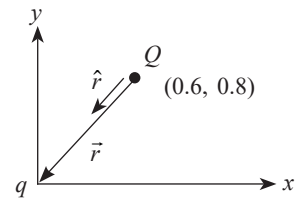
$$= -0.6\hat{i} - 0.8\hat{j}$$

$$\therefore \hat{r} = \frac{\vec{r}}{r} = -0.6\hat{i} - 0.8\hat{j}$$

$$\therefore \text{Required force is } \vec{F} = k \frac{qQ}{r^2} \hat{r}$$

$$= \frac{9 \times 10^9 \times (-2 \times 10^{-6}) (1 \times 10^{-6})}{(1.0)^2} (-0.6\hat{i} - 0.8\hat{j})$$

$$= 1.8 \times 10^{-3} (6\hat{i} + 8\hat{j}) \text{ N}$$



In Short

- (i) Coulomb's law gives force between two point charges. We can use it directly to get force between two small charged bodies kept at large separation.
- (ii) Magnitude of force between two charges is

$$F = \frac{1}{4\pi\epsilon_0\epsilon_r} \frac{|q_1q_2|}{r^2}$$

where $\epsilon_r = 1$ for vacuum and air. For all other medium $\epsilon_r > 1$

(iii) $k = \frac{1}{4\pi\epsilon_0} = 9 \times 10^9 \frac{\text{N-m}^2}{\text{C}^2}$ and $\epsilon_0 = 8.85 \times 10^{-12} \frac{\text{C}^2}{\text{N-m}^2}$

- (iv) Force due to source charge (q) on a test charge (Q) can be written in vector form as

$$\vec{F} = \left(k \frac{qQ}{r^2} \right) \hat{r}$$

where r is distance between the two charges and $\hat{r}$ is a unit vector directed from the source charge to the test charge. While using this equation we must put q and Q with their respective signs.

- (v) Electrostatic force is far stronger than the gravitational force.
- (vi) 1 coulomb is a large charge.



YOUR TURN

Q.7 Write dimensional formula for permittivity of free space (ϵ_0).

Q.8 Force between two charged particles is F_0 when they are kept in vacuum at a separation r . When the charges are immersed in a gaseous medium force between them for separation $\frac{r}{2}$ is still F_0 . Find the relative permittivity of the gaseous medium.

Q.9 A charge Q is divided into two point charges having magnitudes x and $(Q - x)$. Find x which results in maximum force between the charges for a given separation.

Q.10 In a hydrogen atom a proton is fixed and an electron revolves around it in an orbit of radius r . The electrostatic attraction provides the centripetal force. Find speed of the electron. Mass of an electron is m and charge on it is $-e$.

Q.11 A point charge $q_1 = 10 \mu\text{C}$ is kept fixed at $(1, 1) \text{ m}$. Find the force that the charge exerts on another point charge $q_2 = 20 \mu\text{C}$ placed at $(4, 5) \text{ m}$. Write the x and y components of the force.

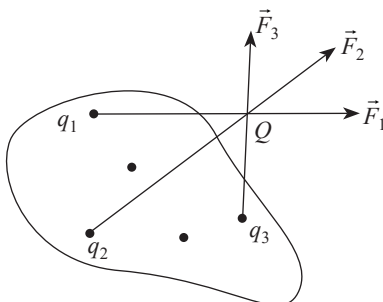
Q.12 Find the smallest possible force between two charged particles kept at a separation of 1 m.

5. PRINCIPLE OF SUPERPOSITION

Force between two charges does not alter if a third charge is brought near them. Consider a set of fixed source charges $q_1, q_2, q_3, \dots$. Force applied by q_1 on a test charge Q is $\vec{F}_1$. This force between q_1 and Q remains same whether $q_2, q_3, \dots$ are present or absent. Similarly, forces by $q_2, q_3, \dots$ on Q are $\vec{F}_2, \vec{F}_3, \dots$ respectively. Net force on Q in presence of all the source charges is given by

$$\vec{F} = \vec{F}_1 + \vec{F}_2 + \vec{F}_3 + \dots \quad (7)$$

This may appear trivial. But scientists have taken the pain of verifying this principle. Imagine how difficult physics



would have been if force between q_1 and Q got changed due to presence of q_2 ! Thanks god, it is not so.

Example 9 Three point charges (q, q and Q) are kept at the vertices of a right angled isosceles triangle. Equal sides of the triangle have length x each. Find the electrostatic force on charge Q .

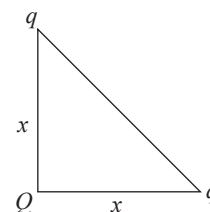
Solution

Concepts

Resultant force on Q is *vector sum* of forces due to the other two charges.

Force due to q on Q has magnitude

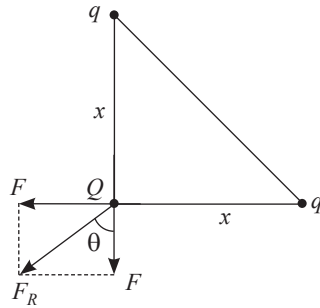
$$F = k \frac{qQ}{x^2}$$



Directions of the two forces are as shown. They are perpendicular to one another. Resultant of these two forces has magnitude:

$$F_R = \sqrt{F^2 + F^2} \\ = \sqrt{2}F = k \frac{\sqrt{2}qQ}{x^2}$$

Directions of $\vec{F}_R$ is as shown with $\theta = 45^\circ$



Example 10 Search the equilibrium position

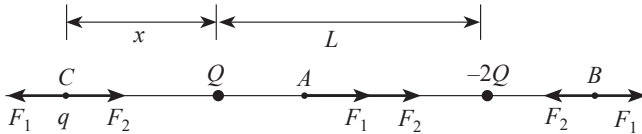
Two point charges Q and $-2Q$ are held fixed at a separation L . Where should we place a third charge q on the line joining the two charges, so that it stays in equilibrium?

Solution

Concepts

For equilibrium of q , vector sum of forces on it due to fixed charges must be zero. The two forces acting on q should have opposite directions and same magnitude.

Magnitude of forces due to Q and $-2Q$ can be equal only if q is held closer to Q .



Let us denote force on q due to Q by F_1 and the force on q due to $-2Q$ by F_2 .

If q is placed at a point A between the two charges, both F_1 and F_2 will be in the same direction and equilibrium will not be possible.

If q is placed at a point B to the right of $-2Q$, then F_1 and F_2 have opposite directions. But in this position magnitude of F_2 is larger than that of F_1 . In this position q is closer to the larger charge. Equilibrium is not possible.

Equilibrium of q is possible only when it is placed to the left of charge Q . Consider q at a distance x from Q at point C.

$$F_1 = k \frac{qQ}{x^2} \quad \text{and} \quad F_2 = k \frac{q(2Q)}{(L+x)^2}$$

For equilibrium, $F_1 = F_2$

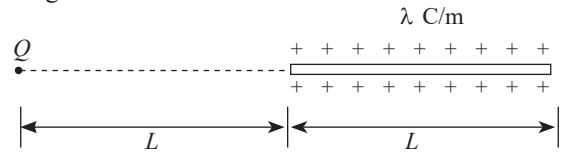
$$\Rightarrow \frac{1}{x^2} = \frac{2}{(L+x)^2} \quad \Rightarrow \quad \frac{L+x}{x} = \sqrt{2}$$

[Ratio of two distances cannot be negative. Therefore, we do not consider $\frac{L+x}{x} = -\sqrt{2}$].

$$\therefore x = \frac{1}{(\sqrt{2}-1)}$$

Example 11 A uniformly charged rod

A thin plastic rod of length L carries charge all along its length. The linear charge density is uniform and is equal to $\lambda \frac{C}{m}$. A point charge Q is kept at a distance L from one end of the rod as shown in figure. Find force on Q due to the charge on the rod.

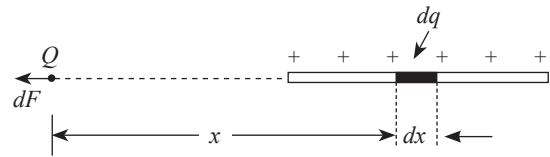


Solution

Concepts

- The rod is made of *plastic*. This ensures that charge on it will not move and linear charge density will remain uniform.
- Don't commit a mistake of assuming the entire charge of the rod at its center. There is no such rule.
- We will have to take small segments of the rod and write force on Q due to each of them. Sum of all these forces is the resultant force. Summation has to be done using integration.

Consider a segment of length dx on the rod at a distance x from the charge Q .



Charge on the segment is $dq = \lambda dx$.

This infinitesimally small charge is like a point charge.

Force on Q due to dq is $dF = k \frac{Qdq}{x^2} = kQ\lambda \frac{dx}{x^2}$.

Forces due to all such elements is in same direction and addition of these forces can be done as

$$F = \int dF = kQ\lambda \int_L^{2L} \frac{dx}{x^2} \\ = kQ\lambda \left[-\frac{1}{x} \right]_L^{2L} \\ = kQ\lambda \left[\frac{1}{L} - \frac{1}{2L} \right] = \frac{kQ\lambda}{2L}$$



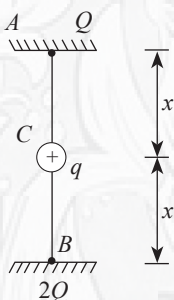
YOUR TURN

Q.13 A regular hexagon has side length x . Five identical point charges are placed on its vertices. A sixth charge is placed at the centre of the hexagon. What is direction of electric force experienced by the sixth charge? Assume all charges to be negative.

What is magnitude of force on the sixth charge if each charge is $-q$?

Q.14 Three charges, q each, are placed at the vertices of an equilateral triangle of side length x . Find the electrostatic force on any one charge.

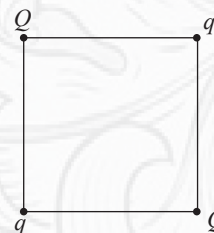
Q.15 A small charged particle (C) having mass m is tied to two strings as shown. A point charge Q is fixed at top end A of the upper string and another point charge $2Q$ is fixed at the lower end B of the other string. All three charges lie on a vertical line and there is no tension in the strings. Find charge q on the particle. Each string has length x .



Q.16 Positive point charges, Q each, are placed at the diagonally opposite corners of a square. Two other identical charges (q each) are placed at the remaining two vertices. In this arrangement of charges, force on Q is found to be zero.

(a) What is sign of q ?

(b) Find $\left| \frac{q}{Q} \right|$.

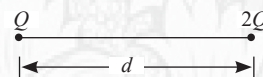


Q.17 Two point charges Q and $2Q$ are held at a separation d . A third charge q is placed on the line joining the two charges such that it remains in equilibrium.

(a) What is sign of q if Q is positive?

(b) What is distance of q from Q ?

(c) Is equilibrium of q stable for small displacements along the line joining the two charges?



6. ELECTRIC FIELD

In the chapter of gravitation we got familiar with the concept of field. The Earth sets up something known as gravitational field in the space surrounding it and this field exerts force on any mass that is there.

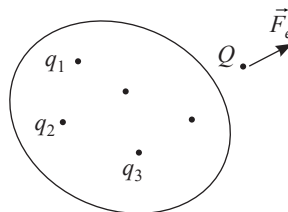
Similarly, it is often more fruitful to describe interaction of charged particles in terms of electric field. Force applied by source charges on a test charge is seen as a two step process. The source charges create something called electric field in the surrounding space and this electric field exerts force on any other charge.

Consider a set of fixed source charges ($q_1, q_2, q_3, \dots$ etc). They exert force on a test charge Q . Now, we will say that the source charges have created an electric field (something which is invisible, has no smell, has no taste) in surrounding space and this field exerts force on Q . Existence of an electric force on Q tells you that there is an electric field. How to tell the strength of electric field at a point? *Electric field intensity* ($\vec{E}$) is a vector quantity that is used to tell strength and direction of field at a point.

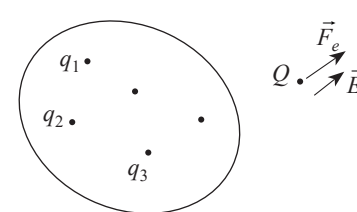
If a small test charge Q feels an electric force $\vec{F}_e$ when placed at a point, the *electric field intensity* at the point is defined as

$$\vec{E} = \frac{\vec{F}_e}{Q} \quad (8)$$

Unit of electric field intensity is N/C. Direction of electric field intensity at a point is in the direction of force experienced by a positive test charge placed at that point.



Till now we thought that source charges exert direct force on test charge Q . Resultant of these forces is $\vec{F}_e$.



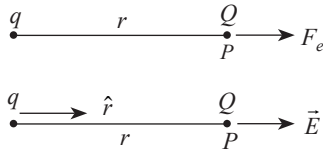
Now we will think differently we will say that all the source charges have together produced an electric field and the field exerts force on $Q \cdot \vec{F}_e = Q\vec{E}$.

While measuring electric field intensity at a point using equation (8), one must ensure that test charge is a *small point charge*. This will ensure that by placing Q at the point we do not disturb the original source charges.

Electric field intensity is often abbreviated as *electric field*.

6.1 Electric field due to a point charge

Consider only a point charge q as the source charge. We wish to write electric field intensity (E) due to this charge at a distance r from it.

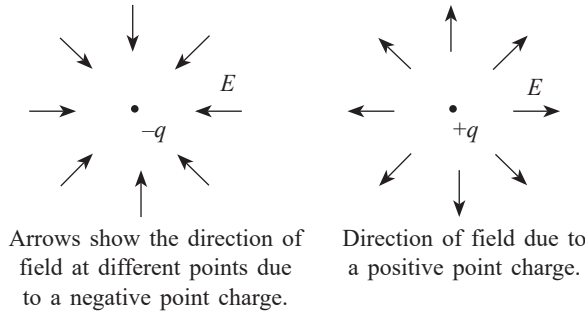


A test charge Q is placed at a point P at a distance r from q . Force experienced by Q is $\vec{F}_e = \left(\frac{1}{4\pi\epsilon_0} \frac{qQ}{r^2} \right) \hat{r}$

$\therefore$ Electric field at P due to q is

$$\vec{E} = \frac{\vec{F}_e}{Q} = \left(\frac{1}{4\pi\epsilon_0} \frac{q}{r^2} \right) \hat{r} \quad (9)$$

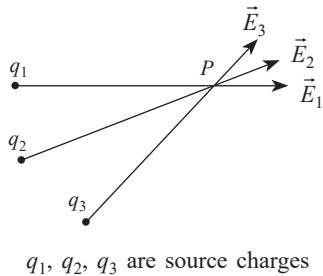
$\hat{r}$ is a unit vector from the source charge q to the point P . Equation (9) tells us that electric field due to a point charge q at a distance r from it is $E = \frac{1}{4\pi\epsilon_0} \frac{q}{r^2}$ and its direction is radially outward if q is positive. For negative q , direction of $\vec{E}$ is radially inwards towards q .



6.2 Electric field obeys superposition principle

Consider an arrangement in which there are number of source charges. Electric field ($\vec{E}$) at a point is obtained by vector addition of electric fields produced due to each of the source charges. Field at point P due to presence of q_1 , q_2 and q_3 (see figure) is

$$\vec{E} = \vec{E}_1 + \vec{E}_2 + \vec{E}_3 \quad (10)$$

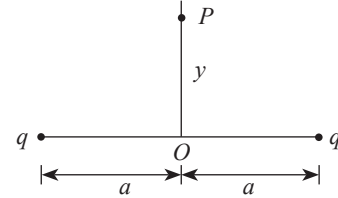


$\vec{E}_1$, $\vec{E}_2$ and $\vec{E}_3$ are electric fields due to q_1 , q_2 and q_3 respectively at P .

Now you are well equipped to calculate electric field due to variety of arrangements of source charges. The examples and discussions to follow will make the procedure more clear.

Example 12 Field on perpendicular bisector of line joining two point charges

Two identical positive charges q are kept at a separation $2a$. P is a point on the perpendicular bisector of the line joining the two charges. Distance OP is y . Find electric field at point P . Approximate your answer for $y \gg a$.



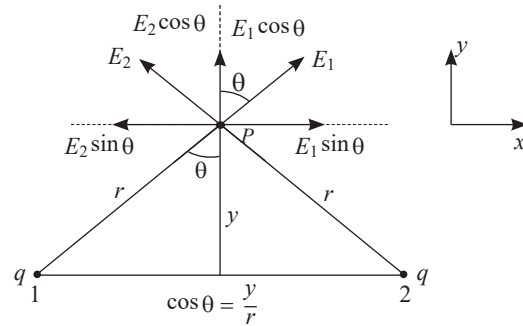
Solution

Concepts

- Field due to a positive charge is directed radially away from it. Magnitude is $E = k \frac{q}{r^2}$.
- Resultant field at P is vector sum of fields produced due to the two point charges.

Field at P due to charge at 1 is

$$E_1 = k \frac{q}{r^2} \text{ where } r = \sqrt{a^2 + y^2}.$$



Field at P due to charge at 2 also has same magnitude.

$$E_2 = k \frac{q}{r^2}$$

Directions of $\vec{E}_1$ and $\vec{E}_2$ are as shown. We have to add $\vec{E}_1$ and $\vec{E}_2$ to get resultant field at P . Components of $\vec{E}_1$ and $\vec{E}_2$ along x direction cancel out. Their components along y direction adds up to give resultant field as

$$E = E_1 \cos \theta + E_2 \cos \theta = 2k \frac{q}{r^2} \cos \theta$$

$$= 2k \frac{q}{r^2} \left(\frac{y}{r} \right) = \frac{2kqy}{r^3}$$

$$\therefore r = (a^2 + y^2)^{\frac{1}{2}}$$

$$\therefore E = 2k \frac{qy}{(a^2 + y^2)^{3/2}} \text{ directed along } y \text{ direction}$$

1.12 Electrostatics and Current Electricity

Note that $k = \frac{1}{4\pi\epsilon_0}$.

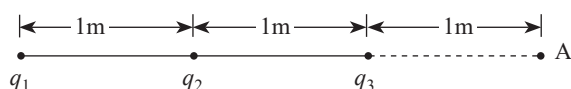
Approximation for the case $y \gg a$ can be done as follows:

$$a^2 + y^2 \simeq y^2$$

$$\therefore E \simeq \frac{2kqy}{(y^2)^{3/2}} = \frac{2kq}{y^2} \quad (i)$$

The answer could have been easily guessed. From a distance y ($\gg a$), the two point charges will appear to be a single point charge of magnitude $2q$ and field due to it will be given by (i).

Example 13 Three charges $q_1 = 4 \text{ nC}$, $q_2 = -2 \text{ nC}$ and $q_3 = 1 \text{ nC}$ are fixed along a straight line as shown in the figure.



- Find electric field at a point A located at a distance of 1 m to the right of q_3 (see figure).
- A charge $Q = 2 \text{ mC}$ is placed at A. Find force on it.
- A charge $Q = -2 \text{ mC}$ is placed at A. Find force on it.

Solution

Concepts

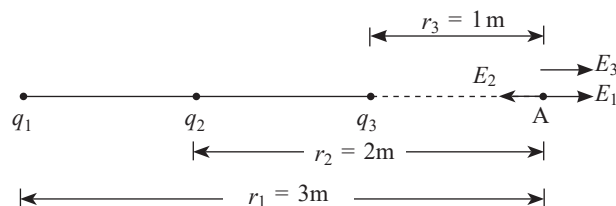
- Field at A is vector sum of fields due to q_1 , q_2 and q_3 .

$$\vec{E} = \vec{E}_1 + \vec{E}_2 + \vec{E}_3$$

- If a charge Q is placed at a point where electric field is E , it experience a force $\vec{F} = Q\vec{E}$.

This means that magnitude of the force is $|QE|$ and its direction is along $\vec{E}$ if Q is positive. Force is opposite to field if Q is negative.

- Field at A due to q_1 :



$$E_1 = k \frac{q_1}{r_1^2} = \frac{9 \times 10^9 \times 4 \times 10^{-9}}{3^2} = 4 \text{ N/C}(\rightarrow)$$

[Field due to a positive charge is radially away from the charge].

Field at A due to q_2 :

$$E_2 = \frac{kq_2}{r_2^2} = \frac{9 \times 10^9 \times 2 \times 10^{-9}}{2^2} = 4.5 \text{ N/C}(\leftarrow)$$

[Field due to negative charge is directed towards the charge].

Field at A due to q_3 :

$$E_3 = \frac{kq_3}{r_3^2} = \frac{9 \times 10^9 \times 1 \times 10^{-9}}{1^2} = 9 \text{ N/C}(\rightarrow)$$

Resultant field at A is

$$E = 4(\rightarrow) + 4.5(\leftarrow) + 9(\rightarrow) = 8.5 \text{ N/C}(\rightarrow)$$

- Force on positive charge will be in the direction of E .

$$F = QE = (2 \times 10^{-3} \text{ C}) \left(8.5 \frac{\text{N}}{\text{C}} \right) = 0.019 \text{ N}(\rightarrow)$$

- Force on negative charge is directed opposite to E

$$|F| = |QE| = 0.019 \text{ N}$$

$$\therefore F = 0.019 \text{ N}(\leftarrow)$$

Note:

- Pay attention to the way we are thinking. We first calculated field at A due to source charges (q_1 , q_2 and q_3) and then we are assuming that the field exerts force on test charge Q .
- Many a times electric field at a point is given without giving information about the source charges. In such situations force on a charge Q placed in the field is written as $\vec{F} = Q\vec{E}$ and we don't care about the source.

Example 14 Electric field along a line having two point charges

Two point charges, q each are held fixed at a separation. (see figure). Plot a graph to show variation of electric field along the line joining the two charges. Take electric field to be positive in rightward direction.

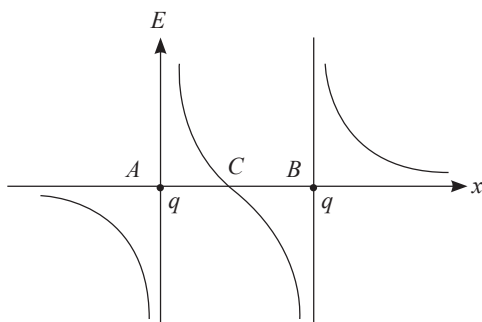


Solution

Concepts

- Field due to a point charge at a distance x from it is $E = k \frac{q}{x^2}$. Direction of field due to a positive charge is radially away.
- When we choose a point very close to a point charge (i.e., $x \rightarrow 0$), the field becomes very large (i.e., $E \rightarrow \infty$). For writing field at a point very close to a point charge, one can always forget about contributions made by other charges.
- When we move far away from the system of the two charges $E \rightarrow 0$.

Let's take E along y axis. E is positive if its direction is in positive x direction. At a point just right to A , field is large and is in positive direction. At a point just left to B , field is again very large and directed in negative x direction. At midpoint C , field is zero as the fields due to two charges are equal and opposite.



To the immediate left of A field is large towards left and decreases to become zero as we move far away from the charges. To the right of B field is large and decreases as we move away. Field is in positive direction.

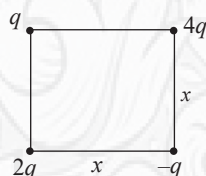
In Short

- (i) An electric field is said to exist at a point in space if a charged particle placed at that point experiences a force that would not be felt by an uncharged particle.
- (ii) If a point charge Q experiences an electric force $\vec{F}_e$ when placed at a point P, we say that there is an electric field at P given by $\vec{E} = \frac{\vec{F}_e}{Q}$.
- (iii) Electric field due to a point charge q at a distance r from it is $E = k \frac{q}{r^2}$.
Field is radially away from the charge if q is positive and the field is directed towards the charge if q is negative.
- (iv) If $q_1, q_2, q_3, \dots$ are fixed source charges then field at a point is vector sum of fields due to individual charges
$$\vec{E} = \vec{E}_1 + \vec{E}_2 + \vec{E}_3 + \dots$$
- (v) A charge Q placed in a field $\vec{E}$, experiences a force $\vec{F} = Q\vec{E}$.
- (vi) Electric field at a point very close to a point charge is very large. At a point far away from the point charge, field is nearly zero.



YOUR TURN

Q.18 Four point charges are kept at the vertices of a square as shown. Side length of the square is x . Find the electric field at the centre of the square.

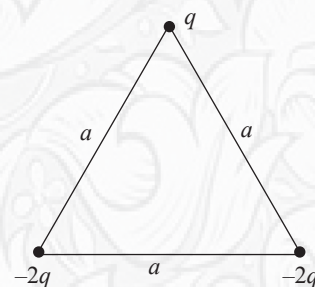


Q.19 Two point charges $q_1 = 5 \text{ nC}$ and $q_2 = 20 \text{ nC}$ are at separation of $d = 2 \text{ m}$. Find the distance of point from q_1 where the electric field is zero.

Q.20 A small droplet of ink in an ink-jet printer carries a charge $q = 1.6 \times 10^{-10}$. It is deflected onto a paper by applying an electric field. The electric force experienced by the droplet is $F = 3.2 \times 10^{-4} \text{ N}$. Find the strength of field.

Q.21 In example 12, at what distance y (from O), the electric field is maximum? Draw a rough graph showing variation of field for $0 \leq y < \infty$.

Q.22 An equilateral triangle has side length a and point charges have been placed at its vertices as shown. Find the magnitude and direction of the electric field at the centroid of the triangle.



Q.23 A point charge $q = -1 \text{ nC}$ is kept at a point having co-ordinates $(0, 1, 2) \text{ m}$. Find electric field due to the charge at a point $(1, 2, 3) \text{ m}$.

Q.24 Twelve equal charges, q , are located at the vertices of a regular 12 sided polygon (for example, one on each numeral of a clock face).

- (a) Find electric field at the center of the polygon.
- (b) One of the charges is moved to the centre of the polygon. Its distance from all other charges is r . Find force acting on it.

6.3 Electric Field due to Continuous Distribution of Charge

In real life it is difficult to get a point charge. We have charged bodies where charge may be distributed on a line (curved or straight), on a surface or in a volume. For example, you can easily think of a long thread carrying charge or a sheet having charge on its surface or a spherical volume occupied by charge.

For finding electric field due to a charged body, we need to mentally divide the body into infinitesimally small charge elements. We have to write field due to all small elements (treating them to be point charges) and add them. Such additions are usually done using integration while remembering that we are adding vectors.

In case of charge distributed on a line, linear charge density (λ) is the charge present on unit length of the line. Its unit is C/m.

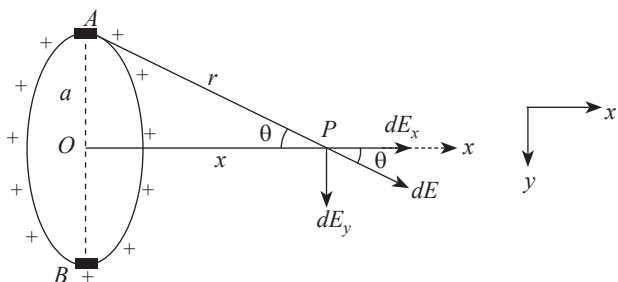
When charge is present on a surface, surface charge density (σ) is a useful quantity and is defined as charge present on unit area. C/m² is unit of σ .

In case of charge present in a three dimensional space, volume charge density or, simply charge density (ρ) represents charge inside a unit volume. Unit of ρ is C/m³.

Below we present few simple cases of continuous charge distribution and show how integration can be used to find electric field. We will take up few more cases after learning Gauss's law, later in this chapter.

6.3.1 Electric field on the axis of a charged ring

Figure shows a uniformly charged ring of radius a having charge Q . P is a point on the axis of the ring at a distance x from its centre O . We wish to find electric field at P due to charge on the ring.



Consider an infinitesimally small segment of the ring at A . Let the element have a charge dq on it. This element can be treated as a point charge and we can write electric field (dE) at P due to this element as

$$dE = k \frac{dq}{r^2}, \text{ where } r = \sqrt{a^2 + x^2}$$

Direction of the field (dE) is along AP if the ring carries a positive charge and its direction is along PA if the ring is negatively charged.

The figure shows two components of the field—along the axis (dE_x) and perpendicular to it (dE_y). Consider an identical charge element at point B that is located exactly opposite to A . This element will produce a field at P having magnitude dE and having its y component in negative y direction (see figure). Therefore, resultant of the two fields has no y component. However the components along x direction add.

We can think of our ring as made up of such pairs of small charges located at diametrically opposite ends. It is easy to see that resultant field is along the axis. Y components of field cancel out in pairs.

Field at P is obtained by adding x components of field due to each element in the ring.

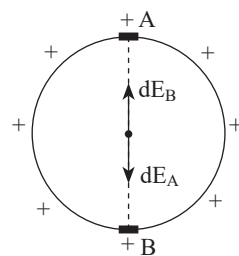
$$\begin{aligned} \therefore E &= \int dE_x = \int dE \cos \theta \\ &\quad \text{All element} \\ &= \frac{k}{r^2} \cos \theta \int dq \end{aligned}$$

Note that value of r is same for all elements. Value of $\cos \theta$ is also constant as we move from one element to another on the ring.

$$\begin{aligned} \therefore E &= \frac{k}{r^2} \cos \theta \cdot Q = \frac{kQ}{r^2} \cdot \frac{x}{r} \quad \left[\because \cos \theta = \frac{x}{r} \right] \\ &= \frac{kQx}{r^3} \\ \Rightarrow E &= k \frac{Qx}{(a^2 + x^2)^{3/2}} \quad [\because r = (a^2 + x^2)^{1/2}] \quad (11) \end{aligned}$$

Note the following points:

- What is field at the centre of the ring? Obviously, it is zero. Field produced due to small identical elements of charge at A and B cancel out. Similarly, all pairs of charges at diametrically opposite ends produce zero field at the centre.



By putting $x = 0$ in equation (11) we get the same result, i.e., $E = 0$.

- What do we expect if $x \gg a$? Looking at the ring from point P , we will hardly see a ring. It will appear like a small point charge if $x \gg a$. Therefore, field at P can be easily guessed to be equal to

$$E = k \frac{Q}{x^2}$$

Does equation (11) predict this result? Let's see.

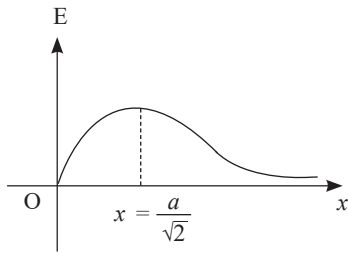
$$E = k \frac{Qx}{(a^2 + x^2)^{3/2}} = k \frac{Qx}{x^3 \left(\frac{a^2}{x^2} + 1 \right)^{3/2}}$$

For $x \gg a$, $\frac{a^2}{x^2} \ll 1$

$$\therefore 1 + \frac{a^2}{x^2} \simeq 1$$

$$\therefore E \simeq k \frac{Q}{x^2}$$

- (iii) Field is zero at $x = 0$ (i.e., at the centre of the ring). It is certainly zero at a large distance $x \rightarrow \infty$. It means that the field increases as we move along the axis starting from the centre, becomes maximum at a point and then decreases to be zero at ∞ .



Variation of electric field along the axis of the ring. x is distance measured from the centre.

The distance (x_0) where the field is maximum can be found by solving the equation

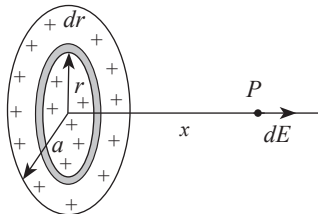
$$\frac{dE}{dx} = 0.$$

Differentiating equation (11) and solving $\frac{dE}{dx} = 0$

$$\text{gives } x_0 = \frac{a}{\sqrt{2}}.$$

6.3.2 Electric field on the axis of a charged disc

Consider a disc of radius a having charge spread uniformly on one of its surfaces. The surface charge density is σ . P is a point on the axis of the disc at a distance x from its centre O . We wish to find electric field at P .



We can consider the entire disc charge to be made up of infinite number or infinitesimally thin concentric rings. One such ring having radius r and width dr has been shown in the figure. Charge on this ring element is

$$dq = (2\pi r \cdot dr)\sigma$$

Using equation (11) we can write field at P due to this ring charge as

$$\begin{aligned} dE &= \frac{k(dq)x}{(r^2 + x^2)^{3/2}} = \frac{1}{4\pi\epsilon_0} \frac{2\pi r dr \cdot \sigma \cdot x}{(r^2 + x^2)^{3/2}} \\ &= \frac{\sigma x}{2\epsilon_0} \frac{r dr}{(x^2 + r^2)^{3/2}} \end{aligned}$$

Direction of field is along the axis as shown. Every ring that we consider in the disc will contribute its field in the same direction. Resultant field at P is obtained by adding contributions due to all such rings.

$$\therefore E = \int dE = \frac{\sigma x}{2\epsilon_0} \int_{r=0}^{r=a} \frac{r dr}{(x^2 + r^2)^{3/2}}$$

Note that x is a constant in the above integration. As we move from one ring to another (i.e., r changes), x remains constant.

On performing the integration we get,

$$E = \frac{\sigma}{2\epsilon_0} \left[1 - \frac{x}{\sqrt{a^2 + x^2}} \right] \quad (12)$$

[Integration can be performed by method of substitution.

Put $z = x^2 + r^2$

$$\frac{dz}{dr} = 2r \Rightarrow dz = 2r dr$$

$$\begin{aligned} \therefore \int_{r=0}^a \frac{r dr}{(x^2 + r^2)^{3/2}} &= \frac{1}{2} \int_{r=0}^a \frac{dz}{z^{3/2}} = - \left[\frac{1}{\sqrt{z}} \right]_{r=0}^a \\ &= - \left[\frac{1}{\sqrt{x^2 + r^2}} \right]_{r=0}^a = \frac{1}{x} - \frac{1}{\sqrt{a^2 + x^2}} \end{aligned}$$

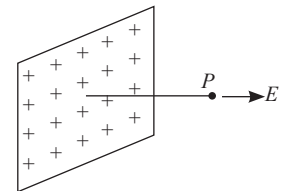
What will be the field at P if radius of the disc is too large ($a \gg x$)?

In this case, the second term inside the bracket in equation (12) becomes extremely small.

$$\frac{a}{\sqrt{a^2 + x^2}} \rightarrow 1$$

$$\therefore E \simeq \frac{\sigma}{2\epsilon_0} \quad (13)$$

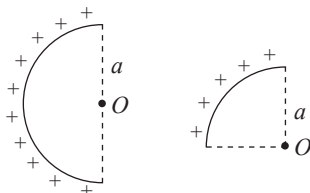
What this essentially means is that electric field in front of a large sheet having uniform charge density is independent of distance (x) from the sheet and has a constant value given by equation (13). In fact, even if a charged sheet is of finite dimensions and you are considering point close to its centre (Not near the edges), the field at the point is given by equation (13).



P is a point close to the sheet somewhere near its centre $E_P = \frac{\sigma}{2\epsilon_0}$.

Example 15 Half ring and quarter ring

A half ring of radius a has a uniformly spread charge along its circumference. The linear charge density is λ . Find the electric field at the centre (O).

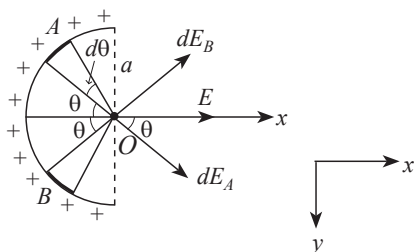


Using the above result find the field at O . Using the above result find the field at O due to a uniformly spread charge on quarter of a circle. Linear charge density is again λ and radius of the circle is a .

Solution**Concepts**

- Due to symmetry, the resultant field is along x axis for the semicircle (see figure given below). You can consider two identical elements on two sides of the x axis to argue for this.
We only need to add x component of field produced due to each element.
- The size of the element is easier to express in terms of its angular width.
- In case of quarter circle $E_x = E_y$ due to symmetry.

Consider an element (A) of angular width $d\theta$ at position θ as shown in figure.



Length of element, $dl = ad\theta$

charge on the element $dq = \lambda dl = \lambda ad\theta$.

This is an infinitesimally small element (fit to be considered a point charge) producing a field at O given by

$$dE_A = k \frac{dq}{a^2} = k \frac{\lambda ad\theta}{a^2} = \frac{k\lambda}{a} d\theta$$

An identical element at B produces a field of same magnitude in the direction shown in figure. Clearly, y components of dE_A and dE_B cancel out and their x components add up.

Picking elements in pairs, we can see that resultant field is along x direction only. To get the field, we need to add x components of field due to all elements.

$$\therefore E = \int dE \cos \theta = \frac{k\lambda}{a} \int_{-\pi/2}^{\pi/2} \cos \theta d\theta$$

[Please take a note of the limits of integration]

$$= \frac{k\lambda}{a} [\sin \theta]_{-\pi/2}^{\pi/2} = \frac{k\lambda}{a} \left[\sin \frac{\pi}{2} - \sin \left(-\frac{\pi}{2} \right) \right]$$

$$E = \frac{2k\lambda}{a} = \frac{\lambda}{2\pi\epsilon_0 a} \quad (i)$$

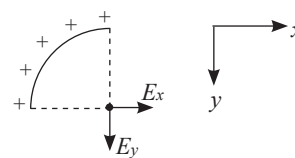
Quarter circle

Two things are important to note:

- Due to symmetry $E_x = E_y$
- E_x is half of the field given by (i)

$$\therefore E_x = E_y = \frac{\lambda}{4\pi\epsilon_0 a}$$

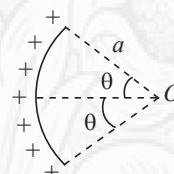
$$\vec{E} = \frac{\lambda}{4\pi\epsilon_0 a} (\hat{i} + \hat{j})$$

**YOUR TURN**

Q.25 In example 15, the half ring has a uniformly distributed charge Q and field at O is E . Find magnitude of field at O in the second diagram where the quarter ring has a uniformly spread charge equal to Q .

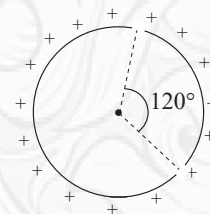
Q.26 A horizontal ring of radius a has a uniformly spread negative charge Q on it. A particle having mass m is released at a height h on the axis of the ring and it is found to stay at rest. Find the charge on the particle. h is height measured from the plane of the ring.

Q.27 A wire is in the shape of an arc of a circle subtending an angle 2θ at the centre as shown. Radius of the arc is a and it carries a uniform linear charge density λ on it. Find electric field at the centre O . How much force will an electron experience if it is placed at O ? In which direction?



Q.28 A particle of mass m and charge $-Q$ is constrained to move along the axis of a ring of radius a . The ring has a uniform linear charge density λ along its periphery. Initially, the particle is at the centre of the ring. It is displaced slightly and released. Show that it will perform SHM and find time period of its oscillation.

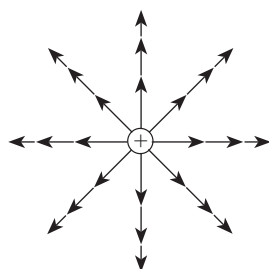
Q.29 A circular wire loop of radius R carries a uniformly distributed charge Q along its circumference. Two identical small segments of length ΔL are cut and removed from the wire. These segments are located at an angular separation of 120° as shown in figure. Find electric field at the centre due to the remaining charge.



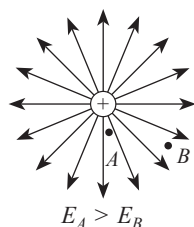
7. ELECTRIC FIELD LINES

Michael Faraday pioneered the use of field lines in visualising a field in a region of space.

Suppose we have been assigned a task of drawing a diagram which can correctly represent the magnitude and direction of electric field produced due to a point charge. After a lot of thinking one of us may come out with a brilliant diagram as shown. The arrows represent the direction of field at any point and their decreasing size tells us that field strength decreases as we move away from the charge.

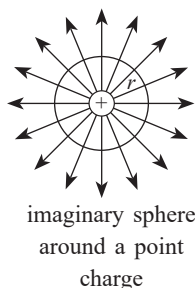


There is an even better idea! We can have a simple drawing as shown in second figure, consisting of radially outwards arrows. Is the information regarding strength of the field lost? No. In this diagram the density of field lines represent the strength of the electric field. Lines are denser (i.e., close together) at A than at B . We conclude that $E_A > E_B$.



There is even deeper significance of the second diagram. The diagram is in agreement with the inverse square law ($E \propto \frac{1}{r^2}$). Consider a point charge q .

To represent its field we draw N lines emanating radially from it in three dimensional space. Draw a sphere of radius r with its centre at q . Surface area of the sphere is $s = 4\pi r^2$. All N lines pierce through this surface. We say that density of lines at the surface of the sphere is



$$n = \frac{N}{s} = \frac{N}{4\pi r^2} \quad [= \text{number of lines per m}^2]$$

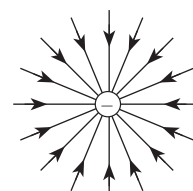
Now, we say that strength of field at a point is proportional to density of lines at the location of the point.

$$\Rightarrow E \propto n \propto \frac{1}{r^2}$$

We draw following conclusions from above discussions:

- (i) The electric field lines are imaginary geometrical construction which help us to visualise the nature of electric field in a region. They have no physical existence.
- (ii) Lines are drawn so that a tangent to a line of force gives the direction of electric field.

- (iii) Lines are drawn so that density of lines at a location is proportional to the magnitude of the field there.



Note that electric field lines due to a negative point charge are also radial pointing towards the charge.

Field lines due to a negative point charge

Following (additional) rule will be helpful when we try to construct electric field lines due to a system of charges.

- (i) Electric field lines are open ended. They have a starting point and a different terminating point. *Electrostatic field lines never form closed loops.*
- (ii) Field lines originate at a positive charge or at infinity and they terminate on a negative charge or at infinity. They never start/end in mid air.

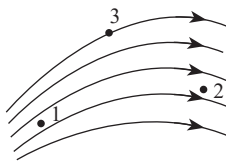
In the above diagrams showing field lines due to a positive point charge, the lines start at the charge and end at infinity (as there is no nearby negative charge).

In the field line diagram for a negative point charge, the lines originate at infinity and terminate on the negative charge.

- (iii) How many lines shall we draw? There is no rule to this. If you have a sharp pencil you can draw many. However, number of lines originating or, terminating at a charge is proportional to the magnitude of the charge. In a field line diagram, if you show N lines originating from a charge q then you must show $2N$ lines terminating at another charge $-2q$ in the same diagram.
- (iv) Two field lines never intersect. Intersection of field lines will imply two different directions of field at a point, which is absurd.

Following examples will make these rules more lucid.

Example 16 A student studies the electric field in a region created by some source charges. To represent the variation of field he draws a field line diagram as shown in the figure.



- At which point is the strength of field higher – 1 or 2?
- What can you say about direction of electric field at point 3?

Solution

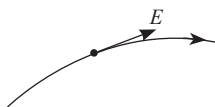
Concepts

Tangent at a point on the field lines gives the direction of field at that point. Density of lines is proportional to magnitude of field.

- Field lines are closer at 1 compared to point 2. Therefore, strength of electric field is higher at 1.

$$E_1 > E_2$$

- Direction of field at point 3 is along the tangent as shown in figure.



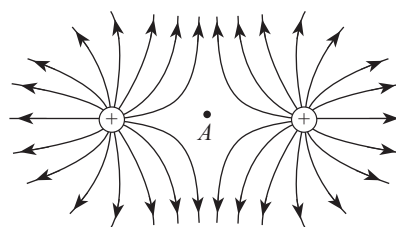
Example 17 Field lines due to two positive point charges

Two point charges, q each, are kept at a separation. Draw a rough sketch of electric field lines due to this system of charge.

Solution

Concepts

- Both charges are positive. Field lines will originate from them and terminate at infinity.
- Both charges have same magnitude. Therefore, equal number of lines must originate from them.
- There is a point exactly midway between the two charges where the field is zero. Density of field lines at this location must be zero.
- If we imagine a plane passing through the two charges, the field lines must be symmetrical on two sides of the plane.
- Close to a point charge, the lines will be radial. This is because field at a point very close to a charge is mainly because of the point charge itself. Other charge is at a relatively large distance and its contribution to the field is negligible.
- At a point that is far away from the system of charges, field lines will be just like the lines due to a point charge $2q$.



Field lines due to two positive point charges. All lines terminate at infinity field at A is zero.

Example 18 Field lines due to a pair of equal and opposite point charges

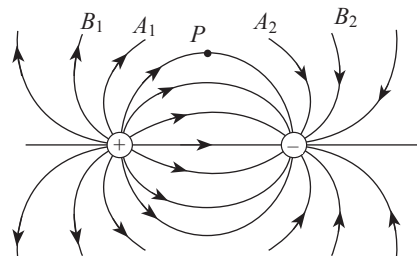
Two point charges, $+q$ and $-q$, are kept at a separation. Draw the electric field lines in the entire space.

Solution

Concepts

- Field lines will originate from $+q$ charge and terminate at $-q$ charge.
- Concepts (iv) and (v) of last example are applicable here also.

Can you find the direction of electric field at a point (P) on perpendicular bisector of the line joining the two charges? Do you think the field line diagram shown here, correctly represents the direction?



All lines start at positive charge and end on negative charge. Line A_1 is same as A_2 and line B_2 is line B_1 itself.

Example 19 Unequal charges

A point charge $+2q$ is kept at a separation from another point charge $-q$. Draw the electric field lines for the system.

Solution

Concepts

- If we draw n lines originating from $2q$ charge, only $\frac{n}{2}$ will terminate on $-q$ charge. Rest of the lines will terminate at infinity.
- If we move far away from the system of charges, the field lines must be similar to those produced by a point charge q .

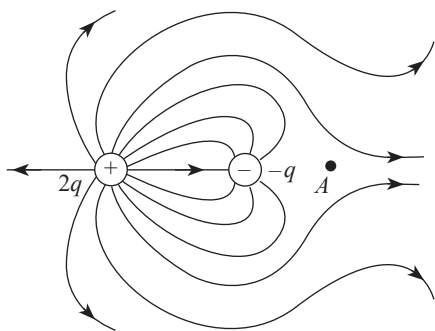


Figure shows 14 lines originating from $2q$ and 7 lines terminating on $-q$. Remaining lines terminate at infinity. At point A field is zero. Line density is zero there.

In Short

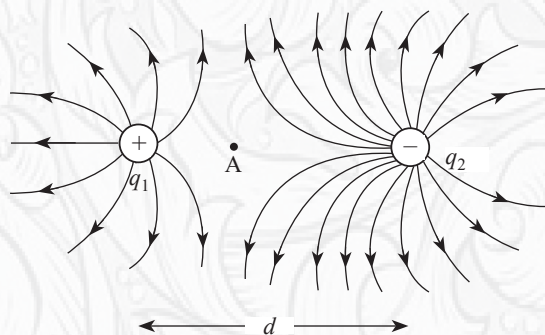
- (i) Electric field lines originate from positive charge or at infinity and end on a negative charge or at infinity.
- (ii) Field lines are radial for an isolated point charge directed away from a positive charge and towards a negative charge.
- (iii) Number of lines at a charge is proportional to its magnitude.
- (iv) Uniform field is represented by equally spaced parallel lines.
- (v) Two field lines never intersect.
- (vi) Absence of field lines in a region indicates that electric field is zero there.



YOUR TURN

Q.30 When electric field is of constant magnitude and has a fixed direction in a region of space, it is said to be uniform. Draw field lines to represent a *uniform electric field*.

Q.31 Two point charges q_1 and q_2 are kept at a separation d . A rough sketch of field lines due to these charges has been shown in figure.

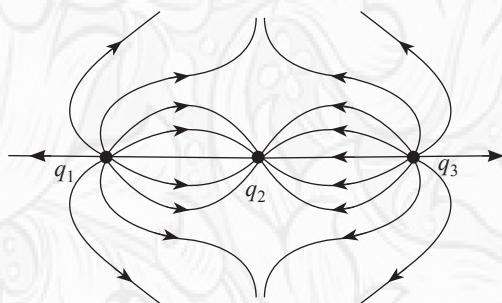


(a) Pay attention to number of lines and find $\frac{q_1}{q_2}$.

(b) No field line pass through point A . Find distance of A from q_1 .

Q.32 Three point charges are placed as shown in figure. Distance of q_2 from q_1 as well as q_3 is d .

(a) What can you say about signs of the three charges?



(b) Is there any relation in magnitudes of the charges?

(c) At distance y from q_2 the electric field is zero. Prove that $4y^6 = (d^2 + y^2)^3$.

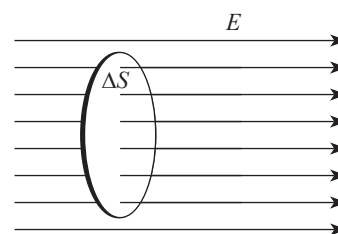
8. ELECTRIC FLUX

Now, we wish to study Gauss's law which helps us in finding electric field in many symmetrical situations in effortless way. To be able to state the law we need to introduce a new physical quantity known as *electric flux*.

Imagine a plane surface of area ΔS placed in a uniform electric field E . For simplicity, we first assume that field is normal to the surface. *Flux of electric field through an area is basically a quantity that gives us an idea about number*

of field lines piercing through the surface. Obviously, larger the area more will be the lines passing through it. Thus, flux is proportional to area (ΔS).

If electric field strength is doubled, the density of field lines doubles and number of lines piercing through the surface doubles. It means flux is proportional to E . Therefore, number of



field lines piercing through the surface is proportional to the product $E\Delta S$.

This quantity is known as *flux of electric field* (ϕ) through the given area ΔS

$$\phi = E\Delta S$$

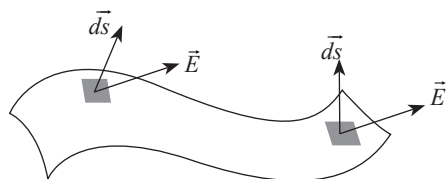
There is one more thing which decides the flux through an area. It is orientation of the area relative to the direction of the field. Consider the surface shown in above diagram to be tilted by an angle θ from its previous position. Now, the number of lines passing through the surface has reduced. Flux has reduced. You can easily argue that number of lines passing through the surface in this new position is same as number of lines passing through projection of area ΔS on a plane that is normal to the direction of the field. Therefore, number of lines piercing through the surface is proportional to E and $\Delta S \cos \theta$. Flux through the surface is defined as $\phi = E\Delta S \cos \theta$.

We draw a vector of magnitude ΔS along the normal to the surface. This is area vector ($\vec{\Delta S}$) for the given surface. Angle between $\vec{\Delta S}$ and $\vec{E}$ in the above figure is θ . Therefore, flux through a plane surface kept in a uniform electric field can be written as

$$\phi = E\Delta S \cos \theta = \vec{E} \cdot \vec{\Delta S} \quad (14)$$

Flux is a scalar and its unit is Nm^2C^{-1} . Being a scalar, addition of flux is simple. If a given surface has two parts and flux through them are ϕ_1 and ϕ_2 , then flux through the surface is $\phi = \phi_1 + \phi_2$. We can use this to find flux when a surface is not plane as well as when the electric field is not uniform.

Consider a curved surface shown in the figure. It is placed in an electric field. To find flux through the surface, we divide the entire surface into infinite number of infinitesimally small areas. We need to write flux through each of these small areas and add them. Mathematically, this means.

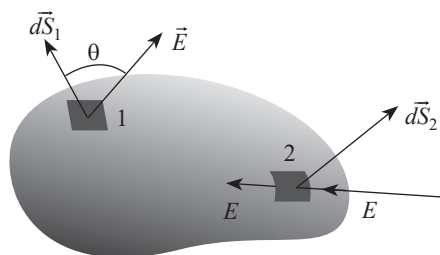


$$\phi = \int \vec{E} \cdot d\vec{S} \quad (15)$$

Integration must be performed so as to cover the entire surface.

In the context of Gauss's law, we will be evaluating electric flux over closed surfaces (e.g., a sphere, a cylinder

with its ends closed, a box shaped surface etc.). A small circle is placed on the sign of integration to indicate that the integration is being performed over a closed surface.



For finding flux over a closed surface, we take the direction of area vector ($d\vec{S}$) along outward normal.

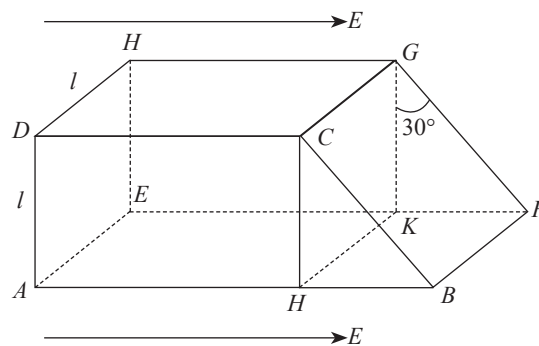
$$\phi = \oint \vec{E} \cdot d\vec{S} \quad (16)$$

While calculating flux over a closed surface we will always consider the direction of area vector to be along outward normal.

Whenever a field line pierces out of a closed surface flux will be positive, and it will be negative when a field line pierces into a closed surface. In the figure shown, flux through small area dS_1 , is positive [$\because \theta$ is acute and $\cos \theta$ is positive]. Flux through small area dS_2 is negative [$\because$ angle between $d\vec{S}_2$ and $\vec{E}$ is obtuse].

Example 20 Flux through a box

A box has the shape as shown in figure. Length AD and DH are l each and the face $BCGF$ is inclined at 30° to the vertical. The box is placed in a uniform electric field E that is perpendicular the face $AEHD$. Find flux of electric field through the box. Also write flux through the face $CBFG$.



Solution

Concepts

- We will calculate flux through each face of the box and add them.
- Box is a closed surface. For a closed surface outward normal is regarded as the direction of area vector.

For face $CDHG$, area vector is vertically up. Therefore, angle between $\vec{E}$ and area vector ($\vec{S}$) is 90° .

$$\therefore \phi_{CDHG} = ES \cos 90^\circ = 0$$

Similarly, $\phi_{ABFE} = \phi_{ABCD} = \phi_{EFGH} = 0$

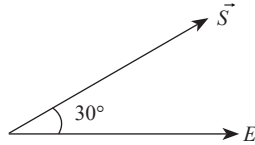
For face $AEHD$, area vector is directed towards left and $\vec{E}$ is towards right.

$$\therefore \theta = 180^\circ$$

$$\therefore \phi_{AEHD} = El^2 \cos 180^\circ = -El^2$$

[Notice that flux is negative whenever electric field lines pierce into a closed surface from outside.]

For face $CBFG$ area vector is inclined at 30° to the horizontal. [You can easily think about the direction of area vector like this: For surface $CHKG$ area vector is horizontal. If you tilt this surface outward by 30° the normal will also tilt by 30° from the horizontal].



For face $CBFG$ $\vec{S}$ is inclined to $\vec{E}$ at 30° .

$$\begin{aligned} \therefore \phi_{CBFG} &= S \cdot E \cos 30^\circ \\ &= (l \cdot l \sec 30^\circ) E \cdot \cos 30^\circ = l^2 E \end{aligned}$$

$$\therefore \phi_{\text{box}} = -El^2 + El^2 = 0$$

Now pay attention to the following points:

- (i) Actually, answer to the question is trivial if you know that the electric flux is basically measure of number of field lines piercing through a surface. And the flux is negative if a line pierces into the surface from outside and is positive if a line goes inside out.

In the given question, any field line going into the box will also come out from it. Therefore, number of lines going in is same as number of lines going out. Thus, flux through the box is zero.

- (ii) While writing flux through the face $CBFG$, you just need to understand that the lines passing through the surface are the same lines which also pass through $CHKG$.

$$\therefore \phi_{CBFG} = \phi_{CHKG} = El^2 \cos 0^\circ = El^2$$

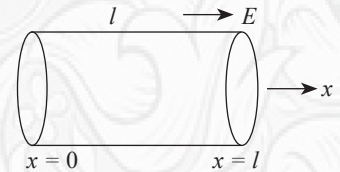


YOUR TURN

Q.33 Consider a circular surface of radius $r = 1\text{ m}$ kept in a uniform electric field $E = 10\text{ N/C}$. The surface of the circle makes an angle of 30° with the direction of the field. Calculate the magnitude of electric flux through the surface.

Q.34 Consider an imaginary cylindrical surface placed in an electric field. The field is along x direction and its magnitude changes with x co-ordinate as $E = ax$. Axis of the

cylinder is parallel to the direction of the field. Cross sectional area of the cylinder is S and its length is l . Find the flux of electric field through the cylinder. One end of the cylinder is at $x = 0$ and the other is at $x = l$.

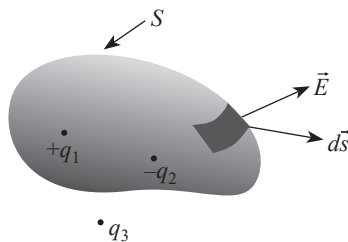


9. GAUSS'S LAW

This law relates the electric flux through a closed surface to the charge present inside it. It can be stated as follows:

The flux of electric field through a closed surface equals the total charge enclosed by the surface divided by ϵ_0

Consider fixed charges q_1 , $-q_2$ and q_3 as shown. An imaginary surface S is drawn so as to enclose q_1 and $-q_2$.



Consider a small patch of area dS on the surface S . Electric field (due to all the charges q_1 , $-q_2$ and q_3) at the location of the patch is E . Flux through this small patch is

$$d\phi = \vec{E} \cdot d\vec{S}$$

Flux through the closed surface is obtained by adding fluxes through all such small patches.

$$\phi = \oint \vec{E} \cdot d\vec{S}$$

The Gauss's law says that

$$\oint \vec{E} \cdot d\vec{S} = \frac{q_{in}}{\epsilon_0} \quad (17)$$

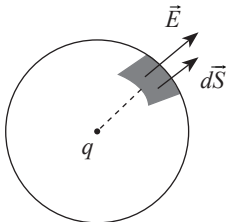
where q_{in} = charge present inside the closed surface. In present example, $q_{in} = q_1 - q_2$.

Proof of the Gauss's Law Using the Coulomb's Law

We will prove the Gauss's law using the Coulomb's law in three steps.

Step 1: Flux due to a point charge inside the surface

Consider a point charge q . Taking it as centre draw a sphere of radius r . Let's take this sphere as our closed surface. Electric field at any point on the surface of the sphere is (given by Coulomb's law)



$$E = \frac{1}{4\pi\epsilon_0} \frac{q}{r^2} \text{ [in radial direction]}$$

A small patch of area dS on the surface has area vector along the radius of the sphere in outward direction. Flux through the small patch will be

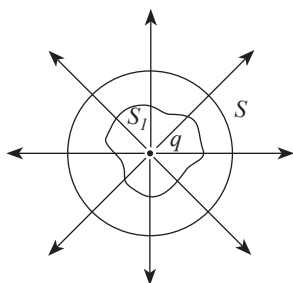
$$d\phi = E dS \cos 0^\circ = \frac{1}{4\pi\epsilon_0} \frac{q}{r^2} dS$$

Flux through the entire sphere is

$$\phi = \oint d\phi = \frac{1}{4\pi\epsilon_0} \frac{q}{r^2} \oint dS$$

$$\Rightarrow \phi = \frac{1}{4\pi\epsilon_0} \frac{q}{r^2} \cdot 4\pi r^2 = \frac{q}{\epsilon_0} \quad (i)$$

Now consider a different surface S_1 enclosing the same charge q . What is flux through this surface? The flux is still given by equation (i), irrespective of the shape of the surface S_1 .



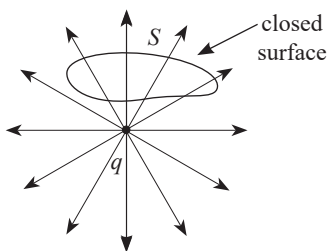
All field lines passing through sphere S also pass through S_1 .

This is because every field line that passes through the sphere S must also pass through the surface S_1 . We are repeatedly stressing that electric flux is proportional to number of field lines passing through a surface.

So, we have proved that flux through a closed surface of any shape is $\frac{q}{\epsilon_0}$ if a point charge q is lying inside it.

Step 2: Flux due to a point charge outside the surface

Consider a single point charge q lying outside a closed surface. The field lines which pierce into the surface will also exit at some other point of the surface. We know that when a line moves into the closed surface, the flux is regarded

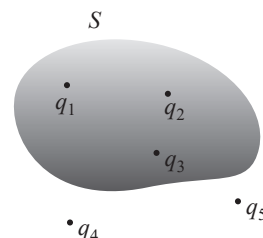


as negative and the flux is taken to be positive at point where it exits the surface. Flux through the surface is zero as number of lines moving into the surface is equal to the number of lines moving out of the surface.

So, we have proved that electric flux through a closed surface due to a point charge placed outside the surface is zero.

Step 3: Flux due to multiple charges

Consider a system of fixed point charges as shown in figure. S is a closed surface, q_1, q_2, q_3 are charges inside the surface and q_4, q_5 are those outside the surface.



While calculating flux, we consider field $\vec{E}$ at a location on the surface which is given by superposition of the electric fields due to $q_1, q_2, \dots, q_5$

$$\vec{E} = \vec{E}_1 + \vec{E}_2 + \vec{E}_3 + \vec{E}_4 + \vec{E}_5$$

where $\vec{E}_i$ is field due to q_i .

Flux through S is

$$\begin{aligned} \phi &= \oint \vec{E} \cdot d\vec{S} = \oint (\vec{E}_1 + \vec{E}_2 + \vec{E}_3 + \vec{E}_4 + \vec{E}_5) \cdot d\vec{S} \\ &= \oint \vec{E}_1 \cdot d\vec{S} + \oint \vec{E}_2 \cdot d\vec{S} + \oint \vec{E}_3 \cdot d\vec{S} + \oint \vec{E}_4 \cdot d\vec{S} + \oint \vec{E}_5 \cdot d\vec{S} \\ &= \frac{q_1}{\epsilon_0} + \frac{q_2}{\epsilon_0} + \frac{q_3}{\epsilon_0} + 0 + 0 \end{aligned}$$

This is because $\oint \vec{E}_i \cdot d\vec{S}$ is flux through S when only q_i is present.

$$\therefore \phi = \frac{1}{\epsilon_0} (q_1 + q_2 + q_3) = \frac{q_{in}}{\epsilon_0}$$

This completes the derivation of the law for all cases.

At the end, I want to make following important comments:

- The closed surface on which the Gauss's law is applied is sometimes known as the *Gaussian surface*.
- The Gaussian surface should never be considered to be passing through a point charge. A point charge shall always be inside the surface or outside it, it should never be on the surface. However, we are free to choose a Gaussian surface passing through a continuous distribution of charge.

Example 21 Charge inside a cube

A point charge q is kept at the centre of a cube.

- Find the flux of electric field through the cube.
- Find the electric flux through one of the faces of the cube.
- Will the flux through a face change if the charge is moved inside the cube? Will the flux through the entire cube also change?

Solution**Concepts**

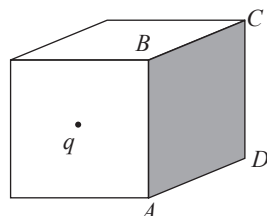
- (i) According to the Gauss's law the electric flux is independent of the shape of the surface enclosing a charge.
- (ii) When the charge is at the centre, all faces are at identical location for the charge. Due to this symmetry, flux through each of the faces must be same.
- (iii) Symmetry gets destroyed when the charge is moved from the centre.

- (a) The six faces of the cube make a closed surface. Flux through this closed surface is given by the Gauss's law.

$$\phi = \frac{q}{\epsilon_0}$$

- (b) Flux through all six faces are equal

$$\therefore \phi_{ABCD} = \frac{\phi}{6} = \frac{q}{6\epsilon_0}$$



- (c) When the charge is moved from the centre of the cube, the flux through the entire cube is still $\frac{q}{\epsilon_0}$. But, now the six faces are not symmetrically located with respect to the charge. Flux through a face may change.

Example 22 Refer to Your Turn Q.34. How much charge is enclosed inside the cylinder described in the question?

Solution**Concepts**

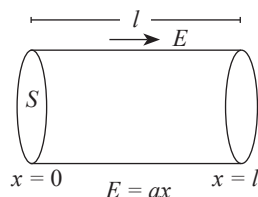
- (i) Flux through the cylinder was calculated in Q.34 to be $\phi = aIS$.
- (ii) According to the Gauss's law there must be positive charge inside the cylinder.

Using Gauss's law for the closed cylindrical surface

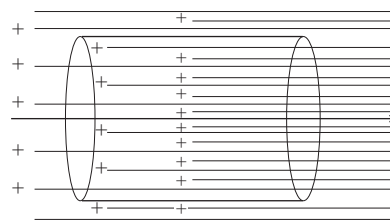
$$\phi = \frac{q_{in}}{\epsilon_0}$$

$$\Rightarrow q_{in} = \epsilon_0 \phi = \epsilon_0 aIS$$

What does this mean? It means that if we wish to have an electric field parallel to x axis which has a linearly increasing magnitude then a cylindrical volume as considered, must have charge within it.



Field lines appear roughly as shown in the figure. New field lines originate at positive charges scattered in space. This makes the density of lines higher as we move along x direction.

**Example 23** Flower vase

Imagine a surface in the shape of a flower vase. A point charge q is placed at the centre of the circular opening.

- (a) Find the electric flux through the curved surface of the flower vase.
- (b) The point charge is moved deep into the vase. Consider the opening to be small. What is the best approximate value for flux through the curved surface that you can give?

**Solution****Concepts**

- (i) Consider an identical vase placed upside down over the given vase. The two vases make a closed surface.
- (ii) When the charge is kept deep inside the vase, very few field lines will escape through the small opening.

- (a) Consider a perfectly closed surface made by two identical vase like surfaces. Flux through the closed surface made by twin vases is given by the Gauss's law as

$$\phi = \frac{q}{\epsilon_0}$$

Due to symmetry, equal number of field lines must pass through both the surfaces.

$$\therefore \text{Flux through given surface} = \frac{\phi}{2} = \frac{1}{2} \frac{q}{\epsilon_0}$$

- (b) When the charge is pushed deep into the vase, very few field lines will escape through the small circular mouth.

$$\therefore \text{Flux through the given surface will be nearly } \frac{q}{\epsilon_0}$$



In Short

- (i) Electric flux (ϕ) through a plane surface (S) placed in a uniform electric field $\vec{E}$ is defined as

$$\phi = \vec{E} \cdot \vec{S} = ES \cos \theta$$

where θ is angle between $\vec{E}$ and area vector ($\vec{S}$). Area vector is taken along the normal to the surface.

- (ii) When the area is curved or the field is non uniform, we calculated flux as

$$\phi = \int \vec{E} \cdot d\vec{S}$$

- (iii) When the flux is calculated over a closed surface, a small circle is drawn on the integral sign to indicate this

$$\phi = \oint \vec{E} \cdot d\vec{S}$$

- (iv) Electric flux is basically measure of number of field lines passing through a surface.

- (v) Flux is zero if field lines are parallel to a surface.

- (vi) The Gauss's law tells us that electric flux through a closed surface is always equal to $\frac{q_{in}}{\epsilon_0}$ where q_{in} is the charge present inside the surface.

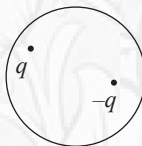
- (vii) A charge outside a closed surface contributes nothing to the flux.



YOUR TURN

Q.35 Two point charges q and $-q$ are placed inside a spherical surface. There is no other charge anywhere else.

- (a) Find flux through the sphere.
 (b) Another point charge Q is placed outside the sphere. Will the electric field at a point on the surface of the sphere change? Will the flux through the spherical surface be non zero now?



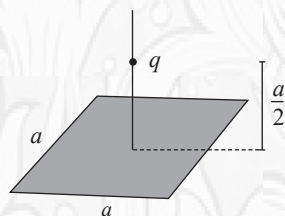
Q.36 A hemisphere of radius R is placed in a uniform electric field of strength E . Find the electric flux through the curved surface if

- (a) The field is parallel to the base
 (b) Perpendicular to the base

Q.37 Electric field in a region is given by $\vec{E} = (12\hat{i} + 16\hat{j})$ N/C. A rectangular surface of area $S = 0.1 \text{ m}^2$ is placed in $Y-Z$ plane. Find flux of electric field through the rectangular surface.

Q.38 Is it right to say that there is no net charge in a region in which electric field is uniform at all points? Explain.

Q.39 There is a horizontal square surface of side length a . A point charge q is placed vertically above its centre at a height $\frac{a}{2}$ from the surface. Find the flux of electric field through the surface.



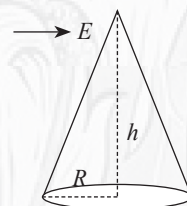
Q.40 A point charge q is located at the centre of a hemispherical surface. Find the flux of electric field through the curved hemispherical surface.



Q.41 A long line has a uniformly spread charge of linear density λ on it. Consider a spherical surface of radius R with its centre at a distance $\frac{R}{2}$ from the line. Find flux of electric field through the sphere.

Q.42 A circular loop has radius $r = 40 \text{ cm}$. It is placed in a uniform electric field. The loop is rotated about a diameter that is perpendicular to the field. In a particular position the flux of the electric field through the loop is found to be maximum and its value is $5.02 \times 10^5 \frac{\text{Nm}^2}{\text{C}}$. Find the magnitude of the electric field.

Q.43 A cone of base radius R and height h is located on a horizontal table. A uniform horizontal field E exists in the region. Determine the flux that enters the cone from its left side.



10. APPLICATION OF GAUSS'S LAW IN CALCULATION OF ELECTRIC FIELD

The Gauss's law is a fundamental law in electrostatics. It is always true. But it may not be always useful in calculation of electric field. It can be used in certain symmetrical situations only, for finding field due to a given charge distribution. There are three kinds of symmetry where the Gauss's law works. Here, we present them one by one.

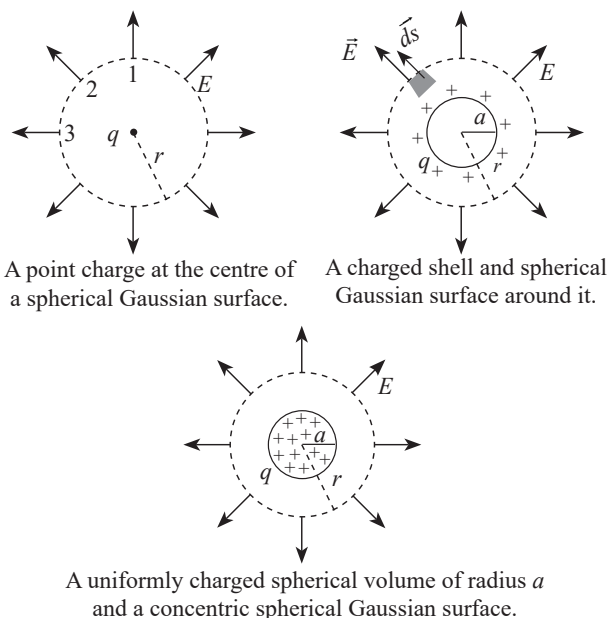
10.1 Spherical Symmetry

Consider following three problems:

- We have to find electric field due to a point charge q at a distance r from it (and assume that we do not know the Coulomb's law!)
- There is a spherical shell of radius a . A charge q is uniformly spread on its surface. We have to find electric field at a point at a distance r from the centre of the shell. The point lies outside the shell ($r > a$).
- A charge q is uniformly distributed in a spherical volume of radius a . We have to find electric field at a point that lies outside the spherical region and is at a distance r ($>a$) from its centre.

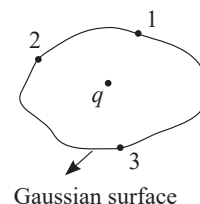
The three situations have been shown in the figure given below. In all three cases we draw an imaginary sphere of radius r passing through point P where we have to find the field. This is our Gaussian surface and its centre is at the point charge (q) or at the centre of the given charged sphere.

We have chosen spherical Gaussian surface because the three situations given here have spherical symmetry. Why will the electric field at point 1 be different from the field at point 2; or from the field at point 3? There is absolutely no reason. The field at all the points on the spherical Gaussian surface (in all three cases) must have the same magnitude.



Also, the field will have identical radial directions at all points on the spherical Gaussian surface. Why the field at 1 will be radial and at 2 in a different direction? Again there is no reason for this to happen.

Now, you must be getting a sense why we picked a spherical Gaussian surface. If we choose any other surface surrounding our charge, the benefit of symmetry is lost. Look at the figure given here. We cannot say that the electric field will be same at points 1, 2 and 3. For writing the flux we will have to assume field as E_1, E_2, E_3 etc. at different points. With so many unknowns we cannot have our answer.



Therefore, we conclude that in all three problems the electric field has same magnitude (say, E) at all points on spherical Gaussian surface and has radial direction.

To write the flux, consider a small patch of area dS on the Gaussian surface. Area vector is radial. Flux through the patch is

$$d\phi = E dS \cos 0^\circ = E dS$$

Flux through the entire surface is

$$\phi = \oint d\phi = E \oint dS = E 4\pi r^2$$

The last thing that we need to look for is the total charge present inside the Gaussian surface. In all three cases this charge is q . Hence, using Gauss's law, we can write

$$\begin{aligned} \phi &= \frac{q_{in}}{\epsilon_0} \\ \Rightarrow E \cdot 4\pi r^2 &= \frac{q}{\epsilon_0} \\ \Rightarrow E &= \frac{1}{4\pi\epsilon_0} \frac{q}{r^2} \end{aligned} \quad (18)$$

This is the common answer to the three problems.

We conclude the followings:

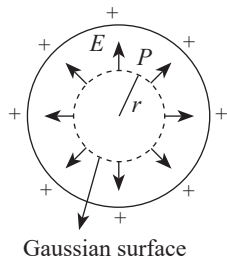
- Equation 18 gives electric field due to a point charge q at a distance r from it. This is Coulomb's law. Yes, we have proved Coulomb's law starting from the Gauss's law!
- The electric field due to a uniformly charged spherical shell or a spherical volume at a point outside it, is identical to the field due to an equal point charge placed at the centre.

We will consider two more problems which have spherical symmetry.

Electric field inside a uniformly charged spherical shell

A spherical shell of radius a has a charge q uniformly spread on its surface. We wish to find electric field due to this charge distribution at a point (P) inside the shell.

Consider a spherical Gaussian surface passing through point P that is concentric to the given shell. Once again symmetry of the situation tells us that electric field should have same magnitude at all points on the Gaussian surface. The direction is radial. Flux through the surface can be written just as we did above.



$$\phi = \oint \vec{E} \cdot d\vec{S} = E \cdot 4\pi r^2$$

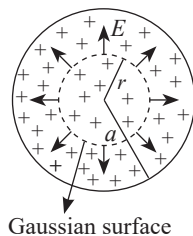
How much charge is there inside the Gaussian surface? There is no charge at all.

$$\therefore E \cdot 4\pi r^2 = \frac{0}{\epsilon_0} \Rightarrow E = 0 \quad (19)$$

Therefore, electric field inside a uniformly charged shell is zero at all points.

Electric field inside a uniformly charged spherical volume

A spherical volume of radius a has a uniform charge density ρ . We are supposed to find electric field at a point inside the sphere at a distance r ($< a$). Draw a spherical Gaussian surface of radius r having its centre at the centre of the charged sphere. Once again due to symmetry we assume that the field is radial at all points and has same magnitude at all points on the Gaussian surface. Let this magnitude be E .



Flux through the Gaussian surface is

$$\phi = \oint \vec{E} \cdot d\vec{S} = E \cdot 4\pi r^2$$

Total charge present inside the Gaussian surface is

$$q_{in} = \rho \cdot \frac{4}{3}\pi r^3$$

Using the Gauss's law

$$\phi = \frac{q_{in}}{\epsilon_0}$$

$$E \cdot 4\pi r^2 = \frac{\rho \cdot \frac{4}{3}\pi r^3}{\epsilon_0}$$

$$\Rightarrow E = \left(\frac{\rho}{3\epsilon_0} \right) r \quad (20)$$

If total charge inside the spherical volume of radius a is Q , then

$$\rho = \frac{Q}{\frac{4}{3}\pi a^3}$$

Equation (20) in terms of Q becomes

$$E = \frac{1}{4\pi\epsilon_0} \frac{Q}{a^3} \cdot r \quad (21)$$

We must take a note of the following points:

- (i) Field at the centre is obtained by putting $r = 0$ in the above equation

$$E_{\text{centre}} = 0.$$

- (ii) As we move from the centre to the surface of the spherical volume, the field increases linearly. Field on the surface is obtained by putting $r = a$ in equation (20)

$$E_{\text{surface}} = \frac{1}{4\pi\epsilon_0} \frac{Q}{a^2}$$

Example 24 Spherically symmetric but variable charge density

A spherical volume of radius a contains a non-uniform charge density which varies with distance x from the centre as $\rho = \rho_0 \frac{a}{x}$, where ρ_0 is a positive constant. Find the electric field at a distance x_0 from the centre of the sphere for

- (a) $0 < x_0 < a$ and
(b) $x_0 > a$.

Also plot a graph showing the variation of electric field with x .

Solution

Concepts

- (i) Though the charge density is non-uniform, it is spherically symmetric. If you draw a sphere of radius x ($< a$), the charge density is same at all points on the surface of the sphere. What is advantage of having a spherical symmetry? We can consider the field to be radial at all points and field will have same magnitude at all the points on the surface of a sphere that is concentric with the given sphere.
- (ii) For using the Gauss's law we need to know the charge contained in a spherical volume of radius x_0 . For finding the electric field at an outside point ($x > a$) we need to know the total charge in the spherical volume of radius a .

Let's first calculate total charge present inside a spherical volume of radius x_0 .

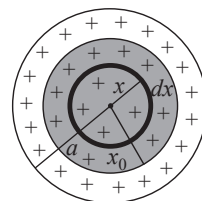
Consider a shell of radius x and thickness dx . Charge in the shell is given by

$$\begin{aligned} dq &= \rho \cdot 4\pi x^2 dx \\ &= 4\pi \rho_0 a x dx \end{aligned}$$

Charge inside sphere of radius x_0 is

$$q = \int_{x=0}^{x_0} dq = 4\pi \rho_0 a \int_0^{x_0} x dx$$

$$\Rightarrow q = 2\pi \rho_0 a x_0^2 \quad (i)$$



Charge inside the spherical volume of radius a is

$$Q = 2\pi\rho_0 a(a)^2 \quad [\text{Putting } x_0 = a \text{ in (i)}]$$

$$= 2\pi\rho_0 a^3 \quad (\text{ii})$$

- (a) Consider a sphere of radius x_0 as our Gaussian surface. Due to symmetry, field is radial at all points and has same magnitude at all points on the surface of the sphere

$$\therefore \phi = E \cdot 4\pi x_0^2$$

$$\phi = \frac{q_{\text{in}}}{\epsilon_0}$$

$$\Rightarrow E \cdot 4\pi x_0^2 = \frac{2\pi\rho_0 a x_0^2}{\epsilon_0}$$

$$\Rightarrow E = \frac{\rho_0 a}{2\epsilon_0}$$

Field has constant magnitude inside the spherical volume of radius a !

- (b) Gaussian surface is a concentric sphere of radius $x_0 > a$

$$\phi = E \cdot 4\pi x_0^2$$

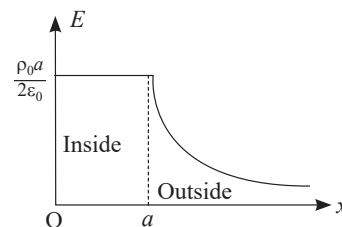
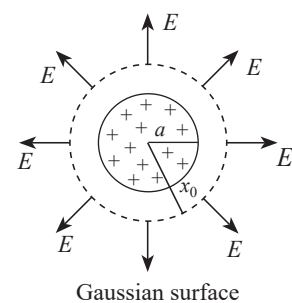
Charge inside the sphere is Q as given by (ii)

$$\therefore \phi = \frac{Q}{\epsilon_0}$$

$$\Rightarrow E \cdot 4\pi x_0^2 = \frac{2\pi\rho_0 a^3}{\epsilon_0}$$

$$\Rightarrow E = \frac{\rho_0 a^3}{2\epsilon_0 x_0^2}$$

Graphical variation of E with x is as shown.



YOUR TURN

Q.44 A spherical shell of radius a has a uniformly distributed charge $+q$ on its surface.

- Plot the variation of electric field (E) with distance x from the centre. Consider $0 \leq x < \infty$.
- How will the field change if charge q is negative?

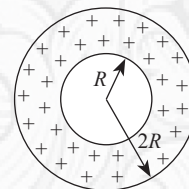
Q.45 A spherical volume of radius a has a uniform charge density and total charge inside it is Q .

- Plot the variation of electric field (E) with distance x from the centre. Consider $0 \leq x < \infty$.
- At what distance from the centre the field is half its value at the surface of spherical volume of radius a ?

Q.46 A spherical volume of radius a has a uniform charge density ρ . Electric field at a distance $\frac{a}{4}$ from the centre is E_1 and the field at a distance $2a$ from the centre is E_2 . Now, the spherical volume is shaved off so that charge remains only inside spherical volume of radius $\frac{a}{2}$. Will the magnitude of field E_1 and E_2 increase or decrease?

Q.47 Imagine two concentric spheres of radii R and $2R$. A uniform charge density ρ exists in the space between the two spheres. Find electric field at a distance x from the centre when

- $0 < x < R$
- $R < x < 2R$
- $x > 2R$.



Q.48 A charge q is uniformly spread in a volume of radius a . Another spherical volume of radius a has a charge $-q$ uniformly spread inside it. The centers of the two spheres are $4a$ apart.

Find electric field at a point which is exactly midway between the centres of the two spheres. Assume that charges on the two spheres do not distort one another.

10.2 Cylindrical Symmetry

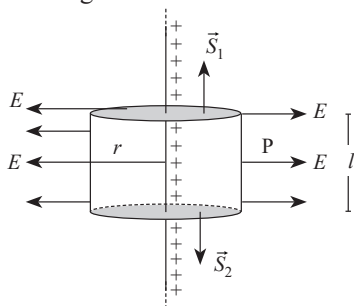
To illustrate the use of Gauss's law in calculating electric field, in situations where we have cylindrical symmetry, we take up following example:

Consider an infinitely long line having charge uniformly spread along its length. The linear charge density is λ . We wish to find electric field due to this line charge at a point (P) located at a perpendicular distance r from it.

Can you guess the direction of electric field at P? Do you think that it should be along direction indicated by arrow 1? If yes, then why it can't be along direction indicated by arrow 2? In fact, the only unique direction of field at point P is along a direction perpendicular to the line charge. It has been marked as direction 3 in the figure. The electric field is along this direction.

You can convince yourself about the direction of the field in a slightly different manner also. Consider two identical charge elements on two sides of line PA marked as 1 and 2 in the figure. These charge elements produce field $d\vec{E}_1$ and $d\vec{E}_2$ respectively. Their components parallel to the line charge (indicated as y direction) cancel out. Their components perpendicular to the line charge (indicated as x direction) add up. Since the line charge is of infinite length, for every charge element above line PA we have another identical element below it. Such pair of elements produce field which is perpendicular to the line charge along x direction.

Now, consider a cylindrical surface passing through P and coaxial with the line charge. Every point on this cylindrical surface is identical to P. Electric field at all the points on this cylindrical surface will have same magnitude, say E . Direction of field at all points on the surface of this cylinder is perpendicular to the line charge. In other words, the field is radial. Thus, we have symmetry over a cylindrical surface around the line charge.



The cylinder with its lids form a closed Gaussian surface.

The cylindrical surface along with its flat circular faces forms a closed surface. Consider it as our Gaussian surface. Flux through our Gaussian surface is sum of fluxes through the curved surface and two flat lids.

The flat lids have area vector ($\vec{S}_1$ and $\vec{S}_2$) that is perpendicular to the direction of electric field. There is no flux through these surfaces.

Flux through the curved cylindrical surface is

$$\phi = \int \vec{E} \cdot d\vec{S} = \int E dS \cos 60^\circ = E \int dS$$

$$\therefore \phi = E 2\pi r l \quad (i)$$

Note that area vector of a small patch on cylindrical surface has radial direction and the electric field is also radial. Angle between them is zero everywhere.

Total flux through closed cylindrical surface is

$$\phi = \oint \vec{E} \cdot d\vec{S} = E \cdot 2\pi r l$$

Charge enclosed inside the cylindrical surface is equal to charge present on line of length l .

$$\therefore q_{in} = \lambda l$$

Using the Gauss's law gives:

$$E \cdot 2\pi r l = \frac{\lambda l}{\epsilon_0}$$

$$\Rightarrow E = \frac{\lambda}{2\pi\epsilon_0 r} \quad (22)$$

Example 25 Field due to a long uniformly charged cylindrical volume

Consider a long cylindrical volume of radius a . The entire volume contains a uniform charge density ρ . Calculate the electric field at a distance r from the axis of the cylinder where

(a) $x > a$ (b) $x < a$.

Draw a graph showing variation of E with x .

Solution

Concepts

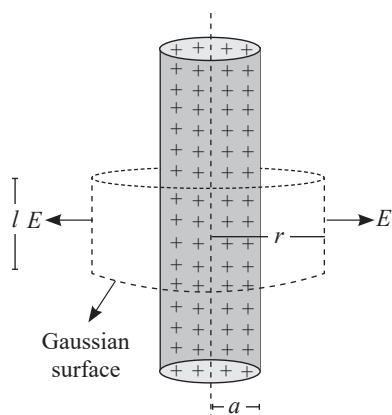
(i) Cylindrical symmetry exists in this question also. Imagine any cylindrical surface coaxial with the given cylinder. All points on the surface will have same strength of field and direction of the field is radial everywhere.

(ii) We will take a cylindrical Gaussian surface with lids.

(a) Gaussian surface is as shown.

Flux through the two flat circular lids of the cylindrical Gaussian surface is zero.

$$\therefore \phi = E \cdot 2\pi r l$$



Charge enclosed inside the Gaussian surface is

$$q_{in} = \rho \cdot (\pi a^2 l)$$

$$\therefore E \cdot 2\pi r l = \frac{\rho \pi a^2 l}{\epsilon_0}$$

$$\Rightarrow E = \frac{\rho a^2}{2\epsilon_0 r} \quad (23)$$

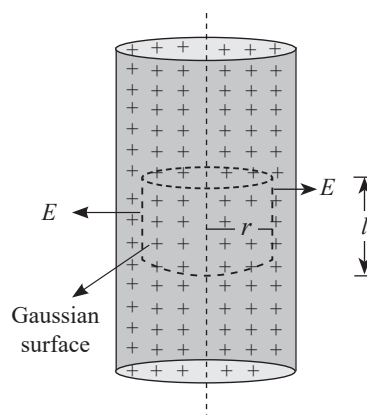
Note: If you take λ as charge present on unit length of the cylinder then, the above expression is exactly same as equation (22). Convince yourself.

- (b) Again we consider a cylindrical Gaussian surface as shown.

$$\phi = 2\pi r l E$$

$$q_{in} = \rho \pi r^2 l$$

$$\therefore \phi = \frac{q_{in}}{\epsilon_0}$$



$$\Rightarrow 2\pi r l E = \frac{\rho \cdot \pi r^2 l}{\epsilon_0}$$

$$\Rightarrow E = \left(\frac{\rho}{2\epsilon_0} \right) r \quad (24)$$

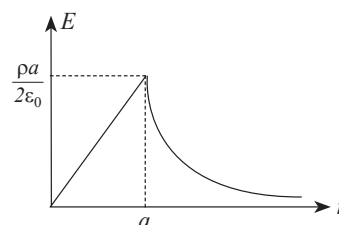
For $r \leq a$, $E \propto r$

and for $r \geq a$, $E \propto \frac{1}{r}$ from equations (i) and (ii).

On the surface ($r = a$) field is

$$E = \frac{\rho a}{2\epsilon_0}$$

The required graph is as shown in the figure.



YOUR TURN

Q.49 A long hollow cylinder has a uniformly distributed charge on its surface. The surface charge density is σ measured in unit of C/m^2 . Radius of the cylinder is a . Find electric field at a point P which is at a distance r from the axis of the cylinder. Consider $r < a$ as well as $r > a$.

Q.50 A long cylinder has a uniform density of negative charge inside it. An electron is released at a point outside the cylinder. Will the electron move with increasing acceleration? Draw a rough graph of speed of the electron with time. Just describe things qualitatively.

Q.51 Find electric field at a point 2 cm away from an infinitely long line charge having linear charge density $2 \mu C/m$.

Q.52 A long cylinder contains a positive charge of linear density λ . An electron revolves around it in a circular path under the influence of electrostatic attraction. Find the kinetic energy of the electron.

10.3 Symmetry on a plane

Gauss's law may be useful in calculation of electric field in situations where we have symmetry on a plane. Consider a large plane sheet of charge with surface charge density (charge on unit area) σ . We have to find electric field E at a point P that is at a distance r from it.

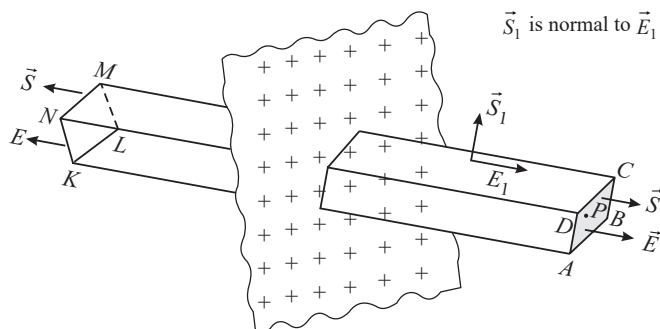
First let's talk about the direction of electric field at P . There is only one unique direction. That is normal to the sheet.

Field at P , or at any other point in front of the sheet, must be directed along the perpendicular to the plane surface. One may also visualise this by choosing ring shaped charge elements on the charged surface. Every ring element (with OP as axis) will produce a field along OP . The entire charged surface can be thought to be made of infinite number of concentric charged rings. Therefore, resultant field at point P is normal to the surface.

There is nothing special about point P . Field at all points in front of the sheet is directed normal to the sheet. Field at a point behind the sheet is directed away from the sheet and is normal to the sheet (see the first figure).

Consider a box shaped Gaussian surface penetrating through the given charged surface (see figure). Point P lies on the front face of the box. The length of the box on two sides of the sheet are equal. Electric field at every point on face $ABCD$ must be same (say E) as all points are at equal distance from the charged sheet.

Therefore, we have symmetry on the plane surface $ABCD$. [One can also consider a cylindrical Gaussian surface piercing through the sheet with point P lying on its flat circular lid].



Since faces $ABCD$ and $KLMN$ are equidistant from the sheet of charge, we must consider the electric field to be same at the two surfaces.

Flux through face $ABCD = ES \cos 0^\circ = ES$

Flux through face $KLMN = ES \cos 0^\circ = ES$

Flux through any other face of the box is zero as area vector is normal to $\vec{E}$.

$\therefore$ Total flux through closed Gaussian surface = $2ES$.
Charge present inside the Gaussian surface is: $q_{in} = \sigma S$.

This is because our Gaussian surface encloses an area S of the charged sheet.

The Gauss's law gives

$$2ES = \frac{\sigma S}{\epsilon_0}$$

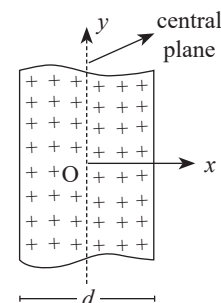
$$\Rightarrow E = \frac{\sigma}{2\epsilon_0} \quad (25)$$

It is important to note that the *field is uniform*. It does not depend on distance of point P from the sheet and direction is normal to the sheet at all locations. Theoretically, all this is true if the sheet is of infinite size. In practice, it is true as long as the dimensions of the sheet are large compared to its distance from P and the point P is not near the edge of the sheet.

Example 26 A slab of charge

Imagine a large slab of thickness d made up of a uniform volume charge density ρ . The dimensions of the slab in yz plane is large (see figure). Find electric field at a distance x from the centre of the slab when

(a) $x < \frac{d}{2}$ (b) $x > \frac{d}{2}$.



Plot a graph showing variation of electric field along x axis.

Solution

Concepts

- Due to symmetry, the electric field is normal to the slab along x axis at all locations. To the right of the central plane the field is in positive x direction. To the left of the central plane, the field is in negative x -direction.
- Field will have same magnitude at all points on a surface that is parallel to yz plane. There is a planar symmetry in this situation.
- Field will be zero at all points on the central plane. It has identical charge distribution on both sides. Field due to charges on the two sides cancel out.

- (a) Consider a cylindrical Gaussian surface with one of its circular face on the central yz plane (where $E = 0$) and the other circular face at a distance x ($< \frac{d}{2}$) from the central plane.

Flux through curved cylindrical surface = 0.

Flux through left circular face = 0.

Flux through right circular face = ES .

Total flux, $\phi = ES$

Charge enclosed inside the cylinder is

$$q_{in} = \rho \cdot Sx$$

Gauss's law gives

$$\phi = \frac{q_{in}}{\epsilon_0}$$

$$ES = \frac{\rho Sx}{\epsilon_0}$$

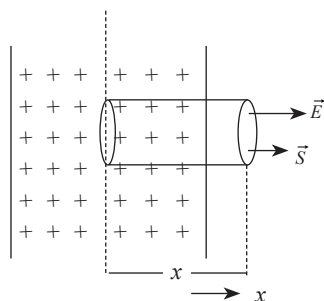
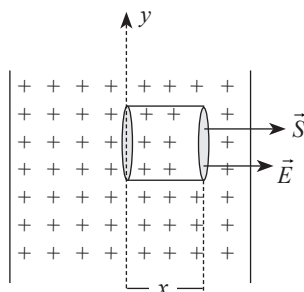
$$\Rightarrow E = \frac{\rho}{\epsilon_0} x$$

- (b) For $x > \frac{d}{2}$, consider a cylindrical Gaussian surface as shown.

Again $\phi = ES$

Charge enclosed is

$$q_{in} = \rho S \frac{d}{2}$$



$$\therefore ES = \frac{\rho Sd}{2\epsilon_0}$$

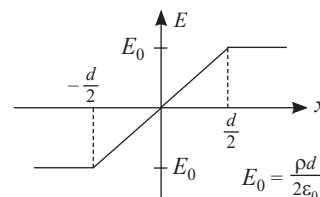
$$\Rightarrow E = \frac{\rho d}{2\epsilon_0} = \text{a constant.}$$

Therefore, magnitude of E varies along the x axis as

$$E = \frac{\rho}{\epsilon_0} x \quad \text{for } 0 \leq |x| \leq \frac{d}{2}$$

$$= \frac{\rho d}{2\epsilon_0} \quad \text{for } |x| \geq \frac{d}{2}$$

Taking the field directed towards left as negative, the graph of E versus x is as shown in the figure.

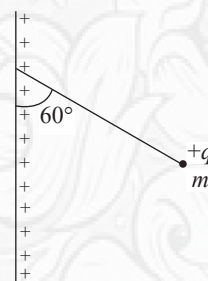


YOUR TURN

Q.53 A large horizontal sheet of charge has a surface charge density σ . A particle of mass m is released at a small height h above the charged surface and it stays at rest.

- Find charge on the particle.
- Discuss the nature of equilibrium of the particle for small displacements in any direction.

Q.54 One end of a thread of length $L = 10$ cm is fixed to a large vertical surface which has a uniform surface charge density σ . There is a small ball of mass $m = 0.01$ kg and charge $q = 4 \mu\text{C}$ at the other end of the thread. In equilibrium the thread makes an angle of 60° with the vertical. Find σ .



In Short

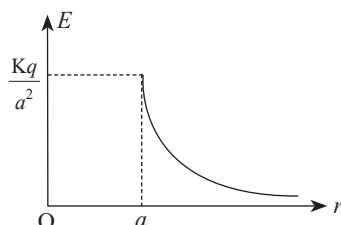
- (i) We can derive Coulomb's law using the Gauss's law.
 (ii) Field due to a uniformly charged spherical shell of radius a containing charge q on it is given by

$$E = \frac{1}{4\pi\epsilon_0} \frac{q}{r^2} \text{ for } r > a$$

$$= 0 \text{ for } r < a$$

where r is distance measured from the centre.

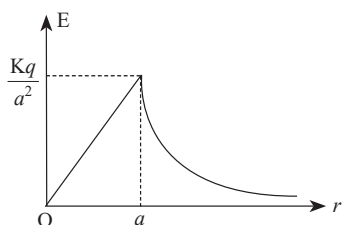
For a point outside the shell we can consider the entire charge as a point charge at the centre of the shell. At inside points the field is zero. Graphical variation of field with distance (r) from the centre is as shown.



- (iii) If we have a spherical volume of radius a having a charge q , the field can be expressed as

$$E = \left(\frac{1}{4\pi\epsilon_0} \frac{q}{a^3} \right) r \text{ for } r \leq a$$

$$= \frac{1}{4\pi\epsilon_0} \frac{q}{r^2} \text{ for } r \geq a$$

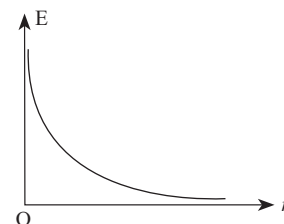


Again the spherical charge behaves like a point charge placed at its centre for all outside points. The graphical variation of the field is as shown.

- (iv) Electric field due to line charge of infinite length having a linear charge density λ is given by

$$E = \frac{\lambda}{2\pi\epsilon_0 r}$$

where r is distance from the line charge. Direction of field is normal to the line (i.e., radial). Variation of E with r is as shown.

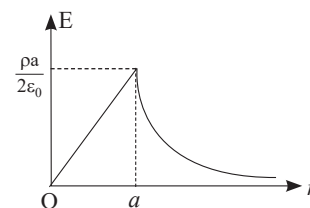


- (v) If we have a long cylindrical shell having λ charge on its unit length, then field at all points inside the cylinder is zero and at outside points the field is given by $E = \frac{\lambda}{2\pi\epsilon_0 r}$

- (vi) If we have a long cylindrical volume of charge having density ρ , then the electric field at a distance r from the axis of the cylinder is

$$E = \left(\frac{\rho}{2\epsilon_0} \right) r \text{ for } r \leq a$$

$$= \frac{\rho a^2}{2\epsilon_0 r} \text{ for } r \geq a$$



where a is radius of the cylindrical volume.

Variation of field (E) with r is as shown in the graph.

- (vii) Electric field in front of a uniformly charged plane surface is uniform. It is normal to the surface and its magnitude is $E = \frac{\sigma}{2\epsilon_0}$.

11. SPHERICAL CHARGE WITH CAVITY

The problem that we are going to consider is a beautiful illustration of use of principle of superposition. A spherical volume of radius a contains a uniform volume charge density ρ . Inside the sphere, there is a spherical region of radius b that is devoid of charge. The centres of the two spheres are l apart. We will try to write electric field inside the cavity as well as outside it.

The charge distribution can be assumed to be composed of a uniformly charged sphere of radius a having charge density ρ and superimposed on it a spherical region of radius b containing charge density $-\rho$. The space having both $+\rho$ and $-\rho$ charge density has no net charge inside it. This is our cavity space.

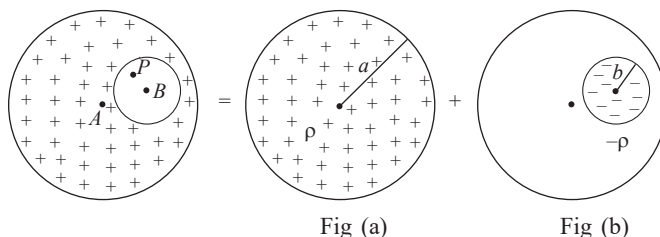


Fig (a)

Fig (b)

Spherical charge with a cavity is equivalent to superposition of two charges: (i) A complete sphere of radius a having charge density $+\rho$ and (ii) A sphere of radius b having charge density $-\rho$. A and B are centres of the two spherical regions.

Suppose we wish to find electric field at a point P inside the cavity. We will write electric field at P due to the positive

charge shown in figure (a). Then we will write the field at P due to the negative charge shown in figure (b). We will add the two field vectorially to get the resultant field at P .

Field due to positive charge at point P is given by equation (20).

$$\vec{E}_+ = \frac{\rho}{3\epsilon_0}(\vec{AP})$$

[Field is directed radially along $\vec{AP}$ and distance r in equation (20) is nothing but AP]

Field due to the negative charge, at P is

$$\vec{E}_- = \frac{\rho}{3\epsilon_0}(\vec{PB})$$

[Field due to negative charge is radially inward.]

$\therefore$ Field at P is

$$\begin{aligned}\vec{E} &= \vec{E}_+ + \vec{E}_- = \frac{\rho}{3\epsilon_0}(\vec{AP} + \vec{PB}) \\ \Rightarrow \vec{E} &= \frac{\rho}{3\epsilon_0} \vec{AB} = \frac{\rho}{3\epsilon_0} \vec{l}\end{aligned}\quad (26)$$

This is an amazing result. Field inside the cavity is uniform. Its magnitude is same at all points inside the cavity and its direction is along the line joining the centres of two spherical regions (i.e., along $\vec{AB}$).

Now consider a point P_1 that is outside the cavity but lies within the charged sphere. Let the position vector of the point with respect to the centre A be $\vec{r}(= \vec{AP}_1)$.

Field at P_1 can be written as

$$\vec{E} = \vec{E}_+ + \vec{E}_-$$

$$\vec{E}_+ = \frac{\rho}{3\epsilon_0} \vec{r}$$

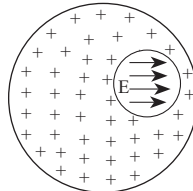
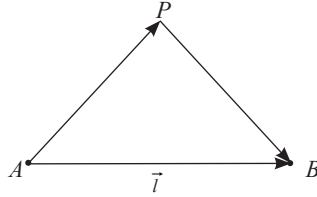
$$\text{and } \vec{E}_- = \frac{1}{4\pi\epsilon_0} \frac{q}{(BP_1)^3} (\vec{BP}_1)$$

$$\text{where } \vec{BP}_1 = \vec{r} - \vec{l}$$

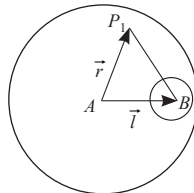
q = negative charge in sphere of radius b

$$= -\frac{4}{3}\pi b^3 \rho$$

$$\therefore \vec{E}_- = \frac{1}{4\pi\epsilon_0} \left(\frac{-\frac{4}{3}\pi b^3 \rho}{|\vec{r} - \vec{l}|^3} \right) (\vec{r} - \vec{l})$$



Field inside the cavity is uniform



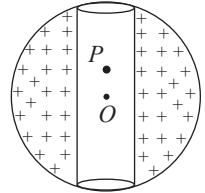
$$= -\frac{\rho}{3\epsilon_0} \left(\frac{b}{|\vec{r} - \vec{l}|} \right)^3 (\vec{r} - \vec{l})$$

$$\therefore \vec{E} = \vec{E}_+ + \vec{E}_- = \frac{\rho}{3\epsilon_0} \left[\vec{r} - \left(\frac{b}{|\vec{r} - \vec{l}|} \right)^3 (\vec{r} - \vec{l}) \right]$$

Proceeding in a similar way one can write the electric field at a point that lies outside the charged sphere.

Example 27 Charged ball with a hole

A spherical region of radius a carries a uniform charge density ρ . There is a cylindrical cavity (region containing no charge) running along a diameter of the sphere. The axis of the cylinder lies along the diameter and radius of the cylinder is $\frac{a}{10}$. Find electric field at



a point P on the axis of the cylindrical cavity. Distance of point P from the centre (O) of the sphere is x ($\ll a$).

Solution

Concepts

- The charge system can be thought to be superposition of two charges: (a) a positively charged sphere having charge density $+\rho$ and (b) a negatively charged cylinder having charge density $-\rho$.
- Equation (20) gives us field inside a uniformly charged sphere. Equation (24) gives field inside a uniformly charged long cylinder.
- Since $x \ll a$, the concerned point is close to the centre of the cylinder and we can assume our cylinder to be long enough for equation (24) to remain valid.

Field at P if there were no cavity and the entire sphere was filled with a uniform charge density ρ

$$E_+ = \frac{\rho}{3\epsilon_0} x \quad (\text{along } \vec{OP})$$

Consider the cylindrical region filled with charge density $-\rho$. Field due to this negative charge on the axis of the cylinder is zero. [$r=0$ in equation 24]

$$\therefore E_- = 0$$

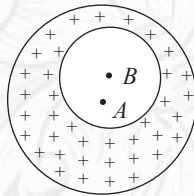
$$\therefore \text{Field at } P \text{ is } E = E_+ + E_-$$

$$= \frac{\rho}{3\epsilon_0} x \quad (\text{along } \vec{OP})$$

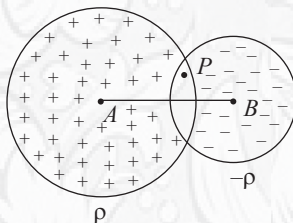


YOUR TURN

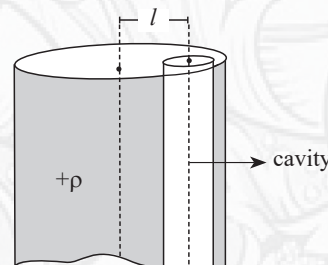
Q.55 A spherical region has a uniform charge density ρ . Centre of the spherical region is point A . A spherical portion of the charge is removed creating a cavity. The cavity region has its centre at B . Find electric Field intensity at points A and B . It is given that distance between A and B is r .



Q.56 Find the electric field intensity $\vec{E}$ in the region of intersection of two spheres which are uniformly charged by unlike charges with the volume densities ρ and $-\rho$. A and B are the centres of the two spheres and $\overline{AB} = \vec{l}$.



Q.57 A cylindrical volume of infinite length has a charge density ρ . Charge is removed from a cylindrical part to form a cavity inside the charged cylinder. The axis of cavity is parallel to the axis of the cylinder and separation between them is l .



In Short

- (i) If there is a spherical cavity (no charge) inside a uniformly charged spherical volume, field inside it is uniform. Direction of field is along the line joining the centre of charged sphere to the centre of cavity. If charge is positive, field is towards the centre of cavity and if the charge is negative, field is directed towards the centre of the charged sphere.

Magnitude of field is $\frac{\rho}{3\epsilon_0}l$, where l is distance between the centre of charged sphere and the centre of the cavity.

- (ii) In fact, if two spherical regions having charge density $+\rho$ and $-\rho$ overlap, the field inside the overlapping region is uniform given by $\frac{\rho\vec{l}}{3\epsilon_0}$. $\vec{l}$ is a vector from the centre of positive charge to the centre of negative charge.

- (iii) For a uniformly charged long cylinder with a cylindrical cavity, field inside the cavity is $\frac{\rho l}{2\epsilon_0}$ if cavity axis is parallel to the cylinder axis.

12. ELECTROSTATIC PROPERTIES OF CONDUCTORS

In an insulator, each electron is attached to a particular atom. When placed in an electric field, these electrons are not able to move. In a metallic conductor, by contrast, one or more electrons per atom are free to roam at will inside the

material. Even a small electric field causes these electrons to move opposite to the direction of electric field. Therefore, a conductor is a material having plenty of free electrons inside it.

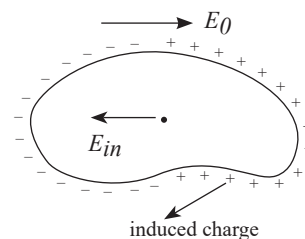
Here, we present few properties and behaviours of conductors under electrostatic condition. We will study few more properties after studying about electric potential in the next chapter.

Electric field is zero inside a conductor

If we are able to maintain an electric field inside a conductor, the field will apply force on electrons and the free electrons will be moving opposite to the direction of the field. We say, there is a current in the conductor. This is a subject matter that we will study later. Here we are interested in discussing *electrostatics* – a situation where charges are static.

If one finds that charges (i.e., free electrons) in a conductor are at rest, he will conclude that electric field at each point inside the 'MEAT' of the conductor is zero. Any presence of field will result in movement of charge.

Suppose there is a uniform electric field $\vec{E}_0$ in a region (created by some source charges). You place a solid piece of a conductor in the field. Free electrons experience a force (see figure) and run towards left. The left face of the conductor becomes negatively charged. The



Field due to induced charge is equal and opposite to external field at each and every point inside the conductor.

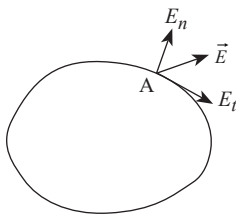
right face gets electron deficient and becomes positively charged. These induced charges on the surface of the conductor produce their own field (E_{in}). The movement of charge in the conductor stops only when E_{in} cancels the external field E_0 at each and every point in the *meat*. This is amazing.

When placed in an external field, the induced charges appear on the surface of a conductor to ensure that resultant field is zero at all points inside it. All this rearrangement of charge takes a fraction of a second. Once the field is zero in the *meat*, there is no more movement of charge. This is electrostatic condition.

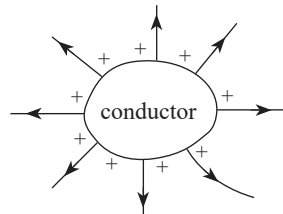
Electric field lines and a conductor

Under electrostatic condition, field inside a conductor is zero. It means density of field lines inside the *meat* of a conductor is zero. No field line can pierce through a conductor.

On the surface of a conductor, field lines (if present) are always normal. A tangential component of field at the surface will mean that the free electrons experience a tangential force causing them to move. For no motion of charge (i.e., for static condition), there should not be any tangential component of electric field.



If direction of field at A is along $\vec{E}$, its tangential component will cause the free electrons to move.



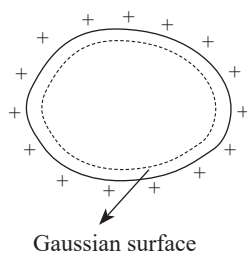
Field lines for a positively charged conductor. Field lines on the surface are normal to the conductor. There are no lines inside the conductor.

Charge resides on surface

If a conductor is given some charge, the charge resides on its surface (i.e., skin). In the *meat* of the conductor there is no charge. This result can be obtained easily by using the Gauss's law.

Consider a piece of conductor that is given some charge q . Under electrostatic condition, the field is zero inside the conductor. If we construct an imaginary Gaussian surface just under the skin of the conductor (see figure), the flux through this surface is zero ($\because E = 0$ everywhere on the surface). Using the Gauss's law we can conclude that charge present inside the closed Gaussian surface is zero.

In fact, if you go on reducing the size of your Gaussian surface. You still get the same answer. There is no charge

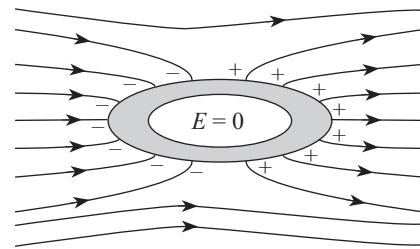


inside any small volume considered in the conductor. Volume charge density is zero everywhere.

Where is the charge located? Obviously, on the surface. This is quite intuitive also. The charges repel each other and spread on the surface (maximum distance apart from one another).

Electrostatic shielding

Think of a piece of conductor having a cavity inside it. This conductor is placed in an external electric field E_0 . We will assume that the external field is uniform, though it makes no difference to the result we are going to obtain even if the field is non-uniform.



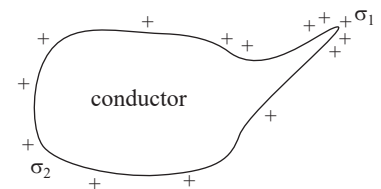
The induced charges on the surface of the conductor ensure that there is no field in the *meat*. Field lines terminate normally on the surface at negative charge and new lines originate at positive charge on the surface. No field line can pierce through the conductor to reach the cavity region. *There is no field inside the cavity.* Thickness of the wall of the conductor between the cavity and the outer surface is of no concern. Even for a small thickness of the wall $E = 0$ inside the cavity. This phenomenon is called *electrostatic shielding*.

If you have some sensitive electronic instrument which gives incorrect readings in presence of an external field, you can keep it inside a metallic box.

If lightning strikes (it happens due to very high electric field) and you are inside your car, you are safe. No electric field can enter the cabin of your car piercing through its metal body. Remember not to touch the metal parts of the car body.

Charge density on the surface

Any charge given to a conductor spreads on its surface. If the conductor is spherical, the charge density on its surface is uniform due to symmetry. For non spherical conductor this surface charge density is not uniform. *Surface charge density on a conductor is inversely proportional to the local radius of curvature of the surface.* At sharp

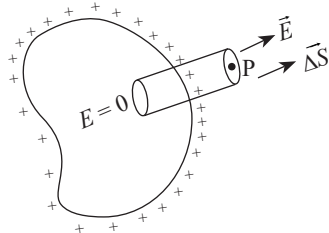


Non uniform charge density on surface of a conductor $\sigma_1 > \sigma_2$. Charge density is high at sharp edges.

edges (i.e., where radius of curvature is small) the charge density is high. We will put argument in favour of this result in next chapter after studying about electric potential.

Electric field near the surface of a conductor.

Charge is spread on a conductor surface. We wish to find electric field at a point P which is very close to the surface. The local charge density (i.e., surface charge density at location close to P) is σ .



Consider a cylindrical Gaussian surface as shown. The flat circular face has point P at its centre and it has a small area ΔS . This circular face is parallel to the conductor surface. The curved cylindrical wall is normal to the conductor surface. The other circular face of the cylinder is inside the conductor.

Since field lines are normal to the surface of a conductor, all field lines are parallel to the curved wall of cylindrical Gaussian surface and they are perpendicular to the flat circular face at P . Field is zero inside a conductor. Hence, field at the other circular face is zero.

$$\therefore \text{Flux through the Gaussian surface is}$$

$$\phi = E \Delta S \cos 0^\circ = E \Delta S$$

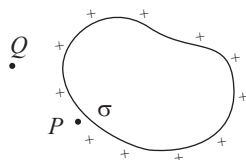
charge enclosed inside the cylinder is $\sigma \Delta S$ which is present on the surface of the conductor.

$$\therefore E \Delta S = \frac{\sigma \Delta S}{\epsilon_0}$$

$$\Rightarrow E = \frac{\sigma}{\epsilon_0} \quad (27)$$

Therefore, we conclude that field near the surface of a conductor is normal to the surface and has a magnitude $\frac{\sigma}{\epsilon_0}$ where σ is the local surface charge density.

Remark: Consider a charged conductor. A charge Q is near the conductor. Surface charge density at a location is σ and we are supposed to write field at a point P very close to this location. The field at P is normal to the surface given by $\frac{\sigma}{\epsilon_0}$. Will the presence of Q make no difference?



Presence of Q certainly makes a difference. In presence of Q value of σ will be different from its value when Q is not there. Therefore, values of field are different at P in presence and absence of Q . However, the expression of electric field is still given by equation (27).

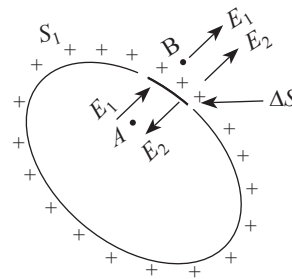
Pressure on the Conductor's Surface

Charges lying on the surface of a conductor repel each other. If you consider a small area ΔS on the surface, it is being repelled outward by the remaining charge. We wish to find force ΔF acting on this small area ΔS .

For calculating this force we need to find electric field at the location of area ΔS that is produced by the charges present on the surface other than the charge present on area ΔS .

The entire conductor surface is mentally divided into two parts – the small patch having area ΔS and the remaining surface S_1 .

First consider a point A that is just inside the surface ΔS . We know that field at this location must be zero, because this is a point in the *meat*. At this point field due to charge on



E_1 is field due to S_1 and E_2 is field due to ΔS . A and B are points very close to the surface.

ΔS is $E_2 = \frac{\sigma}{2\epsilon_0}$ directed normally inwards. Here σ is surface charge density on ΔS . [Point A extremely close to the surface and ΔS appears like a large area].

For field to be zero at A , the charge on S_1 must produce an equal but opposite field. Therefore, field at A due to charge on S_1 is $E_1 = \frac{\sigma}{2\epsilon_0}$ normal to the surface outwards.

Now, points A and B are very close to each other (the drawing is a bit misleading. Points A and B are just inside and just outside the surface) and the field due to S_1 cannot be different at these two points. Therefore, field at B (or, at the surface) due to charge on S_1 is $E_1 = \frac{\sigma}{2\epsilon_0}$ normal to the surface outwards. Charge on ΔS is $\Delta q = \sigma \Delta S$.

Force on Δq due to E_1 is

$$\Delta F = \Delta q E_1 = (\sigma \Delta S) \left(\frac{\sigma}{2\epsilon_0} \right)$$

$$\Rightarrow \Delta F = \frac{\sigma^2}{2\epsilon_0} \Delta S$$

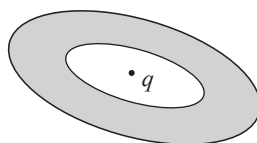
Force per unit area or pressure is

$$P = \frac{\Delta F}{\Delta S} = \frac{\sigma^2}{2\epsilon_0} \quad (28)$$

This is the *electrostatic pressure* that a conductor's surface experiences when charge is put on it. The remarkable thing about the above expression is that it depends on local charge density only.

Example 28 Charge inside cavity

A conducting block has a cavity. There is a point charge inside the cavity.

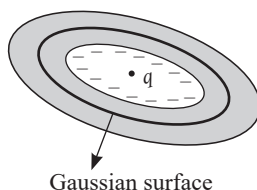


- Find charge induced on the surface of the cavity.
- Draw electric field lines in the entire space.

Solution**Concepts**

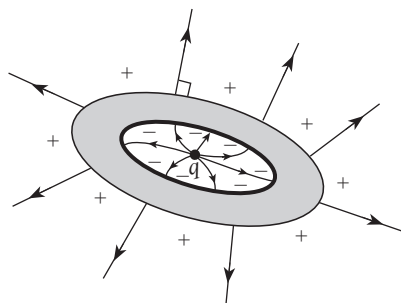
- If q is positive, then it attracts electrons and negative charge is induced on the surface of the cavity.
Conductor is neutral. Therefore, equal amount of positive charge appears on the outer surface.
- Charge appearing on the cavity wall can be found by considering a Gaussian surface through the conductor.
- Field lines are always normal to the surface of the conductor. There are no field lines inside the meat of the conductor.

- Consider a Gaussian surface as shown in the figure. Electric field is zero at each point on the surface. Therefore, flux through the surface is zero. The Gauss's law tells us that total charge inside this surface must be zero.



This leads us to conclude that the cavity wall has exactly $-q$ charge on it.

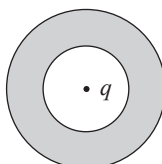
- Since the overall conductor is neutral, $+q$ charge appears on the outer surface.



Field lines starting from $+q$ charge terminate at $-q$ charge on the cavity wall. Lines starting from outer surface of the conductor terminate at infinity.

Example 29 Charge on outer surface does not know what is happening inside cavity

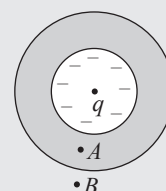
A spherical metal ball has a concentric cavity. A point charge q is placed at the centre as shown.



- Is the induced charge appearing on the cavity wall distributed uniformly? What about the charge on the outer surface of the ball? Is that distributed uniformly?
- The point charge inside the cavity is moved away from the centre. Will the charge distribution on the outer surface of the ball change?

Solution**Concepts**

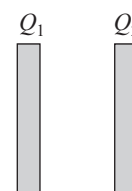
The field lines originating from the point charge terminate on the surface of the cavity. This means that the charge system composed of the point charge q and $-q$ charge on the cavity wall create field inside cavity only. There is no field due to this charge system at any point outside the cavity. The system of $+q$ and $-q$ charge shown in the figure produces no field at point A or B . At B we have field due to charge appearing on outer surface of the ball.



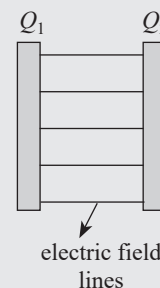
- When point charge q is at the centre, the induced charges on inner as well as outer surface of the conductor are distributed uniformly. This is just because of symmetry.
- If q is moved inside the cavity, the distribution of charge $-q$ on the cavity wall changes. It becomes non uniform. But the system of charge inside cavity ($+q$ and $-q$) still has no influence outside the cavity. The charge on the outer surface never knows about movement of q . Distribution of the charge on outer surface is still uniform.

Example 30 Parallel Plates

Two large and identical metal plates are kept parallel to each other. They are given charge Q_1 and Q_2 . Find the charges on four faces.

**Solution****Concepts**

- When plates are large, we can assume that the electric field produced is normal to the surface. This assumption is reasonable even when the plates are of finite size but separation between them is small and we are interested in electric field in the space between them.
- Electric field in the meat of both the conductors must be zero. Charges will distribute on the four surfaces so as to ensure this.



Consider a box shaped Gaussian surface as shown of this in the figure. Two faces of this surface lies inside metal where $E = 0$. There is no flux through these faces. All other faces of the box are parallel to the electric field. There is no flux through these faces as well. Total flux through the box is zero. According to the Gauss's law, total charge inside this Gaussian surface must be zero. Therefore, the inner faces of the two plates must have equal and opposite charges on them.

The diagram illustrates two parallel vertical plates, labeled Q_1 and Q_2 , representing a parallel plate capacitor. A dashed rectangular box, labeled 'Gaussian surface', is positioned between the plates. An arrow labeled E points from the left plate (Q_1) towards the right plate (Q_2), indicating the direction of the electric field. A label 'Direction of electric field can be from left to right or from right to left' points to the E arrow.

Let charge on the inner wall of the left plate be x . Then charge on the inner surface of the other plate is $-x$. Knowing that the total charge on the left plate is Q_1 , we can write the charge on its left face as $(Q_1 - x)$. Similarly, charge on the outer face of the other plate is $(Q_2 + x)$.

The electric field at a point P inside the metal plate must be zero. E_1 , E_2 , E_3 and E_4 are fields at P due to charges on faces 1, 2, 3, and 4 respectively. Carefully look at their directions.

$$\vec{E}_1 + \vec{E}_2 + \vec{E}_3 + \vec{E}_4 = 0$$

$$\frac{Q_1 - x}{2A\epsilon_0}(\rightarrow) + \frac{x}{2A\epsilon_0}(\leftarrow) + \frac{x}{2A\epsilon_0}(\rightarrow) + \frac{Q_2 + x}{2A\epsilon_0}(\leftarrow) = 0$$

[Field due to charge on a large surface $\frac{\sigma}{2\epsilon_0} = \frac{q}{2A\epsilon_0}$]

$$\Rightarrow Q_1 - x + x = x + Q_2 + x$$

$$\Rightarrow x = \frac{Q_1 - Q_2}{2}$$

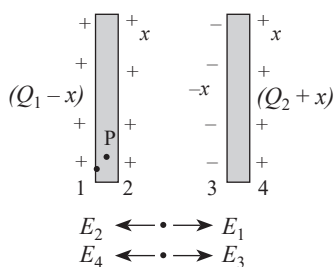
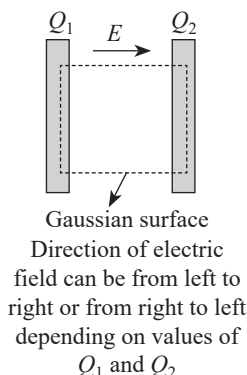
Thus charges on the four faces are

$$q_1 = Q_1 - x = \frac{Q_1 + Q_2}{2}$$

$$q_2 = x = \frac{Q_1 - Q_2}{2}$$

$$q_3 = -x = -\left(\frac{Q_1 - Q_2}{2}\right)$$

$$q_4 = Q_2 + x = \frac{Q_1 + Q_2}{2}$$



$$\begin{array}{ccccccc} & & + & & - & & + \\ & & + & & - & & + \\ \frac{Q_1 + Q_2}{2} & + & \left(\frac{Q_1 - Q_2}{2} \right) & - & & + & \frac{Q_1 + Q_2}{2} \\ & + & & - & & + & \\ & + & - & \left(\frac{Q_1 - Q_2}{2} \right) & + & & \\ & + & & & + & & \end{array}$$

Remarks:

- (i) While deriving equation (i) we could have considered charge Q_2 on the right plate together, rather than dividing it into two parts: $(-x)$ and $(Q_2 + x)$.

$$\frac{Q_1 - x}{2A\epsilon_0}(\rightarrow) + \frac{x}{2A\epsilon_0}(\leftarrow) + \frac{Q_2}{2A\epsilon_0}(\leftarrow) = 0$$

$$\Rightarrow Q_1 - x = x + Q_2$$

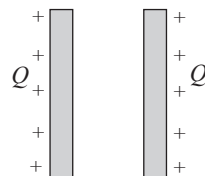
$$\Rightarrow x = \frac{Q_1 - Q_2}{2}$$

Field in front of a large charged sheet does not depend on distance.

- (ii) Let $Q_1 = Q_2 = Q$.

Then $x = 0$.

All charge lies on the outer faces. They repel and move maximum distance from one another.



- (iii) Let $Q_1 = Q$ and $Q_2 = -Q$

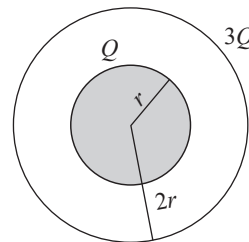
Then $x = Q$

All charge lies on the inner faces. They attract and hold one another. We will study later that such an arrangement of parallel plates having equal and opposite charges is known as a *parallel plate capacitor*.



- (i) **Example 31** *Concentric spheres*

There is a metal ball of radius r having charge Q on it. Concentric to it there is a metal shell of radius $2r$ having charge $3Q$ on it. Find the amount of charge on the inner and the outer surfaces of the shell.



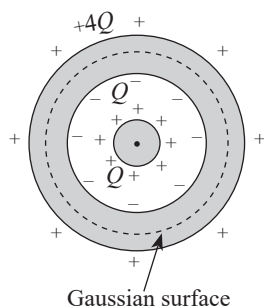
Solution

Concepts

However small, there is some thickness of the metal wall of the outer shell. Inside the wall (in the *meat*) field is zero.

Consider a spherical Gaussian surface passing through the metal of the outer shell. Field at each point on this surface is zero. Flux through this surface is zero. Net charge inside

this surface must be zero. This is possible if the inner wall has a charge $-Q$ on it. The outer wall obviously has $4Q$ charge on it.



The attraction of $+Q$ charge on the free electrons causes them to accumulate on the inner wall of the shell. When this charge becomes $-Q$, it ensures that the field in the *meat* of the conductor is zero.

In Short

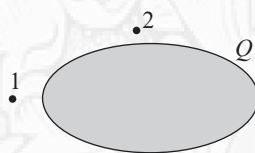
- When charges are static, electric field inside the material of a conductor is zero.
- Any excess charge given to a conductor resides on its surface.
- Electric field lines are normal to the surface of a conductor and are not present inside the material if condition is static.

- Charge density on the surface of a conductor is inversely proportional to local radius of curvature of the surface. Density of charge is high at sharp edges and corners.
- Electric field near the surface of a conductor is $\frac{\sigma}{\epsilon_0}$ in a direction perpendicular to the surface. σ is local surface charge density.
- Force on unit area on the surface of a charged conductor due to repulsion of charges is given by $\frac{\sigma^2}{2\epsilon_0}$ where σ is local surface charge density.
- If charge q is placed inside a cavity in a conductor, induced charge on cavity wall is $-q$.
- If there are two large parallel metal plates carrying charge, the inner faces have equal and opposite charge. If charges on two plates are Q_1 and Q_2 , the inner faces have charge equal to $\pm \left(\frac{Q_1 - Q_2}{2} \right)$.
The outer walls have equal charge equal to $\left(\frac{Q_1 + Q_2}{2} \right)$.

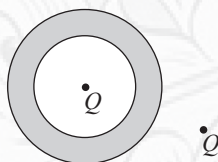


YOUR TURN

Q.58 A conductor has the shape shown in figure. It is given a charge Q . By visual inspection only, tell where the electric field is stronger – at point 1 or at 2. These are two points close to the surface of the conductor.



Q.59 A metal ball has concentric cavity. At the centre of the cavity there is a point charge q .



- What is the shape of electric field lines inside the cavity?
- A charge Q is brought near the ball. How does this affect the field lines inside the cavity?

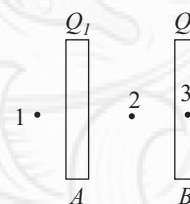
Q.60 A metal ball is placed in a region where exists a uniform electric field. Draw the field lines near the ball.

Q.61 A large isolated metal plate is given a charge Q . Is it necessary that each of the two surfaces has $\frac{Q}{2}$ charge. Here 'isolated' means that the plate is far away from any other charge.

Q.62 Two large metal plates are held parallel to one another. They are given charges Q_1 and Q_2 as shown. Find electric field at points 1, 2, and 3 indicated in the figure for following cases:

- $Q_1 = Q_2 = Q$
- $Q_1 = Q, Q_2 = -Q$
- $Q_1 = Q, Q_2 = 0$

Area of the plates is A . Also show charges on each face.



13. MOTION OF A CHARGED PARTICLE IN AN ELECTRIC FIELD

Force acting on a charged particle in a uniform electric field is:

$$\vec{F} = q\vec{E}$$

This force is in the direction of field when q is positive and is directed opposite to the field when q is negative.

Acceleration of the particle, in absence of any other force is given by

$$\vec{a} = \frac{q}{m} \vec{E}$$

In a uniform field, this acceleration is constant. Therefore, the path of a charged particle in such a field is either a straight line or a parabola. The path is a straight line when the particle is released from rest in a uniform field or when it is projected along the direction of uniform field. The path is parabolic when the initial velocity of the charge makes an angle ($\neq 0^\circ, 180^\circ$) with the direction of the field.

Example 32 A large sheet of charge has surface charge density equal to $\sigma = +8.85 \times 10^{-9} \text{ C/m}^2$. An electron escapes out from the surface travelling perpendicular to the surface at a speed of $5 \times 10^6 \text{ m/s}$.

- How far will the electron travel in front of the surface before stopping for a moment?
- Find the time after which the electron hits back the surface.

Solution

Concepts

- In front of the charged surface, there is a uniform electric field given by $E = \frac{\sigma}{2\epsilon_0}$.
- An electron experiences a force opposite to the electric field, i.e., directed towards the surface. Therefore, it stops after travelling some distance and returns back.

Electric field in front of the surface is

$$E = \frac{\sigma}{2\epsilon_0} = \frac{8.85 \times 10^{-9}}{2 \times 8.85 \times 10^{-12}} = 0.5 \times 10^3 \text{ N/C}$$

Force on the electron, $F = eE$

$$\begin{aligned} \text{Acceleration, } a &= \frac{F}{m} = \frac{e}{m} E \\ &= \frac{1.6 \times 10^{-19}}{9.1 \times 10^{-31}} \times 0.5 \times 10^3 \\ &= 8.8 \times 10^{13} \text{ m/s}^2 \end{aligned}$$

- When the electron moves in front of the surface, it retards. Let's assume that it stops after travelling a distance x

$$\begin{aligned} v^2 &= u^2 + 2ax \\ \Rightarrow 0 &= (5 \times 10^6)^2 - 2(8.8 \times 10^{13})x \\ \Rightarrow x &= 0.15 \text{ m} = 15 \text{ cm} \end{aligned}$$

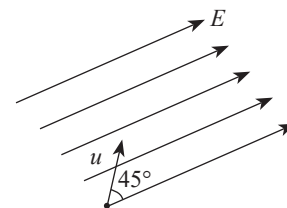
- Time taken in forward motion is given by

$$\begin{aligned} v &= u + at \\ 0 &= 5 \times 10^6 - 8.8 \times 10^{13} t \\ \Rightarrow t &= 5.7 \times 10^{-8} \text{ s} \end{aligned}$$

Time for return journey will be also equal to $5.7 \times 10^{-8} \text{ s}$.

$\therefore$ Time for the electron to return back to the surface is $= 2t = 1.14 \times 10^{-7} \text{ s}$

Example 33 A particle having mass m and charge q is projected in a uniform electric field E with a velocity u making an angle 45° with the direction of electric field. After what time the speed of the particle will be equal to $2u$? Assume no other force apart from the electric force.



Solution

Concepts

Force on the charge is qE and its acceleration is constant equal to $a = \frac{qE}{m}$.

Motion of the particle will be identical to motion of a projectile in uniform gravitational field. The particle will move on a parabola. Velocity component perpendicular to the direction of E will remain unchanged.

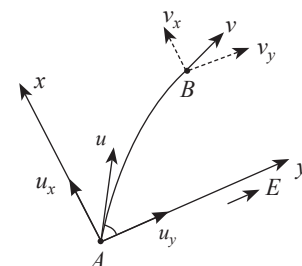
Consider the direction along E to be the y direction.

$$u_x = \frac{u}{\sqrt{2}}, u_y = \frac{u}{\sqrt{2}}$$

$$a_x = 0, a_y = \frac{qE}{m}$$

Let the speed of particle at some point B be equal to $2u$.

$$\begin{aligned} v_y^2 + v_x^2 &= (2u)^2 \\ \Rightarrow v_y^2 + \left(\frac{u}{\sqrt{2}}\right)^2 &= 4u^2 \\ \Rightarrow v_y^2 &= \frac{7}{2}u^2 \\ \Rightarrow v_y &= \sqrt{\frac{7}{2}} \cdot u \end{aligned}$$



Velocity component perpendicular to E ($= u_x$) does not change.

We need to find time in which the y component of velocity changed from u_y to v_y .

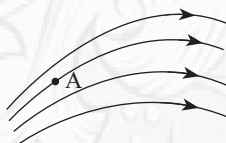
$$\begin{aligned} v_y &= u_y + a_y t \\ \Rightarrow \sqrt{\frac{7}{2}} \cdot u &= \frac{u}{\sqrt{2}} + \frac{qE}{m} t \quad \Rightarrow t = \frac{mu}{qE} \left(\frac{\sqrt{7}-1}{\sqrt{2}} \right) \end{aligned}$$



YOUR TURN

Q.63 A particle of mass m has a charge $-q$ on it. It is projected horizontally and it manages to move with constant velocity. Find the electric field existing in the region.

Q.64 Electric field lines in a region of space is as shown. A charge particle is released at point A . Will it move along a field line?

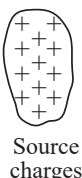


14. CONSERVATIVE NATURE OF COULOMB FORCE

Force applied by a source charge (Q) on another charge (q) is a radial force. It is always directed along the line joining the two charges and depends on distance r only. Such forces are always conservative. Work done by (a conservative) electrostatic force is independent of path and work done in a closed path is zero.

You may revisit the chapter on work, energy and power for further discussions on this.

If a charge q is moved from A to B in an electric field (created by some source charge), work done on it by the electric force is same for all the paths shown in the figure

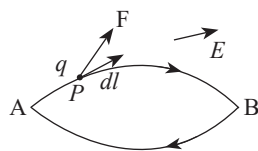


Source charges

$$W_I = W_{II} = W_{III}$$

If the test charge q is moved from A to B along path I and is then brought back to A along the path III, total work done on it by the electric force in the round trip ($A \rightarrow B \rightarrow A$) is zero.

Again consider a test charge q which is moved along a closed path $A \rightarrow B \rightarrow A$. At any point P on the path, electric force on the charge is $\vec{F}_e = q\vec{E}$



In a small displacement $d\vec{l}$, work done by the electric force is

$$dW = \vec{F}_e \cdot d\vec{l} = q\vec{E} \cdot d\vec{l}$$

Work done by the electric force in the round trip is

$$W = q \oint \vec{E} \cdot d\vec{l}$$

Since the electric force is conservative, its work in a closed path is zero. Therefore, we conclude that

$$\oint \vec{E} \cdot d\vec{l} = 0 \quad (29)$$

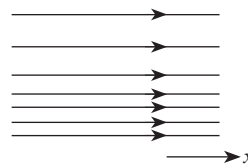
In words, we say that line integration of electrostatic field over a closed path is zero.

Q.65 A small ball has a mass $m = 1\text{ kg}$ and carries a charge $q = 2\text{ }\mu\text{C}$. It is thrown from a horizontal surface at a speed of 20 m/s making an angle of $\theta = 45^\circ$ with the horizontal. The entire space has a horizontal electric field $E = 2 \times 10^7\text{ N/C}$. At what distance from the point of projection will the particle again hit the horizontal surface.

[In a later chapter we will study that electric field has one more source apart from static charges. Such electric fields are not conservative. However, we will not bother ourselves with such things right now].

Example 34 This pattern of electrostatic field lines is not possible

In a region of space electric field is directed along x direction. A student draws field lines as shown. His teacher told him that such field lines are just not possible. Using equation (29) try to guess what the teacher was thinking.

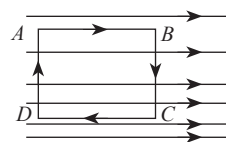


Solution

Concepts

- Density of lines is increasing as we move down. This implies that the magnitude of the electric field increases as we move down.
- In a closed path line integral of E must be zero. If it is non zero, then such field is not possible.

Consider a rectangular closed path $ABCD$ with $AB = L$.



Let Electric field on line $AB = E_1$ and Electric field on line $DC = E_2$. Certainly $E_2 > E_1$.

$$\begin{aligned} \oint \vec{E} \cdot d\vec{l} &= \int_A^B \vec{E} \cdot d\vec{l} + \int_B^C \vec{E} \cdot d\vec{l} + \int_C^D \vec{E} \cdot d\vec{l} + \int_D^A \vec{E} \cdot d\vec{l} \\ &= E_1 \cdot L + 0 + (-E_2 \cdot L) + 0 \end{aligned}$$

While travelling along BC and DA the displacement is perpendicular to the electric field, and the dot product is zero.

$$\therefore \oint \vec{E} \cdot d\vec{l} = (E_1 - E_2)L$$

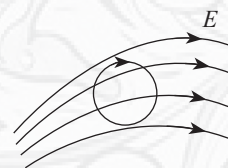
Because $E_1 \neq E_2$, the line integral is not zero. This implies that the field lines shown in the figure are not possible.



YOUR TURN

Q.66 A charge q is moved along a circular path in a non uniform electric field (see figure). In the first half of

the circle work done by the electric field is W . Can you find work done by the electric field in second half of the circle?



In Short

- (i) Electrostatic force is conservative. Work done by electrostatic force on a moving charge is independent of path. And, work done by electrostatic force on a closed path is zero.
- (ii) Line integral of electric field on a closed path is zero $\oint \vec{E} \cdot d\vec{l} = 0$.

15. GAUSS'S LAW IN GRAVITATION

Gravitational force between two point masses also follows inverse square law. Therefore, we can write Gauss's law for gravitational field also.

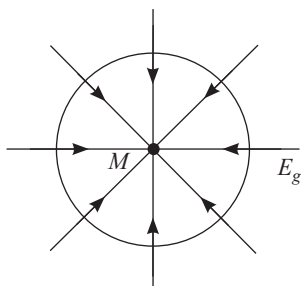
Electric field due to a point charge is $E = \frac{1}{4\pi\epsilon_0} \frac{q}{r^2}$

Gravitational field due to a point mass is $E_g = \frac{GM}{r^2}$

Gravitational field is always directed towards the source point mass (you can compare it with a field produced due to negative charge).

Consider a spherical Gaussian surface surrounding a point mass M . Flux through the surface is

$$\phi_g = \oint \vec{E}_g \cdot d\vec{S} = -E_g \cdot 4\pi r^2$$



Gravitational field due to a point mass spherical surface is the Gaussian surface.

We have taken negative sign as field lines are piercing into the surface and the area vector is taken to be positive in outward direction. If you wish to do away with this negative sign, you can consider area as a vector directed inwards.

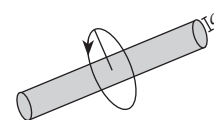
$$\therefore \phi_g = -\frac{GM}{r^2} \cdot 4\pi r^2$$

$$\Rightarrow \phi_g = -(4\pi G)M \quad (30)$$

This is Gauss's law for gravitation.

Example 35 A cylindrical planet

Imagine a peculiar planet that is in the shape of a long cylinder of radius a . Mass per unit length for the planet is μ . A satellite of mass m revolves around it in a circular orbit of radius r . Find its orbital speed.



Solution

Concepts

- (i) The gravitational field due to the planet can be written in the same way as we did for a line charge in section 10.2.

For a line charge of large length

$$E = \frac{\lambda}{2\pi\epsilon_0 r} = \frac{2\lambda}{4\pi\epsilon_0 r}$$

Gravitational field can be written by replacing λ with μ and $\frac{1}{4\pi\epsilon_0}$ with G .

- (ii) Writing the equation for centripetal force will give us the orbital speed of the satellite.

Gravitational field can be obtained using Gauss's law. We can also directly use equation

$$E = \frac{\lambda}{2\pi\epsilon_0 r} = \frac{2\lambda}{4\pi\epsilon_0 r}$$

$$E_g = \frac{2G\mu}{r}$$

For the satellite's circular motion:

$$\frac{mv_0^2}{r} = \frac{2G\mu}{r}$$

$$\Rightarrow v_0 = \sqrt{\frac{2G\mu}{m}}$$



YOUR TURN

Q.67 If the Earth were a flat disc of mass M and radius R , what would have been the acceleration due to gravity somewhere near the centre of the Earth.

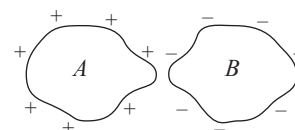
16. DIELECTRIC BREAKDOWN

Dielectrics (insulators) are material which do not have free charge particles. All electrons are bound to their respective atoms (or molecules). When such a material is placed in a region of high electric field, the electrons and the positive nucleus experience forces in opposite directions. An electron may be torn apart from an atom. In a polar molecule, the positive and negative ions may get separated. Usually, the mechanism for breakdown may be complex. But whatever the reason, after breakdown of a dielectric happens, it has plenty of free charged particles. It begins to conduct.

Air is a dielectric. Its breakdown happens when it is subjected to an electric field 3×10^6 N/C or higher. This value of electric field is known as *breakdown strength* of air. Material like rubber, plastic, mica etc. have much higher breakdown strength.

Consider two objects A and B having positive and negative charges respectively. They are brought closer. When gap

between them is small, the air in the gap may suffer breakdown (i.e., get ionized) due to high electric field in the region. Air begins to conduct and charge begins to move from one body to another. Such movement of charge can happen irrespective of the fact whether A and B are conductors or insulators.



Such breakdown of air and jump of charge from one body to another is often seen as spark. Why light is emitted during sparking? I leave this question for you to explore on your own. Possibly you will understand this much better after studying atomic spectrum.

Lightning is just a big spark. Clouds may get charged and create a huge electric field between them and the ground. This causes breakdown of air and charge flows between the cloud and the ground. On an average there are more than 40,000 thunderstorms every day producing more than 8 million lightning flashes every day on our planet.



YOUR TURN

Q.68 Calculate the maximum charge that can be placed on a metal sphere of radius $R = 1.0$ m before the charge

begins to leak into atmosphere due to dielectric breakdown of air.

Miscellaneous Examples

Example 36 Two identical balls, each having density ρ , are suspended from a common point by two insulating strings of equal length. Both the balls have equal charge and can be treated like point charges. In equilibrium, each string makes an angle θ with the vertical. Now the entire system is immersed in a non conducting liquid of density σ . Angle θ does not change. Find the relative permittivity (ϵ_r) of the liquid.

Solution

Concepts

- (i) When the system is in air, forces on a ball are its weight, tension in the string and electrostatic repulsion.

- (ii) When submerged in liquid the electrostatic force reduces by a factor of ϵ_r (there is no change in distance between the balls). One additional force—buoyancy starts acting on the balls.

F = electrostatic repulsion force

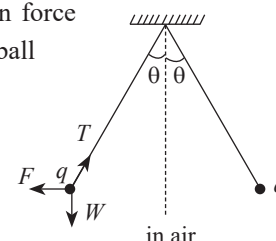
$W = V\rho g$ = weight of a ball
where V is its volume.

T = tension in the string.

For equilibrium:

$$T \sin \theta = F \text{ and}$$

$$T \cos \theta = V\rho g$$



$$\Rightarrow \tan \theta = \frac{F}{V\rho g}$$

Inside the liquid, electrostatic repulsion force becomes

$$F' = \frac{F}{\epsilon_r}$$

Tension charges to $T' \cdot B$ is buoyancy.

For equilibrium,

$$T' \sin \theta = \frac{F}{\epsilon_r} \quad (a)$$

and $T' \cos \theta + B = W$

$$\Rightarrow T' \cos \theta = V\rho g - V\sigma g \quad (b)$$

Taking ratio of (a) and (b)

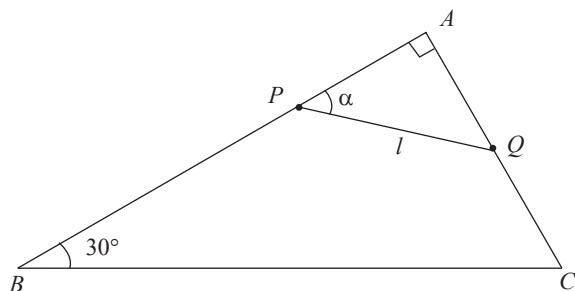
$$\tan \theta = \frac{F}{\epsilon_r V g (\rho - \sigma)} \quad (ii)$$

From (i) and (ii)

$$\frac{F}{V\rho g} = \frac{F}{\epsilon_r V g (\rho - \sigma)}$$

$$\Rightarrow \epsilon_r = \frac{\rho}{\rho - \sigma}$$

Example 37 A rigid wire frame in form of a right angled triangle ABC is set in a vertical plane as shown. The wire frame is made of insulating material. Two beads P and Q have equal mass m and carry charges q_1 and q_2 respectively. They are connected by an insulating cord of length l and are free to slide frictionlessly along the arms AB and AC of the frame. The two beads are in equilibrium in the position shown.



- Find the angle α .
- Find the tension in the cord.
- Find normal force applied by the frame on each bead.

Solution

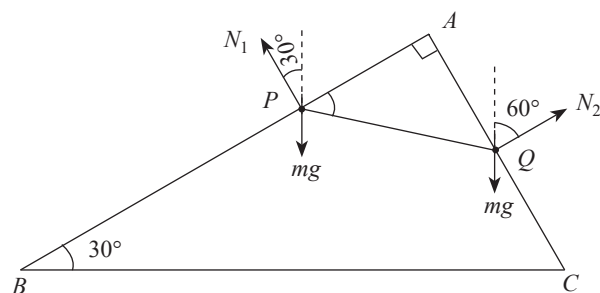
Concepts

- The problem can be solved by drawing the free body diagrams for both the beads and framing two equations for each of them. However, we can

simplify the mathematics to a great extent by first considering the two beads plus the cord as our system. This eliminates the need of considering the tension force and the electrostatic force as both are internal forces.

- To find tension and α , we can consider either P or Q separately.

Considering P and Q (plus the cord) as our system, we have normal reactions (N_1 and N_2) and weight (mg) as our external forces.



For horizontal equilibrium:

$$N_1 \sin 30^\circ = N_2 \sin 60^\circ$$

$$\Rightarrow N_1 = \sqrt{3}N_2 \quad (i)$$

For vertical equilibrium:

$$N_1 \cos 30^\circ + N_2 \cos 60^\circ = 2mg$$

$$\Rightarrow \sqrt{3}N_1 + N_2 = 4mg \quad (ii)$$

Solving (i) and (ii) gives

$$N_1 = \sqrt{3}mg \text{ and } N_2 = mg$$

This is answer to part (c) of the problem. Now we will consider equilibrium of Q .

Forces on Q are:

$$N_2 = mg; \text{ weight} = mg$$

Electrostatic repulsion

$$F = \frac{1}{4\pi\epsilon_0} \frac{q_1 q_2}{l^2}$$

Tension = T

N_2 and mg are forces of equal magnitude inclined at 120° . Their resultant (R) has magnitude = mg and makes an angle of 60° with N_2 .

Resultant of tension and electrostatic force must be equal and opposite to R for equilibrium. Since T and F are two forces along the same line their resultant is simply $T - F$.

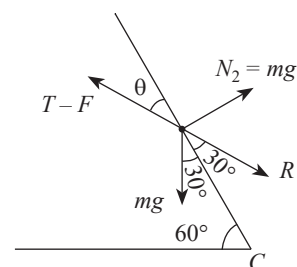
$$\therefore T - F = R$$

$$\Rightarrow T = mg + \frac{1}{4\pi\epsilon_0} \frac{q_1 q_2}{l^2}$$

This is answer to part (b) of the problem.

Obviously $\theta = 30^\circ$

$$\therefore \alpha = 60^\circ$$



Example 38 Three charged particles

Three particles, each of mass m , are suspended from a common point by three insulated massless strings, each having length l . Each of them is given a charge q . The particles repel each other and in equilibrium they settle at the vertices of an equilateral triangle of side length a . Find q .

Solution**Concepts**

- The most important part in this problem is to understand the three dimensional geometry. Due to symmetry, the three particles will lie on a horizontal plane in equilibrium. The point of suspension will be exactly above the centroid of the triangle.
- For any one particle, the tension in string, its weight and electrostatic repulsion by other two charges have zero resultant.

A , B and C are three particles. In the figure, $\triangle ABC$ is an equilibrium triangle. O is the point of suspension and OA , OB and OC are three strings. G is the centroid of the $\triangle ABC$ and OG is a vertical line. Each string makes an angle θ with the vertical.

Electrostatic forces on the particle at A , due to the other two particles are as shown in the figure. The two forces act at an angle of 60° and their resultant is

$$F_0 = \sqrt{F^2 + F^2 + 2FF \cos 60^\circ} = \sqrt{3}F$$

$$\Rightarrow F_0 = \frac{\sqrt{3}}{4\pi\epsilon_0} \frac{q^2}{a^2}$$

The resultant F_0 is horizontal directed along GA . $\triangle OAG$ is a right angled triangle in vertical plane.

For equilibrium of the particle at A we must have:

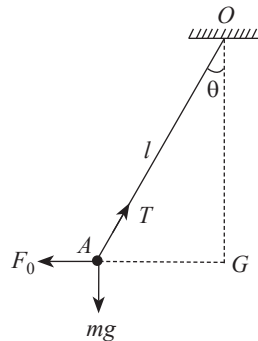
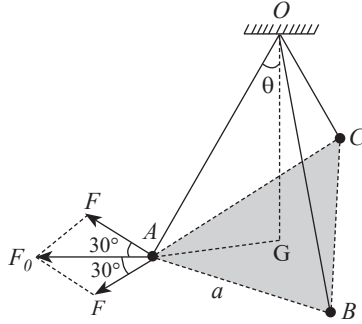
$$T \sin \theta = F_0 \text{ and } T \cos \theta = mg$$

$$\Rightarrow \tan \theta = \frac{F_0}{mg}$$

$$\Rightarrow \tan \theta = \frac{\sqrt{3}}{4\pi\epsilon_0} \frac{q^2}{mg \cdot a^2}$$

Now, we need to express $\tan \theta$ in terms of l and a .

In the equilateral $\triangle ABC$



(i)

$$AG = \frac{2}{3} \left(\frac{\sqrt{3}}{2} a \right) = \frac{a}{\sqrt{3}}$$

$$OG = \sqrt{l^2 - \left(\frac{a}{\sqrt{3}} \right)^2} = \sqrt{l^2 - \frac{a^2}{3}}$$

$$\therefore \tan \theta = \frac{AG}{OG} = \frac{a}{\sqrt{3l^2 - a^2}}$$

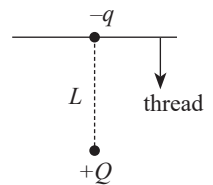
From (i), we get

$$\frac{a}{\sqrt{3l^2 - a^2}} = \frac{\sqrt{3}}{4\pi\epsilon_0} \frac{q^2}{mga^2}$$

$$\Rightarrow q = \left[\frac{4\pi\epsilon_0 mga^3}{\sqrt{9l^2 - 3a^2}} \right]^{\frac{1}{2}}$$

Example 39

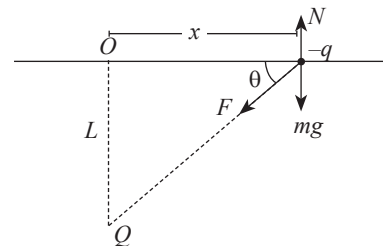
A small bead of mass m has a charge $-q$. It is constrained to move along a horizontal smooth and insulating thread. A positive charge Q lies at a vertical distance L from the thread. The bead is displaced slightly from its equilibrium position and released. Find the time period of oscillation.

**Solution****Concepts**

- This is a routine SHM problem. We need to identify the equilibrium position and then consider the bead at a position displaced from its equilibrium. A component of the Coulomb force provides the necessary restoring force.
- Displacement (x) of the bead is small. This implies that $x \ll L$.

When $-q$ is vertically above Q it is in equilibrium. Consider it at a distance x from its equilibrium. Electrostatic force is

$$F = \frac{1}{4\pi\epsilon_0} \frac{Qq}{(L^2 + x^2)}$$



Restoring force directed towards equilibrium position O is $F \cos \theta$.

$$\therefore ma = -F \cos \theta$$

$$\Rightarrow a = -\frac{Qq}{4\pi\epsilon_0 \cdot m (L^2 + x^2)^{3/2}} x$$

Since $L \gg x$, $\therefore L^2 + x^2 \approx L^2$

$$\Rightarrow a = -\left(\frac{Qq}{4\pi\epsilon_0 \cdot mL^3}\right)x$$

Comparing with the standard equation of SHM: $a = -\omega^2 x$, we get,

$$\omega = \sqrt{\frac{Qq}{4\pi\epsilon_0 mL^3}}$$

$$\Rightarrow T = 2\pi \sqrt{\frac{4\pi\epsilon_0 mL^3}{Qq}}$$

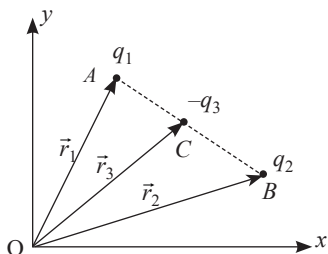
Example 40 Two positive charges q_1 and q_2 are located at points having position vectors $\vec{r}_1$ and $\vec{r}_2$ respectively. A third charge q_3 is negative and remains in equilibrium when placed at a point having position vector $\vec{r}_3$. Find $\vec{r}_3$. Consider only electrostatic forces.

Solution

Concepts

- The answer must be expressed in terms of $\vec{r}_1$, $\vec{r}_2$ and q_1 , q_2 . It is important to express your answer using *vectors* $\vec{r}_1$ and $\vec{r}_2$.
- The third charge must be located somewhere on the line joining q_1 and q_2 .

For equilibrium of q_3 we must have:



$$k \frac{q_1 q_3}{(AC)^2} = k \frac{q_2 q_3}{(BC)^2}$$

$$\Rightarrow \frac{AC}{BC} = \sqrt{\frac{q_1}{q_2}} \quad (i)$$

$$\text{Also } AC + BC = AB \quad (ii)$$

$$\text{Solving: (i) and (ii) gives, } AC = \left(\frac{\sqrt{q_1}}{\sqrt{q_1} + \sqrt{q_2}} \right) AB$$

$$\therefore \vec{AC} = \left(\frac{\sqrt{q_1}}{\sqrt{q_1} + \sqrt{q_2}} \right) \vec{AB} = \left(\frac{\sqrt{q_1}}{\sqrt{q_1} + \sqrt{q_2}} \right) (\vec{r}_2 - \vec{r}_1)$$

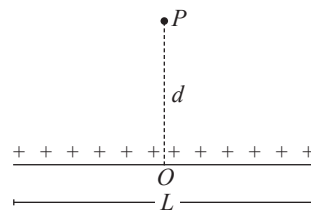
Using triangle rule of vector addition, we can write

$$\begin{aligned} \vec{r}_3 = \vec{r}_1 + \vec{AC} &= \vec{r}_1 + \left(\frac{\sqrt{q_1}}{\sqrt{q_1} + \sqrt{q_2}} \right) (\vec{r}_2 - \vec{r}_1) \\ &= \frac{\vec{r}_1 \sqrt{q_2} + \vec{r}_2 \sqrt{q_1}}{\sqrt{q_1} + \sqrt{q_2}} \end{aligned}$$

Example 41 Field due to a line charge

A line segment of length L carries a uniform linear charge density λ .

- Find the electric field due to the line charge at a point P that lies on the perpendicular bisector of the line at a distance d from it.



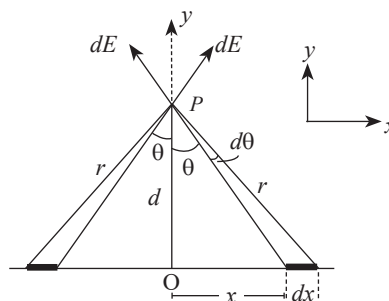
- Approximate the result if $d \gg L$.
- Write the field when $L \rightarrow \infty$.

Solution

Concepts

- Due to symmetry the field will be along OP only.
- We will consider two elements of length dx at a distance x from O . Then we write field at P due to the charge on the element. Component of this field along OP is integrated for getting the resultant field.
- The entire integration can be set up in terms of variable x . But we will express it in terms of variable θ (see figure given below) as it will result in a simpler integration. Those of you who are comfortable with 'integration by substitution method', can try solving the problem with x as variable.

- Consider symmetrically placed pair of elements on the line charge as shown in figure.



$$\frac{x}{d} = \tan \theta$$

$$\Rightarrow \frac{dx}{d\theta} = d \sec^2 \theta$$

$$\Rightarrow dx = d \sec^2 \theta \cdot d\theta$$

= Length of each element.

Charge on each element is, $dq = \lambda dx = \lambda d \sec^2 \theta \cdot d\theta$.
Field at P due to either of the elements has magnitude

$$dE = \frac{1}{4\pi\epsilon_0} \frac{dq}{r^2} = \frac{1}{4\pi\epsilon_0} \frac{\lambda d \sec^2 \theta \cdot d\theta}{(d \sec \theta)^2} = \frac{1}{4\pi\epsilon_0} \frac{\lambda}{d} d\theta$$

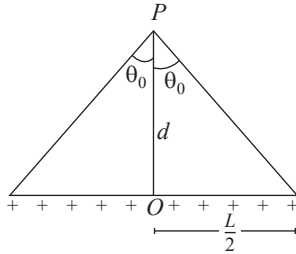
x components of the fields due to the two elements cancel out. The entire line is made of such pair of elements. Hence, there is no field in x direction. To get the field at P we just need to add y component of field due to every element.

$$\therefore E = \int_{-\theta_0}^{\theta_0} dE \cos \theta$$

one can also write the field as

$$E = \int_0^{\theta_0} 2dE \cos \theta$$

You must take note of the limits and understand the difference in thinking while writing the field as in (i) and as in (ii). In (i), $dE \cos \theta$ is component of field in y direction due to one element and therefore, the limit of integration must be applied so as to include all elements from left end of the line to its right end. In (ii), $2dE \cos \theta$ is resultant of fields due to the two elements shown in figure. Therefore, integrating from $\theta = 0$ to $\theta = \theta_0$ includes all elements.



$$\therefore E = \frac{2\lambda}{4\pi\epsilon_0 d} \int_0^{\theta_0} \cos \theta \cdot d\theta = \frac{2\lambda}{4\pi\epsilon_0 d} [\sin \theta]_0^{\theta_0}$$

$$\therefore E = \frac{2\lambda}{4\pi\epsilon_0 d} \sin \theta_0 = \frac{2\lambda}{4\pi\epsilon_0 d} \frac{L/2}{\sqrt{d^2 + \frac{L^2}{4}}}$$

$$\therefore E = \frac{2\lambda L}{4\pi\epsilon_0 d \sqrt{4d^2 + L^2}}$$

(b) If $d \gg L$, $4d^2 + L^2 \approx 4d^2$

$$\therefore E \approx \frac{1}{4\pi\epsilon_0} \frac{\lambda L}{d^2}$$

This is reasonable as the line charge will appear like a point charge from a great distance.

$$(c) E = \frac{2\lambda}{4\pi\epsilon_0 d \sqrt{1 + \frac{4d^2}{L^2}}}$$

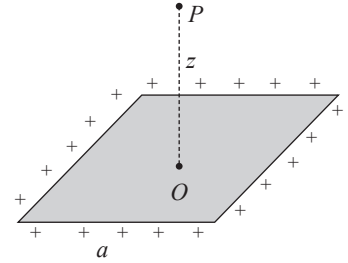
For $L \rightarrow \infty$

$$E = \frac{\lambda}{2\pi\epsilon_0 d}$$

This is the same result that we obtained using the Gauss's law in section 10.2.

Example 42 A square frame

A square loop of side length a has a uniform linear charge density λ along its periphery. The loop lies in a horizontal plane. A particle having charge q is placed exactly above the centre of the square at a height z . The particle stays in equilibrium. Find its mass.



Solution

Concepts

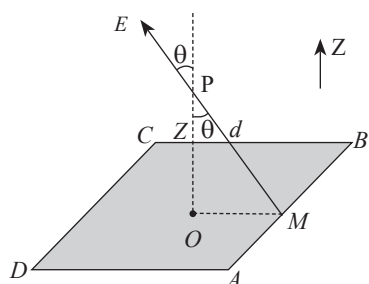
- (i) We will first find electric field due to the square frame at point P . For finding the field we will use the result obtained in the last example.
- (ii) Due to symmetry the field must be along OP . We will find field at P due to charge on one side of the square; take its vertical component and multiply it with 4 to get the resultant field at P .
- (iii) For equilibrium of charged particle at P , the vertical electric force must balance its weight.

Consider side AB of the square. M is mid point of AB and MP is a perpendicular bisector of the side AB .

$$MP = d = \sqrt{z^2 + \frac{a^2}{4}}$$

According to the solution of example 41, the electric field due to AB at P is along MP and its magnitude is:

$$E = \frac{2\lambda a}{4\pi\epsilon_0 d \sqrt{4d^2 + a^2}}$$



z component of E is

$$E_z = E \cos \theta$$

$$= E \cdot \frac{z}{d} = \frac{2\lambda az}{4\pi\epsilon_0 d^2 \sqrt{4d^2 + a^2}}$$

Substituting $d^2 = z^2 + \frac{a^2}{4}$ we get

$$E_z = \frac{2\lambda az}{\pi\epsilon_0 (4z^2 + a^2) \sqrt{(4z^2 + 2a^2)}}$$

For equilibrium of the particle

$$mg = qE_z$$

$$\Rightarrow m = \frac{2q\lambda az}{\pi\epsilon_0 g (4z^2 + a^2) \sqrt{4z^2 + 2a^2}}$$

Example 43 Deflection of charged balls

Two small balls have charges q_1 and q_2 and have same velocity. Both of them enter into a uniform electric field simultaneously. After some time the direction of motion of the first ball changes by 60° and its speed get reduced by half. The direction of motion of the second ball changes by 90° in the same interval.

- How does the speed of second ball change?
 - Find the ratio of specific charges of the two balls.
- Ignore any interaction between the balls.

Solution

Concepts

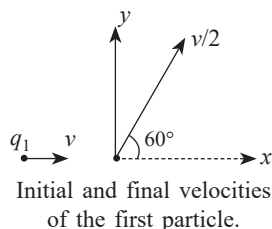
- Specific charge is ratio of charge to mass of a particle.
- Impulse of electric force on each particle is equal to change in momentum of the particle.

Let's assume that both the particles were initially travelling along x direction.

$$\vec{v}_1 = v\hat{i} \text{ and } \vec{v}_2 = v\hat{j}$$

Let the electric field be

$$\vec{E} = E_x\hat{i} + E_y\hat{j}$$



In a time interval Δt , impulse on the first particle is

$$q_1 \vec{E} \Delta t = q_1 \Delta t (E_x \hat{i} + E_y \hat{j})$$

Find velocity of the particle is $\vec{v}_{f1} = \frac{v}{2} \cos 60^\circ \hat{i} + \frac{v}{2} \sin 60^\circ \hat{j}$

$$\Rightarrow \vec{v}_{f1} = \frac{1}{4} v \hat{i} + \frac{\sqrt{3}}{4} v \hat{j}$$

Impulse = change in momentum

$$\Rightarrow q_1 \Delta t (E_x \hat{i} + E_y \hat{j}) = m_1 (\vec{v}_{f1} - \vec{v}_1)$$

$$\Rightarrow q_1 \Delta t (E_x \hat{i} + E_y \hat{j}) = m_1 \left(-\frac{3}{4} v \hat{i} + \frac{\sqrt{3}}{4} v \hat{j} \right)$$

$$\therefore q_1 \Delta t E_x = -\frac{3}{4} m_1 v \quad (a)$$

$$\text{and } q_1 \Delta t E_y = -\frac{\sqrt{3}}{4} m_1 v \quad (b)$$

Similarly, for the second particle we have

$$q_2 \Delta t (E_x \hat{i} + E_y \hat{j}) = m_2 (\vec{v}_{f2} - \vec{v}_2)$$

$$\Rightarrow q_2 \Delta t (E_x \hat{i} + E_y \hat{j}) = m_2 (u_2 \hat{j} - v \hat{i})$$

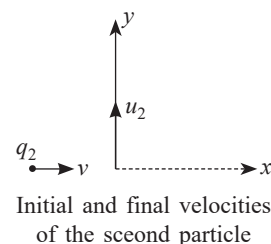
$$\Rightarrow q_2 \Delta t E_x = -m_2 v \quad (c)$$

$$\text{and } q_2 \Delta t E_y = m_2 u_2 \quad (d)$$

Dividing equation (a) by (c) we get

$$\frac{q_1}{q_2} = \frac{3}{4} \frac{m_1}{m_2}$$

$$\Rightarrow \frac{q_1/m_1}{q_2/m_2} = \frac{3}{4}$$



This is the ratio of specific charges of the two particles.

Dividing equation (b) by (d) we get

$$\frac{q_1}{q_2} = \frac{\sqrt{3}}{4} \frac{v}{u_2} \frac{m_1}{m_2}$$

$$\Rightarrow \frac{q_1/m_1}{q_2/m_2} = \frac{\sqrt{3}}{4} \frac{v}{u_2}$$

$$\Rightarrow \frac{3}{4} = \frac{\sqrt{3}}{4} \frac{v}{u_2}$$

$$\Rightarrow u_2 = \frac{v}{\sqrt{3}}$$

Therefore, the final speed of the second particle becomes $\frac{1}{\sqrt{3}}$ times its initial speed.

Example 44 Two point charges $2q$ and $-q$ are placed at a separation. An electric field line is shown to be emanating from $2q$ charge making an angle α with the line joining the two charges. At what angle β (see figure) the line shall be shown terminating at the charge $-q$.



Solution

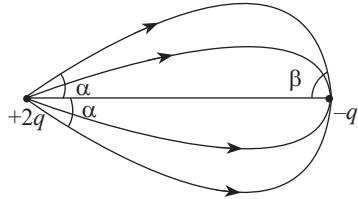
Concepts

- Field lines emanate radially at charge $+2q$. Half of them terminate radially at $-q$. Remaining half terminate at infinity. Very close to a point charge the lines are equally separated from one another (as if there is no other charge nearby).
- A cone of semivertical angle α encloses a solid angle of $\Omega = 2\pi(1 - \cos\alpha)$.

Let N be the number of lines emanating from charge $2q$. $\frac{N}{2}$ lines terminate on $-q$. If we consider a cone of semi vertical angle α with its apex at $+2q$ charge, the cone occupies a solid angle $\Omega = 2\pi(1 - \cos\alpha)$ steradian. A sphere centred at $2q$ charge is also shown in the figure. The sphere subtends a solid angle 4π steradian at the charge. Number of lines contained inside the cone is

$$n_1 = \frac{2\pi(1 - \cos\alpha)}{4\pi} \cdot N$$

Number of lines within angle α at $2q$ = number of lines within angle β at $-q$.



Now consider lines terminating at $-q$ charge. Number of lines terminating at $-q$ which are contained inside a cone of semi vertical angle β is

$$n_2 = \frac{2\pi(1 - \cos\beta)}{4\pi} \cdot \frac{N}{2}$$

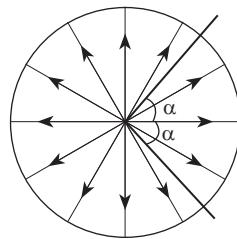
But n_1 must be equal to n_2 (see the first figure).

$$\therefore (1 - \cos\beta) \frac{1}{2} = 1 - \cos\alpha$$

$$\Rightarrow 2\sin^2 \frac{\beta}{2} = 4 \cdot \sin^2 \frac{\alpha}{2}$$

$$\Rightarrow \sin \frac{\beta}{2} = \sqrt{2} \sin \frac{\alpha}{2}$$

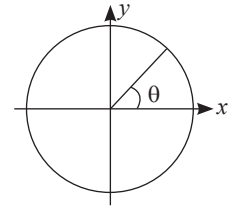
$$\Rightarrow \beta = 2 \cdot \sin^{-1} \left[\sqrt{2} \sin \frac{\alpha}{2} \right]$$



Radial field lines near a point charge.

Example 45 Ring with variable charge

A thin non conducting ring of radius R has linear charge density λ varying as $\lambda = \lambda_0 \cos\theta$, where λ_0 is a constant and θ is angle measured from positive x direction as shown. Find the magnitude of the field at the centre.

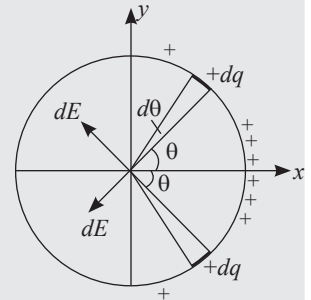


Solution

Concepts

- $\cos\phi$ is positive in 1st and 4th quadrant. It is negative in 2nd and 3rd quadrant. Therefore, charge is positive in 1st and 4th quadrant and is negative in 2nd and 3rd quadrant.

- The charge distribution is not uniform but it is easy to see that the resultant field at the centre is in negative x direction. The elements shown in the figure have equal charges since $\cos(-\theta) = \cos\theta$. Field produced due to them is in negative x direction. Similarly, you can prove that the other half of the ring also produces field in $-x$ direction.



- Contribution made by positive and negative charges will be same. We will find x component of field due to positive charge and double it to get the net field.

Consider a charge element at an angle θ subtending angle $d\theta$ at the centre (see the figure given above).

$$dq = \lambda(Rd\theta) = \lambda_0 R \cos\theta \cdot d\theta$$

Electric field at the centre due to this charge element is

$$dE = \frac{1}{4\pi\epsilon_0} \frac{dq}{R^2} = \frac{1}{4\pi\epsilon_0} \frac{\lambda_0}{R} \cos\theta \cdot d\theta$$

Component of this field in negative x direction is

$$dE_x = dE \cos\theta = \frac{1}{4\pi\epsilon_0} \frac{\lambda_0}{R} \cos^2 \theta d\theta$$

Field due to positive charge is

$$\begin{aligned} E_x &= \int_{\theta=-\pi/2}^{\pi/2} dE_x = \frac{1}{4\pi\epsilon_0} \frac{\lambda_0}{R} \int_{-\pi/2}^{\pi/2} \cos^2 \theta d\theta \\ &= \frac{\lambda_0}{4\pi\epsilon_0 R} \int_{-\pi/2}^{\pi/2} \frac{(1 + \cos 2\theta)}{2} d\theta \end{aligned}$$

$$\begin{aligned}
&= \frac{\lambda_0}{4\pi\epsilon_0 R} \left[\theta + \frac{\sin 2\theta}{2} \right]_{-\pi/2}^{\pi/2} \\
&= \frac{\lambda_0}{8\pi\epsilon_0 R} \left[\left(\frac{\pi}{2} + 0 \right) - \left(-\frac{\pi}{2} + 0 \right) \right] \\
&= \frac{\lambda_0}{8\pi\epsilon_0 R}
\end{aligned}$$

Negative charge also makes an equal contribution. Hence, the field at the centre is

$$E = 2E_x = \frac{\lambda_0}{4\pi\epsilon_0 R} \text{ in negative } x \text{ direction}$$

Example 46 Field inside a uniformly charged spherical shell

We have already proved in section 10.1, using the Gauss's law that electric field inside a uniformly charged spherical shell is zero. Prove this result directly using the Coulomb's law.

Solution

Concepts

- We will consider a small patch of charge on the surface of the shell. This patch subtends a solid angle $\Delta\Omega$ at the point (P) where we want to find the field. A second patch subtending same solid angle at P is considered opposite to the first patch (see figure below).
- We will prove that fields due to charges on these two patch cancel out.
- The entire shell can be divided into number of pairs of such patches.
- Alternately, you can divide the sphere into number of coaxial rings having point P on their axis. Equation (11) can be used to write field at P due to any such ring.

Consider a small patch of area ΔS on the surface of the shell. The patch is at a distance r_1 from point P . Solid angle subtended by ΔS_1 at P is

$$\Delta\Omega = \frac{\Delta S_1}{r_1^2}$$

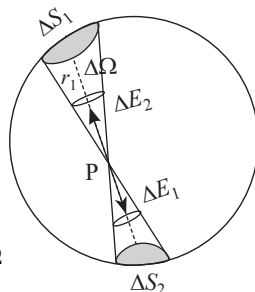
Charge on the patch is

$$\Delta q_1 = \sigma \Delta S_1 = \sigma \Delta\Omega r_1^2$$

Field due to this patch at P is

$$\Delta E_1 = k \frac{\Delta q_1}{r_1^2} = k \frac{\sigma \Delta\Omega r_1^2}{r_1^2} = k \sigma \Delta\Omega$$

Now, we consider a patch exactly opposite to the first patch on the line joining the first patch to P . This patch subtends same solid angle $\Delta\Omega$ at P . [You can perceive it like this: join the points on the periphery of area ΔS_1 to point P



and extend them so that they intersect the spherical surface. The area intercepted by these lines on the spherical surface has area ΔS_2 . This area subtends same solid angle $\Delta\Omega$ at P]

Distance of area ΔS_2 from P is r_2

$$\therefore \Delta\Omega = \frac{\Delta S_2}{r_2^2}$$

$$\Rightarrow \Delta S_2 = \Delta\Omega r_2^2$$

Field due to charge on this patch at P is ΔE_2 directed exactly opposite to ΔE_1 .

$$\Delta E_2 = k \frac{\Delta q_2}{r_2^2} = k \frac{\sigma \cdot \Delta S_2}{r_2^2} = k \sigma \Delta\Omega$$

Thus, ΔE_1 and ΔE_2 cancel out.

Notice that our calculations are exactly correct when $\Delta\Omega \rightarrow 0$ (i.e., when ΔS_1 and ΔS_2 are infinitesimally small). The entire shell is made up of such pair of charges and the resultant field at point P must be zero.

Example 47 A charge distribution generates a radial electric field in the entire space given by

$$\vec{E} = \frac{a}{r^2} e^{-\frac{r}{b}} \hat{r}$$

where r is distance of a point from the origin of the co-ordinates and $\hat{r}$ is a unit vector from the origin to the concerned point. a and b are positive constants. Find the total charge in space giving rise to this field.

Solution

Concepts

Consider a sphere of infinite radius and find flux of electric field through the surface.

Consider a spherical Gaussian surface of radius r centred at the origin. Flux of electric field through the sphere is

$$\phi = E \cdot 4\pi r^2 = \frac{a}{r^2} e^{-r/b} \cdot 4\pi r^2 = 4\pi a e^{-r/b}$$

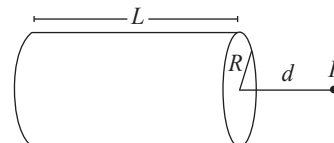
when $r \rightarrow \infty$

$$\phi \rightarrow 0$$

therefore, total charge in the entire space is zero.

Example 48 Field on the axis of a charged cylinder

A cylindrical volume of length L and radius R carries charge with uniform volume density. The total charge is Q . Find the electric field at a point on its axis which is at a distance d from one of its faces. Approximate your result for $d \ll L$, $d \ll R$.



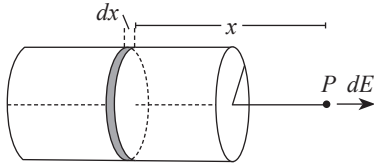
Solution**Concepts**

- (i) The entire cylinder is made of numerous thin discs.
(ii) Equation (12) gives electric field on the axis of a uniformly charged disc at a distance x from its centre

$$E = \frac{\sigma}{2\epsilon_0} \left[1 - \frac{x}{\sqrt{R^2 + x^2}} \right].$$

Here R is radius of the disc and σ is surface charge density.

Consider a disc of thickness dx at a distance x from point P .



Charge on the disc is $\frac{Q}{L} dx$

Charge per unit area for the thin disc is

$$\sigma = \frac{Q dx}{L \cdot \pi R^2}$$

Electric field at P due to the disc charge is

$$dE = \frac{Q}{2\pi\epsilon_0 LR^2} \left[1 - \frac{x}{\sqrt{R^2 + x^2}} \right] dx$$

Total field at P is obtained by adding contributions made by all such discs in the cylinder.

$$\therefore E = \frac{Q}{2\pi\epsilon_0 LR^2} \int_{x=d}^{d+L} \left[1 - \frac{x}{\sqrt{R^2 + x^2}} \right] dx$$

[Integration $\int \frac{xdx}{\sqrt{R^2 + x^2}}$ can be solved by method of substitution as follows:

Put $R^2 + x^2 = t$

$$\Rightarrow 2x \frac{dx}{dt} = 1$$

$$\Rightarrow x dx = \frac{1}{2} dt$$

$$\therefore \int \frac{xdx}{\sqrt{R^2 + x^2}} = \frac{1}{2} \int \frac{dt}{\sqrt{t}} = \sqrt{t} = \sqrt{R^2 + x^2}$$

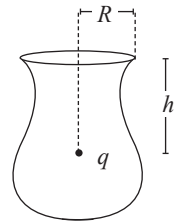
$$\begin{aligned} \therefore E &= \frac{Q}{2\pi\epsilon_0 LR^2} \left[x - \sqrt{R^2 + x^2} \right]_{x=d}^{d+L} \\ &= \frac{Q}{2\pi\epsilon_0 LR^2} \left[\left\{ d+L - \sqrt{R^2 + (d+L)^2} \right\} - \left\{ d - \sqrt{R^2 + d^2} \right\} \right] \end{aligned}$$

$$= \frac{Q}{2\pi\epsilon_0 LR^2} \left[L + \sqrt{R^2 + d^2} - \sqrt{R^2 + (d+L)^2} \right]$$

For $d \ll L$ and $d \ll R$, we can approximate the result as

$$\begin{aligned} E &= \frac{Q}{2\pi\epsilon_0 R^2} \left[1 + \sqrt{\frac{R^2 + d^2}{L^2}} - \sqrt{\frac{R^2 + (d+L)^2}{L^2}} \right] \\ &\approx \frac{Q}{2\pi\epsilon_0 R^2} \left[1 + \frac{R}{L} - \sqrt{\frac{R^2}{L^2} + 1} \right]. \end{aligned}$$

Example 49 A pot has a circular opening of radius R . A point charge q is placed at a depth $h = \sqrt{3}R$ exactly below the centre of the opening of the vessel (see figure). Calculate the electric flux through the surface of the pot.

**Solution****Concepts**

- (i) If we consider a closed surface formed by the vessel covered with a circular lid, the flux through the surface will be $\frac{q}{\epsilon_0}$.
(ii) Flux through the given pot surface = $\frac{q}{\epsilon_0}$ - (flux through the circular face of radius R).
(iii) To find flux through the circular face we need to find what percentage of field lines starting from charge q will pass through the circle of radius R .

Draw a sphere centred at the point charge q . The shaded circle has radius R and is at a distance $h = \sqrt{3}R$ from the charge.

Radius of the sphere is

$$r = \sqrt{R^2 + (\sqrt{3}R)^2} = 2R$$

$$\tan \theta = \frac{R}{h} = \frac{1}{\sqrt{3}}$$

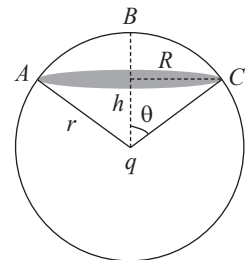
$$\Rightarrow \theta = 30^\circ.$$

Field lines passing through the shaded region are those which are contained in a cone of semi vertical angle θ (or, those which pass through the spherical cap ABC).

Area of cap ABC is given by $s = 2\pi r^2(1 - \cos \theta)$.

Therefore, fraction of field lines passing through the shaded area is

$$\frac{2\pi r^2(1 - \cos \theta)}{4\pi r^2} = \frac{1}{2}(1 - \cos \theta).$$



Alternate way of telling the same thing is: The solid angle enclosed by the cone is given by

$$\Omega = 2\pi(1 - \cos\theta) \quad [\text{Remember this}]$$

total solid angle subtended by a sphere at its centre is 4π steradian.

$\therefore$ fraction of field lines passing through the shaded circle (i.e., through the cone) is

$$\frac{\Omega}{4\pi} = \frac{1}{2}(1 - \cos\theta) = \frac{1}{2}(1 - \cos 30^\circ) = \frac{2 - \sqrt{3}}{4}$$

$$\therefore \text{Flux through the circle is} = \frac{q}{\epsilon_0} \left(\frac{2 - \sqrt{3}}{4} \right)$$

$\therefore$ Required flux through the surface of the pot is

$$\phi = \frac{q}{\epsilon_0} - \frac{q}{\epsilon_0} \left(\frac{2 - \sqrt{3}}{4} \right) = \frac{q}{\epsilon_0} \left(\frac{2 + \sqrt{3}}{4} \right).$$

Example 50 Radial field of constant magnitude

A ball of radius R has a spherically symmetric charge Q distributed in it. The surrounding of the ball is filled with a charge of density $\rho = \frac{\alpha}{r}$, where α is a positive constant and r is distance from the centre of the ball. It is found that the electric field has magnitude that is independent of r at points lying outside the ball. Find the value of Q in terms of α and R .

Solution

Concepts

- We will consider a spherical Gaussian surface of radius r surrounding the ball. Due to symmetry, field has same magnitude at all points on the spherical surface and is radial. Flux through the Gaussian surface is $E \cdot 4\pi r^2$.
- We need to find charge present in the region between the ball's surface and the spherical Gaussian surface. If this charge is q , then the total charge inside the Gaussian surface is $Q + q$.
- For E to be independent of r , all terms containing r (in the expression of E) should add to zero.

Charge in a shell of radius

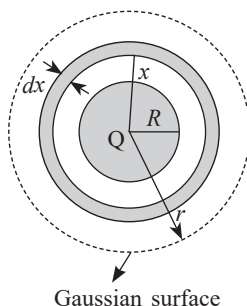
x ($R < x < r$) having thickness dx is

$$dq = \rho \cdot 4\pi x^2 dx$$

$$= 4\pi\alpha \frac{x^2 dx}{x} = 4\pi\alpha x dx$$

Charge in the region between the surface of the ball and radius r is

$$q = 4\pi\alpha \int_R^r x dx = 2\pi\alpha (r^2 - R^2)$$



total charge inside a spherical Gaussian surface of radius r is $Q + q$

$$\therefore E \cdot 4\pi r^2 = \frac{Q + q}{\epsilon_0}$$

$$\Rightarrow E = \frac{Q}{4\pi\epsilon_0 r^2} + \frac{2\pi\alpha (r^2 - R^2)}{4\pi\epsilon_0 r^2}$$

$$= \frac{\alpha}{2\epsilon_0} + \frac{Q - 2\pi\alpha R^2}{4\pi\epsilon_0 r^2}$$

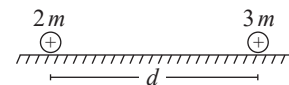
For E to be independent of r , the numerator of the second term in the above expression shall be zero.

$$\therefore Q = 2\pi\alpha R^2$$

Constant value of E is $\frac{\alpha}{2\epsilon_0}$

Example 51 Two charged particles of masses $2m$ and $3m$ have charge $+q$ on each of them. They are placed at a separation d on a smooth horizontal surface and released. The time after which the separation between the charges becomes $3d$ is given by

$$t = \sqrt{\frac{amd}{5Kq^2}} \int_d^{3d} \sqrt{\frac{x}{x-d}} \cdot dx$$



Here $K = \frac{1}{4\pi\epsilon_0}$. Find the value of constant a . Consider only electrostatic interaction.

Solution

Concepts

- The question does not ask you to calculate t . This is because I am assuming that most of you will find it difficult to do the integration at this stage. Those of you who are familiar with this kind of integration can find t .
- Charges exert equal and opposite force on one another and accelerate. However, their acceleration changes as separation changes.
- We will first use $a = v \frac{dv}{dx}$ to find v as function of separation x . Here v is relative speed of separation and a is relative acceleration.
- Then use $v = \frac{dx}{dt}$ to write t as function of x .

Accelerations of mass $2m$ and $3m$ when separation becomes x are

$$a_1 = \frac{Kq^2}{2m \cdot x^2} \quad \text{and} \quad a_2 = \frac{Kq^2}{3m \cdot x^2}$$

Relative acceleration is

$$a = a_1 + a_2 = \frac{Kq^2}{mx^2} \left(\frac{1}{2} + \frac{1}{3} \right) = \frac{5Kq^2}{6mx^2}$$

If v is relative speed of separation, then

$$\begin{aligned} v \frac{dv}{dx} &= \frac{5Kq^2}{6mx^2} \\ \Rightarrow \int_0^v v dv &= \frac{5Kq^2}{6m} \int_d^x \frac{dx}{x^2} \\ \Rightarrow \frac{v^2}{2} &= \frac{5Kq^2}{6m} \left[\frac{1}{d} - \frac{1}{x} \right] \\ \Rightarrow \frac{dx}{dt} &= \sqrt{\frac{5Kq^2}{3m} \left(\frac{1}{d} - \frac{1}{x} \right)} \\ \Rightarrow \sqrt{\frac{3md}{5Kq^2}} \int_d^x \sqrt{\frac{x}{x-d}} dx &= \int_0^t dt \end{aligned}$$

Obviously $a = 3$.

Those of you who can perform the integration will find

$$t = \sqrt{\frac{3md^3}{5Kq^2}} \left[\sqrt{6} + \ln(\sqrt{3} + \sqrt{2}) \right]$$

Example 52 Excess pressure inside a charged soap bubble

A soap bubble has a uniform surface density of charge σ on its outer surface. Its radius is R and surface tension of soap solution is S . Calculate the pressure difference of air inside the bubble with the atmospheric pressure.

Solution

Concepts

When a liquid surface is curved having radius of curvature R , pressure on its concave side is higher than pressure on the convex side. The pressure difference is $\frac{2S}{R}$.

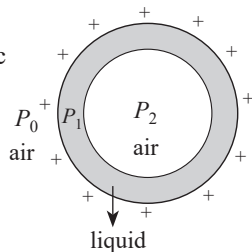
P_1 is pressure inside liquid wall.
Pressure created due to electrostatic repulsion of charge is

$$P_{\text{elect}} = \frac{\sigma^2}{2\epsilon_0}$$

$$\therefore (P_1 + P_{\text{elect}}) - P_0 = \frac{2S}{R}$$

$$\therefore P_1 - P_0 = \frac{2S}{R} - \frac{\sigma^2}{2\epsilon_0} \quad (a)$$

$$\text{and } P_2 - P_1 = \frac{2S}{R} \quad (b)$$



The thickness of bubble wall is greatly exaggerated in the diagram.

Adding (a) and (b) gives

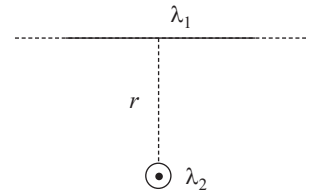
$$\Rightarrow P_2 - P_0 = \frac{4S}{R} - \frac{\sigma^2}{2\epsilon_0}$$

$$\text{Excess pressure inside the bubble} = \frac{4S}{R} - \frac{\sigma^2}{2\epsilon_0}$$

Example 53 Force between two long line charges

Two infinitely long wires having linear charge densities λ_1 and λ_2 are placed perpendicular to one another as shown in figure. Wire 1 is in the plane of the paper and wire 2 is perpendicular to this plane.

The symbol $\odot$ indicates that the wire is going into the plane of the paper. Separation between the two wires is r . Find the force of interaction between them.



Solution

Concepts

- We will write electric field due to charge on wire 2 at a position on the wire 1. Then we will write force on a small segment of wire 1 due to this field. We need to add forces on every element of wire 1.
- From symmetry one can see that force will be perpendicular to a wire.

Consider an element of length dx on wire 1 at a distance x from O . Field produced by wire 2 at this location is

$$E_2 = \frac{\lambda_2}{2\pi\epsilon_0 l} = \frac{\lambda_2}{2\pi\epsilon_0 r \sec \theta}$$

Force on the element is

$$dF = (\lambda_1 dx) E_2 = \frac{\lambda_1 \lambda_2 dx}{2\pi\epsilon_0 r \sec \theta}$$

since resultant force will be along y direction, we need to consider y component of dF only

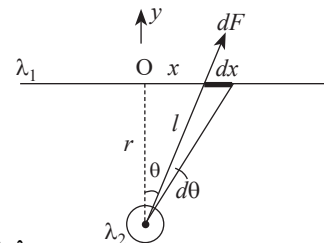
$$dF_y = dF \cos \theta = \frac{\lambda_1 \lambda_2 dx}{2\pi\epsilon_0 r \sec^2 \theta}$$

$$\text{Now } \frac{x}{r} = \tan \theta$$

$$\Rightarrow dx = r \sec^2 \theta \cdot d\theta$$

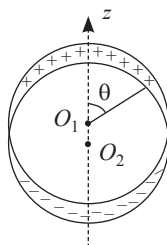
$$\therefore dF_y = \frac{\lambda_1 \lambda_2}{2\pi\epsilon_0} d\theta$$

$$\therefore F_y = \frac{\lambda_1 \lambda_2}{2\pi\epsilon_0} \int_{-\pi/2}^{\pi/2} d\theta = \frac{\lambda_1 \lambda_2}{2\epsilon_0}$$



Example 54 Think of two spherical regions, of radius R each, having their centres on z axis at a small separation l . The two spherical regions have charge densities $+\rho$ and $-\rho$.

- (a) Show that this charge distribution is nearly equivalent to a charge distributed on the surface of a spherical shell having surface charge density $\sigma = \sigma_0 \cos \theta$ (see figure).



O_1 and O_2 are centres of two spherical regions. $O_1 O_2 = l$

- (b) Write the electric field at a point inside a spherical shell having surface charge density of $\sigma = \sigma_0 \cos \theta$.

Solution

Concepts

- (i) In the common region of the two spheres charge density is zero and we know that the field in this cavity region is uniform given by $E = \frac{\rho}{3\epsilon_0} l$.
- (ii) To be able to write surface charge density on the shell (in terms of ρ), we need to find thickness of charge layer at angular position θ . If thickness of charge layer is x then the amount of charge present on unit surface area of the shell is $\sigma = \rho(l \cdot x) = \rho \cdot x$

- (a) $O_1 B = R$
 $O_2 A = R$

Thickness of charge layer is AB .

$$\therefore x = AB = R - O_1 A$$

Drop a perpendicular from O_1 on $O_2 B$. C is foot of the perpendicular. $O_1 A C$ is a right angled Δ with a very small base $O_1 C$.

$$\begin{aligned} \therefore O_1 A &\approx CA = R - O_2 C \\ &= R - l \cos \theta \\ [O_1 O_2 C &\approx \theta] \end{aligned}$$

$$\therefore \text{From (i)} \quad x = R - (R - l \cos \theta) = l \cos \theta$$

Surface charge density at angular position θ is

$$\sigma = \rho(l \cdot x) = \rho l \cos \theta$$

$$\Rightarrow \sigma = \sigma_0 \cos \theta \quad \text{where } \sigma_0 = \rho l$$

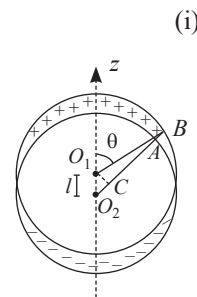
- (b) Now we know that surface charge density of $\sigma = \sigma_0 \cos \theta$ on a shell is equivalent to superposition of two spherical charged volumes having equal and opposite charge densities and their centres separated by a small distance (l).

$$\rho l = \sigma_0 \quad \Rightarrow \quad \rho = \frac{\sigma_0}{l}$$

Field inside the cavity region is (refer to article 11)

$$E = \frac{\rho}{3\epsilon_0} l = \frac{\sigma_0}{3\epsilon_0}$$

Direction is from O_1 and O_2 .



Worksheet 1

1. An electroscope is having two identical gold leaves. It is given a positive charge, causing its two leaves to separate. When an object is brought near the metal disc at the top of the electroscope, the separation increases even further. We can conclude



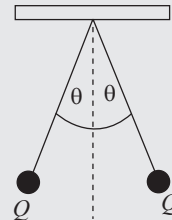
- that the object is positively charged.
 - that the object is electrically neutral.
 - that the object is negatively charged.
 - only that the object is charged.
2. Two point charges $+9e$ and $+e$ are kept 16 cm apart. Where should a third charge q be placed between them so that it is in equilibrium state:
- 24 cm from $+9e$
 - 12 cm from $+9e$
 - 24 cm from $+e$
 - 12 cm from $+e$
3. Given are four arrangements of three fixed point charges. All the charges are of the same magnitude, Q , but they can be either positive or negative as indicated. In each arrangement, a point labeled P has also been identified. A test charge, $+q$, is placed at point P . The charges and point P all lie on a straight line. The distances between adjacent items, either between two charges or between a charge and point P , are same.

-
-
-
-

Correct order of the arrangements in a decreasing order of magnitude of force on P is

- $II > I > III > IV$
- $I > II > III > IV$
- $II > I > IV > III$
- $III > IV > I > II$

4. Two small pith balls, with mass m each, are suspended from same point using two identical insulating threads. When the balls are given equal positive charge Q , they hang in equilibrium as shown. The charge on the left ball is slowly increased from Q to $2Q$. Assume that its mass does not change. Which of the following diagrams best represents the new equilibrium?

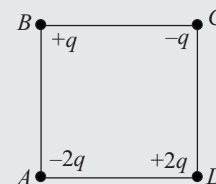


-
-
-
-

5. Two small balls having equal positive charge Q on each are suspended by two insulating strings of equal length L from a hook fixed to a stand. The whole set-up is taken inside a satellite. The angle between the strings and the tension in each of them are respectively given by

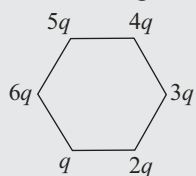
- $180^\circ, \frac{Q^2}{12\pi\epsilon_0 L^2}$
- $120^\circ, \frac{Q^2}{16\pi\epsilon_0 L^2}$
- $180^\circ, \frac{Q^2}{16\pi\epsilon_0 L^2}$
- $90^\circ, \frac{Q^2}{16\pi\epsilon_0 L^2}$

6. Four charges are arranged at the corners of a square $ABCD$, as shown. Force acting on a positive charge kept at the centre of the square is

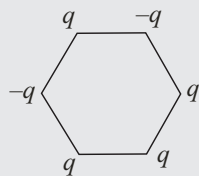


- zero
- along diagonal AC
- along diagonal BD
- perpendicular to the side AB

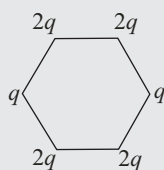
7. Figures below show a regular hexagon with point charges placed at the vertices. In which of the following arrangements the electric field at the centre of the hexagon is zero.



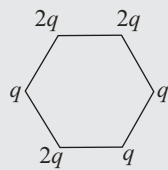
I



II

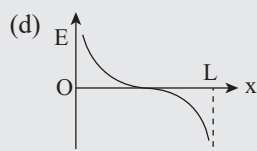
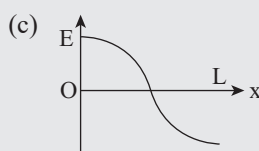
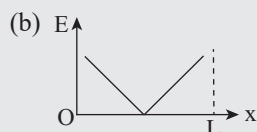
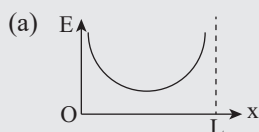


III

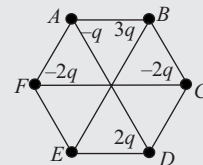


IV

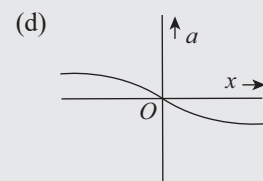
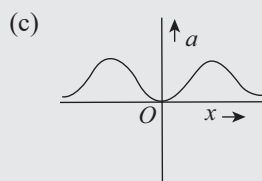
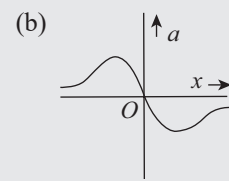
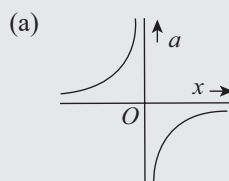
- (a) IV (b) III
(c) I (d) II
8. A point charge $+Q$ is placed at the centroid of an equilateral triangle. When a second charge $+Q$ is placed at a vertex of the triangle, the magnitude of the electrostatic force on the central charge is 0.1 N. The magnitude of the net force on the central charge when a third charge $+Q$ is placed at another vertex of the triangle is:
- (a) zero (b) 0.2 N
(c) 0.12 N (d) 0.1 N
9. A point charge $50\mu\text{C}$ is located in the XY plane at a point of position vector $\vec{r}_0 = (2\hat{i} + 3\hat{j})\text{m}$. What is the electric field at a point of position vector $\vec{r}_1 = (8\hat{i} - 5\hat{j})\text{m}$
- (a) $900(3\hat{i} - 4\hat{j})\text{NC}^{-1}$ (b) $500(3\hat{i} - 4\hat{j})\text{NC}^{-1}$
(c) $300(3\hat{i} + 4\hat{j})\text{NC}^{-1}$ (d) $400(3\hat{i} - 4\hat{j})\text{NC}^{-1}$
10. Two point charges have opposite sign and same magnitude. The positive charge is placed at $x = 0$ and the negative one is placed at $x = L$. The electric field E is plotted against x for values of x from close to zero to slightly less than L . Which of the following best represents the resulting curve?



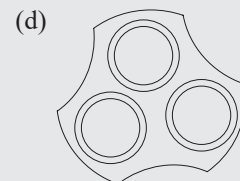
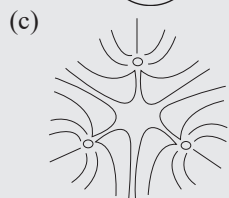
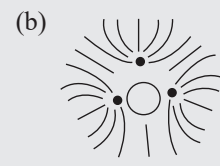
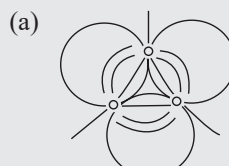
11. Six charges are placed at the corner of a regular hexagon as shown. If an electron is placed at its centre O , force on it will be



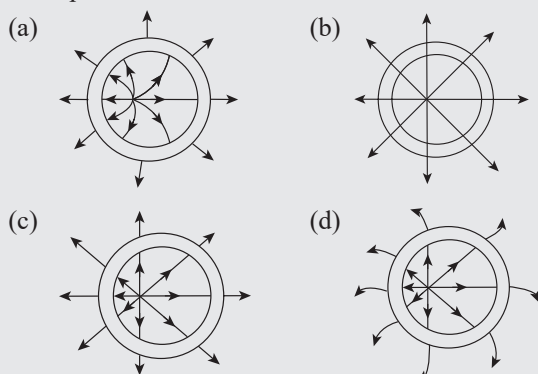
- (a) Zero
(b) Along OF
(c) Along OC
(d) None of these
12. Two identical positive charges are fixed on y -axis, equidistant from origin O . A particle with a negative charge starts on the negative x -axis at a large distance from O and moves along the positive x -axis; passes through O and moves far away from O . Its acceleration a is taken as positive along its direction of motion. The particle's acceleration a is plotted against its x -coordinate. Which of the following best represents the plot?



- (a) execute simple harmonic motion about the origin
(b) move to the origin and remain at rest
(c) move to infinity
(d) execute oscillatory but not simple harmonic motion.
13. Two equal negative charges are fixed at the points $[0, a]$ and $[0, -a]$ on the y -axis. A positive charge Q is released from rest at the point $[2a, 0]$ on the x -axis. The charge Q will
- (a) execute simple harmonic motion about the origin
(b) move to the origin and remain at rest
(c) move to infinity
(d) execute oscillatory but not simple harmonic motion.
14. Three positive charges of equal value q are placed at the vertices of an equilateral triangle. Which of the followings most correctly represents the resulting lines of force



15. A point charge is placed at a point inside a hollow conducting sphere. Which of the following electric field pattern is correct?



16. A nonconducting ring of radius R has uniformly distributed positive charge Q . An arc of the ring, of length d , is removed ($d \ll R$). The electric field at the centre of the ring will now be

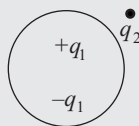
- directed towards the gap, inversely proportional to R^3 .
- directed towards the gap, inversely proportional to R^2 .
- directed away from the gap, inversely proportional to R^3 .
- directed away from the gap, inversely proportional to R^2 .

17. Two small spherical conductors B and C having equal radii and carrying equal charges on them repel each other with a force F when kept apart at some distance. A third spherical conductor A having same radius as that of B , but uncharged, is brought in contact with B , then brought in contact with C and finally removed away from both. The new force of repulsion between B and C is:

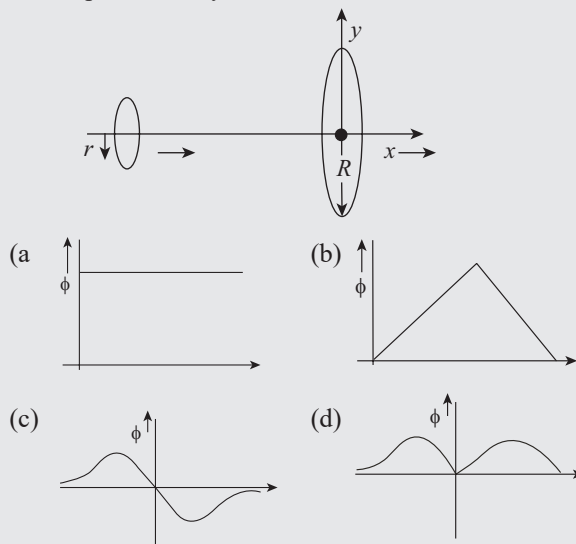
- $\frac{F}{4}$
- $\frac{3F}{4}$
- $\frac{F}{8}$
- $\frac{3F}{8}$

18. Consider the charge configuration and a spherical Gaussian surface as shown in the figure. When calculating the flux of the electric field over the spherical surface, the considered electric field will be due to

- q_2
- only the positive charges
- all the charges
- $+q_1$ and $-q_1$



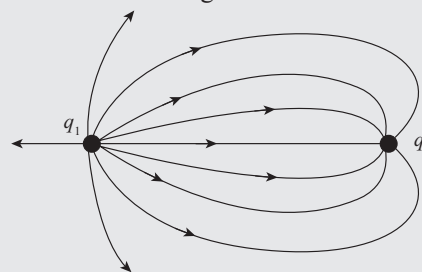
19. A ring of radius R is placed with its centre at the origin and its axis along the x -axis. It carries a uniformly distributed positive charge. A ring of radius r ($\ll R$) is held coaxially with the larger ring. It is moved along the axis. The variation of electrical flux (ϕ) passing through the smaller ring with position is best represented by:



20. A charged particle having some mass is resting in equilibrium at a height H above the centre of a uniformly charged non-conducting horizontal ring of radius R . The force of gravity acts downwards. The equilibrium of the particle will be stable

- for all values of H
- only if $H > \frac{R}{\sqrt{2}}$
- only if $H < \frac{R}{\sqrt{2}}$
- only if $H = \frac{R}{\sqrt{2}}$

21. The figure shows the electric field lines in the vicinity of two point charges. Which one of the following statements concerning this situation is true?

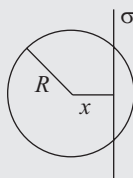


- q_1 is negative and q_2 is positive
- The magnitude of the ratio (q_2/q_1) is less than one
- Both q_1 and q_2 have the same sign of charge
- The electric field is strongest, midway between the charges.

22. Electric field in a region is given by $\vec{E} = \hat{i} + \sqrt{2}\hat{j} + \sqrt{3}\hat{k}$ (in V/m). Electric flux through a surface of area 100 m^2 lying in the xy plane is (in V-m)

(a) 100 (b) 141.4
(c) 173.2 (d) 200

23. An infinite, uniformly charged sheet with surface charge density σ cuts through a spherical Gaussian surface of radius R at a distance x from its center (see figure). The electric flux ϕ through the Gaussian surface is

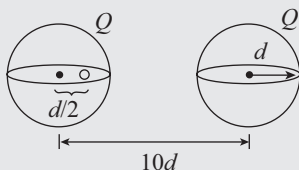


(a) $\frac{\pi R^2 \sigma}{\epsilon_0}$ (b) $\frac{2\pi(R^2 - x^2) \sigma}{\epsilon_0}$
(c) $\frac{\pi(R - x)^2 \sigma}{\epsilon_0}$ (d) $\frac{\pi(R^2 - x^2) \sigma}{\epsilon_0}$

24. In the last problem, assume that the Gaussian surface is a cylinder of radius R and length L . An infinite, uniformly charged sheet with surface charge density σ cuts through the Gaussian surface at a distance x from the axis of the cylinder. The above figure shows the circular cross section of the cylinder. The electric flux through the Gaussian surface is

(a) $\frac{\pi RL \sigma}{\epsilon_0}$ (b) $\frac{2\pi L \sqrt{(R^2 - x^2)} \sigma}{\epsilon_0}$
(c) $\frac{\pi(R - x)^2 \sigma}{\epsilon_0}$ (d) $\frac{\pi(R^2 - x^2) \sigma}{L \epsilon_0}$

25. Two spherical, nonconducting, and very thin shells having uniformly distributed positive charge Q on their surfaces are located a distance $10d$ from each other. Radius of each shell is d . A positive point charge q is placed inside one of the shells at a distance $d/2$ from the center, on the line connecting the centers of the two shells. What is the net force on the charge q ?



(a) $\frac{qQ}{361\pi\epsilon_0 d^2}$ to the left
(b) $\frac{qQ}{361\pi\epsilon_0 d^2}$ to the right
(c) $\frac{362qQ}{361\pi\epsilon_0 d^2}$ to the left
(d) $\frac{360qQ}{361\pi\epsilon_0 d^2}$ to the right

26. A positive charge q is placed in a spherical cavity made in a positively charged sphere. The centres of sphere and cavity are displaced by a small distance $\vec{l}$. Force on charge q is:

(a) in the direction parallel to vector $\vec{l}$
(b) in radial direction
(c) in a direction which depends on the magnitude of charge density in sphere
(d) direction can not be determined.

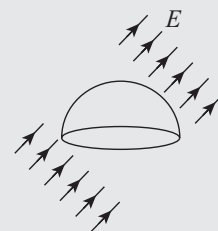
27. A block of mass m is connected to a spring of force constant k . The system is suspended vertically in the gravitational field of the Earth. Its time period is found to be T_1 . Now the block is given a charge q and entire space is filled with an electric field E which is uniform and horizontal. Now the time period of oscillation of the spring block system is T_2 . Somehow the Earth loses its gravity and the system now oscillates with time period T_3 in the electric field. The ratio $T_1 : T_2 : T_3$

(a) is 1 : 1 : 1
(b) will depend on relative strength of gravitational force and electric force on the block.
(c) is 1 : 2 : 0
(d) is 1 : 1 : 1 only if electric force is equal to the gravitational force mg on the block.

28. The maximum electric field at a point on the axis of a uniformly charged ring is E_0 . At how many points on the axis will the magnitude of electric field be $E_0/2$

(a) 1 (b) 2
(c) 3 (d) 4

29. A hemispherical surface of radius R is located in a uniform electric field E that is inclined at 45° to the circular base of the hemisphere. What is the magnitude of the electric flux through the hemispherical surface?



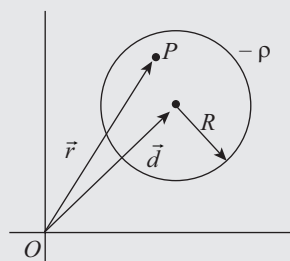
(a) 0 (b) $4\pi R^2 E/3$
(c) $2\pi R^2 E$ (d) $\frac{\pi R^2 E}{\sqrt{2}}$

30. Which of the following is **false** for a closed Gaussian surface?

(a) If the electric field is zero everywhere on the surface, then there can be no net charge enclosed by the surface.
(b) If the total electric flux through the surface is zero, then the total charge enclosed by the surface is zero.

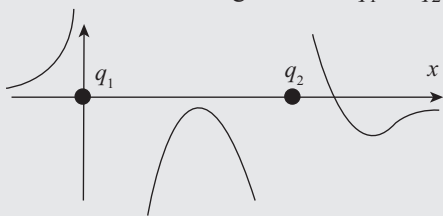
- (c) If the electric field is zero everywhere on the surface, then the total electric flux through the surface is zero.
- (d) If the total electric flux through the surface is zero, then the electric field must be zero everywhere on the surface.

31. A non-conducting sphere of radius R is filled with uniform volume charge density $-\rho$. The center of this sphere is displaced from the origin by $\vec{d}$. The electric field $\vec{E}$ at any point P having position vector $\vec{r}$ inside the sphere is



- (a) $\frac{\rho}{3\epsilon_0} \vec{d}$ (b) $\frac{\rho}{3\epsilon_0} (\vec{r} - \vec{d})$
- (c) $\frac{\rho}{3\epsilon_0} (\vec{d} - \vec{r})$ (d) $\frac{\rho}{3\epsilon_0} (\vec{d} + \vec{r})$

32. Two point charges q_1 and q_2 are held fixed on x -axis and electric field along x -axis is plotted against x . Field in positive x direction has been marked as negative. Choose correct statement about nature and magnitude of q_1 & q_2 .



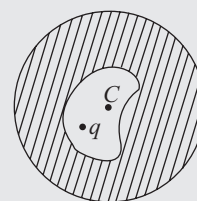
- (a) q_1 +ve, q_2 -ve ; $|q_1| > |q_2|$
- (b) q_1 +ve, q_2 -ve ; $|q_1| < |q_2|$
- (c) q_1 -ve, q_2 +ve ; $|q_1| > |q_2|$
- (d) q_1 -ve, q_2 +ve ; $|q_1| < |q_2|$

33. Volume charge density inside a sphere of radius R is proportional to the distance (x) from the centre and is given by $\rho = \alpha x$, where α is a constant. The strength of electric field at a distance $R/2$ from the centre is

- (a) $\frac{\alpha R^3}{4 \epsilon_0}$ (b) $\frac{\alpha R^2}{4 \epsilon_0}$
- (c) $\frac{\alpha R^2}{3 \epsilon_0}$ (d) none of these

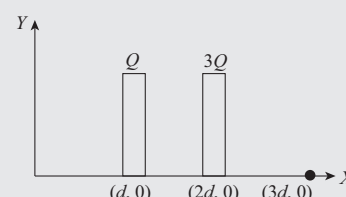
34. The figure shows a charge q placed inside a cavity in an uncharged conductor. Now, if an external electric field is switched on:

- (a) only induced charge on outer surface will redistribute.
- (b) only induced charge on inner surface will redistribute.
- (c) induced charges on outer and inner surface will redistribute.



- (d) force on charge q placed inside the cavity will change.

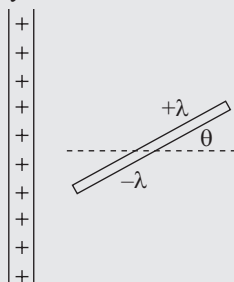
35. Two identical large thin conducting plates are placed parallel to each other as shown in figure. They are carrying charges ' Q ' and ' $3Q$ ' respectively. The variation of electric field as a function of x



- (for $x = 0$ to $x = 3d$) will be best represented by.

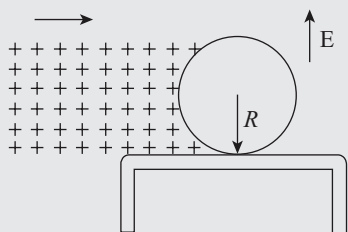
- (a)
- (b)
- (c)
- (d)

36. A large sheet carries uniform surface charge density σ . A non conducting rod of length $2l$ has a linear charge density λ on one half and $-\lambda$ on the second half. The rod is hinged at mid point and makes an angle θ with the normal to the sheet. The torque experienced by the rod is

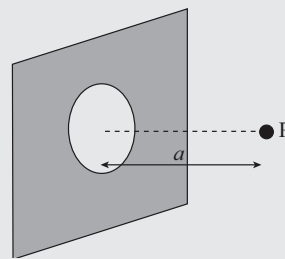


- (a) 0
(b) $\frac{\sigma\lambda l^2}{2\epsilon_0} \sin\theta$
(c) $\frac{\sigma\lambda l^2}{\epsilon_0} \sin\theta$
(d) $\frac{\sigma\lambda l}{2\epsilon_0}$

37. Figure shows a neutral metal ball of mass 50 kg and radius $\frac{2}{\sqrt{\pi}}$ m placed on an insulating horizontal surface. An upward electric field of strength 5×10^5 N/C exists in the space. A stream of light ions is switched on at $t = 0$. The ions are incident on the ball from left side at a speed 2×10^6 m/s. The charge density of ion beam is 5×10^{-12} C/m³. All charges incident on the ball are absorbed. Friction is large enough to prevent any horizontal movement of the ball. Neglect any effect of the electric field on the ion beam. The time in second at which ball will rise up from the surface is

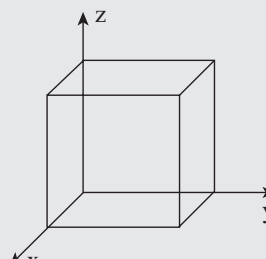


- (a) 25 s
(b) 15 s
(c) 20 s
(d) 5 s
38. An infinitely large non-conducting plane has uniform surface charge density σ . It has a circular hole carved out from it. The electric field at a point which is at a distance ' a ' from the centre of the hole (perpendicular to the plane) is $\frac{\sigma}{2\sqrt{2}\epsilon_0}$. The radius of the hole is:



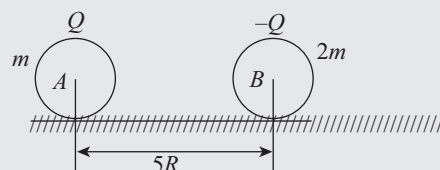
- (a) $2a$
(b) a
(c) $3a$
(d) none of the above.

39. Electric field in a region is given by $\vec{E} = -4x\hat{i} + 6y\hat{j}$. Then the magnitude of charge enclosed in the cube of unit side length, oriented as shown in the diagram, is



- (a) $-2\epsilon_0$
(b) $2\epsilon_0$
(c) $4\epsilon_0$
(d) $-4\epsilon_0$

40. Two spherical shells each of radius R having uniformly distributed charge Q and $-Q$ on their surfaces are released from rest, on a smooth non-conducting horizontal surface, when the distance between their centres is $5R$. The mass of A is m and that of B is $2m$. Consider the electric as well as gravitational interactions between the balls as they move. The charge on one ball affects the distribution of the charge on the other ball as they come closer. The distance travelled by A just by the time the two balls collide is

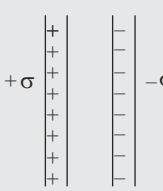


- (a) $2R$
(b) R
(c) $3R$
(d) $15R/7$

Worksheet 2

- A point charge q is placed at the origin. Let $\vec{E}_A$, $\vec{E}_B$ and $\vec{E}_C$ be the electric field at three points A (1, 2, 3), B (1, 1, -1) and C (2, 2, 2) due to charge q . Then select the correct alternative/s
 - $\vec{E}_A \perp \vec{E}_B$
 - $|\vec{E}_B| = 4|\vec{E}_C|$
 - $\vec{E}_A // \vec{E}_B$
 - None of the above
- Two free positive charges $4q$ and q are a distance l apart. A charge Q is placed at a point on the line joining the two charges such that the entire system of three charges remain in equilibrium. Which of the followings is/are true
 - $Q = -\frac{4}{9}q$
 - Q is kept at a distance $\frac{l}{3}$ from q
 - $Q = \frac{4}{9}q$
 - such equilibrium is not possible
- Mid way between the two equal and similar charges, we place a third equal and similar charge. Which of the following statements is/are correct, concerned to the equilibrium along the line joining the charges?
 - The third charge experiences a net force inclined to the line joining the charges
 - The third charge is in stable equilibrium
 - The third charge is in unstable equilibrium
 - The third charge experiences a net force perpendicular to the line joining the charges
- Select the correct statements: (Only force on the particle is due to electric field)
 - A charged particle always moves along the electric line of force.
 - A charged particle may move along the line of force
 - A charge particle never moves along the line of force
 - A charged particle moves along the line of force only if released from rest.
- Two infinite sheets of uniform charge density $+\sigma$ and $-\sigma$ are parallel to each other as shown in the figure.

$+\sigma$



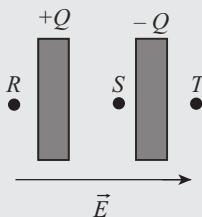
 - Electric field at the points to the left or to the right of the sheets is zero.
 - Electric field at the midpoint between the sheets is zero.
- Electric field at the midpoint of the sheets is σ/ϵ_0 and is directed towards right.
- The force applied by one sheet on the unit area of the other sheet is $\frac{\sigma^2}{2\epsilon_0}$.
- Which of the following statements are correct?
 - Electric field calculated by Gauss's law is the field due to only those charges which are enclosed inside the Gaussian surface.
 - Gauss's law is true only when there is a symmetrical distribution of charge.
 - The value of electric flux through a closed surface depends only on charges enclosed by the surface.
 - A point charge is enclosed inside an irregular surface and there is no charge outside the surface. The flux of field through a finite size patch on the surface must be zero.
- In which of following cases, electric field is uniform?
 - Inside a uniformly charged spherical shell
 - In any cavity inside a uniformly charged sphere
 - In front of an infinite sheet of uniform surface charge density
 - At a distance x from a point charge q .
- Charges Q_1 and Q_2 lie inside and outside, respectively of a closed surface S . Let E be the field at any point on S and ϕ be the total flux of the electric field over S .
 - If Q_1 changes, both E and ϕ will change.
 - If Q_2 changes, E will change but ϕ will not change.
 - If $Q_1 = 0$ and $Q_2 \neq 0$ then $E \neq 0$ but $\phi = 0$.
 - If $Q_1 \neq 0$ and $Q_2 = 0$ then $E = 0$ but $\phi \neq 0$.
- Pick the correct statements.
 - If a point charge is placed off-centre inside an electrically neutral spherical metal shell then induced charge on its inner surface is uniformly distributed.
 - If a point charge is placed off-centre inside an electrically neutral, isolated spherical metal shell, then induced charge on its outer surface is uniformly distributed.

- (c) A uniformly charged non conducting shell attracts or repels a charged particle that is outside the shell as if all the shell's charge were concentrated at the centre of the shell.
- (d) If a charged particle is located inside a uniformly charged nonconducting shell, there is no electrostatic force on the particle due to the shell.

10. A particle of mass m and charge q is thrown in a region where uniform gravitational field and electric field are present. The path of particle

- (a) may be a straight line
(b) may be a circle
(c) may be a parabola
(d) may be a hyperbola

11. Two large thin conducting plates with small gap in between are placed in a uniform electric field ' E ' (perpendicular to the plates). Area of each plate is A and charges $+Q$ and $-Q$ are given to these plates as shown in the figure. R , S and T (see figure) are three points in space, then the



- (a) field at point R is E
(b) Charge on the inner wall of the left plate will be $(Q + \epsilon_0 AE)$
(c) field at point T is $\left(E + \frac{Q}{\epsilon_0 A}\right)$
(d) field at point S is $\left(E + \frac{Q}{A\epsilon_0}\right)$

12. Three charged particles are in equilibrium under their electrostatic forces only

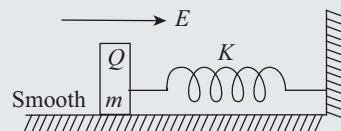
- (a) The particles must be collinear.
(b) All the charges cannot have the same magnitude.
(c) All the charges cannot have the same sign.
(d) The equilibrium is unstable.

13. Two point charges (Q each) are placed at $(0, y)$ and $(0, -y)$. A point charge q of the same polarity can move along X-axis. Then:

- (a) the force on q is maximum at $x = \pm y / \sqrt{2}$
(b) the charge q is in equilibrium at the origin
(c) the charge q performs an oscillatory motion about the origin
(d) for any position of q other than the origin the force is directed away from origin

14. In the figure shown, initially the spring of negligible mass and force constant K is in undeformed state

and the block has zero velocity. Suddenly, a uniform horizontal electric E is switched on. Subsequently

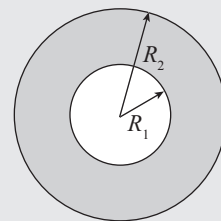


- (a) The maximum speed of the block will be attained when the compression in the spring is $\frac{QE}{K}$
(b) The maximum speed of the block will be attained when the compression in the spring is $\frac{2QE}{K}$
(c) The maximum compression of the spring will be larger than $\frac{QE}{K}$
(d) The maximum compression of the spring will be when the block is in equilibrium.

15. A positively charged particle of mass m and charge q is released from rest at position $(x_0, 0)$ in a space where exists a uniform electric field $E_0 \hat{j}$. Neglect any other force apart from the electric force on the particle.

- (a) The angular momentum of the particle about origin will increase with time
(b) The electric torque on the particle about origin will increase with time
(c) The electric potential energy of the particle goes on increasing
(d) The kinetic energy of the particle increases with time

16. A hollow dielectric sphere, as shown in the figure, has inner and outer radii of R_1 and R_2 respectively. The total charge carried by the sphere is $+Q$. This charge is uniformly distributed in the volume of the dielectric. A point charge Q_0 is placed at the centre and it is seen that the electric field in the region $R_1 < r < R_2$ varies linearly with r .



- (a) Value of Q_0 is $\frac{QR_1^3}{R_2^3 - R_1^3}$
(b) The electric field for $r > R_2$ is given by $\frac{Q + Q_0}{4\pi\epsilon_0 r^2}$
(c) The electric field for $r < R_1$ also varies linearly with r .
(d) None of the above are true.

17. A space is filled with charge with volume density $\rho = \rho_0 e^{-\alpha r^3}$, where ρ_0 and α are positive constants,

r is the distance from the centre of the sphere. The magnitude of the electric field at a distance r from the centre of the sphere is:

(a) $E \approx \frac{\rho_0}{3\epsilon_0\alpha r^2}$ for $\alpha r^3 \gg 1$

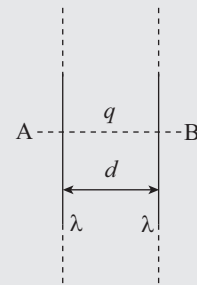
(b) $E \approx \frac{\rho_0 r}{3\epsilon_0}$, for $\alpha r^3 \ll 1$

(c) $E \approx \frac{2\rho_0}{\epsilon_0\alpha r^2}$ for $\alpha r^3 \gg 1$

(d) $E \approx \frac{\rho_0 r}{\epsilon_0}$, for $\alpha r^3 \ll 1$

18. Two infinitely long line charges, having charge density λ each, are parallel to each other and separated by a distance d . A particle having mass m and charge q is placed symmetrically between them such that the two lines and the charge lie in the same

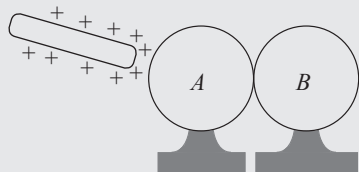
plane. This charge is displaced slightly in the plane along the line AB , which is perpendicular to the line charges, and released. Neglect gravity.



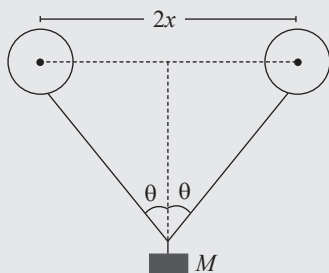
- (a) the motion of the particle will be SHM if $\lambda q > 0$.
 (b) the motion of the particle will be SHM if $\lambda q < 0$.
 (c) in case the motion is SHM the time period is directly proportional to d .
 (d) none of the above.

Worksheet 3

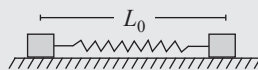
1. Two metal spheres A and B are mounted on insulating stands. They are put in contact. A positively charged rod is brought close to the sphere A .



- (a) The rod is removed and then the spheres are separated. Will the two spheres attract each other?
- (b) The spheres are separated and then the rod is removed. Will the two spheres attract each other?
2. Two copper spheres, each of mass 1 g , are separated by distance of 1 m . One percent of electrons in each of the sphere is removed by some means. Find the force they exert on one another. It is given that molar mass of copper is 63.54 g .
3. Two identical He-filled spherical balloons each carrying a charge q are tied to a mass M with two identical strings and the system floats in air as shown. In equilibrium the centres of the balloons are at separation $2x$. Assume that the charge is spread uniformly on the surface of each balloon.



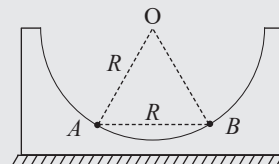
- (a) Find the angle θ that each string makes with vertical.
- (b) Find the volume of each balloon. Take density of helium and air as ρ_{He} and ρ_0 , respectively. Mass of unfilled balloon is negligible.
4. Two identical metal blocks rest on a horizontal surface connected by a conducting spring. Force constant of the spring is K and its unstretched length is L_0 . The horizontal surface is smooth and non conducting. A charge Q is slowly poured over the blocks causing the spring to stretch. The system attains equilibrium when length of the spring is L ($>L_0$). Assuming the blocks to be very small find the value of Q .



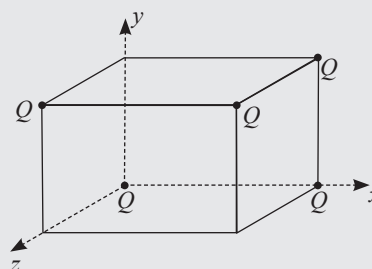
5. A small ball of mass m has a charge q on it. It is constrained to move inside a narrow vertical cylinder as shown. At the base of the cylinder there is a small ball having charge Q . The cylinder is non conducting. The ball of mass m stays in equilibrium at a height L above the charge Q . The ball is displaced by a distance x ($\ll L$) and released. Will the ball oscillate? If yes, find its time period.



6. A and B are two identical small balls having mass m and charge q on each of them. When placed inside a hemispherical bowl of radius R with smooth, non-conducting wall, the balls stay in equilibrium when they are on a horizontal line at separation R as shown in figure. Find charge on each ball.



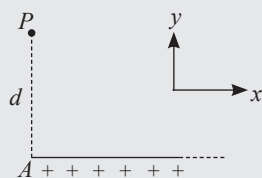
7. Five identical point charges, Q each, are located at the vertices of a cube of side length L as shown in the figure. Find the electrostatic force acting on the charge located at coordinates (L, L, L) .



8. Two identical positive point charges, each Q , are fixed at a separation $2d$. A particle is placed midway between the two fixed charges, having mass m and charge q .
- (a) Show that for a small displacement along the line joining the fixed charges, the charge q executes SHM if it is positive and find its time period.
- (b) Show that for a small displacement perpendicular to the line joining the fixed charges, the charge q executes SHM if it is negative. Find the time period for this oscillation.
9. A thin wire ring has a charge q uniformly spread on its circumference. A point charge Q is placed at the centre of the ring. Find change in tension in the ring. Radius of the ring is r .

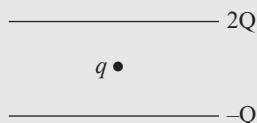
10. A spherical region of radius R contains charge. The volume charge density varies with radial distance (r) from the centre as $\rho = \frac{\rho_0}{R^2} r^2$, where ρ_0 is a positive constant. Find the total charge Q inside the sphere.

11. A line charge extends from point A to ∞ . Linear density of charge on the line is λ . P is a point at a perpendicular distance d from the line. Find the x and y components of electric field at a point P .



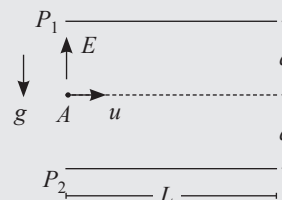
12. Three identical positive charges, Q each are arranged at the vertices of an equilateral triangle. Side length of the triangle is a . Find the electric field intensity at the vertex of a regular tetrahedron of which the triangle is a base.
13. An infinite plane surface in xy plane has a uniform surface charge density σ . A point charge q is moved from a point A (l, l, l) to another point B. Find work done by electric force if co-ordinates of B are
(a) $(2l, 2l, 2l)$ (b) $(2l, 2l, l)$

14. Two metal plates of identical dimensions are placed parallel to one another. Separation between them is small compared to their dimension. The plates are given charges $2Q$ and $-Q$ as shown. Find the acceleration of a particle of mass m having charge q when it is released between the plates. Neglect gravity and take area of each plate to be A .



15. Two identical small metal balls are suspended from a common point using two identical insulating threads. The balls stay in equilibrium in a position where separation between them is a which is very small compared to the lengths of the threads. Due to some unknown reasons, one of the balls loses its entire charge. The two balls collide and then separate to finally settle in a new equilibrium position with separation between them equal to b . Find b in terms of a .
16. Two point charges Q and $4Q$ are fixed at a distance d from one another. Sketch the electric field lines and find the distance of a point from charge Q where the electric field is zero. Such a point is sometimes known as *neutral point*.
17. Two parallel plates have length L each. They are kept horizontal and there is a uniform electric field between them in vertically upwards direction.

Separation between the plates is $2d$. A particle of mass m having charge q enters the region between the plates with a velocity u directed along the length of the plates. The particle enters exactly midway between the plates at point A. The particle just missed the upper plate P_1 and emerged out of the plate region making an angle θ with its original direction of motion. Find u and θ .



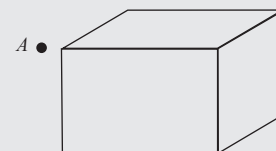
18. In some part of our universe, instead of the Coulomb's law, electric force $\vec{F}$ on a point charge q due to another point charge Q is found to obey the following law

$$\vec{F} = \frac{Qq(1-4r)}{4\pi\epsilon_0 r^3} \vec{r}$$

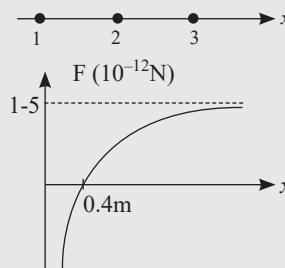
Here $\vec{r}$ is position vector of charge q with respect to Q .

- (a) Write the electric field vector produced due to a point charge Q at a point having position vector $\vec{r}$ with respect to Q .
- (b) Find value of $\oint \vec{E} \cdot d\vec{l}$ in this new world.

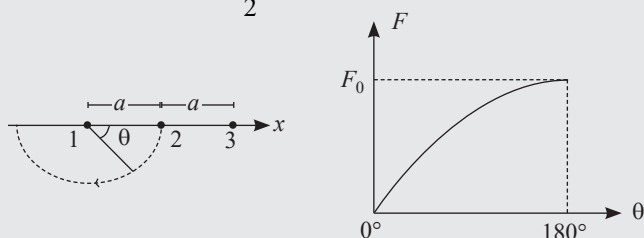
19. A cube of side length L has uniform volume charge density ρ . The electric field at point A, just outside one of the vertices of the cube has magnitude E_0 . Now consider a different cube having side length $2L$ and charge density ρ . Find the magnitude of the electric field just outside the vertex of this new cube.



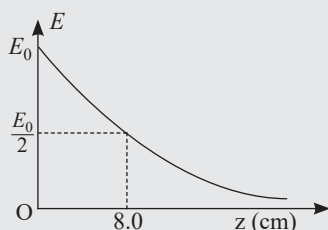
20. Point charges are fixed at points 1 and 2 on the x axis. The charge at 1 has magnitude of $8 \mu\text{C}$. A third particle 3 has charge $q_3 = +7 \mu\text{C}$ and is initially held to the right of point 2 close to it. Now, the particle 3 is moved to right along positive x direction and total force (F) on particle at 2 is measured. Taking 2 to be origin, F is plotted versus position (x) of the third particle. Find charge on particle 2.



21. Three charged particles are kept on x axis at positions 1, 2 and 3 as shown. Separation between two consecutive particles is a . Particles at 1 and 3 are held fixed but the particle at 2 can be moved along a semicircle centred around 1. As the second particle is moved, force on the particle at 1 is recorded. Graph in the diagram shows the variation of net force on 1 with angle θ . Find the value of angle θ for which force on 1 is $\frac{F_0}{2}$.

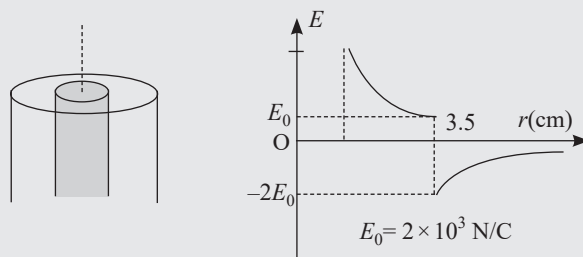


22. A tiny sphere has charge Q on it. A portion of charge x is transferred to a second nearby sphere. The two spheres are fixed and small enough to be treated as particles. When the ratio $\frac{x}{Q}$ is equal to η_0 the force between the two spheres is found to be maximum with a value equal to F_0 .
- Find η_0
 - Find the ratio $\frac{x}{Q}$ for which the force between the spheres is 75% of F_0 .
23. A uniformly charged disc lies in xy plane with its centre at the origin. The strength of electric field (E) changes along the z axis according to the graph shown. Find the radius of the disc.

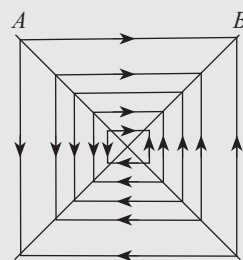


24. A point P is at a distance R from an infinite plane surface having charge density σ . PO is the normal from P on the surface, so that $PO = a$. Calculate the radius of the circle centred at O (and lying in the plane of the charge) which contributes 50% of the total field at P .
25. A long solid cylinder carries uniform charge density on its surface. Coaxial to it there is a hollow cylinder. The outer surface of this cylindrical shell has uniform surface charge density and thickness of the wall is negligible. The graphical plot of electric field versus

the radial distance (r) from the axis is given in the graph. Radially outward field has been considered positive. Find the linear charge density (i.e. charge per unit length) on the shell.



26. Two infinitely large planes A and B intersect at right angles. They carry uniform surface charge densities $+\sigma$ and $-\sigma$. Electric field lines have been drawn for the situation. Give one reason why the diagram depicting the field lines is wrong.



27. A point charge placed at origin is surrounded by a charge distribution. Careful measurements reveal an electric field

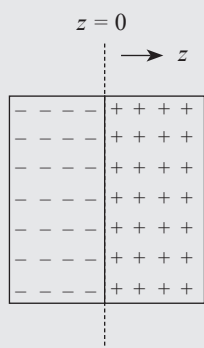
$$\vec{E} = \frac{a}{r^2} \left(1 - \frac{r^3}{R^3} \right) \hat{r} \quad \text{for } r \leq R$$

$$= 0 \quad \text{for } r \geq R$$

where a and R are positive constants and r is distance of a point from the origin. $\hat{r}$ is a unit vector in radial direction directed along the line joining the origin to the concerned point. Find the value of the charge at the origin and explain the distribution of charge surrounding it.

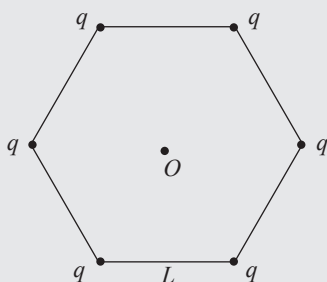
28. A pyramid has square base of side d and four faces which are equilateral triangles. A charge Q is placed at the centre of the base of the pyramid. Find the flux of electric field emerging from one of the triangular faces of the pyramid.
29. A spherical volume of radius R has a charge density $\rho = \rho_0 \frac{r}{R}$ where ρ_0 is a constant and r is distance from the centre of the sphere.
- Find the total charge inside the sphere.
 - Find the electric field at a distance r from the centre of the sphere.
30. Consider two infinite slabs of charge, both of thickness d . The junction plane of the two slabs is $z = 0$ plane. The slab in the region $0 < z < d$ has

a uniform charge density $+\rho_0$ and slab in the region $-d < z < 0$ has a uniform charge density $-\rho_0$. Find electric field in the entire space.

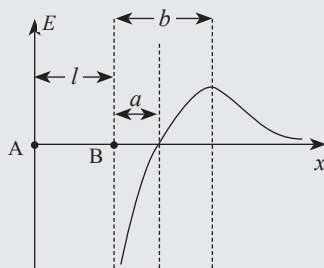


Note: A nearly similar charge develops in the depletion layer of a p - n junction. We will study about it later.

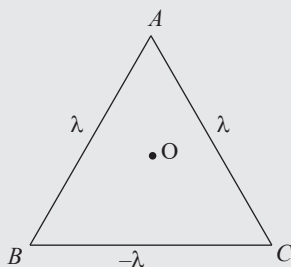
31. Six identical point charges, each q , are placed on the vertices of a regular hexagon of side length L . The hexagon is in a horizontal plane. Find the electric field at a point (P) at a vertical height L above the centre of the hexagon.



32. Two point charges Q_1 and Q_2 are positioned on the x -axis at A and B respectively. Separation between them is l . Electric field produced due to this charge system has been plotted for all points lying to the right of point B . Field is taken to be positive when it is in positive x direction.

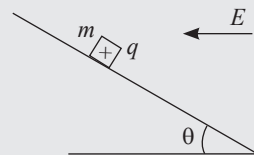


- (a) What are the signs of Q_1 and Q_2 ?
- (b) Find $\left| \frac{Q_1}{Q_2} \right|$ in terms of l and a .
- (c) Find distance b .
33. Three charged rods of length L each are made of insulating material and form an equilateral triangle. They have linear charge densities of λ , λ and $-\lambda$ as shown. Use the result

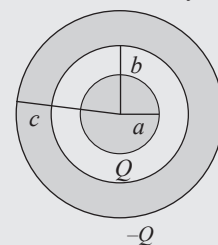


obtained in example 41 to find the electric field at the centroid (O) of the triangle.

34. A block of mass m is on verge of sliding when placed on an incline of inclination angle $\theta = 37^\circ$. The block is given a charge q and a horizontal uniform electric field E is switched on (see figure). Find the range of values of E for which the block does not slide on the incline. $\left[\tan 37^\circ = \frac{3}{4} \right]$.



35. Electric field on the surface of an irregular conductor varies from 56 kNC^{-1} to 112 kNC^{-1} .
- (a) Calculate the local surface charge density at the points on the surface where the radius of curvature of the surface is greatest and where it is smallest.
- (b) Find the ratio of greatest and smallest radius of curvature of the surface.
36. In a particular region of air at an altitude of 500 m above the ground the electric field is directed downward having a magnitude of 100 NC^{-1} . At 600 m above the ground the electric field is 80 NC^{-1} downwards. Calculate the average volume charge density in the layer of air between these two elevations.
37. A solid metal ball of radius a is surrounded by a concentric metal shell of inner and outer radii b and c respectively. The inner ball is given a charge Q and the outer ball is given a charge $-Q$.

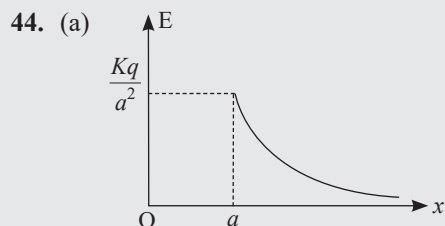


- (a) Find the charge on inner surface of the shell.
- (b) Find the charge on outer surface of the shell.
- (c) Write electric field at a distance r from the centre when
- $r < a$
 - $a < r < b$
 - $b < r < c$
 - $r > c$

Answer Sheet

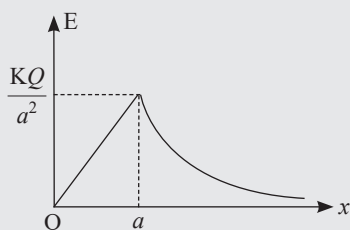
Your Turn

1. Yes
2. 6.25×10^{18}
3. The beam will get wider as charged drops will repel each other.
6. 0.16 C.
7. $[M^{-1}L^{-3}T^4A^2]$
8. 4
9. $x = \frac{Q}{2}$
10. $\frac{e}{\sqrt{4\pi\epsilon_0 mr}}$
11. $F_x = 0.0432 \text{ N}$, $F_y = 0.0576 \text{ N}$
12. $2.3 \times 10^{-28} \text{ N}$
13. Force is directed towards the vertex having no charge. $F = k \frac{q^2}{x^2}$
14. $\frac{\sqrt{3}kq^2}{x^2}$
15. $\frac{mg \cdot x^2}{k \cdot Q}$
16. (a) Negative (b) $\frac{1}{2\sqrt{2}}$
17. (a) Any sign (b) $\frac{d}{\sqrt{2}+1}$ (c) Equilibrium is stable if q is positive and it is unstable if q is negative
18. $\frac{4\sqrt{2}kq}{x^2}$ perpendicular to line joining $2q$ and $-q$
19. $\frac{2}{3}m$
20. $2 \times 10^6 \text{ N/C}$.
21. $y = \frac{a}{\sqrt{2}}$
22. $\frac{9kq}{a^2}$
23. $-\sqrt{3}(\hat{i} + \hat{j} + \hat{k})N/C$
24. (a) zero (b) $\frac{kq^2}{r^2}$
24. $\sqrt{2}E$
25. $q = -\frac{4\pi\epsilon_0 mg(a^2 + h^2)^{3/2}}{Qh}$
26. $E = \frac{\lambda \sin \theta}{2\pi\epsilon_0 a}$; $F_e = \frac{\lambda e \sin \theta}{2\pi\epsilon_0 a}$ towards left.
27. $2\pi\sqrt{\frac{2m\epsilon_0 a^2}{Q\lambda}}$
28. $\frac{Q\Delta L}{8\pi^2\epsilon_0 R^3}$
31. (a) $\frac{1}{2}$ (b) $\frac{d}{\sqrt{2}+1}$
32. (a) q_1 and q_3 are positive and q_2 is negative. (b) $|q_1| = |q_2| = |q_3|$
33. $5\pi \frac{N}{C} m^2$
34. alS
35. (a) 0 (b) yes, no
36. (a) 0 (b) $\pi R^2 E$
37. $1.2 \frac{Nm^2}{C^2}$
38. Yes
39. $\frac{q}{6\epsilon_0}$
40. $\frac{q}{2\epsilon_0}$
41. $\frac{\sqrt{3}\lambda R}{\epsilon_0}$
42. 10^5 N/C
43. ERh



- (b) The direction of electric field outside the shell will be directed towards the centre of the shell when q is negative.

45. (a)

(b) $\frac{a}{2}, \sqrt{2}a$ 46. E_1 remains unchanged and E_2 decreases47. (a) $E = 0$ (b) $E = \frac{\rho}{3\epsilon_0} \left(x - \frac{R^3}{x^2} \right)$ (c) $E = \frac{7\rho}{3\epsilon_0} \frac{R^3}{x^2}$ 48. $\frac{kq}{2a^2}$ 49. $E = 0$ for $r < a$
 $= \frac{\sigma a}{\epsilon_0 \cdot r}$ for $r > a$

50. No.

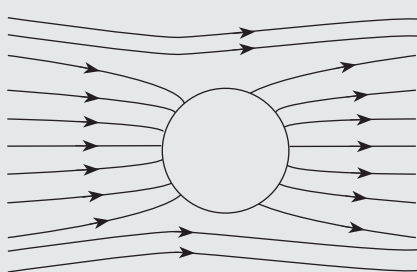
51. $1.8 \times 10^6 \text{ N/C}$ 52. $\frac{\lambda e}{2\pi\epsilon_0}$ 53. (a) $q = \frac{2mg\epsilon_0}{\sigma}$ (b) particle is in neutral equilibrium.54. $\sigma = \frac{2\sqrt{3}\epsilon_0 mg}{q} = 7.5 \times 10^{-7} \text{ C/m}^2$ 55. $E_A = E_B = \frac{\rho r}{3\epsilon_0}$ (along $\overrightarrow{AB}$)56. $\vec{E} = \frac{\rho \vec{l}}{3\epsilon_0}$ 57. $E = \frac{\rho l}{2\epsilon_0}$

58. 1.

59. (a) Radial

(b) No effect.

60.



61. Yes.

62. (a) $E_1 = \frac{Q}{A\epsilon_0}$, $E_2 = 0$, $E_3 = 0$ (b) $E_1 = 0$, $E_2 = \frac{Q}{A\epsilon_0}$, $E_3 = 0$ (c) $E_1 = E_2 = \frac{Q}{2A\epsilon_0}$, $E_3 = 0$ 63. $\frac{mg}{q}$ downward

64. No.

65. 200 m

66. -W

67. $\frac{2GM}{R^2}$ 68. $\frac{1}{3}mC$ **Worksheet 1**

- | | | | | | | | | |
|---------|---------|---------|---------|---------|---------|---------|---------|---------|
| 1. (a) | 2. (b) | 3. (c) | 4. (d) | 5. (c) | 6. (d) | 7. (b) | 8. (d) | 9. (a) |
| 10. (a) | 11. (d) | 12. (b) | 13. (d) | 14. (c) | 15. (a) | 16. (a) | 17. (d) | 18. (c) |
| 19. (c) | 20. (b) | 21. (b) | 22. (c) | 23. (d) | 24. (b) | 25. (a) | 26. (a) | 27. (a) |
| 28. (d) | 29. (d) | 30. (d) | 31. (c) | 32. (c) | 33. (b) | 34. (a) | 35. (c) | 36. (b) |
| 37. (a) | 38. (b) | 39. (b) | 40. (a) | | | | | |

Worksheet 2

1. (a, b)
2. (a, b)
3. (b)
4. (b)
5. (a, c, d)
6. (c)
7. (a, c)
8. (a, b, c)
9. (b, c, d)
10. (a, c)
11. (a, b, d)
12. (a, b, c, d)
13. (a, b, d)
14. (a, c)
15. (a, d)
16. (a, b)
17. (a, b)
18. (a, c)

Worksheet 3

1. (a) No (b) Yes
2. $1.7 \times 10^{15} \text{ N}$
3. (a) $\theta = \tan^{-1} \left[\frac{q^2}{8\pi\epsilon_0 x^2 Mg} \right]$ (b) $\frac{M}{2(\rho_0 - \rho_{\text{He}})}$
4. $Q = 4L\sqrt{\pi\epsilon_0 K(L - L_0)}$
5. $T = 2\pi\sqrt{\frac{L}{2g}}$
6. $q = \left[\frac{4\pi\epsilon_0 mgR^2}{\sqrt{3}} \right]^{\frac{1}{2}}$
7. $\frac{KQ^2}{L^2} \left[\left(1 + \frac{1}{3\sqrt{3}}\right)\hat{i} + \left(\frac{1}{3\sqrt{3}} + \frac{1}{2\sqrt{2}}\right)\hat{j} + \left(1 + \frac{1}{3\sqrt{3}} + \frac{1}{2\sqrt{2}}\right)\hat{k} \right]$
8. (a) $2\pi\sqrt{\frac{\pi\epsilon_0 md^3}{Qq}}$ (b) $2\pi\sqrt{\frac{2\pi\epsilon_0 md^3}{Qq}}$
9. $\frac{qQ}{8\pi^2\epsilon_0 r^2}$
10. $\frac{4}{5}\pi\rho_0 R^3$
11. $|E_x| = |E_y| = \frac{\lambda}{4\pi\epsilon_0 d}$
12. $\frac{\sqrt{3}Q}{2\sqrt{2}\pi\epsilon_0 a^2}$
13. (a) $\frac{q\sigma l}{2\epsilon_0}$ (b) zero
14. $\frac{3Qq}{2A\epsilon_0 m}$
15. $b = \frac{a}{4^{1/3}}$
16. $\frac{d}{3}$
17. $u = L\sqrt{\frac{qE}{2md} - \frac{g}{2d}}; \theta = \tan^{-1} \left(\frac{2d}{L} \right)$
18. (a) $\frac{Q(1-4r)}{4\pi\epsilon_0 r^3} \vec{r}$ (b) zero
19. $E = 2E_0$
20. $3.8 \times 10^{-18} \text{ C}$
21. 60°
22. (a) $\frac{1}{2}$ (b) $\frac{1}{4}, \frac{3}{4}$
23. $\sqrt{192} \text{ cm}$
24. $\sqrt{3}a$
25. $1.17 \times 10^{-8} \text{ C/m}$
26. Density of lines is not proper
27. $q = 4\pi\epsilon_0 a$ at the origin surrounded by a sphere of radius R having a uniform volume charge density $\rho = -\frac{3\epsilon_0 a}{R^3}$
28. $\frac{Q}{8\epsilon_0}$
29. (a) $\rho_0 \pi R^3$ (b) $E = \frac{\rho_0 r^2}{4\epsilon_0 R}$ for $0 \leq r \leq R$ and $E = \frac{\rho_0 R^3}{4\epsilon_0 r^2}$ for $r \geq R$
30. $E = 0$ for $|z| > a$; $\vec{E} = \frac{-\rho_0(d+z)}{\epsilon_0} \hat{k}$ for $-d < z < 0$; $\vec{E} = \frac{-\rho_0(d-z)}{\epsilon_0} \hat{k}$ for $0 < z < d$
31. $\frac{3q}{4\sqrt{2}\pi\epsilon_0 L^2}$
32. (a) Q_1 is positive and Q_2 is negative (b) $\left(\frac{l+a}{a}\right)^2$ (c) $\frac{l}{\left(1+\frac{l}{a}\right)^{2/3} - 1}$
33. $\frac{3\lambda}{\pi\epsilon_0 L}$ along perpendicular bisector of BC
34. $0 \leq E \leq \frac{24 \text{ mg}}{7q}$
35. (a) $496 \text{ nCm}^{-2}, 992 \text{ nCm}^{-2}$ (b) 2:1
36. $1.77 \times 10^{-12} \text{ Cm}^{-3}$
37. (a) $-Q$ (b) zero
(c) (i) zero (ii) $\frac{1}{4\pi\epsilon_0} \frac{Q}{r^2}$ (iii) zero (iv) zero]

Electrostatic Potential

“Don’t confound static electricity with ecstatic eccentricity. One will leave your hair up, the other will live up in the air!”.

—Ana Claudia Antunes

1. INTRODUCTION

We have studied in mechanics that law of conservation of mechanical energy is a powerful tool when it comes to relate speed and position of a body moving under influence of conservative forces. For framing the law of conservation of mechanical energy, we introduced the concept of potential energy.

We studied that the gravitational force and spring force are conservative in nature. We defined potential energy functions for these forces. Potential energy is defined corresponding to a conservative force only. In the last chapter, we have said that the electrostatic force is conservative in nature. We can define potential energy corresponding to this force as well.

In this chapter, we will develop the concept of electric potential using our understanding of potential energy. Actually, effect of a charge in its surrounding can be expressed in two ways:

- (i) by telling the electric field which is a vector, and
- (ii) by telling the electric potential which is a scalar.

Both these quantities are related to one another and knowledge of one can be used to find another.

2. ELECTRICAL POTENTIAL ENERGY

Let us recall the case of potential energy of a book in uniform gravitational field of the Earth. When the book is raised through a height h from point 1 to 2, its gravitational potential energy changes. Change in potential energy is defined as negative of work done by the gravitational force.

$$U_2 - U_1 = -W_{\text{gravity}}^{1 \rightarrow 2}$$

Work done by gravity in this example is $W_{\text{gravity}}^{1 \rightarrow 2} = -mgh$ where m is mass of the book

$$\therefore U_2 - U_1 = mgh$$

If point 1 is on the floor of your room and you assume $U_1 = 0$, then potential energy of the book at height h above the floor is written as

$$U = mgh$$

Actually, this is potential energy of the system of the Earth and the book. But we can associate this potential energy to the book only. Any change in potential energy does not affect the Earth. It practically remains fixed.

The concept of electric potential energy is no different. *In a system of charges, if the position of one or more charges is changed work may be done by the electric forces. We define change in electric potential energy of the system as negative of the work done by the conservative electric forces as the system changes its configuration.*

$$\Delta U = -W_{\text{elect}} \quad (1)$$

2.1 Potential Energy of a Pair of Point Charges

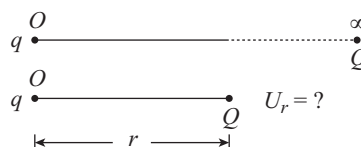
We will find the potential energy function for a simple system of two point charges.

When writing the gravitational potential energy of a book in your room, you often choose the floor to be the reference level where the potential energy of the book is assumed to be zero. It is for convenience. Choosing any other position, say table top, as reference level makes no difference.

When dealing with charges, we assume that the potential energy of a pair of charges is zero when they are at infinite separation (it means when they are widely separated).

Consider a point charge q fixed at point O . Another point charge Q is located at infinity. Potential energy of this system is zero.

$$U_{\infty} = 0.$$



2.2 Electrostatics and Current Electricity

Now, charge Q is brought to a point A at a distance r from q . We want to know what will be potential energy (U_r) now.

According to our definition of change in potential energy (equation (1)), we can write

$$U_r - U_\infty = -W_{\text{elect}}^{\infty \rightarrow r}$$

$$\Rightarrow U_r = -W_{\text{elect}}^{\infty \rightarrow r} \quad (2)$$

Therefore, *potential energy of a system of two point charges is equal to negative of work done by the electric force when charges are moved from infinite separation to their present separation (r).*

We need to find the work done by the electric force when Q is moved from ∞ to r .

Consider Q at some distance x from fixed charge q .

Electric force on it is

$$F_{\text{elect}} = \frac{1}{4\pi\epsilon_0} \frac{qQ}{x^2}$$

Assuming the charges to be positive, this force is towards right in the diagram shown.

Charge Q is pushed further towards q by a small distance dx . Work done by electric force is

$$dW_{\text{elect}} = |F_{\text{elect}}| |-dx| \cos 180^\circ$$

$$= \frac{qQ}{4\pi\epsilon_0 x^2} (-dx) (-1)$$

Note that x is decreasing and dx is a negative quantity. $(-dx)$ is magnitude of the small displacement.

$$\therefore dW_{\text{elect}} = \frac{qQ}{4\pi\epsilon_0} \frac{dx}{x^2}$$

Work done by the electric force when Q is moved from ∞ to separation r is

$$W_{\text{elect}}^{\infty \rightarrow r} = \int_{\infty}^r \frac{qQ}{4\pi\epsilon_0} \frac{dx}{x^2} = \frac{qQ}{4\pi\epsilon_0} \left[\frac{1}{\infty} - \frac{1}{r} \right]$$

$$\therefore W_{\text{elect}}^{\infty \rightarrow r} = -\frac{1}{4\pi\epsilon_0} \frac{qQ}{r} \quad (3)$$

From equation (2), we write the potential energy of two charges at separation r as

$$U_r = \frac{1}{4\pi\epsilon_0} \frac{qQ}{r} \quad (4)$$

Following points must be noted in context of the above derivation:

- Equation (4) gives potential energy of a system of two charges separated by a distance r . It is shared property of two charges. There is no meaning of electrical potential energy of an isolated point charge. However, if one of the two charges is fixed and only

the other charge changes position, we can say that the movable charge possesses the potential energy in the field of the fixed charge.

- While using equation (4) always substitute q and Q with their respective signs, electric potential energy is negative for a pair of point charges having opposite signs.
- In the above derivation, we assumed that charge q was fixed and Q was moved from ∞ to separation r . However, it makes no difference if both charges are moved. The potential energy depends on separation r and not on the spatial location of the charged particles.
- In the derivation, we assumed that q is moved along a straight line. This is not necessary. Work done by electric force is path independent.
- We can also define change in potential energy in terms of work done by external agent in changing the configuration of charged particles. In the above derivation, if we assume that charge Q always *moves slowly (without gaining kinetic energy)*, then the external agent must always apply a force on Q that is equal and opposite to the electric force acting on it. In such a case, work done by the external agent will be same in magnitude but opposite in sign to the work done by the electric force. Equation (2) can be written as

$$U_r = -W_{\text{elect}}^{\infty \rightarrow r} = W_{\text{ext Agt}}^{\infty \rightarrow r}(\text{slowly}) \quad (5)$$

Example 1 Potential energy in hydrogen atom

In a hydrogen atom, a proton and an electron are at a separation of $0.53 \text{ \AA}$. Calculate the electric potential energy of the atom. What is significance of the sign of potential energy?

Solution

Concepts

We must substitute charges with sign in equation (4)

$$U = \frac{1}{4\pi\epsilon_0} \frac{e \cdot (-e)}{r} = \frac{-9 \times 10^9 \times (1.6 \times 10^{-19})^2}{0.53 \times 10^{-10}}$$

$$\Rightarrow U = -4.35 \times 10^{-18} \text{ J}$$

Potential energy is always negative for a pair of positive and negative charge. *A negative value of potential energy is indicative of attractive force between the charges.*

Note: If you assume reference level at your table top then a book kept on the floor of your room has negative gravitational potential energy. A negative value of potential energy tells you that the potential energy is less than its value at the reference level.

Negative potential energy for a pair of charges tells you that potential energy is less than the potential energy when the two charges are at infinite separation.

Example 2 Work done by internal electric forces

Two charges q_1 and q_2 are held at separation r on x axis. They are moved so that they are brought on the y axis at a separation $2r$. Find work done by electric forces during this change in position. (You are supposed to give sum of work done by the electric forces on both the charges).

Solution

Concepts

- (i) $\Delta U = -W_{\text{elect}}$
- (ii) Work done by conservative electric force is path independent. Potential energy depends on separation and not on spatial location of the charges.

$$\text{Initial potential energy, } U_i = \frac{1}{4\pi\epsilon_0} \frac{q_1 q_2}{r}$$

$$\text{Final potential energy } U_f = \frac{1}{4\pi\epsilon_0} \frac{q_1 q_2}{2r}$$

$$\Delta U = U_f - U_i = -\frac{1}{2} \frac{q_1 q_2}{4\pi\epsilon_0 r} = -\frac{q_1 q_2}{8\pi\epsilon_0 r}$$

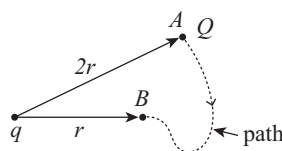
$$\text{As per our definition of change in potential energy,} \\ -W_{\text{elect}}^{i \rightarrow f} = \Delta U$$

$$\therefore W_{\text{elect}}^{i \rightarrow f} = -\Delta U = \frac{q_1 q_2}{8\pi\epsilon_0 r}$$

Note: Whenever internal electric forces do positive work, the potential energy of the system decreases.

Example 3 Work done by external agent

A charge q is fixed. Another charge Q is at point A at a distance $2r$ from q . This charge is moved slowly along the path shown to bring it to point B at a distance r from q . Find work done by the external agent in the process. How much work the external agent must perform to move Q from B to infinity.



Solution

Concepts

When movement is slow (so that the charge does not gain kinetic energy), force by the external agent is always equal and opposite to the electric force. Work done by agent is equal to negative of work done by the electric force.

$$\Delta U = -W_{\text{elect}}^{A \rightarrow B} = W_{\text{ext Agt}}^{A \rightarrow B}(\text{slowly})$$

$$\begin{aligned} W_{\text{ext Agt}}^{A \rightarrow B}(\text{slowly}) &= U_B - U_A \\ &= \frac{1}{4\pi\epsilon_0} \frac{qQ}{r} - \frac{1}{4\pi\epsilon_0} \frac{qQ}{2r} \\ &= \frac{qQ}{8\pi\epsilon_0 r} \end{aligned}$$

This is answer to the first part of the question.

The second part of the question asks us to find the work needed to move Q to infinity. *The question does not explicitly mention that the charge is moved slowly but we will assume that. The intent is not to provide kinetic energy to the charge; we just want to separate the charges.*

$$\begin{aligned} W_{\text{ext Agt}}^{B \rightarrow \infty}(\text{slowly}) &= U_{\infty} - U_B \\ &= 0 - \frac{1}{4\pi\epsilon_0} \frac{qQ}{r} \\ &= -\frac{qQ}{4\pi\epsilon_0 r} \end{aligned}$$

In Short

- (i) Electrostatic force is conservative; therefore, we can define a potential energy corresponding to it.
- (ii) When charges change position in a system of two or more charges, the electric forces may perform work. Change in potential energy of the system is defined as negative of work done by the electric forces. Change in potential energy may also be defined as work done by the external agent in slowly changing the position of the charges
- (iii) For a pair of charges, q and Q , placed at an infinite separation, potential energy is assumed to be zero. When they are at separation r

$$U_r = K \frac{qQ}{r} = \frac{1}{4\pi\epsilon_0} \frac{qQ}{r}$$

Values of the two charges must be substituted with sign in the above expression.

- (iv) Potential energy is negative when force involved is attractive and reference level is taken at infinity. Gravitational potential energy of two masses (m_1 and m_2) at separation r is also negative.

$$U = -\frac{Gm_1 m_2}{r}$$



YOUR TURN

Q.1 Is it right to say that potential energy of a pair of charges decreases when their separation is increased?

Q.2 A charge q is fixed, another charge Q is near to q and is being moved. Is it necessary that the potential energy of the system is changing?

Q.3 A point charge $q_1 = 2\mu\text{C}$ is kept at a distance of $r_1 = 1.0\text{ m}$ from another point charge $q_2 = -1\mu\text{C}$. The two

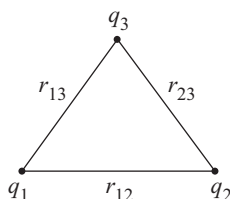
charges are slowly pulled apart to double their separation to $r_2 = 2.0\text{ m}$. In the process, charge q_1 was moved by 0.7 m and q_2 was shifted by 0.3 m .

- Find work done by electric forces in the process.
- Find work done by the external agent in the process.

Q.4 A point charge $q_1 = -1.0\text{ }\mu\text{C}$ is fixed at the origin. A second point charge $q_2 = 2\mu\text{C}$ moves from $(1, 0, 0)\text{m}$ to $(0, 2, 0)\text{m}$. Find work done by electric force on q_2 .

2.2 Potential Energy for a System of More Than Two Charges

Consider a system of three charges arranged as shown in the figure. We wish to write potential energy of this system. For this, we write potential energy of all pairs of charges and add them.



$$U = \frac{1}{4\pi\epsilon_0} \left(\frac{q_1 q_2}{r_{12}} + \frac{q_1 q_3}{r_{13}} + \frac{q_2 q_3}{r_{23}} \right) \quad (6)$$

What does the above expression represent? Consider that charges are far apart from one another. q_1 is held fixed and q_2 is moved to the position shown. Work done by the electric force in the process is W_1 .

Now, q_1 and q_2 are held fixed at their positions and q_3 is moved from infinity to the position shown. Work done by electric forces (due to q_1 and q_2) in movement of q_3 is W_2 . Therefore, total work done by electric forces in assembling the charges is $W_1 + W_2$. The potential energy of the system expressed by equation (6) is negative of work done by the electric forces in assembling the charges. It is equal to $-(W_1 + W_2)$.

If the charges were at rest when at infinity and are at rest when they are in the position shown, then potential energy (U) of the system can also be defined as work done by the external agent in assembling them to their present position.

While writing potential energy for a collection of point charges, one must write potential energy corresponding to each pair but must avoid the mistake of double counting them. If a system has n point charges, the expression of potential energy will have $\frac{n(n-1)}{2}$ terms in it.

Some people deliberately take all permutations (thereby double counting each pair) and add them. Multiplying the result obtained by $\frac{1}{2}$ gives the potential energy.

Example 4 Three equal point charges, q each, are located at the vertices of an equilateral triangle of side length L .

- Calculate electrostatic potential energy of the configuration.
- What charge must be placed at the centroid of the triangle so that the electrostatic potential energy of the system becomes zero?

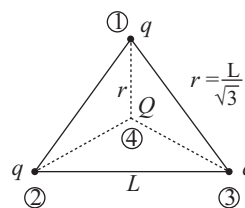
Solution

Concepts

Initially, there are three charges in the system. The expression of potential energy has 3 terms.

After a charge is placed at the centre, now there are $\frac{n(n-1)}{2} = \frac{4 \times 3}{2} = 6$ terms in the expression.

$$\begin{aligned} \text{(a)} \quad U &= U_{12} + U_{13} + U_{23} \\ &= \frac{1}{4\pi\epsilon_0} \frac{q^2}{L} \times 3 \\ &= \frac{3q^2}{4\pi\epsilon_0 L} \end{aligned}$$

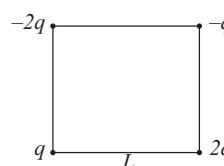


$$\begin{aligned} \text{(b)} \quad &\text{Let the charge at the centroid be } Q. \\ U' &= 0 = (U_{12} + U_{13} + U_{23}) + (U_{41} + U_{42} + U_{43}) \\ \Rightarrow 0 &= \frac{3q^2}{4\pi\epsilon_0 L} + 3 \frac{1}{4\pi\epsilon_0} \frac{qQ}{L/\sqrt{3}} \\ \Rightarrow \sqrt{3} Q &= -q \Rightarrow Q = -\frac{q}{\sqrt{3}} \end{aligned}$$

Example 5 Work done in disassembling

Four point charges are kept at the vertices of a square of side length L as shown in the figure.

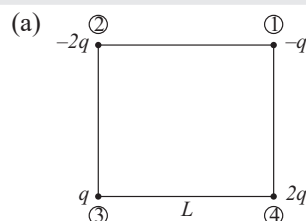
- Calculate work done by the external agent in slowly moving the $-q$ charge to infinity.



- (b) Calculate the work needed in separating all charges from one another by a large distance.

Solution**Concepts**

$$W_{\text{extAgt}} = \Delta U = U_f - U_i$$



Potential energy of the charge system is

$$U_i = U_{12} + U_{13} + U_{14} + U_{23} + U_{24} + U_{34}$$

$$\therefore U_i = \frac{1}{4\pi\epsilon_0} \left[\frac{(-q)(-2q)}{L} + \frac{(-q)(q)}{\sqrt{2}L} + \frac{(-q)(2q)}{L} + \frac{(-2q)(q)}{L} + \frac{(-2q)(2q)}{\sqrt{2}L} + \frac{q(2q)}{L} \right]$$

$$= -\frac{1}{4\pi\epsilon_0} \frac{5}{\sqrt{2}} \frac{q^2}{L}$$

After $-q$ is removed, three charges remain in the system and the potential energy changes to

$$U_f = U_{23} + U_{24} + U_{34} = -\frac{1}{4\pi\epsilon_0} \frac{4q^2}{\sqrt{2}L}$$

$$W_{\text{extAgt}} = \Delta U = U_f - U_i$$

$$\Rightarrow W_{\text{extAgt}} = \frac{q^2}{4\sqrt{2}\pi\epsilon_0 L}$$

- (b) When the complete system is disassembled, the final potential energy is

$$U_f = 0$$

$$W_{\text{extAgt}} = U_f - U_i = \frac{5q^2}{4\sqrt{2}\pi\epsilon_0 L}$$

**YOUR TURN**

Q.5 A point charge $q_1 = -1\mu\text{C}$ is fixed on x axis at $x = 0$. Another point charge $q_2 = 1\mu\text{C}$ is fixed at $x = 1\text{m}$. A third point charge $q_3 = 1\mu\text{C}$ is moved from infinity to $x = 2\text{m}$.

- Find the total potential energy of the system of three charges.
- How much work was done by the external agent in moving the third charge?

Q.6 Four charges $+q, -q, +q$ and $-q$ are placed in the same order on the four consecutive corners of a square of side length a .

- Calculate the potential energy of the system.
- Calculate the work done in interchanging the positions of any two neighbouring charges of opposite sign.

Q.7 Six point charges are held fixed at the vertices of a regular hexagon of side length L . The algebraic sum of the charges is Q and the electric potential energy of the system is U_0 . A seventh point charge q is placed at the centre of the hexagon and the potential energy of the system of seven charges now becomes U . Find $U - U_0$.

In Short

- (i) Potential energy of a system having n number of point charges is

$$U = \frac{1}{4\pi\epsilon_0} \sum \frac{q_i q_j}{r_{ij}} \text{ with } i < j$$

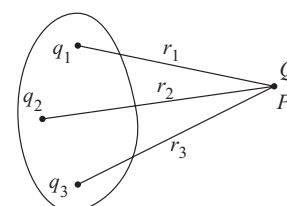
$i < j$ ensures that there is no repetition of the pairs. For example, we choose $q_2 q_3$ but not $q_3 q_2$.

$$\text{Total number of terms are } \frac{n(n-1)}{2}$$

- (ii) Potential energy of a charge collection is equal to work done by an external agent in assembling the charges from infinite separation to the present separation.

3. ELECTRIC POTENTIAL

Consider a group of fixed charges q_1, q_2 and q_3 . Another charge Q is placed at a point P near these charges. The potential energy of this system has two parts:



- (i) Potential energy due to interactions of q_1, q_2 and q_3 amongst themselves.

$$\text{It has terms like } \frac{1}{4\pi\epsilon_0} \frac{q_1 q_2}{r_{12}}.$$

Let's call this energy as U_0 . It is *self-energy* of source charges arising due to their interactions.

- (ii) Potential energy due to interactions of q_1 , q_2 and q_3 with Q . This part of the energy is

$$U = \frac{1}{4\pi\epsilon_0} \left[\frac{q_1 Q}{r_1} + \frac{q_2 Q}{r_2} + \frac{q_3 Q}{r_3} \right]$$

$$= \frac{1}{4\pi\epsilon_0} \left[\frac{q_1}{r_1} + \frac{q_2}{r_2} + \frac{q_3}{r_3} \right] Q$$

When the charge Q is allowed to move (or is moved by an external agent), the potential energy of the charge system may change. Note that U_0 does not change; it is only U which changes. Therefore, change in potential energy can be obtained by knowing the change in U . We can simply forget about U_0 – the internal interaction energy of the fixed source charges.

The energy U can be termed as potential energy of charge Q in the field of source charges q_1 , q_2 and q_3 .

$$U = \frac{1}{4\pi\epsilon_0} \left[\frac{q_1}{r_1} + \frac{q_2}{r_2} + \frac{q_3}{r_3} \right] Q$$

$$\Rightarrow U = VQ \quad (7)$$

$$\text{Here, } V = \frac{1}{4\pi\epsilon_0} \left[\frac{q_1}{r_1} + \frac{q_2}{r_2} + \frac{q_3}{r_3} \right] \quad (8)$$

V is known as electric potential due to source charges at point P . Unit of electric potential is J C^{-1} which is commonly known as volt (V). Potential is a scalar.

Important lesson from the above discussion is that we think of potential energy of a test charge in the field of a given set of fixed source charges in two steps:

- Source charges set up a potential field (V).
- A test charge (Q) kept at a point in this field has potential energy $U = QV$.

We can define electric potential at a point using equation (7) as:

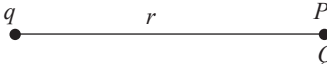
If a test charge Q placed at a point P (in field of some source charges) has electric potential energy U , then potential at the point is defined as

$$V = \frac{U}{Q}$$

3.1 Electric Potential Due to a Point Charge

Consider a fixed point charge q . We wish to write electric potential due to this single point charge at a point P which is at a distance r .

Imagine a test charge Q placed at P . Potential energy of charge Q is

$$U = \frac{1}{4\pi\epsilon_0} \frac{qQ}{r}$$


Potential at P due to our source charge q is

$$V = \frac{U}{Q} = \frac{1}{4\pi\epsilon_0} \frac{q}{r} \quad (9)$$

Well, you may interpret electric potential as potential energy of a unit test charge.

Recall that

$$U = -W_{\text{elect}}^{\infty \rightarrow r} = W_{\text{ext Agt}}^{\infty \rightarrow r}(\text{slowly})$$

$$\therefore V = \frac{U}{Q} = -\frac{W_{\text{elect}}^{\infty \rightarrow r}}{Q} = \frac{W_{\text{ext Agt}}^{\infty \rightarrow r}(\text{slowly})}{Q} \quad (10)$$

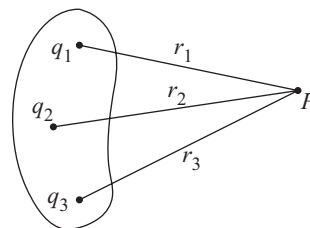
In words, *potential at a point P is work done by an external agent in slowly moving a unit positive test charge from infinity to point P* . Here, we are assuming that potential is zero at infinity.

3.2 Potential Due to Multiple Charges

Electric potential obeys principle of superposition. *Addition of potentials due to each and every source charge gives the resultant potential at a point*. Addition is simple algebraic addition because potential is a scalar.

In the diagram shown, potential at P is given by

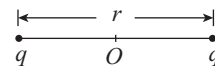
$$V = \frac{1}{4\pi\epsilon_0} \frac{q_1}{r_1} + \frac{1}{4\pi\epsilon_0} \frac{q_2}{r_2} + \frac{1}{4\pi\epsilon_0} \frac{q_3}{r_3} \quad (11)$$



This is nothing but equation (8). Remember to substitute q_1 , q_2 and q_3 with respective signs.

Example 6 Two point charges, $q = 1\mu\text{C}$ each, are kept at a separation $r = 2.0$ m.

- Find the electric potential at mid point O of the line joining the two charges.
- Find work done in moving a unit positive test charge (1C) from infinity to point O .
- What is potential energy of a charge $Q = -2\mu\text{C}$ placed at O ?



Solution

Concepts

- Potential obeys principle of superposition.
- $U = QV$ gives potential energy of a charge Q kept in the field of fixed source charges.

$$(a) V = \frac{1}{4\pi\epsilon_0} \frac{q}{r/2} + \frac{1}{4\pi\epsilon_0} \frac{q}{r/2}$$

[Note that potential is a scalar. Simple algebraic addition is needed]

$$\therefore V = \frac{1}{4\pi\epsilon_0} \frac{4q}{r} = 9 \times 10^9 \times \frac{4 \times 1 \times 10^{-6}}{2} = 18000 \text{ V.}$$

(b) By definition, potential at a point is work done in slowly moving a unit positive test charge from infinity to that point

$\therefore$ Required work done is 18000 J

$$\left[\text{Volt} = \frac{\text{Joule}}{\text{Coulomb}} \right]$$

$$U = QV = (-2 \times 10^{-6}) (18000) = -0.036 \text{ J}$$

(c) In all expressions of potential and potential energy, put values of charges with sign.

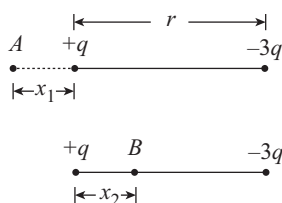
Example 7 Two charges $+q$ and $-3q$ are fixed a distance r apart. Find the points on the line joining the two charges where potential is zero.

Solution

Concepts

- Potential at a point will be $V = V_1 + V_2$ where V_1 and V_2 are potentials due to $+q$ and $-3q$, respectively.
- Potential due to a positive charge is positive and is negative due to a negative charge.

A and B are two possible points where potential can be zero. Potential cannot be zero at a point that lies to the right of $-3q$ charge because the point will be closer to the charge of larger magnitude.



For $V_A = 0$

$$\frac{1}{4\pi\epsilon_0} \frac{q}{x_1} + \frac{1}{4\pi\epsilon_0} \frac{(-3q)}{r+x_1} = 0$$

$$\Rightarrow x_1 = \frac{r}{2}$$

For $V_B = 0$

$$\frac{1}{4\pi\epsilon_0} \frac{q}{x_2} + \frac{1}{4\pi\epsilon_0} \frac{(3q)}{(r-x_2)} = 0$$

$$\Rightarrow x_2 = \frac{r}{4}$$

Example 8 Work done in moving a charge

A set of fixed source charges produce a potential of 100 V at a point A . Potential at a different point B is -50 V. Calculate

the work needed to slowly shift a $Q = 4$ mC charge from point A to B .

Solution

Concepts

- Potential energy associated with a charge Q placed at a point having potential V is $U = QV$.
- $W_{\text{ext Agt}}^{A \rightarrow B} = U_B - U_A = \text{change in potential energy.}$

$$U_A = QV_A$$

$$U_B = QV_B$$

$$\begin{aligned} W_{\text{ext Agt}}^{A \rightarrow B} &= U_B - U_A = Q(V_B - V_A) \\ &= (4 \times 10^{-3} \text{ C}) [-50\text{V} - 100\text{V}] \\ &= -600 \times 10^{-3} \text{ J} = -0.6 \text{ J} \end{aligned}$$

Note that work done by electric force (applied by the source charges on Q) will be 0.6 J as the charge Q moves from A to B .

Example 9 Charges on a hexagon

On the vertices of a regular hexagon point charges $q_1 = -2\mu\text{C}$, $q_2 = +2\mu\text{C}$, $q_3 = +5\mu\text{C}$, $q_4 = -7\mu\text{C}$, $q_5 = +7\mu\text{C}$ and $q_6 = -5\mu\text{C}$ have been placed in the order mentioned.

- Find potential at the centre of the hexagon.
- Position of charges q_2 and q_4 are interchanged. How does the potential at the centre change?
- A charge $Q = +6\mu\text{C}$ is moved from infinity to the centre of the hexagon without changing its speed. How much work is done?

Solution

Concepts

$$V = V_1 + V_2 + \dots + V_6$$

All charges are equidistant from the point where we intend to know the potential.

$$\begin{aligned} (a) V &= \frac{1}{4\pi\epsilon_0} \left[\frac{q_1}{r} + \frac{q_2}{r} + \dots + \frac{q_6}{r} \right] \\ &= \frac{1}{4\pi\epsilon_0 r} [q_1 + q_2 + \dots + q_6] = \frac{1}{4\pi\epsilon_0 r} \sum (q_i) \\ &= \frac{1}{4\pi\epsilon_0 r} [-2 + 2 + 5 - 7 + 7 - 5] = 0 \end{aligned}$$

- When position of charges is changed, the sum $\sum q_i$ is still zero.

$$\therefore V = 0$$

$$\begin{aligned} (c) W_{\text{ext Agt}}^{\infty \rightarrow \text{centre}} &= U_{\infty} - U_{\text{centre}} \\ &= Q(V_{\infty} - V_{\text{centre}}) \\ &= Q(0 - 0) = 0 \end{aligned}$$

Example 10 Variation of potential

The point charges, $+q$ and $-3q$ are held fixed at some separation on the x axis. Draw a rough graph showing the variation of potential on the x axis.

Solution**Concepts**

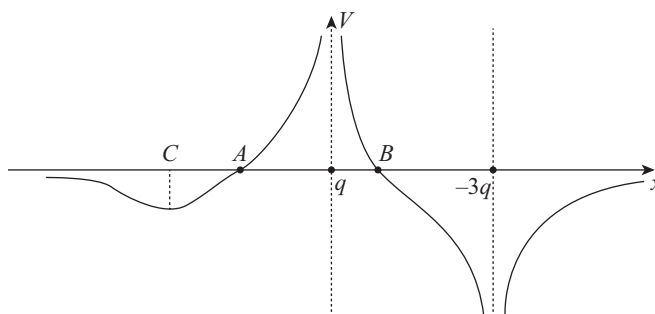
- (i) Refer to example 7. There are two points on x axis where the potential is zero. One point lies to the left of charge q and the other point lies somewhere between the two charges.
- (ii) Potential very close to a point charge is very large.

$$V = \frac{1}{4\pi\epsilon_0} \frac{q}{r}$$

$$V \rightarrow \infty \text{ when } r \rightarrow 0$$
- (iii) Potential very close to a negative charge is a large negative number.
- (iv) Potential to the right of $-3q$ charge is negative everywhere.

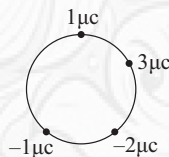
- (v) Potential to the left of charge q is positive close to the charge. It becomes zero at point A and thereafter it will become negative. The magnitude of the potential increases as we move to the left and then decreases to become zero at a large distance.
- (v) At distance large compared to separation of the charges, potential is just equal to potential created by a point charge $-2q$.

The required graph is as shown below. Potential is zero at A and B and has a local minima at C.

**YOUR TURN**

Q.8 Two point charges q and $-q$ are kept fixed at a distance L . Find the electric potential at a point on the perpendicular bisector of the line joining the two charges.

Q.9 Write the electric potential at the centre of the circle shown in the figure. Radius is 1.0 m.



Q.10 A point charge $q = 1\mu\text{C}$ is moved slowly along a curved path from infinity to a point where electric potential (due to some undefined source charges) is 10 V. Find work done by the external agent.

Q.11 Refer to example 9. For the charge arrangement given in the problem, calculate electric potential at a point

which lies at a distance x from the centre of the hexagon on a line perpendicular to the plane of the hexagon and passing through its centre.

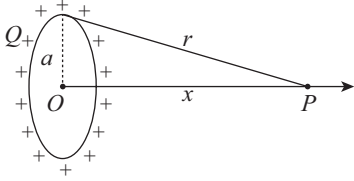
Q.12 A charge $Q = 10^{-3} \text{ C}$ is moved from a point A to point B in an electric field. Work done against the electric field in the process is 1.2 J. Find the difference in potentials at points B and A.

Q.13 Refer to the graph given in solution of example 10. Find the distance of point C from charge q , if distance between the two charges is r .

Q.14 Three point charges $q_1 = 1\mu\text{C}$, $q_2 = -2\mu\text{C}$ and $q_3 = 3\mu\text{C}$ are kept fixed at $(0, 0, 1)\text{m}$, $(0, 2, 0)\text{m}$ and $(3, 0, 0)\text{m}$, respectively. Find work done by the electric forces when a point charge $q_0 = 1\text{mC}$ is moved from infinity to the origin of the co-ordinate system.

3.3. Electric Potential on the Axis of a Charged Ring

A ring of radius a carries charge Q . A point P lies on its axis at a distance x from its centre. We wish to write electric potential at P due to the ring charge.



It is easy to observe that every point on the circumference of the ring is at same distance $r (= \sqrt{a^2 + x^2})$ from the point P . A small charge Δq on the ring produces a potential at P given by

$$\Delta V = \frac{1}{4\pi\epsilon_0} \frac{\Delta q}{r}$$

Resultant potential at P is sum of contributions made by all such charges on the ring.

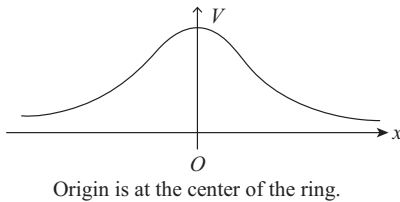
$$\begin{aligned} \therefore V &= \sum \Delta V = \frac{1}{4\pi\epsilon_0 r} \sum \Delta q \\ \Rightarrow V &= \frac{1}{4\pi\epsilon_0} \frac{Q}{r} = \frac{1}{4\pi\epsilon_0} \frac{Q}{\sqrt{a^2 + x^2}} \end{aligned} \quad (12)$$

It makes no difference whether the ring is uniformly charged or not. The potential is always given by equation (12).

To write potential at the centre O of the ring, you can put $x = 0$ in equation (12).

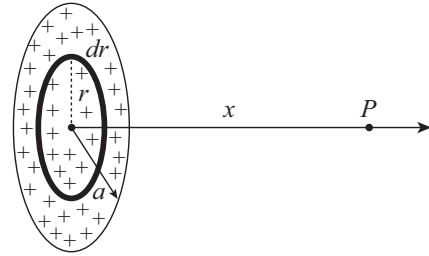
$$V_{\text{centre}} = \frac{1}{4\pi\epsilon_0} \frac{Q}{a} \quad (13)$$

As one moves along the axis of the ring, potential is maximum at the centre and decreases on either side to become zero at large values of x . Graphical variation of potential along the axis of a positively charged ring is as shown.



3.4. Electric Potential on the Axis of a Uniformly Charged Disc

A disc of radius a has charge uniformly spread on its surface. The surface charge density is σ . We wish to find electric potential at a point on the axis of the disc which is at a distance x from the centre of the disc.



We can divide the entire disc charge into a large number of thin circular strips. Consider a strip of radius r and width dr . Charge on this circular ring is

$$dq = (2\pi r dr) \sigma$$

Potential at P due to charge dq can be written using equation (12) as

$$\begin{aligned} dV &= \frac{1}{4\pi\epsilon_0} \frac{dq}{\sqrt{r^2 + x^2}} = \frac{1}{4\pi\epsilon_0} \frac{\sigma(2\pi r dr)}{\sqrt{r^2 + x^2}} \\ &= \frac{\sigma}{2\epsilon_0} \frac{r dr}{\sqrt{r^2 + x^2}} \end{aligned}$$

Potential at P is obtained by summing the contributions made by all such ring charges.

$$\begin{aligned} \therefore V &= \int_{r=0}^a dV = \frac{\sigma}{2\epsilon_0} \int_0^a \frac{r dr}{\sqrt{r^2 + x^2}} \\ \Rightarrow V &= \frac{\sigma}{2\epsilon_0} \left[\sqrt{r^2 + x^2} \right]_0^a \\ \Rightarrow V &= \frac{\sigma}{2\epsilon_0} [\sqrt{a^2 + x^2} - x] \end{aligned} \quad (14)$$

$$[\text{Integration, } I = \int \frac{r dr}{\sqrt{r^2 + x^2}}]$$

$$\text{Let } r^2 + x^2 = t$$

$$\Rightarrow 2r \frac{dr}{dt} + 0 = 1$$

$$\Rightarrow r dr = \frac{1}{2} dt$$

$$\therefore I = \frac{1}{2} \int \frac{dt}{\sqrt{t}} = \sqrt{t} = \sqrt{r^2 + x^2}$$

Note that x is a constant in the integration.]

Notes:

- (i) Potential at the centre of the disc can be found by putting $x = 0$ in the above expression.

$$V_{\text{centre}} = \frac{\sigma a}{2\epsilon_0} \quad (15)$$

- (ii) For a point located at a distance $x \gg a$, we can approximate our result as

$$V = \frac{\sigma}{2\epsilon_0} [(x^2 + a^2)^{1/2} - x]$$

$$= \frac{\sigma x}{2\epsilon_0} \left[\left(1 + \frac{a^2}{x^2} \right)^{\frac{1}{2}} - 1 \right]$$

$$\approx \frac{\sigma x}{2\epsilon_0} \left[1 + \frac{1}{2} \frac{a^2}{x^2} - 1 \right] = \frac{\sigma a^2}{4\epsilon_0 x}$$

[Here, we have used binomial expansion to write $(1+x)^n \approx 1+nx$ for $x \ll 1$]

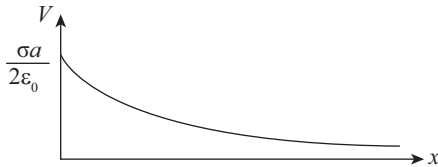
If total charge on the disc is Q , then

$$\sigma = \frac{Q}{\pi a^2}$$

$$\therefore V \approx \frac{1}{4\pi\epsilon_0} \frac{Q}{x}$$

The result is on expected lines. At a faraway point, the disc charge will appear like a point charge.

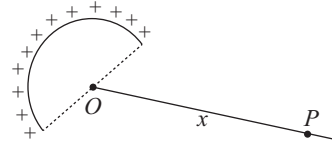
- (iii) The graphical plot of variation of potential with distance x is as shown in the adjoining figure.



Example 11 A half ring

A half ring has a charge Q spread on it. Radius of the ring is a . Write the potential at:

- (a) the centre (O)
 (b) a point P on the axis at a distance x from O .



Solution

Concepts

Every charge point is at distance a from O and at a distance $\sqrt{a^2 + x^2}$ from P .

$$(a) V = \frac{1}{4\pi\epsilon_0} \frac{1}{a} \sum \Delta q = \frac{1}{4\pi\epsilon_0} \frac{Q}{a}$$

$$(b) V = \frac{1}{4\pi\epsilon_0} \frac{1}{\sqrt{a^2 + x^2}} \sum \Delta q = \frac{1}{4\pi\epsilon_0} \frac{Q}{\sqrt{a^2 + x^2}}$$

Example 12 Radially varying charge density

A disc has radius a and carries a total charge Q on it. The surface density of charge varies with distance r from the centre as $\sigma = \sigma_0 \left(\frac{r}{a} \right)$, where σ_0 is a constant.

- (a) Find σ_0 in terms of Q and a .
 (b) Find electric potential at the centre of the disc.

Solution

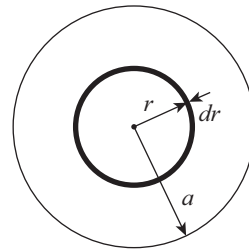
Concepts

- (i) We will find charge on disc with the help of variable charge density expression and integration. Equating the total charge to Q will give the value of σ_0 . Value of σ_0 is such that total charge on the disc is Q .
 (ii) Charge density changes with distance from the centre but it is same everywhere on a ring of radius r . We can divide the disc into number of thin rings and add the potential produced due to each of the rings at the centre.

- (a) Consider a thin ring of thickness dr having radius r . Charge on the ring element is

$$dq = \sigma \cdot 2\pi r dr$$

$$= \frac{2\pi\sigma_0}{a} r^2 dr$$



$\therefore$ Total charge on the disc is

$$Q = \int_{r=0}^a dq = \frac{2\pi\sigma_0}{a} \int_0^a r^2 dr$$

$$\Rightarrow Q = \frac{2\pi\sigma_0 a^2}{3}$$

$$\Rightarrow \sigma_0 = \frac{3Q}{2\pi a^2}$$

- (b) Potential at the centre due to the ring element is

$$dV = \frac{1}{4\pi\epsilon_0} \frac{dq}{r} = \frac{1}{4\pi\epsilon_0} \frac{2\pi\sigma_0}{a} r dr$$

$$\Rightarrow dV = \frac{\sigma_0}{2\epsilon_0 a} r dr$$

Potential at the centre is obtained as

$$V = \int_{r=0}^a dV = \frac{\sigma_0}{2\epsilon_0 a} \int_0^a r dr = \frac{\sigma_0 a}{4\epsilon_0}$$

$$= \frac{3Q}{2\pi a^2} \frac{a}{4\epsilon_0} = \frac{3Q}{8\pi\epsilon_0 a}$$

In Short

- (i) Electric potential at a point is negative of work done by electric force in moving a unit positive test charge from infinity to that point. It is also equal to work done by the external agent in slowly shifting a unit positive test charge from infinity to the point.
- (ii) Potential due to a point charge at a distance r from it is

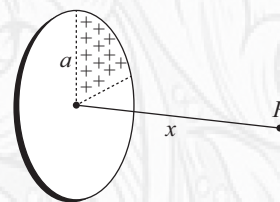
$$V = \frac{1}{4\pi\epsilon_0} \frac{q}{r}$$
- (iii) Potential obeys principle of superposition. If there are number of source charges, then potential at a point is the sum of potential due to individual charges.
- (iv) Potential energy of a charge Q placed at a point where electric potential is V is given by $U = QV$.
- (v) Potential is a scalar. Its unit is JC^{-1} or volt (V).



YOUR TURN

Q.15 A ring has charge $+Q$ and its radius is R . A point charge $-q$ is kept at the centre of the ring. Find work done in moving the charge slowly to infinity, keeping the ring fixed.

Q.16 A disc of radius a has charge Q spread uniformly on quarter of its surface. Find electric potential at a point P on its axis at a distance x from its centre.



4. CONSERVATION OF ENERGY AND WORK-ENERGY THEOREM

We will quickly recapitulate the principle of conservation of mechanical energy and the work-energy theorem. They have wide applications in problem solving in this chapter.

Mechanical energy of a system remains conserved when only conservative forces perform work. Electrostatic force is conservative.

If a charged particle moves under the electric field set up by fixed charges, and no other force performs work on it, sum of its kinetic and potential energy will remain conserved. If the particle moves from initial point (i) to final point (f), we write

$$U_i + K_i = U_f + K_f \quad (16)$$

If the particle is experiencing other conservative forces (e.g., gravity) apart from the electrostatic force, then also we can use the above equation. Now the potential energy (U) is sum of electrostatic and gravitational potential energy.

$$U_{ig} + U_{ie} + K_i = U_{fg} + U_{fe} + K_f \quad (17)$$

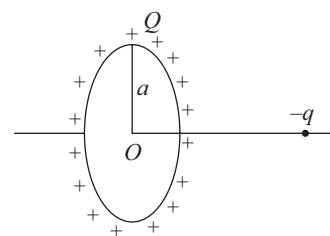
Consider a group of charged particles interacting with one another. Due to their mutual interactions, their position changes. The kinetic energy of the system of the particles may also change. Energy of the system remains conserved if no other dissipative force is present. Equation (16) can still be used with U equal to potential energy of interaction of all pairs of charges and K equal to sum of kinetic energies of all the particles.

Consider a charge moving in a static electric field. It experiences an external force F . The force performs work but it may not be conservative. Now, the initial and final energies of the particle are related by work-energy theorem.

$$W_F = (U_f + K_f) - (U_i + K_i) \quad (18)$$

Example 13 Charge moving on axis of a ring

A uniformly charged ring has radius a and carries a charge Q on it. The ring is fixed in space. A particle having mass m and charge $-q$ is released at a point on the axis of the ring at a distance $x = 2a$ from its centre.



2.12 Electrostatics and Current Electricity

- Will the particle move along the axis?
- Find the speed of the particle when it reaches the centre O .
- Will the particle move to the other side of the ring? Upto what distance will it go on the other side of the ring?

Solution

Concepts

- Ring charge is a fixed source producing a field. Charge $-q$ moves in this field. We can associate a potential energy with $-q$ as $U = -qV$ where V is electric potential due to the ring charge.
- Energy of charge q remains conserved.

$$U_f + K_f = U_i + K_i$$

- Electric field on the axis is away from O . The particle experiences an attractive force towards O along the axis. It will move along the axis.

Note: If the charge were distributed non-uniformly on the ring, the electric field might not be along the axis.

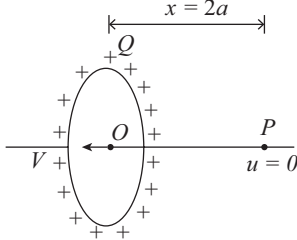
- Potential due to the ring charge at point P is

$$V_P = \frac{1}{4\pi\epsilon_0} \frac{Q}{\sqrt{a^2 + x^2}}$$

$$= \frac{Q}{4\sqrt{5}\pi\epsilon_0 a}$$

Potential at centre O is

$$V_0 = \frac{1}{4\pi\epsilon_0} \frac{Q}{a}$$



Applying conservation of energy for motion of the particle from point P to O :

$$K_0 + U_0 = K_P + U_P$$

$$\Rightarrow \frac{1}{2}mv^2 + (-q)V_0 = 0 + (-q)V_P$$

$$\Rightarrow \frac{1}{2}mv^2 = q(V_0 - V_P)$$

$$\Rightarrow \frac{1}{2}mv^2 = \frac{Qq}{4\pi\epsilon_0 a} \left(1 - \frac{1}{\sqrt{5}}\right)$$

$$\Rightarrow v = \sqrt{\frac{Qq}{2\pi\epsilon_0 am} \left(1 - \frac{1}{\sqrt{5}}\right)}$$

- At O , the particle has velocity directed towards left. It crosses O and moves to the left of the ring. Electric field due to the ring exerts a rightward force on the negative particle and it slows down. The particle will stop exactly at a distance $2a$ on the other side of the ring. This follows from conservation of energy. It is easy to see that the particle will perform oscillations about O .

Example 14 Two particles having masses m and $2m$ carry equal charge q on them. They are held at separation r and released. Assume that no other force acts on them apart from the Coulomb force.

- Find their speeds when separation between them grows to $2r$.
- Find the maximum speed acquired by the particle of mass m .

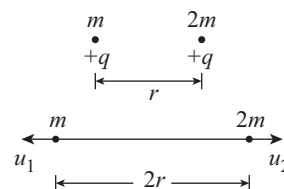
Solution

Concepts

- Particles move under mutual electric forces. Energy of the system is conserved.
- There is no external force acting on the system of two particles. Momentum of the system must also remain conserved.

- Momentum of the system will always remain zero.

Let the velocities of the two particles be u_1 and u_2 in opposite directions when separation between them becomes $2r$.



$$mu_1 = 2mu_2$$

$$\Rightarrow u_1 = 2u_2 \quad (i)$$

Conservation of energy:

$$K_f + U_f = K_i + U_i$$

$$\frac{1}{2}mu_1^2 + \frac{1}{2}(2m)u_2^2 + \frac{1}{4\pi\epsilon_0} \frac{q^2}{2r} = 0 + \frac{1}{4\pi\epsilon_0} \frac{q^2}{r}$$

$$\Rightarrow \frac{1}{2}m(2u_2)^2 + \frac{1}{2}(2m)u_2^2 = \frac{q^2}{8\pi\epsilon_0 r}$$

$$\Rightarrow u_2^2 = \frac{q^2}{24\pi\epsilon_0 mr}$$

$$\Rightarrow u_2 = \frac{q}{\sqrt{24\pi\epsilon_0 mr}}$$

$$u_1 = 2u_2 = \frac{q}{\sqrt{6\pi\epsilon_0 mr}}$$

Notes:

- Decrease in electrostatic potential energy results in increased kinetic energy.
 - Internal forces can change the kinetic energy of a system but they cannot change momentum.
- Speeds of both the particles will be maximum when they are at infinite separation. The entire potential energy gets converted into kinetic energy.

$$K_\infty + U_\infty = K_i + U_i$$

$$\frac{1}{2}mu_1^2 + \frac{1}{2}(2m)u_2^2 + 0 = 0 + \frac{1}{4\pi\epsilon_0} \frac{q^2}{r}$$

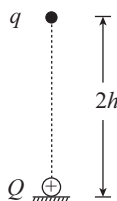
$$\Rightarrow \frac{1}{2}m(2u_2)^2 + mu_2^2 = \frac{q^2}{4\pi\epsilon_0 r}$$

$$\Rightarrow u_2 = \sqrt{\frac{q^2}{12\pi\epsilon_0 mr}}$$

$$\text{and } u_1 = 2u_2 = \sqrt{\frac{q^2}{3\pi\epsilon_0 mr}}$$

Example 15 Two types of potential energies

A small ball having charge Q is fixed on a non-conducting horizontal surface. A particle of mass m having charge q is held at a height $2h$ above it. The particle is released and it stops momentarily after falling through a distance h . Find the value of Q in terms of other given parameters. Also find the height at which the particle can stay in equilibrium.



Solution

Concepts

- There are two conservative forces performing work – gravity and Coulomb force. Energy is conserved. We need to consider two types of potential energy while writing the equation of energy conservation – gravitational potential energy and the electrostatic potential energy.
- At equilibrium position the Coulomb repulsion on the particle balances its weight.

One can guess that equilibrium will lie between the heights h and $2h$. Initially (at $2h$ height), mg is larger than Coulomb force and the particle accelerates. It accelerates till it reaches its equilibrium position. After this, the Coulomb force exceeds mg and the particle retards. It eventually stops at a height h .

Initially, at height $2h$, the particle is at rest. It is also at rest at height h .

$$K_i + U_{ie} + U_{ig} = K_f + U_{fe} + U_{fg}$$

$$\Rightarrow 0 + \frac{1}{4\pi\epsilon_0} \frac{qQ}{2h} + mg(2h) = 0 + \frac{1}{4\pi\epsilon_0} \frac{qQ}{h} + mgh$$

$$\Rightarrow \frac{qQ}{8\pi\epsilon_0 h} = mgh$$

$$\Rightarrow Q = \frac{8\pi\epsilon_0 h^2 mg}{q}$$

This is the answer to the first part.

Let equilibrium height of the particle be h_0

$$\Rightarrow \frac{1}{4\pi\epsilon_0} \frac{qQ}{h_0^2} = mg \Rightarrow h_0^2 = \frac{qQ}{4\pi\epsilon_0 mg}$$

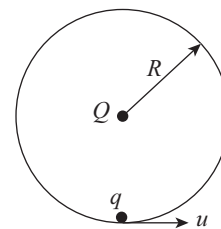
Substituting for Q gives

$$h_0^2 = 2h$$

$$\Rightarrow h_0 = \sqrt{2h}$$

Example 16 Vertical circle

A non-conducting ring of radius $R = 0.5\text{ m}$ is fixed with its plane vertical. A point charge $Q = -10\mu\text{C}$ is fixed at its centre. A particle of mass $m = 0.5\text{ kg}$ carrying charge $q = +8.5\mu\text{C}$ is constrained to move on the smooth inner wall of the ring. The particle is imparted a horizontal velocity u at the lowest point and it manages to complete the circle. Find the smallest value of u for which it does not lose contact at the top most point of the ring.



Solution

Concepts

- Energy of the particle is conserved.
- There are two types of potential energies – gravitational and electrostatic. But the electrostatic potential energy does not change as the particle moves because distance R between the two charges remains constant.
- The particle does not lose contact if normal force applied by the ring wall on it is greater than zero.

At top point B , the equation for motion of the particle is

$$F_e + mg + N = \frac{mv^2}{R}$$

In the critical case, $N = 0$

$$\Rightarrow F_e + mg = \frac{mv^2}{R}$$

$$\Rightarrow \frac{1}{4\pi\epsilon_0} \frac{Qq}{R^2} + mg = \frac{mv^2}{R}$$

$$\Rightarrow 9 \times 10^9 \times \frac{(10 \times 10^{-6})(8.5 \times 10^{-6})}{(0.5)^2} + 0.5 \times 10 = \frac{0.5 \times v^2}{0.5}$$

$$\Rightarrow v^2 = 3.06 \Rightarrow v = 1.75 \text{ ms}^{-1}$$

Using conservation of energy between points A and B

$$K_A + U_{eA} + U_{gA} = K_B + U_{eB} + U_{gB}$$

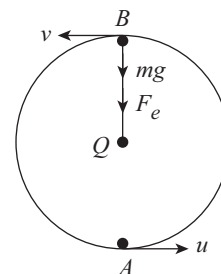
$$[U_{eA} = U_{eB}]$$

$$\therefore \frac{1}{2}mu^2 + 0 = \frac{1}{2}mv^2 + mg(2R)$$

$$\Rightarrow u^2 = v^2 + 4gR$$

$$= 3.06 + 4 \times 10 \times (0.5) = 23.06$$

$$\Rightarrow u = 4.8 \text{ ms}^{-1}$$



Example 17 Size of the gold nucleus

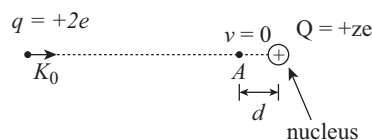
Rutherford performed the famous alpha particle scattering experiment. The experiment gave an upper limit on the size of the gold nucleus.

An alpha particle having kinetic energy $K_0 = 12.3 \times 10^{-13}$ J was headed directly towards a fixed gold nucleus from a large distance. The particle returned back along its line of motion due to strong repulsion from the nucleus. How close to the nucleus must it have reached? This distance was taken as upper limit on the size of the nucleus.

Atomic number of gold is $Z = 79$.

Solution**Concepts**

- Alpha particle is He^{++} . Charge on it is $q = +2e$. Charge on a gold nucleus is $Q = +Ze = +79e$.
- At large distance, potential energy of interaction of alpha particle with gold nucleus is zero.
- The particle stops where its entire kinetic energy gets converted into potential energy.



The particle stops at A and then returns.

$$K_A + U_A = K_\infty + U_\infty$$

$$0 + \frac{1}{4\pi\epsilon_0} \frac{(2e)(79e)}{d} = K_0 + 0$$

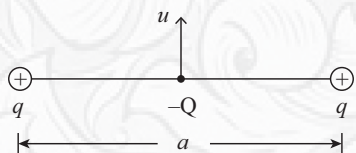
$$\Rightarrow d = \frac{1}{4\pi\epsilon_0} \frac{158e^2}{K_0}$$

$$= 9 \times 10^9 \times \frac{158 \times (1.6 \times 10^{-19})^2}{12.3 \times 10^{-13}} = 2.96 \times 10^{-14} \text{ m}$$

A particle with higher energy will of course get much closer to the nucleus. This experiment established that size of nucleus was even smaller.

**YOUR TURN**

Q.17 Two small balls, having charge $+q$ each, are fixed at a distance a apart. Another small ball having mass m and charge $-q$ is projected from the midpoint of the line joining the two charges. It is projected with a velocity u along the perpendicular bisector of the line joining the two charges.



- Find minimum value of u for which the ball will escape out of the electrostatic field of the fixed charges.

- How will your answer to part (a) change if the ball is projected at a small angle to the perpendicular bisector of the line joining the fixed charges? Assume no other force except the Coulomb force.

Q.18 A particle of mass m and charge q is at rest in free space. Another identical particle is approaching the particle, head on, from a large distance travelling at speed u . Calculate the smallest distance between the particles during their subsequent motion.

5. RELATION BETWEEN ELECTRIC FIELD AND POTENTIAL

Electric field ($\vec{E}$) and potential (V) at a point are two ways of expressing the effect of a source charge in its surrounding. In this section, we will explore the relationship between these two quantities.

Consider a space where an electric field has been set up by some source charges. Suppose we place a test point charge Q at point 1 in this electric field. Electric field at this point is $\vec{E}$ and potential at this point is V . Potential energy of the test charge at point 1 is

$$U = Q \cdot V$$

Now, we displace this charge by an infinitesimal amount $\vec{dr}$ to point 2. Work done on the charge by the electric force in this small displacement is

$$\begin{aligned} dW_{\text{elect}} &= \vec{F}_e \cdot \vec{dr} = (Q\vec{E}) \cdot \vec{dr} \\ &= Q (\vec{E} \cdot \vec{dr}) \end{aligned}$$

As the charge moves from 1 to 2, its potential energy changes and the change is equal to

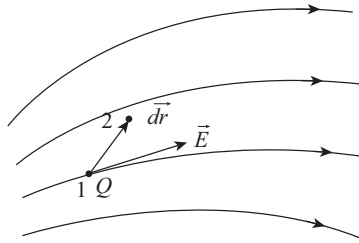
$$\begin{aligned} dU &= -dW_{\text{elect}} \\ dU &= -Q (\vec{E} \cdot \vec{dr}) \\ \frac{dU}{Q} &= -\vec{E} \cdot \vec{dr} \end{aligned}$$

$$\text{But } \frac{dU}{Q} = \frac{U_2 - U_1}{Q} = \frac{U_2}{Q} - \frac{U_1}{Q} = V_2 - V_1 = dV$$

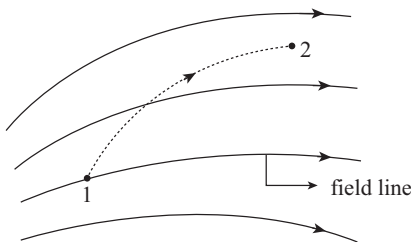
Here dV is difference between potentials at points 2 and 1.

$$\therefore dV = -\vec{E} \cdot \vec{dr} \quad (19)$$

The above equation tells us the change in potential when we make an infinitesimally small displacement at a location where the electric field is $\vec{E}$. If we move from point 1 to 2, which are some finite distance apart in an electric field, then potential difference will be calculated by integrating the above expression.



Charge Q is moved from point 1 to 2 by an infinitesimal displacement $\vec{dr}$ in electric field



The broken line shows a path from point 1 to 2. Potential difference between the two points does not depend on the chosen path.

$$\begin{aligned} \int_{V_1}^{V_2} dV &= - \int_1^2 \vec{E} \cdot \vec{dr} \\ \Rightarrow V_2 - V_1 &= - \int_1^2 \vec{E} \cdot \vec{dr} \\ \Rightarrow \Delta V &= - \int_1^2 \vec{E} \cdot \vec{dr} \quad (20) \end{aligned}$$

The right side of the above expression is nothing but negative of work done by the electric force on a unit positive test charge when it is moved from point 1 to 2. It will be same as (positive of) work done by the external agent in slowly moving the unit charge from point 1 to 2.

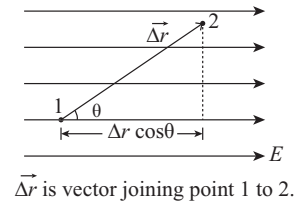
$$\begin{aligned} V_2 - V_1 &= - \int_1^2 \vec{E} \cdot \vec{dr} = -W_{\text{elect}}^{1 \rightarrow 2} \quad (\text{unit positive charge}) \\ &= W_{\text{ext}}^{1 \rightarrow 2} \quad (\text{slowly unit positive charge}) \quad (21) \end{aligned}$$

The value of work done by an electric field is path independent. Therefore, value of integration in equation (20) will not depend on the path chosen between the points 1 and 2.

5.1 Potential Difference in a Uniform Field

Evaluation of integration in equation (20) is very easy when the electric field is uniform.

$$\begin{aligned} \Delta V &= V_2 - V_1 = - \int_1^2 \vec{E} \cdot \vec{dr} \\ &= -\vec{E} \cdot \int_1^2 \vec{dr} \end{aligned}$$



$\vec{dr}$ is vector joining point 1 to 2.

[$\vec{E}$ can be placed outside the integration sign when it is constant. The dot product is maintained. This is not difficult to prove. Try yourself.]

$$\Rightarrow \Delta V = -\vec{E} \cdot \vec{\Delta r} \quad (22)$$

If θ is angle between $\vec{E}$ and displacement vector $\vec{\Delta r}$, then

$$\Delta V = -E \Delta r \cdot \cos \theta$$

It is important to note that path considered for integration may not be a straight line. The value of ΔV is path independent.

Also note that $\Delta r \cos \theta$ is component of displacement in the direction of the electric field (call it r_E).

$$\Delta V = -E r_E \quad (23)$$

Now, come to interpretation of negative sign in the above expression. When angle between $\vec{E}$ and $\vec{\Delta r}$ is acute, ΔV is negative. This means that electric potential drops when displacement has a component in the direction of the field. When one moves against the field (θ is obtuse), ΔV is positive implying an increase in potential. It is just like when you lift a book against gravity, its potential energy increases.

5.2 Potential Difference in Terms of Rectangular Components of Field

Electric field at a point can be expressed as

$$\vec{E} = E_x \hat{i} + E_y \hat{j} + E_z \hat{k}$$

An infinitesimally small displacement can be expressed as

$$d\vec{r} = (dx)\hat{i} + (dy)\hat{j} + (dz)\hat{k}$$

The equation $dV = -\vec{E} \cdot d\vec{r}$ can be expressed as

$$dV = -E_x dx - E_y dy - E_z dz \quad (24)$$

Potential difference between points $B (x_2, y_2, z_2)$ and $A (x_1, y_1, z_1)$ is given by

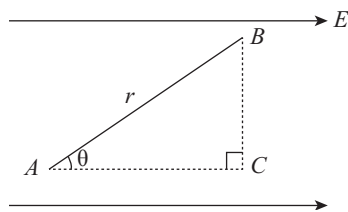
$$\begin{aligned} \int_{V_A}^{V_B} dV &= - \int_{(x_1, y_1, z_1)}^{(x_2, y_2, z_2)} (E_x dx + E_y dy + E_z dz) \\ \Rightarrow V_B - V_A &= - \int_{(x_1, y_1, z_1)}^{(x_2, y_2, z_2)} (E_x dx + E_y dy + E_z dz) \quad (25) \end{aligned}$$

In one-dimensional case, where displacement is along x axis, the above expression reduces to

$$V_B - V_A = - \int_{x_1}^{x_2} E_x dx \quad (26)$$

From the above expressions, it is easy to see that unit of electric field is V/m. It is same as NC^{-1} .

Example 18 A, B and C are three points in a uniform electric field of strength $E = 50 \text{ N/C}$. Distance between A and B is $r = 2.0 \text{ m}$ and the line AB makes an angle of $\theta = 60^\circ$, with the direction of the field (see figure). In the figure, line BC is perpendicular to AC .



- Find potential difference between points B and A .
- Which point has higher potential $-A$ or C ?
- What is the difference in potential at points B and C ?

Solution

Concepts

- $\Delta V = -\vec{E} \cdot \Delta\vec{r}$ for uniform field.
- When $\vec{E}$ and $\Delta\vec{r}$ are perpendicular, $\Delta V = 0$.

$$\begin{aligned} \text{(a)} \quad V_B - V_A &= -\vec{E} \cdot \Delta\vec{r} = -Er \cos 60^\circ \\ &= -50 \times 2 \times \frac{1}{2} = -50 \text{ V} \end{aligned}$$

Potential at B is less than potential at A by 50 V . Displacement in direction of field decreases potential.

- Moving against the field increases potential. Point A has higher potential than C (or, B).
- Line BC is perpendicular to the direction of the electric field.

$$V_B - V_C = -E\Delta r \cos 90^\circ = 0$$

$$\Rightarrow V_B = V_C$$

We conclude that there is no change in potential if one moves perpendicular to the field.

Example 19 In the last example, a charge $Q = -10 \mu\text{C}$ is moved slowly from point A to B . Find work done by the external agent.

Solution

Concepts

Refer to equation (21). Potential difference $V_B - V_A$ is work done by an external agent in slowly moving a unit positive charge from A to B .

$$\begin{aligned} W_{\text{ext}} &= Q (V_B - V_A) \quad [= U_B - U_A] \\ &= (-10 \times 10^{-6}) (-50) = 5 \times 10^{-4} \text{ J} \end{aligned}$$

Example 20 Electric field changes with x coordinate as $\vec{E} = (ax) \hat{i}$. Find difference in potential between points B and A having x coordinates x_2 and x_1 , respectively. If potential at A is assumed to be zero, what is potential at B ?

Solution

Concepts

$$\begin{aligned} V_B - V_A &= - \int_{x_1}^{x_2} E_x dx \\ V_B - V_A &= - \int_{x_1}^{x_2} (ax) dx = -\frac{a}{2} (x_2^2 - x_1^2) \end{aligned}$$

If $V_A = 0$, then

$$V_B = -\frac{a}{2} (x_2^2 - x_1^2)$$

Example 21 The electric field in a two-dimensional space is given by $\vec{E} = (y\hat{i} + x\hat{j})$. Find potential difference between points $A (0, 0)$ and $B (1, 1) \text{ m}$.

Solution

Concepts

$$\text{(i)} \quad V_A - V_B = - \int_B^A \vec{E} \cdot d\vec{r} = - \int_B^A (E_x dx + E_y dy)$$

(ii) If $t = xy$, then

$$\frac{dt}{dx} = 1 \cdot y + x \frac{dy}{dx}$$

$$\Rightarrow dt = ydx + xdy$$

- (iii) One may find such integrations difficult to handle. In that case, you can try to pick an easy path from A to B . Perform your integration along the chosen path. Potential difference is independent of path.

$$\begin{aligned}\vec{E} &= y\hat{i} + x\hat{j} \\ dV &= \vec{E} \cdot d\vec{x} = -(E_x dx + E_y dy) \\ &= -(ydx + xdy) \\ \therefore \int_B^A dV &= -\int_B^A (ydx + xdy) \\ \Rightarrow V_A - V_B &= -\int_B^A dt \text{ where } t = xy \\ &= -[t]_{x_B y_B}^{x_A y_A} = -[0 - 1 \times 1] = 1V.\end{aligned}$$

[It hardly matters whether you take limits of integration from B to A or A to B . When you integrate from A to B , you get $V_B - V_A$]

Alternate Method

Let's choose a path ACB from A to B (see figure).

When we move on x axis from A to C , field at different point is given by

$$\vec{E} = 0\hat{i} + x\hat{j} = x\hat{j}$$

From A to C , field increases with x but it is directed along y direction. We are moving perpendicular to the field. Therefore, potential will not change as we move from A to C .

$$V_A = V_C$$

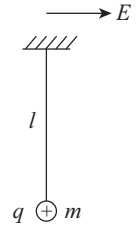
Now, we move from C to B . Electric field at points on this line is

$$\begin{aligned}\vec{E} &= y\hat{i} + 1\hat{j} \quad [\because x = 1 \text{ at all points on line } CB] \\ \therefore \int_C^B dV &= \int_C^B [(E_x dx) + (E_y dy)]\end{aligned}$$

$$\begin{aligned}\Rightarrow V_B - V_C &= -\int_{y=0}^{y=1} 1 \cdot dy \quad [\because dx = 0 \text{ on line } CB] \\ &= -1 \\ \Rightarrow V_B - V_A &= -1 \quad [\because V_C = V_A] \\ \Rightarrow V_A - V_B &= 1 \text{ Volt}\end{aligned}$$

Example 22 A charged pendulum

A pendulum bob is a small ball having mass m and charge q . Length of the string is l . A uniform electric field E is switched on when the pendulum is vertical. The pendulum just managed to reach the horizontal position. Find the magnitude of E .



Solution

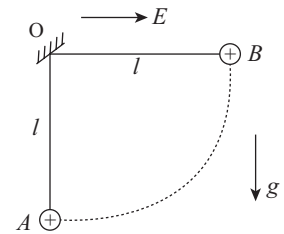
Concepts

- Energy of the charged ball remains conserved.
 $K + U_e + U_g = a \text{ constant}$
- Electrostatic potential energy of a charge q kept at a point having potential V is $U = qV$.
- In a uniform field (E), potential drops by $\Delta V = El$ when we displace in the direction of the field by l .

The pendulum bob first accelerates due to large tangential component of force qE . After a certain point, it begins to retard because tangential component of mg exceeds that of qE .

According to the problem kinetic energy is zero at A as well as B .

$$\begin{aligned}\therefore K_A + U_{Ae} + U_{Ag} &= K_B + U_{Be} + U_{Bg} \\ \Rightarrow 0 + qV_A + 0 &= 0 + qV_B + mgl \\ \Rightarrow q(V_A - V_B) &= mgl \\ \Rightarrow qEl &= mgl \\ E &= \frac{mg}{q}\end{aligned}$$



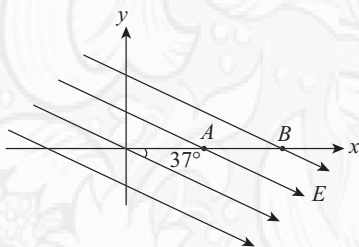


YOUR TURN

Q.19 There is a vertically downward electric field in a room. An electron moves up inside the room. Does the electric potential energy of the electron increase or decrease?

Q.20 A uniform electric field of strength $E = 10 \text{ Vm}^{-1}$ is along positive x direction. Potential is assumed to be zero at the origin. Find potential at $x = -2.0 \text{ m}$.

Q.21 A uniform electric field of strength $E = 20 \text{ V/m}$ is directed as shown in the figure. Find potential difference between points B and A . Distance $AB = 5 \text{ cm}$.



Q.22 A large plane surface has surface charge density of $\sigma = +0.01 \mu\text{C/m}^2$. Points A and B are at perpendicular distance $r_A = 0.5 \text{ m}$ and $r_B = 1.5 \text{ m}$ from the surface (on same side of the surface). Find $V_A - V_B$.

Q.23 A long cylinder has linear charge density $\lambda = 0.01 \mu\text{C/m}$. Its radius is 2 cm . Find the potential difference between a point (A) on its surface and another point (B) at a distance of 4 cm from its axis. Will your answer change if the points A and B do not lie on a single line that is perpendicular to the cylinder?

Q.24 Electric field in a two-dimensional space is given by $\vec{E} = (x\hat{i} + y\hat{j})$. Assume potential at origin to be zero and write potential at a point (x, y) .

Q.25 A block of mass m having charge q is attached to an ideal spring of force constant K as shown in the figure. The system is in equilibrium on a smooth horizontal table. A uniform horizontal electric field E is switched on. Find the maximum compression in the spring caused after this.



5.3 Getting Electric Field From Potential Function

Electric field and potential are related. If we know the variation of potential in a space, we can predict the electric field.

Consider a simple case of electric field being directed along x direction. Potential at some point in this field is V . When you displace by dx , the potential changes to $V + dV$. We have already understood that

$$\begin{aligned} dV &= -E dx \\ \Rightarrow E &= -\frac{dV}{dx} \end{aligned} \quad (27)$$

In words, *electric field is negative of rate of change of potential with position x* . Negative sign is a reminder that field is directed in the direction of decreasing potential. If we displace in positive x direction (dx is positive) and find that potential increases (i.e., dV is positive), then $\frac{dV}{dx}$ is positive. This implies that E is negative, and it has direction along negative x axis.

Now, assume that you have no knowledge about direction or magnitude of electric field but you know how potential changes from point to point in space. How will you find the electric field at a point?

Take three mutually perpendicular directions as x , y and z axes. Make a small displacement dx along x direction. You find that potential changes by dV .

$$dV = -(E_x dx + E_y dy + E_z dz)$$

Since $dy = dz = 0$

$$\therefore dV = -E_x dx$$

$$\Rightarrow E_x = -\frac{dV}{dx}$$

The last expression is differentiation of V with respect to x keeping y and z as constants. In mathematics, we call such differentiations as partial differentiation. Partial differentiations are represented by slightly different symbols.

$$E_x = -\frac{\partial V}{\partial x} \quad (28)$$

Suppose you move in y direction by dy and find that the potential changes by dV . Thinking as above, we can write

$$E_y = -\frac{\partial V}{\partial y} \quad (29)$$

$$\text{Similarly, } E_z = -\frac{\partial V}{\partial z}$$

$$\text{Therefore, } \vec{E} = E_x \hat{i} + E_y \hat{j} + E_z \hat{k}$$

$$= -\left(\frac{\partial V}{\partial x} \hat{i} + \frac{\partial V}{\partial y} \hat{j} + \frac{\partial V}{\partial z} \hat{k} \right) \quad (30)$$

Often, the operator $\left(\frac{\partial}{\partial x}\hat{i} + \frac{\partial}{\partial y}\hat{j} + \frac{\partial}{\partial z}\hat{k}\right)$ is represented by $\vec{\nabla}$ and is known as *gradient operator*.

$$\therefore \vec{E} = -\vec{\nabla} V \quad (31)$$

we say *electric field is negative gradient of potential*.

Example 23 The electric potential at any point (x, y, z) in a three-dimensional space is given by $V = (4x^2)$ volt. Find the electric field at a point $(1, 1, 2)$ m.

Solution

Concepts

$$E_x = -\frac{\partial V}{\partial x}; E_y = -\frac{\partial V}{\partial y}; E_z = -\frac{\partial V}{\partial z}$$

$$E_x = -\frac{\partial V}{\partial x} = -4 \frac{\partial x^2}{\partial x} = -8x$$

$$E_y = -\frac{\partial V}{\partial y} = 0;$$

$$E_z = -\frac{\partial V}{\partial z} = 0$$

$$\therefore \vec{E} = -8x\hat{i}$$

$$\text{At } (1, 1, 2) \text{ m, } \vec{E} = (-8\hat{i}) \text{ Vm}^{-1}$$

Example 24 Electric potential in a two-dimensional space varies as $V = 4xy$. Find the electric field at $(1, 1)$ m.

Solution

$$E_x = -\frac{\partial V}{\partial x} = -4y \frac{\partial x}{\partial x} = -4y \quad [y \text{ is treated as constant}]$$

$$E_y = -\frac{\partial V}{\partial y} = -4x \frac{\partial y}{\partial y} = -4x \quad [x \text{ is treated as constant}]$$

$$\therefore \vec{E} = -4y\hat{i} - 4x\hat{j}$$

$$\text{At } (1, 1), \text{ we have } \vec{E} = (-4\hat{i} - 4\hat{j}) \text{ Vm}^{-1}$$

In Short

- (i) Electric field is a vector which can be obtained if variation of electric potential (a scalar) in space is known.
- (ii) The components of electric field in x , y and z directions are given by

$$E_x = -\frac{\partial V}{\partial x}; E_y = -\frac{\partial V}{\partial y}; E_z = -\frac{\partial V}{\partial z}.$$

- (iii) Knowledge of electric field helps us to calculate potential difference between two points.

$$V_2 - V_1 = -\int_1^2 \vec{E} \cdot d\vec{r}$$

- (iv) In a uniform electric field

$$V_2 - V_1 = \Delta V = -\vec{E} \cdot \Delta \vec{r}$$

Here $\Delta \vec{r}$ is displacement vector directed from point 1 to 2. Put in a simpler way

$$\Delta V = -Er_E$$

Here r_E is component of displacement in the direction of electric field E .

- (v) Change in electrostatic potential energy of a charge Q when it moves from a point having potential V_1 to another point having potential V_2 is

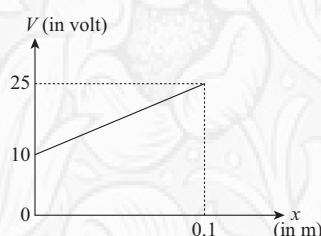
$$\Delta U = Q(V_2 - V_1).$$

- (vi) Conservation of energy and work energy theorem can be used as usual when a charged body moves through an electric field. We just need to include electrostatic potential energy in our equations.



YOUR TURN

Q.26 Variation of electric potential at points along x axis has been graphed in figure. Find the x component of electric field at $x = 0.05$ m.



Q.27 Electric potential at a point is zero. Can we say that field must be zero at the point?

Q.28 Electric potential varies in a space as $V = x^2 + y^2 + z^2$ (in volt). Find the component of electric field parallel to xy plane at a point $(1, 2, 3)$ m.

6. EQUIPOTENTIAL SURFACES

A surface obtained by joining points at same electric potential is known as an equipotential surface. In three-dimensional space, equipotential surfaces are described by

$$V(x, y, z) = \text{a constant}$$

In two-dimension, we get curves of constant potentials. These are known as equipotential curves.

Consider an isolated point charge. Potential at all the points on the surface of a sphere of radius r centred at q is same. This sphere is an equipotential surface. A 10 V equipotential surface is a sphere of radius r such that

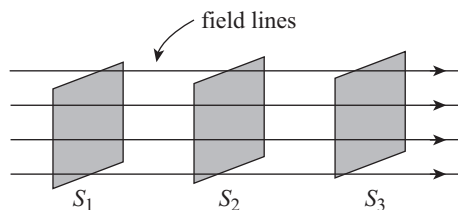
$$\frac{1}{4\pi\epsilon_0} \frac{q}{r} = 10 \text{ V}$$

Electric field lines are always perpendicular to the equipotential surfaces. In the figure shown, radial field lines are normal to the spherical equipotentials. When one moves perpendicular to the direction of electric field, the electric potential does not change.

Consider a uniform electric field. Field lines are equally spaced and parallel to one another. The equipotentials are plane surfaces perpendicular to the field lines. If V_1 , V_2 and V_3 are potentials on the three surfaces shown in the figure, then

$$V_1 > V_2 > V_3$$

This is because the potential decreases in the direction of the field.



S_1 , S_2 and S_3 are three different equipotential surfaces.

Example 25 Equation of an equipotential

A point charge $q = 1.0 \mu\text{C}$ is fixed at the origin of a cartesian co-ordinate system. Write the equation of an equipotential surface having potential 1000 V.

Solution

Concepts

Equipotential surface (a sphere) is locus of a point having potential equal to 1000 V.

Consider a point (x, y, z) .

Distance of the point from the point charge is

$$r = \sqrt{x^2 + y^2 + z^2}$$

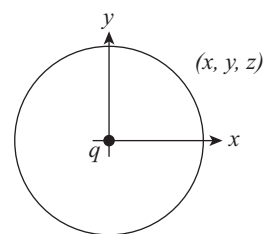
$$\frac{1}{4\pi\epsilon_0} \frac{q}{r} = 1000$$

$$\Rightarrow 9 \times 10^9 \times \frac{1 \times 10^{-6}}{r} = 10^3$$

$$\Rightarrow r = 9 \text{ m}$$

$$\therefore x^2 + y^2 + z^2 = 81$$

This is the desired equation.



Example 26 Two point charges

Two equal and opposite point charges are kept at a separation. Draw the equipotential curves in a plane of the two charges.

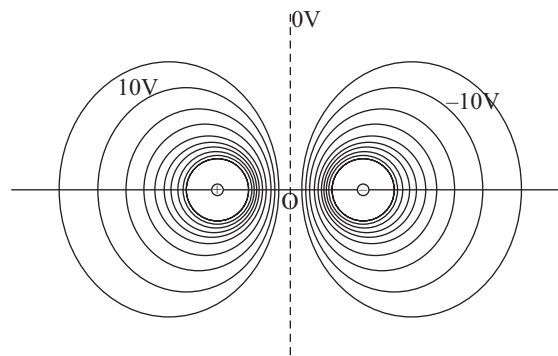
Solution

Concepts

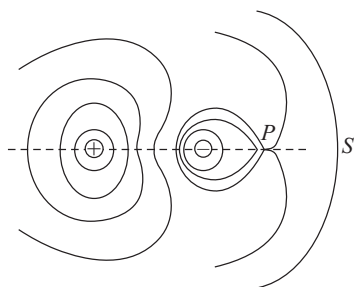
- Equipotentials are perpendicular to field lines. You can have a look at the electric field lines due to a pair of equal and opposite point charges. Refer to the last chapter for this.
- Very close to a point charge, equipotentials are definitely spherical (circular in a two-dimensional diagram). This is because effect of any distant charge will be negligible.
- Perpendicular bisector of the line joining two charges is a line of zero potential.
- Potential near the midpoint O is close to zero. At a large distance from O , the potential will be close to zero.

Field lines have not been shown in the diagram for the sake of clarity. I have marked a 10 V (a random number) equipotential curve and a -10 V curve for comparison.

The exact shape of the curves is quite complicated.



Example 27 The figure shows few equipotential surfaces produced due to two point charges of opposite signs.



- Which charge has higher magnitude?
- What is significance of point P ?
- Is the potential of surface S positive or negative?

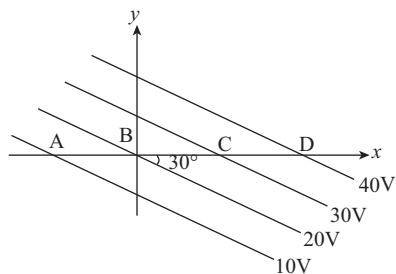
Solution

Concepts

- Field lines are normal to the equipotentials. There is no unique direction for electric field at P ; therefore, $E = 0$ at P .
- At large distances, equipotentials are spheres. From a large distance, the charge system appears like a point charge.

- Electric field at point P is zero. Field due to the two charges must be equal and opposite at P . This is possible only if $|Q_+| > |Q_-|$ because the positive charge is at larger distance from P .
- $E = 0$ at P .
- S is a surface that will pass through some point to the left of positive charge (it will be a closed surface enclosing both the charges). Any point to the left of positive charge has a positive potential. Therefore, potential on surface S is positive.

Example 28 Figure shows equipotential lines in xy plane. Distances AB , BC and CD are all equal to 4 cm. Find the magnitude and direction of electric field in the region.



Solution

Concepts

- Electric field lines are perpendicular to equipotentials.
- Field points in the direction of decreasing potential.
- $|E| = \left| \frac{\Delta V}{\Delta r} \right|$ where Δr is perpendicular distance between two equipotentials having a potential difference of ΔV .

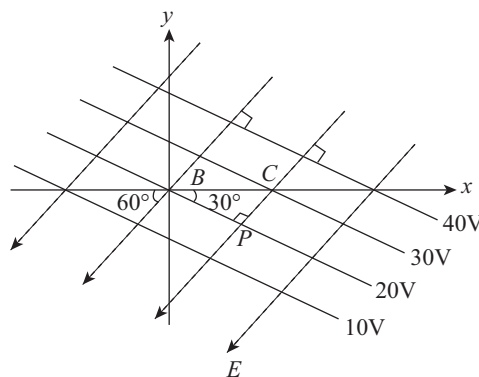
Electric field lines are directed perpendicular to the given equipotential lines. Their direction is as indicated in the figure given below.

Perpendicular distance between two consecutive equipotentials given in the figure is

$$\Delta r = 4 \sin 30^\circ = 2 \text{ cm}$$

This implies that when we move 2 cm in the direction of the field, the potential drops by 10 V.

$$|E| = \left| \frac{\Delta V}{\Delta r} \right| = \frac{10V}{2\text{cm}} = 5 \text{ Vcm}^{-1}$$



$$\text{Distance } PC = \Delta r = 4 \sin 30^\circ = 2 \text{ cm.}$$

Broken lines represent electric field lines.

Obviously, direction of electric field makes 60° angle with negative x direction as shown.

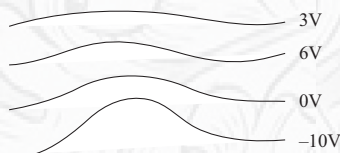
In Short

- A surface obtained by joining points having same potential is known as an equipotential surface.
- Field lines are always perpendicular to the equipotentials.



YOUR TURN

Q.28 The diagram shows two-dimensional equipotential curves. Draw the field lines.



Q.30 Two point charges, $+q$ each, are held at a separation. Draw equipotential curves in a plane containing the two charges.

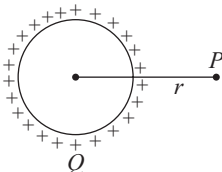
Q.31 A charge q is moved from a point A to another point B . Both points lie on an equipotential surface. What is work done by the electric field?

Q.32 What will be shape of equipotential surfaces around a long straight line charge having uniform charge density.

Q.33 Two point charges $+q$ and $-2q$ are kept on x axis at $x = -a$ and $x = a$, respectively. Show that the shape of an equipotential curve in xy plane representing a potential of zero is a circle. Find its radius.

7. POTENTIAL DUE TO A UNIFORMLY CHARGED SHELL

Consider a shell of radius R having charge Q uniformly spread on its surface. We wish to write electric potential at a point P which is at a distance r from its centre ($r \geq R$). Potential at the point P is work done by an external agent in slowly shifting a unit positive test charge from infinity to point P (refer to equation (10)). If we replace the shell with a point charge Q placed at its centre and then move a unit charge from infinity to P , the work done will remain same. This is because the shell applies same force on our test charge as does the point charge kept at its centre.



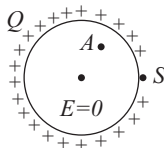
It means that potential due to the shell at an outside point is same as potential due to a point charge placed at the centre of the shell.

$$V = \frac{1}{4\pi\epsilon_0} \frac{Q}{r} \quad \text{for } r \geq R \quad (32)$$

Potential at the surface is

$$V_{\text{surface}} = \frac{1}{4\pi\epsilon_0} \frac{Q}{R} \quad (33)$$

What is potential at an inside point A ? Recall that there is no electric field inside the shell. It means no work is done if a charge is slowly moved from a point S on the surface to the point A .



Potential at $A = W_{\text{extAgt}(\text{slowly unit charge})}^{\infty \rightarrow A}$

$$= W_{\text{extAgt}(\text{slowly unit charge})}^{\infty \rightarrow S} + W_{\text{extAgt}(\text{slowly unit charge})}^{S \rightarrow A}$$

$$\Rightarrow V_A = V_S + 0$$

$$\Rightarrow V_A = V_S$$

Therefore, *electric potential at an inside point of the shell is same as its surface potential.*

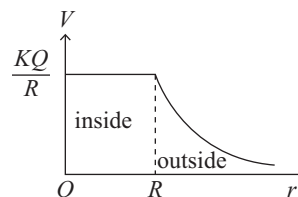
$$V = V_{\text{surface}} = \frac{1}{4\pi\epsilon_0} \frac{Q}{R} \quad \text{for } r \leq R \quad (34)$$

One can also simply say that

$$V_A - V_S = -\int_S^A \vec{E} \cdot d\vec{l} = 0$$

$$\Rightarrow V_A = V_S$$

Graphical variation of the electric potential (V) with distance (r) from the centre of the shell has been shown in the figure.



8. ELECTRIC POTENTIAL DUE TO A UNIFORM SPHERICAL VOLUME CHARGE

A spherical volume of radius R has a uniform volume charge density. Total charge in the spherical region is Q .

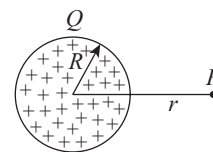
For outside points ($r \geq R$), it behaves like a point charge placed at its centre.

Just like we argued for a spherical shell

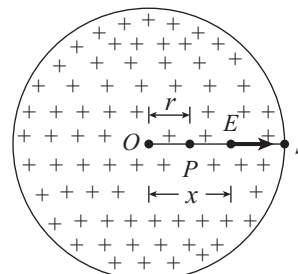
of charge, we can say that electric potential at point P is

$$V = \frac{1}{4\pi\epsilon_0} \frac{Q}{r} \quad \text{for } r \geq R \quad (35)$$

$$V_{\text{surface}} = \frac{1}{4\pi\epsilon_0} \frac{Q}{R} \quad (36)$$



Now, we will calculate potential at a point inside the spherical region ($r < R$). P is a point inside the sphere at a distance r from its centre. S is a point on the surface of the sphere. We will find potential



difference between S and P using equation (20). Because we know V_S , we will get V_P .

Electric field at a distance x ($< R$) from the centre is

$$E = \left(\frac{1}{4\pi\epsilon_0} \frac{Q}{R^3} \right) \cdot x$$

$$\therefore \int_P^S dV = - \int_{x=r}^{x=R} E \cdot dx$$

$$\Rightarrow V_S - V_P = - \frac{1}{4\pi\epsilon_0} \frac{Q}{R^3} \int_r^R x dx$$

$$\Rightarrow \frac{1}{4\pi\epsilon_0} \frac{Q}{R} - V_P = - \frac{1}{4\pi\epsilon_0} \frac{Q}{R^3} \left[\frac{R^2}{2} - \frac{r^2}{2} \right]$$

Rearranging the terms, we get

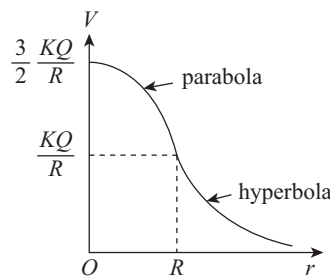
$$V_P = \frac{1}{4\pi\epsilon_0} \frac{Q}{R^3} \left[\frac{3}{2} R^2 - \frac{1}{2} r^2 \right] \quad (37)$$

Potential at the centre is obtained by substituting $r = 0$ in the above equation

$$\therefore V_{\text{centre}} = \frac{3}{2} \frac{1}{4\pi\epsilon_0} \frac{Q}{R} = \frac{3}{2} V_{\text{surface}} \quad (38)$$

Variation of potential (V) with distance r from the centre of the sphere has been graphed in the figure.

Note that for $r \leq R$, the equation (37) tells that V is a quadratic function of r . Graph of V versus r is a parabola. For $r \geq R$, equation (35) tells us that the graph of V versus r is a hyperbola.



In Short

- A uniformly distributed charge on a spherical surface or a spherical volume behaves like a point charge kept at the centre for the purpose of writing field and potential at outside points.
- Electric field inside a uniformly charged shell is zero. This does not mean that potential is zero at an inside point. What it means is that potential does not change when one moves from a point on the surface of the shell to an inside point.

The whole shell is at one single potential.

- Electric potential is maximum at the centre of a uniformly charged spherical volume. It is 1.5 times the surface potential.

- Variation of electric potential inside a uniformly charged sphere of radius R is given by

$$V = \frac{Q}{4\pi\epsilon_0 R^3} \left[\frac{3}{2} R^2 - \frac{1}{2} r^2 \right] = \frac{\rho}{3\epsilon_0} \left[\frac{3}{2} R^2 - \frac{1}{2} r^2 \right]$$

Example 29 Solid conducting ball

A solid metal ball of radius R is given a charge Q . Find potential at a point at a distance $\frac{R}{2}$ from its centre.

Solution

Concepts

Though the ball is solid, charge will reside on its surface as it is conducting.

Charge is spread on the surface of the ball. It will behave like a uniformly charged shell.

$$\therefore V_{\text{inside}} = V_{\text{surface}} = \frac{1}{4\pi\epsilon_0} \frac{Q}{R}$$

in fact, the complete ball is at one single potential.

Example 30 Escape speed

A conducting ball of radius R has a charge Q on it. An electron (charge = e , mass = m) escapes from its surface with speed u . Find the smallest value of u such that the electron does not return back and escapes to infinity.

Solution

Concepts

A ball on the surface of the Earth has negative potential energy $\left(= -\frac{GMm}{R} \right)$. It can escape the gravity only when it is given enough kinetic energy to make its total energy zero.

Similarly, an electron on the surface of a positively charged sphere has negative potential energy. It can escape only when its total energy is made greater than or equal to zero.

Potential of the ball is

$$V = \frac{1}{4\pi\epsilon_0} \frac{Q}{R}$$

Potential energy of an electron on the surface of the ball is

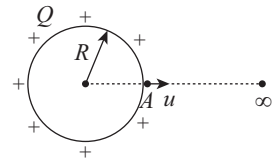
$$U = (-e)V = -\frac{1}{4\pi\epsilon_0} \frac{Qe}{R}$$

For the electron to just manage to reach infinity, we can consider its kinetic energy to be zero at infinity.

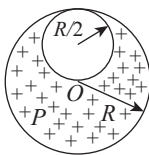
$$K_A + U_A = K_\infty + U_\infty$$

$$\frac{1}{2} mu^2 - \frac{1}{4\pi\epsilon_0} \frac{Qe}{R} = 0 + 0$$

$$\Rightarrow u = \sqrt{\frac{Qe}{2\pi\epsilon_0 mR}}$$



Example 31 A spherical volume of radius R has a uniform charge density $+\rho$ except for a cavity region of radius $\frac{R}{2}$ lying inside it as shown. Find the electric potential at the centre O of the spherical region.



Solution

Concepts

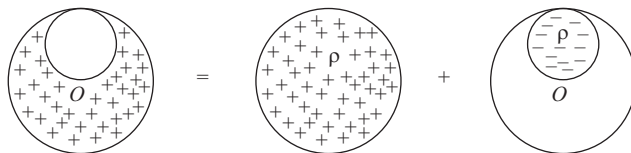
- (i) Potential on the surface of a uniformly charged spherical volume is

$$V = \frac{1}{4\pi\epsilon_0} \frac{Q}{R} = \frac{1}{4\pi\epsilon_0} \frac{\rho \cdot \frac{4}{3}\pi R^3}{R} = \frac{\rho}{3\epsilon_0} R^2$$

- (ii) Potential at the centre of a uniformly charged spherical volume is

$$V = \frac{3}{2} \frac{1}{4\pi\epsilon_0} \frac{Q}{R} = \frac{1}{2} \frac{\rho}{\epsilon_0} R^2$$

- (iii) The charge distribution given in the problem is superposition of a uniformly distributed volume charge of density $+\rho$ in a sphere of radius R and a uniformly distributed volume charge of density $-\rho$ in a sphere of radius $R/2$.



$$V_0 = \frac{1}{2} \frac{\rho}{\epsilon_0} R^2 - \frac{\rho}{3\epsilon_0} \left(\frac{R}{2}\right)^2$$

$$= \frac{5}{12} \frac{\rho R^2}{\epsilon_0}$$



YOUR TURN

Q.34 A spherical volume of radius R has a uniform charge density and contains a total charge Q . Concentric to this, there is a spherical shell having charge $-Q$. Shell has radius $2R$ and spread of charge is uniform on its surface.

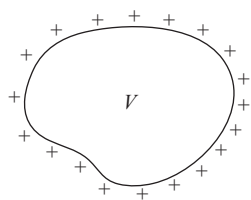
- Find electric potential at the centre.
- Find potential at a point at a distance $1.5R$ from the centre.

- Find potential at a point that is at a distance $3R$ from the centre.

Q.35 A hemispherical volume of radius R has a uniform volume charge density. Total charge in the volume is Q . Find potential at its centre.

9. POTENTIAL OF A CONDUCTOR

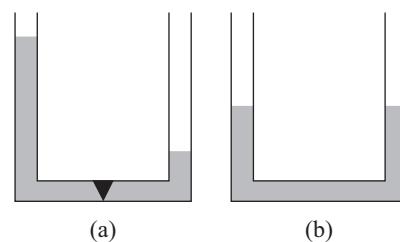
We know that electric field lines, if any, are perpendicular to the surface of a conductor and there is no electric field inside the material of a conductor. Therefore, potential does not change when we move on the surface of a conductor or when we move from one point to another in the material. A piece of conductor is at one single potential under electrostatic conditions.



The entire conductor is at a single potential (V).

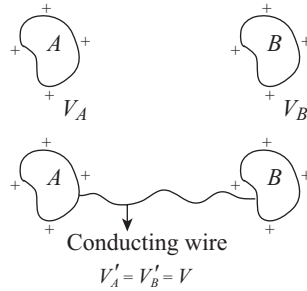
9.1 Charge Sharing

Figure (a) shows a U-shaped tube having a liquid filled to different heights in the two arms. A plug at the bottom separates the two parts. Gravitational potential (i.e., gravitational potential energy per unit mass. It is essentially equal to $g \cdot h$, where h is height from the reference level) at the surface of the left arm is higher than the potential at the surface of the other arm. Now, the plug is removed. Liquid flows from the region of high potential



to the region of low potential. Equilibrium (static situation) is reached when the potential (i.e., height) on both sides becomes same. This analogy can be used to understand charge flow between two conductors.

Consider a conductor A at a potential V_A , higher than potential V_B of another conductor B. The conductors are far apart and charge on one does not have any appreciable effect on the potential of the other. The two conductors are joined using a conducting wire. Charge begins to move through the wire. We say there is a current through the wire. The current stops when potentials of the two conductors become equal. Therefore, under static condition (no movement of charge taking place), both conductors will have same potential.



Actually, existence of a potential difference between the two conductors means there is an electric field between them. This field forces the charge to move through the connecting wire. When charges are so distributed that there is no field anywhere in the material of the wire, the current stops.

Consider two metallic balls of radii r_1 and r_2 charged to potentials V_1 and V_2 , respectively. The balls are far apart. If Q_1 and Q_2 are charges on them, then

$$V_1 = \frac{1}{4\pi\epsilon_0} \frac{Q_1}{r_1} \Rightarrow Q_1 = 4\pi\epsilon_0 r_1 V_1$$

and similarly, $Q_2 = 4\pi\epsilon_0 r_2 V_2$

The balls are connected by a conducting wire. They share charge and attain a final common potential V .

Final charges on the two balls are q_1 and q_2 , respectively

$$V = \frac{1}{4\pi\epsilon_0} \frac{q_1}{r_1} = \frac{1}{4\pi\epsilon_0} \frac{q_2}{r_2}$$

$$\Rightarrow q_1 = 4\pi\epsilon_0 r_1 V, \quad q_2 = 4\pi\epsilon_0 r_2 V$$

from the principle of conservation of charge, we have

$$q_1 + q_2 = Q_1 + Q_2$$

$$\Rightarrow 4\pi\epsilon_0 (r_1 + r_2)V = 4\pi\epsilon_0 (r_1 V_1 + r_2 V_2)$$

$$\Rightarrow V = \frac{r_1 V_1 + r_2 V_2}{r_1 + r_2} \quad (39)$$

9.2 Surface Charge Density

A spherical conductor of radius r is at potential V . Charge on it is

$$Q = 4\pi\epsilon_0 r V$$

Surface charge density is

$$\sigma = \frac{Q}{4\pi r^2} = \frac{\epsilon_0 V}{r}$$

For a given potential, charge density on the surface is inversely proportional to the radius of the conductor. If two spherical conductors at same potential have radii r_1 and r_2 , then ratio of surface charge density on them is

$$\frac{\sigma_1}{\sigma_2} = \frac{r_2}{r_1} \quad (40)$$

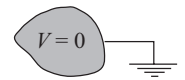
If we have a non-spherical conduction (at some potential V), then density of surface charge is higher at locations of smaller radius of curvature. At sharp corners, charge density is very high.

9.3 Grounding a Conductor

The Earth is a conductor of enormous size. For all real life calculations, we ignore any charge on the Earth and its potential is regarded as zero.

If a small quantity of charge is given to the Earth or taken away from it, its potential does not change much and we continue to take its potential to be zero. If you add a bucket of water to an ocean, the sea level does not change.

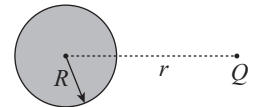
When we connect a conductor to the Earth (using a conducting wire), the conductor is said to be *earthed* or *grounded*. Figure shows the symbol used to indicate grounding. For a real-life conductor, the Earth acts as an infinitely large source or sink of charge. Charge flows from the conductor to the Earth or from the Earth to the conductor so that its potential becomes equal to zero.



An earthed conductor.

Example 32 Point charge in front of a conducting ball

A neutral conducting ball of radius R is kept in front of a fixed point charge $+Q$. Distance of the point charge from the centre of the ball is r . Find the potential of the ball.



Solution

Concepts

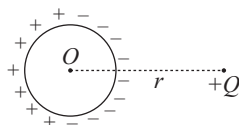
- Induced charges will appear on the surface of the ball. Face of the ball facing $+Q$ charge will become negatively charged and the rear face will become positive. But sum of induced charges on the surface of the ball is zero.
- Potential of the entire ball (a conductor) is same. It is easy to find potential at the centre of the ball and it will be the potential of the entire ball.

Induced charge is not uniformly distributed. But potential due to induced charge at the centre is zero.

$$V_{\text{in}} = \sum \frac{1}{4\pi\epsilon_0} \frac{\Delta q_{\text{in}}}{R} = \frac{1}{4\pi\epsilon_0 R} \sum \Delta q_{\text{in}} = 0$$

Where Δq_{in} is a small part of induced charge.

Potential at the centre (O) is only because of Q .



$$\therefore V_{\text{centre}} = \frac{1}{4\pi\epsilon_0} \frac{Q}{r}$$

this is the potential of the entire ball.

Example 33 A metal ball of radius $r_1 = 10$ cm is charged to a potential of $V_1 = 20$ V. Another distant conducting ball of radius $r_2 = 5$ cm is charged to a potential $V_2 = 40$ V. They are connected by a copper wire.

- In which direction, the electrons in the copper wire will move?
- After the charge flow stops, find charges on the two spheres.

Solution

Concepts

- Electric field is directed from the ball at higher potential to the ball at lower potential. Positive charge moves in the direction of field. Here, electrons are charge carriers and they move opposite to the field.

$$(ii) \text{ Final common potential is } V = \frac{r_1 V_1 + r_2 V_2}{r_1 + r_2}$$

- The electrons move from sphere at lower potential V_1 towards the sphere at higher potential V_2 .
- Final common potential is

$$V = \frac{r_1 V_1 + r_2 V_2}{r_1 + r_2} = \frac{(10\text{cm})(20V) + (5\text{cm})(40V)}{10\text{cm} + 5\text{cm}} \\ = \frac{80}{3} V$$

$\therefore$ Final charge on the ball of radius r_1 is given by

$$\frac{1}{4\pi\epsilon_0} \frac{q_1}{r_1} = V \Rightarrow q_1 = 4\pi\epsilon_0 r_1 V \\ \Rightarrow q_1 = \frac{1}{9 \times 10^9} \times 0.1 \times \frac{80}{3} \\ = 2.96 \times 10^{-10} \text{ C.}$$

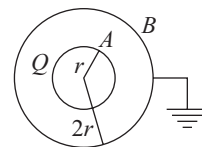
Similarly, $q_2 = 4\pi\epsilon_0 r_2 V$

$$= \frac{1}{9 \times 10^9} \times 0.05 \times \frac{80}{3}$$

$$= 1.48 \times 10^{-10} \text{ C.}$$

Example 34 Two concentric metal balls A and B have radii r and $2r$, respectively. Ball A is given a charge Q and ball B is grounded.

- Find charge on ball B.
- Find charge on inner and outer surfaces of ball B.



Solution

Concepts

- Potential of grounded ball is zero. Earth will supply it any amount of charge to make its potential zero.
- A Gaussian surface passing entirely through the material (the wall) of shell B will help you in finding charge on the inner wall of B.

- Let charge on B be q . This charge is supplied by the Earth to ensure that potential of B becomes zero.

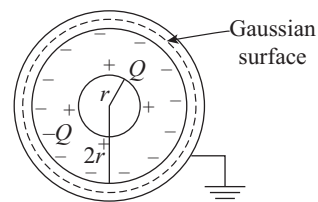
$$V_B = 0 \\ \Rightarrow V_{BQ} + V_{Bq} = 0 \\ \Rightarrow \frac{1}{4\pi\epsilon_0} \frac{Q}{2r} + \frac{1}{4\pi\epsilon_0} \frac{q}{2r} = 0$$

the first term is potential of B due to charge Q on A, and the second term is potential of B due to its own charge q .

$$\Rightarrow q = -Q$$

Thus, shell B has charge equal to $-Q$.

- Consider a Gaussian surface (shown in broken line) running through the material of shell B. [The wall of B is actually thin but for clarity of the diagram, we have shown it a bit thick]. Flux through the Gaussian surface is zero since $E = 0$ inside metal.



$\therefore$ Charge enclosed by the Gaussian surface = 0.

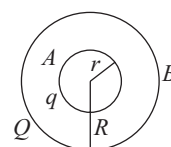
Therefore, inner wall of the shell has $-Q$ charge.

It means there is no charge on the outer wall of the shell, as it is having a total charge of $-Q$ only.

Example 35 Concentric charged shells

Two concentric conducting balls of radii r and R have charges q and Q on them.

- Find potential difference between them.
- What will happen if the two balls are connected using a copper wire?



Solution**Concepts**

- (i) We can write potential of A and that of B . Taking the difference will give the potential difference.
- (ii) When two conductors are connected, they share charge till potential difference between them becomes zero.

$$(a) \quad V_A = V_{Aq} + V_{AQ} = \frac{1}{4\pi\epsilon_0} \left[\frac{q}{r} + \frac{Q}{R} \right]$$

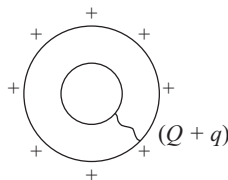
$$V_B = V_{Bq} + V_{BQ} = \frac{1}{4\pi\epsilon_0} \left[\frac{q}{R} + \frac{Q}{R} \right]$$

Note that potential due to charge Q (on B) is same at all inside points and the surface of shell B .

$$\therefore V_A - V_B = \frac{q}{4\pi\epsilon_0} \left[\frac{1}{r} - \frac{1}{R} \right] \quad (i)$$

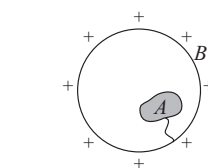
This is an amazing result. *The potential difference between the two shells depends only on the charge (q) on the inner shell and not on the charge (Q) on the outer shell!*

- (b) $V_A - V_B$ can become zero in only one way. That is, charge on the inner shell should become zero (see equation (i)). After the two shells are connected, the entire charge of the inner shell will move on to the outer shell. Inner shell will have no charge.



The result is not very difficult to guess. After the two balls are connected by a conducting wire, the entire system will behave like a conducting ball of radius R . The charge will reside on the outer surface only.

If the inner conductor A is not spherical or not concentric to B , then also the entire charge will reside on the outer wall of B after the two conductors are connected.



Entire charge given to $A + B$ system will remain on the outer surface of B .

In Short

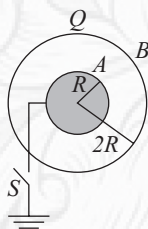
- (i) Under electrostatic condition, a piece of conductor has all its point at the same potential.
- (ii) When two conductors are connected by a conducting wire or are put in contact, they share charge till potential of both of them becomes equal.
- (iii) The Earth is a huge conductor at zero potential.
- (iv) A conductor connected to the Earth attains zero potential by way of sharing charge with the Earth.
- (v) A conductor A is kept inside a hollow conductor B and the two are connected using a wire. Any charge given to the system will reside on the outer surface of B .
- (vi) Sum of induced charges on the surface of a neutral conductor is zero. For a spherical conductor, such induced charges produce zero potential at the centre.

**YOUR TURN**

Q.36 An uncharged metal ball of radius $r = 2\text{ cm}$ is connected to a distant metal ball of radius $R = 4\text{ cm}$ having a charge $Q = 10\text{ nC}$. Find the final charge on the smaller ball.

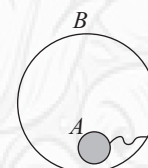
Q.37 Two balls, one big and the other small, are at same potential. You connect them by a wire. Which ball will lose charge?

Q.38 A metal ball (A) of radius R is surrounded by a concentric conducting shell (B) of radius $2R$. There is a wire connecting A to the Earth through a switch S . The wire passes through a small hole in shell B . Initially, the switch is open and B is given a charge Q .



- (a) What is potential of A ?
- (b) How much charge flows from the earth to the conductor A after the switch is closed?

Q.39 A metal ball A is surrounded by a spherical shell B . The two balls are given a total charge Q and are connected by a wire. Find potential of A if radii of the two balls are r and $2r$, respectively.

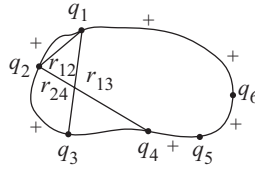


Q.40 Two concentric metallic shells have radii r and $2r$ and charges on them are q and Q , respectively. Charge Q is doubled. How is potential difference between the two shells affected?

10. ELECTROSTATIC SELF-ENERGY

Consider a charged body. The charge on it can be divided into number of small point like charges $- q_1, q_2, q_3, q_4, \dots$

The sum of potential energies of interaction of each pair of charges is termed as *electrostatic self-energy* of the body.



$$U = \frac{1}{4\pi\epsilon_0} \left[\frac{q_1 q_2}{r_{12}} + \frac{q_1 q_3}{r_{13}} + \dots + \frac{q_2 q_3}{r_{23}} + \frac{q_2 q_4}{r_{24}} + \dots + \dots \right]$$

We can write this as

$$U = \frac{1}{2} \frac{1}{4\pi\epsilon_0} \sum \frac{q_i q_j}{r_{ij}} \quad (41)$$

The factor $\frac{1}{2}$ is to account for the fact that we have taken every pair twice in our summation.

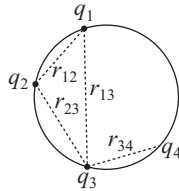
This energy is same as work done in assembling the charge on the body by bringing it in small pieces from infinity.

This potential energy of the charge system changes if the arrangement of the charge gets changed.

10.1 Electrostatic Potential Energy of a Uniformly Charged, Thin Spherical Shell

Consider a uniformly charged, thin shell of radius R having charge Q . We wish to find the self-energy of interactions of charge on the shell. We can think in two slightly different ways.

Method 1: Let's say the entire charge on the shell is made of number of point charges $q_1, q_2, q_3, \dots$



$$U = \frac{1}{2} \cdot \frac{1}{4\pi\epsilon_0} \left[\left(\frac{q_1 q_2}{r_{12}} + \frac{q_1 q_3}{r_{13}} + \frac{q_1 q_4}{r_{14}} + \dots \right) + \left(\frac{q_2 q_1}{r_{21}} + \frac{q_2 q_3}{r_{23}} + \frac{q_2 q_4}{r_{24}} + \dots \right) + \left(\frac{q_3 q_1}{r_{31}} + \frac{q_3 q_2}{r_{32}} + \dots \right) + \dots \right]$$

$$= \frac{1}{2} \cdot \frac{1}{4\pi\epsilon_0} \left[q_1 \left(\frac{q_2}{r_{12}} + \frac{q_3}{r_{13}} + \frac{q_4}{r_{14}} + \dots \right) + q_2 \left(\frac{q_1}{r_{21}} + \frac{q_3}{r_{23}} + \frac{q_4}{r_{24}} + \dots \right) + q_3 \left(\frac{q_1}{r_{31}} + \frac{q_2}{r_{32}} + \dots \right) + \dots \right]$$

$\frac{1}{4\pi\epsilon_0} \left(\frac{q_2}{r_{12}} + \frac{q_3}{r_{13}} + \frac{q_4}{r_{14}} + \dots \right)$ essentially represents the potential at the location of charge q_1 . This is potential of the shell when $q_1, q_2, q_3, \dots$ are infinitesimally small point charges.

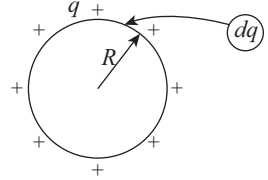
$$\begin{aligned} \therefore U &= \frac{1}{2} [q_1 V + q_2 V + q_3 V + \dots] \\ &= \frac{1}{2} V(q_1 + q_2 + q_3 + \dots) \\ &= \frac{1}{2} QV \end{aligned}$$

$$\text{But } V = \frac{1}{4\pi\epsilon_0} \frac{Q}{R}$$

$$\therefore U = \frac{Q^2}{8\pi\epsilon_0 R} \quad (42)$$

Method 2: The electrostatic potential energy of the charged shell is total work done by the external agent in bringing charges from infinity (in small bits) and putting them on the shell.

Consider a neutral shell. An external agent brings a small charge from a faraway point and spreads it uniformly on the shell. She again brings a small instalment of charge and puts it on the shell. This way she has piled up some charge q on the shell. Potential at the surface is



$$V = \frac{1}{4\pi\epsilon_0} \frac{q}{R}$$

work done in bringing next instalment of charge dq is

$$dW = V(dq) = \frac{1}{4\pi\epsilon_0} \frac{q dq}{R}$$

Total work done in assembling a charge Q on the shell is

$$W = \int_0^Q dW = \frac{1}{4\pi\epsilon_0 R} \int_0^Q q dq = \frac{Q^2}{8\pi\epsilon_0 R}$$

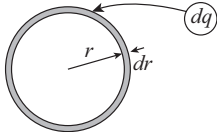
this is the required potential energy.

10.2 Electrostatic Potential Energy of a Uniformly Charged Spherical Volume

We will find the potential energy of a uniformly charged spherical volume of radius R , having charge Q , by the second method described in the last section.

Consider that an external agent brings charges in infinitesimally small instalments and spreads it so as to grow the radius of sphere keeping density of charge uniform.

At some instant, charge has been assembled up to a radius r . Charge on this sphere of radius r is

$$q = \rho \cdot \frac{4}{3}\pi r^3 = \frac{Q}{\frac{4}{3}\pi R^3} \cdot \frac{4}{3}\pi r^3$$


A charge dq transferred from infinity is used to increase the radius of the sphere from r to $r + dr$.

$$\Rightarrow q = \frac{Q}{R^3} \cdot r^3$$

Potential at the surface is

$$V = \frac{1}{4\pi\epsilon_0} \frac{q}{r} = \frac{1}{4\pi\epsilon_0} \frac{Q}{R^3} \cdot r^2$$

Next instalment of charge brought from infinity is dq which increases the radius of the charged volume by dr .

$$dq = (4\pi r^2 \cdot dr) \rho = (4\pi r^2 \cdot dr) \frac{Q}{\frac{4}{3}\pi R^3} = \frac{3Q}{R^3} r^2 dr$$

Work done in bringing this charge dq to the surface of the sphere (from infinity) is

$$dW = V(dq) = \frac{3Q^2}{4\pi\epsilon_0 R^6} r^4 dr$$

Work done in growing the sphere from zero radius to radius R is

$$W = \int_0^R dW = \frac{3Q^2}{4\pi\epsilon_0 R^6} \int_0^R r^4 dr = \frac{3Q^2}{20\pi\epsilon_0 R}$$

$\therefore$ Electrostatic potential energy is

$$U = \frac{3Q^2}{20\pi\epsilon_0 R} \quad (43)$$

Example 36 Heat generation

A spherical metal ball of radius r has charge Q on it. There is another identical ball at a large separation. This ball is neutral. Now, the two balls are connected using a metal wire. Find the amount of heat generated.

Solution

Concepts

- After the two balls are connected, charge gets shared. Finally, both of them will have $\frac{Q}{2}$ charge on them. This will make their potential same.
- When the charges in a system are allowed to move on their own, the electrostatic self-energy of the system always decreases. The loss in electrostatic potential energy appears as heat and electromagnetic radiations.
- As long as charge is moving from one ball to another, there is electric current. Electric current in a conductor leads to heating of the conductor.

Electrostatic self-energy of the charged ball is

$$U_i = \frac{Q^2}{8\pi\epsilon_0 r} \quad [\text{using equation (42)}]$$

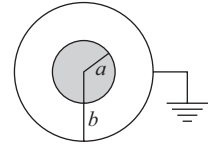
After both the balls are connected, charge on both of them is $\frac{Q}{2}$. Hence, final potential energy is

$$U_f = \frac{\left(\frac{Q}{2}\right)^2}{8\pi\epsilon_0 r} \times 2 = \frac{Q^2}{16\pi\epsilon_0 r}$$

As the two balls are far apart, interaction energy of the two balls can be neglected.

$$\begin{aligned} \text{Heat generated} &= U_i - U_f \\ &= \frac{Q^2}{16\pi\epsilon_0 r} \end{aligned}$$

Example 37 A metal ball of radius a carries a charge Q . A grounded metal shell of radius $b(>a)$ is concentric to the ball. Find the electrostatic potential energy of the system of the two spheres.



Solution

Concepts

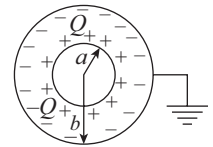
- Refer to example (34). Charge on the outer shell will be $-Q$ and it will reside on its inner wall.
- The potential energy of the system has three terms:
 - Potential energy of charged ball of radius a .
 - Potential energy of charge on the inner wall of the shell of radius b .
 - Potential energy of interaction of charges on the two balls.

Self-energy of the inner ball is

$$U_a = \frac{Q^2}{8\pi\epsilon_0 a}$$

Self-energy of the outer ball is

$$U_b = \frac{(-Q)^2}{8\pi\epsilon_0 b} = \frac{Q^2}{8\pi\epsilon_0 b}$$



To write potential energy of interaction of $+Q$ and $-Q$ charge, we can think that $-Q$ charge is located in the field of $+Q$ charge. Therefore,

$U_{ab} = V(-Q)$, where V is potential due to $+Q$ charge at the location of $-Q$ charge.

$$\therefore U_{ab} = \frac{1}{4\pi\epsilon_0} \frac{Q}{b} (-Q) = -\frac{Q^2}{4\pi\epsilon_0 b}$$

$\therefore$ Total potential energy of the system is

$$\begin{aligned} U &= U_a + U_b + U_{ab} \\ &= \frac{Q^2}{4\pi\epsilon_0} \left[\frac{1}{2a} + \frac{1}{2b} - \frac{1}{b} \right] = \frac{Q^2}{8\pi\epsilon_0} \left[\frac{1}{a} - \frac{1}{b} \right] \end{aligned}$$

In Short

- (i) Interaction energy of all charges amongst themselves is often termed as self-energy of a charge system.
- (ii) If the source charges are fixed and a test charge moves in the electric field created by them, the self-energy of source charges is not something very relevant. Remember, we are always concerned with change in potential energy. In such situation, the energy of interaction of the source charges with the test charge is the only relevant quantity.
- (iii) Electrostatic self-energy of a uniformly charged shell having radius R and charge Q is

$$U = \frac{3Q^2}{8\pi\epsilon_0 R}$$

- (iv) For a uniformly charged spherical volume, the self-energy is

$$U = \frac{3Q^2}{20\pi\epsilon_0 R}$$

- (v) Whenever charges in a system move under their mutual interactions, the system loses potential energy. As a simple example, consider two positive charges placed close to one another and released. The charges repel and move away. The potential energy decreases and kinetic energy increases.



YOUR TURN

Q.41 A point charge Q is placed at the centre of a spherical metal shell of radius R . The shell is grounded. Find the electrostatic energy of the system. The point charge is slowly

moved to infinitely (through a small hole in the shell). Find gain in potential energy in the process.

11. ACCELERATING AND RETARDING POTENTIAL DIFFERENCE

Consider two plates A and B maintained at a potential difference V

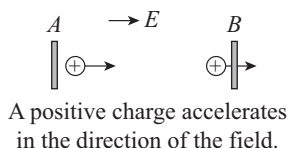
$$V = V_A - V_B$$

For creating such potential difference, we can give some charges to the two plates. For example, if A is given a positive charge and B is given a negative charge, the plate A will be at a higher potential. [Later we will learn that by connecting the two plates to the terminals of a battery, we can create a potential difference between them].

If $V_A > V_B$, then there is an electric field directed from A to B between the plates. If you release a positively charged particle near plate A , it experiences a force towards B . It gains kinetic energy during its motion from A to B . A potential difference between A and B is being used to accelerate a charge. We say that the charge has been accelerated through a potential difference of V volt.

Using conservation of energy for the charge (q):

$$\begin{aligned} K_A + U_A &= K_B + U_B \\ \Rightarrow K_B - K_A &= U_A - U_B \\ &= q(V_A - V_B) \\ \Delta K &= qV. \end{aligned} \quad (44)$$



Kinetic energy of a charge increases by qV when it is accelerated through a potential difference of V volt.

If the same charge were projected from B towards A , it retards. We say it is subjected to a *retarding potential difference*. In this case, it will lose kinetic energy $\Delta K = qV$ as it moves from B to A .

A negative charge will accelerate while moving from B to A and retard while moving from A to B , if $V_A > V_B$.

Example 38 Electronvolt (eV)

Kinetic energy gained by an electron when it is accelerated through a potential difference of 1 V is defined as a unit of energy known as *electronvolt*. Write value of 1eV in joule.

Solution**Concepts**

$$\Delta K = qV$$

KE gained by an electron when it is accelerated through a potential difference of 1V is

$$\begin{aligned} \Delta K &= (1.6 \times 10^{-19} \text{ C}) (1\text{V}) \\ &= 1.6 \times 10^{-19} \text{ J} \end{aligned}$$

This amount of energy is called 1 eV.

$$\therefore 1 \text{ eV} = 1.6 \times 10^{-19} \text{ J}.$$

Unit 'electron volt' is very useful while dealing with subatomic particles because a joule is a very large unit of energy for such particles.



YOUR TURN

Q.42 An electron is accelerated through a potential difference of 200 V. Find gain in its Kinetic energy in eV.

Q.43 An alpha particle is accelerated through a potential difference of 1000 V. Find gain in its kinetic energy.

12. ELECTRIC DIPOLE

A pair of equal and opposite charges ($+q$ and $-q$) separated by a small distance ($2d$) constitutes an electric dipole.

The line joining the two charges is known as the axis of the dipole.

Characteristic property of an electric dipole is its *electric*

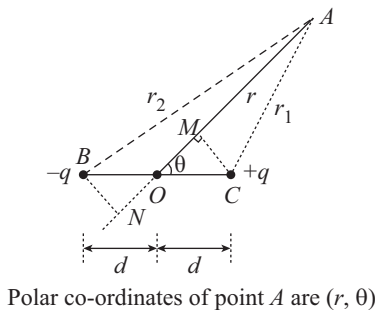
dipole moment ($\vec{P}$). It is a vector having magnitude $q(2d)$ and is directed from negative charge towards the positive charge.

$$P = q(2d) \quad (45)$$

Dipole moment is a characteristic property because the electric field and potential at a point due to an electric dipole does not depend on q and $2d$ separately, rather it depends on their product. This is what we will observe in the discussions to follow.

12.1 Potential Due to a Dipole

Consider the centre (O) of the dipole to be the origin of our coordinate system. We will find potential at a point A which is at a distance r from O . The line OA makes an angle θ with the axis of the dipole. In fact, (r, θ) are polar co-ordinates of point A with respect to O .



Polar co-ordinates of point A are (r, θ)

We could have considered Cartesian system as well, but the results obtained in polar coordinate system are simpler to express and remember.

Have a look at the diagram and keep in mind that we are considering point A such that $r \gg d$. CM and BN are perpendiculars on line OA .

$\triangle ACM$ is a right angle triangle having a base (CM) very small compared to the other sides.

$$\therefore AC \simeq AM = OA - OM$$

$$\Rightarrow r_1 \simeq r - OM = r - d \cos \theta$$

Similarly, in $\triangle ABN$:

$$r_2 \simeq AN = r + ON = r + d \cos \theta$$

Potential at A is sum of potentials due to $+q$ and $-q$ charge.

$$\begin{aligned} \therefore V &= V_+ + V_- \\ &= \frac{1}{4\pi\epsilon_0} \frac{q}{r_1} + \frac{1}{4\pi\epsilon_0} \frac{(-q)}{r_2} \\ &= \frac{q}{4\pi\epsilon_0} \left[\frac{1}{r - d \cos \theta} - \frac{1}{r + d \cos \theta} \right] \\ &= \frac{1}{4\pi\epsilon_0} \frac{q \cdot 2d \cdot \cos \theta}{r^2 - d^2 \cos^2 \theta} \end{aligned}$$

Since $r \gg d$ and $q(2d) = P$

$$\therefore V = \frac{1}{4\pi\epsilon_0} \frac{P \cos \theta}{r^2} \quad (46)$$

In general, any charge distribution that produces electric potential given by the above equation is known as an electric dipole. The direction from which the angle θ is to be measured so as to get the above expression for potential is taken as the direction of dipole moment vector ($\vec{P}$).

12.2 Electric Field Due to a Dipole

In section 5.3, we have learnt that electric field is equal to the negative gradient of electric potential. We will use this to find electric field due to dipole at a point A .

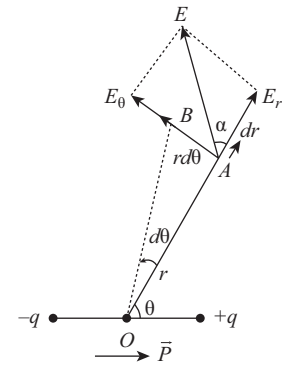
We will find two perpendicular components of the field at A . E_r is component of field along OA (i.e., along $\vec{r}$) and E_θ is the component of field in a direction perpendicular to OA .

If we displace by a very small amount dr (along OA) at A and potential changes by dV , then

$$E_r = -\frac{dV}{dr}$$

To indicate that it is only r which has been changed and θ has been held constant, we use symbol of partial differentiation

$$\therefore E_r = -\frac{\partial V}{\partial r} = -\frac{\partial}{\partial r} \left(\frac{1}{4\pi\epsilon_0} \frac{P \cos \theta}{r^2} \right)$$



$$= -\frac{P \cos \theta}{4\pi\epsilon_0} \frac{\partial}{\partial r} \left(\frac{1}{r^2} \right)$$

$$\Rightarrow E_r = \frac{1}{4\pi\epsilon_0} \frac{2P \cos \theta}{r^3} \quad (47)$$

Now, consider a small displacement in a direction perpendicular to OA . For obtaining such a displacement, we can keep r constant and rotate OA by a very small angle $d\theta$. In figure, AB is such a small displacement. Obviously,

$$AB = r d\theta$$

If moving from A to B changes the potential by dV , then

$$E_\theta = -\frac{dV}{(AB)} = -\frac{dV}{r d\theta}$$

Since r is held constant, we write:

$$E_\theta = -\frac{1}{r} \frac{\partial V}{\partial \theta}$$

$$= -\frac{1}{r} \frac{\partial}{\partial \theta} \left[\frac{1}{4\pi\epsilon_0} \frac{P \cos \theta}{r^2} \right]$$

$$= -\frac{1}{4\pi\epsilon_0} \frac{P}{r^3} \frac{\partial}{\partial \theta} (\cos \theta)$$

$$\therefore E_\theta = \frac{1}{4\pi\epsilon_0} \frac{P \sin \theta}{r^3} \quad (48)$$

The resultant electric field (E) at the point A is:

$$E = \sqrt{E_r^2 + E_\theta^2} = \frac{P}{4\pi\epsilon_0 r^3} \sqrt{4 \cos^2 \theta + \sin^2 \theta}$$

$$\Rightarrow E = \frac{1}{4\pi\epsilon_0} \frac{P}{r^3} \sqrt{1 + 3 \cos^2 \theta} \quad (49)$$

The electric field at A makes an angle α with the radial line OA which is given by

$$\tan \alpha = \frac{E_\theta}{E_r} = \frac{1}{2} \tan \theta$$

$$\Rightarrow \alpha = \tan^{-1} \left(\frac{1}{2} \tan \theta \right) \quad (50)$$

12.3 Field and Potential on Axis and Equator

Consider point A on the axis of a dipole. Such a position is often referred to as *end-on position*. Obviously, $\theta = 0$ for such a position.

Potential and field at point A are obtained by substituting $\theta = 0$ in the equations obtained above.

$$V = \frac{1}{4\pi\epsilon_0} \frac{P \cos \theta}{r^2}$$

$$\therefore V = \frac{1}{4\pi\epsilon_0} \frac{P}{r^2} \quad (51)$$

$$\text{and } E_r = \frac{1}{4\pi\epsilon_0} \frac{2P}{r^3} \quad (52)$$

Clearly $E_\theta = 0$.

The electric field is along the axis and its magnitude is given by equation (52).

When point A is on the perpendicular bisector of the dipole, we say that the point is on the *equator of the dipole* or it is at *broadside on position*. Such position is characterized by $\theta = 90^\circ$.

$$\therefore V = \frac{1}{4\pi\epsilon_0} \frac{P \cos 90^\circ}{r^2} = 0 \quad (53)$$

$$E_r = 0 \text{ and}$$

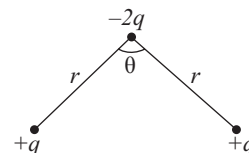
$$E_\theta = \frac{1}{4\pi\epsilon_0} \frac{P}{r^3} \quad (54)$$

Electric field at A has magnitude given by equation (54) and its direction is anti parallel to $\vec{P}$.

All the formulae derived above are valid only for the case $r \gg d$.

Example 39 Dipole moment of a water molecule

In a water molecule, two hydrogen atoms form bonds with one oxygen atom. Due to difference in electronegativity, the atoms acquire net charge on them. Assume that charge on each H atom is $+q$ and that on the oxygen atom is $-2q$. Bond length is r . The angle between the two bonds is θ . Find electric dipole moment of a water molecule.



Solution

Concepts

- The given system is equivalent to two dipoles inclined at an angle θ .
- Dipole moments of each dipole when added, according to the rules of vector addition, give the dipole moment of the system.

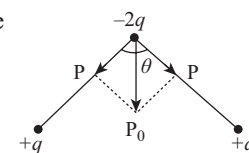
There are two dipoles. Each dipole has dipole moment

$$P = q \cdot r$$

Directions of the two dipoles are as shown. Resultant dipole moment is

$$P_0 = \sqrt{P^2 + P^2 + 2PP \cos \theta} = 2P \cos \left(\frac{\theta}{2} \right)$$

Direction of P_0 bisects the angle between the two bonds.



Example 40 One dipole can be replaced with two

Electric field at an axial position of a dipole is given by equation (52) and the field at an equatorial position is given by equation (54). Using these two results, find the electric field at a general point A having coordinates (r, θ) .

Solution**Concepts**

We can resolve dipole moment vector $\vec{P}$ into two perpendicular components and assume that there are two dipoles.

We resolve the dipole moment vector ($\vec{P}$) into two components along OA and perpendicular to it.

The components are

$$P_1 = P \cos \theta \text{ and}$$

$$P_2 = P \sin \theta$$

A is a point on axial position for P_1 . Field due to P_1 at A is along OA given by

$$E_1 = \frac{1}{4\pi\epsilon_0} \frac{2P_1}{r^3} = \frac{1}{4\pi\epsilon_0} \frac{2P \cos \theta}{r^3}$$

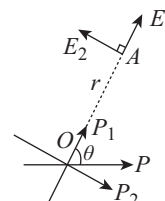
A is a point on equatorial position for P_2 . Field due to P_2 at A is anti parallel to P_2 (i.e., perpendicular to OA) and has a magnitude

$$E_2 = \frac{1}{4\pi\epsilon_0} \frac{P_2}{r^3} = \frac{1}{4\pi\epsilon_0} \frac{P \sin \theta}{r^3}$$

Resultant field at A is vector sum of fields due to the two component dipoles.

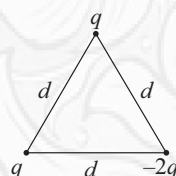
$$\therefore E = \sqrt{E_1^2 + E_2^2} = \frac{1}{4\pi\epsilon_0} \frac{P}{r^3} \sqrt{1 + 3 \cos^2 \theta}$$

This is exactly same as equation (49).

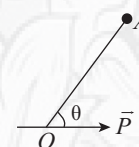
**YOUR TURN**

Q.44 In a HCl molecule, equal and opposite charges of magnitude 3.4×10^{-20} C lie on the two atoms. Bond length is 1.0×10^{-10} m. Find the dipole moment of the molecule.

Q.45 Find electric dipole moment of the charge configuration shown in the figure.



Q.46 Figure shows a dipole and a faraway point A . The electric field at A makes an angle of 45° with OA . Find the angular position (θ) of point A with respect to the dipole axis.

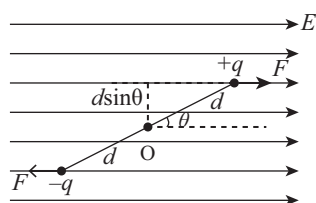


Q.47 At a point on the axis of an electric dipole, the strength of field is E_0 . Distance of the point from the dipole is r which is large compared to the length of the dipole. Find the magnitude of electric field at a distance $2r$ along the perpendicular bisector.

13. ELECTRIC DIPOLE IN AN ELECTRIC FIELD

For simplicity, consider a dipole placed in a uniform field E . The axis of the dipole makes an angle θ with the direction of the electric field. The two charges, $+q$ and $-q$, experience force $F = qE$ in opposite directions.

Therefore, the dipole experiences a zero net force. If a dipole is released in a uniform field its centre of mass will not accelerate.



The two forces on the dipole balance out. But they form a torque.

However, the pair of forces acting on the dipole forms a couple. Torque due to any one force (F) about the centre O is:

$$(\text{Force}) \times (\text{perpendicular distance of the line of the force from } O) = F \cdot d \sin \theta$$

Here, $2d$ is length of the dipole. In the diagram shown, the sense of this torque is clockwise. The other force also produces an equal torque in the same sense. Therefore, the total torque on the dipole is:

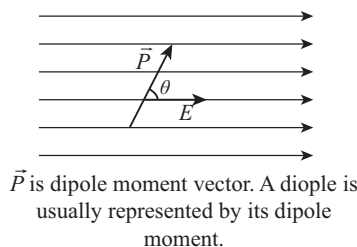
$$T = 2Fd \sin \theta = 2qEd \sin \theta = q(2d)E \cdot \sin \theta$$

$$\therefore T = PE \sin \theta \quad (55)$$

In vector notation, we can write the expression of the torque on the dipole as

$$\vec{\tau} = \vec{P} \times \vec{E} \quad (56)$$

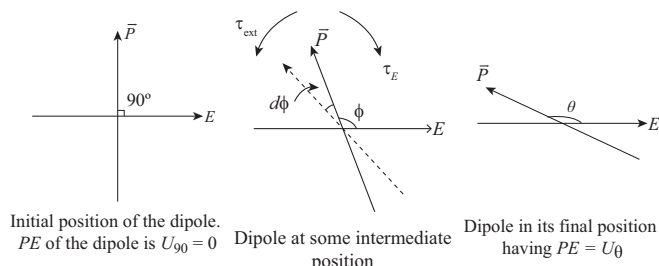
In the diagram shown, direction of $\vec{P} \times \vec{E}$ is into the plane of the figure. Recall that torque is an axial vector. If you curl your right hand fingers in clockwise sense (from $\vec{P}$ to $\vec{E}$) your thumb will point into the plane of the figure. Such a torque will have an effect to rotate the dipole in clockwise sense.



Dipole will experience no torque and will be in equilibrium when $\theta = 0^\circ$ or, 180° .

13.1 Potential Energy of a Dipole in a Uniform Field

When a ball is at some height you know that it has potential energy. If you release the ball, it is able to gain kinetic energy. The dipole, shown in the diagrams above, also has a potential energy. If you release it from the position shown, it starts to spin (due to the torque acting on it) and gains kinetic energy.



If you assume the potential energy of a ball to be zero when it is on the floor of a room, its potential energy at height h is equal to work done by you in slowly raising it to that height from the floor. Similarly, potential energy of a dipole in an electric field is work done by an external agent in slowly rotating the dipole from the reference position (where we assume $U = 0$) to its present position.

We usually consider the potential energy to be zero when the dipole makes an angle of 90° with the direction of the electric field. The dipole is rotated slowly from this position to a new position where it makes an angle θ with the field direction. To find work done in the process, consider the dipole at some intermediate stage where it makes an angle ϕ with the direction of the field.

Electric torque is $\tau_E = PE \sin\phi$ (clockwise)

To rotate the dipole slowly, torque by the external agent must be equal and opposite.

$$\therefore \tau_{\text{ext}} = PE \sin\phi \text{ (anticlockwise)}$$

Work done in rotating the dipole further by $d\phi$ is

$$dW_{\text{ext}} = \tau_{\text{ext}} \cdot d\phi = PE \sin\phi \cdot d\phi$$

Work done in rotating the dipole from 90° to θ is

$$\begin{aligned} W_{\text{ext}} &= PE \int_{90^\circ}^{\theta} \sin\phi \cdot d\phi = -PE [\cos\phi]_{90^\circ}^{\theta} \\ &= -PE \cos\theta \end{aligned}$$

Thus, potential energy of a dipole kept in a uniform field making an angle θ with the direction of the field is

$$U_\theta = -PE \cos\theta = -\vec{P} \cdot \vec{E} \quad (57)$$

13.2 Dipole in a Non-uniform Field

Consider an electric field along x direction. The magnitude of the field changes with position x . A short dipole of length Δx is placed in the field with its axis along the x direction (see figure). The negative pole of the dipole is at co-ordinate x where the field is E . Force on this negative end is

$$F_1 = Eq \text{ (}\leftarrow\text{)}$$

The positive end of the dipole is at $x + \Delta x$ where the electric field has strength $E + \Delta E$. Force on positive end of the dipole is

$$F_2 = (E + \Delta E)q \text{ (}\rightarrow\text{)}$$

Resultant force on the dipole is

$$F = \Delta Eq \text{ (}\rightarrow\text{)}$$

If Δx is small, we can express ΔE as:

$$\Delta E \simeq \left(\frac{dE}{dx} \right) \Delta x$$

$$\therefore F = \left(\frac{dE}{dx} \right) (q\Delta x) = \left(\frac{dE}{dx} \right) P \quad (58)$$

The above expression can be used to find force on the dipole when E is known as a function of x .

In general, a dipole placed in a non-uniform field will experience a force as well as a torque.

Example 41 Oscillation in a uniform field

An electric dipole, having dipole moment P , is placed in a uniform electric field E .

- Identify the stable equilibrium position of the dipole.
- Calculate the time period of small oscillations of the dipole about its stable equilibrium position. It may be assumed that the dipole consists of two point masses, m each, at separation $2d$.

Solution

Concepts

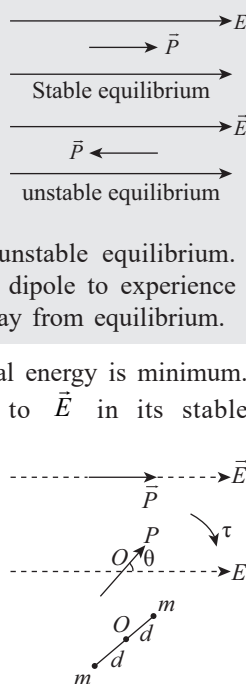
A position where net force and torque are zero on the dipole is its equilibrium position. There are two such equilibrium positions. When $\vec{P}$ is parallel to $\vec{E}$ (i.e., when $\theta = 0^\circ$) torque on the dipole is zero. Potential energy of the dipole in this position is minimum ($U = -PE \cos 0^\circ$).

This is position of stable equilibrium. If you rotate the dipole slightly and release it, the torque due to electric field will bring it back to its equilibrium position.

When $\vec{P}$ is anti parallel to $\vec{E}$ (i.e., when $\theta = 180^\circ$) torque on the dipole is zero and it is in equilibrium. Potential energy is maximum at this position ($U = -PE \cos 180^\circ$). This is position of unstable equilibrium. A slight disturbance will cause the dipole to experience a torque that will take it further away from equilibrium.

- (a) In stable equilibrium, potential energy is minimum. Dipole is aligned parallel to $\vec{E}$ in its stable equilibrium position.

- (b) The dipole is rotated slightly from its equilibrium position and released. In the diagram shown, a line perpendicular to the plane of the figure and passing through centre (O) of the dipole is the rotation axis.



Moment of inertia about the rotation axis is

$$I = md^2 + md^2 = 2md^2.$$

When dipole is at an angular displacement θ from its equilibrium position, the electric torque on it is

$$\tau = PE \sin \theta \simeq PE\theta$$

[$\sin \theta \simeq \theta$, for small values of θ]

This is restoring torque trying to bring the dipole back to its equilibrium position.

$$\therefore I \frac{d^2 \theta}{dt^2} = -PE\theta$$

$$\Rightarrow \frac{d^2 \theta}{dt^2} = -\left(\frac{PE}{I}\right)\theta \quad \vec{E}$$

This is standard equation of SHM with

$$\omega^2 = \frac{PE}{I} \quad \Rightarrow \quad \omega = \sqrt{\frac{PE}{I}}$$

$$\Rightarrow T = 2\pi \sqrt{\frac{I}{PE}} = 2\pi \sqrt{\frac{2md^2}{PE}}$$



YOUR TURN

Q.48 A dipole consists of two point charges $3.2 \times 10^{-19}\text{C}$ and $-3.2 \times 10^{-19}\text{C}$ separated by a distance $2 \times 10^{-8}\text{m}$. The dipole is in stable equilibrium position in an electric field $E = 10^6 \text{Vm}^{-1}$. Calculate the work needed to rotate the dipole by 60° .

Q.49 An electric dipole has dipole moment P and its moment of inertia about the perpendicular bisector (passing through the centre of mass) of the dipole axis is I . It is released in a uniform electric field E from a position where dipole axis is perpendicular to the direction of the field. Find the angular speed of the dipole when it gets parallel to the field.

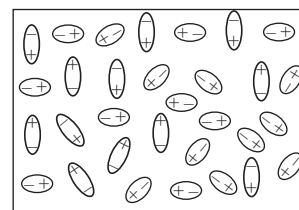
14. DIELECTRIC IN ELECTRIC FIELD

A dielectric is a non-conductor. It does not have free electrons. Molecules like H_2O , NH_3 , etc. have separation of positive and negative charges in them (i.e., they have poles). These molecules have a dipole moment and are said to be *polar molecules*. Molecules like O_2 , He , etc. do not have such separation of positive and negative charges. Their dipole moment is zero. They are *non-polar molecules*.

When you pick a piece of a polar dielectric material, you will not notice any separation of charge. There are a

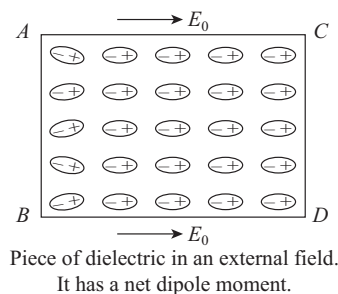
large number of molecules even in a small piece of material and their thermal motion causes them to be in random orientations. We do not see any separation of charge. There is no dipole moment of the piece.

Now place the piece of material in an electric field. Individual molecular dipoles experience torque and tend to get aligned parallel to the



A piece of polar dielectric material has no dipole moment.

electric field. The alignment is not perfect due to thermal agitation. Still one can see separation of charge on the surface of the material. Faces AB and CD of the rectangular dielectric slab acquire negative and positive charge, respectively. These are induced charges, also known as *bound charges*. These charges are not free to move as in metals; they are bound in a molecule. Now the piece of material has a non-zero dipole moment (P). We say that the material is polarized.



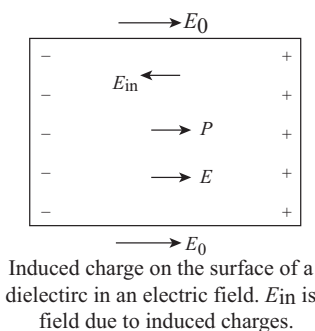
The induced charges produce their own electric field (E_{in}) inside the material. This field weakens the field inside the material. Assume that the rectangular block shown in the figure is large. In that case induced charges will create a field (E_{in}) in the direction shown. Resultant field inside the dielectric material is

$$E = E_0 - E_{in} \quad (59)$$

here E_0 is the external field.

The resultant field E is in the direction of external field E_0 and has a smaller magnitude.

$$E = \frac{E_0}{K} \quad (60)$$



Here, K (>1) is known as dielectric constant of the material (or, its relative permittivity).

Using (59) and (60), we can write

$$E_{in} = E_0 \left[1 - \frac{1}{K} \right] \quad (61)$$

When a non-polar dielectric is kept in an external field, its molecules get polarized. The field exerts forces in opposite directions on the positive and negative charges of a molecule causing a separation between them. Once polarized, the material will behave just like a polar material.

In the above discussion, we assumed a rectangular dielectric block. However, the electric field inside a dielectric gets reduced by a factor K (dielectric constant) whatever be its shape and this happens due to induced charges creating their own field.

When a point charge is kept inside an infinite dielectric medium, polarization of the dielectric medium takes place. Field created by the polarization charges reduce the electric field due to the point charge. Field at a distance r from a point charge q kept in a dielectric medium is

$$E = \frac{1}{4\pi\epsilon_0 K} \frac{q}{r^2} \quad (62)$$

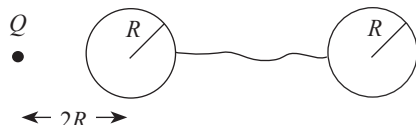
If a test charge Q is placed at a distance r from q , it experiences a force

$$F = \frac{1}{4\pi\epsilon_0 K} \frac{qQ}{r^2}$$

This is the equation we wrote in section 4.1 in the last chapter. Once again, it is being reiterated that field given by equation (62) is due to combined effect of charge q and the polarization charges appearing in the medium.

Miscellaneous Examples

Example 42 Two neutral conducting balls, each of radius R , are kept at a large separation connected by a conducting wire. A point charge Q is placed at a distance of $2R$ from the centre of one of the balls. Calculate the total amount of charge appearing on the other ball.



Solution

Concepts

- Potential of both the balls must be same.
- If q is total charge on the surface of a conducting ball, its contribution to the potential at the centre

of the ball is $\frac{Kq}{R}$. This does not depend on how the charge is distributed on the surface.

- Potential calculated at the centre of a ball will be the potential of the entire ball as it is conducting.

Let charge on right side ball be q .

Charge on the surface of left ball will be $-q$. This is because net charge on the two balls is zero.

Potential at the centre of the left ball will be:

$$V_1 = \frac{KQ}{2R} + \frac{K(-q)}{R} \quad (a)$$

Charge on the other ball will have no effect on this potential as the two balls are at a large separation. V_1 is potential of the left ball.

Potential at the centre of the right ball is:

$$V_2 = \frac{Kq}{R} \quad (b)$$

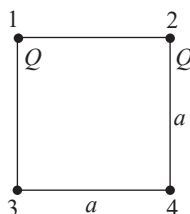
and this is the potential of the entire ball

$$V_1 = V_2$$

$$\Rightarrow \frac{KQ}{2R} - \frac{Kq}{R} = \frac{Kq}{R}$$

$$\Rightarrow \frac{Q}{4} = q$$

Example 43 Four identical balls, each of mass m , are connected using four insulating threads to form a square of side length a . The system lies on a smooth horizontal table. Balls 1 and 2 are given charge Q and the thread connecting them is cut. Find maximum speed of the two balls after the string is cut.



Solution

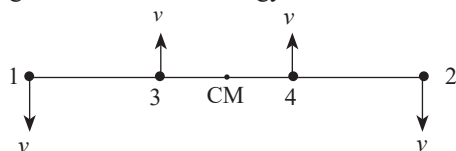
Concepts

- After the system is released, there is no external force on the system. Centre of mass of the system will not move.
- Balls 1 and 2 repel each other and distance between them increases. When their separation is maximum, PE of the system is least. At this point, the kinetic energy of the system is maximum.
- Due to symmetry, balls 3 and 4 will move identically in y direction.
- To conserve momentum (i.e., to keep it equal to zero), the velocities of the balls must be as shown in the figure below when the balls fall in a straight line. In this position, separation between 1 and 2 is maximum.
- Thinking in terms of force, you must be able to see that the ball 1 (or, 2) is having a tangential component of electrostatic force acting on it which is accelerating it.

This tangential force becomes zero when the balls are in a straight line as shown below. This is the point where speed of 1 (or, 2) is maximum.

Let speed of all balls when they are in a straight line = v .

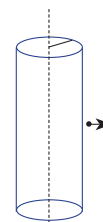
Applying conservation of energy:



$$\frac{1}{2}mv^2 \times 4 + K \frac{Q^2}{3a} = K \frac{Q^2}{a}$$

$$\Rightarrow v = \sqrt{\frac{KQ^2}{3ma}}$$

Example 44 A long cylinder of radius a is having a uniform surface charge density σ . An electron (charge $= -e$, mass $= m$) shoots out of the cylinder surface moving radially outward with speed u . It is able to move to a distance r from the axis of the cylinder. Find u .



Solution

Concepts

- The electron moves in the electric field of the cylinder and its energy is conserved.
- The equation $\int_{V_1}^{V_2} dV = - \int_1^2 E dr$ can be used to write potential difference between a point on the surface of the cylinder and another at a distance r from the axis.

Electric field at a distance x ($>a$) from the axis is

$$E = \frac{\lambda}{2\pi\epsilon_0 x} = \frac{\sigma \cdot 2\pi R \cdot 1}{2\pi\epsilon_0 x} = \frac{\sigma R_0}{\epsilon_0 x}$$

Let V_0 be potential on the surface and V be the potential at a distance r from the axis.

$$\begin{aligned} V - V_0 &= - \int_a^r E dx = - \frac{\sigma R_0}{\epsilon_0} \int_a^r \frac{dx}{x} \\ &= - \frac{\sigma R_0}{\epsilon_0} \ln \frac{r}{a} \end{aligned}$$

Using conservation of energy for motion of the electron

$$\begin{aligned} \frac{1}{2}mu^2 + (-e)V_0 &= 0 + (-e)V \\ \Rightarrow \frac{1}{2}mu^2 &= e(V_0 - V) \\ \Rightarrow \frac{1}{2}mu^2 &= \frac{e\sigma R_0}{\epsilon_0} \ln \left(\frac{r}{a} \right) \\ \Rightarrow u &= \sqrt{\frac{2e\sigma R_0}{m\epsilon_0} \ln \left(\frac{r}{a} \right)} \end{aligned}$$

Example 45 Two rings at small separation

Two rings, A and B , have radius R and carry charge $+Q$ and $-Q$, respectively. The charge is distributed uniformly on the circumference of each ring. Both the rings have their planes

parallel to yz plane and their centres are at $(b, 0, 0)$ and $(-b, 0, 0)$, respectively. Find electric potential at point $P(x, 0, 0)$. It is given that $b \ll x$.

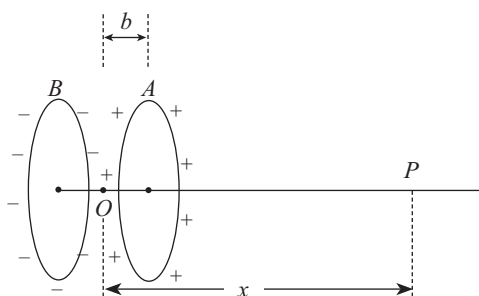
Solution**Concepts**

- (i) If the centre of a charged ring is at the origin, the electric field at a distance x on its axis is

$$E = \frac{KQx}{(R^2 + x^2)^{3/2}}$$

- (ii) If one moves a small distance Δx along the axis, the change in potential will be $\Delta V = -E\Delta x$.

Let, V = potential at P due to ring A when it is at $x = 0$.



When the ring A is moved by a small distance ($\Delta x = b$) towards point P , the potential at P becomes

$$V_A = V + |\Delta V| \quad (i)$$

where $|\Delta V| = E\Delta x = Eb$

Similarly, potential at P will be $-V$ if the ring B was placed at $x = 0$.

when ring B is moved by $\Delta x = b$ away from P , potential due to it at P becomes

$$V_B = -(V - |\Delta V|) \quad (ii)$$

Resultant potential at P is

$$V_P = V_A + V_B = V + |\Delta V| - (V - |\Delta V|)$$

$$= 2|\Delta V| = 2Eb$$

$$= 2 \frac{kQxb}{(R^2 + x^2)^{3/2}}$$

Example 46 Vertical circle in uniform electric field

A small ball of mass $m = 0.6$ kg carrying a charge $q = 80 \mu\text{C}$ is connected to a light thread of length $l = 30\text{cm}$. The other end of the thread is fixed. A horizontal electric field $E = 10^5 \text{ NC}^{-1}$ exists in the space. The ball is given a speed and

it moves in a vertical circle. Find the speed of the ball at the top point of the circle if it is just able to complete the circle without the string getting slack at any point [$g = 10 \text{ ms}^{-2}$].

Solution**Concepts**

- (i) Apart from the string tension, the ball experiences two forces mg and qE . Resultant of these two forces is

$$F = \sqrt{(mg)^2 + (qE)^2}$$

The particle is moving in an environment where we can regard the effective free fall acceleration to be

$$g_{\text{eff}} = \frac{F}{m} = \sqrt{g^2 + \left(\frac{qE}{m}\right)^2}$$

Direction of g_{eff} makes an angle θ with vertical where

$$\tan \theta = \frac{qE}{mg}$$

- (ii) Tension will be maximum and minimum when the particle (on its circular path) lies on the diameter along the direction of g_{eff} .

$$\begin{aligned} g_{\text{eff}} &= \sqrt{g^2 + \left(\frac{qE}{m}\right)^2} \\ &= \sqrt{10^2 + \left(\frac{80 \times 10^{-6} \times 10^5}{0.6}\right)^2} \\ &= \frac{50}{3} \text{ ms}^{-2} \end{aligned}$$

Direction of g_{eff} makes an angle θ with the vertical where

$$\tan \theta = \frac{qE}{mg} = \frac{4}{3}$$

$$\Rightarrow \theta = 53^\circ$$

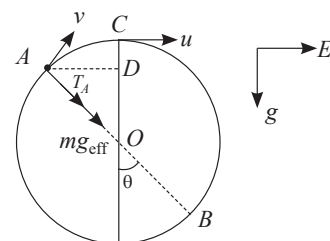
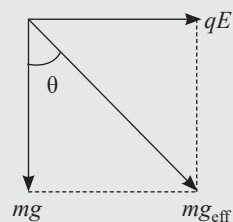
Tension will be maximum when the ball is at B and it will be minimum when the ball is at A . For completing the circle, tension at A should be

$$T_A \geq 0$$

In critical case, $T_A = 0$

$$\Rightarrow \frac{mv^2}{l} = mg_{\text{eff}} \Rightarrow v = \sqrt{lg_{\text{eff}}}$$

$$\Rightarrow v = \sqrt{0.3 \times \frac{50}{3}} = \sqrt{5} \text{ ms}^{-1}$$



To get the speed at C , we will use conservation of energy between points A and C .

$$\begin{aligned}
 K_A + U_{gA} + U_{eA} &= K_B + U_{gC} + U_{eC} \\
 \Rightarrow \frac{1}{2}mv^2 + (U_{eA} - U_{eC}) &= \frac{1}{2}mu^2 + (U_{gC} - U_{gA}) \\
 \Rightarrow \frac{1}{2}mv^2 + (V_A - V_C)q &= \frac{1}{2}mu^2 + mgl(1 - \cos\theta) \\
 \Rightarrow \frac{1}{2}mv^2 + (E \cdot l \cdot \sin\theta)q &= \frac{1}{2}mu^2 + mgl(1 - \cos\theta) \\
 \Rightarrow \frac{1}{2} \times 0.6 \times 5 + 10^5 \times 0.3 \times \frac{4}{5} \times 80 \times 10^{-6} \\
 &= \frac{1}{2} \times 0.6 \times u^2 + 0.6 \times 10 \times 0.3 \left(1 - \frac{3}{5}\right) \\
 \Rightarrow u &= 3 \text{ ms}^{-1}
 \end{aligned}$$

Example 47 A non-conducting rod of mass M and length L is hinged at one of its ends so that it is free to rotate in a vertical plane. Positive charge of linear density λ is spread uniformly on the rod and the rod is held in horizontal position. A uniform electric field E is switched on in horizontal direction and the rod is released from rest. The rod again comes to rest in vertical position. Find E .

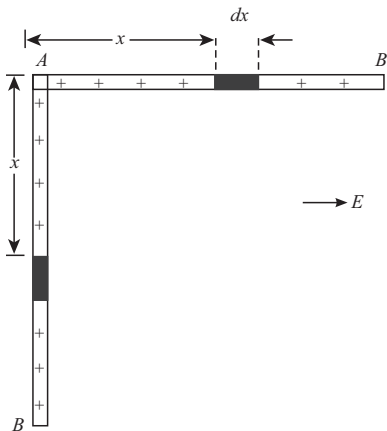


Solution

Concepts

- When a positive charge moves against an electric field, its electric potential energy increases. As the rod gets vertical, the electric potential energy increases. This comes at the cost of decrease in gravitational potential energy.
- Increase in potential when one moves a distance r against a uniform field (E) is $\Delta V = Er$.

Charge on length dx of the rod is, $dq = \lambda dx$.



When the rod gets vertical, this charge moves a distance x against the field. Increase in its electric potential energy is

$$dU = (dq)(Ex) = \lambda Ex dx$$

Increase in electric potential energy of the entire rod is

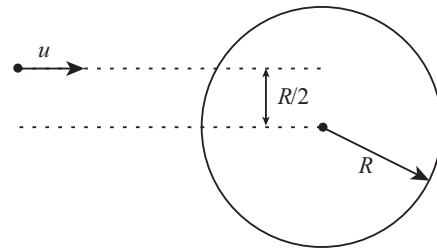
$$\Delta U = \lambda E \int_0^L x dx = \frac{1}{2} \lambda EL^2$$

This must be equal to loss in gravitational potential energy of the rod.

$$\begin{aligned}
 \therefore \frac{1}{2} \lambda EL^2 &= Mg \frac{L}{2} \\
 \Rightarrow E &= \frac{Mg}{\lambda L}
 \end{aligned}$$

Example 48 Particle misses the sphere

A fixed non-conducting spherical shell of radius R has a charge Q spread uniformly on its surface.



A particle having mass m and charge q is projected towards the shell from a large distance at a speed u . Impact parameter is $\frac{R}{2}$ (see figure). The particle passes the shell grazing its surface. Find the projection speed u . Neglect gravity.

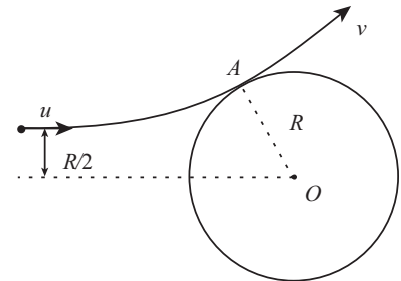
Solution

Concepts

- Force acting on the particle is always along a radial line from the centre of the shell. Therefore, angular momentum of the particle is conserved about the centre of the shell.
- The particle grazes the surface of the shell. This means that its velocity is tangential to the surface when it gets near the shell.
- Energy of the particle is conserved.

Let the particle graze the sphere at point A where its speed is v .

Conservation of angular momentum about centre (O) gives



$$mu \frac{R}{2} = mvR$$

$$\Rightarrow v = \frac{u}{2}$$

Conservation of energy gives:

$$K_A + U_A = K_\infty + U_\infty$$

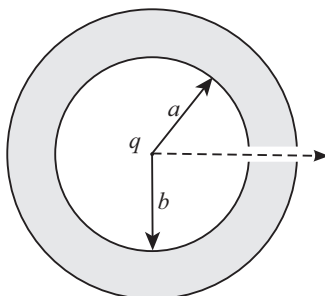
$$\Rightarrow \frac{1}{2}mv^2 + K \frac{Qq}{R} = \frac{1}{2}mu^2 + 0$$

$$\Rightarrow \frac{3}{8}mu^2 = K \frac{Qq}{R}$$

$$\left[K = \frac{1}{4\pi\epsilon_0} \right]$$

$$\Rightarrow u = \sqrt{\frac{2Qq}{3\pi\epsilon_0 mR}}$$

Example 49 A metal shell has inner and outer radii a and b , respectively. A point charge q is at its centre. The point charge is slowly moved to infinity through a small hole in the shell. Find the change in electrostatic energy of the system in the process.



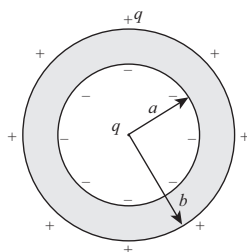
Solution

Concepts

- A charge $-q$ is induced on the inner wall of the shell. The outer wall has $+q$ charge on it.
- The total electrostatic energy of the system is self-energy of two spherical charge plus the interaction energy of three charges.
- When q is taken to infinity, energy of system becomes zero.

Initial potential energy of the system is

U_i = (self-energy of charge on inner wall) + (self-energy of charge on the outer wall) + (interaction energy of point charge with charge on inner wall) + (interaction energy of point charge with charge on outer wall) + (interaction energy of charge on two walls)



$$\begin{aligned} &= \frac{k(-q)^2}{2a} + \frac{kq^2}{2b} + \frac{kq(-q)}{a} + \frac{kqq}{b} + \frac{k(-q)q}{b} \\ &= \frac{kq^2}{2} \left[\frac{1}{b} - \frac{1}{a} \right] \end{aligned}$$

When charge is moved to infinity, the final potential energy is $U_f = 0$.

$$\therefore \Delta U = U_f - U_i = \frac{kq^2}{2} \left(\frac{1}{a} - \frac{1}{b} \right)$$

Example 50 Potential at the edge of a disc

A disc of radius R has a uniform surface charge density σ . Find potential at a point on the edge of the disc.

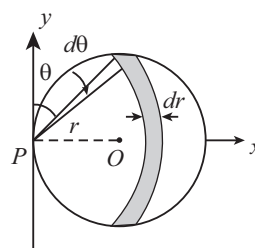
Solution

Concepts

We can divide the disc into segments of thin rings centred at the concerned point on the edge of the disc.

We wish to find potential at a point P on the edge of the charged disc.

We draw a circular arc of radius r centred at P . Then we draw another arc of radius $r + dr$. Consider the ring-shaped band between these two arcs (i.e., the shaded part in the diagram).



From the diagram, one can easily conclude that

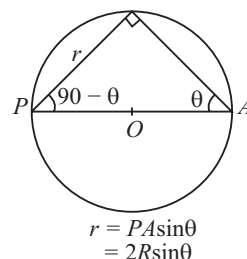
$$r = 2R \sin \theta$$

$$\Rightarrow \frac{dr}{d\theta} = 2R \cos \theta$$

$$\Rightarrow dr = 2R \cos \theta \cdot d\theta$$

Charge on the shaded ring is

$$\begin{aligned} dq &= \sigma \cdot dA \\ &= \sigma r 2 \left(\frac{\pi}{2} - \theta \right) dr \end{aligned}$$



[$\because$ arc length = (radius)(angle)]

Potential at P due to charge dq on the ring element is

$$\begin{aligned} dV &= \frac{1}{4\pi\epsilon_0} \frac{dq}{r} = \frac{1}{4\pi\epsilon_0} \sigma (\pi - 2\theta) dr \\ &= \frac{\sigma}{4\pi\epsilon_0} (\pi - 2\theta) 2R \cos \theta \cdot d\theta \end{aligned}$$

Potential at P can be obtained by adding contributions made by all such rings.

$$\begin{aligned} \therefore V &= \frac{\sigma R}{2\pi\epsilon_0} \int_0^{\frac{\pi}{2}} (\pi - 2\theta) \cos \theta d\theta \\ &= \frac{\sigma R}{2\pi\epsilon_0} \pi \left[\sin \theta \right]_0^{\frac{\pi}{2}} - \frac{\sigma R}{\pi\epsilon_0} \int_0^{\frac{\pi}{2}} \theta \cos \theta d\theta \\ &= \frac{\sigma R}{2\epsilon_0} - \frac{\sigma R}{\pi\epsilon_0} \left[\theta \int \cos \theta d\theta - \int 1 \cdot \sin \theta d\theta \right]_0^{\frac{\pi}{2}} \end{aligned}$$

Here, we have used integration by parts.

$$\begin{aligned}\therefore V &= \frac{\sigma R}{2\epsilon_0} - \frac{\sigma R}{\pi\epsilon_0} \left[\theta \sin \theta + \cos \theta \right]_0^{\frac{\pi}{2}} \\ &= \frac{\sigma R}{2\epsilon_0} - \frac{\sigma R}{\pi\epsilon_0} \left[\frac{\pi}{2} - 1 \right] = \frac{\sigma R}{\pi\epsilon_0}\end{aligned}$$

Example 51 A conducting sphere S_1 of radius r is attached to an insulating handle. Another conducting sphere S_2 of radius R is mounted on an insulating stand. S_2 is initially uncharged. S_1 is given a charge Q , brought into contact with S_2 and removed. S_1 is recharged such that the charge on it is again Q . It is again brought into contact with S_2 and removed. This procedure is repeated n times.

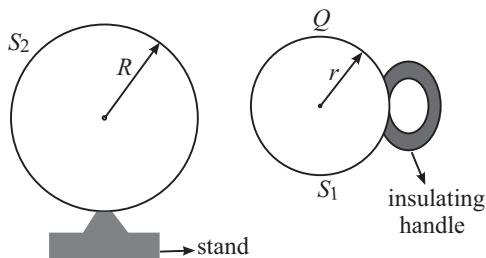
- Find the electrostatic energy of S_2 after n such contacts with S_1 .
- What is the limiting value of this energy as $n \rightarrow \infty$.

Solution

Concepts

- Charge is transferred from S_1 to S_2 till both of them attain same potential, during each contact.
- The lost charge is replenished on S_1 and it is again put into contact with S_2 . It is easy to guess that after infinite contacts, the potential of S_2 will be same as potential of S_1 having charge Q on it.
- 'Insulating handle' and 'insulating stand' ensure that charge does not leak out of the system comprising of S_1 and S_2 .

- After a contact, let charge on the two spheres be q_R and q_r .



Their potential will be same.

$$\therefore k \frac{q_R}{R} = k \frac{q_r}{r} \Rightarrow \frac{q_R}{R} = \frac{q_r}{r} \quad (i)$$

After first contact, the charge Q gets divided in the ratio $\frac{R}{r}$ on the two spheres. Therefore, charge on S_2 after first contact is

$$q_{R1} = \frac{QR}{(R+r)}$$

Before second contact, S_1 has charge Q and S_2 has charge q_{R1} . Total charge $(Q + q_{R1})$ again gets divided in the ratio $\frac{R}{r}$. Therefore, charge on S_2 after second contact is

$$\begin{aligned}q_{R2} &= (Q + q_{R1}) \left(\frac{R}{R+r} \right) \\ &= Q \left[1 + \frac{R}{R+r} \right] \left(\frac{R}{R+r} \right) \\ &= Q [\eta + \eta^2], \quad \text{where } \eta = \frac{R}{R+r}\end{aligned}$$

After n contacts

$$\begin{aligned}q_{Rn} &= Q [\eta + \eta^2 + \eta^3 + \dots + \eta^n] \\ &= Q \eta \left[\frac{1 - \eta^n}{1 - \eta} \right] = \frac{QR}{r} \left[1 - \left(\frac{R}{R+r} \right)^n \right]\end{aligned}$$

Energy of S_2 is

$$U_2 = \frac{(q_{Rn})^2}{8\pi\epsilon_0 R} = \frac{Q^2 R}{8\pi\epsilon_0 r^2} \left[1 - \left(\frac{R}{R+r} \right)^n \right]^2$$

- When $n \rightarrow \infty$, $\left(\frac{R}{R+r} \right)^n \rightarrow 0$
[Because $\frac{R}{R+r}$ is a fraction]

$$\therefore \text{After infinite contacts, } U_2 = \frac{Q^2 R}{8\pi\epsilon_0 r^2}$$

This answer is easy to guess as potential of S_2 will become equal to potential of S_1 with a charge Q on it.

$$V = k \frac{q}{R} = k \frac{Q}{r}$$

$$\Rightarrow q = \frac{QR}{r}$$

where q = charge on S_2 after infinite contacts.

$$\therefore U_2 = \frac{q^2}{8\pi\epsilon_0 R} = \frac{Q^2 R}{8\pi\epsilon_0 r^2}$$

Example 52 A non-conducting disc of radius a and uniform positive surface charge density σ is placed on ground with its axis vertical. A particle of mass m and positive charge q is dropped, along the axis of the disc, from a height H with zero initial velocity.

The particle has specific charge $\frac{q}{m} = \frac{4\epsilon_0 g}{\sigma}$.

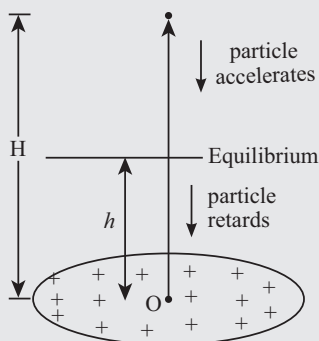
- Find the value of H if the particle just reaches the disc.

- (b) Sketch the potential energy of the particle as a function of its height and find its equilibrium position.

Solution**Concepts**

- (i) Potential on the axis of a uniformly charged disc is given by equation (14).
(ii) The particle has two kinds of potential energy-gravitational and electric.
(iii) Initially, mg is greater than electrostatic force on the particle and it begins to accelerate. Its kinetic energy increases and the total potential energy decreases. At a certain point, the electric force becomes equal to mg . The particle continues to fall and now the electric force exceeds mg . The particle retards and stops just before reaching the disc.

- (iv) After equilibrium position, the kinetic energy decreases and the potential energy increases. Equilibrium position can be found by using $\frac{dU}{dx} = 0$.



- (a) Potential due to the disc at height H is

$$V_1 = \frac{\sigma}{2\epsilon_0} \left[\sqrt{a^2 + H^2} - H \right]$$

Potential at the centre of the disc is

$$V_0 = \frac{\sigma a}{2\epsilon_0}$$

Applying conservation of energy for the particle:

$$K_H + U_{eH} + U_{gH} = K_O + U_{eO} + U_{gO}$$

$$\Rightarrow 0 + qV_1 + mgH = 0 + qV_0 + 0$$

$$\Rightarrow mgH = q[V_0 - V_1]$$

$$\Rightarrow gH = \frac{q}{m} \left[\frac{\sigma a}{2\epsilon_0} - \frac{\sigma}{2\epsilon_0} (\sqrt{a^2 + H^2} - H) \right]$$

$$\Rightarrow gH = \frac{4\epsilon_0 g}{\sigma} \frac{\sigma}{2\epsilon_0} \left[a - \sqrt{a^2 + H^2} + H \right]$$

$$\Rightarrow H = 2a + 2H - 2\sqrt{a^2 + H^2}$$

$$\Rightarrow 2\sqrt{a^2 + H^2} = H + 2a$$

$$\Rightarrow 4a^2 + 4H^2 = H^2 + 4a^2 + 4aH$$

$$\Rightarrow H = \frac{4a}{3}$$

- (b) The equilibrium position can be identified with the help of potential energy function.

Potential energy of the particle at a height x from the disc is

$$U = mgx + qV$$

$$= mgx + \frac{q\sigma}{2\epsilon_0} \left[\sqrt{a^2 + x^2} - x \right] \quad (i)$$

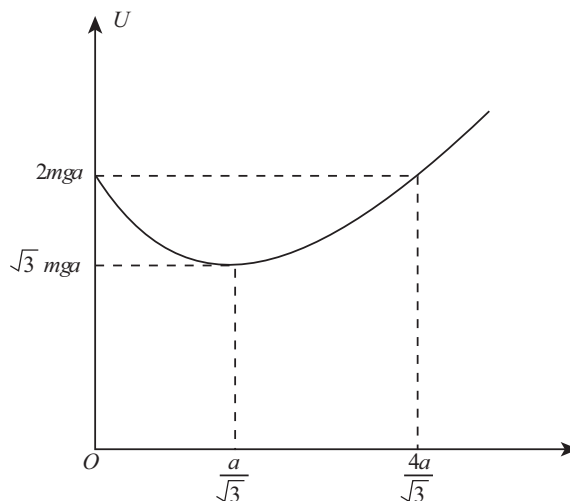
At equilibrium, $\frac{dU}{dx} = 0$

$$\Rightarrow mg + \frac{q\sigma}{2\epsilon_0} \left[\frac{x}{\sqrt{a^2 + x^2}} - 1 \right] = 0$$

$$\Rightarrow \frac{1}{2} + \frac{x}{\sqrt{a^2 + x^2}} - 1 = 0 \quad \left[\because mg = \frac{q\sigma}{4\epsilon_0} \right]$$

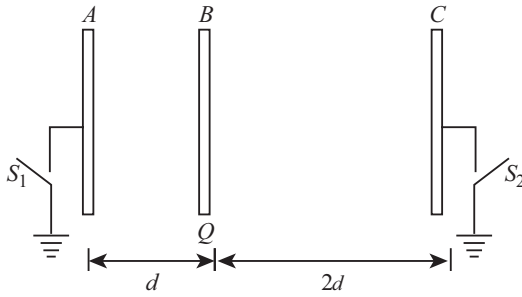
$$\Rightarrow x = \frac{a}{\sqrt{3}}$$

Value of potential energy at height H , $\frac{a}{\sqrt{3}}$ and 0 can be obtained by putting the values of x in equation (i). The graph of U versus x looks as shown below.

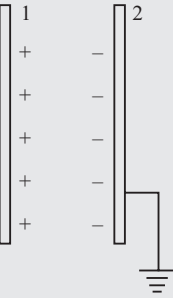
**Example 53** Three parallel plates

Three identical large metal plates, A , B and C are kept parallel to each other as shown. The middle plate B is given a charge Q .

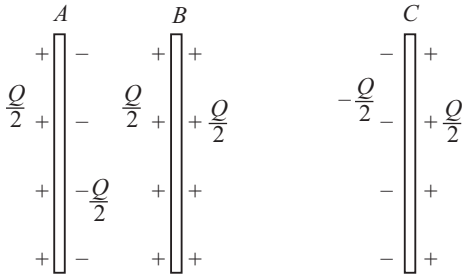
- (a) Show the charge distribution on the surfaces of each plate.
(b) Switch S_1 is closed. Now show the charge distribution on all surfaces.
(c) Switch S_1 and S_2 both are closed. Find the charge on plate A .

**Solution****Concepts**

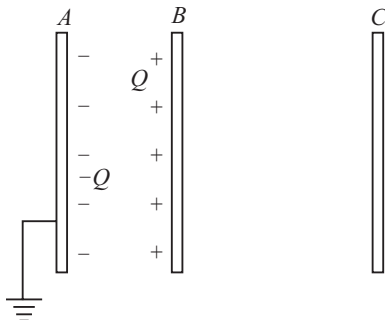
- (i) For a system of two parallel plates, with one grounded, the outer walls cannot have any charge. If there is charge on outer wall of 2, it will imply that there is an electric field to the right of plate 2. But there should not be any such field as potential of 2 is zero.
- (ii) In the system of two plates the inner walls must have equal and opposite charges. This is necessary to ensure that electric field is zero inside the metal.
- (iii) With S_1 and S_2 both closed, the potential difference $V_B - V_A = V_B - V_C$. This is because both $V_A = V_C = 0$.



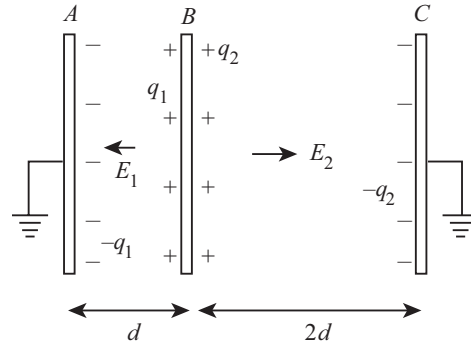
- (a) In this case, the charge distribution is as shown below. The separation between the plates is irrelevant as long as the dimensions of the plates are large compared to the separation.



- (b) With S_1 closed, there cannot be charge on the outer wall of A. The charge distribution is as shown.



- (c) With S_1 and S_2 both closed, let the charge distribution be as shown.



$$E_1 = \frac{q_1}{A\epsilon_0} \quad \text{and} \quad E_2 = \frac{q_2}{A\epsilon_0}$$

$$V_B - V_A = E_1 d \quad \text{and} \quad V_B - V_C = E_2 (2d)$$

$$\Rightarrow E_1 d = E_2 (2d)$$

$$\Rightarrow \frac{q_1 d}{A\epsilon_0} = \frac{q_2 (2d)}{A\epsilon_0}$$

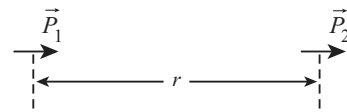
$$\Rightarrow \frac{q_1}{q_2} = 2$$

$$\text{Also, } q_1 + q_2 = Q$$

$$\therefore q_1 = \frac{2Q}{3}$$

Example 54 Force between two dipoles

Two short electric dipoles are kept along a straight line as shown. Separation between them is r . Their dipole moments are P_1 and P_2 . Find the force applied by one dipole on the other.

**Solution****Concepts**

- (i) Field due to a dipole on axial position is given by

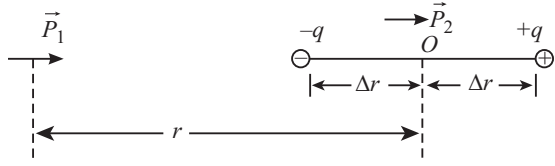
$$E = \frac{1}{4\pi\epsilon_0} \frac{2p}{r^3}$$

- (ii) Change in field, when r is changed by a small amount, can be obtained as:

$$\Delta E \simeq \left(\frac{dE}{dr} \right) \Delta r$$

Field due to first dipole at the centre (O) of the other dipole is

$$E = \frac{1}{4\pi\epsilon_0} \frac{2p_1}{r^3}$$



Change in field when r is made to change by Δr is

$$\Delta E = \left(\frac{dE}{dr} \right) \Delta r = \frac{1}{4\pi\epsilon_0} \left(\frac{-6P_1}{r^4} \right) \Delta r$$

We consider the second dipole to be made of $-q$ and $+q$ charge kept at a separation $2\Delta r$.

Field at position of $-q$ is $E_1 = E + |\Delta E|$

Field at position of $+q$ is $E_2 = E - |\Delta E|$

Force on the second dipole is

$$F = qE_1 - qE_2 (\leftarrow)$$

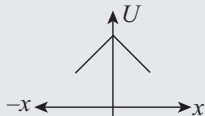
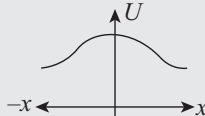
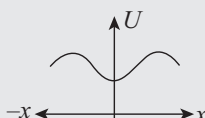
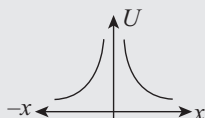
$$\Rightarrow F = 2q|\Delta E|$$

$$= \frac{6p_1(2q\Delta r)}{4\pi\epsilon_0 r^4}$$

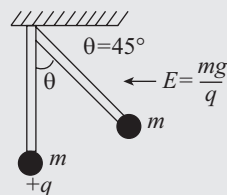
$$= \frac{6P_1P_2}{4\pi\epsilon_0 r^4}$$

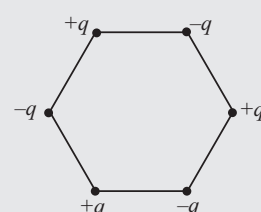
Note that by considering only the magnitude of ΔE (i.e., $|\Delta E|$), we have done away with keeping track of sign. It is obvious that the field at $-q$ is higher than at O and is lower at $+q$ than at O .

Worksheet 1

- Which of the following is same as volt:
 - Erg per cm
 - joule per coulomb
 - Erg per ampere
 - Newton/(coulomb $\times$ m²)
- An infinite nonconducting sheet of charge has a surface charge density of 10^{-7} C/m². The separation between two equipotential surfaces near the sheet whose potential differ by 5V is
 - 0.88 cm
 - 0.88 mm
 - 0.88 m
 - 5×10^{-7} m
- Four equal charges $+q$ are placed at four corners of a square in yz plane which has its centre at the origin. The electrostatic potential energy of a fifth charge $+Q$ varies on x -axis as:
 - 
 - 
 - 
 - 
- Two identical thin rings, each of radius R are coaxially placed at distance R apart. Q_1 and Q_2 coulomb are respectively the charges uniformly spread on the two rings. The work done in moving a charge q from the centre of the first ring to that of the other is
 - zero
 - $(Q_2 - Q_1)(\sqrt{2} - 1) / (\sqrt{2} \cdot 4\pi\epsilon_0 R)$
 - $q\sqrt{2}(Q_1 + Q_2) / 4\pi\epsilon_0 R$
 - $q(Q_1 - Q_2)(\sqrt{2} + 1) / (\sqrt{2} \cdot 4\pi\epsilon_0 R)$
- Two positively charged particles, X and Y , have equal mass but unequal charges. They are free to move and are initially far away from each other. With Y at rest, X begins to move towards it with initial velocity u . After a long time
 - X will stop, Y will move with velocity u .
 - both move with velocities $u/2$ each.
 - X will stop, Y will move with velocity $< u$.
 - both will move with velocities $< u/2$.
- In a space a uniform horizontal electric field exist as shown in figure. Strength of the field is $E = (mg)/q$. A light rod of length l is free to rotate in vertical plane about its hinged upper end. A particle of mass m is attached at the lower end of the rod. The particle is

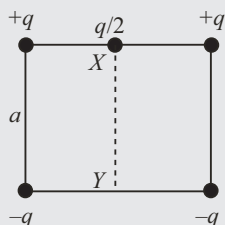
released from the position shown in figure. Find the angular velocity of the rod when it becomes vertical



- $\sqrt{\frac{g}{l}}$
 - $\sqrt{\frac{2g}{l}}$
 - $\sqrt{\frac{3g}{l}}$
 - $\sqrt{\frac{5g}{l}}$
- Two identical particles of mass m carry a charge Q each. Initially one is at rest on a smooth horizontal plane and the other is projected along the plane directly towards first particle from a large distance with speed v . The closest distance of approach is
 - $\frac{1}{4\pi\epsilon_0} \frac{Q^2}{mv}$
 - $\frac{1}{4\pi\epsilon_0} \frac{4Q^2}{mv^2}$
 - $\frac{1}{4\pi\epsilon_0} \frac{2Q^2}{mv^2}$
 - $\frac{1}{4\pi\epsilon_0} \frac{3Q^2}{mv^2}$
 - The diagram shows a small bead of mass m carrying charge q . The bead can freely move on a smooth fixed ring placed on a smooth horizontal plane. In the same plane a charge $+Q$ has also been fixed as shown. The potential at the point P due to $+Q$ is V . The velocity with which the bead should be projected from the point P so that it can complete a circle should be greater than
 - $\sqrt{\frac{6qV}{m}}$
 - $\sqrt{\frac{qV}{m}}$
 - $\sqrt{\frac{3qV}{m}}$
 - none
 - Six charges of magnitude $+q$ and $-q$ are fixed at the corners of a regular hexagon of edge length a as shown in the figure. The electrostatic interaction energy of the charged particles is:
 

- (a) $\frac{q^2}{\pi\epsilon_0 a} \left[\frac{\sqrt{3}}{8} - \frac{15}{4} \right]$ (b) $\frac{q^2}{\pi\epsilon_0 a} \left[\frac{\sqrt{3}}{2} - \frac{9}{4} \right]$
 (c) $\frac{q^2}{\pi\epsilon_0 a} \left[\frac{\sqrt{3}}{4} - \frac{15}{2} \right]$ (d) $\frac{q^2}{\pi\epsilon_0 a} \left[\frac{\sqrt{3}}{2} - \frac{15}{8} \right]$

10. Four point charges are placed at the vertices of a square. X and Y are mid points of the opposite sides of the square. Work done in shifting a charge $q/2$ from a point X to a point Y in the diagram shown in figure is:



- (a) $\frac{8Kq^2}{a} \left[\frac{\sqrt{5}-1}{\sqrt{5}} \right]$ (b) $\frac{4Kq^2}{a} \left[\frac{1-\sqrt{5}}{\sqrt{5}} \right]$
 (c) $\frac{4Kq^2}{a} \left[\frac{\sqrt{5}-1}{\sqrt{5}} \right]$ (d) $\frac{4Kq^2}{a} \left[\frac{1-\sqrt{5}}{\sqrt{5}} \right]$

11. A charged particle of charge Q is held fixed and another charged particle of mass m and charge q (of the same sign) is released from a distance r . The impulse of the force exerted by the external agent on the fixed charge by the time distance between Q and q becomes $2r$ is

- (a) $\sqrt{\frac{Qq}{4\pi\epsilon_0 mr}}$ (b) $\sqrt{\frac{Qqm}{4\pi\epsilon_0 r}}$
 (c) $\sqrt{\frac{Qqm}{\pi\epsilon_0 r}}$ (d) $\sqrt{\frac{Qqm}{2\pi\epsilon_0 r}}$

12. In a uniform electric field, the potential is 10V at the origin of coordinates, and 8V at each of the points (1, 0, 0), (0, 1, 0) and (0, 0, 1). The potential at the point (1, 1, 1) will be

- (a) 0 (b) 4 V
 (c) 8 V (d) 10 V

13. In a regular polygon of n sides, each corner is at a distance r from the centre. Identical charges are placed at $(n-1)$ corners. At the centre, the magnitude of electric field intensity is E and the magnitude of potential is V . The ratio V/E has magnitude.

- (a) $r n$ (b) $r(n-1)$
 (c) $(n-1)/r$ (d) $r(n-1)/n$

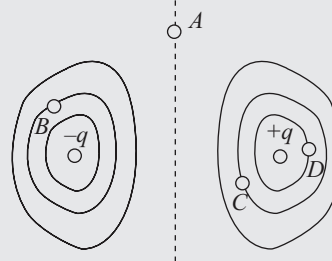
14. The equation of an equipotential line in an electric field is $y = 2x$, then the electric field strength vector at (1, 2) may be

- (a) $4\hat{i} + 3\hat{j}$ (b) $4\hat{i} + 8\hat{j}$
 (c) $8\hat{i} + 4\hat{j}$ (d) $-8\hat{i} + 4\hat{j}$

15. A charge 3 coulomb experiences a force 3000 N when placed in a uniform electric field. The potential difference between two points separated by a distance of 1 cm along the field lines is

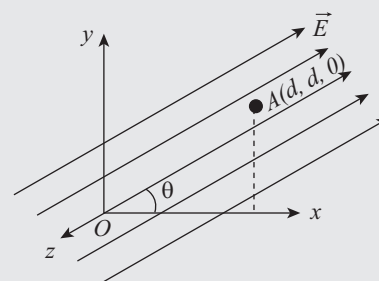
- (a) 10 V (b) 90 V
 (c) 1000 V (d) 9000V

16. Figure shows equi-potential surfaces for a two charge system. At which of the labeled points will an electron have the highest potential energy?



- (a) Point A (b) Point B
 (c) Point C (d) Point D

17. A uniform electric field having strength $\vec{E}$ is existing in x - y plane as shown in figure. Find the potential difference between the origin O and point $A(d, d, 0)$



- (a) $Ed(\cos\theta + \sin\theta)$ (b) $-Ed(\sin\theta - \cos\theta)$
 (c) $\sqrt{2} Ed$ (d) none of these

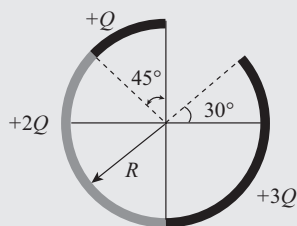
18. In a certain region of space, the potential is given by: $V = k[2x^2 - y^2 + z^2]$. The electric field at the point (1, 1, 1) has magnitude =

- (a) $k\sqrt{6}$ (b) $2k\sqrt{6}$
 (c) $2k\sqrt{3}$ (d) $4k\sqrt{3}$

19. Uniform electric field of magnitude 100 V/m in space is directed along the line $y = 3 + x$. Find the potential difference between points $A(3, 1)$ & $B(1, 3)$

- (a) 100 V (b) $200\sqrt{2}$ V
 (c) 200 V (d) 0

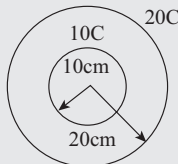
20. Figure shows three circular arcs, each of radius R and total charge as indicated. The net electric potential at the centre of curvature is:



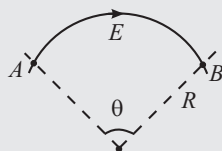
- (a) $\frac{Q}{2\pi\epsilon_0 R}$ (b) $\frac{Q}{4\pi\epsilon_0 R}$
 (c) $\frac{2Q}{\pi\epsilon_0 R}$ (d) $\frac{Q}{\pi\epsilon_0 R}$
21. Electric field in a region is given by $E_x = -2x^3$ N/C. Potential of the point (2, 4) is taken as zero. The potential at the point (1, -2) is

- (a) -7.5 V (b) 7.5×10^3 V
 (c) -15 V (d) 15 V

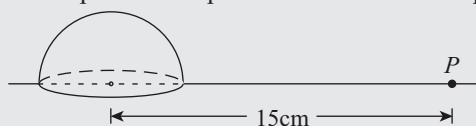
22. Two concentric uniformly charged spheres of radius 10 cm & 20 cm are arranged as shown in the figure. Potential difference between the spheres is:



- (a) 4.5×10^{11} V (b) 2.7×10^{11} V
 (c) 0 (d) none of these
23. Electric field exist in a region of space. Assume that a field line is in the shape of a circular arc. The magnitude of electric field intensity is same at all points on this curve and is equal to E . If the potential at A is V , then the potential at B is:



- (a) $V - ER\theta$ (b) $V - 2ER \sin \frac{\theta}{2}$
 (c) $V + ER\theta$ (d) $V + 2ER \sin \frac{\theta}{2}$
24. Figure shows a solid hemisphere with a charge of 5 nC distributed uniformly throughout its volume. P is a point located in the plane of the base of the hemisphere. The point is at a distance of 15 cm from the the centre of curvature of the hemisphere. The electric potential at point P due to the hemisphere, is:



- (a) 150 V (b) 300 V
 (c) 450 V (d) 600 V

25. A and B are two points on the axis and the perpendicular bisector, respectively of an electric dipole. A and B are far away from the dipole and at equal distance from it. The field at A and B are $\vec{E}_A$ and $\vec{E}_B$.

- (a) $\vec{E}_A = \vec{E}_B$
 (b) $\vec{E}_A = 2\vec{E}_B$
 (c) $\vec{E}_A = -2\vec{E}_B$
 (d) $|E_B| = \frac{1}{2}|E_A|$, and $\vec{E}_B$ is perpendicular to $\vec{E}_A$

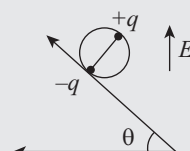
26. Two small dipoles of dipole moments $p\hat{k}$ and $\frac{p}{2}\hat{k}$ are located at (0, 0, 0) & (1 m, 0, 2 m) respectively. The resultant electric field due to the two dipoles at the point (1 m, 0, 0) is:

- (A) $\frac{9p}{32\pi\epsilon_0}\hat{k}$ (B*) $\frac{-7p}{32\pi\epsilon_0}\hat{k}$
 (C) $\frac{7p}{32\pi\epsilon_0}\hat{k}$ (d) none of these

27. A dipole of dipole moment p is kept at the centre of a ring of radius R and charge Q . The dipole moment has direction along the axis of the ring. The resultant force on the ring due to the dipole is:

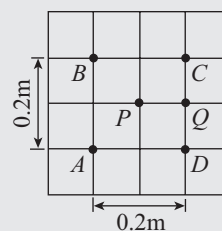
- (a) zero
 (b) in a direction parallel to the dipole moment
 (c) in a direction opposite to the dipole moment
 (d) in a direction opposite to the dipole moment only if the charge is uniformly distributed on the ring.

28. A uniform wheel having mass m has charges $+q$ and $-q$ at diametrically opposite points. It remains in equilibrium on a rough inclined plane in the presence of uniform vertical electric field E as shown in the figure. The magnitude of E is



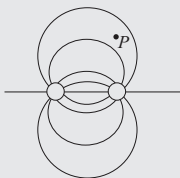
- (a) $\frac{mg}{q}$ (b) $\frac{mg}{2q}$
 (c) $\frac{mg \tan \theta}{2q}$ (d) none

29. A, B, C, D, P and Q are points in a uniform electric field. The potentials at these points are $V(A) = 2$ volt, $V(P) = V(B) = V(D) = 5$ volt, $V(C) = 8$ volt. The electric field at P is



- (a) 10 Vm^{-1} along PQ
 (b) $15\sqrt{2} \text{ V m}^{-1}$ along PA
 (c) 5 V m^{-1} along PC
 (d) 5 V m^{-1} along PA

30. Figure shows the electric field lines around an electric dipole. Which of the arrows best represents the electric field at point P ?



- (a) (b) (c) (d)

31. The dipole moment of a system of charge $+q$ distributed uniformly on an arc of radius R subtending an angle $\pi/2$ at its centre where another charge $-q$ is placed is:

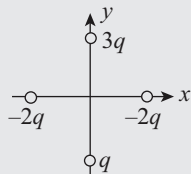
- (a) $\frac{2\sqrt{2}qR}{\pi}$ (b) $\frac{\sqrt{2}qR}{\pi}$
 (c) $\frac{qR}{\pi}$ (d) $\frac{2qR}{\pi}$

32. Point P lies on the axis of a dipole. If the dipole is rotated by 90° anticlockwise, the electric field vector $\vec{E}$ at P will rotate by

- (a) 90° clockwise (b) 180° clockwise
 (c) 90° anti clockwise (d) none

33. 4 charges are placed each at a distance ' a ' from the origin. The dipole moment of configuration is

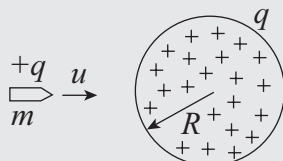
- (a) $2qa\hat{j}$
 (b) $3qa\hat{j}$
 (c) $2aq[\hat{i} + \hat{j}]$
 (d) none



34. A solid sphere of radius R is charged uniformly. At what distance from its surface is the electrostatic potential half of the potential at the centre?

- (a) R (b) $R/2$
 (c) $R/3$ (d) $2R$

35. A bullet of mass m and charge q is fired towards a solid uniformly charged sphere of radius R and total charge $+q$. If it strikes the surface of sphere with speed u , find the minimum speed u so that it can penetrate through the sphere.



(Neglect all resistance forces and friction acting on the bullet except electrostatic forces)

- (a) $\frac{q}{\sqrt{2\pi\epsilon_0 mR}}$ (b) $\frac{q}{\sqrt{4\pi\epsilon_0 mR}}$
 (c) $\frac{q}{\sqrt{8\pi\epsilon_0 mR}}$ (d) $\frac{\sqrt{3} q}{\sqrt{4\pi\epsilon_0 mR}}$

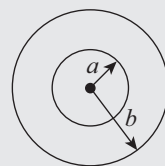
36. n small drops of same size are charged to V volts each. If they coalesce to form a single large drop, then its potential will be

- (a) V/n (b) Vn
 (c) $Vn^{1/3}$ (d) $Vn^{2/3}$

37. A hollow metal sphere of radius 5 cm is charged such that the potential on its surface is 10 V. The potential at the centre of the sphere is

- (a) 0 V
 (b) 10 V
 (c) same as at a point 5 cm away from the surface outside sphere.
 (d) same as at a point 25 cm away from the surface.

38. In the shown figure, the two concentric metal spheres have some charge on them. The electric potential of the inner metal sphere is 10 volt and that of the outer shell is 5 volt. The potential at the centre will be:



- (a) 10 volt (b) 5 volt
 (c) 15 volt (d) 0

39. Three concentric metallic spherical shells A , B and C of radii a , b and c ($a < b < c$) have surface charge densities $-\sigma$, $+\sigma$, and $-\sigma$ respectively. The potential of shell A is:

- (a) $(\sigma/\epsilon_0)[a + b - c]$ (b) $(\sigma/\epsilon_0)[a - b + c]$
 (c) $(\sigma/\epsilon_0)[b - a - c]$ (d) none

40. Two identical small conducting spheres, having charges of opposite sign, attract each other with a force of 0.108 N when separated by 0.5 m. The spheres are connected by a conducting wire, which is then removed, and thereafter, they repel each other with a force of 0.036 N. The initial charges on the spheres are

- (a) $\pm 5 \times 10^{-6} \text{ C}$ and $\mp 15 \times 10^{-6} \text{ C}$
 (b) $\pm 1.0 \times 10^{-6} \text{ C}$ and $\mp 3.0 \times 10^{-6} \text{ C}$
 (c) $\pm 2.0 \times 10^{-6} \text{ C}$ and $\mp 6.0 \times 10^{-6} \text{ C}$
 (d) $\pm 0.5 \times 10^{-6} \text{ C}$ and $\mp 1.5 \times 10^{-6} \text{ C}$

41. A non-conducting ring of radius 0.5 m carries a total charge of $1.111 \times 10^{-10} \text{ C}$ distributed non-uniformly on its circumference producing an electric field $\vec{E}$ everywhere in space. The value of the line integral

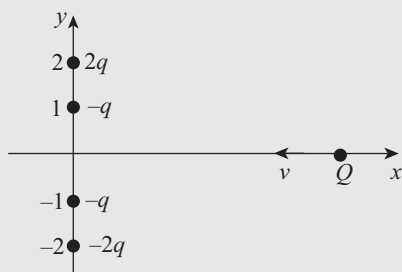
$$\int_{\ell=0}^{\ell=\infty} -\vec{E} \cdot d\vec{\ell} \quad (\ell=0 \text{ being centre of the ring}) \text{ in volts is:}$$

- (a) +2 (b) -1
(c) -2 (d) zero

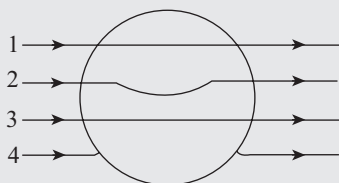
42. There are four concentric shells A , B , C and D of radii a , $2a$, $3a$ and $4a$ respectively. Shells B and D are given charges $+q$ and $-q$ respectively. Shell C is now earthed. The potential difference $V_A - V_C$ is:

- (a) $\frac{Kq}{2a}$ (b) $\frac{Kq}{3a}$
(c) $\frac{Kq}{4a}$ (d) $\frac{Kq}{6a}$

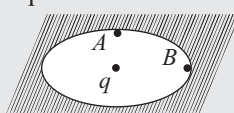
43. Four charges are rigidly fixed along the Y axis as shown. A positive charge approaches the system along the X axis with initial speed just enough to cross the origin. Then its total energy at the origin is



- (a) zero (b) positive
(c) negative (d) data insufficient
44. A number of spherical conductors of different radii have same potential. Then the surface charge density on them
- (a) is proportional to their radii
(b) is inversely proportional to their radii
(c) are equal
(d) is proportional to square of their radii
45. A metallic solid sphere is placed in a uniform electric field. An electric line of force is most appropriately represented by (see figure)



- (a) 1 (b) 2
(c) 3 (d) 4
46. An ellipsoidal cavity is carved within a perfect conductor. A positive charge q is placed at the center of the cavity. The points A & B are on the cavity surface as shown in the figure. Then:

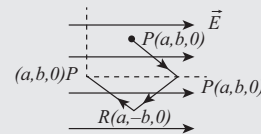


- (a) electric field near A in the cavity = electric field near B in the cavity
(b) charge density at A = charge density at B
(c) potential at A = potential at B
(d) total electric flux through the surface of the cavity is q/ϵ_0 .

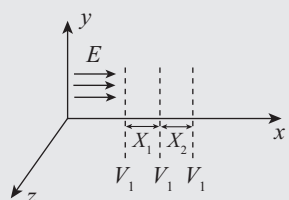
47. A non uniformly charged ring is kept near an uncharged conducting solid sphere. The distance between their centres (which are on the same line normal to the plane of the ring) is 3m and their radius is 4m . If total charge on the ring is $1\mu\text{C}$ then the potential of the sphere will be:

- (a) 2.25 kV (b) 3 kV
(c) 1.8 kV (d) cannot be determined

48. A point charge q moves from point P to point S along the path $PQRS$ in a uniform electric field $\vec{E}$ pointing parallel to the positive direction of the x -axis. The coordinates of the points P , Q , R and S are $(a, b, 0)$, $(2a, 0, 0)$, $(a, -b, 0)$ and $(0, 0, 0)$ respectively. The work done by the field in the above process is given by the expression

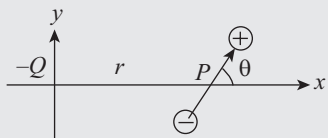


- (a) qaE (b) $-qaE$
(c) $q(\sqrt{a^2+b^2})E$ (d) $3qE\sqrt{a^2+b^2}$
49. In an electric field the potential at a point is given by $V = \frac{343}{r}$. Here, r is distance of the point from the origin. The electric field at $\vec{r} = 3\hat{i} + 2\hat{j} + 6\hat{k}$ is:
- (a) $21\hat{i} + 14\hat{j} + 42\hat{k}$ (b) $3\hat{i} + 2\hat{j} + 6\hat{k}$
(c) $\frac{1}{7}(3\hat{i} + 2\hat{j} + 6\hat{k})$ (d) $-(3\hat{i} + 2\hat{j} + 6\hat{k})$
50. An electron travelling in an uniform electric field passes from an equipotential surface of potential V_1 to another equipotential surface of higher potential V_2 . Then
- (a) No change takes place in velocity component parallel to the first equipotential surface.
(b) Direction of its motion remains unchanged but speed increases.
(c) Direction of its motion may change but speed must decrease.
(d) Decrease in kinetic energy is proportional to $\sqrt{V_2 - V_1}$.
51. In an electric field shown in figure three equipotential surfaces are shown. If the variation of electric field is $E = 2x^2\text{V/m}$, and given that $V_1 - V_2 = V_2 - V_3$, then we have



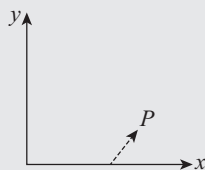
- (a) $x_1 = x_2$ (b) $x_1 > x_2$
 (c) $x_2 > x_1$ (d) data insufficient

52. A point negative charge $-Q$ is placed at a distance r from a dipole with dipole moment P as shown in figure. The x component of force acting on the charge $-Q$ is



- (a) $-\frac{PKQ}{r} \cos \theta$ (b) $\frac{PKQ}{r} \cos \theta$
 (c) $-\frac{2PKQ}{r^3} \cos \theta$ (d) $\frac{2PKQ}{r^3} \cos \theta$

53. A small electric dipole $\vec{P}$ is placed on the x -axis at the point $(1, 0)$. The dipole vector forms an angle of 30° with the x -axis. A non uniform electric field has been applied in the region given by the vector $\vec{E} = x^2\hat{i} + y^2\hat{j}$. The force acting on the dipole is most appropriately given by:

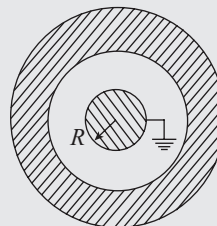


- (a) $2\vec{P} \cos 30^\circ (\hat{i} + 2\hat{j})$ (b) $2\vec{P} \cos 30^\circ (\hat{i})$
 (c) $2\vec{P} \cos 30^\circ (2\hat{j})$ (d) None
54. An electrically isolated hollow (initially uncharged), conducting sphere has a small positively charged ball suspended by an insulating rod from its inside surface, (see diagram). This causes the inner surface of the sphere to become negatively charged. When the ball is placed at the centre inside the sphere the electric field outside the conducting sphere is approximately

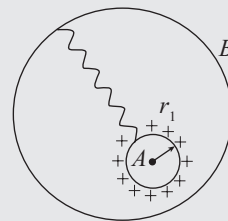


- (a) zero
 (b) the same as if the sphere wasn't there
 (c) twice what it would be if the sphere wasn't there
 (d) equal in magnitude but opposite in direction to what it would be if the sphere wasn't there

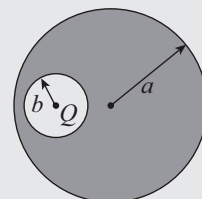
55. A conducting sphere of radius R and a concentric thick spherical shell of inner radius $2R$ and outer radius $3R$ is shown in figure. A charge $+10Q$ is given to the shell and inner sphere is earthed. Then charge on the inner sphere is



- (a) $-4Q$ (b) $-10Q$
 (c) zero (d) none
56. A metal sphere A of radius r_1 charged to a potential ϕ_1 is enveloped by a thin walled conducting spherical shell B of radius r_2 . The potential ϕ_2 of the sphere A after it is connected by a thin wire to the shell B will be

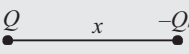


- (a) $\phi_1 \frac{r_1}{r_2}$ (b) $\phi_1 \left(\frac{r_2}{r_1} \right)$
 (c) $\phi_1 \left(1 - \frac{r_1}{r_2} \right)$ (d) $\phi_1 \left(\frac{r_1 r_2}{r_1 + r_2} \right)$
57. A point charge ' Q ' is placed at the centre of a spherical cavity of radius ' b ' carved inside a solid conducting sphere of radius ' a '. Then total energy of the system is: $\left[k = \frac{1}{4\pi\epsilon_0} \right]$



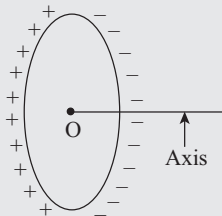
- (a) $\frac{kQ^2}{2a} - \frac{kQ^2}{2b}$ (b) $\frac{kQ^2}{2a} + \frac{kQ^2}{2b}$
 (c) $\frac{kQ^2}{a} + \frac{kQ^2}{b}$ (d) $\frac{kQ^2}{a} - \frac{kQ^2}{b}$

Worksheet 2

- Which of the followings are true?
 - When a negative charge is released in an electric field, it moves toward a position of lower electric potential and higher potential energy
 - When a positive charge is released in an electric field, it moves toward a position of lower electric potential and higher potential energy
 - When a negative charge is released in an electric field, it moves toward a position of higher electric potential and lower potential energy
 - When a positive charge is released in an electric field, it moves toward a position of lower electric potential and lower potential energy
- Two positively charged particles X and Y are initially far away from each other and at rest. X begins to move towards Y with some initial velocity. The total momentum and energy of the system are p and E .
 - If Y is fixed, both p and E are conserved.
 - If Y is fixed, E is conserved, but not p .
 - If both are free to move, p is conserved but not E .
 - If both are free, E is conserved, but not p .
- A spherical shell with an inner radius ' a ' and an outer radius ' b ' is made of conducting material. A point charge $+Q$ is placed at the centre of the spherical shell and a total charge $-q$ is given to the shell.
 - Charge on the inner and outer surfaces of the shell are $-Q$ and $-q$, respectively.
 - Charge on the inner and outer surfaces of the shell are $-Q$ and $-q + Q$, respectively.
 - The electrostatic potential at a distance R ($a < R < b$) from the centre of the shell is $K \frac{Q-q}{b}$ (where $K = \frac{1}{4\pi\epsilon_0}$)
 - No force is needed to hold the point charge $+Q$ at the centre, but a force will be needed if we wish to keep it at any other point inside the cavity.
- Two charged particles having charges Q and $-Q/4$ are separated by a distance x . Which of the followings are correct in this regard
 
 - potential is zero at a point on the axis which is at a distance $x/3$ on the right side of the charge $-Q/4$
 - potential is zero at a point on the axis which is $x/5$ on the left side of the charge $-Q/4$
 - electric field is zero at a point on the axis which is at a distance x on the right side of the charge $-Q/4$
 - the two charges are projected directly away from one another at a relative speed v . When the separation between them again becomes x , the relative speed between them is higher than v .
- An electric charge 10^{-8} C is placed at the point (4m, 7m, 2m). At the point (1m, 3m, 2m), the electric
 - potential will be 18 V
 - field has no Y-component
 - field will be along Z-axis
 - potential will be 1.8 V
- Three point charges Q , $4Q$ and $16Q$ are placed on a straight line 9 cm long. Charges are placed in such a way that the system has minimum potential energy. Then
 - $4Q$ and $16Q$ must be at the ends and Q at a distance of 3 cm from the $16Q$.
 - $4Q$ and $16Q$ must be at the ends and Q at a distance of 6 cm from the $16Q$.
 - Electric field at the position of Q is zero.
 - Electric field at the position of Q is $\frac{Q}{4\pi\epsilon_0}$.
- Potential at a point A is 3 volt and at a point B is 7 volt, an electron is moving towards A from B .
 - It must have some K.E. at B to reach A
 - It need not have any K.E. at B to reach A
 - to reach A it must have more than or equal to 4 eV K.E. at B .
 - when it will reach A , it will have K.E. more than or at least equal to 4 eV if it was released from rest at B .
- A conducting sphere of radius r has a charge. Then
 - The charge is uniformly distributed over its surface, if there is an external electric field.
 - Distribution of charge over its surface will be non uniform if no external electric field exist in space.

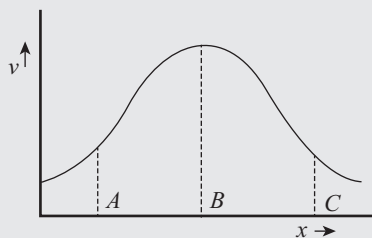
- (c) Electric field strength inside the sphere will be equal to zero only when no external electric field exists.
- (d) Potential at every point of the sphere must be same.

9. The figure shows a nonconducting ring which has positive and negative charge non uniformly distributed on it such that the total charge is zero.



Which of the following statements is/are true?

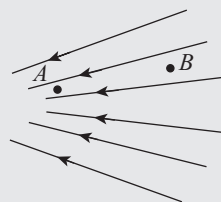
- (a) The potential at all the points on the axis will be zero.
- (b) The electric field at all the points on the axis will be zero.
- (c) The direction of electric field at all points on the axis will be along the axis.
- (d) If the ring is placed inside a uniform external electric field then net force acting on the ring would be zero.
10. At distance of 5 cm and 10 cm outwards from the surface of a uniformly charged solid sphere, the potentials are 100 V and 75 V respectively. Then
- (a) potential at its surface is 150 V.
- (b) the charge on the sphere is $(5/3) \times 10^{-10} \text{ C}$.
- (c) the electric field on the surface is 1500 V/m.
- (d) the electric potential at its centre is 225 V.
11. Four identical charges are placed at the points (1, 0, 0), (0, 1, 0), (-1, 0, 0) and (0, -1, 0).
- (a) The potential at the origin is zero.
- (b) The field at the origin is zero.
- (c) The potential at all points on the z-axis, other than the origin, is zero.
- (d) The field at all points on the z-axis, other than the origin acts along the z-axis.
12. Variation of electrostatic potential along x-direction is shown in the graph. The correct statement about electric field is



- (a) x component at point B is maximum
- (b) x component at point A is towards positive x-axis.

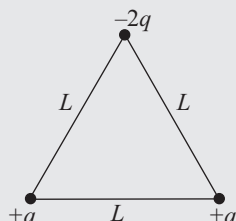
- (c) x component at point C is along negative x-axis
- (d) x component at point C is along positive x-axis

13. A particle of charge $1 \mu\text{C}$ & mass 1 g moving with a velocity of 4 m/s is subjected to a uniform electric field of magnitude 300 V/m for 10 sec. Then its final speed cannot be:
- (a) 0.5 m/s (b) 4 m/s
- (c) 3 m/s (d) 6 m/s
14. A proton and a deuteron are initially at rest and are accelerated through the same potential difference. Which of the following are true concerning the final properties of the two particles?
- (a) They have different speeds
- (b) They have same momentum
- (c) They have same kinetic energy
- (d) none of these
15. Particle A having positive charge is moving directly head-on towards initially stationary positively charged particle B. At the instant when A and B are closest together
- (a) the momenta of A and B must be equal
- (b) the velocities of A and B must be equal
- (c) B would have gained less kinetic energy than A would have lost.
- (d) B would have gained the same momentum as A would have lost
16. Which of the following is true for the figure showing electric lines of force? (E is electrical field, V is potential)



- (a) $E_A > E_B$ (b) $E_B > E_A$
- (c) $V_A > V_B$ (d) $V_B > V_A$
17. If we use permittivity ϵ , resistance R , gravitational constant G and voltage V as fundamental physical quantities, then
- (a) [angular displacement] = $\epsilon^0 R^0 G^0 V^0$
- (b) [Velocity] = $\epsilon^{-1} R^{-1} G^0 V^0$
- (c) [dipole moment] = $\epsilon^1 R^0 G^0 V^1$
- (d) [force] = $\epsilon^1 R^0 G^0 V^2$
18. Units of electric flux are
- (a) $\frac{\text{Nm}^2}{\text{C}^2}$ (b) $\frac{\text{N}}{\text{C}^2 \text{m}^2}$
- (c) volt-m (d) Volt- m^3

19. Three point charges are placed at the corners of an equilateral triangle of side L as shown in the figure.



- (a) The potential at the centroid of the triangle is zero.
 (b) The electric field at the centroid of the triangle is zero.
 (c) The dipole moment of the system is $\sqrt{2} qL$
 (d) The dipole moment of the system is $\sqrt{3} qL$.

20. An electric dipole is placed at the centre of a sphere. Mark the correct answer

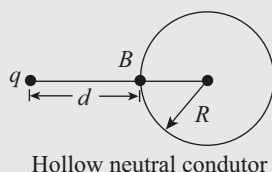
- (a) the flux of the electric field through the sphere is zero
 (b) the electric field is zero at every point of the sphere.
 (c) the electric potential is zero everywhere on the sphere.
 (d) the electric potential is zero on a circle on the surface.

21. For the situation shown in the figure below (assume $r \gg$ length of dipole) mark out the correct statement(s).



- (a) Force acting on the dipole is zero
 (b) Force acting on the dipole is approximately $\frac{pQ}{4\pi\epsilon_0 r^3}$ and is acting upward
 (c) Torque acting on the dipole is $\frac{pQ}{4\pi\epsilon_0 r^2}$ in clockwise direction
 (d) Torque acting on the dipole is $\frac{pQ}{4\pi\epsilon_0 r^2}$ in anti-clockwise direction

22. For the situation shown in the figure below, mark out the correct statement(s)

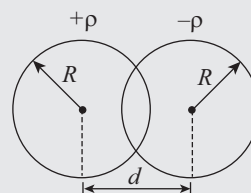


Hollow neutral conductor

- (a) Potential of the conductor is $\frac{q}{4\pi\epsilon_0(d+R)}$
 (b) Potential of the conductor is $\frac{q}{4\pi\epsilon_0 d}$

- (c) Potential of the conductor can't be determined as nature of distribution of induced charges is not known
 (d) Potential at point B due to induced charges is $\frac{-qR}{4\pi\epsilon_0(d+R)d}$

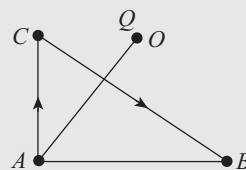
23. There are two non-conducting spheres having uniform volume charge densities ρ and $-\rho$. Both spheres have equal radius R . The spheres are now laid down such that they overlap as shown in the figure.



- (A*) The electric field $\vec{E}$ in the overlap region is $\frac{\rho}{3\epsilon_0} \vec{d}$

- (b) The potential difference ΔV between the centers of the two spheres for $d = R$ is $\frac{\rho}{3\epsilon_0} d^2$
 (c) The potential difference ΔV between the centers of the two spheres for $d = R$ is $\frac{\rho}{\epsilon_0} d^2$ non uniform
 (d) None

24. There is a fixed positive charge Q at O and A and B are points equidistant from O . A positive charge $+q$ is taken slowly by an external agent from A to B along the line AC and then along the line CB .



- (a) The total work done on the charge is zero
 (b) The work done by the electrostatic force from A to C is negative
 (c) The work done by the electrostatic force from C to B is positive
 (d) The work done by electrostatic force in taking the charge from A to B is dependent on the actual path

25. You are moving a negative charge q at a small constant speed away from a uniformly charged non-conducting spherical shell on which resides a negative charge Q . The electrostatic field due to Q is E . Let U be the total energy of the system, W_a be the work done by the force (F_a) you exert on q and

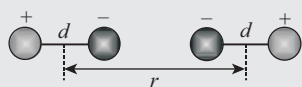
W_E the work done by the electrostatic force (F_E) on q . Then, as q is being moved,

- (a) $W_a = -W_E$, therefore U remains constant
- (b) $F_a = -F_E$
- (c) U increases
- (d) U decreases

26. Two large conducting sheets are kept parallel to each other as shown. Plate 1 is given a charge Q_1 and the plate 2 is given a charge Q_2 . Magnitude of Q_1 is greater than the magnitude of Q_2 . The potential difference $V_1 - V_2$ between the two plates is

- (a) positive if both Q_1 and Q_2 are positive
- (b) positive if both Q_1 and Q_2 are negative
- (c) positive if Q_1 is positive and Q_2 is negative
- (d) none of these

27. We have two electric dipoles. Each dipole consists of two equal and opposite point charges at the ends of an insulating rod of length d . The dipoles sit along the x -axis a distance r apart ($r \gg d$), oriented as shown in figure. The dipole on the left:

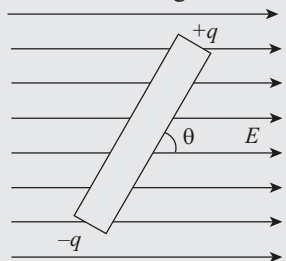


- (a) will feel a force to the left.
- (b) will feel a force to the right.
- (c) will feel a torque trying to make it rotate counterclockwise.
- (d) will feel no torque

28. Imagine that a dipole is at the centre of a spherical surface. If magnitude of electric field at a certain point on the surface of sphere is 10 N/C, then which of the following cannot be the magnitude of electric field anywhere on the surface of sphere

- (a) 4 N/C
- (b) 8 N/C
- (c) 16 N/C
- (d) 32 N/C

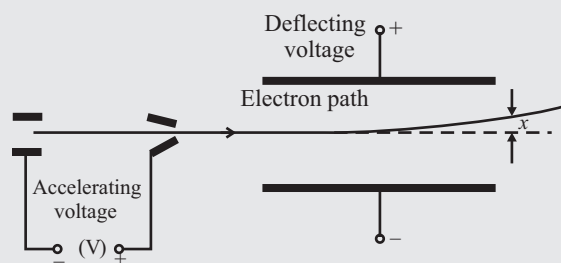
29. A small electric dipole is placed in a uniform electric field as shown in the diagram.



Considering the situation above, choose the correct statement(s):

- (a) The torque on the dipole points into the plane of the paper.
- (b) If allowed to rotate freely about its center, the dipole would initially swing counter-clockwise.
- (c) Work done by the electric field on the dipole, in rotating it from $\theta = 90^\circ$ to $\theta = 30^\circ$ is positive.
- (d) The potential energy of the dipole is maximum when the electric field is perpendicular to the dipole moment.

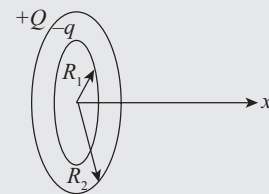
30. The diagram shows an electron gun and a pair of parallel deflecting plates. The electron gun accelerates the electrons through a potential difference V_0 . A narrow beam of accelerated electrons enter into the space between two parallel deflecting plates, while moving parallel to the plates. The potential difference between the two deflecting plates is V . The deflection of the electron beam as it leaves the parallel plates is x . (Do not consider gravity.)



Which of the following will change the deflection x of the beam as it leaves the parallel plates

- (a) increasing the accelerating voltage, V_0
- (b) increasing the deflecting voltage, V
- (c) increasing the distance between the electron gun and the deflecting plates
- (d) increasing the distance between the two deflecting plates

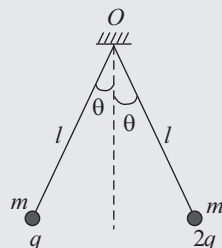
31. Two concentric rings of radii $R_1 = \sqrt{6}$ m and $R_2 = 4$ m are placed in y - z plane with their centres at the origin. They have uniformly distributed charge $-q$ and $+Q = 2\sqrt{2}q$, respectively. Consider the electrostatic potential to be zero at infinity. Then



- (a) The electric potential is zero at origin.
- (b) The electric field intensity is zero at $x = 2$ m.
- (c) A positive charged particle disturbed slightly from the origin along the x -axis will restore back to origin.
- (d) Potential is maximum on the x -axis at $x = 0$.

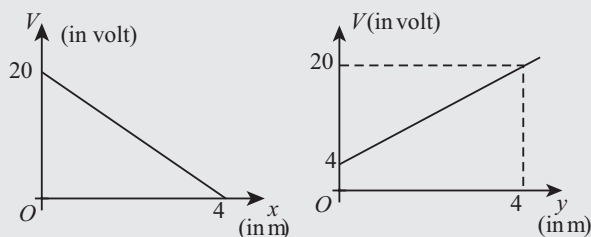
Worksheet 3

1. Two small balls of mass m are suspended at point O using two identical insulating threads of length l . Charge on one ball is q and that on the other is $2q$. Both are pushed towards one another using insulating sticks so as to keep the threads inclined at $\theta_1 = 30^\circ$ to the vertical. The balls are released simultaneously from this position. The string holding the ball with charge $2q$ reaches the largest deflection angle $\theta_2 = 60^\circ$ with vertical.



- (a) Find the largest deflection angle for the other thread.
(b) Find value of charge q .
2. Four point charges, q each, are held at the vertices of a square of side length x . One of the charges is slowly moved to infinity and then one another charge is moved so that the three charges now reside on the vertices of an equilateral triangle of side length x . Find work done by the external agent in the entire process.
3. The variation of electric potential in a two-dimensional xy plane has been shown in the two graphs. Find acceleration of a charge $-q$ immediately after it is released at a point $(2, 3)\text{m}$.

Mass of the particle is m . Given $\frac{q}{m} = 2 \text{ C/kg}$



4. A conducting sphere of radius R has charge Q on it. A point charge $-Q$ is placed at a distance $3R$ from its centre. Draw an equipotential surface very close to the surface of the sphere.
5. In a two-dimensional space, electric field is given as $\vec{E} = (2xy)\hat{i} + (x^2)\hat{j}$. Write the equation of an equipotential curve having potential V_0 . Assume the potential to be zero at $(0, 0)$.

6. A cubical volume has uniform charge density. Electric potential at the centre of the cube is V_0 . Find the potential at the centre of a cube of twice the side length and half the charge density of the previous cube.

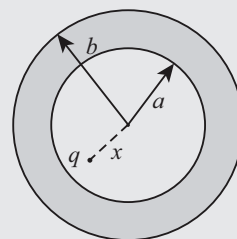
7. In a region of space, the electric potential varies with x co-ordinate only as $V = -ax^2 + b$. Here, a and b are positive constants. Find the volume charge density in space.

8. In an electrostatic field, the electric potential has a local maxima at point M and a local minima at point N .

- (a) Are either (or both) of these points equilibrium points for a negative charge? If yes, are they position of stable equilibrium?

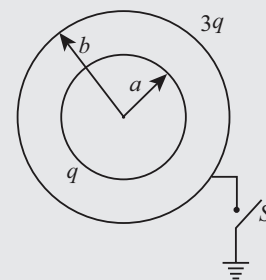
- (b) Assume that there is no other local maxima or minima of electric potential between the points M and N . An electron at N is gently pushed towards M . Will the electron reach point M ?

9. A conducting spherical shell has inner radius a and outer radius b . A point charge q is placed at a distance x ($< a$) from the centre inside the cavity. Find



- (a) Electric potential at the centre of the shell.
(b) Electric potential at a point outside the sphere at a distance r from the centre ($r > b$).

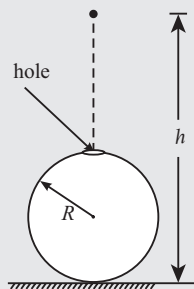
10. Two concentric metallic shells have radii a and b and charge on them are q and $3q$, respectively. Switch S is closed to ground the outer shell.



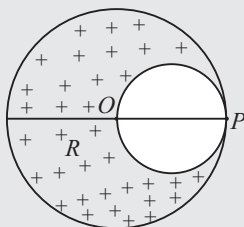
- (a) By what factor does the potential difference between the two shells change after the switch is closed?
(b) Find potential of the inner shell before and after the switch is closed.
(c) Find the quantity of charge that flows through S after the switch is closed.

11. Two solid metal spheres have radii R and $2R$. Charge density on each of them is σ . They are far apart. They are connected by a thin conducting wire. Find the new charge density on the smaller sphere.

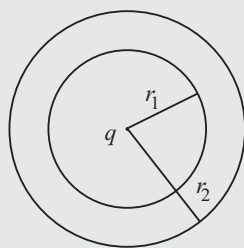
12. A hollow conducting sphere of radius R is fixed on an insulating horizontal surface. There is a small hole at the top of the sphere. Small charged drops of mercury fall from a height h and enter the sphere through the hole. Each mercury drop has mass m and charge q . How many drops can enter the sphere? Assume that a nearby drop does not cause any change in charge distribution on the spherical ball.



13. A disc of radius R has a hole of radius $\frac{R}{2}$ passing through its centre O (see figure). The remaining disc surface carries a uniform surface charge of density σ . Find potential at point P (where the circumference of the original disc and the hole meet).

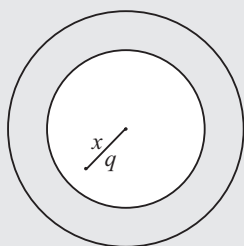


14. Figure shows two equipotential surfaces around a point charge q . The difference in radii of two surfaces is $\Delta r = r_2 - r_1$ for a potential difference of ΔV between the two surfaces. As we move away from the charge, will Δr increase or decrease for a fixed value of ΔV ?



15. A long cylindrical plastic rod has radius a , and it contains a uniform surface charge density σ . Potential at its surface is assumed to be V_0 . Find potential at a distance $2a$ from its axis.

16. A conducting hollow sphere has inner radius R_1 and outer radius R_2 . A point charge q is placed at a distance x from the centre inside the sphere as shown. Find



- (a) Potential at the centre of the sphere.
(b) Potential at a point at a distance x from the centre where $R_1 < x < R_2$.

- (c) Potential at a point at a distance x from the centre where $x > R_2$.

17. Two fixed charges $-2Q$ and Q are located at the points with coordinates $(-3a, 0)$ and $(+3a, 0)$, respectively in the xy plane.

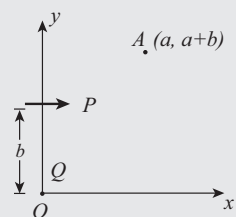
- (a) Show that the locus of all the points in xy plane where the electric potential is zero, is a circle.
(b) Write the variation of potential along the x axis.
(c) If a particle of charge $+q$ starts from rest at the centre of the circle, show by short qualitative argument that the particle eventually crosses the circle. Find its speed when it does so.

18. Two point charges $+q$ and $-2q$ are kept at a separation. Draw a graph showing the variation of the electric potential along the line joining the two charges.

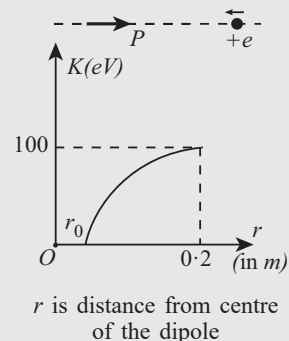
19. Two concentric metal shells have radii R and $2R$ and charge on them is q and $-2q$, respectively. Plot the variation of electric potential with distance from the centre. Using the graph for potential and knowing that $E = -\frac{dV}{dr}$, plot the variation of electric field also. Neglect the thickness of shells.

20. A short dipole of dipole moment $\vec{P} = (2\hat{i} - 3\hat{j} + 4\hat{k})$ cm is placed at a point $A(2, -3, 1)$ m. Find electric potential due to this dipole at a point $B(4, -1, 0)$ m.

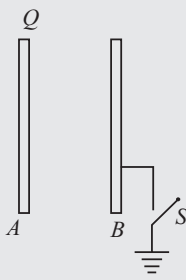
21. A short electric dipole is kept at $(0, b)$. Dipole moment of the dipole is $P\hat{i}$. A point charge Q lies at the origin (O) . Electric field at a point $A(a, a+b)$ is zero. Find relationship between a and b .



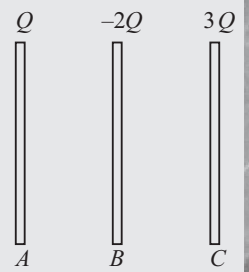
22. A short electric dipole is fixed and a proton approaches the dipole axis (see figure). At large distance from the dipole, the kinetic energy of the proton was $K_0 = 300$ eV. Graph shows the variation of kinetic energy of the proton at points close to the dipole. Find the magnitude of the dipole moment of the dipole. Also find the distance r_0 .



23. Two large metal plates are kept parallel to one another. Plate A is given a charge Q and plate B is neutral. Potential difference between the two plates is V_0 . Now switch S is closed and the plate B gets grounded. Find the new potential difference between the plates.



24. Three identical metal plates are kept parallel to one another as shown. Charges given to the plates are as indicated. Find charge on the inner wall of plate A . What is potential difference between A and B ?



Answer Sheet

Your Turn

1. No
2. No
3. (a) -9mJ (b) 9mJ
4. -9mJ
5. (a) -4.5 mJ (b) 4.5 mJ
6. (a) $\frac{\sqrt{2}(1-2\sqrt{2})}{4\pi\epsilon_0} \frac{q^2}{a}$ (b) $\frac{(4-2\sqrt{2})}{4\pi\epsilon_0} \frac{q^2}{a}$
7. $\frac{1}{4\pi\epsilon_0} \frac{qQ}{a}$
8. 0
9. 9000 V
10. 10^{-5} J
11. 0
12. 1200 V
13. $\frac{r}{\sqrt{3}-1}$
14. -9 J
15. $\frac{Qq}{4\pi\epsilon_0 R}$
16. $\frac{Q}{2\pi\epsilon_0 a^2} (\sqrt{a^2+x^2}-x)$
17. (a) $u = \sqrt{\frac{2qQ}{\pi\epsilon_0 ma}}$ (b) No change
18. $\frac{q^2}{\pi\epsilon_0 mu^2}$
19. Decrease
20. 20 V
21. $V_B - V_A = -0.8\text{ V}$
22. 565 V
23. $V_A - V_B = 124.7\text{ V}$
24. $V = -\frac{1}{2}(x^2 + y^2)$
25. $\frac{2qE}{K}$
26. -150 Vm^{-1}
27. No
31. Zero.
32. Cylindrical
33. $\frac{4}{3}a$
34. (a) $\frac{Q}{8\pi\epsilon_0 R}$ (b) $-\frac{Q}{12\pi\epsilon_0 R}$ (c) 0
35. $\frac{1}{4\pi\epsilon_0} \frac{Q}{R}$
36. $\frac{10}{3}\text{ nC}$
37. There will be no movement of the charge as both the balls are at same potential.
38. (a) $\frac{Q}{8\pi\epsilon_0 R}$ (b) $\frac{-Q}{2}$
39. $\frac{1}{4\pi\epsilon_0} \frac{Q}{2r}$
40. No change
41. $U = -\frac{Q^2}{8\pi\epsilon_0 R}; \Delta U_{\text{gain}} = \frac{Q^2}{8\pi\epsilon_0 R}$
42. 200 eV
43. 2000 eV
44. $3.4 \times 10^{-20}\text{ Cm}$
45. $\sqrt{3} qd$ along the bisector of the angle at $-2q$.
46. $\tan^{-1}(2)$
47. $\frac{E_0}{16}$
48. $3.2 \times 10^{-21}\text{ J}$
49. $\sqrt{\frac{2PE}{I}}$

Worksheet 1

1. (b)
2. (b)
3. (b)
4. (b)
5. (a)
6. (b)
7. (b)
8. (a)
9. (d)
10. (d)
11. (b)
12. (b)
13. (b)
14. (d)
15. (a)
16. (b)
17. (a)
18. (b)
19. (d)
20. (a)
21. (a)
22. (a)
23. (a)
24. (b)
25. c)
26. (b)
27. (c)
28. (b)
29. (b)
30. (b)
31. (a)
32. (a)
33. (a)
34. (c)
35. (b)
36. (d)
37. (b)
38. (a)
39. (c)
40. (b)
41. (a)
42. (d)
43. (b)
44. (b)
45. (d)
46. (c)
47. (c)
48. (b)
49. (b)
50. (a)
51. (b)
52. (c)
53. (b)
54. (b)
55. (a)
56. (a)
57. (a)

Worksheet 2

1. (c, d)
2. (b)
3. (b, c, d)
4. (a, b, c)
5. (a)
6. (b, c)
7. (a, c)
8. (d)
9. (a, d)
10. (a, c, d)
11. (b, d)
12. (d)
13. (a)
14. (a, c)
15. (b, c, d)
16. (a, d)
17. (a, b, d)
18. (c)
19. (a, d)
20. (a, d)
21. (b, c)
22. (a, d)
23. (a, b)
24. (a, b, c)
25. (b, d)
26. (a, c)
27. (a, d)
28. (a, d)
29. (a, c)
30. (a, b, d)
31. (b, c)

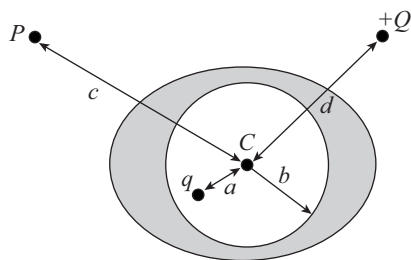
Worksheet 3

1. (a) 60° (b) $l\sqrt{2\sqrt{3}\pi\epsilon_0 mg}$
2. $W = -\frac{Kq^2}{a}(1+\sqrt{2})$
3. $(-10\hat{i} + 8\hat{j})ms^{-2}$
4. Equipotential surface is spherical.
5. $x^2y = -V_0$
6. $2V_0$
7. $\rho = 2a\epsilon_0$
8. (a) Yes, point M is position of stable equilibrium. (b) yes
9. (a) $kq\left[\frac{1}{x} - \frac{1}{a} + \frac{1}{b}\right]$ (b) $V_r = \frac{kq}{r}$
10. (a) No change (b) $kq\left(\frac{1}{a} + \frac{3}{b}\right); kq\left(\frac{1}{a} - \frac{1}{b}\right)$ (c) Charge $-4q$ flows from the ground to the shell
11. $\frac{5}{3}\sigma$
12. $n = \frac{4\pi\epsilon_0 mgR(h-R)}{q^2}$
13. $\frac{\sigma R}{2\pi\epsilon_0}$
14. Increase
15. $V_0 - \frac{a\sigma}{\epsilon_0} \ln 2$
16. (a) $\frac{kq}{x} - \frac{kq}{R_1} + \frac{kq}{R_2}$ (b) $\frac{kq}{R_2}$ (c) $\frac{kq}{x}$
17. (b) $V = kQ\left[\frac{2}{x+3a} - \frac{1}{x-3a}\right]$ for $x < -3a$
 $= kQ\left[\frac{1}{3a-x} - \frac{2}{3a+x}\right]$ for $-3a < x < 3a$
 $= kQ\left[\frac{1}{x-3a} - \frac{2}{x+3a}\right]$ for $x > 3a$
 (c) $\sqrt{\frac{qQ}{8\pi\epsilon_0 ma}}$
20. $-2 \times 10^9 \text{ V}$
21. $b = 2a$
22. $P = 8.9 \times 10^{-10} \text{ Cm}; r_0 = 0.16\text{m}$
23. $2 V_0$
24. zero, zero.

Miscellaneous Problems on Chapters 1 & 2

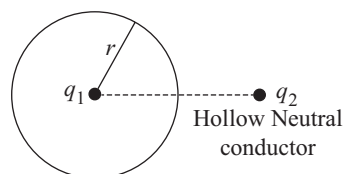
MATCH THE COLUMN

1. In the figure shown, the conductor is uncharged and a charge q is placed inside a spherical cavity at a distance ' a ' from its centre (C). Point P and charge $+Q$ are as shown. a, b, c, d are known distances.



Column-I		Column-II	
(A)	Electric field at P due to induced charges on the inner surface of cavity is	(P)	zero
(B)	Electric potential at P due to charges on the inner surface of cavity and q	(Q)	non-zero
(C)	Electric field at C due to induced charges on the outer surface of conductor and Q	(R)	value can be stated with the given data
(D)	Electric potential at C due to induced charges on the inner surface of cavity	(S)	finite but its value cannot be stated from the given data

2. A point charge q_1 is kept at the centre of a hollow neutral conductor. Another point charge q_2 is kept outside it as shown. Match the entries in Column-I with entries in Column-II.



Column-I		Column-II	
(A)	In the situation shown	(P)	Distribution of charge on inner surface of conductor is uniform
(B)	If outside charge is not present	(Q)	Distribution of charge on inner surface of conductor is non-uniform
(C)	If we displace the outside charge while the inside charge remains at centre	(R)	Distribution of charge on outer surface of conductor is uniform
(D)	If the inside charge is displaced by small amount from centre then	(S)	Distribution of charge on outer surface of conductor is non-uniform

3.2 Electrostatics and Current Electricity

3. Column-I shows graphs of electric potential V versus x and y in a certain region for four situations. Column-II gives the angle which the electric field vector may make with positive x -direction.

Column-I		Column-II	
(A)		(P)	$0 \leq \theta < 45^\circ$
(B)		(Q)	$45^\circ \leq \theta < 90^\circ$
(C)		(R)	$90^\circ \leq \theta < 135^\circ$
(D)		(S)	$135^\circ \leq \theta \leq 180^\circ$

4. Column-I gives certain situations involving two thin conducting shells connected by a conducting wire via a key K . In all situations one sphere has net charge $+q$ and other sphere has no net charge. After the key K is pressed, Column-II gives some resulting effects. Match the figures in Column-I with the statements in Column-II.

Column-I		Column-II	
(A)		(P)	charge flows through connecting wire
(B)		(Q)	Potential energy of system of spheres decreases

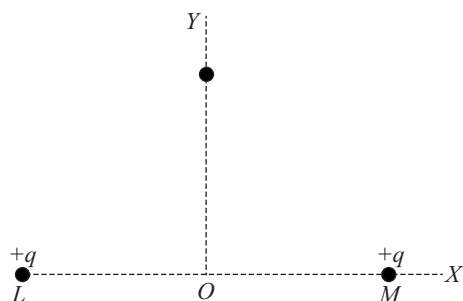
(C)		(R)	No heat is produced
(D)		(S)	The sphere I has no charge after equilibrium is reached
		(T)	charge does not flows through connecting wire

5. In each situation of column-I, some charge distributions are given. The electrostatic potential energy and its nature is given in column-II. Match situation in column-I with the corresponding results in column-II.

Column-I		Column-II	
(A)	A thin shell of radius a and having a charge $-Q$ uniformly distributed over its surface as shown 	(P)	$\frac{1}{8\pi\epsilon_0} \frac{Q^2}{a}$ in magnitude
(B)	A thin shell of radius $\frac{5a}{2}$ and having a charge $-Q$ uniformly distributed over its surface and a point charge $-Q$ placed at its centre as shown. 	(Q)	$\frac{3}{20\pi\epsilon_0} \frac{Q^2}{a}$ in magnitude

(C)	A solid sphere of radius a and having a charge $-Q$ uniformly distributed throughout its volume as shown	(R)	$\frac{2}{5\pi\epsilon_0} \frac{Q^2}{a}$ in magnitude
(D)	A solid sphere of radius a and having a charge $-Q$ uniformly distributed throughout its volume. The solid sphere is surrounded by a concentric thin uniformly charged spherical shell of radius $2a$ and carrying charge $-Q$	(S)	Positive in sign
		(T)	zero

6. In the figure shown, charges, each $+q$, are fixed at L and M . O is the mid-point of line LM . X and Y axes are as shown. Consider the situations given in Column-I and match them with the information in Column-II:



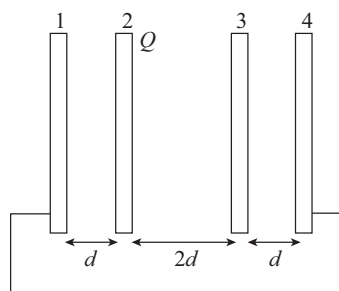
Column-I		Column-II	
(A)	Let us place a charge $+q$ at O , displace it slightly along X -axis and release. Assume that it is allowed to move only along X -axis. At position O	(P)	force on the charge is zero
(B)	Place a charge $-q$ at O . Displace it slightly along X -axis and release. Assume that it is allowed to move only along X -axis. At position O	(Q)	potential energy of the system is maximum

(C)	Place a charge $+q$ at O . Displace it slightly along Y -axis and release. Assume that it is allowed to move only along Y -axis. At position O	(R)	potential energy of the system is minimum
(D)	Place a charge $-q$ at O . Displace it slightly along Y -axis and release. Assume that it is allowed to move only along Y -axis. At position O	(S)	the charge is in equilibrium

PASSAGE BASED PROBLEMS

Passage 1

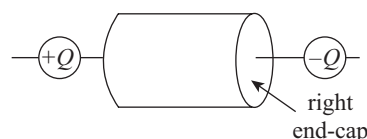
Four metallic plates are placed as shown in the figure. Plate 2 is given a charge Q whereas all other plates are uncharged. Plates 1 and 4 are joined together. The area of each plate is same.



- The charge appearing on the right side of plate 3 is
 - zero
 - $+Q/4$
 - $-3Q/4$
 - $Q/2$
- The charge appearing on right side of plate 4 is
 - zero
 - $-Q/4$
 - $-3Q/4$
 - $+Q/2$
- The potential difference between plates 1 and 2 is
 - $\frac{3}{2} \frac{Qd}{\epsilon_0 A}$
 - $\frac{Qd}{\epsilon_0 A}$
 - $\frac{3}{4} \frac{Qd}{\epsilon_0 A}$
 - $\frac{3Qd}{\epsilon_0 A}$

Passage 2

In the figure shown, positive and negative charges of equal magnitude lie along the symmetry axis of a cylinder. The distance from the positive charge to the left end-cap of the cylinder is the same as the distance from the negative charge to the right end -cap.



4. The flux of the electric field through the closed cylinder is
- (a) 0 (b) $+Q/\epsilon_0$
(c) $+2Q/\epsilon_0$ (d) $-Q/\epsilon_0$
5. The sign of the flux through the right end-cap of the cylinder is
- (a) Positive
(b) Negative
(c) There is no flux through the right end-cap

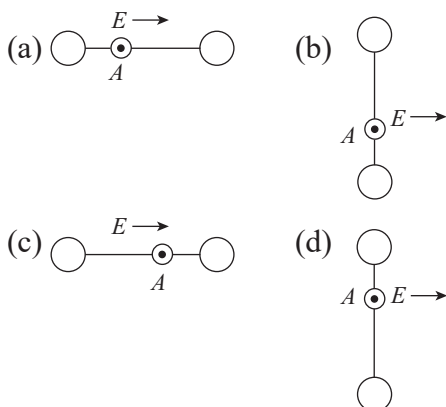
Passage 3

There is an insulator rod of length L having negligible mass. Two small balls of mass m , each having electric charge Q , are attached to its ends. The rod can rotate in the horizontal plane around a vertical axis at $L/4$ distance from one of its ends.

6. At first the rod is in unstable equilibrium in a horizontal uniform electric field of strength E . Then we gently displace it from this position. Determine the maximum velocity attained by the ball which is closer to the axis in the subsequent motion.

- (a) $\sqrt{\frac{2QEL}{m}}$ (b) $\sqrt{\frac{2QEL}{5m}}$
(c) $\sqrt{\frac{QEL}{5m}}$ (d) $\sqrt{\frac{4QEL}{5m}}$

7. In what position is the rod to be set so that if displaced a little from that position it begins a harmonic oscillation about the axis A ?

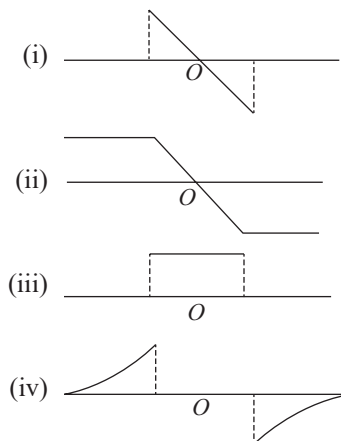


8. What is the time period of the SHM as mentioned in the above question?

- (a) $2\pi\sqrt{\frac{mL}{QE}}$ (b) $2\pi\sqrt{\frac{2mL}{3QE}}$
(c) $2\pi\sqrt{\frac{5mL}{QE}}$ (d) $2\pi\sqrt{\frac{5mL}{4QE}}$

Passage 4

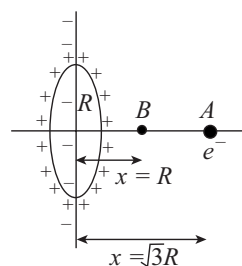
Two **very large** parallel disks of charge have their centers on the x -axis and their planes perpendicular to the x -axis. The disk that intersects the x axis at $x = -R$ has uniform positive surface charge density $+\sigma$; the disk that intersects the x axis at $x = +R$ has uniform negative surface charge density $-\sigma$.



9. Which graph best represents the plot of the x -component of the electric field vector on the x -axis?
- (a) (i) (b) (ii)
(c) (iii) (d) (iv)
10. Which graph best represents the plot of the electric potential (V) as a function of x (treating $V = 0$ at $x = 0$)?
- (a) (i) (b) (ii)
(c) (iii) (d) (iv)

Passage 5

There is a uniformly charged ring having radius R . An infinite line charge (charge per unit length $-\lambda$) is placed along a diameter of the ring (in gravity free space). Total charge on the ring is $Q = 4\sqrt{2}\lambda R$. An electron of mass m is released from rest on the axis of the ring at a distance $x = \sqrt{3}R$ from its centre.



11. Magnitude of initial acceleration of the electron is

- (a) $\frac{e\lambda}{\pi\epsilon_0 mR} \left(\frac{3-2\sqrt{2}}{4\sqrt{6}} \right)$
(b) $\frac{e\lambda}{\pi\epsilon_0 mR} \left(\frac{3+2\sqrt{2}}{4\sqrt{6}} \right)$
(c) $\frac{e\lambda}{\pi\epsilon_0 mR} \left(\frac{3+2\sqrt{2}}{4\sqrt{3}} \right)$
(d) none

12. The distance from centre of ring on the axis where the net force on the electron is zero is

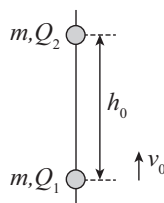
(a) $2R$ (b) $\sqrt{2} R$
(c) R (d) none of these

13. Potential difference between points A ($x = \sqrt{3} R$) and B ($x = R$) i.e. ($V_A - V_B$) is

(a) $-\frac{\lambda}{\pi\epsilon_0} \left[\left(1 - \frac{1}{\sqrt{2}} \right) - \frac{\ln 3}{4} \right]$
(b) $\frac{\lambda}{\pi\epsilon_0} \left[\left(1 - \frac{1}{\sqrt{2}} \right) - \frac{\ln 3}{4} \right]$
(c) $-\frac{\lambda}{\pi\epsilon_0} \left[\left(1 + \frac{1}{\sqrt{2}} \right) - \frac{\ln 3}{4} \right]$
(d) none of these

Passage 6

At the lower end of a vertically positioned insulator bar shown in the figure there is a pearl of the mass m having a charge Q_1 . Above it, at a height h_0 , there is another pearl with the same mass m and electric charge Q_2 resting in equilibrium. The pearls can move along the bar without friction. At time $t = 0$, we kick the lower pearl and it starts upwards at a velocity of $u = 2 \text{ m/s}$. [$g = 10 \text{ m/s}^2$]



14. The speed of the centre of mass of the system of pearls at time $t = 0.1 \text{ s}$ is

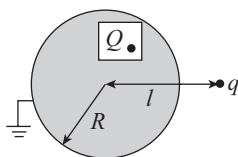
(a) 2 m/s (b) zero
(c) 1 m/s (d) dependent on value of h_0

15. v_0 is the speed of the pearls when they have equal velocity. The rise in centre of mass of the system of two pearls when they attain velocity v_0 is

(a) $\frac{u^2}{8g} - \frac{v_0^2}{2g}$ (b) $\frac{u^2}{2g} - \frac{v_0^2}{2g}$
(c) $\frac{u^2}{4g} - \frac{v_0^2}{3g}$ (d) $\frac{u^2}{8g} - \frac{v_0^2}{4g}$

Passage 7

There is a cubical cavity inside a conducting sphere of radius R . A positive point charge Q is placed at the centre of the cube and another positive charge q is placed at a distance l ($> R$) from the centre of the sphere. The sphere is earthed



16. Charge induced on the inner surface of cavity is

(a) $-Q$, uniformly distributed
(b) $-Q$, non-uniformly distributed
(c) $-(Q + q)$ non-uniformly distributed
(d) none

17. Net charge on the outer surface of conducting sphere is

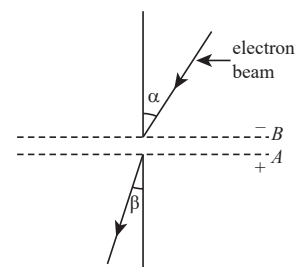
(a) $+Q$ (b) $Q - qR/l$
(c) $-qR/l$ (d) none

18. Potential at a point inside the cavity is

(a) zero (b) positive
(c) negative (d) cannot be said

Passage 8

A and B are two finely woven wire meshes that are kept parallel to one another at a small separation. A potential difference V is applied across them with A as positive and B as negative. A narrow electron beam is incident on B at an angle of incidence α . Electrons in the beam have speed u . Electron beam emerges from A at an angle β as shown.



Assume that electric field between A and B is uniform and elsewhere it is zero. Also assume that wire in the mesh does not interfere with electrons. Let us define $\frac{\sin \alpha}{\sin \beta} = \mu$ as refractive index of the system of wire mesh.

[m = mass of an electron, $-e$ = charge on an electron]

19. The value of μ is

(a) independent of angle of incidence α
(b) independent of speed μ
(c) independent of mass of the electron m
(d) independent of potential difference V

20. The potential difference V is doubled and the kinetic energy of the incident electrons is also doubled. This will

(a) half the value of μ
(b) double the value of μ
(c) increase μ by a factor of $2\sqrt{2}$
(d) not change the value of μ

Passage 9

Two point charges $+9Q$ and $-Q$ are fixed on the x axis at $x = -a$ and $x = a$, respectively. A point charge q of mass m performs oscillations of small amplitude A about a point

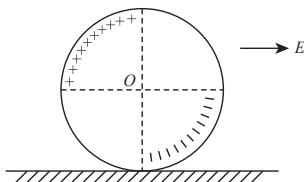
3.6 Electrostatics and Current Electricity

$x = x_0$ on the x axis. Ignore all forces on q apart from the electrostatic force.

21. The sign of charge q
- must be positive
 - must be negative
 - may be positive or negative
 - depends on value of a
22. For $A \ll a$, the time period of oscillation is given as
- $2\pi\sqrt{\frac{3\pi\epsilon_0 md^3}{|q|Q}}$
 - $2\pi\sqrt{\frac{\pi\epsilon_0 md^3}{|q|Q}}$
 - $\pi\sqrt{\frac{\pi\epsilon_0 md^3}{|q|Q}}$
 - $2\pi\sqrt{\frac{\pi\epsilon_0 md^3}{3|q|Q}}$
23. Speed of the oscillating charge at $x = x_0$ is
- $A\sqrt{\frac{Q|q|}{2\pi\epsilon_0 md^3}}$
 - $A\sqrt{\frac{Q|q|}{3\pi\epsilon_0 md^3}}$
 - $\sqrt{\frac{AQ|q|}{2\pi\epsilon_0 md^3}}$
 - $A\frac{Q|q|}{q^2\epsilon_0 md^2}$

Passage 10

A non-conducting ring of radius R and mass m is given charge on its two quadrants as shown. The linear charge density is $+\lambda$ and $-\lambda$. It is placed on a rough non-conducting plane as shown. A uniform horizontal electric field E is switched on. The ring starts rolling without sliding immediately after the field is switched on. Answer the following question for the time immediately after the field is switched on.



24. Torque of electric force on the ring about its centre (O) is
- $2\lambda R^2 E$
 - $\frac{1}{2}\lambda R^2 E$
 - $\frac{1}{4}\lambda R^2 E$
 - $\lambda R^2 E$
25. Friction force acting on the ring is
- $\lambda R E$ towards right
 - $\lambda R E$ towards left
 - $2\lambda R E$ towards right
 - $\frac{1}{4}\lambda R E$ towards left

Passage 11

Two small identical balls (1 and 2) are placed on a smooth horizontal surface. The balls are connected with an ideal

spring. Ball 2 is held fixed and the other ball is given a small speed. Ball 1 oscillates with frequency f_0 . Now both the balls are given same charge and the length of the spring becomes n times its original length in equilibrium position. When Ball 1 is disturbed slightly from equilibrium, the oscillation frequency is found to be f . The spring is non-conducting and its force constant is K .

26. If q is charge given to each of the balls and l is natural length of the spring, then

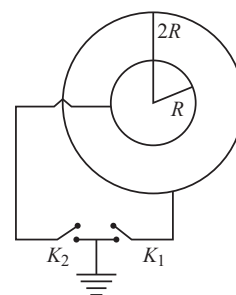
- $l^3 = \frac{q^2}{4\pi\epsilon_0 n^2 (n-1)k}$
- $l^3 = \frac{q^2}{\pi\epsilon_0 n^2 (n-1)k}$
- $l^3 = \frac{4q^2}{\pi\epsilon_0 n^2 k}$
- $l^3 = \frac{q^2}{4\pi\epsilon_0 (n-1)^3 k}$

27. With $n = 2$, $\frac{f}{f_0}$ is equal to

- $\frac{1}{\sqrt{2}}$
- $\frac{1}{\sqrt{3}}$
- $\sqrt{2}$
- 1

Passage 12

Two concentric metallic shells have radii R and $2R$ as shown in the figure. A charge q is given to the inner sphere. Now, key K_1 is closed and then opened. After this, key K_2 is closed and opened. The process is repeated n time and finally K_2 remains closed.

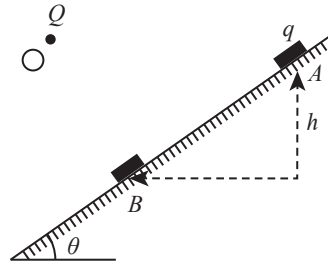


28. After one event involving closer and opening of keys K_1 and K_2 , charge on the inner sphere is
- q
 - $-\frac{q}{2}$
 - $\frac{q}{2}$
 - zero
29. The final potential of the outer shell is

- $\frac{-q}{4\pi\epsilon_0 2^{n+1} \cdot R}$
- $\frac{q}{4\pi\epsilon_0 2^{n+1} \cdot R}$
- $\frac{-q}{4\pi\epsilon_0 2^{n-1} \cdot R}$
- $\frac{q}{4\pi\epsilon_0 2^n \cdot R}$

Passage 13

A small block carrying positive charge q is held at rest on a rough incline. A negative point charge Q is fixed at a point O in the vertical plane of the figure. Coefficient of friction between the block and the incline is μ . When released, the block acquires a speed v_0 after sliding down on the incline through a vertical height h .



F_1 represents the magnitude of component of electric force parallel to the incline (acting on the block), and F_2 is magnitude of electric force perpendicular to the incline.

30. If ds is infinitesimally small displacement of the block down the incline, then the value of

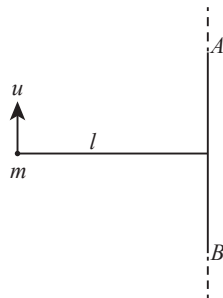
$$\int_A^B F_1 ds + \int_A^B \mu F_2 ds \text{ is}$$

- (a) $\frac{1}{2}mv_0^2 - mgh + \mu mgh \cot \theta$
 (b) $\frac{1}{2}mv_0^2 - mgh - \mu mgh \tan \theta$
 (c) $mgh - \frac{1}{2}mv_0^2 + \mu mgh \tan \theta$
 (d) $mgh + \frac{1}{2}mv_0^2 - \mu mgh \cot \theta$
31. If the block was given a negative charge of same magnitude ($-q$), what speed would it acquire after sliding from A to B ?

- (a) $\sqrt{2gh(1 - \mu \cot \theta) - v_0^2}$
 (b) $\sqrt{4gh(1 - \mu \cot \theta) - v_0^2}$
 (c) $\sqrt{2gh(1 - \mu \tan \theta) - 2v_0^2}$
 (d) Cannot be determined

Passage 14

AB is a vertical rigid thin infinite line carrying a linear charge density λ . A particle having mass m and charge q is connected to the line by means of a light, insulating and inextensible string of length l . The particle is projected vertically up with velocity u with the string horizontal and fully stretched. The string gets slack when the particle rotates by $\theta = 45^\circ$.



32. The speed (v) of the particle at the instant when the string gets slack is given by

- (a) $\frac{mv^2}{l} = mg - \frac{\sqrt{2}q\lambda}{\pi\epsilon_0 l}$
 (b) $\frac{mv^2}{l} = \frac{mg}{\sqrt{2}} - \frac{q\lambda}{\sqrt{2}\pi\epsilon_0 l}$
 (c) $\frac{mv^2}{l} = \frac{mg}{\sqrt{2}} - \frac{q\lambda}{2\sqrt{2}\pi\epsilon_0 l}$
 (d) $\frac{mv^2}{l} = \frac{mg}{\sqrt{2}} - \frac{q\lambda}{2\pi\epsilon_0 l}$

33. Work done by the electric force on charge q by the time q rotates by 45° is

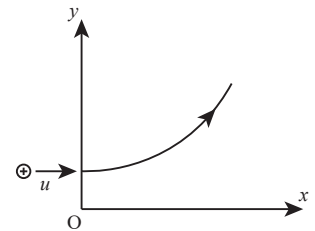
- (a) $-\frac{q\lambda \ln 2}{4\pi\epsilon_0}$ (b) $-\frac{q\lambda \ln 2}{2\pi\epsilon_0}$
 (c) $\frac{2q\lambda \ln 2}{\pi\epsilon_0}$ (d) $\frac{q\lambda \ln 2}{4\pi\epsilon_0}$

34. Value of u is

- (a) $\sqrt{\frac{gl}{\sqrt{2}} - \frac{q\lambda(1 - \ln 2)}{2\pi\epsilon_0 m}}$
 (b) $\sqrt{\frac{3}{\sqrt{2}}gl - \frac{q\lambda(1 - \ln 2)}{2\pi\epsilon_0 m}}$
 (c) $\sqrt{\frac{\sqrt{3}gl}{2} - \frac{q\lambda(1 + \ln 2)}{2\pi\epsilon_0 m}}$
 (d) $\sqrt{\frac{\sqrt{3}gl}{\sqrt{2}} + \frac{q\lambda(1 + \ln 2)}{2\pi\epsilon_0 m}}$

Passage 15

Volume charge density in space $y > 0$ varies as $\rho = ay$. A particle of mass m carries a charge $+q$. It enters the region with velocity $\vec{u} = u\hat{i}$ at point $(0, y_0)$. We neglect gravity and assume the electric field to be zero at $y = 0$. The electric field exists in y direction only.



35. The electric field at a point having y co-ordinate y is

- (a) $\frac{ay^2}{\epsilon_0}$ (b) $\frac{2ay^2}{\epsilon_0}$
 (c) $\frac{ay^2}{2\epsilon_0}$ (d) $-\frac{2ay^2}{\epsilon_0}$

36. The y component of velocity of the particle when its y co-ordinate is

(a) $\sqrt{\frac{qa}{3m\epsilon_0}}(y^3 - y_0^3)$ (b) $\sqrt{\frac{qa}{m\epsilon_0}}(y^2 - y_0^2)$
 (c) $\sqrt{\frac{2qa}{3m\epsilon_0}}(y^3 - y_0^3)$ (d) $\sqrt{\frac{qa}{m\epsilon_0}}(y^2 - y_0^2)$

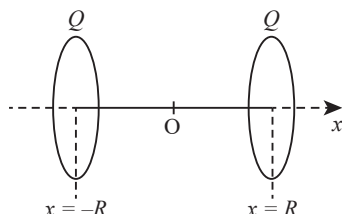
37. Slope of the trajectory of the particle as a function of y is

(a) $\sqrt{\frac{qa}{m\epsilon_0 u^2}}(y^3 - y_0^3)$
 (b) $\sqrt{\frac{qa}{3m\epsilon_0 u^2}}(y^3 - y_0^3)$
 (c) $\sqrt{\frac{qa}{m\epsilon_0 u}}(y^2 - y_0^2)$
 (d) $\sqrt{\frac{2qa}{m\epsilon_0 u}}(y^2 - y_0^2)$

Passage 16

Two charged rings of radius R , having charge Q each, are placed parallel to yz plane with their centres at $(-R, 0)$ and $(R, 0)$. Minimum value of potential at a point on the x -axis for $-R < x < R$ is V_0 .

At a point on the x axis, very close to the origin, the potential is $V = V_0 + ax^2$.



38. Value of V_0 is

(a) $\frac{\sqrt{2}Q}{\pi\epsilon_0 R}$ (b) $\frac{\sqrt{2}Q}{2\pi\epsilon_0 R}$
 (c) $\frac{Q}{\sqrt{2}\pi\epsilon_0 R}$ (d) $\frac{\sqrt{2}Q}{4\pi\epsilon_0 R}$

39. Value of a is

(a) $\frac{3Q}{16\pi\epsilon_0 \sqrt{2}R^3}$ (b) $\frac{Q}{16\pi\epsilon_0 R^3}$
 (c) $\frac{Q}{8\pi\epsilon_0 R^3}$ (d) $\frac{\sqrt{2}Q}{4\pi\epsilon_0 R^3}$

40. A charge particle having charge $+q$ is placed at the origin. It is displaced by a small distance along the x -axis and released. It will perform

- (a) oscillations but not SHM
 (b) SHM
 (c) non-oscillatory motion
 (d) a motion with continuously increasing acceleration

Passage 17

A point charge q_0 is fixed at the origin. A short electric dipole, having dipole moment $\vec{P}$, is placed along the x -axis, at a far away point from the origin. The dipole moment vector is directed along the positive x -direction. When the dipole is released, it gets attracted towards the point charge. A small disturbance causes the dipole to oscillate in the xy plane with the x -axis as its mean position. However, the energy of oscillation is a very small fraction of the translational kinetic energy acquired by the dipole as it reaches a finite distance x from the point charge. Mass of the dipole is M .

41. The speed of the centre of mass of the dipole when it is at a distance x from the origin is

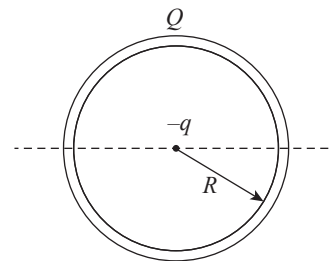
(a) $\sqrt{\frac{q_0 P}{2\pi\epsilon_0 M x^2}}$ (b) $\sqrt{\frac{q_0 P}{\pi\epsilon_0 M x^2}}$
 (c) $\frac{1}{2} \sqrt{\frac{q_0 P}{\pi\epsilon_0 M x^2}}$ (d) $\sqrt{\frac{2q_0 P}{\pi\epsilon_0 M x^2}}$

42. The time period of oscillations of the dipole, as it approaches the origin

- (a) goes on increasing
 (b) goes on decreasing
 (c) remains constant
 (d) cannot be predicted

Passage 18

A thin metallic shell of radius R is given a charge Q . A point charge $-q$ is placed at its centre. If the shell is divided into two identical hemispheres, the two parts remain in equilibrium touching one another.



43. The magnitude of charge (q) at the centre must be such that

(a) $q \geq \frac{Q}{2}$ (b) $q < \frac{Q}{2}$
 (c) $q \geq \frac{Q}{4\pi}$ (d) $q \geq \frac{Q}{2\pi}$

44. The surface charge density on the outer surface of the shell, when the two hemispherical parts do not exert force on one another and stay in equilibrium, is

(a) $\frac{q}{2\pi R^2}$ (b) $\frac{3q}{4\pi R^2}$
 (c) $\frac{q}{4\pi R^2}$ (d) zero

Answer Sheet

Match the Columns

1. (A) Q, S (B) P, R (C) P R (D) Q,R
2. (A) P, S (B) P, R (C) P, S (D) Q, S
3. (A) S (B) P (C) R (D) Q
4. (A) P, Q (B) P, Q (C) P, Q, S (D) R, S, T
5. (A) P, S (B) Q, S (C) Q, S (D) S
6. (A) P, R, S (B) P, Q, S (C) P, Q, S (D) P, R, S

Passage Based Problems

- | | | | | | | | | |
|---------|---------|---------|---------|---------|---------|---------|---------|---------|
| 1. (b) | 2. (d) | 3. (c) | 4. (a) | 5. (a) | 6. (c) | 7. (a) | 8. (d) | 9. (c) |
| 10. (b) | 11. (a) | 12. (c) | 13. (a) | 14. (b) | 15. (a) | 16. (b) | 17. (c) | 18. (b) |
| 19. (a) | 20. (d) | 21. (b) | 22. (a) | 23. (b) | 24. (a) | 25. (a) | 26. (a) | 27. (c) |
| 28. (c) | 29. (a) | 30. (a) | 31. (b) | 32. (d) | 33. (a) | 34. (b) | 35. (c) | 36. (a) |
| 37. (b) | 38. (d) | 39. (a) | 40. (b) | 41. (a) | 42. (b) | 43. (a) | 44. (c) | |

Capacitors

“The history of electricity yields a well-known truth – that the physicist shirking measurement only plays, different from children only in the nature of his game and the construction of his toys”.

—Franz Karl Achard

1. INTRODUCTION

A capacitor is an energy storage device. They are used extensively in electronic circuits. Capacitors are particularly important in AC circuits. However, in this chapter, we will study about their behaviour in DC circuits.

In the beginning, we will discuss about the charge holding capacity of a conductor and then proceed to discuss about capacitors.

2. CAPACITANCE OF A CONDUCTOR

Capacitance (or, capacity) of a conductor is measure of its ability to hold charge at a given potential.

Consider an isolated conductor. If we manage to go on adding charge to it, its potential will keep on increasing. It is not possible to add charge to an isolated conductor while keeping its potential constant. In fact, charge (Q) on an isolated conductor is directly proportional to its potential (V).

$$Q \propto V$$

$$\Rightarrow Q = CV,$$

where C is a constant, known as the *capacitance* of the conductor. The capacitance (C) depends on the geometry of the conductor.

The above equation can be written as

$$C = \frac{Q}{V} \quad (1)$$

SI unit of capacitance is $\frac{C}{V}$, commonly known as farad (F).

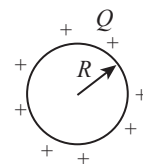
2.1 Capacitance of a spherical conductor

Consider a spherical conductor of radius R having charge Q on it. Potential of the conductor is

$$V = \frac{1}{4\pi\epsilon_0} \frac{Q}{R}$$

$$\Rightarrow \frac{Q}{V} = 4\pi\epsilon_0 R$$

$$\Rightarrow C = 4\pi\epsilon_0 R \quad (2)$$



Spherical conductor in vacuum

For other geometrical shapes, it may be very difficult to calculate the capacitance. However, capacitance of a conductor can be measured.

Example 1 One farad is too large a capacitance

Equation (2) suggests that capacitance of a spherical conductor increases with radius. The largest conductor that we can imagine is our earth (radius = 6400 km). Calculate the capacitance of the earth.

Solution

Concepts

For spherical conductors, $C = 4\pi\epsilon_0 R$

$$\begin{aligned} C &= 4\pi\epsilon_0 R = \frac{1}{9 \times 10^9} \times 6400 \times 10^3 \\ &= 711 \times 10^{-6} \text{ F} \\ &= 711 \mu\text{F} \end{aligned}$$

Capacity of such a huge conductor is only 711 μF (less than 1mF). Therefore, we conclude that 1F is a very large unit of capacitance. More practical units are μF , nF, pF, etc.



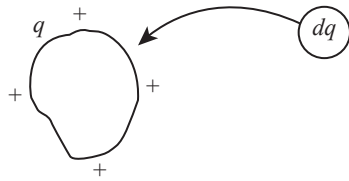
YOUR TURN

Q.1 Capacitance of a conductor is defined as $C = \frac{Q}{V}$. If the charge on a conductor is doubled, will its capacitance also double?

Q.2 A conductor has capacitance of $2 \mu\text{F}$. How much charge shall be placed on it to make its potential equal to 1000 V ?

2.2 Energy of a charged conductor

In the previous chapter, we calculated the electrostatic energy of a charged spherical conductor. Now we will generalize it for a conductor of any shape. We will express the energy in terms of the capacitance of the conductor.



Work is done in moving a charge dq from ∞ on to a charged conductor.

Consider a conductor being charged by placing charges in small instalments. At some intermediate stage, it has charge q on it. Its potential is

$$V = \frac{q}{C}$$

where C is the capacitance of the conductor. Now an infinitesimally small charge (dq) is moved from infinity to the conductor. Work done in the process is

$$dW = V \cdot dq = \frac{q}{C} dq$$

Total work done in piling a charge Q on the conductor is

$$W = \int dW = \frac{1}{C} \int_0^Q q dq$$

$$\Rightarrow W = \frac{Q^2}{2C}$$

This work done is the electrostatic energy of the charge system on the conductor.

If V is final potential of the conductor, then $Q = CV$ and we can write:

$$U = \frac{Q^2}{2C} = \frac{1}{2} CV^2 \quad (3)$$

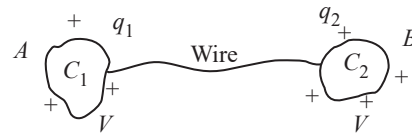
3. CHARGE SHARING BETWEEN TWO CONDUCTORS

We know that a temperature difference between the two ends of a conducting rod causes heat to flow from the hotter end to the other. Temperature difference is the reason behind heat

flow. Similarly, a potential difference between the ends of a conductor causes charges flow (i.e., a current). The current persists as long as the potential difference is maintained.

Consider two conductors A and B having capacitance C_1 and C_2 , respectively. Charges on them are Q_1 and Q_2 . If they are far apart (so that charge on one does not alter the potential of other), their potentials are given as

$$V_1 = \frac{Q_1}{C_1} \quad \text{and} \quad V_2 = \frac{Q_2}{C_2}$$



When connected, the two conductors acquire same potential.

Now the two conductors are connected using a metal wire. Potential difference between the ends of the wire causes charge to flow. Charge moves from one conductor to another (through the wire) till both of them are at same potential. Let the final charge on the two conductors be q_1 and q_2 . The final common potential of the two conductors is:

$$V = \frac{q_1}{C_1} = \frac{q_2}{C_2}$$

$$\Rightarrow \frac{q_1}{q_2} = \frac{C_1}{C_2} \quad (a)$$

From conservation of charge:

$$q_1 + q_2 = Q_1 + Q_2 \quad (b)$$

solving (a) and (b) gives

$$q_1 = \left(\frac{C_1}{C_1 + C_2} \right) (Q_1 + Q_2)$$

$$q_2 = \left(\frac{C_2}{C_1 + C_2} \right) (Q_1 + Q_2)$$

Final common potential is

$$V = \frac{q_1}{C_1} = \frac{Q_1 + Q_2}{C_1 + C_2}$$

$$\Rightarrow V = \frac{C_1 V_1 + C_2 V_2}{C_1 + C_2} \quad (4)$$

Loss of electrostatic energy

Whenever charge gets redistributed on a conductor or between two conductors, there is loss in electrostatic energy. Energy is lost in the form of heat and electromagnetic radiation.

Let us calculate loss in energy of the system of two conductors (mentioned above) after they are connected. Initial energy of the system is

$$U_i = \frac{1}{2} C_1 V_1^2 + \frac{1}{2} C_2 V_2^2$$

Final energy, after redistribution of the charge, is

$$U_f = \frac{1}{2} C_1 V^2 + \frac{1}{2} C_2 V^2$$

$$= \frac{1}{2} (C_1 + C_2) \left(\frac{C_1 V_1 + C_2 V_2}{C_1 + C_2} \right)^2 \quad [\text{using (4)}]$$

Loss in electrostatic energy is

$$\Delta U_{\text{loss}} = U_i - U_f$$

$$= \frac{1}{2} C_1 V_1^2 + \frac{1}{2} C_2 V_2^2 - \frac{1}{2} (C_1 + C_2) \left(\frac{C_1 V_1 + C_2 V_2}{C_1 + C_2} \right)^2$$

Algebraic simplification gives

$$\Delta U_{\text{loss}} = \frac{1}{2} \left(\frac{C_1 C_2}{C_1 + C_2} \right) (V_1 - V_2)^2 \quad (5)$$

Example 2 Two conductors have capacitance $C_1 = 1 \mu\text{F}$ and $C_2 = 2 \mu\text{F}$. Each of them carries charge $Q = 10 \mu\text{C}$. They are placed at a large separation and connected by a conducting wire.

- Find the final common potential of the two conductors.
- Find the final charges on the two conductors.

- Find the amount of heat generated after the two conductors are connected.

Solution

Concepts

- We will first calculate the final common potential (V) using equation (4).
- Final charges on the two conductors are obtained as $q_1 = C_1 V$ and $q_2 = C_2 V$.
- Heat generated is equal to loss in electrostatic energy given by equation (5).

- Final common potential is

$$V = \frac{C_1 V_1 + C_2 V_2}{C_1 + C_2} = \frac{Q_1 + Q_2}{C_1 + C_2}$$

$$= \frac{(10+10)\mu\text{C}}{(1+2)\mu\text{F}} = \frac{20}{3} \text{ V}$$

- Final charges on the two conductors are:

$$q_1 = C_1 V = (1\mu\text{F}) \left(\frac{20}{3} \text{ V} \right) = \frac{20}{3} \mu\text{C}$$

$$q_2 = C_2 V = (2\mu\text{F}) \left(\frac{20}{3} \text{ V} \right) = \frac{40}{3} \mu\text{C}$$

- Heat generated = loss in electrostatic energy

$$= \frac{1}{2} \frac{C_1 C_2}{C_1 + C_2} (V_1 - V_2)^2$$

Initial potentials of the two conductors are:

$$V_1 = \frac{Q}{C_1} = \frac{10\mu\text{C}}{1\mu\text{F}} = 10\text{V}$$

$$V_2 = \frac{Q}{C_2} = \frac{10\mu\text{C}}{2\mu\text{F}} = 5\text{V}$$

$$\therefore \text{Heat} = \frac{1}{2} \frac{(1\mu\text{F})(2\mu\text{F})}{(1+2)\mu\text{F}} (10 - 5)^2$$

$$= \frac{25}{3} \mu\text{J}$$



YOUR TURN

Q.3 Two conductors having capacitance $C_1 = 2 \mu\text{F}$ and $C_2 = 4 \mu\text{F}$ have charge $Q_1 = 20 \mu\text{C}$ and $Q_2 = 40 \mu\text{C}$, respectively. They are far apart. Find heat that is produced when they are connected using a metal wire.

Q.4 The radius of a spherical conductor A is twice that of another spherical conductor B . Conductor A is given a

charge $-20 \mu\text{C}$ and the conductor B is given a charge $+20 \mu\text{C}$. The two conductors placed far apart are connected using a conducting wire. Find the final charge on A .

Q.5 In the last question, only A is given a charge of $-20 \mu\text{C}$ and no charge is given to B . Find final charge on A , after the two conductors are connected.

In Short

- Capacitance of an isolated conductor is $C = \frac{Q}{V}$, where Q is charge on the conductor and V is its potential.
- Capacitance of a conductor depends on its shape and size.
- Unit of capacitance is $\frac{C}{V}$, known as farad (F). Practical units are μF , nF , pF , etc.
- Capacitance of a spherical conductor of radius R is $4\pi\epsilon_0 R$.
- When two conductors at different potentials are connected, charge flows between them till the potential difference becomes zero.
- When two isolated conductors of capacitance C_1 and C_2 and potentials V_1 and V_2 , respectively are connected, their final common potential is

$$V = \frac{C_1 V_1 + C_2 V_2}{C_1 + C_2}$$

Electrostatic energy dissipated as heat in the process is

$$\Delta U_{\text{loss}} = \frac{C_1 C_2}{2(C_1 + C_2)} (V_1 - V_2)^2$$

4. CAPACITOR

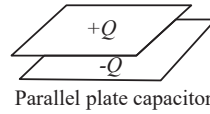
Two isolated conductors kept at a separation make a capacitor. These conductors, no matter what their geometry, are known as plates of the capacitor. One of the conductors is given a positive charge and the other is given an equal negative charge. The charges on the two plates attract and keep each other secured. The charge ($+Q$), on the positive plate, is called the *charge on the capacitor*. If V_+ and V_- are potentials of the two plates, then potential difference between them is $V = V_+ - V_-$.

For a capacitor, charge on the capacitor is always proportional to the potential difference between the plates.

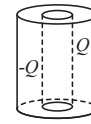
$$\begin{aligned} Q &\propto V \\ \Rightarrow Q &= CV \end{aligned} \quad (6)$$

The proportionality constant (C) is known as *capacitance of the capacitor*. It depends on shape, size, placement of the conductors and the medium between them. Unit of C is farad (F).

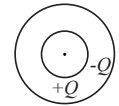
Capacitors have different types depending on the geometry of the plates. A capacitor made by placing two identical metal plates parallel to each other is known as *parallel plate capacitor*. Two co axial metal cylinders make a *cylindrical capacitor* and two concentric conducting spheres form a *spherical capacitor*. We will focus mainly on parallel plate capacitor.



Parallel plate capacitor



Cylindrical capacitor

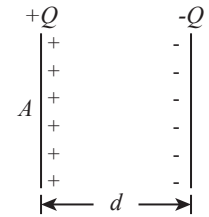


Spherical capacitor

4.1 Capacitance of a parallel plate capacitor

Two large and identical metal plates kept parallel to one another with a small separation between them makes a parallel plate capacitor. The plates can be circular, rectangular square or of any other shape. The separation (d) between the plates is small compared to the dimensions of the plates.

Let us assume that there is no medium between the plates. One of the plates is given a charge $+Q$ and the other one is given a charge $-Q$. The charges will attract one another and stay on the inner wall of the two plates (one can refer back to Chapter 1 for more details on this).



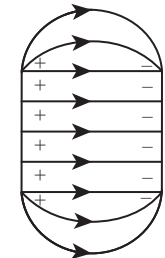
Parallel plate capacitor

The charge density on each surface has magnitude

$$\sigma = \frac{Q}{A},$$

where A is area of each plate.

Since the plates are large and separation between them is small, the electric field between them will be uniform and perpendicular to the plates except for a small region near the edges.



Actual field lines

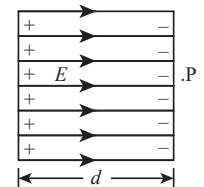
Actually, there will be fringing of the field lines near the edges. But we will ignore this small effect in our discussions and assume that field lines are normal to the plates everywhere.

Electric field between the plates is directed from positive to negative plate and its magnitude can be written as

$$\begin{aligned} E &= \frac{\sigma}{2\epsilon_0} + \frac{\sigma}{2\epsilon_0} \\ \Rightarrow E &= \frac{\sigma}{\epsilon_0} \end{aligned} \quad (7)$$

Electric field outside the plates (at a point like P) is zero. Potential difference between the two plates is

$$\begin{aligned} V &= Ed = \frac{\sigma}{\epsilon_0} d \\ \Rightarrow V &= \frac{Qd}{A\epsilon_0} \end{aligned}$$



Idealised field lines neglecting edge effect

∴ The capacitance of the parallel plate capacitor is

$$C = \frac{Q}{V} = \frac{\epsilon_0 A}{d} \quad (8)$$

It is being re-emphasized that the capacitance depends on geometry. We will soon learn that it also depends on the insulating medium between the plates of the capacitor.

Example 3 A parallel plate capacitor has square plates having side length 10 cm. Gap between the plates is 1 mm.

- Find the capacitance of the capacitor.
- Find potential difference between the plates if charge given to the capacitor is 0.35 nC.

Solution

Concepts

$$(i) C = \frac{\epsilon_0 A}{d} \quad (ii) V = \frac{Q}{C}$$

$$(a) C = \frac{\epsilon_0 A}{d} = \frac{8.85 \times 10^{-12} \times (0.1 \times 0.1)}{10^{-3}} \\ = 8.85 \times 10^{-11} F = 88.5 \text{ pF}$$

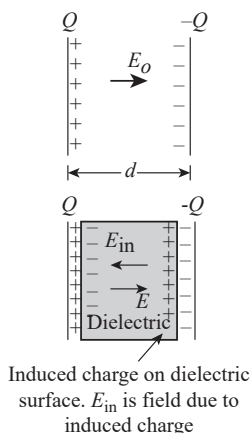
$$(b) V = \frac{Q}{C} = \frac{0.35 \times 10^{-9}}{8.85 \times 10^{-11}} = 3.95 \text{ V}$$

4.2 Parallel plate capacitor with dielectric

The two plates of a capacitor shall be separated by an insulating (dielectric) medium. When we fill nothing between the plates, there is air and it is a dielectric. However, there is always a risk of two plates falling on one another when there is air between them. Also, breakdown of air occurs when the electric field between the plates exceeds $3 \times 10^6 \text{ V/m}$. The electrons are ripped apart from the molecules of air and it gets ionised. The air begins to conduct when electric field exceeds $3 \times 10^6 \text{ V/m}$. This limits the quantity of charge that we can place on the capacitor. To get rid of these difficulties, we fill the space between the plates with dielectric material like paper, glass, castor oil, etc. These materials can tolerate very high electric fields before they breakdown. As we shall see now, filling of space between the plates of a capacitor with a dielectric increases the capacitance.

Consider an ideal parallel plate capacitor having charge Q . With no medium between the plates, field is

$$E_0 = \frac{\sigma}{\epsilon_0} = \frac{Q}{A\epsilon_0}$$



Now imagine the same capacitor with space between its plates filled completely with a dielectric slab. Again charge Q is given to the capacitor. The dielectric gets polarized in the presence of electric field E_0 . The induced charges on its two faces are as shown in the figure. The induced charges create their own field E_{in} that is opposite to E_0 . Therefore, resultant field in the space between the plates gets reduced. The field is

$$E = \frac{E_0}{K} = \frac{Q}{AK\epsilon_0} \quad (9)$$

where K is known as the dielectric constant of the material, which is a *dimensionless number and greater than 1*.

Potential difference between the plates is

$$V = E \cdot d = \frac{Qd}{AK\epsilon_0}$$

$$\therefore \text{Capacitance is } C = \frac{Q}{V} = \frac{K\epsilon_0 A}{d} \quad (10)$$

Therefore, capacitance of the capacitor becomes K times after the space between the plates is filled with a dielectric of constant K .

We can also write the surface density of induced charge on the dielectric slab. The resultant field (E) is difference of magnitudes of field (E_0) created by charges on metal plates and the field (E_{in}) created by charges induced on dielectric walls.

$$E = E_0 - E_{in} \\ \text{or, } \frac{E_0}{K} = E_0 - E_{in} \\ \text{or, } E_{in} = E_0 \left[1 - \frac{1}{K} \right] \\ \text{or, } \frac{\sigma_{in}}{\epsilon_0} = \frac{\sigma}{\epsilon_0} \left[1 - \frac{1}{K} \right] \\ \text{or, } \sigma_{in} = \sigma \left[1 - \frac{1}{K} \right] \quad (11)$$

Here, σ is surface charge density on metal plates.

Example 4 A parallel plate capacitor has plate area 20 cm^2 and separation between the plates is 1.0 mm.

- Find the maximum charge that can be placed on the capacitor when air fills the space between the plates.
- The space between the plates is filled with glass. Now, calculate the maximum charge that can be placed on the capacitor.

It is given that breakdown strength of air and glass are $3 \times 10^6 \text{ V/m}$ and $30 \times 10^6 \text{ V/m}$, respectively. Dielectric constant of the glass is 5.

Solution

Concepts

When charge on a capacitor is increased, electric field between the plates increases. When the field exceeds the breakdown strength, the dielectric begins to conduct. We say that the capacitor has broken down or it is punctured. Now, the positive and negative charges on the two plates cannot be kept separated.

- (a) Maximum allowed electric field is 3×10^6 V/m.

$$\therefore \frac{\sigma}{\epsilon_0} = 3 \times 10^6$$

where σ is maximum charge density on the plates.

$$\Rightarrow \sigma = 8.85 \times 10^{-12} \times 3 \times 10^6$$

$$= 2.65 \times 10^{-5} \text{ C/m}^2$$

$$\therefore Q_{\max} = \sigma A = 2.65 \times 10^{-5} \times 20 \times 10^{-4}$$

$$= 5.3 \times 10^{-8} \text{ C}$$

- (b) Field between the plates is $\frac{\sigma}{K\epsilon_0}$

Maximum σ is obtained as:

$$\frac{\sigma}{K\epsilon_0} = 30 \times 10^6$$

$$\Rightarrow \sigma = 30 \times 10^6 \times 5 \times 8.85 \times 10^{-12}$$

$$= 1.33 \times 10^{-3} \text{ C/m}^2$$

$$\therefore Q_{\max} = \sigma \cdot A = 1.33 \times 10^{-3} \times 20 \times 10^{-4}$$

$$= 2.66 \times 10^{-6} \text{ C}$$

4.3 Parallel plate capacitor with multiple dielectrics

In the figure, A and B are metal plates and space between them is filled with three dielectrics. Dielectric walls are parallel to the metal plates; x_1 , x_2 and x_3 are thicknesses of the dielectrics having dielectric constants K_1 , K_2 and K_3 , respectively.

Assume that a charge Q is given to the capacitor. Had there been no dielectrics, the electric field between the plates would have been

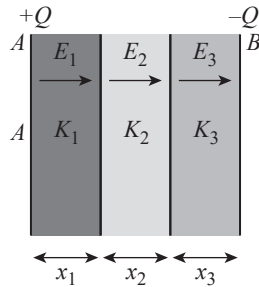
$$E_0 = \frac{\sigma}{\epsilon_0} = \frac{Q}{A\epsilon_0}$$

Field inside the three dielectrics will be:

$$E_1 = \frac{E_0}{K_1}; E_2 = \frac{E_0}{K_2}; E_3 = \frac{E_0}{K_3}$$

Potential difference between the plates A and B will be:

$$V = E_1 x_1 + E_2 x_2 + E_3 x_3$$



$$= E_0 \left[\frac{x_1}{K_1} + \frac{x_2}{K_2} + \frac{x_3}{K_3} \right]$$

$$\text{or, } V = \frac{Q}{A\epsilon_0} \left[\frac{x_1}{K_1} + \frac{x_2}{K_2} + \frac{x_3}{K_3} \right]$$

$$\therefore C = \frac{Q}{V} = \frac{\epsilon_0 A}{\frac{x_1}{K_1} + \frac{x_2}{K_2} + \frac{x_3}{K_3}} \quad (12)$$

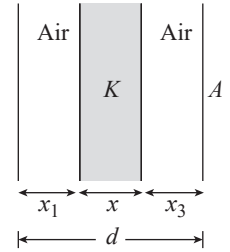
If space between the plates is partly filled with a dielectric of constant K , then the capacitance can be written as

$$C = \frac{\epsilon_0 A}{\frac{x_1}{1} + \frac{x}{K} + \frac{x_3}{1}}$$

Since dielectric constant of air is nearly 1.

$$\therefore C = \frac{\epsilon_0 A}{\frac{x}{K} + (x_1 + x_3)}$$

$$C = \frac{\epsilon_0 A}{\frac{x}{K} + (d - x)} \quad (13)$$



where d is separation between the plates.

It is important to note that the capacitance depends on thickness x of the dielectric and is independent of the position of the dielectric (i.e., the capacitance does not depend on whether the dielectric is closer to a plate or is kept exactly at the middle).

4.4 Capacitance of a spherical capacitor

Two concentric metallic spheres have radii a and b and they carry charges $+Q$ and $-Q$, respectively. One way of doing this can be to ground the outer sphere and give $+Q$ charge to the inner one. The charge ($-Q$) on the outer sphere will reside on its inner wall. The potential of inner and outer spheres can be written as

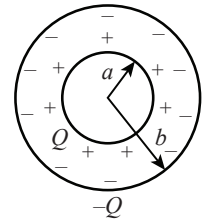
$$V_{\text{inner}} = \frac{1}{4\pi\epsilon_0} \left[\frac{Q}{a} - \frac{Q}{b} \right]$$

$$V_{\text{outer}} = \frac{1}{4\pi\epsilon_0} \left[\frac{Q}{a} - \frac{Q}{b} \right] = 0$$

$\therefore$ Potential difference between the inner and the outer spheres will be

$$V = V_{\text{inner}} - V_{\text{outer}}$$

$$\text{or, } V = \frac{Q}{4\pi\epsilon_0} \left[\frac{1}{a} - \frac{1}{b} \right]$$



$$\begin{aligned} \text{or, } \frac{Q}{V} &= \frac{4\pi\epsilon_0 ab}{(b-a)} \\ \text{or, } C &= \frac{4\pi\epsilon_0 ab}{(b-a)} \end{aligned} \quad (14)$$

4.5 Capacitance of a cylindrical capacitor

Consider two co-axial metallic cylinders having radii a and b . Length of the cylinder (L) is large compared to the separation between them. It is a cylindrical capacitor. Suppose a charge Q is placed on the inner cylinder and a negative charge $-Q$ is placed on the outer cylinder. The electric field in the region between the two cylinders will be radial when this gap is small compared to the length of the cylinders. Using the Gauss's law, we can calculate this electric field. At a distance x ($a < x < b$) from the axis, the field is

$$E = \frac{\lambda}{2\pi\epsilon_0 x} = \frac{Q}{2\pi\epsilon_0 Lx} \quad (15)$$

where $\lambda = \frac{Q}{L}$ = linear charge density on the inner cylinder.

The potential difference between the cylinders can be calculated as

$$dV = -E dx$$

$$\int_{V_{\text{inner}}}^{V_{\text{outer}}} dV = -\frac{Q}{2\pi\epsilon_0 L} \int_a^b \frac{dx}{x}$$

$$\text{or, } V_{\text{outer}} - V_{\text{inner}} = -\frac{Q}{2\pi\epsilon_0 L} \ln\left(\frac{b}{a}\right)$$

$$\text{or, } V_{\text{inner}} - V_{\text{outer}} = \frac{Q}{2\pi\epsilon_0 L} \ln\left(\frac{b}{a}\right)$$

$$\text{or, } V = \frac{Q}{2\pi\epsilon_0 L} \ln\left(\frac{b}{a}\right) \quad (16)$$

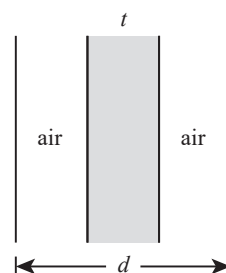
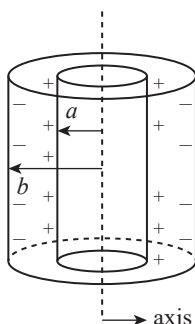
$$\text{or, } C = \frac{Q}{V} = \frac{2\pi\epsilon_0 L}{\ln\left(\frac{b}{a}\right)} \quad (17)$$

Example 5 A parallel plate capacitor has plate area A and separation between them is d . A metal sheet of thickness t is inserted between the plates. What is the capacitance, now?

Solution

Concepts

Electric field inside the metal plate will be zero. Dielectric constant of a conductor is infinity. Now using equation (13) will give the capacitance.



$$C = \frac{\epsilon_0 A}{\frac{t}{\infty} + (d-t)} = \frac{\epsilon_0 A}{d-t}$$

Example 6 A cylindrical capacitor has inner and outer cylinders of radii a and b , respectively. Length of the capacitor is L . The space between the two cylinders is filled with an insulator having dielectric constant K . Find the capacitance of the capacitor.

Solution

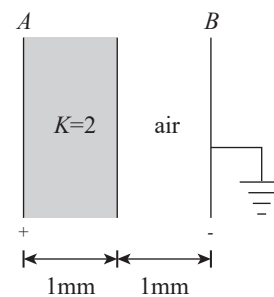
Concepts

The electric field at points between the two cylinders will be $\frac{1}{K}$ times given by equation (15).

The potential difference between the two conductors will be $\frac{1}{K}$ times given by equation (16). Therefore, capacitance will be K times given by equation (17).

$$\therefore C = \frac{2\pi K \epsilon_0 L}{\ln\left(\frac{b}{a}\right)}$$

Example 7 A parallel plate capacitor has plate separation $d = 2$ mm. Half the space between the plates is filled with a dielectric of constant $K = 2$ and the remaining half has air. One of the plates is earthed as shown. Area of each plate is 600 cm^2 . Plate A is given a charge $Q = 35.4 \text{ nC}$.



- Find potential difference between the plates of the capacitor.
- Plot the variation of electric field as one moves from plate A to B along a line perpendicular to the plates.
- Plot the variation of electric potential as one moves from A to B along a line perpendicular to the plates.

Solution**Concepts**

- (i) Equation (13) can be used to find capacitance.
- (ii) We can find potential difference between the plates using $V = \frac{Q}{C}$
- (iii) If the electric field in the air is E_0 , then the field in the dielectric is $\frac{E_0}{K}$.

Capacitance of the capacitor is

$$C = \frac{\epsilon_0 A}{\frac{d}{2K} + \frac{d}{2}} = \frac{4\epsilon_0 A}{3d}$$

$$\text{or, } C = \frac{4 \times 8.85 \times 10^{-12} \times 600 \times 10^{-4}}{3 \times 2 \times 10^{-3}} = 3.54 \times 10^{-10} \text{ F}$$

- (a) When charge Q is given to plate A , a charge $-Q$ induces on B . The potential difference between the plates is

$$V = \frac{Q}{C} = \frac{35.4 \times 10^{-9}}{3.54 \times 10^{-10}} = 100 \text{ V}$$

- (b) If E_0 is electric field in air, then field in dielectric is $\frac{E_0}{2}$. Potential difference between A and B can be expressed as

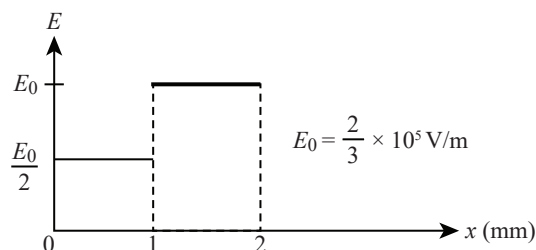
$$V = \frac{E_0}{2} \cdot \frac{d}{2} + E_0 \cdot \frac{d}{2} = \frac{3E_0 d}{4}$$

$$\text{or, } 100 = \frac{3(E_0)(2 \times 10^{-3})}{4}$$

$$\text{or, } E_0 = \frac{2}{3} \times 10^5 \text{ V/m.}$$

We take origin at plate A and x axis along a line perpendicular to A . Plate B is at $x = 2$ mm.

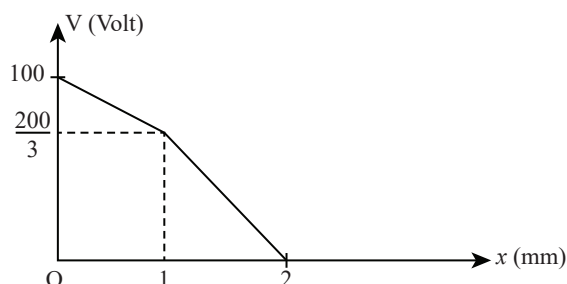
The graphical variation of E versus x is as shown.



- (c) As one moves from A to B , the potential drops. Rate of fall of potential is equal to magnitude of the electric field. If the drop in potential is V_1 when one moves a distance $\frac{d}{2} = 1$ mm from A to B , then the drop in potential is $2V_1$ as one moves in air a distance of $\frac{d}{2} = 1$ mm.

$$\therefore V_1 + 2V_1 = 100 \quad \Rightarrow \quad V_1 = \frac{100}{3} \text{ V.}$$

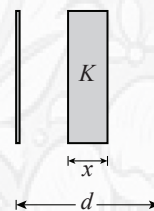
A potential drop of $\frac{100}{3}$ V occurs in the dielectric and the remaining drop of $2V_1 = \frac{200}{3}$ V takes place in air.

**YOUR TURN**

Q.6 A parallel plate capacitor has plate area $A = 20 \text{ cm}^2$ and separation between the plates $d = 1$ mm. It is given charge so that potential difference between the plates becomes 50 V. Find electric field in the region between the plates.

Q.7 In the last question, what will be potential difference between the plates if a dielectric of constant 5 is inserted between the plates so as to occupy the entire space between the plates.

Q.8 A parallel plate capacitor has separation between the plates d and has a capacitance C . Now a dielectric of thickness x is inserted between the plates as shown. Dielectric constant is $K = 3$. Find the ratio $\frac{x}{d}$ if the capacitance is found to be $2C$ after insertion of the dielectric.



Q.9 A spherical dielectric ball of radius 20 cm has a concentric cavity of radius 10 cm. The cavity wall and the outer surface of the ball are metal coated. The outer surface is grounded. Find the capacitance between the coated surfaces. Dielectric constant of the material is $K = 3.0$.

5. ENERGY STORED IN A CAPACITOR

Electrostatic energy of a charge system is the work done in assembling the charge system.

Consider a parallel plate capacitor of capacitance C . Initially, both the plates are neutral. Now, assume that an external agency begins to move positive charge from plate B to plate A in small instalments. The plate A will get positively charged and B will get an equal amount of negative charge on it. At certain intermediate stage, let the charge on A and B be $+q$ and $-q$, respectively. Plate A is at a potential higher than B . The potential difference is

$$V = \frac{q}{C}$$

Work needed to shift an infinitesimally small charge dq from B to A is

$$dW = V \cdot dq = \frac{1}{C} q \, d \cdot q$$

Total work done in piling up a charge Q on A (and therefore putting $-Q$ on B) is

$$W = \int dW = \frac{1}{C} \int_0^Q q \, dq$$

$$\text{or, } W = \frac{Q^2}{2C}$$

This is the electrostatic energy stored in the capacitor.

$$U = \frac{Q^2}{2C} = \frac{1}{2} CV^2 \quad (18)$$

Here, V is the final potential difference between the plates so that $Q = CV$.

Who is the external agent moving charge from one plate to another? In most of the cases, this external agent is a battery. Soon we will learn about this.

Energy stored in a capacitor is given by equation (18), even if the capacitor is spherical or cylindrical or of any other geometry.

6. ENERGY DENSITY IN AN ELECTRIC FIELD

A capacitor of capacitance C having charge Q on it stores an energy $U = \frac{Q^2}{2C}$ in it. Where is this energy? Modern view is that this energy is stored in the electric field present in the space between the plates. Thinking like this helps us in developing further theories in physics. It is believed that a region containing electric field has energy stored in it. Energy contained in unit volume of the region is known as *energy density* in the electric field.

The electric field between the plates of a capacitor is

$$E = \frac{\sigma}{\epsilon_0}$$

Volume of the space where the electric field exists is Ad where A is plate area and d is the plate separation.

Therefore, energy density in the electric field is

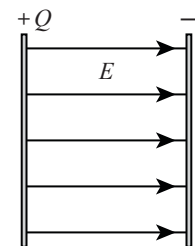
$$u = \frac{U}{A \cdot d} = \frac{Q^2}{2CA \cdot d}$$

$$\text{or, } u = \frac{(\sigma A)^2}{2 \epsilon_0 \frac{A}{d} A d} = \frac{1}{2} \frac{\sigma^2}{\epsilon_0}$$

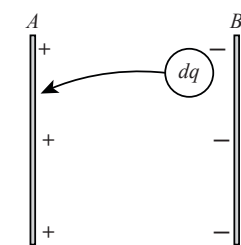
$$\text{or, } u = \frac{1}{2} \epsilon_0 \left(\frac{\sigma}{\epsilon_0} \right)^2$$

$$\text{or, } u = \frac{1}{2} \epsilon_0 E^2 \quad (19)$$

Although the above equation has been derived for a specific example of electric field between the plates of a capacitor, the result is true in general. If there is electric field anywhere in space, a unit volume of the space contains an amount of energy given by equation (19). Unit of energy density (u) is Jm^{-3} .



Energy is stored in the electric field between the plates



When moving a small charge (dq) from B to A one has to do work against the electric force

Example 8 Energy of a charged metallic sphere

A metal ball of radius R is given a charge Q . It creates an electric field in the space outside it. The electrostatic energy associated with the ball can be thought to be stored in this electric field. Calculate the energy stored in the electric field outside the ball.

Solution

Concepts

- (i) We have already calculated the energy of a charged spherical shell in Chapter 2. We know that

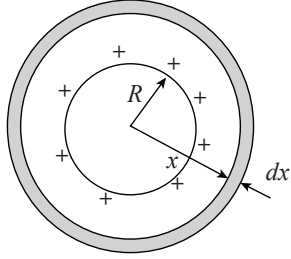
$$U = \frac{Q^2}{8\pi\epsilon_0 R} = \frac{Q^2}{2C}$$

This question asks us to take a different approach to establish this relation.

- (ii) The electric field extends from the surface of the sphere to infinity. Energy is stored in this field.
- (iii) We will consider a thin spherical region between radius x and $x + dx$ and write energy contained in the region. Integration will be used to add energy contained in all such regions from $x = R$ to $x = \infty$.

4.10 Electrostatics and Current Electricity

Electric field at a distance x ($>R$) from the centre of the sphere is $E = \frac{Q}{4\pi\epsilon_0 x^2}$



Energy per unit volume at a point at a distance x is

$$u = \frac{1}{2} \epsilon_0 E^2 = \frac{Q^2}{32\pi^2 \epsilon_0 x^4}$$

Energy contained in the shaded region is

$$dU = u \cdot 4\pi x^2 dx = \frac{Q^2}{8\pi\epsilon_0} \frac{dx}{x^2}$$

$\therefore$ Energy contained in the entire space outside the sphere is

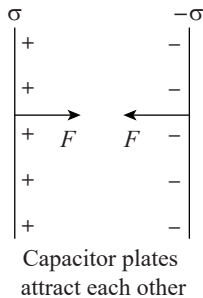
$$U = \int_{x=R}^{\infty} dU = \frac{Q^2}{8\pi\epsilon_0} \int_R^{\infty} \frac{dx}{x^2}$$

$$\text{or, } U = \frac{Q^2}{8\pi\epsilon_0 R}$$

7. FORCE BETWEEN PLATES OF A PARALLEL PLATE CAPACITOR

The oppositely charged capacitor plates attract one another. The electric field created by the positively charged plate is

$$E_+ = \frac{\sigma}{2\epsilon_0} = \frac{Q}{2A\epsilon_0}$$



This field exerts force on the negative plate. Force is given by

$$F = QE_+ = \frac{Q^2}{2A\epsilon_0} \quad (20)$$

Example 9 Work done in increasing the plate separation

A parallel plate capacitor has plate area A and separation between the plates is d . The capacitor is given a charge Q . Calculate the work done in slowly moving the plates to a separation $2d$. How does the energy stored in the capacitor change?

Solution

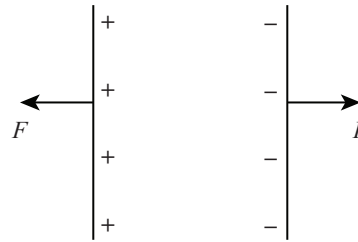
Concepts

The force between the plates of a parallel plate capacitor does not depend on plate separation (d). Therefore, force needed to pull apart the plates is a constant force.

The force between the two plates is

$$F = \frac{Q^2}{2A\epsilon_0}$$

The external agent must apply force F in outward direction to separate the plates. Force is constant, therefore work done is simply a product of force and displacement.



Force by the external agent

$$W = F \cdot d = \frac{Q^2 d}{2A\epsilon_0} = \frac{Q^2}{2C}$$

[Here C is the initial capacitance].

The energy stored in the capacitor increases exactly by W .

$$\Delta U = \frac{1}{2} \frac{Q^2}{\frac{C}{2}} - \frac{1}{2} \frac{Q^2}{C} = \frac{Q^2}{2C}$$

Actually, the volume of space where electric field exists has doubled. This leads to increase in stored energy.

In Short

- (i) Two isolated conductors placed at a separation make a capacitor. When charge on one of them is $+Q$ and on the other one is $-Q$, the capacitor is said to carry a charge Q . The two conductors are called plates of the capacitor.

- (ii) Capacitance of a capacitor is $C = \frac{Q}{V}$

where V is potential difference between the plates.

- (iii) A parallel plate capacitor has two large plates placed at a small separation. Capacitance of a parallel plate capacitor (with air or nothing between the plates) is

$$C = \frac{\epsilon_0 A}{d}$$

Here, A is plate area and d is plate separation.

- (iv) With the entire space between the plates filled with an insulator of dielectric constant K , the capacitance

$$\text{is } C = \frac{K\epsilon_0 A}{d}.$$

- (v) A parallel plate capacitor with multiple layers of dielectrics filling the space between the plates has capacitance

$$C = \frac{\epsilon_0 A}{\frac{x_1}{K_1} + \frac{x_2}{K_2} + \frac{x_3}{K_3} + \dots}$$

Here, $x_1, x_2, x_3, \dots$ are widths of the dielectric slabs having constant $K_1, K_2, K_3, \dots$

- (vi) Energy stored in a charged capacitor is

$$U = \frac{1}{2} CV^2 = \frac{Q^2}{2C}$$

- (vii) The energy is thought to be stored in any electric field. Energy stored in unit volume of space where there is an electric field E is given by

$$u = \frac{1}{2} \epsilon_0 E^2$$

This quantity is called energy density in an electric field.

- (viii) Force between plates of a parallel plate capacitor is

$$F = \frac{Q^2}{2A\epsilon_0}$$



YOUR TURN

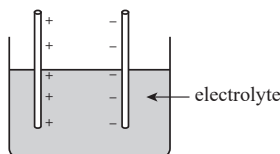
Q.10 A parallel plate capacitor has plate separation of 2 mm and it carries some charge so that the energy density in the electric field between the plates is 0.01 Jm^{-3} . How will

the energy density in the space between the plates change if the plate separation is decreased to 1 mm?

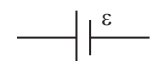
8. CELL (OR, BATTERY)

We have already stated that a cell is a device that can charge a capacitor. In fact, a cell is needed to drive current in an electric circuit [more about it in chapters to follow]. Here, we will briefly discuss few important things regarding a cell, which will help us get started.

In a typical electrochemical cell, two electrodes made up of different metals are held dipped in an electrolyte solution. Due to chemical reaction, positive charge gets deposited on an electrode and the other one acquires an equal amount of negative charge. A potential difference is developed between two electrodes. This potential difference is known as the



An electrochemical cell. The potential difference between the electrodes depends on the choice of material



symbol of a cell

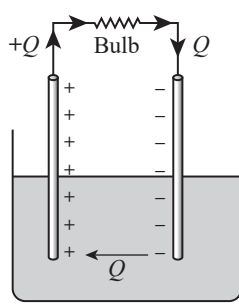
electromotive force (emf) of the cell. The emf of a cell is not decided by its size. It is decided by the choice of the material of the electrodes. We will denote emf by ϵ or V .

The total charge on a cell always remains zero. If the positive terminal of a cell supplies a charge $+Q$ when connected in a circuit, then the negative terminal must also supply a charge $-Q$. The other way of saying the same thing is to say that the $+Q$ charge supplied by the positive terminal of the cell must return back to it through its negative terminal.

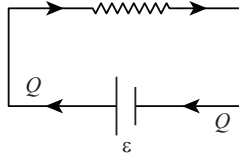
Consider a bulb connected to a cell. One end of the bulb filament (a resistance) is at the potential of the positive terminal of the cell and its other end is at the potential of the negative electrode. It means there is a potential difference across the ends of the filament. Positive charge flows from the high potential end to the low potential end. As long as the cell is alive, it maintains this potential difference and the charge keeps flowing. Such continuous flow of charge is known as current. We will study about electric current in Chapter 5.

4.12 Electrostatics and Current Electricity

When charge $+Q$ moves from the positive terminal to the negative terminal of the cell (through the bulb), quantity of charge on both electrodes reduces. Will the charges on the electrodes disappear soon? Answer is no! The electrolyte solution keeps replenishing the charge on the two electrodes and thereby maintaining the potential difference between the terminals. Effectively the solution moves $+Q$ charge from the negative electrode to the positive one.



Charge Q supplied by the positive terminal returns into the cell through its negative terminal.



A cell does positive work when positive charge Q flows through it as shown

Symbolic representation of the above circuit has been shown. When a charge Q flows through the cell (as shown), the chemical energy of the cell gets reduced. The chemical energy is used to move charge $+Q$ from the low potential negative terminal to the high potential positive terminal inside the electrolyte solution.

Work done by a cell of emf ϵ when a charge Q flows through it, as shown in the above diagram, is given by

$$W_{\text{cell}} = \epsilon Q \quad (21)$$

If we have a rechargeable cell and we manage to flow positive charge Q through it in reverse direction (see figure), then the charge loses electric potential energy as it moves from the high potential terminal to the low potential terminal. This energy lost by the charge is gained as chemical energy. The cell gets recharged. We say work done by the cell is negative. Work done on the cell is positive

$$W_{\text{cell}} = -\epsilon Q \quad (22)$$

$$W_{\text{on cell}} = \epsilon Q \quad (23)$$

A group of cells is known as a battery. A group of cells is used for getting higher emf or a longer life.

Example 10 A cell has an emf of 6 V. How much chemical energy the cell loses when a charge $Q = 2\mu\text{C}$ flows out from its positive terminal and re-enters it through its negative terminal.

Solution

Concepts

The cell is getting discharged. Loss in chemical energy = work done by the cell.

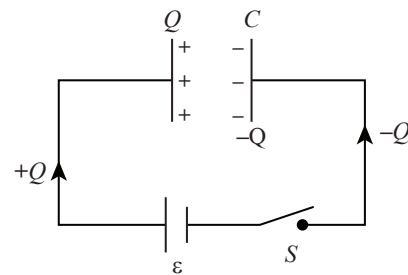
According to the question, a charge Q flows through the cell as shown (circuit has not been shown).



$$\begin{aligned} \text{Loss in chemical energy} &= W_{\text{cell}} = Q \cdot \epsilon \\ &= (2\mu\text{C})(6\text{V}) \\ &= 12\mu\text{J} \end{aligned}$$

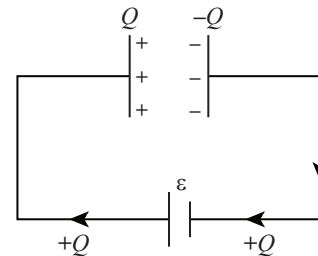
9. CHARGING A CAPACITOR WITH A CELL

The figure shows a capacitor (capacitance = C) connected to a cell (emf = ϵ) through a switch (S). What we have shown is a symbol of a capacitor and it could be a capacitor of any type—parallel plate or spherical or cylindrical. An alternative symbol used to represent a capacitor has also been shown.



Alternative symbol of a capacitor

As soon as the switch (S) is closed, the plates of the capacitor get connected to the terminals of the cell. A potential difference ϵ is established between the plates. This is achieved by the battery supplying charge to the capacitor plates. A charge $+Q$ moves out of the positive terminal of the battery and sits on the left plate of the capacitor. At the same time, a charge $-Q$ moves out of the negative terminal and sits on the other plate. One may also think that the battery sucks $+Q$ charge from the right plate and puts it on the left plate. Both approaches are exactly same. Charge on the capacitor is $Q = C\epsilon$.



One may think that the cell moves $+Q$ charge from one plate to another.

Work done by the cell in the process of charging the capacitor is

$$W = Q \cdot \varepsilon = (C \varepsilon) \varepsilon = C \varepsilon^2 \quad (23)$$

Energy stored in the charged capacitor is

$$U = \frac{1}{2} C \varepsilon^2$$

Energy lost during the charging process is

$$\Delta U_{\text{loss}} = W - U = \frac{1}{2} C \varepsilon^2 \quad (24)$$

Half of the work done by the cell is lost. Where does this energy go? This energy is lost as heat and electromagnetic radiation (i.e., light). As the switch is closed, charge rushes through the connecting wires. There is a current. This causes heating. The charging of a capacitor takes place in a small time interval after the switch is closed (it depends on resistance of the circuit, as we will learn later). After the capacitor is charged, everything is static. Charges do not move. This is electrostatic condition. At present, we are interested in discussing this electrostatic condition only.

After charging the capacitor, if the battery is disconnected, the charges on the capacitor plates will remain there. Now this charged capacitor is a device having electric energy stored in it. If we wish, we can use this energy.

Example 11 *A capacitor can give energy*

A capacitor of capacitance C is charged using a cell of emf V . The cell is disconnected and the capacitor is connected to another cell of emf $\frac{V}{2}$. The positive plate of the capacitor is connected to the positive terminal of the cell and its negative plate is connected to the negative terminal.

- Find the energy lost by the capacitor after it is connected to the second cell.
- Find the energy gained by the cell, assuming it to be an ideal rechargeable cell.

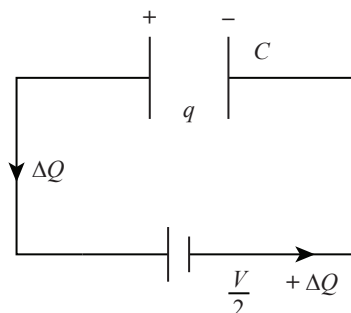
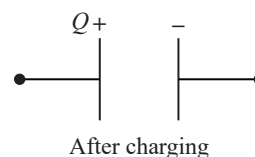
Solution

Concepts

- Charge on the capacitor is $Q = CV$, after it is connected to a cell of emf V . The charge remains on the plate even after the cell is disconnected.
- When connected to the cell of emf $\frac{V}{2}$, the potential difference between the plates becomes $\frac{V}{2}$. For this reason, the capacitor sheds charge.
- The capacitor loses energy and the cell gains energy.

Charge on the capacitor after it is disconnected with the first cell is

$$Q = CV$$



Energy stored in it is

$$U_1 = \frac{1}{2} C V^2$$

After it is connected to the second cell, charge on it is

$$q = C \frac{V}{2}$$

Charge outflow from the capacitor is

$$\Delta Q = Q - q = \frac{CV}{2}$$

We can think that positive charge ΔQ flows out of the positive plate, crosses the cell and moves to the negative plate. This reduces the magnitude of charge on the negative plate by ΔQ . Alternatively, we may think that the positive plate sheds a charge ΔQ and the negative plate also throws out a charge $-\Delta Q$. Both approaches are perfectly fine.

Energy stored in the capacitor is

$$U_2 = \frac{1}{2} C \left(\frac{V}{2} \right)^2 = \frac{1}{8} C V^2$$

- Energy lost by the capacitor is

$$U_1 - U_2 = \frac{3}{8} C V^2$$

- Energy gained by the cell (or, work done on the cell) is

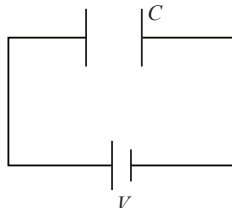
$$\Delta U = \Delta Q \cdot \frac{V}{2} = \frac{CV}{2} \cdot \frac{V}{2} = \frac{1}{4} C V^2$$

Note: The capacitor lost more energy than the energy gained by the cell. The rest was lost as heat.

$$\text{Heat produced} = \frac{3}{8} C V^2 - \frac{1}{4} C V^2 = \frac{C V^2}{8}$$

Example 12 Changing plate separation of a capacitor connected to a cell.

A parallel plate capacitor has capacitance C and is connected to a cell of emf V . The separation between the plates is doubled by slowly pulling them apart. The capacitor remains connected to the cell.



- Calculate change in the energy stored in the capacitor.
- Calculate work done by the cell during the process of separation of the plates.
- How much heat is dissipated when the plates are being separated?
- Calculate work done by the external agent in doubling the separation.

Solution

Concepts

- As the plate separation increases, the capacitance decreases. Therefore, the charge on the capacitor plates keeps on decreasing.
- As the capacitor loses charge, the cell gets recharged.
- Plates are moved apart slowly. This means that the capacitor loses charge very slowly. Actually, current in the circuit is almost zero.

$$(a) U_i = \frac{1}{2} CV^2$$

$$U_f = \frac{1}{2} \frac{C}{2} V^2$$

$$\therefore U = U_f - U_i = -\frac{1}{4} CV^2$$

Negative sign indicates that there is decrease in energy stored in the capacitor.

- Initial charge on the capacitor is $Q_i = CV$

$$\text{Final charge on the capacitor is } Q_f = \frac{C}{2} V$$

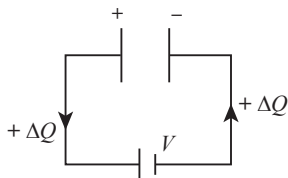
$$\text{Charge flow, } \Delta Q = \frac{CV}{2}$$

$$W_{\text{cell}} = -\Delta QV$$

$$= -\frac{CV^2}{2}$$

Negative sign indicates that work has been done on the cell. The cell has gained energy.

- As the charge moves very slowly, current in the circuit is nearly zero. No heat is dissipated in such a situation. This is important and we should remember this.



- If we assume the capacitor as our system, then there are two external agencies that can perform work on it—

- The cell.
- The agent pulling the plates.

$$\therefore W_{\text{cell}} + W_{\text{extAgt}} = \Delta U$$

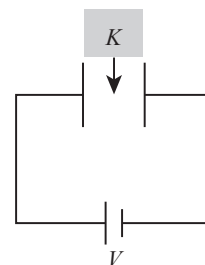
$$\Rightarrow -\frac{CV^2}{2} + W_{\text{extAgt}} = -\frac{CV^2}{4}$$

$$W_{\text{extAgt}} = \frac{CV^2}{4}$$

In simple language, work done by the agent plus the loss in energy of the capacitor $\left(\frac{CV^2}{4} + \frac{CV^2}{4} = \frac{CV^2}{2} \right)$ is equal to gain in energy of the cell.

Example 13 Inserting a dielectric inside a capacitor connected to a cell.

A parallel plate capacitor having capacitance C is connected to a cell having emf V . A dielectric block is slowly inserted between the plates so as to occupy the complete space between the plates.



- Find work done by the cell during the process the dielectric slab is being inserted.
- Find increase in electric energy stored in the capacitor.
- Find work done by the external agent in inserting the dielectric. Neglect gravity.

Solution

Concepts

- Capacitance increases after the dielectric is inserted. Therefore, charge on the capacitor increases.
- The cell does positive work in supplying charge to the capacitor.
- No heat is dissipated as the process is slow.
- $W_{\text{cell}} + W_{\text{extAgt}} = \Delta U$

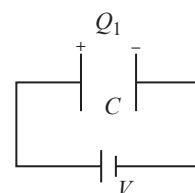
- Initial charge on the capacitor is:

$$Q_1 = CV$$

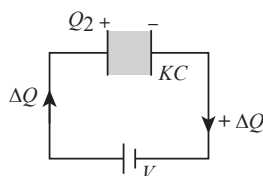
$$\text{Capacitance after the dielectric is inserted} = KC$$

$\therefore$ Charge on capacitor after the insertion of the dielectric is:

$$\begin{aligned} Q_2 &= KC \cdot V \\ W_{\text{cell}} &= \Delta Q \cdot V \\ &= (Q_2 - Q_1)V \\ &= (K - 1)CV^2 \end{aligned}$$



$$\begin{aligned}
 \text{(b)} \quad \Delta U &= U_2 - U_1 \\
 &= \frac{1}{2}(KC)V^2 - \frac{1}{2}CV^2 \\
 &= \frac{1}{2}(K-1)CV^2
 \end{aligned}$$

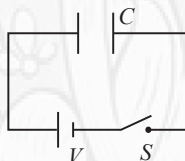


$$\begin{aligned}
 \text{(c)} \quad W_{\text{extAgt}} + W_{\text{cell}} &= \Delta U \\
 \Rightarrow W_{\text{extAgt}} &= \Delta U - W_{\text{cell}} \\
 &= \frac{1}{2}(K-1)CV^2 - (K-1)CV^2 \\
 &= -(K-1)CV^2
 \end{aligned}$$



YOUR TURN

Q.11 A capacitor is connected to a cell through a switch. After the switch is put on, $50\mu\text{J}$ of heat is dissipated in the circuit. What is loss in chemical energy of the cell?



Q.12 In the last question, find the emf of the cell if charge on the capacitor after the switch is closed is $5\mu\text{C}$ and heat dissipated is $50\mu\text{J}$.

Q.13 A capacitor of capacitance C is charged using a cell of emf V and the cell is disconnected. This charged capacitor is now connected across another cell of emf $2V$. Positive and

negative plates are connected to the positive and negative terminals of the cell, respectively. Find the ratio of work done by the two cells used to charge the capacitor.

Q.14 A capacitor of capacitance $10\mu\text{F}$ is connected to a 20V cell. The separation between the plates is reduced by half during a slow process.

- Find work done by the cell during the process.
- Find work done by the external agent during the process.
- Find the ratio of energy density in electric field between the capacitor plates, before and after the plate separation is changed.

In Short

- A cell is a neutral entity. If a charge Q flows out of a terminal of a cell, then the same charge Q must return back into the cell through the other terminal (in other words, a charge $-Q$ flows out of the other terminal).
- Work done by a cell when a charge $+Q$ flows out of its positive terminal (and $-Q$ flows out of the negative terminal) is $W = QV$. This is actually loss in stored energy of the cell.
If charge $+Q$ enters into the positive terminal (and $-Q$ flows into the negative terminal) of the cell, work done by the cell is $W = -QV$. Actually the cell gets recharged and gains energy equal to QV .
- When charge moves rapidly in a circuit, there is current and heat is generated.
- When a capacitor is connected to a cell, the potential difference between the plates becomes equal to the emf (V) of the cell. As long as the capacitor remains connected to the cell, potential difference between the plates remains V , no matter whether the plate separation is changed or a dielectric is moved in/out of the capacitor.
- While charging a capacitor, 50% of the work done by the cell is lost as heat and radiation.
- When an external agent is involved in slowly changing the plate separation or moving a dielectric slab in/out of the capacitor, as it remains connected to a cell, the energy equation can be written as

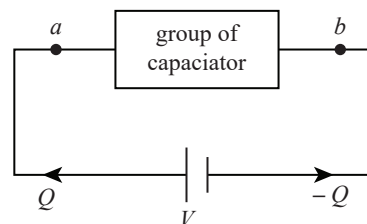
$$W_{\text{extAgt}} + W_{\text{cell}} = \Delta U \quad (25)$$

Here, ΔU is change in energy of the capacitor.

10. GROUPING OF CAPACITORS

Two or more capacitors can be connected in many ways. Consider a battery of emf V connected across two points (a and b) to a group of capacitors. The battery supplies a charge Q to the capacitors group. We say that the equivalent capacitance of the group of capacitors is

$$C_0 = \frac{Q}{V} \quad (26)$$



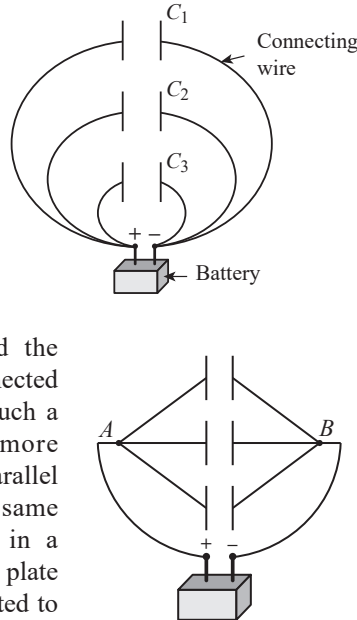
The box contains a group of capacitors.
The battery supplies a charge Q to this group.

A combination of capacitors can always be replaced with a single equivalent capacitance in any circuit. *Energy stored in the equivalent capacitor* ($U = \frac{1}{2}C_0V^2$) is equal to sum of energy stored in individual capacitors in the group.

We will first study about the two common methods of grouping of capacitors – parallel and series grouping.

10.1 Parallel combination of capacitors

Suppose we have three capacitors (number of capacitors can be even more). We wish to apply same potential difference across all of them. We can connect them all in *parallel* to a single battery. The figure shows that one plate of each capacitor is connected to the positive terminal of a battery and the other plates have been connected to the negative terminal. Such a grouping of two or more capacitors is known as parallel grouping. However, the same connection can be made in a slightly different way. One plate of each capacitor is connected to a fixed point A and the other plates are connected to another fixed point B . Now the battery is connected across the two points A and B . Points A and B acquire the potential of the battery electrodes. Therefore, potential difference between A and B is same as emf of the battery. All capacitors have same potential difference across them.



We will show the circuit of above type in form of a better looking diagram as shown in the figure. Now, let us find the equivalent capacitance of three capacitors connected in parallel. Potential difference across each capacitor is V . Therefore, charges on them will be

$$Q_1 = C_1 V$$

$$Q_2 = C_2 V$$

$$Q_3 = C_3 V$$

Charge supplied by the battery to the group is

$$Q = Q_1 + Q_2 + Q_3$$

$$\text{or, } Q = (C_1 + C_2 + C_3)V$$

$$\text{or, } \frac{Q}{V} = C_1 + C_2 + C_3$$

$$\text{or, } C_0 = C_1 + C_2 + C_3 \quad (26)$$

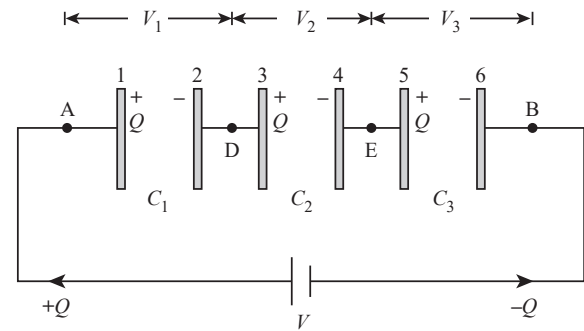
C_0 is the equivalent capacitance of the group of three capacitors connected in parallel.

Charge Q supplied by the positive terminal of the battery gets divided into three plates as Q_1 , Q_2 and Q_3 . The battery also supplies charge $-Q$ out of its negative terminal. This

charge gets distributed on the three plates as $-Q_1$, $-Q_2$ and $-Q_3$. The opposite plates of a capacitor always have equal and opposite charge.

10.2 Series combination of capacitors

The diagram shows three capacitors connected in series. The capacitances are C_1 , C_2 and C_3 . When a battery of emf V is connected across A and B , its positive terminal supplies a charge $+Q$ and the negative terminal supplies a charge $-Q$. Charge $+Q$ rests on plate 1 and $-Q$ goes to plate 6. The positive charge on plate 1 induces a negative charge $-Q$ on plate 2. Charge $+Q$ on plate 1 and $-Q$ on 2 attract one another and reside on the inner faces of the plates. Plates 2 and 3 are connected to one another but isolated from the outside objects. The sum of charges on these two plates must always be zero.



All capacitors in series have same charge on them

Therefore, a charge $+Q$ appears on plate 3. This charge attracts electrons in plate 4 and induces $-Q$ charge on it. Plates 4 and 5 make an isolated system and hence a charge $+Q$ appears on plate 5. Plate 6 already has charge $-Q$. This way, *all capacitors connected in series acquire same charge*.

Now, we will find the equivalent capacitance of this series combination. As one moves from point A to D (i.e., from plate 1 to 2), there is a drop in potential equal to

$$V_1 = \frac{Q}{C_1}$$

This is nothing but potential difference between the plates of the first capacitor. Whenever we move from the positive to the negative plate of a capacitor, the potential will decrease.

Similarly, the potential drop when one moves from D to E is $V_2 = \frac{Q}{C_2}$ and the potential drop when one moves from E to B is $V_3 = \frac{Q}{C_3}$

Total drop in potential when one moves from A to B is V (equal to the emf of the battery)

$$\therefore V = V_1 + V_2 + V_3$$

$$\text{or, } V = \frac{Q}{C_1} + \frac{Q}{C_2} + \frac{Q}{C_3}$$

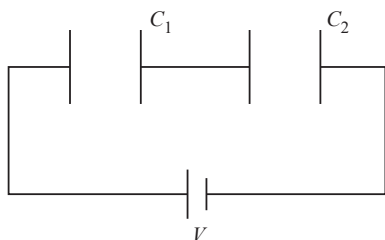
$$\text{or, } \frac{V}{Q} = \frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3}$$

The equivalent capacitance is $C_0 = \frac{Q}{V}$. Therefore,

$$\frac{1}{C_0} = \frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3} \quad (27)$$

Example 14 Potential difference across capacitors in series.

Two capacitors having capacitance C_1 and C_2 are connected in series to a cell of emf V . Find the charge on each capacitor and potential difference across each of them.

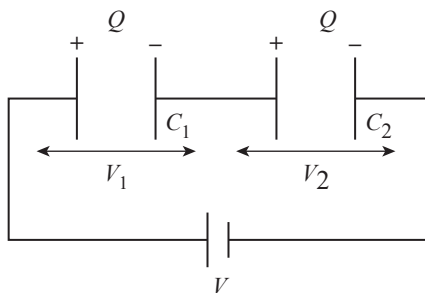


Solution

Concepts

We will find the equivalent capacitance (C_0). Charge supplied by the cell will be $Q = C_0 V$. In series connection, all capacitors will have same charge Q .

Equivalent capacitance is given by



$$\frac{1}{C_0} = \frac{1}{C_1} + \frac{1}{C_2}$$

$$\text{or, } C_0 = \frac{C_1 C_2}{C_1 + C_2}$$

Charge supplied by the cell is

$$Q = C_0 V = \frac{C_1 C_2 V}{C_1 + C_2}$$

Both the capacitors have same charge Q on them. The potential difference across them are

$$V_1 = \frac{Q}{C_1} = \left(\frac{C_2}{C_1 + C_2} \right) V$$

$$V_2 = \frac{Q}{C_2} = \left(\frac{C_1}{C_1 + C_2} \right) V$$

Alternate Method

One can observe that the potential difference across a capacitor is inversely proportional to its capacitance.

$$V_1 = \frac{Q}{C_1} \quad \text{and} \quad V_2 = \frac{Q}{C_2}$$

$$\text{or, } \frac{V_1}{V_2} = \frac{C_2}{C_1} \quad (i)$$

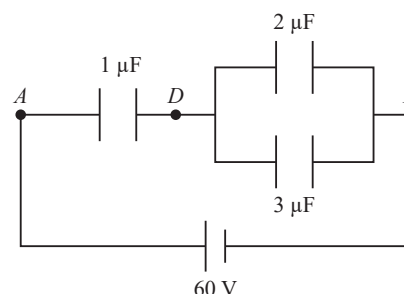
$$\text{Also } V_1 + V_2 = V \quad (ii)$$

Solving (i) and (ii) gives

$$V_1 = \left(\frac{C_2}{C_1 + C_2} \right) V \quad \text{and} \quad V_2 = \left(\frac{C_1}{C_1 + C_2} \right) V \quad (28)$$

Example 15 In the circuit shown in the figure, find

- equivalent capacitance between points A and B,
- charge on each of the three capacitors.



Solution

Concepts

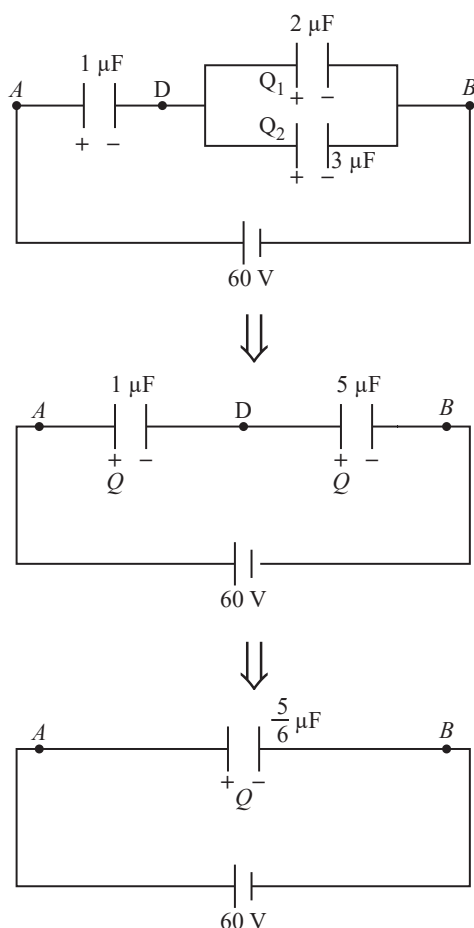
- We will find the equivalent capacitance in two steps. Note that $2\mu\text{F}$ and $3\mu\text{F}$ capacitors are in parallel and their equivalent is in series with $1\mu\text{F}$.
- Charge on the equivalent capacitor will be $Q = C_0 V$. To find charge on capacitors, we will work in reverse from the equivalent circuit to the given circuit as explained below.

- We can replace the $2\mu\text{F}$ and $3\mu\text{F}$ capacitors with a single capacitor between points D and B. This single capacitor has capacitance

$$2 + 3 = 5\mu\text{F}$$

Now, $1\mu\text{F}$ and $5\mu\text{F}$ capacitors are in series. Their equivalent capacitance is

$$C_0 = \frac{(1\mu\text{F})(5\mu\text{F})}{(1+5)\mu\text{F}} = \frac{5}{6}\mu\text{F}$$



This capacitor is connected to a cell of emf 60V.

∴ Charge on it is

$$Q = C_0 V = \frac{5}{6} \mu\text{F} (60\text{V})$$

$$= 50\mu\text{C}$$

(b) The two capacitors ($1\mu\text{F}$ and $5\mu\text{F}$ shown in the second figure) will have charge Q on each of them.

Potential difference between D and B will be

$$V_{DB} = \frac{Q}{5\mu\text{F}} = \frac{50\mu\text{C}}{5\mu\text{C}} = 10\text{V}$$

[Obviously, $V_{AD} = 50\text{V}$]

Now, look at the first diagram. The potential difference V_{DB} is known. Therefore, we can write charge on each of the two capacitors in parallel.

Charge on $2\mu\text{F}$ capacitor is

$$Q_1 = (V_{DB})(2\mu\text{F})$$

$$= (10\text{V})(2\mu\text{F}) = 20\mu\text{C}$$

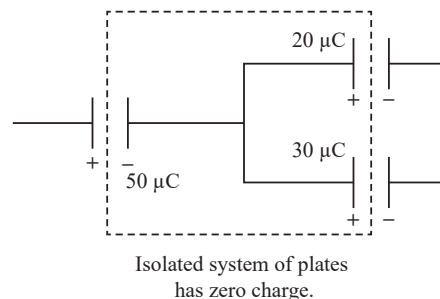
Charge on $3\mu\text{F}$ capacitor is

$$Q_2 = (10\text{V})(3\mu\text{F})$$

$$= 30\mu\text{C}$$

We have already decided that the charge on $1\mu\text{F}$ capacitor is $50\mu\text{C}$.

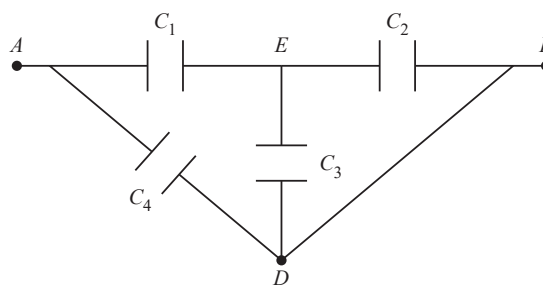
Note: The negative plate of $1\mu\text{F}$ capacitor and the two positive plates of the other capacitors make an isolated system. They can share charge among them but they cannot receive charge from outside. Sum of charges on these three plates must be zero.



$$-50\mu\text{C} + 20\mu\text{C} + 30\mu\text{C} = 0$$

The negative terminal of the battery supplies $-50\mu\text{C}$ charge that gets distributed on the two negative plates of $2\mu\text{F}$ and $3\mu\text{F}$ capacitors. The $50\mu\text{C}$ charge gets divided so that the potential difference across both the capacitors is same.

Example 16 A capacitor network has four capacitors. All capacitance are equal; $C_1 = C_2 = C_3 = C_4 = C$. Find the equivalent capacitance between points A and B .



Solution

Concepts

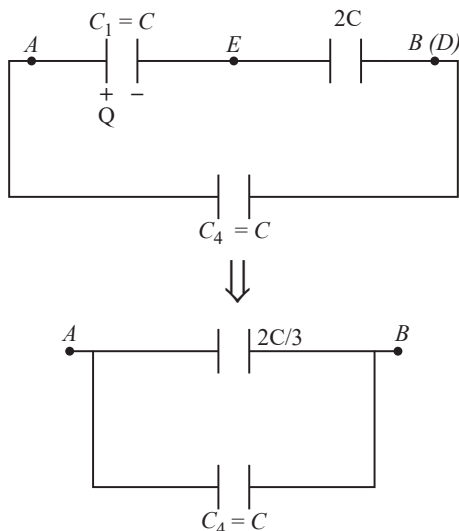
- A common mistake is to consider C_1 and C_2 in series. There is branching at E (i.e., another branch of the circuit joins the wire connecting C_1 and C_2). In such a case, charges on C_1 and C_2 may not be same. Similarly, C_1 and C_4 are not in series. Battery is connected across A and B and one can see a branch going out from A (to the battery).
- All points on the connecting wire BD are at same potential. In fact, B and D are same points. [Similarly, the three plates connected to point E are at same potential]. We must remember that *all points connected by a conducting wire (with no battery, capacitor or resistance in between) are at same potential*.

- (iii) Capacitors C_2 and C_3 are in parallel. One plate of both the capacitors is connected to E and their other plates are connected to B (or D).

C_2 and C_3 are in parallel. Their equivalent is

$$C_2 + C_3 = C + C = 2C$$

We can replace C_2 and C_3 with a single capacitor between E and B and redraw the circuit as shown. Note that C_4 is connected between A and B (B and D are same points).



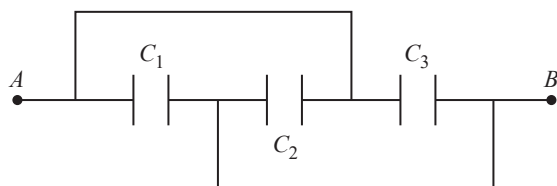
Now, C_1 and $2C$ are in series. Their equivalent is

$$\frac{C(2C)}{C+2C} = \frac{2C}{3}$$

Finally, $\frac{2C}{3}$ and C_4 are in parallel. The equivalent is

$$C_0 = \frac{2C}{3} + C = \frac{5C}{3}$$

Example 17 Find the equivalent capacitance between points A and B for the network shown.

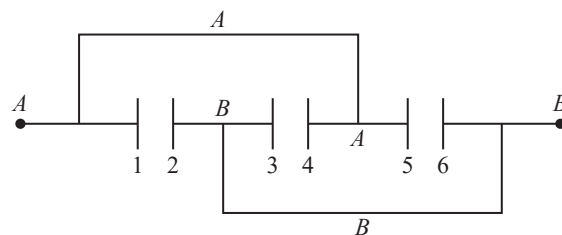


Solution

Concepts

All points connected directly by a conducting wire are at same potential.

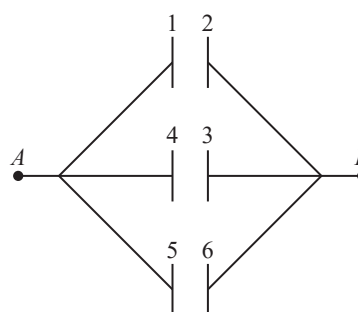
All the points marked as A in the figure are same points. Similarly, the points marked as B are same points.



Plates 1, 4 and 5 are connected to point A and the points 2, 3 and 6 are connected to point B . Therefore, the three capacitors are in parallel.

The equivalent capacitance is

$$C_0 = C_1 + C_2 + C_3$$



Example 18 Breakdown in series connection

A capacitor is marked as $10\mu\text{F}$, 100V (it means the breakdown voltage for the capacitor is 100V). Another capacitor is marked $20\mu\text{F}$, 100V . The two capacitors are connected in series. Find the maximum potential difference that can be applied across this series combination so that none of the capacitors is punctured.

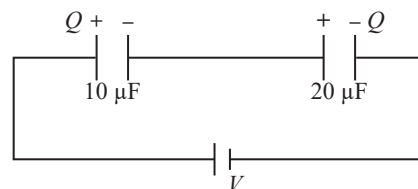
Solution

Concepts

In series combination, charge will be same on the two capacitors. This charge should not exceed the maximum allowed charge on both the capacitors.

Maximum allowed charge on the first capacitor is

$$(Q_1)_{\max} = C_1 V_{\max} = (10\mu\text{F})(100\text{V}) = 1000 \mu\text{C}$$



Maximum allowed charge on the second capacitor is

$$\begin{aligned} (Q_2)_{\max} &= C_2 V_{\max} \\ &= (20\mu\text{F})(100\text{V}) \\ &= 2000 \mu\text{C} \end{aligned}$$

4.20 Electrostatics and Current Electricity

When in series, the two capacitors have same charge. This charge should not exceed $1000\mu\text{C}$ if both the capacitors are to be safe.

$$\therefore Q_{\max} = 1000\mu\text{C}$$

Potential difference across two capacitors will be

$$V_1 = \frac{1000\mu\text{C}}{10\mu\text{F}} = 100\text{ V}$$

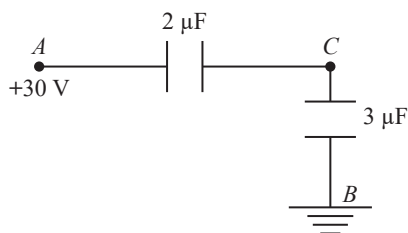
$$V_2 = \frac{1000\mu\text{C}}{20\mu\text{F}} = 50\text{ V}$$

$\therefore$ Applied maximum potential difference is

$$V = V_1 + V_2 = 150\text{ V}$$

Example 19 *This is not an open circuit*

In the arrangement shown, point B is grounded and point A is maintained at $+30\text{V}$. Find charge on $2\mu\text{F}$ capacitor. Find potential of point C .

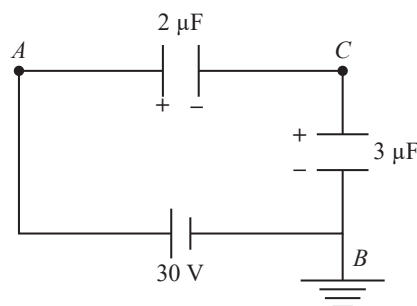


Solution

Concepts

Apparently the circuit is open. But it is not so. Potential at B is zero and to keep the potential of A at 30V above B , we will need a battery. Actually, there is a battery (or any other agency which can maintain a potential difference) between A and B .

For convenience, we can redraw the circuit as shown. Equivalent capacitance between A and B is



$$\frac{2 \times 3}{2 + 3} = \frac{6}{5} \mu\text{F}$$

$$\text{Charge supplied by the battery is } Q = \left(\frac{6}{5} \mu\text{F} \right) (30\text{ V})$$

$$= 36\text{ }\mu\text{C}.$$

Both capacitors have charge Q on them. Potential difference between A and C is

$$V_A - V_C = \frac{36\mu\text{C}}{2\mu\text{F}} = 18\text{ V}$$

Because $V_A = 30\text{ V}$

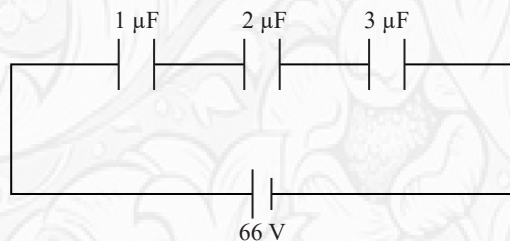
$$\therefore 30 - V_C = 18$$

$$\text{or, } V_C = 12\text{ V}$$



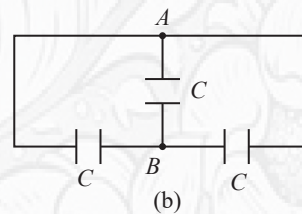
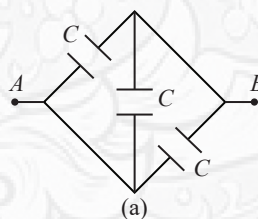
YOUR TURN

Q.15 Find charge on each of the three capacitors shown in the figure. What is potential difference across each of them?



Q.16 An engineer has two capacitors. Using one or both of them, he is able to obtain capacitance values of $25\mu\text{F}$, $20\mu\text{F}$, $5\mu\text{F}$ and $4\mu\text{F}$. What is the capacitance of the two capacitors?

Q.17 Find equivalent capacitance between points A and B in the following circuits.

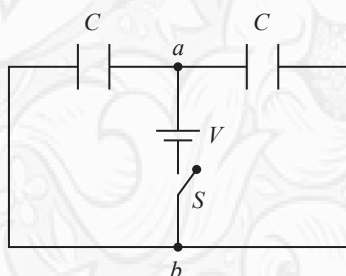


Q.18 In the circuit shown in the figure, capacitance of each capacitor is C . After this the switch is closed.

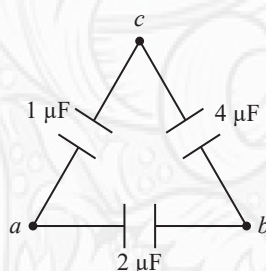
(a) Find the charge that flows through the cell

(b) Find the electric energy stored in the two capacitors

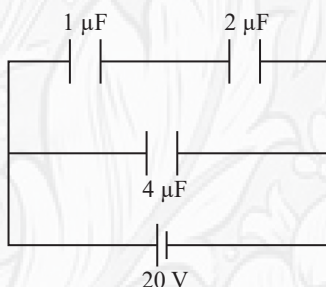
(c) Find heat generated in the circuit.



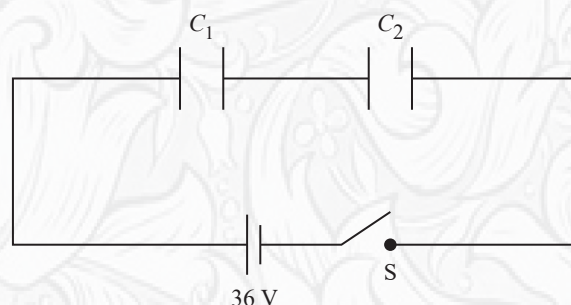
Q.19 In the network shown in the figure, the equivalent capacitance between points a and b is C_1 and the equivalent capacitance between the points a and c is C_2 . Find $\frac{C_1}{C_2}$



Q.20 In the circuit shown, how will the charge on $2\mu\text{F}$ capacitor change if the $4\mu\text{F}$ capacitor is removed? Find potential difference across $2\mu\text{F}$ capacitor.

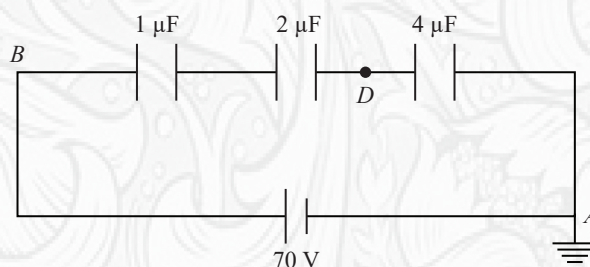


Q.21 Capacitor C_1 has capacitance $1\mu\text{F}$ and its breakdown voltage is 20 V . Capacitor C_2 has capacitance $2\mu\text{F}$ and its breakdown voltage is also 20V .



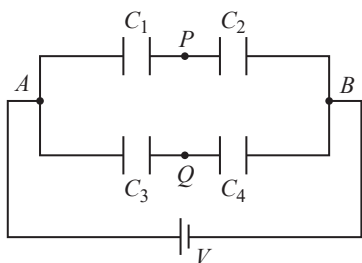
Which capacitor will breakdown after the switch S is closed?

Q.22 In the circuit shown, find potential of point D .



11. BALANCED WHEATSTONE BRIDGE OF CAPACITORS

Consider the circuit shown in the figure. The two parallel branches are independent of one another. The potential difference across the series combination of C_1 and C_2 is V . The potential difference across C_1 is



$$V_1 = V_A - V_P$$

$$= \left(\frac{C_2}{C_1 + C_2} \right) V \quad \text{[using equation (28)]}$$

Similarly, the potential difference across the series combination of C_3 and C_4 is V . The potential difference across C_3 is

$$V_3 = V_A - V_Q = \left(\frac{C_4}{C_3 + C_4} \right) V$$

Points P and Q will be at same potential if

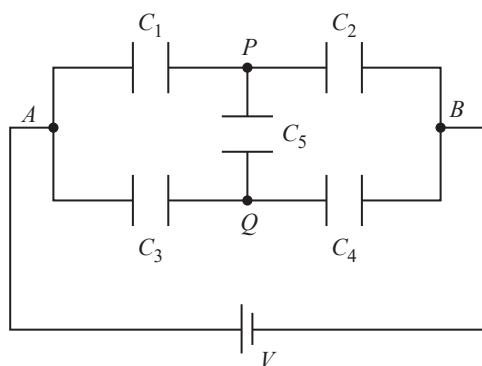
$$V_1 = V_3$$

$$\text{or, } \frac{C_2 V}{C_1 + C_2} = \frac{C_4 V}{C_3 + C_4}$$

$$\text{or, } \frac{C_1 + C_2}{C_2} = \frac{C_3 + C_4}{C_4}$$

$$\text{or, } \frac{C_1}{C_2} = \frac{C_3}{C_4} \quad (29)$$

Therefore, we conclude that when condition (29) is satisfied, the potentials at the points P and Q are same. Under such condition, if we connect a capacitor (C_5) between points P and Q , it will have no charge on it. Its presence or absence will make no difference to the circuit.



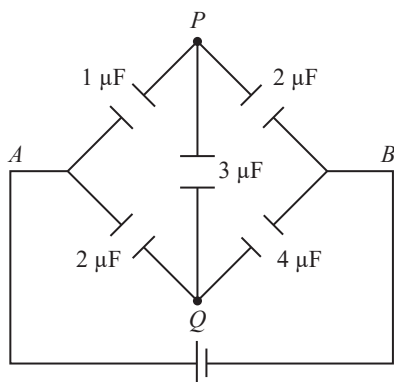
Wheatstone bridge

The circuit consisting of five capacitors as shown in the figure is known as *Wheatstone bridge* of capacitors. The bridge is said to be balanced when there is no charge on C_5 (i.e., when $V_P = V_Q$). The necessary condition for a balanced bridge is given by equation (29), i.e.,

$$\frac{C_1}{C_2} = \frac{C_3}{C_4}$$

In general, a Wheatstone bridge is a complex circuit (no two capacitors are in series or parallel). But when the ratio $C_1:C_2$ is same as $C_3:C_4$, the bridge is balanced. In this situation, there is no charge on C_5 and it can be removed from the circuit.

Example 20 Find the equivalent capacitance of the network between points A and B .



Solution

Concepts

- In a balanced Wheatstone bridge, the fifth capacitor (between P and Q) can be removed.
- All points having same potential can be tied together in any electric circuit. It will make no difference to the circuit. We have used this in the alternative method given below.

The given circuit is a Wheatstone bridge. It is balanced because

$$\frac{1\mu F}{2\mu F} = \frac{2\mu F}{4\mu F}$$

The $3\mu F$ capacitor has no charge ($V_P = V_Q$). We can remove it and redraw the circuit as shown.

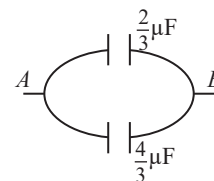
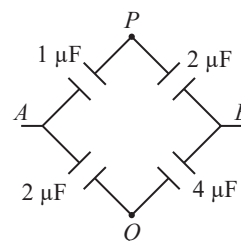
Now, $1\mu F$ and $2\mu F$ capacitor in series have an equivalent of

$$\frac{1 \times 2}{1 + 2} = \frac{2}{3} \mu F$$

The $2\mu F$ and $4\mu F$ capacitors in series have an equivalent of $\frac{4}{3} \mu F$.

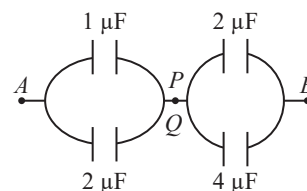
Equivalent of the two capacitors in parallel is

$$C_0 = \frac{2}{3} + \frac{4}{3} = \frac{6}{3} = 2\mu F$$



Alternate Method

Points P and Q are equipotential points. We can redraw the circuit by joining them together.



Points P and Q can be joined together as they are at same potential.

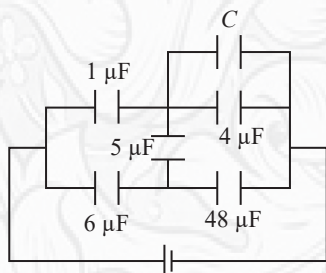
In this circuit, we have $1\mu F$ and $2\mu F$ capacitors in parallel having an equivalent of $3\mu F$. Note that $2\mu F$ and $4\mu F$ capacitors in parallel have an equivalent of $6\mu F$. Now, $3\mu F$ and $6\mu F$ are in series.

$$C_0 = \frac{3 \times 6}{3 + 6} = 2\mu F$$

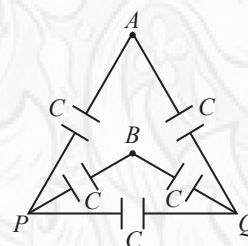


YOUR TURN

Q.23 For what value of capacitance C , there will be no charge on $5\mu\text{F}$ capacitor?



Q.24 Find the equivalent capacitance between points A and B for the network shown in the figure.



In Short

- All capacitors in parallel have same potential difference across them.
- All capacitors in parallel have charges proportional to their capacitance.
- All capacitors in series have same charge on them. Potential difference across each capacitor in series is inversely proportional to the capacitance.
- If a group of capacitors draws a charge Q when connected to a cell of emf V , their equivalent capacitance is $C_0 = \frac{Q}{V}$.

- For n capacitors in parallel, the equivalent capacitance is given by

$$C_0 = C_1 + C_2 + C_3 + \dots + C_n$$

- For n capacitors in series, the equivalent capacitance is given by

$$\frac{1}{C_0} = \frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3} + \dots + \frac{1}{C_n}$$

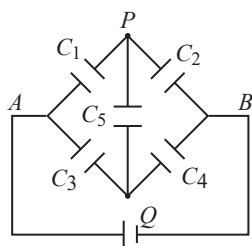
- Equivalent of n number of identical capacitors (each of capacitance C) connected in series is $C_0 = \frac{C}{n}$.

- The equivalent capacitance of capacitors in series is always smaller than all individual capacitances.

- A Wheatstone bridge is arrangement of five capacitors as shown. It is balanced when

$$\frac{C_1}{C_2} = \frac{C_3}{C_4}$$

In a balanced bridge, $V_P = V_Q$ and C_5 has no charge on it.

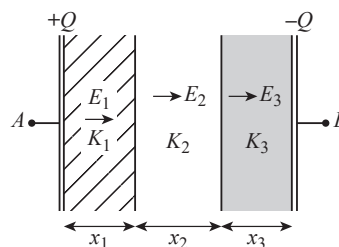


12. USE OF SERIES-PARALLEL GROUPING TO FIND CAPACITANCE OF A CAPACITOR FILLED WITH MULTIPLE DIELECTRICS

Capacitance of a parallel plate capacitor filled with multiple dielectrics can be calculated using series-parallel grouping.

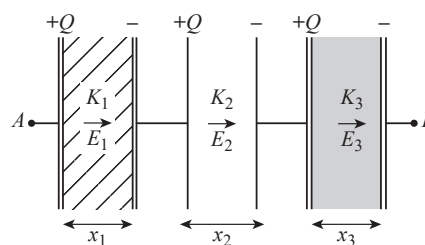
12.1 Using series grouping

Consider the problem that we discussed in section 4.3. Plate area of the capacitor is A and the space between them is filled using three dielectrics of thickness x_1 , x_2 and x_3 having constants K_1 , K_2 and K_3 , respectively. We can consider this capacitor to be made of three capacitors connected in series as shown in the figure.



If we give a charge Q to the original capacitor, the electric field inside the three dielectrics will be

$$E_1 = \frac{Q}{AK_1\epsilon_0}, E_2 = \frac{Q}{AK_2\epsilon_0} \text{ and } E_3 = \frac{Q}{AK_3\epsilon_0}$$



In the figure showing three capacitors in series, if a charge Q is given to the system, the electric field in the space between the plates of the three capacitors will be given by above expressions only. Therefore, for a charge Q , we get same potential difference between A and B in both the diagrams. Obviously, the effective capacitance $\left(\frac{Q}{V_{AB}}\right)$ is same in both cases.

Capacitances of the three capacitors shown in the second figure are

$$C_1 = \frac{K_1 \epsilon_0 A}{x_1}, C_2 = \frac{K_2 \epsilon_0 A}{x_2} \text{ and } C_3 = \frac{K_3 \epsilon_0 A}{x_3}$$

Capacitance of the original capacitor can be calculated as

$$\frac{1}{C} = \frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3}$$

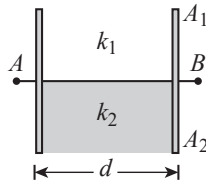
$$\text{or, } \frac{1}{C} = \frac{1}{\epsilon_0 A} \left[\frac{x_1}{K_1} + \frac{x_2}{K_2} + \frac{x_3}{K_3} \right]$$

$$\text{or, } C = \frac{\epsilon_0 A}{\frac{x_1}{K_1} + \frac{x_2}{K_2} + \frac{x_3}{K_3}}$$

This is same as equation (12).

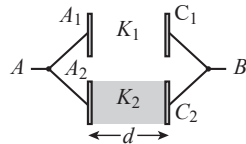
12.2 Using parallel grouping

A parallel plate capacitor has the space between its plates filled with two dielectrics as shown. Dielectric of constant K_1 fills the space between the plates covering area A_1 of the plates. The other dielectric of constant K_2 fills the remaining space covering area A_2 of the two plates.



The capacitor can be assumed to be a parallel combination of two capacitors. Capacitances of the two capacitors are given as

$$C_1 = \frac{K_1 \epsilon_0 A_1}{d} \text{ and } C_2 = \frac{K_2 \epsilon_0 A_2}{d}$$



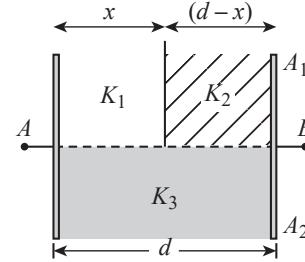
The capacitance of the original capacitor is

$$C = C_1 + C_2 = \frac{K_1 \epsilon_0 A_1}{d} + \frac{K_2 \epsilon_0 A_2}{d}$$

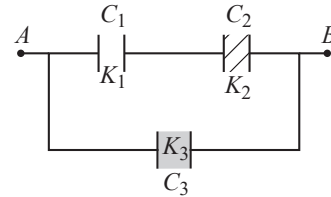
When a battery is connected across A and B , the two capacitors (C_1 and C_2) will acquire different charges, in general. The charge density on area A_1 and A_2 of a plate will also be different.

12.3 Mixed grouping

The space between the plates of a parallel plate capacitor has been filled using three different dielectrics as shown in the figure.



We can find the capacitance by assuming a capacitor (C_3) to be in parallel with two capacitors (C_1 and C_2) in series.



$$C_1 = \frac{K_1 \epsilon_0 A_1}{x}, C_2 = \frac{K_2 \epsilon_0 A_1}{d-x},$$

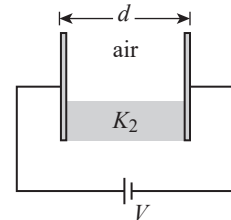
$$C_3 = \frac{K_3 \epsilon_0 A_2}{d}$$

The capacitance of the given capacitor is

$$C = \frac{C_1 C_2}{C_1 + C_2} + C_3$$

Example 21 Charge density on plate

A parallel plate capacitor has one-third of the space between the plates filled with a dielectric constant $K = 2$, as shown in the figure. The capacitor is connected to a cell.



Find the ratio of charge densities on the part of its plate facing air and the part in contact with the dielectric.

Solution

Concepts

The electric field between the plates (inside air as well as dielectric) is $E = \frac{V}{d}$. One can also see this as two capacitors in parallel.

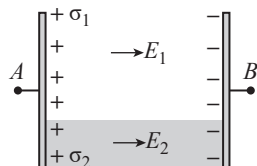
Let the charge density on the upper part of the plate (where there is air) be σ_1 and the charge density on the lower part (where there is dielectric) be σ_2 .

$$E_1 = \frac{\sigma_1}{\epsilon_0} \text{ and } E_2 = \frac{\sigma_2}{K\epsilon_0}$$

But $E_1 = E_2$

$$\text{or, } \frac{\sigma_1}{\epsilon_0} = \frac{\sigma_2}{K\epsilon_0}$$

$$\text{or, } \frac{\sigma_1}{\sigma_2} = \frac{1}{K} = \frac{1}{2}$$



Example 22 *This is not very simple!*

Three different dielectric materials (constants K_1 , K_2 and K_3) have been used to fill the space between the plates of a parallel plate capacitor as shown in the figure. Each of the slabs having constants K_1 and K_2 cover half the plate area.

Find the capacitance. Given $K_1 = K$, $K_2 = 2K$ and $K_3 = 3K$.

Solution

Concepts

- (i) One can commit a mistake here. Taking two capacitors (filled with dielectrics of constants K_1 and K_2) in parallel and combining their equivalent in series with the third capacitor (filled with dielectric of constant K_3) is wrong. Two capacitors will be in parallel only if potential difference across them is same. In this question, the charge density on two halves of the capacitor plate will not be same. The electric field inside the dielectrics of constant K_1 and K_2 may not be same, in general. The field inside two halves of dielectric slab having constant K_3 will also be different if the charge densities in the upper half and lower half of the plates are different.

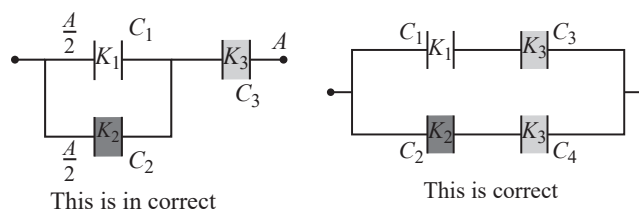
Therefore, walking a distance d in the first dielectric will not result in same potential drop as walking a distance d in the second dielectric.

- (ii) We will have to take the entire capacitor as combination of four capacitors. The upper half has two capacitors in series and the lower half has other two capacitors in series.

The first diagram shows the incorrect combination of capacitors. The capacitors C_1 and C_2 cannot be taken in parallel as we are not sure that they have same potential difference across them.

The correct representation has been shown in the second figure.

$$C_1 = \frac{K_1 \epsilon_0 A/2}{d} = \frac{K \epsilon_0 A}{2d}$$



$$C_2 = \frac{K_2 \epsilon_0 A/2}{d} = \frac{2K \epsilon_0 A}{2d} = 2C_1$$

$$C_3 = \frac{K_3 \epsilon_0 A/2}{d} = \frac{3K \epsilon_0 A}{2d} = 3C_1$$

$$C_4 = \frac{K_3 \epsilon_0 A/2}{d} = 3C_1$$

$$\text{Equivalent of } C_1 \text{ and } C_3 \text{ in series is } \frac{C_1(3C_1)}{C_1 + 3C_1} = \frac{3C_1}{4}$$

$$\text{Equivalent of } C_2 \text{ and } C_4 \text{ in series is } \frac{(2C_1)(3C_1)}{2C_1 + 3C_1} = \frac{6C_1}{5}$$

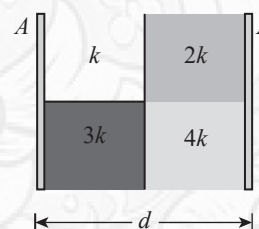
$$\frac{3C_1}{4} \text{ and } \frac{6C_1}{5} \text{ in parallel have a resultant}$$

$$C = \frac{3C_1}{4} + \frac{6C_1}{5} = \frac{39}{20}C_1 = \frac{39K\epsilon_0 A}{40d}$$



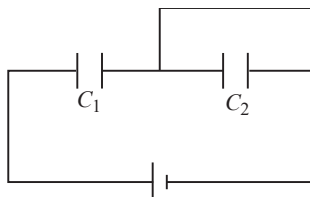
YOUR TURN

Q.25 Find the capacitance of the capacitor shown in the figure. Plate area is A and separation between the plates is d . Four dielectric blocks are equal in size.

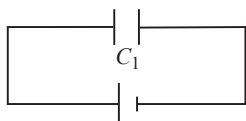


13. SHORT CIRCUITED CAPACITOR

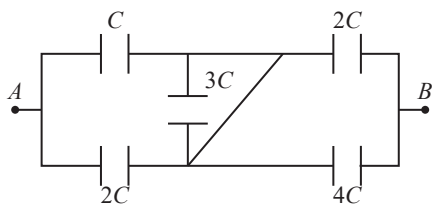
When the two plates of a capacitor are connected by a conducting wire, it is said to be shorted. There is no potential difference between the plates of such a capacitor and it will have no charge. In the figure shown, the capacitor C_2 is short circuited. It has no charge. In fact, it has no role in the circuit. We can remove it and redraw the circuit as shown in the second diagram.



capacitor C_2 is shorted. We can redraw the circuit as shown below



Example 23 Find the equivalent capacitance between points A and B in the figure shown.

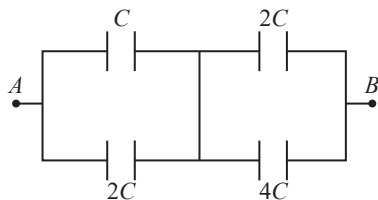


Solution

Concepts

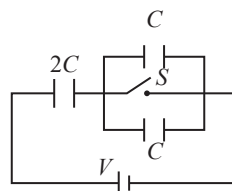
The capacitor of capacitance $3C$ is shorted. We can remove it.

After removing $3C$, the circuit can be redrawn as shown. C and $2C$ in parallel make $3C$. $2C$ and $4C$ in parallel make $6C$. $3C$ and $6C$ in series make



$$C_0 = \frac{(3C)(6C)}{3C + 6C} = 2C$$

Example 24 In the circuit shown in the figure, calculate the work done by the cell after the switch S is closed.

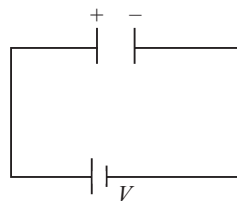


Solution

Concepts

- We will find charge on capacitor having capacitance $2C$ when the switch is open.
- We will find charge on $2C$ after the switch is closed. Closing the switch results in short circuits in both the capacitors in parallel.
- By looking at charge on the positive plate of capacitor having capacitance $2C$, we will decide how much charge flows through the cell after the switch is closed.

When switch is open, the two capacitors in parallel have an equivalent of $2C$. This $2C$ capacitance is in series with the other $2C$ capacitance. The equivalent is C . Charge supplied by the battery is $Q_1 = CV$. The positive plate of the capacitor of capacitance $2C$ has charge $Q = CV$.



After closing the switch

After closing the switch, the two parallel capacitors get shorted. The circuit has only capacitance $2C$.

Charge on the capacitor is

$$Q_2 = 2CV.$$

Charge on the positive plate has increased by

$$\Delta Q = Q_2 - Q_1 = CV.$$

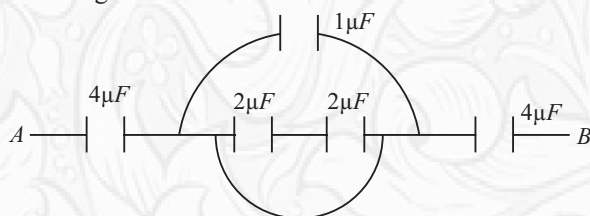
This charge (on the positive plate of $2C$) can only come through the battery. Therefore, the battery supplies $\Delta Q = CV$ charge after ' S ' is closed.

$$W_{\text{cell}} = \Delta Q \cdot V = CV^2$$

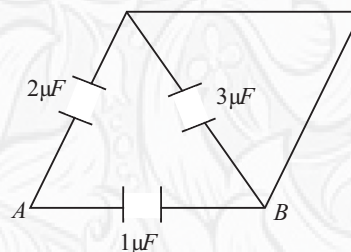


YOUR TURN

Q.26 Find equivalent capacitance between points A and B in the figure shown.

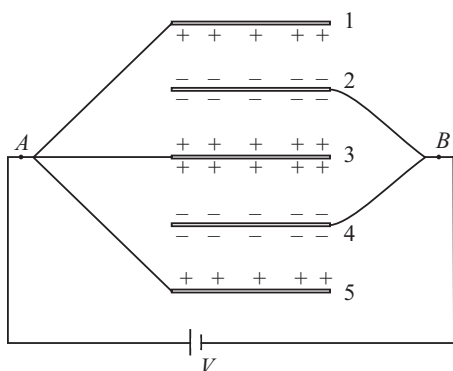


Q.27 Find equivalent capacitance between points A and B in the circuit shown.



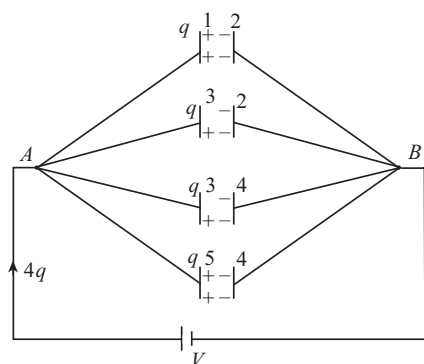
14. MULTI PLATE CAPACITOR

The figure shows a multi plate capacitor in which there are five identical metal plates. Each plate has area A and two successive plates are at separation d . Plates 1, 3 and 5 are connected together at A and the plates 2 and 4 are connected together at B . Consider a battery connected across A and B .



The arrangement consists of four parallel plate capacitors connected in parallel. There is a capacitor formed by plates 1 and 2, a capacitor formed by the plates 2 and 3, a capacitor formed by 3 and 4 and a capacitor between plate 4 and plate 5. Each capacitor has capacitance $C = \frac{\epsilon_0 A}{d}$.

We can redraw the diagram as shown to realize that the four capacitors are in parallel.



Note that we have drawn plates 1, 3 and 5 on one side so that they can be easily connected.

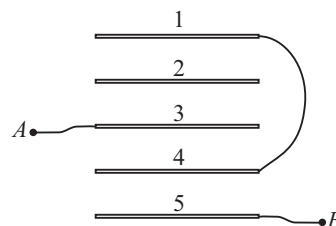
The capacitance of the entire system between A and B is $4C = \frac{4\epsilon_0 A}{d}$

Each capacitor will acquire a charge $q = CV = \frac{\epsilon_0 AV}{d}$.

Therefore, plate 1 has charge q (see the first figure) and plate 2 has a total charge $-2q$ on it. In the second figure, plate 2 occurs twice – once with plate 1 and then with plate 3. Similarly, one can understand that plate 3 has a total $2q$ charge and plate 4 has $-2q$ charge on it. Charge on the plate 5 is q .

Example 25 A bit tricky connections

Five identical metal plates are kept parallel to one another with a gap d between two consecutive plates. Area of each plate is A . Plates 1 and 4 are connected using a conducting wire.



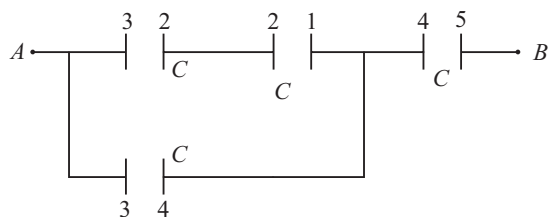
Find equivalent capacitance between A and B .

Solution

Concepts

- (i) There are 4 capacitors.
- (ii) Plates 1 and 4 are at same potential.
- (iii) Plate 2 is neutral and its two faces will have equal and opposite charge.
- (iv) Keeping above facts in mind we will redraw the arrangement of 4 capacitors. It will help to keep plate 3 (i.e. A) on extreme left and B (i.e., plate 5) on extreme right of the diagram.

Capacitance of a capacitor formed between two consecutive plates is $C = \frac{\epsilon_0 A}{d}$



Four capacitors are connected as shown in the figure. We have placed the plates 4 and 1 so that they can be connected easily.

C and C in series make $\frac{C}{2}$.

$\frac{C}{2}$ and C in parallel make $\frac{3C}{2}$.

$\frac{3C}{2}$ and C in series make $\frac{3C}{5}$.

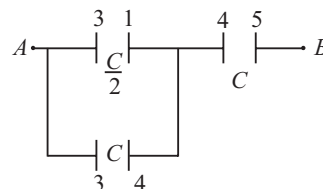
$\therefore$ Capacitance is $\frac{3\epsilon_0 A}{5d}$.

Alternate Method

Plate 2 is not connected anywhere. We can think that there is a capacitor between 1 and 3 and a thin metal plate (2) is kept in between. Referring to Example 5, we can write capacitance between 1 and 3 as

$$\frac{\epsilon_0 A}{2d} = \frac{C}{2}$$

The presence of plate 2 makes no difference to capacitance between 1 and 3 (thickness of 2 is negligible). So we have only 3 capacitors connected as shown.

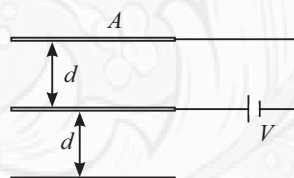
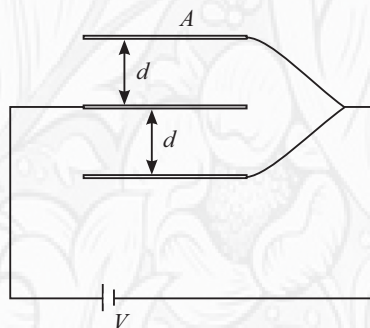


Equivalent is again $\frac{3C}{5}$.



YOUR TURN

Q.28 Find the charge supplied by the cell to the plate systems shown in two figures.



15. CHARGE SHARING BETWEEN TWO CAPACITORS

Capacitor A has capacitance C_1 and has been charged using a cell of emf V_1 . Another capacitor B , having capacitance C_2 , has been charged using a cell of emf V_2 . The capacitors are disconnected from the respective cells. Now the capacitors are connected to each other joining their positive plates (1 and 3) together and the negative plates (2 and 4) together. The capacitors will share charge so that potential difference across both of them becomes same (say, V).

If the final charges on the two capacitors are q_1 and q_2 , then

$$q_1 + q_2 = Q_1 + Q_2$$

$$\text{or, } q_1 + q_2 = C_1 V_1 + C_2 V_2 \quad (\text{i})$$

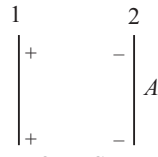
This follows from the principle of conservation of charge. Total charge on plates 1 and 3 remain $Q_1 + Q_2$ and that on plates 2 and 4 must be $-(Q_1 + Q_2)$.

$$\text{Now, } V = \frac{q_1}{C_1} = \frac{q_2}{C_2}$$

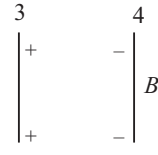
$$\text{or, } \frac{q_1}{q_2} = \frac{C_1}{C_2} \quad (\text{ii})$$

Solving equations (i) and (ii) gives,

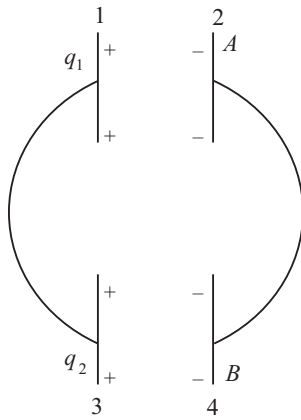
$$q_1 = \frac{(C_1 V_1 + C_2 V_2) C_1}{C_1 + C_2}, \quad q_2 = \frac{(C_1 V_1 + C_2 V_2) C_2}{C_1 + C_2}$$



$$Q_1 = C_1 V_1$$



$$Q_2 = C_2 V_2$$



Potential difference across both the capacitors will be same, after they are connected.

The final common potential difference is

$$V = \frac{q_1}{C_1} = \frac{q_2}{C_2} = \frac{C_1 V_1 + C_2 V_2}{C_1 + C_2} \quad (30)$$

As the charge moves, there will be energy loss in the form of heat and light. Loss in energy is

$$\begin{aligned} \Delta U_{\text{loss}} &= U_i - U_f \\ &= \left[\frac{1}{2} C_1 V_1^2 + \frac{1}{2} C_2 V_2^2 \right] - \left[\frac{1}{2} (C_1 + C_2) V^2 \right] \end{aligned}$$

Substituting for V and simplifying gives

$$\Delta U_{\text{loss}} = \frac{C_1 C_2}{2(C_1 + C_2)} (V_1 - V_2)^2 \quad (31)$$

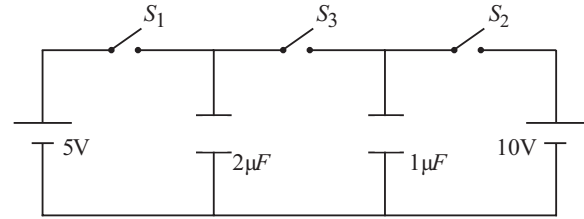
If positive plate of one capacitor is connected to the negative plate of other (i.e., plate 1 is connected to 4 and 2 is connected to 3), then the total charge available for distribution will be $|Q_1 - Q_2|$ and equation (i) will change to

$$q_1 + q_2 = |C_1 V_1 - C_2 V_2|$$

The final common potential difference will be

$$V = \frac{|C_1 V_1 - C_2 V_2|}{C_1 + C_2} \quad (32)$$

Example 26 In the circuit shown, switches S_1 and S_2 are closed and S_3 is open. Now, S_1 and S_2 are opened and thereafter S_3 is closed.



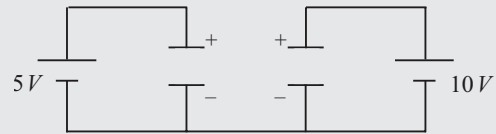
(a) Find final potential difference across $2 \mu\text{F}$ capacitor.

(b) Find heat generated after S_3 is closed.

Solution

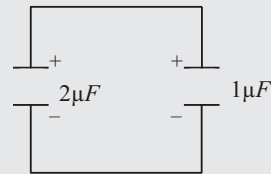
Concepts

- (i) When S_3 is open, the circuit has two independent loops. Note that $2 \mu\text{F}$ capacitor is charged using 5V cell and the $1 \mu\text{F}$ capacitor is charged using a 10V cell.



S_1, S_2 closed and S_3 open

- (ii) When S_1 and S_2 are opened, the capacitors get detached from the cells. By closing S_3 , we are connecting the two capacitors in parallel to one another.



S_3 closed and other switches open

- (a) Final common potential difference is given by equation (30)

$$V = \frac{C_1 V_1 + C_2 V_2}{C_1 + C_2} = \frac{2 \times 5 + 1 \times 10}{2 + 1} = \frac{20}{3} \text{ V}$$

- (b) Energy stored in capacitors before S_3 is closed is

$$U_i = \frac{1}{2} C_1 V_1^2 + \frac{1}{2} C_2 V_2^2 = \frac{1}{2} \times 2 \times 5^2 + \frac{1}{2} \times 1 \times 10^2 = 75 \mu\text{J}.$$

Energy in capacitors after S_3 is closed is

$$U_f = \frac{1}{2} C_1 V^2 + \frac{1}{2} C_2 V^2 = \frac{1}{2} (2 + 1) \left(\frac{20}{3} \right)^2 = \frac{200}{3} \mu\text{J}$$

$$\begin{aligned} \therefore \text{Heat generated} &= U_i - U_f = 75 - \frac{200}{3} \\ &= 8.33 \mu\text{J} \end{aligned}$$



YOUR TURN

Q.29 A $2\mu\text{F}$ capacitor has a charge $20\mu\text{C}$ on it. Another capacitor of capacitance $3\mu\text{F}$ has a charge of $30\mu\text{C}$ on it. The two capacitors are connected in parallel with like plates together. Find heat generated after they are connected.

Q.30 Two parallel plate capacitors A and B having capacitance of $1\mu\text{F}$ and $5\mu\text{F}$ are charged separately to a potential difference of 100V . Now the positive plate of A is connected to the negative plate of B and the negative plate of A to the positive plate of B . Find final charge on A .

In Short

- When the space between the plates of a capacitor is occupied by multiple dielectrics, we can consider the capacitor as series – parallel combination of many capacitors.
- A capacitor is said to be short circuited if its plates are connected by a plain conducting wire. A short circuited capacitor has no charge. We can remove it from the circuit.
- If n number of identical plates, each of area A , are arranged parallel to one another and the alternate plates are connected together to form two terminals, then the system is equivalent to $(n-1)$ capacitors in parallel. The capacitance of the system is $(n-1)C$, where, $C = \frac{\epsilon_0 A}{d}$. d is separation between two consecutive plates.
- When two charged capacitors are connected in parallel, final common potential difference across them is

$$V = \frac{C_1 V_1 + C_2 V_2}{C_1 + C_2}, \text{ when like plates are put together}$$

$$= \frac{|C_1 V_1 - C_2 V_2|}{C_1 + C_2} \text{ when unlike plates are connected together.}$$

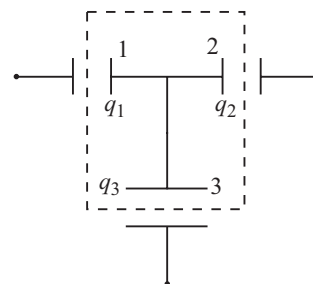
16. COMPLEX CAPACITOR CIRCUITS

For a capacitor circuit having no series parallel grouping, solution is often done using two laws known as *Kirchhoff's laws*.

Kirchhoff's first law (the junction law)

This law is basically law of conservation of charge. In the context of capacitor circuits, we can state it as:

When a capacitor circuit is connected to a battery, a system of plates forming an isolated junction will continue to have zero charge. They can share charge among themselves but total charge on them will be zero.



$$q_1 + q_2 + q_3 = 0$$

The system of plates 1, 2 and 3 is isolated from the outside world

The figure shows a part of a circuit. Plates 1, 2 and 3 form an isolated junction. No charge can enter this junction from anywhere. Therefore, sum of charges on these three plates is zero.

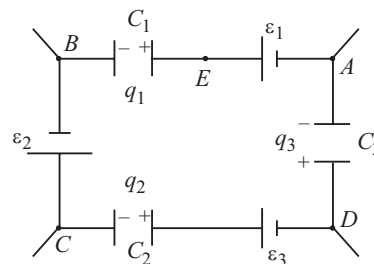
$$q_1 + q_2 + q_3 = 0$$

Kirchhoff's second law (the loop law or voltage law)

This law can be stated as:

The algebraic sum of potential differences across all elements in a closed electric loop is zero.

Consider a closed loop in an electric circuit having capacitors and cells. The diagram shown is not the complete circuit, it is just one loop in a circuit.



Let the potential of point A be V_A . We will start from A and move anticlockwise (one can also go clockwise) along the loop to return back to point A . We will go on adding any potential gain and subtracting a potential drop as we move along the loop.

When we move from A and cross the cell of emf ε_1 , the potential increases by ε_1 and the potential of point E is $V_A + \varepsilon_1$. As we move from E to B , the potential drops by $\frac{q_1}{C_1}$ (as we move from positive to negative plate of the capacitor) and the potential of point B is $V_A + \varepsilon_1 - \frac{q_1}{C_1}$. In this fashion, we go on adding and subtracting potential gains and drops and return back to A . The equation that we get is

$$V_A + \varepsilon_1 - \frac{q_1}{C_1} + \varepsilon_2 + \frac{q_2}{C_2} - \varepsilon_3 - \frac{q_3}{C_3} = V_A$$

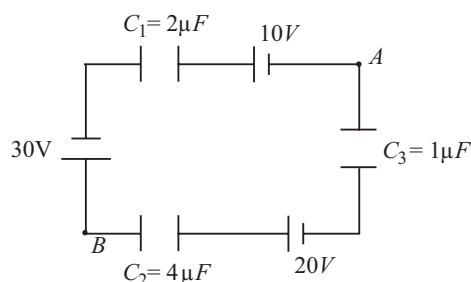
$$\text{or, } \varepsilon_1 - \frac{q_1}{C_1} + \varepsilon_2 + \frac{q_2}{C_2} - \varepsilon_3 - \frac{q_3}{C_3} = 0$$

The left hand side of the last equation is sum of potential differences across all elements (gains taken as positive and potential drops as negative) in the loop. Their sum is zero. This is Kirchhoff's loop law. One can also take potential drops as positive and gains as negative. It will also give the same equation as above.

The above two laws are very powerful and can be used to solve any circuit problem. Now, we will take a series of examples to illustrate the process.

Example 27 Single loop circuit

A circuit has three cells and three capacitors as shown.



Find

- Potential difference between the points A and B .
- Potential of the point A if point B is grounded.

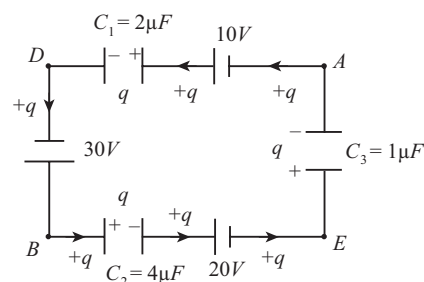
Solution

Concepts

- For knowing the potential difference between A and B , we need to know the charge on capacitors.
- We will assume charge on capacitors and then apply Kirchhoff's voltage law (kVL) in the loop to find the value of charge.
- In a single loop circuit, all capacitors will acquire same charge.

- Assume that a charge $+q$ flows out of the positive terminal of the 30V cell and sits on a plate of capacitor C_2 . Obviously, a charge $-q$ moves out of the negative terminal of the 30V cell and sits on a plate of capacitor C_1 . (One can also say that $+q$ charge

moves out of the plate of C_1 and enters the cell through its negative terminal). The positive charge on a plate of C_2 induces $-q$ charge on the opposite plate. This charge is supplied from the 20V cell (or we can say that $+q$ charge moves into the positive terminal of the cell). Same charge $+q$ moves out of the negative terminal of the 20V cell and sits on a plate of C_3 . This induces $-q$ charge on the opposite plate. $+q$ charge exits from this plate and enters the 10V cell. $+q$ charge exits from the positive terminal of 10V cell and sits on a plate of C_1 . This completes the charge distribution. We see that all the capacitors have same charge on them.



We will start from point A and move anticlockwise along the loop to apply kVL. We will take potential gains as positive and drops as negative.

$$10 - \frac{q}{2} + 30 - \frac{q}{4} - 20 - \frac{q}{1} = 0 \quad (i)$$

$\downarrow$ cell (gain) $\downarrow$ drop in C_1 $\downarrow$ drop in C_2 $\downarrow$ drop in C_3 $\downarrow$ cell (drop)
 $\downarrow$ cell (gain) $\downarrow$ cell (gain)

$$\text{or, } q \left(\frac{1}{2} + \frac{1}{4} + \frac{1}{1} \right) = 10 + 30 - 20$$

$$\text{or, } q = \frac{10 + 30 - 20}{\frac{1}{2} + \frac{1}{4} + \frac{1}{1}} \quad (ii)$$

$$= \frac{80}{7} \mu C$$

Note that $\left(\frac{1}{2} + \frac{1}{4} + \frac{1}{1} \right) = \frac{1}{C_0}$, where C_0 is series equivalent of the three capacitors and $(10 + 30 - 20)$ volt is the net emf in the loop. Equation (ii) tells us that in a single loop circuit we can find charge on the capacitors by

$$q = C_0 \varepsilon_0$$

where C_0 = series equivalent of all capacitors,

and ε_0 = net emf in the loop.

For finding ε_0 , go around the loop just looking at the potential gains and drops across the cells (forget about the capacitors).

One more important thing must be noted. While writing equation (i), we were having in mind that q is in unit of μC . This is because the capacitance has been written in unit of μF .

To find potential difference between points A and B , we will move from A to B (via any path) keeping track of gain/drop in potential. Let us move along ADB .

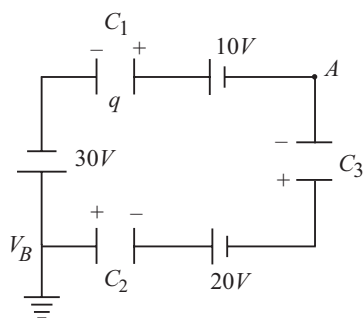
$$V_A + \underset{\substack{\downarrow \\ \text{gain} \\ \text{in cell}}}{10} - \underset{\substack{\downarrow \\ \text{drop in} \\ C_1}}{\frac{q}{2}} + \underset{\substack{\downarrow \\ \text{gain in} \\ \text{cell}}}{30} = V_B$$

$$\text{or, } V_A + 10 - \frac{80}{7 \times 2} + 30 = V_B$$

$$\text{or, } V_A - V_B = \frac{40}{7} - 40 = -\frac{240}{7} \text{ V} \quad (\text{iii})$$

Even if you move along AEB , we will get the same result.

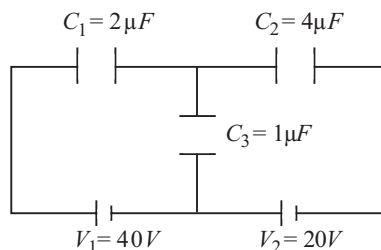
(b) When B is grounded, $V_B = 0$



From (iii)

$$V_A = -\frac{240}{7} \text{ V.}$$

Example 28 No series, no parallel. Use Kirchhoff's laws. Calculate charge on each capacitor in the circuit shown in the figure.



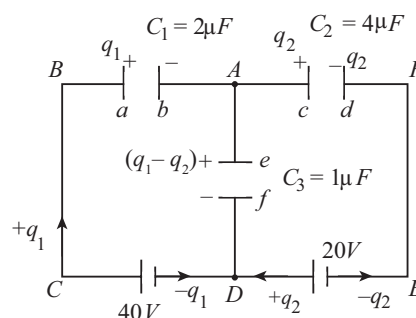
Solution

Concepts

- (i) No two capacitors are in series or parallel. The circuit is not a balanced Wheatstone bridge. We will use the Kirchhoff's laws to solve such a circuit.
- (ii) First we will assume charges on the capacitors (q_1, q_2 , etc.). By use of first law, we can keep the number of unknowns to a minimum. While assuming charges, we must keep in mind that if a battery supplies $+q$ charge from one terminal, it will also supply $-q$ from its other terminal.
- (iii) We will frame two equations using the second law (KVL). For this, we can choose any two closed loops in the circuit (the circuit has three loops – two smaller ones and an outer one).

Distributing charge

Let the 40V cell supply a charge q_1 . Plate a gets $+q_1$ charge. $-q_1$ charge is induced on plate b .



Let the 20V cell supply a charge q_2 as shown. The plate d acquires $-q_2$ charge and $+q_2$ is induced on plate c .

Plates b, c and e form an isolated system. Sum of charges on these three plates will be zero. If q is charge on e , then

$$q - q_1 + q_2 = 0$$

$$\text{or, } q = q_1 - q_2$$

A charge $-(q_1 - q_2)$ is induced on plate f . Let us make a quick check if our charge distribution is correct or not. Plate f receives charge from the two cells. Charge supplied by the cells (which goes to plate f) is $-q_1 + q_2$, which is exactly $-q$. This confirms that our charge distribution is correct.

The process of charge distribution given above is not unique. We can assign charges to capacitors thinking in a different way. It is OK as long as we do not violate any basic principle.

Writing loop equations

Consider loop $ABCD$. We will start from point A and move anticlockwise. Potential gains will be taken positive and drops will be taken negative.

$$\frac{q_1}{C_1} - 40 + \frac{q_1 - q_2}{C_3} = 0$$

$$\text{or, } \frac{q_1}{2} - 40 + \frac{q_1 - q_2}{1} = 0$$

$$\text{or, } 3q_1 - 2q_2 = 80 \quad (\text{i})$$

Now, consider loop $ADEF$. We will start from A and move anticlockwise.

$$-\frac{q_1 - q_2}{C_3} - 20 + \frac{q_2}{C_2} = 0$$

$$\text{or, } -\frac{q_1 - q_2}{1} - 20 + \frac{q_2}{4} = 0$$

$$\text{or, } -4q_1 + 5q_2 = 80 \quad (\text{ii})$$

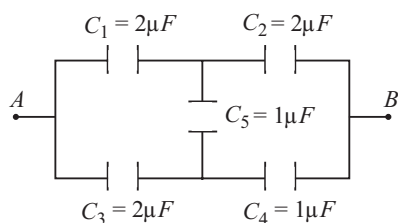
Solving (i) and (ii) gives

$$q_1 = 80\mu\text{C} \text{ and } q_2 = 80\mu\text{C}$$

Charge on C_3 is $q_1 - q_2 = 0$.

Example 29 Finding equivalent capacitance (general procedure)

Find the equivalent capacitance between points A and B for the circuit shown in the figure.



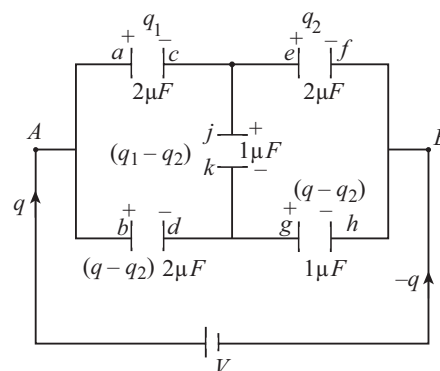
Solution

Concepts

- (i) The circuit is a Wheatstone bridge but it is not balanced because $\frac{C_1}{C_2} \neq \frac{C_3}{C_4}$.
- (ii) We will assume that a cell of emf V is connected across A and B and it supplies charge q to the circuit. Our aim is to find the equivalent capacitance, i.e., the ratio $\frac{q}{V}$.
- (iii) We will distribute the charge on the capacitors keeping the number of unknowns (q_1, q_2 , etc.) at minimum.
- (iv) We will write loop equations and express q_1, q_2 , etc. in terms of q .
- (v) Then we will walk from point A to B and write $V_A - V_B$. $V_A - V_B$ is nothing but emf (V) of the cell. From here, we will get the ratio $\frac{q}{V}$.

Charge distribution

Consider a cell of emf V connected across A and B . It supplies a charge q .



We assume that q_1 charge goes to plate a and remaining $(q - q_1)$ goes to plate b . From the negative terminal, the cell supplies a charge $-q$ and it enters into the circuit through point B . Out of this negative q charge, we assume that q_2 goes to plate f and the remaining negative charge $(q - q_2)$ goes to plate h . Charge induced on the opposite plates of a capacitor will have same magnitude and opposite polarity. We can write charges on plates c, d, e and g .

Now, plates c, e and j make an isolated system. If charge on j is q_0 , then

$$q_0 + (-q_1) + q_2 = 0$$

$$\text{or, } q_0 = q_1 - q_2$$

Loop equations

Our next target is to express q_1 and q_2 in terms of q . For this, we frame two loop equations.

For loop having capacitors C_1, C_3 and C_5 , the equation is

$$\frac{q_1}{2} - \frac{q - q_1}{2} + \frac{q_1 - q_2}{1} = 0$$

$$\text{or, } 4q_1 - 2q_2 = q \quad (\text{i})$$

For the loop having capacitors C_2, C_5 and C_4 , we can write:

$$\frac{q_2}{2} - \frac{q_1 - q_2}{1} - \frac{q - q_2}{1} = 0$$

$$\text{or, } 5q_2 - 2q_1 = 2q \quad (\text{ii})$$

Solving (i) and (ii) gives

$$q_1 = \left(\frac{9}{16}\right)q, \quad q_2 = \left(\frac{5}{8}\right)q$$

Equivalent capacitance

Let us write potential difference between the points A and B moving via C_1 and C_2 .

4.34 Electrostatics and Current Electricity

$$V_A - \frac{q_1}{C_1} - \frac{q_2}{C_2} = V_B$$

$$\text{or, } V_A - V_B = \left(\frac{9}{16 \times 2} \right) q + \left(\frac{5}{8 \times 2} \right) q$$

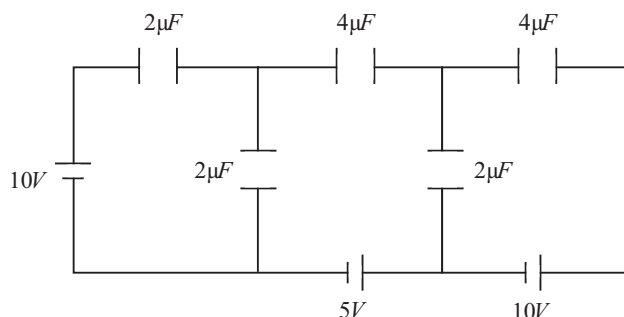
$$\text{or, } V = \left(\frac{19}{32} \right) q$$

$$\text{or, } \frac{q}{V} = \frac{32}{19}$$

$$\text{or, } C_0 = \frac{32}{19} \mu F$$

Example 30 Different way of using Kirchhoff's laws

Find charges on the capacitors shown in the circuit.

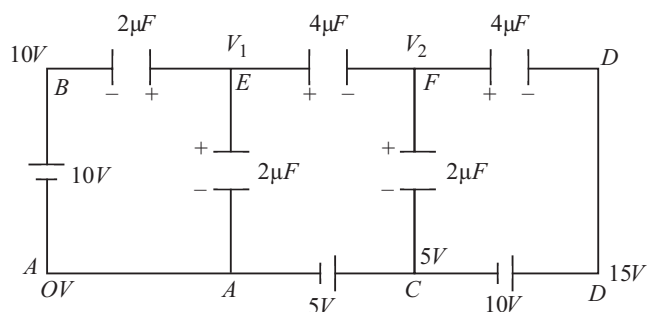


Solution

Concepts

- We will assume potential of a point in the circuit to be zero. Then we write potential of all points that can be decided by visual inspection.
- The remaining junctions are assigned potentials like V_1 , V_2 , etc.
- Now, charge on every capacitor can be written in terms of V_1 , V_2 , etc.
- Equations are framed using the facts that sum of charges on the plates forming an isolated system is zero.

Let potential of point A be 0. Potential of points B , C and D are 10V, 5V and 15V, respectively. [Just look at the batteries to conclude this.]



Let potentials of E and F be V_1 and V_2 , respectively.

Now, we have assigned potential to all the points in the circuit.

Potential difference across the $2\mu F$ capacitor between E and B is $V_1 - 10$. Charge on it will be $2(V_1 - 10)$. Here, we have assumed the plate connected to E to be positive.

Similarly, charge on $4\mu F$ capacitor between E and F is $4(V_1 - V_2)$ and the charge on $2\mu F$ capacitor between E and A is $2(V_1 - 0)$.

Sum of charges on the three plates connected to E must be zero.

$$2(V_1 - 10) + 4(V_1 - V_2) + 2(V_1 - 0) = 0$$

$$\text{or, } 2V_1 - V_2 = 5 \quad (i)$$

Similarly, one can write an equation for the three plates connected to F .

$$-4(V_1 - V_2) + 4(V_2 - 15) + 2(V_2 - 5) = 0$$

$$\text{or, } -2V_1 + 5V_2 = 35 \quad (ii)$$

Solving (i) and (ii)

$$V_1 = 7.5V, \quad V_2 = 10V$$

Charges on the capacitors will be potential difference across them multiplied with respective capacitance.

Charge on capacitor between B and E is

$$= (2\mu F)(10 - 7.5)V = 5\mu C.$$

Plate connected to B will be positive (our assumed polarity in figure is wrong. But it makes no difference to our solution).

Charge on capacitor between E and F is $4 \times 2.5 = 10\mu C$.

Polarity will be opposite to that shown.

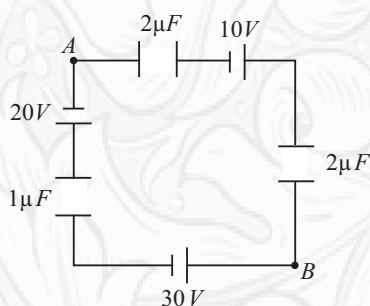
Charge on capacitor between E and A is $2 \times 7.5 = 15\mu C$

Charge on capacitor between F and C is $2 \times 5 = 10\mu C$.

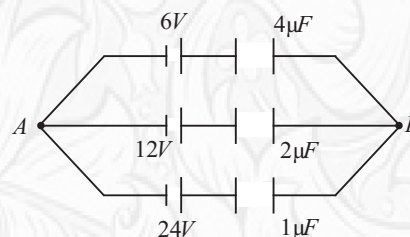


YOUR TURN

Q.31 Find the potential difference between the points A and B in the figure shown.



Q.32 Find the potential difference between the points B and A shown in the figure.

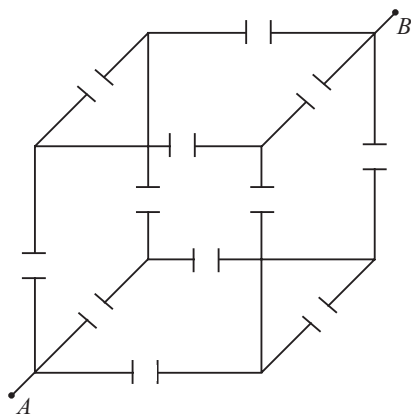


17. SYMMETRY IN CIRCUITS

If we can spot some symmetry in an electric circuit, it can be exploited to make the solution of a problem much easier. There are various kinds of symmetries that can exist in a circuit. We will illustrate the use of symmetry through the following examples.

Example 31 A cube of capacitors

Twelve identical capacitors (each of capacitance C) are placed along the sides of a cube as shown. Find the equivalent capacitance between the diagonally opposite points A and B .



Solution

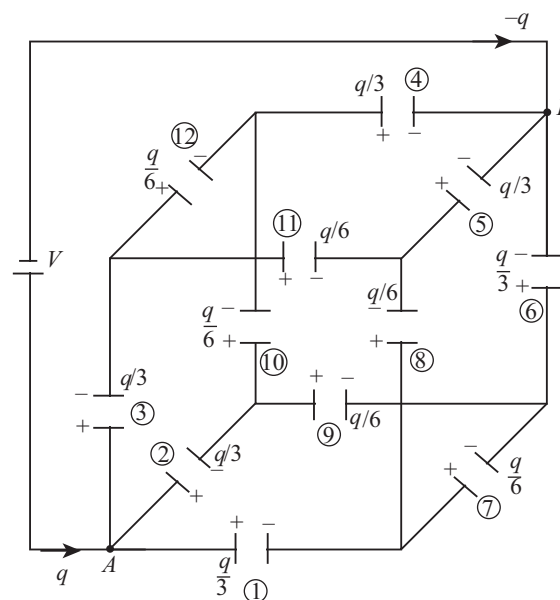
Concepts

- Assume a battery of emf V connected across the points A and B . If the battery supplies a charge q to the network, the equivalent capacitance is $\frac{q}{V}$.
- Charge q that enters at A will get equally divided on the plates of three capacitors (1, 2 and 3 shown in figure). This is because of symmetry. There is no difference in positions of these three capacitors. Even starting from point B , you will have to take identical paths to reach any of these capacitors.

Similarly, charge $-q$ that enters at B must get equally divided amongst the plates of the capacitors 4, 5 and 6.

The negative plate of capacitor 1 gets its charge from the plates of capacitors 7 and 8. Capacitors 7 and 8 are also at identical locations. We cannot differentiate between them. Therefore, positive plates of 7 and 8 contribute equally to the negative plate of 1.

Taking symmetry into account, the charge distribution has been shown. Charge on capacitors 1, 2, 3, 4, 5 and 6 is $\frac{q}{3}$.



Plates of capacitors 1, 7 and 8 form an isolated system. Total charge on them must be zero.

Charge on the positive plates of 7 and 8 must be $\frac{q}{6}$.

All other capacitors will also have $\frac{q}{6}$ charge on them.

Go back to Example 28. We had to assume unknowns like q_1, q_2 to complete the charge distribution. In the present problem, the symmetry has greatly simplified the situation. There are no unknowns like q_1, q_2 etc.

We will find potential difference between A and B moving along capacitors 1, 7 and 6.

$$V_A - \frac{q}{3C} - \frac{q}{6C} - \frac{q}{3C} = V_B$$

$$\text{or, } V_A - V_B = \frac{5q}{6C}$$

$$\text{or, } V = \frac{5q}{6C}$$

$$\text{or, } \frac{q}{V} = \frac{6C}{5}$$

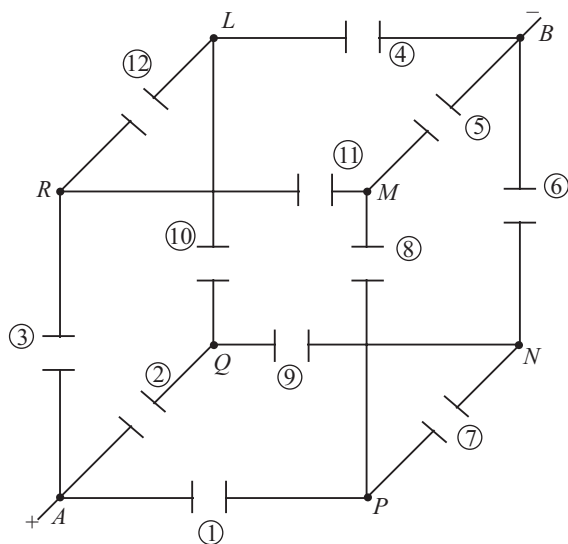
$$\text{or, } C_0 = \frac{6C}{5}$$

Alternate Method

Concepts

We can identify points having equal potential. Two or more points at same potential can be joined together. This will simplify the circuit.

Why the potentials at P, Q and R will be different? There is absolutely no reason for this. These points are symmetrically located with respect to terminal A as well as with respect to B .

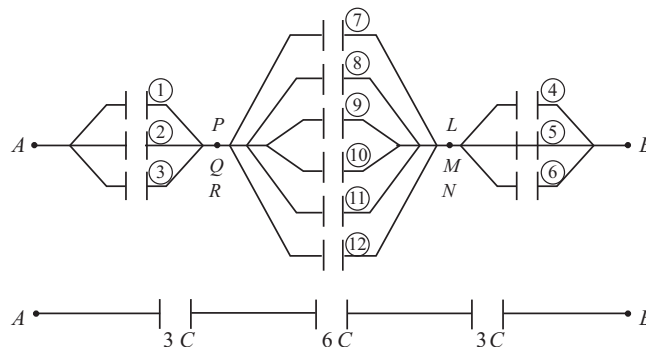


P, Q and R have same potential. L, M and N have same potential.

Similarly, the points L, M and N are identical and they must be at same potential.

Note: A point like P and L are not identical. P is closer to the positive terminal A , whereas L is closer to the negative terminal B .

We will redraw the circuit keeping points P, Q and R together. Points L, M and N are also tied together. It will be easier to complete the diagram keeping the points A and B at the extreme ends.

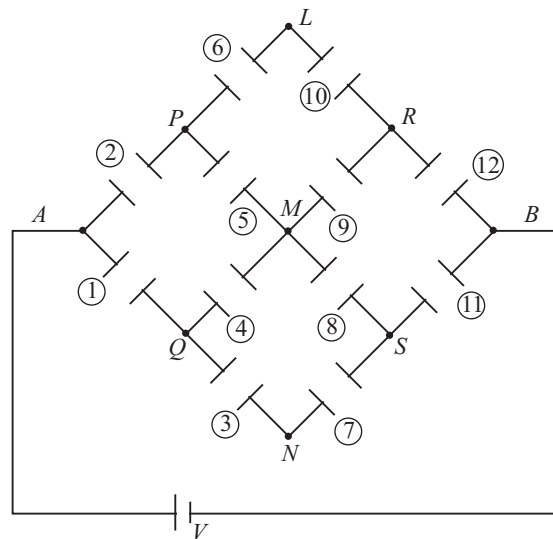


Equivalent capacitance is given by

$$\frac{1}{C_0} = \frac{1}{3C} + \frac{1}{6C} + \frac{1}{3C}$$

$$\text{or, } C_0 = \frac{6C}{5}$$

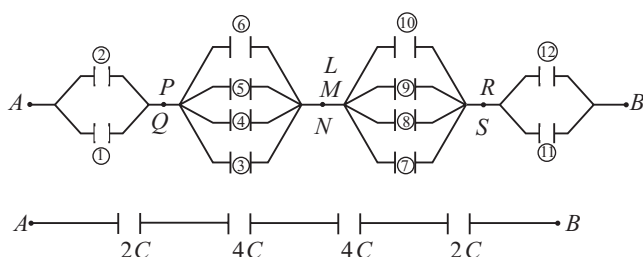
Example 32 In the circuit shown in the figure, each capacitor has capacitance C . Find the equivalent capacitance between the points A and B .



Solution**Concepts**

- (i) Line joining the terminals (AB) is a line of symmetry. It divides the circuit in exactly identical halves. In such situations, points lying on a perpendicular drawn to AB are at same potentials. Points P and Q are at same potential. Points L, M and N must be at same potential. Similarly, the points R and S are at same potential.
- (ii) We can join the points at the same potential.

We draw the circuit after joining P and Q , joining L, M and N together and points R and S together.



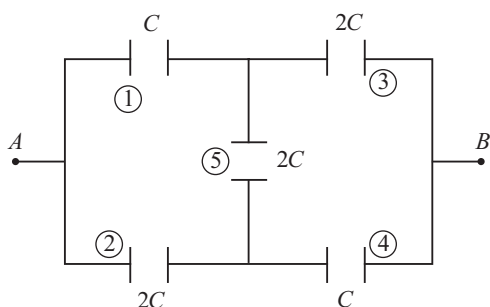
Equivalent capacitance is given by

$$\frac{1}{C_0} = \frac{1}{2C} + \frac{1}{4C} + \frac{1}{4C} + \frac{1}{2C}$$

$$\text{or, } C_0 = \frac{2C}{3}$$

Example 33 Wheatstone bridge again!

Find the equivalent capacitance between the points A and B shown in the figure.

**Solution****Concepts**

- (i) This is a Wheatstone bridge but it is not balanced. In example 28, we have given a general method for finding the equivalent capacitance for such a circuit.
- In this problem, we can reduce our work by observing that there is some symmetry in the circuit.

- (ii) Look into the circuit from side A . Now, look at it from the side B . There is no difference.

If a battery is connected and charge $+Q$ enters at A , it gets distributed on the capacitors 1 and 2. If 1 receives a charge q , then 2 must receive $Q - q$. Now, think from the other end B . A charge $-Q$ enters at B . Obviously, the capacitor 4 will receive $-q$ and 3 will receive $-(Q - q)$.

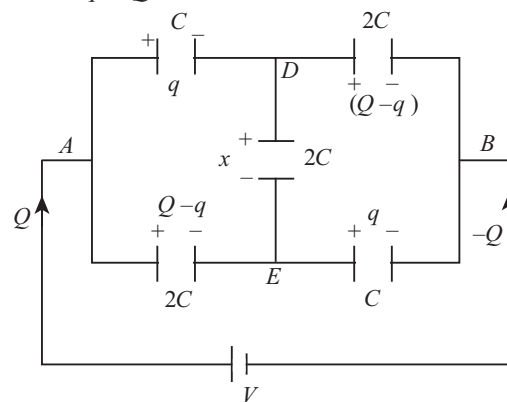
- (iii) We will need to write just one loop equation (in example 28 we wrote 2 equations) to express q in terms of Q .

- (iv) Then we will write $V_A - V_B (=V)$ and the equivalent capacitance will be $\frac{Q}{V}$.

The charge distribution has been shown in the figure. To write charge on the central capacitor (5), we must note that three plates connected at D form an isolated system. If charge on upper plate of capacitor 5 is x , then

$$x - q + Q - q = 0$$

$$\text{or, } x = 2q - Q$$



We will write loop equation in the loop $ADEA$. Let us start from A and move clockwise. We will write potential gain as positive and drop as negative.

$$-\frac{q}{C} - \frac{2q - Q}{2C} + \frac{Q - q}{2C} = 0$$

$$\text{or, } q = \frac{2Q}{5}$$

Now, we write $V_A - V_B$ moving along ADB .

$$V_A - \frac{q}{C} - \frac{Q - q}{2C} = V_B$$

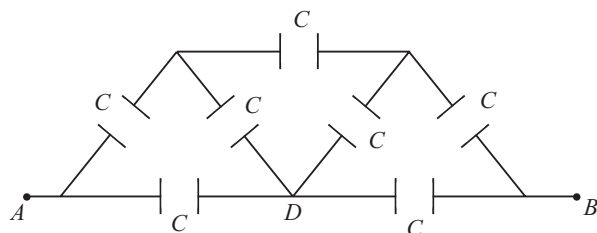
$$\text{or, } V_A - V_B = \frac{7Q}{10C} \quad \left[\text{putting } q = \frac{2Q}{5} \right]$$

$$\text{or, } V = \frac{7Q}{10C}$$

$$\text{or, } C_0 = \frac{Q}{V} = \frac{10C}{7}$$

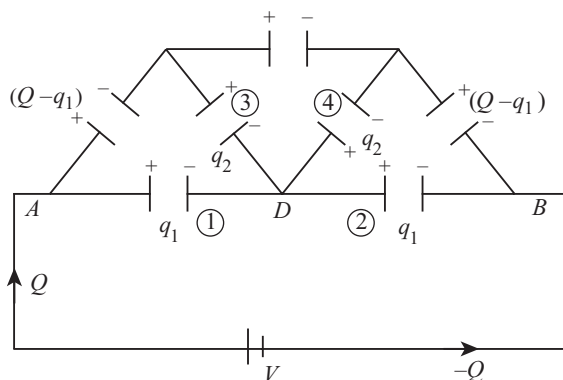
Example 34 False junction

Find the equivalent capacitance of the system between the points A and B .

**Solution****Concepts**

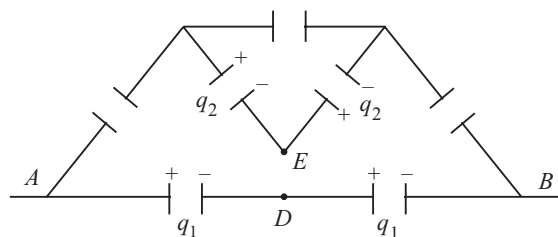
- This circuit also has a kind of symmetry as seen in the last example. Charge $+Q$ entering at A will divide in a way similar to division of charge $-Q$ entering at B .
- As explained below, the junction at D is a false junction. Charge on any capacitor will not change even if the junction is opened. Actually, it is reverse of what we did in example 31. If two points are tied together, we can separate them if they remain at same potential even after separation.

Charge distribution could be as shown. Because the circuit looks identical when looking from A or B , we conclude that the capacitors 1 and 2 must have same charge and 3 and 4 must also have equal charge. Focus on polarity of the plates.

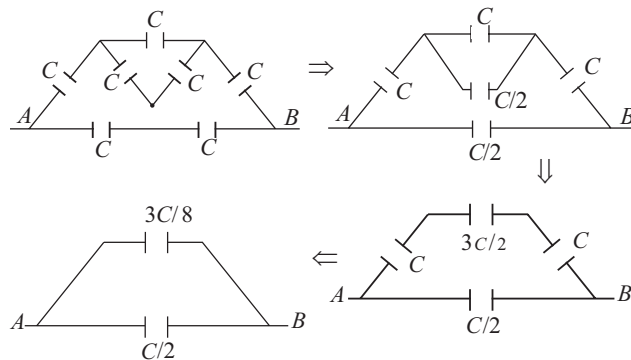


Though there is a junction at D , the plates of the capacitors 1 and 2 (joined at D) have net zero charge and the plates of 3 and 4 (connected at D) separately have a net zero charge on them. The capacitors 1 and 2 behave as if they are in series and not connected to 3 and 4. If we open the junction at D (see figure), it will make no difference to our charge distribution.

If we were given the circuit as shown in the next figure, we could easily point out that line DE divides the circuit into two symmetrical halves. D and E must be equipotential points. They can be joined together. Therefore, the two circuits are all the same.



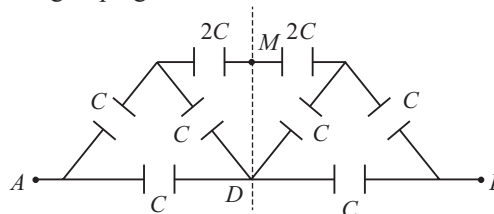
After opening the junction, the equivalent capacitance can be found using series parallel grouping.



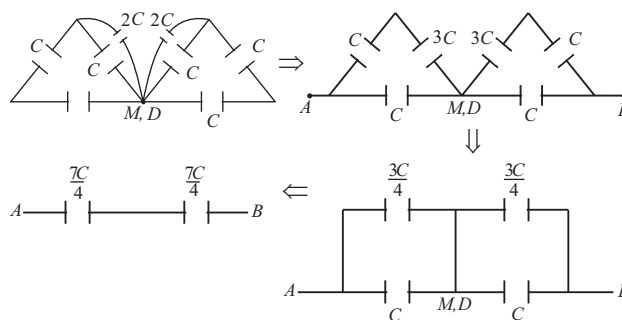
The equivalent is $\frac{7C}{8}$

Alternate Method

We have redrawn the circuit after replacing the capacitor at the top by two capacitors (each of capacitance $2C$) in series. One can easily see that the line MD divides the circuit in two symmetrical halves and the potential of point M must be same as that of D . We can merge points M and D and the equivalent can be found by series-parallel grouping. This has been demonstrated below.



Note: If line AB is not a line of symmetry, but its perpendicular bisector divides the circuit into two identical halves, then all point on this perpendicular bisector are equipotential.



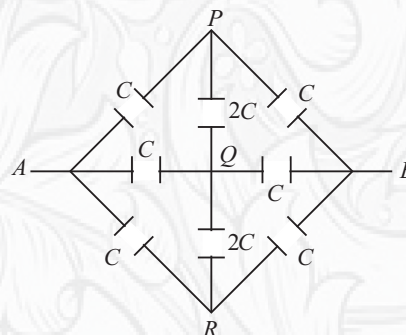
Equivalent is obviously $\frac{7C}{8}$.



YOUR TURN

Q.33 Solve the question in Example 32 by identifying a false junction and opening that junction.

Q.34 Find the equivalent capacitance between the points A and B .

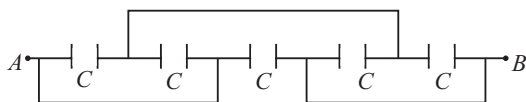


In Short

- (i) In a single loop capacitor circuit, charge on each capacitor is given by $q = C_0 \varepsilon_0$, where C_0 = effective capacitance calculated assuming all the capacitors in series. ε_0 is net emf of the loop.
- (ii) We can solve complex capacitor circuits by use of Kirchhoff's laws. The first law is basically law of conservation of charge. For an isolated part of a circuit, total charge is zero. The second law (also known as loop law or Kirchhoff's voltage law) states that algebraic sum of potential gains (or drops) across each element in an electrical loop is always zero.
- (iii) When we spot points at identical locations (seen with respect to both the terminals) in a circuit, we can treat them as equipotential points. These points can be joined together. Any capacitor between such two points shall be removed.
- (iv) If the line AB , joining the two terminals A and B of a circuit, divides the entire circuit in exactly two symmetrical halves, then all the points lying on a perpendicular drawn to line AB are at same potential. These points can be connected together.
- (v) If the perpendicular bisector of line AB (A and B are the two terminals) divides the circuit in exactly two identical halves, then all points on this perpendicular bisector are at same potential.
- (vi) If charge $+q$ entering a terminal of a circuit divides into $q_1, q_2, q_3 \dots$ etc. along different branches, then charge $-q$ entering the other terminal will also divide into $-q_1, -q_2, -q_3 \dots$ etc., if both $+q$ and $-q$ charge see identical circuit ahead.

Miscellaneous Examples

Example 35 Determine the effective capacitance between points A and B in the figure shown.

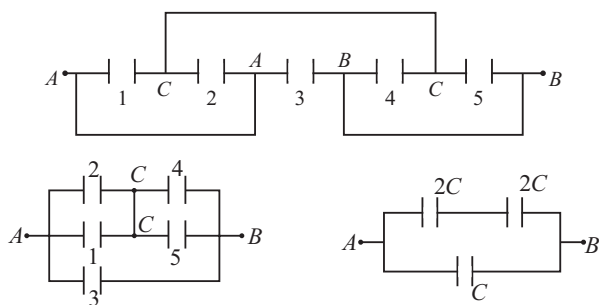


Solution

Concepts

It is easy to make mistakes while simplifying a complex circuit. To avoid mistakes, we should label all the capacitors, e.g. with numbers. We should also label all the nodes. Give same label (name) to all the nodes that are connected by a plain wire. Then, start redrawing the circuit by marking one node (A) and drawing all the capacitors, which are connected to it. Next, select another node and draw the capacitors, which are attached to that node. Repeat the process until you redraw the entire circuit.

In the first drawing, we have labelled the capacitors and the nodes. Nodes connected by a plain wire have been assigned the same name (A , B and C).



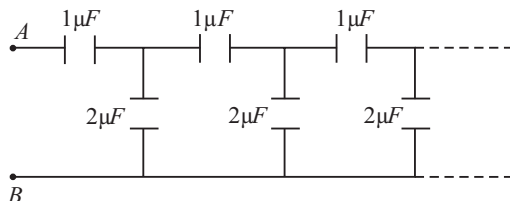
In the second diagram, we have redrawn the circuit. We start by drawing node A , which is connected to one plate of each of the capacitors 1, 2 and 3. The other plates of 1 and 2 are connected to C . The second plate of 3 is directly connected to B . The other two capacitors 4 and 5 are in parallel between the points C and B .

The circuit can be reduced to the one shown in the third diagram. Equivalent capacitance is

$$C_0 = 2C.$$

Example 36 Infinite ladder of capacitors

An infinite network of capacitors is made up of $1\mu\text{F}$ and $2\mu\text{F}$ capacitors as shown in the figure.



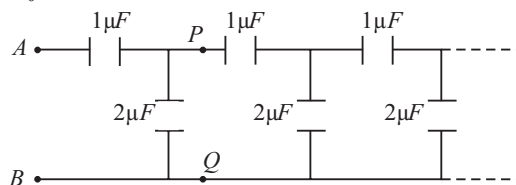
- Find the equivalent capacitance between the points A and B .
- Find the potential difference across first $2\mu\text{F}$ capacitor when a 20V cell is connected across A and B .

Solution

Concepts

We will not be able to recognize any difference in the circuit if the first set of $1\mu\text{F}$ and $2\mu\text{F}$ capacitors are removed. Therefore, the equivalent capacitance will not change if the first set of $1\mu\text{F}$ and $2\mu\text{F}$ capacitors are removed.

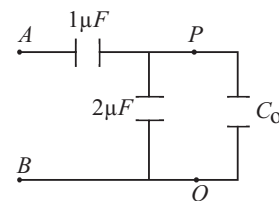
- Let the equivalent capacitance across A and B be C_0 . If we remove the first set of $1\mu\text{F}$ and $2\mu\text{F}$ capacitors and the circuit starts from the points P and Q , then equivalent capacitance across P and Q will also be C_0 .



Therefore, we can replace the entire network to the right of PQ by their equivalent C_0 and redraw the circuit as shown. Now, the equivalent of network of three capacitors across A and B is still C_0 .

[It is like writing

$$y = x^{1+x+x+\dots\infty} \text{ as } y = x^y]$$



Equivalent of C_0 and $2\mu\text{F}$ in parallel is $(C_0 + 2)$. Now, $1\mu\text{F}$ and $(C_0 + 2)$ in series gives the equivalent which is again C_0 .

$$\therefore C_0 = \frac{(C_0 + 2) \cdot 1}{C_0 + 2 + 1}$$

$$\text{or, } C_0^2 + 2C_0 - 2 = 0$$

$$\text{or, } C_0 = \frac{-2 \pm \sqrt{4+8}}{2} = -1 \pm \sqrt{3}$$

Since capacitance cannot be negative, hence

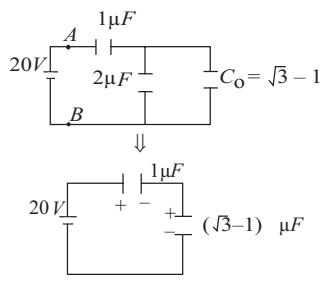
$$C_0 = (\sqrt{3} - 1)\mu\text{F}$$

- (b) The circuit is effectively two capacitors of $1\mu\text{F}$ and $(\sqrt{3}+1)\mu\text{F}$ connected in series to a 20V cell.

Potential difference across the $(\sqrt{3}+1)\mu\text{F}$ capacitor is

$$V_2 = \frac{C_1}{C_1 + C_2} V$$

$$= \frac{1 \times 20}{\sqrt{3} + 1 + 1}$$

$$= \frac{20}{\sqrt{3} + 2} = 5.36\text{V}$$


This will be the potential difference across the first $2\mu\text{F}$ capacitor.

Example 37 An isolated conductor, initially free of charge, is charged by repeated contacts with a charged metal plate. The plate is replenished to a charge Q after each contact. The conductor acquires a charge q after first contact with the plate. Prove that the maximum charge that can be given to the conductor in this way is $\frac{Qq}{Q-q}$

Solution

Concepts

After every contact, the conductor and the plate will be at the same potential. Charge on each of them will be in ratio of their capacitance.

Let C_1 be the capacitance of the plate and C_2 that of the conductor. The charge on the conductor after the first contact is q . Charge on the plate will be $Q - q$.

$$\therefore \frac{q}{Q-q} = \frac{C_2}{C_1} \quad (i)$$

After many contacts, the potential of the conductor will become same as that of the plate (having charge Q). After this, no more charge will flow from the plate to the conductor in the next contact.

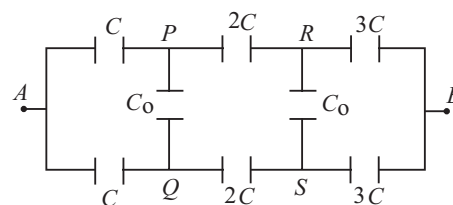
$$V_{\text{conductor}} = V_{\text{plate}}$$

$$\text{or, } \frac{q_{\text{max}}}{C_2} = \frac{Q}{C_1}$$

$$\text{or, } q_{\text{max}} = \left(\frac{C_2}{C_1} \right) Q = \frac{qQ}{Q-q} \quad [\text{using (i)}]$$

Example 38 Extended Wheatstone bridge

Find the equivalent capacitance of the network between the points A and B .



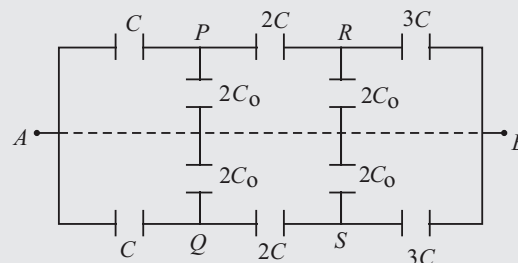
Solution

Concepts

Points that are at identical locations (when seen from end A and also when seen from the end B) will be at same potential. P and Q are such points. R and S are another pair of such points.

Another way to look at it is to replace each capacitor with capacitance C_0 with two capacitors in series (each of capacitance $2C_0$). Now the line AB divides the circuit in two symmetrical halves. Line PQ is perpendicular to AB . Therefore, $V_P = V_Q$. Similarly, line RS is perpendicular to AB and hence

$V_R = V_S$. (see the figure given below)



Yet another way of looking will be to remove the capacitance C_0 . Then it becomes clear that $V_P = V_Q$ and $V_R = V_S$ when a potential difference is applied across A and B .

$$V_P = V_Q \text{ and } V_R = V_S$$

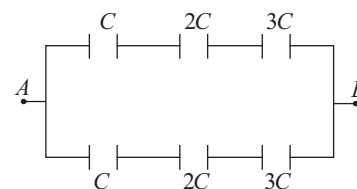
The capacitors having capacitance C_0 are having no charge. They can be removed.

Series equivalent of C , $2C$ and $3C$ is

$$\frac{1}{C_0} = \frac{1}{C} + \frac{1}{2C} + \frac{1}{3C}$$

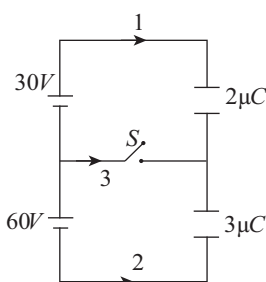
$$\text{or, } C_0 = \frac{6C}{11}$$

$$\text{Equivalent of two parallel branches is } = 2C_0 = \frac{12C}{11}.$$



Example 39 In the circuit shown, after the switch S is closed, find

- the charge flow in the paths 1, 2 and 3 in the indicated directions.
- the heat generated in the circuit.



Solution

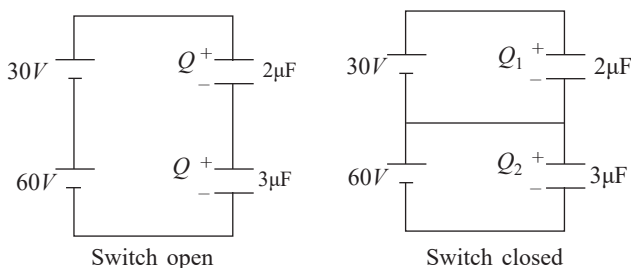
Concepts

- We will find the charge on the capacitors with the switch open and then with the switch closed. By looking at the initial and final charges on the capacitor plates, we will be able to decide about the movement of charge in the circuit.
- Work done by the cells = (change in energy of capacitors) + (Heat generated)..

- The figures shown below depict the situation before the switch is closed and after it is closed.

When the switch is open, the two capacitors are in series connected to a 90V supply.

$$\text{Equivalent capacitance } C_0 = \frac{2 \times 3}{2 + 3} = \frac{6}{5} \mu\text{F}.$$



Charge on each capacitor is $Q = C_0 \cdot V = \frac{6}{5} \times 90 = 108 \mu\text{C}$.

After the switch is closed, the potential difference across $2 \mu\text{F}$ capacitor is 30V and the potential difference across $3 \mu\text{F}$ capacitor is 60V. Charges on them are

$$Q_1 = (30)(2) = 60 \mu\text{C}$$

$$Q_2 = (60)(3) = 180 \mu\text{C}.$$

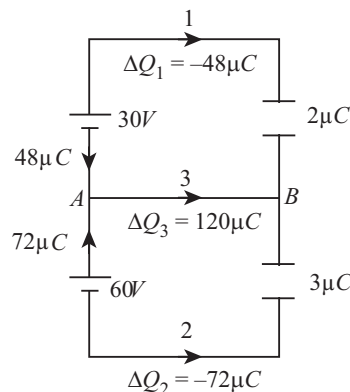
Look at the upper plate of the $2 \mu\text{F}$ capacitor. Initially, it is having a charge $Q = +108 \mu\text{C}$. After the switch is closed, it has charge $Q_1 = +60 \mu\text{C}$.

Charge can reach or move out from this plate only via path 1. Therefore, a charge $\Delta Q_1 = -48 \mu\text{C}$ must have flown in the indicated direction along path 1.

Now, look at the lower plate of $3 \mu\text{F}$ capacitor. Charge on it changed from $-108 \mu\text{C}$ to $-180 \mu\text{C}$. Charge

can reach this plate only via path 2. Therefore, a charge $\Delta Q_2 = -72 \mu\text{C}$ flows through this path in the indicated direction.

Look at the two plates connected at B . Together, these two plates had charge $= -108 + 108 = 0$. After closing the switch, the two plates have a total charge $= -60 + 180 = +120 \mu\text{C}$. This charge could have reached these plates only through path AB . Hence, charge flow $\Delta Q_3 = +120 \mu\text{C}$.



- Initial energy stored in the capacitors is

$$U_i = \frac{1}{2} C_0 V^2 = \frac{1}{2} \times \frac{6}{5} \times (90)^2 = 4860 \mu\text{J}$$

Final energy in the capacitors is

$$U_f = \frac{1}{2} (2)(30)^2 + \frac{1}{2} (3)(60)^2 = 6300 \mu\text{J}$$

Work done by the cells is

$$W = (72)(60) - (48)(30) = 2880 \mu\text{J}.$$

$$W = (U_f - U_i) + \text{Heat}$$

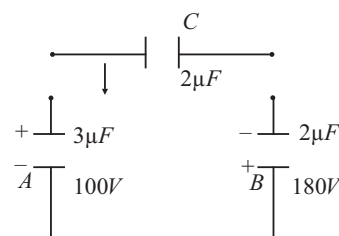
$$\text{or, Heat} = 2880 - (6300 - 4860)$$

$$= 1440 \mu\text{J}$$

$$= 1.44 \text{ mJ}.$$

Example 40 Two capacitors A and B with capacities $3 \mu\text{F}$ and $2 \mu\text{F}$ are charged to a potential difference of 100V and 180V, respectively. One plate of each capacitor is connected as shown. One lead of both the capacitors is free. The upper plate of A is positive and that of B is negative. An uncharged $2 \mu\text{F}$ capacitor C , with lead wires, falls and completes the circuit. Find:

- the final charge on the three capacitors
- the amount of electrostatic energy stored in the system before and after completion of the circuit.



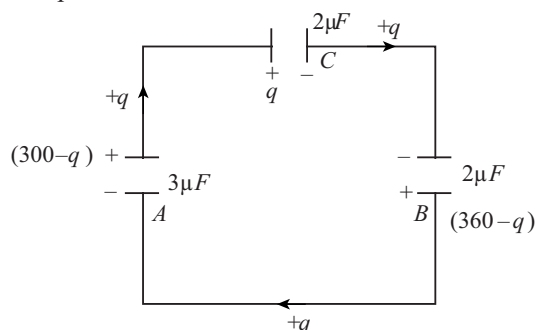
Solution**Concepts**

- (i) One common mistake is to consider the capacitors A and B in series. The capacitors are in series only when they have same charge on them.
- (ii) We will find final charge using Kirchhoff's laws.

- (a) Initial charge on capacitor A is $q_1 = 300 \mu\text{C}$.

Charge on capacitor B is $q_2 = 360 \mu\text{C}$.

Let charge q flow in the loop (as shown) after the capacitor C is connected.



[Actually, we can think that A loses charge q . The positive plate of A gives charge q to C and its negative plate gives $-q$ charge to B (or, receives $+q$ from B). Charge on B also reduces by q]

Applying Kirchhoff's loop law:

$$\frac{300 - q}{3} - \frac{q}{2} + \frac{360 - q}{2} = 0$$

or, $q = 210 \mu\text{C}$.

$\therefore$ Final charges on the three capacitors are

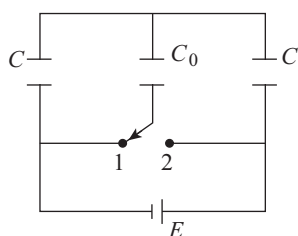
$$q_A = 300 - 210 = 90 \mu\text{C}$$

$$q_B = 360 - 210 = 150 \mu\text{C}$$

$$q_C = 210 \mu\text{C}.$$

$$\begin{aligned} \text{(b)} \quad U_i &= \frac{1}{2}(3)(100)^2 + \frac{1}{2}(2)(180)^2 \\ &= 15,000 + 32,400 = 47,400 \mu\text{J} = 47.4 \text{ mJ} \\ U_f &= \frac{(90)^2}{2 \times 3} + \frac{(150)^2}{2 \times 2} + \frac{(210)^2}{2 \times 2} \\ &= 18,000 \mu\text{J} = 18 \text{ mJ}. \end{aligned}$$

Example 41 In the circuit shown, find the heat generated after the switch is shifted from position 1 to position 2.

**Solution****Concepts**

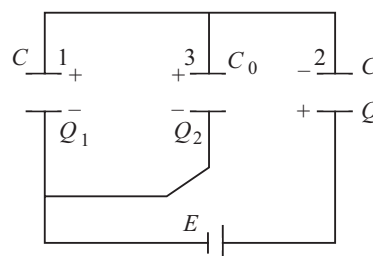
- (i) We need to find charges on all capacitors with switch in position 1 and then in position 2. From this, we will find the flow of charge through the cell.

- (ii) Heat = $W_{\text{cell}} + U_i - U_f$

U_i and U_f are initial and final energies stored in the capacitor system.

When the switch is in position 1, capacitors 1 and 3 are in parallel. Their equivalent is $(C + C_0)$. $(C + C_0)$ is in series with C . Thus, the equivalent capacitance is

$$\frac{C(C + C_0)}{C + C + C_0}$$



$$\therefore \text{Charge given by the cell is } Q = \frac{C(C + C_0)E}{2C + C_0}$$

Capacitor 2 has charge Q . Charge Q gets divided on 1 and 3 such that potential difference across them is same.

$$\frac{Q_1}{C} = \frac{Q_2}{C_0} \quad \text{(i)}$$

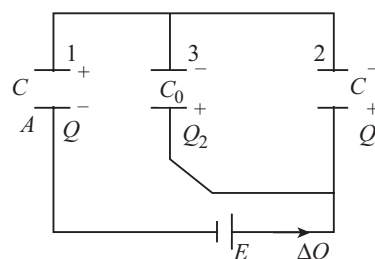
$$\text{And } Q_1 + Q_2 = Q \quad \text{(ii)}$$

$$\text{From (i) and (ii), } Q_1 = \frac{CQ}{C + C_0} = \frac{C^2 E}{2C + C_0}$$

$$\text{and } Q_2 = \frac{C_0 Q}{C + C_0} = \frac{CC_0 E}{2C + C_0}$$

Polarity of the plates are as shown.

After the switch is moved to position 2, the circuit is similar to the earlier circuit and with a careful observation one can write the charges on the capacitors. The charges are as shown. Obviously, total energy stored in the capacitor system has not changed.



4.44 Electrostatics and Current Electricity

$$U_i - U_f = 0.$$

Look at plate A of capacitor 1. Initial charge on it is $-Q_1$.

After the switch is shifted, charge on the plate is $-Q$.

Change in charge on plate A is

$$\begin{aligned} -[Q - Q_1] &= -\left[\frac{C(C+C_0)E}{2C+C_0} - \frac{C^2E}{2C+C_0} \right] \\ &= -\frac{CC_0E}{2C+C_0} \end{aligned}$$

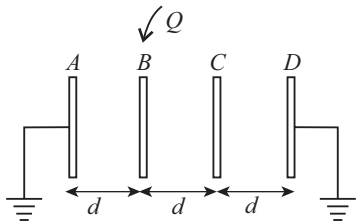
This charge comes to plate A from the negative terminal of the cell. Therefore, the charge flow through the cell is

$$\Delta Q = \frac{CC_0E}{2C+C_0}$$

$$\therefore W_{\text{cell}} = E \cdot \Delta Q = \frac{CC_0E^2}{2C+C_0}$$

$$\therefore \text{Heat} = W_{\text{cell}} = \frac{CC_0E^2}{2C+C_0}$$

Example 42 Four identical large plates are kept parallel to one another. d is separation between the two consecutive plates. The extreme plates are grounded as shown. Area of each plate is S . Plate B is given a charge Q . Find the potential difference between the plates B and C .



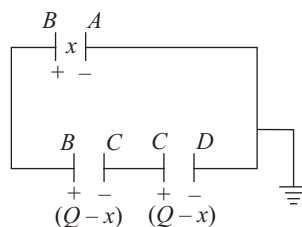
Solution

Concepts

- The problem can be solved easily by considering the system as combination of three capacitors.
- Plate A and D can acquire any amount of charge as they are grounded.
- Total charge on plate C is zero.

We can assume the plate system as combination of three capacitors as shown.

We have assumed a charge x on the capacitor formed between B and A . It means the left face of B has x charge. Obviously, its other face (forming a capacitor with C) has $(Q-x)$ charge. Net charge on two faces of C adds to zero.



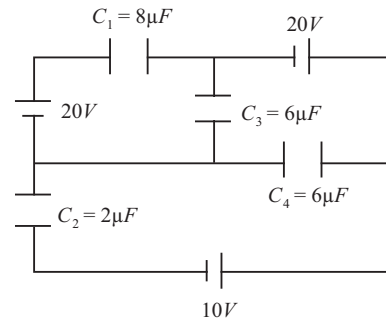
$$C = \frac{\epsilon_0 S}{d} = \text{capacitance of one capacitor.}$$

$$\frac{x}{C} = \frac{Q-x}{C/2}; \quad \Rightarrow \quad x = \frac{2Q}{3}$$

$\therefore$ Charge on the capacitor between B and C is $\frac{Q}{3}$.

$$\therefore V_{BC} = \frac{Q}{3C} = \frac{Qd}{3\epsilon_0 S}.$$

Example 43 Find the charge on capacitor C_1 shown in the figure.

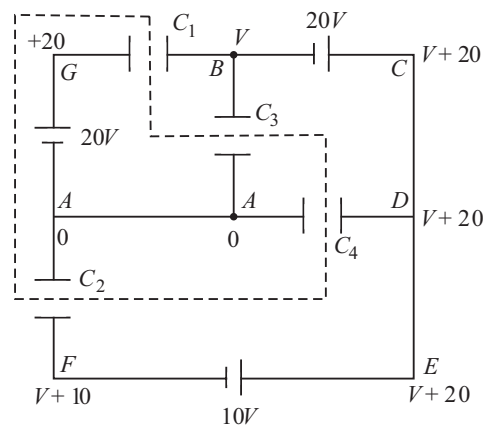


Solution

Concepts

- The problem can be solved using Kirchhoff's law. But we will use the method illustrated in Example 30. This will greatly simplify the mathematics.
- An isolated system will have a net zero charge on it. An isolated system may also include a cell inside it.

We will assume $V_A = 0$ and $V_B = V$.



Obviously, potential at C is $V+20$. C , D and E are same points. Potential at F is 10V lower than the potential at E , i.e., $V+10$. Potential at G is 20V more than V_A .

Thus, we have marked potential at all points in the circuit. The part of the circuit inside the loop shown in broken line is an isolated system. Net charge on this system must be zero.

Charge on left plate of $C_1 = (20 - V) \cdot 8$

Charge on lower plate of $C_3 = (0 - V) \cdot 6$

Charge on left plate of $C_4 = [0 - (V + 20)] \cdot 6$

Charge on the upper plate of $C_2 = [0 - (V + 10)] \cdot 2$

Sum of all these charges must be zero.

$$\therefore (20 - V) \cdot 8 - V \cdot 6 - (V + 20) \cdot 6 - (V + 10) \cdot 2 = 0$$

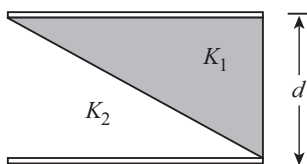
$$\text{or, } V = \frac{10}{11}V$$

$\therefore$ Charge on C_1 is $= (20 - V) \cdot 8$

$$= (20 - \frac{10}{11}) \cdot 8$$

$$= \frac{1680}{11} \mu\text{C}.$$

Example 44 A parallel plate capacitor has plate area A and separation d . The space between the plates is filled with two wedges of dielectric constants K_1 and K_2 as shown. Find the capacitance.

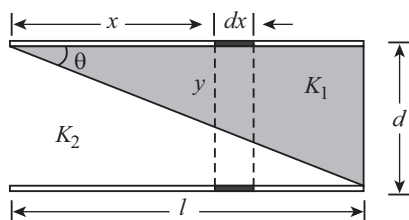


Solution

Concepts

Think of a thin strip in a metal plate and a corresponding parallel strip in the other plate. These thin strips make a capacitor with the space between them filled with two dielectrics. The entire capacitor is made of many such thin capacitor, all connected in parallel.

Let the length of each plate be l and its breadth (perpendicular to the plane of the diagram) be b .



Consider a strip of width dx as shown in the diagram. The capacitance of capacitor between these two small strips is

$$\begin{aligned} dc &= \frac{\epsilon_0 b dx}{\frac{y}{K_1} + \frac{d-y}{K_2}} \\ &= \frac{\epsilon_0 b dx}{\frac{x \tan \theta}{K_1} + \frac{d - x \tan \theta}{K_2}} \\ &= \frac{K_1 K_2 \epsilon_0 b dx}{d K_1 + (K_2 - K_1) \tan \theta \cdot x} \end{aligned}$$

$$= \frac{K_1 K_2 \epsilon_0 b dx}{d \cdot K_1 + (K_2 - K_1) \frac{d}{l} \cdot x}$$

$$= \frac{K_1 K_2 \epsilon_0 b l dx}{d [l K_1 + (K_2 - K_1) \cdot x]}$$

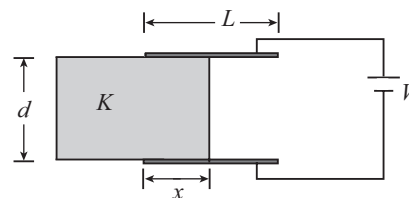
$$= \frac{K_1 K_2 \epsilon_0 A}{d} \frac{dx}{l K_1 + (K_2 - K_1) x}$$

All small capacitors are in parallel. Their equivalent is found by adding their capacitances.

$$\begin{aligned} \therefore C &= \frac{K_1 K_2 \epsilon_0 A}{d} \int_0^l \frac{dx}{l K_1 + (K_2 - K_1) x} \\ &= \frac{K_1 K_2 \epsilon_0 A}{d (K_2 - K_1)} \left[\ln [l K_1 + (K_2 - K_1) x] \right]_0^l \\ &= \frac{K_1 K_2 \epsilon_0 A}{d (K_2 - K_1)} \left[\ln (K_2 l) - \ln (K_1 l) \right] \\ &= \frac{K_1 K_2 \epsilon_0 A}{d (K_2 - K_1)} \ln \left(\frac{K_2}{K_1} \right) \end{aligned}$$

Example 45 Force on dielectric slab

Figure shows a parallel plate capacitor with plates of width b and length l . Separation between the plates is d . The plates are held fixed and connected to a cell of emf V . A dielectric slab of thickness d and width b is slowly inserted between the plates. Find the force experienced by the dielectric slab due to the electric field between the plates when a length x of the slab is inside the plates. The dielectric constant is K . How much force does the external agent apply on the dielectric slab?

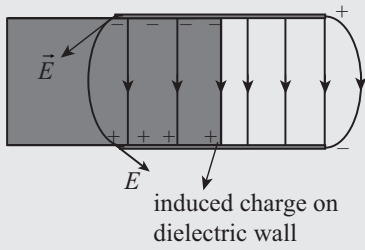


Solution

Concepts

- (i) Why will the dielectric slab experience any force?

The electric field lines are curved near the edges of the capacitor plates. Tangent on these lines gives the direction of electric field due to charge on the plates. When a dielectric is inserted, charge is induced on its wall. These induced charges experience force due to the field (E) created by charge on the plates. You can justify that this force on the dielectric slab will be directed into the capacitor.



- (ii) It is difficult to know the exact value of induced charge and electric field near the edges. Therefore, direct calculation of the force will be very difficult. This force will be calculated using work energy method.

Capacitance, when a length x of the dielectric is inside the capacitor is

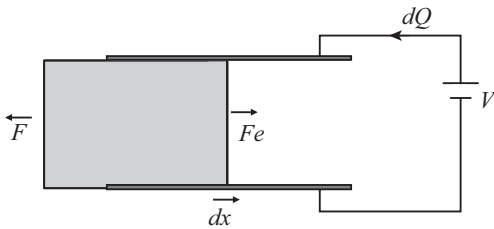
C = Parallel combination of capacitance of the dielectric-filled part and the air-filled part.

$$\begin{aligned}
 &= \frac{K\epsilon_0 bx}{d} + \frac{\epsilon_0 b(l-x)}{d} \\
 &= \frac{\epsilon_0 b}{d} [l + x(K-1)] \quad (i)
 \end{aligned}$$

If the slab is further inserted by dx , change in capacitance is

$$dC = \frac{\epsilon_0 b(K-1)}{d} dx \quad (ii)$$

If the slab is being moved very slowly (so that its KE does not change), then there must be an external agent applying force F on the slab in outward direction. This force should be equal to the force applied by the electric force (F_e) on the slab.



As the slab moves, capacitance increases and additional charge flows to the capacitor.

$$Q = CV$$

When the capacitance changes (due to movement of the slab by dx), the additional charge flow is

$$dQ = V \cdot dC$$

Work done by the cell is $dW_{\text{cell}} = VdQ = V^2 dC$.

Change in energy stored in the capacitor can be written as

$$U = \frac{1}{2} CV^2$$

$$\Rightarrow dU = \frac{1}{2} V^2 dC$$

Using work energy theorem, we can write

$$dW_{\text{cell}} + dW_{\text{ext}} = dU \quad [\because \text{change in } KE = 0]$$

$$\text{or, } V^2 dC - F \cdot dx = \frac{1}{2} V^2 dC$$

$$\text{or, } F = \frac{1}{2} V^2 \frac{dC}{dx} = \frac{\epsilon_0 b(K-1)V^2}{2d} \quad [\text{using (ii)}]$$

Force is independent of x .

Example 46 Force on dielectric slab (battery disconnected)

Answer the last question if the capacitor is given a charge Q and battery is disconnected before the dielectric gets inserted.

Solution

Concepts

Battery is not there. It means work done by the battery is not there. Charge on the capacitor is constant.

$$\text{As in last example, } \frac{dC}{dx} = \frac{\epsilon_0 b(K-1)}{d}$$

Energy stored in the capacitor when slab is inserted to a length x is

$$U = \frac{Q^2}{2C}$$

$$\text{or, } \frac{dU}{dx} = -\frac{Q^2}{2C^2} \frac{dC}{dx}$$

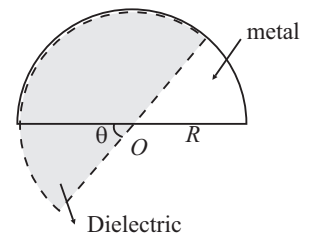
$$\text{or, } -F = -\frac{Q^2}{2C^2} \frac{dC}{dx} \quad [F \text{ is the electric force}]$$

$$\begin{aligned}
 \text{or, } F &= \frac{Q^2}{2 \left[\frac{\epsilon_0 b}{d} [l + x(K-1)] \right]^2} \frac{\epsilon_0 b(K-1)}{d} \\
 &= \frac{Q^2 \cdot d(K-1)}{2\epsilon_0 b [l + x(K-1)]^2}
 \end{aligned}$$

This force is dependent on x .

Example 47 Torque on dielectric slab.

A variable capacitor consists of two parallel stationary semi-circular plates of radius R at separation d . A dielectric plate is inserted between the plates so as to occupy the complete space. The dielectric disc can rotate about the centre of curvature O .



Dielectric constant is K . The capacitor is kept connected to a cell of emf V . Find the torque experienced by the dielectric slab due to the electric field when it is in the position shown (a part of the dielectric is outside the metal plates).

Solution

Concepts

- This question is conceptually similar to Example 45. We will write the capacitance and potential energy for the capacitor in position shown. Now, we find change in energy stored and work done by the cell when θ is increased by $d\theta$. Applying work energy theorem will give the desired result.
- Work done by a torque in angular displacement $d\theta$ is $dW = \tau d\theta$.

In the position shown, the capacitor is a parallel combination of an air capacitor and a dielectric capacitor.

$$C = \frac{K\epsilon_0 \left(\frac{\pi R^2}{2} - \frac{\theta R^2}{2} \right)}{d} + \frac{\epsilon_0 \frac{\theta R^2}{2}}{d}$$

$$\text{or, } C = \frac{\epsilon_0 R^2}{2d} [K\pi + (1-K)\theta] \quad (\text{i})$$

When the dielectric is further rotated by $d\theta$, the capacitance changes by

$$dC = \frac{\epsilon_0 R^2}{2d} (1-K) d\theta$$

$$\text{Energy stored } U = \frac{1}{2} CV^2$$

$$dU = \frac{1}{2} V^2 dC$$

Charge on capacitor $Q = CV$

$$dQ = V dC$$

Work done by the cell when capacitance changes by dC is

$$dW_{\text{cell}} = V \cdot dQ = V^2 dC.$$

In slowly turning the dielectric disc by $d\theta$, let the external agent apply a torque τ using work energy theorem.

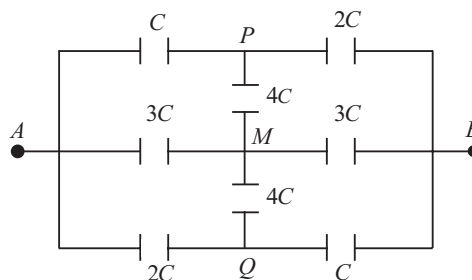
$$dW_{\text{ext}} + dW_{\text{cell}} = dU$$

$$\text{or, } \tau d\theta + V^2 dC = \frac{1}{2} V^2 dC$$

$$\text{or, } \tau = -\frac{1}{2} V^2 \frac{dC}{d\theta}$$

$$= -\frac{V^2 \epsilon_0 R^2}{4d} (1-K) = \frac{V^2 \epsilon_0 R^2}{4d} (K-1)$$

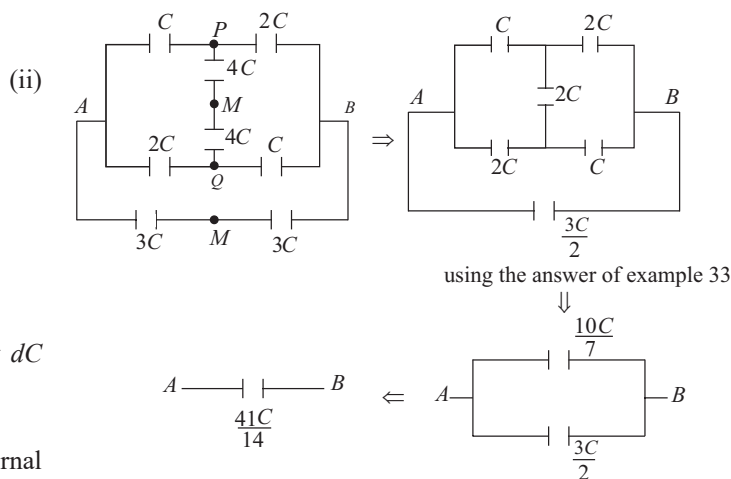
Example 48 Find the equivalent capacitance of the circuit between the points A and B .



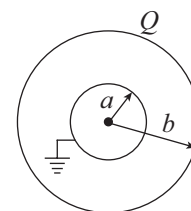
Solution

Concepts

- Refer to Example 34. Due to symmetry, the charge on two $3C$ capacitance will be same as if they are in series. For similar reason, the charge on two $4C$ capacitance will also be same. The junction at M is a false junction. We can consider the path AMB to be a separate parallel path from A to B .
- Refer to Example 33. After putting two $3C$ capacitances in a separate parallel path from A to B , the remaining circuit is exactly same as the one given in Example 33.



Example 49 Two concentric metallic spherical shells have radii a and $b(>a)$. The inner shell is grounded and the outer shell is given a charge Q . Find the capacitance of the system.



Solution

Concepts

- Charge will be induced on the inner shell so as to make its potential zero.
- The inner surface of the outer shell will have a charge equal and opposite to that on the inner shell.

- (iii) The outer wall of the larger shell will also have some charge.
- (iv) There is a capacitor formed between the inner shell and the inner wall of the outer shell. There is another capacitor between the outer wall of the bigger shell and infinity (think of a very large shell having equal and opposite charge of that on the outer wall of shell of radius b). Inner ball is at zero potential and the large ball (at ∞) is also at zero potential. Therefore, these two capacitors are in parallel.

Capacitance of the spherical capacitor formed between the inner wall of larger shell and the smaller shell is

$$C_1 = \frac{4\pi\epsilon_0 ab}{b-a}$$

Capacitance between outer shell and infinity is

$$C_2 = 4\pi\epsilon_0 b$$

$\therefore$ Equivalent capacitance, $C = C_1 + C_2$

$$= 4\pi\epsilon_0 b \left[\frac{a}{b-a} + 1 \right] = \frac{4\pi\epsilon_0 b^2}{(b-a)}$$

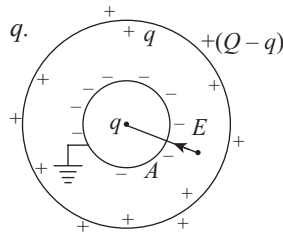
Alternative

Let charge on the inner ball be q .

$$V_{\text{inner}} = 0$$

$$\text{or, } \frac{1}{4\pi\epsilon_0} \frac{q}{a} + \frac{1}{4\pi\epsilon_0} \frac{Q}{b} = 0$$

$$\text{or, } q = -\frac{aQ}{b}$$



The electric field at a radial distance x from the centre of the system ($a < x < b$) is

$$E = \frac{1}{4\pi\epsilon_0} \frac{|q|}{x^2} \text{ directed radially inwards}$$

Potential difference between the two shells can be obtained as

$$dV = -\vec{E} \cdot d\vec{x}$$

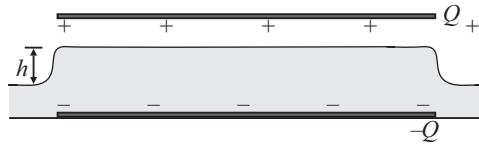
$$\text{or, } \int_{V_A}^{V_B} dV = - \int_a^b E dx \cos 180^\circ$$

$$\text{or, } V_B - V_A = \frac{|q|}{4\pi\epsilon_0} \int_a^b \frac{dx}{x^2} = \frac{|q|}{4\pi\epsilon_0} \left(\frac{1}{a} - \frac{1}{b} \right)$$

$$\text{or, } V = \frac{Q}{4\pi\epsilon_0} \frac{(b-a)}{b^2}$$

$$\text{or, } C = \frac{Q}{V} = \frac{4\pi\epsilon_0 b^2}{(b-a)}$$

Example 50 A parallel plate capacitor is placed such that the plates are horizontal and the lower plate is dipped into a liquid of dielectric constant K and density ρ . The capacitor is given a charge Q . It is found that the liquid surface inside the capacitor is h above the outside liquid surface. Find h in equilibrium position. The plate area of the capacitor is A and neglect edge effect.



Solution

Concepts

- The weight of the elevated liquid part inside the plates is balanced by electrostatic force acting on it.
- Charge is induced on the upper as well as lower dielectric surface. Value of induced charge can be written using equation (11).
- The upper surface of the liquid gets a negative charge induced on it and its lower surface (in contact with the lower plate) has a positive charge induced on it.

Charge induced on the upper liquid surface is

$$Q_{\text{in}} = -Q \left[1 - \frac{1}{K} \right]$$

Lower surface of liquid has same magnitude of positive charge induced on it. We can say that the effective charge on the lower plate is

$$-Q + Q \left[1 - \frac{1}{K} \right] = -\frac{Q}{K}$$

The electric field at the location of top surface of the liquid created due to the charge on the upper plate and the effective charge on the lower plate is

$$E = \frac{Q}{2A\epsilon_0} + \frac{Q}{2KA\epsilon_0} = \frac{(K+1)Q}{2A\epsilon_0 K} (\downarrow)$$

Force on the negative charge on the upper surface of the liquid due to this field is

$$F = Q_{\text{in}} E = Q \left(1 - \frac{1}{K} \right) \frac{(K+1)Q}{2A\epsilon_0 K} = \frac{(K^2-1)Q^2}{2A\epsilon_0 K^2} (\uparrow)$$

Weight of the liquid that is raised inside the plates is

$$W = hA\rho g$$

For equilibrium,

$$hA\rho g = \frac{(K^2-1)Q^2}{2A\epsilon_0 K^2}$$

$$\text{or, } h = \frac{(K^2-1)Q^2}{2A^2 K^2 \epsilon_0 \rho g}$$

Worksheet 1

1. 64 identical mercury drops combine to form one bigger drop. The capacitance of the bigger drop, as compared to that of a smaller drop will be-

(a) 8 times (b) 64 times
(c) 4 times (d) 16 times

2. A parallel plate capacitor is charged and then isolated from the battery. On increasing the plate separation—

Charge	Potential	Capacity
--------	-----------	----------

(a) remains constant	remains constant	decreases
(b) remains constant	increases	decreases
(c) remains constant	decreases	increases
(d) increases	increases	decreases

3. On increasing the plate separation of a charged isolated capacitor, its energy

(a) increases (b) decreases
(c) remains unchanged (d) None of the above

4. Two isolated charged metallic conductors of capacitance C_1 and C_2 having charges Q_1 and Q_2 , respectively are connected to each other. After connection, there is:

(a) No change in the energy of the system
(b) An increase in the energy of the system
(c) Always a decrease in the energy of the system
(d) A decrease in energy of the system except when $q_1 C_2 = q_2 C_1$

5. The capacitance of a parallel plate condenser is C_0 . If a dielectric of relative permittivity ϵ_r and thickness equal to one fourth the plate separation is placed between the plates, then its capacity becomes C . Then value of $\frac{C}{C_0}$ will be-

(a) $\frac{4\epsilon_r}{3\epsilon_r + 1}$	(b) $\frac{4\epsilon_r}{3\epsilon_r + 1}$
(c) $\frac{3\epsilon_r}{2\epsilon_r + 1}$	(d) $\frac{2\epsilon_r}{\epsilon_r + 1}$

6. Three capacitors 2 μF , 3 μF and 5 μF can withstand voltages of 3V, 2V and 1V respectively. Their series combination can withstand a maximum voltage equal to

(a) 5 Volts (b) (31/6) Volts
(c) (26/5) Volts (d) None

7. A parallel plate capacitor has a plate separation of 0.01 mm and use a dielectric (whose dielectric strength

is 19KV/mm) as an insulator. The maximum potential difference that can be applied to the terminals of the capacitor is

(a) 190V (b) 290V
(c) 95V (d) 350V

8. A parallel plate capacitor has an electric field of 10^5 V/m between the plates. If the charge on the capacitor plate is $1\mu\text{C}$, then the force on each capacitor plate is

(a) 0.1 N (b) 0.05 N
(c) 0.02 N (d) 0.01 N

9. A 40 μF capacitor in a defibrillator is charged to 3000 V. The energy stored in the capacitor is sent through a patient during a pulse of duration 2 ms. The power delivered to the patient is:-

(a) 45kW (b) 90kW
(c) 180kW (d) 360k

10. A capacitor is connected to a battery. The force of attraction between the plates when the separation between them is halved

(a) remains the same
(b) becomes eight times
(c) becomes four times
(d) becomes two times

11. A conducting body 1 has some initial charge Q , and its capacitance is C . There are two other conducting bodies, 2 and 3, having capacitances: $C_2 = 2C$ and $C_3 \rightarrow \infty$. Bodies 2 and 3 are initially uncharged. Body 2 is touched with body 1. Then, body 2 is removed from body 1 and touched with body 3, and then removed. This process is repeated N times. Then, the charge on body 1 at the end must be

(a) $Q/3^N$ (b) $Q/3^{N-1}$
(c) Q/N^3 (d) None

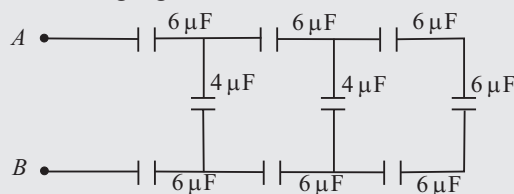
12. A capacitor stores $60\mu\text{C}$ charge when connected across a battery. When the gap between the plates is filled with a dielectric, a charge of $120\mu\text{C}$ flows through the battery. The dielectric constant of the material inserted is:

(a) 1 (b) 2
(c) 3 (d) none

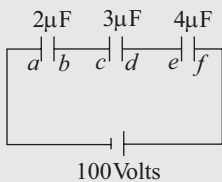
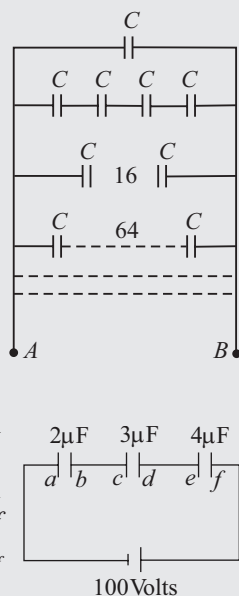
13. You are given 64 capacitors each of capacitance 6 mF. How would you connect them, so that the total capacity is again 6 mF

- (a) 16 in series and 4 in parallels
 (b) 64 in series
 (c) a group of 8 in series and 8 groups in parallel
 (d) none of these
14. Three capacitors each of capacity of 6 mF are to be connected in such a way that the effective capacitance is 9 mF. This can be done by
- (a) connecting all of them in series
 (b) connecting them in parallels
 (c) two in series and one in parallel
 (d) two in parallel and one in series
15. Two metal plates form a parallel plate condenser. The distance between the plates is d . A metal sheet of thickness $d/2$ and of same area as the plates is inserted completely between the plates. The ratio of capacitance before the insertion of the metal sheet to that after insertion is
- (a) 1: 1 (b) 1: 2
 (c) 2: 1 (d) None

16. The resultant capacitance between A and B in the following figure is.



- (a) 1 μF (b) 6 μF
 (c) 2 μF (d) 4 μF
17. An infinite number of identical capacitors, each of capacitance $1\mu\text{F}$ are connected in infinite number of rows having 1, 4, 16, 64, 256,..... capacitors respectively as indicated in the fig. If these rows are connected in parallel, the equivalent capacitance between point A and B will be:
- (a) $2/3 \mu\text{F}$
 (b) $1/3 \mu\text{F}$
 (c) $4/3 \mu\text{F}$
 (d) $5/3 \mu\text{F}$
18. Three capacitors are connected to a D.C. source of 100 volt. If the charge accumulated on plates of a, b, c, d, e and f are q_a, q_b, q_c, q_d, q_e and q_f respectively, then



(a) $q_b + q_d + d_f = \frac{100}{9}C$

(b) $q_b + q_d + d_f = 0$

(c) $q_a + q_c + d_e = 50C$

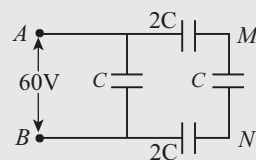
(d) $q_b = q_d = q_f$

19. A conducting sphere of radius 10cm is given a charge of $10 \mu\text{F}$. Another uncharged sphere of radius 20cm is allowed to touch it for some time. After that if the spheres are separated, then surface density of charges, on the spheres will be in the ratio of
- (a) 1:4 (b) 1:3
 (c) 2:1 (d) 1:1
20. A capacitor of capacity 5 mF is charged to a potential difference of 100 V. The battery is disconnected, and then the capacitor is connected in parallel to an uncharged capacitor of capacitance C . The potential difference measured across this combination is 25 V. The capacitance C is

- (a) 15 mF (b) 10 mF
 (c) 5 mF (d) 20 mF

21. Two identical capacitors that are initially uncharged are connected in series across a 10 V battery. If only one of these capacitors is connected across this 10 V battery, the energy stored in it is E_1 . The total energy stored in the capacitors when the series combination is connected to the battery is
- (a) $4 E_1$ (b) $2 E_1$
 (c) E_1 (d) $E_1/2$

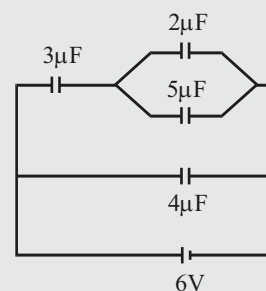
22. In the circuit shown, a potential difference of 60V is applied across AB . The potential difference between the points M and N is



- (a) 10 V (b) 15 V
 (c) 20 V (d) 30 V

23. In the circuit shown in figure, the ratio of charges on $5\mu\text{F}$ and $4\mu\text{F}$ capacitor is:

- (a) $4/5$
 (b) $3/5$
 (c) $3/8$
 (d) $1/2$



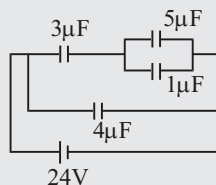
24. The minimum number of capacitors, each of $3\ \mu\text{F}$, required to make a circuit with an equivalent capacitance $1.8\ \mu\text{F}$ is

(a) 3 (b) 4
(c) 5 (d) 6

25. There are large number of identical capacitors rated $8\ \mu\text{F}$, $250\ \text{V}$. The minimum number of capacitors required to form a network having equivalent capacitance $16\ \mu\text{F}$ with a peak voltage rating of $1000\ \text{V}$ is:

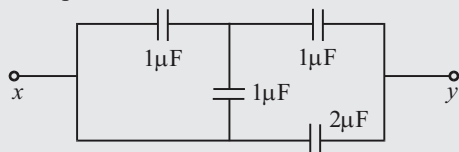
(a) 2 (b) 4
(c) 16 (d) 32

26. In the circuit shown, the energy stored in $1\ \mu\text{F}$ capacitor is



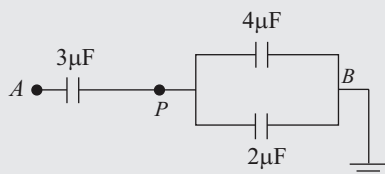
(a) $40\ \mu\text{J}$ (b) $64\ \mu\text{J}$
(c) $32\ \mu\text{J}$ (d) none

27. In the given diagram, the equivalent capacitance across points X and Y is:



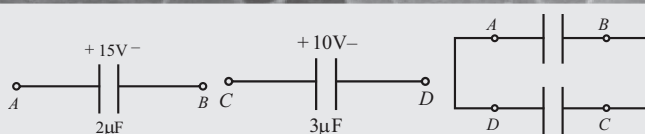
(A) $\frac{5}{6}\ \mu\text{F}$ (b) $\frac{7}{6}\ \mu\text{F}$
(c) $\frac{8}{3}\ \mu\text{F}$ (d) $1\ \mu\text{F}$

28. In the figure a potential of $+1200\text{V}$ is given to point A and point B is earthed, what is the potential at the point P :



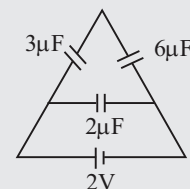
(a) 100V (b) 200V
(c) 400V (d) 600V

29. In the first two figures initial status of two charged capacitors have been shown. The third diagram shows how the two capacitors are connected to each other. Which of the followings is incorrect about the circuit shown in the third diagram?



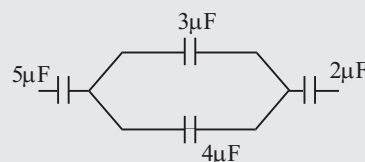
(a) Final charge on each capacitor will be zero
(b) Final total electrical energy of the capacitors will be zero
(c) Total charge flow from A to D is $30\ \mu\text{C}$
(d) Total charge flow from A to D is $-30\ \mu\text{C}$

30. The total energy stored in the condenser system shown in the figure will be—



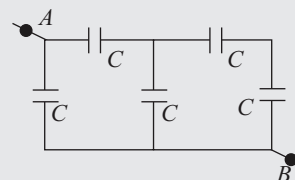
(a) $2 \times 10^{-6}\ \text{J}$
(b) $4 \times 10^{-6}\ \text{J}$
(c) $8 \times 10^{-6}\ \text{J}$
(d) $16 \times 10^{-6}\ \text{J}$

31. If charge on left plate of the $5\ \mu\text{F}$ capacitor in the circuit segment shown in the figure is $-20\ \mu\text{C}$, the charge on the right plate of $3\ \mu\text{F}$ capacitor is



(a) $+8.57\ \mu\text{C}$ (b) $-8.57\ \mu\text{C}$
(c) $+11.42\ \mu\text{C}$ (d) $-11.42\ \mu\text{C}$

32. What is the equivalent capacitance of the system of capacitors between A & B

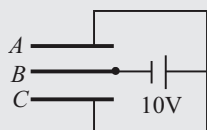


(a) $\frac{7}{6}\ \text{C}$ (b) $1.6\ \text{C}$
(c) C (d) None

33. Two capacitors having capacitances $8\ \mu\text{F}$ and $16\ \mu\text{F}$ have breaking voltages $20\ \text{V}$ and $80\ \text{V}$. They are combined in series. The maximum energy we can store in the combination is

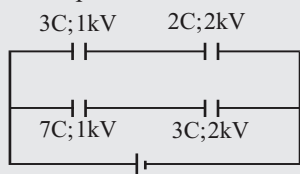
(a) $2400\ \mu\text{J}$ (b) $200\ \mu\text{J}$
(c) $1280\ \mu\text{J}$ (d) $1600\ \mu\text{J}$

34. Three plates A , B and C each of area $0.1\ \text{m}^2$ are separated by $0.885\ \text{mm}$ from each other as shown in the figure. A $10\ \text{V}$ battery is used to charge the system. The energy stored in the system is

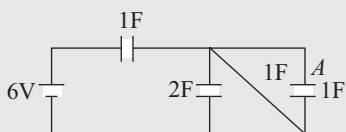


- (a) $1 \mu\text{J}$ (b) $10^{-1} \mu\text{J}$
 (c) $10^{-2} \mu\text{J}$ (d) $10^{-3} \mu\text{J}$
35. A capacitor of capacitance C is initially charged to a potential difference of V volt. Now it is connected to a battery of $2V$ volt with opposite polarity. The ratio of heat generated to the final energy stored in the capacitor will be

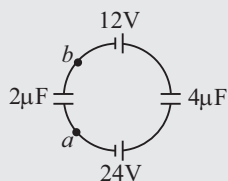
- (a) 1.75 (b) 2.25
 (c) 2.5 (d) $1/2$
36. The diagram shows four capacitors with capacitances and break down voltages as mentioned. What should be the maximum value of the external emf source such that no capacitor breaks down?



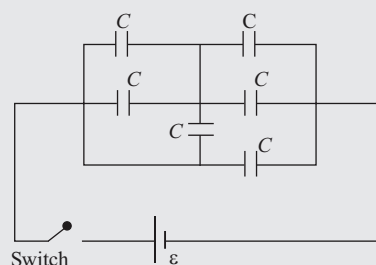
- (a) 2.5 kV (b) $10 / 3$ kV
 (c) 3 kV (d) 1 kV
37. In the combination of capacitors shown in figure, the potential difference across the plates of the capacitor A will be:



- (a) 4.8 V (b) 6 V
 (c) 1.2 V (d) 2.4 V
38. Find the potential difference $V_a - V_b$ between the points a and b shown in the figure:

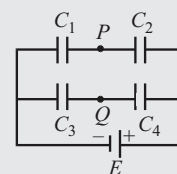


- (a) 8V
 (b) -0.25V
 (c) 0.50 V
 (d) -8V
39. Six capacitors each of capacitance ' C ' are connected as shown in the figure and initially all the capacitors are uncharged. Now a battery of emf $= \epsilon$ is connected. How much charge will flow through the battery if the switch is put on



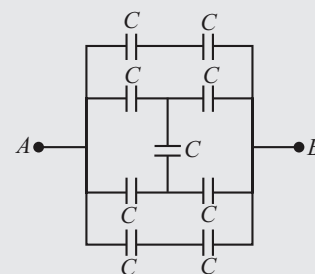
- (a) $\frac{9C\epsilon}{5}$ (b) $\frac{11C\epsilon}{5}$
 (c) $\frac{13C\epsilon}{5}$ (d) $\frac{7C\epsilon}{5}$
40. The potential difference between the points P and Q in the adjoining circuit will be-

- (a) $\frac{(C_1C_4 - C_2C_3)E}{(C_1 + C_3)(C_2 + C_4)}$
 (b) $\frac{C_2C_3E}{C_1C_2(C_3 + C_4)}$
 (c) $\frac{(C_2C_3 - C_1C_4)E}{(C_1 + C_2)(C_3 + C_4)}$
 (d) $\frac{(C_2C_3 - C_1C_4)E}{(C_1 + C_2 + C_3 + C_4)}$

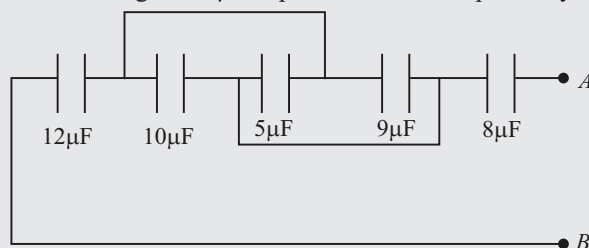


41. The resultant capacity between the points A and B in the shown circuit will be-

- (a) C
 (b) $\frac{C}{2}$
 (c) $2C$
 (d) $3C$

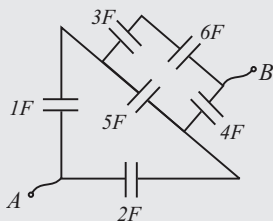


42. Five capacitors together with their capacitances are shown in the adjoining figure. The potential difference between the points A and B is 60 volt. The equivalent capacitance between the points A and B and charge on $5\mu\text{F}$ capacitor will be respectively -

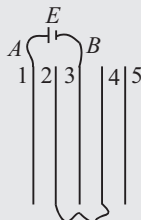


- (a) $44 \mu\text{F}$, $30 \mu\text{C}$ (b) $16 \mu\text{F}$, $150 \mu\text{C}$
 (c) $15 \mu\text{F}$, $200 \mu\text{C}$ (d) $4 \mu\text{F}$, $50 \mu\text{C}$

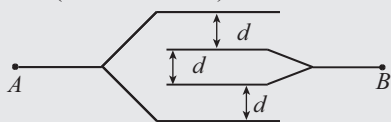
43. In the figure shown, the equivalent capacitance between 'A' and 'B' is:



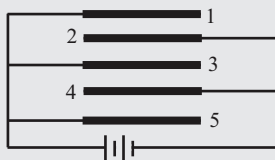
- (a) 3.75 F (b) 2 F
(c) 21 F (d) 16 F
44. Five conducting parallel plates having area A and separation between them d , are placed as shown in the figure. Plate number 2 and 4 are connected by a wire and between points A and B , a cell of emf E is connected. The charge supplied by the cell is



- (a) $\frac{3 \epsilon_0 A E}{4 d}$ (b) $\frac{2 \epsilon_0 A E}{3 d}$
(c) $\frac{4 \epsilon_0 A E}{d}$ (d) $\frac{\epsilon_0 A E}{2 d}$
45. Four metallic plates are arranged as shown in the figure. If d is the distance between each plate, then capacitance of the given system between points A and B is (Given $d \ll A$)



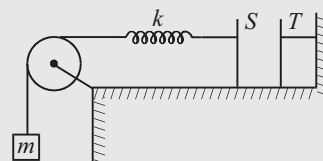
- (a) $\frac{\epsilon_0 A}{d}$ (b) $\frac{2 \epsilon_0 A}{d}$
(c) $\frac{3 \epsilon_0 A}{d}$ (d) $\frac{4 \epsilon_0 A}{d}$
46. Five identical plates are connected across a battery as shown:



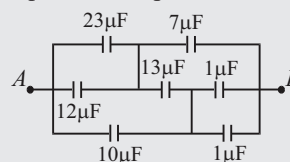
If the charge on plate 1 is $+q$, then the charges on the plates 2, 3, 4 and 5 are

- (a) $-q, +q, -q, +q$ (b) $-2q, +2q, -2q, +q$
(c) $-q, +2q, -2q, +q$ (d) None of the above
47. The plates S and T of an uncharged parallel plate capacitor are connected across a battery. The battery is then disconnected and the charged plates are now

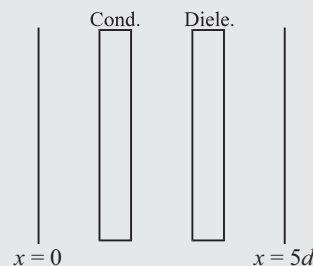
connected in a system as shown in the figure. The system shown is in equilibrium. All the strings are insulating and massless. The magnitude of charge on the capacitor is: [Area of plates = A]



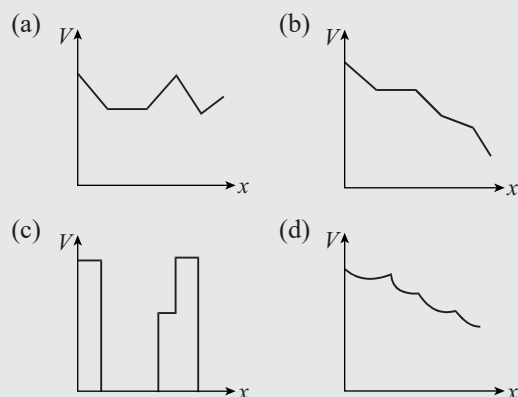
- (a) $\sqrt{2mgA\epsilon_0}$ (b) $\sqrt{\frac{4mgA\epsilon_0}{k}}$
(c) $\sqrt{mgA\epsilon_0}$ (d) $\sqrt{\frac{2mgA\epsilon_0}{k}}$
48. Find the equivalent capacitance across A & B



- (a) $\frac{28}{3} \mu F$ (b) $\frac{15}{2} \mu F$
(c) $15 \mu F$ (d) none
49. The distance between plates of a parallel plate capacitor is $5d$. Let the positively charged plate be at $x = 0$ and negatively charged plate be at $x = 5d$. Two slabs - one of conductor and other of a dielectric of equal thickness d - are inserted between the plates as shown in figure.



Potential versus x graph will look like:



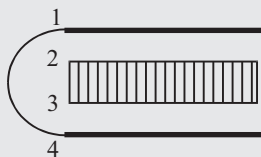
50. The distance between the plates of a charged parallel plate capacitor is 5 cm and electric field between the plates is 200 Vcm^{-1} . An uncharged metal bar of width 2 cm is fully immersed into the capacitor. The length of the metal bar is same as that of the plate of the capacitor. The voltage across capacitor after the immersion of the bar is

(a) zero (b) 400 V
(c) 600 V (d) 100 V

51. Condenser A has a capacity of $15 \mu\text{F}$ when it is filled with a medium of dielectric constant 15. Another condenser B has a capacity $1 \mu\text{F}$ with air between the plates. Both are charged separately by a battery of 100V. After charging, both are connected in parallel (like plates together) without the battery and the dielectric material removed. The common potential difference now is

(a) 400V (b) 800V
(c) 1200V (d) 1600V

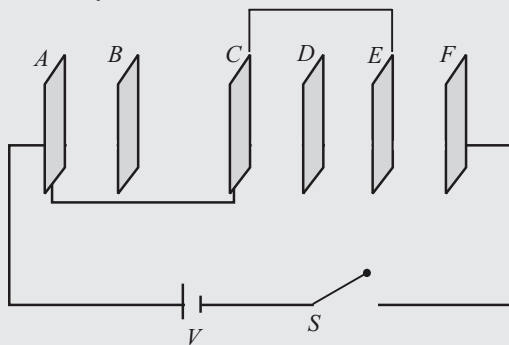
52. Four identical plates 1, 2, 3 and 4 are placed parallel to each other at equal distance as shown in the figure. Plates 1 and 4 are joined together and the space between 2 and 3 is filled with a dielectric of dielectric constant $k = 2$. The capacitance of the system between 1 and 3 & 2 and 4 are C_1 and C_2 respectively.



The ratio $\frac{C_1}{C_2}$ is:

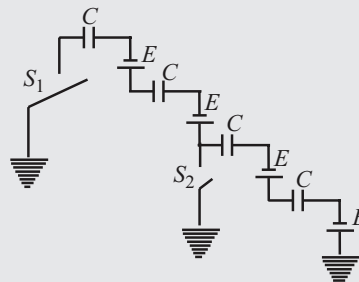
(a) $\frac{5}{3}$ (b) 1
(c) $\frac{3}{5}$ (d) $\frac{5}{7}$

53. In the figure shown A, B, C, D, E and F are conducting plates, each of area A . Any two consecutive plates are separated by a distance d . The net energy stored in the system after the switch S is closed is:



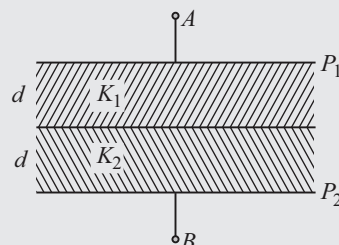
(a) $\frac{3\varepsilon_0 A}{2d} V^2$ (b) $\frac{5\varepsilon_0 A}{12d} V^2$
(c) $\frac{\varepsilon_0 A}{2d} V^2$ (d) $\frac{\varepsilon_0 A}{d} V^2$

54. In the given circuit, all the capacitors are initially uncharged. After closing the switch S_1 for a long time suddenly S_2 is also closed and kept closed for a long time. Total heat produced after closing S_2 will be:



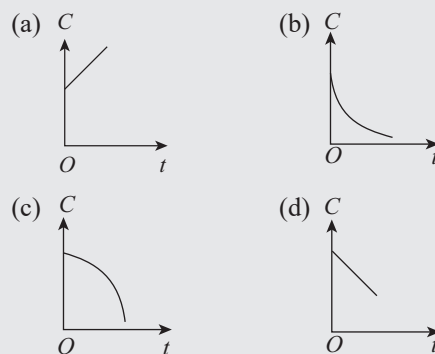
(a) $4 C\varepsilon^2$ (b) $\frac{1}{2} C\varepsilon^2$
(c) $2 C\varepsilon^2$ (d) 0

55. In the figure shown P_1 and P_2 are two conducting plates having charges of equal magnitude and opposite sign. Two dielectrics of dielectric constant K_1 and K_2 fill the space between the plates as shown in the figure. The ratio of electrical energy in 1st dielectric to that in the 2nd dielectric is

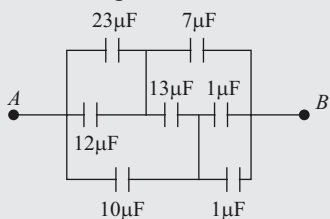


(a) 1 : 1 (b) $K_1 : K_2$
(c) $K_2 : K_1$ (d) $K_2^2 : K_1^2$

56. A dielectric slab is placed between the plates of a parallel plate capacitor. The capacitor is charged and the battery is disconnected. The dielectric slab is pulled out with a constant speed. Which of the following is correct for the variation of capacitance with time

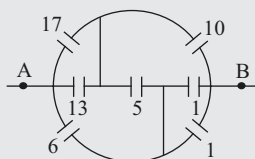


57. The equivalent capacitance across A & B is



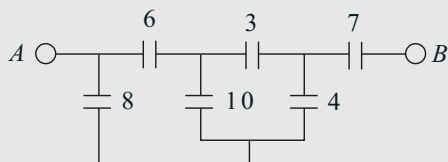
- (a) $\frac{28}{3} \mu\text{F}$ (b) $\frac{15}{2} \mu\text{F}$
(c) $15 \mu\text{F}$ (d) None of these

58. The equivalent capacitance across AB (all capacitance in μF) is



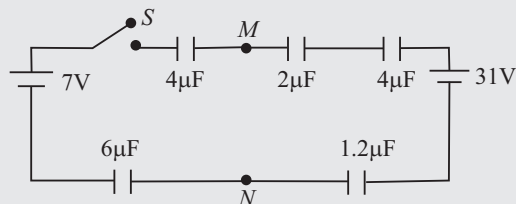
- (a) $6 \mu\text{F}$ (b) $9 \mu\text{F}$
(c) $15 \mu\text{F}$ (d) None of the above

59. In the circuit diagram shown all the capacitors are in μF . The equivalent capacitance between points A and B is: (in μF)



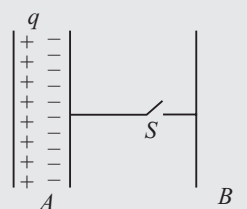
- (a) $14/5$ (b) $7/5$
(c) $3/7$ (d) None of these

60. Five capacitors are connected as shown in fig. Initially S is opened and all capacitors are uncharged. Now S is closed and steady state is obtained. The p.d. between the points M and N will be -



- (a) 12 V (b) 10 V
(c) 4 V (d) 3 V

61. Consider the situation shown in the figure. The capacitor A has a charge q on it whereas B is uncharged. The charge appearing on the capacitor B a long time after the switch is closed is:



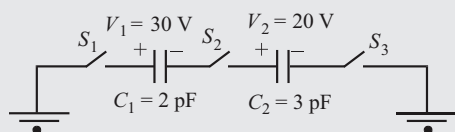
- (a) Zero (b) $q/2$
(c) q (d) $2q$

Worksheet 2

1. The plates of a air core parallel plate capacitor are connected to a voltage source. Now, a dielectric of dielectric constant K is inserted to fill the whole space between the plates while the voltage source remains connected to the capacitor.

- the energy stored in the capacitor will become K -fold
- the electric field inside the capacitor will become $1/K$ times the original field
- the force of attraction between the plates will increase to K^2 - times
- the charge on the capacitor will increase to K -times

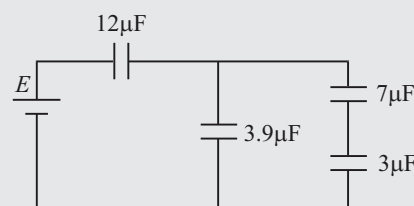
2. For the circuit shown, which of the following statements are true?



- with S_1 closed, $V_1 = 15$ V, $V_2 = 20$ V
 - with S_3 closed, $V_1 = V_2 = 25$ V
 - with S_1 & S_2 closed, $V_1 = V_2 = 0$
 - with S_1 & S_2 closed, $V_1 = 30$ V, $V_2 = 20$ V
3. A parallel plate capacitor of capacitance $10 \mu\text{F}$ is connected to a cell of emf 10 volt and fully charged. Now a dielectric slab ($k = 3$) of thickness equal to the gap between the plates, is very slowly inserted to completely fill in the gap, keeping the cell connected. During the filling process:
- the increase in charge on the capacitor is $200 \mu\text{C}$.
 - the heat produced is zero.
 - energy supplied by the cell = increase in stored potential energy + work done on the person who is filling the dielectric slab.
 - energy supplied by the cell = increase in stored potential energy + work done on the person who is filling the dielectric slab + heat produced.
4. A capacitor C is charged to a potential difference V and battery is disconnected. After this the capacitor plates are brought close slowly by some distance. Now, the battery is connected again and the plates are slowly pulled apart to restore the original separation.
- work is done by external agent in the entire process is positive

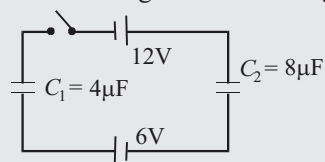
- the battery gains energy in the entire process
- force between the capacitor plates is highest when the plates are closest
- heat is generated in the process

5. Four capacitors and a battery are connected as shown. The potential drop across the $7 \mu\text{F}$ capacitor is 6 V. Then the:

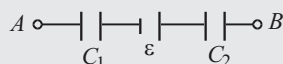


- potential difference across the $3 \mu\text{F}$ capacitor is 10 V
 - charge on the $3 \mu\text{F}$ capacitor is $42 \mu\text{C}$
 - e.m.f. of the battery is 30 V
 - potential difference across the $12 \mu\text{F}$ capacitor is 10 V.
6. The capacitance of a parallel plate capacitor is C when the region between the plate has air. This region is now filled with a dielectric slab of dielectric constant k . The capacitor is connected to a cell of emf V , and the slab is slowly taken out
- charge density varies but remains uniform on a plate of the capacitor when the dielectric slab is being moved out.
 - Charge density on the part of the plate that is exposed to air remains same as the dielectric is being pulled out
 - energy $V^2 C(k - 1)$ is absorbed by the cell.
 - the energy stored in the capacitor is reduced by $V^2 C(k - 1)$
7. In the circuit shown, each capacitor has a capacitance C . The emf of the cell is E . If the switch S is closed
-
- positive charge will flow out of the positive terminal of the cell
 - positive charge will enter the positive terminal of the cell
 - the amount of charge flowing through the cell will be CE .
 - the amount of charge flowing through the cell will be $4/3 CE$.

8. In the circuit shown, initially, the two capacitors - C_1 and C_2 - are uncharged. After closing the switch



- (a) The charge on C_2 is greater than on C_1
 (b) The charge on C_1 and C_2 are the same
 (c) The potential drops across C_1 and C_2 are the same
 (d) The potential drops across C_2 is greater than that across C_1
9. A part of a circuit, shown in the figure, consists of a battery of emf 10 V and two capacitance $C_1 = 1.0 \mu\text{F}$ and $C_2 = 2.0 \mu\text{F}$. The potential difference $V_A - V_B$ is 5V



- (a) charge on capacitor C_1 is equal to charge on capacitor C_2
 (b) Voltage across capacitor C_1 is 5V.
 (c) Voltage across capacitor C_2 is 10V
 (d) Energy stored in capacitor C_1 is two times the energy stored in capacitor C_2 .
10. A parallel plate capacitor of plate area A and plate separation d is charged to potential difference V and then the battery is disconnected. A slab of dielectric constant K is then slowly inserted between the plates of the capacitor so as to fill the space between the plates. If Q , E and W denote the magnitude of charge on each plate, the electric field between the plates (after the slab is inserted) and the work done on the system, respectively in the process of inserting the slab, then

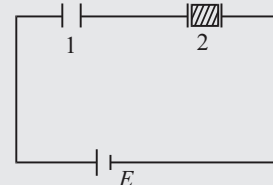
(a) $Q = \frac{\epsilon_0 AV}{d}$ (b) $Q = \frac{\epsilon_0 KAV}{d}$
 (c) $E = \frac{V}{Kd}$ (d) $W = -\frac{\epsilon_0 AV^2}{2d} \left(1 - \frac{1}{K} \right)$

11. A parallel-plate capacitor is connected to a cell. Its positive plate A and its negative plate B have charges $+Q$ and $-Q$ respectively. A third plate C, identical to A and B, with charge $+Q$, is now introduced midway between A and B, parallel to them. Which of the following are correct?

- (a) The charge on the inner face of B is now $-\frac{3Q}{2}$
 (b) There is no change in the potential difference between A and B.

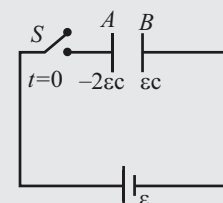
- (c) The potential difference between A and C is one-third of the potential difference between B and C.
 (d) The charge on the inner face of A is now $\frac{Q}{2}$.

12. Two identical capacitors 1 and 2 are connected in series to a battery as shown in figure. Capacitor 2 contains a dielectric slab of dielectric constant k as shown. Q_1 and Q_2 are the charges stored in the capacitors. Now the dielectric slab is removed and the corresponding charges are Q'_1 and Q'_2 . Then



(a) $\frac{Q'_1}{Q_1} = \frac{k+1}{k}$ (b) $\frac{Q'_2}{Q_2} = \frac{k+1}{2}$
 (c) $\frac{Q'_2}{Q_2} = \frac{k+1}{2k}$ (d) $\frac{Q'_1}{Q_1} = \frac{k}{2}$

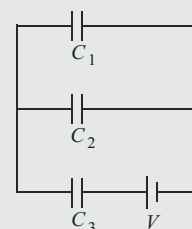
13. A parallel plate capacitor of capacitance ' C ' carries charges on its plates as shown in the figure. Now at $t = 0$, the switch ' S ' is closed so as to connect it to a cell of emf ϵ .



Select the correct alternative(s) for this circuit diagram.

- (a) In steady state the charges on the outer surfaces of plates 'A' and 'B' will be same in magnitude and sign.
 (b) In steady state the charges on the outer surfaces of plates 'A' and 'B' will be same in magnitude and opposite in sign.
 (c) In steady state the charges on the inner surfaces of the plates 'A' and 'B' will be same in magnitude and opposite in sign.
 (d) The work done by the cell by the time steady state is reached is $\frac{5\epsilon^2 C}{2}$.

14. Which of the followings are true with respect to the circuit shown.

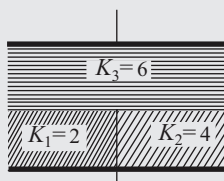


- (a) The potential difference across C_3 is $\frac{(C_1 + C_2)V}{C_1 + C_2 + C_3}$
 (b) The potential difference across C_3 will increase if a dielectric slab is inserted in C_1 .
 (c) If C_3 is short circuited, the electric field between the plates of C_1 will increase.
 (d) None of the above.

15. Two parallel plate capacitors, with capacities C and $2C$, are joined in parallel. These are charged by applying a potential difference V . The battery is now removed and a slab of dielectric constant K is slowly inserted between the plates of the capacitor having capacitance C .

- (a) The potential difference across each capacitor becomes $\frac{V}{K+2}$
- (b) There is loss in stored energy in the capacitor system
- (c) A lot of heat is produced when the dielectric is inserted
- (d) None of the above

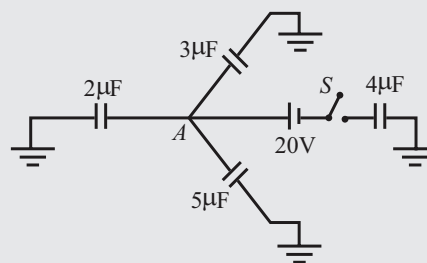
16. A parallel plate air capacitor has capacitance C . It is filled by dielectric slabs as shown in figure. The lower two dielectrics occupy equal space. The capacitor is connected to a battery.



- (a) The electric field is same everywhere inside the dielectric of dielectric constant K_3 .
- (b) The capacitance can be calculated by considering two capacitors, filled with slabs of dielectric constants K_1 and K_2 , in parallel; connected to a third capacitor, filled with dielectric of constant K_3 , in series.

- (c) The electric field inside the two smaller dielectrics is same
- (d) None of the above.

17. Three capacitors of $2\mu\text{F}$, $3\mu\text{F}$ and $5\mu\text{F}$ are individually charged with batteries of emf's 5V , 20V and 10V , respectively. After disconnecting from the voltage sources, these capacitors are connected as shown in the figure with their positive polarity plates connected to A and their negative polarity plates being earthed. Now, a battery of 20V and an uncharged capacitor of $4\mu\text{F}$ capacitance are connected to the junction A with a switch S , as shown. The switch is closed.

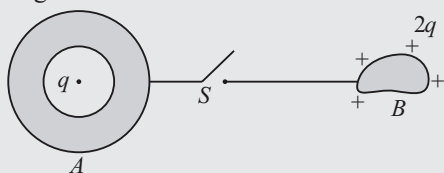


- (a) the potential of the junction A becomes $100/7\text{ V}$
- (b) final charges on the capacitance of lowest capacitance is $200\text{ }\mu\text{C}$
- (c) the total charge flow to the ground after closing the switch is $200\text{ }\mu\text{C}$
- (d) the charge flow to the ground after closing the switch is zero.

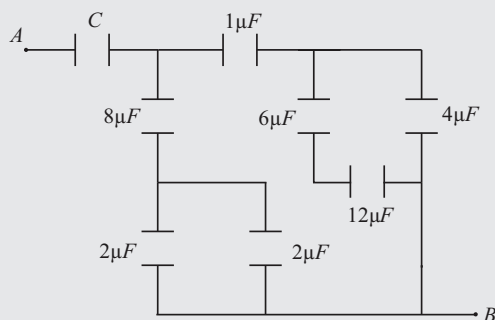
Worksheet 3

- Which has higher capacitance – a copper conductor or, an aluminium conductor of identical geometry?
 - Two conductors have identical surfaces. However, one of them has a cavity inside it. Which of the two has higher capacitance?
- A is a neutral metal ball of radius $R = 1.0\text{m}$. It has a concentric spherical cavity of radius $r = 0.2\text{m}$. A point charge q is kept at the centre of the sphere. At a large distance, there is another conductor B having capacitance $\left(\frac{1}{18}\right)nF$. Conductor B carries a charge

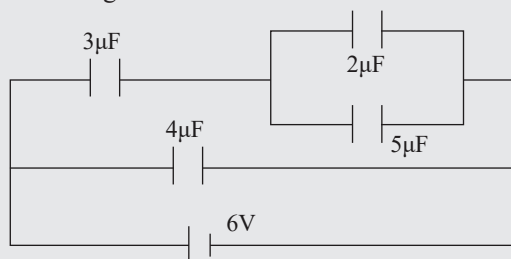
$2q$. Switch S is closed to connect the two conductors through a metal wire.



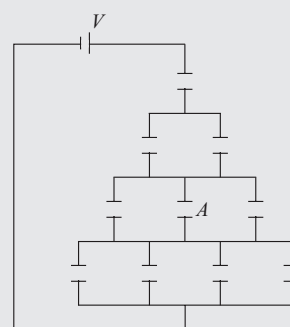
- Find final charge on the outer surface of A .
 - If $q = 2nC$, find heat generated in the system after S is closed.
- An air-filled parallel plate capacitor is needed that can store $24\text{ }\mu\text{C}$ of charge when potential difference between its plate is 1200 V . Find the minimum plate area of such a capacitor. The dielectric strength of air is $3 \times 10^6\text{ V/m}$.
 - For the circuit shown in the figure, the equivalent capacitance between points A and B is $1\text{ }\mu\text{F}$. Find the capacitance C .



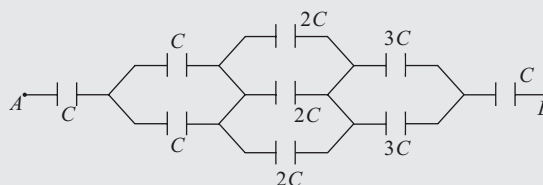
- Find the charge on $5\text{ }\mu\text{F}$ capacitor in the circuit shown in the figure.



- In the network shown, each capacitor has capacitance C .

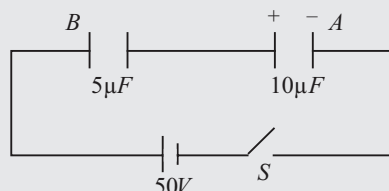


- Find charge supplied by the cell to the network
 - Find charge on the capacitor marked as A .
- Find the equivalent capacitance between the points A and B for the network shown.



- A capacitor of capacitance C_0 is charged to a potential difference of V_0 . It is disconnected from the cell and now connected to an uncharged capacitor in parallel. The charge on the capacitor reduces by 50%. Find heat produced in the circuit after the two capacitors are connected.
- A capacitor of large capacitance C_0 is charged to a potential difference V_0 and then isolated from the source. A capacitor of small capacitance C is connected in parallel to C_0 so as to charge it. The small capacitor is removed, discharged and again connected to the large capacitor to get it charged. The process is repeated n times. The potential of the large capacitor falls to V . Find the value of C .
- An air-filled parallel plate capacitor has plate area of 200 cm^2 and plate separation of 4 mm . Calculate the percentage change in capacitance if a layer of varnish ($k = 3$) of thickness 0.1 mm is applied on the inner surface of each plate.
- Capacitors A and B are connected in series to a 100 V supply and the potential difference across them was found to be 60 V and 40 V , respectively. A third capacitor of $2\text{ }\mu\text{F}$ capacitance is connected in parallel with A and the potential difference across B becomes 90 V . Find the capacitance of A and B .

12. Capacitor A has capacitance of $10\ \mu\text{F}$ and has been charged using a cell of emf 15V . It is disconnected from the cell and put in a circuit with another capacitor B (capacitance $5\ \mu\text{F}$) and a cell (emf 50V) as shown.



Find the new potential difference across A after the switch S is closed.

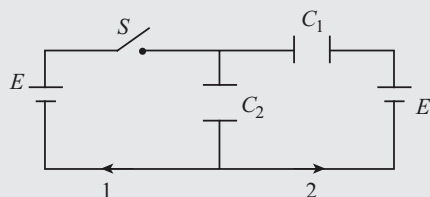
13. Four identical air-filled capacitors have plate separation $d = 0.7\text{ cm}$. They are connected in series.
- (a) Find the maximum potential difference that can be applied across this combination without puncturing the capacitors. Breakdown strength of air is $3 \times 10^4 \text{Vcm}^{-1}$.

- (b) One of the capacitors is filled with glass. Now find the maximum permissible voltage so that none of the capacitors gets punctured. The relative permittivity of glass is $k = 7$ and its breakdown field is $E_g = 6.3 \times 10^4 \text{Vcm}^{-1}$.

14. Three capacitors of capacities $1\ \mu\text{F}$, $2\ \mu\text{F}$ and $3\ \mu\text{F}$ are charged to 10V , 20V and 30V potential difference, respectively. The cells are disconnected. The positive plates of the first two capacitors and the negative plate of the third capacitor are connected. The remaining plates are connected together separately.

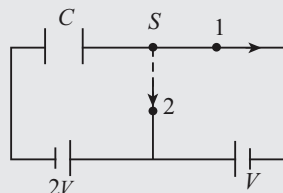
Find the final charge on the three capacitors.

15. In the circuit shown in the figure, switch S is open. Calculate the amount of charge that will flow in the wire 1 and 2 in the indicated directions after the switch S is closed.

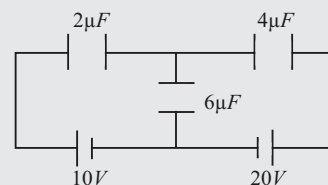


16. In the circuit shown in the figure, S is a two-way switch.

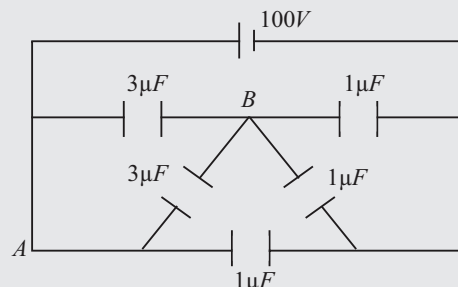
Initially, it is in position 1. Find heat generated in the circuit after the switch is shifted to position 2.



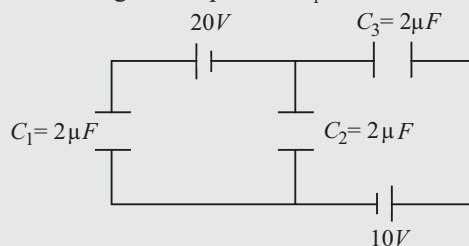
17. Find the charges on the three capacitors shown in the figure.



18. Find potential difference between the points A and B in the circuit shown in the figure.

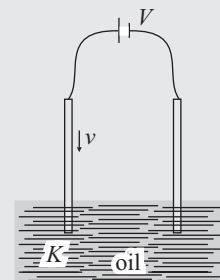


19. Find charge on capacitor C_1 shown in the circuit.

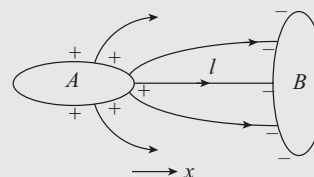


20. A parallel plate air capacitor has square plates of side length $L = 1.0\text{ m}$ and plate separation $d = 0.02\text{ m}$.

It is connected to a power supply of emf $V = 500\text{ V}$. The plates are held vertical and are lowered into an oil tank at constant speed of $v = 0.01\text{ m/s}$. The dielectric constant of the oil is $K = 11$. Find the rate of flow of charge (i.e., the current) through the power source.



21. Two conductors A and B carry equal and opposite charge (Q) on them. The electric field along the straight line l (along x direction) changes with distance x from the surface of A as $E = \frac{Q}{n}(a + bx^2)$, where n , a and b are positive constants. Find the capacitance. Length of line l is d .

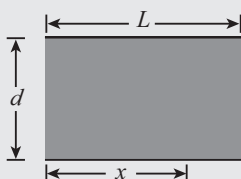


22. A parallel plate air capacitor has capacitance C_0 . The space between the plates is filled with a dielectric slab having dielectric constant K . It is then charged using a cell of emf V . Now the cell is disconnected. An external agent slowly pulls out the dielectric slab and then slowly pulls apart the plates to double their separation. Find the total work done by the agent.

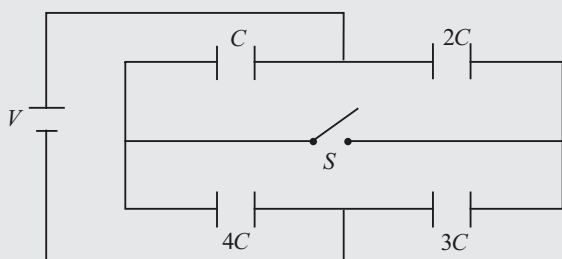
23. A parallel plate capacitor has square plates of side length L , kept at a separation d . The gap between the plates is filled with a dielectric of dielectric constant K , which varies parallel to an edge as

$$K = K_0 + ax$$

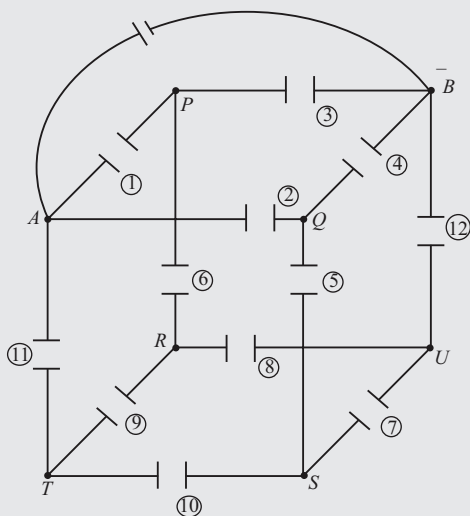
where K_0 and a are constants and x is distance from one end. Find the capacitance.



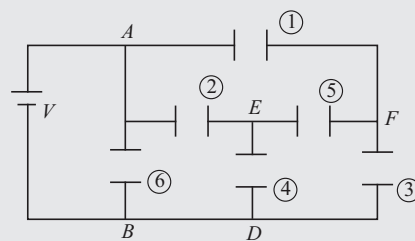
24. In the circuit shown, how much charge will flow through the switch S after it is closed?



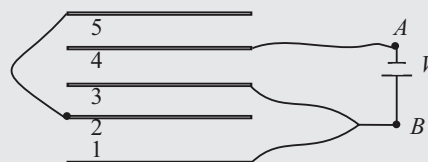
25. Twelve capacitors, each of capacitance C , are placed along the sides of a cubical network. Find the equivalent capacitance between points A and B .



26. All capacitors shown in the circuit have same capacitance C . Find charge on capacitors 6 and 1. Is there a capacitor having no charge on it? Which one is it?



27. Five identical plates are fixed parallel to one another. Each has area A and separation between two consecutive plates is d . Plates are connected using conducting wires as shown. The cell has an emf V . How much heat will be generated if the plates 1 and 2 are also connected using a wire?



28. A point charge q is held fixed at a point O . Find the electrostatic energy stored in space outside a sphere of radius R centred at O .

29. A metal sphere has radius R . It is given some charge. The electrostatic energy stored in a concentric spherical region of radius a is same as the energy stored outside the region of radius a . Find a in terms of R .

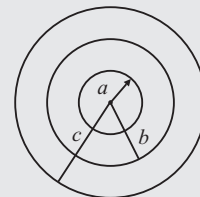
30. A long cylindrical pipe has radius a . Charge is spread uniformly on it and charge on unit length of the pipe is λ . Calculate the electrostatic energy stored in a cylinder of radius b ($b > a$) and length l , that is coaxial with the cylindrical pipe.

31. An isolated plate capacitor has plate area A and charge on it is Q . The plates of the capacitors are pulled apart so as to separate them by a large distance. Take $A = 40 \text{ cm}^2$ and $Q = 10 \text{ } \mu\text{C}$ and calculate the energy stored in a cubical volume of edge 1.0 cm near the positive plate.

32. Two large identical metal plates are well separated and carry charges Q and $2Q$. The plates are brought closer so as to form a parallel plate capacitor of capacitance C . Find the potential difference between the two plates.

33. Three coaxial thin metal cylinders have radii a , b and c . Length of each cylinder is L ($L \gg c$).

Find the capacitance of the system between the inner and the outer cylinder.



Answer Sheet

Your Turn

- | | | | | |
|--|-------------------|---|------------------------------------|-------------------------------|
| 1. No. | 2. 2 mC | 3. Zero. | 4. Zero | 5. $-\frac{40}{3}\mu\text{C}$ |
| 6. 50,000 Vm^{-1} | 7. 10V | 8. $\frac{3}{4}$ | 9. $6.7 \times 10^{-11} \text{ F}$ | |
| 10. Energy density will not change | | 11. 100 μJ | 12. 20V | 13. $\frac{1}{2}$ |
| 14. (a) 4 mJ (b) -2mJ (c) $\frac{1}{4}$ | | | | |
| 15. Charge on every capacitor is 36 μC . $V_1 = 36\text{V}$, $V_2 = 18\text{V}$, $V_3 = 12\text{V}$ | | | | |
| 16. 5 μF and 20 μF | 17. (a) 3C (b) 3C | 18. (a) 2CV (b) CV ² (c) CV ² | | 19. $\frac{6}{5}$ |
| 20. $\frac{20}{3}\text{V}$. Removal of 4 μF capacitor makes no difference | | | 21. C_1 | 22. 10 V |
| 23. 4 μF | 24. C | 25. $\frac{50 k\epsilon_0 A}{21 d}$ | 26. 2 μF | 27. 3 μF |
| 28. $\frac{2\epsilon_0 AV}{d}$ | 29. 0 | 30. $\frac{200}{3}\mu\text{C}$ | 31. -30V | 32. 10.3V |
| 34. $\frac{3C}{2}$ | | | | |

Worksheet 1

- | | | | | | | | | |
|---------|---------|---------|---------|---------|---------|---------|---------|---------|
| 1. (c) | 2. (b) | 3. (a) | 4. (d) | 5. (b) | 6. (b) | 7. (a) | 8. (b) | 9. (b) |
| 10. (c) | 11. (a) | 12. (c) | 13. (c) | 14. (c) | 15. (b) | 16. (c) | 17. (c) | 18. (d) |
| 19. (c) | 20. (a) | 21. (d) | 22. (d) | 23. (c) | 24. (b) | 25. (d) | 26. (c) | 27. (c) |
| 28. (c) | 29. (d) | 30. (c) | 31. (a) | 32. (b) | 33. (a) | 34. (b) | 35. (b) | 36. (a) |
| 37. (c) | 38. (d) | 39. (b) | 40. (c) | 41. (c) | 42. (d) | 43. (b) | 44. (b) | 45. (b) |
| 46. (b) | 47. (a) | 48. (b) | 49. (b) | 50. (c) | 51. (b) | 52. (b) | 53. (d) | 54. (d) |
| 55. (d) | 56. (c) | 57. (b) | 58. (b) | 59. (a) | 60. (a) | 61. (a) | | |

Worksheet 2

- | | | | | | |
|---------------|---------|-----------------|---------------|------------------|-----------|
| 1. (a, c, d) | 2. (d) | 3. (a, b, c, d) | 4. (a, c, d) | 5. (b, c, d) | 6. (b, c) |
| 7. (a, d) | 8. (b) | 9. (a, d) | 10. (a, c, d) | 11. (a, b, c, d) | 12. (c) |
| 13. (a, c, d) | 14. (a) | 15. (b) | 16. (d) | 17. (a, d) | |

Worksheet 3

1. (a) Both have same capacitance. (b) Both have same capacitance. 2. (a) $2q$ (b) 54nJ 3. 0.9 m^2
4. $\frac{32}{23}\mu\text{F}$ 5. $9\mu\text{C}$ 6. (a) 10 CV (b) $\frac{10\text{ CV}}{3}$ 7. $\frac{6C}{17}$ 8. $\frac{1}{4}C_0V_0^2$
9. $C = \left(\frac{V_0}{V}\right)^{\frac{1}{n}} - 1$ 10. 3.46% 11. $0.16\mu\text{F}$ and $0.24\mu\text{F}$ 12. $\frac{80}{3}V$
13. (a) $8.4 \times 10^4\text{ V}$ (b) $6.6 \times 10^4\text{ V}$ 14. $\frac{20}{3}\mu\text{C}$, $\frac{40}{3}\mu\text{C}$ and $20\mu\text{C}$
15. $\Delta q_1 = C_2E$; $\Delta q_2 = -\left(\frac{C_1C_2}{C_1+C_2}\right)E$ 16. $\frac{1}{2}CV^2$ 17. $\frac{10}{3}\mu\text{C}$, $\frac{140}{3}\mu\text{C}$, $50\mu\text{C}$
18. 25V 20. $2.2 \times 10^{-8}\text{ Cs}^{-1}$. 21. $\frac{3n}{3ad+bd^3}$ 22. $C_0V^2K\left(K-\frac{1}{2}\right)$
23. $\frac{\epsilon_0L^2}{d}\left(k_0+\frac{La}{2}\right)$ 24. $\frac{CV}{2}$ 25. $\frac{4C}{3}$ 26. $q_6 = CV$, $q_1 = \frac{CV}{2}$, capacitor 5 has no charge
27. $\frac{CV^2}{6}$ 28. $\frac{Q^2}{8\pi\epsilon_0R}$ 29. $a = 2R$ 30. $\frac{\lambda^2l}{4\pi\epsilon_0}\ln\left(\frac{b}{a}\right)$ 31. 0.087 J
32. $\frac{Q}{2C}$ 33. $\frac{2\pi\epsilon_0L}{\ln\left(\frac{c}{a}\right)}$

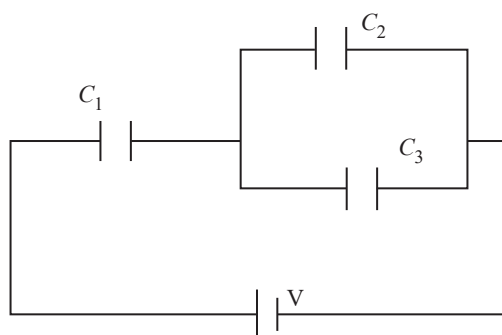
Miscellaneous Problems on Chapter 4

MATCH THE COLUMN

1. Column-I mentions the experimental steps performed with a parallel plate capacitor of capacitance C . Match the questions asked in Column-I with the energy values given in Column-II.

Column-I		Column-II	
(A)	The capacitor is charged using a cell of emf V . The capacitor is disconnected from the cell and its positive plate is connected to the negative terminal of a cell of emf V . The negative terminal of the capacitor is connected to the positive terminal of the cell. Heat dissipated after the capacitor is connected to the second cell is	(P)	$\frac{1}{2} CV^2$
(B)	The capacitor is charged using a cell of emf V . With the cell connected, the plate separation of the capacitor is doubled. Loss in energy stored in the electric field between the capacitor plates is	(Q)	0
(C)	The capacitor is charged to a potential difference of V . The cell is disconnected. A dielectric of dielectric constant 2 is inserted between the plates. Change in the stored energy in the capacitor is	(R)	$\frac{1}{4} CV^2$
(D)	A capacitor charged to a potential difference V is short circuited after being disconnected from the cell. Loss in energy stored is	(S)	$2CV^2$

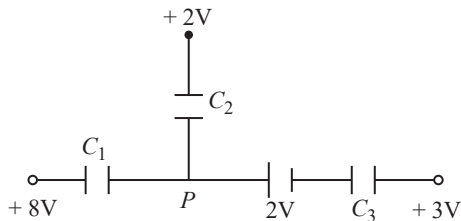
2. For the circuit shown in the figure, few changes have been mentioned in Column-I. Match these with the results given in Column-II.



Column-I		Column-II	
(A)	Plate separation of C_1 is reduced. The ratio of energy stored in C_1 and C_2	(P)	remains unchanged
(B)	A dielectric is inserted between the plates of C_2 . The ratio of electric field between the plates of C_1 and C_3	(Q)	decreases
(C)	The capacitor C_3 is short circuited. The ratio of energy stored in C_2 and C_1	(R)	becomes zero
(D)	A dielectric is inserted between the plates of C_1 . The ratio of charge on C_1 and C_2	(S)	increases

5.2 Electrostatics and Current Electricity

3. The figure shows a part of a large capacitor circuit. Each capacitor has capacitance $2\mu F$. Match the quantities in Column-I to their correct values given in Column-II.

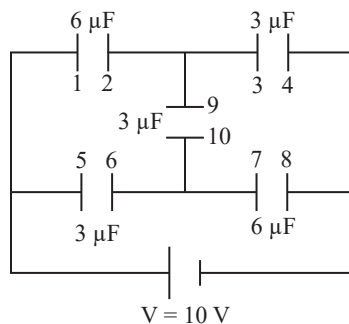


Column-I		Column-II	
(A)	Charge on the plate of C_1 connected at P , in μC , is	(P)	0
(B)	Charge (in μC) on the plate of C_3 at $3V$ potential is	(Q)	-6
(C)	The energy stored in C_2 in μJ is	(R)	9
(D)	The potential at point P (in volt) is	(S)	5

4. Three metal plates are held parallel to one another such that capacitance between the adjacent plates is $100nF$. In an experiment X , a charge of $2\mu C$ is given to the middle plates B . In another experiment Y , charge of $2\mu C$ is given to the plate A , instead of B . Now match the quantities given in Column-I with those in Column-II.

Column-I		Column-II	
(A)	Charge on the inner surface of the plate A is (in μC)	(P)	10 in experiment X
(B)	Magnitude of potential difference between the plates A and B is (in volt)	(Q)	1 in experiment Y
(C)	Potential difference between the plates B and C is (in volt)	(R)	-1 in experiment X
(D)	Charge on the inner surface of plate C is (in μC)	(S)	10 in experiment Y

5. In the circuit shown in the figure, the plates of the different capacitors have been numbered and their capacitance has been indicated. With reference to the circuit, match the entries in Column-I with those in Column-II.



Column-I		Column-II	
(A)	The sum of charges is zero for the plates	(P)	1, 4, 5, and 8
(B)	The sum of charges on the plates 1 and 5 is same as the sum of charges on the plates	(Q)	1 and 7
(C)	A pair of plates having equal charge	(R)	9 and 8
(D)	The potential difference between plates 1 and 10 is same as the potential difference between the Plates	(S)	3 and 4
		(T)	3 and 7

6. Column-I shows capacitor circuits and Column-II gives the value of equivalent capacitance between points A and B . Match the circuits in Column-I with their equivalent capacitance given in Column-II.

Column-I		Column-II	
(A)		(P)	$\frac{18}{5} \mu F$
(B)		(Q)	$1 \mu F$

(C)	(R) $\frac{27}{34} \mu F$
(D)	(S) $3 \mu F$

PASSAGE BASED PROBLEMS

The figures show two infinite ladder network of capacitors. The equivalent capacitance of the first network between A and B is C_1 and the equivalent of the second network between D and E is C_2 .

(a) $\frac{1}{2}$ (b) 2

2. Value of C_1 is

(c) $\frac{\sqrt{17}-1}{4}\mu F$

(d) $\frac{\sqrt{17}+1}{4}\mu F$

Passage 2

A large conducting plate of surface area A and small thickness d is placed perpendicular to a uniform electric field E . (Neglect the edge effects.)

3. The electrostatic interaction energy of the charges induced on the two surfaces of the metal plate is

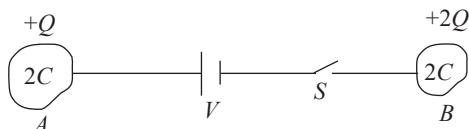
- (a) $\frac{1}{2}\epsilon_0 E^2 A d$ (b) $\epsilon_0 E^2 A d$
(c) $2\epsilon_0 E^2 A d$ (d) zero

4. Heat dissipated in the metal plate after the external electric field is switched off is

- (a) $\epsilon_0 E^2 A d$ (b) $\frac{1}{2}\epsilon_0 E^2 A d$
(c) $\epsilon_0 E^2 A d$ (d) zero

Passage 3

Two conductors A and B have capacities of $2C$ and C , respectively. They are kept a large distance apart with a cell of emf V and a switch (S) connecting them. Initially, the conductors A and B are given charges Q and $2Q$, respectively, and the switch is open (see the figure). Now, the switch is closed.



5. The final charge on A will be

- (a) $Q - \frac{2CV}{3}$ (b) $2Q + \frac{2CV}{3}$
(c) $Q + \frac{CV}{3}$ (d) $2Q - \frac{CV}{3}$

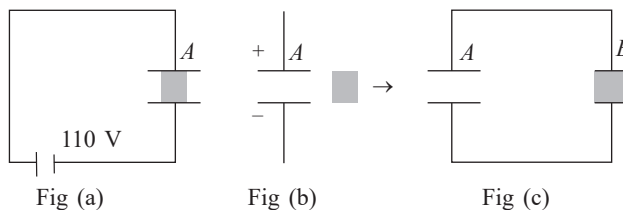
6. Heat generated after S is closed is

- (a) $QV + CV^2 + \frac{Q^2}{C}$ (b) $QV + CV^2 + \frac{3Q^2}{4C}$
(c) $QV - CV^2 + \frac{3Q^2}{4C}$ (d) $QV + \frac{1}{3}CV^2 + \frac{3Q^2}{4C}$

Passage 4

Two parallel plate capacitors A and B have the same separation $d = 8.85 \times 10^{-4}$ m between the plates. The plate areas of A and B are 0.04 m^2 and 0.02 m^2 , respectively. A slab of dielectric constant $K = 9$ has dimensions such that it can exactly fill the space between the plates of capacitor B . Initially, the dielectric slab is placed inside A (as shown in Figure [a]). A is then connected to a cell of emf 110 V. Energy stored in A in this position is U_1 . Now, the cell is disconnected (Figure [b]) and an external agent slowly takes

out the slab from A . Work done by the external agent in the process is W . The same dielectric slab is now placed inside the uncharged capacitor B so as to fill it completely. Now, A and B are connected in parallel (Figure (c)). Energy stored in the two capacitor systems is now U_2 .



7. The ratio $\frac{W}{U_1}$ is nearly equal to

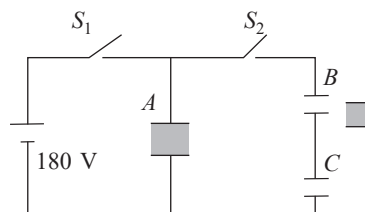
- (a) 3.0 (b) 2.0
(c) 4.0 (d) 1.0

8. Value of $U_1 - U_2$ is nearly equal to

- (a) $0.08 \times 10^{-3} \text{ J}$ (b) $0.8 \times 10^{-5} \text{ J}$
(c) $0.11 \times 10^{-5} \text{ J}$ (d) zero

Passage 5

In the figure shown, capacitor A has capacitance $C_1 = 2\mu\text{F}$ when it is filled with a dielectric slab of dielectric constant $K = 2$. B and C are air capacitors having capacitances $C_2 = 3\mu\text{F}$ and $C_3 = 6\mu\text{F}$, respectively. First, the switch S_1 is closed and then opened after some time. Then S_2 is closed. After some times S_2 is opened. Then slab inside A is removed and another slab of dielectric constant $K = 2$ is inserted inside B to fill its entire space. Finally the switch S_2 is closed again.



9. Final charge on the upper plate of capacitor A is

- (a) $-90\mu\text{C}$ (b) $45\mu\text{C}$
(c) $-45\mu\text{C}$ (d) $90\mu\text{C}$

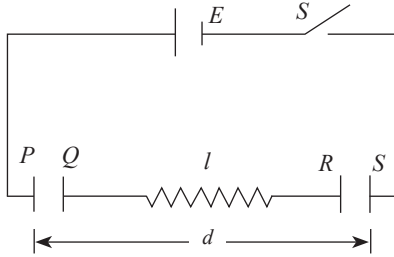
10. Heat generated in the circuit after S_2 is closed finally is

- (a) 5.4 mJ (b) 2.7 mJ
(c) 9.4 mJ (d) zero

Passage 6

Two identical parallel plate air capacitors have plate area A . They are connected in series by a conducting spring of natural length l . Plates P and S are fixed at a separation d , but the plates Q and R can move frictionlessly. The switch

is closed to connect the two capacitors to a cell of emf E . When equilibrium is established, the plate separation of each capacitor is same as the extension in the spring.



11. Charge (q) on each capacitor is

- (a) $\frac{3\epsilon_0 AE}{d-l}$ (b) $\frac{3\epsilon_0 AE}{2(d-l)}$
 (c) $\frac{\epsilon_0 AE}{2(d-l)}$ (d) $\frac{5\epsilon_0 AE}{(d-l)}$

12. The spring constant k is equal to

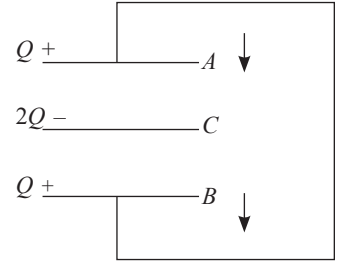
- (a) $\frac{27\epsilon_0 AE^2}{(d-l)^2}$ (b) $\frac{27\epsilon_0 AE^2}{8(d-l)^3}$
 (c) $\frac{9\epsilon_0 AE^2}{(d-l)^3}$ (d) $\frac{25\epsilon_0 AE^2}{8(d-l)^3}$

Passage 7

The identical plates A , B and C are held parallel to one another. Each plate has area A and separation between A and C and that between B and C is d .

Charges on the plates A , B and C are $+Q$, $+Q$ and $-2Q$, respectively. At time $t = 0$,

the plates A and B both start moving with constant speed u . A moves towards C and B moves away from C . The plate C remains fixed.



13. The magnitude of charge on the upper face of plate C

- (a) increases (b) decreases
 (c) remains $-Q$ (d) remains $-2Q$

14. The rate of change of charge on plate A is

- (a) $\frac{Qu}{2d}$ (b) $\frac{2Qu}{d}$
 (c) $\frac{Qu}{d}$ (d) zero

Answer Sheet

Match the Columns

- | | |
|--|---|
| 1. $A \rightarrow S$; $B \rightarrow R$; $C \rightarrow P$; $D \rightarrow P$ | 2. $A \rightarrow Q$; $B \rightarrow S$; $C \rightarrow R$; $D \rightarrow P$ |
| 3. $A \rightarrow Q$; $B \rightarrow P$; $C \rightarrow R$; $D \rightarrow S$ | 4. $A \rightarrow R, Q$; $B \rightarrow P, S$; $C \rightarrow P, S$; $D \rightarrow R$ |
| 5. $A \rightarrow P, S$; $B \rightarrow T$; $C \rightarrow Q$; $D \rightarrow R, Q$ | 6. $A \rightarrow Q$; $B \rightarrow R$; $C \rightarrow Q$; $D \rightarrow P$ |
| 7. $A \rightarrow S$ $B \rightarrow P$ $C \rightarrow Q$ $D \rightarrow R$ | |

Passage Based Problems

- | | | | | | | | | |
|---------|---------|---------|---------|---------|--------|--------|--------|--------|
| 1. (b) | 2. (c) | 3. (a) | 4. (b) | 5. (b) | 6. (d) | 7. (c) | 8. (c) | 9. (d) |
| 10. (a) | 11. (b) | 12. (b) | 13. (a) | 14. (c) | | | | |

Electric Current and Resistance

“God is the electricity and we are the lamps.”

—Marinne Williamson

1. INTRODUCTION

Till now our discussion was mainly focused on static charge. But it is the moving charge that is most important for modern life. Flow of charge is known as electric current and this is what causes bulbs to glow, our TV, computers, mobiles and many such appliances to function. Moreover, it powers all our industrial motors.

In this chapter we will focus on learning the mechanism of current flow in a conductor and about the property of resistance. We will also discuss about heating effect of current and learn how to solve circuit related problems.

2. ELECTRIC CURRENT

Flow of electric charge is known as electric current. Quantitatively, the rate at which charge flows across a given cross sectional area is defined as the current through the area.

Consider an imaginary area ΔA held in a stream of charged particles. If ΔQ charge crosses the area in time Δt , the average current through the area is defined as

$$I_{av} = \frac{\Delta Q}{\Delta t} \quad (1)$$

In the limiting case of $\Delta t \rightarrow 0$, the instantaneous current is written as

$$I = \frac{dQ}{dt} \quad (2)$$

Unit of electric current is C/s, known as ampere (A). An ampere is one of the seven fundamental units in SI system. Common currents that we encounter in our daily life ranges

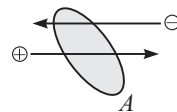
from nano amperes (in our nerves) to mega amperes (in lightning).

It is a standard convention to take the direction of electric current in the direction of flow of positive charge. If current arises due to movement of negative charges (e.g., electrons), the direction of current is assumed to be opposite to the direction of flow of the electrons.

Current may be caused due to movement of charged particles in free space or due to movement of positive and negative ions in an electrolyte solution. But in this chapter we will mainly discuss current flow inside a conductor that happens due to movement of electrons.

Example 1 Consider an imaginary area A

Positive charge $+2C$ flows past the area from left to right in a time interval of 4s. In the same interval negative charge $-2C$ flows through the area from right to left. Find the current through the area.



Solution

Concepts

- (i) Current is in the direction of flow of positive charge.
- (ii) A charge flow $-Q$ from right to left is same as a charge flow of $+Q$ from left to right.

$$I_{av} = \frac{\Delta Q}{\Delta t} = \frac{4C}{4s} = 1 \text{ ampere.}$$

Example 2 Electric current has a direction. Is current a vector?

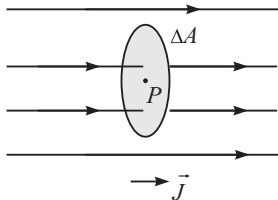
Solution

Vector are physical quantities which are added according to the triangle rule. Electric current does not obey the triangle rule of addition. Hence it is not a vector.

3. CURRENT DENSITY

Current density ($\vec{j}$) is a vector quantity. It is useful in microscopic description of electric current in a conductor, as we shall learn shortly.

Suppose charge is flowing in certain region. To define current density at a point P , we consider a small area ΔA surrounding the point. The area is perpendicular to the flow of the charge. If ΔI is current through the area ΔA , then the average current density is defined as



$$j_{av} = \frac{\Delta I}{\Delta A} \quad (3)$$

The current density at point P will be defined as

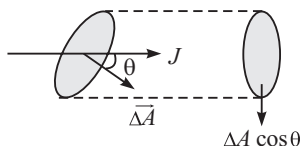
$$j = \lim_{\Delta A \rightarrow 0} \frac{\Delta I}{\Delta A} = \frac{dI}{dA} \quad (4)$$

The direction of current density vector is taken in the direction of the current.

When the area ΔA is not perpendicular to the direction of charge flow, the current density is defined as

$$j = \frac{\Delta I}{\Delta A \cos \theta}$$

where θ is angle between the direction of the current and the area vector (i.e., normal to the area).



$$\text{or, } \Delta I = j \Delta A \cos \theta$$

$$\text{or, } \Delta I = \vec{j} \cdot \vec{\Delta A} \quad (5)$$

Actually, $\Delta A \cos \theta$ is projection of area ΔA in a direction perpendicular to the charge flow.

When the current density is non uniform over the area, we can write current through the area as

$$I = \int \vec{j} \cdot \vec{dA} \quad (6)$$

Example 3 A beam of electrons has a cross sectional area of 2mm^2 . A total of 1.2×10^{17} electrons pass through the cross section in 1 second. Find the current density in the beam.

Solution

Concepts

First we will find the electric current through the cross section and then using equation (3) we will find current density.

Total charge crossing the cross section in $\Delta t = 1\text{s}$ is

$$\Delta q = ne = 1.2 \times 10^{17} \times 1.6 \times 10^{-19} = 1.92 \times 10^{-2} \text{ C}$$

$$\therefore \text{current } I = \frac{\Delta q}{\Delta t} = \frac{1.92 \times 10^{-2} \text{ C}}{1\text{s}} = 1.92 \times 10^{-2} \text{ A}$$

The direction of current is opposite to the direction of flow of the negative electrons.

Current density is

$$j = \frac{I}{A} = \frac{1.92 \times 10^{-2} \text{ A}}{2 \times 10^{-6} \text{ m}^2} = 9.6 \times 10^3 \text{ A/m}^2.$$

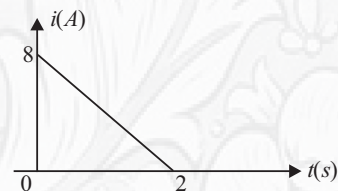


YOUR TURN

Q.1 A constant current of 2A passes a wire for 10s . Find the total charge flowing through the wire in 10s . If charge carriers are electrons, how many electrons passed the wire in 10s ?

Q.2 In Bohr's model of hydrogen atom, an electron is revolving in a circular orbit of radius $r = 5.3 \times 10^{-11} \text{ m}$ at a speed of $v = 2.2 \times 10^6 \text{ m/s}$. Find the current in the orbit.

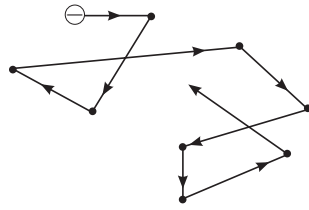
Q.3 Current in a wire decreases linearly from 8A to zero in 2s . Find the amount of charge that flows through the wire in 2s .



Q.4 Current through a given area is changing with time as $i = (2.0\text{A}) + (1.0 \text{ A/s})t$, where t is time in second. Find the total charge flow through the area in 4s .

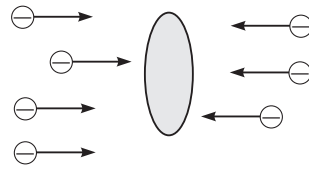
4. ELECTRIC CURRENT IN CONDUCTORS

A conductor is a material having large number of free electrons. These electrons are not bound to any atom and wander inside the material. At usual room temperature, the free electrons move at very high speed of the order of hundreds of kilometers per second. They collide frequently with the fixed atoms (ions, in fact). The speed of an electron and direction of its motion changes randomly at every collision. A typical electron moves on a zig-zag path. But this random motion of a large population of free electrons does not produce any current. The number of electrons passing through an area ΔA from one side very nearly equals the number crossing from the other side, in a given interval of time. The electric current through the area ΔA is zero.



Path of typical free electron inside a conductor

Think of a box having large number of honey bees inside it. Now, imagine the mid section of the box. A large number of randomly moving bees will cross the section from left to right and nearly the same number will cross in opposite direction. The net flow of bees through the section is nearly zero. There is no “bee current”.



Number of electrons crossing the area from left to right is nearly same as the number of electrons crossing from right to left

Now, consider a situation where we have created an electric field ($\vec{E}$) inside a conductor. Each free electron experiences an electric force ($\vec{F}_e = -e\vec{E}$) opposite to the electric field. Note that all the electrons experience force in the same direction. Due to this force the electrons acquire a small average speed directed opposite to the field. This small average speed of the free electrons is known as *drift speed*.

In absence of the field an electron will end up almost at the same location, where it started, after a long time. But in presence of the field it will end up some distance ahead of its original position, after a long time. This may not be true of all the electrons, but will be true for most of them. This slow drift of electrons produces a current through a cross section. Actually, the entire “cloud of electrons” has started moving though their random motion continues. Recall our box filled with bees. This box begins to move slowly. The honey bees continue their random motion but the entire group now has an ordered motion. This ordered motion (which we are calling as *drift*) produces a bee current through any cross section. If the entire box moves through a section, all bees have crossed that section and certainly there is a bee current.

The drift speed (v_d) of electrons in a conductor is typically small ($\approx 0.01 \text{ mm s}^{-1}$) compared to the random thermal speed

of the electrons. But this small drift speed is responsible for current. Let's estimate this drift speed.

4.1 Drift Speed

The average time that an electron spends between two successive collisions inside a conductor is known as its *relaxation time* (τ). The relaxation time for a given metal is fixed at a given temperature. When there is an electric field in the conductor, it gets a chance to accelerate the electrons during this relaxation time.

Let the velocity of a given electron immediately after collision be $\vec{v}_i$. Its velocity just before the next collision is $\vec{v}_f = \vec{v}_i + \vec{a}t$ where t = time interval between two collisions.

$$\text{or, } \vec{v}_f = \vec{v}_i - \frac{e\vec{E}}{m}t \quad \text{where acceleration } \vec{a} = -\frac{e\vec{E}}{m}$$

We take average over all time intervals

$$\langle \vec{v}_f \rangle = \langle \vec{v}_i \rangle - \frac{e\vec{E}}{m} \langle t \rangle$$

In a typical collision, an electron completely loses its memory and starts fresh in random direction with random speed. $\langle \vec{v}_i \rangle$ is average of velocities of an electron after every collision, over a long period of time. Obviously, it will be zero.

The average velocity of an electron, averaged over a long time is

$$\langle \vec{v}_f \rangle = -\frac{e\vec{E}}{m} \tau \quad [\langle t \rangle = \tau = \text{relaxation time}]$$

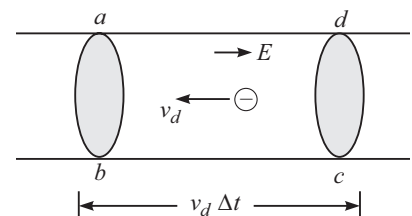
This is drift velocity. It is opposite to $\vec{E}$. Magnitude of drift velocity (i.e., drift speed) is

$$v_d = \frac{eE}{m} \tau \quad (7)$$

Moreover, if we measure the drift speeds of all electrons at a given time, we find that their average is also given by equation (7).

4.2 Relation between Current and Drift Speed

Random thermal motion of electrons cause no current. For the time being it is better to forget about this motion. It is the ordered drift that causes current in a conductor. Consider a cylindrical conductor having cross sectional area A . An electric field E is established inside it. The electrons acquire a drift speed v_d given by equation (7).



All free electrons inside the volume abcd will cross the section ab in time interval Δt .

Consider a volume (abcd) as shown. The total number of free electrons in this volume is

$$N = n(Av_d \Delta t)$$

6.4 Electrostatics and Current Electricity

where n = number of free electrons per unit volume in the conductor and $(Av_d\Delta t)$ is the volume of the cylindrical segment of length $v_d\Delta t$.

N number of electrons (all moving at speed = v_d) will cross the section ab in next Δt interval of time.

$\therefore$ Charge flow through section ab in time Δt is

$$\Delta q = N.e = (nAv_d\Delta t)e$$

$\therefore$ Current through section ab is

$$i = \frac{\Delta q}{\Delta t} = nev_d A \quad (8)$$

Current density at the cross section ab is

$$j = \frac{i}{A} = nev_d \quad (9)$$

Example 4 Small drift speed but large current

Number of free electrons in unit volume of copper is 10^{29}m^{-3} . A copper wire of cross section 2mm^2 carries a current of 5A. Find the drift speed of the electrons in the wire.

Solution

Concepts

(i) $j = nev_d$ and $i = nev_d A$

(ii) Even for small values of v_d , we can get good current because n is a large number.

$$i = nev_d A$$

$$\begin{aligned} \text{or, } v_d &= \frac{i}{neA} = \frac{5}{10^{29} \times 1.6 \times 10^{-19} \times 2 \times 10^{-6}} \\ &= 1.56 \times 10^{-4} \text{ ms}^{-1} = 0.156 \text{ mms}^{-1} \end{aligned}$$

5. OHM'S LAW

We know that current density at a location inside a conductor is given as

$$j = nev_d$$

Using equation (7), we can write

$$j = ne \left(\frac{eE}{m} \tau \right) = \left(\frac{ne^2 \tau}{m} \right) E$$

$$\text{or, } j = \sigma E \quad (10)$$

where $\sigma = \frac{ne^2 \tau}{m}$ is known as *electrical conductivity* of the material. *The electrical conductivity depends on the material of the conductor and its temperature.* For a given material, at a given temperature, σ is a constant. A material having higher value of σ is said to be a better conductor.

Inverse of σ is defined as resistivity of the material (ρ).

$$\rho = \frac{1}{\sigma} = \frac{m}{ne^2 \tau} \quad (11)$$

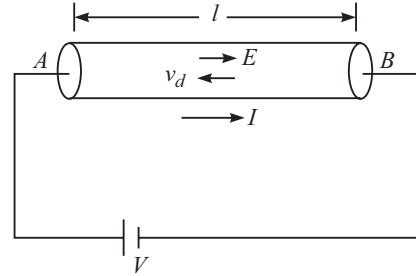
Equation (10) can be written as

$$j = \frac{E}{\rho} \quad (12)$$

Equation (10) or (12) is known as *Ohm's law in microscopic form*. It tells us that current density at a point inside a conductor is proportional to the electric field at that point.

An electric field is essential to cause the free electrons in a conductor to move. How do we establish an electric field inside a conductor? We can do that by applying a potential difference across a conductor. Consider a cylindrical conductor of cross section A and length l . We connect a battery to apply a potential difference V across it. The electric field inside the conductor is $E = \frac{V}{l}$. Ohm's law

given by equation (12) can be written as



$$j = \frac{E}{\rho}$$

$$\text{or, } \frac{I}{A} = \frac{V}{l\rho} \quad \text{or, } V = \left(\frac{\rho l}{A} \right) I$$

$$\text{or, } V = RI \quad (13)$$

where $R = \frac{\rho l}{A}$ is known as resistance of the given piece of the conductor.

Equation (13) is commonly used form of Ohm's law. It says that current in a conductor is directly proportional to the potential difference applied across it.

The ratio $\frac{V}{I} = R$ is resistance of the conductor across two points between which the potential difference is applied.

Unit of resistance $\frac{\text{volt}}{\text{ampere}}$ is known as ohm.

Unit of resistivity (ρ) is $\Omega\text{-m}$ and that of conductivity is $(\Omega\text{-m})^{-1}$.

6. RESISTANCE

The property of resistance (R) of a conductor decides the value of current through it, when a potential difference is applied across it.

$$I = \frac{V}{R}$$

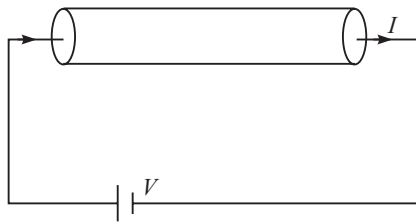


Fig. (a)

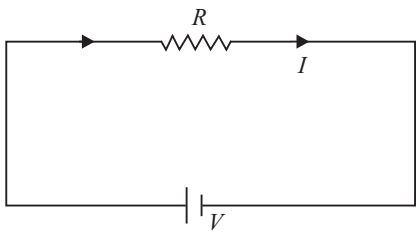


Fig. (b)

The arrangement in Fig (a) is symbolically represented as in Fig (b).

The property of resistance is indicated by the symbol shown in the Fig. (b).

As already stated, the property of resistance of a piece of conductor depends on its size, shape and material. For a conductor of uniform cross section (A), resistance (R) is given by

$$R = \frac{\rho l}{A} = \frac{l}{\sigma A} \quad (14)$$

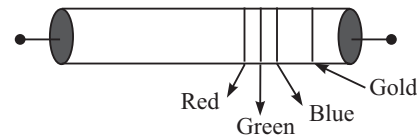
Where l is length of the conductor and ρ is resistivity of the material. $\sigma = \frac{l}{\rho}$ is conductivity of the material.

A specimen of conducting material, having a resistance of desired value is known as a *resistor*. We can make resistors using wires (of alloys like nichrome, manganin or constantan). Modern day electronic circuits use resistors made of carbon. They are made from a mixture of powdered carbon and an insulating material, usually ceramic. They are in shape of a cylinder with two leads for connection. The value of resistance is read using four coloured-bands, marked on the surface of the cylinder. Each colour stands for a digit (as given in the table). The first two bands indicate the two significant digits, the third band gives the power of the multiplier. The last band is either golden or silver in colour. A golden band indicates 5% tolerance and a silver band indicates 10% tolerance.

Colour	Digit	Multiplier	Tolerance
Black	0	1	
Brown	1	10	
Red	2	10^2	
Orange	3	10^3	
Yellow	4	10^4	
Green	5	10^5	
Blue	6	10^6	
Violet	7	10^7	
Gray	8	10^8	
White	9	10^9	
Gold		0.1	5%
Silver		0.01	10%

For the resistor shown in the figure, we can write the value of resistance as

$$\begin{aligned} R &= (25 \times 10^6 \pm 5\%) \Omega \\ &= (25 \pm 5\%) \text{M}\Omega \end{aligned}$$



Example 5 Copper and aluminium

What length of copper wire will have same resistance as 1 m long aluminium wire of same cross section. The resistivities for the two metals are:

$$\rho_{\text{cu}} = 1.7 \times 10^{-8} \Omega\text{-m} \text{ and } \rho_{\text{Al}} = 2.6 \times 10^{-8} \Omega\text{-m}$$

Solution

Concepts

$$R = \frac{\rho l}{A}$$

$$\text{For copper, } R_{\text{cu}} = \frac{\rho_{\text{cu}} l_{\text{cu}}}{A}$$

$$\text{For aluminium, } R_{\text{al}} = \frac{\rho_{\text{al}} l_{\text{al}}}{A}$$

$$\text{Given } R_{\text{cu}} = R_{\text{al}}$$

$$\Rightarrow \rho_{\text{cu}} l_{\text{cu}} = \rho_{\text{al}} l_{\text{al}}$$

$$\text{or, } l_{\text{cu}} = \frac{2.6 \times 10^{-8}}{1.7 \times 10^{-8}} \times 1 = 1.53 \text{ m}$$

Note: Copper has less resistivity than aluminium means it is better conductor. But copper is costlier too!

Example 6 A metal bar has a rectangular cross section. Find the ratio of its resistance when a battery is connected to it as shown in Fig (a) and as shown in Fig (b).

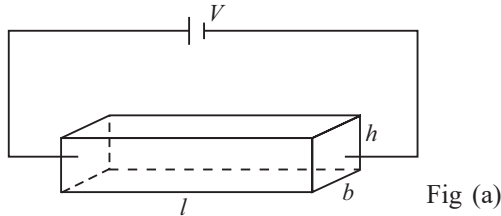


Fig (a)

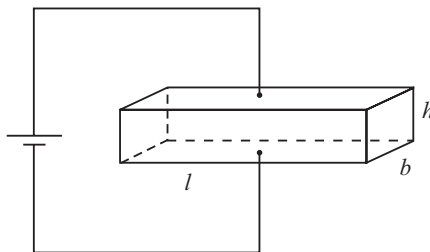


Fig (b)

Solution**Concepts**

- (i) ρ is same in both cases. Its value remains same if the material remains same.
- (ii) In the expression $R = \frac{\rho l}{A}$, l is the length of the conductor in the direction of current flow and A is area of cross section perpendicular to it.

In Figure (a), $R_1 = \frac{\rho l}{bh}$ [$\because A = bh$]

In Figure (b), $R_2 = \frac{\rho h}{lb}$ [$\because A = lb$]

$$\therefore \frac{R_1}{R_2} = \frac{l^2}{h^2}$$

6.1 Temperature Dependence of Resistivity

With increase in temperature, the speed of random motion of free electrons increases. Electrons collide more frequently with the fixed atoms. In other words, the relaxation time (τ) decreases. According to equation (11), the resistivity increases. For small temperature changes, we can write

$$\rho = \rho_0 (1 + \alpha \Delta\theta) \quad (15)$$

Here ρ_0 = Resistivity at temperature θ_0

ρ = Resistivity at temperature θ

$\Delta\theta = \theta - \theta_0$ = change in temperature

α is known as temperature coefficient of resistivity for the material. Its unit is $(^\circ\text{C})^{-1}$ or K^{-1} .

Value of α for copper is nearly 0.0039 K^{-1} and for gold it is 0.0034 K^{-1} , at room temperature. In fact, value of α also changes with temperature. We will mostly ignore this change in our discussions.

Resistance of a piece of conductor is $R = \rho \frac{l}{A}$. For small temperature changes we will assume that change in the ratio $\frac{l}{A}$, due to thermal expansion, is small. Therefore, $R \propto \rho$ and we can write

$$R = R_0 (1 + \alpha \Delta\theta) \quad (16)$$

Here R_0 = Resistance of the specimen at temperature θ_0

R = Resistance of the specimen at temperature θ

$\Delta\theta = \theta - \theta_0$ = change in temperature

α = Temperature coefficient of resistivity

Example 7

Resistance of a conducting wire is $10.0 \, \Omega$ at 20°C . The wire is dipped in a hot insulating liquid and its resistance rises to $11.5 \, \Omega$. Find the temperature of the liquid. The temperature coefficient of resistivity for the material of the wire is $\alpha = 3.8 \times 10^{-3}/^\circ\text{C}$.

Solution**Concepts**

$$R = R_0 (1 + \alpha \Delta\theta)$$

$$R = R_0 (1 + \alpha \Delta\theta)$$

$$\text{or, } 11.5 = 10.0 (1 + 3.8 \times 10^{-3} \cdot \Delta\theta)$$

$$\text{or, } \Delta\theta = 39.47^\circ\text{C}$$

$$\therefore \theta = \theta_0 + \Delta\theta = 20 + 39.47 = 59.47^\circ\text{C}$$

6.2 Different Type of Material

Based on their electrical conductivities, we can classify material into four groups:

- (i) **Conductors:** These are material with high conductivity (low resistivity). A typical conductor copper has a resistivity of $1.7 \times 10^{-8} \, \Omega\text{-m}$. Resistivity of conductors increase with rise in temperature, i.e., their conductivity decreases.
- (ii) **Insulators:** Materials like plastic, ceramic, rubber etc. Are insulators. They have very high resistivity (i.e., poor conductivity). A typical value may be $\rho = 5 \times 10^{17} \, \Omega\text{-m}$. Resistivity of an insulator is not affected much by small changes in temperature. In simple language, when a potential difference is applied across a wire made of plastic, the current will be nearly zero. We say that the plastic does not conduct.
- (iii) **Semiconductors:** A material like Silicon has a resistivity of nearly $650 \, \Omega\text{-m}$ at room temperature.

This is not very small like conductors nor it is very high like insulators. Such materials are called semiconductors and are widely used in electronic device. Unlike conductors, their resistivity decreases with rise in temperature. We will learn about the reasons in a separate chapter on semiconductors.

Current (i) through a device (known as p - n junction) made of semiconducting material changes with the applied potential difference (V) in non linear fashion as shown in Figure (a). Such material which do not obey Ohm's law ($I \propto V$) are known as *non-ohmic* material.

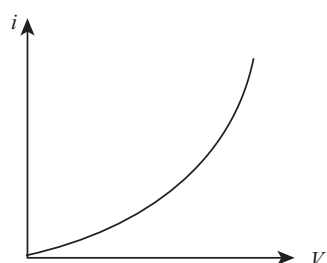


Fig. (a)

current-voltage graph for a p - n junction made of semiconductor. It does not follow ohm's law.

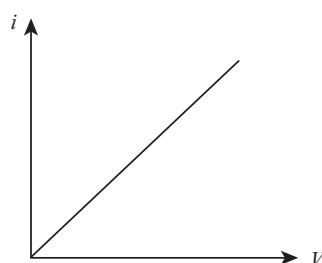


Fig (b)

current-voltage graph for a material that follows ohm's law.

- (iv) **Superconductors:** A material having zero resistivity is known as a superconductor. No such material is known at room temperature. But there are certain materials for which the resistivity suddenly becomes zero below a certain temperature. This temperature is called critical temperature. Material below this temperature is superconductor. Hydrogen sulfide (H_2S) under very high pressure has been found to be superconducting below 203 K ($= -70^\circ\text{C}$) temperature. It is one of the highest temperature superconductor known till date.

If a current is established in a superconducting loop, it can persist for long time without any emf. Superconductors are used to create very strong magnetic fields. They find applications in particle physics, very fast computers, maglev trains etc.



YOUR TURN

Q.5 A conducting wire carries a current I . If the electric field at each point inside the conductor gets doubled, how will the drift speed of electrons change? What will be the affect on the current?

Q.6 A copper wire is thin and long. Another copper wire is thick and short. Both have same volume. Which one has higher resistance?

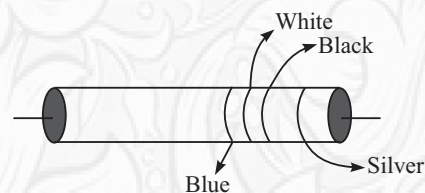
Q.7 A metal wire has resistance R . It is stretched to double its length. Find the resistance of the new wire.

Q.8 A metal wire is stretched to make it 0.2% longer. Find percentage change in its resistance.

Q.9 What will be the mass of a cylindrical copper wire 2.0m long having a resistance of 0.2Ω . The resistivity of copper is $\rho = 1.7 \times 10^{-8} \Omega\text{-m}$ and its density is $d = 9 \times 10^3 \text{ kg/m}^3$.

Q.10 Resistance of a copper and an iron wire at 0°C are 8.2Ω and 7.8Ω respectively. Find the temperature at which both of them have same resistance. Neglect thermal expansion and take temperature coefficient of resistivities as: $\alpha_{\text{cu}} = 4 \times 10^{-3} \text{ K}^{-1}$ and $\alpha_{\text{Fe}} = 5 \times 10^{-3} \text{ K}^{-1}$

Q.11 The colour bands on a carbon resistor is as shown. Find the resistance.



Q.12 Calculate the electric field in a copper wire of cross sectioned area 4 mm^2 carrying a current of 1.0 A. The resistivity of copper is $\rho = 1.7 \times 10^{-8} \Omega\text{-m}$.

In Short

- (i) Rate of flow of charge through a cross section is known as current through the section. Its unit is Cs^{-1} known as ampere (A).

- (ii) Current per unit area, through an area held normal to the direction of charge flow, is known as current density.

$$j = \frac{I}{A}$$

Current density is a vector, in the direction of positive charge flow.

- (iii) When an electric field is set up in a conductor, it causes the entire cloud of free electrons to drift slowly in a direction opposite to itself. The average speed acquired by the electrons (not thinking of their random thermal motion) is known as drift speed (v_d). Drift speed is responsible for current in a conductor.

- (iv) Drift speed is given by $v_d = \frac{eE}{m}\tau$, where e , m , τ and

E are charge on an electron, mass of an electron, the relaxation time and the applied electric field, respectively.

- (v) The relaxation time is average time between two successive collisions of an electron. It decreases with rise in temperature.

- (vi) $j = nev_d$

Here, n = number of free electrons in a unit volume of the conductor.

- (vii) $j = \sigma E$, where $\sigma = \frac{ne^2\tau}{m}$ is a constant for a given

material at a given temperature. σ is known as conductivity of the material. $\rho = \frac{1}{\sigma}$ is known as resistivity of the material.

- (viii) $j = \sigma E$, is same as $V = RI$ and is known as Ohm's law. Here, V is potential difference applied across a conductor and I is current in it. R is known as resistance of the conductor.

- (ix) Resistance of a conductor of uniform cross sectional area (A) and length l is given by $R = \frac{\rho l}{A}$

- (x) Resistivity of a material changes with temperature (θ) according to the relation $\rho = \rho_0 (1 + \alpha\Delta\theta)$ for small temperature changes.

If we neglect the change in physical dimensions (length and area) of a piece of conductor, its resistance changes with temperature as

$$R = R_0 (1 + \alpha\Delta\theta).$$

Here, α is known as temperature coefficient of resistance.

7. HEATING EFFECT OF CURRENT

A current flows through a resistor when a potential difference is applied across it. The current (I) when potential difference is V , is given by the Ohm's law as

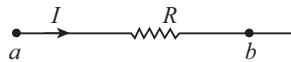
$$I = \frac{V}{R}$$

The current is always directed from the high potential end to the low potential end. In the figure shown $V_a > V_b$ and $V_a - V_b = V$.

Charge flow in time interval Δt is $\Delta q = I\Delta t$. This amount of charge moves from a point (a) at higher potential to another point (b) at lower potential. The charge loses electric energy. Amount of energy lost by the charge is

$$\begin{aligned}\Delta U_{\text{loss}} &= \Delta q V_a - \Delta q V_b = \Delta q (V_a - V_b) \\ &= \Delta q V = I\Delta t V\end{aligned}$$

$$\text{or, } \Delta U_{\text{loss}} = (IV) \Delta t \quad (17)$$



Current in a resistor always flows from high potential end to the low potential end.

Where does this energy go? It gets converted into heat and light in the resistor. The electric field in the resistor accelerates free electrons. They gain energy only to lose it when they collide with an atom. Huge number of electrons, each losing tiny bit of energy, contribute to a substantial loss in energy which appears as heat and light. Total heat generated in time Δt is given by equation (17).

Rate of heat loss, known as power dissipated, in the resistor is

$$P = \frac{\Delta U_{\text{loss}}}{\Delta t} = IV \quad (18)$$

Using Ohm's law, we can write the above expression as

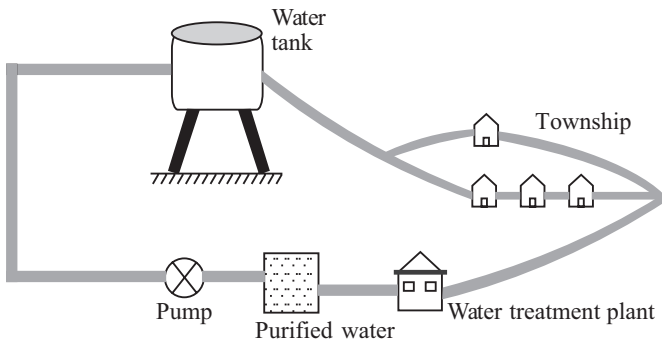
$$P = I^2 R = \frac{V^2}{R} \quad (19)$$

Unit of power is Js^{-1} also known as watt (W).

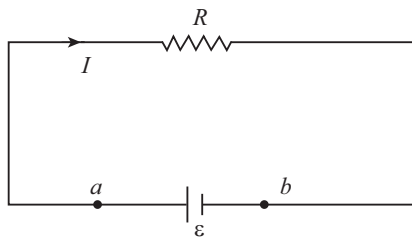
In nutshell, a resistor is a device which converts electrical energy into heat energy. In practice, every wire in a circuit has some resistance and produces heat when there is a current in it.

7.1 Battery as a Source of Energy

Consider the water supply system shown in the figure. An overhead tank stores water at a great height. Water in it has a large gravitational potential energy. When valves are opened, water reaches the homes in the township. The water discharged from the houses goes to a water treatment plant, where it gets purified. A water pump lifts the water back to the water tank. The water which rushes out of the tank, gains speed as it loses its gravitational potential energy. But it loses energy all the way due to friction. The water cannot climb back to the tank all by itself. It has lost all its energy (which got converted into heat due to friction) in the ‘circuit’ by the time it gets purified. (In practice it may lose its energy much before this stage and we may need a pump between the township and the treatment plant). The pump eats electrical energy and provides the necessary energy to the water to keep the ‘water current’ flowing.



A battery is like the water pump in the above example. It is a ‘charge pump’. It provides energy to charge by ‘pumping’ it from low potential to high potential. In an electric circuit this energy is lost by the charge as heat when it travels through a resistance. A battery uses chemical energy to increase the electrical potential energy of the charge.



Electromotive force (emf) of a battery can be defined as the amount of work the battery does in moving a unit positive charge from its negative terminal to its positive terminal. It is same as the potential difference across its terminal when it is not in a circuit.

In the circuit shown in the figure, emf of the battery is ε and the current in the circuit is I . Amount of charge that moves from the negative terminal (b) to the positive terminal (a) in time Δt is

$$\Delta q = I\Delta t.$$

Battery does work in moving this charge from a point of low potential to a point of higher potential.

$$W_{\text{batt}} = \varepsilon \Delta q = \varepsilon I \Delta t$$

This is same as loss in chemical energy of the battery. Power supplied by the battery is the rate at which the battery does work.

$$P_{\text{batt}} = \frac{W_{\text{batt}}}{\Delta t} = \varepsilon I \quad (20)$$

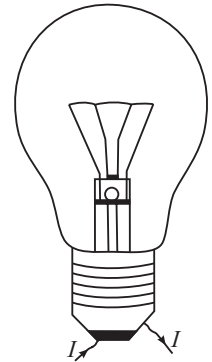
When the battery is ideal (i.e., it itself does not get heated), this power delivered by it is converted into heat by the resistor.

$$P_{\text{batt}} = P_R = \varepsilon I = \frac{\varepsilon^2}{R} = I^2 R.$$

Charge loses its energy in crossing the resistor and the battery again ‘pumps’ it to higher energy. The battery keeps on losing energy and the resistor keeps converting it into heat and light. The above electric circuit is a means of transferring energy from the battery to the resistor.

7.2 Incandescent Bulb

An incandescent light bulb has a tungsten made wire filament inside a glass bulb filled with inert gas. When current is passed through the filament it gets heated and emits light. These bulbs are usually inefficient in converting the consumed energy into light. Less than 5% energy gets converted into visible light and rest is dissipated as heat. This is the reason they are getting replaced with CFLs and LED bulbs.



Tungsten has high melting point and is therefore suitable for making incandescent bulbs. However, tungsten keeps on evaporating and the filament gets thinner and thinner with time. Filling the bulb with an inert gas reduces the rate of evaporation.

A bulb having specification 100W, 220V consumes 100W power when connected to a source of 220V. It consumes less power (and glows dim) when applied potential difference is less than 220V. Its filament is likely to blow out (i.e., melt) when applied potential difference is larger than 220 V.

7.3 Electric Heater

An electric heater finds many applications such as cooking, water heating, industrial heating, room heating etc. The resistor used to produce heat is commonly known as *heating element*. It is usually made of nichrome – an alloy of nickel and chromium. The element can be in the shape of a coiled wire, a ribbon or a strip. Nichrome is an ideal material due

6.10 Electrostatics and Current Electricity

to its high resistivity and also because it forms an adherent layer of chromium oxide when it gets heated for the first time. Material beneath this layer is protected from getting oxidized.

A 220 V, 1000W heater will consume 1000 W power when connected to a power supply of 220 V.

7.4 Fuse

A fuse is a safety device. It is a short wire designed to melt and break the circuit in the event of excessive current. A fuse is always connected in series with the component to be protected from overcurrent. A fuse wire has some resistance. When a current flows, it gets heated. It is designed to melt when excessive current flows through it. A fuse must have a small resistance so as to contribute negligible extra resistance to the circuit. This is largely accomplished by making the fuse wire as short as practically possible. A fuse wire of certain material and cross section will blow at a certain current no matter how long it is. A fuse is rated in amperes. A 5A fuse means it will blow out if the current exceeds 5A.

Modern day circuits also use *circuit breakers* which are specially designed switches which open when current through them exceeds a limit.

Example 8 Less resistance, more power

Which bulb has higher resistance – a 60W, 220V bulb or a 100W, 220V bulb?

Solution

Concepts

- A common mistake is to use $P = I^2 R$ and hence $P \propto R$. This is wrong simply because the two bulbs will not have same current when connected to 220V supply.
- Use $P = \frac{V^2}{R}$, and hence $P \propto \frac{1}{R}$, because both the bulbs are connected to 220V supply.

$$P = \frac{V^2}{R} \Rightarrow R = \frac{V^2}{P}$$

For both the bulbs V is same. Therefore, R will be higher for the bulb which has lower value of P .

$$\therefore R_{60} > R_{100}$$

Example 9 An old bulb glows dim

The brightness of an incandescent bulb goes on decreasing with prolonged use. Explain.

Solution

Tungsten evaporates with time. You can see it as black substance on the glass wall. The filament gets thinner with time. This increases the resistance ($R = \frac{\rho l}{A}$) of the filament.

The power consumed by the bulb decreases because $P = \frac{V^2}{R}$. Therefore, the bulb gives less light.

Example 10 Electric kettle

An electric kettle takes 3A current from a 210V supply and brings 1 liter of water from 20°C to 80°C in 10 minutes. Find the efficiency of the kettle.

The specific heat capacity of water is $s = 4.2 \text{ Jg}^{-1}(\text{°C})^{-1}$.

Solution

Concepts

- Efficiency refers to the fraction of electrical power consumed which got utilised for heating of water. Some energy will always leak out as radiation as well as get conducted to the atmosphere through the body of the kettle.
- Heat absorbed by water $\Delta Q = ms\Delta\theta$.

Heat absorbed by the water

$$\begin{aligned}\Delta Q &= ms\Delta\theta \\ &= (1000\text{g})(4.2 \text{ Jg}^{-1}(\text{°C})^{-1})(60\text{°C}) \\ &= 2.52 \times 10^5 \text{ J.}\end{aligned}$$

Electrical power consumed, $P = VI = 210 \times 3 = 630\text{W}$.

Electrical energy consumed in 10 minute is

$$E = \rho\Delta t = 630 \times 10 \times 60 = 3.78 \times 10^5 \text{ J}$$

$$\text{Efficiency, } \eta = \frac{\Delta Q}{E} = \frac{2.52 \times 10^5}{3.78 \times 10^5} = 0.67$$

We can say that the efficiency is 67%.

Example 11 Kilowatt hour (kWh)

The heating element of a heater has resistance of 100Ω . It is used for 8 hour, attached to a source of 200V. How much energy does the heater consume in unit of kWh?

Solution

Concepts

1 kWh is energy consumed in 1 hour if rate of energy consumption (i.e., power) is 1 kW.

Energy in kWh = (power in kW) (time in hour)

$$\begin{aligned}\text{Power of the heater is } P &= \frac{V^2}{R} = \frac{(200)^2}{100} = 400\text{W} \\ &= 0.4 \text{ kW}\end{aligned}$$

$$\begin{aligned}\therefore \text{Energy consumed} &= (P)(t) \\ &= (0.4 \text{ kW})(8 \text{ hour}) = 3.2 \text{ kWh.}\end{aligned}$$



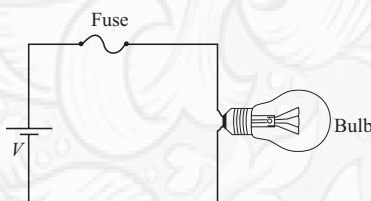
YOUR TURN

Q.13 Two wires of same material having ratio of lengths 1 : 2 and ratio of cross sectional areas 2 : 1 are connected one by one to the same voltage supply. Power dissipated in first wire is 20 W. Find the power dissipated in the second wire.

Q.14 A 100 W bulb is designed to operate on a potential difference of 230V.

- Find the resistance of the filament of the bulb.
- Find the current drawn by the bulb when connected to a 230 V supply.
- Find power consumed when the bulb is connected to a 200V cell.

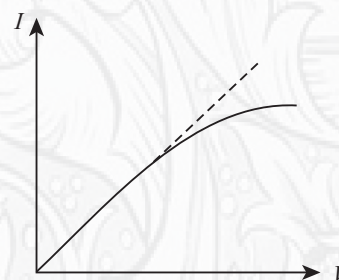
Q.15 A bulb is connected to a 100 V source. There is a 5A fuse in the circuit. Find the maximum wattage of the bulb that will



not blow out the fuse. Neglect resistance of the fuse and connecting wires.

Q.16 What will be change in power output of an electric heater if length of its element is increased 10% keeping everything else fixed?

Q.17 The I - V characteristic (i.e., current versus voltage graph) for a wire resistance is observed to deviate from straight line for higher values of current (see figure). Why?



In Short

- When a current I is given by a cell of emf ϵ , the power supplied by the cell is ϵI . This is the rate at which a chemical cell loses energy.
- When a current flows in a resistor, heat is dissipated. Rate of heat dissipation (i.e., power) in the resistor is given by

$$P = I^2 R = \frac{V^2}{R} = VI$$

Here, V is potential difference across R and I is current.

- An incandescent bulb has a filament which is nothing but a resistor. On passing current through it, it emits heat and light.
- Higher wattage bulb (designed to operate on a given voltage) have lesser resistance.
- With time the filament of a bulb gets thinner and its resistance increases.
- Electric energy consumption is often expressed in unit of kilowatt hour (kWh). 1 kWh is the amount of energy consumed when energy is consumed at a rate of 1 kW ($= 1000 \text{ J s}^{-1}$) for 1 hour.
- An electric fuse is a safety device used to break the circuit when the current exceeds a specified limit.

8. KIRCHHOFF'S LAWS

The following two laws, named after German physicist G.R. Kirchhoff, are very powerful tools in solving any circuit related problem.

Junction Law (Kirchhoff's first law)

In any electric circuit the sum of the currents entering a junction must be equal to the sum of the currents leaving that junction.

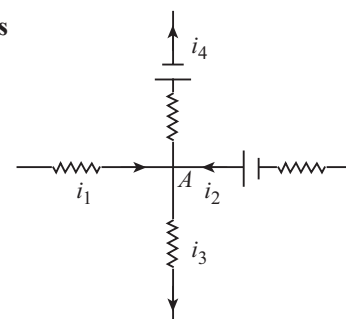
The figure shows a part of an electric circuit with a junction at point A . The junction law says that

$$i_1 + i_2 = i_3 + i_4$$

This rule is nothing but a statement of the *principle of conservation of charge* for a steady flow. Charge cannot buildup or get depleted at a junction.

Loop Law or Voltage Law (Kirchhoff's Second Law)

The algebraic sum of the changes in potential across all elements in a complete traversal of an electric loop of a circuit must be zero. [This is exactly same as what we learnt in the chapter on capacitors.]



i_1 and i_2 are the currents entering the junction and i_3 and i_4 are the currents leaving the junction.

6.12 Electrostatics and Current Electricity

Suppose A is a point at a height h above the floor in your room. You move a ball along a zig-zag path in the room and bring it back to A . The elevation of the ball has not changed. The algebraic sum of changes in elevation that you encountered must be zero as you return to A .

Similarly, when you traverse an electric loop, the potential may rise and fall across various elements. When you return to your starting point, the sum of all changes is zero.

Consider the loop shown in the figure. It is part of some circuit. We start from point A and travel clockwise to again reach back at point A , while keeping track of potential changes. If V_A is potential of point A , the potential drops by $i_1 R_1$ as we cross the resistance R_1 to reach point B . Then the potential increases by ε_1 as we move to point C . It again increases by $i_2 R_2$ as we cross the resistance R_2 to reach point D and then drops by ε_2 as we are back to A . We write

$$V_A - i_1 R_1 + \varepsilon_1 + i_2 R_2 - \varepsilon_2 = V_A$$

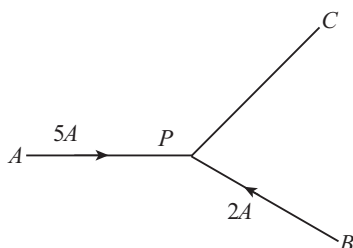
$$\text{or, } -i_1 R_1 + \varepsilon_1 + i_2 R_2 - \varepsilon_2 = 0$$

The last equation tells us that algebraic sum of all potential changes in an electric loop adds up to zero. While writing the loop equation, we must keep following points in mind:

- All points connected by plain conducting wires will be assumed to be at same potential. Resistance of a connecting wire shown in a circuit will be assumed to be zero.
- When we move through a resistance in the direction of the current, the potential drops by iR and when we move against the current it increases by iR . Actually, current in a resistance is always from high potential end to low potential end.

We will return to use of Kirchhoff's laws in solving circuit related problems, later in this chapter. Here we present two examples to re-emphasize the above points.

Example 12 P is a junction point in an electric circuit. Find the direction and magnitude of current in the branch PC .



Solution

Concepts

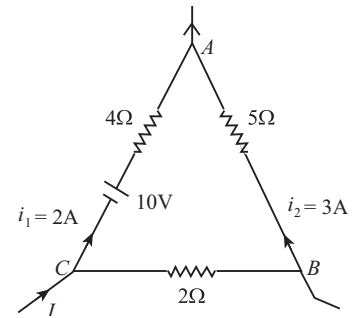
Sum of current going towards P = sum of current going out of P

Sum of currents towards $P = 5A + 2A = 7A$.

Therefore, $7A$ current must flow out of P along PC .

Example 13

The diagram shows a closed path in an electrical circuit. Find the current in 2Ω resistor. Find current I shown in the figure.



Solution

Concepts

Use of Kirchhoff's loop law will give the answer.

Let $V_B - V_C = V$ be the potential difference across 2Ω resistor.

We will start from A and move clockwise in the loop to go back at A , while keeping track of potential

$$V_A + 5 \times 3 - V + 10 - 2 \times 4 = V_A$$

$$\text{or, } 15 - V + 10 - 8 = 0$$

$$\text{or, } V = 17 \text{ V}$$

Potential drops by 17 V from B to C . It means current is directed from B to C and its value is

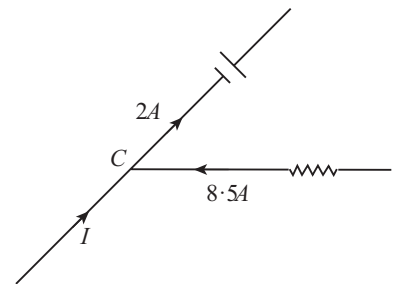
$$i = \frac{17V}{2\Omega} = 8.5A$$

At junction C ,

$$I + 8.5 = 2$$

$$\text{or, } I = -6.5A.$$

The negative sign indicates that the direction of current I shown in the figure is incorrect and actually it is 6.5 A current flowing out of C .



9. RESISTORS IN SERIES AND PARALLEL

Resistances can be grouped in many different ways. Series and parallel grouping are the two most common ways of grouping the resistances.

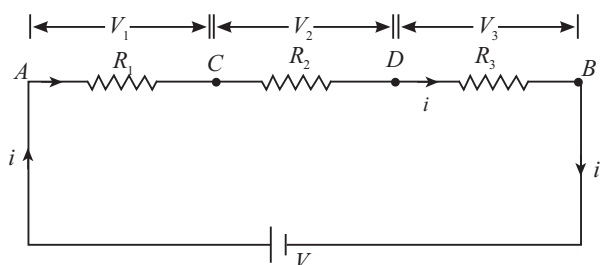
A group of resistors, when given a potential difference V across its two terminals, draws a current I . We define the equivalent of the combination as

$$R_{eq} = \frac{V}{I}$$

Single resistance (R_{eq}) draws the same current as the given combination when the same potential difference is applied across it.

9.1 Series Combination

The figure shows three resistors, having resistances R_1 , R_2 and R_3 , connected in series. A and B are terminals of this group. This group can be connected in any circuit using terminals A and B . If a connection is also made at C or D (or both), then the three resistors are no longer in series.



In series combination current is same in all resistors

The figure shows a potential difference V applied across the group using an ideal cell. A current i passes through all the resistors. Resistors in series always carry same current.

The potential drop across the three resistors are

$$\left. \begin{aligned} V_1 &= V_A - V_C = iR_1 \\ V_2 &= V_C - V_D = iR_2 \\ V_3 &= V_D - V_B = iR_3 \end{aligned} \right\} \quad (a)$$

Adding the above equations gives

$$V_A - V_B = i(R_1 + R_2 + R_3)$$

$$\text{or, } V = i(R_1 + R_2 + R_3)$$

$$\text{or, } \frac{V}{i} = R_1 + R_2 + R_3 \quad (b)$$

$$\text{or, } R_{eq} = R_1 + R_2 + R_3$$

In, general for any number of resistors in series, we have

$$R_{eq} = R_1 + R_2 + R_3 + R_4 + \dots \quad (21)$$

From set of equations in (a) it is clear that the total potential drop between A and B , gets divided into three potential drops across R_1 , R_2 and R_3

$$V_1 + V_2 + V_3 = V \quad (22)$$

$$\text{Also, } V_1 : V_2 : V_3 = R_1 : R_2 : R_3 \quad (23)$$

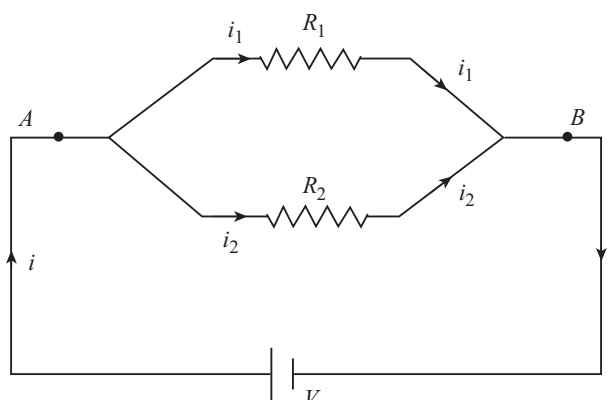
From (22) and (23)

$$V_1 = \left(\frac{R_1}{R_1 + R_2 + R_3} \right) V; \quad V_2 = \left(\frac{R_2}{R_1 + R_2 + R_3} \right) V;$$

$$V_3 = \left(\frac{R_3}{R_1 + R_2 + R_3} \right) V$$

9.2 Parallel Combination

The diagram shows two resistors in parallel. One end of the two resistors have been connected to a common point A and their other terminals are connected to another points B . When a potential difference is applied between A and B , it is experienced by both the resistors. In figure, an ideal cell of emf V has been used to apply a potential difference across A and B .



In parallel grouping, each resistor experiences same potential difference.

In parallel grouping, the potential difference across each resistor is same though the current in them may be different. Let the current through R_1 be i_1 and the current in R_2 be i_2 . The current supplied by the cell is

$$i = i_1 + i_2 \quad (a)$$

Since potential difference across each resistor is V , therefore

$$i_1 = \frac{V}{R_1} \quad \text{and} \quad i_2 = \frac{V}{R_2}$$

$$\therefore \text{ From (a) } i = \frac{V}{R_1} + \frac{V}{R_2}$$

$$\text{or, } \frac{i}{V} = \frac{1}{R_1} + \frac{1}{R_2}$$

From the definition of equivalent resistance $R_{eq} = \frac{V}{i}$.

$$\therefore \frac{1}{R_{eq}} = \frac{1}{R_1} + \frac{1}{R_2}$$

6.14 Electrostatics and Current Electricity

The general result for any number of resistances in parallel is

$$\frac{1}{R_{eq}} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} + \dots \quad (24)$$

The current i supplied to the combination gets divided between R_1 and R_2 such that potential difference across both of them remains same.

$$i_1 R_1 = i_2 R_2 = V$$

$$\text{or, } \frac{i_1}{i_2} = \frac{R_2}{R_1} \quad (25)$$

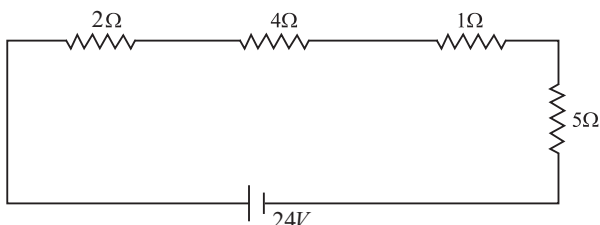
The current gets divided in inverse ratio of resistances.

Solving (a) and (25) gives

$$i_1 = \left(\frac{R_2}{R_1 + R_2} \right) i \quad \text{and} \quad i_2 = \left(\frac{R_1}{R_1 + R_2} \right) i \quad (26)$$

The most important thing to realize about parallel connection is that the parallel paths are independent of one another. If you remove the resistance R_2 , the resistance R_1 will see no change. Current in it will still be i_1 .

Example 14 Four resistors are connected to a 24V cell as shown. Find the current in the circuit and potential difference across 2 Ω resistor.



Solution

Concepts

(i) All resistors are in series. $R_{eq} = R_1 + R_2 + R_3 + R_4$

(ii) Current $i = \frac{V}{R_{eq}}$. Current is same in all resistors.

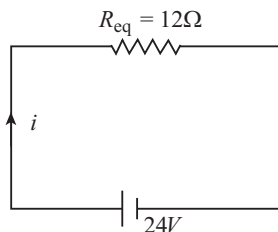
$$R_{eq} = 2 + 4 + 1 + 5 = 12 \Omega$$

Current supplied by the cell to the equivalent resistance is

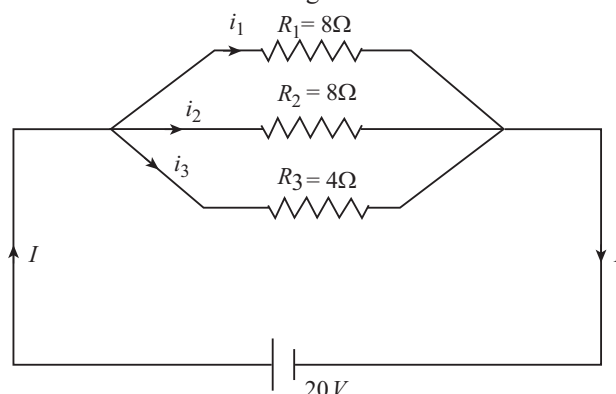
$$i = \frac{24V}{12\Omega} = 2A$$

This same current $i = 2A$ flows in the original circuit also. Potential difference across 2 Ω resistor is

$$V_1 = iR = (2A)(2\Omega) = 4V.$$



Example 15 Three resistors are connected in parallel to a 20V cell as shown in the figure.



- Find the equivalent resistance of the combination.
- Find current I .
- Find current i_1 through R_1 .
- Find power dissipated in R_2 .
- How will the current i_1 change if the resistor R_3 burns out?

Solution

Concepts

$$(i) \quad \frac{1}{R_{eq}} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}$$

(ii) Each parallel path is independent of another.

$$(a) \quad \frac{1}{R_{eq}} = \frac{1}{8} + \frac{1}{8} + \frac{1}{4} \Rightarrow R_{eq} = 2\Omega$$

$$(b) \quad I = \frac{V}{R_{eq}} = \frac{20}{2} = 10A$$

(c) Potential difference across R_1 is 20V.

$$\therefore \text{ using Ohm's law } i_1 = \frac{20V}{8\Omega} = 2.5A$$

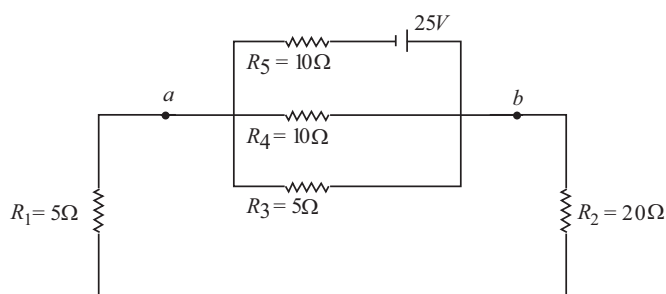
(d) Power dissipated in R_2 is

$$P_2 = \frac{V^2}{R_2} = \frac{20^2}{8} = 50W$$

(e) Removal of R_3 will have no effect on i_1 or i_2 . But I will get reduced as $I = i_1 + i_2$.

Example 16 Consider the circuit shown in the figure. Find

- Potential difference between points a and b .
- Current in 20 Ω resistor.

**Solution****Concepts**

- First we will reduce the circuit by using series-parallel grouping and find the current supplied by the cell.
- Potential difference across resistances in series are in the ratio of the resistances.
- Current gets divided into parallel paths in inverse ratio of the resistances.

R_1 and R_2 are in series. Their equivalent is

$$R_{12} = R_1 + R_2 = 25\Omega.$$

Now, R_{12} , R_3 and R_4 are in parallel. Their equivalent is given by

$$\frac{1}{R_0} = \frac{1}{25} + \frac{1}{5} + \frac{1}{10}$$

or, $R_0 = \frac{50}{17} \Omega$

R_0 and R_5 are in series, connected to a 25 V cell.

$$\therefore I = \frac{25V}{\left(\frac{50}{17} + 10\right)\Omega} = 1.93A$$

- (a) The 25 V potential drop gets divided between R_0 and R_5 in ratio of the resistances. Because we have already calculated the current (I), we can directly write

$$V_{ab} = R_0 I = \left(\frac{50}{17}\right)(1.93) = 5.68V$$

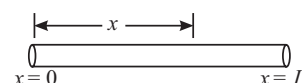
- (b) Look at the first diagram. The potential difference $V_{ab} = 5.68V$ is common to R_4 , R_3 and R_{12} . Therefore, current in R_{12} is

$$i = \frac{5.68V}{25\Omega} = 0.23A$$

This current (i) is there in R_2 as well as R_1 .

Example 17 Resistance of a wire with variable resistivity

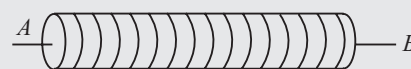
A cylindrical wire has cross sectional area A and its length is L . Its resistivity changes with distance (x) from one of



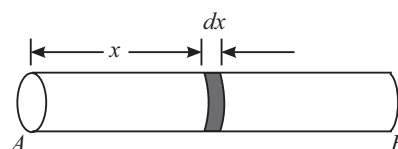
its ends as $\rho = \rho_0 \left(1 + \frac{x}{L}\right)$. Find the resistance of the wire.

Solution**Concepts**

The wire can be thought to be made up of large number of thin discs. All these discs are stacked in series between the end points A and B. We need to add resistances of all such discs to get the resistance of the wire.



Consider a disc shaped element in the wire. It is at a distance x from the end A and its thickness is dx . Resistance of this disc shaped element is



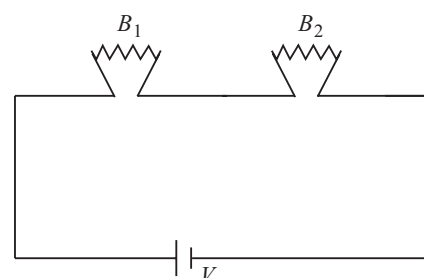
$$dR = \frac{\rho dx}{A} = \frac{\rho_0}{A} \left(1 + \frac{x}{L}\right) dx$$

$\therefore$ Resistance of the wire will be

$$R = \int dR = \frac{\rho_0}{A} \int_0^L \left(1 + \frac{x}{L}\right) dx = \frac{\rho_0}{A} \left[L + \frac{L}{2}\right] = \frac{3\rho_0 L}{2A}$$

Example 18 Which bulb is brighter?

Bulb B_1 (100W, 200V) is connected in series with another bulb B_2 (60W, 200V). The bulbs in series are connected to a power supply of $V = 200$ volt.



6.16 Electrostatics and Current Electricity

- (a) Which bulb will be brighter?
 (b) Calculate the power dissipated in the bulb B_1 .

Solution

Concepts

- (i) A bulb will give rated power output if it is connected to a rated voltage supply. Both the bulbs, in the given problem, will have a potential difference less than 200V across them. Both will give less light than their rated capacity.
 (ii) Bulbs are in series. Same current flows through both of them.
 (iii) Refer to Example 8.

- (a) From Example 8, $R_{60} > R_{100}$.
 Since current (I) is same in both the bulbs
 $P_1 = I^2 R_{100}$ and $P_2 = I^2 R_{60}$
 Obviously, $P_2 > P_1$.

The bulb B2 will give more light.

$$(b) \quad P = \frac{V^2}{R} \quad \Rightarrow \quad R = \frac{V^2}{P}$$

Resistances of the two bulbs are

$$R_1 = \frac{(200)^2}{100} = 400\Omega$$

$$R_2 = \frac{(200)^2}{60} = \frac{2000}{3}\Omega$$

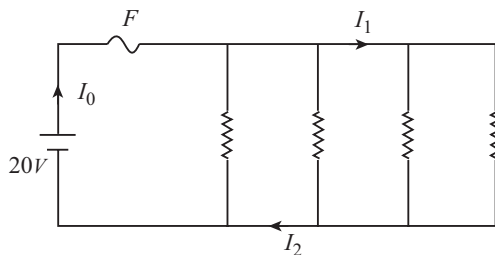
$$\text{Total resistance } R = R_1 + R_2 = \frac{3200}{3}\Omega$$

$$\text{Current } I = \frac{V}{R} = \frac{200}{\frac{3200}{3}} = \frac{3}{16}A$$

$$\therefore P_1 = I^2 R_1 = \left(\frac{3}{16}\right)^2 \times 400 = 14.06 \text{ W.}$$

Example 19 Connect bulbs in parallel to keep them independent

The circuit shows four identical bulbs connected in parallel to an ideal 20 V source. F is a fuse of rating 6.5 A. Resistance of each bulb is 20Ω .



- (a) Find the values of currents I_0 , I_1 and I_2 shown in the figure.
 (b) How many more bulbs can be connected in parallel without blowing the fuse?

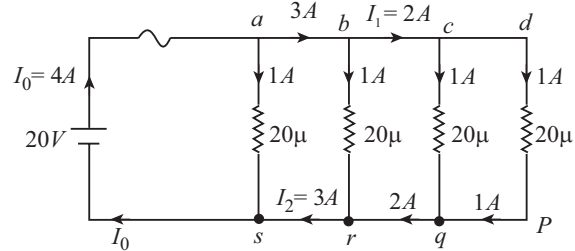
Solution

Concepts

- (i) By connecting bulbs in parallel we ensure that all of them get a potential difference of 20V across them. The parallel paths are independent. Even if one of the bulb fuses, the others will keep glowing. In your house, all bulbs and fans are connected in parallel.
 (ii) Kirchhoff's junction law can be used to write currents like I_1 and I_2 .
 (iii) Addition of more bulbs will decrease the overall resistance and hence I_0 will increase. If I_0 increases beyond 6.5A, the fuse blows out.

- (a) Current in each resistor can be calculated using ohm's law

$$i = \frac{V}{R} = \frac{20V}{20\Omega} = 1A$$



The battery supplies $I_0 = 4A$ current. At junction a, the current branches into two parts – 1A and 3A (see figure). At junction b the 3A current branches into two parts – 1A and 2A. Thus $I_1 = 2A$.

Now, I_1 further branches at junction c into two currents of 1A each.

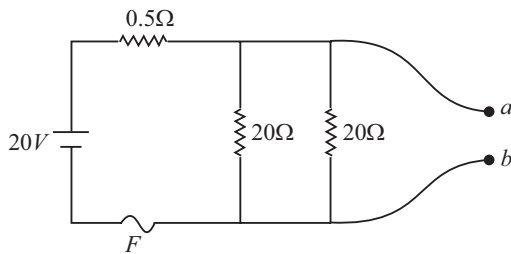
At junction q, the 1A currents join to become one 2A current. At junction r, the 2A current joins another 1A current to become 3A in branch rs. This is I_2 .

- (b) Adding one bulb increases I_0 by 1A. Therefore, we can add two more bulbs so that $I_0 = 6A$. If the seventh bulb is added, current I_0 will become 7A and the fuse will blow out.

Note: By adding more and more bulbs in parallel we increases the battery current. The battery loses its chemical energy at a much faster rate.

Example 20 Short circuit

Two bulbs, each of 20Ω resistance, are connected in parallel to a 20V ideal cell. The lead wires (main connecting wires coming out of the battery terminals) have resistance of 0.5Ω .

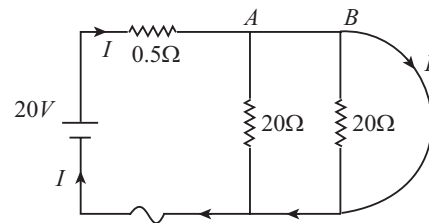


- (a) By mistake, someone connects the loose ends *a* and *b*. The parallel path thus created has negligible resistance. Will the bulb become brighter?
- (b) What do you think is the role of fuse (*F*) with regard to the above mentioned event of *a* and *b* getting connected?

Solution

- (a) When the ends *a* and *b* are joined, it creates a path of zero resistance. Now, 20Ω , 20Ω and 0Ω resistances are in parallel. Current always prefers path of least resistance (it means current is higher in a parallel path of lesser resistance). The current I does not divide at *A* or *B* and gets routed through zero resistance path. The bulbs will stop glowing. We say, they have been *short circuited*. A device is said to be

shorted if its terminals are connected using a zero resistance wire.



There is no current in the bulbs after they have been short circuited

- (b) Due to short circuit, the effective resistance faced by the current is only 0.5Ω .

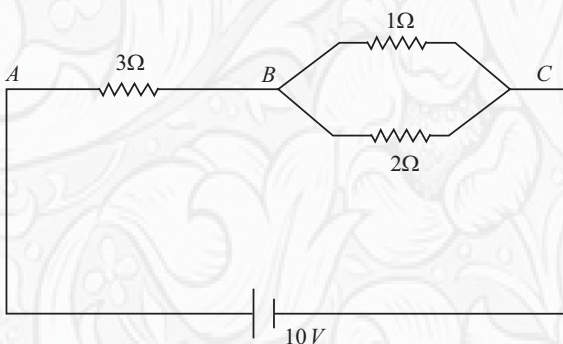
$$I = \frac{20V}{0.5\Omega} = 40A$$

The current becomes large due to short circuiting. The battery gets overloaded (i.e., it has to do a lot of work) and the lead wires get overheated. They may catch fire. To prevent such mishaps, fuse *F* has been placed in the circuit. It blows out and disconnects the battery when current increases beyond a limit due to events like short circuiting.

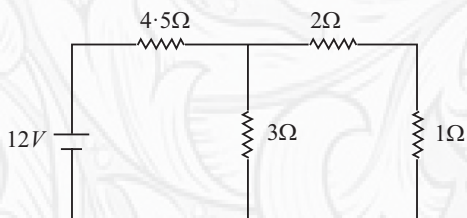


YOUR TURN

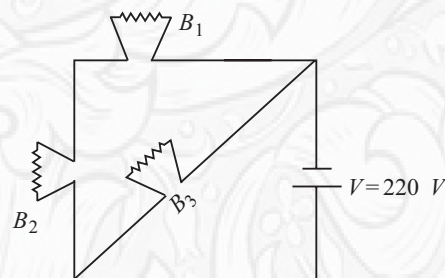
- Q.18** In the circuit shown in the figure, find the current in 1Ω resistor and potential difference across 3Ω resistor.



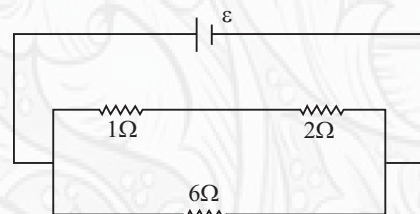
- Q.19** In the circuit shown, find the current in 3Ω resistor.



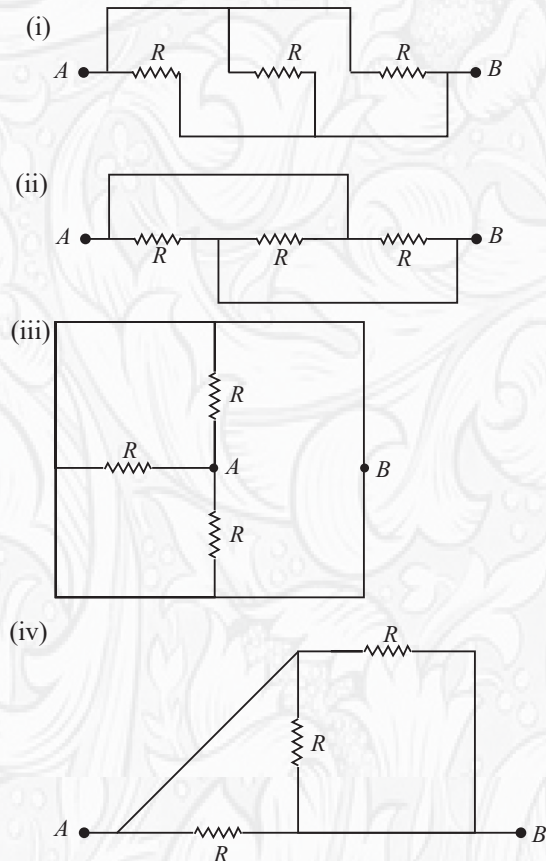
- Q.20** Three bulbs are connected to a 220V supply as shown. B_1 and B_3 are 60W, 220V bulbs and B_2 is a 100W, 220V bulb. Which bulb is brightest? Which bulb is least bright?



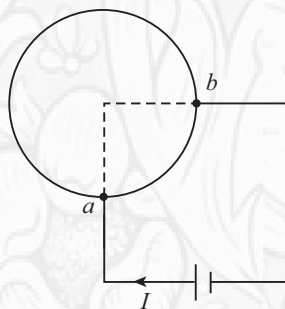
- Q.21** In the circuit shown, the potential difference across 2Ω resistor is 12V. Find the emf (ϵ) of the cell and the current in 6Ω resistor.



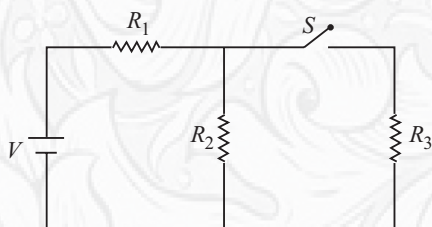
Q.22 Find the equivalent resistance between the points A and B for each of the networks given below.



Q.23 A wire having length 2m has resistance of 20Ω . It is bent into a circle. A $20V$ cell is connected across points a and b as shown. Find the current I through the cell

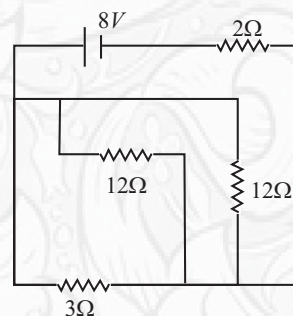


Q.24 In the circuit shown, V_2 is potential difference across R_2 with switch S open. After the switch is closed, the potential difference across R_2 becomes V_2' .

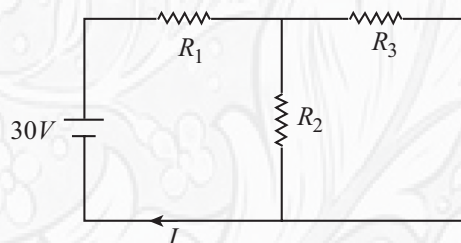
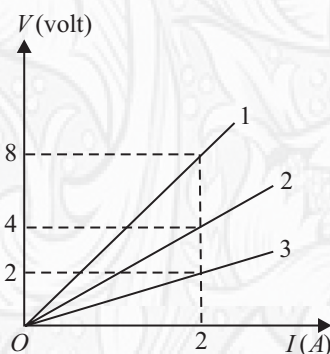


Which one is larger V_2 or V_2' ?

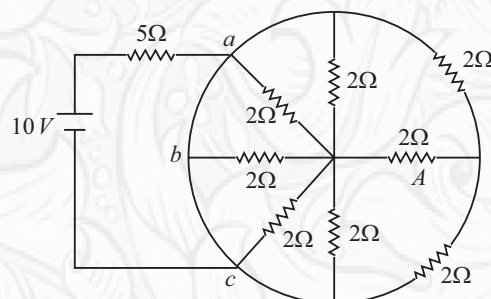
Q.25 In the circuit shown, find the cumulative heat dissipated in all the resistors in 30 seconds.



Q.26 The $V - I$ characteristic of three resistors R_1 , R_2 and R_3 are as shown in figure. These three resistors are connected to a $30V$ cell as shown in the second figure. Find current I .



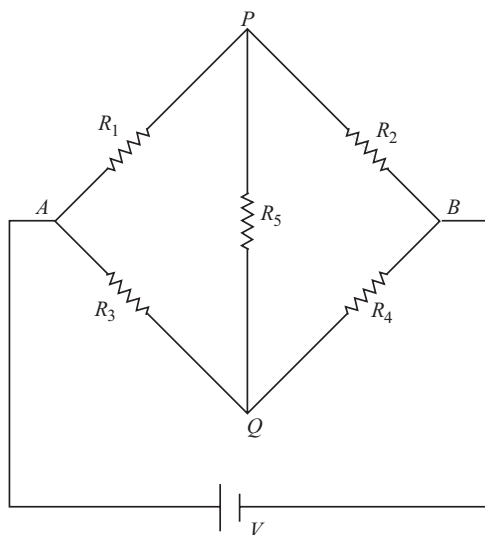
Q.27 In the circuit shown, find current in 5Ω resistor and the 2Ω resistor marked as A .



Q.28 You are having 20 resistors of 2Ω each. What is the maximum and minimum resistance that you can create by combining them?

10. WHEATSTONE BRIDGE OF RESISTORS

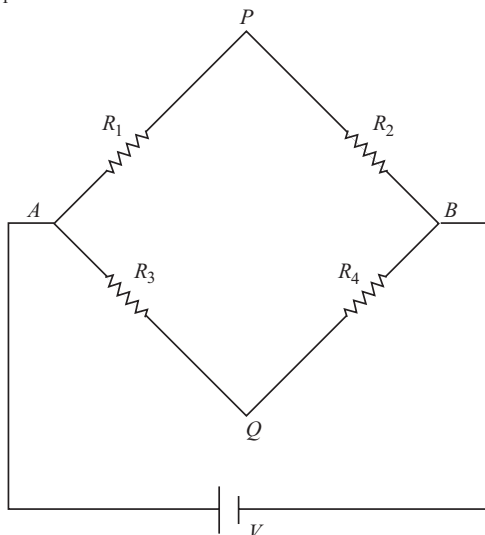
We have already discussed the Wheatstone bridge circuit formed by capacitors. The concept is more or less similar with resistors.



The circuit shown in the figure is known as Wheatstone bridge. In general, it is a complex circuit and we need to use Kirchhoff's loop law to find current in any branch. We will take up this question later on. Right now we will focus on a special situation when there is no current in resistance R_5 . The bridge is said to be balanced in such situation. A Wheatstone bridge is balanced when the following condition is satisfied.

$$\frac{R_1}{R_2} = \frac{R_3}{R_4} \quad (27)$$

To understand this, consider the circuit with R_5 removed. There are two independent parallel paths. Consider the one with resistors R_1 and R_2 in series. The potential difference across R_1 is



$$V_A - V_P = \left(\frac{R_1}{R_1 + R_2} \right) V$$

$$\text{or, } V_A - V_P = \frac{V}{1 + R_2 / R_1} \quad (i)$$

Similarly, consider the series combination of R_3 and R_4 . The potential difference V gets divided across the two resistors in the ratio of the resistances.

$$\therefore V_A - V_Q = \left(\frac{R_3}{R_3 + R_4} \right) V$$

$$\text{or, } V_A - V_Q = \frac{V}{1 + \frac{R_4}{R_3}} \quad (ii)$$

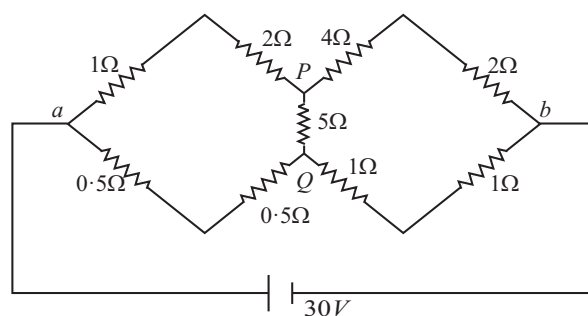
From (i) and (ii) $V_A - V_P = V_A - V_Q$ if

$$\frac{R_1}{R_2} = \frac{R_3}{R_4}$$

It means $V_P = V_Q$, when equation (27) holds. Now, if you go and connect a resistance (R_5) between the points P and Q , nothing will change! There will be no current in R_5 and the circuit is same with or without R_5 .

In the next chapter we will see how the principle of wheatstone bridge can be exploited in electrical measurements.

Example 21 Find the current supplied by the cell in the circuit shown. Also find the current in 4Ω resistance.



Solution

Concepts

- When we see a Wheatstone bridge circuit, we must check whether it is balanced or not.
- In a balanced bridge, the resistance between the two equipotential points can be removed. If we want, we can also join the equipotential points together.

1Ω and 2Ω , between a and P , are in series.

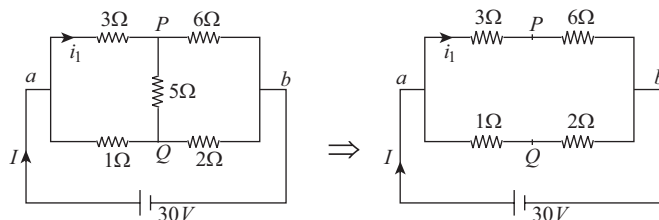
4Ω and 2Ω , between P and b , are in series.

0.5Ω and 0.5Ω , between a and Q , are in series.

1Ω and 1Ω , between Q and b , are in series.

6.20 Electrostatics and Current Electricity

The circuit can be redrawn as shown below. It is a wheatstone bridge and it is balanced since $\frac{3\Omega}{6\Omega} = \frac{1\Omega}{2\Omega}$



There will be no current in 5Ω resistance. We can remove it from the circuit.

The equivalent resistance between points a and b is

$R = (3+6)\Omega$ in parallel with $(1+2)\Omega$

$$= \frac{9 \times 3}{9 + 3} = \frac{9}{4} \Omega$$

$\therefore$

$$I = \frac{30V}{\frac{9}{4}\Omega} = \frac{40}{3} A$$

Current i_1 in the upper branch is

$$i_1 = \frac{30V}{9\Omega} = \frac{10}{3} A$$

This is the current in given 4Ω resistor.

Note: If we want, we can redraw the circuit in a different way by joining the equipotential points P and Q together.

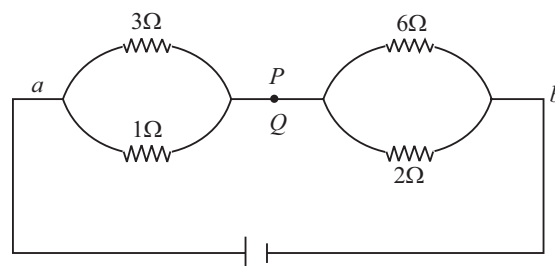
The equivalent resistance is

$$R = \frac{3 \times 1}{3 + 1} + \frac{6 \times 2}{6 + 2}$$

$$= \frac{9}{4} \Omega$$

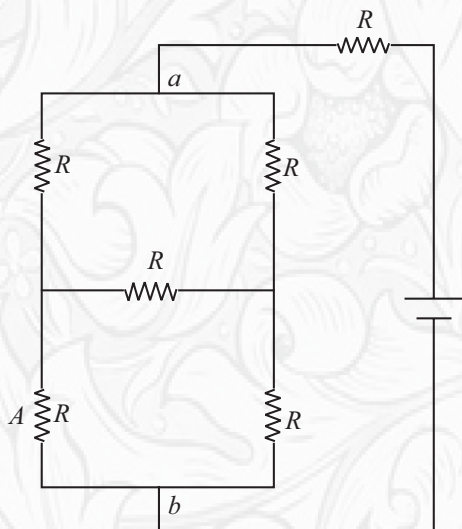
(Same as (i))

(i)

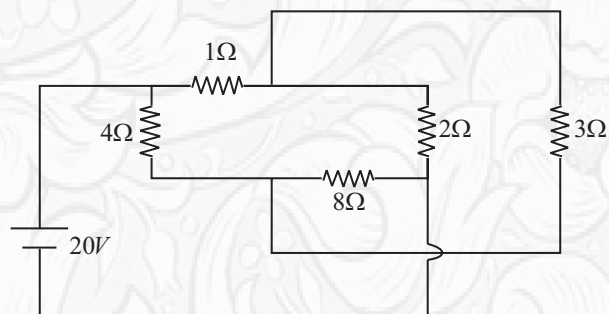


YOUR TURN

Q.29 In the circuit shown in the figure, find the potential difference across the resistor marked as A .



Q.30 In the circuit shown in the figure, find current through the 3Ω resistor and also through the cell.



In Short

- (i) Current is same in all resistances in series connection.
- (ii) The potential difference across resistors in series is in the ratio of respective resistances. If two resistances R_1 and R_2 are in series and a potential difference V is applied across the combination, then the potential drop across the two resistors are:

$$V_1 = \left(\frac{R_1}{R_1 + R_2} \right) V; \quad V_2 = \left(\frac{R_2}{R_1 + R_2} \right) V$$

- (iii) In parallel connection, all resistors have same potential difference across them.
- (iv) The current in resistances connected in parallel is in inverse ratio of the resistances. If i is the current that gets divided into two parallel resistances R_1 and R_2 , then currents in the two resistors are given by

$$i_1 = \left(\frac{R_2}{R_1 + R_2} \right) i \quad \text{and} \quad i_2 = \left(\frac{R_1}{R_1 + R_2} \right) i$$

- (v) Equivalent of two resistances in parallel is always less than both the individual resistances.
- (vi) The equivalent of n resistors in parallel, each having resistance R , has resistance given by $R_{eq} = \frac{R}{n}$
- (vii) If the positive and negative terminal of a device is connected using a wire of low resistance, it is said to be short circuited.
- (viii) A short circuit may cause excessive current in a circuit. To prevent damage, we often use fuse.
- (ix) In a balanced wheatstone bridge, we can remove the central resistance.
- (x) Total electric current approaching a junction in an electric circuit is equal to total electric current going out of the junction. This is Kirchhoff's first law.
- (xi) According to the Kirchhoff's second law (also called the Kirchhoff's voltage law or loop law), the sum of potential differences across various circuit elements in a closed electrical loop is always zero.

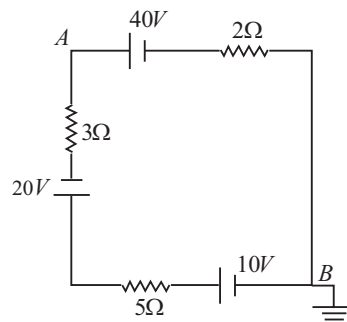
11. COMPLEX CIRCUITS USING RESISTANCES

We may encounter complex electric circuits which cannot be reduced using series or parallel groupings. Solution of such circuits fundamentally involves use of Kirchhoff's laws. There may be circuits which can be simplified using symmetry arguments. The following examples will clarify the procedure in various cases. It is advised that you also go through the circuit solution techniques given in the chapter of capacitor. Most of the concepts and methods that we will

use here are similar to those learnt while solving circuits involving capacitors.

Example 22 Single loop circuit

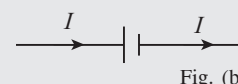
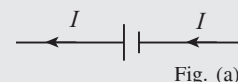
Find the potential of point A shown in the circuit.



Solution

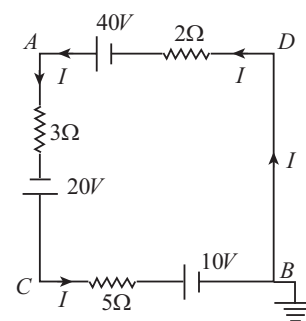
Concepts

- (i) Point B is grounded. It means $V_B = 0$.
- (ii) For knowing $V_A - V_B$, we need to know the current through the resistors.
- (iii) For finding the current we will use Kirchhoff's loop law.
- (iv) The current flowing out of one terminal of a cell must return back to it through the other terminal. It is absolutely fine if there is a current through a cell as shown in Fig (b).



There must be same current (I) throughout the loop. Whatever charge moves out of one terminal of the 40V cell, must return to it through the other terminal. This is same for all the cells. This is possible if there is same current throughout the loop.

We will start from point A in the loop and move anticlockwise along the loop to return back to A. We will keep track of rise and fall in potential all the way.



$$V_A - 3I + 20 - 5I - 10 - 2I + 40 = V_A \quad (i)$$

$$\text{or, } -3I + 20 - 5I - 10 - 2I + 40 = 0 \quad (ii)$$

This expression is Kirchhoff's loop law – sum of potential differences across every element in a loop is zero. One can always write (ii) directly without writing equation (i).

$$\therefore I = \frac{50}{10} = 5A$$

To find $V_A - V_B$, we will move from A to B. Let's move along ACB (you can also move along ADB).

6.22 Electrostatics and Current Electricity

$$V_A - 3 \times 5 + 20 - 5 \times 5 - 10 = V_B$$

or, $V_A - V_B = 30V$

$$\therefore V_B = 0$$

$$\therefore V_A = 30V.$$

Note: (i) For a single loop circuit, a quick way of writing

current is to use $I = \frac{E_{\text{net}}}{R_{\text{net}}}$.

Here, E_{net} = Sum of all emfs (with sign)

R_{net} = Sum of all resistances in the loop as if they are in series.

While writing E_{net} , move along the loop keeping track of emfs. In this example

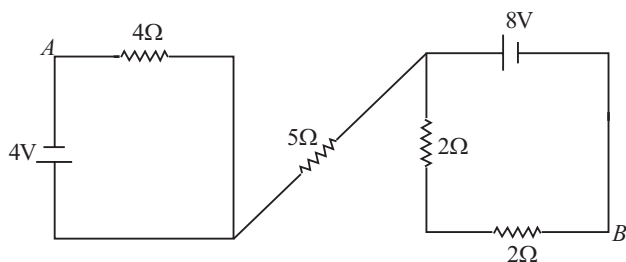
$$E_{\text{net}} = 40 + 20 - 10 = 50V$$

$$R_{\text{net}} = 2 + 3 + 5 = 10\Omega$$

(2) Had we considered the current I to be in clockwise sense, the value of I would turn out to be $I = -5A$. The negative sign will tell you that assumed direction for the current is not correct.

Example 23 Connected but independent loops

Find the potential difference between the points A and B in the shown figure.



Solution

Concepts

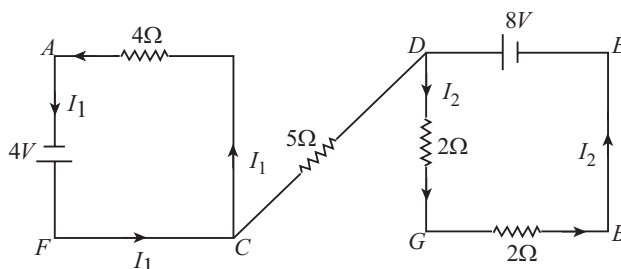
A current exists in a loop. A current cannot exist in a path if there is no return path for it. There will be no current in 5Ω resistor.

For a moment assume that the current I_1 divides into two parts at junction C . A part i goes through the 5Ω resistor. Then only $I_1 - i$ current can return to the $4V$ cell through the 4Ω resistor. This never happens. Current going out of a terminal of a cell must return to it through the other terminal. It means that current I_1 cannot divide into parts at C .

The two loops are independent. Current in the left loop is

$$I_1 = \frac{4V}{4\Omega} = 1A$$

And current in the right loop is $I_2 = \frac{8V}{4\Omega} = 2A$



There is no current in 5Ω resistor.

To write $V_A - V_B$ we can take a very simple route $AFCDEB$.

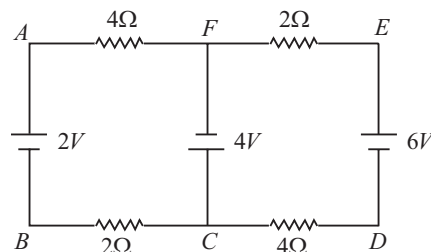
$$V_A + 4 - 8 = V_B$$

or, $V_A - V_B = 4V.$

Any other route (like $AFCDGB$) will give the same result.

Example 24 Using Kirchhoff's laws to solve a circuit problem

In the circuit shown in the figure, find the potential difference between points B and E .



Solution

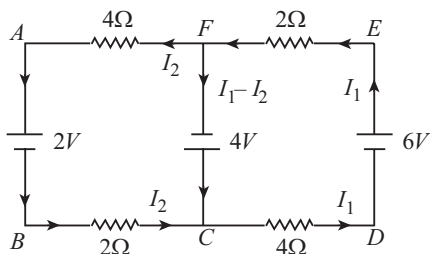
Concepts

- If we move from B to E , we encounter resistors in the path. To be able to write the potential difference we must know the current in the resistors.
- We will assume currents (I_1 , I_2 etc) in various branches of the circuit. To keep the unknown currents minimum in number, we use Kirchhoff's junction law.
- We write equations using Kirchhoff's loop law. In the present circuit we have 3 possible loops (2 smaller loops and 1 bigger loop). However, we should write equations for two loops only; the third equation is not independent and can be established using the two equations.
- Solution of the equations gives the values of unknown currents.

First let's do the current distribution in the circuit. We assume a current I_1 flowing out of the positive terminal of $6V$ cell.

This current flows through 2Ω resistor and branches at junction F . Let current in branch FA be I_2 . Then the current

in path FC must be $I_1 - I_2$. This follows from the junction law. At junction C , use of junction law tells us that current in branch CD will be $I_2 + (I_1 - I_2) = I_1$. This is perfect, as current I_1 must return to the 6V cell.



Now, we have two unknowns – I_1 and I_2 and we must set up two equations for finding them. We will write two loop equations.

Loop $ABCFA$: We start from A and move anticlockwise.

$$-2 - 2I_2 - 4 - 4I_2 = 0$$

$$\text{or, } I_2 = -1\text{A}$$

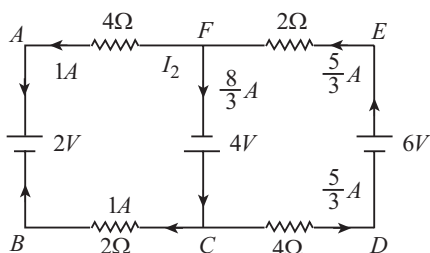
The negative sign indicates that the direction of current I_2 must be opposite to that shown in the figure above.

Loop $CDEFC$: Starting from C and moving anticlockwise we can write:

$$-4I_1 + 6 - 2I_1 + 4 = 0$$

$$\text{or, } I_1 = \frac{5}{3}\text{A}$$

Now, we move from B to E via path $BCDE$.

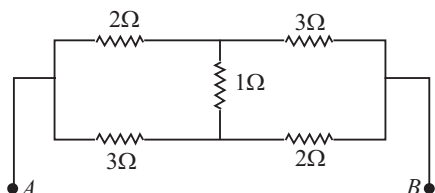


$$V_B + 2 \times 1 - 4 \times \frac{5}{3} + 6 = V_E$$

$$\text{or, } V_B - V_E = -\frac{4}{3} \text{ volt.}$$

Example 25 Symmetrical circuit when looked through either terminals

Find the equivalent resistance of the network between points A and B .



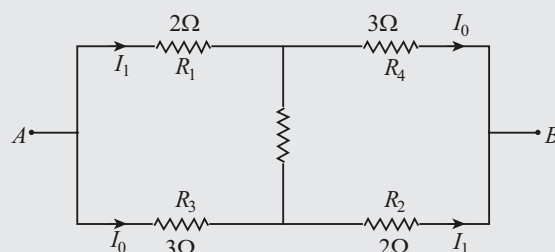
Solution

Concepts

- (i) If we apply a potential difference V across the terminals A and B and the current drawn by the network is I , then the equivalent resistance is

$$R_{eq} = \frac{V}{I}$$

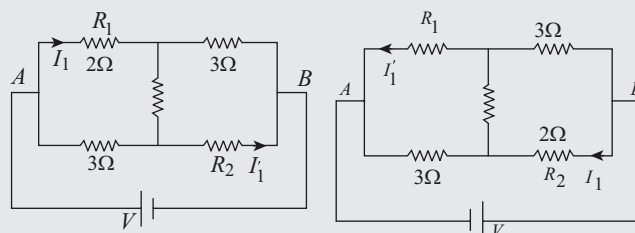
- (ii) The circuit looks exactly same when looked into from the end A or when looked through the end B . Consider that the terminal A is connected to the positive plate of a battery and B is connected to its negative plate. Let current in 2Ω resistor (R_1) be I_1 . Then current in the 2Ω resistor (R_2) will also be I_1 as shown. Similarly, current in two 3Ω resistors must also be same.



The current in R_1 and R_2 must be same. Current will also be same in R_3 and R_4 .

I will try to argue this in a slightly different way. Assume that current in R_1 is I_1 and in R_2 is I'_1 when a battery is connected with its positive plate connected to A . What will happen if we connect the same battery with polarity reversed, i.e., the positive plate connected to B ? If the only battery in the circuit has been reversed, the current in every branch will reverse without change in magnitude. It means that current in R_1 will be I_1 and that in R_2 will be I'_1 with directions reversed. But it is easy to realize that the circuit is essentially same from both sides and the current in R_2 must be I_1 .

$$\therefore I'_1 = I_1$$

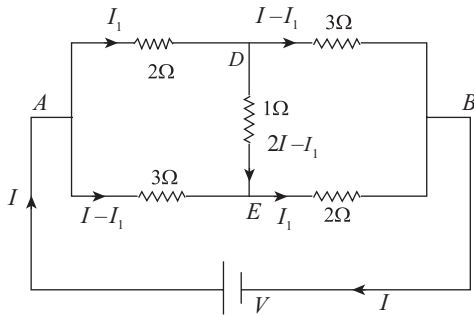


Reversal of polarity of the battery should not change the magnitude of the current in any branch. Only the direction of current shall reverse. Hence $I'_1 = I_1$.

Let the current supplied by the cell be I . The current branches into 2Ω and 3Ω resistors. Let the respective currents be I_1 and $I - I_1$. From symmetry the current in other 2Ω resistor is I_1 and that in the other 3Ω resistor is $I - I_1$. If current in branch DE is x , then using junction law at D gives

$$x + I - I_1 = I_1$$

$$\text{or, } x = 2I_1 - I.$$



This completes the current distribution. To express I_1 in terms of I , we will write loop equation in the loop $ADEA$.

$$-2I_1 - (2I_1 - I)(1) + (I - I_1)(3) = 0$$

$$\text{or, } I_1 = \frac{4}{7}I$$

Now, we will express V in terms of I . For this we will walk from A to B along ADB .

$$V_A - 2I_1 - 3(I - I_1) = V_B$$

$$\text{or, } V_A - V_B = 3I - I_1$$

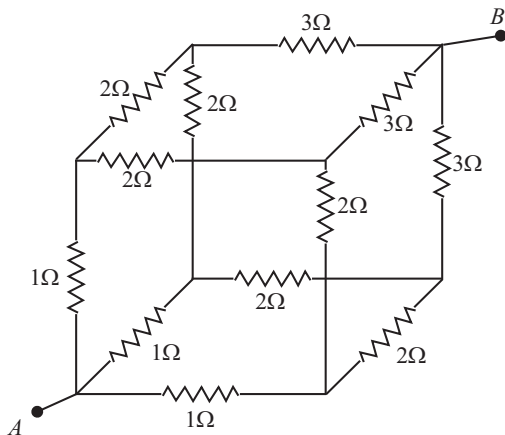
$$\text{or, } V = 3I - \frac{4I}{7}$$

$$\text{or, } V = \frac{17I}{7}$$

$$\text{or, } R_{eq} = \frac{V}{I} = \frac{17}{7}\Omega$$

Example 26 A cube of resistors

Find the equivalent resistance between the points A and B for the network shown in the figure.



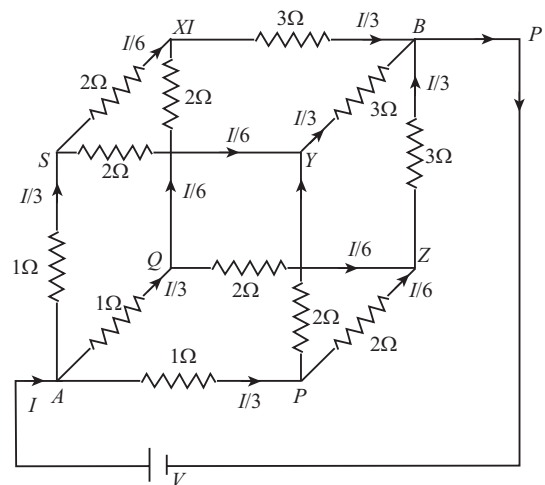
Solution

Concepts

- Assume that current supplied by a V volt battery connected across A and B is I . Then the equivalent resistance is $R_{eq} = \frac{V}{I}$.
- We need to express $V (= V_A - V_B)$ in terms of I to get the equivalent resistance.
- All paths from A to B are identical. Using this we can divide current I in various branches.
- We can also identify equipotential points in the network and join them together to convert the entire network into a series-parallel grouping.

Current I enters the network at A . It gets divided in three identical paths AP , AQ and AS .

In each path the current is $I/3$.



At junction P , current $\frac{I}{3}$ divides along two identical paths

PY and PZ . Obviously, current along both these paths will be $\frac{I}{6}$. Similarly, current divides at junctions Q and S . The current in the three identical paths connecting at B must be $I/3$.

Now, we move along the path $APZB$ to write $V_A - V_B$

$$V_A - \frac{I}{3}(1) - \frac{I}{6}(2) - \frac{I}{3}(3) = V_B$$

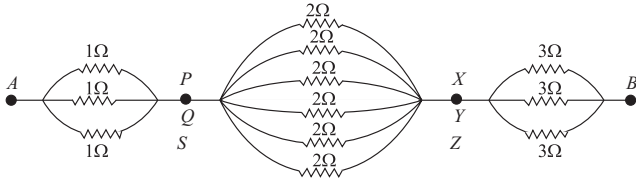
$$\text{or, } V_A - V_B = \frac{5I}{3}$$

$$\text{or, } V = \frac{5I}{3}$$

$$\text{or, } R_{eq} = \frac{V}{I} = \frac{5}{3}\Omega$$

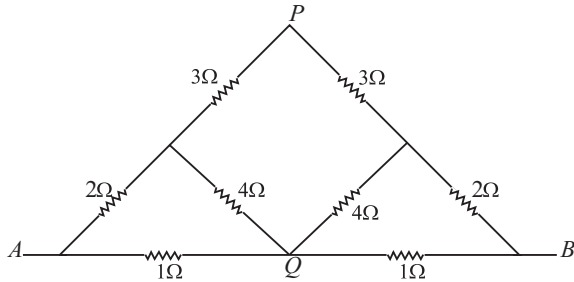
Alternate Method

Points P , Q and S are symmetrically located with respect to A as well as B . They must have same potential. We can join them together. Similarly, points X , Y and Z are equipotential. We can also join them together. The circuit can be redrawn as shown below:



$$\text{The equivalent resistance is } R_{eq} = \frac{1}{3} + \frac{2}{6} + \frac{3}{3} = \frac{5}{3} \Omega$$

Example 27 In the network shown, find the equivalent resistance between the points A and B .

**Solution****Concepts**

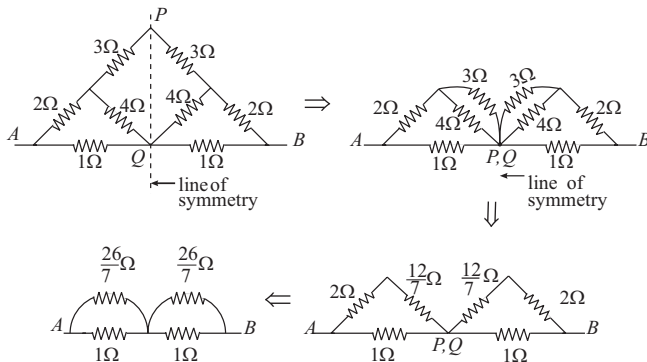
- (i) A and B are terminals. Line PQ is perpendicular bisector of AB and the circuit on two sides of line PQ are identical. The perpendicular bisector of AB is line of symmetry.

In such a case all points on the perpendicular bisector (PQ) are at same potential.

Here, P and Q are at same potential they can be joined together.

- (ii) After connecting P to Q , the resistors form a series parallel grouping.

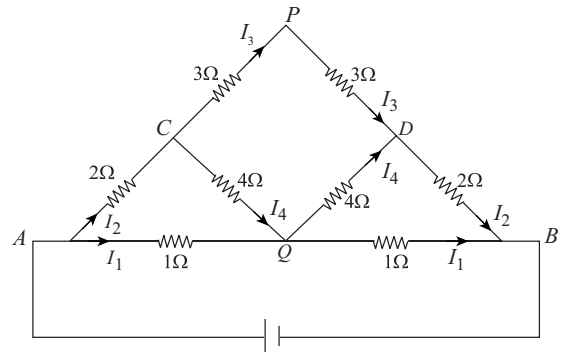
Points P and Q are equipotential. After joining them the circuit can be reduced as shown below.



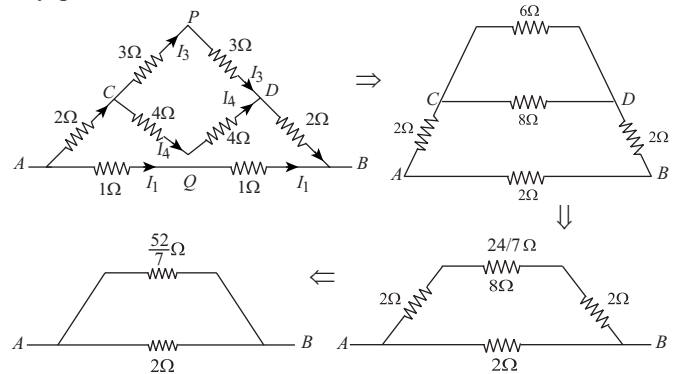
$$\text{The equivalent resistance is } R_{eq} = \frac{52}{33} \Omega$$

Alternate Method

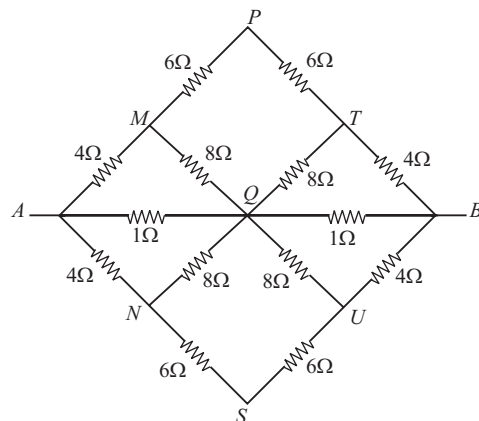
The circuit is symmetrical from side A as well as B . [Refer to Example 26.] We can say that current in both 1Ω resistors must be same. Currents in the two 2Ω resistors must also be same. Similar is the case for 3Ω and 4Ω resistors.



Look at the junction Q . Current I_1 , in branch AQ , passes straight into QB as if there is no junction. Similarly, current I_4 in CQ passes to QD . The junction at Q , if opened (as shown below) will make no difference to the current in any path.



Example 28 Find equivalent resistance between the points A and B for the network shown in the figure.

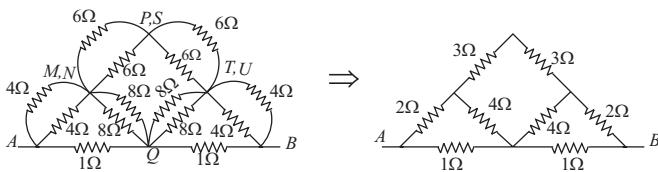


Solution**Concepts**

Terminals are A and B and line AB is a line of symmetry (In the last question, AB was not the line of symmetry). All points lying on a line perpendicular to AB will be at the same potential.

Points M and N are equipotential; P , Q and S are equipotential. Points T and U are also at same potential. Equipotential points can be joined together.

After connecting the equipotential points, the circuit can be redrawn as shown below. In effect, the circuit has been folded about line AB .



The final circuit is same as that given in the last example.

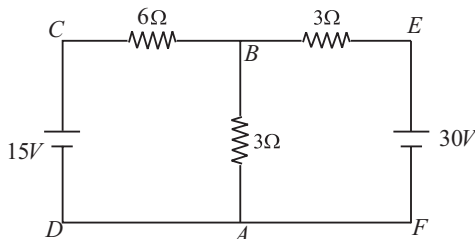
$$\therefore R_{eq} = \frac{52}{33} \Omega.$$

11.1 Another Way to Use Kirchhoff's Laws

The strategy we have adopted till now has been to assume current in different branches of a circuit (keeping in mind the first law) and then frame equations using the loop law. These equations were then solved to find the unknown currents.

However, we can adopt a reverse strategy. We will assume the potentials at different junctions in a circuit as V_1 , V_2 , V_3 etc. Then using the Kirchhoff's first law we can frame equations. These equations can be used to find the values of V_1 , V_2 , V_3 etc. The following example will help you understand the method.

Example 29 In the circuit shown, find the current in the branch AB .

**Solution****Concepts**

- (i) We will assume potential of a point (say, A) to be zero. Then we will try to write potential of other points. If we do not know the potential of a point, we will assign it an unknown value.

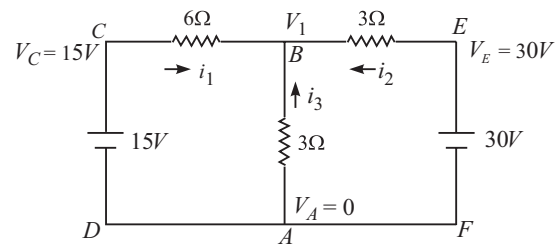
- (ii) Using the potential difference between two points we can write currents in branches like CB , EB and AB .

The total current in all three branches- directed towards B - must be zero.

$$\text{Let } V_A = 0 (= V_D = V_F)$$

There is a 15V cell between the points C and A . Therefore, it is easy to see that $V_C = 15V$.

$$\text{Similarly, } V_E = 30V.$$



We do not know the potential of point B . Let this potential be V_1 .

$$\text{Current in branch } CB \text{ will be } i_1 = \frac{V_C - V_1}{6}$$

$$\text{or, } i_1 = \frac{15 - V_1}{6}$$

$$\text{Similarly, current from } E \text{ to } B \text{ is } i_2 = \frac{30 - V_1}{3}$$

$$\text{And the current in the branch } AB \text{ is } i_3 = \frac{0 - V_1}{3}$$

[Note: We have assumed all currents to be towards B .]

From Kirchhoff's junction law

$$i_1 + i_2 + i_3 = 0$$

$$\frac{15 - V_1}{6} + \frac{30 - V_1}{3} + \frac{0 - V_1}{3} = 0$$

$$\text{or, } V_1 = 15V.$$

Potential of the point B is 15V.

$$\therefore i_1 = \frac{15 - V_1}{6} = 0$$

$$i_2 = \frac{30 - V_1}{3} = \frac{15}{3} = 5A$$

$$\text{And, } i_3 = \frac{0 - V_1}{3} = -5A$$

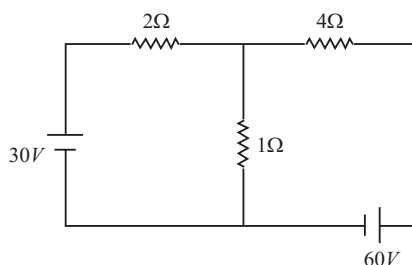
$\therefore$ Current i_3 in the branch is directed from B to A and its magnitude is 5A.

11.2 Superposition Theorem for Cells

Consider an electric circuit with multiple emf sources. The current in any branch can be found by determining currents in that particular branch produced by each emf source acting alone, with all other emfs shorted. The total current in the branch is the algebraic sum of currents due to individual emf sources.

The following example will illustrate the procedure.

Example 30 Find current in the 1Ω resistance shown in the circuit.



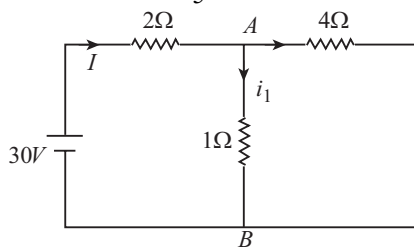
Solution

Concepts

- (i) We can solve the problem using Kirchhoff's law. Here we will use superposition theorem to illustrate its use.
- (ii) First we will short circuit the 60V emf (i.e., replace it with a copper wire). With only 30V emf source we will find current (i_1) in the 1Ω resistance. Then we will short the 30V source and assume only 60V source in the circuit. We again find the current (i_2) in the 1Ω resistance.
- (iii) If i_1 and i_2 both are in same direction, the current in 1Ω resistance when both cells are there is given by $i_1 + i_2$.
If i_1 and i_2 are opposite in direction, then the current in 1Ω resistor is difference of i_1 and i_2 .

The circuit, with 60V source short circuited, is as shown. 4Ω and 1Ω are in parallel.

Their equivalent is $\frac{4}{5}\Omega$.



The $\frac{4}{5}\Omega$ resistance in series with 2Ω resistance gives

$$2 + \frac{4}{5} = \frac{14}{5}\Omega$$

$$\text{The current } I \text{ is } I = \frac{30}{14/5} = \frac{75}{7}\text{A}$$

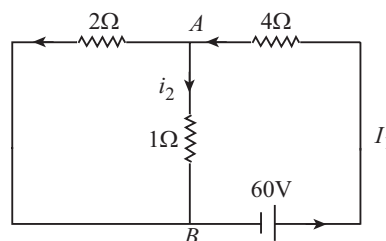
This current branches at A in two parallel paths. The current in 1Ω resistance is

$$i_1 = \left(\frac{4}{1+4} \right) I = \frac{4}{5} \times \frac{75}{7} = \frac{60}{7}\text{A}.$$

In the second step, we short circuit the 30V emf and consider only 60V emf in the circuit.

$$R_{eq} = \frac{2 \times 1}{2+1} + 4 = \frac{14}{3}\Omega$$

$$\therefore I_1 = \frac{60}{R_{eq}} = \frac{90}{7}\text{A}$$



$$i_2 = \left(\frac{2}{2+1} \right) I_1 = \frac{2}{3} \times \frac{90}{7} = \frac{60}{7}\text{A}$$

$\therefore$ When both sources are there, the current in branch AB will be

$$i_1 + i_2 = \frac{120}{7}\text{A}$$

In Short

- (i) In a single loop circuit, current is given by

$$I = \frac{E_{net}}{R_{net}}$$

Here, E_{net} = total emf in the loop.

R_{net} = sum of all resistances in the loop.

- (ii) Current going out of a terminal of a cell must return back into it through its other terminal.
- (iii) If a circuit looks identical when looked into from either of its terminals, then the current going into the circuit gets divided into branches in a way similar to the way it comes out through the branches at the other terminal.
- (iv) Equipotential points in a circuit can be connected together. This does not alter the circuit.
- (v) If A and B are terminals of a network of resistors and the line AB divides the circuit into two symmetrical halves, then all the points on a perpendicular to AB will be at same potential.

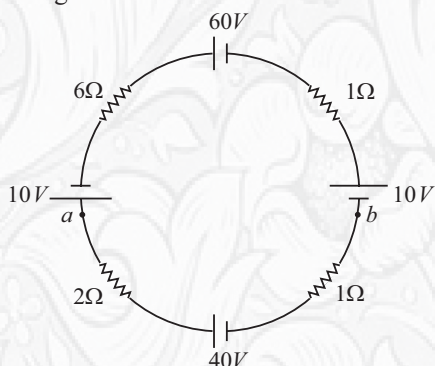
- (vi) If A and B are terminals of a resistor network and the perpendicular bisector of line AB divides the circuit into two identical halves, then all the points on the perpendicular bisector are at same potential.
- (vii) When a circuit has multiple cells, we can use superposition theorem to find current in a branch. We draw the circuit with one cell only (others shorted)

and find the current in the branch. We repeat the process with all other cells connected individually in the circuit. If $i_1, i_2, i_3, \dots$ are currents in the branch with each individual cell in the circuit, the actual current when all cells are there is $i = i_1 + i_2 + i_3 + \dots$. For using this rule you must ensure that no two emfs are directly connected in parallel.

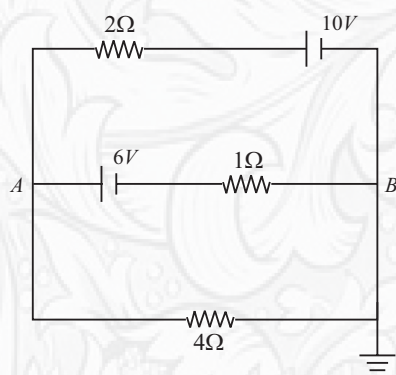


YOUR TURN

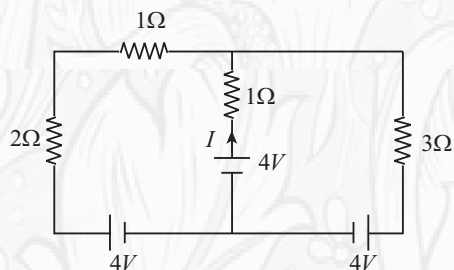
Q.31 Find the potential difference between the points a and b in the figure.



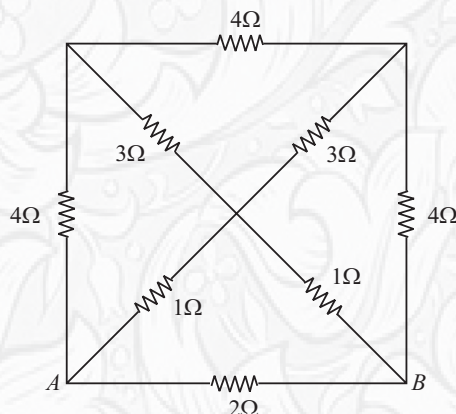
Q.32 Find the potential of point A shown in the figure.



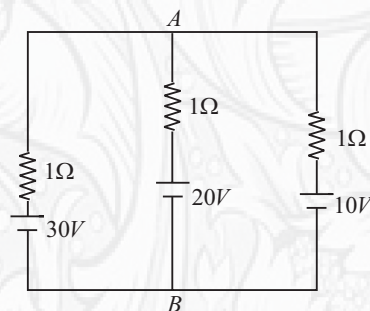
Q.33 Find current I in the circuit shown.



Q.34 Find the equivalent resistance between the points A and B for the network shown in the figure.



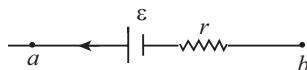
Q.35 Use superposition theorem to find current in the branch AB of the circuit.



12. REAL CELLS

Till now we treated all our cells to be a source of emf, without any resistance. This is something very ideal. In practice, all our cells have some resistance. When a current flows through them the material of the cell offers a resistance. There is a potential drop in this resistance. The potential difference across the terminals of a cell is not exactly equal to its emf when there is a current in the cell. Besides, heat is dissipated in this resistance and the cell itself gets heated.

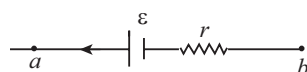
In a circuit, we represent a real cell by a source of emf in series with a resistance.



The potential difference between the terminals a and b of the cell is same as its emf (ϵ) when there is no current in the cell. Now, let's investigate how the potential difference (V_{ab}) between the terminals changes in various cases.

(1) A cell getting discharged

When a cell is giving energy to a circuit, current (I) through it is as shown. The potential difference between the terminals can be written as



$$V_b - Ir + \epsilon = V_a$$

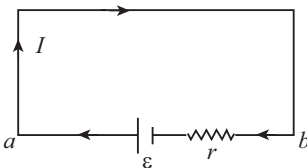
$$\text{or, } V_a - V_b = \epsilon - Ir \quad (28)$$

The potential difference ($V_{ab} = V_a - V_b$) is certainly less than emf of the cell. We may even record a zero potential difference between the terminals if the current is sufficiently strong. Can we have a situation where $V_a < V_b$? I leave this for you to wonder.

(2) A short circuited cell

The terminals of a cell have been connected using a conducting wire.

In this case the potential difference across the terminals is zero.



$$V_b - Ir + \epsilon = V_a$$

$$\text{But } Ir = \epsilon$$

$$\therefore V_a - V_b = 0$$

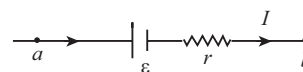
If you short circuit a dry cell using a copper wire, current will be large because of small internal resistance of the cell.

$$I = \frac{\epsilon}{r} \rightarrow \text{large}$$

The cell will lose chemical energy at a rapid pace and soon get discharged. The entire chemical energy will get converted into heat.

(3) A cell getting charged

A rechargeable cell can gain energy if current is forced through it in reverse direction.



This is what happens when you charge the battery of your mobile phone. This can happen if a potential difference (V_{ab}) is applied to the cell which is larger than the emf (ϵ).

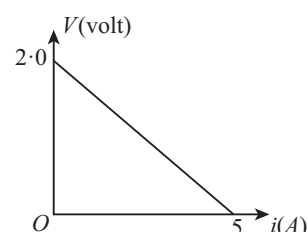
$$V_a - \epsilon - Ir = V_b$$

$$\text{or, } V_a - V_b = \epsilon + Ir \quad (29)$$

Note: The internal resistance of a chemical cell usually increases with rise in temperature.

Example 31 V - i graph for a discharging real cell

The potential difference (V) between the positive and negative terminals of a cell changes with the current (i) through it as shown in the graph. Find the internal resistance of the cell.



Solution

Concepts

- The potential difference between the terminals is same as emf of the cell when there is no current through it.
- With rise in current through the cell, the potential difference across the terminals decreases linearly according to the equation (28). It becomes zero when $\epsilon = ir$.

When $i = 0$, the graph shows that the potential difference between the terminals is $V = 2.0$ volt. This must be the emf of the cell.

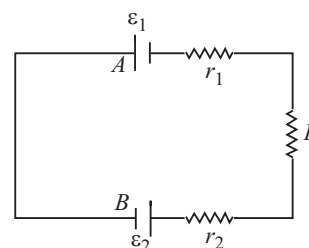
$$\therefore \epsilon = 2.0 \text{ volt.}$$

When $i = 5$ A, the potential difference becomes zero.

$$\therefore \epsilon = ir$$

$$\text{or, } r = \frac{2}{5} = 0.4 \, \Omega.$$

Example 32 In the circuit shown in the figure, cell A has emf $\epsilon_1 = 2.0$ V and internal resistance $r_1 = 1.0 \, \Omega$. The other cell B has emf $\epsilon_2 = 8.0$ V and internal resistance $r_2 = 1.0 \, \Omega$. Find the value of resistance R so that the



6.30 Electrostatics and Current Electricity

potential difference across cell A is zero. At what rate is the cell B losing energy in this case?

Solution

Concepts

- (i) This is a single loop circuit, hence $i = \frac{\mathcal{E}_{\text{net}}}{R_{\text{net}}}$
- (ii) The potential difference across the cell A is zero when $ir_1 = \mathcal{E}_1$

$$\text{Current in the loop } i = \frac{\mathcal{E}_1 + \mathcal{E}_2}{r_1 + r_2 + R} = \frac{2 + 8}{1 + 1 + R}$$

$$\text{or, } i = \frac{10}{2 + R} \quad (\text{i})$$

Terminals of the cell A are a and b . Potential difference between them is

$$V_{ab} = \mathcal{E}_1 - ir_1$$

This is zero when $ir_1 = \mathcal{E}_1$

$$\text{or, } i = \frac{2}{1} = 2.0 \text{ A}$$

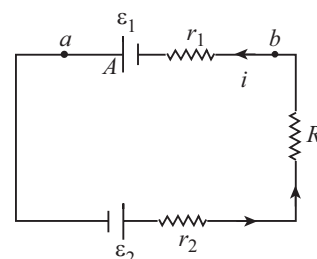
$$\therefore \text{From (i) } \frac{10}{2 + R} = 2$$

$$\text{or, } R = 3 \Omega$$

Rate of energy loss by cell B is

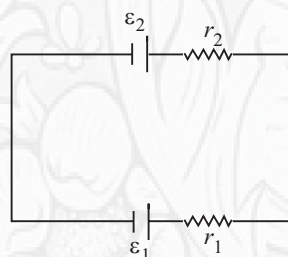
$$= \mathcal{E}_2 i = 8 \times 2 = 16 \text{ W}$$

This is the rate at which the cell is losing its stored energy. A part of it is getting wasted in heating itself as the internal resistance dissipates heat.

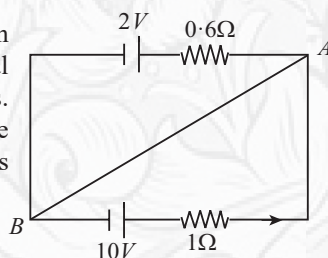


YOUR TURN

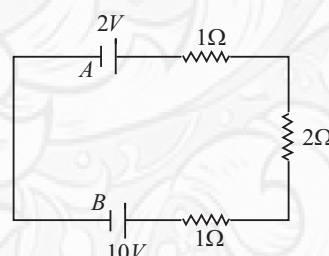
Q.36 The two cells of emfs $\mathcal{E}_1 = 10\text{V}$ and $\mathcal{E}_2 = 2\text{V}$ have internal resistances $r_1 = 1.0 \Omega$ and $r_2 = 0.6 \Omega$ respectively. Find the potential difference across the terminals of both the cells.



Q.37 In the figure shown 0.6Ω and 1Ω are internal resistances of the two cells. Find the potential difference across their terminals. What is current in branch AB ?



Q.38 In the circuit shown, which cell is getting recharged? What is potential difference across its terminals? The 1Ω resistances are the internal resistances of the cells.



12.1 Capacity of a Battery (The Ampere Hour)

How long a battery will last depends on its ampere-hour rating. Your mobile phone battery has its rating expressed in mAh (milli ampere hour). It expresses the quantity of charge that a battery can supply. A battery having rating of 3000 mAh, can (ideally) give 3000mA current for one hour or 100 mA current for 30 h.

Example 33 A mobile phone battery

A mobile phone battery has a rating of 3000 mAh.

It lasts for 90 minute when someone watches a video on it.

- (a) How much average current the battery supplied during the 90 minute period.
- (b) The emf of the battery is 3.7 V. How much energy did it supply during the 90 minute period?

Assume that the emf remains constant.

Solution

Concepts

- (i) 3000 mAh is capacity of the battery. It is equal to current times time.
- (ii) Power supplied = VI

(a) $I \Delta t = 3000 \text{ mAh}$

or, $I (1.5 \text{ h}) = 3000 \text{ mAh}$

or, $I = 2000 \text{ mA} = 2 \text{ A}$

(b) Power supplied, $P = VI$

$$\begin{aligned} \text{Energy supplied, } E &= \rho \Delta t = V (I \Delta t) \\ &= (3.7 \text{ V})(3000 \text{ mAh}) \\ &= (3.7 \text{ V})(3 \text{ A} \times 3600 \text{ s}) \\ &= 3.996 \times 10^4 \simeq 4 \times 10^4 \text{ J} \end{aligned}$$

12.2 Maximum power transfer

Suppose you have a battery of emf ε and internal resistance r . You are trying to design a heater which will dissipate maximum power when connected to this battery. What should be resistance of your heater? We will try to answer this question here.

Assume an external resistance R connected to the battery. The current in the circuit will be

$$I = \frac{\varepsilon}{R+r}$$

Power dissipated in R is

$$P_R = I^2 R = \left(\frac{\varepsilon}{R+r} \right)^2 R \quad (\text{a})$$

For different values of R , we get different values for P_R . There is a value of R for which P_R is maximum. This value of R can be calculated using calculus as:

$$\frac{dP_R}{dR} = \varepsilon^2 \left[\frac{(R+r)^2 - R \cdot 2(R+r)}{(R+r)^4} \right] = 0$$

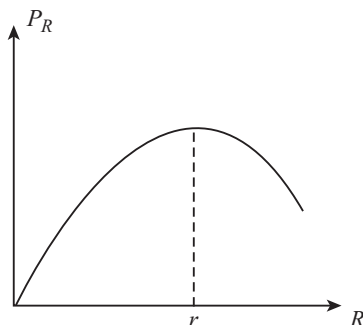
$$\text{or, } R = r \quad (30)$$

It means maximum power is dissipated in R when its value is equal to the internal resistance of the cell. The maximum power dissipated in R can be obtained by substituting $R = r$ in (a)

$$P_{\max} = \frac{\varepsilon^2}{4r} \quad (31)$$

This is the maximum 'useful' power that we can draw from a cell of emf ε and internal resistance r .

The variation of P_R with R has been roughly depicted in the graph shown. In general, there are two values of R for which you get same amount of power dissipated in it.



Variation of power dissipated in R versus R .

Example 34 You have a 12V battery having 1Ω internal resistance.

- What is the maximum power that an electric heater can dissipate when connected to this battery?
- What could be the resistance values of a heater that dissipates 18 W power when connected to this battery? Which resistance value will you prefer and why?

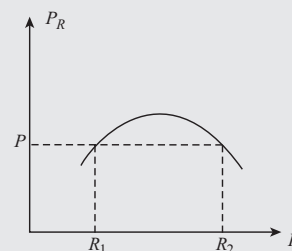
Solution

Concepts

- Power dissipated in the external resistance will be maximum when $r = R$ and the maximum power is

$$\text{given by } P_{\max} = \frac{\varepsilon^2}{4r}.$$

- Except for $R = r$, we have two values of R which dissipate same power when connected to a cell. See the given graph. R_1 and R_2 are two values of external resistances which dissipate same power when connected to the cell.



$$(a) P_{\max} = \frac{\varepsilon^2}{4r} = \frac{12^2}{4 \times 1} = 36 \text{ W}$$

$$(b) P_R = I^2 R = \left(\frac{\varepsilon}{R+r} \right)^2 R$$

$$\text{or, } 18 = \frac{12^2 R}{(R+1)^2}$$

$$\text{or, } (R+1)^2 = 8R$$

$$\Rightarrow R^2 - 6R + 1 = 0$$

$$\text{or, } R = \frac{6 \pm \sqrt{36-4}}{2} = 3 \pm \sqrt{8}$$

$$\text{or, } R = 5.83 \Omega, 0.17 \Omega$$

So one can use a heater having resistance 5.83Ω or, 0.17Ω . However, we should prefer the 5.83Ω heater because it will ensure smaller current in the circuit. Smaller the current, lesser is the rate at which the battery is losing energy. In other words, for higher value of R there is lesser heat loss in the internal resistance of the battery. The battery is less strained and will last longer.



YOUR TURN

Q.39 Your mobile phone battery is rated in mAh but the UPS attached to your computer is rated in VA (volt-ampere). Why? Think and do your own research.

Q.40 A battery is rated 300 Ah. It is used to supply 2.5 A current for 20 h. After this it is used to supply 5 A current for 20h. Now, it is put to use where 10A current

is being drawn from it. How many hours more the battery is going to last.

Q.41 A 6V torch battery has an internal resistance of $1\ \Omega$. What maximum power output can we expect from a bulb connected to it? If such a bulb is connected to the battery, what will be the potential difference across the terminals of the battery?

12.3 Series Grouping of Cells

A group of cells is known as a battery. To get a battery of high emf, we usually put a number of cells in series. The figure shows three cells in series. r_1 , r_2 and r_3 are the internal resistances. A and B are the terminals of the battery thus formed. An external resistance R has been connected to the terminals of the battery. The battery supplies a current I to the external resistance.

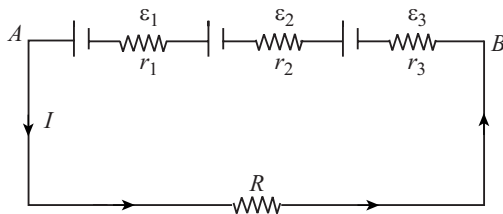


Figure (a)

If we assume that the battery so formed has an effective emf ε_0 and internal resistance r_0 , then

$$I = \frac{\varepsilon_0}{R + r_0} \quad (i)$$

Figure (b) shows the equivalent battery.

Let's go back to the circuit in figure (a) and write the value of current. This is a single loop circuit.

$$\therefore I = \frac{\varepsilon_{net}}{R_{net}} = \frac{\varepsilon_1 + \varepsilon_2 + \varepsilon_3}{R + (r_1 + r_2 + r_3)} \quad (ii)$$

Comparing (i) and (ii), we can write

$$\varepsilon_0 = \varepsilon_1 + \varepsilon_2 + \varepsilon_3 \quad (32)$$

$$\text{and } r_0 = r_1 + r_2 + r_3 \quad (33)$$

The above two equations give the effective emf and internal resistance of the battery formed by joining three cells in series. The same logic can be extended to any number of cells.

If there are n identical cells in series, then the

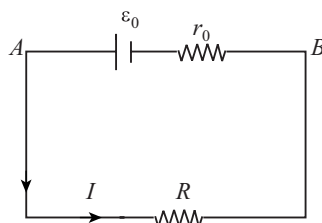


Figure (b): Equivalent battery

effective emf and internal resistance of the battery formed is given by

$$\varepsilon_0 = n\varepsilon$$

$$r_0 = nr$$

Here, ε and r are emf and internal resistance of each cell.

12.4 Parallel Grouping of Cells

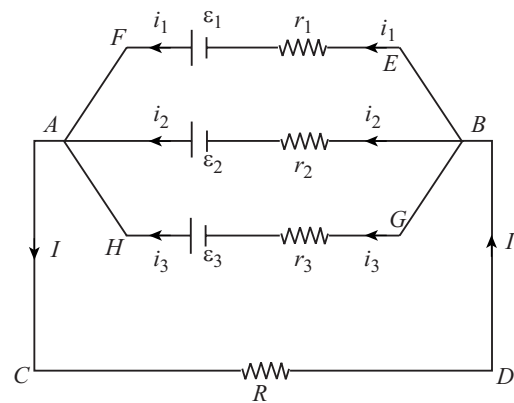
Cells may be grouped in parallel to get a battery of large capacity and small internal resistance. Consider three cells connected in parallel. Terminals of the battery so formed are A and B. An external resistance R is connected to the battery.

Current in the three cells are i_1 , i_2 and i_3 and the current in the external resistance R is

$$I = i_1 + i_2 + i_3 \quad (a)$$

Let's write loop equation in the loop ACDBEFA.

$$IR + i_1 r_1 = \varepsilon_1$$



$$\text{or, } I \frac{R}{r_1} + i_1 = \frac{\varepsilon_1}{r_1} \quad (b)$$

Similar equations can be framed for loops ACDBA and ACDBGHA. The equations are

$$I \frac{R}{r_2} + i_2 = \frac{\varepsilon_2}{r_2} \quad (c)$$

$$I \frac{R}{r_3} + i_3 = \frac{\varepsilon_3}{r_3} \quad (d)$$

Adding (b), (c) and (d) we get

$$IR \left(\frac{1}{r_1} + \frac{1}{r_2} + \frac{1}{r_3} \right) + (i_1 + i_2 + i_3) = \left(\frac{\varepsilon_1}{r_1} + \frac{\varepsilon_2}{r_2} + \frac{\varepsilon_3}{r_3} \right)$$

$$\text{or, } IR \sum \frac{1}{r} + I = \sum \frac{\varepsilon}{r}$$

$$[\text{from (a): } i_1 + i_2 + i_3 = I]$$

$$\text{or, } I = \frac{\sum \frac{\varepsilon}{r}}{R \sum \frac{1}{r} + 1}$$

Dividing the numerator and the denominator by $\sum \frac{1}{r}$, we get

$$I = \frac{\left(\sum \frac{\varepsilon}{r} \right) / \left(\sum \frac{1}{r} \right)}{R + 1 / \left(\sum \frac{1}{r} \right)} \quad (\text{e})$$

If ε_0 is the effective emf of the battery and r_0 is the effective resistance, then

$$I = \frac{\varepsilon_0}{R + r_0} \quad (\text{f})$$

Comparing (e) and (f), we can write

$$\varepsilon_0 = \frac{\sum \varepsilon / r}{\sum 1/r} = \frac{\frac{\varepsilon_1}{r_1} + \frac{\varepsilon_2}{r_2} + \frac{\varepsilon_3}{r_3}}{\frac{1}{r_1} + \frac{1}{r_2} + \frac{1}{r_3}} \quad (34)$$

$$\text{and } r_0 = \frac{1}{\sum \frac{1}{r}}$$

$$\text{or, } \frac{1}{r_0} = \sum \frac{1}{r} = \frac{1}{r_1} + \frac{1}{r_2} + \frac{1}{r_3} \quad (35)$$

The resistance of the battery is the parallel equivalent of r_1 , r_2 and r_3 .

The above equations can be extended for any number of cells. For n identical cells each of emf ε and internal resistance r_1 connected in parallel,

$$\varepsilon_0 = \varepsilon \quad (36)$$

$$\text{and } r_0 = \frac{r}{n} \quad (37)$$

Example 35 Mixed grouping of cells.

A battery contains n parallel branches, each having m identical cells in series. The emf and internal resistance of each cell is ε and r .

(a) Find the effective emf and internal resistance of the battery.

(b) A resistance R is powered by this battery. Find value of R so that maximum amount of power is dissipated in it.

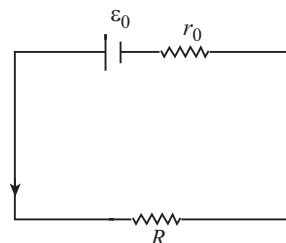
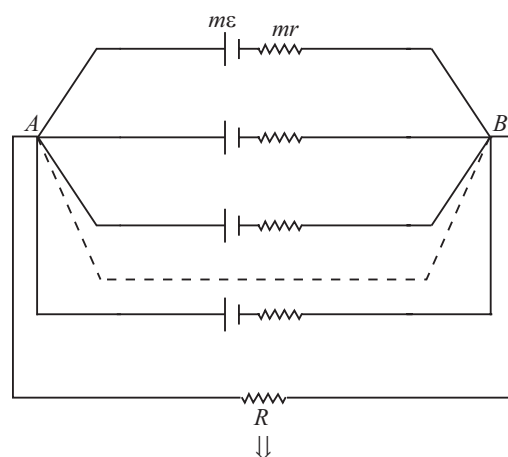
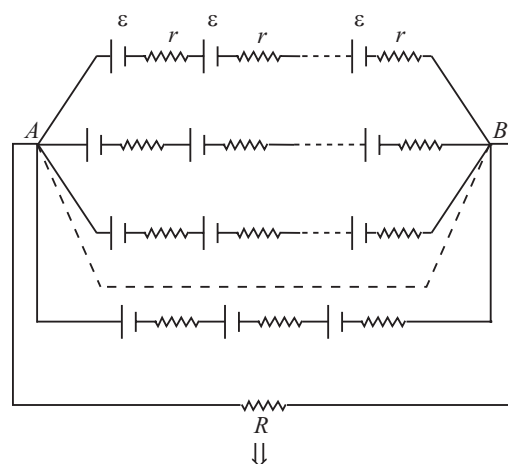
Solution

Concepts

(i) Each row of the cells can be replaced with one single equivalent cell using equations (32) and (33). Then we can use equations (36) and (37) to get the effective emf and internal resistance.

(ii) The second part of the question is direct application of the maximum power transfer theorem.

(a) m cells in series can be replaced with a single cell of emf $m\varepsilon$ and internal resistance mr . Now, we have n identical cells (each of emf $m\varepsilon$ and internal resistance mr) connected in parallel. The effective emf and internal resistance of the battery is



$$\varepsilon_0 = m\varepsilon$$

$$r_0 = \frac{mr}{n}$$

(b) Power transferred to R will be maximum when

$$R = r_0 = \frac{mr}{n}$$

Example 36 *A mistake*

In the battery explained in example 35, by mistake, one cell was connected with its polarity reversed. Find the current supplied by such a battery to an external resistance R .

Solution

Concepts

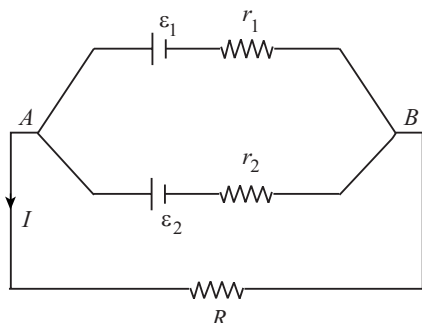
- (i) The branch in which a cell has been placed with polarity reversed, will have $(m-1)$ cells in additive polarity and one opposite to them. Net emf in the branch will be
- $$(m-1)\varepsilon - \varepsilon = (m-2)\varepsilon.$$
- (ii) However, the internal resistance of the branch will still be mr .

The equivalent of $(n-1)$ correct rows in parallel will be a cell with emf

$$\varepsilon_1 = m\varepsilon$$

and internal resistance

$$r_1 = \frac{mr}{n-1}$$



The row having a wrongly connected cell has an effective emf

$$\varepsilon_2 = (m-2)\varepsilon$$

and internal resistance

$$r_2 = mr$$

The emf and internal resistance of the effective battery is given by equations (34) and (35) as

$$\begin{aligned} \varepsilon_0 &= \frac{\varepsilon_1 + \varepsilon_2}{\frac{1}{r_1} + \frac{1}{r_2}} = \frac{\frac{m\varepsilon(n-1)}{mr} + \frac{(m-2)\varepsilon}{mr}}{\frac{n-1}{mr} + \frac{1}{mr}} \\ &= \frac{[m(n-1) + (m-2)]\varepsilon}{n} = \left(\frac{mn-2}{n}\right)\varepsilon \end{aligned}$$

$$\frac{1}{r_0} = \frac{1}{r_1} + \frac{1}{r_2} = \frac{n-1}{mr} + \frac{1}{mr} = \frac{n}{mr}$$

$$\text{or, } r_0 = \frac{mr}{n}$$

∴ Current supplied to the external resistance R is

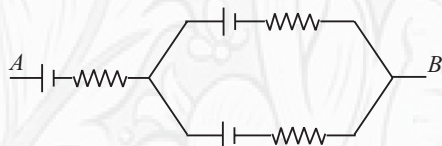
$$I = \frac{\varepsilon_0}{R + r_0} = \frac{\left(\frac{mn-2}{n}\right)\varepsilon}{R + \frac{mr}{n}}$$

$$\text{or, } I = \frac{(mn-2)\varepsilon}{nR + mr}$$



YOUR TURN

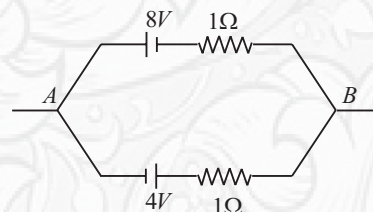
Q.42 Three identical cells have been connected as shown to form a battery. A and B are terminals of the battery. Emf of each cell is 1.5 V and internal resistance is 0.5 Ω. Find the effective emf and internal resistance of the battery so formed.



Q.43 40 identical cells, each of emf 1.5 V and internal resistance 1 Ω are to be grouped so as to give us a battery

of 6 V emf and least internal resistance. What is the internal resistance of the battery?

Q.44 A battery is formed by combining two cells as shown. Find the emf and internal resistance of the battery across the terminals A and B.



In Short

- (i) All real life cells have some internal resistance (r).
- (ii) When there is a current in a cell, the potential difference across its terminals is not same as its emf.
- (iii) For a cell getting discharged the potential difference between its terminal decreases linearly with the current flowing through it.

$$V = \varepsilon - ir$$

- (iv) When $ir = \varepsilon$, the potential difference between the terminals becomes zero. This is what happens when a cell is short circuited.
- (v) For a cell getting charged, the potential difference between the terminals is larger than its emf.

$$V = \varepsilon + ir$$

- (vi) Power transferred by a cell to an external resistance is maximum when the external resistance is equal to the internal resistance of the cell. Maximum power that can get dissipated in an external resistance connected to a cell of emf ε and internal resistance r is

$$P_{\max} = \frac{\varepsilon^2}{4r}$$

- (vii) For n cells, each of emf ε and resistance r , connected in series, the effective emf and resistance are

$$\varepsilon_0 = n\varepsilon \quad \text{and} \quad r_0 = nr$$

- (viii) For n cells each of emf ε and resistance r , connected in parallel, the effective emf and resistance are

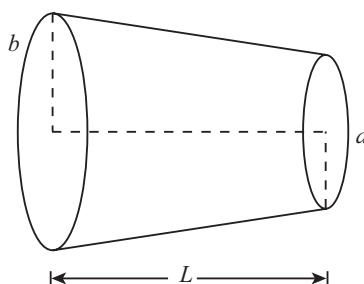
$$\varepsilon_0 = \varepsilon \quad \text{and} \quad r_0 = \frac{r}{n}$$

Miscellaneous Examples

Example 37 Conductor with changing cross section.

A conductor is in the shape of a truncated cone of length L . the cross sectional radii at the two ends are a and b . material has resistivity ρ .

When a potential difference is applied across the two ends, a current flows. Assume that current is distributed uniformly across any section of the cone.



- (a) Find resistance of the conductor.
- (b) Is the electric field same at all points inside the conductor?
- (c) Is the drift speed of electrons same at all points inside the conductor?

Solution**Concepts**

- (i) We will consider the conductor to be made of number of thin discs. Adding the resistances of each disc will give us the total resistance.
- (ii) Electric current must be same at every cross section. Otherwise it will mean that either the charge is continuously getting deposited or depleted in a region inside the conductor.

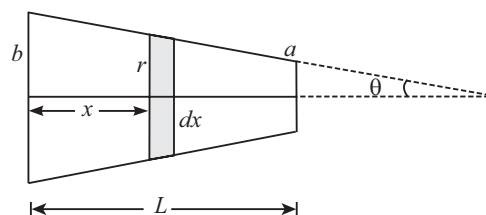
Current density, $j = \frac{I}{A}$ will be different at cross sections having different areas.

- (a) Consider a thin disc of radius r at a distance x from the left end. Thickness of the disc is dx .

From the figure

$$\tan \theta = \frac{b-r}{x} = \frac{r-a}{L-x}$$

$$\text{or, } r = b + \frac{(a-b)x}{L}$$



Resistance of the disc element is

$$dR = \frac{\rho dx}{\pi r^2} = \frac{\rho dx}{\pi \left[b + \frac{(a-b)x}{L} \right]^2}$$

Resistance of the conductor is

$$\begin{aligned} R &= \int dR = \frac{\rho}{\pi} \int_0^L \frac{dx}{\left[b + \frac{(a-b)x}{L} \right]^2} \\ &= \frac{\rho L}{\pi ab} \end{aligned}$$

$$\text{Note: } \int \frac{dx}{(\beta + \alpha x)^2} = -\frac{1}{\alpha(\beta + \alpha x)}$$

- (b) Current is same at all sections. However, the cross sectional area is decreasing from left and right.

Therefore, the current density $\left(J = \frac{I}{A}\right)$ increases from left to right.

Microscopic form of Ohm's law is $J = \sigma E$.

If J increases from left to right, then E must also increase from left to right.

- (c) $J = nev_d$

As J is increasing from left to right, the drift speed v_d must also be increasing from left to right.

It is like speed of water flow must increase with decreasing cross section of the pipe to maintain same water current at every cross section.

Example 38 At 0°C , resistance of a conductor B is n times the resistance of conductor A . The temperature coefficient of resistance of conductors A and B are α_A and α_B , respectively. Find the temperature coefficient of the resistance of series combination of A and B .

Solution

Concepts

We must express the resistance of series combination at some temperature θ as

$$R_s = R_{so} [1 + \alpha \theta]$$

Where R_{so} = resistance of series combination at 0°C

R_s = resistance of series combination at temperature θ .

The constant α in the above expression will be the effective temperature coefficient.

Let resistance of A and B at 0°C be R_0 and nR_0 respectively. The series combination has a resistance

$$R_{so} = R_0 + nR_0 = (1 + n)R_0$$

At temperature θ ,

$$R_A = R_0[1 + \alpha_A\theta] \text{ and } R_B = nR_0[1 + \alpha_B\theta]$$

The resistance of the series combination at temperature θ is

$$R_s = R_A + R_B = R_0[1 + \alpha_A\theta] + nR_0[1 + \alpha_B\theta]$$

$$\text{or, } R_s = (R_0 + nR_0) + (\alpha_A + n\alpha_B)R_0\theta$$

$$\text{or, } R_s = R_{so} + R_{so}\left(\frac{\alpha_A + n\alpha_B}{1 + n}\right)\theta$$

$$\text{or, } R_s = R_{so}\left[1 + \frac{\alpha_A + n\alpha_B}{1 + n}\theta\right]$$

Therefore, the effective temperature coefficient of the resistance is

$$\alpha = \frac{\alpha_A + n\alpha_B}{1 + n}$$

Example 39 A rod of length L and cross section area A lies along the x axis between $x = 0$ and $x = L$. The material obeys ohm's law, but its resistivity varies along its length as

$$\rho = \rho_0 e^{-\frac{x}{L}}$$

The end of the rod at $x = 0$ is at potential V_0 and the other end at $x = L$ is grounded.

- Find the resistance of the rod
- Find current in the rod
- Find potential along the rod as function of x .

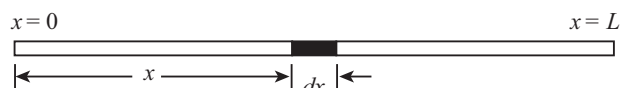
Solution

Concepts

- We can think that the rod is made of large number of small segments connected in series.
- Potential difference between any two points in the rod = (current) (resistance of the rod between the two points).

- Resistance of a small segment of length dx will be

$$dR = \frac{\rho dx}{A} = \frac{\rho_0}{A} e^{-\frac{x}{L}} \cdot dx$$



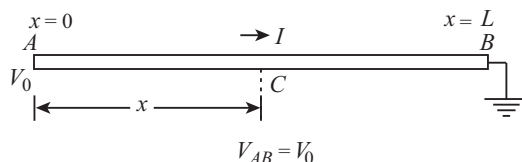
Resistance of the rod = sum of resistance of all small segments.

$$\begin{aligned} \therefore R &= \int dR = \frac{\rho_0}{A} \int_0^L e^{-\frac{x}{L}} \cdot dx \\ &= -\frac{\rho_0 L}{A} \left[e^{-\frac{x}{L}} \right]_0^L \\ &= \frac{\rho_0 L}{A} (1 - e^{-1}) \end{aligned} \quad (i)$$

$$(b) I = \frac{V}{R} = \frac{V_0 A}{\rho_0 L} \left(\frac{e}{e-1} \right)$$

- From (i), the resistance of length x of the rod is

$$R_x = \frac{\rho_0 L}{A} \left(1 - e^{-\frac{x}{L}} \right)$$



$$\therefore V_A - V_C = IR_x = \frac{\rho_0 L}{A} \left(1 - e^{-\frac{x}{L}}\right) \cdot \frac{V_0 A}{\rho_0 L} \left(\frac{e}{e-1}\right)$$

$$\text{or, } V_0 - V_C = V_0 \left(\frac{e}{e-1}\right) \left(1 - e^{-\frac{x}{L}}\right)$$

$$\text{or, } V_C = V_0 - V_0 \left(\frac{e}{e-1}\right) \left(1 - e^{-\frac{x}{L}}\right)$$

$$= V_0 \left(\frac{e^{-\frac{x}{L}} - 1}{e-1}\right)$$

Example 40 Current rating of a fuse is independent of its length

A fuse wire is made of a material of resistivity ρ and has a cross sectional radius r . Assume that the fuse wire loses heat through its curved cylindrical surface only and that the rate of heat loss is proportional to the temperature difference between the fuse wire and the surrounding. θ is the excess temperature of the fuse wire over the surrounding temperature in steady state when a current I flows through it. Show that θ does not depend on length of the wire and is directly proportional to r^{-3} .

Solution

Concepts

In steady state, the rate of heat loss from the surface of the fuse wire will be equal to the rate of heat generation in it. Here, steady state refers to a situation when the temperature of the wire becomes constant.

$$\text{Rate of heat generation, } H = I^2 R = I^2 \frac{\rho l}{\pi r^2}$$

When temperature difference between the wire and the surrounding is θ , the rate of heat loss is given by

$$H_1 \propto (2\pi r l) \theta$$

$$\text{or, } H_1 = k(2\pi r l) \theta$$

Where k is some constant.

In steady state $H_1 = H$

$$\text{or, } k \cdot 2\pi r l \theta = I^2 \frac{\rho l}{\pi r^2} \Rightarrow \theta = \frac{I^2 \rho}{2\pi^2 k r^3}$$

$\therefore \theta$ is independent of length and $\theta \propto r^{-3}$

The material of the fuse will melt for a typical value of temperature. As I increases, θ also increases. When I reaches a specified value, the fuse melts. Obviously, the length of fuse wire is irrelevant in this context.

Example 41 Charge accumulation at junction

Two wires of same cross section but different conductivities form a joint. Area of cross section of the wires is A and their conductivities are σ_1 and σ_2 . A steady current I is made to flow through the wires. Find the amount of charge accumulated at the junction.



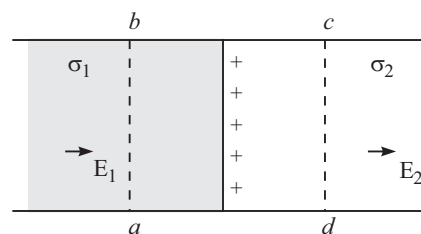
Solution

Concepts

- Current density is same in both the conductors. But they have different conductivities and electric field in them must be different.
- The difference in electric field on two sides of the junction arises due to charge accumulation at the junction.
- Amount of charge accumulated at the junction can be calculated using the Gauss's law.

$$J = \sigma_1 E_1 = \sigma_2 E_2$$

$$\text{or, } E_2 = \frac{\sigma_1}{\sigma_2} E_1 \quad \dots(i)$$



Consider a cylindrical surface ($abcd$) as shown. Electric flux through this surface is

$$\phi = E_2 A - E_1 A = (E_2 - E_1) A$$

If q is charge accumulated at the junction, then using the Gauss's law we can write

$$\phi = \frac{q}{\epsilon_0}$$

$$\text{or, } (E_2 - E_1) A = \frac{q}{\epsilon_0}$$

$$\text{or, } q = A \epsilon_0 (E_2 - E_1)$$

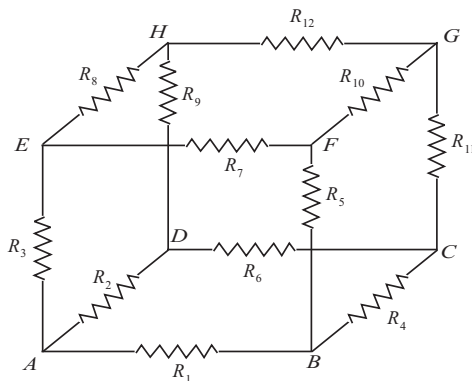
$$= A \epsilon_0 E_1 \left(\frac{\sigma_1}{\sigma_2} - 1\right) \quad [\text{using (i)}]$$

$$= \epsilon_0 A E_1 \left(\frac{1}{\sigma_2} - \frac{1}{\sigma_1}\right)$$

$$= \epsilon_0 I \left(\frac{1}{\sigma_2} - \frac{1}{\sigma_1}\right) \quad [\because I = A j = A E_1 \sigma_1]$$

Example 42 Cube of resistors, again

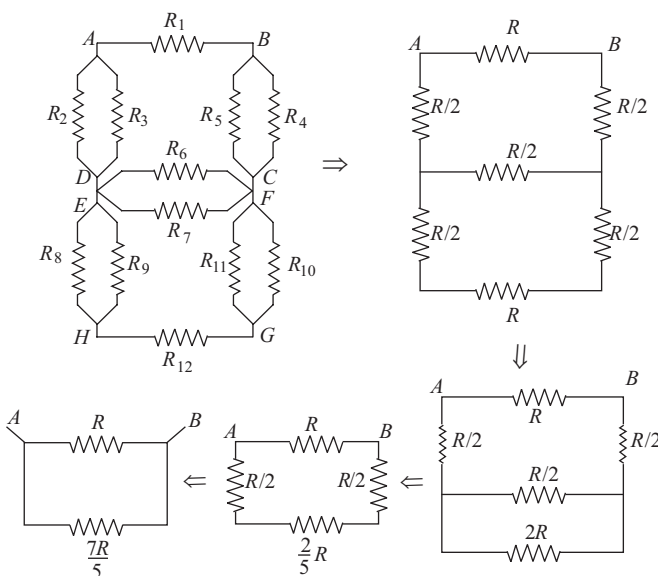
Twelve resistors each of resistance R are connected together so that each one of them lies along an edge of a cube. Find the equivalent resistance between points A and B .

**Solution****Concepts**

Points E and D must be at same potential when a battery is connected between A and B . To understand this you must see that path to D and E from A are identical. Also, path to D and E from B are identical.

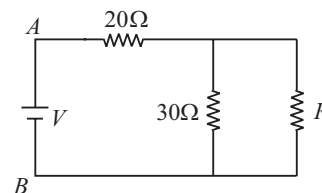
Similarly, F and C are equipotential points.

We connect E and D together, and F and C together. The circuit has been redrawn after this.



The equivalent resistance is $\frac{7R}{5}$.

Example 43 In the circuit shown in figure, a small change in value of resistance R does not cause any variation in the power dissipated in it. What is the value of R ?

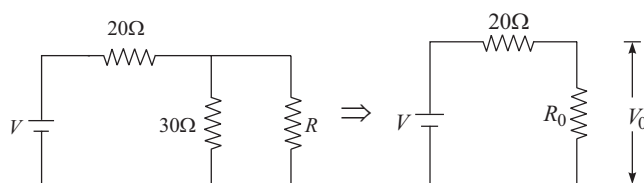
**Solution****Concepts**

If P is power dissipated in R , then $\frac{dP}{dR} = 0$

The equivalent of 30Ω and R is $\frac{30R}{30+R} = R_0$ (say)

The potential difference across R_0 is

$$V_0 = \frac{R_0 V}{20 + R_0}$$



Power generated in R is

$$P = \frac{V_0^2}{R} = \frac{R_0^2 V^2}{(20 + R_0)^2 R}$$

$$\Rightarrow P = \frac{\left(\frac{30R}{30+R}\right)^2 V^2}{\left(20 + \frac{30R}{30+R}\right)^2 R} = \frac{900RV^2}{(50R + 600)^2}$$

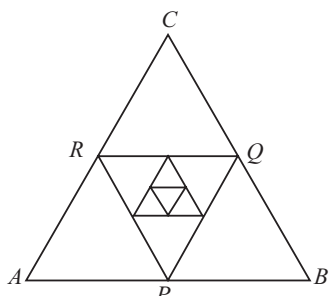
For P to be constant, $\frac{dP}{dR} = 0$

$$\text{or, } \frac{(50R + 600)^2 (900V^2) - (1800)(50)(RV^2)(50R + 600)}{(50R + 600)^4} = 0$$

or, $R = 12 \Omega$

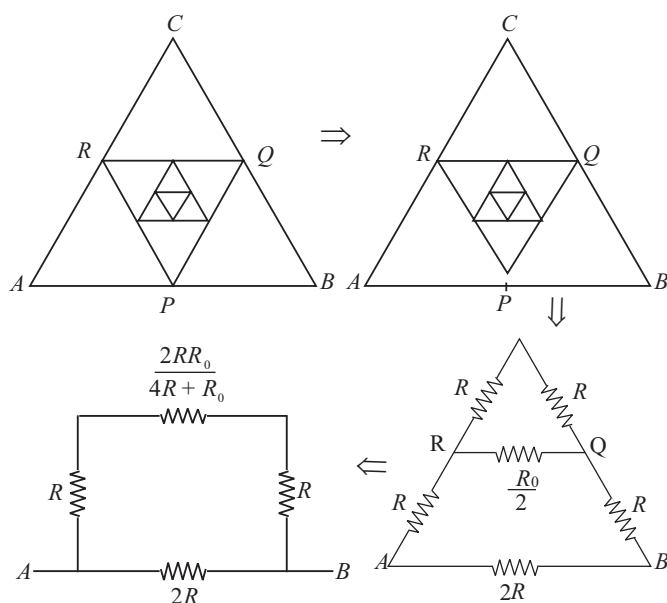
Example 44 Infinite number of embedded triangles

In the diagram shown, the number of successively embedded equilateral triangles tends to infinity. Every triangle has its side length half of the preceding larger triangle. All triangles are made of same uniform wire having resistance λ per meter length. Side length of the outermost triangle is a find the equivalent resistance between points A and B .

**Solution****Concepts**

- (i) Let the equivalent resistance across A and B be R_0 . If we remove the outer triangle, then the resistance of the remaining network about two vertices (like R and Q) will be $\frac{R_0}{2}$. This is because the network remains identical, with every resistance halved. You will not be able to find any difference between the original network and the network obtained after removal of the outer triangle except for the fact that every resistance has been halved.
- (ii) The circuit is symmetrical when looked from A or B . Current in AP will be same as current in PB . The junction at P can be opened.

Let $\lambda \frac{a}{2} = R$



The equivalent resistance across R_0 can be expressed as:

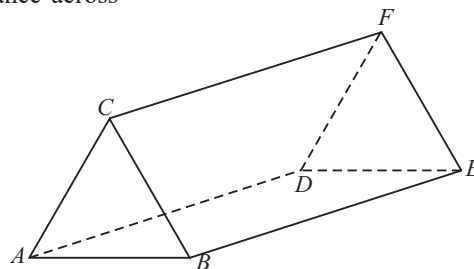
$$R_0 = \frac{\left(\frac{2RR_0}{4R + R_0} + 2R \right) 2R}{\frac{2RR_0}{4R + R_0} + 2R + 2R}$$

$$\text{or, } 3R_0^2 + 4RR_0 - 8R^2 = 0$$

$$\text{or, } R_0 = \left(\frac{2\sqrt{7}-2}{3} \right) R = \left(\frac{2\sqrt{7}-2}{6} \right) \lambda a$$

Example 45 *A prism of resistors*

A wire prism has the shape shown in the figure. Each side of the prism has resistance R . Calculate the equivalent resistance across

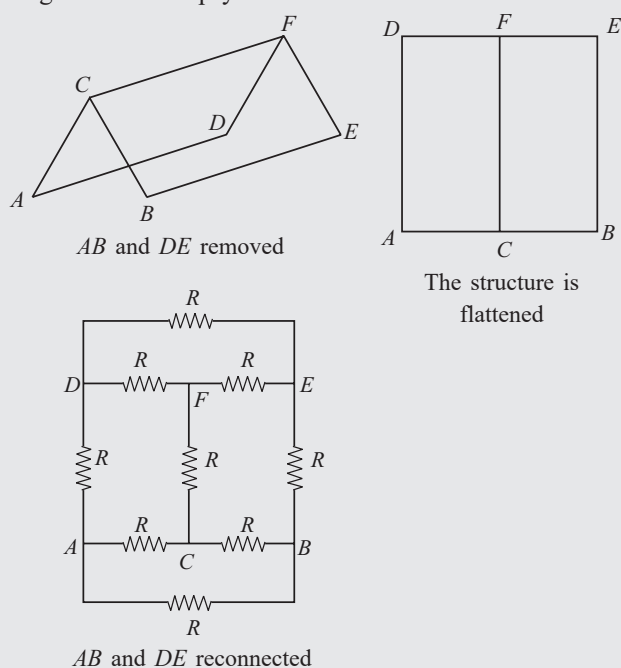


- (a) Points A and B .
(b) Points C and F .

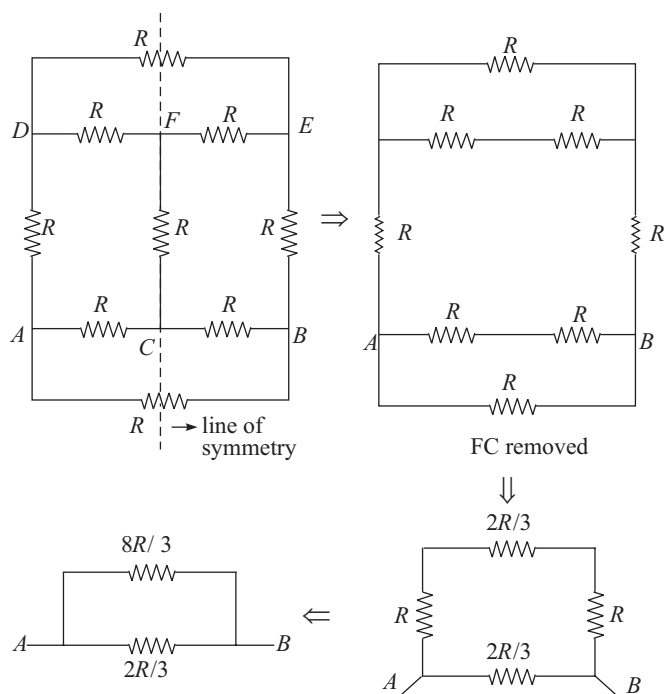
Solution**Concepts**

Though there are many ways to visualize symmetry in this problem, one simple way is to draw the diagram in two dimension.

To do this, remove AB and DE and flatten the network. Now connect back the branches AB and DE . The following diagrams will help you understand this.

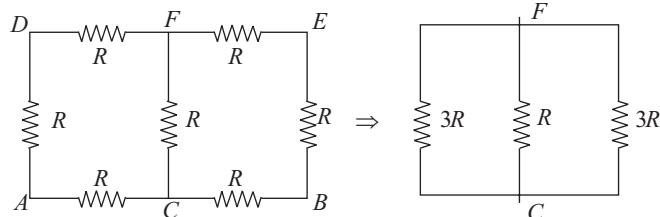


- (a) When a battery is connected across A and B , the line FC divides the circuit in two identical halves. F and C are equipotential points. There will be no current in path FC . We can remove it.



The equivalent resistance is $\frac{8}{15}R$.

(b) When the cell is connected across A and C , there is no difference in points D and E . also, A and B are exactly similar. We can remove the resistance directly connecting D to E and that connecting A to B .

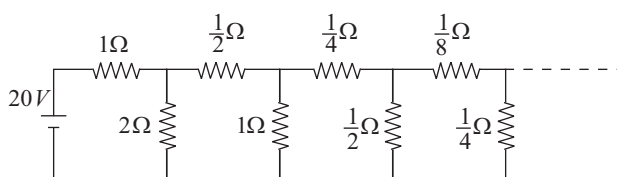


Equivalent resistance (R_0) is given by

$$\frac{1}{R_0} = \frac{1}{3R} + \frac{1}{R} + \frac{1}{3R}$$

$$\text{or, } R_0 = \frac{3R}{5}$$

Example 46 In the infinite ladder network shown in the figure, find the current in 2Ω resistor.



Solution

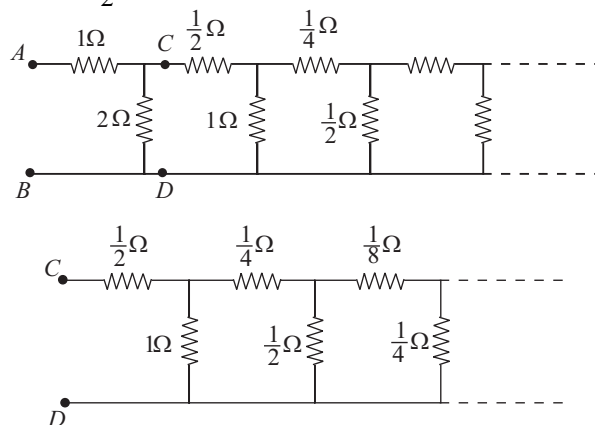
Concepts

- In every ladder the values of resistances is $\frac{1}{2}$ the values in preceding ladder.
- If one of the ladder (nearest to the cell) is removed, the network of resistors remains exactly same with every resistance value halved.
- If equivalent resistance of a network is R_0 and its every resistance is halved, the equivalent resistance becomes $\frac{R_0}{2}$.

Let the equivalent resistance between A and B be R_0 .

Now, remove one link and consider the resistor network between C and D . The network is exactly like the original one with resistance values halved. Equivalent of this network is $\frac{R_0}{2}$.

In the last diagram, we have redrawn the circuit after replacing the entire network to the right of CD with its equivalent $\frac{R_0}{2}$.



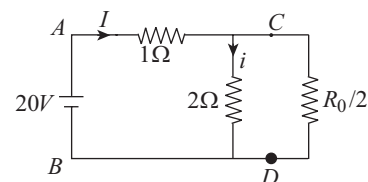
The two networks are exactly same apart from the fact that every resistance in second network is half of the corresponding resistance in the first network.

Obviously,

$$\frac{\frac{R_0}{2} \cdot 2}{\frac{R_0}{2} + 2} + 1 = R_0$$

$$\text{or } R_0^2 + R_0 - 4 = 0$$

$$\text{or, } R_0 = \frac{-1 \pm \sqrt{1+16}}{2} \approx 1.56 \Omega$$



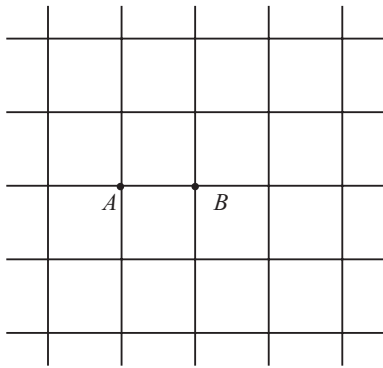
$$\therefore I = \frac{20}{1.56} = 12.82 \text{ A}$$

Current I divides in parallel branches having resistances 2Ω and $\frac{R_0}{2} = 0.78 \Omega$

$$\therefore i = \frac{0.78}{2+0.78} \times 12.82 = 3.6 \text{ A}$$

Example 47 *Tricky use of superposition principle*

An infinite wire grid has identical square units. Each side of a square has resistance R . Find the equivalent resistance between two adjacent points A and B .



Solution

Concepts

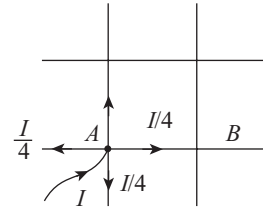
If a current enters at A , it goes on dividing at junctions. [At a large distance from A , current in a branch will be small.] Now this must return and exit through B .

We will first assume a current I entering at A (forget about the exiting current for a moment) and divide it into 4 identical branches at A .

Then we will assume a current I leaving the network at B . We will again use symmetry to write current in the four branches joining at B .

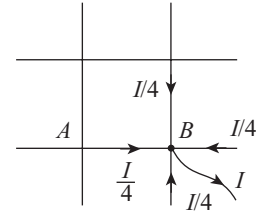
Superposing the in current and out current can let us know the true current in path AB .

Consider current I entering at A (forget the outgoing current at B). due to symmetry it will divide equally in 4 paths at A . Current in path AB is $\frac{I}{4}$.



The current goes on distributing to infinity. Then there must be a return current also.

Currents returning in various paths will eventually join at B and exit as total current I . Again due to symmetry, the current in 4 paths joining at B must be same. Hence, current in path AB is $\frac{I}{4}$.



The exact current in any path will be superposition of the currents in two situations.

$$\therefore \text{Actual current in path } AB = \frac{I}{4} + \frac{I}{4} = \frac{I}{2}$$

It means that current I divides equally in path AB and the remaining circuit. This can happen only when the resistance of the remaining circuit between A and B is same as that of path AB .

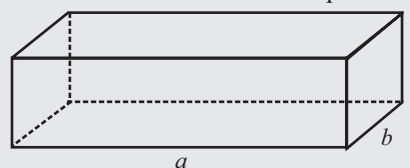
$\therefore$ We can assume that the entire network between A and B is parallel combination of two equal resistances R .

$$\therefore R_{\text{eq}} = \frac{R}{2}$$

Worksheet 1

- In the presence of an applied electric field ($\vec{E}$) in a metallic conductor
 - The electrons move in the direction of $\vec{E}$.
 - The electrons move in a direction opposite to $\vec{E}$.
 - The electrons may move in any direction randomly, but slowly drift in the direction of $\vec{E}$.
 - The electrons move randomly but slowly drift in a direction opposite to $\vec{E}$.
- The drift velocity of electrons in a conducting wire is of the order of 1mm/s, yet the bulb glows almost instantaneously after the switch is put on. This is because
 - The random speed of electrons is very high, of the order of 10^6 m/s
 - The electrons transfer their energy very quickly through collisions
 - Electric field is set up in the wire very quickly, producing a current everywhere in the wire, almost instantaneously
 - The drift speed is very high initially.
- In the Bohr's model of hydrogen atom the electron moves around the nucleus in a circular orbit of radius 5×10^{-11} m with time period 1.5×10^{-16} s. The current associated with the electron motion is
 - Zero
 - 1.6×10^{-19} A
 - 0.17 A
 - 1.07×10^{-3} A
- A uniform wire of resistance 100Ω is melted and recast in a wire of length double that original. What would be the resistance of the wire?
 - 400Ω
 - 200Ω
 - 800Ω
 - 100Ω
- If the length of the filament of a heater is reduced by 10%, the power of the heater will
 - increase by about 9%
 - increase by about 11%
 - increase by about 19%
 - increase by about 10%
- A quantity X is given by $\epsilon_0 L \frac{\Delta V}{\Delta t}$ where ϵ_0 is the permittivity of free space, L is a length, ΔV is a potential difference and Δt is a time interval. The dimensional formula for X is the same as that of:
 - Resistance
 - Charge
 - Voltage
 - Current

- Consider a wire of length 4m and cross-sectional area 1 mm^2 carrying a current of 2A. If each cubic metre of the material contains 10^{29} free electrons, find the average time taken by an electron to cross the length of the wire.
 - 1.2×10^4 sec
 - 3.2×10^4 sec
 - 3.2×10^3 sec
 - 1.2×10^3 sec
- Masses of three wires of copper are in the ratio of 1: 3: 5 and their lengths are in the ratio of 5: 3: 1. The ratio of their electrical resistances are:
 - 1 : 3 : 5
 - 5 : 3 : 1
 - 1 : 15 : 125
 - 125 : 15 : 1
- A block is in cuboidal shape. Its longest edge is twice its shortest edge. The ratio of the maximum to minimum resistance between parallel faces is:

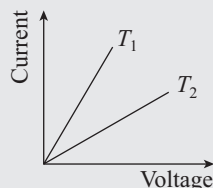


- 2
 - 4
 - 8
 - indeterminate unless the length of the third edge is specified.
- Two wires each of radius of cross section r but of different materials are connected together end to end (in series). The densities of charge carriers in the two wires are in the ratio 1 : 4. When a current is made to flow in the wires, the drift velocity of electrons in the two wires will be in the ratio:
 - 1 : 2
 - 2 : 1
 - 4 : 1
 - 1 : 4
 - In a wire of cross-section radius r , free electrons travel with drift velocity v when a current I flows through the wire. What is the current in another wire of half the radius and of the same material when the drift velocity is $2v$?
 - $2I$
 - I
 - $I/2$
 - $I/4$
 - An insulating pipe of cross-section area 'A' contains an electrolyte which has two types of ions— their charges being $-e$ and $+e$. A potential difference applied between the ends of the pipe result in the drifting of the two types of ions, having drift speed $= v$ ($-ve$ ion) and $v/4$ ($+ve$ ion). Both ions have

the same number per unit volume $= n$. The current flowing through the pipe is

- (a) $nev A/2$ (b) $nev A/4$
(c) $5nev A/4$ (d) $3nev A/2$

13. The current in a metallic conductor is plotted against voltage at two different temperatures T_1 and T_2 . Which is correct

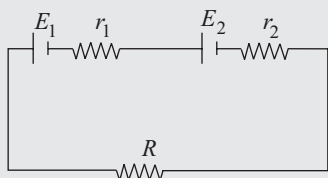


- (a) $T_1 > T_2$
(b) $T_1 < T_2$
(c) $T_1 = T_2$
(d) none

14. A storage battery with internal resistance 1Ω is connected to a charger. The charger needs a charging voltage of 12.5 V when current through it is 0.5 A . The emf of the storage battery is:

- (a) 13 Volts (b) 12.5 Volts
(c) 12 Volts (d) 11.5 Volts

15. The internal resistances of the two batteries shown in the figure are r_1 and r_2 . Under what condition current passing through the resistance R can be increased by short circuiting the battery of emf E_2 .



- (a) $E_2 r_1 > E_1 (R + r_2)$ (b) $E_1 r_2 > E_2 (R + r_1)$
(c) $E_2 r_2 > E_1 (R + r_2)$ (d) $E_1 r_1 > E_2 (R + r_1)$

16. One end of a Nichrome wire of length $2L$ and cross-sectional area A is attached to an end of another Nichrome wire of length L and cross-sectional area $2A$. If the free end of the longer wire is at an electric potential of 8.0 volts , and the free end of the shorter wire is at an electric potential of 1.0 volt , the potential at the junction of the two wires is equal to

- (a) 2.4 V (b) 3.2 V
(c) 4.5 V (d) 5.6 V

17. An electric fan and a heater are marked as 100 watt , 220 volt and 1000 watt , 220 volt respectively. The resistance of heater is:

- (a) Zero
(b) Greater than that of fan
(c) Lesser than that of fan
(d) Equal to that of fan

18. In an electric kettle 1 litre water boils in 15 minute after switching on the power supply. If the length of heating wire is decreased to $2/3$ of the original length,

then the time taken in minutes at the same electrical voltage for boiling the water will be-

- (a) 15 (b) 12
(c) 10 (d) 8

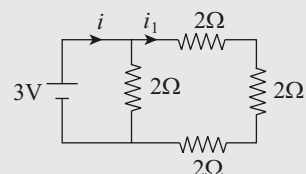
19. An electrical cable of copper has just one wire of radius 9 mm . Its resistance is 5Ω . This single copper wire of the cable is replaced by 6 different well insulated copper wires each of radius 3 mm . The total resistance of the cable will now be equal to

- (a) 7.5Ω (b) 45Ω
(c) 90Ω (d) 270Ω

20. When a wire of uniform cross-section A , length l and resistance R is bent into a complete circle, resistance between any two of diametrically opposite points will be

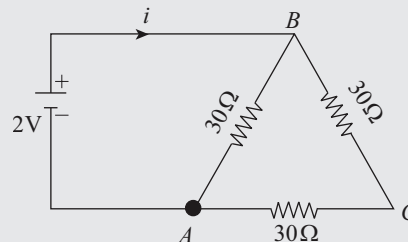
- (a) $R/4$ (b) $R/8$
(c) $4R$ (d) $R/2$

21. What are the currents (i) and (i_1) in the circuit shown in figure:



- (a) 2 A , 0.5 A
(b) 1.2 A , 1 A
(c) 1 A , 0.5 A
(d) 0.5 A , 0.25 A

22. The current i in the circuit shown in figure is -

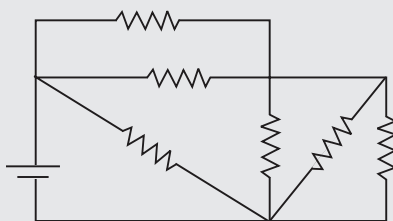


- (a) $\frac{1}{45}\text{ amp.}$ (b) $\frac{1}{15}\text{ amp.}$
(c) $\frac{1}{10}\text{ amp.}$ (d) $\frac{1}{5}\text{ amp.}$

23. If 2 bulbs rated 2.5 W , 110 V and 100 W , 110 V are connected in series to a 220 V supply then

- (a) 2.5 W bulb is likely to fuse
(b) 100 W bulb is likely to fuse
(c) both are likely to fuse
(d) both will not fuse

24. In the figure shown each resistor is of 20Ω and the cell has emf 10 volt with negligible internal resistance. Then rate of joule heating in the circuit is (in watts)

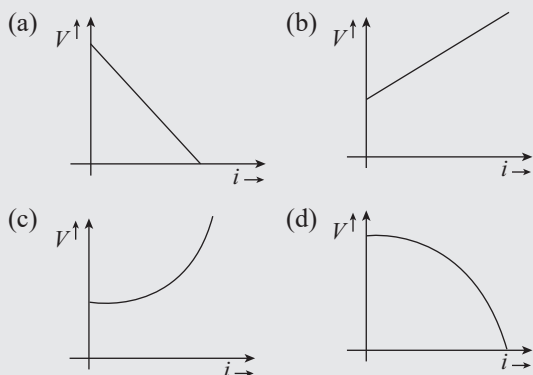


- (a) $100/11$ (b) $10000/11$
(c) 11 (d) None of these

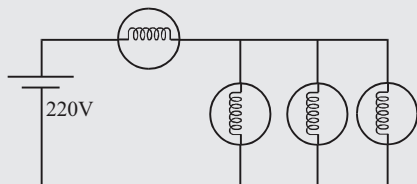
25. Power generated across a uniform wire connected across a voltage supply is H . If the wire is cut into n equal parts and all the parts are connected in parallel across the same supply, the total power generated in the wire is

- (a) $\frac{H}{n^2}$ (b) $n^2 H$
(c) nH (d) $\frac{H}{n}$

26. If internal resistance of a cell is proportional to current drawn from the cell. Then the best representation of terminal potential difference of a cell with current drawn from cell will be:



27. Four identical bulbs each rated 100 watt, 220 volts are connected across a battery as shown. The electric power consumed by the bulb that is farthest from the battery is:



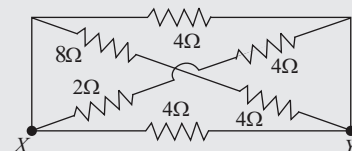
- (a) 6.25 W (b) 12.5 W
(c) 50 W (d) 25 W

28. A resistor of resistance R is connected to a cell of internal resistance 5Ω . The value of R is varied from 1Ω to 5Ω . The power consumed by R :

- (a) increases continuously
(b) decreases continuously

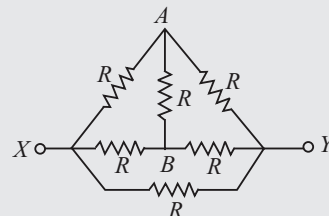
- (c) first decreases then increases
(d) first increases then decreases.

29. For the circuit shown in the figure, the value of total resistance between points X and Y is-



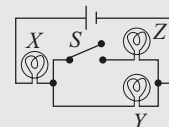
- (a) 2Ω
(b) 4Ω
(c) $4/3 \Omega$
(d) $2/3 \Omega$

30. An ideal cell is connected between points X and Y of the circuit shown in the figure. Current in the resistance at the lowest position in the diagram is I . The current in all other resistances in the circuit is-



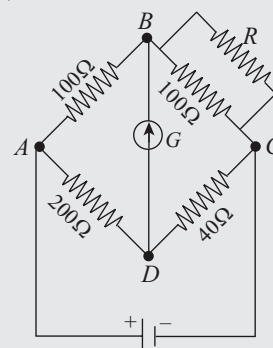
- (a) $I/2$ or zero
(b) I or $I/2$
(c) $I/2$ or $I/4$
(d) $2I$

31. X , Y and Z , shown in figure, are identical electrical lamps. which of the following changes to the brightnesses of the lamps occur when switch S is closed?



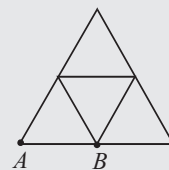
- (a) X stays the same, Y decreases
(b) X increases, Y decreases
(c) X increases, Y stays the same
(d) X decreases, Y increases

32. The given Wheatstone bridge is showing no deflection in the galvanometer joined between the points B and D (Figure). Calculate the value of R .



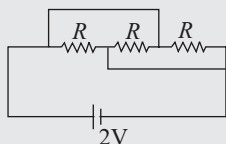
- (a) 25Ω (b) 50Ω
(c) 40Ω (d) 100Ω

33. In the diagram, resistance of each segment of the wire is R . Equivalent resistance across terminals A and B is



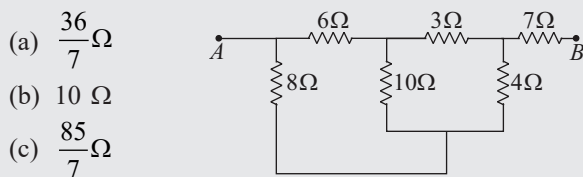
- (a) $\frac{11R}{7}$ (b) $\frac{18R}{11}$
(c) $\frac{7R}{11}$ (d) $\frac{11R}{18}$

34. Three equal resistance, each of R ohm, are connected as shown in figure. A battery of 2 volts and internal resistance 0.1 ohm is connected across the circuit. Calculate the value of R for which the heat generated in the external circuit is maximum.



- (a) 0.1Ω (b) 0.2Ω
(c) 0.3Ω (d) 0.4Ω
35. A wire of resistance 0.1 ohm cm^{-1} is bent to form a square $ABCD$ of side 10 cm . A similar wire is connected between the corners B and D to form the diagonal BD . If a 2V battery of negligible internal resistance is connected across A and C , the total power dissipated is

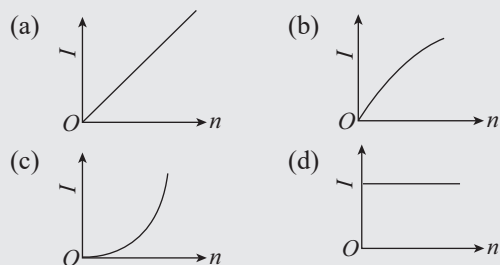
- (a) 3 W (b) 4 W
(c) 0.4 W (d) 0.04 W
36. The equivalent resistance between the points A and B is:



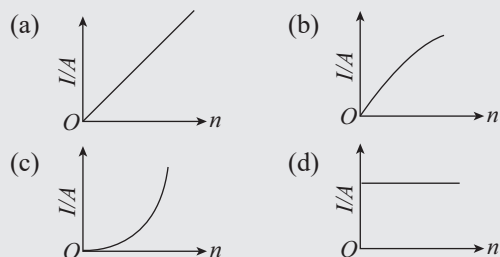
- (a) $\frac{36}{7} \Omega$
(b) 10Ω
(c) $\frac{85}{7} \Omega$
(d) none of these
37. Two non ideal batteries are connected in parallel. Consider the following two statements:
(I) The equivalent emf is smaller than either of the two emfs.
(II) The equivalent internal resistance is smaller than either of the two internal resistances.
(a) Both I and II are correct
(b) I is correct but II is wrong
(c) II is correct but I is wrong
(d) Each of I and II is wrong.
38. 12 cells are connected in series and are kept in a closed box. Some of the cells are wrongly connected with their polarity reversed. This battery is connected in series with an ammeter and two cells (these two cells are in series). All cells are identical. The current is 3 A when the external cells support this battery and is 2 A when the cells oppose the battery. How many cells in the battery are wrongly connected?
- (a) one (b) two
(c) three (d) none

39. A battery consists of a variable number n of identical cells, having internal resistance, connected in series. The terminals of the battery are short circuited and the current I measured.

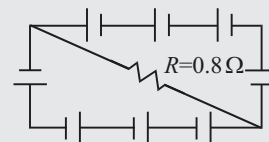
Which one of the graph below shows the relationship between I and n ?



40. In previous problem, if the cells had been connected in parallel (instead of in series) which of the graphs shows the relationship between total current I and n ?



41. A circuit is comprised of eight identical batteries and a resistor $R = 0.8 \Omega$. Each battery has an emf of 1.0 V and internal resistance of 0.2Ω . The voltage difference across any of the battery is



- (a) 0.5V (b) 1.0V
(c) 0 V (d) 2 V
42. A wire of cross-section area A , length L_1 , resistivity ρ_1 and temperature coefficient of resistivity α_1 is connected to a second wire of length L_2 , resistivity ρ_2 , temperature coefficient of resistivity α_2 and the same area A , so that the two wires carry same current. Total resistance R is independent of temperature for small temperature changes if (Thermal expansion effect is negligible)
- (a) $\alpha_1 = -\alpha_2$ (b) $\rho_1 L_1 \alpha_1 + \rho_2 L_2 \alpha_2 = 0$
(c) $L_1 \alpha_1 + L_2 \alpha_2 = 0$ (d) None

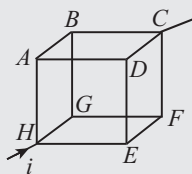
43. The diagram shows a wire frame in the shape of a cube. The current i enters at H and leaves at C.

$$\text{If } i_{AB} = \frac{i}{6}, i_{DC} = \frac{2i}{3}, i_{HA} = \frac{i}{2},$$

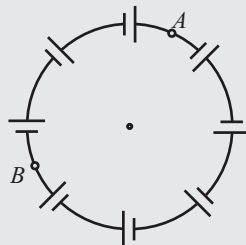
$$i_{GF} = \frac{i}{6}, i_{HE} = \frac{i}{6}, \text{ choose the}$$

branch in which current is zero

- (a) BG (b) FC
(c) ED (d) none

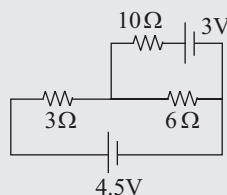


44. N sources with different emf's are connected as shown in figure. The emf's of the sources are proportional to their internal resistances, i.e. $E = \alpha R$, where α is a constant. The connecting wires have negligible resistance. The potential difference between points A and B dividing the circuit in n and $N - n$ sources is



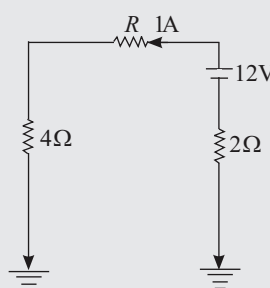
- (a) 0 (b) $nE/2$
(c) NE (d) $(N - n)E$

45. Find the current through the $10\ \Omega$ resistor shown in figure

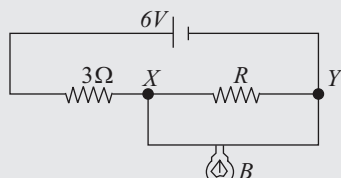


46. In the circuit shown in figure the value of R is-

- (a) $8\ \Omega$
(b) $6\ \Omega$
(c) $10\ \Omega$
(d) $12\ \Omega$

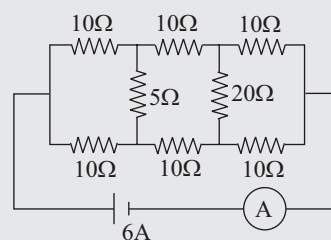


47. In the circuit shown, the bulb is rated as 1.5V, 0.45W. If bulb glows with full intensity then what will be the equivalent resistance between X and Y



- (a) $0.45\ \Omega$ (b) $1\ \Omega$
(c) $3\ \Omega$ (d) $5\ \Omega$

48. The current measured by the ammeter in the circuit shown in figure is



- (a) 2 A (b) 1 A
(c) 4 A (d) 0.4 A

49. A resistance R carries a current i in steady state. The power lost to the surroundings is $\lambda(\theta - \theta_0)$. Here, λ is a constant, θ is temperature of the resistance and θ_0 is the temperature of the atmosphere. The coefficient of linear expansion is α . The strain in the resistance developed after the current is passed through it is:

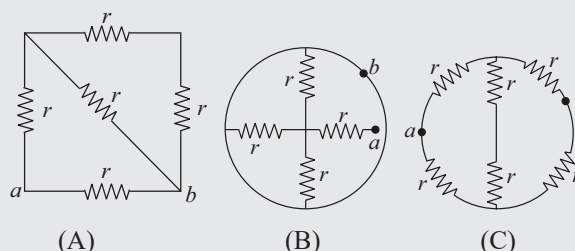
(a) $\frac{\alpha}{\lambda} i^2 R$

(b) $\alpha \lambda i R$

(c) $\frac{\alpha i^2 R}{2\lambda}$

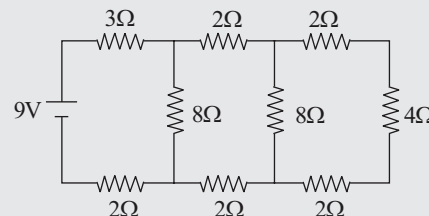
- (d) proportional to the length of the resistance wire

50. The equivalent resistance of the networks shown in figure between the points a and b is largest in



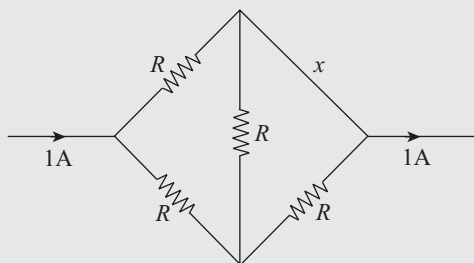
- (a) A (b) B
(c) C (d) equal in all

51. In the circuit shown in figure the current through

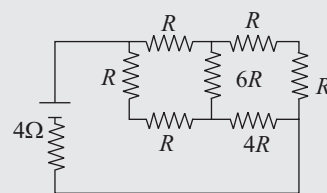


- (a) the $3\ \Omega$ resistor is 0.50 A
(b) the $3\ \Omega$ resistor is 0.25 A
(c) the $4\ \Omega$ resistor is 0.50 A
(d) the $4\ \Omega$ resistor is 0.25 A

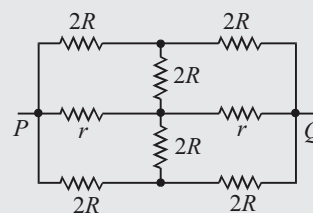
52. In the circuit shown, $R = 1\Omega$ and the branch x is resistance free. The current in the branch x is:



- (a) 1A (b) 0
(c) $\frac{2}{3}A$ (d) None of these
53. Two wires of equal diameters of resistivities ρ_1 and ρ_2 and lengths x_1 and x_2 are joined in series. The equivalent resistivity of the combination is -
- (a) $\frac{\rho_1 x_1 + \rho_2 x_2}{x_1 + x_2}$ (b) $\frac{\rho_1 x_2 - \rho_2 x_1}{x_1 - x_2}$
(c) $\frac{\rho_1 x_2 + \rho_2 x_1}{x_1 + x_2}$ (d) $\frac{\rho_1 x_1 + \rho_2 x_2}{\rho_1 + \rho_2}$
54. A battery of internal resistance 4Ω is connected to the network of resistance as shown. In order that maximum power can be delivered to the network, the value of R in ohm should be-



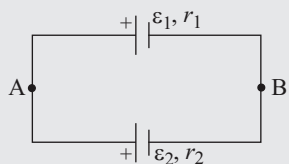
- (a) $\frac{4}{9}$ (b) 2
(c) $\frac{8}{3}$ (d) 18
55. A wire of length L and three identical cells of negligible internal resistances are connected in series. Due to the current, the temperature of the wire is raised by ΔT in a time t . N number of similar cells are now connected in series with a wire of the same material and cross-section but of length $2L$. The temperature of the wire is raised by the same amount ΔT in the same time t . The value of N is: (Assume that all heat dissipated is retained by the wire)
- (a) 4 (b) 6
(c) 8 (d) 9
56. The effective resistance between the points P and Q of the electrical circuit shown in the figure is



- (a) $2Rr / (R + r)$ (b) $8R(R + r)/(3R + r)$
(c) $2r + 4R$ (d) $5R/2 + 2r$

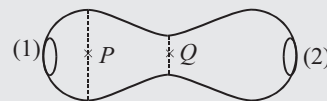
Worksheet 2

- When no current is passed through a conductor
 - the free electrons do not move.
 - the average speed of a free electron over a large period of time is zero.
 - the average velocity of a free electron over a large period of time is zero.
 - the average of the velocities of all the free electrons at an instant is zero.
- A current passes through a copper wire of non-uniform cross-section. Which of the following quantities are independent of the cross-sectional area?
 - charge crossing through a section in a given time interval
 - drift speed
 - current density
 - free-electron density.
- A bulb is connected to an ideal battery of emf 30 V so that the resulting current is 10 mA. When the bulb is connected to a 220 V ideal source, the current is 50 mA in steady state. Choose the correct alternative
 - In the first case, the resistance of the bulb is 3 k Ω and in second case, it is 4.4 k Ω .
 - It is not possible since ohm's law is not followed.
 - The filament of the bulb is hotter when it is connected to 220 V source.
 - None of these.
- Two cells of unequal emfs ε_1 and ε_2 , and internal resistances r_1 and r_2 are joined as shown. They are rechargeable chemical cells. V_A and V_B are the potentials at A and B respectively.

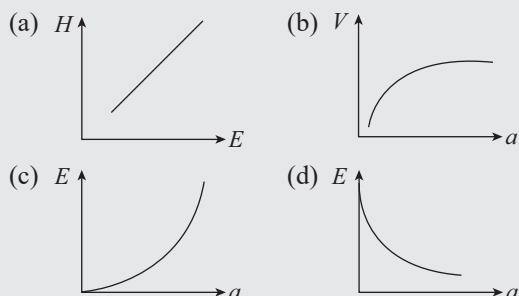
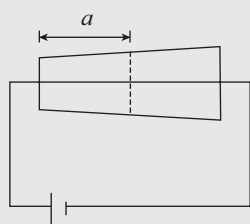


- One cell will supply energy to the other
- The potential difference across both the cells will be equal
- The rate of loss in chemical energy of one cell is more than the rate of gain in chemical energy of the other.
- $V_A - V_B = \frac{(\varepsilon_1 r_2 + \varepsilon_2 r_1)}{r_1 + r_2}$

- Choose the correct alternatives
 - The internal resistance of a car battery decreases with rise in temperature
 - It is more economical to transmit electric power at high voltage and low current rather than at low voltage and high current because heat loss is proportional to square of current.
 - The heating coil of an electric iron is enclosed in mica sheets because mica is a bad conductor of heat and good conductor of electricity
 - The heating coil of an electric iron is enclosed in mica sheets because mica is a good conductor of heat and bad conductor of electricity.
- A metallic conductor of irregular cross-section is as shown in the figure. A constant potential difference is applied across the ends (1) and (2). The cross sectional area at P and Q are A_1 and A_2 respectively. S_1 is a small length of the wire at P and S_2 is a segment of same length at Q.



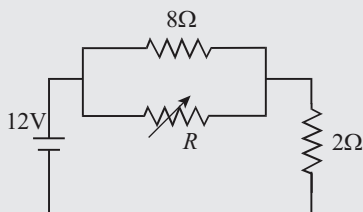
- the current at the cross-section P equals the current at the cross-section Q
 - the electric field intensity at P is less than that at Q.
 - the rate of heat generated in S_1 and S_2 are in ratio $A_2 : A_1$
 - the rate of heat generated per unit volume in S_1 and S_2 are in ratio $A_2 : A_1$
- The charge flowing in a conductor of resistance R varies with time as $Q = at - bt^2$. Then,
 - The current decreases exponentially with time
 - Total charge flow before the current becomes zero is $\frac{a^2}{4b}$
 - The current falls to zero after time $t = \frac{a}{2b}$
 - Heat generated in the conductor by the time the current falls to zero is $\frac{a^3}{6b}$
 - The electric field intensity, potential and rate of heat generation per unit length are E , V and H at a cross section distant a from left end of a conical shaped conductor connected to a battery of emf E , as shown. Choose the correct graphs:



9. Two fuses are rated at 10A and 5 A. The two fuse wires have equal but small resistance. If they are joined

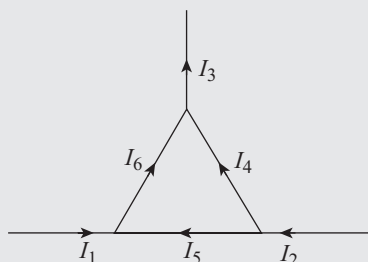
- (a) in parallel, the combination acts as a fuse of rating 15A
- (b) in parallel, the combination acts as a fuse of rating 10A
- (c) in series, the combination acts as a fuse of rating 15A.
- (d) in series, the combination acts as a fuse of rating 5A.

10. The value of the resistance R in figure is adjusted such that power dissipated in the 2Ω resistor is maximum. Under this condition



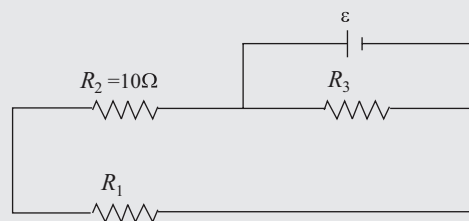
- (a) $R = 0$
- (b) $R = 8\Omega$
- (c) power dissipated in the 2Ω resistor is 72 W.
- (d) power dissipated in the 2Ω resistor is 8 W.

11. The diagram below shows a junction with currents labeled I_1 to I_6 . Which of the following statements are correct?

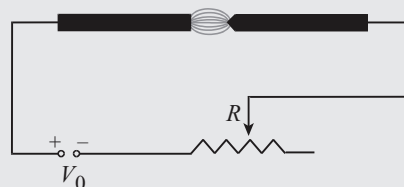


- (a) $I_5 + I_3 = I_6 + I_2$
- (b) $I_1 + I_2 = I_6 + I_4$
- (c) $I_4 + I_3 = I_6$
- (d) $I_2 = I_6 + I_4$

12. The power dissipated in resistor R_3 shown in the figure is 15W. The current recorded in R_1 is 500 mA and the potential difference across it is 10V. The battery is ideal.



- (a) The value of resistance R_1 is 20Ω
 - (b) The value of resistance R_3 is 15Ω
 - (c) The emf ε of the supply battery is 25 V
 - (d) The emf ε of the supply battery is 15 V
13. The carbon arc light consists of an arc between carbon electrodes in air. It was invented by Humphry Davy in the first decade of the 1800s. The circuit of a carbon arc lamp is shown in the figure. To ignite the lamp the carbon electrodes are touched together, then they are slowly drawn apart. The voltage across the gap between the carbon rods as a function of the current can be given by the following formula: $V = A + B/I$, where A and B are constants.



The voltage of the battery V_0 can be considered constant.

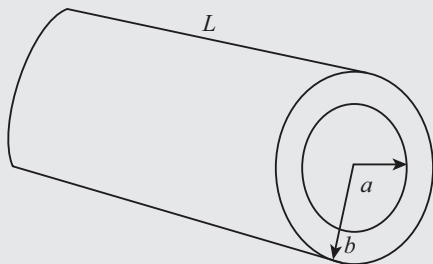
Given that: $A = 40$ V, $B = 20$ W, $V_0 = 100$ V

What is the maximum resistance of the rheostat $R_{\max}$, that should be connected in series to ignite the lamp.

- (a) 15Ω
- (b) 15Ω
- (c) 15Ω
- (d) 45Ω

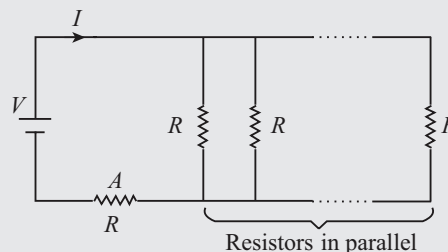
Worksheet 3

1. A copper wire of cross section 4 mm^2 carries a 10A current. Find the drift speed of electrons in the wire. It is given that density of copper is 9 gcc^{-1} and its molar mass is 63.5 g. Assume that every copper atom contributes one free electron.
2. An electric heater uses nichrome for its heating element. When a very small current is passed through it, its resistance is found to be 75.3Ω at 27°C . When the heater is connected to a 230 V supply, the current settles, after some time to a steady value of 2.68 A. Find the temperature of the nichrome element in the steady state. The temperature coefficient of resistivity of the nichrome has an average value of $\alpha = 1.7 \times 10^{-4} (\text{C}^\circ)^{-1}$.
The thermal expansion of the nichrome element can be neglected
3. Wire A and B have their mass, length, density and resistivity in the ratios 1 : 1, 1 : 2, 1 : 3 and 2 : 1, respectively. When connected to same power supply the rate of heat dissipation in wire A is 10W. Find the rate of heat dissipation in wire B.
4. A hollow cylindrical conductor has inner radius a and outer radius b . Its length is L . It is made of a material of resistivity ρ .

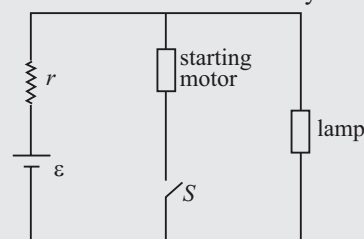


- (a) A potential difference is applied between the ends of the cylinder and causes a current to flow along the length of the cylinder. What is resistance of the cylinder?
 - (b) A potential difference is applied between the inner and outer surfaces and the current flows radially. What is resistance of the cylinder in this case?
5. Sea water can conduct electricity chiefly due to presence of Na^+ and Cl^- ions. Density of each of these ions is about $3 \times 10^{20} \text{ cm}^{-3}$. Resistivity of the sea water is $25 \Omega \text{ cm}$. A plastic tube 2m long is filled with sea water and connected to a 12 V battery with the help of electrodes at the two ends. Find the drift speed of the ions.

6. A copper wire has length l_0 and cross sectional area A_0 . Its resistance is R_0 . Find percentage change in its length, area and resistance if temperature is increased by 1°C . It is given that the coefficient of linear expansion for copper is $\alpha = 1.7 \times 10^{-5} (\text{C}^\circ)^{-1}$ and the thermal coefficient for resistivity is $\alpha_R = 3.9 \times 10^{-3} (\text{C}^\circ)^{-1}$.
7. A wire is made by joining two pieces having temperature coefficient of resistivity $\alpha_1 = -0.5 \times 10^{-3} (\text{C}^\circ)^{-1}$ and $\alpha_2 = 4 \times 10^{-3} (\text{C}^\circ)^{-1}$. The wire has a resistance of 30Ω at 0°C and its resistance does not change for small variations in temperature. Find the resistance of each piece at 0°C .
8. A copper wire of length L and radius r is electroplated with nickel till its radius becomes R . Resistivity of copper and nickel are ρ_C and ρ_N respectively. Calculate the resistance of the electroplated wire.
9. A spherical region of radius R contains charge q inside it. The charge q changes with time t as $q = q_0 e^{-kt}$, where k and q_0 are constants. Find the current density on the surface of the sphere at $t = 0$, assuming it to be uniform.
10. n -resistors are connected in parallel. This group is connected to an ideal cell with another resistor (A) in series. All resistances in the circuit are identical. Current through the cell is I . When one more resistor is connected in parallel to the group of n resistors in parallel, the current I changes by $\frac{5}{6}\%$. Find n .

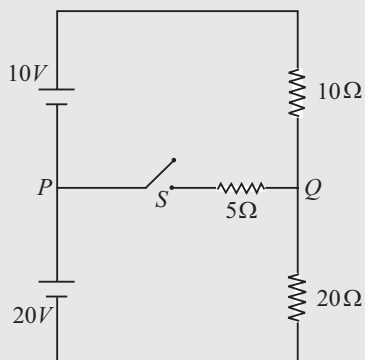


11. A car battery has emf ε and internal resistance $r = 0.05 \Omega$. When the light of the car is switched on, current through the lamp is 5.0 A and the voltage across the terminals of the battery is 12.0 V.

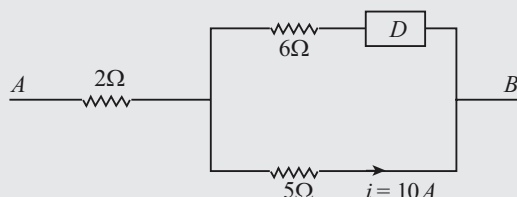


Now, the starter motor of the car is switched on by closing the switch S . The current in the lamp drops to 4.5 A (and the light gets dim). Find the emf of the battery and the current through the starting motor when the lights are on.

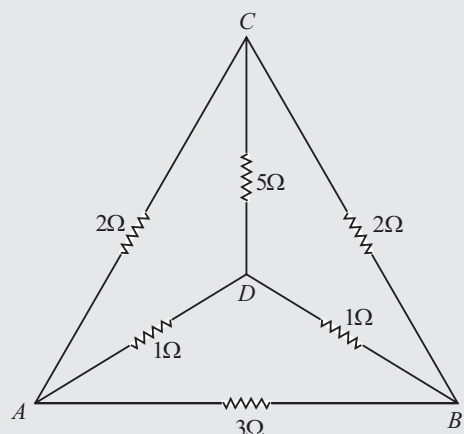
12. Find the change in power dissipated in the $10\ \Omega$ resistor after the switch S is closed.



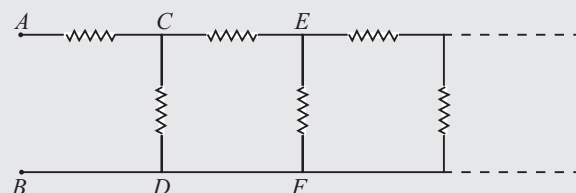
13. The diagram shows a part of an electric circuit. The potential difference between the points A and B is 80V and the current in $5\ \Omega$ resistor is 10A . There is an unknown device D inside the box shown. Is the device absorbing energy or supplying energy to the circuit, and at what rate?



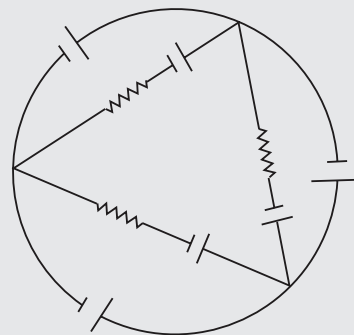
14. For the resistor network shown in the figure, find the equivalent resistance between points (a) A and B , (b) C and D .



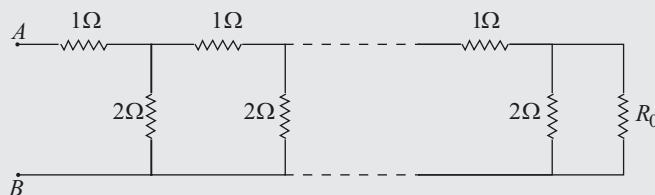
15. Find the effective resistance between points A and B of an infinite chain of resistors shown in the figure. Each resistor has resistance R .



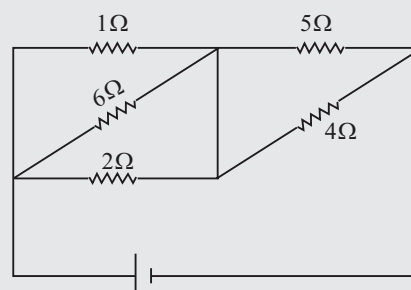
16. In the circuit shown, all cells are nearly ideal and have an emf of $\varepsilon = 2\text{V}$. Each resistance is $4\ \Omega$. Find the current in each of the resistances.



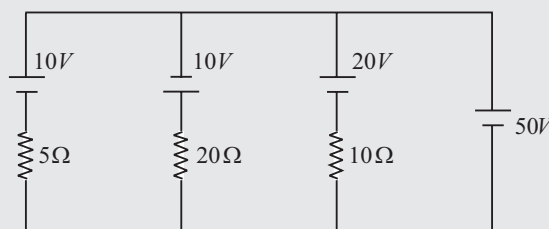
17. A chain of resistances has been shown in the figure. For what value of resistance R_0 , the equivalent resistance between A and B is independent of number of chain links in the network?



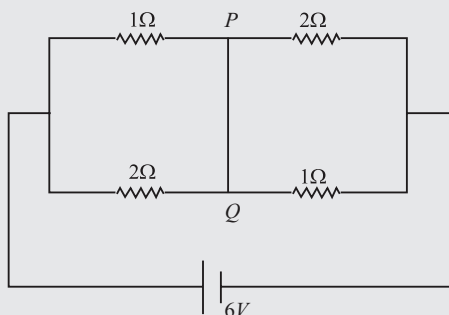
18. In the circuit shown in the figure, identify the resistor in which maximum power is dissipated.



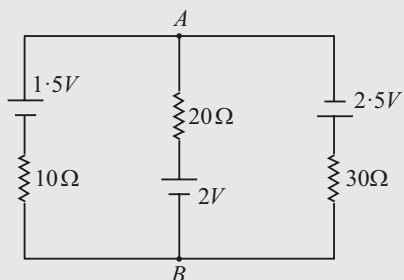
19. In the circuit shown in the figure, find the rate of energy loss for 50V cell and 10V cell.



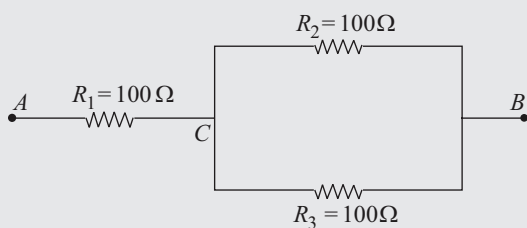
20. In the circuit shown find the current in the path PQ .



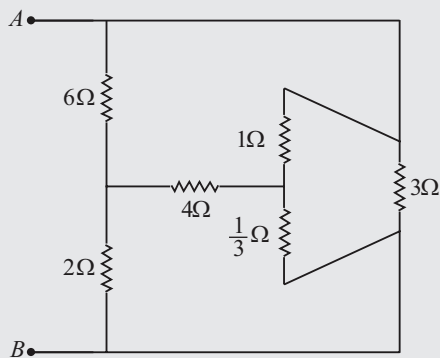
21. Using the formula for parallel combination of cells, find the potential difference between the points A and B in the figure shown.



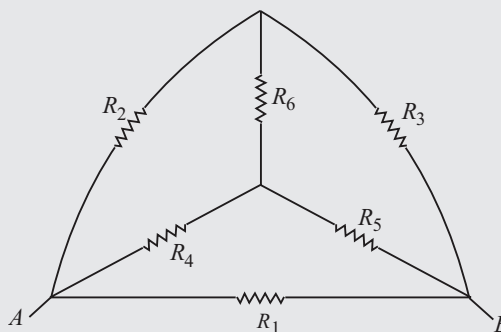
22. Three $100\ \Omega$ resistors are connected as shown. The maximum power dissipation allowed in each of the three resistors is $36\ \text{W}$.



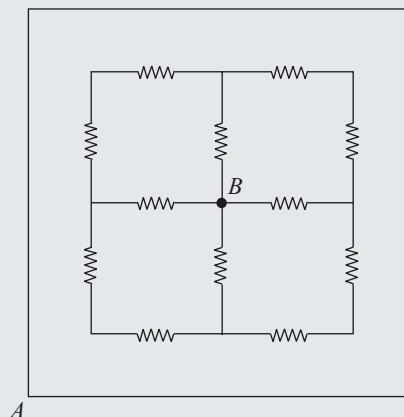
- (a) Find the maximum voltage (V_0) permissible between the terminals A and B .
 (b) Find the total power dissipated in the circuit when the maximum allowed voltage V_0 is applied.
23. Find the equivalent resistance of the given network between the points A and B .



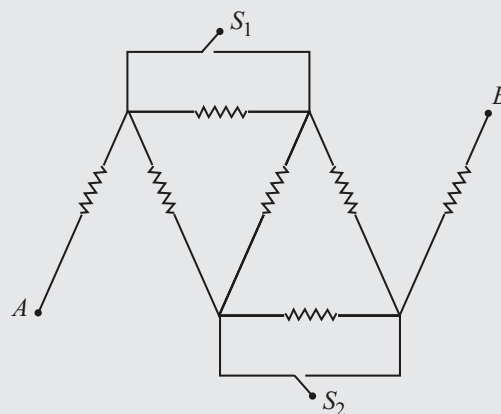
24. In the network shown, each resistor has resistance R . Find the equivalent resistance between the points A and B .



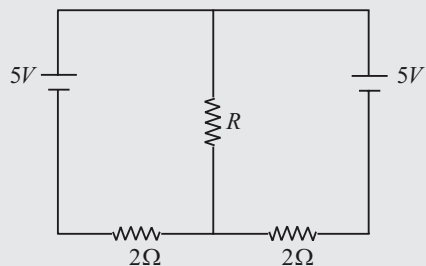
25. In the circuit shown in the figure, resistance of each resistor is R . Find the equivalent resistance of the network between the points A and B .



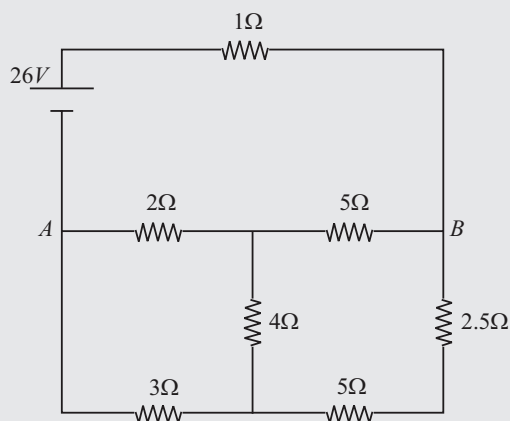
26. Each resistance, in the network shown, has value R . With both the switches open, the equivalent resistance between the points A and B is R_0 . Find the equivalent resistance between the points A and B after the two switches are closed.



27. Find value of R so that the power dissipated in it is maximum.

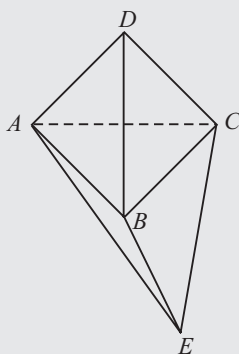


28. Find the potential difference between the points A and B in the figure shown.



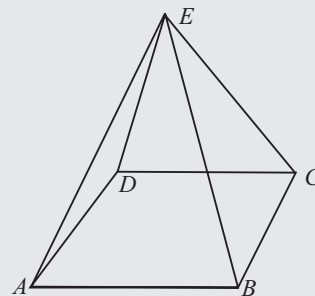
29. Figure shows a triangular bipyramid made of wire. Each arm has resistance R . Find the equivalent resistance between

- (a) The points A and C .
(b) The points D and E .

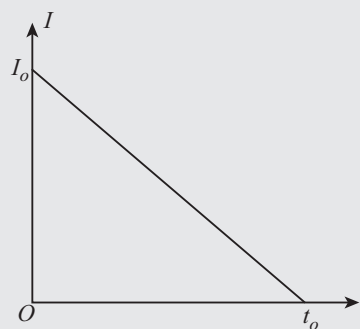


30. Figure shows a wire frame in the shape of a square pyramid. Each side has a resistance R .

- (a) Find current in path EC when an ideal cell of emf V is connected across AB .
(b) Find current in path EA when an ideal cell is connected across AC .



31. A charge q is passed through a resistance R such that the current decreases linearly to zero in time t_0 . Find the heat generated in the resistance.



Answer Sheet

Your Turn

1. 20C, 1.25×10^{20}
2. 1.06 mA
3. 8C
4. 16C.
5. Both will get doubled.
6. The one that is thin and long
7. 4R
8. 0.4%
9. 3.1g
10. 62.5°C
11. $(69 \pm 10\%) \Omega$
12. $4.25 \times 10^{-3} \text{ Vm}^{-1}$
13. 5W
14. (a) 529 Ω , (b) $\frac{10}{23} \text{ A}$ (c) 75.6 W
15. 500 W
16. Power output decreases 9%
17. Due to heating
18. $\frac{20}{11} \text{ A}$, $\frac{90}{11} \text{ V}$
19. 1A
20. B_3 is brightest and B_2 is least bright
21. 18V, 3A
22. $\frac{R}{3}$ in all cases
23. $\frac{16}{3} \text{ A}$
24. V_2 is larger
25. 480 J
26. $\frac{45}{7} \text{ A}$
27. 2A and zero
28. 40 Ω , 0.1 Ω
29. $\frac{V}{4}$
30. zero, $\frac{25}{3} \text{ A}$
31. 52V
32. $\frac{44}{7} \text{ V}$
33. zero
34. $\frac{52}{57} \Omega$
35. Zero
36. 5V
37. zero, $\frac{40}{3} \text{ A}$
38. cell A is getting recharged. $V_A = 4\text{V}$
40. 15h
41. 9W, 3V
42. 3.0 V, 0.75 Ω
43. 0.4 Ω
44. 2 V, 0.5 Ω

Worksheet 1

1. (d)
2. (c)
3. (d)
4. (a)
5. (b)
6. (d)
7. (b)
8. (d)
9. (b)
10. (c)
11. (c)
12. (c)
13. (b)
14. (a)
15. (b)
16. (a)
17. (c)
18. (c)
19. (a)
20. (a)
21. (a)
22. (c)
23. (a)
24. (c)
25. (b)
26. (d)
27. (a)
28. (a)
29. (c)
30. (a)
31. (b)
32. (a)
33. (d)
34. (c)
35. (b)
36. (c)
37. (c)
38. (a)
39. (d)
40. (a)
41. (c)
42. (b)
43. (b)
44. (a)
45. (a)
46. (b)
47. (b)
48. (d)
49. (a)
50. (b)
51. (d)
52. (d)
53. (a)
54. (b)
55. (b)
56. (a)

Worksheet 2

1. (c, d)
2. (a, d)
3. (a, c)
4. (a, b, c, d)
5. (a, b, d)
6. (a, b, c)
7. (b, c, d)
8. (a, d)
9. (b, d)
10. (a, c)
11. (a, b)
12. (a, b, d)
13. (d)

Worksheet 3

1. $1.8 \times 10^{-4} \frac{m}{s}$
2. 849°C
3. $\frac{5}{3} \text{ W}$
4. (a) $\frac{\rho L}{\pi(b^2 - a^2)}$ (b) $\frac{\rho}{2\pi L} \ln\left(\frac{b}{a}\right)$
5. $2.5 \times 10^{-5} \text{ cm s}^{-1}$
6. Percentage change in length = 0.0017%
Percentage change in area = 0.0034%
Percentage change in resistance = 0.39%
7. $\frac{10}{3} \Omega$ and $\frac{80}{3} \Omega$
8. $\frac{L}{\pi \left[\frac{r^2}{\rho_C} + \frac{R^2 - r^2}{\rho_N} \right]}$
9. $\frac{q_0 k}{4\pi R^2}$ in radial direction
10. $n = 10$
11. 12.25 V, 24.5A
12. No charge
13. Absorbing energy at a rate 100 W
14. (a) $\frac{12}{13} \Omega$, (b) $\frac{15}{13} \Omega$
15. $\left(\frac{1 + \sqrt{5}}{2} \right) R$
16. Zero
17. 2Ω
18. 4Ω
19. 700 W, 30 W
21. $\frac{10}{11} V$
22. (a) 90 V (b) 54 W
23. $\frac{24}{29} \Omega$
24. $\frac{R}{2}$
25. $\frac{3R}{8}$
26. $\frac{7R_0}{9}$
27. 1Ω
28. $V_A - V_B = -21 \text{ V}$
29. (a) $\frac{2R}{5}$, (b) $\frac{2R}{3}$
30. (a) $\frac{V}{8R}$ (b) $\frac{V}{2R}$
31. $\frac{4 q^2 R}{3 t_0}$

Basic Electrical Measurement

“An experiment is a question which science poses to Nature, and a measurement is the recording of Nature’s answer.”

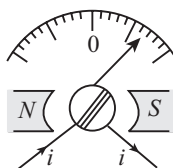
—Max Planck

1. INTRODUCTION

In this chapter, we will learn basic methods to measure current, voltage and resistance. We will discuss measuring instruments like ammeter, voltmeter, meter bridge, compact Wheatstone bridge and potentiometer. One essential component in nearly all measuring instruments is a galvanometer.

2. GALVANOMETER

A galvanometer is a device which can detect electric current. It has a coil placed between the poles of a magnet. When a current is passed through the coil, it experiences a torque and rotates. There is a pointer attached to the coil. This pointer moves, with the coil, over a scale. The pointer deflects towards left if the current is passed in one direction and towards right if the direction of current through the coil is reversed.



A coil wrapped on a cylinder is placed between poles of a magnet.

We will discuss the construction and working of a galvanometer, in detail, in a different chapter. Right now, it is sufficient to learn that it is a device in which a pointer moves on a scale when a current is passed. The pointer can move either way (though we can also make galvanometers in which deflection is permitted in one direction only). Terminals of a galvanometer are not marked + and –. Higher the current, higher is the deflection of the pointer.

It is a very sensitive instrument and can take only small currents. The value of current in the galvanometer coil for which the pointer deflects to its extreme position is known as its *full scale deflection current* (i_g). Another important parameter of a galvanometer, essential in design of various instruments using a galvanometer, is the resistance of its coil. We will call this as *resistance of the galvanometer* (R_g).

When we say that we have a 20Ω , 2mA galvanometer, we are essentially saying that the resistance of our galvanometer is $R_g = 20\Omega$ and its full scale deflection current is $i_g = 2\text{mA}$.

A sensitive galvanometer can even detect a current as small as 10^{-9}A .

Example 1 When a potential difference of 100 mV is applied to the terminals of a galvanometer, the pointer shows full scale deflection of 4mA. Find the resistance of the galvanometer.

Solution

$$R_g = \frac{V}{i} = \frac{100\text{mV}}{4\text{mA}} = 25\Omega.$$

3. AMMETER

Currents in different circuits may vary widely. We cannot use one single device to measure all currents. An ammeter is a device used to measure the current in a circuit, and for different purposes, we need ammeters of different ranges. For example if we have to measure a current of the order of 0.1 A, we may use an ammeter of 0–1 A range. Similarly, for measuring current of the order of 1.0A, we may choose an ammeter of 0–10 A range. By making a small modification in a galvanometer, we can get an ammeter of desired range.

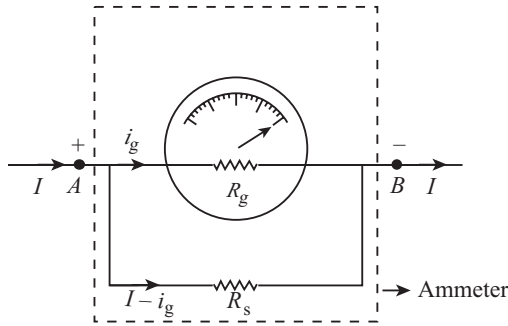
We take a galvanometer in which deflection of the coil is permitted in one direction only. Current is permitted in one direction only; it should always enter through the terminal marked as +. Let’s assume that the resistance of the galvanometer is R_g and the full scale deflection current is i_g . We wish to design an ammeter which can read any current from zero to I . For this, the pointer should show full scale deflection when current through the ammeter is I . This can happen if we add a small resistance in parallel to the galvanometer coil. This small resistance is known as *shunt*.

7.2 Electrostatics and Current Electricity

The figure given below shows the desired ammeter made by connecting a shunt resistance (R_s) in parallel with the galvanometer coil. When a current I is passed through the ammeter, it gets divided into two parallel paths. We have chosen R_s such that exactly i_g current passes through the galvanometer coil and rest of the current ($I - i_g$) passes through R_s . Obviously,

$$(I - i_g)R_s = i_g R_g$$

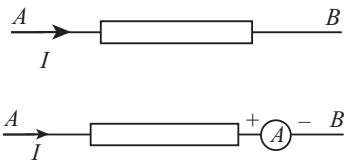
$$\text{or, } R_s = \frac{i_g R_g}{I - i_g} \quad (1)$$



A and B are + and - terminals of the ammeter.

By supplying known currents to the ammeter (in the range zero to I), we can graduate the scale. For example, we supply 1A current and mark the position of the pointer as 1.0A. Similarly, we supply a known current 2A and mark 2.0A at the position of the pointer and so on. Now, we have an ammeter ready to measure any current in the range 0 to I .

An ammeter is always connected in series in the segment of a circuit where we wish to measure current. The current should always enter through the terminal marked as + in the ammeter. The figure below shows a segment AB of a circuit in which we wish to measure current. The ammeter should be connected in series as shown.



This is how we should place our ammeter to measure current in branch AB.

As soon as an ammeter is inserted in a branch in an electrical circuit, it increases the resistance of the branch. This leads to a drop in the value of the current. Therefore, *the reading of a real-life ammeter will always be less than the true value of the current.* However, we try to keep the resistance of an ammeter as small as possible. Smaller the resistance of the ammeter, smaller will be the change in current when the ammeter is added to the circuit. *Ideally, we wish our ammeter to have zero resistance.*

Effective resistance of the ammeter described above is given by

$$\frac{1}{R_A} = \frac{1}{R_g} + \frac{1}{R_s}$$

$$\text{or, } R_A = \frac{R_g R_s}{R_g + R_s} \quad (2)$$

When $R_s \ll R_g$, we have $R_A \simeq R_s$.

Example 2 Convert a galvanometer to an ammeter.

A 20Ω , 1.0 mA galvanometer is used to make an ammeter of range 0 to 100.0 mA.

- Find the value of shunt resistance needed.
- Find the resistance of the ammeter.
- Find the current through the shunt resistance when the ammeter is used to measure a current of 50.0 mA.

Solution

Concepts

- Value of shunt resistance can be calculated by direct use of equation (1).
- R_A can be calculated by obtaining the equivalent of R_g and R_s in parallel.
- The current is always divided between the coil and the shunt so that the potential difference across both of them is the same.

Full scale deflection current, $i_g = 1.0$ mA

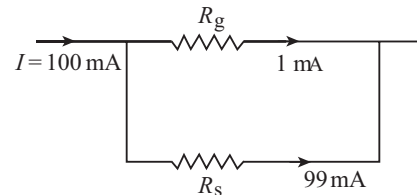
Resistance of the galvanometer coil, $R_g = 20 \Omega$

The desired ammeter shall measure up to $I = 100.0$ mA.

$$\begin{aligned} \text{(a) } \therefore R_s &= \frac{i_g R_g}{I - i_g} \\ &= \frac{1 \times 20}{100 - 1} \simeq \frac{20}{100} = 0.2 \Omega \end{aligned}$$

Here, we have used the approximation $I - i_g = 100 - 1 \simeq 100$ for the ease of calculation.

Note: When a current of 100 mA is fed to the ammeter, only 1mA passes through the galvanometer coil and remaining 99mA is bypassed through the shunt.



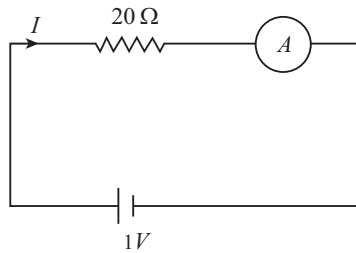
$$\begin{aligned} \text{(b) } R_A &= \frac{R_g R_s}{R_g + R_s} = \frac{R_s}{1 + \frac{R_s}{R_g}} \\ &= \frac{0.2}{1 + \frac{0.2}{20}} \approx 0.2 \Omega \end{aligned}$$

Again we have used $1 + \frac{0.2}{20} \simeq 1$ for ease of calculation.

- (c) The current will always get divided between R_g and R_s in inverse ratio of the resistance. When 50.0 mA current is passed through the ammeter, 0.5mA will pass through the galvanometer coil (and it will show half the full scale deflection) and remaining 49.5 mA will pass through R_s .

Example 3 Ammeter reading is less than true reading

The ammeter described in example 2 is used to measure current I in the circuit shown in the figure. Calculate the percentage difference between the true current (i.e. current before insertion of the ammeter) and the reading of the ammeter.



Solution

Concepts

After insertion of the ammeter, the resistance increases and the current drops.

Current, without ammeter in the circuit, is $I = \frac{1V}{20\Omega} = 0.05 \text{ A} = 50 \text{ mA}$

From the last example, resistance of the ammeter is $R_A = 0.2\Omega$. With ammeter in the circuit the current is

$$i = \frac{1}{20 + 0.2} \simeq 0.0495 \text{ A} = 49.5 \text{ mA}.$$

$$\begin{aligned} \text{Percentage change} &= \frac{I - i}{I} \times 100 \\ &= \frac{50 - 49.5}{50} \times 100 = 1\% \end{aligned}$$



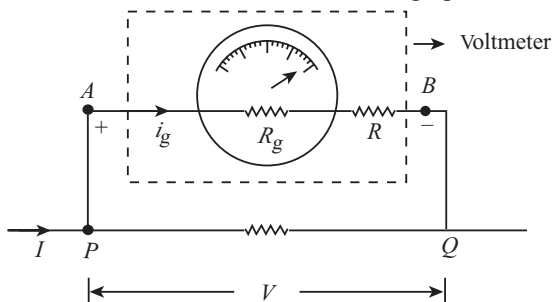
YOUR TURN

- Q.1** What should be the resistance of an ideal ammeter which does not disturb the circuit when connected into it?
- Q.2** How will you make an ammeter of range 0 to 10A, using a galvanometer having full scale deflection

current of 1mA and coil resistance of 20Ω. Compare your answer with the result obtained in example 2. What is your observation?

4. VOLTMETER

A voltmeter is an instrument used to measure potential difference between two points in an electric circuit. We can connect a high resistance in series with a galvanometer coil to convert it into a voltmeter. The figure given below shows a voltmeter with terminals A and B . Terminals are marked $+$ and $-$, respectively. If we desire to measure the potential difference between points P and Q in a circuit, the voltmeter is connected in parallel across P and Q . The positive terminal of the voltmeter must be connected to the high potential end.



A high resistance R is connected in series with the galvanometer coil to convert it into a voltmeter.

In the figure, R is the high resistance connected in series with the galvanometer coil. Suppose we wish to make a voltmeter that can measure from 0 to V volt. We need to choose R such that the current through the galvanometer is i_g (full scale deflection current) when the potential difference across the terminals is V . This will ensure that the pointer is at extreme deflection when potential difference V is applied across the voltmeter.

$$i_g (R_g + R) = V$$

$$\text{or,} \quad R = \frac{V}{i_g} - R_g \quad (3)$$

The scale of the instrument can be calibrated by applying known potential differences and marking the positions of the pointer.

The effective resistance of the voltmeter is

$$R_v = R + R_g \quad (4)$$

When a voltmeter is connected in a circuit, it disturbs the circuit. Some part of the circuit current (see the previous figure) is diverted through the voltmeter. Therefore, the

7.4 Electrostatics and Current Electricity

potential difference to be measured (across P and Q) drops. The voltmeter will always record this reduced potential difference. Therefore, the reading of a voltmeter is always less than the true value. *An ideal voltmeter is one which has an infinite resistance so that it does not alter the current in the original circuit.*

Example 4 Convert a galvanometer to a voltmeter

A galvanometer has 50 divisions on its scale. Deflection of its pointer increases by 1 division when current is increased by 0.5 mA. Resistance of the galvanometer is 10Ω . How will you design a voltmeter by using this galvanometer which can measure voltages from 0 to 50 V?

Solution

Concepts

- Full scale deflection current (i_g) is the current needed to cause the pointer deflect by 50 divisions.
- Using equation (3) will give the answer:

$$i_g = 50 \times 0.5 = 25\text{mA}.$$

$$R_g = 10\Omega$$

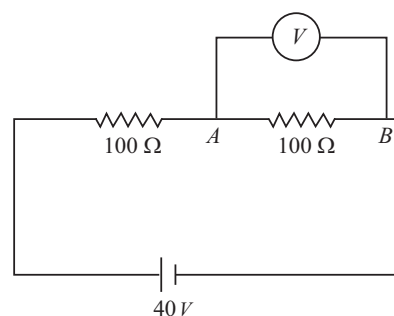
Resistance (R) to be connected in series to convert the galvanometer into a voltmeter of desired range is

$$R = \frac{V}{i_g} - R_g = \frac{50\text{V}}{25\text{mA}} - 10$$

$$= 2000 - 10 = 1990\Omega.$$

Example 1 Voltmeter reading is less than true reading

A voltmeter has resistance $R_v = 2000\Omega$. It is used to measure potential difference between points A and B in the circuit shown. Find the percentage difference in the true value and the recorded value.



Solution

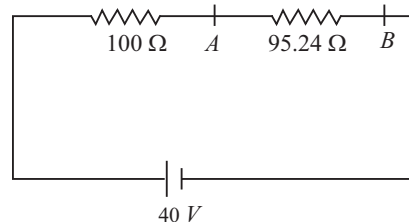
Concepts

- The effective resistance between A and B is parallel combination of 100Ω and 2000Ω .
- Potential drop across A and B gets changed due to insertion of the voltmeter.

Higher the resistance of the voltmeter (with respect to 100Ω), lesser will be the effect.

$$R_{AB} = \frac{100 \times 2000}{100 + 2000} = 95.24\Omega$$

$$V_{AB} = \left(\frac{95.24}{100 + 95.24} \right) \times 40 = 19.51\text{ V}$$



When there is no voltmeter, the true potential difference across A and B is 20V. Percentage difference in true value and recorded value is

$$\frac{20 - 19.51}{20} \times 100 = 2.45\%$$



YOUR TURN

Q.3 Calculate the resistance that should be added to a 1Ω , 5mA galvanometer so as to convert it into a voltmeter of range 0–5V.

Q.4 A cheap voltmeter with internal resistance 30Ω is connected across the terminals of a cell of emf 1.5V and internal resistance 5Ω . Find the reading of the voltmeter.

In Short

- (i) A galvanometer is a sensitive device which can handle small currents only. The deflection of the pointer is proportional to the current provided to the galvanometer.
- (ii) Important specifications of a galvanometer are its full scale deflection current (i_g) and its resistance (R_g).
- (iii) A galvanometer can be converted into an ammeter by adding a small shunt resistance in parallel to the galvanometer and recalibrating the galvanometer scale. To make an ammeter which can measure current upto I , the value of shunt resistance should be

$$R_s = \frac{i_g R_g}{I - i_g}.$$

- (iv) An ideal ammeter should have zero resistance.
- (v) Reading of a real-life ammeter is always less than the true current in the circuit before the introduction of the ammeter.
- (vi) A galvanometer can be converted into a voltmeter by adding a large resistance in series with the galvanometer and recalibrating the scale. To make a voltmeter that can read upto V volt, the resistance to be added should be

$$R = \frac{V}{i_g} - R_g.$$

- (vii) An ideal voltmeter must have infinite resistance.
- (viii) Reading of a real voltmeter is always less than the true potential difference before the introduction of the meter.

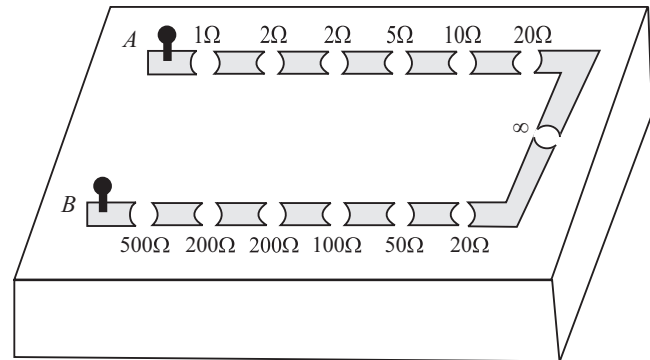
5. RESISTANCE BOX

A resistance box gives you an option of choosing a resistance of desired value. A typical resistance box has a thick conducting strip laid over a wooden box.

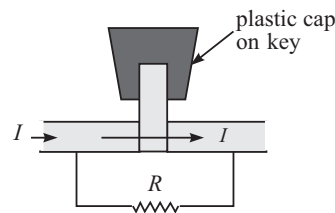
There are slots at regular intervals on this strip. These slots have tight fitting conducting keys. Each key has a plastic cap. Holding these caps, one can take out the keys or push them back into the slots. When all keys are in place, the entire strip is a continuous conductor with negligible resistance. When you plug out a key (say, the one which is in a slot marked as $5\ \Omega$), you create a gap. But inside the box, there is a $5\ \Omega$ resistance bridging the gap. Therefore, the whole strip acts like a resistance of $5\ \Omega$. If you take out keys from the slots marked as $10\ \Omega$ and $5\ \Omega$, then the entire arrangement will act like a resistance of $15\ \Omega$ ($10\ \Omega$ and $5\ \Omega$ in series) between the terminals of the resistance box.

The figure given below depicts a resistance box (with all keys removed). The figure also explains how a resistance

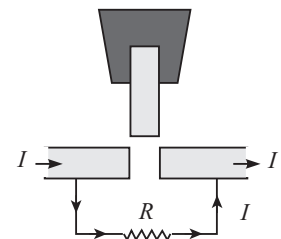
gets activated when a key is removed. The resistance box may have a key marked as infinity. If you remove this key, a permanent gap is created as there is no resistance beneath the key.



A and B are two terminals of the resistance box



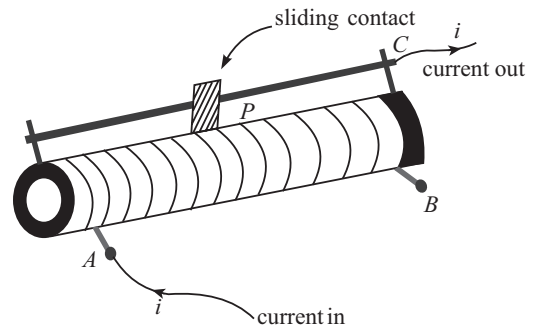
Key plugged in: The effective resistance is zero. Thick conductor offers almost zero resistance to current I .



Key taken out: The effective resistance seen by the current is I .

6. RHEOSTAT

Resistors are the most omnipresent components in electric circuits. When we need a variable resistor, a rheostat is a preferred device. A common type of rheostat has a long wire wrapped on a cylinder. The two ends of the wire are connected to the terminals A and B . A sliding contact can touch the wire at a desired location. This sliding contact is connected to terminal C (see the figure).



The rheostat is connected in a circuit using terminal C and any one of the terminals A and B . If you connect the

rheostat using terminals A and C , then the resistance of the wire between A and P is being used in the circuit. When the contact is moved towards the end B , the length of the wire inside the circuit increases. Therefore, the value of resistance offered by the device to the circuit increases. When the contact is moved towards the end A , the resistance offered by it decreases.

Actually, it is nothing but a long wire which has been wrapped on a cylinder to accommodate it in a small space. Symbol of a rheostat in an electric circuit is as shown in the figure.



These two symbols are used for variable resistance in a circuit.

7. VERIFYING OHM'S LAW AND MEASURING A RESISTANCE

The figure shows a circuit used to verify Ohm's law. A resistor R is connected to a power source through a switch (S) and a rheostat.

An ammeter (A) in the circuit measures current through R , and a voltmeter (V) measures the potential difference across R . We will assume that the meters are ideal.

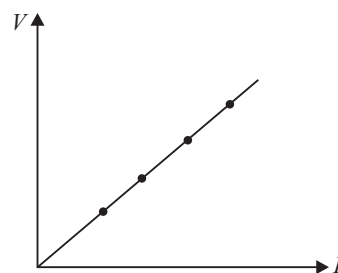
The switch is put on and the readings of the ammeter and the voltmeter are

recorded. Let the readings be I_1 and V_1 . The ratio $\frac{V_1}{I_1}$ gives

the value of resistance (call it R_1). Now, we can change the current in the circuit by changing the rheostat resistance. [If you do not have a rheostat in the circuit, everytime you wish to change the current, you will need to change the source itself or insert additional resistance into the circuit.]

Again record the readings (I_2 and V_2) of the ammeter and the voltmeter. The ratio $\frac{V_2}{I_2}$ gives resistance (call it R_2). This way we measure the resistance few times. If $R_1 = R_2 = R_3 = \dots$, then it implies that Ohm's law is correct.

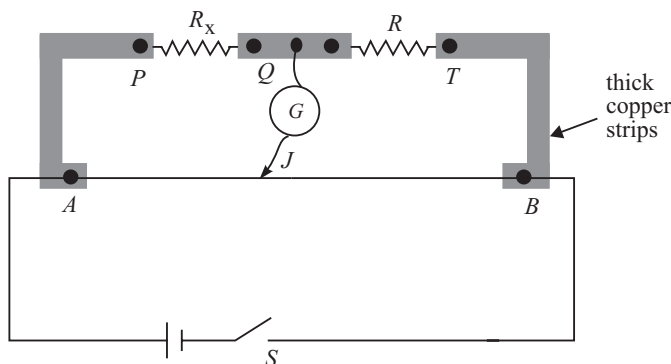
One can also plot a graph of V vs I using the set of data obtained. A straight line graph confirms the validity of Ohm's law (i.e. $V \propto I$).



In practice, the values $R_1, R_2, R_3, \dots$ will be slightly different from one another. The resistance R should be taken as average of $R_1, R_2, R_3, \dots$.

8. METER BRIDGE

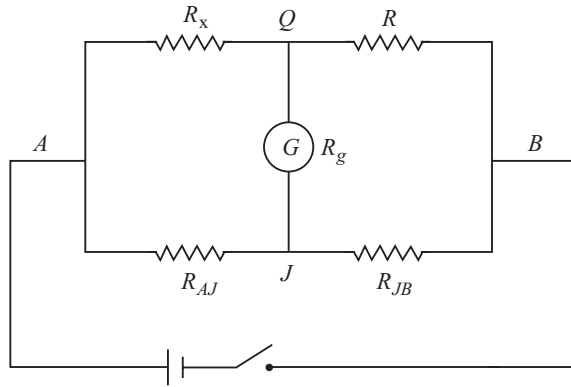
A meter bridge works on the principle of a Wheatstone bridge and is used to measure an unknown resistance. It has a 1 m long wire stretched on a wooden platform. The wire is usually made of an alloy called Manganin due to its high resistivity and low temperature coefficient. In the figure, AB is the meter bridge wire. Thick copper strips are used to provide resistance-free paths, wherever possible.



There are two slots – one between points P and Q and the other between points Q and R . We connect the unknown resistance (R_x) in the first slot and a known resistance (R) is connected in the second slot. One terminal of a galvanometer (G) is also connected at point Q . The other terminal of the galvanometer is connected to a sliding contact ("jockey") on the wire AB . There is a meter scale beneath the wire AB and any length on the wire can be read easily.

The jockey divides the wire AB into two segments. The resistances of the segments AJ and JB are in ratio of their lengths. The circuit is a Wheatstone bridge. The fifth resistance is the galvanometer itself.

The power supply is switched on. The sliding contact is moved on the wire to a position such that the galvanometer shows zero deflection. In this position the Wheatstone bridge is balanced. Length AJ is measured.



Let it be x cm.

$$\frac{R_x}{R} = \frac{R_{AJ}}{R_{JB}}$$

Since $\frac{R_{AJ}}{R_{JB}} = \frac{x}{(100-x)}$

$$\therefore \frac{R_x}{R} = \frac{x}{100-x}$$

or, $R_x = R \left(\frac{x}{100-x} \right)$ (5)

End correction

The length of wire AB is 100 cm, but while making connections at A and B , there will be some extra length used for making the connection. In simple language, the effective length of AB will be slightly higher than 100 cm. The end corrections can be measured using two known resistances R_1 and R_2 in place of R_x and R . If the galvanometer shows zero reading when $AJ = x_1$, then

$$\frac{R_1}{R_2} = \frac{x_1 + a}{100 - x_1 + b} \quad (i)$$

where a and b are the end corrections at the ends A and B , respectively.

Position of R_1 and R_2 is interchanged and this time the null point is obtained for $AJ = x_2$.

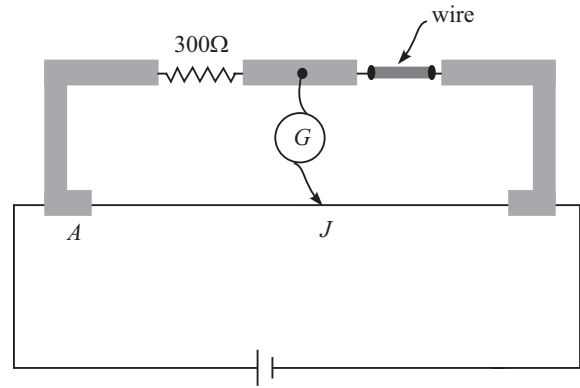
$$\frac{R_2}{R_1} = \frac{x_2 + a}{100 - x_2 + b} \quad (ii)$$

Equations (i) and (ii) can be solved to get the values of the end corrections.

Example 6 Specific resistance of material

A wire has diameter $d = 1$ mm and its length is $L = 31.4$ cm. This wire is connected in one of the slots of a meter bridge and a known resistance of 300Ω is placed in the other slot.

Under balanced condition, the length AJ is measured to be 75.0 cm. Find the specific resistance of the material of the wire. Ignore end corrections.



Solution

Concepts

- The resistance of the wire can be calculated using the meter bridge principle.
- $R = \frac{\rho L}{A}$ can be used to find ρ .

Let R be the resistance of the wire.

When the bridge is balanced:

$$\frac{300}{R} = \frac{x}{100-x}$$

or, $\frac{300}{R} = \frac{75}{100-75}$

or, $R = 100\Omega$.

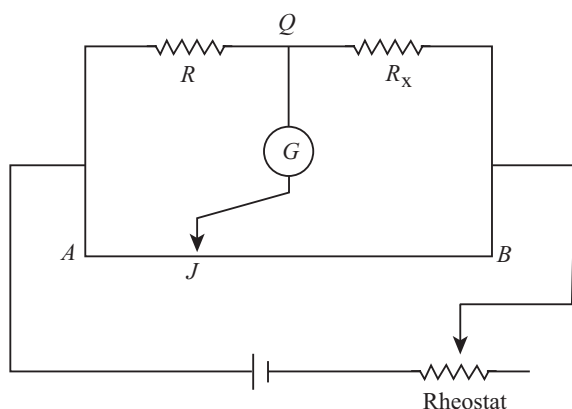
For the wire $R = \frac{\rho L}{A}$

or, $\rho = \frac{AR}{L} = \frac{\pi d^2 \cdot R}{4L}$

$$= \frac{3.14 \times (10^{-3})^2 \times 100}{4 \times 0.314} = 2.5 \times 10^{-4} \Omega\text{-m.}$$

Example 7 In the meter bridge circuit, R is a known resistance and R_x is unknown. When the jockey is at J (distance $AJ = x$), the galvanometer shows zero reading.

- How will the value of x change if the rheostat resistance is increased?
- When a 10Ω resistance is connected in parallel to R , the null point shifts. Will the x be larger or smaller now? If the change in value of x is 10cm, find R_x . The original value of x was 20cm.

**Solution****Concepts**

- When we add a resistance in parallel to R , the effective resistance between the points A and Q reduces.
- By changing the rheostat resistance, we can change the potential difference across A and B . But this has no bearing on the null point. The null point gets decided by the ratio of the resistances.

(a) Value of x will not change.

$$(b) \frac{R}{R_x} = \frac{x}{100-x}$$

When a 10Ω resistance is added in parallel to R , the effective resistance between A and Q reduces. Therefore, x will also reduce.

It means that the distance AJ reduces from 20cm to 10cm after the resistance is added.

$$\frac{R}{R_x} = \frac{20}{100-20} \Rightarrow R_x = 4R \quad (i)$$

After adding 10Ω resistance,

$$\frac{(10)(R)}{\frac{10+R}{R_x}} = \frac{10}{100-10}$$

$$\text{or, } \left(\frac{R}{10+R} \right) \times 90 = R_x$$

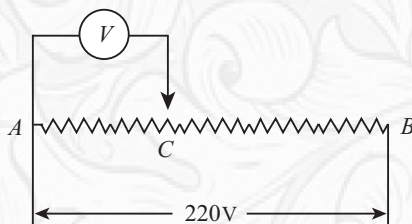
$$\text{or, } \frac{90R}{10+R} = 4R \quad [\text{using (i)}]$$

$$\text{or, } R = \frac{50}{4} \Omega$$

$$\therefore R_x = 4R = 50\Omega.$$

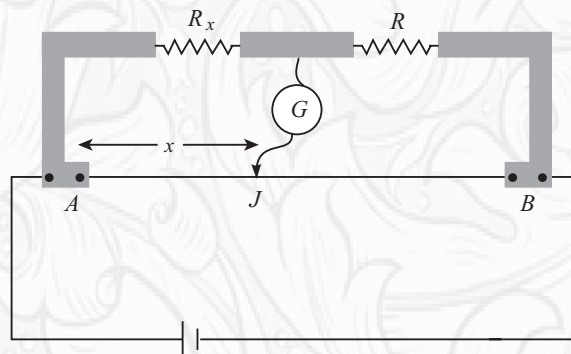
**YOUR TURN**

Q.5 AB is a $12k\Omega$ rheostat. A potential difference of 220V is maintained across it. A voltmeter is connected to it as shown. The length AC is $\left(\frac{1}{4}\right)^{th}$ the length of the rheostat.



- Find the reading of the voltmeter assuming it to be ideal.
- Find the reading of the voltmeter if its resistance is $6k\Omega$.

Q.6 In the meter bridge circuit shown in the figure, R_x is an unknown resistance and value of resistance R is 100Ω . The balance point is obtained for length $x = 40\text{cm}$.

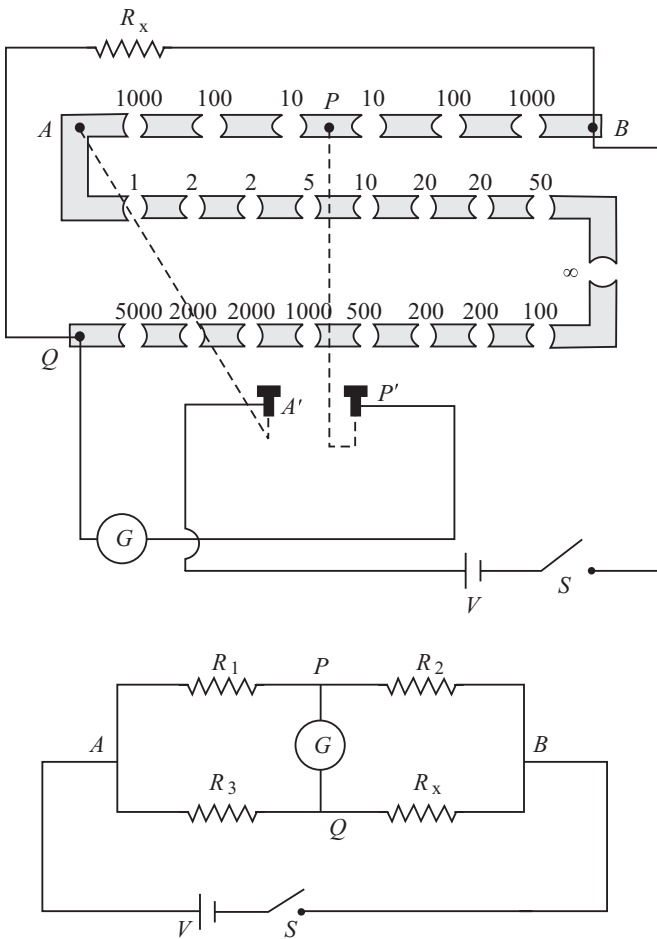


- Find R_x assuming ideal conditions.
- Find R_x assuming an end correction of 4cm at end A and 2cm at end B .

9. COMPACT WHEATSTONE BRIDGE (POST OFFICE BOX)

A post office (P.O.) box is an electrical fault detecting device that was used in telecommunication industry and was usually found in post offices in old times. The device used the principle of the Wheatstone bridge. A similar looking compact Wheatstone bridge is used in school laboratories to measure an unknown resistance.

Similar to a resistance box, it has slotted conducting strip pasted on a wooden box. There are conducting keys in each slot and removal of a key introduces a resistance.



First have a look at the second figure showing a Wheatstone bridge with an unknown resistance R_x . We will make exactly same circuit using the P.O. box. We choose resistance R_1 between A and P. If we remove the 10 Ω key, we have chosen $R_1 = 10\Omega$ and if we have removed the 100 Ω key, we have chosen $R_1 = 100\Omega$.

Similarly, we choose R_2 by removing a key between P and B in the box. The arms AP and PB are known as ratio arms of the bridge. The two rows shown between A and Q are for choosing the resistance R_3 . You can assign any integral value to R_3 from zero to 11,110 Ω . And of course you can

choose R_3 to be infinity by simply removing the key marked as infinity. The resistance to be measured (R_x) is connected as the 4th resistance of the Wheatstone bridge between Q and B. Now we want to connect a galvanometer between the points P and Q. For the ease of making connections, there is a key P' in front of the box which is internally connected to P. Anything that you want to connect to P should be connected at P'. The galvanometer is connected between the points P' and Q. Similarly, anything that is to be connected to A shall be actually connected to the key at A'. A' is internally connected to A. We connect a battery and a switch (S) between A' and B. We have completed a Wheatstone bridge circuit.

Before we start our measurements, we wish to be doubly sure that our connections are correct. To check this, set $R_1 = R_2 = 10\Omega$ and $R_3 = 0$. Put on the switch. Notice the deflection in the galvanometer. Now, switch off the circuit, make $R_3 = \infty$ and switch on again. If the deflection in the galvanometer is in a direction opposite to the earlier deflection, our connections are likely to be correct. Now, we are ready for measurements.

Take $R_1 = R_2 = 10\Omega$ and try with different values of R_3 till you notice near zero deflection in the galvanometer. Most of the time, you will not get exact zero deflection. For example, you may notice that the galvanometer shows a small deflection in one direction when $R_3 = 63\Omega$ and shows a small deflection in other direction when $R_3 = 64\Omega$. For exact balance, value of R_3 required is somewhere between 63.0 Ω and 64.0 Ω . But we can only have integral values of R_3 ! So all we can say is that value of R_3 required for balance

is somewhere between 63 Ω and 64 Ω . Since $\frac{R_x}{R_3} = \frac{R_2}{R_1} = 1$,

we can say that R_x lies between 63.0 Ω and 64.0 Ω .

Now, change R_1 to 100 Ω and leave R_2 at 10 Ω . Again try finding R_3 for which you get a balance. You may again find deflections in two directions for, say, $R_3 = 634\Omega$ and 635 Ω .

This time $\frac{R_x}{R_3} = \frac{R_2}{R_1} = \frac{1}{10}$. Hence, $63.4 < R_x < 63.5\Omega$.

Now, take $R_1 = 1000\Omega$ and $R_2 = 10\Omega$. Repeat the exercise. The following table summarizes the readings obtained in a typical experiment and the value of R_x . Note that the value of R_x is very precise. It is difficult to measure the value of a resistance to such a precision using a device such as a meter bridge.

R_1	R_2	R_1/R_2	R_3 for balance	R_x
10 Ω	10 Ω	1	63 – 64	63.0 – 64.0 Ω
100 Ω	10 Ω	10	634 – 635	63.4 – 63.5 Ω
1000 Ω	10 Ω	100	6347 – 6348	63.47 – 63.48 Ω

Readings in a P.O. box experiment. Measured value of R_x will be reported as 63.475 Ω .



YOUR TURN

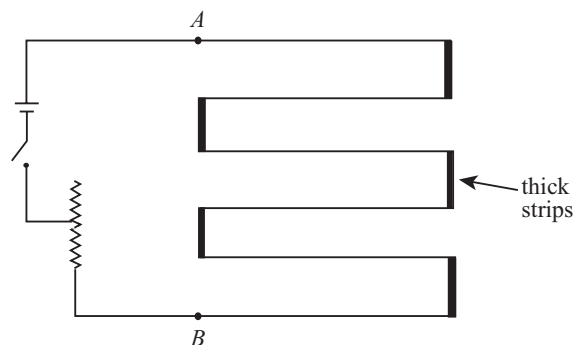
Q.7 A student used a P.O. box to measure an unknown resistance. He tabulated his measurements and calculations as shown below. Fill in the missing data. What is reported value of unknown resistance R_x ?

R_1	R_2	R_3 for balance (Ω)	R_x (Ω)
10Ω	10Ω	212–213	
	100Ω	2122–2123	

10. POTENTIOMETER

This device works like an ideal voltmeter. It measures potential difference between two points in a circuit without disturbing it.

The device has a long uniform wire AB fixed on a wooden platform. The wire is 5–10 m long. For convenience, separate pieces of wire, each 1 m long, are fixed parallel to one another on a platform. The wires are joined to each other by thick copper strips so that the combination acts as a single wire. In diagrams to follow, we will show the wire AB as a straight wire. A strong battery is connected across A and B through a switch and a rheostat. When the switch is put on, a current flows and a potential difference (V_0) is established between A and B . Potential falls as one moves along the wire from end A to B . The rate of fall of the potential (with respect to distance moved from A) is known as potential gradient along the wire.



This potentiometer has 6 segment of 1m each. Effectively, it is a wire AB of Length 6m.

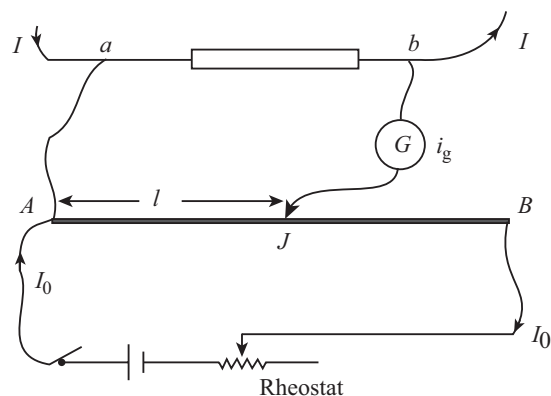
$$\text{Potential gradient} = \frac{V_0}{L} \text{ (V/m)} \quad (6)$$

where L = length of wire AB .

If one moves a distance l along the wire, the potential changes by $\frac{V_0}{L}l$.

Consider that we have been asked to measure potential difference between points a and b of an electric circuit. It is known to us that the point a is at higher potential. We connect the point a to the terminal A of the potentiometer.

The point b is connected to a sliding contact J (which can slide on wire AB) through a galvanometer G . In general, the galvanometer will show some deflection. It means that there is current in path bJ as well as aA . The original current between a and b will change when the potentiometer gets connected. Now, our task is to find a position of sliding contact J on the wire for which there is no current in the galvanometer. This point is known as null point or balance point. When current (i_g) through the galvanometer is zero, it means that $V_a - V_b = V_A - V_J$. In this position, the current I_0 in the potentiometer circuit does not branch at A . The entire current is in the wire AB . Similarly, current I in the external circuit does not branch at a to enter into the potentiometer wire. Therefore, we have not disturbed the original circuit and the potential gradient along wire AB is the same as its value when no circuit was connected to the potentiometer.



a and b are two points in an external circuit AB is the potentiometer wire.

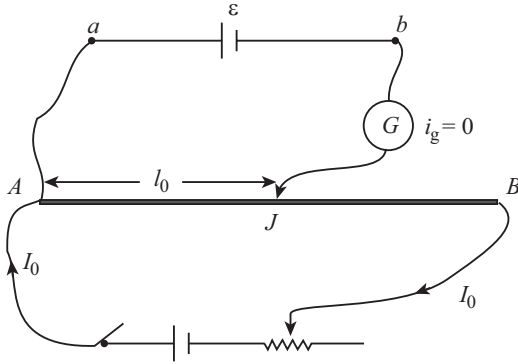
$$\therefore V_a - V_b = V_A - V_J = \frac{V_0}{L}l \quad (7)$$

where l = length AJ in balance position.

We have measured $V_a - V_b$ without disturbing the original circuit, provided we know the value of the potential gradient $\frac{V_0}{L}$.

It is easy to measure $\frac{V_0}{L}$. Don't disturb the potentiometer circuit (i.e. do not change the rheostat setting or the battery) and connect a cell of known emf (ϵ) across the terminals of

the potentiometer circuit. This means to replace the external circuit by a cell as shown in the figure. Again search for the balance point. Let the length AJ be equal to l_0 when the galvanometer shows zero deflection. Note that the current in the potentiometer circuit is still I_0 as we have not changed its settings. The potential difference across A and B is still V_0 ; the potential gradient is still $\frac{V_0}{L}$.



In balanced position:

$$V_a - V_b = V_A - V_J$$

$$\text{or, } \quad \varepsilon = \frac{V_0}{L} l_0 \quad (8)$$

$$\text{or, } \quad \frac{V_0}{L} = \frac{\varepsilon_0}{l_0} \quad (9)$$

Therefore, equation (7) can now be written as

$$\frac{V_a - V_b}{\varepsilon} = \frac{l}{l_0} \quad (10)$$

$$\text{or, } \quad V_a - V_b = \varepsilon \left(\frac{l}{l_0} \right)$$

We can simply remember that the potential difference between the terminals of a potentiometer (points a and b in above discussions) is proportional to the balancing length l on the potentiometer wire.

10.1 Comparing emfs of Two Cells

The figure shows a cell of emf ε_1 connected to the potentiometer. The balancing length is $AJ = l_1$. We can write:

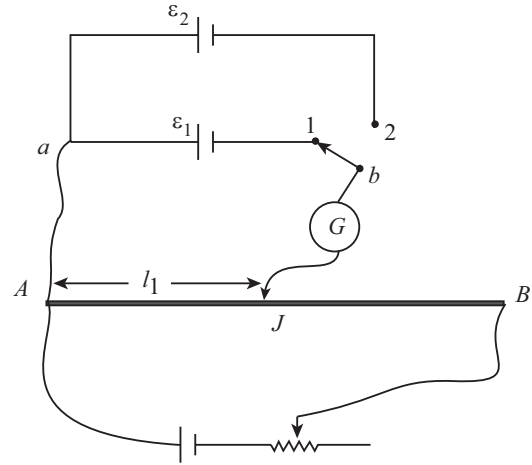
$$\varepsilon_1 \propto l_1 \quad (a)$$

Now, the switch at b is shifted to position 2 and balance point is obtained. Let the balancing length be l_2 this time.

$$\varepsilon_2 \propto l_2 \quad (b)$$

From (a) and (b),

$$\frac{\varepsilon_1}{\varepsilon_2} = \frac{l_1}{l_2} \quad (11)$$

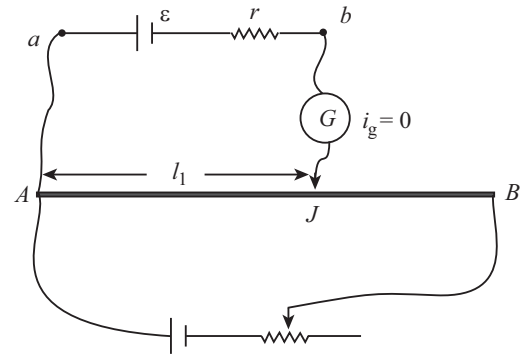


10.2 Measuring Internal Resistance of a Cell

A potentiometer can be used to measure internal resistance of a given cell. First we connect the cell to the potentiometer and obtain the balance point. Let the balancing length be l_1 .

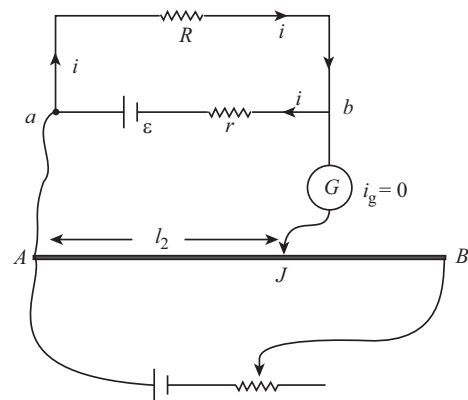
$$\varepsilon \propto l_1 \quad (a)$$

Note that in balanced position, there is no current through the cell and the potential difference between its terminal is the same as its emf.



No current flows through the cell in balance position.

$$V_{ab} = \varepsilon$$



A known resistance R is connected in parallel to the cell. There is a current through the cell.

$$V_{ab} < \varepsilon$$

7.12 Electrostatics and Current Electricity

Now, a known resistance R is connected in parallel to the cell. Again we search for the balance point. Let $AJ = l_2$ when the galvanometer reading is zero. In this position, there is a current i through the cell and resistance R .

$$i = \frac{\varepsilon}{R + r}$$

Potential difference between the terminals a and b is

$$V_{ab} = iR = \frac{\varepsilon R}{R + r}$$

Obviously, $V_{ab} \propto l_2$

$$\text{or, } \frac{\varepsilon R}{R + r} \propto l_2 \quad (b)$$

From (a) and (b),

$$\frac{R}{R + r} = \frac{l_2}{l_1}$$

$$\text{or, } r = R \left(\frac{l_1}{l_2} - 1 \right) \quad (12)$$

Example 8 Why there is a rheostat in the potentiometer circuit?

Suppose you are measuring potential difference between two points using a potentiometer. You move your jockey from one end of the wire to the other end and throughout you found the galvanometer showing deflection in one direction only. What will you do to obtain a balance point?

Solution

This can happen if the potential difference to be measured is higher than the potential difference V_0 across the potentiometer wire. For example, the potential difference V_0 is 20V and you are trying to measure a potential difference of 25V.

You need to decrease the rheostat resistance so that the potential difference V_0 across the potentiometer wire increases and then again try obtaining a balance point.

Example 9 Sensitivity of potentiometer

A potentiometer wire has a length $L = 5.0$ m and potential difference across it is 100 V. The scale used to measure the balancing length can measure upto 1mm accuracy.

- What is the theoretical value of the smallest potential difference that can be measured using this potentiometer?
- In a circuit, potential difference between points a and b is 20.00 V and the potential difference between points c and d is 20.01 V. Can we differentiate between these two potential differences using the above described potentiometer?
- If the length of the potentiometer wire was $L = 10.0$ m with 100 V potential difference across it, can

it differentiate between two potential differences of 20.00 V and 20.01 V?

Solution

Concepts

- The measuring scale can differentiate between 1.462 m and 1.463 m, but it cannot differentiate between 1.4625 m and 1.4626 m.

- The smaller the potential gradient $\left(\frac{V_0}{L} \right)$, lesser

will be the change in potential difference when one moves by 1 mm along the wire. Therefore, a potentiometer wire with smaller potential gradient is much better to differentiate between two close potential differences. The solution of this problem will make this point more clear.

$$\begin{aligned} \text{(a) Potential gradient} &= \frac{V_0}{L} \\ &= \frac{100 \text{ V}}{5000 \text{ mm}} = 0.02 \text{ Vmm}^{-1}. \end{aligned}$$

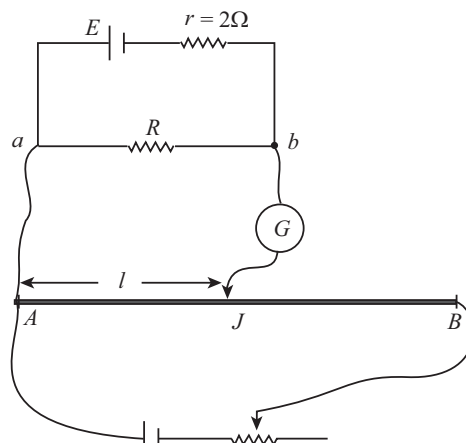
When one moves 1 mm along the wire, the potential changes by 0.02 V. Our scale cannot measure any distance less than 1 mm. Hence, the smallest potential difference that can be measured is 0.02 V.

- A potential difference smaller than 0.02 V cannot be measured. Our potentiometer will not be able to differentiate between 20.00 V and 20.01 V.
- Potential gradient $= \frac{100 \text{ V}}{10,000 \text{ mm}} = 0.01 \text{ Vmm}^{-1}$.

Obviously, this potentiometer can differentiate between 20.00 V and 20.01 V.

Note: We say that the second potentiometer (described above) is more sensitive than the first one. *The large length of potentiometer wire ensures a better sensitivity.*

Example 10 AB is a potentiometer wire. With resistance $R = 10\Omega$, the balance is obtained for $l_1 = 50$ cm. Now, the resistance is replaced with an unknown resistance R_x . The balance is obtained for $l_2 = 30$ cm. Find R_x .



Solution**Concepts**

In balanced condition, the current in the loop having E , R and r will not get diverted into the potentiometer circuit.

$$V_{ab} \propto l.$$

Current in the upper loop having E , R and r is

$$i = \frac{E}{R+r}$$

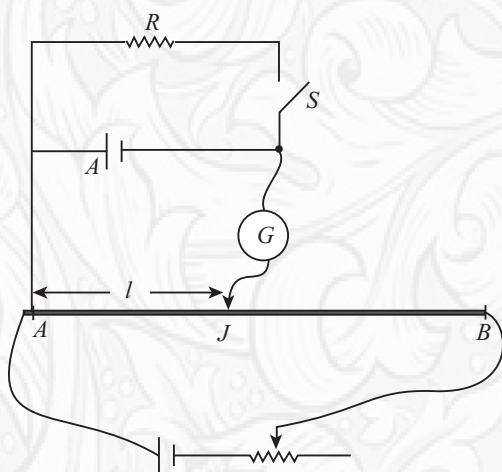
$$\therefore V_{ab} = iR = \frac{ER}{R+r}$$

$$V_{ab} \propto l_1$$

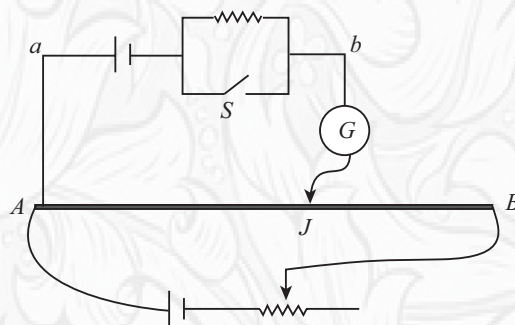
$$\text{or, } \frac{ER}{R+r} \propto l_1 \quad (\text{a})$$

**YOUR TURN**

Q.8 The figure shows a set up for measuring the internal resistance of cell A using a potentiometer. With switch S open, the balancing length is 76.3 cm. When the switch S is closed, the balance point shifts to a length $AJ = 64.8$ cm. Find the internal resistance of the cell (given $R = 9.5\Omega$).



Q.9 In an experiment using a potentiometer, the null point is at J (see the figure). After switch S is closed, in which direction will the null point move?



Q.10 A potentiometer wire has length of 500 cm and resistance of 10Ω . The cell connected to power it has an emf of 10V and the rheostat is set to a resistance r . A cell of emf 20 mV is balanced against a length of 100 cm of the potentiometer wire. Find r .

In Short

- (i) A meter bridge and a P.O. box are devices used to measure an unknown resistance. Both are based on the principle of Wheatstone bridge. However, a P.O. box gives a more precise measurement.
- (ii) A potentiometer behaves like an ideal voltmeter.
- (iii) If V_0 is potential difference across the potentiometer wire and L is its length, then $\frac{V_0}{L}$ is termed as potential gradient.

- (iv) In balanced condition, in a potentiometer, the potential difference being measured is

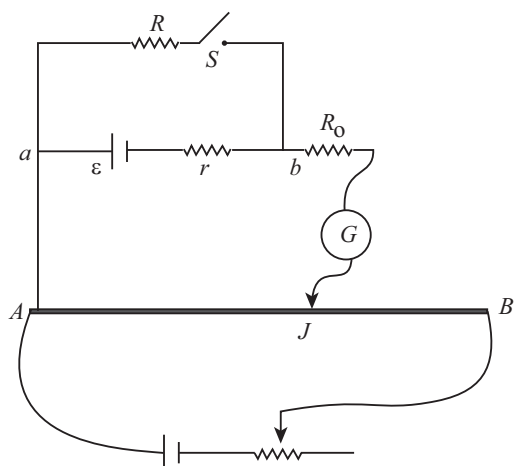
$$V_{ab} = \left(\frac{V_0}{L} \right) l,$$

where l = distance of the jockey from the positive end of the potentiometer wire.

- (v) Smaller the potential gradient $\left(\frac{V_0}{L} \right)$, higher is the sensitivity of a potentiometer.

Miscellaneous Examples

Example 11 Internal resistance of a cell is to be measured using a potentiometer. A physics professor advises his student to include a large resistance R_0 in the circuit, connected in series with the galvanometer. What is the professor thinking? Is this going to affect the balance point?

**Solution**

When the jockey is placed far away from the balance point, a large current may flow through the galvanometer. This may damage it. A large resistance in series with galvanometer will ensure that there is never a large current in the path.

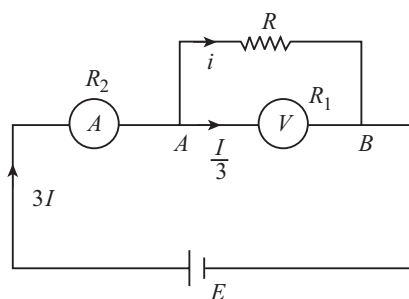
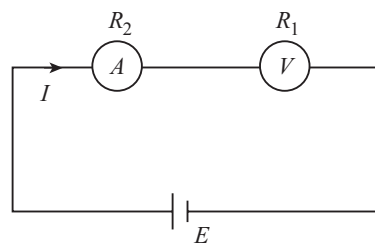
When the jockey is at balance point, there is no current in the galvanometer. There is no current in R_0 as well. There is no potential drop across R_0 . Its presence does not alter the position of balance.

Example 12 A voltmeter of resistance R_1 and an ammeter of resistance R_2 are connected in series across a battery of negligible internal resistance. When a resistance R is connected in parallel to the voltmeter, the ammeter reading becomes 3 times while that of the voltmeter reduces to one-third. Find R_1 and R_2 in terms of R .

Solution**Concepts**

A voltmeter has large resistance. When another resistance is connected in parallel, it reduces the overall resistance. [The equivalent of two resistances in parallel is less than both the resistances.]

Therefore, the ammeter reading increases. However, the current sees a lesser resistance at the voltmeter-resistor combination. The question says that overall there is drop in potential difference across the voltmeter.



In the first case, the current in the circuit is

$$I = \frac{E}{R_1 + R_2} \quad \text{.(a)}$$

In the second case, the current increased to $3I$. This current gets divided at A into two parts. Since the potential difference between A and B became one-third, the current through the voltmeter must be $\frac{I}{3}$.

$$\text{Current through } R \text{ is } i = 3I - \frac{I}{3} = \frac{8I}{3}$$

$$\text{Now, } V_{AB} = \frac{I}{3} R_1 = \frac{8I}{3} R$$

$$\text{or, } R_1 = 8R$$

Now, potential drop across ammeter + potential drop across voltmeter = E

$$\therefore 3IR_2 + \frac{I}{3} R_1 = E$$

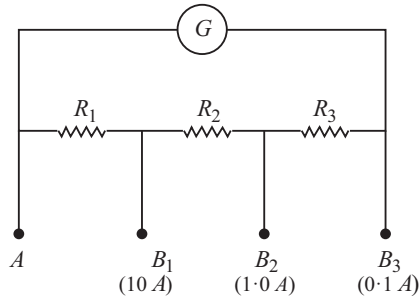
$$\text{or, } 3IR_2 + \frac{I}{3} R_1 = I (R_1 + R_2)$$

$$\text{or, } 2R_2 = \frac{2}{3} R_1$$

$$\text{or, } R_2 = \frac{8}{3} R$$

Example 13 The galvanometer shown in the figure has a resistance $R_g = 50\Omega$ and its full scale deflection current is $i_g = 1 \text{ mA}$. It is to be converted into an ammeter which can be adjusted for three different ranges – (i) 0 to 10A, (ii) 0 to 1.0 A and (iii) 0 to 0.1 A.

For this we take the positive terminal of the ammeter to be A and its negative terminal is chosen from B_1 , B_2 and B_3 . Find the resistances R_1 , R_2 and R_3 .

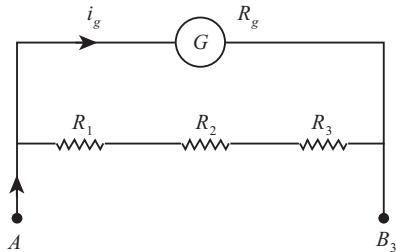


Solution

Concepts

- (i) The range is higher for smaller shunt.
- (ii) When B_3 is taken as negative terminal, $(R_1 + R_2 + R_3)$ is the shunt. The shunt is largest and hence the range (0 to 0.1A) is the smallest.
- (iii) When B_1 is taken as negative terminal, R_g , R_3 and R_2 are in series. R_1 is connected as a shunt in parallel to them.
- (iv) When B_2 is the terminal, R_g and R_3 are in series. $R_1 + R_2$ is in parallel as a shunt.

For 0 to 0.1 A range, B_3 is the negative terminal. When current $I = 0.1\text{A}$, the current through galvanometer is $i_g = 1 \text{ mA}$.



$$\therefore (I - i_g)(R_1 + R_2 + R_3) = i_g R_g$$

$$\text{or, } R_1 + R_2 + R_3 = \frac{10^{-3} \times 50}{0.1 - 10^{-3}} = \frac{50}{99} \quad (i)$$

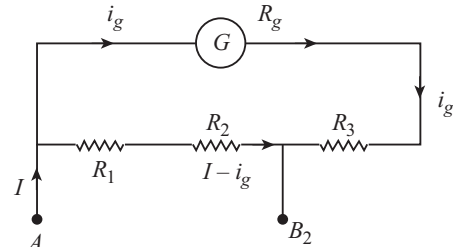
For range 1A, B_2 is the negative terminal. $(R_1 + R_2)$ is in parallel with $(R_g + R_3)$.

$$\therefore i_g(R_g + R_3) = (I - i_g)(R_1 + R_2)$$

$$\text{or, } i_g(R_1 + R_2 + R_3 + R_g) = I(R_1 + R_2)$$

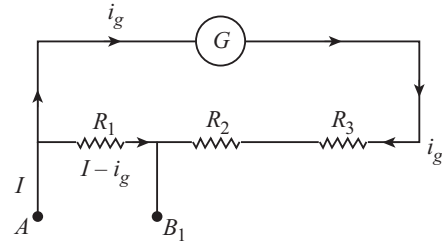
using (i) we get

$$R_1 + R_2 = \frac{10^{-3} \left(\frac{50}{99} + 50 \right)}{1}$$



$$\text{or, } R_1 + R_2 = \frac{5}{99} \quad (ii)$$

Similarly, for range 10A, $(R_g + R_2 + R_3)$ is in parallel with R_1 .



$$i_g(R_g + R_2 + R_3) = (I - i_g)R_1$$

$$\text{or, } R_1 = \frac{i_g(R_1 + R_2 + R_3 + R_g)}{I}$$

$$\text{or, } R_1 = \frac{1}{198} \Omega \quad (iii)$$

Putting this in (ii) gives R_2 :

$$R_2 = \frac{5}{99} - \frac{1}{198} = \frac{1}{22} \Omega$$

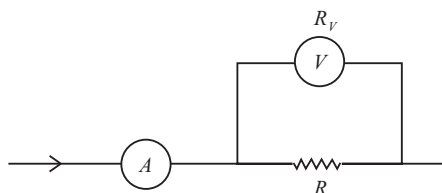
Substituting for R_1 and R_2 in (i) gives

$$R_3 = \frac{15}{33} \Omega.$$

$$\therefore R_1 = \frac{1}{198} \Omega, R_2 = \frac{1}{22} \Omega \text{ and } R_3 = \frac{15}{33} \Omega.$$

7.16 Electrostatics and Current Electricity

Example 14 To measure the value of a resistance R , a voltmeter, having resistance R_v , is connected in parallel to the resistance and an ammeter is connected in series to the combination. The readings of the ammeter and the voltmeter are I_0 and V_0 , respectively. What is value of R ? Discuss the result for $R_v \gg \frac{V_0}{I_0}$.



Solution

Concepts

Since the voltmeter has a finite resistance, the current I_0 will get divided between the voltmeter and the resistor.

Let the current through the voltmeter be I_1 . The current in R will be $I_0 - I_1$.

$$I_1 R_v = (I_0 - I_1) R = V_0 \text{ (measured potential difference)}$$

$$\therefore \left(I_0 - \frac{V_0}{R_v} \right) R = V_0 \quad \left[\because I_1 = \frac{V_0}{R_v} \right]$$

$$\text{or,} \quad R = \frac{V_0}{I_0 - \frac{V_0}{R_v}}$$

To understand the situation when $R_v \gg \frac{V_0}{I_0}$, let's write the above expression as

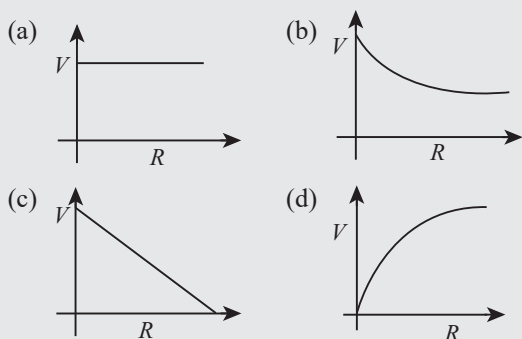
$$R = \frac{V_0 / I_0}{1 - \left(\frac{V_0}{I_0} \right) \left(\frac{1}{R_v} \right)}$$

$$\text{or,} \quad R \approx \frac{V_0}{I_0} \quad \left[\text{since,} \left(\frac{V_0}{I_0} \right) \left(\frac{1}{R_v} \right) \approx 0 \right].$$

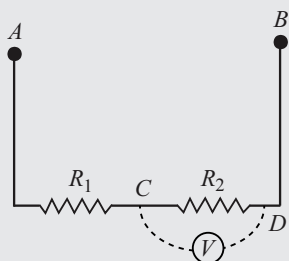
Worksheet 1

- Which of the followings is true regarding a good ammeter
 - It has very small resistance
 - It has a reasonably high resistance
 - It is designed to change the current in the circuit when connected
 - Resistance of the ammeter is larger than its shunt
- An ammeter having resistance of $0.01\ \Omega$ is shorted using a $1\ \Omega$ resistance. This will
 - increase the range of the ammeter
 - decrease the range of the ammeter
 - will harm the ammeter
 - will increase the resistance of the ammeter

- A cell of emf E has an internal resistance r and is connected to a rheostat. When resistance R of rheostat is changed, correct graph of potential difference across it is



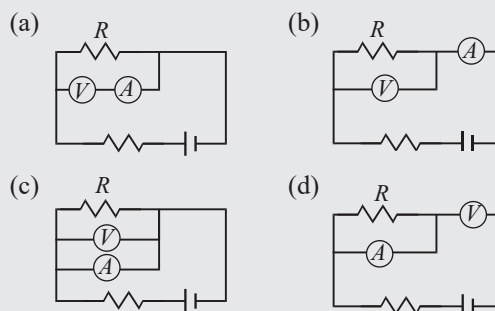
- Resistances R_1 and R_2 each $60\ \Omega$ are connected in series as shown in figure. The Potential difference between A and B is kept $120\ \text{V}$. What will be the reading of voltmeter connected between the point C and D if resistance of the voltmeter is $120\ \Omega$?



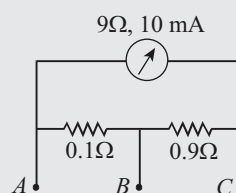
- $48\ \text{V}$
 - $24\ \text{V}$
 - $40\ \text{V}$
 - None
- In a galvanometer, the deflection becomes half when the galvanometer is shunted by a $20\ \Omega$ resistor but given the same current. The galvanometer resistance is

- $5\ \Omega$
- $10\ \Omega$
- $40\ \Omega$
- $20\ \Omega$

- A galvanometer has a resistance of $20\ \Omega$ and reads full-scale when $0.2\ \text{V}$ is applied across it. To convert it into a $10\ \text{A}$ ammeter, the galvanometer coil should have a
 - $0.01\ \Omega$ resistor connected across it
 - $0.02\ \Omega$ resistor connected across it
 - $200\ \Omega$ resistor connected in series with it
 - $2000\ \Omega$ resistor connected in series with it
- Which of the following wiring diagrams could be used to experimentally determine R using ohm's law? Assume an ideal voltmeter and an ideal ammeter.

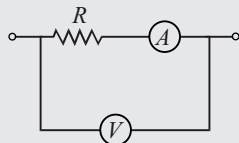


- A milliammeter of range $10\ \text{mA}$ and resistance $9\ \Omega$ is joined in a circuit as shown. The metre gives full-scale deflection for current I when A and B are used as its terminals, i.e., current I enters at A and leaves at B (C is left isolated). The value of I is

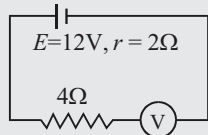


- $100\ \text{mA}$
 - $900\ \text{mA}$
 - $1\ \text{A}$
 - $1.1\ \text{A}$
- A galvanometer coil has a resistance $90\ \Omega$ and full scale deflection current $10\ \text{mA}$. A $910\ \Omega$ resistance is connected in series with the galvanometer to make a voltmeter. If the least count of the voltmeter is $0.1\ \text{V}$, the number of divisions on its scale is
 - 90
 - 91
 - 100
 - none

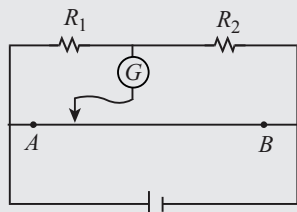
10. In the circuit shown the resistance of voltmeter is 10,000 ohm and that of ammeter is 20 ohm. The ammeter reading is 0.10 A and voltmeter reading is 12 V. R is equal to



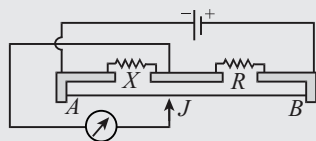
- (a) 122 Ω (b) 140 Ω
(c) 116 Ω (d) 100 Ω
11. By mistake, a student places a nearly ideal voltmeter V in series with a resistance in a circuit as shown. The voltmeter reading will be near to



- (a) 0 (b) 4V
(c) 6V (d) 12V
12. In the figure shown AB is a uniform meter bridge wire. For a given values of R_1 and R_2 the balance point is at 40 cm from A . When R_2 is shunted by a resistance of 10 Ω , balance point shifts to 50 cm from A . R_1 and R_2 are:

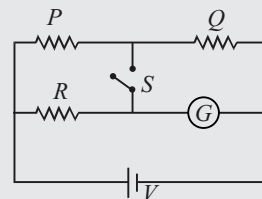


- (a) $\frac{10}{3} \Omega$, 5 Ω (b) 20 Ω , 30 Ω
(c) 10 Ω , 15 Ω (d) 5 Ω , $\frac{15}{2} \Omega$
13. The figure shows a metre-bridge circuit, with $AB = 100$ cm, $X = 12 \Omega$ and $R = 18 \Omega$, and the jockey J in the position of balance. If R is now made 8 Ω , through what distance will J have to be moved to obtain balance?



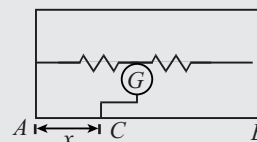
- (a) 10 cm (b) 20 cm
(c) 30 cm (d) 40 cm

14. In the circuit shown, $P \neq R$, the reading of the galvanometer (G) is same with switch S open or closed. I_R , I_G , I_P and I_Q represent current through R , G , P and Q respectively. Then

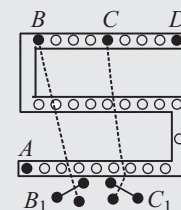


- (a) $I_R = I_G$ (b) $I_P = I_G$
(c) $I_Q = I_G$ (d) $I_Q = I_R$

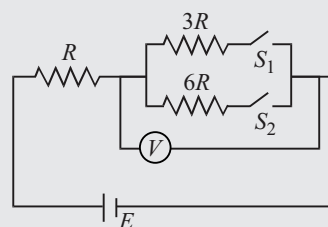
15. In the given circuit, no current is passing through the galvanometer and length AC is x . If the cross-sectional diameter of AB is doubled then for no deflection in the galvanometer the value of AC would be



- (a) x (b) $x/2$
(c) $2x$ (d) None
16. For the post office box arrangement to determine the value of unknown resistance, the unknown resistance should be connected between

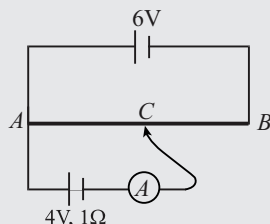


- (a) B and C
(b) C and D
(c) A and D
(d) B_1 and C_1
17. In the circuit shown in figure reading of voltmeter is V_1 when only S_1 is closed, reading of voltmeter is V_2 when only S_2 is closed. The reading of voltmeter is V_3 when both S_1 and S_2 are closed. Then



- (a) $V_2 > V_1 > V_3$
(b) $V_3 > V_2 > V_1$
(c) $V_3 > V_1 > V_2$
(d) $V_1 > V_2 > V_3$

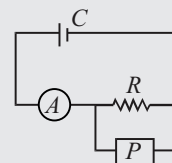
18. A 6 V battery of negligible internal resistance is connected across a uniform wire of length 1 m. The positive terminal of another battery of emf 4V and internal resistance $1\ \Omega$ is joined to the point A as shown in figure. The ammeter shows zero deflection when the jockey touches the wire at the point C. Length AC is equal to



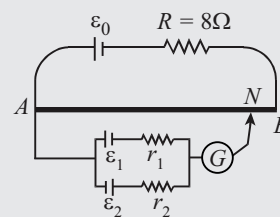
- (a) $2/3$ m
(b) $1/3$ m
(c) $3/5$ m
(d) $1/2$ m
19. A potentiometer wire has length 10 m and resistance $10\ \Omega$. It is connected to a battery of EMF 11 volt and internal resistance $1\ \Omega$, then the potential gradient in the wire is
- (a) 10 V/m
(b) 1 V/m
(c) 0.1 V/m
(d) none
20. The length of a potentiometer wire is l . A cell of emf E is balanced at a length $l/3$ from the positive end of the wire. The length of the wire is increased by $l/2$, keeping potential difference across it the same. At what distance, from the positive end, will the same cell give a balance point.

- (a) $\frac{2l}{3}$
(b) $\frac{l}{2}$
(c) $\frac{l}{6}$
(d) $\frac{4l}{3}$

21. An ideal ammeter A of finite resistance, and a resistor R are joined in series to an ideal cell C . Reading of the ammeter is i . Now the ammeter is replaced with an ammeter of finite resistance. A potentiometer P is joined in parallel to R and potential difference across it is measured. The ammeter reading is I_0 and the potentiometer reading is V_0 . P is now replaced by a voltmeter of finite resistance. The ammeter reading now is I and the voltmeter reading is V .



- (a) $i > I > I_0, V < V_0$
(b) $i < I > I_0, V = V_0$
(c) $I = I_0, V < V_0$
(d) $i < I_0, V = V_0$
22. A battery of emf $E_0 = 12$ V is connected across a 4m long uniform wire having resistance $4\ \Omega/\text{m}$. The cells of emfs $\varepsilon_1 = 2$ V and $\varepsilon_2 = 4$ V having internal resistance $2\ \Omega$ and $6\ \Omega$ respectively, are connected as shown in the figure. If galvanometer shows no deflection at the point N, the distance of point N from the point A is equal to

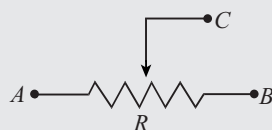


- (a) $\frac{1}{6}$ m
(b) $\frac{1}{3}$ m
(c) 25 cm
(d) 50 cm

Worksheet 2

1. A battery is to be connected so that the shown rheostat will behave like a potential divider. Which of the followings are correct?

- (a) The battery can be connected across A and B and the output can be taken across A and C .



- (b) The battery can be connected across A and B and the output can be taken across C and B .
 (c) The battery can be connected across A and C and the output can be taken across A and B .
 (d) The battery can be connected across C and B and the output can be taken across A and B .

2. An experiment is to be designed for verification of ohm's law. The experiment is to be designed using a source of variable D.C. voltage, a main resistance of $100\ \Omega$, two identical galvanometers and two resistances of values $10^6\ \Omega$ and $10^{-3}\ \Omega$ respectively.

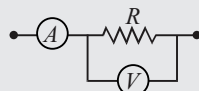
- (a) One of the galvanometers, with a resistance in series, must be connected in parallel to the main resistance.
 (b) One of the galvanometers, with a resistance in series, must be connected in series to the main resistance.
 (c) One of the galvanometers, with a resistance in parallel, must be connected in parallel to the main resistance.
 (d) The galvanometer connected in series with the main resistance must have the smaller resistance connected in parallel to it.

3. A galvanometer may be converted into an ammeter or a voltmeter. In which of the following cases the resistance of the device will be the largest?

- (a) an ammeter of range 10A
 (b) a voltmeter of range 5 V
 (c) an ammeter of range 5 A
 (d) a voltmeter of range 10 V

4. In the circuit shown the readings of ammeter and voltmeter are 4A and 20V, respectively. If the meters are non ideal, then R is:

- (a) certainly $5\ \Omega$
 (b) may be less than $5\ \Omega$
 (c) greater than $5\ \Omega$
 (d) between $4\ \Omega$ & $5\ \Omega$



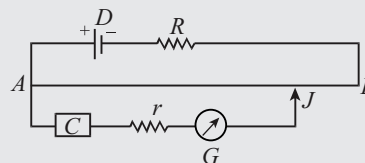
5. A micrometer has a resistance of $100\ \Omega$ and a full scale range of $50\ \mu\text{A}$. It can be used as a voltmeter or a higher range ammeter provided a resistance is added to it. Pick the correct range and resistance combination(s).

- (a) 50 V range with $10\ \text{k}\Omega$ resistance in series.
 (b) 10 V range with $200\ \text{k}\Omega$ resistance in series.
 (c) 5 mA range with $1\ \Omega$ resistance in parallel.
 (d) 10 mA range with $1\ \text{k}\Omega$ resistance in parallel.

6. In a potentiometer wire experiment the emf of a battery in the primary circuit is 20V and its internal resistance is $5\ \Omega$. There is a rheostat in series with the battery and the potentiometer wire, whose resistance can be varied from $120\ \Omega$ to $170\ \Omega$. Resistance of the potentiometer wire is $75\ \Omega$. The following potential differences can be measured using this potentiometer.

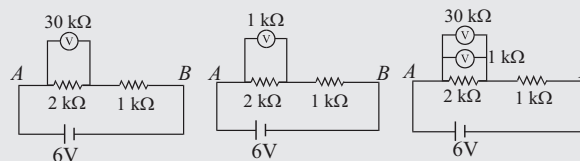
- (a) 5V (b) 6V
 (c) 7V (d) 8V

7. In the given potentiometer circuit, the resistance of the potentiometer wire AB is R_0 . D is the cell in the primary circuit of the potentiometer. C is a cell of internal resistance r . The galvanometer G does not give zero deflection for any position of the jockey J . Which of the following cannot be a reason for this?



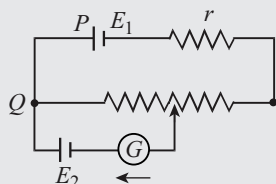
- (a) $r > R_0$
 (b) $R \gg R_0$
 (c) emf of $C >$ emf of D
 (d) The negative terminal of C is connected to A .

8. A series combination of a $2\ \text{k}\Omega$ resistor and a $1\ \text{k}\Omega$ resistor, is connected across a battery of emf 6V and negligible internal resistance. The potential drop, across the $2\ \text{k}\Omega$ resistor is measured by (i) a $30\ \text{k}\Omega$ voltmeter (ii) a $1\ \text{k}\Omega$ voltmeter and (iii) both these voltmeters connected in parallel across it. If the voltmeter readings in the three cases are V_1 , V_2 and V_3 respectively.

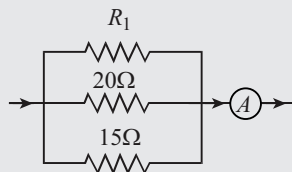


- (a) $V_1 > V_2 > V_3$
- (b) $V_1 < V_2 > V_3$
- (c) if the potential drop were measured across the series combination of the $2\text{k}\Omega$ and the $1\text{k}\Omega$ resistor i.e., across the points A and B , then $V_1 > V_2 > V_3$
- (d) if the potential drop were measured across the series combination of the $2\text{k}\Omega$ and the $1\text{k}\Omega$ resistor i.e., across the points A and B , then $V_1 = V_2 = V_3$

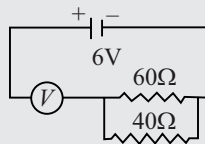
9. In the potentiometer circuit of given figure the galvanometer reveals a current in the direction shown wherever the sliding contact (Jockey) touches the wire. This could be caused by



- (a) E_1 being too low
 - (b) r being too high
 - (c) a break in PQ
 - (d) E_2 being too low
10. In the given circuit, the current through the 20Ω resistance is 0.3 ampere while the ideal ammetre reads 0.8 ampere.

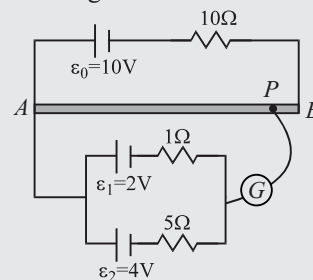


- (a) the reading of ammeter will be less than 0.3 A if it is a non ideal ammeter
 - (b) the value of R_1 is 40 ohms
 - (c) the value of R_1 is 50 ohms
 - (d) the value of R_1 is 60 ohms
11. The measurement of voltmeter in the circuit shown is:
- (a) 6 V if the voltmeter is ideal
 - (b) 3.4 V if the voltmeter is ideal



- (c) Less than 4.0 V if the voltmeter has finite resistance
- (d) Less than 6.0 V if the voltmeter has finite resistance

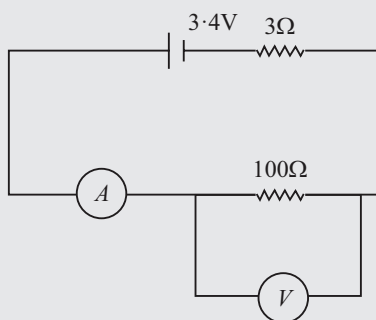
12. A battery of emf $\epsilon_0 = 10\text{ V}$ is connected across a 1 m long uniform wire having resistance 10Ω . Two cells of emf $\epsilon_1 = 2\text{ V}$ and $\epsilon_2 = 4\text{ V}$ having internal resistances 1Ω and 5Ω respectively are connected as shown in the figure.



- (a) The galvanometer cannot show zero deflection for any position of point P
 - (b) If a galvanometer shows no deflection at the point P , the distance of point P from the point A is $7/15\text{ m}$.
 - (c) If the cell of emf ϵ_2 is replaced with a cell identical to the $\epsilon_1 = 2\text{ V}$ cell, then the galvanometer shows no deflection for a point P such that distance $PA = 0.4\text{ m}$
 - (d) None of the above.
13. Which of the followings are true regarding a non ideal ammeter and a non ideal voltmeter?
- (a) Ammeter shows the true value of current flowing through it.
 - (b) Voltmeter shows the true potential difference across its terminals.
 - (c) Ammeter shows a reading that is less than the value of current in the path before insertion of the ammeter
 - (d) Voltmeter shows a reading that is less than the value of potential difference that existed between the terminals before insertion of the voltmeter

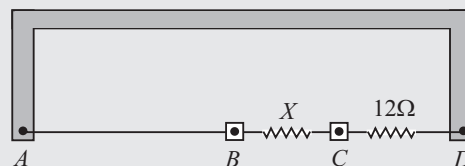
Worksheet 3

1. A certain circuit with resistance $R = 100\Omega$ is powered by a DC source. An ammeter with an internal resistance $r = 1\Omega$ is inserted into the circuit and shows a reading of 5A. What was the current before the ammeter was inserted?
2. The scale of a galvanometer is divided into 150 equal divisions. The galvanometer has a current sensitivity of 10 divisions per mA and voltage sensitivity of 2 divisions per mV. What changes will you recommend to convert this galvanometer into
 - (a) an ammeter that can read 0.6A per division?
 - (b) a voltmeter that can read 1V per division?
3. In the circuit shown, the ammeter has a resistance of 2Ω . The reading of the ammeter is 0.04 A.



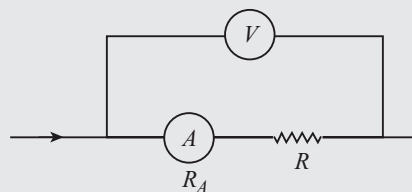
- (a) Find the reading of the voltmeter. What is the resistance of the voltmeter?
 - (b) Had the voltmeter been an ideal one? What would have been its reading?
4. A galvanometer has a current sensitivity of 1 mA per division. A shunt of 5Ω is connected to it and the galvanometer is put into a circuit in series with a 500Ω resistance connected to a cell of internal resistance 0.5Ω . The galvanometer shows a deflection of 5 divisions. When the shunt resistance is replaced with another resistance of 25Ω , the galvanometer shows 20 deflections. Find the emf of the cell and the resistance of the galvanometer.
 5. In a meter bridge, the wire consists of two parts – the left part of length 30 cm having cross-sectional radius r and the right part of length 70 cm having cross-sectional radius $2r$. Both parts are made of same material. Where will the null point occur if the resistances in the left and right gaps are 5Ω and 8Ω , respectively?
 6. Consider the P.O. box described in section 9. Find the maximum and minimum values of the unknown resistance R_x which can be measured using the P.O. box.

7. A thin uniform wire AB of length 1 m, an unknown resistance X and a resistance of 12Ω are connected by thick conducting strips, as shown. A battery and a galvanometer (with a sliding jockey connected to it) are also available.



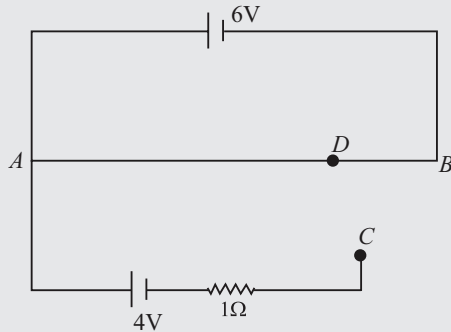
Connections are to be made to measure the unknown resistance X using the principle of Wheatstone bridge.

- (a) Are there positive and negative terminals on a galvanometer?
 - (b) Copy the figure and show the battery and the galvanometer (with jockey) connected at appropriate points.
 - (c) After appropriate connections are made, it is found that no deflection occurs in the galvanometer when the sliding jockey touches the wire at a distance of 60 cm from A . Obtain the value of X .
8. To measure the resistance R of a resistor, an ammeter of resistance R_A is placed in series with the resistor and the voltmeter is placed across the series combination as shown. The readings of the ammeter and the voltmeter are I_0 and V_0 , respectively.



- (a) Find R .
 - (b) Approximate the value of R when $\frac{V_0}{I_0} \gg R_A$.
9. What resistance r should be used to shunt a galvanometer with an internal resistance $R = 10k\Omega$ to reduce its sensitivity 50 times?
 10. In a potentiometer experiment, no deflection is observed in the galvanometer when a cell is balanced across 52 cm length of the potentiometer wire. If the cell is shunted by a resistance of 5Ω , a balance is obtained when the cell is connected across 0.4 m length of the potentiometer wire. Find the internal resistance of the cell.

11. AB is a uniform wire of length 100 cm. The positive terminal of a 6V battery (negligible internal resistance) is connected to the end A , and the negative terminal is connected to the end B of the wire. The positive terminal of another 4V cell (having internal resistance of 1Ω) is connected to A . Assume the potential of point B to be zero.



- What is the potential of points A and C ?
- At which point D on wire AB is the potential same as that at C ?
- Points C and D are connected using a wire. How much current flows in the wire?
- If the 4V cell is replaced by a 7.5 V cell, what will be your answers to the above questions?

Answer Sheet

Your Turn

1. Zero 2. Add $2\text{m}\Omega$ shunt. For making ammeter of higher range, we need smaller shunt.
 3. 999Ω 4. 1.29V 5. (a) 55V ; (b) 40V 6. (a) 66.7Ω ; (b) 71Ω .

7.

R_1	R_1	R_1	R_1
10Ω	10Ω	212–213	212.0–213.0
1000Ω	100Ω	2122–2123	212.2–212.3

Reported value is $R_x = 212.25\Omega$

8. 1.68Ω 9. It will not move. 10. 990Ω

Worksheet 1

1. (a) 2. (a) 3. (d) 4. (a) 5. (d) 6. (b) 7. (b) 8. (c) 9. (c)
 10. (d) 11. (d) 12. (a) 13. (b) 14. (a) 15. (a) 16. (c) 17. (a) 18. (a)
 19. (b) 20. (b) 21. (a) 22. (c)

Worksheet 2

1. (a, b) 2. (a, c, d) 3. (d) 4. (c) 5. (b, c) 6. (a, b, c)
 7. (a) 8. (a, d) 9. (a, b, c) 10. (d) 11. (a, d) 12. (b, c)
 13. (a, b, c, d)

Worksheet 3

1. 5.05A 2. (a) Add a shunt $R_s = 8.3 \times 10^{-4}\Omega$ (b) Add a resistance in series, $R = 9995\Omega$
 3. (a) 3.2V , 400Ω ; (b) 3.24V 4. 45V , 83.3Ω 5. 18.27cm
 6. $1.111\text{M}\Omega$, $10\text{m}\Omega$ 7. (a) No; (c) 8Ω 8. (a) $R = \frac{V_0}{I_0} - R_A$; (b) $R \simeq \frac{V_0}{I_0}$
 9. 204Ω 10. 1.5Ω
 11. (a) $V_A = 6\text{V}$, $V_C = 2\text{V}$;
 (b) 66.67cm from A ;
 (c) Zero;
 (d) $V_A = 6\text{V}$, $V_C = -1.5\text{V}$, No point D is possible where $V_C = V_D$.

Transient RC Circuits

“I think character is permanent, and issues are transient”.

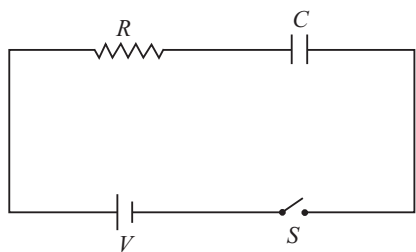
— James Stockdale

1. INTRODUCTION

When we put on or put off a switch in an electric circuit (or make any other sudden change in the circuit), there is a small time, preceding the steady state, when the current is not constant. The small interval of time during which the current settles down to a steady value is known as a *transient state*. In this chapter, we will study about the behaviour of an electric circuit, containing resistors and capacitors, during this transient phase. In the following chapter, after studying about inductors, we will discuss about other kinds of transient circuits.

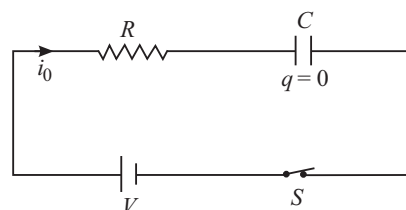
2. CHARGING OF A CAPACITOR

A capacitor, having capacitance C , is to be charged using a cell of emf V . If the circuit has no resistance, the moment we put on the switch the capacitor will get charged without any time loss. In practice, every circuit has some resistance. Presence of a resistance makes the instantaneous charging of a capacitor impossible. Capacitor getting a charge ($q = CV$) immediately after the switch is closed implies that there is an infinite current. This is just not possible in the presence of a resistance. The process of charging of the capacitor will take some time. We will study how the charge on the capacitor varies with time.



2.1 Condition immediately after the switch is closed

Let us assume that the switch ‘S’ shown in the circuit is closed at time $t = 0$. The phrase ‘immediately (or, just) after the switch is closed’ refers to a time $t = 0.0000\dots\dots 001$ s. We will denote this time as $t = 0^+$.



Circuit at $t = 0^+$

At time $t = 0^+$, there is no charge on the capacitor but the charge has begun to move, that is, there is a current. Can we write the value of current at this moment? Yes. There is no potential difference across the capacitor ($\because q = 0$) and the entire drop of V volt must happen across R .

$$\therefore i_0 R = V$$

$$\text{or, } i_0 = \frac{V}{R} \quad (i)$$

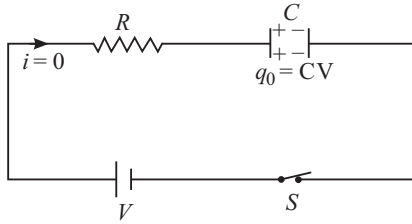
At time $t = 0^+$, charge (q) on the capacitor is zero but the rate of change of q is high $\left(\frac{dq}{dt} = i_0 = \frac{V}{R}\right)$.

2.2 Condition after a long time

What is status of the circuit a long time ($t = \infty$) after the switch is closed? The charge on the capacitor keeps growing after the switch is closed. When the potential difference $\left(V_c = \frac{q}{C}\right)$ across the capacitor becomes equal to V , there

8.2 Electrostatics and Current Electricity

should not be any potential drop across R . It means current in the circuit is zero. Once a capacitor gets fully charged, it will not allow a direct current to flow through it. After all, the two plates are separated by an insulator. How can there be any current?



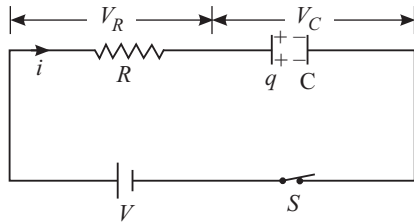
Circuit at $t = \infty$

Therefore, we conclude that after a long time charge on the capacitor is $q_0 = CV$ and the current in the circuit is zero. Now, everything is static. We say that the circuit is in *steady state*.

2.3 Dependence of charge and current on time

We have discussed the state of the circuit at time $t = 0^+$ and $t = \infty$, after the switch is closed at $t = 0$. Now, we will try to express the charge on the capacitor and the current in the circuit as a function of time.

Let the charge on the capacitor be q and current in the circuit be i at time t .



Circuit at time t

$$V_R + V_C = V$$

$$\text{or, } iR + \frac{q}{C} = V$$

$$\text{or, } R \frac{dq}{dt} + \frac{q}{C} = V \quad \left[\because i = \frac{dq}{dt} \right]$$

$$\text{or, } R \frac{dq}{dt} = V - \frac{q}{C}$$

$$\text{or, } \int_0^q \frac{dq}{V - \frac{q}{C}} = \frac{1}{R} \int_0^t dt$$

$$\text{or, } -C \left[\ln \left(V - \frac{q}{C} \right) \right]_0^q = \frac{t}{R}$$

$$\text{or, } \ln \left(V - \frac{q}{C} \right) - \ln V = -\frac{t}{RC}$$

$$\text{or, } \ln \left(\frac{V - \frac{q}{C}}{V} \right) = -\frac{t}{RC}$$

$$\text{or, } 1 - \frac{q}{CV} = e^{-t/RC}$$

$$\text{or, } q = CV (1 - e^{-t/RC})$$

$$\text{or, } q = q_0 (1 - e^{-t/\tau}), \quad (2)$$

$$\text{where } q_0 = CV = \text{final charge on the capacitor and } \tau = RC \quad (3)$$

τ is known as *time constant* of the circuit. Unit of time constant is second.

We can check the validity of equation (2) under two extreme conditions:

$$(i) \text{ when } t = 0^+, q = q_0 (1 - 1) = 0$$

$$(ii) \text{ when } t \rightarrow \infty, q = q_0 (1 - 0) = q_0$$

To find current, we can differentiate equation (2) with respect to time.

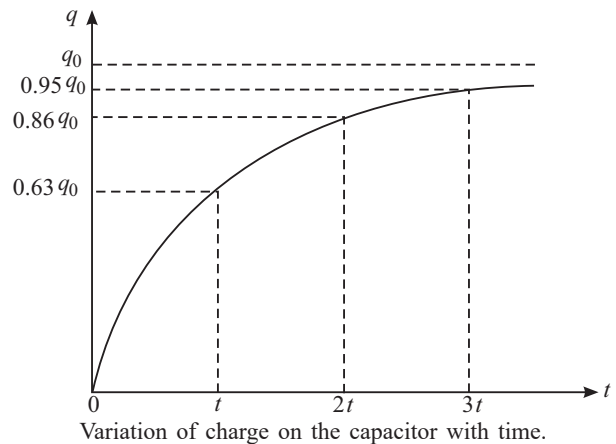
$$i = \frac{dq}{dt} = \frac{q_0}{\tau} e^{-t/\tau}$$

$$\text{or, } i = \frac{CV}{CR} e^{-t/\tau} = \frac{V}{R} e^{-t/\tau}$$

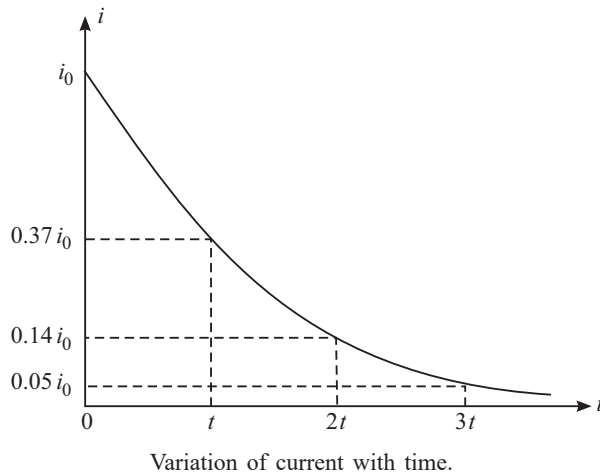
$$\text{or, } i = i_0 e^{-t/\tau}, \quad (4)$$

where $i_0 = \frac{V}{R}$ = current immediately after the switch is closed.

The graphical plots of equations (2) and (4) have been shown in the figure below. Theoretically, it takes infinite time for the capacitor to get fully charged.



But, practically it takes a time equal to few time constants for a capacitor to get fully charged. Values of charge on the capacitor at $t = \tau, 2\tau$ and 3τ have been shown in the figure. They can be calculated as:



charge at time $t = \tau$ is

$$\begin{aligned} q &= q_0 (1 - e^{-1}) \\ &= q_0 (1 - 0.37) \quad [\because e^{-1} \simeq 0.37] \\ &= 0.63 q_0 \end{aligned}$$

charge at time $t = 2\tau$ is

$$q = q_0 (1 - e^{-2}) \simeq 0.86 q_0$$

Similarly, $q \simeq 0.95 q_0$ at $t = 3\tau$.

For all practical purposes, we can deem the capacitor to be fully charged after 5 or 6 time constants. Values of current at $t = \tau$, 2τ and 3τ have also been indicated in the graph.

Example 1 Significance of time constant

A capacitor of capacitance $C = 10\mu\text{F}$ is to be charged using a cell of emf $V = 10\text{V}$. Resistance of the circuit is $R = 10\text{k}\Omega$. The circuit is switched on at $t = 0$.

- Find the time at which 63% charging of the capacitor will be completed.
- Find the time at which current in the circuit is 37% of its initial value.
- How will answer to (a) and (b) change if the resistance in the circuit is changed to $R = 20\text{k}\Omega$.
- Draw a graph showing variation in charge (q) on the capacitor with time (t), for the two cases – when $R = 10\text{k}\Omega$ and $R = 20\text{k}\Omega$.

Solution

Concepts

- Note that 63% charging takes place in one time constant. The current in the circuit decays by 63% (becomes equal to 37%) in one time constant.
- Higher the time constant, slower is the charging.

- Time constant of the circuit is

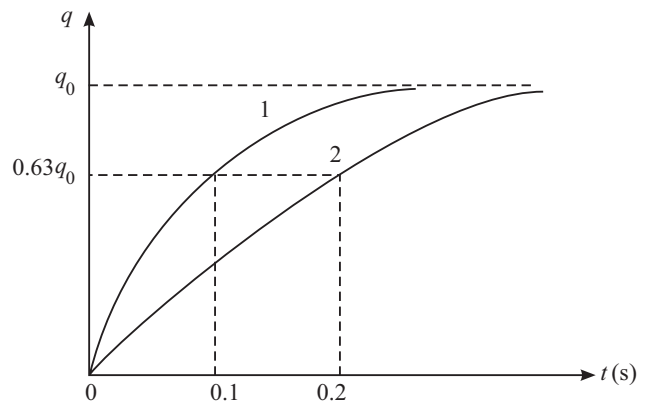
$$\tau = RC = 10 \times 10^3 \times 10 \times 10^{-6} = 0.1 \text{ s}$$

63% charging will get completed in time $t = \tau = 0.1 \text{ s}$

- Current becomes 37% after $t = \tau = 0.1 \text{ s}$

- When R is doubled, the time constant gets doubled. New value of time constant is 0.2 s . Charge becomes $0.63 q_0$ and the current becomes $0.37 i_0$ at time $t = 0.2 \text{ s}$.

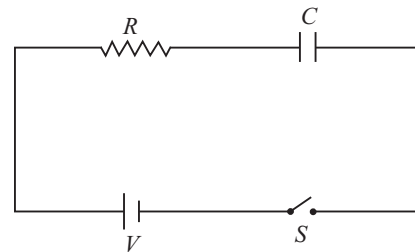
(d)



In the figure, graph 1 is for the circuit with $R = 10\text{k}\Omega$ and graph 2 is for the circuit with $R = 20\text{k}\Omega$.

Higher resistance implies higher time constant and hence slower charging.

Example 2 In the circuit shown in the figure, switch S is closed at time $t = 0$.



Find the time when

- Potential difference across R is $\frac{V}{2}$
- Energy stored in the capacitor is $\frac{1}{4}$ of its maximum value.

Solution

Concepts

The problem is based on direct use of equations (4) and (2).

- $$V_R = iR$$

$$\text{or, } \frac{V}{2} = iR$$

$$\text{or, } i = \frac{V}{2R} = \frac{i_0}{2}$$

We have to find the time when current in the circuit becomes $\frac{i_0}{2}$.

8.4 Electrostatics and Current Electricity

$$i = i_0 e^{-t/\tau}$$

$$\frac{i_0}{2} = i_0 e^{-t/\tau}$$

or, $e^{-t/\tau} = \frac{1}{2}$ (i)

or, $-\frac{t}{\tau} \ln e = -\ln\left(\frac{1}{2}\right)$

or, $t = \tau(\ln 2)$ [$\because \ln e = 1$]

$= (\ln 2) (RC)$

$= (0.693) RC$ [$\ln 2 = 0.693$]

(b) Maximum energy stored in the capacitor (when it is fully charged) is $U_{\max} = \frac{q_0^2}{2C}$.

Energy stored in $\frac{1}{4} U_{\max}$ when charge on the capacitor is $q = \frac{q_0}{2}$.

Using equation (2), we get:

$$q = q_0 (1 - e^{-t/\tau})$$

or, $\frac{q_0}{2} = q_0 (1 - e^{-t/\tau})$

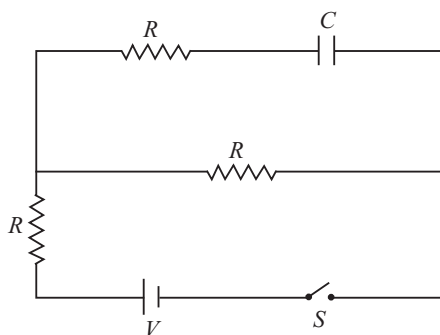
or, $e^{-t/\tau} = \frac{1}{2}$

This equation is same as (i)

$$\therefore t = (\ln 2) (RC) = (0.693) RC.$$

Example 3 Status of circuit at $t = 0^+$ and $t \rightarrow \infty$

In the circuit shown, each of the three resistors has resistance 10Ω and emf of the cell is $20V$. Capacitance of the capacitor is $C = 5\mu F$.



- Find the current through the cell immediately after the switch is closed.
- Find the current through the cell a long time after the switch is closed.
- Find the steady state charge on the capacitor.

Solution

Concepts

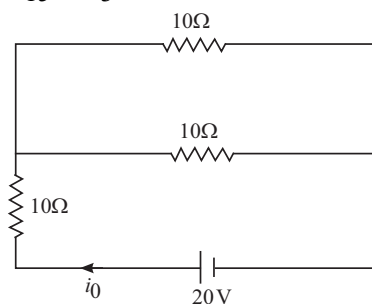
- At $t = 0^+$, the capacitor has no charge. An uncharged capacitor is like zero resistance as far as calculation of current is concerned.
- At $t = \infty$, the capacitor is fully charged and no current can flow through it. For the purpose of calculation of current, a fully charged capacitor is like an infinite resistance.

- (a) At $t = 0^+$, the effective circuit is as shown in the figure. Equivalent resistance is

$$R = 15\Omega$$

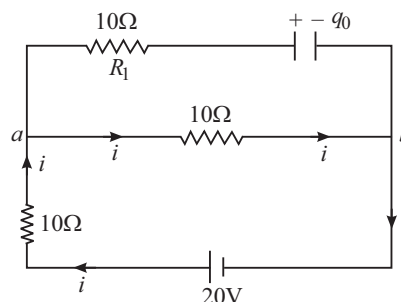
Current through the cell is

$$i_0 = \frac{V}{R} = \frac{20}{15} = \frac{4}{3} A$$



- (b) At $t = \infty$, there is no current in the branch having the capacitor.

$$\therefore i = \frac{20V}{(10+10)\Omega} = 1A$$



- (c) Steady state refers to $t = \infty$, when the capacitor is fully charged and nothing is changing in the circuit. In this state, the potential difference between the points a and b in the circuit is

$$V_{ab} = (10\Omega) (1A) = 10V$$

The same potential difference must exist across the capacitor since the potential drop across R_1 is zero (see figure). There is no current in R_1 .

$$\therefore V_{ab} = \frac{q_0}{C}$$

or, $q_0 = (10V) (5\mu F) = 50\mu C$

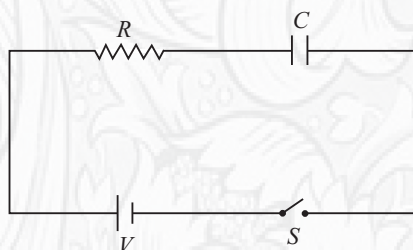


YOUR TURN

Q.1 A capacitor of capacitance $C = 4\mu\text{F}$ is to be charged using a battery of emf $V = 20\text{V}$. Resistance of the circuit is $R = 10\text{k}\Omega$. The switch is closed at time $t = 0$.

- Find the final steady state charge on the capacitor.
- Find the current in the circuit, immediately after the switch is closed.
- Find the current in the circuit at $t = 0.08\text{ s}$.
- Find the potential difference across the capacitor at $t = 0.08\text{ s}$.

Q.2 In the circuit shown, the switch is closed at $t = 0$. Find the time when potential difference across R is one-third of the potential difference across C .

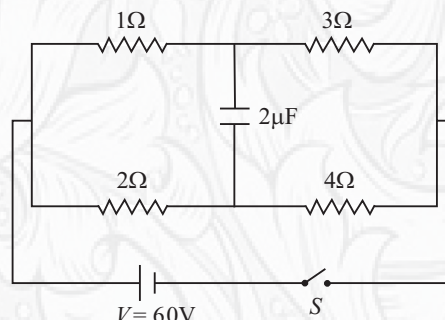


Q.3 In the circuit shown, the switch is closed at $t = 0$.

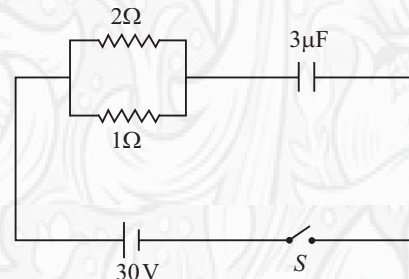
- Find the current through the cell, immediately after the switch is closed.

(b) Find the current through the cell, a long time after the switch is closed.

(c) Find charge on the capacitor after a long time.

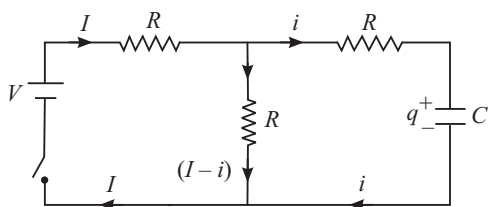


Q.4 In the circuit shown, the switch is closed at $t = 0$. Find current through $2\text{k}\Omega$ resistor at $t = 2\text{ms}$.



3. CHARGING OF A CAPACITOR IN A COMPLEX NETWORK OF RESISTORS

A capacitor is to be charged in the circuit shown. The switch (S) is closed at $t = 0$. At time t , charge on the capacitor is q



and currents in various branches are as shown in the figure.

Using Kirchhoff's loop law in the left loop, we can write

$$IR + (I - i)R = V$$

$$\text{or, } I = \frac{V}{2R} + \frac{i}{2} \quad (\text{i})$$

Using the loop law in bigger loop gives (one may choose the right small loop as well):

$$IR + iR + \frac{q}{C} = V$$

Substituting for I using (i)

$$\frac{V}{2} + \frac{iR}{2} + iR + \frac{q}{C} = V$$

$$\text{or, } \frac{3R}{2}i = \frac{V}{2} - \frac{q}{C}$$

$$\text{or, } \frac{3R}{2} \frac{dq}{dt} = \frac{V}{2} - \frac{q}{C}$$

$$\text{or, } \int_0^q \frac{dq}{\frac{V}{2} - \frac{q}{C}} = \frac{2}{3R} \int_0^t dt$$

$$\text{or, } \left[\ln \left(\frac{V}{2} - \frac{q}{C} \right) \right]_0^q = -\frac{2t}{3RC}$$

$$\text{or, } \ln \left(\frac{V}{2} - \frac{q}{C} \right) - \ln \left(\frac{V}{2} \right) = -\frac{2t}{3RC}$$

$$\text{or, } 1 - \frac{2q}{CV} = e^{-\frac{2t}{3RC}}$$

$$\text{or, } q = \frac{CV}{2} \left[1 - e^{-\frac{2t}{3RC}} \right]$$

$$\left[\because i = \frac{dq}{dt} \right]$$

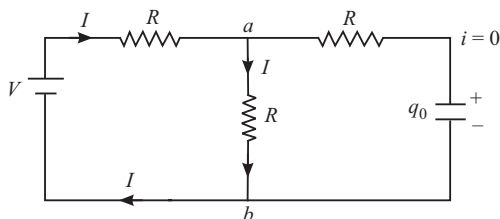
8.6 Electrostatics and Current Electricity

$$\text{or, } q = q_0[1 - e^{-t/\tau}]$$

$$\text{where } q_0 = \frac{CV}{2} \text{ and } \tau = \frac{3RC}{2}$$

The above equation is same as equation (2). Thus, the charging of a capacitor always follows equation (2) and we just need to figure out the values of q_0 and τ . If we can do this, we will not need to do the elaborate mathematics as done above.

Finding q_0 : Consider the circuit in the steady state (i.e., at $t = \infty$). The above circuit has been redrawn to depict the situation at $t = \infty$. There is no current through the capacitor.



At $t = \infty$, there is no current through the capacitor.

$$I = \frac{V}{2R}$$

$$\therefore V_{ab} = IR = \frac{V}{2}$$

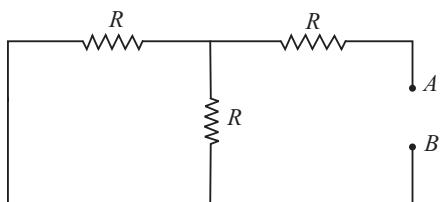
Therefore, the potential difference across the capacitor is

$$V_{ab} = \frac{V}{2}.$$

$\therefore$ Final charge on the capacitor is

$$q_0 = \frac{CV}{2}$$

Finding τ : The effective resistance that the capacitor sees during its charging is obtained by finding the equivalent resistance across the capacitor plates after short circuit in the cell. The figure shows the relevant circuit. The capacitor is between the points A and B . The equivalent resistance between A and B is



The cell has been short circuited. Capacitor is between A and B . We need to find equivalent resistance between A and B .

$$R_0 = \frac{R}{2} + R = \frac{3R}{2}$$

(ii) Time constant for charging of the capacitor is given by

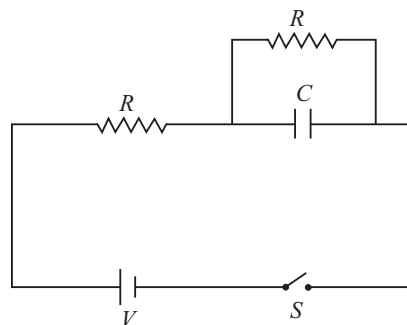
$$\tau = R_0 C = \frac{3R}{2} C.$$

After finding q_0 and τ , we can write variation in charge on the capacitor with time using equation (2).

$$\begin{aligned} q &= q_0 [1 - e^{-t/\tau}] \\ &= \frac{CV}{2} \left[1 - e^{-\frac{2t}{3RC}} \right] \end{aligned}$$

This is same as equation (ii) obtained after detailed mathematics.

Example 4 Write the variation in charge on the capacitor plate as a function of time after the switch (S) is closed at $t = 0$.



Solution

Concepts

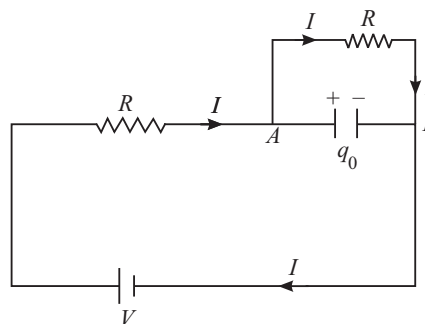
- (i) Final charge (q_0) on the capacitor can be calculated considering the circuit in its steady state.
- (ii) Time constant $\tau = R_0 C$, where R_0 is resistance across the capacitor plates when the cell is shorted.

In steady state, there is no current through the capacitor.

$$I = \frac{V}{2R}$$

$$\therefore V_{AB} = IR = \frac{V}{2}$$

$$\therefore q_0 = CV_{AB} = \frac{VC}{2}$$



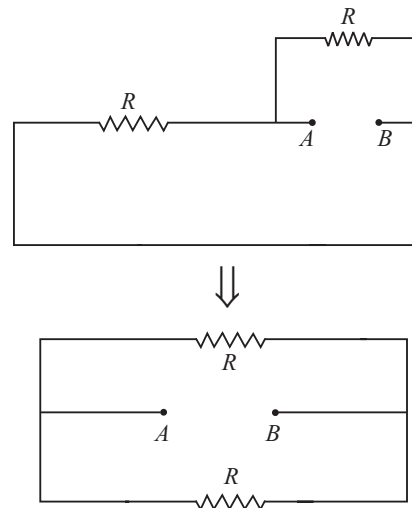
Now, we will find the time constant. The circuit, after shorting the cell, is as shown in the figure. The resistance across A and B is

$$R_0 = \frac{R}{2}$$

$$\therefore \tau = R_0 C = \frac{RC}{2}$$

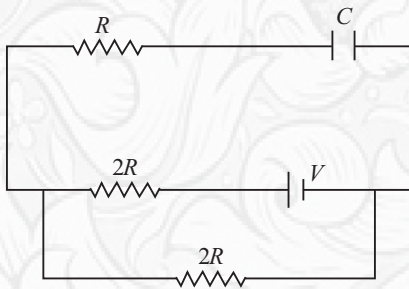
Variation in charge on the capacitor plate with time can be expressed as

$$q = q_0 [1 - e^{-t/\tau}] = \frac{CV}{2} \left[1 - e^{-\frac{2t}{3RC}} \right]$$



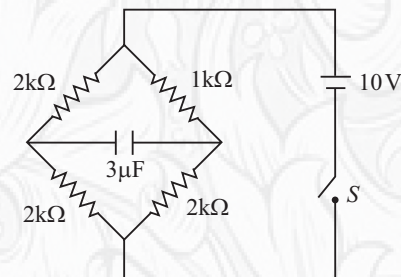
YOUR TURN

Q.5 Find the time constant for charging of the capacitor as shown in the figure.



Q.6 Switch (S) shown in the circuit is closed at $t = 0$.

- Find potential difference across the capacitor at $t = \infty$.
- Write variation in potential difference across the capacitor as a function of time.



In Short

- In the steady state, a capacitor does not allow DC current to pass through it.
- An uncharged capacitor is like a zero resistance for the purpose of calculation of electric current.
- When a cell is connected to charge a capacitor through a resistance, charge on the capacitor changes with time as

$$q = q_0 (1 - e^{-t/\tau})$$
 where $q_0 = CV$ = final charge on the capacitor and $\tau = RC$ = time constant.
- Theoretically, a capacitor will get completely charged in infinite time. Practically, we can assume it to be completely charged after few time constants.

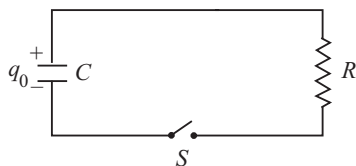
- Smaller the value of τ , faster is the charging.
- Charging current varies with time as

$$i = i_0 e^{-t/\tau}$$
 where $i_0 = \frac{V}{R}$ = initial current.
- In one time constant, 63% charging gets completed.
- When a capacitor is charged in a complex resistor network, the time constant is $\tau = R_0 C$. R_0 is equivalent resistance of the network between the capacitor plates after the cell is shorted.

4. DISCHARGING OF A CAPACITOR

A capacitor, of capacitance C , is charged using a cell of emf V_0 . Charge on the capacitor is $q_0 = CV_0$. This capacitor is to be discharged through a resistance R . The switch (S) is closed at time $t = 0$. The capacitor begins to discharge. Immediately after closing the switch, the potential difference

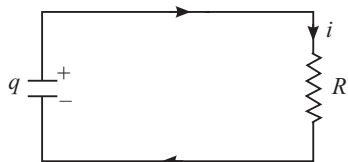
across R is $V_0 \left(= \frac{q_0}{C} \right)$. Current is obviously $i_0 = \frac{V_0}{R}$.



The charged capacitor acts like a voltage source for the resistance. As the charge on the capacitor plates decrease, the potential difference between the plates decreases. Therefore, current in the circuit goes on decreasing. Ultimately, the capacitor will get completely discharged.

Let the charge on the capacitor be q and current in the circuit be i at time t .

$$i = -\frac{dq}{dt}$$



Negative sign is necessary in the above equation as q is decreasing with time.

Potential difference across R must be same as the potential difference across C at any instant.

$$iR = \frac{q}{C}$$

$$\text{or, } -\frac{dq}{dt} = \frac{q}{RC}$$

$$\text{or, } \int_{q_0}^q \frac{dq}{q} = -\frac{1}{RC} \int_0^t dt$$

$$\text{or, } [\ln q]_{q_0}^q = -\frac{t}{RC}$$

$$\text{or, } \ln q - \ln q_0 = -\frac{t}{RC}$$

$$\text{or, } \ln \left(\frac{q}{q_0} \right) = -\frac{t}{RC}$$

$$\text{or, } q = q_0 e^{-t/RC}$$

$$\text{or, } q = q_0 e^{-t/\tau} \quad (5)$$

$$\text{or, } \frac{q}{C} = \frac{q_0}{C} e^{-t/\tau}$$

$$\text{or, } V = V_0 e^{-t/\tau} \quad (6)$$

Here, V is potential difference across C (or R) at any time t .

Current at time t is

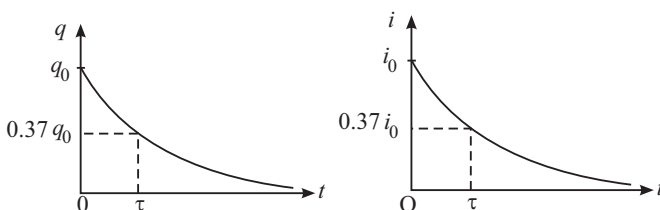
$$i = -\frac{dq}{dt} = \frac{q_0}{\tau} e^{-t/\tau}$$

$$\text{or, } i = \frac{CV_0}{CR} e^{-t/\tau} = \frac{V_0}{R} e^{-t/\tau}$$

$$\text{or, } i = i_0 e^{-t/\tau} \quad (7)$$

$$\text{where } i_0 = \frac{V_0}{R} = \text{initial current} \quad (8)$$

Variation in q and i with time has been shown in the graphs given below.



Example 5 A $10\mu\text{F}$ capacitor is charged using a 20V cell and then the cell is disconnected. The capacitor is connected to a 100Ω resistor.

- Find the heat dissipated in the resistor.
- Find the maximum rate of heat dissipation in the resistor.

Solution

Concepts

- The entire energy stored in the capacitor gets dissipated as heat.
- Rate of heat dissipation ($= i^2 R$) is maximum when the current is maximum.

- Energy stored in the capacitor is $U = \frac{1}{2} CV^2$

$$\text{or, } U = \frac{1}{2} (10\mu\text{F}) (20\text{ V})^2 = 2000\mu\text{J} = 2\text{mJ}.$$

This entire energy gets dissipated in the resistance.

- Maximum power $= i_0^2 \cdot R$, where i_0 is initial current.

$$i_0 = \frac{20\text{V}}{100\Omega} = \frac{1}{5}\text{A}$$

$$\therefore P_{\max} = \left(\frac{1}{5} \right)^2 \times 100 = 4\text{W}$$

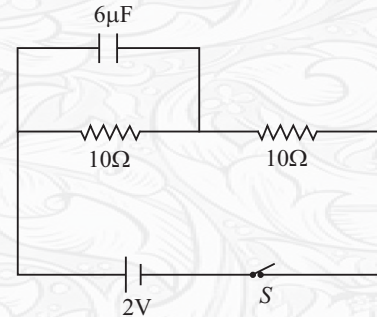


YOUR TURN

Q.7 How many time constants will elapse before the current in a discharging RC circuit drops to half its initial value?

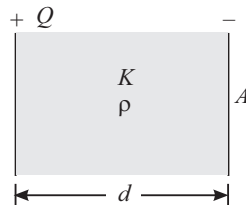
Q.8 The circuit shown in the figure is in the steady state. The switch is opened at $t = 0$.

- How much heat is dissipated after the switch is opened?
- At what time the rate of loss of charge on the capacitor plate is 0.05A ?



5. LEAKAGE IN A CAPACITOR

Consider a parallel plate capacitor having plate area A and plate separation d . The space between the plates is filled with a medium of dielectric constant K . This medium is an insulator and has a very high resistivity (ρ). But it is not a perfect insulator. It will conduct. A very small current will begin to flow from the positive plate to the negative plate (through the dielectric medium) after the charged capacitor is disconnected from the battery. After a long period of time, the capacitor will get discharged on its own. We say that the capacitor is leaking.

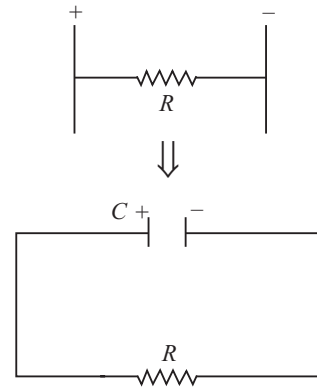


Resistance of the dielectric block is $R = \rho \frac{d}{A}$.

Effectively, we can consider that the two plates of the charged capacitor are connected through a resistance R .

The charge left on the capacitor after time t will be:

$$Q = Q_0 e^{-t/\tau},$$



where Q_0 = initial charge.

$$\tau = RC$$

$$R = \frac{\rho d}{A} \text{ and } C = \frac{K\epsilon_0 A}{d}$$

Leakage current at time t is

$$i = \frac{dQ}{dt} = i_0 e^{-t/\tau},$$

where i_0 = initial current = $\frac{Q_0}{RC} = \frac{V_0}{R}$.

V_0 is the initial potential difference across the capacitor.



YOUR TURN

Q.9 A parallel plate capacitor has plate area A and separation between the plates equal to d . The material

filled in the space between the plates has conductivity σ and dielectric constant K . Write the value of leakage current when charge on the capacitor is Q .

In Short

- (i) When a charged capacitor is discharged through a resistance, the energy stored in the capacitor gets dissipated as heat.
- (ii) In a discharging capacitor circuit, the charge on the capacitor and the current in the circuit varies with time as

$$q = q_0 e^{-t/\tau}$$

$$i = i_0 e^{-t/\tau}$$

where $\tau = RC$, q_0 = initial charge on the capacitor

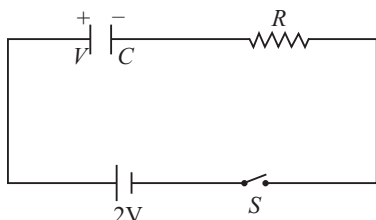
$$\text{and } i_0 = \frac{q_0}{RC}.$$

- (iii) A capacitor keeps leaking (i.e., getting discharged) through the dielectric between the plates. Practically, this leakage is very slow because of very high resistance of the dielectric material.

For solving problems on leakage current, one can show a resistance (equal to the resistance of the dielectric slab) connected in parallel to the capacitor.

Miscellaneous Examples

Example 6 A capacitor of capacitance C is charged using a cell of emf V . Now, this charged capacitor is connected to another cell of emf $2V$ through a resistance R . This connection is completed by putting on the switch S at time $t = 0$. Express the charge on the capacitor as a function of time.



Solution

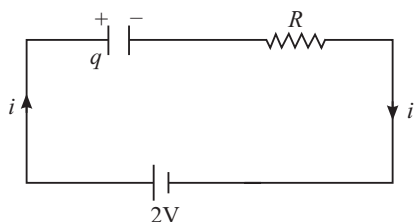
Concepts

The capacitor will continue to receive charge till potential difference across it becomes $2V$. We will set up a differential equation and then solve it to get q as a function of t .

Initial charge on the capacitor is $q_0 = CV$.

Let charge on the capacitor at time t be q and the current in the circuit be i .

$$i = \frac{dq}{dt} \quad (i)$$



Using Kirchhoff's loop law, we can write:

$$iR + \frac{q}{C} = 2V$$

$$\text{or, } R \frac{dq}{dt} + \frac{q}{C} = 2V$$

$$\text{or, } \int_{q_0}^q \frac{dq}{2V - \frac{q}{C}} = \frac{1}{R} \int_0^t dt$$

$$\text{or, } \ln\left(2V - \frac{q}{C}\right) - \ln\left(2V - \frac{q_0}{C}\right) = -\frac{t}{RC}$$

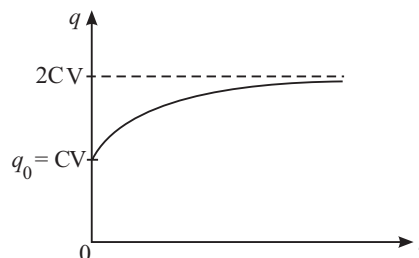
$$\text{or, } \ln\left(2V - \frac{q}{C}\right) - \ln V = -\frac{t}{RC} \quad [\because q_0 = CV]$$

$$\text{or, } \ln\left(2 - \frac{q}{CV}\right) = -\frac{t}{CV}$$

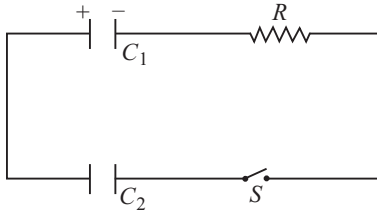
$$\text{or, } 2 - \frac{q}{CV} = e^{-t/RC}$$

$$\text{or, } q = CV(2 - e^{-t/RC})$$

The graphical plot of q versus t is as shown in the figure.



Example 7 A capacitor having capacitance $C_1 = C$ is charged to a potential difference V . It is disconnected from the cell and arranged in another circuit with a resistance R and capacitance $C_2 = C$ as shown in the figure. Find the charge on capacitor C_1 at time t after the switch S is closed at $t = 0$. Find total heat dissipated.


Solution
Concepts

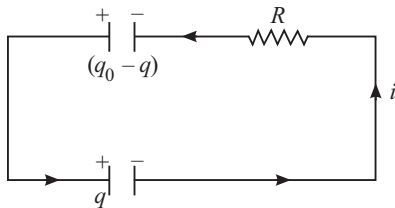
- (i) Loss in charge of C_1 will be gain in charge of C_2 .
- (ii) Current in the circuit will be equal to rate of increase in charge on C_2 .
- (iii) Keeping the above points in mind, differential equation can be written for the circuit at time t . Solution of this equation will give charge as a function of time.
- (iv) Energy loss in the capacitor system = Heat dissipated in R .

Let charge on C_2 at time t be q .

Charge on C_1 at time t will be $q_0 - q = CV - q$.

Current in the circuit is:

$$i = \frac{dq}{dt}$$



Using Kirchhoff's loop law, we can write

$$\frac{q}{C} + Ri = \frac{q_0 - q}{C}$$

$$\text{or, } RC \frac{dq}{dt} = q_0 - 2q$$

$$\text{or, } \int_0^q \frac{dq}{q_0 - 2q} = \frac{1}{RC} \int_0^t dt$$

$$\text{or, } [\ln(q_0 - 2q)]_0^q = -\frac{2t}{RC}$$

$$\text{or, } \ln(q_0 - 2q) - \ln q_0 = -\frac{2t}{RC}$$

$$\text{or, } \ln\left(1 - \frac{2q}{q_0}\right) = -\frac{2t}{RC}$$

$$\text{or, } q = \frac{q_0}{2} (1 - e^{-\frac{2t}{RC}})$$

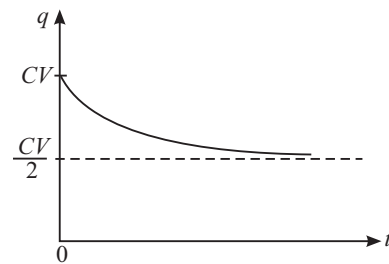
$$\text{or, } q = \frac{CV}{2} (1 - e^{-\frac{2t}{RC}}) \quad [\because q_0 = CV]$$

Charge on C_1 at time t is

$$q_1 = q_0 - q = \frac{CV}{2} (1 + e^{-\frac{2t}{RC}})$$

Graphical plot of this equation is as shown.

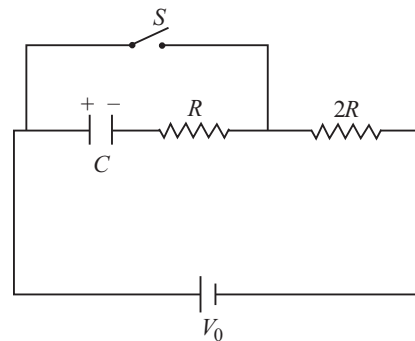
$$\text{At } t \rightarrow \infty, q = \frac{CV}{2}$$



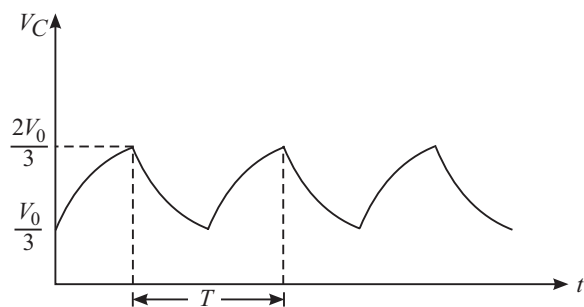
After a long time, both capacitors will have same charge and potential difference across each of them will be $\frac{V}{2}$.

$$\therefore \text{Heat} = U_i - U_f$$

$$= \frac{1}{2} CV^2 - \left[\frac{1}{2} C \left(\frac{V}{2} \right)^2 + \frac{1}{2} C \left(\frac{V}{2} \right)^2 \right] = \frac{1}{4} CV^2$$

Example 8 A voltage controlled switch


The switch (S) shown in the figure is a voltage-controlled switch. It is designed to close when the potential difference across the capacitor becomes $V_C > \frac{2}{3} V_0$ and opens when $V_C < \frac{V_0}{3}$. The potential difference across the capacitor is recorded and its variation with time (t) is found to be as shown in the figure. Find the time interval (T) between the two successive peak values of V_C .

**Solution****Concepts**

- (i) Consider the situation when a potential difference across the capacitor is $\frac{2V_0}{3}$. At this instant, the switch closes and the capacitor begins to discharge through R . Time constant of the discharging capacitor is RC .

This is because the potential difference across R is same as the potential difference across C and the entire analysis of the discharging capacitor will be same as if it is discharging through R .

- (ii) When the potential difference across the capacitor falls to $\frac{V_0}{3}$, the switch opens. Now the capacitor

begins to recharge with a time constant $3RC$ and supply voltage R_0 .

- (iii) When the voltage across C rises to $\frac{2V_0}{3}$, the switch closes again and the whole process gets repeated.

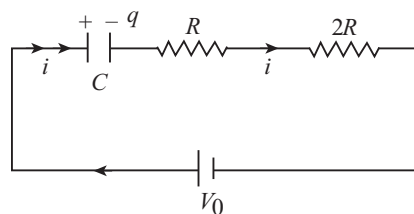
The time required for the voltage across the capacitor to fall from $\frac{2V_0}{3}$ to $\frac{V_0}{3}$ is given by

$$\frac{V_0}{3} = \frac{2V_0}{3} e^{-t/\tau} \text{ where } \tau = RC$$

$$\text{or, } e^{-t/\tau} = \frac{1}{2}; \text{ or, } t = (\ln 2) \tau = (\ln 2) RC \quad (a)$$

Now, we need to find the time needed for the capacitor to get charged from a voltage of $\frac{V_0}{3}$ to $\frac{2V_0}{3}$. The effective circuit is as shown in the figure. For simplicity, let us take $t = 0$, when $V_C = \frac{V_0}{3}$ and the switch opens.

$$i(3R) + \frac{q}{C} = V_0$$



$$\text{or, } 3R \frac{dq}{dt} = V_0 - \frac{q}{C}$$

$$\text{or, } \int_{q_0}^q \frac{dq}{V_0 - \frac{q}{C}} = \frac{1}{3R} \int_0^t dt$$

$$\text{or, } \ln \left(V_0 - \frac{q}{C} \right) - \ln \left(V_0 - \frac{q_0}{C} \right) = -\frac{t}{3RC}$$

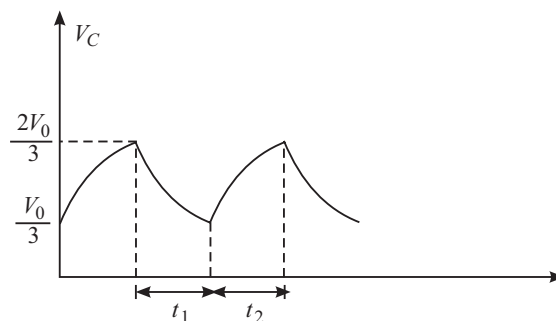
$$\text{At } t = 0, q_0 = C \frac{V_0}{3}.$$

$$\text{We need to find time when } q = \frac{2CV_0}{3}$$

$$\therefore \ln \left(V_0 - \frac{2V_0}{3} \right) - \ln \left(V_0 - \frac{V_0}{3} \right) = -\frac{t}{3RC}$$

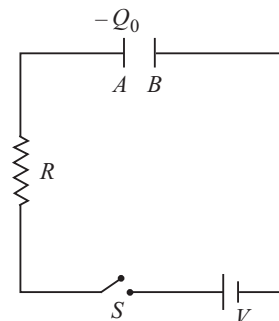
$$\text{or, } t = 3RC(\ln 2) \quad (b)$$

The required time is sum of time given by (a) and (b)



$$\tau = t_1 + t_2 = (\ln 2) RC + (3 \ln 2) RC = (4 \ln 2) RC$$

Example 9 Figure shows an RC circuit with a parallel plate capacitor. Before switching the circuit, the plate A of the capacitor has a net charge $-Q_0$ on it, while plate B has no net charge on it. Now, at $t = 0$, the circuit is switched on. How much time will elapse before the net charge on plate A becomes zero? Take



$$C = 1\mu\text{F}, \quad Q_0 = 1\text{mC}, \quad V = 1000\text{V},$$

$$R = \left(\frac{2}{\ln 3} \right) \times 10^6 \Omega.$$

Solution
Concepts

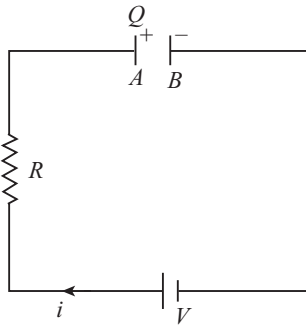
- (i) If the battery supplies charge q , the plate A will have $-Q_0 + q$ charge on it and plate B will have $-q$ charge on it.

- (ii) The inner walls of the plates A and B will have

$$\text{charge } \pm \left(\frac{-Q_0 + q + q}{2} \right) = \pm \left(\frac{2q - Q_0}{2} \right)$$

If charge supplied by the cell is q in time t , then the inner walls of the plates A and B have charges $\pm Q$.

$$\text{Here, } Q = \frac{2q - Q_0}{2}.$$



The potential difference between the plates depends on the charge on the inner walls only.

$$V_C = \frac{Q}{C} = \frac{2q - Q_0}{2C}$$

The loop equation is given as

$$iR + V_C = V$$

$$\text{or, } R \frac{dq}{dt} + \frac{2q - Q_0}{2C} = V$$

$$\text{or, } \int_0^{Q_0} \frac{dq}{2VC + Q_0 - 2q} = \frac{1}{2RC} \int_0^t dt$$

[when $q = Q_0$, total charge on A will become zero]

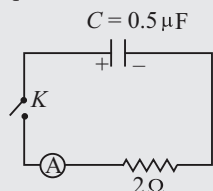
$$\text{or, } \ln \left(\frac{2VC + Q_0 - 2Q_0}{2VC + Q_0} \right) = -\frac{t}{RC}$$

$$\text{or, } t = RC \ln \left(\frac{2VC + Q_0}{2VC - Q_0} \right)$$

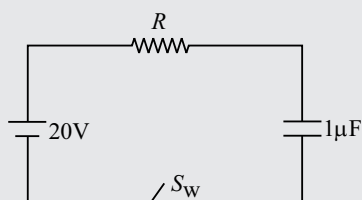
Substituting the numerical values gives $t = 2\text{s}$.

Worksheet 1

1. A charged capacitor is allowed to discharge through a resistor by closing the key at the instant $t = 0$. At the instant $t = (\ln 4)\mu\text{s}$, the reading of the ammeter falls to half the initial value. The resistance of the ammeter is equal to—



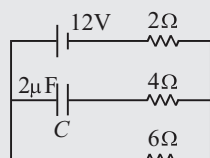
- (a) $1\text{M}\Omega$ (b) 1Ω
(c) 2Ω (d) $2\text{M}\Omega$
2. Figure shows an uncharged capacitor connected in series with a resistor R and ideal battery. At $t = 0$, switch S_w is closed and it is observed that when charge on capacitor is 75% of its maximum value the current in the circuit is 1A. The value of R is



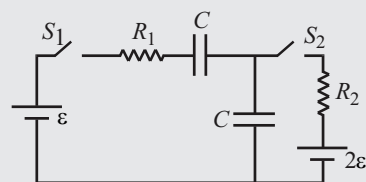
- (a) 1Ω (b) 10Ω
(c) 5Ω (d) can't be determined
3. A capacitor of capacitance C is charged by a battery. There is a resistance R in series with the capacitor. The time after which potential difference across the resistor becomes n times to that across the capacitor is

- (a) $RC \ln\left(\frac{1+n}{n}\right)$ (b) $RC \ln\left(\frac{n}{1+n}\right)$
(c) $RC \ln\left(\frac{n}{n-1}\right)$ (d) $RC \ln(1+n)$

4. Charge on the capacitor in the given circuit in steady state condition is:

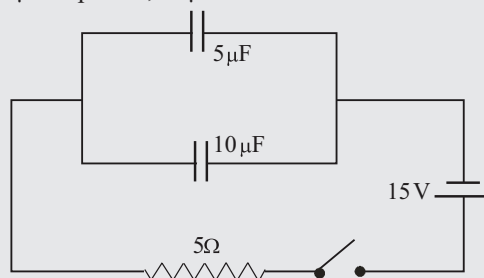


- (a) $12\mu\text{C}$ (b) $15\mu\text{C}$
(c) $18\mu\text{C}$ (d) $6\mu\text{C}$
5. In the circuit shown, switch S_2 is kept closed for a long time. Now, S_1 is also closed. Just after that instant the current through S_1 is:



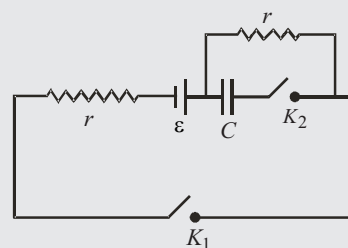
- (a) $\frac{\varepsilon}{R_1}$ towards right (b) $\frac{\varepsilon}{R_1}$ towards left
(c) zero (d) $\frac{2\varepsilon}{R_1}$

6. In the shown circuit, the current flowing through the $5\mu\text{F}$ capacitor, $75\mu\text{s}$ after the switch is closed, will be

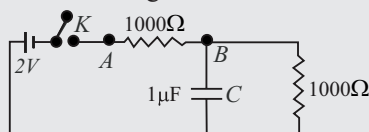


- (a) $\frac{e-1}{e}$ A (b) e^{-1} A
(c) $\frac{2(e-1)}{e}$ A (d) none of the above

7. Keys K_1 is kept closed for a long time, in the shown circuit. Now, K_2 is also closed. The ratio of current through K_1 immediately after closing K_2 and a long time after that, is:

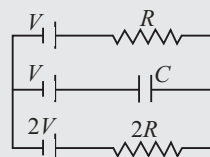


- (a) 2 : 1 (b) 1 : 2
(c) 1 : 1 (d) infinite
8. When the key K is pressed at time $t = 0$, which of the following statements about the current I in the resistor AB of the given circuit is true:



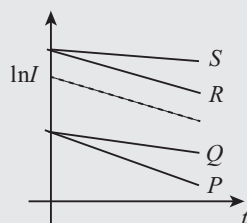
- (a) $I = 2\text{mA}$ at all t
 (b) I oscillates between 1 mA and 2mA
 (c) $I = 1\text{mA}$ at all t
 (d) At $t = 0$, $I = 2\text{mA}$ and with time it goes to 1mA

9. In the given circuit, with steady current, the potential drop across the capacitor must be:



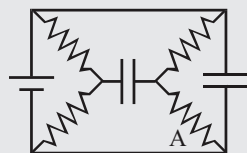
- (a) V (b) $V/2$
 (c) $V/3$ (d) $2V/3$

10. A capacitor is charged using an external battery with a resistance x in series. The dashed line shows the variation of $\ln I$ with respect to time. If the resistance is changed to $2x$, the new graph will be:



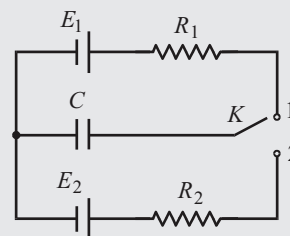
- (a) P (b) Q
 (c) R (d) S

11. Each resistor in the given circuit has a resistance of $2\text{M}\Omega$ and each capacitor has capacitance of $1\mu\text{F}$. The battery voltage is 3V . The voltage across resistor 'A', when the circuit is in steady state, is:



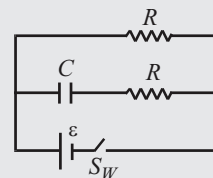
- (a) 0 V (b) 0.5 V
 (c) 0.75 V (d) 1.5 V

12. The key K in the figure is a two way switch. It is connected in turn to each of the two contacts over short identical time intervals so that the change in the charge on the capacitor over each connection is small. The final charge q_0 on the capacitor is:



- (a) $\frac{(E_1 R_2 + E_2 R_1) C}{R_1 + R_2}$ (b) $\frac{(E_1 E_2) C}{E_1 + E_2}$
 (c) $\frac{(E_1 R_1 + E_2 R_2) C}{R_1 + R_2}$ (d) none of these

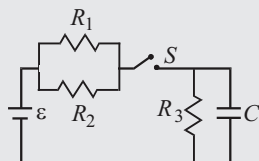
13. The capacitor shown in the circuit is uncharged. At $t = 0$ the switch S_w is closed. Charge on capacitor when the current through battery becomes 50% of its maximum value is:



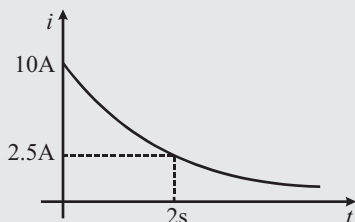
- (a) $\frac{C\varepsilon}{3}$ (b) $\frac{C\varepsilon}{2}$
 (c) $\frac{C\varepsilon}{4}$ (d) $C\varepsilon$

Worksheet 2

1. The circuit shown in the figure consists of a battery of emf $\varepsilon = 10\text{ V}$; a capacitor of capacitance $C = 1.0\text{ }\mu\text{F}$ and three resistors of resistances $R_1 = 2\Omega$, $R_2 = 2\Omega$ and $R_3 = 1\Omega$. Initially the capacitor is completely uncharged and the switch S is open. The switch S is closed at $t = 0$.



- The current through resistor R_3 at the moment the switch is closed is zero.
 - The current through resistor R_3 , a long time after the switch is closed is 5 A .
 - The ratio of current through R_1 and R_2 is always constant.
 - The maximum charge on the capacitor during the operation is $5\mu\text{C}$.
2. The figure shows, a graph of the current in a discharging circuit of a capacitor through a resistor of resistance $10\text{ }\Omega$.

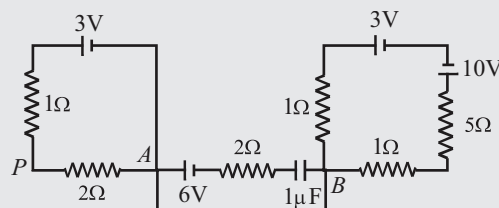


- The initial potential difference across the capacitor is 100 volt .
 - The capacitance of the capacitor is $\frac{1}{10\ln 2}\text{ F}$.
 - The total heat produced in the circuit will be $\frac{500}{\ln 2}\text{ joules}$.
 - The thermal power in the resistor will decrease with a time constant $\frac{1}{2\ln 2}\text{ second}$.
3. Capacitor C_1 of capacitance 1 micro-farad and capacitor C_2 of capacitance 2 microfarad are separately charged fully by a common battery. The two capacitors are then separately allowed to discharge through equal resistors at time $t = 0$.
- The current in each of the two discharging circuits is zero at $t = 0$
 - The currents in the two discharging circuits are equal at $t = 0$

- The currents in the two discharging circuits at $t = 0$ are unequal

- Capacitor C_1 , loses 50% of its initial charge sooner than C_2 loses 50% of its initial charge.

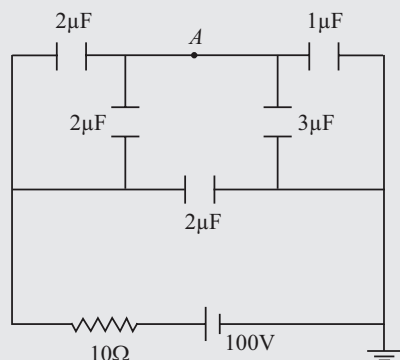
4. Consider the circuit shown in the figure and choose the correct alternatives.



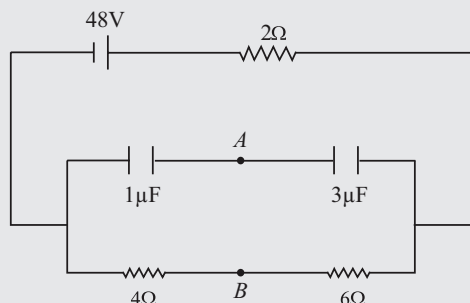
- Charge on the capacitor in steady state is $6\mu\text{C}$
 - Current through 6 V cell when the capacitor is completely uncharged is 3 A
 - The potential difference between points P and A is 1 V
 - None of the above.
5. Which of the followings are true for charging of a capacitor in a RC circuit?
- The rate of energy storage in the capacitor increases with time
 - The rate of energy storage in the capacitor is maximum initially and thereafter decreases with time
 - The rate of energy storage in the capacitor first increases and then it decreases with time
 - The rate of heat generation in the circuit is maximum initially and then goes on decreasing with time.
6. Which of the followings are true for discharging of a capacitor in a RC circuit?
- The rate of energy loss of the capacitor increases with time
 - The rate of energy loss of the capacitor decreases with time
 - The rate of energy loss of the capacitor first increases and then it decreases with time
 - The rate of heat generation in the circuit is maximum initially and then goes on decreasing with time.

Worksheet 3

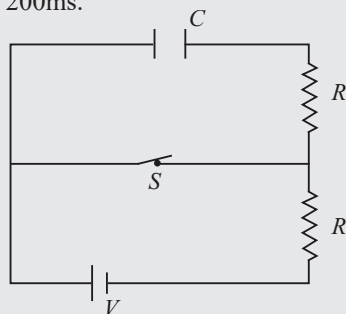
1. The circuit shown in figure is in the steady state. Find the potential of point A.



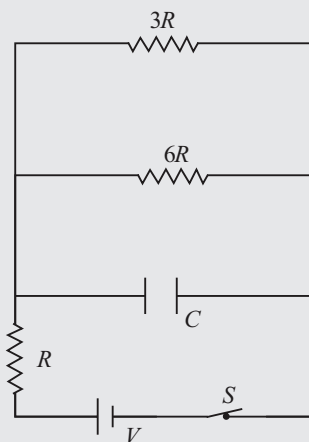
2. The circuit shown in the figure is in the steady state. Find the potential difference between the points A and B.



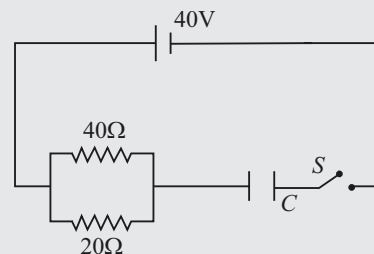
3. A capacitor of capacitance $C = 10\mu\text{F}$ is connected to a resistance $R = 2\Omega$ and a battery of negligible internal resistance having an emf of $V = 5\text{ V}$. $20\mu\text{s}$ after completing the circuit, find
- power delivered by the battery.
 - rate of heat dissipation.
 - rate of energy storage in the capacitor.
4. A capacitor is charged to 100 V . The cell is disconnected. Now, the capacitor is connected to a resistance. It is found that the potential difference across the capacitor drops to 10V after time $t_0 = 100\text{ms}$. Find the potential difference across the capacitor at time $t = 200\text{ms}$.
5. In the circuit shown, the switch (S) is held closed for a long time and then it is opened at $t = 0$. Find charge on the capacitor at time t .



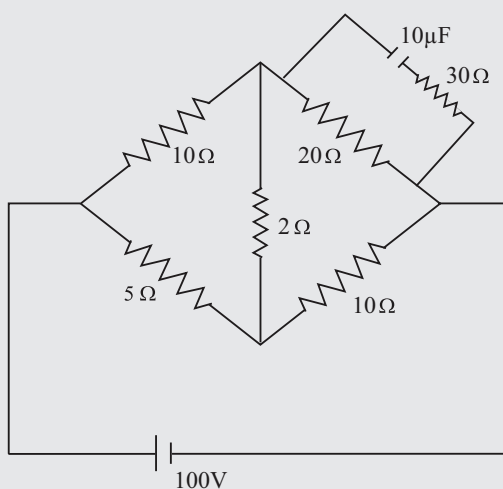
6. The circuit shown in the figure is in the steady state. Find heat dissipated in the resistance $3R$ after the switch (S) is opened.



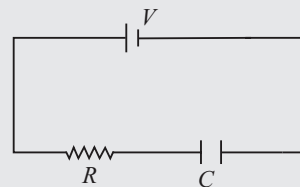
7. A charged capacitor, having capacitance $C = 2.0\mu\text{F}$, is disconnected from the cell. Now, the plates of the capacitors are connected using a thin wire. The electric field between the plates drops to one-third of its initial value in 4.4ms . Find the resistance of the thin wire [Take $\ln 3 \approx 1.1$].
8. A parallel plate capacitor is filled completely with a material having dielectric constant $K = 5.01$ and electrical conductivity $\sigma = 7.4 \times 10^{-12} (\Omega\text{m})^{-1}$. Charge on the capacitor at time $t = 0$ was found to be $q_0 = 8.86\text{ mC}$. Calculate the leakage current through the dielectric at $t = 12\text{ s}$ [$\epsilon_0 = 8.86 \times 10^{-12}\text{ SI unit}$].
9. The capacitor shown in the figure has a capacitance $C = 10\mu\text{F}$. After the switch is closed, how much heat will get dissipated in 20Ω resistor till the capacitor gets fully charged.



10. Find the charge on the capacitor in the steady state.



11. An air-filled parallel plate capacitor has capacitance C . It is connected to an ideal source of emf V through a resistance R . A dielectric slab of dielectric constant K is inserted between the capacitor plates to occupy the entire space. The slab is quickly pulled out after the steady state is reached. Find the heat generated in the circuit before the steady state is reached again.



Answer Sheet

Your Turn

1. (a) $80\mu\text{C}$, (b) 2mA , (c) 0.28 mA , (d) 17.2 V 2. $(2\ln 2)RC$ 3. (a) $\frac{126}{5}A$, (b) $25A$, (c) $10\mu\text{C}$
4. 5.55 mA 5. $2RC$ 6. (a) $\frac{5}{3}\text{ V}$, (b) $\frac{5}{3}\left(1 - e^{-\frac{t}{0.005}}\right)$
7. $\ln 2 = 0.69$ 8. (a) $3\mu\text{J}$, (b) $4.14 \times 10^{-5}\text{s}$ 9. $\frac{Q\sigma}{K\epsilon_0}$

Worksheet 1

1. (c) 2. (c) 3. (a) 4. (c) 5. (b) 6. (b) 7. (a) 8. (d) 9. (c)
10. (b) 11. (d) 12. (a) 13. (d)

Worksheet 2

1. (a, b, c, d) 2. (a, b, c, d) 3. (b, d) 4. (a, b) 5. (c, d) 6. (b, d)

Worksheet 3

1. 50 V 3. (a) 4.625W , (b) 1.71W , (c) 2.91W 4. 1V 5. $q = (CV) e^{-\frac{t}{2RC}}$ 6. $\frac{4}{27}CV^2$
7. $2\text{k}\Omega$ 8. 0.2 mA 9. $\frac{16}{3}\text{mJ}$ 10. $\frac{2}{3}\text{mC}$ 11. $\frac{1}{2}CV^2(K-1)^2$

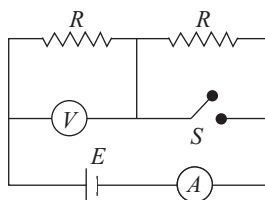
Miscellaneous Problems on Chapters 6–8

MATCH THE COLUMN

1. Match the physical quantities with their units

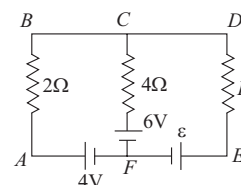
Column-I		Column-II	
(A)	Electric current	(P)	(farad)(volt) / second
(B)	Current density	(Q)	$\frac{\text{kg-m}}{(\text{second}) \times (\text{newton})}$
(C)	Resistivity	(R)	volt/ Ω
(D)	Time constant	(S)	$\Omega - \text{m}$
		(T)	$\frac{(\text{volt})(\text{m})(\text{second})}{\text{coulomb}}$
		(U)	A/m^2

2. In the circuit shown in figure, battery, ammeter and voltmeter are ideal and the switch S is initially closed. When switch S is opened, match the parameters of column-I with the effects in column-II



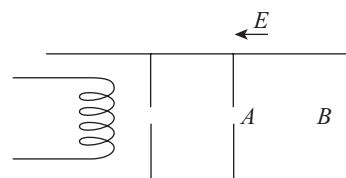
Column-I		Column-II	
(A)	Equivalent resistance across the battery	(P)	remains same
(B)	Power dissipated by left resistance R	(Q)	increases
(C)	Voltmeter reading	(R)	decreases
(D)	Ammeter reading	(S)	becomes zero

3. A circuit is shown in figure. R is a non zero variable and finite resistance. e is some unknown emf with polarities as shown. Match the columns



Column-I		Column-II	
(A)	Current passing through 4Ω resistance can be zero	(P)	possible if $e = 6\text{V}$
(B)	Current passing through 4Ω resistance can be from F to C	(Q)	possible if $e > 6\text{V}$
(C)	Current passing through 4Ω resistance can be from C to F	(R)	possible if $e < 6\text{V}$
(D)	Current passing through 2Ω resistance will be from B to A	(S)	possible for any positive or zero value of e

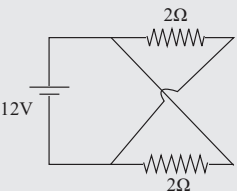
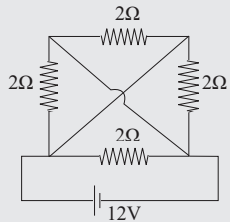
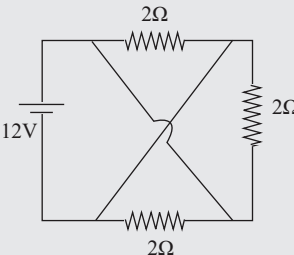
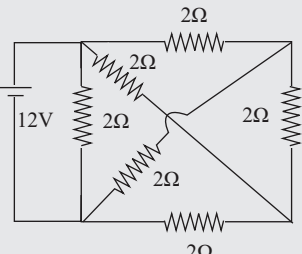
4. Electrons are emitted by a hot filament and are accelerated by an electric field as shown in figure. The two stops at the left ensure that the electron beam has a uniform cross-section. Match the entries of column-I with column-II as electrons move from A to B :



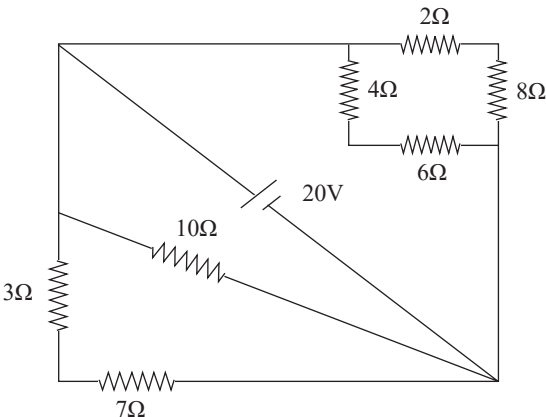
Column-I		Column-II	
(A)	Speed of an electron	(P)	Increases
(B)	Number of free electrons per unit volume	(Q)	Decreases
(C)	Current density	(R)	Remains same

9.2 Electrostatics and Current Electricity

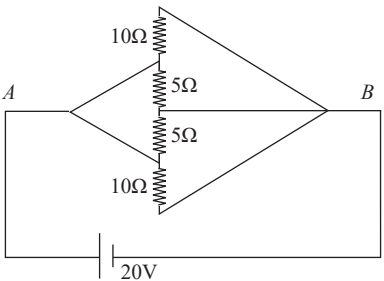
5. Column-I shows some circuits. Column-II gives current in ampere. Match the battery current in the figures given in column-I to their correct values in column-II.

Column-I		Column-II	
(A)		(P)	18A
(B)		(Q)	24A
(C)		(R)	6A
(D)		(S)	12 A

6. Consider the following two electric circuits and match the entries in column-I with those in column-II.



Circuit (X)

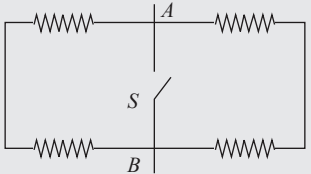


Circuit (Y)

Column-I		Column-II	
(A)	$\frac{\text{Battery current in circuit (X)}}{\text{Battery current in circuit (Y)}}$	(P)	The ratio will not change if one 5Ω resistor is interchanged with a 10Ω resistor in circuit (Y)
(B)	$\frac{\text{Current in } 2\Omega \text{ resistance in circuit (X)}}{\text{Current in } 10\Omega \text{ resistance in circuit (Y)}}$	(Q)	$\frac{1}{2}$
(C)	$\frac{\text{Current in } 7\Omega \text{ resistance in circuit (X)}}{\text{Current in } 10\Omega \text{ resistance in circuit (Y)}}$	(R)	1
(D)	$\frac{\text{Power dissipated in circuit (X)}}{\text{Power dissipated in circuit (Y)}}$	(S)	$\frac{2}{3}$

7. Fill in the blanks in column-I with appropriate choices from column-II.

Column-I		Column-II	
(A)	Two identical non-ideal cells of emf x each, when combined to form a battery, may act as a battery of emf _____	(P)	$0.5x$
(B)	Two identical fuses rated (in ampere) at x , when combined together may act as a fuse of rating _____	(Q)	x
(C)	Four identical bulbs (each of power x , when given a V volt supply) are arranged as shown. A cell of emf V is connected across A and B . The combined power output of the network can be _____. The switch (S) may be open or closed	(R)	$2x$



(D)	A voltmeter can measure voltage up to x volt. A resistance is connected in parallel to the voltmeter. The resistance has value equal to the resistance of the voltmeter. The new voltmeter so formed can read up to _____ volt	(S)	0
-----	--	-----	---

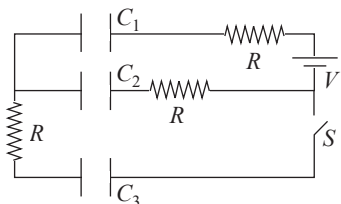
8. A capacitor is connected in series to a resistance R . This series RC group is connected to a cell of emf $2V$ and the capacitor is charged to saturation. In another experiment, the RC group is first connected to a cell of emf V and after the capacitor gets fully charged, it is connected to a cell of emf $2V$. The circuit is allowed to attain a steady state.

Column-I		Column-II	
(A)	Saturation charge on the capacitor	(P)	is higher in the first case
(B)	Maximum energy stored in the capacitor	(Q)	is higher in the second case
(C)	Energy supplied by the cell/s	(R)	is same in both cases
(D)	Heat generated in the resistor	(S)	is half in the second case

9. In column-I, some physical quantities have been given for a current carrying wire. Column-II lists the parameters on which the quantities in column-I may depend. Match them.

Column-I		Column-II	
(A)	Electrical conductivity	(P)	Cross-sectional area of the conductor
(B)	Heat generated in a unit volume of a conductor	(Q)	Temperature of the conductor
(C)	Current density	(R)	Material of the conductor
(D)	Current	(S)	Electric field inside the conductor

10. In the circuit shown, all capacitors have same capacitance $C_1 = C_2 = C_3 = C$. Switch S is closed at $t = 0$. Match the quantities in column-I with their values in column-II for a long period after S is closed.



Column-I		Column-II	
(A)	Loss in energy stored in C_1	(P)	$\frac{1}{6}CV^2$
(B)	Loss in energy stored in C_2	(Q)	$\frac{1}{18}CV^2$
(C)	Gain in energy stored in C_3	(R)	$-\frac{7}{72}CV^2$
(D)	Loss in chemical energy of the cell	(S)	$\frac{5}{72}CV^2$

PASSAGE BASED PROBLEMS

Passage 1

An accumulator of emf 2 volt and negligible internal resistance is connected across a uniform wire of length 10m and resistance 30Ω . The appropriate terminals of a cell of emf 1.5 volt and internal resistance 1Ω is connected to one end of the wire, and the other terminal of the cell is connected through a sensitive galvanometer to a slider on the wire.

- What length of the wire will be required to produce zero deflection of the galvanometer?
 - 7.5 m
 - 8.75 m
 - 6.25 m
 - 4.0 m
- What will be the balancing length when a coil of resistance 5Ω is placed in series with the accumulator.
 - 7.5 m
 - 8.75 m
 - 6.25 m
 - 4.0 m

Passage 2

The current I through a rod of a certain metallic oxide is given by $I = 0.2V^{1/2}$ where V is the potential difference across it. The rod is connected in series with a resistance to a 6 volt battery of negligible internal resistance. What value should the series resistance have so that:

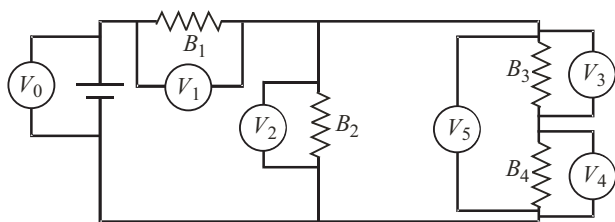
- The current in the circuit is 0.4 A
 - 2 Ohm
 - 1 Ohm
 - 5 Ohm
 - 4 Ohm
- The power dissipated in the rod is twice that dissipated in the resistance
 - 2 Ohm
 - 1 Ohm
 - 5 Ohm
 - 4 Ohm

Passage 3

B_1 , B_2 , B_3 and B_4 are identical light bulbs. There are six voltmeters connected to the circuit as shown. All voltmeters are connected so that they display positive voltages. Assume that the voltmeters have very large resistances and

9.4 Electrostatics and Current Electricity

practically, they do not affect the circuit. The filament of the bulb B_2 burns out, and reading of voltmeter V_1 changes from V to V' .



5. Which voltmeter(s) would show zero reading?

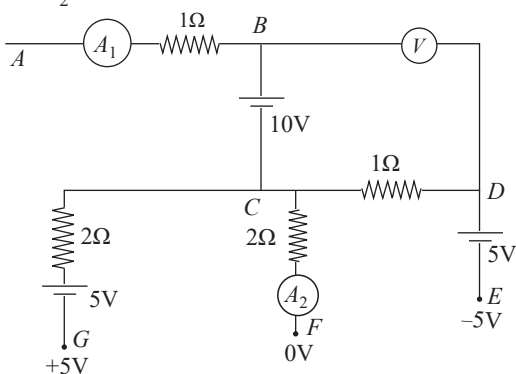
- (a) none would read zero
- (b) only V_2
- (c) only V_3 and V_4
- (d) only V_3 , V_4 and V_5

6. Which of the following is true?

- (a) $V' < V/2$
- (b) $2V > V' > V$
- (c) $V' = V$
- (d) $V/2 < V' < V$

Passage 4

In the circuit shown in the figure, the terminals E , F and G are maintained at potentials $-5V$, $0V$ and $+5V$, respectively. The ammeters and voltmeters are ideal. The reading in ammeter A_2 is zero.



7. Voltmeter reading is

- (a) 0V
- (b) 10V
- (c) 5V
- (d) 15V

8. The reading of ammeter A_1 is

- (a) 0A
- (b) 10A
- (c) 5A
- (d) 2A

9. Potential of the terminal A is

- (a) 0V
- (b) 10V
- (c) 2V
- (d) 5V

Passage 5

A cylindrical conductor is connected to an ideal battery and a steady state is reached. Temperature of the conductor exceeds the ambient temperature by $\Delta T_1 = 6^\circ\text{C}$.

Now, half the length of the conductor is cut and only the remaining half is connected to the battery. This time, in

steady state the temperature of the conductor exceeds the ambient temperature by ΔT_2 . Neglect any change in resistivity due to temperature change and assume that heat is lost by the cylindrical conductor from its curved cylindrical surface only. The rate of heat loss is proportional to the temperature difference between the conductor and the surrounding.

10. In steady state, the power lost by the cell is same

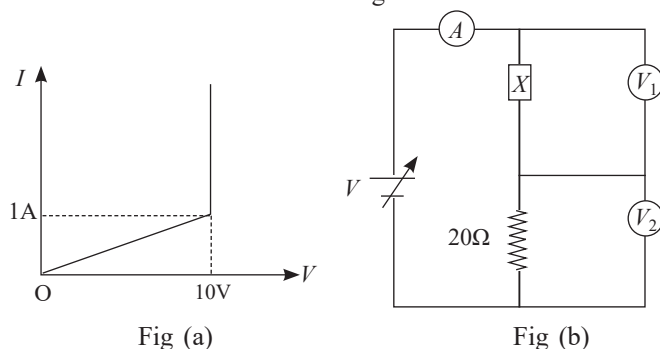
- (a) as rate of net energy loss by the cylindrical conductor surface to the surrounding.
- (b) as rate of energy absorbed by the conductor surface from the surrounding.
- (c) for the longer and the shorter conductors.
- (d) None of the above

11. Value of ΔT_2 is

- (a) 6°C
- (b) 12°C
- (c) 60°C
- (d) 24°C

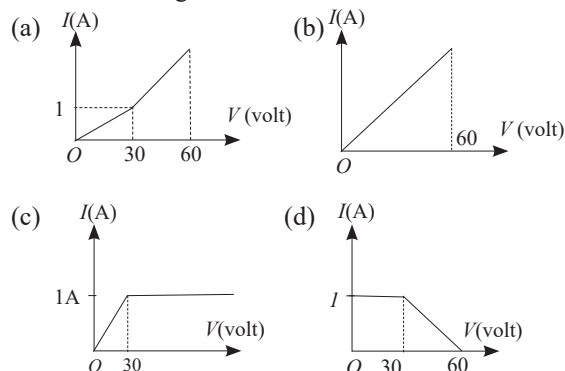
Passage 6

The current voltage characteristic of a device X is as shown in figure (a). The device is connected in series to a 20Ω resistor and then a variable voltage.

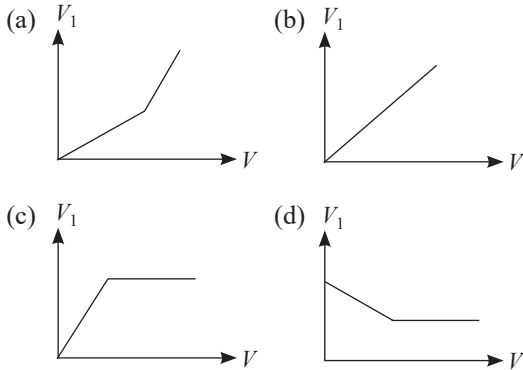


Source is connected across the combination. The source voltage can be varied between 0 and 60V. The current supplied by the source is measured using ammeter A and the voltages across X and the 20Ω resistance are measured using voltmeters V_1 and V_2 , respectively. The ammeter and the voltmeters are ideal.

12. Reading of the ammeter (I) is plotted against the source voltage (V). The correct graph is given as shown in the figure.

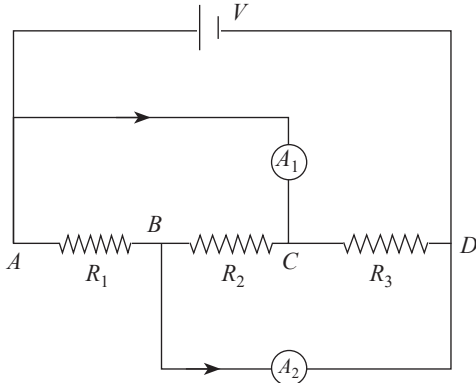


13. Reading of voltmeter V_1 is plotted against the source voltage. The correct graph will be



Passage 7

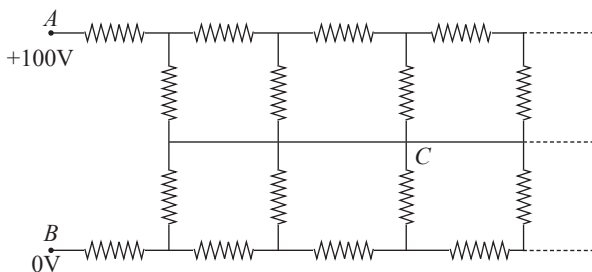
In the circuit shown in the figure, the readings of the ideal ammeters A_1 and A_2 are 0.2A and 0.3A , respectively. It is observed that if positions of resistors R_2 and R_3 are interchanged, the readings of the ammeters do not change. The battery is ideal.



14. The current through the battery is
 (a) 0.3A (b) 0.5A
 (c) 0.35A (d) 0.4A
15. If $R_1 = 2\Omega$, then R_2 is equal to
 (a) 1Ω (b) 4Ω
 (c) 1.5Ω (d) 2Ω

Passage 8

Each resistance in the circuit shown is r . The circuit extends to right up to infinity. Points A and B are maintained at potentials 100V and 0V , respectively.



16. Potential of point C will be

- (a) $\frac{100}{3}\text{V}$ (b) 50V
 (c) $\frac{200}{3}\text{V}$ (d) $\frac{50}{3}\text{V}$

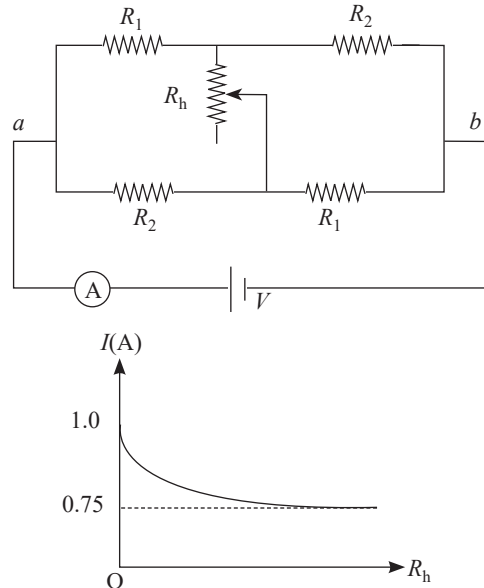
17. The equivalent resistance between A and B will be

- (a) $(\sqrt{3}-1)r$ (b) $(\sqrt{5}-1)r$
 (c) $(\sqrt{5}+1)r$ (d) $\left(\frac{\sqrt{5}+1}{2}\right)r$

Passage 9

The circuit shown in the figure has ideal battery of emf $V = 3\text{V}$. The resistance $R_1 > R_2$.

The resistance R_h of the rheostat is varied from zero to a large value and the ammeter reading (I) is recorded. The graph of I versus R_h is shown in the figure.



18. The equivalent resistance of the network between a and b , when R_h is zero, is

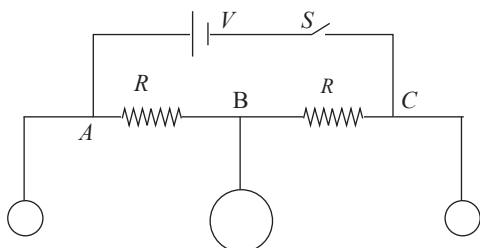
- (a) $\frac{2R_1R_2}{R_1+R_2}$ (b) $2(R_1+R_2)$
 (c) $\sqrt{\frac{R_1^2R_2^2}{R_1^2+R_1R_2}}$ (d) $\frac{R_1R_2}{R_1+R_2}$

19. Value of R_1 is nearly

- (a) 3Ω (b) 2Ω
 (c) 6Ω (d) 4Ω

Passage 10

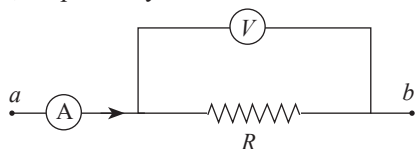
Two resistances, R each, are connected in series to an ideal cell of emf V . Three metal balls are connected to the terminals A , B and C , as shown in the figure. The central ball has radius b and the other two balls have same radius a . The balls are far apart from one another. Neglect the capacitance of everything other than the balls. The switch (S) is closed and the circuit attains steady state.



20. Sum of the charges on the three balls is
 (a) $4\pi\epsilon_0(2a+b)V$ (b) $2\pi\epsilon_0(a+b)V$
 (c) $2\pi\epsilon_0(2a+b)V$ (d) 0
21. Charge on the ball connected to A is
 (a) $2\pi\epsilon_0 aV$ (b) $4\pi\epsilon_0 aV$
 (c) $\pi\epsilon_0(a+b)V$ (d) 0

Passage 11

To measure the value of resistance R , an ammeter (A) is connected in series to it and a voltmeter (V) is connected in parallel. When a potential difference is applied between the points a and b , the readings of the ammeter and voltmeter are I and V , respectively.



Measured value of resistance is $R_m = \frac{V}{i}$.

The ammeter and voltmeter have resistance R_A and R_V , respectively.

22. The true value of resistance (R), if calculated, will
 (a) depend on R_A as well as R_V
 (b) not depend on R_A
 (c) be equal to R_m
 (d) be smaller than R_m

23. True value of R is given by

- (a) $\frac{1}{R} = \frac{1}{R_m} - \frac{1}{R_V}$ (b) $\frac{1}{R} = \frac{1}{R_m} - \frac{1}{R_V} + \frac{1}{R_A}$
 (c) $\frac{1}{R} = \frac{R_V - R_A}{R_m R_A}$ (d) $\frac{1}{R} = \frac{R_V - R_A}{R_m R_V}$

Passage 12

The circuit (a) contains resistors of resistances R and $2R$ and a cell of emf V . It also has a bulb (B_1) of resistance R . Another circuit (b) has an identical bulb (B_2) connected to a cell of emf E with a resistance R_0 in series. When the switch (S) in circuit (a) is closed, the bulb (B_1) glows with the same brightness as bulb (B_2) in the circuit (b). Using circuit analysis, it can be shown that this is possible if:

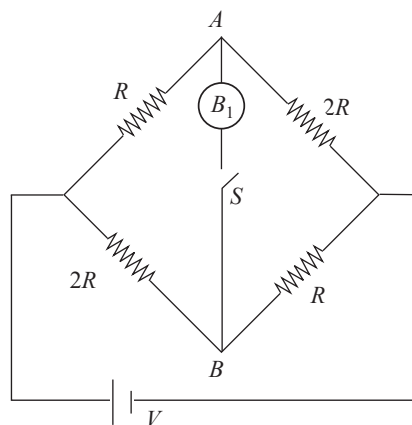
- (i) R_0 is chosen to have the value of resistance, between A and B in circuit (a) when the switch (S) is open and the battery is replaced by a copper wire link,
 (ii) ϵ is chosen equal to the potential difference between A and B when the switch (S) is open.

24. Value of R_0 is

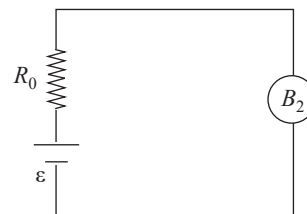
- (a) $\frac{2R}{3}$ (b) $\frac{R}{3}$
 (c) $\frac{4R}{3}$ (d) R

25. Current (I) in bulb B_2 is

- (a) $\frac{V}{7R}$ (b) $\frac{V}{R}$
 (c) $\frac{2V}{7R}$ (d) $\frac{3V}{7R}$



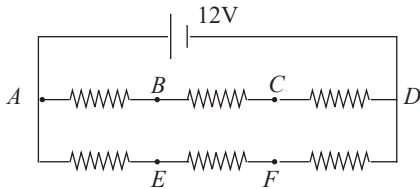
Circuit (a)



Circuit (b)

Passage 13

Each resistor, in the circuit shown, has a resistance of 4Ω . The cell has an emf of 12V and negligible internal resistance. When a resistance R_1 is connected between the points A and F , there is no deflection in a galvanometer connected between B and F . When the same resistance R_1 is connected between D and E , a galvanometer between C and E records a current i .



26. Value of R_1 is

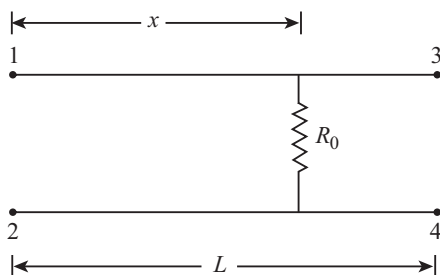
- (a) $\frac{2}{3}\Omega$ (b) $\frac{8}{3}\Omega$
(c) 1Ω (d) $\frac{1}{3}\Omega$

27. Value of i is

- (a) 0 (b) 1A
(c) 2A (d) 1.2 A

Passage 14

An underground cable has two parallel wires each of length $L = 1.0\text{ km}$. At some unknown distance x from the terminals 1 and 2, the cable developed a fault. The fault can be characterised by a resistance R_0 between the parallel lines as shown in the figure. An engineer decides to find the distance x , from the terminals 1 and 2, where the fault is located. For this, he measured the resistance between the terminals 1 and 2 and found it to be $R_1 = 400\Omega$. Then he measured the resistance between the terminals 3 and 4 to be $R_2 = 450\Omega$. Thereafter, the terminals 3 and 4 were connected using a wire of negligible resistance and the measured resistance between 1 and 2 now becomes $R_2 = 75\Omega$. Assume that the resistance R_0 of the fault is large compared to the resistance of the cables. (This will give an approximate value of x which can help locate the fault.)



28. Resistance (R) of the total length, $2L$, of the lines is nearly equal to

- (a) 25Ω (b) 400Ω
(c) 500Ω (d) 75Ω

29. Value of x is

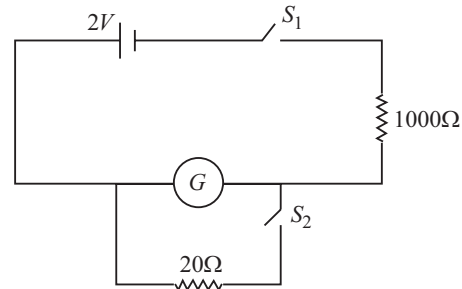
- (a) $\frac{1}{6}\text{ km}$ (b) $\frac{1}{2}\text{ km}$
(c) $\frac{1}{3}\text{ km}$ (d) $\frac{1}{4}\text{ km}$

30. Value of R_0 is

- (a) 300Ω (b) 3000Ω
(c) 387.5Ω (d) 4000Ω

Passage 15

A galvanometer is to be modified to be used as a voltmeter. It was decided to measure the internal resistance and the figure of merit of the galvanometer using the circuit arrangement shown in the figure. The figure of merit can be defined as change in current in the galvanometer for deflection of its pointer to change by 1 division. The given galvanometer has 20 divisions. Its resistance is small compared to 1000Ω .



With the switch S_2 open and S_1 closed, the galvanometer shows full-scale deflection. When S_2 is also closed, the deflection is halved.

31. Figure of merit of the galvanometer is

- (a) 0.1 mA/div (b) 0.2 mA/div
(c) 0.15 mA/div (d) 0.05 mA/div

32. To convert the given galvanometer into a voltmeter of range $0\text{--}4.5\text{V}$, the resistance to be connected in series is

- (a) 2000Ω (b) 2230Ω
(c) 4300Ω (d) 4500Ω

Passage 16

Consider two different ideal ammeters in which angular deflection of the needle is proportional to the current flowing through it. The two ammeters are connected in series in a circuit as shown in figure (a). Deflections in them are α_1 and α_2 . R is a known resistance and X is an unknown resistance. In another experiment, the two ammeters are connected in

parallel into a circuit as shown in figure (b). This time the deflections in them are β_1 and β_2 .

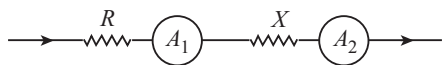


Fig (a)

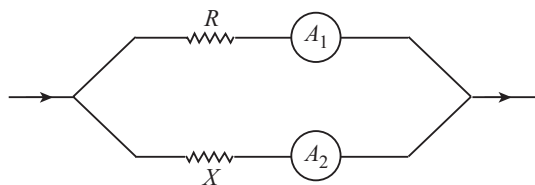


Fig (b)

33. If $\alpha_1 = 2\alpha_2$, then

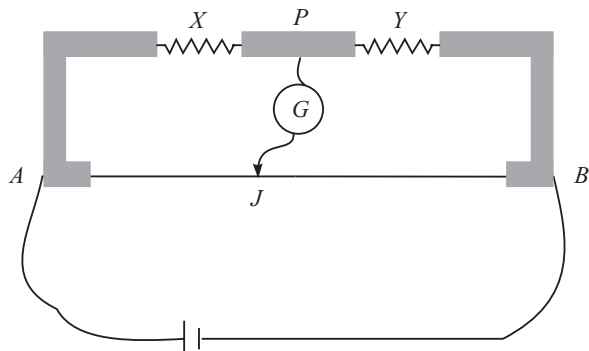
- (a) $\beta_1 = \beta_2$ if $R = X$
- (b) $\beta_1 = 2\beta_2$ if $R = 2X$
- (c) $R = X$ if $\beta_1 = 2\beta_2$
- (d) $R = \frac{X}{2}$ if $\beta_1 = 2\beta_2$

34. Value of R_x is

- (a) $\frac{R\alpha_2\beta_1}{\alpha_1\beta_2}$
- (b) $R \frac{\alpha_1\beta_2}{\alpha_2\beta_1}$
- (c) $\frac{R\alpha_1\alpha_2}{\beta_1\beta_2}$
- (d) $\frac{R\alpha_1\beta_1}{\alpha_2\beta_2}$

Passage 17

AB is a uniform wire of a meter bridge made of constantan. X is a standard 4Ω resistance and Y is a resistance made of a metal wire. With the wire Y immersed in melting ice, the null point is found to be at a distance of 40 cm from point A .



35. When the wire Y is heated, the null point

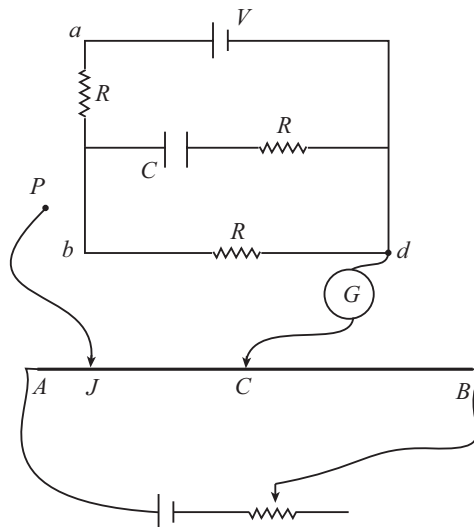
- (a) moves towards A
- (b) moves away from A
- (c) will not move
- (d) none of the above

36. Wire Y is heated to 100°C and a 100Ω resistance is connected in parallel to it. The null point remains at a distance of 40 cm from A . The temperature coefficient of resistivity for the material of the wire Y is

- (a) 6.4K^{-1}
- (b) $6.4 \times 10^{-1}\text{K}^{-1}$
- (c) $6.2 \times 10^{-3}\text{K}^{-1}$
- (d) $6.4 \times 10^{-4}\text{K}^{-1}$

Passage 18

AB is a potentiometer wire of length $L = 4\text{m}$. C is midpoint of the wire. Point C is connected to a point d in an external circuit through a galvanometer. A jockey (J) can slide from A to B on the wire. The terminal P is connected to point a in the circuit and the galvanometer shows 0 deflection when the jockey is placed at a distance of 60 cm to the left of point C .



37. When the terminal P is connected to point b in the circuit, the galvanometer shows 0 deflection when jockey is placed

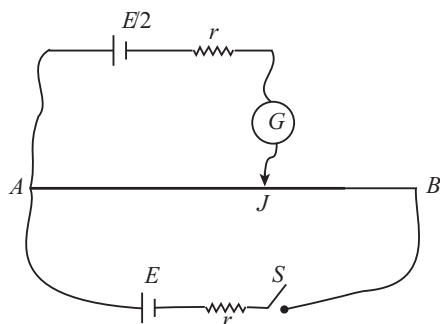
- (a) to the right of C
- (b) to the left of C
- (c) cannot be said unless emf of the cell in the circuit is known.
- (d) cannot be said unless value of capacitance is known.

38. With P connected to b , the galvanometer shows 0 deflection when jockey is at a distance of

- (a) 30 cm from C
- (b) 20 cm from C
- (c) 60 cm from C
- (d) cannot be said unless value of capacitance is known.

Passage 19

AB is a 600 cm long potentiometer wire. It is powered using a cell of emf E and internal resistance r . Potential difference across the ends A and B is found to be $\frac{15}{16}E$ when switch S is put on. A cell of emf $\frac{E}{2}$ and internal resistance r is being balanced on the potentiometer wire.



39. During the experiment, the jockey (J) is made to touch the potentiometer wire at a distance of 40 cm from the end B . In this position, the current through the galvanometer will be

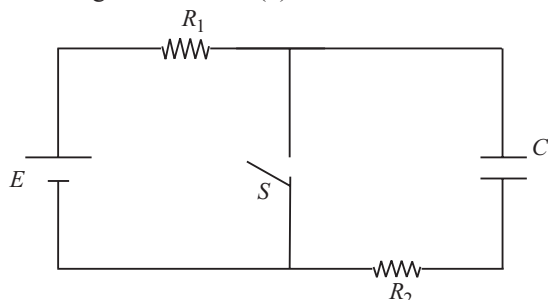
- (a) $\frac{E}{29r}$ (b) $\frac{2E}{29r}$
(c) $\frac{E}{58r}$ (d) $\frac{13E}{58r}$

40. J_0 is a point on the wire. When the jockey is placed to the left of the point J_0 , deflection in the galvanometer is to the right. When the jockey is placed to the right of J_0 , deflection in the galvanometer is to the left. The distance AJ_0 is

- (a) 320 cm (b) 120 cm
(c) 180 cm (d) 200 cm

Passage 20

In the circuit shown in the figure, the switch (S) has been kept open for a long time. At time $t = 0$, it is suddenly closed. At any time, t current through the cell is i and the current through this switch (S) is I .



41. Value of i , immediately after the switch is closed, is

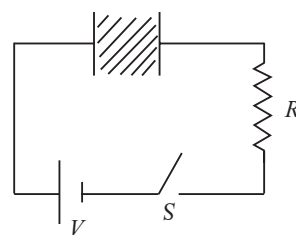
- (a) $\frac{ER_1}{R_2(R_1 + R_2)}$ (b) $\frac{ER_2}{R_1(R_1 + R_2)}$
(c) $\frac{E}{R_1}$ (d) $\frac{E}{R_1 + R_2}$

42. Value of I at time t is

- (a) $\frac{E}{R_1} + \frac{E}{R_2} e^{-\frac{t}{R_2 C}}$ (b) $\frac{E}{R_1} + \frac{E}{R_2} e^{-\frac{t}{(R_1 + R_2)C}}$
(c) $\frac{E}{R_2} \left[1 + e^{-\frac{(R_1 + R_2)t}{R_1 R_2 C}} \right]$ (d) None of the above

Passage 21

A parallel plate capacitor has space between its plates filled with a partially conducting medium. The capacitance of the capacitor is C and the medium between the plates has a resistance R_0 . This capacitor is connected to a cell of emf



V through a resistance R and a switch S . The ratio $\frac{R}{R_0}$ is equal to β . The switch is closed at $t = 0$.

43. The time at which 63% of the charging of the capacitor gets completed is nearly

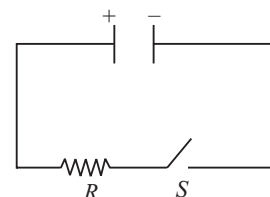
- (a) βRC (b) $\frac{RC}{\beta}$
(c) $(\beta + 1)RC$ (d) $\frac{RC}{\beta + 1}$

44. Charge (q) on the capacitor as a function of time is given by

- (a) $\frac{VC}{1 + \beta} \left[1 - e^{-\frac{(\beta + 1)t}{RC}} \right]$ (b) $\frac{\beta VC}{1 + \beta} \left[1 - e^{-\frac{\beta t}{RC}} \right]$
(c) $\frac{VC}{1 + \beta} \left[1 - e^{-\frac{\beta t}{RC}} \right]$ (d) None of the above.

Passage 22

An air core parallel plate capacitor has capacitance C_0 and separation between its plates is d_0 . It is given a charge Q_0 and connected to a bulb of resistance R through a switch S . The switch is closed at time $t = 0$ and the separation



between the capacitor plate is changed as $d = \frac{d_0}{(1+t)}$ (for $0 \leq t \leq 1$ second). For $t > 1$ second, the plate separation is held fixed at $d = \frac{d_0}{2}$.

45. Charge on the capacitor at $t = 1$ second is

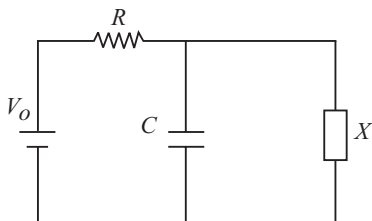
- (a) $(2)^{\frac{1}{RC_0}} Q_0$ (b) $\frac{Q_0}{(2)^{\frac{1}{RC_0}}}$
 (c) $\frac{Q_0}{2}$ (d) None of the above

46. Charge on the capacitor at a time $t > 1$ second is

- (a) $\frac{Q_0}{(2)^{\frac{1}{RC_0}}} e^{-\frac{(t-1)}{2RC_0}}$ (b) $\frac{Q_0}{(2)^{\frac{1}{RC_0}}} e^{-\frac{(t-1)}{RC_0}}$
 (c) $\frac{Q_0}{2} e^{-\frac{(t-1)}{2RC_0}}$ (d) None of the above

Passage 23

Device X , in the shown circuit, has the property that if it is initially nonconducting, it remains nonconducting until the voltage across it rises to a value $\frac{V_0}{2}$. Once the device conducts, the capacitor rapidly (almost instantaneously) discharges through X so that the voltage across it drops nearly to zero. After this, the device returns to its nonconducting state.

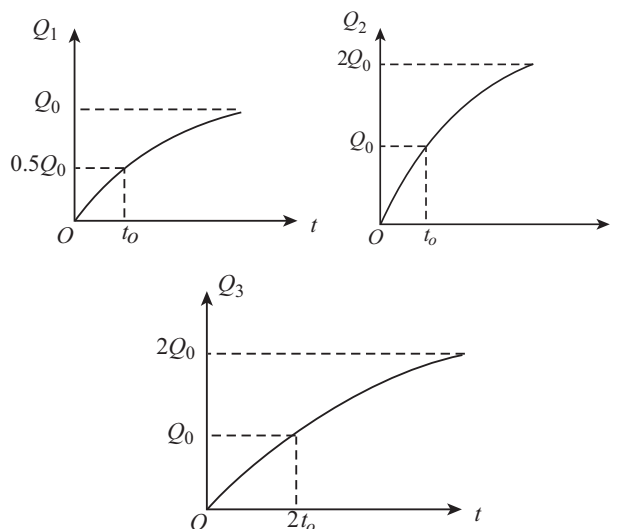


47. The potential difference measured across X is
 (a) Periodic
 (b) Periodic with a square waveform
 (c) Periodic with sinusoidal waveform
 (d) None of the above
48. The time period of the voltage waveform across X (V_x) is
 (a) RC (b) $(\ln 2)RC$
 (c) $\frac{RC}{\ln 2}$ (d) $(2\ln 2)RC$

Passage 24

Three capacitors A , B and C have capacitance C_1 , C_2 and C_3 , respectively. They are charged using cells of emfs V_1 , V_2 and

V_3 , respectively. All three circuits have same resistance. The variation of charge Q_1 , Q_2 and Q_3 on the three capacitors with time has been shown in the figure.



49. The relation among V_1 , V_2 , and V_3 is certainly

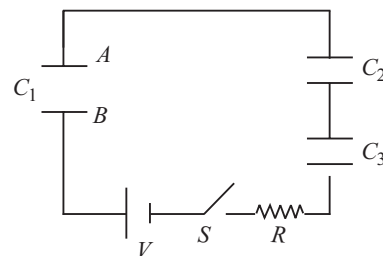
- (a) $V_1 = V_2 = V_3$ (b) $2V_1 = V_2 = 2V_3$
 (c) $V_1 = 2V_2 = 2V_3$ (d) $2V_1 = V_2 = V_3$

50. The correct relationship among the capacitance C_1 , C_2 and C_3 is

- (a) $2C_1 = C_2 = C_3$ (b) $C_1 = 2C_2 = 2C_3$
 (c) $C_1 = C_2 = 2C_3$ (d) $2C_1 = 2C_2 = C_3$

Passage 25

A capacitor C_1 , having capacitance C is charged using a cell of emf V . Now, the charged capacitor is inserted into a circuit as shown in the figure. The capacitors C_2 and C_3 each have capacitance $2C$. The resistor has resistance R .



When the plate A of capacitor C_1 is positive and the switch S is closed, the initial current in the circuit is i_0 and the final steady state charge on C_2 is q_0 . When the plate B of the capacitor C_1 is positive, the initial current after S is closed is observed to be I_0 and the final steady state charge on C_3 is found to be Q_0 .

51. The value of i_0 is

- (a) $\frac{V}{R}$ (b) $\frac{V}{2R}$
 (c) $\frac{3V}{R}$ (d) $\frac{2V}{R}$

52. The value of q_0 is

- (a) $\frac{CV}{2}$ (b) $2CV$
 (c) $\frac{3CV}{2}$ (d) CV

53. The value of I_0 is

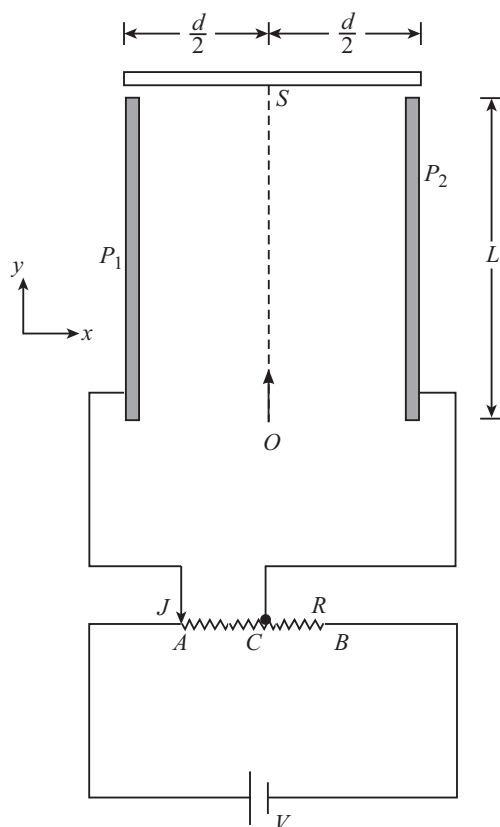
- (a) 0 (b) $\frac{2V}{R}$
 (c) $\frac{V}{R}$ (d) $\frac{V}{2R}$

54. The value of Q_0 is

- (a) 0 (b) $\frac{CV}{2}$
 (c) $\frac{3CV}{2}$ (d) CV

Passage 26

P_1 and P_2 are two identical parallel plates separated by a distance d . Length of the plates is L . At the upper ends of the plates, there is a horizontal screen with S as its midpoint. The plate P_2 is connected to the midpoint C of a rheostat (R). The other plate P_1 is connected to the rheostat through a sliding jockey J . A cell of emf V is connected across the rheostat as shown in the figure.



A source at O projects electrons at a fixed speed along the line OS . With the jockey placed at A , when an electron is fired from O , it travels along the straight line OS and hits the screen in time t_0 . We can neglect the gravity.

55. The jockey is held at A and an electron is projected from O at time $t = 0$. At time $t = \frac{t_0}{2}$, the jockey is suddenly (without any time loss) moved to end B of the rheostat. The electron hits the screen.

- (a) at S (b) to the left of S
 (c) to the right of S (d) cannot be predicted

56. In the situation described in the previous problem, the electron hits the screen.

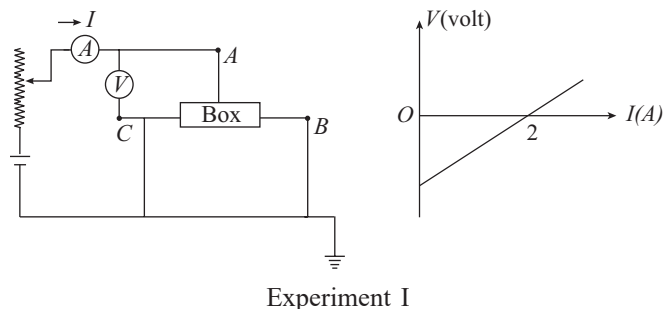
- (a) normally
 (b) making 30° with the screen
 (c) making 45° with the screen
 (d) making 60° with the screen

57. The jockey is held at C and an electron is fired. The jockey is moved from C to A and then back to C in time t_0 . The jockey was moved with constant speed. The velocity with which the electron hits the screen is

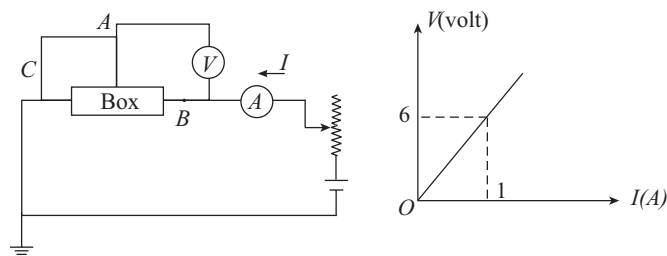
- (a) $-\left(\frac{eVt_0}{4md}\right)\hat{i} + \left(\frac{L}{t_0}\right)\hat{j}$ (b) $\frac{L}{t_0}\hat{j}$
 (c) $\left(-\frac{eVt_0}{8md}\right)\hat{i} + \left(\frac{L}{t_0}\right)\hat{j}$ (d) None

Passage 27

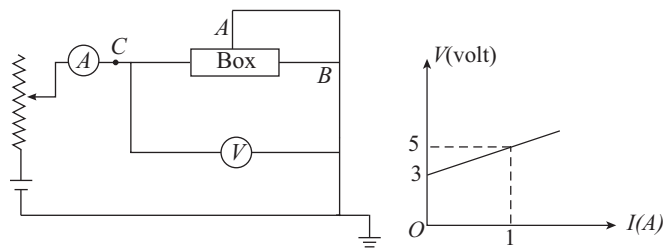
A box contains an electric circuit as shown in the figure. It has three terminals A , B and C . Three different experiments were performed with the box by connecting a battery, an ammeter and a voltmeter. Measurements of these experiments have been expressed in the form of a graph. Ammeter and voltmeter are ideal.



Experiment I

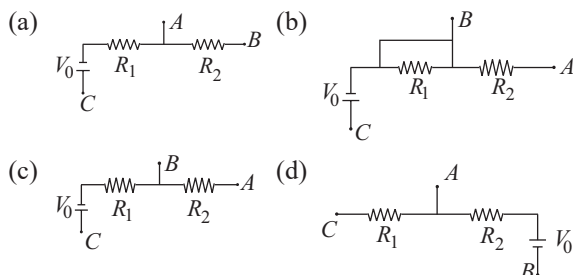


Experiment II



Experiment III

58. The circuit inside the box can be



59. In the circuit inside the box (shown in previous question), the value of R_1 is

- (a) 3Ω (b) 2Ω
(c) 6Ω (d) 1Ω

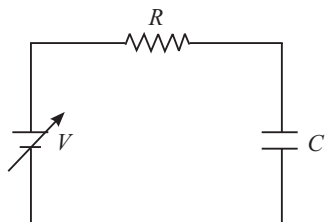
60. In the circuit inside the box, the value of R_2 is

- (a) 2Ω (b) 3Ω
(c) 6Ω (d) $1-5\Omega$

Passage 28

A capacitor (capacitance = C) is connected to a variable dc voltage source through a resistance (R). The voltage of the source changes with time and is zero at $t = 0$. At time t , the charge on the capacitor is Q . Using Kirchhoff's loop law, we can write

$$\frac{dQ}{dt} + \frac{Q}{RC} = \frac{V}{R}$$



This is a differential equation that can be solved by multiplying both sides by $e^{\frac{t}{RC}}$.

61. The potential difference across the capacitor can be expressed as

$$V_C = \frac{e^{-\frac{t}{RC}}}{aRC} \int V e^{\frac{t}{RC}} \cdot dt$$

The value of constant a is

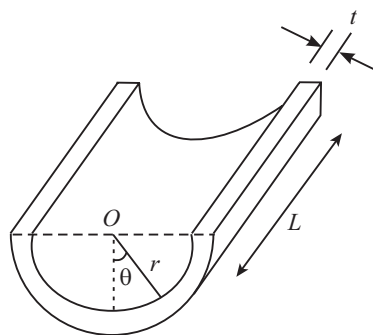
- (a) 1 (b) 2
(c) 4 (d) $\frac{1}{2}$

62. When the time t is much shorter than RC , the potential difference across the capacitor can be approximately written as $V_C =$

- (a) $\frac{1}{RC} \int V dt$ (b) $\frac{1}{2RC} \int V dt$
(c) $\frac{2}{RC} \int V dt$ (d) $\frac{1}{4RC} \int V dt$

Passage 29

A conductor of length L has the shape of a half cylinder of radius r . Thickness of the sheet used to make the conductor is t . The conductivity of the material varies with the angle θ as $\sigma = \sigma_0 \cos \theta$. A battery of emf V and negligible internal resistance is connected across the two semicircular ends.



63. Which of the followings is true regarding the current density (j)?

- (a) Current density is same at all points on the semicircular cross-section.
(b) Current density varies on the semicircular cross-section, decreasing with θ .
(c) Current density varies on the semicircular cross-section, decreasing with θ .
(d) Current density is same at every point in the conductor.

64. Current flowing through the battery is

- (a) $\frac{\sigma_0 V r t}{L}$ (b) $\frac{\sigma_0 V L}{t r}$
(c) $\frac{2\sigma_0 V L t}{r}$ (d) $\frac{2\sigma_0 V r t}{L}$

Answer Sheet

Match the Columns

1. (A) P, R (B) U (C) S, T (D) Q
2. (A) Q (B) R (C) R (D) R
3. (A) Q (B) P, Q, R, S (C) Q, (D) P, Q, R
4. (A) P (B) Q (C) R
5. (A) S (B) Q (C) P (D) S
6. (A) P, S (B) P, R (C) P, R (D) P, R
7. (A) Q, R (B) Q, R (C) Q, S (D) Q
8. (A) R (B) R (C) P (D) P, S
9. (A) Q, R (B) Q, R, S (C) Q, R, S (D) P, Q, R, S
10. (A) R (B) S (C) Q (D) P

Passage Based Problems

- | | | | | | | | | |
|---------|---------|---------|---------|---------|---------|---------|---------|---------|
| 1. (a) | 2. (b) | 3. (c) | 4. (c) | 5. (a) | 6. (d) | 7. (b) | 8. (c) | 9. (d) |
| 10. (a) | 11. (d) | 12. (a) | 13. (c) | 14. (d) | 15. (b) | 16. (b) | 17. (c) | 18. (a) |
| 19. (c) | 20. (d) | 21. (a) | 22. (b) | 23. (a) | 24. (c) | 25. (a) | 26. (b) | 27. (a) |
| 28. (d) | 29. (a) | 30. (c) | 31. (a) | 32. (b) | 33. (c) | 34. (a) | 35. (b) | 36. (d) |
| 37. (b) | 38. (a) | 39. (d) | 40. (a) | 41. (c) | 42. (a) | 43. (d) | 44. (a) | 45. (b) |
| 46. (a) | 47. (a) | 48. (b) | 49. (b) | 50. (d) | 51. (d) | 52. (d) | 53. (a) | 54. (a) |
| 55. (b) | 56. (a) | 57. (a) | 58. (a) | 59. (b) | 60. (c) | 61. (a) | 62. (a) | 63. (c) |
| 64. (d) | | | | | | | | |

Error and Measurement

“It is better to be roughly right than precisely wrong.”

—Alan Greenspan

1. INTRODUCTION

No measurement can be assumed to be perfect. There is always a chance of an error creeping into our measurement due to various factors. When we use measured quantities in a formula to calculate the value of some other quantity, the calculated quantity will also have error. This is known as *propagation of error*. For example, when we measure the side length L of a square and use the formula $A = L^2$ to calculate its area, the value of area that we obtain is not free of error. The error has propagated into the value of A because there was an error in the value of L . In this chapter, we will study how to estimate the propagation of error. But before that we must learn how to express our measurements.

In the end, we will also study about two basic measuring instruments in a laboratory – vernier caliper and screw gauge.

2. REPORTING A SINGLE MEASUREMENT

When we measure a quantity, we believe that there exists a true value of the quantity. We may never know this true value exactly, but we try to find this true value to the best of our ability with the resources available. We may make multiple measurements of a quantity using the same method, or even make measurements by different methods. Using statistics, we can report a value that is the *best estimate* according to us. But we are never absolutely confident that the value being reported by us is the true value. Despite every effort that we make, there is always some *uncertainty* in the reported value. Therefore, we always report our measurement as

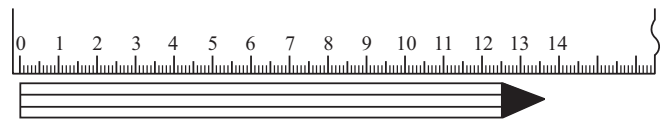
$$\text{measured value} = \text{best estimate} \pm \text{uncertainty} \quad (1)$$

In our discussions, we will use the terms *uncertainty* and *error* interchangeably, although they have slightly different meaning.

We will first learn about significant digits in a single measurement and then develop a simple method of finding the *true value (best estimate)* of a quantity and the uncertainty involved in the reported value.

2.1 Uncertainty in a single measurement

Suppose you are measuring the length of your pencil using a ruler. The ruler has a millimeter scale marked on it. It means that the gap between two nearest divisions is 1 mm. We say that the *least count (LC)* of this scale is 1 mm.



Look at the diagram. We can be absolutely sure that the length of the pencil is larger than 13 cm. But how much larger? A little bit of estimation (guess) is involved in answering this. Let us say that we make a judgment that the length of the pencil is 13.7 cm. Are we absolutely sure? Of course, not. We are doubtful about the digit 7. We felt as if it could be 13.6 cm, then we felt that it is somewhat larger than 13.7 cm. In the end, we made our best estimate, that is 13.7 cm. *We report all certain digits and the first doubtful digit in a measurement.* In the present example, 1 and 3 are digits about which we are fully confident and 7 is the doubtful digit. Our measurement has 3 *significant digits (or, significant figures)*.

$$\text{Length} = \underbrace{13}_{\substack{\text{confident} \\ \text{digits}}} \cdot \underbrace{7}_{\substack{\text{doubtful} \\ \text{digits}}} \text{ cm} \quad [3 \text{ significant digit}]$$

No sincere physics student shall report the above measurement as 13.76 cm. When we are not sure about first digit after decimal, how can we say anything about the next digit? It makes no sense.

As we can see in the above example that *the uncertainty in a single measurement is limited by the precision and accuracy of the measuring instrument, besides other factors like skill of the experimenter*. There could be some uncertainty in the above measurement due to parallax! There is no prescribed rule for determining the uncertainty in a measurement. It is the experimenter who is the best judge of the uncertainty in a measurement. In our discussions, we will mostly discard any uncertainty due to experimental conditions and experimenter's skill and assume that it is the measuring instrument, which decides the amount of uncertainty. *Unless stated otherwise, we will take the LC of the measuring instrument to be the maximum uncertainty in a measurement*. The above measurement of the length shall be expressed as

$$L = (13.7 \pm 0.1) \text{ cm}$$

Note: Even if we report the length as $L = 13.7 \text{ cm}$, it is understood that 7 is a doubtful digit and we consider the uncertainty to be 0.1 cm in this measurement.

2.2 Rules to identify significant figures

We have said that significant figures comprise digits that are known reliably and one last digit that is doubtful. There are prescribed rules to identify significant figures in a reported value. Here are the rules:

- (1) Consider that mass of a gold ring is reported as 5.026 g. Can we drop any of the non-zero digits in the reported value? No. It will change the mass of the ring. Therefore, the rule is:

All non-zero digits are significant figures in a reported value. There are 4 significant figures in the above value.

- (2) Can we drop '0' lying between 5 and 2 in the mass of the ring considered above? No. Dropping '0' will change the mass of the ring. Therefore, the second rule is:

All zeros between two non-zero digits are significant irrespective of the position of the decimal.

- (3) Let us express the mass of the ring in kilogram. The value will be 0.005026 kg. Just by changing the unit of a reported value, we cannot claim that more number of digits are known reliably. The measuring instrument remains same, and its LC remains same. The number of significant digits should also remain same. Therefore, the next rule is:

"If the value is less than 1, then all zeros before the first non-zero digit are not significant."

- (4) If we express the above mass in milligram, it will be 5026 mg. When expressed in microgram, the same would be 5,026,000 μg . If we consider the trailing zeros to be significant, it will amount to believing

that we know 6 digits confidently and the last zero is doubtful. In a way we are saying that we have used an instrument that can measure a mass like 5,026,434 μg or 2,134,621 μg . This is not true. Our reported value 5,026,000 μg still has only 4 significant digits. Therefore, rule number 4 is:

"The trailing zeros in a non-decimal number are not significant."

- (5) Now, assume that a different ring weighed on the same scale has a mass of 5.020 g. The scale is same, and the LC is same. The last zero is a legitimate digit in our measurement. It is significant. But one may treat it as not significant after reading the above rules. We need a method to differentiate between a trailing '0', which is the result of a measurement, from the '0' in earlier example which was introduced as a result of unit change. Therefore, the next rule is:

All trailing zeros, in a number with decimal, are significant.

We often need to write our measurements in *scientific notation* in order to correctly identify the number of significant digits, using rule number 5.

Suppose we measure the width of our study table to be 500 cm using a ruler marked in centimeters. If we simply write length = 500 cm, people will assume that only 5 is the significant digit and the two zeros are due to unit change. They will think that we used a meter stick with no marking on it. We laid the stick across the table and estimated that the width is nearly half the length of the stick. It means the measured length was 0.5 m (5 is the only significant digit and it is doubtful! After all we cannot be exactly sure where lies the midpoint of the stick). But this is far from the truth. We have used a more precise scale that could read upto centimeter. To convey this, we should report our measurement as

$$\text{length} = 5.00 \times 10^2 \text{ cm}$$

$$\text{or, length} = 5.00 \times 10^3 \text{ mm}$$

$$\text{or, length} = 5.00 \times 10^{-1} \text{ m}$$

The methodology used to express a value as above is known as *scientific notation*. We always write one digit before the decimal. The power of 10 does not change the number of significant digits. All the above values (written in cm, mm and m) have 3 significant digits.

Example 1 *Significant digits depend on the experimental condition and the experimenter, besides the LC of a measuring instrument.*

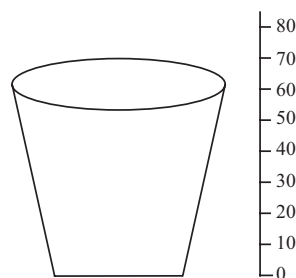
Suppose we have a meter stick graduated at every millimeter. How many significant digits will be there in a length measurement with this scale? Assume that the measured length is smaller than a meter.

Solution

Suppose we are interested in knowing the approximate height of a tapering bucket. Due to parallax, we may not be able to judge the height to the nearest centimeter. You decided to judge it to the nearest 10 cm. We may measure the height as

$$h = 0.7 \text{ m}$$

This measurement has only 1 significant digit. Note that by writing $h = 70 \text{ cm}$, we cannot claim that there are two significant digits. Will we be able to distinguish between the two buckets having heights of 70 cm and 71 cm, with the measurement technique that we are using? No. We are not confident about 0 or 1. In fact, we are in doubt about the digit 7 itself.



An estimate of height of a bucket may involve judging the height upto nearest 10 cm

When we use the same scale to measure the length of a cloth piece, we will be satisfied measuring it to the nearest centimeter. Trying to measure it upto millimeter may be futile as the edges of the cloth piece may be uneven and the length may also vary slightly as we move along the width. In such case, we will use the scale as a centimeter scale and may report our measurement as 63 cm. This measurement has 2 significant figures.

Now, go back to the diagram given in the beginning of Section 2. The length of the pencil was reported as 13.7 cm. This measurement has 3 significant digits.



Finally, assume that we decide to measure the length of a metal part of a machine in order to judge whether it will fit properly or not. We measured the length after taking all necessary precautions and concluded that it is certainly larger than 10.5 cm. We made an estimate that the length is 10.55 cm. Here, the last digit '5' is doubtful and our measurement has 4 significant digits. (see the figure given above)

We see that the measurements done using the given scale can have 1, 2, 3 or 4 significant digits.

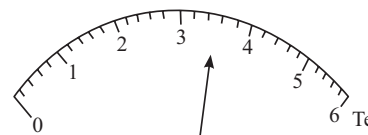
Example 2 Write number of significant digits in reported values:

- | | |
|----------------------------|---------------|
| (a) 2008 | (b) 200.8 |
| (c) 200.80 | (d) 0.0020080 |
| (e) 2.080×10^{-3} | |

Solution

- All non-zero digits and the zeros between them are significant.
Number of significant digits in 2008 is 4.
- From the same rule as stated above, the number of significant digits is 4 in 200.8
- Trailing zeros in a decimal number are significant. The value 200.80 has 5 significant digits.
- In 0.0020080, the leading zeros are not significant. There are 5 significant digits.
- The number 2.080×10^{-3} is in scientific notation. All 4 digits (2, 0, 8 and 0) are significant.

Example 3 A weigh bridge, used to weigh small trucks, has its scale and pointer as shown in the figure. The scale measures in ton. When a truck arrives on the bridge, the pointer deflects to the position shown. Express the reading in kilogram. How many significant digits are there in our measurement?

**Solution****Concepts**

- Significant digits are those digits that are reliable and the first uncertain digit.
- Change in unit does not affect the number of significant digits in a measurement.

The LC of the scale is 0.2 Te (interval between 1 and 2 is divided in 5 equal parts). We are sure that the reading is larger than 3 and smaller than 4. There is some judgement involved in telling how large above 3. My best estimate is that the reading is 3.6 Te. (We may differ and say that it is 3.7 or 3.5 or 3.8 Te). The last digit 6 is doubtful. We are not allowed to add any more digit.

There are 2 significant digits in our measurement. In kilogram, the measurement must be written as $3.6 \times 10^3 \text{ kg}$.

Suggest a simple way to improve the scale shown in the figure.

Example 4 In the last example, what is the uncertainty in the measurement? How will you report your measurement?

Solution**Concepts**

The maximum uncertainty = least count of the instrument, assuming that extreme precaution has been taken to eliminate any error due to experimental conditions and methods.

Uncertainty = 0.2 Te

∴ Reported measurement will be (3.6 ± 0.2) Te

This implies that we are claiming that the weight of the truck is between 3.4 Te to 3.8 Te.

Example 5 Exact numbers

You count the number of students in your class as 42. How many significant digits are there in this measurement?

Solution

There are infinite number of significant digits in such an exact number. Number of students is 42.000000000 We can go on adding many more zeros. We never doubt whether the 8th digit is zero or 1. We know it confidently that it is zero.



YOUR TURN

Q.1 You are having a wrist watch having hour, minute and second hands. What is the LC of the watch? If the second hand is removed, what is the least count, now?

Q.2 In 1849 Louis Fizeau measured the speed of light to be 313,300 km/s. Today, the known value is 299792.458 km/s. Should we conclude that the method used by Fizeau, for measuring the speed, was wrong?

Q.3 Write the number of significant digits in the followings

- (a) 0.03 (b) 121.340 (c) 0.001200
(d) 500 (e) 6.000 (f) 6.0×10^{-5}
(g) 6.000×10^8

Q.4 You measured the length of a stick to be 500 mm using a scale graduated at every millimeter. How will you report your measurement so that people understand the true nature (i.e., precision) of the scale that you have used.

3. ERRORS IN MEASUREMENT

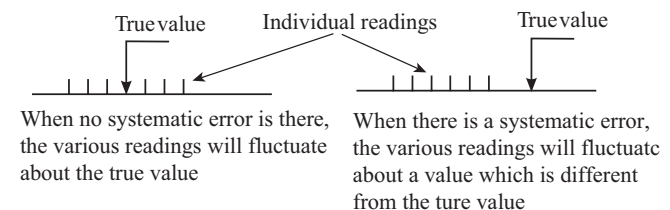
Measurement errors can be classified as either *random* or *systematic*.

Random Errors: If you measure the length of your pencil 5 times using a ruler, you may not always get the same values. Your measurements may be 12.34 cm, 12.34 cm, 12.32 cm, 12.33 cm and 12.35 cm. It means that the measured values fluctuate randomly about the true value. No specific reason can be assigned to random errors. Random errors cannot be eliminated completely. Their effect can be reduced by recording a measurement multiple times. Random errors are sometimes positive and sometimes negative. Thus, the average of a large number of measurements gets close to the true value.

Systematic Errors: There are the errors which arise due to specific reasons and can be eliminated if the reasons get identified. These errors tend to be in one direction and cannot be reduced by increasing the number of observations and taking average.

For example, consider a steel meter scale designed to give correct measurements at 20°C. A user might be using it on a day when the ambient temperature is 40°C. The scale will show a reading which is less than the true value. But if we are able to identify the source of error, we can eliminate it.

Go inside a room where the temperature has been maintained at 20°C!



Common sources of error in a physics laboratory

- (i) **Environmental factors:** The working environment may cause errors in the experimental measurements due to factors like vibrations, changes in temperature, noise etc. Such errors can be systematic or random.
- (ii) **Instrument resolution:** All instrument have finite precision. We cannot use a meter stick to differentiate between lengths of two objects if the difference is 0.1 mm. We need to use a more precise instrument known as vernier.

The *Null or the balance method* that we use for measuring unknown resistance in the experiments of meter bridge or P.O. Box is an excellent method to counter the error arising due to LC of the instrument. In these experiments, we record a measurement when

the galvanometer shows null deflection. The LC of the galvanometer is irrelevant. We eliminated this source of error. However, the error remained while measuring the length of the wire.

- (iii) **Instrument calibration:** We should always check the calibration of an instrument before taking readings.
- (iv) **Zero offset:** We must check the zero reading of an instrument. If possible we should re-zero it. If not, then we should record the zero offset so that all the readings can be corrected later.

Suppose, a weighing scale shows 30 g when nothing is placed over it and there is no provision to reset it to zero. In this case we must subtract 30 g from all the subsequent readings.

- (v) **Parallax:** This is a common cause of error that is there when there is some distance between the scale and the pointer. If your eye is not exactly aligned with the pointer and the scale, the reading may be high or low.
- (vi) **Neglecting a factor:** Error arise due to neglect of important factors. While measuring acceleration due to gravity, you neglect air resistance. This will result in error. While performing electrical measurements, you tend to neglect the resistance of connecting wires. This leads to error.
- (vii) **Personal errors:** Lag time, carelessness, poor technique or bias on the part of the experimenter are some other causes for errors.

As a rule, personal errors are excluded in error analysis and it is assumed that experimental result was obtained by following correct procedures.

4. ESTIMATING UNCERTAINTY IN MULTIPLE MEASUREMENTS

Suppose that you want to measure the diameter of a rod using a vernier caliper. You take every possible care to eliminate systematic errors. For example, you record the zero error and make due corrections in every reading that you take. Still you cannot be sure that the value you measure is the true value. The best thing you can do is repeat the measurement several times (at different locations of the rod) and average the result. This average can be regarded as the *best estimate of the true value*. Generally, higher the number of measurements, the better is our estimate. Let's assume that the chart shown below shows the readings that you obtained.

Observation number (<i>i</i>)	Recorded diameter of the rod (x_i) in cm
1	$x_1 = 2.43$
2	$x_2 = 2.42$
3	$x_3 = 2.40$

Observation number (<i>i</i>)	Recorded diameter of the rod (x_i) in cm
4	$x_4 = 2.42$
5	$x_5 = 2.43$
6	$x_6 = 2.41$
7	$x_7 = 2.44$
8	$x_8 = 2.39$
9	$x_9 = 2.42$
10	$x_{10} = 2.44$

True value of the diameter can be calculated as the mean of all recorded values

$$\text{Mean (Average), } \bar{x} = \frac{x_1 + x_2 + \dots + x_N}{N} \quad (2)$$

In the present example

$$\bar{x} = \frac{2.43 + 2.42 + 2.40 + 2.42 + 2.43 + 2.41 + 2.44 + 2.39 + 2.42 + 2.44}{10}$$

$$\text{or, } \bar{x} = 2.42 \text{ cm}$$

This average value still has some uncertainty in it. One simple way to express the uncertainty is to use *average deviation* ($\bar{\delta}$). To find this we calculate the deviation in each measurement from the true value and take the average.

$$\bar{\delta} = \frac{|x_1 - \bar{x}| + |x_2 - \bar{x}| + |x_3 - \bar{x}| + \dots + |x_N - \bar{x}|}{N} \quad (3)$$

The following table illustrates the calculation of average deviation ($\bar{\delta}$) for the data set obtained above.

Observation number	Recorded diameter of the rod (x_i) in cm	Deviation from mean $ x_i - \bar{x} $
1	$x_1 = 2.43$	$ 2.43 - 2.42 = 0.01$
2	$x_2 = 2.42$	$ 2.42 - 2.42 = 0.00$
3	$x_3 = 2.40$	$ 2.40 - 2.42 = 0.02$
4	$x_4 = 2.42$	$ 2.42 - 2.42 = 0.00$
5	$x_5 = 2.43$	$ 2.43 - 2.42 = 0.01$
6	$x_6 = 2.41$	$ 2.41 - 2.42 = 0.01$
7	$x_7 = 2.44$	$ 2.44 - 2.42 = 0.02$
8	$x_8 = 2.39$	$ 2.39 - 2.42 = 0.03$
9	$x_9 = 2.42$	$ 2.42 - 2.42 = 0.00$
10	$x_{10} = 2.44$	$ 2.44 - 2.42 = 0.02$

Mean deviation is

$$\bar{\delta} = \frac{0.01 + 0.00 + 0.02 + 0.00 + 0.01 + 0.01 + 0.02 + 0.03 + 0.00 + 0.02}{10}$$

$$= 0.012 \text{ cm}$$

$$\simeq 0.01 \text{ cm}$$

Thus, the diameter of the rod can be expressed as

$$D = (2.42 \pm 0.01) \text{ cm}$$

However, the standard deviation is the most common way to express the uncertainty. Let's learn how to calculate the standard deviation of a data set.

4.1 Standard deviation

Follow the steps given below to calculate the standard deviation for a sample of N measurements.

1. Find the mean of the data set by adding all the data and dividing the sum by N .

$$\bar{x} = \frac{x_1 + x_2 + \dots + x_N}{N} = \frac{\sum x_i}{N}$$

2. Obtain deviation of each measurement from the mean. Deviation in the i^{th} measurement is

$$\delta_i = x_i - \bar{x}$$

3. Square each of the deviations to get $\delta_i^2 = (x_i - \bar{x})^2$
4. Sum the square of all deviations and divide by $(N-1)$.
5. Take the square root of the number obtained after step 4. This is the standard deviation of the given data set.

The formula for standard deviation can be written as

$$\sigma = \sqrt{\frac{1}{(N-1)} \sum_{i=1}^N (x_i - \bar{x})^2} \quad (4)$$

The following table explains the procedure for calculating the standard deviation in the example that we have been discussing.

Observation number (i)	x_i	$\delta_i = x_i - \bar{x}$	$\delta_i^2 = (x_i - \bar{x})^2$
1	2.43	$2.43 - 2.42 = 0.01$	$(0.01)^2 = 0.0001$
2	2.42	$2.42 - 2.42 = 0.00$	$(0.00)^2 = 0.0000$
3	2.40	$2.40 - 2.42 = -0.02$	$(-0.02)^2 = 0.0004$
4	2.42	$2.42 - 2.42 = 0.00$	$(0.00)^2 = 0.0000$
5	2.43	$2.43 - 2.42 = 0.01$	$(0.01)^2 = 0.0001$
6	2.41	$2.41 - 2.42 = -0.01$	$(-0.01)^2 = 0.0001$
7	2.44	$2.41 - 2.42 = 0.02$	$(0.02)^2 = 0.0004$
8	2.39	$2.39 - 2.42 = -0.03$	$(0.03)^2 = 0.0009$
9	2.42	$2.42 - 2.42 = 0.00$	$(0.00)^2 = 0.0000$
10	2.44	$2.44 - 2.42 = 0.02$	$(0.02)^2 = 0.0004$
			SUM = 0.0024

$$\sum_{i=1}^N (x_i - \bar{x})^2 = 0.0024$$

$$\therefore \sigma = \sqrt{\frac{0.0024}{10-1}} = 0.0163 \approx 0.02$$

Significance of standard deviation

The best value of the measured quantity is the mean value $\bar{x}$. But we cannot be 100% sure that the value is exactly $\bar{x}$. If the standard deviation is σ , then

- (i) there is a 68% chance that the true value lies in the range $\bar{x} \pm \sigma$.
- (ii) there is about 95% chance that the true value lies in the range $\bar{x} \pm 2\sigma$, and
- (iii) there is about 99.7% chance that the true value lies in the range $\bar{x} \pm 3\sigma$.

In the above example, we can say that there is a 68% chance that the true diameter of the wire is

$$D = \bar{x} \pm \sigma = (2.42 \pm 0.02) \text{ cm}$$

We are 95% sure that the diameter is

$$D = \bar{x} \pm 2\sigma = (2.42 \pm 0.04) \text{ cm}$$



YOUR TURN

Q.5 You are standing on a weighing scale. You look vertically down on the scale and record a reading of 70.0 kg. The pointer moves from your left side to your right side when you stand on the scale. Your friend is standing on the ground to your right. Will he record your weight as 70.0 kg, less than that or more than that? Explain.

Q.6 When an ammeter is not connected in a circuit its pointer is not at zero. It shows a reading of 0.01A. This

is known as zero error. When you use it in an experiment to measure a current its reading is 12.34A. What value of current will you report?

Q.7 In a certain experiment you measure a current thrice. The readings are 2.0A, 2.4A and 2.2A. How will you express the value of current taking the mean deviation as a measure of uncertainty?

4.2 Fractional and percentage uncertainty

Fractional uncertainty is defined as uncertainty divided by the true value (i.e., the average value). If a reported measurement is $x \pm \Delta x$, then the fractional uncertainty is:

$$\text{Fractional uncertainty} = \frac{\text{Uncertainty}}{\text{True value}} = \frac{\Delta x}{x} \quad (5)$$

The percentage uncertainty is

$$\text{Percentage uncertainty} = \frac{\Delta x}{x} \times 100 \quad (6)$$

The fractional uncertainty is dimensionless and is important in calculation of propagation of uncertainty, as we shall study shortly.

Example 6 Measurement of a length is reported as (5.0 ± 0.1) cm. Find the fractional and percentage error (or, uncertainty).

Solution

$$\text{Fractional error} = \frac{\Delta x}{x} = \frac{0.1}{5.0} = 0.02$$

$$\text{Percentage error} = 0.02 \times 100 = 2\%$$

The measurement can also be reported as $(5.0 \pm 2\%)$ cm



YOUR TURN

Q.8 An experimenter expressed his measurement of a current as $(4.0 \pm 5\%)$ A. If one more reading of the current is taken, what can be the measured value?

In Short

- Measured value = Best estimate $\pm$ Uncertainty
- Best estimate is taken to be the average of many readings that we take. It is regarded as the true value.
- Repeating the measurement many times and taking the average, reduces the effect of random error.
- While recording a measurement we must write all the reliable digits and the first uncertain digit. These digits are known as significant figures. Number of significant figures tells us about the precision of the measurement.
- Changing the unit of a measured quantity does not change the number of significant digits.
- LC of the measuring instrument is taken as maximum uncertainty in a measurement assuming that there is no systematic error.
- When we make repeated measurements, the uncertainty can be taken as average deviation. The average deviation is calculated as

$$\bar{\delta} = \frac{|x_1 - \bar{x}| + |x_2 - \bar{x}| + \dots + |x_N - \bar{x}|}{N}$$

Here, $x_1, x_2 \dots x_N$ are the readings and $\bar{x}$ is their average (i.e., the true value).

Chance of the quantity lying in the range $\bar{x} \pm \bar{\delta}$ is nearly 50%.

- A much better way of estimating the uncertainty is standard deviation (δ).

$$\delta = \sqrt{\frac{1}{(N-1)} \sum_{i=1}^N (x_i - \bar{x})^2}$$

There is 68% chance that the true value of the measured quantity lies in the range $\bar{x} \pm \sigma$ and there is nearly 95% chance that the quantity falls in the range $\bar{x} \pm 2\sigma$.

- A quantity expressed as $x \pm \Delta x$, can also be expressed as $x \pm \frac{\Delta x}{x} \times 100\%$.

$\frac{\Delta x}{x}$ is known as fractional uncertainty and $\frac{\Delta x}{x} \times 100$ is called the percentage uncertainty.

5. SIGNIFICANT FIGURES IN ALGEBRAIC OPERATIONS

Till now we have studied how to record a measurement keeping significant figures in mind and how to report uncertainty in measurement of a physical quantity.

Most of the time we will be required to measure one or more than one quantity and do some algebraic operations with them to calculate a new quantity. For example, we need to measure the length and breadth of a rectangle to calculate its area using the relation $A = lb$. How will we decide about the number of significant digits in a calculated quantity like A ? This is what we intend to discuss in this section. In the next section we will discuss how to estimate the uncertainty in a calculated quantity, like A , when uncertainties in l and b are known.

5.1 Multiplication and Division

The number of significant digits in a product or quotient is same as the smallest number of significant figures in any of the original numbers.

Assume that the current in a resistor is $I = 2.2$ A when the applied potential difference is $V = 12.35$ V. Arithmetic gives the value of resistance as:

$$R = \frac{V}{I} = \frac{12.35}{2.2} = 5.613636 \dots \Omega$$

However, all the digits in this answer cannot be significant. How can we have so many reliable digits in the value of R when number of significant digits in the value of the current is only 2 and the number of significant digits in the value of voltage is 4. In such cases we report our answer only upto 2 significant digits

$$\therefore \frac{12.35 \text{ (four significant digits)}}{2.2 \text{ (two significant digits)}} = 5.613636 \dots$$

$$= 5.6 \Omega$$

(Rounded to 2 significant digits)

5.2 Addition and Subtraction

In addition and subtraction, the result should be rounded off to the last decimal place of the least precise number.

In simple language, if A has 2 digits after decimal and B has 3 digits after decimal, then $A \pm B$ will have only 2 digits after decimal. Following example will clarify the rule.

$$\begin{array}{rcccl} \underbrace{225.64}_{\text{Two digits after decimal}} & + & \underbrace{28}_{\text{No digits after decimal}} & = & 253.64 = \underbrace{254}_{\text{No digits after decimal}} \\ \\ \underbrace{2260.5}_{\text{1 digit after decimal}} & + & \underbrace{0.006}_{\text{3 digits after decimal}} & = & 2260.506 = \underbrace{2260.5}_{\text{1 digit after decimal}} \end{array}$$

5.3 Rules for Rounding off

Following rules are used to round off a number to the required precision.

- (i) If the digit to the right of the cut off digit is less than 5, then the cut off digit is retained without any change. However, if the digit to be dropped is more than 5, then the cut off digit is increased by 1.

For example, 6.34 is rounded off to 6.3 if we want only two significant digits. 6.37 is rounded off to 6.4 to two digits.

- (ii) If the digit to be dropped is 5 followed by a digit other than zero, then the preceding digit is increased by 1.

4.352 is rounded off as 4.4 to two significant digits.

- (iii) If the digit to be dropped is 5 followed by no digit or zero, then the preceding digit is

- (a) Left unchanged if it is even
(b) increased by 1 if it is odd.

3.350 or 3.35 will be rounded off as 3.4 to two digits.

3.250 or 3.25 will become 3.2 after being rounded off to two digits.

Note: If a calculated number is to be used in further calculations, it is good practice to retain one extra digit. This reduces rounding errors.

Example 7 Radius of a circle was measured to be 1.3×10^{-2} m. Find its area. Assume that value of $\pi = 3.14$ is known exactly.

Solution

Concepts

- (i) Value of π is exact, means there are infinite number of significant digits in it.
(ii) Radius is known upto 2 significant digits. Hence area will also be reported to two significant digits.

$$\begin{aligned} A &= \pi r^2 = 3.14 \times (1.3 \times 10^{-2}) \times (1.3 \times 10^{-2}) \\ &= 5.3066 \times 10^{-4} \\ &= 5.3 \times 10^{-4} \text{ m}^2 \end{aligned}$$

Example 8 A cube has a side length of $a = 1.2$ cm and its density is $d = 7.86$ g/cm³. A ball of mass $m = 3.21$ g is attached to the cube. What is the combined mass of the system.

Solution

Concepts

- (i) Volume of the cube is $V = a^3$ and its mass is $M = V \cdot d = a^3 \cdot d$

Number of significant digits in value of a is 2 and that in the value of d is 3. Only two digits in the

value of M are significant. However, we are going to use this in further calculation. Therefore, we will retain one extra digit.

(ii) Total mass = $M + m$

Since, M is known to two digits only, our answer cannot be more precise than that.

$$\begin{aligned}\text{Mass of the cube, } M &= a^3 \cdot d = (1.2)^3 \times 7.86 = 13.582 \\ &= 13.6 \text{ g (Rounded off to 3 digits)}\end{aligned}$$

Here we have retained one extra digit to minimize the rounding error.

$$\begin{aligned}\text{Total mass} &= 13.6 + 3.21 = 16.81 \text{ g} \\ &= 16.8 \text{ g (rounded off to 1 digit after decimal)}\end{aligned}$$

But finally our answer shall not have more than two significant digits.

$$\therefore \text{Total mass} = 17 \text{ g}$$



YOUR TURN

Q.9 Round off the following numbers to three significant digits:

- (a) 12362 (b) 19.845 (c) 23.750 (d) 45.650

Q.10 Mass of a basket is 3.3 kg. Two objects of masses 3.15 g and 23.28 g are added to it. Find the total mass of the basket to the correct number of significant digits.

Q.11 The length, width and thickness of a rectangular block are measured as 120.5 cm, 25.0 cm and 10 cm respectively. Find its volume.

6. PROPAGATION OF UNCERTAINTY (ERROR)

Let us assume that a quantity A depends on two quantities x and y .

$$A = f(x, y)$$

x and y are measured and then A is calculated. The errors in x and y may not have the same effect on the error in A . Final error in A depends on the way the errors in x and y combine through mathematical operations. We will always try to find out the maximum possible error in the calculated quantity A .

6.1 Error in sum or difference of quantities

Suppose $A = x + y$ (i)

Measured value of x is $x \pm \Delta x$ and the measured value of y is $y \pm \Delta y$. Here, Δx and Δy represent the maximum possible errors in the measured value of x and y .

What is the true value of A ? To the best of our knowledge it is $(x + y)$ [the sum of true values of x and y]. Now, we wish to estimate the maximum possible error in this value of A .

If the value of x , that we are considering as true value, deviates from the true value by Δx and similarly, the value of y that we are assuming to be its true value, deviates from the actual true value by Δy , then the maximum error in value of A will be given by

$$\Delta A = \Delta x + \Delta y$$

The reported value of A shall be $A \pm \Delta A$.

Now, suppose $A = x - y$ (ii)

True value of A is calculated as true value of x minus the true value of y . But how much uncertainty could be there in this value? Let's assume that the value of x , accepted by us as its true value, is actually higher by Δx compared to the true value known to the God. It may happen that value of y that we are considering as true is less by Δy compared to its actual value (again known only to the God!). In such a situation, the value of A calculated using (ii) will differ from its real value by $(\Delta x + \Delta y)$. Therefore, the maximum error in A is again given by

$$\Delta A = \Delta x + \Delta y$$

Therefore, we conclude that

$$\Delta A = \Delta x + \Delta y \quad (7)$$

When $A = x \pm y$

Let's take a numerical example to make this clear. Assume that the true values of x and y (known to the God) are 5.4 and 1.2 respectively. The measured values could be

$$x = 5.5 \pm 0.1 \text{ and}$$

$$y = 1.1 \pm 0.1$$

[We have assumed that it is one of our worst days as an experimenter and we have highest possible errors in our measured true values]

Now, we will calculate the value of $A = x - y$ as

$$A = 5.5 - 1.1 = 4.4$$

But the true value of A , known to the God is $5.4 - 1.2 = 4.2$. Therefore, our value deviates from actual by 0.2 ($=\Delta x + \Delta y = 0.1 + 0.1$).

Maximum possible error in A is $\Delta A = \Delta x + \Delta y$ and NOT $\Delta x - \Delta y$.

6.2 Error in product of quantities

Consider $A = Kx^a y^b$

Here, K , a and b are constants positive or negative. We wish to know the maximum possible error in A when the errors in x and y are Δx and Δy , respectively. One easy way of doing this is to use calculus.

We will calculate the amount by which value of A will deviate when the values of x and y deviate (from their true values) by some small amounts. Differentiate the above equation with respect to x

$$\frac{dA}{dx} \simeq K \left[y^b \cdot a \cdot x^{a-1} + x^a \cdot b \cdot y^{b-1} \frac{dy}{dx} \right]$$

$$\text{or, } dA = Kx^a y^b \left[a \frac{dx}{x} + b \frac{dy}{y} \right]$$

The above equation tells us the change in value of A when the values of x and y change by infinitesimally small amounts dx and dy . For finite changes (Δx and Δy) the above equation will be approximately correct as long as Δx and Δy are small.

$$\therefore \Delta A = Kx^a y^b \left[a \frac{\Delta x}{x} + b \frac{\Delta y}{y} \right]$$

$$\text{or, } \Delta A = A \left[a \frac{\Delta x}{x} + b \frac{\Delta y}{y} \right]$$

$$\text{or, } \frac{\Delta A}{A} = a \frac{\Delta x}{x} + b \frac{\Delta y}{y}$$

If Δx and Δy are maximum errors in x and y , then the maximum error in A is given by

$$\frac{\Delta A}{A} = \left| a \frac{\Delta x}{x} \right| + \left| b \frac{\Delta y}{y} \right| \quad (8)$$

Irrespective of the signs of a and b , we must always add the errors. This has been explained in section 6.1.

The above equation can be remembered as:

When $A = Kx^a y^b$

$$\left\{ \begin{array}{l} \text{fractional} \\ \text{error in } A \end{array} \right\} = |a| \left\{ \begin{array}{l} \text{fractional} \\ \text{error in } x \end{array} \right\} + |b| \left\{ \begin{array}{l} \text{fractional} \\ \text{error in } y \end{array} \right\}$$

Also remember that fractional error multiplied with 100 gives percentage error.

Few important corollaries of the above result are:

(i) When $A = xy$, then

$$\frac{\Delta A}{A} = \frac{\Delta x}{x} + \frac{\Delta y}{y} \quad (9)$$

(ii) When $A = \frac{x}{y}$, then

$$\frac{\Delta A}{A} = \frac{\Delta x}{x} + \frac{\Delta y}{y} \quad (10)$$

The fractional errors in x and y add when $A = xy$ or $A = \frac{x}{y}$

In terms of percentage errors we can write the above expression as

$$\frac{\Delta A}{A} \times 100 = \frac{\Delta x}{x} \times 100 + \frac{\Delta y}{y} \times 100$$

$$\Rightarrow \% \text{ error in } A = \% \text{ error in } x + \% \text{ error in } y$$

(iii) When $A = Kxy$, or $A = K \frac{x}{y}$

(K is a constant), then

$$\frac{\Delta A}{A} = \frac{\Delta x}{x} + \frac{\Delta y}{y}$$

(iv) When $A = Kx^a$

$$\frac{\Delta A}{A} = |a| \frac{\Delta x}{x}$$

$$\% \text{ error in } A = |a| (\% \text{ error in } x)$$

Example 9 In an experiment designed to measure a resistance, the voltage across the resistor and the current through it were measured to be $V = (5.00 \pm 0.10)$ volt and $I = (2.00 \pm 0.01)$ A. Find the value of resistance and the maximum possible error in its value.

Solution

Concepts

(i) Best known values of V and I are 5.00 V and 2.00 A, respectively.

$$(ii) R = \frac{V}{I} \Rightarrow R = V^1 I^{-1}$$

$$\begin{aligned} \Rightarrow \frac{\Delta R}{R} &= 1 \cdot \frac{\Delta V}{V} + |-1| \frac{\Delta I}{I} \\ &= \frac{\Delta V}{V} + \frac{\Delta I}{I} \end{aligned}$$

$$\text{Best value of resistance } R = \frac{5.00 \text{ V}}{2.00 \text{ A}} = 2.50 \Omega$$

Maximum fractional error in value of R is

$$\frac{\Delta R}{R} = \frac{\Delta V}{V} + \frac{\Delta I}{I} = \frac{0.10}{5.00} + \frac{0.01}{2.00}$$

$$\begin{aligned}
 &= 0.02 + 0.005 = 0.025 \\
 \therefore \Delta R &= (0.025)R = 0.025 \times 2.5 \\
 &= 0.0625 \Omega \\
 &\approx 0.06 \Omega \\
 \therefore \text{Value of } R \text{ will be reported as } R &= 2.50 \Omega \pm 0.06 \Omega
 \end{aligned}$$

Note: Percentage error in R is $\frac{\Delta R}{R} \times 100 = 0.025 \times 100 = 2.5\%$

$$\therefore R = (2.50 \pm 2.5\%) \Omega$$

Example 10 Resistance of a wire is measured to be $R = 2.05 \Omega \pm 2\%$. Its diameter and length are measured to be $d = 1.00 \text{ mm} \pm 1\%$ and $l = 70.0 \text{ cm} \pm 1.5\%$. These observed quantities are used to calculate the resistivity of the material of the wire. Find its value with maximum percentage error. Take $\pi = 3.14$ (exact).

Solution

Concepts

$$(i) R = \frac{\rho l}{A} = \frac{\rho l}{\pi/4 d^2}$$

$$\Rightarrow \rho = \frac{\pi R d^2}{4 l} = K R^1 d^2 l^{-1}$$

$$(ii) \frac{\Delta \rho}{\rho} = 1 \left(\frac{\Delta R}{R} \right) + 2 \left(\frac{\Delta d}{d} \right) + 1 \left(\frac{\Delta l}{l} \right)$$

$$\rho = \frac{\pi R d^2}{4 l} = \frac{3.14}{4} \times \frac{2.05 \times (1.00 \times 10^{-3})^2}{70.0 \times 10^{-2}}$$

$$= 2.299 \times 10^{-6} \Omega\text{-m}$$

$$\approx 2.30 \times 10^{-6} \Omega\text{-m}$$

$$\frac{\Delta \rho}{\rho} \times 100 = \frac{\Delta R}{R} \times 100 + 2 \left(\frac{\Delta d}{d} \times 100 \right) + \left(\frac{\Delta l}{l} \times 100 \right)$$

$$= 2\% + 2(1\%) + 1.5\%$$

$$= 5.5\%$$

Value of ρ will be reported as $\rho = (2.30 \times 10^{-6} \pm 5.5\%) \Omega$

Example 11 Errors in different quantities affect the end result differently

In the last question, you have an option to reduce error in measurement of one of the three quantities R , d and l which of the three quantities will you choose?

Solution

$$\frac{\Delta \rho}{\rho} = \frac{\Delta R}{R} + 2 \left(\frac{\Delta d}{d} \right) + \frac{\Delta l}{l}$$

The coefficient of $\left(\frac{\Delta d}{d} \right)$ is 2. Coefficient for $\frac{\Delta R}{R}$ and $\frac{\Delta l}{l}$ is 1. Error in measurement of d has maximum effect on the error in ρ . We would like to reduce the error in measurement of d .

Example 12 Resistances in Parallel

Two resistances are measured to be $R_1 = (4 \pm 0.2) \Omega$ and $R_2 = (12 \pm 0.2) \Omega$. Calculate the equivalent resistance when they are connected in parallel. Also find the percentage error in the calculated value. Equivalent resistance was calculated

$$\text{using } R = \frac{R_1 R_2}{R_1 + R_2}$$

Solution

Concepts

$$(i) \text{ Let } R_1 + R_2 = R_0$$

$$\Delta R_0 = \Delta R_1 + \Delta R_2$$

$$(ii) R = \frac{R_1 R_2}{R_0}$$

$$\Rightarrow \frac{\Delta R}{R} = \frac{\Delta R_1}{R_1} + \frac{\Delta R_2}{R_2} + \frac{\Delta R_0}{R_0}$$

$$R_0 = R_1 + R_2 = 16 \Omega$$

$$\Rightarrow \Delta R_0 = \Delta R_1 + \Delta R_2 = 0.2 + 0.2 = 0.4 \Omega$$

$$\begin{aligned} \text{Now, } \frac{\Delta R}{R} &= \frac{\Delta R_1}{R_1} + \frac{\Delta R_2}{R_2} + \frac{\Delta R_0}{R_0} \\ &= \frac{0.2}{4} + \frac{0.2}{12} + \frac{0.4}{16} \end{aligned}$$

$$= 0.05 + 0.02 + 0.02 = 0.09$$

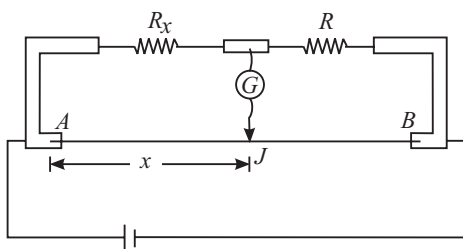
$$\therefore \frac{\Delta R}{R} \times 100 = 0.09 \times 100 = 9\%$$

$$R = \frac{R_1 R_2}{R_0} = \frac{4 \times 12}{16} = 3 \Omega$$

$$\therefore R = (3 \pm 9\%) \Omega$$

Example 13 Minimizing error in meter bridge experiment

In a meter bridge experiment, we want to measure an unknown resistance R_x . R is a known resistance. Value of R is supposed to be known exactly. When the galvanometer reads zero, the sliding jockey is at J such that $AJ = x$. The scale used to measure x has a LC of Δx and therefore, the maximum error in reported value of x is Δx . Show that when J is at the midpoint of the meter bridge wire AB , the error in the value of R_x is least.

**Solution****Concepts**

$$(i) R_x = R \left(\frac{x}{L-x} \right) = R \frac{x}{y}$$

where $L = 100$ cm is the length of the meter bridge wire. And $y = L - x$.

(ii) Error in x = error in $y = \Delta x$

(iii) We need to find out the condition when the error in R_x will be smallest for a given maximum error in x .

$$R_x = R \frac{x}{y}$$

$$\frac{\Delta R_x}{R_x} = \frac{\Delta x}{x} + \frac{\Delta y}{y}$$

$$\text{or, } \eta = \frac{\Delta x}{x} + \frac{\Delta x}{y} \quad [\because \Delta y = \Delta x]$$

η = fractional error in R_x

$$\therefore \eta = \Delta x \left(\frac{y+x}{xy} \right) = \Delta x \frac{L}{xy}$$

$$\text{or, } \eta = \frac{\Delta x \cdot L}{x(L-x)}$$

Δx and L are fixed quantities η is minimum when $z = x(L-x)$ is maximum.

This happens when $\frac{dz}{dx} = 0$

$$\Rightarrow L - 2x = 0$$

$$\Rightarrow x = \frac{L}{2}$$

This implies that the error in R_x is least when the balance point is near the middle of the meter bridge wire.

**YOUR TURN**

Q.12 Time period of a simple pendulum is given by

$$T = 2\pi \sqrt{\frac{l}{g}}. \text{ Values of } \pi \text{ and } g \text{ are known exactly and are}$$

equal to 3.14 and 9.80 ms^{-2} . Length of the simple pendulum is measured to be $l = (1.00 \pm 0.01) \text{ m}$. Find the time period and percentage error in its value.

Q.13 Length of a simple pendulum is measured to be $l = (1.00 \pm 0.01) \text{ m}$ and its time period is measured as $T = (2.00 \pm 0.05) \text{ s}$. Find the value of acceleration due to gravity at the place of experiment. It is given that $\pi = 3.14$ (exact).

Q.14 The error in measurement of momentum of a body is 0.2% and its mass is known exactly to be equal to 4.2 kg. Find the percentage error in calculated value of kinetic energy of the body.

Q.15 $A = \frac{x^2 y^3}{z^5}$ x , y and z are measured and A is calculated. Which of these three must be measured more accurately to minimize error in A .

In Short

- In addition and subtraction, the result is reported to the same number of decimal places as are present in the number having the least number of decimal places.
- In multiplication and division the result is expressed to the same number of significant figures as that of the original number with minimum number of significant figures.

- If $A = x \pm y$ and x and y have measured values of $x \pm \Delta x$ and $y \pm \Delta y$, then the maximum error in the value of A is $\Delta A = \Delta x + \Delta y$

- If $A = Kx^a y^b$, then error in value of A is given by

$$\frac{\Delta A}{A} = \left| a \frac{\Delta x}{x} \right| + \left| b \frac{\Delta y}{y} \right|$$

$$\% \text{ error in } A = |a| (\% \text{ error in } x) + |b| (\% \text{ error in } y)$$

7. SENSITIVITY OF A MEASURING INSTRUMENT

Suppose you are measuring a potential difference, which is of the order of few volts, using a voltmeter having a range of 0–20 kV. The voltmeter scale has 200 divisions on its scale. You will never be able to record a reading. The voltmeter will show a reading of 0 V even when the applied voltage is 2 V or 5 V. Each division of the voltmeter corresponds to 100 V. The deflection of the pointer for a 10 V potential difference is unlikely to be noticeable. We say that the given voltmeter has a poor sensitivity.

The *sensitivity of a measuring device is its ability to distinguish between two close values of a quantity*. Using an ordinary millimeter scale we cannot differentiate between the lengths of two sticks which are 14.36 cm and 14.35 cm long. The scale is not sensitive enough. A scale with a vernier attached to it can differentiate between two lengths differing by 0.01 cm.

Sensitivity of a voltmeter can be defined as the number of divisions that the pointer moves through when the applied potential difference is changed by unity. If the pointer moves Δn divisions when the voltage is changed by ΔV , then the sensitivity can be defined as

$$s = \frac{\Delta n}{\Delta V}$$

In a potentiometer, we use a long wire. This increases the sensitivity of the instrument. Suppose the potential difference applied across a potentiometer wire of length L is V_0 . The potential gradient along the length of the wire is $\frac{V_0}{L}$. This potential gradient is small for a large value of L . It means we have to move a long distance along the wire for every 1 V change in potential. This increases the sensitivity. Decreasing V_0 also decreases the sensitivity.

Example 14 Making a more sensitive potentiometer

Consider a potentiometer A in which wire length is 4 m and the potential difference applied across it is $V_0 = 100$ V. A

millimeter scale is used to measure the length along the wire. Can this potentiometer be used to distinguish between emfs of two cells which are $\varepsilon_1 = 10.00$ and $\varepsilon_2 = 10.01$ V?

Now, consider a potentiometer B in which the wire length is 10 m and the potential difference across it is $V'_0 = 20$ V. Again a millimeter scale is used to measure the length. Can this potentiometer distinguish between the emfs of two cells which are $\varepsilon_1 = 10.00$ V and $\varepsilon_2 = 10.01$ V?

Solution

Concepts

We can detect the movement of jockey when it is at least 1 mm. Any movement below 1 mm can not be detected.

A potentiometer can differentiate between $\varepsilon_1 = 10.00$ V and $\varepsilon_2 = 10.01$ V only when the movement of the jockey corresponding to a potential difference of 0.01 V is more than 1 mm.

In potentiometer A , the potential gradient is

$$\frac{V_0}{L} = \frac{100 \text{ V}}{4000 \text{ mm}} = 0.025 \text{ V/mm}$$

It means, the jockey will shift by 1 mm (in balanced position) when the potential difference to be measured changes by 0.025 V. In the given problem $\Delta V = 0.01$ V. For this the shift in position of the jockey will be

$$\Delta l = \frac{0.01}{0.025} \text{ mm} = 0.4 \text{ mm}$$

This cannot be detected. Hence, the potentiometer A will fail to differentiate between ε_1 and ε_2 .

In potentiometer B , the potential gradient is

$$\frac{V'_0}{L'} = \frac{20 \text{ V}}{10000} = 0.002 \text{ V/mm}$$

When the measured potential difference changes by 0.01 V, the balance point shifts by 5 mm. This can be easily detected. This potentiometer has better sensitivity than A and will be able to measure ε_1 and ε_2 as two different values.

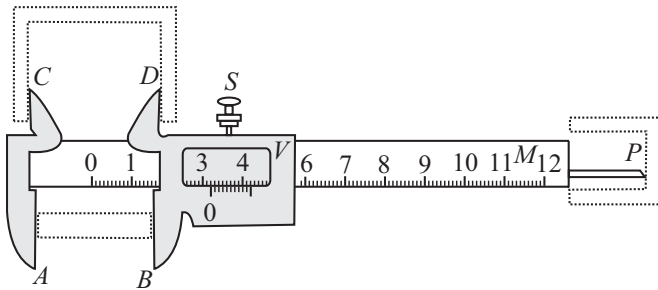


YOUR TURN

Q.16 Why do we need weighing scales of different ranges for weighing ornaments and groceries?

8. VERNIER CALIPER

This device is used to measure a length with a precision of 0.1 mm. A common type of vernier caliper is as shown. The steel strip M is graduated in cm and mm. There is another strip V which can slide on M . It can be fixed at any position with the help of screw S . A scale is marked on this strip and is called as vernier scale. Each division on this scale is slightly smaller than 1 mm such that 10 divisions of this scale has length of 9 mm. The depth probe P is connected to strip V and moves with the strip. If we wish to measure the diameter of a ball, we will place the ball between the jaws A and B . If we wish to measure the internal diameter of a pipe, the jaws C and D are inserted inside the pipe. To measure the depth of a cylinder, the probe P is inserted inside it. Ideally, when the jaws A and B touch one another, the zero of vernier scale should coincide with the zero of the main scale. If that is not the case we say that the vernier has a zero error.



- The main scale (M) is mm scale.
- 10 division of vernier scale (V) = 9 division of main scale M .
- AB is used to measure external diameter
- CD is used to measure internal diameter.
- P is used to measure depth.

8.1 Measurement with a Vernier caliper

Fig. (a) shows a vernier caliper with its jaws closed. The zero of the vernier scale coincide with zero of the main scale. Also, 10 divisions on the vernier scale have same length as 9 divisions on the main scale.

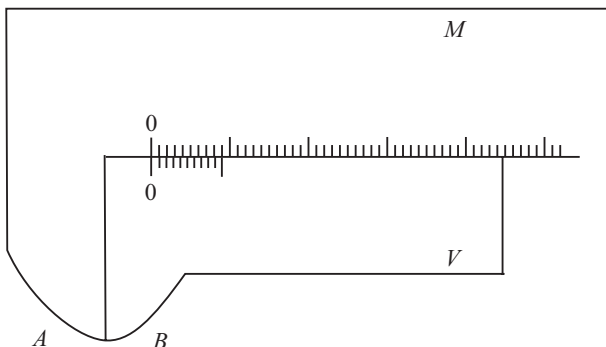


Fig. (a)

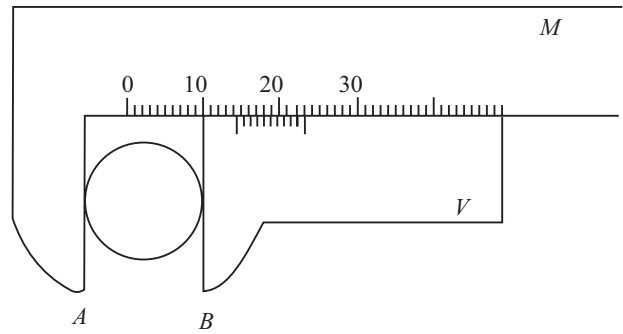


Fig. (b)

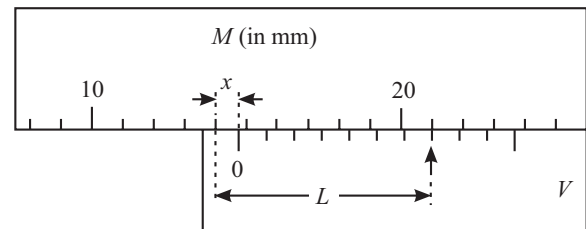


Fig. (c) Magnified view of the scales in Fig. (b)

$\therefore$ 1 vernier scale division = 0.9 times 1 main scale division

$$\begin{aligned} 1 \text{ VSD} &= 0.9 \text{ MSD} \\ &= 0.9 \text{ mm} \end{aligned}$$

Let's assume that we wish to measure the diameter of a marble. It is placed between the jaws as shown in Fig. (b). The vernier strip (V) slides on the main strip (M) by a distance that is equal to the diameter of the marble. We will measure the distance that the zero of the vernier has travelled on the main scale.

The magnified view of the two scales has been shown in Fig. (c). Certainly, the reading is larger than 14 mm. Using the vernier we can find how much larger the reading is above 14 mm. This extra length has been shown as x .

As shown in the diagram, the 7th vernier mark (indicated by $\uparrow$ in the diagram) coincides with a main scale. Looking at the main scale, we can write

$L = 7 \text{ MSD}$ and looking at the vernier scale we can write

$$L = x + 7 \text{ VSD}$$

$$\therefore x + 7 \text{ VSD} = 7 \text{ MSD}$$

$$\text{or } x = 7 (1 \text{ MSD} - 1 \text{ VSD})$$

$$\text{or } x = 7(\text{LC})$$

Here, 1 MSD – 1 VSD is defined as the least count (LC) of the vernier caliper. For our vernier

$$\text{LC} = 1 \text{ mm} - 0.9 \text{ mm} = 0.1 \text{ mm}$$

$$\therefore x = 0.7 \text{ mm}$$

Therefore, diameter of the marble is

$$d = 14 \text{ mm} + 0.7 \text{ mm} = 14.7 \text{ mm}$$

The procedure of taking a reading with a vernier caliper can be summarized as:

1. If n VSD have same length as $(n - 1)$ MSD, then

$$1 \text{ VSD} = \left(\frac{n-1}{n} \right) \text{ MSD}$$

$$\text{LC} = 1 \text{ MSD} - 1 \text{ VSD} = \left[1 - \left(\frac{n-1}{n} \right) \right] \text{ MSD}$$

$$\text{or, } \text{LC} = \frac{1}{n} \text{ MSD}$$

2. After the object is placed between the jaws, the main scale reading is
 N = main scale reading just to the left of the zero on the vernier scale.
3. Look carefully to find which mark on the vernier scale is coinciding with a mark on the main scale. If it is the n th division on the vernier scale, which is coinciding with a main scale mark, then

$$\text{Total reading} = N + n(\text{LC})$$

8.2 Zero error and correction

Close the jaws of the vernier caliper such that the jaws A and B (and the straight edges of C and D) touch one another. If the zero of the vernier coincides with the zero of the main scale, we say there is no zero error.

If the zero of the vernier is to the right of the zero of the main scale (see Fig. (b)), we say that the instrument has a positive zero error. Such an instrument will always show an excess value of the measured length by an amount x_0 (see Fig.) We must subtract this length from any reading that we take using such an instrument.

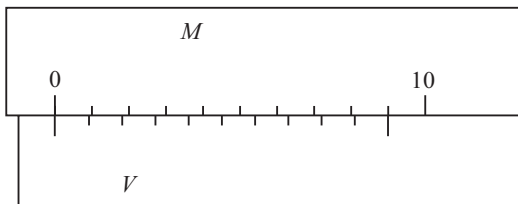


Fig. (a) No zero error. Zeros of both scales coincide

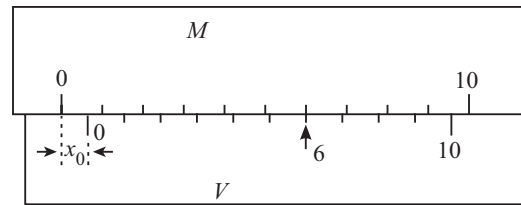


Fig. (b) Positive zero error = x_0 . 6th vernier division is coinciding with a main scale division.

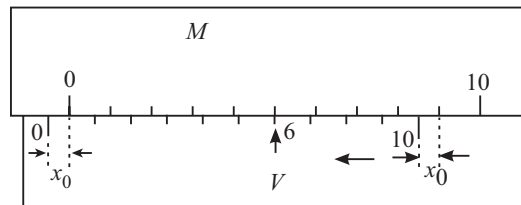


Fig. (c) Negative zero error. 4th vernier division, counted to left starting from the 10 marked on vernier, coincides with a main scale division.

To measure the zero error (x_0), see which mark on the vernier coincides with a mark on the main scale.

In Fig. (b), the 6th vernier division coincides with a main scale division.

$$\therefore x_0 + 6 \text{ VSD} = 6 \text{ MSD}$$

$$\text{or } x_0 = 6(\text{MSD} - \text{VSD})$$

$$= 6(\text{LC}) = 0.6 \text{ mm}$$

In the figure shown, the zero error is +0.6 mm. *This must be subtracted* from any reading that we take using this vernier.

Now, consider a vernier caliper with its scale as shown in Fig. (c) when the jaws are closed. The zero of the vernier is to the left of the zero of the main scale. We say that the instrument has a negative zero error equal to $-x_0$. For finding the value of x_0 , note which vernier division from right end (i.e., the last mark 10 on the vernier) matches the main scale mark. In the shown diagram, it is the 4th vernier division from the right that coincides with a main scale division. You can prove yourself that

$$x_0 = 4 (\text{LC}) = 0.4 \text{ mm}$$

We say that zero error is -0.4 mm . *We must add 0.4 mm to any subsequent reading taken using this vernier caliper.*



YOUR TURN

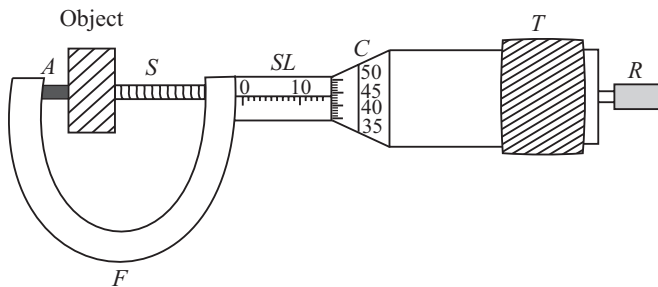
Q.17 When there is no object between the closed jaws of a vernier caliper, the zero of the vernier is slightly to the right of the main scale and the 6th vernier division is coinciding with a main scale division. Find the zero error in the vernier.

Q.18 The vernier mentioned in the last question is used to measure the thickness of a plate. The main scale reading is 43 mm and the 2nd vernier division coincides with a main scale division. What is the plate thickness.

Q.19 When no object is there between the jaws, the zero of a vernier lies to the left of the zero of the main scale. The 7th vernier division coincides with a main scale division. When a wire is placed between the jaws, the zero of vernier is slightly right of 4 mm mark on the main scale. Again the 7th vernier division is coinciding with a main scale division. Find the diameter of the wire.

9. SCREW GAUGE OR MICROMETER

A common vernier has a LC of 0.1 mm. If we need even greater precision we can use a screw gauge. A common type of screw gauge has a LC of 0.01 mm. A screw gauge has been shown in the figure below. It consists of a carefully machined screw (S) attached to a circular scale (C). A linear scale (on the sleeve SL) helps us in recording the forward or backward movement of the screw.



S = Spindle (the screw) T = Thimble
 R = Ratchet A = Anvil
 SL = Sleeve C = Circular scale
 F = Frame

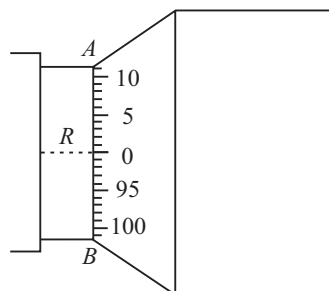
The distance moved by the screw in one turn is known as the *pitch* of the screw. The commonly available screw gauge has a screw with a pitch $p = 1$ mm. When the screw advances by 1 mm, the circular scale makes one full rotation. The circular scale helps us to record the fractions of the turns, and the linear scale enable us to record the whole number of turns.

Consider that the circular scale has 100 divisions and the pitch of the screw is $p = 1$ mm. This implies that if the circular scale rotates through 1 division (i.e., $\frac{1}{100}$ th of a complete rotation), the screw advances by $\left(\frac{1}{100}\right)$ mm. This is the *LC* of the instrument

$$LC = \frac{\text{Pitch}}{\text{Number of divisions on circular scale}}$$

For the screw gauge that we are considering, $LC = 0.01$ mm.

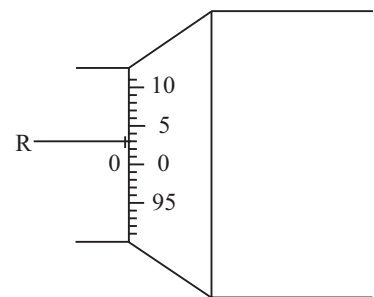
Note that the linear scale is marked on sleeve (SL) and the circular scale turns on this sleeve. When the spindle touches the anvil (A), the circular scale should ideally appear as shown in the figure. The edge of the circular scale is exactly at the zero of the



No zero error. The edge (AB) of the circular scale coincides with zero of the main scale and zero of circular scale is on the reference line (R)

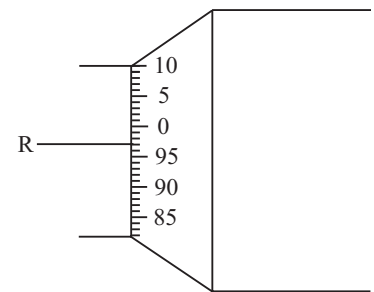
main scale and the circular scale zero coincides with the reference line on the sleeve.

When the spindle touches the anvil and the scales appear as shown in the figure, we say that the instrument has a *positive zero error*. It is showing an excess reading of $3(LC) = +0.03$ mm. We must subtract this amount from any subsequent readings.



Positive zero error. The edge of the circular scale is slightly to the right of zero on the main scale. Third division of circular scale coincides with the reference line.

A screw gauge with negative zero error is also shown. This instrument will always read 0.03 mm less than the true value. We say zero error is $-3(LC) = -0.03$ mm. We must add 0.03 mm to all subsequent readings taken using this instrument.



Negative zero error. The zero of main scale is not visible. Zero of the circular scale is 3 divisions above the reference line.

When an object is placed between the anvil and the spindle, the screw retracts by an amount that is equal to the thickness of the object. Look at the first figure in this section. What is the thickness of the object? The main scale tells you that the screw must have moved back by a distance slightly larger than 14 mm. The circular scale reading tells you exactly how much more (than 14 mm) it has moved. The reading is recorded as:

Main scale reading (MSR) = 14 mm

Circular scale reading (CSR) = 44

[The 44th division on circular scale is coinciding with the reference line]

$$\begin{aligned}
 \therefore \text{Thickness of the object} &= \text{MSR} + \text{CSR} (LC) \\
 &= 14 + 44 \times 0.01 \\
 &= 14.44 \text{ mm}
 \end{aligned}$$

If there is a zero error we must correct the above thickness.

Example 15 A screw gauge has $\frac{1}{2}$ mm marks on its main scale. Its circular scale has 50 divisions. When the spindle touches the anvil, the scale appear as shown. With a glass plate between the anvil and the spindle, the scale appear as shown in Fig. (b). Find the thickness of the plate. Pitch of the screw is $\frac{1}{2}$ mm.

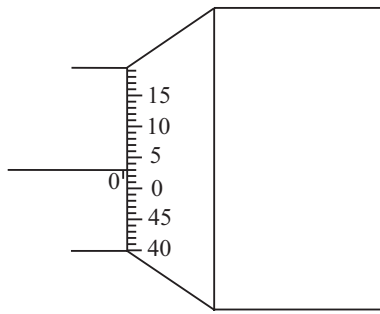


Fig. (a) No object

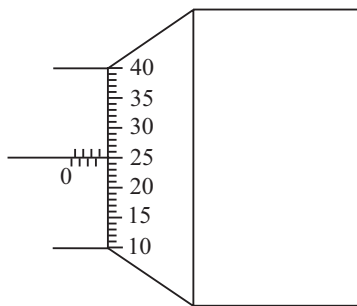


Fig. (b) With object between jaws

Solution**Concepts**

- (i) The main scale has marks at $\frac{1}{2}$ mm separations. The consecutive marks are on opposite sides of the reference line for clarity.
- (ii) The instrument has positive zero error. It is showing some reading when there is no object between the jaws. This zero error must be subtracted from the final reading.

$$\begin{aligned}\text{Least count} &= \frac{\text{Pitch}}{\text{Number of divisions on circular scale}} \\ &= \frac{0.5 \text{ mm}}{50} = 0.01 \text{ mm}\end{aligned}$$

$$\text{Zero error} = +(\text{CSR}) (\text{LC}) = 3 \times 0.01 = 0.03 \text{ mm}$$

Reading in Fig. (b) is:

$$\text{MSR} = 3.5 \text{ mm}$$

$$\text{CSR} = 25$$

$$\therefore \text{Recorded value} = 3.5 \text{ mm} + 25 (\text{LC}) = 3.75 \text{ mm}$$

$$\text{Thickness of the plate} = 3.75 - 0.03 = 3.72 \text{ mm}$$

**YOUR TURN**

Q.20 A screw gauge has a screw of pitch 2 mm and number of divisions on its circular scale is 50. Find its LC.

Q.21 When the spindle of a screw gauge is touching the anvil, the zero on the main scale is not visible and the 60th mark on the circular scale is coinciding with the reference

line. The pitch is 1 mm and total divisions on the circular scale are 100. Find the zero error.

Q.22 When a plate is placed between the jaws of the screw gauge mentioned in previous question, the main scale reading is 12 mm and the 28th mark on the circular scale is at the reference line. Find the thickness of the plate.

10. PRECISION AND ACCURACY

A meter scale can measure a length as 2.4 cm. The same length can be measured by a vernier to be equal to 2.42 cm and a screw gauge can measure it as 2.426 cm. We say that the measurement by the screw gauge is most precise and that by the meter scale is least precise. The smaller the LC of a measuring instrument, better is its precision.

But we cannot say that the measurement 2.426 cm is most accurate. It may be, it may not be. What if the God tells you that the true length is 2.401 cm. The measurement made by the meter scale (though least precise) is closest to the true value. It is the most accurate measurement.

Miscellaneous Examples

Example 16 *More oscillations, less error*

Time period of a simple pendulum is given by $T = 2\pi\sqrt{\frac{l}{g}}$.

By measuring the length (l) of the pendulum and its time period (T), we can calculate the value of acceleration due to gravity (g). The length of the pendulum was measured to be $l = (1.00 \pm 0.01)$ m. To measure the time period, the pendulum was set into oscillations and the stop watch was put on. The time of 20 oscillations was recorded as 40.24 s. The stop watch has a LC of 0.01 s. Find the value of g and the percentage error in its calculated value.

Solution
Concepts

- (i) The error in measurement of a time span using the mentioned stop watch will be ± 0.01 s, whether we use it to measure the time of one oscillation or 20 oscillations.
- (ii) If t is measured time for n oscillations, then the time period is $T = \frac{t}{n}$.

$$\therefore \Delta T = \frac{\Delta t}{n}$$

$$g = 4\pi^2 \frac{l}{T^2}$$

$$\text{and } T = \frac{t}{n} = \frac{40.24}{20} = 2.012 \text{ s}$$

$$\therefore g = 4 \times 3.14^2 \times \frac{1.00}{(2.012)^2} = 9.742 \text{ ms}^{-2}$$

$$\text{Now, } g = K \frac{l}{T^2} = K l^1 T^{-2} \quad [\text{where } K = 4\pi^2]$$

$$\therefore \frac{\Delta g}{g} \times 100 = \left(\frac{\Delta l}{l} + 2 \frac{\Delta T}{T} \right) \times 100 \quad (i)$$

But what is ΔT ?

$$T = \frac{t}{n} \Rightarrow \Delta T = \frac{\Delta t}{n} = \frac{0.01}{20} = 0.0005$$

[Note that ΔT will get even smaller if we measure the time for 30 oscillations]

$$\therefore \frac{\Delta g}{g} \times 100 = \left(\frac{0.01}{1.00} + 2 \times \frac{0.0005}{2.012} \right) \times 100$$

$$= 1.00 + 0.049 = 1.05\%$$

$$\therefore g = (9.742 \pm 1.05\%) \text{ ms}^{-2}$$

Example 17 *Large error*

The displacement of a particle starting from rest and moving under a constant acceleration is given by the formula

$s = \frac{1}{2}at^2$. Value of the acceleration is known exactly but the time was measured with an uncertainty of 30%. What will be error in the value of s ?

Solution
Concepts

$$(i) \quad s = \frac{1}{2}at^2 \Rightarrow s = Kt^2$$

It is wrong to use $\frac{\Delta s}{s} = 2 \frac{\Delta t}{t}$ because this estimation of error is nearly correct when error in t is small.

- (ii) We just need to find out, using simple arithmetic, how the value of s will deviate when error in t is 30%.

Let the true value of t be t_0 . The true value of s will

$$\text{be } s_0 = Kt_0^2 \quad \left[K = \frac{1}{2}a \right]$$

If the value of t that we have measured is 30% above the true value t_0 , then value of s that we obtain is

$$s_1 = K(1.3t_0)^2 = 1.69s_0$$

Error in value of s is 69%

If the value of t that we have measured is 30% lower than the true value t_0 , then the value of s that we obtain is

$$s_2 = K(0.7t_0)^2 = 0.49s_0$$

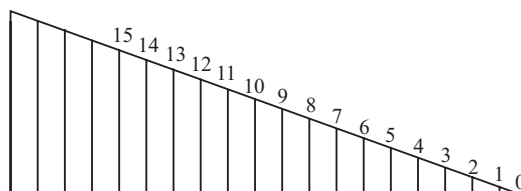
Error in the value of s is 51%

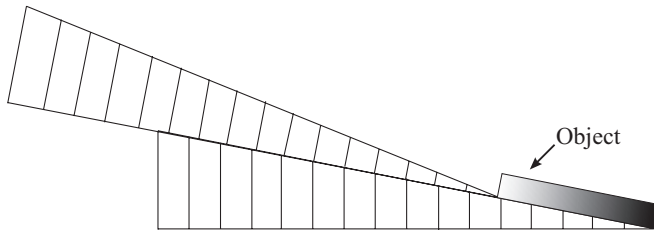
Therefore, the error in the value of s can be anywhere between -51% to $+69\%$.

Example 18 *Figure shows a simple vernier scale. The*

wedge shown has an inclination angle of $\cos^{-1}\left(\frac{14}{15}\right)$.

Marking on the hypotenuse is in mm. Two identical wedges are used to measure the length of an object as shown in the second figure. Find the measured length.



**Solution****Concepts**

- (i) $\frac{\text{Length of one division on base}}{\text{Length of one division on incline}} = \cos \theta = \frac{14}{15}$
- (ii) The divisions on the incline act as main scale and those on the base act as vernier scale divisions.

Length of one division on base (i.e., vernier)

$$= \frac{14}{15} (\text{length of one main scale division})$$

$$\text{Least count} = 1 \text{ MSD} - 1 \text{ VSD} = 1 \text{ mm} - \frac{14}{15} \text{ mm}$$

In the figure shown:

Main scale reading = 5 mm

5th vernier division coincides with a main scale division

$$\therefore \text{Vernier reading} = 5 \times \text{LC}$$

$$= 5 \times \frac{1}{15} = \frac{1}{3} \text{ mm} = 0.33 \text{ mm}$$

$$\therefore \text{Length of the object is } 5 \text{ mm} + 0.33 \text{ mm} = 5.33 \text{ mm}$$

Example 19 A vernier caliper has 19 main scale divisions equal to 20 vernier scale divisions. With the jaws closed, the

zero of vernier is slightly to the left of the main scale zero and the 12th vernier division is coinciding with a main scale division. When a plate is placed between the jaws, the zero of vernier lies slightly to the right of 16 mm mark on the main scale and the 15th vernier division coincides with a main scale division. Find the thickness of the plate.

Solution**Concepts**

- (i) Least count = 1 MSD – 1 VSD
- (ii) The instrument has negative zero error.

$$1 \text{ VSD} = \frac{19}{20} \text{ mm}$$

$$\therefore \text{LC} = 1 \text{ mm} - \frac{19}{20} \text{ mm} = \frac{1}{20} \text{ mm} = 0.05 \text{ mm}$$

When jaws are closed, the 12th division coincides with a main scale division, but the zero of vernier is to the left of main scale zero. We must count from the opposite end of the vernier scale to find that the 8th division (starting from the rightmost mark) coincides with a main scale marking.

$$\therefore \text{Zero error} = -8(\text{LC}) = -8 \times 0.05 = -0.40 \text{ mm}$$

With plate between the jaws, the reading is

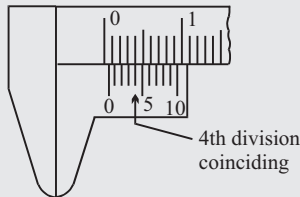
$$16 \text{ mm} + 15(\text{LC}) = 16 + 0.75 = 16.75 \text{ mm}$$

Correction applied for the zero error gives the plate thickness as

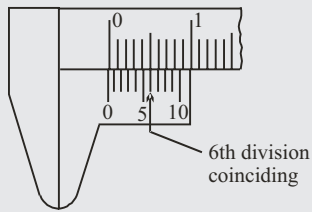
$$t = 16.75 + 0.40 = 17.15 \text{ mm}$$

Worksheet 1

- The edge of a cube is $a = 1.2 \times 10^{-2}$ m. Then its volume will be recorded as:
 - $1.7 \times 10^{-6} \text{ m}^3$
 - $1.70 \times 10^{-6} \text{ m}^3$
 - $1.70 \times 10^{-7} \text{ m}^3$
 - $1.78 \times 10^{-6} \text{ m}^3$
- In a vernier calipers, n divisions of its main scale match with $(n + 1)$ divisions on its vernier scale. Each division of the main scale is a units. Using the vernier principle, its least count will be
 - $an/(n + 1)$
 - $a/(n + 1)$
 - $n/(n + 1)$
 - $an/(n + 2)$
- A wire has a mass 0.3 ± 0.003 g, radius 0.5 ± 0.005 mm and length 6 ± 0.06 cm. The maximum percentage error in the measurement of its density is
 - 1
 - 2
 - 3
 - 4
- A vernier calipers having 1 main scale division = 0.1 cm is designed to have a least count of 0.02 cm. If n be the number of divisions on vernier scale and m be the length of vernier scale, then
 - $n = 10, m = 0.5$ cm
 - $n = 9, m = 0.4$ cm
 - $n = 10, m = 0.8$ cm
 - $n = 10, m = 0.2$ cm
- In a Vernier Calipers (VC), N divisions of the main scale coincide with $N + m$ divisions of the vernier scale. What is the value of m for which the instrument has minimum least count?
 - 1
 - N
 - Infinity
 - $N/2$
- Which of the followings is true
 - Most precise measurement is the most accurate
 - Most accurate measurement is the most precise
 - Least precise measurement may be the most accurate
 - Measurement using a meter scale is more precise than a vernier caliper
- Which of the followings has highest number of significant digits
 - 2.8312
 - 2.0012
 - 2.01010
 - 0.000002
- A measurement of 834.551 rounded off to 4 digits is
 - 834.5
 - 835.0
 - 834.6
 - 834.56
- A measurement of 834.501 rounded off to 4 digits is
 - 834.5
 - 834.5
 - 834.6
 - 834.56
- A measurement of 834.50 rounded off to 3 digits is
 - 834
 - 834.5
 - 833
 - 835
- The time period of oscillation of simple pendulum is given by $T = 2\pi\sqrt{l/g}$. What is the accuracy in the determination of 'g' if 10 cm length is known to 1 mm accuracy and 0.6 s time period is measured from time of 100 oscillations with a watch of 1 sec. resolution:
 - $\pm 5.3\%$
 - $\pm 6.3\%$
 - $\pm 4.3\%$
 - $\pm 2.9\%$
- The measure of radius of a sphere is $(4.22 \pm 2\%)$ cm. The volume of the sphere is:
 - $(315 \pm 6\%)$
 - $(315 \pm 2\%)$
 - $(315 \pm 4\%)$
 - $(315 \pm 5\%)$
- The coefficient of viscosity (η) of a liquid by the method of flow through a capillary tube is given by the formula $\eta = \frac{\pi R^4 P}{8 l Q}$; where R = radius of the capillary tube, l = Length of the tube, P = Pressure difference between its ends, and Q = volume of liquid flowing per second. Which measurement needs to be made most accurate for least error in value of (η)?
 - P
 - Q
 - l
 - R
- The vernier scale shown has its 10 division equal to 9 divisions on the main scale. When the jaws are closed the scale appears as shown. The zero error is

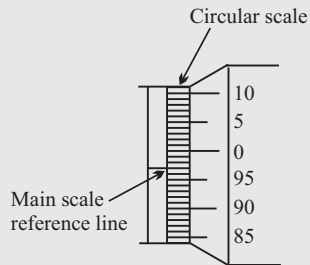


 - +0.4mm
 - 0.4mm
 - 0.6mm
 - +4.0mm
- The vernier scale shown has its 10 division equal to 9 divisions on the main scale. When the jaws are closed the scale appears as shown. The zero error is



- (a) +0.4mm (b) -0.4mm
(c) -0.6mm (d) +4.0mm

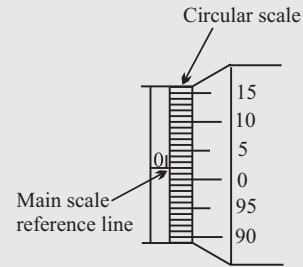
16. The circular scale of the screw gauge shown has 100 divisions and the pitch of the screw is 1 mm. When the jaws are closed the scale appears as shown. The zero error is



- (a) +0.3mm (b) -0.03mm
(c) -0.097mm (d) +0.03mm

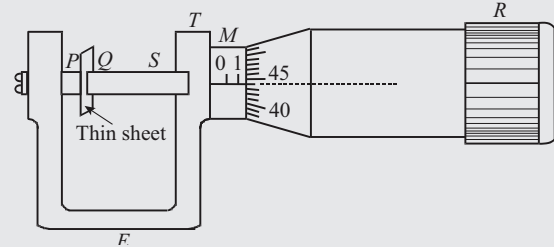
17. The circular scale of the screw gauge shown has 100 divisions and the pitch of the screw is 1 mm. When

the jaws are closed the scale appears as shown. The zero error is



- (a) -0.002 cm (b) +0.002 cm
(c) -0.98 mm (d) +0.02 mm

18. The screw gauge mentioned in Q.17 is used to measure the thickness of a thin sheet and the scale appears as shown. The thickness reported will be



- (a) +2.402 cm (b) 2.42 cm
(c) 2.42 mm (d) 2.402 mm

Worksheet 2

1. The main scale of a Vernier calipers is calibrated in mm and 19 divisions of main scale are equal in length to 20 divisions of Vernier scale. In measuring the diameter of a cylinder by this instrument, the main scale reads 35 divisions and 4th division of Vernier scale coincides with a main scale division.
 - (a) The least count of the Vernier caliper is 0.005 cm
 - (b) Radius of the cylinder is 35.20 cm
 - (a) The least count of the Vernier caliper is 0.05 cm
 - (b) Radius of the cylinder is 35.2 cm
2. Which of the followings are correct?
 - (a) Measurement using a P.O. Box is more precise than the measurement using a meter bridge.
 - (b) Measurement of the weight of a straw using a weighing machine having a range of 0 to 50 g with 100 divisions on its scale will be more accurate than measurement of its weight using a machine having a range of 0 to 500 g having 100 divisions on its scale.
 - (c) Measurement of the weight of a straw using a weighing machine having a range of 0 to 50g with 100 divisions on its scale *can be* accurate than measurement of its weight using a machine having a range of 0 to 500 g having 100 divisions on its scale.
 - (d) None of the above.
3. A galvanometer has 50 divisions on its scale. Resistance of the galvanometer is R .
 - (a) Its sensitivity towards measurement of a current decreases if a resistance smaller than R is connected in parallel to it.
 - (b) Its sensitivity towards measurement of a current increases if a resistance greater than R is connected in parallel to it.
 - (c) Its sensitivity towards measurement of a current increases if a resistance smaller than R is connected in series to it.
 - (d) Its sensitivity towards measurement of a current increases if a resistance greater than R is connected in series to it.
4. The uncertainty in measured potential difference using a potentiometer
 - (a) increases with increase in applied potential difference across the potentiometer wire.
 - (b) increases with decrease in applied potential difference across the potentiometer wire.
 - (c) increases with increase in number of divisions on the galvanometer scale.
 - (d) decreases with increase in number of divisions on the galvanometer scale

Worksheet 3

1. The length of a rod measured in an experiment are recorded as 2.50 m, 2.54 m, 2.49 m, 2.58 m, 2.57 m and 2.49 m.

- (a) Find the mean length.
- (b) Find the mean deviation.
- (c) Consider the mean deviation to be the maximum uncertainty and find the percentage uncertainty.

2. When a mass M is suspended from a wire of length L and increase in its length is l , then the Young's modulus of the material of the wire is given by

$$Y = \frac{MgL}{\pi r^2 l}$$

Here, r is radius of the wire.

In one experiment, the recorded values were:

$L = 2.890$ m, $M = 3.00$ kg, $l = 0.87$ cm and the diameter of the wire $d = 0.041$ cm

Find the maximum percentage error in the calculated value of Y .

3. Evaluate the following with due regards to the significant digits

(a) $\sqrt{6.5 - 6.32}$ (b) $\frac{2.91 \times 0.3842}{0.08}$

4. The main scale of a vernier caliper reads in millimeter and its vernier scale is divided into 10 divisions which coincide with 9 divisions of the main scale. When the two jaws touch each other, the zero of the vernier lies to the right of the zero of the main scale and the seventh division of the vernier scale coincides with a main scale division. When the outer jaws are held tightly inside a hollow cylinder, the zero of the vernier lies slightly to the right of 3.2 cm mark on the

main scale and the fourth vernier division coincides with a main scale division. What is the measured internal diameter of the cylinder.

5. A vernier caliper has a negative zero error. With the jaws closed, the 6th division of vernier coincides with a main scale division. A bar of known length 4.32 cm is placed between the jaws. In this position, which division of the vernier will coincide with a main scale division?
6. The pitch of a screw gauge is 1 mm and there are 50 divisions on its circular scale. When nothing is kept between the jaws, the 44th circular scale division coincides with the reference line and zero of the main scale is not visible. When a plate is placed between the jaws, the main scale reads 3 divisions and the circular scale reads 26 divisions. Find the thickness of the plate.
7. A quantity A is given by $A = x \cos \theta$. The quantities x and θ are measured as $x = 2.0 \pm 0.1$ and $\theta = (37 \pm 1)^\circ$. Find A . Take $\cos 37^\circ = 0.80$.
8. A screw gauge has 100 divisions on its circular scale and the pitch of the screw is 1 mm. While measuring the diameter of a wire, the linear scale reads 1 mm and 47th division on the circular scale coincides with the reference line. The length of the wire is 5.6 cm. Find the curved surface area of the wire in appropriate number of significant digits.
9. You measured the distance between your house and your school using the odometer in your motor bike. You repeated the measurement 5 times. The values reported by you are 3.4 km, 3.1 km, 3.6 km, 3.7 km and 3.1 km. State the distance with 95% confidence.

Answer Sheet

Your Turn

- 1 s, 1 minute
- No. We should conclude that today's methods/instruments are more precise.
- (a) 1 (b) 6 (c) 4 (d) 1 (e) 4 (f) 2 (g) 4
- 5.00×10^2 mm
- Less than 70.0 kg
- 12.33A
- (2.2 ± 0.1) A
- Measured value is likely to be between 3.8A and 4.2A
- (a) 12400 (b) 19.8 (c) 23.8 (d) 45.6
- 3.3 kg
- 30000 cm^3
- $(2.00 \pm 0.5\%)$ s
- $(9.86 \pm 6\%) \text{ ms}^{-2}$
- 0.4%
- z
- +0.6 mm
- 42.6 mm
- 5.0 mm
- 0.04 mm
- 0.40 mm
- 12.68 mm

Worksheet 1

- | | | | | | | | | |
|---------|---------|---------|---------|---------|---------|---------|---------|---------|
| 1. (a) | 2. (b) | 3. (d) | 4. (c) | 5. (a) | 6. (c) | 7. (c) | 8. (c) | 9. (b) |
| 10. (a) | 11. (c) | 12. (a) | 13. (d) | 14. (a) | 15. (b) | 16. (b) | 17. (b) | 18. (c) |

Worksheet 2

- | | | | |
|-----------|-----------|--------|--------|
| 1. (a, b) | 2. (a, c) | 3. (a) | 4. (a) |
|-----------|-----------|--------|--------|

Worksheet 3

- | | |
|-----------------------------------|-----------------------|
| 1. (a) 2.53 m (b) 0.035 (c) 1.38% | 2. 6.40% |
| 3. (a) 0.4 (b) 14 | 4. 3.17 cm |
| 5. 8th | 6. 3.64 mm |
| 7. 1.6 ± 0.1 | 8. 2.6 cm^2 |
| 9. (3.4 ± 0.56) km | |

Miscellaneous Problems on Chapter 10

MATCH THE COLUMN

- 1 Two quantities x and y are measured to be $x \pm \Delta x$ and $y \pm \Delta y$. Three quantities A , B and C are calculated as $A = x + y$, $B = x - y$ and $C = xy$.

Column-I		Column-II	
(A)	Uncertainty in A is	(P)	Higher than that in B
(B)	Uncertainty in B is	(Q)	Higher than that in C
(C)	Uncertainty in $\frac{A}{B}$ is	(R)	Higher than that in A
(D)	Uncertainty in AB is	(S)	Higher than that in C

Similarly, if $A = \frac{ab}{xy}$, then uncertainty in A expressed in terms of uncertainties in a , b , x and y is better expressed as

$$\Delta A = \sqrt{\left(\frac{\Delta a}{a}\right)^2 + \left(\frac{\Delta b}{b}\right)^2 + \left(\frac{\Delta x}{x}\right)^2 + \left(\frac{\Delta y}{y}\right)^2}$$

- Two distances x and y were measured to be $x = (400 \pm 5)$ cm and $y = (300 \pm 1)$ cm. A good estimate of uncertainty in $L = x + y$ is
 - 6 cm
 - 5.1 cm
 - 25 cm
 - 4 cm
- A bird flies a distance $x = (120 \pm 3)$ m in a time interval $t = (20.0 \pm 1.2)$ s. The average speed of the bird can be expressed as
 - $(6.0 \pm 6.5\%)$ m/s
 - $(6.0 \pm 3.6\%)$ m/s
 - $(6.0 \pm 8.5\%)$ m/s
 - $(6.0 \pm 7.2\%)$ m/s

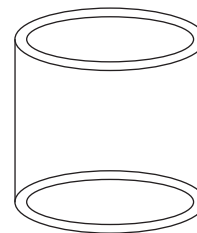
PASSAGE BASED PROBLEMS

Passage 1

When experimental results are reported, they are accompanied by an estimate of the experimental error, called the uncertainty. This uncertainty indicates how reliable the experimenter believes the results to be. Suppose the quantities x and y were measured to be $x_0 \pm \Delta x$ and $y_0 \pm \Delta y$. The maximum uncertainty in a quantity $A = x + y$ is $(\Delta x + \Delta y)$. But the uncertainty obtained this way in A is a slight overestimate. If the errors in x and y are uncorrelated, there is very small chance that errors in both x and y will be positive and maximum simultaneously. It will be more realistic to estimate the error in A as $\Delta A = \sqrt{(\Delta x)^2 + (\Delta y)^2}$. This method of calculating uncertainty in A puts more weight on the larger uncertainty.

Passage 2

A group of students were given a metal cylinder and asked to measure the volume of metal in the cylinder. The cylinder is open at both ends. The students decided to measure the inner diameter d , outer diameter D and the length L of the cylinder. For all the three measurements, they used a vernier calliper. With the jaws closed, the zero of the vernier was found to be to the right of the zero of the main scale. The 4th vernier division coincided with a main scale mark. The cylinder was placed between the jaws to measure its outer diameter. Three readings (at different locations) were taken. The readings recorded were



- (i) Main scale reading (MSR) = 50 mm
Vernier scale reading (VSR) = 5

- (ii) MSR = 51 mm
VSR = 0

- (iii) MSR = 50 mm
VSR = 8

Here, VSR refers to the division of vernier that coincides with a main scale division.

Now, the jaws used to measure the inner diameter were inserted inside the cylinder and made to touch the cylinder walls at diametrically opposite ends. The readings recorded were

- (i) MSR = 49 mm
VSR = 4

- (ii) MSR = 48 mm
VSR = 8

- (iii) MSR = 4 mm
VSR = 6

The length was measured only once and the reading was MSR = 60 cm and VSR = 2.

3. The uncertainty (mean deviation) in the measurement of outer diameter is nearly

- (a) 0.2 mm (b) 0.4 mm
(c) 0.3 mm (d) 0.02 mm

4. The thickness of the metal in the cylinder is

- (a) (2.5 ± 0.05) mm
(b) (2.5 ± 0.5) mm
(c) (1.5 ± 0.5) mm
(d) (1.0 ± 0.02) mm

5. The length of the cylinder is

- (a) (59.8 ± 0.1) mm
(b) (59.2 ± 0.2) mm
(c) (60.2 ± 0.1) mm
(d) (59.8 ± 0.1) mm

Answer Sheet

Match the Columns

1. (A) S (B) S (C) P, Q, R, S (D) P, Q, R, S

Passage Based Problems

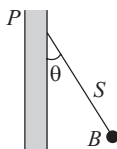
1. (b) 2. (a) 3. (a) 4. (c) 5. (a)

Previous Years' JEE Questions

ELECTROSTATICS

AIEEE/JEE Main Questions

1. A charged ball B hangs from a silk thread S , which makes an angle θ with a large charged conducting sheet P , as shown in the figure. The surface charge density σ of the sheet is proportional to:



[AIEEE 2005]

- (a) $\sin\theta$ (b) $\tan\theta$
(c) $\cos\theta$ (d) $\cot\theta$

2. Two thin wire rings each having a radius R are placed at a distance d apart with their axes coinciding. The charges on the two rings are $+q$ and $-q$. The potential difference between the centres of the two rings is:

[AIEEE 2005]

- (a) $\frac{Q}{2\pi\epsilon_0} \left[\frac{1}{R} - \frac{1}{\sqrt{R^2 + d^2}} \right]$
(b) $\frac{Q}{4\pi\epsilon_0} \left[\frac{1}{R} - \frac{1}{\sqrt{R^2 + d^2}} \right]$
(c) $\frac{QR}{4\pi\epsilon_0 d^2}$

(d) Zero

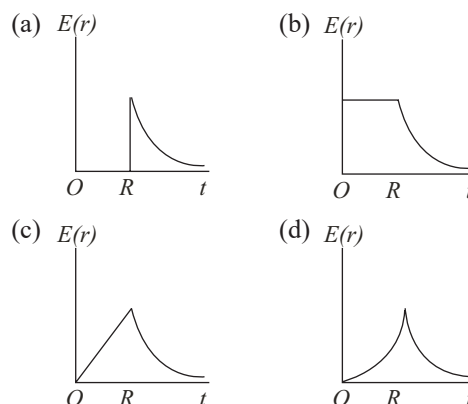
3. Two point charges $+8q$ and $-2q$ are located at $x = 0$ and $x = L$ respectively. The location of a point on the x -axis at which the net electric field due to these two point charges is zero is: [AIEEE 2005]

- (a) $8L$ (b) $4L$
(c) $2L$ (d) $L/4$

4. A thin spherical shell of radius R has charge Q spread uniformly over its surface. Which of the following graphs most closely represents the electric field $E(r)$

produced by the shell in the range $0 \leq r < \infty$, where r is the distance from the centre of the shell?

[AIEEE 2008]



5. Let $\rho(r) = (Qr/\pi R^4)$ be the charge density distribution for a solid sphere of radius R and total charge Q . For a point 'P' inside the sphere at distance r_1 from the centre of the sphere, the magnitude of electric field is [AIEEE 2009]

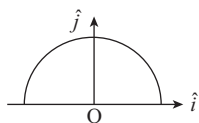
- (a) 0 (b) $\frac{Q}{4\pi\epsilon_0 r_1^2}$
(c) $\frac{Qr_1^2}{4\pi\epsilon_0 R^4}$ (d) $\frac{Qr_1^2}{3\pi\epsilon_0 R^4}$

6. Two points P and Q are maintained at the potentials of 10V and $-4V$, respectively. The work done in moving 100 electrons from P to Q is:

[AIEEE 2009]

- (a) $-9.60 \times 10^{-17} \text{ J}$ (b) $9.60 \times 10^{-17} \text{ J}$
(c) $-2.24 \times 10^{-16} \text{ J}$ (d) $2.24 \times 10^{-16} \text{ J}$

7. A thin semi-circular ring of radius r has a positive charge q distributed uniformly over it. The net field $\vec{E}$ at the centre O is: [AIEEE 2010]



- (a) $\frac{q}{2\pi^2\epsilon_0 r^2} \hat{j}$ (b) $\frac{q}{4\pi^2\epsilon_0 r^2} \hat{j}$
 (c) $-\frac{q}{4\pi^2\epsilon_0 r^2} \hat{j}$ (d) $-\frac{q}{2\pi^2\epsilon_0 r^2} \hat{j}$

8. Let there be a spherically symmetric charge distribution with charge density varying as $\rho(r) = \rho_0 \left(\frac{5}{4} - \frac{r}{R} \right)$ upto $r = R$, and $\rho(r) = 0$ for $r > R$, where r is the distance from the origin. The electric field at a distance r ($r < R$) from the origin is given by:

[AIEEE 2010]

- (a) $\frac{\rho_0 r}{3\epsilon_0} \left(\frac{5}{4} - \frac{r}{R} \right)$ (b) $\frac{4\pi\rho_0 r}{3\epsilon_0} \left(\frac{5}{4} - \frac{r}{R} \right)$
 (c) $\frac{\rho_0 r}{4\epsilon_0} \left(\frac{5}{4} - \frac{r}{R} \right)$ (d) $\frac{4\rho_0 r}{3\epsilon_0} \left(\frac{5}{4} - \frac{r}{R} \right)$

9. Two identical charged spheres are suspended by strings of equal lengths. The strings make an angle of 30° with each other. When suspended in a liquid of density 0.8 g cm^{-3} , the angle remains the same. If density of the material of the sphere is 1.6 g cm^{-3} , the dielectric constant of the liquid is:

[AIEEE 2010]

- (a) 1 (b) 4
 (c) 3 (d) 2

10. The electrostatics potential inside a charged spherical ball is given by $\phi = ar^2 + b$ where r is the distance from the centre; a, b are constants. Then the charge density inside the ball is:

[AIEEE 2011]

- (a) $-24\pi\epsilon_0$ (b) $-6a\epsilon_0$
 (c) $-24\pi a\epsilon_0 r$ (d) $-6a\epsilon_0 r$

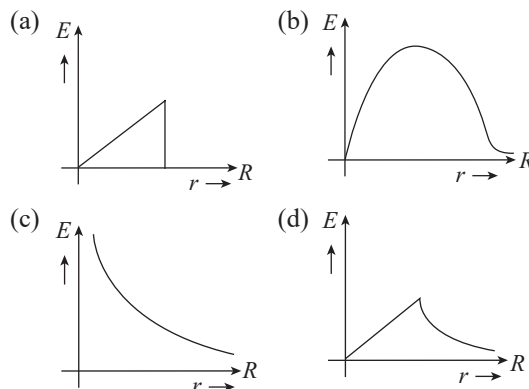
11. Two identical charged spheres suspended from a common point by two massless strings of length l are initially a distance d ($d \ll l$) apart because of their mutual repulsion. The charge begins to leak slowly from both the spheres at a constant rate. As a result the charges approach each other with a velocity v . The velocity v expressed as a function of distance x between them is:

[AIEEE 2011]

- (a) $v \propto x^{1/2}$ (b) $v \propto x$
 (c) $v \propto x^{-1/2}$ (d) $v \propto x^{-1}$

12. In a uniformly charged sphere of total charge Q and radius R , the electric field E is plotted as a function of distance from the centre. The graph which would correspond to the above will be:

[AIEEE 2012]



13. Assume that an electric field $\vec{E} = 30x^2 \hat{i}$ exists in space. Then the potential difference $V_A - V_0$, where V_0 is the potential at the origin and V_A the potential at $x = 2\text{m}$ is:

[JEE Main 2014]

- (a) 120V (b) -120V
 (c) -80V (d) 80V

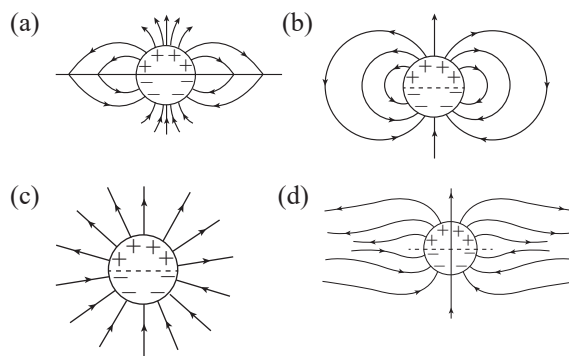
14. A uniformly charged solid sphere of radius R has potential V_0 (measured with respect to ∞) on its surface. For this sphere the equipotential surfaces with potentials $\frac{3V_0}{2}, \frac{5V_0}{4}, \frac{3V_0}{4}$ and $\frac{V_0}{4}$ have radius R_1, R_2, R_3 and R_4 respectively. Then

[JEE Main 2015]

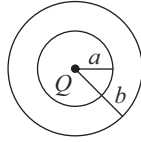
- (a) $2R < R_4$
 (b) $R_1 = 0$ and $R_2 > (R_4 - R_3)$
 (c) $R_1 \neq 0$ and $(R_2 - R_1) > (R_4 - R_3)$
 (d) $R_1 = 0$ and $R_2 < (R_4 - R_3)$

15. A long cylindrical shell carries positive surface charge σ in the upper half and negative surface charge $-\sigma$ in the lower half. The electric field lines around the cylinder will look like figure given in: (figures are schematic and not drawn to scale)

[JEE Main 2015]



16. The region between two concentric spheres of radii 'a' and 'b', respectively (see figure), has volume charge density $\rho = \frac{A}{r}$, where A is a constant and r is the distance from the centre. At the centre of the spheres is a point charge Q . The value of A such that the electric field in the region between the spheres will be constant, is: [JEE Main 2016]

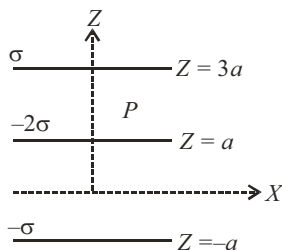


- (a) $\frac{Q}{2\pi a^2}$ (b) $\frac{Q}{2\pi(b^2 - a^2)}$
 (c) $\frac{2Q}{\pi(a^2 - b^2)}$ (d) $\frac{2Q}{\pi a^2}$
17. An electric dipole has a fixed dipole moment $\vec{p}$, which makes angle θ with respect to x-axis. When subjected to an electric field $\vec{E}_1 = E\hat{i}$, it experiences a torque $\vec{T}_1 = \tau\hat{k}$. When subjected to another electric field $\vec{E}_2 = \sqrt{3}E_1\hat{j}$ it experiences a torque $\vec{T}_2 = -\vec{T}_1$. The angle θ is: [JEE Main 2017]
- (a) 30° (b) 45°
 (c) 60° (d) 90°
18. Three concentric metal shells A, B and C of respective radii a , b and c ($a < b < c$) have surface charge densities $+\sigma$, $-\sigma$ and $+\sigma$ respectively. The potential of shell B is: [JEE Main 2018]

- (a) $\frac{\sigma}{\epsilon_0} \left[\frac{a^2 - b^2}{a} + c \right]$ (b) $\frac{\sigma}{\epsilon_0} \left[\frac{a^2 - b^2}{b} + c \right]$
 (c) $\frac{\sigma}{\epsilon_0} \left[\frac{b^2 - c^2}{b} + a \right]$ (d) $\frac{\sigma}{\epsilon_0} \left[\frac{b^2 - c^2}{c} + a \right]$

IIT-JEE / JEE Advanced Questions

19. Three infinitely long charge sheets are placed as shown in figure. The electric field at point P is [IIT-JEE 2005]

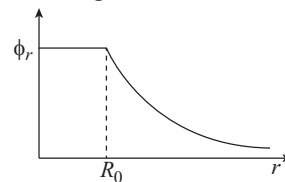


- (a) $\frac{2\sigma}{\epsilon_0} \hat{k}$ (b) $\frac{4\sigma}{\epsilon_0} \hat{k}$
 (c) $-\frac{2\sigma}{\epsilon_0} \hat{k}$ (d) $-\frac{4\sigma}{\epsilon_0} \hat{k}$

20. A conducting liquid bubble of radius a and thickness t ($t \ll a$) is charged to potential V . If the bubble collapses to a droplet, find the potential on the droplet. [IIT-JEE 2005]

- (a) $V \left[\frac{a}{3t} \right]^{1/3}$ (b) $V \left[\frac{a}{3t} \right]^{2/3}$
 (c) $V \left[\frac{2a}{3t} \right]^{2/3}$ (d) $V \left[\frac{2a}{3t} \right]^{1/3}$

21. The electrostatic potential (ϕ_r) of a spherical symmetric system, kept at origin, is shown in the adjacent figure, and given as



$$\phi_r = \frac{q}{4\pi\epsilon_0 r} \quad (r \geq R_0)$$

$$\phi_r = \frac{q}{4\pi\epsilon_0 R_0} \quad (r \leq R_0)$$

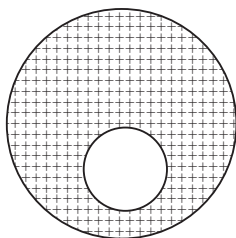
Which of the following option(s) is/are correct?

- (a) For spherical region $r \leq R_0$, total electrostatic energy stored is zero.
 (b) Within $r = 2R_0$, total charge is q .
 (c) There will be no charge anywhere except at $r = R_0$.
 (d) Electric field is discontinuous at $r = R_0$. [IIT-JEE 2006]
22. Consider a neutral conducting sphere. A positive point charge is placed outside the sphere. The net charge on the sphere is then, [IIT-JEE 2007]
- (a) negative and distributed uniformly over the surface of the sphere
 (b) negative and appears only at the point on the sphere closest to the point charge
 (c) negative and distributed non-uniformly over the entire surface of the sphere
 (d) zero

23. A long, hollow conducting cylinder is kept coaxially inside another long, hollow conducting cylinder of larger radius. Both the cylinders are initially electrically neutral. [IIT-JEE 2007]

- A potential difference appears between the two cylinders when a charge density is given to the inner cylinder.
- A potential difference appears between the two cylinders when a charge density is given to the outer cylinder.
- No potential difference appears between the two cylinders when a uniform line charge is kept along the axis of the cylinders.
- No potential difference appears between the two cylinders when same charge density is given to both the cylinders.

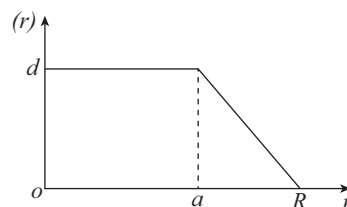
24. A spherical portion has been removed from a solid sphere having a charge distributed uniformly in its volume as shown in the figure. The electric field inside the emptied space is [IIT-JEE 2007]



- zero everywhere
 - non-zero and uniform
 - non-uniform
 - zero only at its center
25. Positive and negative point charges of equal magnitude are kept at $\left(0, 0, \frac{a}{2}\right)$ and $\left(0, 0, \frac{-a}{2}\right)$, respectively. The work done by the electric field when another positive point charge is moved from $(-a, 0, 0)$ to $(0, a, 0)$ is [IIT-JEE 2007]
- positive
 - negative
 - zero
 - depends on the path connecting the initial and final positions

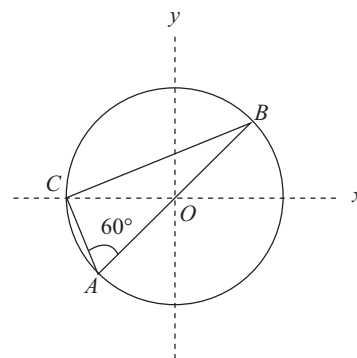
Q. 26 to 28 are based on following passage:

The nuclear charge (Ze) is non-uniformly distributed within a nucleus of radius R . The charge density $\rho(r)$ [charge per unit volume] is dependent only on the radial distance r from the centre of the nucleus as shown in figure. The electric field is only along the radial direction [IIT-JEE 2008]



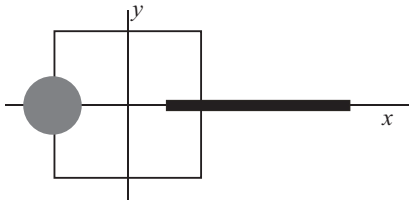
26. The electric field at $r = R$ is
- independent of a
 - directly proportional to a
 - directly proportional to a^2
 - inversely proportional to a
27. For $a = 0$, the value of d (maximum value of ρ as shown in the figure) is
- $\frac{3Ze}{4\pi R^3}$
 - $\frac{3Ze}{\pi R^3}$
 - $\frac{4Ze}{3\pi R^3}$
 - $\frac{Ze}{3\pi R^3}$
28. The electric field within the nucleus is generally observed to be linearly dependent on r . This implies
- $a = 0$
 - $a = \frac{R}{2}$
 - $a = R$
 - $a = \frac{2R}{3}$
29. Consider a system of three charges $\frac{q}{3}$, $\frac{q}{3}$ and $-\frac{2q}{3}$

placed at points A , B and C , respectively, as shown in the figure. Take O to be the centre of the circle of radius R and angle $CAB = 60^\circ$. [IIT-JEE 2008]



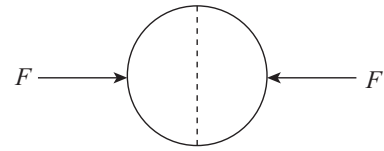
- The electric field at point O is $\frac{q}{8\pi\epsilon_0 R^2}$ directed along the negative x -axis
- The potential energy of the system is zero
- The magnitude of the force between the charges at C and B is $\frac{q^2}{54\pi\epsilon_0 R^2}$
- The potential at point O is $\frac{q}{12\pi\epsilon_0 R}$

30. A solid sphere of radius R has a charge Q distributed in its volume with a charge density $\rho = kr^a$, where k and a are constants and r is the distance from its centre. If the electric field at $r = \frac{R}{2}$ is $\frac{1}{8}$ times that at $r = R$, find the value of a . [IIT-JEE 2009]
31. A disk of radius $a/4$ having a uniformly distributed charge $6C$ is placed in the x - y plane with centre at $(-a/2, 0, 0)$. A rod of length a carrying a uniformly distributed charge $8C$ is placed on the x -axis from $x = a/4$ to $x = 5a/4$. Two point charges $-7C$ and $3C$ are placed at $(a/4, -a/4, 0)$ and $(-3a/4, 3a/4, 0)$, respectively. Consider a cubical surface formed by six surfaces $x = \pm a/2$, $y = \pm a/2$, $z = \pm a/2$. The electric flux through this cubical surface is [IIT-JEE 2009]

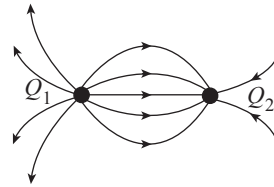


- (a) $\frac{-2C}{\epsilon_0}$ (b) $\frac{2C}{\epsilon_0}$
 (c) $\frac{10C}{\epsilon_0}$ (d) $\frac{12C}{\epsilon_0}$
32. Three concentric metallic spherical shells of radii R , $2R$, $3R$ are given charges Q_1 , Q_2 , Q_3 respectively. It is found that the surface charge densities on the outer surface of the shells are equal. Then, the ratio of the charges given to the shells, $Q_1 : Q_2 : Q_3$ is [IIT-JEE 2009]
- (a) $1 : 2 : 3$ (b) $1 : 3 : 5$
 (c) $1 : 4 : 9$ (d) $1 : 8 : 18$
33. A tiny spherical oil drop carrying a net charge q is balanced in still air with a vertical uniform electric field of strength $\frac{81\pi}{7} \times 10^5 \text{ Vm}^{-1}$. When the field is switched off, the drop is observed to fall with terminal velocity $2 \times 10^{-3} \text{ ms}^{-1}$. Given $g = 9.8 \text{ ms}^{-2}$, viscosity of the air $= 1.8 \times 10^{-5} \text{ Nsm}^{-2}$ and the density of oil $= 900 \text{ kgm}^{-3}$, the magnitude of q is: [IIT-JEE 2010]
- (a) $1.6 \times 10^{-19} \text{ C}$ (b) $3.2 \times 10^{-19} \text{ C}$
 (c) $4.8 \times 10^{-19} \text{ C}$ (d) $8.0 \times 10^{-19} \text{ C}$
34. A uniformly charged thin spherical shell of radius R carries uniform surface charge density of σ per unit area. It is made of two hemispherical shells, held

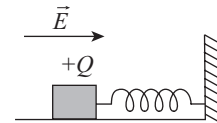
together by pressing them with force F (see figure). F is proportional to: [IIT-JEE 2010]



- (a) $\frac{1}{\epsilon_0} \sigma^2 R^2$ (b) $\frac{1}{\epsilon_0} \sigma^2 R$
 (c) $\frac{1}{\epsilon_0} \frac{\sigma^2}{R}$ (d) $\frac{1}{\epsilon_0} \frac{\sigma^2}{R^2}$
35. A few electric field lines for a system of two charges Q_1 and Q_2 fixed at two different points on the x -axis are shown in the figure. These lines suggest that: [IIT-JEE 2010]



- (a) $|Q_1| > |Q_2|$
 (b) $|Q_1| < |Q_2|$
 (c) at a finite distance to the left of Q_1 the electric field is zero
 (d) at a finite distance to the right of Q_2 the electric field is zero
36. A wooden block performs SHM on a frictionless surface with frequency, ν_0 . The block carries a charge $+Q$ on its surface. If now a uniform electric field $\vec{E}$ is switched-on as shown, then the SHM of the block will be: [IIT-JEE 2011]



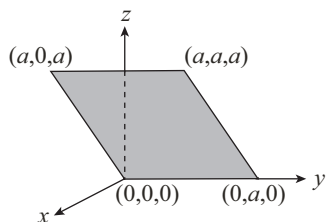
- (a) of the same frequency and with shifted mean position.
 (b) of the same frequency and with the same mean position.
 (c) of changed frequency and with shifted mean position.
 (d) of changed frequency and with the same mean position.
37. Four point charges, each of $+q$, are rigidly fixed at the four corners of a square planar soap film of side ' a '. The surface tension of the soap film is γ . The

system of charges and planar film are in equilibrium,

and $a = k \left[\frac{q^2}{\gamma} \right]^{1/N}$, where 'k' is a constant. Then N is

[IIT-JEE 2011]

38. Consider an electric field $\vec{E} = E_0 \hat{x}$, where E_0 is a constant. The flux through the shaded area (as shown in the figure) due to this field is: [IIT-JEE 2011]



- (a) $2E_0a^2$ (b) $\sqrt{2}E_0a^2$
(c) E_0a^2 (d) $\frac{E_0a^2}{\sqrt{2}}$

39. A spherical metal shell A of radius R_A and a solid metal sphere B of radius $R_B (< R_A)$ are kept far apart and each is given charge '+Q'. Now they are connected by a thin metal wire. Then:

[IIT-JEE 2011]

- (a) $E_A^{\text{inside}} = 0$ (b) $Q_A > Q_B$
(c) $\frac{\sigma_A}{\sigma_B} = \frac{R_B}{R_A}$ (d) $E_A^{\text{on surface}} < E_B^{\text{on surface}}$

40. Which of the following statement(s) is/are correct?

[IIT-JEE 2011]

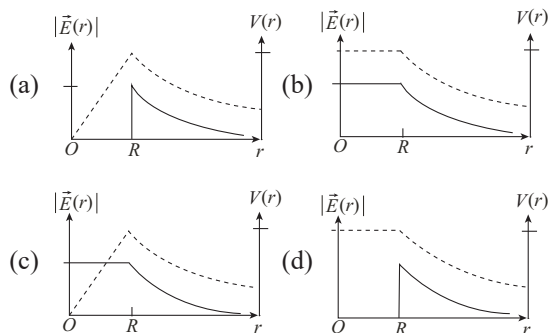
- (a) If the electric field due to a point charge varies as $r^{-2.5}$ instead of r^{-2} , then the Gauss's law will still be valid.
(b) The Gauss's law can be used to calculate the field distribution around an electric dipole.
(c) If the electric field between two point charges is zero somewhere, then the sign of the two charges is the same.
(d) The work done by the external force in moving a unit positive charge from point A at potential V_A to point B at potential V_B is $(V_B - V_A)$.

41. Two large vertical and parallel metal plates having a separation of 1 cm are connected to a DC voltage source of potential difference X. A proton is released at rest midway between the two plates. It is found to move at 45° to the vertical just after release. Then X is nearly

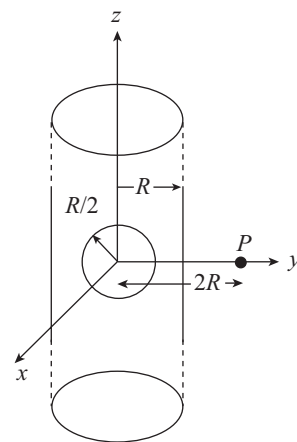
[IIT-JEE 2012]

- (a) 1×10^{-5} V (b) 1×10^{-7} V
(c) 1×10^{-9} V (d) 1×10^{-10} V

42. Consider a thin spherical shell of radius R with its centre at the origin, carrying uniform positive surface charge density. The variation of the magnitude of the electric field $|\vec{E}(r)|$ and the electric potential V(r) with the distance r from the centre, is best represented by which graph? [IIT-JEE 2012]



43. An infinitely long solid cylinder of radius R has a uniform volume charge density ρ . It has a spherical cavity of radius $R/2$ with its centre on the axis of the cylinder, as shown in the figure. The magnitude of the electric field at the point P, which is at a distance $2R$ from the axis of the cylinder, is given by the expression $\frac{23\rho R}{16k\epsilon_0}$. The

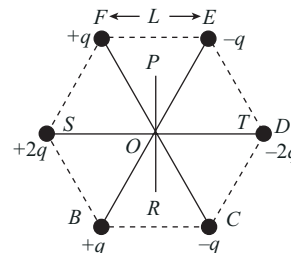


value of k is:

[IIT-JEE 2012]

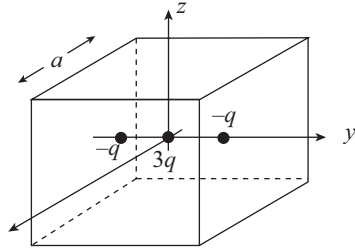
44. Six point charges are kept at the vertices of a regular hexagon of side L and centre O, as shown in the figure. Given that $K = \frac{1}{4\pi\epsilon_0} \frac{q}{L^2}$, which of the following statement (s) is (are) correct?

[IIT-JEE 2012]

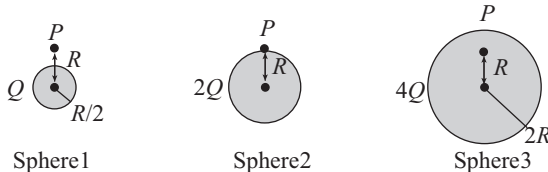


- (a) The electric field at O is $6K$ along OD.
(b) The potential at all points on the line PR is same.
(c) The potential at O is zero.
(d) The potential at all points on the line ST is same.

45. A cubical region of side a has its centre at the origin. It encloses three fixed point charges, $-q$ at $(0, -a/4, 0)$, $+3q$ at $(0, 0, 0)$ and $-q$ at $(0, +a/4, 0)$. Choose the correct option(s) [IIT-JEE 2012]



- (a) The net electric flux crossing the plane $x = +a/2$ is equal to the net electric flux crossing the plane $x = -a/2$
- (b) The net electric flux crossing the plane $y = +a/2$ is more than the net electric flux crossing the plane $y = -a/2$
- (c) The net electric flux crossing the entire region is $\frac{q}{\epsilon_0}$.
- (d) The net electric flux crossing the plane $z = +a/2$ is equal to the net electric flux crossing the plane $z = -a/2$
46. Two non-conducting solid spheres of radii R and $2R$, having uniform volume charge densities σ_1 and σ_2 respectively, touch each other. The net electric field at a distance $2R$ from the centre of the smaller sphere, along the line joining the centres of the spheres, is zero. The ratio $\frac{\rho_1}{\rho_2}$ can be: [JEE Advance 2013]
- (a) -4 (b) $-\frac{32}{25}$
- (c) $\frac{32}{25}$ (d) 4
47. Charges Q , $2Q$ and $4Q$ are uniformly distributed in three dielectric solid spheres 1, 2 and 3 of radii $R/2$, R and $2R$ respectively, as shown in figure. If magnitudes of the electric fields at point P at a distance R from the centre of spheres 1, 2 and 3 are E_1 , E_2 and E_3 respectively, then

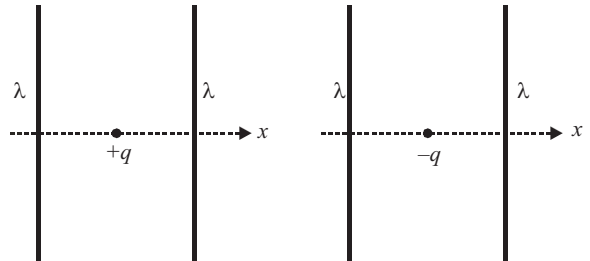


- (a) $E_1 > E_2 > E_3$ (b) $E_3 > E_1 > E_2$
- (c) $E_2 > E_1 > E_3$ (d) $E_3 > E_2 > E_1$

48. Let $E_1(r)$, $E_2(r)$ and $E_3(r)$ be the respective electric fields at a distance r from a point charge Q , an infinitely long wire with constant linear charge density λ and an infinite plane with uniform surface charge density σ . If $E_1(r_0) = E_2(r_0) = E_3(r_0)$ at a given distance r_0 then [JEE Advanced 2014]

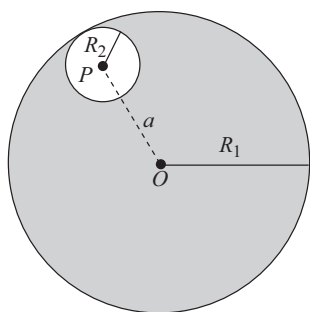
- (a) $Q = 4\sigma\pi r_0^2$
- (b) $r_0 = \frac{\lambda}{2\pi\sigma}$
- (c) $E_1(r_0/2) = 2E_2(r_0/2)$
- (d) $E_2(r_0)/2 = 4E_3(r_0/2)$

49. The figures below depict two situations in which two infinitely long static line charges of constant positive line charge density λ are kept parallel to each other. In their resulting electric field, point charges q and $-q$ are kept in equilibrium between them. The point charges are confined to move in the x -direction only. If they are given a small displacement about their equilibrium positions, then the correct statement(s) is (are) [JEE Advanced 2015]



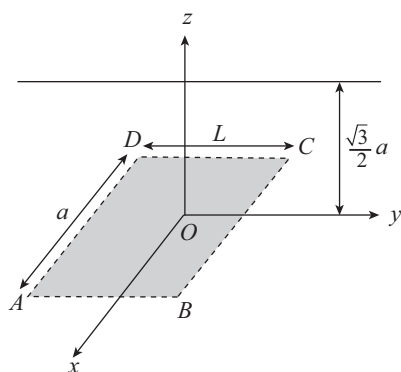
- (a) Both charges execute simple harmonic motion.
- (b) Both charges will continue moving in the direction of their displacement
- (c) Charge $+q$ executes simple harmonic motion while charge $-q$ continues moving in the direction of its displacement.
- (d) Charge $-q$ executes simple harmonic motion while charge $+q$ continues moving in the direction of its displacement.
50. Consider a uniform spherical charge distribution of radius R_1 centred at the origin O . In this distribution, a spherical cavity of radius R_2 , centred at P with distance $OP = a = R_1 - R_2$ (see figure) is made. If the electric field inside the cavity at position $\vec{r}$ is $\vec{E}(\vec{r})$, then the correct statement(s) is (are)

[JEE Advanced 2015]



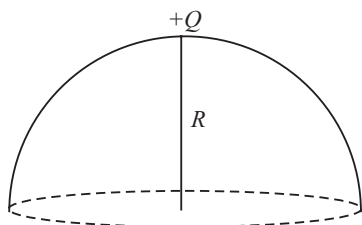
- (a) $\vec{E}$ is uniform, its magnitude is independent of R_2 but its direction depends on $\vec{r}$
- (b) $\vec{E}$ is uniform, its magnitude depends on R_2 and its direction depends on $\vec{r}$
- (c) $\vec{E}$ is uniform, its magnitude is independent of a but its direction depends on $\vec{a}$
- (d) $\vec{E}$ is uniform and both its magnitude and direction depends on $\vec{a}$
51. An infinitely long uniform line charge distribution of charge per unit length λ lies parallel to the y-axis in the y-z plane at $z = \frac{\sqrt{3}}{2}a$ (see figure). If the magnitude of the flux of the electric field through the rectangular surface ABCD lying in the x-y plane with its centre at the origin is $\frac{\lambda L}{n\epsilon_0}$ (ϵ_0 = permittivity of free space), then the value of n is:

[JEE Advanced 2015]



52. A point charge $+Q$ is placed just outside an imaginary hemispherical surface of radius R as shown in the figure. Which of the following statements is/are correct?

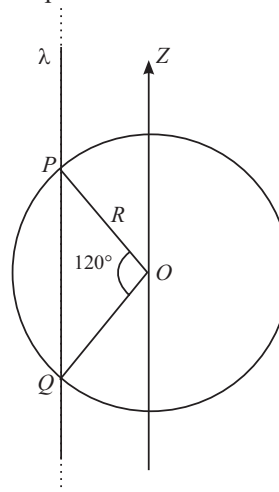
[JEE Advanced 2017]



- (a) The electric flux passing the curved surface of the hemisphere is $-\frac{Q}{2\epsilon_0}\left(1 - \frac{1}{\sqrt{2}}\right)$
- (b) Total flux through the curved and the flat surfaces is $\frac{Q}{\epsilon_0}$
- (c) The component of the electric field normal to the flat surface is constant over the surface.
- (d) The circumference of the flat surface is an equipotential.
53. An infinitely long thin non-conducting wire is parallel to the z-axis and carries a uniform line charge density λ . It pierces a thin non-conducting spherical shell of radius R in such a way that the arc PQ subtends an angle 120° at the centre O of the spherical shell, as shown in the figure. The permittivity of free space is ϵ_0 . Which of the following statements is (are) true?

[JEE Advanced 2018]

- (a) The electric flux through the shell is $\sqrt{3}R\lambda/\epsilon_0$
- (b) The z-component of the electric field is zero at all the points on the surface of the shell
- (c) The electric flux through the shell is $\sqrt{2}R\lambda/\epsilon_0$
- (d) The electric field is normal to the surface of the shell at all points



54. A particle, of mass 10^{-3} kg and charge 1.0 C, is initially at rest. At time $t = 0$, the particle comes under the influence of an electric field $\vec{E}(t) = E_0 \sin \omega t \hat{i}$ where $E_0 = 1.0 \text{ NC}^{-1}$ and $\omega = 10^3 \text{ rad s}^{-1}$. Consider the effect of only the electrical force on the particle. Then the maximum speed, in ms^{-1} , attained by the particle at subsequent times is _____.

[JEE Advanced 2018]

55. The electric field E is measured at a point $P(0, 0, d)$ generated due to various charge distributions

and the dependence of E on d is found to be different for different charge distributions. List-I contains different relations between E and d . List-II describes different electric charge distributions, along with their locations. Match the functions in List-I with the related charge distributions in List-II.

[JEE Advanced 2018]

List-I	List-II
P. E is independent of d	1. A point charge Q at the origin
Q. $E \propto 1/d$	2. A small dipole with point charges Q at $(0, 0, l)$ and $-Q$ at $(0, 0, -l)$ Take $2l \ll d$
R. $E \propto 1/d^2$	3. An infinite line charge coincident with the x-axis, with uniform linear charge density λ .
S. $E \propto 1/d^3$	4. Two infinite wires carrying uniform linear charge density parallel to the x-axis. The one along $(y = 0, z = l)$ has a charge density $+\lambda$ and the one along $(y = 0, z = -l)$ has a charge density $-\lambda$. Take $2l \ll d$
	5. Infinite plane charge coincident with the xy-plane with uniform surface charge density.

- (a) $P \rightarrow 5; Q \rightarrow 3, 4; R \rightarrow 1; S \rightarrow 2$
 (b) $P \rightarrow 5; Q \rightarrow 3; R \rightarrow 1, 4; S \rightarrow 2$
 (c) $P \rightarrow 5; Q \rightarrow 3; R \rightarrow 1, 2; S \rightarrow 4$
 (d) $P \rightarrow 4; Q \rightarrow 2, 3; R \rightarrow 1; S \rightarrow 5$

CAPACITOR

AIEEE / JEE Main Questions

1. A fully charged capacitor has a capacitance ' C '. It is discharged through a small coil of resistance wire embedded in a thermally insulated block of specific heat capacity ' s ' and mass ' m '. If the temperature of the block is raised by ' ΔT ', The potential difference ' V ' across the capacitance is: [AIEEE 2005]

- (a) $\frac{ms\Delta T}{C}$ (b) $\sqrt{\frac{2ms\Delta T}{C}}$
 (c) $\sqrt{\frac{2mC\Delta T}{s}}$ (d) $\frac{mC\Delta T}{s}$

2. A parallel plate capacitor is made by stacking n equally spaced plates connected alternately. If the capacitance between any two plates is C then the resultant capacitance is: [AIEEE 2005]

- (a) C (b) nC
 (c) $(n-1)C$ (d) $(n+1)C$

3. A parallel plate capacitor with air between the plates has a capacitance of 9 pF. The separation between its plates is ' d '. The space between the plates is now filled with two dielectrics. One of the dielectrics has dielectric constant $k_1 = 3$ and thickness $\frac{d}{3}$ while the other one has dielectric constant $k_2 = 6$ and thickness $\frac{2d}{3}$. Capacitance of the capacitor is now [AIEEE 2008]

- (a) 1.8pF (b) 45pF
 (c) 40.5pF (d) 20.25pF

4. Two capacitors C_1 and C_2 are charged to 120V and 200V respectively. It is found that by connecting them together the potential on each one can be made zero. Then: [JEE Main 2013]

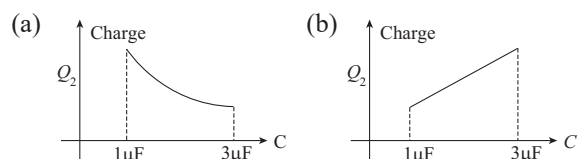
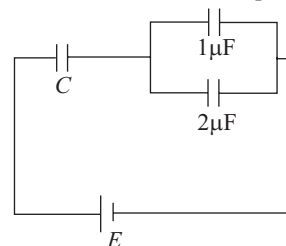
- (a) $3C_1 = 5C_2$ (b) $3C_1 + 5C_2 = 0$
 (c) $9C_1 + 4C_2$ (d) $5C_1 = 3C_2$

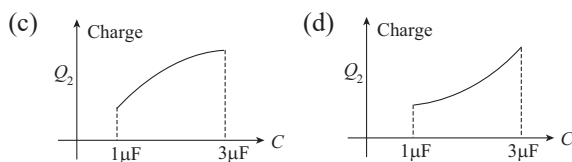
5. A parallel plate capacitor is made of two circular plates separated by a distance of 5 mm and with a dielectric of dielectric constant 2.2 between them. When the electric field in the dielectric is 3×10^4 V/m, the charge density of the positive plate will be close to: [JEE Main 2014]

- (a) $6 \times 10^{-7} \text{ C/m}^2$ (b) $3 \times 10^{-7} \text{ C/m}^2$
 (c) $3 \times 10^4 \text{ C/m}^2$ (d) $6 \times 10^4 \text{ C/m}^2$

6. In the given circuit, charge Q_2 on the $2\mu\text{F}$ capacitor changes as C is varied from $1\mu\text{F}$ to $3\mu\text{F}$. Q_2 as a function of ' C ' is given properly by: (figures are drawn schematically and are not to scale)

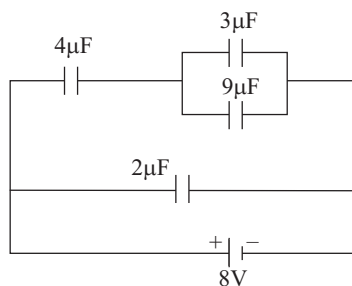
[JEE Main 2015]





7. A combination of capacitors is set up as shown in the figure. The magnitude of the electric field, due to a point charge Q (having a charge equal to the sum of the charges on the $4\mu\text{F}$ and $9\mu\text{F}$ capacitors), at a point distant 30 m from it, would equal:

[JEE Main 2016]



- (a) 240 N/C (b) 360 N/C
(c) 420 N/C (d) 480 N/C
8. A capacitance of $2\mu\text{F}$ is required in an electrical circuit across a potential difference of 1.0 kV. A large number of $1\mu\text{F}$ capacitors are available which can withstand a potential difference of not more than 300V. The minimum number of capacitors required to achieve this is:

[JEE Main 2017]

- (a) 2 (b) 16
(c) 24 (d) 32
9. A parallel plate capacitor of capacitance 90 pF is connected to a battery of emf 20V. If a dielectric material of dielectric constant $K = \frac{5}{3}$ is inserted between the plates, the magnitude of the induced charge will be:

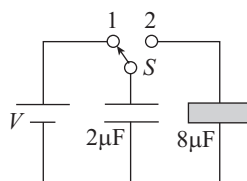
[JEE Main 2018]

- (a) 1.2 nC (b) 0.3 nC
(c) 2.4 nC (d) 0.9 nC

IIT-JEE / JEE Advanced Questions

10. A $2\mu\text{F}$ capacitor is charged as shown in figure. The percentage of its stored energy dissipated after the switch S is turned to position 2 is:

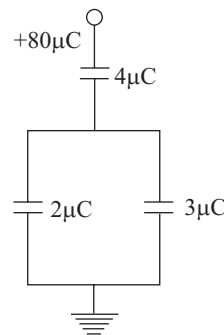
[IIT-JEE 2011]



- (a) 0% (b) 20%
(c) 75% (d) 80%

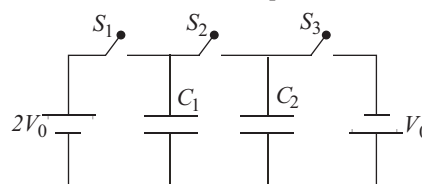
11. In the given circuit, a charge of $+80\mu\text{C}$ is given to the upper plate of the $4\mu\text{F}$ capacitor. Then in the steady state, the charge on the upper plate of the $3\mu\text{F}$ capacitor is:

[IIT-JEE 2012]



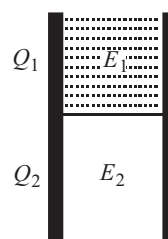
- (a) $+32\mu\text{C}$ (b) $+40\mu\text{C}$
(c) $+48\mu\text{C}$ (d) $+80\mu\text{C}$
12. In the circuit shown in the figure, there are two parallel plate capacitors each of capacitance C . The switch S_1 is pressed first to fully charge the capacitor C_1 and then released. The switch S_2 is then pressed to charge the capacitor C_2 . After some time, S_2 is released and then S_3 is pressed. After some time.

[JEE Advanced 2013]



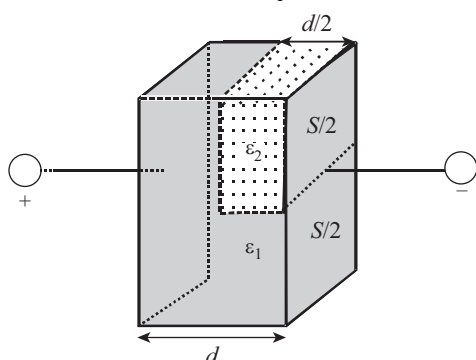
- (a) the charge on the upper plate of C_1 is $2CV_0$
(b) the charge on the upper plate of C_1 is CV_0
(c) the charge on the upper plate of C_2 is 0
(d) the charge on the upper plate of C_2 is $-CV_0$
13. A parallel plate capacitor has a dielectric slab of dielectric constant K between its plates that covers $1/3$ of the area of its plates, as shown in the figure. The total capacitance of the capacitor is C while that of the portion with dielectric in between is C_1 . When the capacitor is charged, the plate area covered by the dielectric gets charge Q_1 and the rest of the area gets charge Q_2 . The electric field in the dielectric is E_1 and that in the other portion is E_2 . Choose the correct option/options, ignoring edge effects.

[JEE Advanced 2014]



- (a) $\frac{E_1}{E_2} = 1$ (b) $\frac{E_1}{E_2} = \frac{1}{K}$
 (c) $\frac{Q_1}{Q_2} = \frac{3}{K}$ (d) $\frac{C}{C_1} = \frac{2+K}{K}$

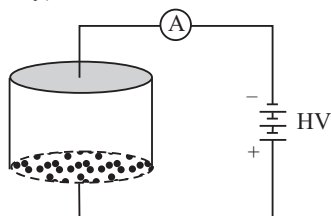
14. A parallel plate capacitor having plates of area S and plate separation d has capacitance C_1 in air. When two dielectrics of different relative permittivities ($\epsilon_1 = 2$ and $\epsilon_2 = 4$) are introduced between the two plates as shown in the figure, the capacitance becomes C_2 . The ratio $\frac{C_2}{C_1}$ is [JEE Advanced 2015]



- (a) 6/5 (b) 5/3
 (c) 7/5 (d) 7/3

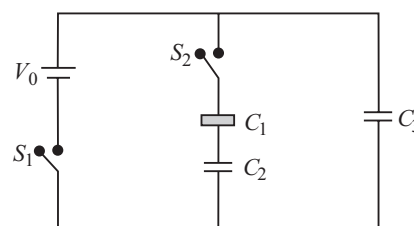
Q. 15 and 16 are based on following Paragraph
 [JEE Advanced 2016]

Consider an evacuated cylindrical chamber of height h having rigid conducting plates at the ends and an insulating curved surface as shown in the figure. A number of spherical balls made of a light weight and soft material and coated with a conducting material are placed on the bottom plate. The balls have radius $r \ll h$. Now a high voltage source (HV) is connected across the conducting plates such that the bottom plate is at $+V_0$ and the top plate at $-V_0$. Due to their conducting surface, the balls will get charged, will become equipotential with the plate and are repelled by it. The balls will eventually collide with the top plate, where the coefficient of restitution can be taken to be zero to the soft nature of the material of the balls. The electric field in the chamber can be considered to be that of a parallel plate capacitor. Assume that there are no collisions between the balls and the interaction between them is negligible. (Ignore gravity)



15. Which one of the following statements is correct?
 (a) The balls will bounce back to the bottom plate carrying the same charge they went up with
 (b) The balls will stick to the top plate and remain there
 (c) The balls will execute simple harmonic motion between the two plates
 (d) The balls will bounce back to the bottom plate carrying the opposite charge they went up with
16. The average current in the steady state registered by the ammeter in the circuit will be
 (a) proportional to the potential V_0
 (b) zero
 (c) proportional to V_0^2
 (d) proportional to $V_0^{1/2}$

17. Three identical capacitors C_1 , C_2 and C_3 have a capacitance of $1.0 \mu\text{F}$ each and they are uncharged initially. They are connected in a circuit as shown in the figure and C_1 is then filled completely with a dielectric material of relative permittivity ϵ_r . The cell has electromotive force (emf) $V_0 = 8\text{V}$. First the switch S_1 is closed while the switch S_2 is kept open. When the capacitor C_3 is fully charged, S_1 is opened and S_2 is closed simultaneously. When all the capacitors reach equilibrium, the charge on C_3 is found to be $5 \mu\text{C}$. The value of $\epsilon_r =$ _____.
 [JEE Advanced 2018]

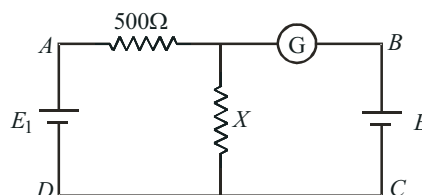


CURRENT, ELECTRICAL MEASUREMENTS AND RC CIRCUITS

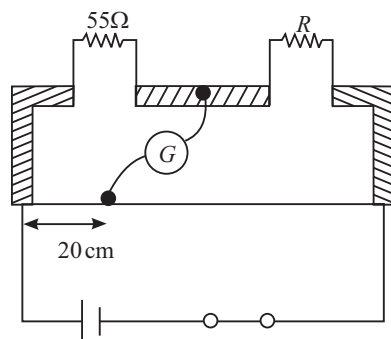
AIEEE / JEE Main Questions

1. In the adjoining circuit, the battery E_1 has an e.m.f. of 12 volt and zero internal resistance while the battery E has an e.m.f. of 2 volt. If the galvanometer G reads zero, then the value of the resistance X in ohm is:

[AIEEE 2005]



- (a) 10 (b) 100
(c) 500 (d) 200
2. Two sources of equal e.m.f. are connected to an external resistance R . The internal resistances of the two source are R_1 and R_2 ($R_2 > R_1$). If the potential difference across the source having internal resistance R_2 is zero, then: [AIEEE 2005]
- (a) $R = \frac{R_1 R_2}{(R_1 + R_2)}$ (b) $R = \frac{R_1 R_2}{(R_2 - R_1)}$
(c) $R = \frac{R_2 \times (R_1 + R_2)}{(R_2 - R_1)}$ (d) $R = R_2 - R_1$
3. An energy source will supply a constant current into the load if its internal resistance is:
(a) Zero
(b) None zero but less than the resistance of the load
(c) Equal to the resistance of the load
(d) Very large as compared to the load resistance [AIEEE 2005]
4. A moving coil galvanometer has 150 equal divisions. Its current sensitivity is 10 divisions per milliampere and voltage sensitivity is 2 divisions per millivolt. In order that each division reads 1 volt, the resistance in ohms needed to be connected in series with the coil will be: [AIEEE 2005]
- (a) 99995 (b) 9995
(c) 10^3 (d) 10^5
5. Shown in the figure is a meter-bridge set up with null deflection in the galvanometer. The value of unknown resistor R is [AIEEE 2008]

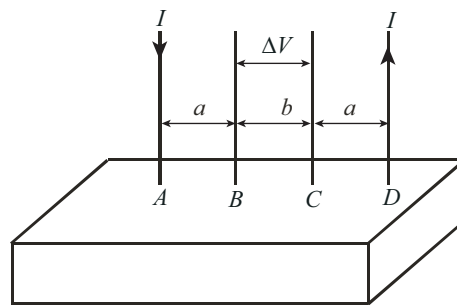


- (a) 13.75Ω (b) 220Ω
(c) 110Ω (d) 55Ω

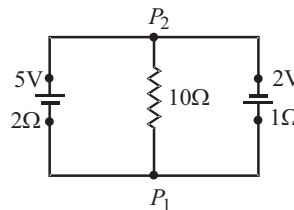
Q. 6 and 7 are based on following passage:

Consider a block of conducting material of resistivity ' ρ ' shown in the figure. Current ' I ' enters at ' A ' and leaves from ' D '. We apply superposition principle to find voltage ' ΔV ' developed between ' B ' and ' C '. The calculation is done in the following steps:

[AIEEE 2008]



- (i) Take current ' I ' entering from ' A ' and assume it to spread over a hemispherical surface in the block.
(ii) Calculate field $E(r)$ at distance ' r ' from A by using Ohm's law $E = \rho j$, where j is the current per unit area at ' r '.
(iii) From the ' r ' dependence of $E(r)$, obtain the potential $V(r)$ at r .
(iv) Repeat (i), (ii) and (iii) for current ' I ' leaving ' D ' and superpose results for ' A ' and ' D '.
6. ΔV measured between B and C is:
(a) $\frac{\rho I}{\pi a} - \frac{\rho I}{\pi(a+b)}$ (b) $\frac{\rho I}{a} - \frac{\rho I}{(a+b)}$
(c) $\frac{\rho I}{2\pi a} - \frac{\rho I}{2\pi(a+b)}$ (d) $\frac{\rho I}{2\pi(a-b)}$
7. For current entering at A , the electric field at a distance ' r ' from A is:
(a) $\frac{\rho I}{8\pi r^2}$ (b) $\frac{\rho I}{r^2}$
(c) $\frac{\rho I}{2\pi r^2}$ (d) $\frac{\rho I}{4\pi r^2}$
8. A 5V battery with internal resistance 2Ω and a 2V battery with internal resistance 1Ω are connected to a 10Ω resistor as shown in figure. The current in the 10Ω resistor is: [AIEEE 2008]

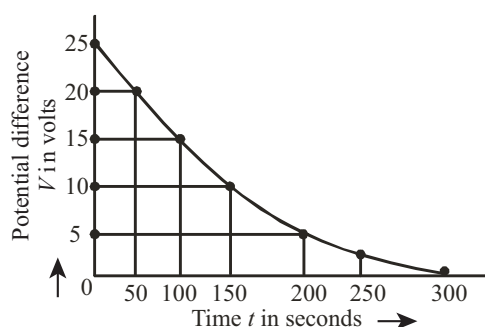


- (a) $0.27 A$ P_2 to P_1 (b) $0.03 A$ P_1 to P_2
(c) $0.03 A$ P_2 to P_1 (d) $0.27 A$ P_1 to P_2
9. Let C be the capacitance of a capacitor discharging through a resistor R . Suppose t_1 is the time taken for the energy stored in the capacitor to reduce to half its initial value and t_2 is the time taken for the charge to reduce to one-fourth its initial value. Then the ratio t_1/t_2 will be: [AIEEE 2010]

- (a) 2 (b) 1
(c) 1/2 (d) 1/4
10. Two conductors have the same resistance at 0°C but their temperature coefficients of resistance are α_1 and α_2 . The respective temperature coefficients of their series and parallel combinations are nearly:

[AIEEE 2010]

- (a) $\frac{\alpha_1 + \alpha_2}{2}, \frac{\alpha_1 + \alpha_2}{2}$ (b) $\frac{\alpha_1 + \alpha_2}{2}, \alpha_1 + \alpha_2$
(c) $\alpha_1 + \alpha_2, \frac{\alpha_1 + \alpha_2}{2}$ (d) $\alpha_1 + \alpha_2, \frac{\alpha_1 \alpha_2}{\alpha_1 + \alpha_2}$
11. A resistor ' R ' and $2\mu\text{F}$ capacitor in series is connected through a switch to 200V direct supply. Across the capacitor is a neon bulb that lights up at 120V. Calculate the value of R to make the bulb light up 5s after the switch has been closed. ($\log_{10} 2.5 = 0.4$): [AIEEE 2011]
- (a) $2.7 \times 10^6 \Omega$ (b) $3.3 \times 10^7 \Omega$
(c) $1.3 \times 10^4 \Omega$ (d) $1.7 \times 10^5 \Omega$
12. Two electric bulbs marked 25W – 220V and 100W – 220V are connected in series to a 440V supply. Which of the bulbs will fuse? [AIEEE 2012]
- (a) neither (b) both
(c) 100W (d) 25W
13. The figure shows an experimental plot for discharging of a capacitor in an R - C circuit. The time constant τ of this circuit lies between: [AIEEE 2012]

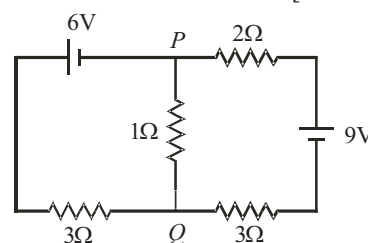


- (a) 100 sec and 150 sec
(b) 150 sec and 200 sec
(c) 0 sec and 50 sec
(d) 50 sec and 100 sec
14. The supply voltage to a room is 120V. The resistance of the lead wires is 6Ω . A 60W bulb is already switched on. What is the decrease of voltage across the bulb, when a 240W heater is switched on in parallel to the bulb? [JEE Main 2013]
- (a) 2.9 Volt (b) 13.3 Volt
(c) 10.04 Volt (d) Zero Volt
15. In a large building, there are 15 bulbs of 40W, 5 bulbs of 100W, 5 fans of 80W and 1 heater of 1kW. The

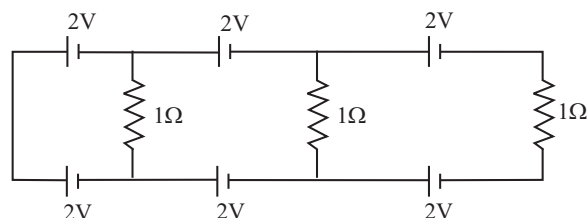
voltage of the electric mains is 220V. The minimum capacity of the main fuse of the building will be:

[JEE Main 2014]

- (a) 8A (b) 10A
(c) 12A (d) 14A
16. In the circuit shown, the current in the 1Ω resistor is: [JEE Main 2015]

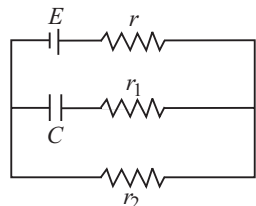


- (a) 0.13A from P to Q
(b) 1.3A from P to Q
(c) 0A
(d) 0.13A from Q to P
17. When 5V potential difference is applied across a wire of length 0.1m, the drift speed of electrons is $2.5 \times 10^{-4} \text{ ms}^{-1}$. If the electron density in the wire is $8 \times 10^{28} \text{ m}^{-3}$, the resistivity of the material is close to: [JEE Main 2015]
- (a) $1.6 \times 10^{-5} \Omega\text{m}$ (b) $1.6 \times 10^{-8} \Omega\text{m}$
(c) $1.6 \times 10^{-7} \Omega\text{m}$ (d) $1.6 \times 10^{-6} \Omega\text{m}$
18. A galvanometer having a coil resistance of 100Ω gives a full scale deflection, when a current of 1 mA is passed through it. The value of the resistance, which can convert this galvanometer into ammeter giving a full scale deflection for a current of 10A, is: [JEE Main 2016]
- (a) 0.01Ω (b) 2Ω
(c) 0.1Ω (d) 3Ω
19. In the circuit shown the current in each resistance is: [JEE Main 2017]



- (a) 1A (b) 0.25A
(c) 0.5A (d) 0A
20. When a current of 5 mA is passed through a galvanometer having a coil of resistance 15Ω , it shows full scale deflection. The value of the resistance to be put in series with the galvanometer to convert it into a voltmeter of range 0 – 10V is: [JEE Main 2017]

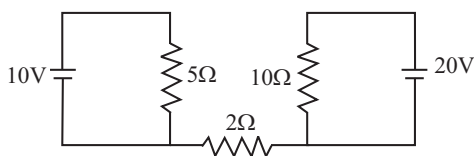
- (a) $1.985 \times 10^3 \Omega$ (b) $2.045 \times 10^3 \Omega$
 (c) $2.535 \times 10^3 \Omega$ (d) $4.005 \times 10^3 \Omega$
21. In the given circuit diagram when the current reaches steady state in the circuit, the charge on the capacitor of capacitance C will be: [JEE Main 2017]



- (a) CE (b) $CE \frac{r_1}{(r_2 + r)}$
 (c) $CE \frac{r_2}{(r + r_2)}$ (d) $CE \frac{r_1}{(r_1 + r)}$
22. Two batteries with e.m.f. 12V and 13V are connected in parallel across a load resistor of 10Ω . The internal resistances of the two batteries are 1Ω and 2Ω respectively. The voltage across the load lies between: [JEE Main 2018]
 (a) 11.6V and 11.7V (b) 11.5V and 11.6V
 (c) 11.4V and 11.5V (d) 11.7V and 11.8V
23. In a potentiometer experiment, it is found that no current passes through the galvanometer when the terminals of the cell are connected across 52 cm length of the potentiometer wire. If the cell is shunted by a resistance of 5Ω , a balance is found when the cell is connected across 40 cm of the wire. Find the internal resistance of the cell. [JEE Main 2018]
 (a) 1Ω (b) 1.5Ω
 (c) 2Ω (d) 2.5Ω
24. On interchanging the resistances, the balance point of a meter bridge shifts to the left by 10 cm. The resistance of their series combination is $1\text{ k}\Omega$. How much was the resistance on the left slot before interchanging the resistances? [JEE Main 2018]
 (a) 990Ω (b) 505Ω
 (c) 550Ω (d) 910Ω

IIT-JEE / JEE Advanced Questions

25. Find out the value of current through 2Ω resistance for the given circuit. [IIT-JEE 2005]



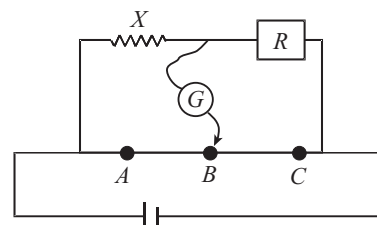
- (a) Zero (b) 2A
 (c) 5A (d) 4A

26. A moving coil galvanometer of resistance 100Ω is used as an ammeter using a resistance 0.1Ω . The maximum deflection current in the galvanometer is $100\mu\text{A}$. Find the minimum current in the circuit so that the ammeter shows maximum deflection. [IIT-JEE 2005]

- (a) 100.1 mA (b) 1000.1 mA
 (c) 10.01 mA (d) 1.01 mA
27. An ideal gas is filled in a closed rigid and thermally insulated container. A coil of 100Ω resistor carrying current 1 A for 5 minutes supplies heat to the gas. The change in internal energy of the gas is [IIT-JEE 2005]

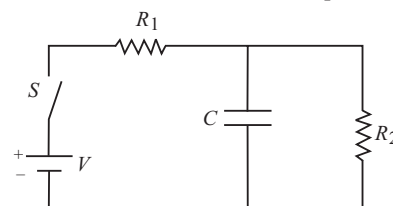
- (a) 10 kJ (b) 30 kJ
 (c) 20 kJ (d) 0 kJ
28. A $4\mu\text{F}$ capacitor and a resistance of $2.5\text{ M}\Omega$ are in series with 12V battery. Find the time after which the potential difference across the capacitor is 3 times the potential difference across the resistor. [Given $\ln(2) = 0.693$] [IIT-JEE 2005]

- (a) 13.86s (b) 6.93s
 (c) 7s (d) 14s
29. The circuit shown in the figure is used to measure the value of an unknown resistance X by comparing it with three different known resistances, $R = R_1$, $R = R_2$ and $R = R_3$. The corresponding null points are obtained at A , B and C . Find which of the above will give the most accurate reading and why? [IIT-JEE 2005]



Q.No. 30 and 31 are based on following passage:

In the given circuit, the switch S is closed at time $t = 0$. The charge Q on the capacitor at any instant t is given by $Q(t) = Q_0(1 - e^{-\alpha t})$. Find the value of Q_0 and α in terms of given parameters as shown in the circuit. [IIT-JEE 2005]



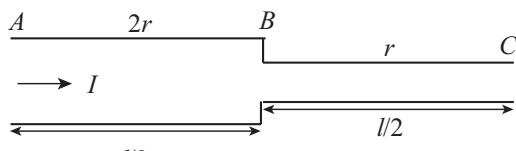
30. The Value of Q_0 is

- (a) $\frac{CVR_2}{R_1 + R_2}$ (b) $\frac{CVR_1}{R_1 + R_2}$
 (c) $\frac{CVR_1}{R_1 - R_2}$ (d) $\frac{CVR_1}{R_2 - R_1}$

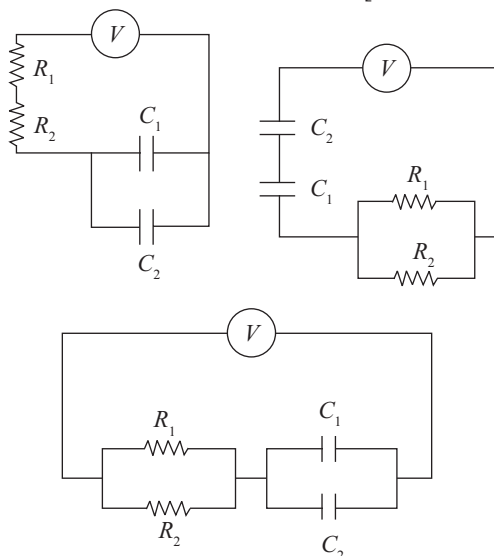
31. The value of α is

- (a) $\frac{R_1 - R_2}{CR_1R_2}$ (b) $\frac{R_1 + R_2}{CR_1R_2}$
 (c) $\frac{R_2 - R_1}{CR_1R_2}$ (d) $\frac{R_2 - R_1}{CR_1}$

32. If current I is flowing through a straight wire ABC . The cross sectional radius of the two segments and their lengths are as mentioned in the diagram. Both the segments are made of same material. The correct relationship is: [IIT-JEE 2006]



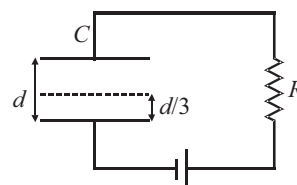
- (a) $V_{AB} = 2V_{BC}$
 (b) Power across BC is 4 times the power across AB .
 (c) Current densities in AB and BC are equal
 (d) Electric field due to current inside AB and BC are equal
33. In the three RC circuits shown below, $R_1 = 1 \Omega$, $R_2 = 2 \Omega$, $C_1 = 4 \text{ mF}$ and $C_2 = 2 \text{ mF}$. The time constant (in millisecond) for the given RC circuits in correct order are: [IIT-JEE 2006]



- (a) 18, 4, 8/9 (b) 18, 8/9, 4
 (c) 4, 18, 8/9 (d) 4, 8/9, 18

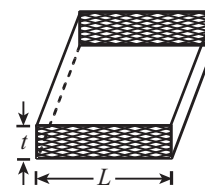
34. A parallel plate capacitor C with plates of unit area and separation d is filled with a liquid of dielectric constant $K = 2$. The level of liquid is $d/3$ initially. Suppose the liquid level decreases at a constant speed V , the time constant as a function of time t is:

[IIT-JEE 2008]

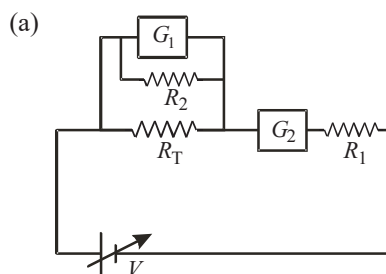


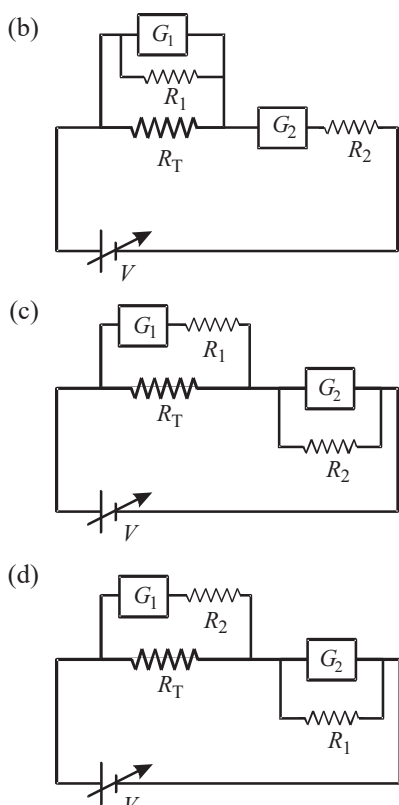
- (a) $\frac{6\epsilon_0 R}{5d + 3Vt}$ (b) $\frac{(15d + 9Vt)\epsilon_0 R}{2d^2 - 3dVt - 9V^2t^2}$
 (c) $\frac{6\epsilon_0 R}{5d - 3Vt}$ (d) $\frac{(15d - 9Vt)\epsilon_0 R}{2d^2 + 3dVt - 9V^2t^2}$

35. Consider a thin square sheet of side L and thickness t , made of a material of resistivity ρ . The resistance between two opposite faces, shown by the shaded areas in the figure is: [IIT-JEE 2010]



- (a) directly proportional to L
 (b) directly proportional to t
 (c) independent of L
 (d) independent of t
36. To verify Ohm's law, a student is provided with a test resistor R_T , a high resistance R_1 , a small resistance R_2 , two identical galvanometers G_1 and G_2 , and a variable voltage source V . The correct circuit to carry out the experiment is: [IIT-JEE 2010]



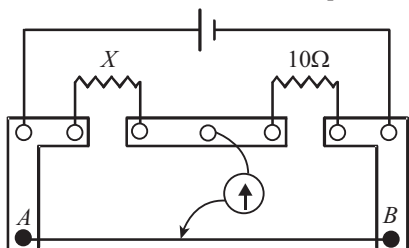


37. Incandescent bulbs are designed by keeping in mind that the resistance of their filament increases with the increase in temperature. If at room temperature, 100W, 60W and 40W bulbs have filament resistances R_{100} , R_{60} and R_{40} , respectively, the relation between these resistances is: [IIT-JEE 2010]

(a) $\frac{1}{R_{100}} = \frac{1}{R_{40}} + \frac{1}{R_{60}}$ (b) $R_{100} = R_{40} + R_{60}$
 (c) $R_{100} > R_{40} > R_{60}$ (d) $\frac{1}{R_{100}} > \frac{1}{R_{60}} > \frac{1}{R_{40}}$

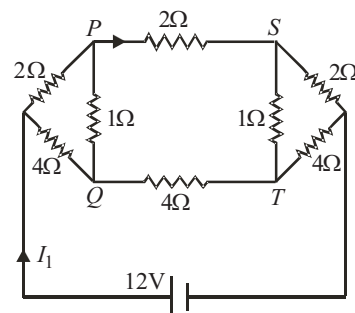
38. A meter bridge is set-up as shown, to determine an unknown resistance ' X ' using a standard 10Ω resistor. The galvanometer shows null point when tapping-key is at 52 cm mark. The end-corrections are 1 cm and 2 cm respectively for the ends A and B . The determined value of ' X ' is:

[IIT-JEE 2011]



- (a) 10.2Ω (b) 10.6Ω
 (c) 10.8Ω (d) 11.1Ω

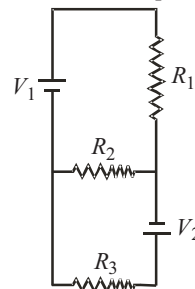
39. For the resistance network shown in the figure, choose the correct option(s): [IIT-JEE 2012]



- (a) The current through PQ is zero
 (b) $I_1 = 3A$
 (c) The potential at S is less than that at Q
 (d) $I_2 = 2A$
40. Heater of an electric kettle is made of a wire of length L and diameter d . It takes 4 minutes to raise the temperature of 0.5 kg water by 40 K. This heater is replaced by a new heater having two wires of the same material, each of length L and diameter $2d$. The way these wires are connected is given in the options. How much time in minutes will it take to raise the temperature of the same amount of water by 40K?

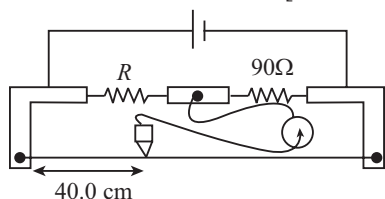
[JEE Advanced 2014]

- (a) 4 if wires are in parallel
 (b) 2 if wires are in series
 (c) 1 if wires are in series
 (d) 0.5 if wires are in parallel
41. Two ideal batteries of emf V_1 and V_2 and three resistances R_1 , R_2 and R_3 are connected as shown in the figure. The current in resistance R_2 would be zero if [JEE Advanced 2014]



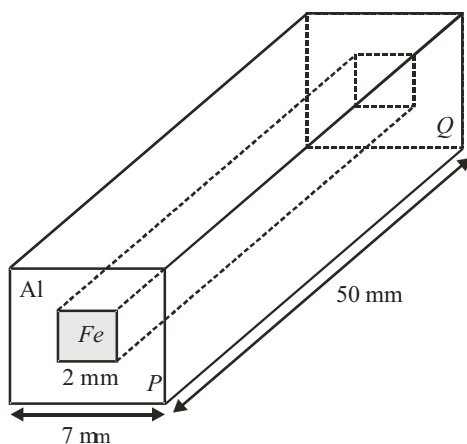
- (a) $V_1 = V_2$ and $R_1 = R_2 = R_3$
 (b) $V_1 = V_2$ and $R_1 = 2R_2 = R_3$
 (c) $V_1 = 2V_2$ and $2R_1 = 2R_2 = R_3$
 (d) $2V_1 = V_2$ and $2R_1 = R_2 = R_3$

42. During an experiment with a metre bridge, the galvanometer shows a null point when the jockey is pressed at 40.0 cm using a standard resistance of 90Ω , as shown in the figure. The least count of the scale used in the metre bridge is 1 mm. The unknown resistance is [JEE Advanced 2014]



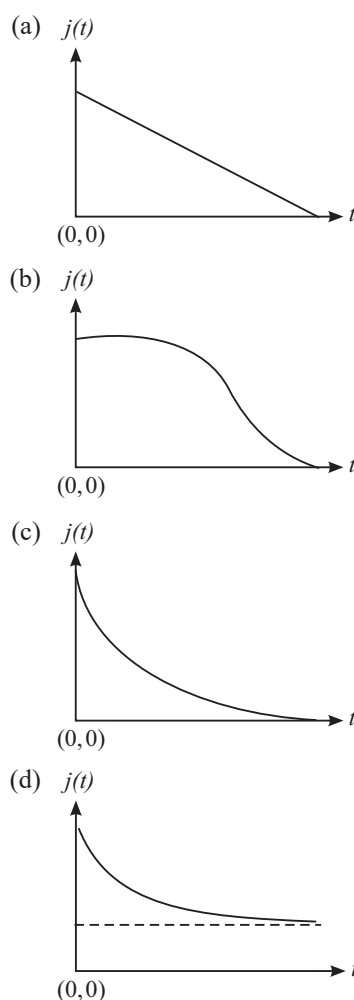
- (a) $60 \pm 0.15\Omega$ (b) $135 \pm 0.56\Omega$
 (c) $60 \pm 0.25\Omega$ (d) $135 \pm 0.23\Omega$
43. In an aluminum (Al) bar of square cross section, a square hole is drilled and is filled with iron (Fe) as shown in the figure. The electrical resistivities of Al and Fe are $2.7 \times 10^{-8}\Omega\text{m}$ and $1.0 \times 10^{-7}\Omega\text{m}$, respectively. The electrical resistance between the two faces P and Q of the composite bar is:

[JEE Advanced 2015]



- (a) $\frac{2475}{64}\mu\Omega$ (b) $\frac{1875}{64}\mu\Omega$
 (c) $\frac{1875}{49}\mu\Omega$ (d) $\frac{2475}{132}\mu\Omega$
44. An infinite line charge of uniform electric charge density λ lies along the axis of an electrically conducting infinite cylindrical shell of radius R . At time $t = 0$, the space inside the cylinder is filled with a material of permittivity ϵ and electrical conductivity σ . The electrical conduction in the material follows Ohm's law. Which one of the following graphs best describes the subsequent variation of the magnitude of current density $j(t)$ at any point in the material?

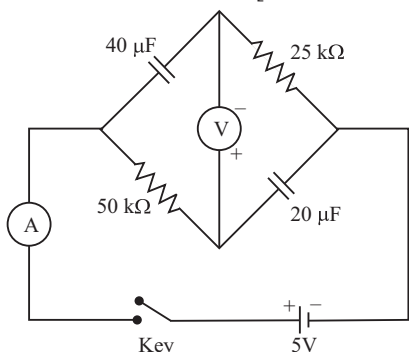
[JEE Advanced 2016]



45. An incandescent bulb has a thin filament of tungsten that is heated to high temperature by passing an electric current. The hot filament emits black-body radiation. The filament is observed to break up at random locations after a sufficiently long time of operation due to non-uniform evaporation of tungsten from the filament. If the bulb is powered at constant voltage, which of the following statement(s) is (are) true? [JEE Advanced 2016]

- (a) The temperature distribution over the filament is uniform
 (b) The resistance over small sections of the filament decreases with time
 (c) The filament emits more light at higher band of frequencies before it breaks up
 (d) The filament consumes less electrical power towards the end of the life of the bulb

46. In the circuit shown below, the key is pressed at time $t = 0$. Which of the following statement(s) is (are) true? [JEE Advanced 2016]



- (a) The voltmeter displays -5 V as soon as the key is pressed, and displays $+5$ V after a long time
 (b) The voltmeter will display 0 V at time $t = \ln 2$ seconds
 (c) The current in the ammeter becomes $1/e$ of the initial value after 1 second
 (d) The current in the ammeter becomes zero after a long time.
47. Consider two identical galvanometers and two identical resistors with resistance R . If the internal resistance of the galvanometers $R_g < R/2$, which of the following statement(s) about any one of the galvanometers is(are) true? [JEE (Adv) 2016]
- (a) The maximum voltage range is obtained when all the components are connected in series
 (b) The maximum voltage range is obtained when the two resistors and one galvanometer are connected in series, and the second galvanometer is connected in parallel to the first galvanometer
 (c) The maximum current range is obtained when all the components are connected in parallel
 (d) The maximum current range is obtained when the two galvanometers are connected in series and the combination is connected in parallel with both the resistors

Q.48 and 49 are based on the following paragraph
 [JEE Advanced 2017]

Consider a simple RC circuit as shown in figure 1.

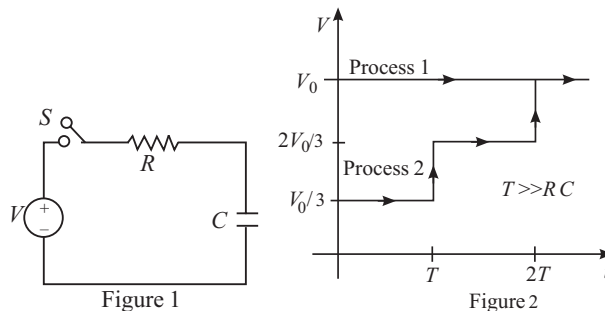
Process 1: In the circuit the switch S is closed at $t = 0$ and the capacitor is fully charged to voltage V_0 (i.e., charging continues for time $T \gg RC$). In the process some dissipation (E_D) occurs across the resistance R . The amount of energy finally stored in the fully charged capacitor is E_C .

Process 2: In a different process the voltage is first set to $\frac{V_0}{3}$ and maintained for a charging time

$T \gg RC$. Then the voltage is raised to $\frac{2V_0}{3}$ without

discharging the capacitor and again maintained for a time $T \gg RC$. The process is repeated one more time by raising the voltage to V_0 and the capacitor is charged to the same final voltage V_0 as in Process 1.

These two processes are depicted in Figure 2.



48. In process 1, the energy stored in the capacitor E_C and heat dissipated across resistance E_D are related by:
- (a) $E_C = E_D$ (b) $E_C = E_D \ln 2$
 (c) $E_C = \frac{1}{2} E_D$ (d) $E_C = 2E_D$
49. In Process 2, total energy dissipated across the resistance E_D is:
- (a) $E_D = \frac{1}{2} CV_0^2$ (b) $E_D = 3 \left(\frac{1}{2} CV_0^2 \right)$
 (c) $E_D = \frac{1}{3} \left(\frac{1}{2} CV_0^2 \right)$ (d) $E_D = 3 CV_0^2$

ERROR AND MEASUREMENT

AIEEE / JEE Main Questions

- Two full turns of the circular scale of a screw gauge cover a distance of 1mm on its main scale. The total number of divisions on the circular scale is 50. Further, it is found that the screw gauge has a zero error of -0.03 mm. While measuring the diameter of a thin wire, a student notes the main scale reading of 3 mm and the number of circular scale divisions in line with the main scale as 35. The diameter of the wire is: [AIEEE 2008]
 (a) 3.32 mm (b) 3.73 mm
 (c) 3.67 mm (d) 3.38 mm
- In an experiment the angles are required to be measured using an instrument. 29 divisions of the main scale exactly coincide with the 30 divisions of

the vernier scale. If the smallest division of the main scale is half-a-degree (0.5°), then the least count of the instrument is: [AIEEE 2009]

- (a) half minute (b) one degree
(c) half degree (d) one minute

3. The respective number of significant figures for the numbers 23.023, 0.0003 and 2.1×10^{-3} are

[AIEEE 2010]

- (a) 5, 1, 2 (b) 5, 1, 5
(c) 5, 5, 2 (d) 4, 4, 2

4. A screw gauge gives the following reading when used to measure the diameter of a wire.

Main scale reading: 0 mm

Circular scale reading: 52 division

Given that 1 mm on main scale corresponds to 100 divisions of the circular scale.

The diameter of wire from the above data is:

[AIEEE 2011]

- (a) 0.52 cm (b) 0.052 cm
(c) 0.026 cm (d) 0.005 cm

5. If 400Ω of resistance is made by adding four 100Ω resistances of tolerance 5%, then the tolerance of the combination is: [AIEEE 2011]

- (a) 5% (b) 10 %
(c) 15 % (d) 20 %

6. Resistance of a given wire is obtained by measuring the current flowing in it and the voltage difference applied across it. If the percentage errors in the measurement of the current and the voltage difference are 3% each, then error in the value of resistance of the wire is: [AIEEE 2012]

- (a) 3% (b) 6%
(c) zero (d) 1%

7. The current voltage relation of diode is given by $I = (e^{1000V/T} - 1) \text{ mA}$, where the applied voltage V is in volts and the temperature T is in degree Kelvin. If a student makes an error $\pm 0.01 \text{ V}$ measuring voltage while measuring the current of 5 mA at 300 K, what will be the error in the value of current in mA?

[JEE Main 2014]

- (a) 0.2mA (b) 0.02mA
(c) 0.5mA (d) 0.05mA

8. The period of oscillation of a simple pendulum is $T = 2\pi\sqrt{\frac{L}{g}}$. Measured value of L is 20.0 cm known to 1 mm accuracy and time for 100 oscillations of the pendulum is found to be 90s using a wrist watch of 1s resolution. The accuracy in the determination of g is: [JEE Main 2015]

- (a) 5% (b) 2%
(c) 3% (d) 1%

9. A student measures the time period of 100 oscillations of a simple pendulum four times. The data set is 90s, 91s, 95s and 92s. If the minimum division in the measuring clock is 1s, then the reported mean time should be: [JEE Main 2016]

- (a) $92 \pm 2\text{s}$ (b) $92 \pm 5.0\text{s}$
(c) $92 \pm 1.8\text{s}$ (d) $92 \pm 3\text{s}$

10. A screw gauge with a pitch of 0.5 mm and a circular scale with 50 divisions is used to measure the thickness of a thin sheet of Aluminium. Before starting the measurement, it is found that when the two jaws of the screw gauge are brought in contact, the 45th division coincides with the main scale line and that the zero of the main scale is barely visible. What is the thickness of the sheet if the main scale reading is 0.5 mm and the 25th division coincides with the main scale line? [JEE Main 2016]

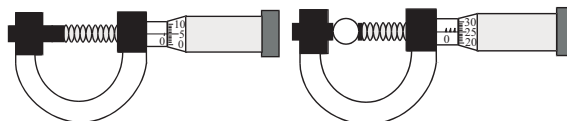
- (a) 0.75 mm (b) 0.80 mm
(c) 0.70 mm (d) 0.50 mm

IIT-JEE / JEE Advanced Questions

11. Side of a cube is measured with the help of vernier calliper. Main scale reading is 10 mm and vernier scale reading is 1. It is known that 9 M.S.D. = 10 V.S.D.. Mass of the cube is 2.735 g. Find density of the cube upto appropriate significant figure.

[JEE 2005]

12. The number of circular divisions on the shown screw gauge is 50. It moves 0.5 mm on main scale for one complete rotation. The main scale has divisions marked at 0.5mm. With the jaws closed the zero on the main scale is visible and the circular scale has its 5th division coinciding with the reference line. With a ball between the jaws the main scale reading is 2 divisions as shown. The circular scale reads as shown. The diameter of the ball is: [JEE 2006]



- (a) 2.25 mm (b) 2.20 mm
(c) 1.20 mm (d) 1.25 mm

13. A student performs an experiment for determination of acceleration due to gravity using a simple pendulum. The formula used is $g = \frac{4\pi^2 l}{T^2}$. $l \approx 1\text{m}$ and he commits an error of Δl in measurement of length. For the experiment he takes the time of n oscillations with

a stop watch of least count ΔT . For which of the following data, the measurement of g will be most accurate?

Δl	ΔT	n
(a) 5 mm	0.2 sec	10
(b) 5 mm	0.2 sec	20
(c) 5 mm	0.1 sec	10
(d) 1 mm	0.1 sec	50

[IIT-JEE 2006]

14. A student performs an experiment to determine the Young's modulus of a wire, exactly 2 m long, by Searle's method. In a particular reading, the student measures the extension in the length of the wire to be 0.8 mm with an uncertainty of ± 0.05 mm at a load of exactly 1.0 kg. The student also measures the diameter of the wire to be 0.4 mm with an uncertainty of ± 0.01 mm. Take $g = 9.8 \text{ m/s}^2$ (exact). The Young's modulus obtained from the reading is:

[IIT-JEE 2007]

- (a) $(2.0 \pm 0.3) \times 10^{11} \text{ N/m}^2$
 (b) $(2.0 \pm 0.2) \times 10^{11} \text{ N/m}^2$
 (c) $(2.0 \pm 0.1) \times 10^{11} \text{ N/m}^2$
 (d) $(2.0 \pm 0.05) \times 10^{11} \text{ N/m}^2$

15. Student I, II and III perform an experiment for measuring the acceleration due to gravity (g) using a simple pendulum. They use different lengths of the pendulum and /or record time for different number of oscillations. The observations are shown in the table.

Least count for length = 0.1 cm

Least count for time = 0.1 s

Student	Length of the pendulum (cm)	Number of oscillations (n)	Total time for (n) oscillations (s)	Time period (s)
I	64.0	8	128.0	16.0
II	64.0	4	64.0	16.0
III	20.0	4	36.0	9.0

If E_I , E_{II} and E_{III} are the percentage errors in g , i.e., $\left(\frac{\Delta g}{g} \times 100\right)$ for students I, II and III, respectively,

[IIT-JEE 2008]

- (a) $E_I = 0$ (b) E_I is minimum
 (c) $E_I = E_{II}$ (d) E_{II} is maximum

16. A vernier calipers has 1 mm marks on the main scale. It has 20 equal division on the Vernier scale which match with 16 main scale divisions. For this Vernier caliper, the least count is:

[IIT-JEE 2010]

- (a) 0.02 mm (b) 0.05 mm
 (c) 0.1 mm (d) 0.2 mm

17. The density of a solid ball is to be determined in an experiment. The diameter of the ball is measured with a screw gauge, whose pitch is 0.5 mm and there are 50 divisions on the circular scale. The reading on the main scale is 2.5 mm and that on the circular scale is 20 divisions. If the measured mass of the ball has a relative error of 2%, the relative percentage error in the density is [IIT-JEE 2011]

- (a) 0.9% (b) 2.4%
 (c) 3.1% (d) 4.2%

18. In the determination of Young's modulus $\left(Y = \frac{4MLg}{\pi d^2 l}\right)$ by using Searle's method, a wire of

length $L = 2$ m and diameter $d = 0.5$ mm is used. For a load $M = 2.5$ kg, an extension $l = 0.25$ mm in the length of the wire is observed. Quantities d and l are measured using a screw gauge and a micrometer, respectively. They have the same pitch of 0.5 mm. The number of divisions on their circular scale is 100. The contributions to the maximum probable error of the Y measurement [IIT-JEE 2012]

- (a) due to the errors in the measurements of d and l are the same.
 (b) due to the error in the measurement of d is twice that due to the error in the measurement of l .
 (c) due to the error in the measurement of l is twice that due to the error in the measurement of d .
 (d) due to the error in the measurement of d is four times that due to the error in the measurement of l .

19. The diameter of a cylinder is measured using a vernier callipers with no zero error. It is found that the zero of the vernier scale lies between 5.10 cm and 5.15 cm of the main scale. The vernier scale has 50 division equivalent to 2.45 cm. The 24th division of the vernier scale exactly coincides with one of the main scale divisions. The diameter of the cylinder is: [IIT-JEE 2013]

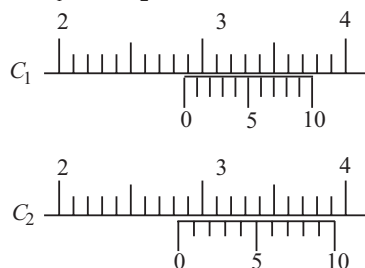
- (a) 5.112 cm (b) 5.124 cm
 (c) 5.136 cm (d) 5.148 cm

20. The energy of a system as a function of time t is given as $E(t) = A^2 \exp(-\alpha t)$, where $\alpha = 0.2 \text{ s}^{-1}$. The measurement of A has an error of 1.25%. If the error in the measurement of time is 1.50%, the percentage error in the value of $E(t)$ at $t = 5$ s is

[JEE Advanced 2015]

21. There are two Vernier calipers both of which have 1 cm divided into 10 equal divisions on the main scale. The Vernier scale of one of the calipers (C_1)

has 10 equal divisions that correspond to 9 main scale divisions. The Vernier scale of the other caliper (C_2) has 10 equal divisions that correspond to 11 main scale divisions. The readings of the two calipers are shown in the figure. The measured values (in cm) by calipers C_1 and C_2 , respectively, are



- (a) 2.87 and 2.83 (b) 2.85 and 2.82
 (c) 2.87 and 2.87 (d) 2.87 and 2.86

[JEE Advanced 2016]

22. In an experiment to determine the acceleration due to gravity g , the formula used for the time period of a periodic motion is $T = 2\pi\sqrt{\frac{7(R-r)}{5g}}$. The values of R and r measured to be (60 ± 1) mm and (10 ± 1) mm, respectively. In five successive measurements, the time period is found to be 0.52 s, 0.56 s, 0.57 s, 0.54 s and 0.59 s. The least count of the watch used for the measurement of time period is 0.01 s. Which of the following statement(s) is (are) true?

[JEE Advanced 2016]

- (a) The error in the measurement of r is 10%
 (b) The error in the measurement of T is 3.57%
 (c) The error in the measurement of T is 2%
 (d) The error in the determined value of g is 11%
23. A person measures the depth of a well by measuring the time interval between dropping a stone and receiving the sound of impact with the bottom of the well. The error in his measurement of time is $\delta T = 0.01$ seconds and he measures the depth of the well to be $L = 20$ meters. Take the acceleration due to gravity $g = 10 \text{ ms}^{-2}$ and the velocity of sound is 300 ms^{-1} . Then the fractional error in the measurement, $\delta L/L$, is closest to

[JEE Advanced 2017]

- (a) 0.2% (b) 1%
 (c) 3% (d) 5%

Q.24 and 25 are based on the following paragraph

[JEE Advanced 2018]

If the measurement errors in all the independent quantities are known, then it is possible to determine the error in any dependent quantity. This is done by the use of series expansion and truncating the

expansion at the first power of the error. For example, consider the relation $z = x/y$. If the errors in x , y and z are Δx , Δy and Δz , respectively, then

$$z + \Delta z = \frac{x + \Delta x}{y + \Delta y} = \frac{x}{y} \left(1 \pm \frac{\Delta x}{x} \right) \left(1 \pm \frac{\Delta y}{y} \right)^{-1}$$

The series expansion for $\left(1 \pm \frac{\Delta y}{y} \right)^{-1}$, to first power in $\Delta y/y$, is $1 \mp \left(\frac{\Delta y}{y} \right)$. The relative errors in independent

variables are always added. So the error in z will be

$$\Delta z = z \left(\frac{\Delta x}{x} + \frac{\Delta y}{y} \right)$$

The above derivation makes the assumption that $\Delta x/x \ll 1$, $\Delta y/y \ll 1$. Therefore, the higher powers of these quantities are neglected.

24. Consider the ratio $r = \frac{(1-a)}{(1+a)}$ to be determined by

measuring a dimensionless quantity a . If the error in the measurement of a is Δa ($\Delta a/a \ll 1$), then what is the error Δr in determining r ?

- (a) $\frac{\Delta a}{(1+a)^2}$ (b) $\frac{2\Delta a}{(1+a)^2}$
 (c) $\frac{2\Delta a}{(1-a^2)}$ (d) $\frac{2a\Delta a}{(1-a^2)}$

25. In an experiment the initial number of radioactive nuclei is 3000. It is found that 1000 ± 40 nuclei decayed in the first 1.0 s. For $|x| \ll 1$, $\ln(1+x) = x$ up to first power in x . The error $\Delta\lambda$ in the determination of the decay constant λ in s^{-1} , is

- (a) 0.04 (b) 0.03
 (c) 0.02 (d) 0.01

26. A steel wire of diameter 0.5 mm and Young's modulus $2 \times 10^{11} \text{ Nm}^{-2}$ carries a load of mass M . The length of the wire with the load is 1.0 m. A vernier scale with 10 divisions is attached to the end of this wire. Next to the steel wire is a reference wire to which a main scale, of least count 1.0 mm, is attached. The 10 divisions of the vernier scale correspond to 9 divisions of the main scale. Initially, the zero of vernier scale coincides with the zero of main scale. If the load on the steel wire is increased by 1.2 kg, the vernier scale division which coincides with a main scale division is _____. Take $g = 10 \text{ ms}^{-2}$ and $\pi = 3.2$

[JEE Advanced 2018]

Answer Sheet

Electrostatics

- | | | | | | | | | |
|-----------|---------|---------------|-----------|-----------|---------|-----------|-------------|-------------|
| 1. (b) | 2. (a) | 3. (c) | 4. (a) | 5. (c) | 6. (d) | 7. (d) | 8. (c) | 9. (d) |
| 10. (b) | 11. (c) | 12. (d) | 13. (c) | 14. (a,d) | 15. (b) | 16. (a) | 17. (c) | 18. (b) |
| 19. (c) | 20. (a) | 21. (a,b,c,d) | 22. (d) | 23. (a) | 24. (b) | 25. (c) | 26. (a) | 27. (b) |
| 28. (c) | 29. (c) | 30. (2) | 31. (a) | 32. (b) | 33. (d) | 34. (a) | 35. (a,d) | 36. (a) |
| 37. (3) | 38. (c) | 39. (a,b,c,d) | 40. (c,d) | 41. (c) | 42. (d) | 43. (6) | 44. (a,b,c) | 45. (a,c,d) |
| 46. (b,d) | 47. (c) | 48. (c) | 49. (c) | 50. (d) | 51. (3) | 52. (a,d) | 53. (a,b) | 54. (2.00) |
| 55. (b) | | | | | | | | |

Capacitor

- | | | | | | | | | |
|---------|---------|-----------|-----------|---------|---------|---------|-----------|--------|
| 1. (b) | 2. (c) | 3. (c) | 4. (a) | 5. (a) | 6. (c) | 7. (c) | 8. (d) | 9. (a) |
| 10. (d) | 11. (c) | 12. (b,d) | 13. (a,d) | 14. (d) | 15. (d) | 16. (c) | 17. (1.5) | |

Current, Electrical Measurements and RC Circuits

- | | | | | | | | | |
|-------------|-----------|---------------|-----------|-------------|---------|---------|---------|-----------|
| 1. (b) | 2. (d) | 3. (d) | 4. (b) | 5. (b) | 6. (a) | 7. (c) | 8. (c) | 9. (d) |
| 10. (a) | 11. (a) | 12. (d) | 13. (b) | 14. (c) | 15. (c) | 16. (d) | 17. (a) | 18. (a) |
| 19. (d) | 20. (a) | 21. (c) | 22. (b) | 23. (b) | 24. (c) | 25. (a) | 26. (a) | 27. (b) |
| 28. (a) | 29. (B) | 30. (a) | 31. (b) | 32. (b) | 33. (b) | 34. (a) | 35. (c) | 36. (c) |
| 37. (d) | 38. (b) | 39. (a,b,c,d) | 40. (b,d) | 41. (a,b,d) | 42. (c) | 43. (b) | 44. (c) | 45. (c,d) |
| 46. (a,b,c) | 47. (a,c) | 48. (a) | 49. (c) | | | | | |

Measurement and Error

- | | | | | | | | | |
|---------|----------------|----------|---------|-------------|---------|---------|---------|---------|
| 1. (d) | 2. (d) | 3. (a) | 4. (b) | 5. (a) | 6. (b) | 7. (a) | 8. (c) | 9. (a) |
| 10. (b) | 11. (2.66g/cc) | | 12. (c) | 13. (d) | 14. (b) | 15. (b) | 16. (d) | 17. (c) |
| 18. (a) | 19. (b) | 20. (4%) | 21. (a) | 22. (a,b,d) | 23. (b) | 24. (b) | 25. (c) | 26. (3) |



Solutions

1. Electrostatic Field
2. Electrostatic Potential
3. Miscellaneous Problems on Chapters 1 & 2
4. Capacitors
5. Miscellaneous Problems on Chapter 4
6. Electric Current and Resistance
7. Basic Electrical Measurement
8. Transient RC Circuits
9. Miscellaneous Problems on Chapters 6–8
10. Error and Measurement
11. Miscellaneous Problems on Chapter 10
12. Previous Years' JEE Questions

Chapter 1 Electrostatic Field



Your Turn

1. Body loses electrons and electrons have mass. Therefore, mass of the body decreases.
2. $n = \frac{q}{e} = \frac{1}{1.6 \times 10^{-19}} = 6.25 \times 10^{18}$
4. As the tanker moves its body acquires charge due to continuous rubbing by dust and air. If not removed, this static charge can cause spark and hence fire. The metallic chain ensures that all such charge flows to the ground.
5. Some charge from the comb gets transferred to paper pieces. Both have net negative charge after this. They repel.
6. Number of electrons removed is

$$n = 0.01\% \text{ of } 10^{22} = \frac{0.01}{100} \times 10^{22} = 10^{18}$$

Since the body loses electrons, it will acquire positive charge

$$Q = +ne = 10^{18} \times 1.6 \times 10^{-19} \text{C} = 0.16 \text{C}.$$

$$7. F = \frac{1}{4\pi\epsilon_0} \frac{qQ}{r^2}$$

$$\therefore [\epsilon_0] = \frac{[qQ]}{[F][r^2]} = \frac{[AT]^2}{[M^1 L^1 T^{-2}][L^2]}$$

[$\therefore$ Coulomb = ampere $\times$ second]

$$\therefore [\epsilon_0] = [M^{-1} L^{-3} T^4 A^2]$$

$$8. F_0 = \frac{1}{4\pi\epsilon_0} \frac{qQ}{r^2} \text{ in vacuum}$$

$$F_0 = \frac{1}{4\pi\epsilon_0\epsilon_r} \frac{qQ}{(r/2)^2} \text{ in medium}$$

$$\Rightarrow F_0 = \frac{4}{\epsilon_r} F_0$$

$$\Rightarrow \epsilon_r = 4$$

$$9. F = k \frac{x(q-x)}{r^2}$$

F is maximum when $y = x(Q-x)$ is maximum

$$\Rightarrow \frac{dy}{dx} = 0 \Rightarrow Q - 2x = 0$$

$$\Rightarrow x = \frac{Q}{2}$$

$$10. \frac{mv^2}{r} = \frac{1}{4\pi\epsilon_0} \frac{(e)(e)}{r^2} \Rightarrow v = \sqrt{\frac{e^2}{4\pi\epsilon_0 mr}}$$

11. Vector connecting q_1 to q_2 is

$$\vec{r} = (4-1)\hat{i} + (5-1)\hat{j} = 3\hat{i} + 4\hat{j}$$

$$\text{Unit vector along } \vec{r} \text{ is } \hat{r} = \frac{3}{5}\hat{i} + \frac{4}{5}\hat{j}$$

Distance between the two charges is $r = |\vec{r}| = 5\text{m}$

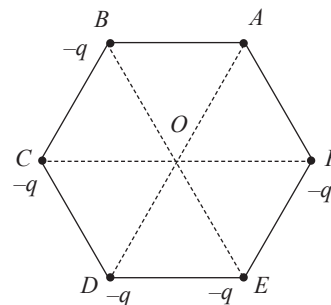
$$\begin{aligned} \therefore \vec{F} &= k \frac{q_1 q_2}{r^2} \hat{r} \\ &= 9 \times 10^9 \times \frac{(10 \times 10^{-6}) \times (20 \times 10^{-6})}{5^2} \left(\frac{3}{5}\hat{i} + \frac{4}{5}\hat{j} \right) \\ &= 1.44 \times 10^{-2} (3\hat{i} + 4\hat{j}) \\ &= (0.0432\hat{i} + 0.0576\hat{j}) \text{N} \end{aligned}$$

12. Smallest charge occurring in nature is e .

Force will be minimum when both the charges are e .

$$\begin{aligned} F &= \frac{1}{4\pi\epsilon_0} \frac{e \cdot e}{r^2} \\ &= 9 \times 10^9 \times \frac{(1.6 \times 10^{-19})^2}{1^2} = 2.304 \times 10^{-28} \text{N} \end{aligned}$$

13. Vertex A does not have a charge. Had there been a charge at A also, the net force on the central charge would have been zero.



$$\vec{F}_A + \vec{F}_B + \vec{F}_C + \vec{F}_D + \vec{F}_E + \vec{F}_F = 0$$

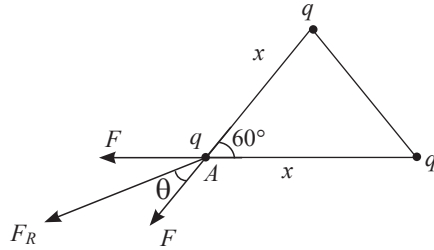
$$\therefore \vec{F}_B + \vec{F}_C + \vec{F}_D + \vec{F}_E + \vec{F}_F = -\vec{F}_A$$

$$\therefore \vec{F}_{\text{required}} = -\vec{F}_A$$

$\vec{F}_A$ is along $\overrightarrow{OA}$. Hence, the required force is along

$$\overrightarrow{OA} \text{ magnitude of force} = |\vec{F}_A| = \frac{kq^2}{x^2}.$$

14. Force on charge at A due to each of the other two charges has magnitude $F = k \frac{q^2}{x^2}$.



The two forces are inclined at 60° and their resultant is

$$F_R = \sqrt{F^2 + F^2 + 2FF \cos 60^\circ} = \sqrt{3}F = \frac{\sqrt{3}kq^2}{x^2}$$

The resultant is in the direction shown with $\theta = 30^\circ$

15. With tension in the strings equal to zero, the charge particle is in equilibrium

$$F_A + mg = F_B$$

$$\Rightarrow k \frac{qQ}{x^2} + mg = k \frac{q(2Q)}{x^2}$$

$$\Rightarrow k \frac{qQ}{x^2} = mg$$

$$\Rightarrow q = \frac{mg \cdot x^2}{k \cdot Q}$$

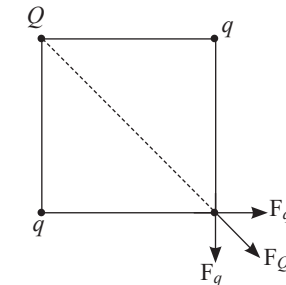
16. If q is also positive, equilibrium of Q will not be possible. All 3 forces on Q will be outward and their resultant cannot be zero.

Equilibrium of Q is possible only when q is negative. In this case direction of the three forces on Q is as shown in second figure.

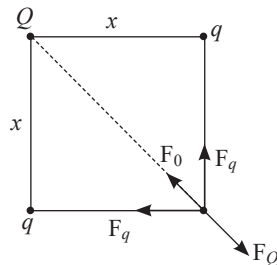
$$|F_q| = k \frac{qQ}{x^2}$$

$$|F_Q| = k \frac{Q^2}{(\sqrt{2}x)^2} = \frac{kQ^2}{2x^2}$$

Resultant of two perpendicular forces (shown as F_q) is F_0



Resultant cannot be zero if q is positive.



$$F_0 = \sqrt{2} |F_q|$$

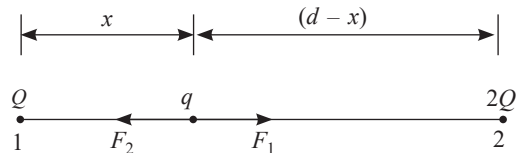
For equilibrium:

$$F_Q = F_0$$

$$\Rightarrow \frac{kQ^2}{2x^2} = \frac{\sqrt{2}kqQ}{x^2}$$

$$\Rightarrow q = \frac{Q}{2\sqrt{2}}$$

17. Irrespective of the sign of q , equilibrium is possible when it is placed at a particular point between the two charges. If q is placed to the left of Q or to the right of $2Q$, it will experience both forces in the same direction. Equilibrium will not be possible.



Let q be at a distance x from Q in equilibrium position.

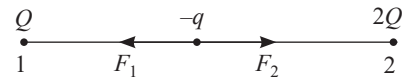
$$F_1 = F_2$$

$$k \frac{qQ}{x^2} = k \frac{q(2Q)}{(d-x)^2}$$

$$\Rightarrow \frac{d-x}{x} = \sqrt{2}$$

$$\Rightarrow x = \frac{d}{\sqrt{2} + 1}$$

Assume that q is positive. If it is displaced slightly towards Q , F_1 will increase and F_2 will decrease. This will create a restoring force. q is in stable equilibrium.



If q is negative and is pushed towards Q , the force F_1 increases and F_2 decreases to create a resultant force towards left. q will move further towards Q . Equilibrium is unstable.

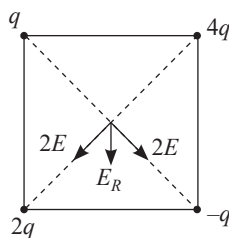
18. Distance of each charge from centre is $r = \frac{\sqrt{2}x}{2} = \frac{x}{\sqrt{2}}$

Field due to q at the centre is

$$E = \frac{kq}{r^2} \text{ (directed towards } -q\text{)}$$

Field due to $-q$ is also E directed towards itself. These two fields add to give a resultant of $2E$ towards $-q$. Field due to $2q$ is $2E$ towards $4q$ and field due to $4q$ is $4E$ towards $2q$. Resultant of

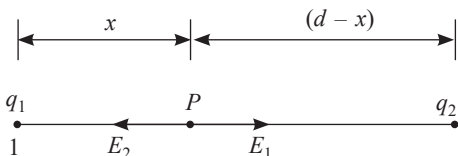
these two fields is $2E$ directed towards charge $2q$ (see figure).



$$\begin{aligned}\text{Resultant field is } E_R &= \sqrt{(2E)^2 + (2E)^2} = 2\sqrt{2}E \\ &= 2\sqrt{2} \frac{kq}{r^2} = 4\sqrt{2} \frac{kq}{x^2}\end{aligned}$$

Direction of the resultant field is perpendicular to the line joining $2q$ and $-q$.

19. One can easily see that the point where $E = 0$ will lie somewhere between the two charges. Fields E_1 and E_2 have opposite directions and can cancel out.



For zero field at P :

$$E_1 = E_2$$

$$k \frac{q_1}{x^2} = k \frac{q_2}{(d-x)^2}$$

$$\Rightarrow \left(\frac{d-x}{x} \right)^2 = \frac{q_2}{q_1}$$

$$\Rightarrow \left(\frac{d-x}{x} \right)^2 = 4$$

$$\Rightarrow \frac{d-x}{x} = 2$$

$$\Rightarrow x = \frac{d}{3} = \frac{2}{3}m$$

$$20. E = \frac{F}{q} = \frac{3.2 \times 10^{-4} N}{1.6 \times 10^{-10} C} = 2 \times 10^6 \text{ N/C}$$

21. From example 12:

$$E = 2kq \frac{y}{(a^2 + y^2)^{3/2}} \quad (i)$$

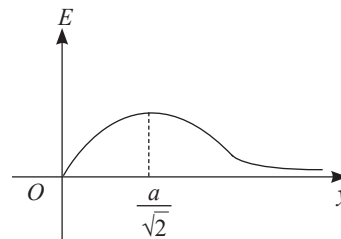
E is zero at $y = 0$ and it must be zero when $y \rightarrow \infty$. Hence it is obvious that field increases as we move from O along y axis, becomes maximum at a point and then decreases to zero. E is maximum at a point

$$\text{where } \frac{dE}{dy} = 0$$

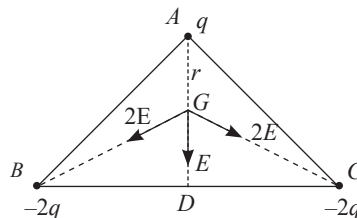
Differentiating (i) and equating it to zero gives

$$y = \frac{a}{\sqrt{2}}$$

Graph of E versus y is as shown.



22. Distance of a vertex from the centroid is $r = \frac{a}{\sqrt{3}}$



Field at the centroid (G) due to charge at A is along AD .

$$\text{Its magnitude is } E = k \frac{q}{r^2} = \frac{3kq}{a^2}$$

Field at G due to other two charges have magnitude $2E$ and are directed along GB and GC .

$$\angle BGC = 120^\circ$$

Therefore, resultant of fields due to negative charges is $2E$ along GD

$$\therefore \text{Resultant field at } G = 2E + E = 3E = \frac{3kq}{a^2}$$

Direction of the field is along GD .

23. Vector joining the source charge to the concerned point is

$$\vec{r} = (1-0)\hat{i} + (2-1)\hat{j} + (3-2)\hat{k} = \hat{i} + \hat{j} + \hat{k}$$

$$\text{Distance of point from the charge, } r = |\vec{r}| = \sqrt{3}m$$

$$\text{Unit vector along } \vec{r} \text{ is } \hat{r} = \frac{1}{\sqrt{3}}(\hat{i} + \hat{j} + \hat{k})$$

$$\therefore \vec{E} = k \frac{q}{r^2} \hat{r}$$

$$= \frac{9 \times 10^9 \times (-1 \times 10^9)}{3} \frac{1}{\sqrt{3}} (\hat{i} + \hat{j} + \hat{k}) \text{ N/C}$$

24. (a) Field due to charges at opposite corners cancel out. Hence resultant is zero.

- (b) If charge is removed from numeral 12 on the face of the clock, the field at the centre due to charge at 6 will remain unbalanced. Field due to all other charges cancel out in pair.

$\therefore$ Field at the centre is $E = \frac{kq}{r^2}$ towards numeral 12

$\therefore$ Force on the charge at the centre is $F = Eq = \frac{kq^2}{r^2}$ towards numeral 12.

25. For quarter ring $E_x = E_y = E$

Please note that in example 15, charge density (and NOT the charge) was same for half ring and quarter ring.

26. Field due to the ring is downward at the location of the particle

$$E = \frac{kQh}{(a^2 + h^2)^{3/2}} \quad \text{from equation (11)}$$

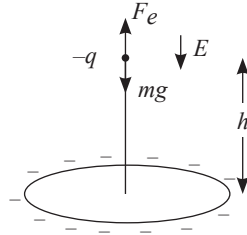
Electric force on the particle must be upward to balance its weight. This means that the particle carries a negative charge ($-q$).

$$F_e = mg \Rightarrow qE = mg$$

$$q \frac{kQh}{(a^2 + h^2)^{3/2}} = mg$$

$$\therefore q = \frac{4\pi\epsilon_0 mg(a^2 + h^2)^{3/2}}{Q \cdot h}$$

q is negative.



27. In example 15, limits of integration will change. The limits will be $-\theta$ to θ (in place of $-\frac{\pi}{2}$ to $\frac{\pi}{2}$)

28. Charge on the ring is $q = \lambda \cdot 2\pi a$.

Field due to ring at a point on its axis at a distance x from its centre is given by equation (11)

$$E = \frac{1}{4\pi\epsilon_0} \frac{\lambda \cdot 2\pi ax}{(a^2 + x^2)^{3/2}} = \frac{\lambda a}{2\epsilon_0} \frac{x}{(a^2 + x^2)^{3/2}}$$

$$\text{when } x \ll a; E \simeq \frac{\lambda a}{2\epsilon_0} \frac{x}{a^3} = \frac{\lambda}{2\epsilon_0 a^2} x$$

At the centre of the ring the particle is in equilibrium. When displaced by a small amount x , it experiences an unbalanced force towards the centre of the ring. This is the restoring force.

$$\therefore m \frac{d^2 x}{dt^2} = -QE$$

$$\Rightarrow \frac{d^2 x}{dt^2} = -\frac{Q\lambda}{2m\epsilon_0 a^2} x$$

$\therefore$ Motion is SHM and angular frequency is

$$\omega = \sqrt{\frac{Q\lambda}{2m\epsilon_0 a^2}}$$

$\therefore$ Time period is $T = \frac{2\pi}{\omega} = 2\pi \sqrt{\frac{2m\epsilon_0 a^2}{Q\lambda}}$

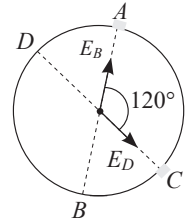
29. When the ring is intact, field at the centre is zero due to cancellation of fields produced by identical segments at diametrically opposite ends.

When a small segment at A is removed, field due to an identical segment at B remains unbalanced, (shown as E_B). Similarly, removing the segment at C , leaves the field due to segment at D unbalanced (shown as E_D)

$$E_B = E_D = \frac{1}{4\pi\epsilon_0} \frac{\Delta q}{R^2}$$

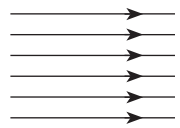
$$\text{and, } \Delta q = \frac{Q\Delta L}{2\pi R} \quad (\text{a small charge})$$

$$\therefore E_B = E_D = \frac{Q\Delta L}{8\pi^2 \epsilon_0 R^3}$$



Resultant of E_B and E_D is along the line bisecting the angle between E_B and E_D and its magnitude is same as E_B .

30. Constant magnitude means density of lines is same everywhere. Lines must be parallel to indicate no change in direction when we move from one point to another. Therefore, a uniform field is represented by equally spaced parallel lines.



31. (a) Figure shows 9 lines originating from q_1 and 18 lines origination from q_2 . Hence, $q_2 = 2q_1$

(b) At point A , $E = 0$

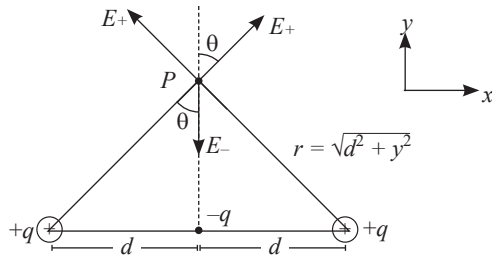
$$\therefore k \frac{q_1}{q_2} = k \frac{q_2}{(d-x)^2} \Rightarrow \frac{1}{x^2} = \frac{2}{(d-x)^2}$$

$$\Rightarrow \frac{d-x}{x} = \sqrt{2} \Rightarrow x = \frac{d}{(\sqrt{2}+1)}$$

32. (a) Field lines originate at positive charges and terminate at negative charges. Therefore, q_1 and q_3 are positive and q_2 is negative.

(b) Number of lines starting/terminating at all three charges is same (there are 10 lines for each charge). Hence all three have same magnitude.

(c) Field can be zero at a point P on Y axis as shown in figure.



E_+ denotes field due to positive charges and E_- is field due to negative charge. The x components of two E_+ field cancels out. For field to have zero y component we must have

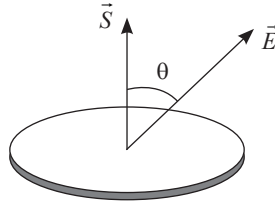
$$2E_+ \cos \theta = E_-$$

$$\Rightarrow 2k \frac{q}{(d^2 + y^2)} \frac{y}{\sqrt{d^2 + y^2}} = k \frac{q}{y^2}$$

$$\Rightarrow 2y^3 = (d^2 + y^2)^{3/2}$$

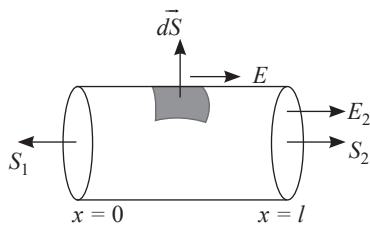
$$\Rightarrow 4y^6 = (d^2 + y^2)^3$$

33. Consider the area vector in shown direction (one can consider it to be in opposite direction as well). Angle between $\vec{E}$ and $\vec{S}$ is $\theta = 60^\circ$ (surface makes 30° with E).



$$\therefore \phi = ES \cos 60^\circ = 10 \times [\pi \cdot (1)^2] \times \frac{1}{2} = 5\pi \frac{\text{Nm}^2}{\text{C}}$$

34. There are three surface in the cylinder – one curved cylindrical surface and two flat circular surfaces. Let's first find the flux through the curved surface



Take an infinitesimally small area dS on the surface. Area vector $\vec{dS}$ is normal to the surface but electric field ($\vec{E}$) is parallel to the axis

$$\therefore d\phi = EdS \cos 90^\circ = 0$$

This is true for each small area of the cylindrical surface. Therefore, flux through the curved surface is zero.

At $x = 0$, electric field is zero. Therefore, flux through flat surface S_1 is zero.

At $x = l$, $E = al$

Area vector ($\vec{S}_2$) of the flat face is along x direction as shown.

$$\therefore \phi = ES \cos 0^\circ = alS$$

$$\therefore \text{Total flux is } \phi = a \cdot lS$$

35. (a) $q_{\text{in}} = 0$

$$\therefore \phi = 0$$

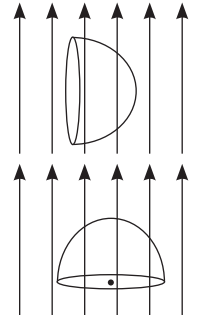
- (b) After Q is placed, field (E) will change but the flux through the surface still remains zero as it depends only on the charge inside the sphere.

36. (a) Field line which pierce into the curved surface they also penetrate out from the surface.

$$\therefore \phi = 0$$

- (b) Number of lines penetrating through the circular base = number of lines which pass through the curved surface.

$$\therefore \text{Flux through the curved surface} = \text{flux through the circular base} = E\pi R^2$$

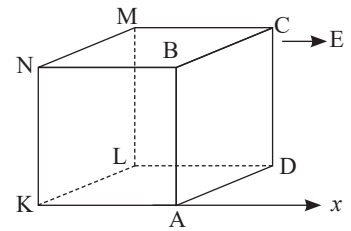


37. Normal to the surface is along x direction. Area vector can be written as:

$$\vec{S} = 0.1\hat{i}$$

$$\therefore \phi = \vec{E} \cdot \vec{S} = 12 \times 0.1 = 1.2 \frac{\text{Nm}^2}{\text{C}^2}$$

38. Consider field to be along x direction. Imagine a cubical surface as shown. Flux through the cube is zero as number of field lines going in is same as number of lines going out. From the Gauss's law, net charge inside the cube will be zero.



The above result remains valid even when the cube is made smaller. Therefore, there must not be any charge in the region.

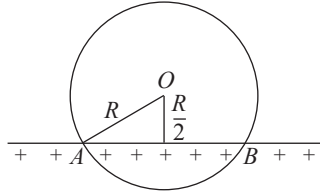
39. You can imagine a cube of side length a with q at its centre. The given surface is one face of the cube.

40. Imagine a complete sphere with q at centre. Flux through the entire sphere = $\frac{q}{\epsilon_0}$

$$\text{Flux through a hemisphere} = \frac{q}{2\epsilon_0}$$

41. All we need to look at is the total charge that is present inside the sphere. For this we need to know the length of the line segment inside the sphere.

$$AB = 2\sqrt{R^2 - \frac{R^2}{4}} = \sqrt{3}R$$



$\therefore$ Charge inside the sphere is $q_{in} = \lambda \cdot \sqrt{3}R$

$$\therefore \phi = \frac{q_{in}}{\epsilon_0} = \frac{\sqrt{3}\lambda R}{\epsilon_0}$$

42. The flux is maximum when angle between the area vector and the field direction is 0° .

$$\Rightarrow ES \cos 0^\circ = 5.02 \times 10^5$$

$$\Rightarrow E[3.14(0.4)^2] = 5.02 \times 10^5$$

$$\Rightarrow E = 10^6 \text{ N/C}$$

43. Imagine a triangle of base length $2R$ and height h that is held vertical normal to the field E . Flux through this triangle is

$$\phi = ES \cos 0^\circ = E\left(\frac{1}{2}2R \cdot h\right) = ERh$$

Number of lines passing through such a triangle will be same as the number of lines entering the cone.

Therefore, the required answer is $\phi = ERh$

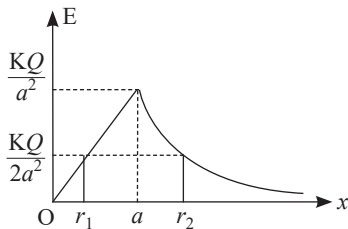
44. $E = 0$ for $0 \leq x \leq a$

$$= \frac{1}{4\pi\epsilon_0} \frac{q}{x^2} \text{ for } x > a$$

The graph is as given in the answer sheet.

45. (a) $E = \left(\frac{kQ}{a^3}\right)x$ for $0 \leq x \leq a$

$$= \frac{KQ}{x^2} \text{ for } x \geq a$$



[Using Equation (21) and (18)].

For $0 \leq x \leq a$ the graph is linear. For $x > a$ the electric field falls off as $E \propto \frac{1}{x^2}$. At the surface the field is maximum, its value being $E_{\text{surface}} = \frac{kQ}{a^2}$.

- (b) It is obvious from the graph that field will be half its maximum (at the surface) for two distances from the centre.

At distance $r_1 (< a)$ from the centre

$$E = \left(k \frac{Q}{a^3}\right)(r_1) = \frac{E_{\text{surface}}}{2}$$

$$\Rightarrow \frac{kQ}{a^3} r_1 = \frac{kQ}{2a^2} \Rightarrow r_1 = \frac{a}{2}$$

At distance $r_2 (> a)$ from the centre

$$E = \frac{kQ}{r_2^2} = \frac{E_{\text{surface}}}{2}$$

$$\Rightarrow \frac{kQ}{r_2^2} = \frac{kQ}{2a^2} \Rightarrow r_2 = \sqrt{2}a$$

46. When charge is distributed with spherical symmetry, the field at a point at a distance x from the centre depends only on the quantity of charge inside the sphere of radius x .

When charge in the region $\frac{a}{2} < x < a$ is removed,

it will make no difference to the field at $x = \frac{a}{4}$ but the field at $x = 2a$ will decrease.

47. Consider three Gaussian spheres shown as S_1 , S_2 and S_3 . Due to symmetry we can assume that field is radial everywhere.

- (a) For Gaussian surface S_1

$$E \cdot 4\pi x^2 = 0 \quad [\because \text{Charge inside } S_1 \text{ is zero}]$$

$$\Rightarrow E = 0$$

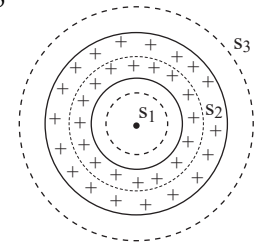
- (b) For Gaussian surface S_2 :

$$q_{in} = \rho \left(\frac{4}{3}\pi x^3 - \frac{4}{3}\pi R^3 \right) = \frac{4}{3}\pi \rho (x^3 - R^3)$$

$$\therefore E \cdot 4\pi x^2 = \frac{q_{in}}{\epsilon_0}$$

$$\Rightarrow E \cdot 4\pi x^2 = \frac{4\pi \rho}{3\epsilon_0} (x^3 - R^3)$$

$$\Rightarrow E = \frac{\rho}{3\epsilon_0} \left(x - \frac{R^3}{x^2} \right)$$



Sphere shown in broken lines are Gaussian surfaces.

- (c) For Gaussian surface S_3 :

$$q_{in} = \rho \left[\frac{4}{3}\pi (2R)^3 - \frac{4}{3}\pi R^3 \right] = \frac{28\pi R^3 \rho}{3}$$

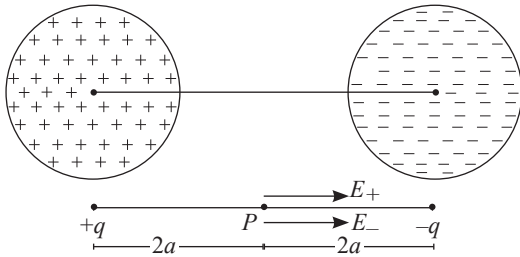
$$\therefore E \cdot 4\pi x^2 = \frac{28\pi R^3 \rho}{3\epsilon_0}$$

$$\Rightarrow E = \frac{7\rho R^3}{3\epsilon_0 x^2}$$

48. For finding field at an outside point we can replace a spherically symmetric charge distribution with an equal point charge at its centre.

Field at P is:

$$\begin{aligned} E &= E_+ + E_- \\ &= \frac{kq}{(2a)^2} + \frac{kq}{(2a)^2} \\ &= \frac{kq}{2a^2} \end{aligned}$$



49. Consider co-axial cylinder of length l as your Gaussian surface

$$\phi = E \cdot 2\pi r l$$

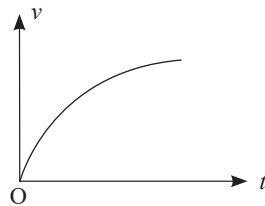
For $r < a$, $q_{\text{in}} = 0$

$$\Rightarrow E = 0$$

For $r > a$, $q_{\text{in}} = 2\pi a l \cdot \sigma$

$$\therefore E \cdot 2\pi r l = \frac{2\pi a l \sigma}{\epsilon_0} \Rightarrow E = \frac{\sigma a}{\epsilon_0 r}$$

50. Field is radially inward towards the axis of the cylinder. Negative electron will experience a radially outward force. Field decreases with distance r as given by equation (23). Force on electron decreases. Its acceleration goes on decreasing. At a large distance it becomes zero.



51. $E = \frac{\lambda}{2\pi\epsilon_0 r}$

$$\frac{1}{2\pi\epsilon_0} = 2 \times 9 \times 10^9 = 18 \times 10^9 \frac{\text{Nm}^2}{\text{C}^2}$$

$$\therefore E = 18 \times 10^9 \times \frac{2 \times 10^{-6}}{0.02} = 18 \times 10^5 \text{ N/C}$$

52. Field is radially outward. A negative electron experiences an inward force that provides the necessary centripetal acceleration

$$\frac{mv^2}{r} = e \cdot E \Rightarrow \frac{1}{2}mv^2 = erE$$

$$\therefore KE = e \cdot r \frac{\lambda}{2\pi\epsilon_0 r} = \frac{\lambda e}{2\pi\epsilon_0}$$

The answer is independent of r !

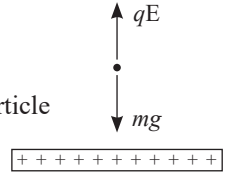
53. (a) Field above the surface is uniform.

$$E = \frac{\sigma}{2\epsilon_0}$$

For equilibrium of the particle

$$qE = mg$$

$$\Rightarrow q = \frac{mg}{E} = \frac{2mg\epsilon_0}{\sigma} \quad [\text{independent of } h]$$



- (b) When the particle is moved a little, neither the electric force nor the gravitational force changes. Net force on it remains zero. The particle is in a state of neutral equilibrium.

54. Electric field due to charge on the vertical surface is

$$E = \frac{\sigma}{2\epsilon_0}$$

Forces on the charged ball are:

Tension (T), weight (mg) and electrostatic force (qE).

For equilibrium in horizontal and vertical directions we must have

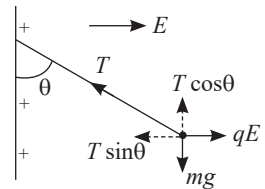
$$T \sin \theta = qE \quad \text{and} \quad T \cos \theta = mg$$

$$\Rightarrow \tan \theta = \frac{qE}{mg} \Rightarrow \tan 60^\circ = \frac{q\sigma}{2\epsilon_0 mg}$$

$$\Rightarrow \sigma = \frac{2\sqrt{3}\epsilon_0 mg}{q}$$

Substituting for ϵ_0 , m , g and q gives

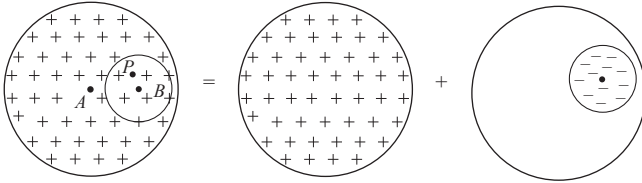
$$\sigma = 7.5 \times 10^{-7} \text{ C/m}^2$$



55. Answer is given by equation 26. Size of the cavity has no influence on the result.

56. Proceed as we did to find equation (26).

57. Imagine the charge system as superposition of a positively charged cylinder and a negatively charged cylinder. Figure shows the cross section.

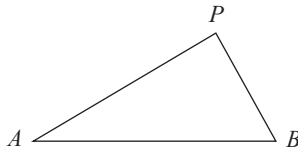


Field at P inside cavity is given by

$$\vec{E} = \vec{E}_+ + \vec{E}_-$$

$$\vec{E}_+ = \frac{\rho}{2\epsilon_0}(\overline{AP}) \quad [\text{using equation 24}]$$

$$\begin{aligned} \vec{E}_- &= \frac{-\rho}{2\epsilon_0}(\overline{BP}) \\ &= \frac{\rho}{2\epsilon_0}(\overline{PB}) \end{aligned}$$



$$\begin{aligned} \therefore \vec{E} &= \frac{\rho}{2\epsilon_0}(\overline{AP} + \overline{PB}) \\ &= \frac{\rho}{2\epsilon_0}(\overline{AB}) = \frac{\rho}{2\epsilon_0}\vec{l} \end{aligned}$$

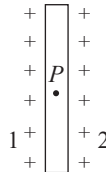
58. Radius of curvature is smaller for the conductor surface near 1 as compared to 2.

$\therefore$ Charge density is higher at 1.

$$\therefore E_1 > E_2 \quad \left[\because E = \frac{\sigma}{\epsilon_0} \right]$$

59. An outside charge will have no effect inside the cavity. The charge distribution on outer surface will change when Q is brought in. This will change the field lines emanating from outer surface of the ball. But all this does not change anything inside the cavity (electrostatic shielding).

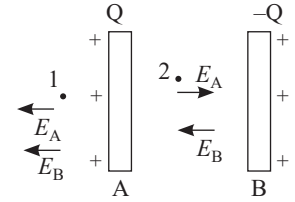
61. Equal charge on the two surfaces will ensure that the field in the material of the plate is zero. Field at P is zero if both faces 1 and 2 have equal charge.



62. Field at point 3 is zero in all cases as the point is inside a conductor.

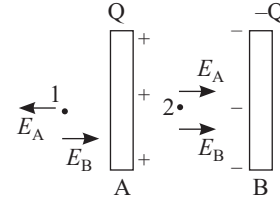
(a) Charge distribution is as shown (refer to remarks in example 30).

$$\begin{aligned} E_1 &= E_A + E_B \\ &= \frac{Q}{2A\epsilon_0} + \frac{Q}{2A\epsilon_0} \\ &= \frac{Q}{A\epsilon_0} \end{aligned}$$



$$E_2 = E_A - E_B = 0$$

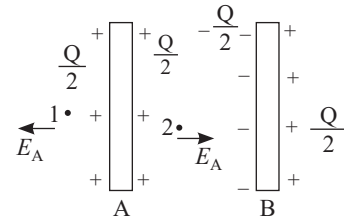
(b) Charge distribution is as shown.



$$E_1 = E_A - E_B = 0$$

$$E_2 = E_A + E_B = \frac{Q}{A\epsilon_0}$$

(c) Charges on the surfaces are as shown.



Charge on plate B does not create any field at 1 or 2. (charge on the plate B is zero)

$$\therefore E_1 = E_A = \frac{Q}{2A\epsilon_0}$$

$$E_2 = E_A = \frac{Q}{2A\epsilon_0}$$

63. For net force to be zero, the electric force on the particle must be upward. This implies that the electric field is downward.

$$qE = mg$$

$$\Rightarrow E = \frac{mg}{q}$$

64. Tangent to a field line is the direction of acceleration of the particle but it is not the direction of velocity.

$$65. F_x = qE$$

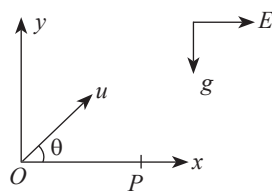
$$\Rightarrow a_x = \frac{qE}{m}(\rightarrow) = 40 \text{ ms}^{-2}$$

$$F_y = mg \Rightarrow a_y = g (\downarrow)$$

The particle is projected from O and lands on the horizontal surface at P . Time of flight is given by considering motion in y direction.

$$y = u_y t - \frac{1}{2} a_y t^2$$

$$0 = \frac{u}{\sqrt{2}} t - \frac{1}{2} g t^2$$

$$\Rightarrow t = \frac{\sqrt{2}u}{g} = 2\sqrt{2} \text{ s}$$


Distance OP is given by

$$OP = u_x t + \frac{1}{2} a_x t^2$$

$$= \frac{20}{\sqrt{2}} (2\sqrt{2}) + \frac{1}{2} (40) (2\sqrt{2})^2 = 200 \text{ m}$$

66. Work done by electric force on a charge in a closed path is zero.

67. $\sigma_M = \frac{M}{\pi R^2}$

For a charged sheet $E = \frac{\sigma}{2\epsilon_0}$

$$\Rightarrow E = \frac{2\pi\sigma}{2\pi\epsilon_0}$$

Replacing $\frac{1}{2\pi\epsilon_0}$ with G and charge density σ with surface mass density σ_M we get the gravitational field.

$$E_g = G \cdot 2\pi\sigma_M = G \cdot 2\pi \cdot \frac{M}{\pi R^2}$$

$$\Rightarrow E_g = \frac{2GM}{R^2}$$

68. Charge begins to leak when field near the surface of the sphere becomes equal to $3 \times 10^6 \text{ N/C}$

$$\therefore K \frac{q}{R^2} = 3 \times 10^6$$

$$\Rightarrow q = \frac{3 \times 10^6 \times 1^2}{9 \times 10^9} = \frac{1}{3} \times 10^{-3} \text{ C}$$



Worksheet 1

1. When the object is positively charged, it attracts electrons and induces negative charge on the metal disc. The gold leaves supply electrons and become more positive. They repel even strongly.

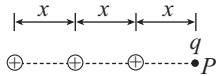
2. $K \frac{q(9e)}{x^2} = K \frac{q(e)}{(16-x)^2}$



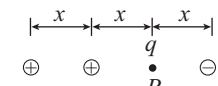
$$\Rightarrow \frac{16-x}{x} = \frac{1}{3}$$

$$\Rightarrow x = 12 \text{ cm}$$

3. Let the separation between two consecutive items be x . Charges are at distance x , $2x$, or $3x$ from P . Let the magnitude of force between a charge Q and q at separation x be F .

$$\Rightarrow \frac{1}{4\pi\epsilon_0} \frac{Qq}{x^2} = F$$


In arrangement I, force on charge at P is

$$F_I = F + \frac{F}{4} + \frac{F}{9} = \frac{49F}{36}$$


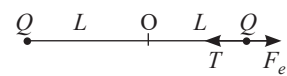
In arrangement II, force on charge at P is

$$F_{II} = F + F + \frac{F}{4} = \frac{9F}{4}$$

Similarly, one can write for other cases as well. Pay attention to direction of the forces.

4. Both balls apply equal and opposite forces on one another. When one of the charges is doubled, the electric force gets doubled on both of them. The two strings must make equal angles with the vertical.
5. Inside a satellite the effective value of acceleration due to gravity is zero. Therefore, the two charges repel each other and angle between the two strings becomes 180° .

For one ball

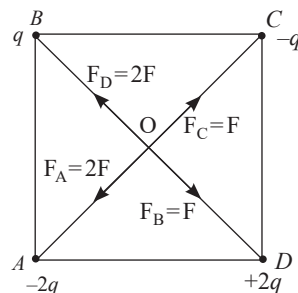


$$T = F_e$$

$$= \frac{1}{4\pi\epsilon_0} \frac{Q^2}{(2L)^2} = \frac{Q^2}{16\pi\epsilon_0 L^2}$$

O is point of suspension

6. Force applied by charge at C on the charge Q placed at the centre is represented by F_C and similar symbols have been used for other forces.



If $F_C = F$,

then $F_A = 2F$, $F_B = F$, $F_D = 2F$

Resultant of F_A and F_C is along OA and has magnitude F . Resultant of F_D and F_B is along

OB and has magnitude F . Resultant force will be parallel to DA (i.e. perpendicular to AB).

7. Electric fields due to charges at opposite corners cancel out in arrangement III.
8. Two forces of same magnitude at 120° angle have a resultant equal to either of the forces.
9. Vector joining the point charge to the concerned point is $\vec{r} = \vec{r}_1 - \vec{r}_0 = 6\hat{i} - 8\hat{j}$

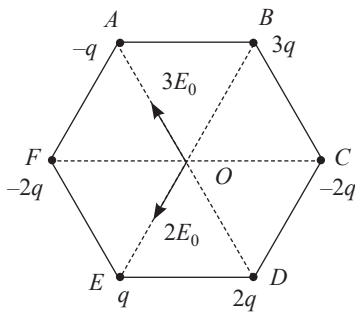
$$|\vec{r}| = \sqrt{6^2 + 8^2} = 10 \text{ m}$$

$$\begin{aligned} \therefore \vec{E} &= k \frac{q}{|\vec{r}|^3} \cdot \vec{r} = \frac{9 \times 10^9 \times 50 \times 10^{-6}}{10^3} (6\hat{i} - 8\hat{j}) \\ &= 450(6\hat{i} - 8\hat{j}) \text{ N/C} \\ &= 900(3\hat{i} - 4\hat{j}) \text{ N/C} \end{aligned}$$

From $x = 0$ to $x = L$, both charges produce field in positive x direction. Therefore, field is always positive between the two charges.

Field is very large near to point charges. Obviously, field becomes minimum somewhere between the charges. Since both charges have same magnitude, the minimum of the field will be at $x = \frac{L}{2}$ only.

11. Electric field due to charges at B and E are oppositely directed. If magnitude of field at the centre due to charge at E is E_0 , then magnitude of field due to charge at B will be $3E_0$. Their resultant is $2E_0$ along OE .

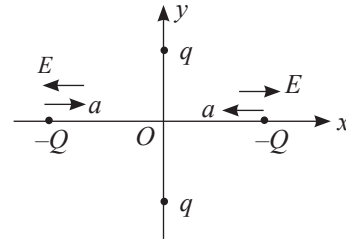


Field produced due to the charges at F and C cancel out. Field due to charges at A and D adds and the resultant is $3E_0$ along OA .

Resultant of $2E_0$ and $3E_0$ field will be some where between OF and OA .

12. Due to charges q and q on the y axis, electric field is zero at O . Field is also zero at large distance from O . This tells us that the field is zero at O , increases to some maximum value (as one moves along x axis away from O) and then again decreases to

zero. Acceleration will be proportional to the field. Directions of field (E) and acceleration (a) are as indicated in the diagram.



13. Refer to the above solution of Q.12.

Charge Q will get acceleration towards O . As it crosses O , direction of its acceleration reverses. It retards and stops. Then it again moves towards O . Motion is oscillatory but not SHM because restoring force is not proportional to x .

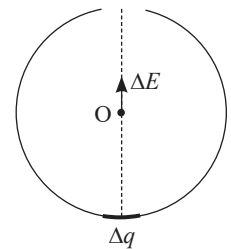
14. Electric field lines never form closed loops hence (B) and (D) are incorrect.
Field lines cannot originate and terminate at the charges of same sign.
15. Field lines must be normal to the surface of a conductor.

16. Charge per unit length is $\lambda = \frac{Q}{2\pi R}$

Charge on arc of length d is $\Delta q = \lambda d = \frac{Qd}{2\pi R}$

Field produced by an identical arc at diametrically opposite end remains unbalanced and field at the centre is

$$\Delta E = K \frac{\Delta q}{R^2} = K \frac{Qd}{2\pi R^3}$$



17. $F = k \frac{Q^2}{r^2}$

When A is placed in contact with B , charge on both of them becomes $q_A = q_B = \frac{Q}{2}$

Now A (having charge $\frac{Q}{2}$) is placed in contact with

C (having charge Q). Charge on C becomes

$$q_c = \frac{Q + Q/2}{2} = \frac{3Q}{4}$$

$$\therefore F_{BC} = K \frac{q_B q_c}{r^2} = K \frac{\left(\frac{Q}{2}\right) \left(\frac{3Q}{4}\right)}{r^2} = \frac{3}{8} F$$

19. The electric field over the entire area of the smaller ring can be assumed to be uniform due to its very small size. If the smaller ring is at co-ordinate x where the field due to the bigger ring is E , then the flux through it is

$$\phi = E \cdot S = \pi r^2 E$$

$$\therefore \phi \propto E$$

Variation of flux is same as variation of field along the axis.

Flux changes sign after crossing the origin because the direction of field changes.

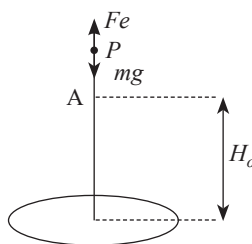
20. Field on the axis of ring is maximum at $H_0 = \frac{R}{\sqrt{2}}$

Above H_0 , as we move up, the field decreases. Consider a particle at P to be in equilibrium.

$$F_e = mg \text{ at } P.$$

When the particle is pushed up, F_e decreases and mg becomes a stronger force. This produces a restoring force back towards P .

When the particle is pushed a little bit downward from P , the electric force increases [$\because E$ is decreasing when we move above A . It means E is increasing when we move towards A]. This increase in F_e pushes the particle back to P . One can show that if a particle is in equilibrium at a point below A , it will not be in stable equilibrium.



21. Field lines emanate at a positive charge and terminate on a negative charge. Therefore, q_1 is positive and q_2 is negative.

More lines originate at q_1 and lesser number of lines terminate at q_2 .

$$\therefore |q_1| > |q_2|$$

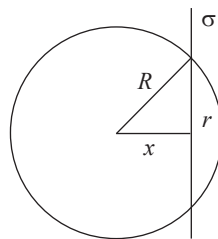
$$22. \vec{s} = (100m^2) \hat{k}$$

$$\vec{E} = (\hat{i} + \sqrt{2}\hat{j} + \sqrt{3}\hat{k}) \text{ Vm}^{-1}$$

$$\therefore \phi = \vec{E} \cdot \vec{S} = 100\sqrt{3} \text{ V}\cdot\text{m} = 173.2 \text{ V}\cdot\text{m}$$

$$23. \phi = \frac{q_{in}}{\epsilon_0}$$

We need to find charge inside the spherical Gaussian surface. The infinite plane surface intersects the spherical surface and a circular part of radius r of the charged surface is trapped inside the sphere.



$$r = \sqrt{R^2 - x^2}$$

$$\text{area of circle} = \pi r^2 = \pi(R^2 - x^2)$$

$$\therefore q_{in} = \sigma \cdot \pi(R^2 - x^2)$$

$$\therefore \phi = \frac{\pi\sigma(R^2 - x^2)}{\epsilon_0}$$

24. A rectangular area of the charged surface is inside the Gaussian cylinder. Area of rectangular surface is (see figure in last solution)

$$S = L \cdot (2r) = 2L\pi\sqrt{R^2 - x^2}$$

$$\therefore \phi = \frac{\sigma \cdot S}{\epsilon_0} = \frac{2\pi\sigma L\sqrt{R^2 - x^2}}{\epsilon_0}$$

25. Uniformly distributed charge on spherical surface produces no field at inside points. Therefore, no force acts on q due to the shell inside which it is located.

The other shell can be treated as a point charge placed at its centre.

$$\therefore F = \frac{1}{4\pi\epsilon_0} \frac{qQ}{\left(10d - \frac{d}{2}\right)^2} = \frac{qQ}{361\pi\epsilon_0 d^2} \text{ to left.}$$

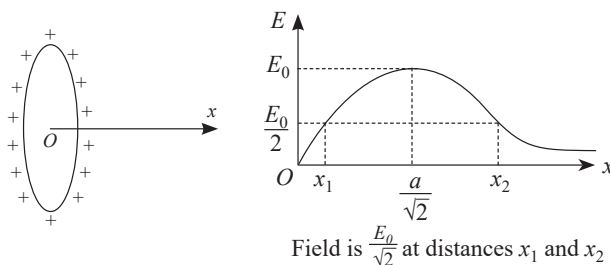
26. Field inside the cavity is uniform given by $\vec{E} = \frac{\rho}{3\epsilon_0} \vec{l}$. Therefore, force on q is along $\vec{l}$.

27. Time period of oscillation of a spring block system depends only on m and k

$$T = 2\pi\sqrt{\frac{m}{k}}$$

28. Variation of electric field on the axis of a charged ring of radius a is as shown in the adjoining graph.

Obviously, field is $\frac{E_0}{2}$ at two points on the right side of the ring (at distances x_1 and x_2 from the centre).



Similarly, two points exist to the left of the ring. Overall there are four such points.

29. Flux through the closed hemisphere (i.e., circular base plus curved surface) is zero.

∴ Outgoing flux through the hemisphere surface
= Flux incoming through the circular base

$$= (\pi R^2)(E) \cos 45^\circ = \frac{\pi}{\sqrt{2}} R^2 E$$

31. Position vector of point P relative to the centre of the sphere is

$$\vec{l} = \vec{r} - \vec{d}$$

$$\text{Field at } P \text{ is } \vec{E} = \frac{-\rho}{3\epsilon_0} \vec{l} = \frac{\rho}{3\epsilon_0} (\vec{d} - \vec{r})$$

32. Field to the left of charge q_1 is positive (directed to right). It means q_1 is negative. Field at a point close to q_2 and right of it is positive. This indicates that q_2 is positive. At large distance on positive x axis the field is negative (that is directed towards left). This indicates that $(q_1 + q_2)$ is a negative number.

$$\therefore |q_1| > |q_2|$$

33. Charge present inside a sphere of radius r ($< R$) is

$$q = \int_0^r \rho \cdot 4\pi x^2 dx = 4\pi \rho \int_0^r x^3 dx = \pi \rho r^4$$

$$\text{For } r = \frac{R}{2}$$

$$q = \frac{\pi \rho R^4}{16}$$

Considering a spherical Gaussian surface of radius $R/2$ we can write

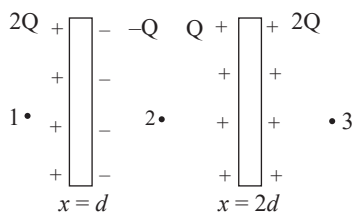
$$E \cdot 4\pi \left(\frac{R}{2}\right)^2 = \frac{\pi \rho R^4}{16\epsilon_0}$$

$$E = \frac{\rho R^2}{4\epsilon_0}$$

34. External field has no effect inside the cavity.

35. Charges on outer faces of the two plates are

$$\frac{Q_1 + Q_2}{2} = 2Q$$



Inner walls have charges $-Q$ and $+Q$ as shown.
Field at a point like 1 is

$$E_1 = \frac{2Q}{2A\epsilon_0} - \frac{Q}{2A\epsilon_0} + \frac{Q}{2A\epsilon_0} + \frac{2Q}{2A\epsilon_0} = \frac{4Q}{2A\epsilon_0} (\leftarrow)$$

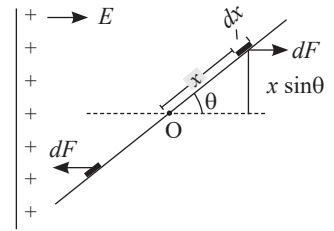
$$\text{Similarly, } E_2 = \frac{2Q}{2A\epsilon_0} (\leftarrow)$$

$$E_3 = E_1 = \frac{4Q}{2A\epsilon_0} (\rightarrow)$$

36. Field in front of the charged sheet is uniform and its magnitude is

$$E = \frac{\sigma}{2\epsilon_0}$$

Consider a segment of length dx in the rod at a distance x from its centre O . Force on this element is



$$dF = (dq)E = (\lambda dx) \frac{\sigma}{2\epsilon_0} = \left(\frac{\lambda \sigma}{2\epsilon_0} \right) dx$$

A similar element of the rod on other side experiences an equal force in the opposite direction. Therefore, one can see that the rod will experience no net force though it experiences torque.

Torque about O due to dF is

$$d\tau = (x \sin \theta)(dF) = \left(\frac{\lambda \sigma}{2\epsilon_0} \sin \theta \right) x dx$$

∴ Torque on one half of the rod is

$$\tau = \left(\frac{\lambda \sigma}{2\epsilon_0} \sin \theta \right) \int_0^l x dx = \frac{\lambda \sigma \sin \theta}{4\epsilon_0} l^2$$

$$\text{Total torque is } = 2\tau = \frac{\lambda \sigma \sin \theta}{4\epsilon_0} l^2$$

37. The ball will rise when electric force on it becomes larger than its weight

$$qE \geq mg$$

$$\Rightarrow q = \frac{mg}{E} = \frac{50 \times 10}{5 \times 10^5} = 10^{-3} \text{ C.}$$

Charge incident on the ball in time t is $\rho \cdot (\pi r^2 \cdot v) t$

$$\therefore t \rho \pi r^2 v = 10^{-3} \Rightarrow t = \frac{10^{-3}}{\rho \pi r^2 v}$$

$$\Rightarrow t = \frac{10^{-3}}{5 \times 10^{-12} \times \pi \times \left(\frac{2}{\sqrt{\pi}}\right)^2 \times 2 \times 10^6} = 25 \text{ s}$$

38. E = Field due to infinite sheet – Field due to a disc of radius r

$$\text{or, } \frac{\sigma}{2\sqrt{2}\epsilon_0} = \frac{\sigma}{2\epsilon_0} - \frac{\sigma}{2\epsilon_0} \left[1 - \frac{a}{\sqrt{a^2 + r^2}} \right]$$

[Refer to equation (12)]

$$\text{or, } \frac{1}{\sqrt{2}} = 1 - 1 + \frac{1}{\sqrt{a^2 + r^2}}$$

$$\text{or, } a^2 + r^2 = 2a^2 \Rightarrow r = a$$

39. Flux through the left face of cube is

$$\phi_1 = (-4xi + 6yj)(-1j) = -6y = 0 \quad [\because y = 0]$$

Flux through the right faces is

$$\phi_2 = (-4xi + 6yj)(1j) = 6y = 6 \quad [\because y = 1]$$

Flux through back face is

$$\phi_3 = (-4xi + 6yj)(-1i) = 4x = 0 \quad [\because x = 0]$$

Flux through the front face is

$$\phi_4 = (-4xi + 6yj)(1i) = -4x = -4 \quad [\because x = 1]$$

There is no flux through the other two faces

$$\therefore \phi = \phi_1 + \phi_2 + \phi_3 + \phi_4 = 2$$

Gauss's law tells us that

$$q = 2\epsilon_0$$

40. If the ball of mass m moves a distance x to left, the other ball must have moved a distance $(3R - x)$ so that they collide. All interactions are internal. The COM of the system shall not get displaced. Therefore,
- $$mx = 2m(3R - x)$$
- $$x = 2R.$$



Worksheet 2

$$1. \vec{E}_A = K \frac{q}{r_A^3} (\vec{r}_A) = Kq \frac{(\hat{i} + 2\hat{j} + 3\hat{k})}{(1^2 + 2^2 + 3^2)^{\frac{3}{2}}}$$

$$= \frac{Kq}{14\sqrt{14}} (\hat{i} + 2\hat{j} + 3\hat{k})$$

$$\vec{E}_B = K \frac{q}{r_B^3} (\vec{r}_B)$$

$$= Kq \frac{(\hat{i} + \hat{j} - \hat{k})}{(1^2 + 1^2 + 1^2)^{\frac{3}{2}}} = \frac{Kq}{3\sqrt{3}} (\hat{i} + \hat{j} - \hat{k})$$

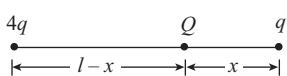
$$\vec{E}_C = \frac{Kq}{r_C^3} (\vec{r}_C) = \frac{Kq(2\hat{i} + 2\hat{j} + 2\hat{k})}{(2^2 + 2^2 + 2^2)^{\frac{3}{2}}}$$

$$= \frac{Kq}{4(3\sqrt{3})} (\hat{i} + \hat{j} + \hat{k})$$

$$\vec{E}_A \cdot \vec{E}_B = 0 \quad \therefore \vec{E}_A \perp \vec{E}_B$$

$$|\vec{E}_B| = 4|\vec{E}_C|$$

2. For Q to be in equilibrium

$$\frac{KqQ}{x^2} = k \frac{4q \cdot Q}{(l-x)^2}$$


$$\Rightarrow \frac{l-x}{x} = 2$$

$$\Rightarrow x = \frac{l}{3}$$

For equilibrium of Q , its sign is not important. But for equilibrium of q (or $4q$), Q must have opposite sign of q . Consider equilibrium of q .

Attraction of Q = Repulsion of $4q$.

$$\text{or, } \frac{K|Q|q}{x^2} = \frac{Kq(4q)}{l^2}$$

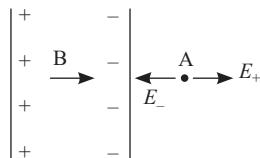
$$\text{or, } |Q| = 4q \left(\frac{x}{l} \right)^2 = 4q \left(\frac{1}{3} \right)^2 = \frac{4q}{9}$$

But sign of Q is opposite to that of q

$$\therefore Q = -\frac{4q}{9}$$

4. Charge will move along the field line only when the field is uniform and it is projected along the direction of $\vec{E}$ or is released from rest.
5. At a point like A , the fields due to the positive and negative charges are oppositely directed.

At a point like B , both the fields are in same direction

$$E_B = \frac{\sigma}{2\epsilon_0} + \frac{\sigma}{2\epsilon_0} = \frac{\sigma}{\epsilon_0}$$


Field due to the positive sheet (at the location of the negative sheet) is

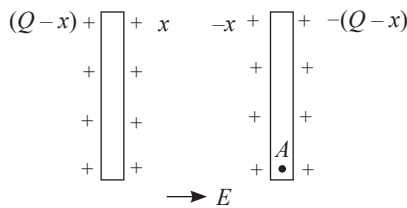
$$E_+ = \frac{\sigma}{2\epsilon_0}$$

S.16 Electrostatics and Current Electricity

This field exerts a force on the negative plate. Force per unit area is = Force on charge $-\sigma \cdot 1$

$$F = \sigma E_+ = \frac{\sigma^2}{2\epsilon_0}$$

10. Net force on the particle is $\vec{F} = q\vec{E} + m\vec{g}$. This is a constant. It means that the particle will experience a constant acceleration. If its initial velocity is along the acceleration, it will move on a straight line. If its initial velocity makes an angle with the acceleration, the path is parabolic.
11. The inner walls of the plates must have equal and opposite charges on them. Let the values be x and $-x$ as shown. The two outer walls have charges $(Q-x)$ and $-(Q-x)$.



The resultant field at a point A inside the metal must be zero

$$E + \frac{Q-x}{2A\epsilon_0} + \frac{x}{2A\epsilon_0} - \frac{x}{2A\epsilon_0} + \frac{Q-x}{2A\epsilon_0} = 0$$

$$\Rightarrow 2A\epsilon_0 E + 2Q = 2x$$

$$\Rightarrow x = Q + A\epsilon_0 E$$

Field at a point like R , due to charges on the four walls are zero. Therefore, field at R is equal to the external field E .

Field at point S is

$$\begin{aligned} E_S &= E + \frac{Q-x}{2A\epsilon_0} + \frac{x}{2A\epsilon_0} + \frac{x}{2A\epsilon_0} + \frac{Q-x}{2A\epsilon_0} \\ &= E + \frac{Q}{A\epsilon_0} \end{aligned}$$

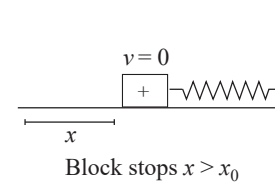
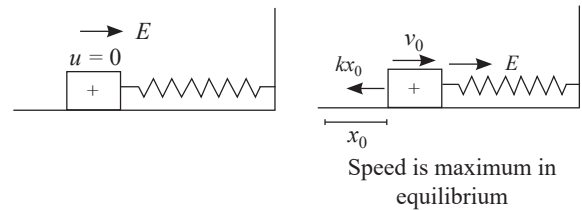
Note: For writing field at points like R or S we need not find the charge on the 4 surfaces, separately. Charge $+Q$ will produce a field at R towards left which will get cancelled by the field due to charge $-Q$ on the other plate. How the charges are divided on the two surface of a plate will make no difference to our calculation.

We can think as if there are two parallel plates having charges $+Q$ and $-Q$ and write the field at S due to these plates as

$$\frac{Q}{2A\epsilon_0} + \frac{Q}{2A\epsilon_0} = \frac{Q}{A\epsilon_0}$$

$$\therefore \text{Net field at } S \text{ is } E + \frac{Q}{A\epsilon_0}$$

12. Refer to Your Turn Q.17.
13. Refer to example 12 and Your Turn Q.21.
14. Initially, there is only an electric force on the block. It accelerates to the right. Spring gets compressed but the block accelerates till the electric force is larger than the spring force. At some point the spring force balances the electric force and after this the spring force exceeds the electric force. This retards the block and it stops. Speed of the block is maximum when it reaches the equilibrium position.



$$Kx_0 = QE$$

$$\Rightarrow x_0 = \frac{QE}{K}$$

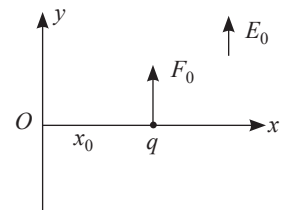
15. The particle experiences a constant force in y direction. Its momentum goes on increasing. Its angular momentum about O is

$$L = mv \cdot x_0$$

Therefore, L also increases.

The torque due to electric force about O remains constant

$$\tau = F_0 \cdot x_0$$



16. Charge density in the region $R_1 < r < R_2$ is

$$\rho = \frac{Q}{\frac{4}{3}\pi(R_2^3 - R_1^3)} = \frac{3Q}{4\pi(R_2^3 - R_1^3)}$$

If we fill the cavity (i.e., the region $r < R_1$) with charge density ρ , the entire ball will have uniform charge density and the field will vary linearly inside it.

However, the question asks us to keep a point charge Q_0 at the centre. We will keep a point charge

$$Q_0 = \rho \cdot \frac{4}{3} \pi R_1^3 = \frac{Q R_1^3}{R_2^3 - R_1^3} \text{ and the field in the region}$$

$R_1 < r < R_2$ will still behave as if the entire thing is a uniformly charged sphere.

With a point charge at the centre, the field inside the cavity will be inversely proportional to square of r .

17. Charge enclosed in a spherical volume of radius r is

$$Q = \int_0^r \rho \cdot \pi x^2 dx = 4\pi \rho_0 \int_0^r e^{-\alpha x^3} \cdot x^2 \cdot dx$$

For integration, put $-\alpha x^3 = t$

$$\Rightarrow -3\alpha x^2 \frac{dx}{dt} = 1$$

$$\Rightarrow x^2 dx = -\frac{1}{3\alpha} \cdot dt$$

$$\therefore \text{Integration } \int e^{-\alpha x^3} x^2 dx = -\frac{1}{3\alpha} \int e^t dt = -\frac{e^t}{3\alpha}$$

$$\begin{aligned} \therefore Q &= -\frac{4\pi\rho_0}{3\alpha} \left[e^{-\alpha x^3} \right]_0^r = -\frac{4\pi\rho_0}{3\alpha} \left[e^{-\alpha r^3} - 1 \right] \\ &= \frac{4\pi\rho_0}{3\alpha} \left[1 - e^{-\alpha r^3} \right] \end{aligned}$$

$$\text{when } \alpha r^3 \gg 1, e^{-\alpha r^3} = \frac{1}{e^{\alpha r^3}} \rightarrow 0$$

$$\therefore Q = \frac{4\pi\rho_0}{3\alpha}$$

$$\therefore E = \frac{1}{4\pi\epsilon_0} \frac{Q}{r^2} = \frac{\rho_0}{3\epsilon_0 \alpha r^2}$$

$$\text{The expansion of } e^{-x} = 1 - \frac{x}{1!} + \frac{x^2}{2!} - \frac{x^3}{3!} + \dots$$

when x is very small compared to 1 we can write $e^{-x} \approx 1 - x$

$$\therefore Q = \frac{4\pi\rho_0}{3\alpha} \left[1 - (1 - \alpha r^3) \right] = \frac{4\pi\rho_0 r^3}{3}$$

$$\text{Now, } E = \frac{1}{4\pi\epsilon_0} \frac{Q}{r^2} = \frac{\rho_0 r}{3\epsilon_0}$$

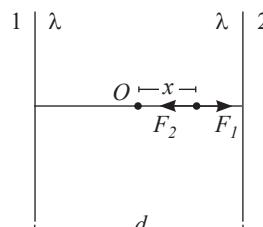
Note: For the case $\alpha r^3 \ll 1$, we can also think in slightly different manner.

$$e^{-\alpha r^3} \approx e^{-0} = 1$$

$$\therefore \rho \approx \rho_0$$

$$\therefore E = \frac{\rho r}{3\epsilon_0}$$

18. O is the equilibrium position for the charge. Consider it at a position x displaced from O .



$$ma = F_1 - F_2$$

$$= q \frac{\lambda}{2\pi\epsilon_0 \left(\frac{d}{2} + x \right)} - q \frac{\lambda}{2\pi\epsilon_0 \left(\frac{d}{2} - x \right)}$$

$$\therefore a = \frac{q\lambda}{2\pi\epsilon_0 m} \frac{(-2x)}{\left(\frac{d^2}{4} - x^2 \right)}$$

$$\text{For } x \ll d; a = -\left(\frac{4q\lambda}{\pi\epsilon_0 m d^2} \right) x$$

The motion is SHM if $q\lambda$ is a positive number.



Worksheet 3

- When the rod is brought close, negative charge induce on the surface of A . B acquires positive charge.
 - When the rod is removed, the charge separation on A and B disappear. Both the spheres are neutral with no charge on their surfaces. They do not apply any force on one another.
 - When the spheres are separated in presence of the rod, A retains negative charge and B retains positive charge on its surface. Therefore, they attract.

$$2. \text{ Number of Cu atoms in 1 g sphere} = \frac{1}{63.54} \times 6.02 \times 10^{23}$$

Number of electrons in 1 g Cu sphere

$$= \frac{1}{63.54} \times 6.02 \times 10^{23} \times 29$$

Number of electrons removed

$$= \frac{1}{100} \times \frac{1}{63.54} \times 6.02 \times 10^{23} \times 29$$

Charge on each sphere;

$$q = \frac{6.02 \times 10^{23} \times 29}{100 \times 63.54} \times 1.6 \times 10^{-19} \text{ C} = 429.8 \text{ C}$$

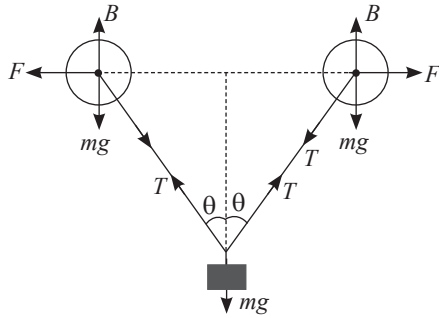
$$\therefore F = k \frac{q \cdot q}{r^2} = \frac{9 \times 10^9 \times (429.8)^2}{1^2} = 1.7 \times 10^{15} \text{ N}$$

3. Let T = tension in each string

F = electrostatic repulsion

B = buoyancy on each balloon

mg = weight of each balloon



(a) For equilibrium of mass M :

$$2T \cos \theta = Mg \quad (i)$$

For horizontal equilibrium of a balloon:

$$T \sin \theta = F$$

$$\Rightarrow T \sin \theta = \frac{1}{4\pi\epsilon_0} \frac{q^2}{(2x)^2} \quad (ii)$$

From (i) and (ii)

$$\tan \theta = \frac{q^2}{8\pi\epsilon_0 Mg x^2}$$

(b) For vertical equilibrium of a balloon

$$T \cos \theta + mg = B$$

$$\Rightarrow \frac{Mg}{2} + V \rho_{\text{He}} g = V \rho_0 g \quad [\text{using (i)}]$$

$$\Rightarrow V = \frac{Mg}{2(\rho_0 - \rho_{\text{He}})}$$

4. The spring is conducting. Therefore, the charge Q divides equally on both the blocks.

In equilibrium, the spring force acting on each block balances the electrostatic force acting on it.

$$\frac{1}{4\pi\epsilon_0} \frac{Q/2 \cdot Q/2}{L^2} = K(L - L_0)$$

$$\Rightarrow Q = 4L\sqrt{\pi\epsilon_0 K(L - L_0)}$$

$$5. \text{ In equilibrium : } mg = \frac{Qq}{4\pi\epsilon_0 L^2} \quad (i)$$

Let's consider the ball at a distance x above its equilibrium position.

$$ma = \frac{Qq}{4\pi\epsilon_0 (L+x)^2} - mg$$

$$\Rightarrow ma = \frac{Qq}{4\pi\epsilon_0 L^2} \left(1 + \frac{x}{L}\right)^{-2} - mg$$

$$\text{For } \frac{x}{L} \ll 1; \left(1 + \frac{x}{L}\right)^{-2} \approx 1 - \frac{2x}{L}$$

$$\therefore ma = \frac{Qq}{4\pi\epsilon_0 L^2} \left(1 - \frac{2x}{L}\right) - mg$$

$$\Rightarrow ma = mg \left(1 - \frac{2x}{L}\right) - mg \quad [\text{using (i)}]$$

$$\Rightarrow a = -\left(\frac{2g}{L}\right)x$$

Therefore, motion of the ball is SHM.

$$\omega = \sqrt{\frac{2g}{L}}$$

$$\Rightarrow T = 2\pi\sqrt{\frac{L}{2g}}$$

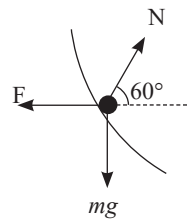
6. For equilibrium of a ball

$$N \sin 60^\circ = mg \text{ and } N \cos 60^\circ = F$$

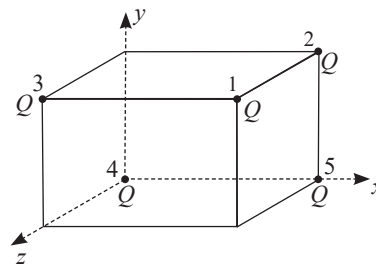
$$\therefore \tan 60^\circ = \frac{mg}{F}$$

$$\Rightarrow \sqrt{3} = \frac{mg}{\frac{q^2}{4\pi\epsilon_0 R^2}}$$

$$\Rightarrow q = \left[\frac{4\pi\epsilon_0 mg R^2}{\sqrt{3}} \right]^{\frac{1}{2}}$$



7. We are supposed to find force on charge at vertex 1:



$$\vec{F}_{12} = K \frac{Q^2}{|\vec{r}_1 - \vec{r}_2|^3} (\vec{r}_1 - \vec{r}_2)$$

$$= K \frac{Q^2}{L^3} (L\hat{k})$$

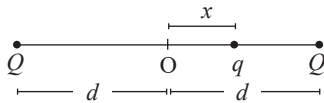
$$\vec{F}_{13} = K \frac{Q^2}{|\vec{r}_1 - \vec{r}_3|^3} (\vec{r}_1 - \vec{r}_3) = K \frac{Q^2}{L^3} (L\hat{i})$$

$$\vec{F}_{14} = K \frac{Q^2}{|\vec{r}_1 - \vec{r}_4|^3} (\vec{r}_1 - \vec{r}_4) = K \frac{Q^2}{(\sqrt{3}L)^3} (L\hat{i} + L\hat{j} + L\hat{k})$$

$$\vec{F}_{15} = K \frac{Q^2}{|\vec{r}_1 - \vec{r}_5|^3} (\vec{r}_1 - \vec{r}_5) = K \frac{Q^2}{(\sqrt{2}L)^3} (L\hat{j} + L\hat{k})$$

$$\begin{aligned} \therefore \vec{F} &= \vec{F}_{12} + \vec{F}_{13} + \vec{F}_{14} + \vec{F}_{15} \\ &= \frac{KQ^2}{L^2} \left[\left(1 + \frac{1}{3\sqrt{3}}\right)\hat{i} + \left(\frac{1}{3\sqrt{3}} + \frac{1}{2\sqrt{2}}\right)\hat{j} + \left(1 + \frac{1}{3\sqrt{3}} + \frac{1}{2\sqrt{2}}\right)\hat{k} \right] \end{aligned}$$

8. (a) O is midpoint of the line joining the fixed charges. Charge q is in equilibrium at O . consider it at a displacement x from O .



$$\begin{aligned} ma &= \frac{1}{4\pi\epsilon_0} \frac{Qq}{(d+x)^2} - \frac{1}{4\pi\epsilon_0} \frac{Qq}{(d-x)^2} \\ &\quad \text{[Rightward force is positive]} \\ &= \frac{Qq}{4\pi\epsilon_0} \left[\frac{-4dx}{(d^2 - x^2)^2} \right] \end{aligned}$$

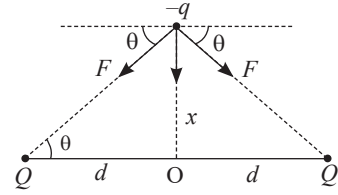
For $x \ll d$; $d^2 - x^2 \approx d^2$

$$\therefore a = -\left(\frac{Qq}{\pi\epsilon_0 d^3 m} \right) x \quad \text{[SHM]}$$

$$\therefore w = \sqrt{\frac{Qq}{\pi\epsilon_0 m d^3}}$$

$$\Rightarrow T = 2\pi \sqrt{\frac{\pi\epsilon_0 m d^3}{Qq}}$$

- (b) When q is negative, it is still in equilibrium at O . When displaced by a distance x , along a line perpendicular to the line joining the fixed charge, it experiences a restoring force towards O given by



$$2F \sin \theta = 2 \frac{Qq}{4\pi\epsilon_0 (d^2 + x^2)} \cdot \frac{x}{\sqrt{d^2 + x^2}}$$

$$= \frac{Qq \cdot x}{2\pi\epsilon_0 (d^2 + x^2)^{3/2}}$$

$$\therefore ma = -\frac{Qq \cdot x}{2\pi\epsilon_0 (d^2 + x^2)^{3/2}}$$

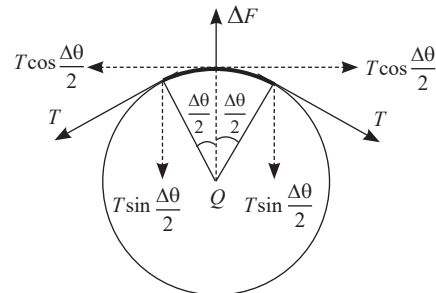
$$\Rightarrow ma = -\left(\frac{Qq}{2\pi\epsilon_0 d^3} \right) x$$

$$\Rightarrow a = -\left(\frac{Qq}{2\pi\epsilon_0 d^3 m} \right) x$$

$$\therefore T = 2\pi \sqrt{\frac{2\pi\epsilon_0 m d^3}{Qq}}$$

9. Charge per unit length

$$\lambda = \frac{q}{2\pi r}$$



Charge on an element subtending an angle $\Delta\theta$ at the centre is

$$\Delta q = \lambda (r\Delta\theta) = \frac{q}{2\pi r} \cdot r\Delta\theta$$

$$\Rightarrow \Delta q = \frac{q\Delta\theta}{2\pi}$$

$$\text{Electric repulsion on } \Delta q \text{ is } \Delta F = \frac{1}{4\pi\epsilon_0} \frac{Q\Delta q}{r^2}$$

$$\Rightarrow \Delta F = \frac{Q \cdot q \cdot \Delta\theta}{8\pi^2 \epsilon_0 r^2}$$

Let the increase in tension in the ring after Q is placed be T .

Tension acting at the two ends of our element has resultant

$$2T \sin \frac{\Delta\theta}{2} \approx 2T \frac{\Delta\theta}{2} = T \Delta\theta$$

[$\therefore$ for small angle $\sin\theta \rightarrow \theta$]

For equilibrium of the element considered

$$T \Delta\theta = \frac{Q \cdot q \cdot \Delta\theta}{8\pi^2 \epsilon_0 r^2}$$

$$\Rightarrow T = \frac{Q \cdot q}{8\pi^2 \epsilon_0 r^2}$$

10. Consider a shell of radius r and thickness dr . Charge in the shell is $dq = \rho(4\pi r^2 dr)$

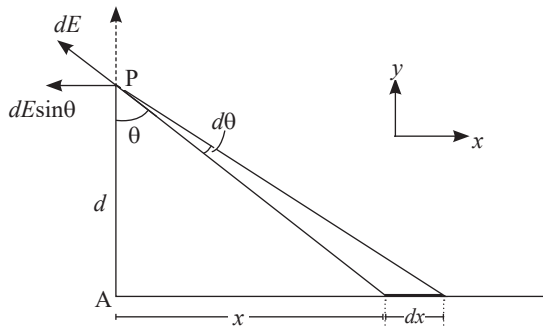
$$= \frac{4\pi\rho_0}{R^2} r^4 dr$$

$$\therefore Q = \frac{4\pi\rho_0}{R^2} \int_0^R r^4 dr$$

$$= \frac{4}{5} \pi \rho_0 R^3$$

11. Had it been a line charge extending to infinity on both sides, the y component of field would be $E_y = \frac{\lambda}{2\pi\epsilon_0 d}$, as obtained in section 10.2 using the Gauss's law. Obviously, the field will become half when half the line is removed. Hence, for the given situation $E_y = \frac{\lambda}{4\pi\epsilon_0 d}$

For finding E_x we need to consider a small element on the line as we did in example 39.



Using the result obtained in example 39. We can write

$$dE = \frac{1}{4\pi\epsilon_0} \frac{\lambda}{d} d\theta$$

$$\therefore dE_x = \frac{1}{4\pi\epsilon_0} \frac{\lambda}{d} \sin\theta d\theta$$

[In negative x direction]

$$\therefore E_x = -\frac{\lambda}{4\pi\epsilon_0 d} \int_0^{90^\circ} \sin\theta d\theta$$

As the line is of infinite length, θ changes from 0° to 90°

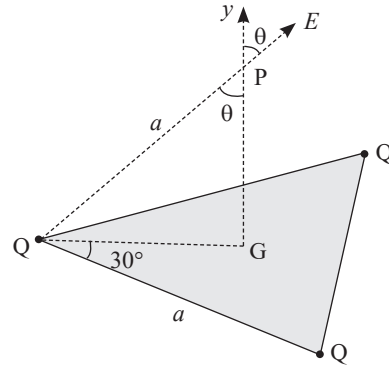
$$E_x = -\frac{\lambda}{4\pi\epsilon_0 d} [\cos\theta]_{0^\circ}^{90^\circ} = \frac{\lambda}{4\pi\epsilon_0 d} \quad \text{In negative } x \text{ direction}$$

Note: $|E_x| = |E_y|$

Direction of the field at point P makes 45° with the line AP

12. Due to symmetry field at P must be along y direction. Magnitude of field due to any one charge

$$\text{is } E = \frac{1}{4\pi\epsilon_0} \frac{Q}{a^2}$$



$$E_y = E \cos\theta = \frac{1}{4\pi\epsilon_0} \frac{Q}{a^2} \cos\theta$$

$$QG = \frac{a}{2 \cos 30^\circ} = \frac{a}{\sqrt{3}}$$

$$PG = \sqrt{a^2 - \frac{a^2}{3}} = \sqrt{\frac{2}{3}} \cdot a$$

$$\therefore \cos\theta = \frac{PG}{a} = \sqrt{\frac{2}{3}}$$

$$\therefore E_y = \sqrt{\frac{2}{3}} \cdot \frac{1}{4\pi\epsilon_0} \frac{Q}{a^2}$$

$$\text{Resultant field in } y \text{ direction is } E_0 = 3E_y = \frac{\sqrt{6}}{4\pi\epsilon_0} \frac{Q}{a^2}$$

13. Field in front of the surface is $E = \frac{\sigma}{2\epsilon_0}$ directed along z direction.

Force on the charge is $F = qE = \frac{q\sigma}{2\epsilon_0}$ in z direction

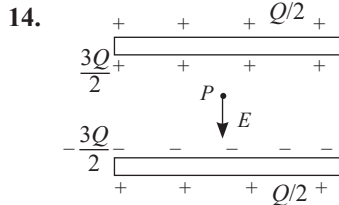
or $\vec{F} = \frac{q\sigma}{2\epsilon_0} \hat{k}$

(a) Displacement $\vec{r} = l(\hat{i} + \hat{j} + \hat{k})$

$\therefore W = \vec{F} \cdot \vec{r} = \frac{q\sigma l}{2\epsilon_0}$

(b) Displacement $\vec{r} = l(\hat{i} + \hat{j} + 0\hat{k})$

$\therefore W = \vec{F} \cdot \vec{r} = 0$



Refer to example 30. Charge on inner walls of the two plates are

$$\pm \left[\frac{2Q - (-Q)}{2} \right] = \pm \frac{3Q}{2}$$

Charge density on the inner walls is

$$\pm \sigma = \pm \frac{3Q}{2A}$$

Electric field at P due to charges on outer faces cancel out. Field at P is produced due to charges on inner faces.

$$E = \frac{\sigma}{2\epsilon_0} + \frac{\sigma}{2\epsilon_0}$$

$$= \frac{\sigma}{\epsilon_0} = \frac{3Q}{2A\epsilon_0}$$

Force on charge q is $F = qE = \frac{3Qq}{2A\epsilon_0}$

Acceleration of charge q is, $a = \frac{F}{m} = \frac{3Qq}{2A\epsilon_0 m}$

15. For equilibrium:

$$T \sin \theta = F$$

$$\Rightarrow T \sin \theta = k \frac{q^2}{a^2} \text{ and } T \cos \theta = mg$$

$$\Rightarrow \tan \theta = k \frac{q^2}{mg \cdot a^2}$$

For small separation, θ is small

$$\therefore \tan \theta \approx \sin \theta = \frac{a}{2l}$$

$$\therefore \frac{a}{2l} = k \frac{q^2}{mg \cdot a^2}$$

$$\Rightarrow a^3 = \frac{2klq^2}{mg}$$

$$\therefore a \propto q^{2/3}$$

After one ball gets discharged and the two balls collide, both of them will have equal charge $\frac{q}{2}$ on them.

$\therefore$ Separation between the balls (b) will be given by

$$\frac{b}{a} = \frac{(q/2)^{2/3}}{(q)^{2/3}} = \left(\frac{1}{2} \right)^{2/3}$$

$$\Rightarrow b = \frac{a}{4^{1/3}}$$

17. Force on the particle in y direction is $F_y = qE - mg$. Acceleration of the particle in y direction is

$$a = \frac{F_y}{m} = \frac{qE}{m} - g$$

Initially, $u_x = u$ and $u_y = 0$

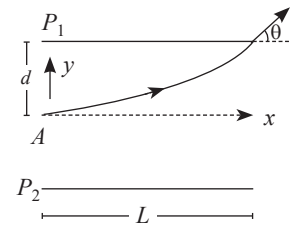
There is no acceleration in x direction and u_x does not change.

The particle just missed the upper plate means that by the time it move a distance d in y direction, it moved a distance L in x directions also.

$$y = u_y t + \frac{1}{2} a_y t^2$$

$$\Rightarrow d = \frac{1}{2} a t^2$$

$$\Rightarrow t = \sqrt{\frac{2d}{a}}$$



In this time the particle moved a distance L in x direction.

$$\Rightarrow u_x \cdot t = L$$

$$\Rightarrow u \sqrt{\frac{2d}{a}} = L$$

$$\Rightarrow u = L \sqrt{\frac{a}{2d}} = L \sqrt{\frac{qE}{2md} - \frac{g}{2d}}$$

At the time of leaving the plate region, components of velocity are:

$$v_x = u$$

$$v_y^2 = 2a \cdot d$$

$$\Rightarrow v_y = \sqrt{\frac{2qEd}{m} - 2g \cdot d}$$

$$\therefore \tan \theta = \frac{v_y}{v_x} = \frac{\sqrt{2ad}}{L\sqrt{\frac{a}{2d}}}$$

$$= \frac{2d}{L}$$

18. (a) $\vec{E} = \frac{\vec{F}}{q}$

(b) The force is radial. Such forces are conservative.

$$\therefore \oint \vec{E} \cdot d\vec{l} = 0$$

19. We need not attempt doing some complicated integration to solve this problem. It can be solved using dimensional analysis.

$$E_0 \propto \frac{Q_{\text{cube}}}{L^2} = \frac{\rho L^3}{L^2} = \rho L$$

When L is doubled, the electric field also doubles.

20. At $x \rightarrow \infty$, force on the charge at 2 is in positive x direction equal to $F = 1.5 \times 10^{-12}$ N.

When q_3 is at infinity, force on 2 is due to q_1 only. It is a force of repulsion. Therefore, q_1 and q_2 have same sign. When q_3 is close to q_2 net force on q_2 is negative (i.e. towards left). This implies that q_3 also repels q_2 . Therefore, q_2 has same sign as q_3 . All charges are positive.

When $x \rightarrow \infty$, force on q_2 due to q_3 is zero. It experiences force only due to q_1 .

$$\therefore k \frac{q_1 q_2}{r^2} = 1.5 \times 10^{-12} \quad (i)$$

[where r is separation between 1 and 2]

For $x = 0.4$ m, force on q_2 is zero. This means both q_1 and q_2 apply same force on it.

$$k \frac{q_2 q_3}{(0.4)^2} = k \frac{q_1 q_2}{r^2}$$

$$\Rightarrow k \frac{q_2 q_3}{(0.4)^2} = 1.5 \times 10^{-12}$$

$$\Rightarrow \frac{(9 \times 10^9)(q_2)(7 \times 10^{-6})}{0.16} = 1.5 \times 10^{-12}$$

$$\Rightarrow q_2 = \frac{1.5 \times 0.16}{9 \times 7} \times 10^{-15} = 3.8 \times 10^{-18} \text{ C}$$

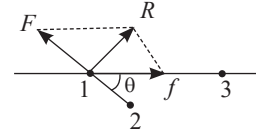
21. For $\theta = 0^\circ$, force on 1 is zero. If force between 1 and 3 is attractive, then it must be a repulsive force between 1 and 2. For equilibrium at $\theta = 0^\circ$

$$|F_{13}| = |F_{12}| = f \quad (\text{say})$$

When q_2 moves by 180° , both the forces on 1 are in same direction

$$\therefore 2f = F_0$$

$$\Rightarrow f = \frac{F_0}{2}$$



Note that magnitude of F_{12} does not change as 2 is moved along the semicircle.

When 2 is at position θ , resultant force is R as shown in the diagram.

Obviously, $R = f = \frac{F_0}{2}$ when $\theta = 60^\circ$.

22. (a) Refer to Your Turn Q.9.

Force is maximum when Q is divided into two equal parts.

$$(b) F_0 = k \frac{\frac{Q}{2} \frac{Q}{2}}{r^2} = \frac{kQ^2}{4r^2}$$

$$\text{For } F = \frac{3}{4} F_0$$

$$\Rightarrow k \frac{x(Q-x)}{r^2} = \frac{3}{4} \cdot \frac{kQ^2}{4r^2}$$

$$\Rightarrow 16x(Q-x) = 3Q^2$$

$$\Rightarrow 16 \frac{x}{Q} - 16 \left(\frac{x}{Q} \right)^2 = 3$$

$$\text{Let } \frac{x}{Q} = \eta$$

$$16\eta^2 - 16\eta + 3 = 0$$

$$\Rightarrow 16\eta^2 - 12\eta - 4\eta + 3 = 0$$

$$\Rightarrow 4\eta(4\eta - 3) - 1(4\eta - 3) = 0$$

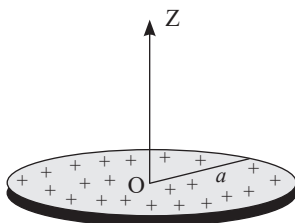
$$\Rightarrow \eta = \frac{1}{4}, \frac{3}{4}$$

23. Refer to equation (12) in section 6.3.2. Field at a distance z from the centre of the disc on its axis is

$$E = \frac{\sigma}{2\epsilon_0} \left[1 - \frac{z}{\sqrt{a^2 + z^2}} \right]$$

E is maximum at $z = 0$

$$\therefore E_0 = \frac{\sigma}{2\epsilon_0}$$



At $z = 8$ cm, field is $\frac{E_0}{2} = \frac{\sigma}{4\epsilon_0}$.

$$\therefore \frac{\sigma}{4\epsilon_0} = \frac{\sigma}{2\epsilon_0} \left[1 - \frac{8}{\sqrt{a^2 + 8^2}} \right]$$

$$\Rightarrow \frac{8}{\sqrt{a^2 + 8^2}} = \frac{1}{2}$$

$$\Rightarrow a^2 + 8^2 = 16^2$$

$$\Rightarrow a = \sqrt{192} \text{ cm}$$

24. Field at P due to charge on infinite plane is

$$E = \frac{\sigma}{2\epsilon_0}$$

According to the problem, the charge present inside a circular surface of radius r contributes half the field

$$E_{\text{circle}} = \frac{\sigma}{2\epsilon_0}$$

$$\Rightarrow \frac{\sigma}{2\epsilon_0} \left[1 - \frac{a^2}{\sqrt{r^2 + a^2}} \right] = \frac{\sigma}{4\epsilon_0} \quad [\text{using equation 12}]$$

$$\Rightarrow 1 - \frac{a^2}{\sqrt{r^2 + a^2}} = \frac{1}{2}$$

$$\Rightarrow r = \sqrt{3}a$$

25. Let linear charge density on inner cylinder be λ_1 and that on outer cylinder be λ_2 .

$R = 3.5$ cm is radius of the outer cylinder.

Just inside the wall of outer cylinder the field can be obtained using the Gauss's law. It is

$$\frac{\lambda_1}{2\pi\epsilon_0 r} = 2 \times 10^3$$

$$\Rightarrow \lambda_1 = r \cdot 4\pi\epsilon_0 \times 10^3$$

Just outside the outer cylinder the field is negative and it is given by

$$\frac{\lambda_1 + \lambda_2}{2\pi\epsilon_0 r} = -4 \times 10^3$$

$$\Rightarrow \lambda_2 = -r \cdot 8\pi\epsilon_0 \times 10^3 - \lambda_1$$

$$= -r \cdot 12\pi\epsilon_0 \times 10^3$$

$$= -\left(\frac{3.5}{100}\right) \times 3 \times \frac{1}{9 \times 10^9} \times 10^3 = -1.17 \times 10^{-8} \text{ C/m}$$

26. Field strength in from of an infinite sheet of charge is independent of distance. Density of lines in any quadrant shall be uniform.

27. Field is zero at all points outside spherical region of radius R . This can happen only when total charge inside the spherical region is zero and the charge distribution is spherically symmetric.

For $r \leq R$

$$E = \frac{a}{r^2} \left(1 - \frac{r^3}{R^3} \right) = \frac{a}{r^2} - \left(\frac{a}{R^3} \right) r$$

The first term in the expression of the field can arise due to the point charge at the centre and the second term can be contributed by a uniform density of negative charge in the region $r \leq R$.

If q = charge at the origin then

$$\frac{1}{4\pi\epsilon_0} \frac{q}{r^2} = \frac{a}{r^2}$$

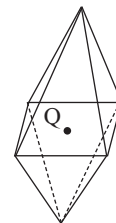
$$\Rightarrow q = 4\pi\epsilon_0 \cdot a$$

If charge density surrounding the point charge is $-\rho$ then

$$\frac{\rho}{3\epsilon_0} r = \frac{a}{R^3} \cdot r$$

$$\Rightarrow \rho = \frac{3\epsilon_0 a}{R^3}$$

28. Consider an 8 faced closed surface consisting of two identical pyramids with the charge at the centre. The total flux through the closed surface is $\frac{Q}{\epsilon_0}$.



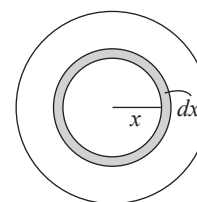
Since each of the 8 faces are identical for the charge, hence flux through one of the faces is $\frac{1}{8} \cdot \frac{Q}{\epsilon_0}$.

29. (a) Charge in a thin shell of thickness dx and radius x is

$$dq = 4\pi x^2 \cdot dx \cdot \rho$$

$$= (4\pi x^2 \cdot dx) \left(\rho_0 \frac{x}{R} \right)$$

$$= \frac{4\pi\rho_0}{R} x^3 \cdot dx$$



Charge inside a sphere of radius R is

$$q = \frac{4\pi\rho_0}{R} \int_0^R x^3 dx = \pi\rho_0 R^3$$

- (b) Consider a point (P) at distance r from the centre such that $r < R$. Draw a spherical Gaussian surface through P . Due to spherical symmetry, field is radial at all points and will have same magnitude at all points on the spherical Gaussian surface.

Flux through the Gaussian surface is

$$\phi = \oint \vec{E} \cdot d\vec{s} = E \cdot 4\pi r^2$$

Charge enclosed inside the spherical region of radius r is calculated as in part (a)

$$q_r = \frac{4\pi\rho_0}{R} \int_0^r x^3 dx = \frac{\pi\rho_0}{R} r^4$$

Using Gauss's law

$$\phi = \frac{q_r}{\epsilon_0}$$

$$\Rightarrow E \cdot 4\pi r^2 = \frac{\pi\rho_0 r^4}{\epsilon_0 R}$$

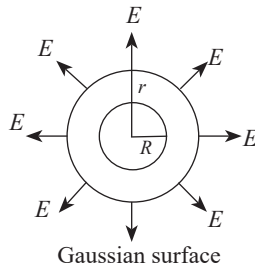
$$\Rightarrow E = \frac{\rho_0 r^2}{4\epsilon_0 R}$$

For $r > R$, we again choose a Gaussian sphere as shown

$$\therefore E \cdot 4\pi r^2 = \frac{q}{\epsilon_0}$$

$$\Rightarrow E \cdot 4\pi r^2 = \frac{\pi\rho_0 R^3}{\epsilon_0}$$

$$\Rightarrow E = \frac{\rho_0 R^3}{4\epsilon_0 r^2}$$



30. Outside both slabs, the field due to positive and negative slabs are equal and opposite. They cancel out and the resultant field is zero

$$\therefore E = 0 \text{ for } |z| > d$$

For finding field inside the positively charge region, consider a Gaussian surface as shown. It is in the shape of a cylinder. Let the field at the inner circular face of the cylinder be E in positive z direction. Flux through the cylinder is

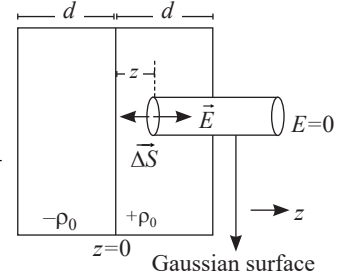
$$\phi = E \cdot \Delta S \cos 180^\circ = -E\Delta S$$

Charge inside the cylinder is

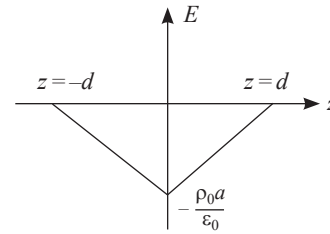
$$q_{in} = \rho_0 \Delta S (d - z)$$

$$\therefore -E\Delta S = \frac{\rho_0 \Delta S (d - z)}{\epsilon_0}$$

$$\Rightarrow E = -\frac{\rho_0 (d - z)}{\epsilon_0}$$



Negative sign indicates field is in negative z direction. Similarly, one can find field in the region $-d < z < 0$. Graphical variation of the field with z is as shown.



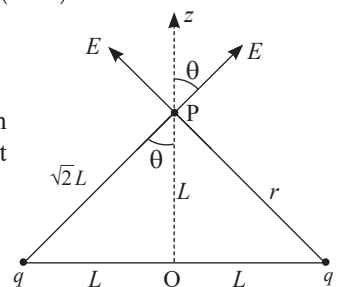
31. Let the hexagon be in xy plane. Field at P is along z direction. This is because a pair of charges at diagonally opposite corners will create a field in z direction only. There are three such pairs. Consider a pair of charges at opposite corners. Field at P due to this pair is

$$= 2E \cos \theta = 2 \frac{1}{4\pi\epsilon_0} \frac{q}{(\sqrt{2}L)^2} \frac{L}{(\sqrt{2}L)}$$

$$= \frac{q}{4\sqrt{2}\pi\epsilon_0 L^2}$$

There are three such pairs. Hence resultant field at P is

$$E_0 = \frac{3q}{4\sqrt{2}\pi\epsilon_0 L^2}$$



32. (a) Field at a point very close to Q_2 , and right to it is in negative x direction. This shows that Q_2 is negative. As we move a great distance to the right the field turns positive. This will be possible only if Q_1 is positive.

- (b) $E = 0$ at a distance a to the right of Q_2

$$\Rightarrow K \frac{|Q_1|}{(l+a)^2} = K \frac{|Q_2|}{a^2}$$

$$\Rightarrow \left| \frac{Q_1}{Q_2} \right| = \left(\frac{l+a}{a} \right)^2$$

- (c) At a distance a to the right of Q_2 , the field is locally maximum. Field at a distance x to the right of Q_2 is

$$E = K \frac{|Q_1|}{(l+x)^2} - K \frac{|Q_2|}{x^2}$$

$$\frac{dE}{dx} = -\frac{2K|Q_1|}{(l+x)^3} + \frac{2K|Q_2|}{x^3}$$

$$\text{At } x = b, \frac{dE}{dx} = 0$$

$$\Rightarrow \frac{|Q_1|}{(l+b)^3} = \frac{|Q_2|}{b^3}$$

$$\Rightarrow \left(\frac{l+b}{b}\right)^3 = \left(\frac{l+a}{a}\right)^2$$

$$\Rightarrow \frac{l}{b} + 1 = \left(\frac{l+a}{a}\right)^{2/3}$$

$$\Rightarrow \frac{l}{b} = \left(1 + \frac{l}{a}\right)^{2/3} - 1$$

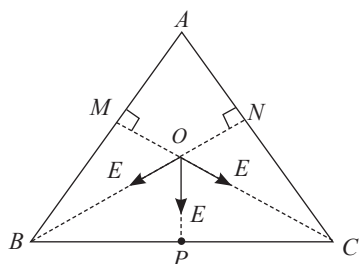
$$\Rightarrow b = \frac{l}{\left(1 + \frac{l}{a}\right)^{2/3} - 1}$$

33. Perpendicular distance of point O from a rod is

$$d = \frac{L}{2\sqrt{3}}$$

Electric field due to line charge AB is along MO and is given by (see the result obtained in example 41)

$$\begin{aligned} E &= \frac{2\lambda L}{4\pi\epsilon_0 d \sqrt{4d^2 + L^2}} \\ &= \frac{2\lambda L}{4\pi\epsilon_0 \frac{L}{2\sqrt{3}} \sqrt{4\frac{L^2}{12} + L^2}} \\ &= \frac{3\lambda}{2\pi\epsilon_0 L} \end{aligned}$$



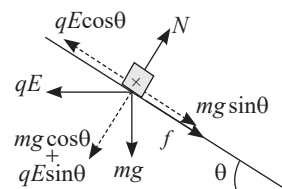
Field due to AC has same magnitude along NO . It has magnitude equal to E . Field due to AB and AC

are inclined at 120° . Their resultant is along OP and has a magnitude E .

Field due to negative charge on BC is also having magnitude E directed along OP . Therefore, the

resultant field is $2E = \frac{3\lambda}{\pi\epsilon_0 L}$ along OP .

34. Coefficient of friction, $\mu = \tan 37^\circ = 0.75$



After the field is switched on, the block experiences a leftward electric force qE . Due to this force, normal reaction increases and hence the limiting friction increase. There is no question of the block sliding down the incline. It can only go up the incline when $qE \cos \theta$ becomes larger than the sum of $mg \sin \theta$ and limiting friction.

$$N = mg \cos \theta + qE \sin \theta = \frac{4}{5}mg + \frac{3}{5}qE$$

The block will begin to move up the plane if

$$qE \cos \theta > mg \sin \theta + \mu N$$

$$qE \cdot \frac{4}{5} > \frac{3}{5}mg + \frac{3}{4} \left(\frac{4}{5}mg + \frac{3}{5}qE \right)$$

$$\Rightarrow 4qE > 3mg + 3mg + 2.25qE$$

$$\Rightarrow 1.75qE > 6mg$$

$$\Rightarrow E > \frac{6mg}{1.75q} = \frac{24mg}{7q}$$

$\therefore$ Range of E for which the block does not move

$$\text{is } 0 \leq E \leq \frac{24mg}{7q}$$

35. Surface charge density (and hence E) is maximum where radius of curvature (R) is smallest. And σ is minimum where R is maximum.

$$(a) E = \frac{\sigma}{\epsilon_0}$$

$$\therefore E_{\min} = \frac{\sigma_{\min}}{\epsilon_0}$$

$$\begin{aligned} \Rightarrow \sigma_{\min} &= \epsilon_0 E_{\min} \\ &= 8.85 \times 10^{-12} \times 56 \times 10^3 \\ &= 496 \text{ nC m}^{-2} \end{aligned}$$

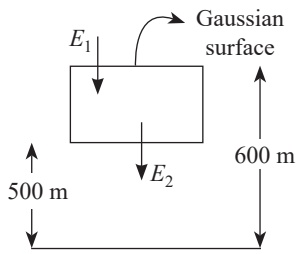
and $E_{\max} = \frac{\sigma_{\max}}{\epsilon_0}$

$$\Rightarrow \sigma_{\max} = \epsilon_0 E_{\max} = 992 \text{ nCm}^{-2}$$

- (b) The location where σ is maximum has smallest radius of curvature and vice versa. Also, the charge density at a location on the surface is in inverse proportion to the radius of curvature

$$\therefore \frac{R_{\max}}{R_{\min}} = \frac{\sigma_{\max}}{\sigma_{\min}} = \frac{E_{\max}}{E_{\min}} = \frac{2}{1}$$

36. Consider a box shaped Gaussian surface with horizontal surfaces having area A and lying at heights of 500 m and 600 m

$$\begin{aligned} \phi &= -E_1 A + E_2 A \\ &= -80A + 100A = 20A \\ \therefore \frac{q_{\text{in}}}{\epsilon_0} &= 20A \\ \Rightarrow \frac{\rho \cdot A(100)}{\epsilon_0} &= 20A \\ \Rightarrow \rho &= \frac{\epsilon_0}{5} = 1.77 \times 10^{-12} \text{ cm}^{-3} \end{aligned}$$


The net charge inside the box is positive.

37. Take a Gaussian surface passing through the material of the metallic shell. Flux through this surface is zero. Therefore, net field inside this Gaussian surface is zero.

$\therefore$ Charge on the inner wall of the shell is $-Q$.

Chapter 2 Electrostatic Potential



Your Turn

1. $U = K \frac{qQ}{r}$

Depending on signs of q and Q , U can be positive or negative. When U is positive (i.e., both charges have same sign and they repel), it decreases when r increases. But, when U is negative (charges are of opposite signs), its value increases when r is increased.

2. If Q is moved on a circle (or a sphere) centred at q , the separation between the charges does not change. In this case, potential energy does not change.

3. $U_1 = K \frac{q_1 q_2}{r_1} = 9 \times 10^9 \frac{(2 \times 10^{-6})(-1 \times 10^{-6})}{1}$
 $= -0.018 \text{ J}$

$$U_2 = K \frac{q_1 q_2}{r_2} = -0.009 \text{ J}$$

(a) $W_{\text{elect}} = -\Delta U = -(U_2 - U_1) = U_1 - U_2$
 $= -0.009 \text{ J}$
 $= -9 \text{ mJ}$

(b) $W_{\text{extAgt}} = \Delta U = 9 \text{ mJ}$

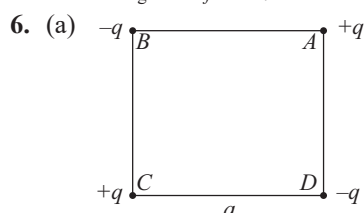
4. $W_{\text{elect}} = -\Delta U = U_1 - U_2 = \frac{q_1 q_2}{4\pi\epsilon_0} \left(\frac{1}{1} - \frac{1}{2} \right)$
 $= -9 \times 10^{-3} \text{ J.}$

5. (a) $U = \frac{1}{4\pi\epsilon_0} \left[\frac{q_3 q_2}{r_{32}} + \frac{q_3 q_1}{r_{31}} + \frac{q_2 q_1}{r_{21}} \right]$
 $= 9 \times 10^{-9} \left[\frac{(1)(1)}{1} + \frac{(1)(-1)}{2} + \frac{(1)(-1)}{1} \right] \times 10^{-12}$
 $= -4.5 \times 10^{-3} \text{ J}$

(b) $U_i = \frac{1}{4\pi\epsilon_0} \frac{q_2 q_1}{r_{21}} = 9 \times 10^9 \frac{(-1 \times 10^{-6})(1 \times 10^{-6})}{1}$
 $= -9 \times 10^{-3} \text{ J}$

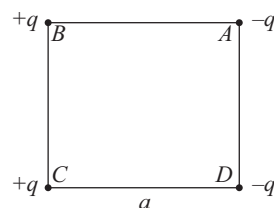
$$U_f = -4.5 \times 10^{-3} \text{ J} \quad [\text{as calculated in part (a)}]$$

$$\therefore W_{\text{extAgt}} = U_f - U_i = 4.5 \times 10^{-3} \text{ J}$$



(a) $U_i = \frac{1}{4\pi\epsilon_0} \left[\frac{q(-q)}{a} + \frac{(q)(q)}{\sqrt{2}a} + \frac{(q)(-q)}{a} + \frac{(-q)(q)}{a} \right]$
 $+ \frac{(-q)(-q)}{\sqrt{2}a} + \frac{(q)(-q)}{a}$
 $= \frac{\sqrt{2}(1-2\sqrt{2})}{4\pi\epsilon_0} \frac{q^2}{a}$

- (b) Position of charges at A and B are interchanged. Potential energy of the system is



$$U_f = \frac{1}{4\pi\epsilon_0} \left[\frac{(-q)(q)}{a} + \frac{(-q)(q)}{\sqrt{2}a} + \frac{(-q)(-q)}{a} \right]$$

$$+ \frac{(q)(q)}{a} + \frac{(q)(-q)}{\sqrt{2}a} + \frac{(q)(-q)}{a}$$

$$= -\frac{\sqrt{2}}{4\pi\epsilon_0} \frac{q^2}{a}$$

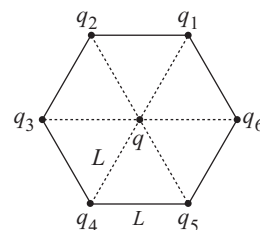
$$W_{\text{extAgt}} = U_f - U_i = \frac{\sqrt{2}}{4\pi\epsilon_0} \frac{q^2}{a} [-1 - (1-2\sqrt{2})]$$

$$= \frac{(4-2\sqrt{2})}{4\pi\epsilon_0} \frac{q^2}{a}$$

7. Initially, the six charges form $\frac{n(n-1)}{2} = \frac{6 \times 5}{2} = 15$ pairs and the total energy is U_0 .

After introduction of the seventh charge, total number of terms in the expression of potential energy will be $\frac{7 \times 6}{2} = 21$.

Actually, the 15 terms remain unchanged and six more terms get added due to interaction of the charge at the centre with the six charges at the vertices.

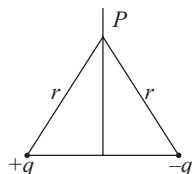


$$\therefore U = U_0 + \frac{1}{4\pi\epsilon_0} \frac{q}{a} (q_1 + q_2 + q_3 + q_4 + q_5 + q_6)$$

$$= U_0 + \frac{1}{4\pi\epsilon_0} \frac{qQ}{a}$$

$$\therefore U - U_0 = \frac{1}{4\pi\epsilon_0} \frac{qQ}{a}$$

8. A point P on the perpendicular bisector is equidistant from the two charges.



$$\therefore V = K \frac{q}{r} + K \frac{(-q)}{r} = 0$$

9. $V = \frac{1}{4\pi\epsilon_0 r} [q_1 + q_2 + q_3 + q_4]$

$$= \frac{9 \times 10^9}{1} [1 + 3 - 2 - 1] \times 10^{-6}$$

$$= 9000 \text{ V}$$

10. $W_{\text{ext Agt}}^{\infty \rightarrow P} = U_P - U_{\infty} = q (V_P - V_{\infty})$

$$= 1 \times 10^{-6} (10 - 0)$$

$$= 10^{-5} \text{ J}$$

11. All charges are equidistant from the given point.

$$V = \frac{1}{4\pi\epsilon_0 r} \sum q_i = 0$$

12. Work done against the electric field = -(work done by the electric field)

$$\therefore W_{\text{against field}} = -W_{\text{elect}}^{A \rightarrow B} = \Delta U$$

$$\therefore W_{\text{against field}} = U_B - U_A = Q (V_B - V_A)$$

$$\Rightarrow 1.2 \text{ J} = (10^{-3} \text{ C}) (V_B - V_A)$$

$$\Rightarrow 1200 \text{ V} = V_B - V_C$$

13. Potential at a point at a distance x to the left of charge q is

$$V = \frac{1}{4\pi\epsilon_0} \left[\frac{q}{x} - \frac{3q}{r+x} \right]$$

At C, $\frac{dV}{dx} = 0$

$$\Rightarrow -\frac{1}{x^2} + \frac{3}{(r+x)^2} = 0$$

$$\Rightarrow \frac{r+x}{x} = \sqrt{3}$$

$$\Rightarrow x = \frac{r}{(\sqrt{3}-1)}$$

14. Potential at the origin due to q_1 , q_2 and q_3 is

$$V = \frac{1}{4\pi\epsilon_0} \left[\frac{q_1}{r_1} + \frac{q_2}{r_2} + \frac{q_3}{r_3} \right]$$

$$= 9 \times 10^9 \left[\frac{1}{1} - \frac{2}{2} + \frac{3}{3} \right] \times 10^{-6} = 9 \times 10^3 \text{ V}$$

$$-W_{\text{elect}}^{\infty \rightarrow 0} = \Delta U = U_{\text{origin}} - U_{\infty}$$

$$\therefore -W_{\text{elect}}^{\infty \rightarrow 0} = -U_{\text{origin}} = -q_0 V$$

$$= -(1 \times 10^{-3})(9 \times 10^3) = -9 \text{ J}$$

15. $V_{\text{centre}} = \frac{1}{4\pi\epsilon_0} \frac{Q}{R}$

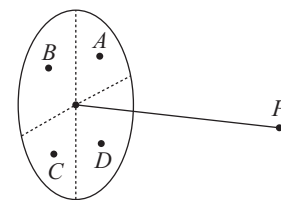
Potential energy of the charge kept at the centre of the ring is

$$U = -q \left(\frac{1}{4\pi\epsilon_0} \cdot \frac{Q}{R} \right)$$

$$W_{\text{ext Agt}} = U_f - U_i = 0 + \frac{1}{4\pi\epsilon_0} \frac{Qq}{R}$$

16. If charge Q is spread uniformly on the entire disc, it will produce same potential at P . Figure shows a uniformly charged disc.

If charges at points B , C and D are shifted to point A , potential at P does not change. Charges at A , B , C and D are equidistant from P .



We can similarly, shift charges from three quadrants of the disc to put them on one quadrant only. This does not change the potential.

For a disc having charge Q ,

$$\sigma = \frac{Q}{\pi a^2}$$

Equation (14) gives: $V = \frac{Q}{2\pi\epsilon_0 a^2} [\sqrt{a^2 + x^2} - x]$

This is the required answer.

17. (a) Energy of the projected ball remains conserved as it moves to infinity.

$$K_i + U_i = K_{\infty} + U_{\infty} \quad (i)$$

For just getting out of the electric field of fixed charges, we can take $K_{\infty} = 0$

Obviously, $U_{\infty} = 0$

Initial potential energy is $U_i = (-Q) V$, where V is potential due to fixed charges at the location of $-Q$.

$$\therefore U_i = (-Q) \left[\frac{1}{4\pi\epsilon_0} \frac{q}{a/2} + \frac{1}{4\pi\epsilon_0} \frac{q}{a/2} \right] = -\frac{qQ}{\pi\epsilon_0 a}$$

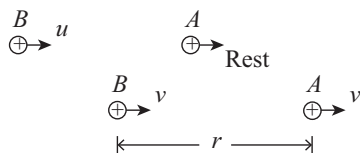
From (i)

$$\frac{1}{2}mu^2 - \frac{qQ}{\pi\epsilon_0 a} = 0$$

$$\Rightarrow u = \sqrt{\frac{2qQ}{\pi\epsilon_0 ma}}$$

(b) Change in direction of motion will not change our answer. The particle will travel on a curved path but will eventually escape if speed of projection is equal to or higher than the answer obtained above.

18. When particle B approaches A , they repel each other. As a result, A begins to accelerate and B retards. After some time, both will have same velocity (v). At this point in time, the separation (r) will be smallest. After this, due to repulsion, velocity of A will become larger than velocity of B , and the separation will begin to rise.



During the entire interaction of particles, momentum of the system remains conserved.

$$\therefore mv + mv = mu \Rightarrow v = \frac{u}{2}$$

v is speed of both particles when they are closest to each other. Energy conservation gives:

$$\frac{1}{2}mv^2 + \frac{1}{2}mv^2 + \frac{1}{4\pi\epsilon_0} \frac{q^2}{r} = \frac{1}{2}mu^2 + 0$$

$$\Rightarrow \frac{1}{4\pi\epsilon_0} \frac{q^2}{r} = \frac{1}{4}mu^2 \quad \left[\text{putting } v = \frac{u}{2} \right]$$

$$\Rightarrow r = \frac{q^2}{\pi\epsilon_0 mu^2}$$

19. Going up against the field increases potential. Electron's potential energy decreases as it is negatively charged.

Or else, you can simply think like this:

Electric force on the electron is upward. The electric force performs positive work as the electron goes up. Hence, its potential energy drops.

20. Moving opposite to the field increases potential. Potential increases by $|E \cdot x| = |10 \times 2| = 20$ V as one moves from the origin to $x = -2$ m.

Put more formally,

$$V_2 - V_1 = -E \cdot \Delta x$$

$$\Rightarrow V_2 - 0 = -(10)(-2)$$

[Displacement Δx is in negative x direction]

$$= 20 \text{ V}$$

21. $\Delta r = AB = 0.05$ m

Angle between $\vec{E}$ and $\Delta \vec{r}$ is $\theta = 37^\circ$

$$\therefore V_B - V_A = -\vec{E} \cdot \Delta \vec{r} = -E \Delta r \cos 37^\circ$$

$$= -20 \times 0.05 \times \frac{4}{5} = -0.8 \text{ V}$$

22. Electric field is uniform in front of the surface.

$$E = \frac{\sigma}{2\epsilon_0} = \frac{0.01 \times 10^{-6}}{2 \times 8.85 \times 10^{-12}} = 565 \text{ Vm}^{-1}$$

If we move from A to B in the direction of field, the potential drops.

$$\therefore V_A - V_B = |E \cdot \Delta r| = 565 \times (1.5 - 0.5) = 565 \text{ V}$$

23. Electric field varies with distance x from the axis as (for points outside the cylinder)

$$E = \frac{\lambda}{2\pi\epsilon_0 x} = \frac{2\lambda}{4\pi\epsilon_0 x}$$

$$= \frac{2 \times 0.01 \times 10^{-6} \times 9 \times 10^9}{x}$$

$$= \frac{180}{x} \text{ Vm}^{-1}$$

Moving from A to B , we can write

$$\int_A^B dV = - \int_A^B E dx$$

$$\Rightarrow V_B - V_A = -180 \int_{x=2\text{cm}}^{x=4\text{cm}} \frac{dx}{x}$$

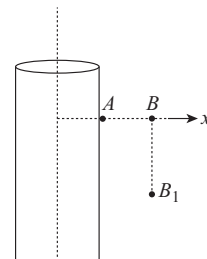
$$\Rightarrow V_B - V_A = -180 [\ln x]_2^4 = -180 \ln 2$$

Value of $\ln 2$ is 0.693

$$\therefore V_B - V_A = -124.7 \text{ V}$$

$$\therefore V_A - V_B = 124.7 \text{ V}$$

If point B is taken at B_1 (see figure) then also our answer will not change. When we move from B to B_1 , there is no change in electric potential as we are moving perpendicular to the direction of electric field.



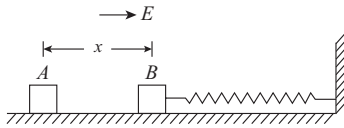
24. $\vec{E} = (x\hat{i} + y\hat{j})$

$$dV = -6(E_x dx + E_y dy) = -(x dx + y dy)$$

$$\Rightarrow \int_0^V dV = - \left[\int_{x=0}^x x dx + \int_{y=0}^y y dy \right]$$

$$\Rightarrow V = -\frac{1}{2}(x^2 + y^2)$$

25. Let maximum compression be x .



$$K_A + U_A = K_B + U_B$$

$$\Rightarrow 0 + qV_A = 0 + qV_B + \frac{1}{2}Kx^2$$

$$\Rightarrow \frac{1}{2}Kx^2 = q(V_A - V_B)$$

$$\Rightarrow \frac{1}{2}Kx^2 = qE \cdot x$$

$$\Rightarrow x = \frac{2qE}{K}$$

26. $E_x = -\frac{dV}{dx} = -(\text{slope of } V \text{ versus } x \text{ graph})$

$$= -\left(\frac{25-10}{0.1}\right) = -150 \text{ Vm}^{-1}$$

Negative sign indicates that the field is directed in negative x direction.

27. $E = -\frac{dV}{dr}$

V may be zero at a point but rate of change of V may not be zero.

28. $E_x = -\frac{\partial V}{\partial x} = -2x$

$$E_y = -\frac{\partial V}{\partial y} = -2y$$

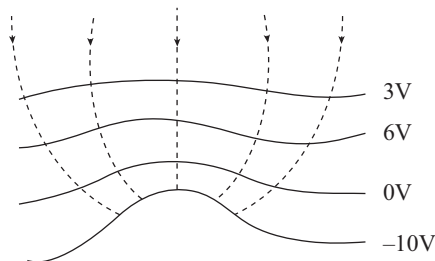
Magnitude of field parallel to xy plane is

$$E_{xy} = \sqrt{E_x^2 + E_y^2} = \sqrt{4x^2 + 4y^2}$$

at $(1, 2, 3)$

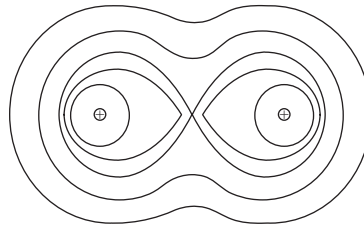
$$E_{xy} = \sqrt{4(1)^2 + 4(2)^2} = 2\sqrt{5} \text{ Vm}^{-1}$$

29.



Broken lines are electric field lines. They are perpendicular to the equipotentials directed towards the region of decreasing potential.

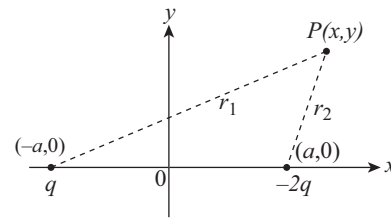
30.



32. A cylinder having line charge along its axis is an equipotential surface. Radial field lines are normal to the cylindrical surface.

33. Potential at a point $P(x, y)$ is

$$V = \frac{1}{4\pi\epsilon_0} \frac{q}{r_1} + \frac{1}{4\pi\epsilon_0} \frac{(-2q)}{r_2}$$



$$r_1 = \sqrt{(x+a)^2 + y^2}$$

$$r_2 = \sqrt{(x-a)^2 + y^2}$$

$V = 0$ when

$$\frac{1}{r_1} = \frac{2}{r_2}$$

$$\Rightarrow \frac{1}{r_1^2} = \frac{4}{r_2^2}$$

$$\Rightarrow r_2^2 = 4r_1^2$$

$$\Rightarrow (x-a)^2 + y^2 = 4[(x+a)^2 + y^2]$$

$$\Rightarrow 3x^2 + 3y^2 + 3a^2 + 10ax = 0$$

$$\Rightarrow x^2 + y^2 + a^2 + \frac{10}{3}ax = 0$$

$$\Rightarrow \left(x + \frac{5}{3}a\right)^2 + y^2 + \left(a^2 - \frac{25}{9}a^2\right) = 0$$

$$\Rightarrow \left(x + \frac{5}{3}a\right)^2 + y^2 = \frac{16}{9}a^2$$

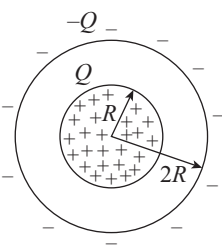
This is equation of a circle with centre at $\left(-\frac{5a}{3}, 0\right)$

and radius $= \frac{4}{3}a$.

$$34. (a) V_{\text{centre}} = \frac{3}{2} \frac{1}{4\pi\epsilon_0} \frac{Q}{R} + \frac{1}{4\pi\epsilon_0} \frac{(-Q)}{R}$$

$$= \frac{Q}{8\pi\epsilon_0 R}$$

Note that the potential due to $-Q$ charge at all inside points is equal to $\frac{1}{4\pi\epsilon_0} \frac{(-Q)}{R}$



$$(b) V = \frac{1}{4\pi\epsilon_0} \frac{Q}{1.5R} + \frac{1}{4\pi\epsilon_0} \frac{(-Q)}{R} = -\frac{Q}{12\pi\epsilon_0 R}$$

$$(c) V = \frac{1}{4\pi\epsilon_0} \frac{Q}{3R} + \frac{1}{4\pi\epsilon_0} \frac{(-Q)}{3R} = 0$$

35. Transfer half the charge on another hemisphere so as to complete a sphere. Now you have a complete sphere of half the charge density but total charge Q . Potential at the centre will be same for this sphere and the given hemisphere. Distance of charges from the centre has not changed.

$$36. \text{ Final common potential } V = \frac{V_1 r_1 + V_2 r_2}{r_1 + r_2} = 0 + \frac{V_2 R}{r + R}$$

$$= \frac{\frac{Q}{4\pi\epsilon_0 R} \cdot R}{r + R} = \left(\frac{Q}{r + R} \right) \frac{1}{4\pi\epsilon_0}$$

$$\therefore q_1 = 4\pi\epsilon_0 r \cdot V$$

$$= r \cdot \frac{Q}{r + R}$$

$$= \frac{2 \times 10}{2 + 4} nC = \frac{10}{3} nC$$

Or, more simply we can proceed like this:

Potential of a ball $\propto \frac{Q}{r}$

Finally, both balls have same potential.

$$\therefore \frac{q_1}{r} = \frac{q_2}{R} \Rightarrow \frac{q_1}{q_2} = \frac{r}{R}$$

We need to divide total charge Q into two parts in the ratio 1:2.

$$\therefore q_1 = \frac{Q}{1+2} = \frac{Q}{3} \text{ and } q_2 = \frac{Q \cdot 2}{1+2} = \frac{2Q}{3}$$

$$38. (a) V_A = \frac{1}{4\pi\epsilon_0} \frac{Q}{2R}$$

- (b) Let charge on A after closing the switch is q . Its potential must be zero.

$$\therefore \frac{1}{4\pi\epsilon_0} \frac{q}{R} + \frac{1}{4\pi\epsilon_0} \frac{Q}{2R} = 0$$

$$\Rightarrow q = -\frac{Q}{2}$$

$\therefore$ The Earth supplies a charge $-\frac{Q}{2}$ to A .

39. Charge resides on outer surface of B . Potential due to this charge at all inside points is same as surface potential.

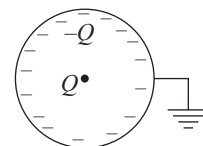
40. The potential difference depends on charge on the inner shell only.

41. Charge $-Q$ is induced on the inner wall of the shell. Outer wall has no charge.

[Refer to example (34).]

$$U = \frac{(-Q)^2}{8\pi\epsilon_0 R} + \left(\frac{1}{4\pi\epsilon_0} \frac{Q}{R} \right) (-Q)$$

$$U = -\frac{Q^2}{8\pi\epsilon_0 R}$$



When the charge is moved to infinity, there will be no charge on the shell as well. Potential energy of the system becomes zero.

$$\therefore \Delta U_{\text{gain}} = \frac{Q^2}{8\pi\epsilon_0 R}$$

43. Charge on alpha particle is $q = +2e$

$$\therefore \Delta K = (2e)(1000 \text{ V})$$

$$= 2[e (1 \text{ V})] \times 1000$$

$$= 2000 \text{ eV.}$$

[$\therefore$ (e) (1 V) = 1 electronvolt energy]

44. $P = q \cdot r = (3.4 \times 10^{-20} \text{ C}) (1 \times 10^{-10} \text{ m})$
- $$= 3.4 \times 10^{-30} \text{ cm.}$$

$$45. P = \sqrt{(qd)^2 + (qd)^2 + 2(qd)(qd)\cos 60^\circ} = \sqrt{3} qd$$

$$46. \tan \alpha = \frac{1}{2} \tan \theta$$

$$\Rightarrow \tan 45^\circ = \frac{1}{2} \tan \theta \Rightarrow \theta = \tan^{-1}(2)$$

$$47. E_0 = \frac{1}{4\pi\epsilon_0} \frac{2P}{r^3}$$

Field on the perpendicular bisector is

$$E = \frac{1}{4\pi\epsilon_0} \frac{P}{(2r)^3} = \frac{E_0}{16}$$

$$48. P = q(2d) = 3.2 \times 10^{-19} \times 2 \times 10^{-8} = 6.4 \times 10^{-27} \text{ cm}$$

In stable equilibrium position $\theta = 0^\circ$.

$$\begin{aligned} W &= U_{60} - U_0 \\ &= -PE \cos 60^\circ - (-PE \cos 0^\circ) = PE (1 - \cos 60^\circ) \\ &= \frac{1}{2} PE = \frac{1}{2} \times 6.4 \times 10^{-27} \times 10^6 = 3.2 \times 10^{-21} \text{ J.} \end{aligned}$$

49. Energy conservation gives

$$K_0 + U_0 = K_{90} + U_{90}$$

$$\frac{1}{2} I \omega^2 + (-PE \cos 0^\circ) = 0 - PE \cos 90^\circ$$

$$\Rightarrow \omega = \sqrt{\frac{2PE}{I}}$$



Worksheet 1

2. Uniform field near the sheet is $E = \frac{\sigma}{2\epsilon_0}$

$$\Rightarrow E = \frac{10^{-7}}{2 \times 8.85 \times 10^{-12}} = 5.65 \times 10^3 \text{ Vm}^{-1}$$

If d is distance between the two equipotential surfaces

$$Ed = \Delta V$$

$$\Rightarrow d = \frac{5V}{5.65 \times 10^3 \text{ Vm}^{-1}}$$

$$= 0.88 \times 10^{-3} \text{ m}$$

3. Potential at a point P due to charges at the vertices of the square is

$$V = 4 \frac{kq}{r} = \frac{4kq}{\sqrt{x^2 + a^2}}$$

[a = half of diagonal]

Therefore, V is maximum at $x = 0$ and decreases as one moves away on either side of the x axis

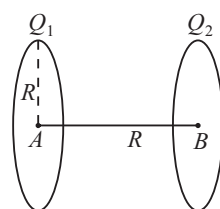
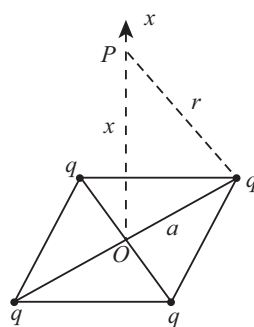
$$U = QV.$$

Thus, U also decreases as one moves away from the origin.

4. $V_A = K \frac{Q_1}{R} + K \frac{Q_2}{\sqrt{R^2 + R^2}}$

$$= K \frac{Q_1}{R} + K \frac{Q_2}{\sqrt{2}R}$$

$$V_B = \frac{KQ_1}{\sqrt{2}R} + \frac{KQ_2}{R}$$



$$W = q(V_B - V_A)$$

$$= \frac{K(\sqrt{2} - 1)}{\sqrt{2}R} [Q_2 - Q_1]$$

5. Due to repulsion, Y gains speed and X slows down. When X stops, Y must have speed u for the momentum to remain conserved.

6. As the rod becomes vertical, the particle loses its gravitational as well as electrostatic potential energy.

$$\text{Loss in gravitational } PE = mgl \left(1 - \frac{1}{\sqrt{2}} \right)$$

$$\text{Loss in electrostatic energy} = qE \frac{l}{\sqrt{2}} = \frac{mgl}{\sqrt{2}}$$

$$[\because qE = mg]$$

Recall that when we move in the direction of a uniform electric field by a distance x , the potential drops by Ex .

$$\therefore \frac{1}{2} mu^2 = mgl \left(1 - \frac{1}{\sqrt{2}} \right) + mg \frac{l}{\sqrt{2}}$$

$$\Rightarrow u = \sqrt{2gl}$$

$$\text{Angular speed, } \omega = \frac{u}{l} = \sqrt{\frac{2g}{l}}$$

7. At closest distance, the velocity (u) of the two particles will be same. Momentum conservation gives:

$$mu + mu = mv$$

$$\Rightarrow u = \frac{v}{2}$$

Energy conservation gives:

$$2 \frac{1}{2} mu^2 + \frac{1}{4\pi\epsilon_0} \frac{Q^2}{r} = \frac{1}{2} mv^2$$

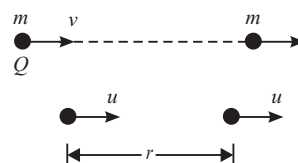
$$\Rightarrow \frac{1}{4\pi\epsilon_0} \frac{Q^2}{r} = \frac{1}{4} mv^2$$

$$\Rightarrow r = \frac{4Q^2}{4\pi\epsilon_0 mv^2}$$

8. As the bead moves from P to M , its distance from fixed charge Q decreases; therefore, its potential energy increases. It has smallest speed at M . For completing the circle, KE at M must be > 0 .

$$\therefore KE_M + K \frac{qQ}{a} = \frac{1}{2} mu^2 + K \frac{Qq}{4a}$$

$$\text{It is given that } K \frac{Q}{4a} = V$$



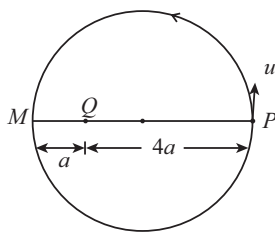
$$\therefore KE_M + 4qV = \frac{1}{2}mu^2 + qV$$

$$KE_M = \frac{1}{2}mu^2 - 3qV$$

$$KE_M > 0$$

$$\Rightarrow \frac{1}{2}mu^2 > 3qV$$

$$\Rightarrow u > \sqrt{\frac{6qV}{m}}$$



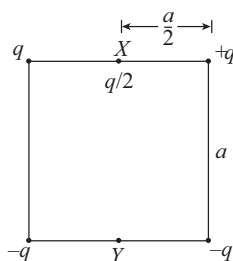
9. There are ${}^6C_2 = 15$ pairs of charges.

There are 6 pairs of $(+q, -q)$ with charges at separation a . There are 3 pairs of $(+q, -q)$ with charges at separation $2a$. There are 6 pairs of $(+q, +q$ or $-q, -q)$ with charges at separation $\sqrt{3}a$.

$$\therefore U = 6k \frac{q(-q)}{a} + 3k \frac{q(-q)}{2a} + 6k \frac{q \cdot q}{\sqrt{3}a}$$

$$= \frac{q^2}{\pi\epsilon_0 a} \left(\frac{\sqrt{3}}{2} - \frac{15}{8} \right)$$

10.



Potential at X due to charges at the vertices is

$$V_x = K \frac{q}{\left(\frac{a}{2}\right)} + K \frac{q}{\left(\frac{a}{2}\right)} + \frac{K(-q)}{\sqrt{a^2 + \frac{a^2}{4}}} + \frac{K(-q)}{\sqrt{a^2 + \frac{a^2}{4}}}$$

$$= \frac{4Kq}{a} \left(1 - \frac{1}{\sqrt{5}} \right)$$

$$\text{Similarly, } V_y = \frac{4Kq}{a} \left(-1 + \frac{1}{\sqrt{5}} \right)$$

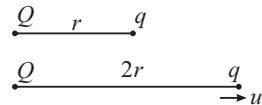
Work done in moving charge $\frac{q}{2}$ from X to Y = change in potential energy

$$= \frac{q}{2} [V_y - V_x]$$

$$= \frac{4Kq^2}{a} \left(\frac{1 - \sqrt{5}}{\sqrt{5}} \right)$$

11. Some external agent has to apply force on Q to keep it fixed. Energy conservation gives

$$\frac{1}{2}mu^2 + k \frac{qQ}{2r} = k \frac{qQ}{r}$$



$$\Rightarrow u^2 = \frac{kqQ}{mr}$$

$$\Rightarrow u = \sqrt{\frac{qQ}{4\pi\epsilon_0 mr}}$$

Internal forces can change KE of a system but they cannot cause the momentum to change. The force applied by the external agent on Q is the cause for change in momentum of the system of $Q + q$.

$\therefore$ Required impulse = change in momentum of the system

$$= mu = \sqrt{\frac{Qqm}{4\pi\epsilon_0 r}}$$

12. Moving 1m in x, y or z direction causes the potential to drop by 2V.

$$\therefore E_x = E_y = E_z = 2 \text{ Vm}^{-1}$$

When one moves from $(0, 0, 0)$ to $(1, 1, 1)$, the potential drops by

$$E_x 1 + E_y 1 + E_z 1 = 2 \times 1 + 2 \times 1 + 2 \times 1 = 6V$$

$\therefore$ Potential at $(1,1,1)$ is $10 - 6 = 4V$.

13. Had there been charges at all corners, the field at the centre would have been 0. With charge missing at one of the corners, field at the centre will be

$$E = \frac{kq}{r^2}$$

$$\text{Potential at the centre is } V = \frac{(n-1)kq}{r}$$

$$\therefore \frac{V}{E} = (n-1)r$$

14. A vector along the line $y = 2x$ can be

$$\vec{r} = \hat{i} + 2\hat{j}$$

The electric field vector must be perpendicular to $\vec{r}$
It means $\vec{E} \cdot \vec{r} = 0 = 0$.

Among the given options $\vec{E} = -8\hat{i} + 4\hat{j}$ fulfils this condition.

15. $E = \frac{F}{Q} = 1000 \text{ NC}^{-1}$

$$\Delta V = E \Delta r = 1000 \times \frac{1}{100} = 10 \text{ V}$$

16. PE of an electron at a point is $U = (-e) V$. At point B , the electric potential is negative. At all other indicated points, it is zero or positive. Therefore, PE of the electron is highest at point B .

17. $\vec{E} = (E \cos \theta) \hat{i} + (E \sin \theta) \hat{j}$

$$\vec{r} = d\hat{i} + d\hat{j}$$

$$\therefore V_A - V_0 = -\vec{E} \cdot \vec{r} = -Ed (\cos \theta + \sin \theta)$$

$$\Rightarrow V_0 - V_A = Ed (\cos \theta + \sin \theta)$$

18. $V = k[2x^2 - y^2 + z^2]$

$$E_x = -\frac{\partial V}{\partial x} = -4kx$$

$$E_y = -\frac{\partial V}{\partial y} = 2ky$$

$$E_z = -\frac{\partial V}{\partial z} = 2kz$$

At $(1, 1, 1)$, the field is

$$\vec{E} = -(4k)\hat{i} + (2k)\hat{j} + (2k)\hat{k}$$

$$\therefore E = k\sqrt{4^2 + 2^2 + 2^2} = 2\sqrt{6}k$$

19. A unit vector along the given line is

$$\hat{n} = \frac{1}{\sqrt{2}}(\hat{i} + \hat{j})$$

[The given line is parallel to a line $y = x$]

$$\therefore \vec{E} = \frac{100}{\sqrt{2}}(\hat{i} + \hat{j}) \text{ Vm}^{-1}$$

$$\vec{r} = \overline{AB} = (-2\hat{i} + 2\hat{j}) \text{ m}$$

$$\vec{E} \cdot \vec{r} = 0$$

$$\therefore \Delta V = 0.$$

20. A total $3Q - 2Q + Q = 2Q$ charge is at a distance R from the centre.

$$\therefore V = \frac{1}{4\pi\epsilon_0} \frac{2Q}{R} = \frac{Q}{2\pi\epsilon_0 R}$$

21. $E_x = -2x^3$

$$\int_0^V dV = -\int_{x=2}^{x=1} E_x dx$$

$$V = 2 \left[\frac{x^4}{4} \right]_2^1 = \frac{1}{2} - \frac{16}{2} = -7.5 \text{ V.}$$

22. Potential of the inner sphere

$$V_1 = k \frac{q_1}{r_1} + k \frac{q_2}{r_2}$$

Potential of outer sphere

$$V_2 = k \frac{q_1}{r_2} + k \frac{q_2}{r_2}$$

$$\therefore V_1 - V_2 = kq_1 \left(\frac{1}{r_1} - \frac{1}{r_2} \right)$$

$$= 9 \times 10^9 \times 10 \left[\frac{100}{10} - \frac{100}{20} \right]$$

$$= 4.5 \times 10^{11} \text{ V}$$

23. The potential difference between A and B will be independent of path taken to perform the following integration. We will move along the given circular curve itself.

$$V_B - V_A = -\int_A^B \vec{E} \cdot d\vec{l}$$

$$V_B - V_A = -\int_A^B E dl \cos 0^\circ$$

$$= -E \int_A^B dl$$

$$\Rightarrow V_B = V - ER\theta$$

24. If you spread the charge on a complete sphere (uniformly), the potential at P will not change.

$$\therefore V = k \frac{Q}{(0.15)} = 9 \times 10^9 \times \frac{5 \times 10^{-9}}{0.15} = 300 \text{ V.}$$

26. Field due to the first dipole at the given point is

$$\vec{E}_1 = k \frac{P}{(1)^3} (-\hat{k})$$

$$= kP (-\hat{k})$$

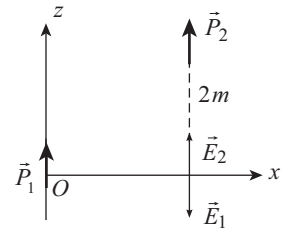
Field due to the second dipole is

$$\vec{E}_2 = k \frac{2 \cdot P/2}{(2)^3} (\hat{k})$$

$$= k \frac{P}{8} (\hat{k})$$

$$\therefore \text{Resultant is } \vec{E} = kP \left[\frac{1}{8} - 1 \right] (\hat{k})$$

$$= -\frac{7P}{32\pi\epsilon_0} (\hat{k})$$



27. Each point on the ring is on a perpendicular bisector to the given dipole. The electric field at each point on the ring is directed opposite to the dipole moment. Force on any small segment of the ring is directed opposite to the dipole moment.

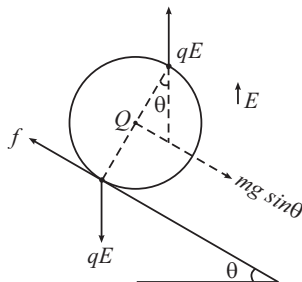
28. The electric forces on the two charges have zero resultant but produce torque. For equilibrium, friction, $f = mg \sin \theta$

For rotational equilibrium:

$$2qER \sin \theta = f R$$

$$2qE \sin \theta = mg \sin \theta$$

$$\Rightarrow E = \frac{mg}{2q}$$



29. BPD is an equipotential line.

APC is perpendicular to line BPD. Therefore, line APC is a field line. Electric field is directed along PA (in the direction of decreasing field).

$$|E| = \left| \frac{\Delta V}{\Delta r} \right| = \frac{3}{0.1\sqrt{2}} = 15\sqrt{2} \text{ Vm}^{-1}$$

30. Field lines are emanating from $+q$ and terminating at $-q$.

31. Charge on a segment subtending $d\theta$ at the centre of curvature is

$$dq = \lambda (Rd\theta) = \frac{q}{\frac{\pi R}{2}} \cdot Rd\theta$$

$$= \frac{2q}{\pi} d\theta$$

A $(-dq)$ charge at the centre and $(+dq)$ charge on the circumference form a dipole. Dipole moment is

$$\vec{dp} = P \cos \theta \hat{j} + P \sin \theta \hat{i}$$

$$= R \cdot dq (\cos \theta \hat{j} + \sin \theta \hat{i})$$

$$= \frac{2qR}{\pi} (\cos \theta \hat{j} + \sin \theta \hat{i}) d\theta$$

Dipole moment of the entire arrangement is

$$\vec{P} = \frac{2qR}{\pi} \left[\int_0^{\pi/2} \cos \theta \cdot d\theta \hat{j} + \int_0^{\pi/2} \sin \theta \cdot d\theta \hat{i} \right]$$

$$= \frac{2qR}{\pi} (\hat{j} + \hat{i})$$

$$\therefore |\vec{P}| = \frac{2\sqrt{2}qR}{\pi}$$

- 32.

33. Charge $-q$ at A and $+q$ at B make a dipole. The dipole moment is

$$P_1 = qd \left[-\frac{i}{\sqrt{2}} - \frac{j}{\sqrt{2}} \right]$$

where $d = \sqrt{2}a$

$$\therefore \vec{P}_1 = qa(-\hat{i} - \hat{j})$$

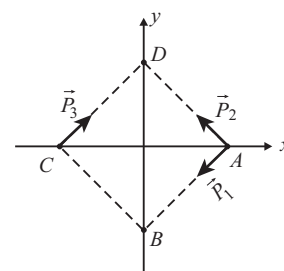
Charge $(-q)$ at A and $(+q)$ at D make another dipole having dipole moment

$$\vec{P}_2 = qa(-\hat{i} + \hat{j})$$

Charge $(-2q)$ at C and $(+2q)$ at D make another dipole having dipole moment

$$\vec{P}_3 = 2qa(\hat{i} + \hat{j})$$

$$\therefore \vec{P} = \vec{P}_1 + \vec{P}_2 + \vec{P}_3 = 2qa \hat{j}$$



34. Potential at surface $= V_0 = \frac{KQ}{R}$

$$\text{Potential at centre} = \frac{3}{2} V_0$$

Potential decreases continuously as we move out from the centre. The point where potential is half that at the centre ($= \frac{3}{4} V_0$) must be outside the sphere. Let this point be at a distance x from the centre.

$$K \frac{Q}{x} = \frac{3}{4} \frac{KQ}{R}$$

$$\Rightarrow x = \frac{4}{3} R$$

Distance of the point from the surface is $x - R = \frac{R}{3}$.

35. The bullet slows down till it reaches the centre of the sphere. After crossing the centre, the electric field accelerates it to the right. The bullet needs to have sufficient speed to reach the centre.

$$KE_{\text{surface}} + U_{\text{surface}} = KE_{\text{centre}} + U_{\text{centre}}$$

$$\frac{1}{2} mu^2 + \frac{1}{4\pi\epsilon_0} \frac{q \cdot q}{R} = 0 + \frac{3}{2} \frac{1}{4\pi\epsilon_0} \frac{q \cdot q}{R}$$

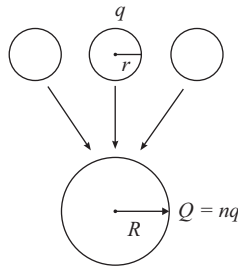
$$\Rightarrow u = \frac{q}{\sqrt{4\pi\epsilon_0 m R}}$$

36. Let each small drop has radius r and charge q on it.

$$V = K \frac{q}{r} \quad (i)$$

When n drops combine to form a large drop, the volume of liquid remains conserved. The radius of the large drop is given by

$$\frac{4}{3} \pi R^3 = n \frac{4}{3} \pi r^3 \Rightarrow R = n^{1/3} r$$



Charge on the large drop is $Q = nq$.

Potential of the large drop is

$$V' = \frac{kQ}{R} = \frac{k nq}{n^{1/3} \cdot r}$$

$$\Rightarrow V' = n^{2/3} V \quad [\text{using (i)}]$$

38. There is no field inside the inner sphere. Hence, the electric potential at all points inside the smaller sphere is same as that of the sphere itself. The answer is 10 V.

$$\begin{aligned} 39. V_A &= k \frac{q_A}{a} + k \frac{q_B}{b} + k \frac{q_C}{c} \\ &= \frac{1}{4\pi\epsilon_0} \left[\frac{4\pi a^2(-\sigma)}{a} + \frac{4\pi b^2(\sigma)}{b} + \frac{4\pi c^2(-\sigma)}{c} \right] \\ &= \frac{\sigma}{\epsilon_0} [-a + b - c] \end{aligned}$$

40. After the spheres are connected, both of them will have equal charge on them (say q) so that their potentials are same.

$$F = k \frac{q q}{(0.5)^2} \Rightarrow 0.036 = \frac{9 \times 10^9}{0.25} q^2$$

$$\Rightarrow q = 10^{-6} \text{ C} \quad (\text{i})$$

If original charge on the two spheres was q_1 and $-q_2$, then

$$0.108 = 9 \times 10^9 \frac{q_1 \cdot q_2}{(0.5)^2}$$

$$\Rightarrow q_1 \cdot q_2 = 3 \times 10^{-12} \quad (\text{ii})$$

$$\text{From (i)} \quad q_1 - q_2 = 2q = 2 \times 10^{-6} \quad (\text{iii})$$

Solving (ii) and (iii) for q_1 and q_2 gives

$$q_1 = 3 \times 10^{-6} \text{ C and } q_2 = 1 \times 10^{-6} \text{ C.}$$

$\therefore$ Charges on the spheres could be $3 \times 10^{-6} \text{ C}$ and $-1 \times 10^{-6} \text{ C}$. One needs to understand that q [given by (i)] can also be negative. Therefore, the charges can also be $-3 \times 10^{-6} \text{ C}$ and $1 \times 10^{-6} \text{ C}$.

$$\begin{aligned} 41. -\int_{\infty}^0 \vec{E} \cdot d\vec{l} &= V_{\text{centre}} - V_{\infty} \\ &= \frac{kQ}{R} = \frac{9 \times 10^9 \times 1.111 \times 10^{-10}}{0.5} = 2 \text{ JC}^{-1} \end{aligned}$$

42. Let charge on C be Q .

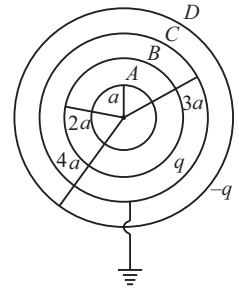
$$V_c = 0$$

$$\Rightarrow k \frac{q}{3a} + k \frac{Q}{3a} + k \frac{(-q)}{4a} = 0$$

$$\Rightarrow Q = -\frac{q}{4}$$

$$\begin{aligned} V_A &= k \frac{q}{2a} + k \frac{Q}{3a} + k \frac{(-q)}{4a} \\ &= \frac{kq}{a} \left[\frac{1}{2} - \frac{1}{12} - \frac{1}{4} \right] = \frac{kq}{6a} \end{aligned}$$

$$\therefore V_A - V_C = \frac{kq}{6a}$$



43. At points close to the origin, the field will be directed towards the origin on the axis. This is due to larger effect of negative charges owing to their smaller distance. The projected charge is accelerated towards the origin when it is near to the origin. Its speed cannot be zero at the origin.

$$\begin{aligned} 44. V &= k \frac{\sigma \cdot 4\pi r^2}{r} \\ \Rightarrow \sigma &\propto \frac{1}{r} \end{aligned}$$

45. Electric field line cannot be inside a metal. Field lines must be normal to the metal surface.

46. Radius of curvature of the surface is smaller at B . Surface charge density at B is larger than that at A [charge is induced on the metal wall].

$$\text{Field near a metal surface is } E = \frac{\sigma}{\epsilon_0}$$

$$\therefore E_B > E_A$$

The entire conductor is at one potential.

$$\therefore V_A = V_B$$

No field line can pierce through the cavity wall. Hence flux is zero.

47. Potential at the centre of the sphere due to induced charge on its surface is zero. Potential is entirely due to the charge on the ring.

$$\begin{aligned} \therefore V &= k \frac{q}{\sqrt{R^2 + x^2}} = 9 \times 10^9 \times \frac{10^{-6}}{\sqrt{4^2 + 3^2}} \\ &= 1.8 \times 10^3 \text{ V} \end{aligned}$$

48. Field is uniform. Work done by it is

$$W = q\vec{E} \cdot \Delta\vec{r}$$

$$= q(E\hat{i})(-a\hat{i} - b\hat{j} + o\hat{k}) = -qEa$$

The path is not important.

$$\Delta\vec{r} = \vec{r}_s - \vec{r}_p = -a\hat{i} - b\hat{j} + o\hat{k}$$

49. For a point (x, y, z) , $r = \sqrt{x^2 + y^2 + z^2}$

$$\therefore V = \frac{343}{\sqrt{x^2 + y^2 + z^2}}$$

$$\therefore E_x = -\frac{\partial V}{\partial x} = \frac{343x}{(x^2 + y^2 + z^2)^{3/2}}$$

$$\text{at } \vec{r} = 3\hat{i} + 2\hat{j} + 6\hat{k}; E_x = \frac{343 \times 3}{(3^2 + 2^2 + 6^2)^{3/2}} = 3$$

Similarly, find E_y and E_z .

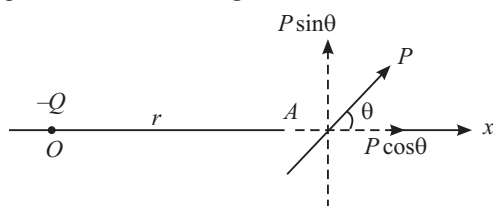
50. Equipotential surfaces are plane parallel surfaces normal to the electric field.

There is no force on the electron, parallel to the equipotential surfaces. Velocity component of the electron parallel to the equipotential surfaces cannot change.

51. Higher value of E implies faster rate of change of potential. The field is higher between the 2nd and the 3rd surface compared to the field between the 1st and the 2nd surface.

$$\therefore x_2 < x_1$$

52. We can replace the single dipole with two perpendicular dipoles as shown. At point O , the dipole $p \sin \theta$ produces a field along y direction. Only $p \cos \theta$ has field along x direction.



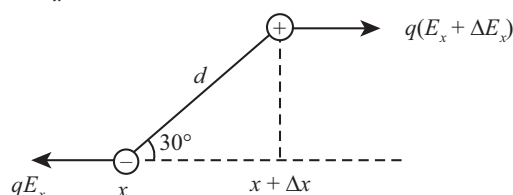
$$E_x = k \frac{2P \cos \theta}{r^3}$$

$\therefore$ Force on $-Q$ is

$$F_x = -\frac{2PkQ \cos \theta}{r^3}$$

53. We will first calculate the x component of force on the dipole. x component of field at a point having x co-ordinate x is

$$E_x = x^2$$



Field at $(x + \Delta x)$ is $(E_x + \Delta E_x)$ where

$$\Delta E_x = \left(\frac{dE_x}{dx} \right) \Delta x = 2x (d \cos 30^\circ) = 2xd \cos 30^\circ$$

$$\therefore F_x = q(E_x + \Delta E_x) - qE_x = q\Delta E_x = 2qxd \cos 30^\circ = [2P \cos 30^\circ] x$$

At $x = 1$

$$F_x = 2P \cos 30^\circ$$

Similarly, $F_y = [2P \sin 30^\circ] y$

At $y = 0$, $F_y = 0$.

55. Let charge on the inner sphere be q . Charge on inner wall of the shell will be $-q$. Charge on the outer surface of the shell will be $Q + q$.

Potential of the inner sphere is zero.

$$\Rightarrow k \frac{q}{R} + k \frac{-q}{2R} + k \left(\frac{10Q + q}{3R} \right) = 0$$

$$\Rightarrow q - \frac{q}{2} + \frac{10Q}{3} + \frac{q}{3} = 0$$

$$\Rightarrow q = -4Q$$

56. Charge on inner sphere = q .

$$\phi_1 = k \frac{q}{r_1} \Rightarrow q = \frac{\phi_1 r_1}{k}$$

When the two spheres are connected, the charge moves to the outer sphere. Potential of the inner sphere is same as that of the outer ball.

$$\therefore \phi_2 = k \frac{q}{r_2} = \phi_1 \frac{r_1}{r_2}$$

57. Charge on cavity wall = $-Q$.

Charge on the outer surface of the sphere = Q .

Charge is uniformly distributed on the cavity surface as well as the outer surface.

The energy of the system has self-energy of the spherical charges on the two surfaces and interaction energy of the three charges.

$$\begin{aligned} U &= \frac{(-Q)^2}{8\pi\epsilon_0 b} + \frac{Q^2}{8\pi\epsilon_0 a} + \frac{1}{4\pi\epsilon_0} \frac{(-Q)}{b} Q + \frac{1}{4\pi\epsilon_0} \frac{Q}{a} Q \\ &\quad + \frac{1}{4\pi\epsilon_0} \frac{Q}{a} (-Q) \\ &= \frac{Q^2}{4\pi\epsilon_0} \left[\frac{1}{2b} + \frac{1}{2a} - \frac{1}{b} + \frac{1}{a} - \frac{1}{a} \right] \\ &= \frac{Q^2}{4\pi\epsilon_0} \left[\frac{1}{2a} - \frac{1}{2b} \right] \end{aligned}$$

Note: The 4th term in the above expression was written using the following logic. Spherical charge

Q on the outer wall creates potential $V = k \frac{Q}{a}$ at all inside points. Point charge Q will have interaction energy QV with this charge. Similarly, you can think of the 5th and the 3rd term.



Worksheet 2

2. If a charge is held fixed, an external force acts on it. Therefore, momentum of the system is not conserved. However, the external force does not perform work in keeping a charge fixed. Therefore, energy of the system will remain conserved.

3. Induced charge on the inner surface of the shell is $-Q$. Total charge on the shell is $-q$. Therefore, the outer surface of the shell has $-q + Q$ charge on it.

The point charge and the charge on the inner surface, together, have no effect outside the cavity region. The potential at the outer surface of the shell is due to outer charge only.

$$V = k \frac{(Q - q)}{b}$$

The entire shell (a conductor) is at same potential. Charge is distributed uniformly on both surfaces of the shell. These charges produce no field inside the cavity region. There is no force on the point charge. When the point charge is not at the centre, charge induced on the inner wall of the shell is non-uniform. This non-uniform charge will exert a net force on the point charge.

4. For option (D), we need to recall that kinetic energy of a two particle system can be written as

$$K = \frac{1}{2} M V_{\text{cm}}^2 + \frac{1}{2} \mu v_{\text{rel}}^2$$

When the particles are back to separation x , the kinetic energy of the system must be same as its original kinetic energy, as there is no change in potential energy. In absence of an external force, V_{cm} does not change and hence v_{rel} cannot change.

5. Charge is at point A (4, 7, 2) and the given point is B (1, 3, 2)

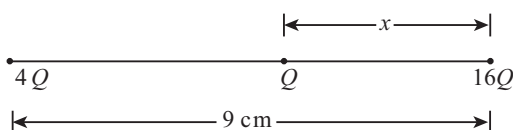
$$\vec{r} = \vec{AB} = -3\hat{i} - 4\hat{j} + 0\hat{k}$$

The electric field will be along $\vec{r}$. It will have x as well as y components.

$$V = k \frac{Q}{r} = 9 \times 10^9 \times \frac{10^{-8}}{\sqrt{3^2 + 4^2}} = 18 \text{ V}$$

6. The charges of highest magnitude must be placed at farthest distance, to ensure minimum potential energy

$$U = k \frac{Q(4Q)}{(9-x)} + k \frac{Q(16Q)}{x} + k \frac{(4Q)(16Q)}{9}$$



$$U \text{ is minimum when } \frac{dU}{dx} = 0$$

$$\Rightarrow 4kQ^2 \frac{d}{dx} \left[\frac{1}{9-x} + \frac{4}{x} + \frac{16}{9} \right] = 0$$

$$\Rightarrow \frac{1}{(9-x)^2} - \frac{4}{x^2} = 0$$

$$\Rightarrow \frac{x}{9-x} = 2$$

$$\Rightarrow x = 6 \text{ cm.}$$

One can easily verify that the electric field due to $4Q$ and $16Q$ at the location of Q is zero.

7. PE of the electron at A is $U_A = -3 \text{ eV}$

$$PE \text{ at } B \text{ is } U_B = -7 \text{ eV}$$

Since $U_A > U_B$, the electron must have KE at A to be able to reach B .

$$K_A + U_A = K_B + U_B$$

$$\Rightarrow K_B - K_A = U_A - U_B = 4 \text{ eV}$$

For the electron to just reach point A , $K_A = 0$. Minimum KE needed at B is 4 eV.

10. Let radius of the sphere be R and charge on it be Q .

$$k \frac{Q}{(R+0.05)} = 100 \quad \text{(i)}$$

$$\text{and } k \frac{Q}{(R+0.10)} = 75 \quad \text{(ii)}$$

$$\Rightarrow \frac{R+0.10}{R+0.05} = \frac{4}{3}$$

$$\Rightarrow R = 0.1 \text{ m} = 10 \text{ cm}$$

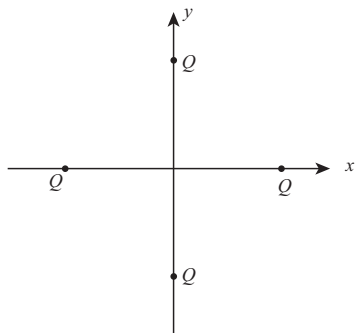
$$V_{\text{surface}} = \frac{kQ}{R} = \frac{kQ}{0.10} = 100 \times \frac{0.15}{0.10} \quad [\text{using (i)}]$$

$$= 150 \text{ V}$$

$$V_{\text{centre}} = \frac{3}{2} V_{\text{surface}} = 225 \text{ V}$$

$$E_{\text{surface}} = \frac{V_{\text{surface}}}{R} = \frac{150}{0.1} = 1500 \text{ Vm}^{-1}$$

11. The charges are in xy plane. They lie on vertices of a square.



12. $E = -\frac{dV}{dx}$

$\therefore E_B = 0$

$\frac{dV}{dx}$ is positive at A and hence E_A is negative.

$\frac{dV}{dx}$ is negative at C and hence E_C is positive.

13. Electric force, $F = qE = 10^{-6} \times 300 = 3 \times 10^{-4}$ N.
Impulse of the electric force $= F\Delta t = 3 \times 10^{-3}$ Ns.
Magnitude of change in momentum $\Delta P = 3 \times 10^{-3}$ Ns.
Initial momentum $P_i = mu = 10^{-3} \times 4 = 4 \times 10^{-3}$ Ns.
Direction of ΔP depends on the direction of the field.
Final momentum $\vec{P}_f = \vec{P}_i + \Delta \vec{P}$

Maximum magnitude of $\vec{P}_f$ is $P_i + \Delta P = 7 \times 10^{-3}$ Ns.

Minimum magnitude of $\vec{P}_f$ is $P_i - \Delta P = 1 \times 10^{-3}$ Ns.

$\therefore v_{\max} = \frac{7 \times 10^{-3}}{m} = 7 \text{ ms}^{-1}$

and $v_{\min} = \frac{1 \times 10^{-3}}{m} = 1 \text{ ms}^{-1}$

Therefore, a speed of 0.5 ms^{-1} is not possible.

14. Deuteron is a deuterium nucleus. Charge on both the proton and the deuteron is same. They have same KE after being accelerated through same potential difference

$K = qV$.

The two particles have different mass. Therefore, they have different speeds and different momenta.

15. In absence of external force, momentum is conserved. Loss in momentum of one particle must be equal to the gain in momentum of the other particle.

But KE is not constant. As the particles come closer, the PE increases. Therefore, KE of the system decreases. Loss in KE of A must be larger than gain in KE of B , so that overall there is loss in KE .

17. Unit of ϵ_0 is $\frac{C^2}{Nm^2}$

Unit of resistance (R) is $\frac{\text{Volt}}{\text{Amp}} = \frac{Js}{C^2} = \frac{Nms}{C^2}$

Unit of G is $\frac{Nm^2}{kg^2}$

Unit of potential (V) is $\frac{J}{C} = \frac{Nm}{C}$

Angular displacement is dimensionless. Hence (A) is correct.

$\epsilon_0^{-1} R^{-1} G^\circ V^\circ = \frac{Nm^2}{C^2} \cdot \frac{C^2}{Nms} = \frac{m}{s}$

Therefore, (b) is also correct.

Similarly, check for (c) and (d).

18. Flux $= E \cdot S$

Unit of flux is $\frac{V}{m} \cdot m^2 = Vm$.

19. The system can be treated as made of two dipoles inclined at 60° . Dipole moment of each dipole is $P = qL$.

$\therefore$ Resultant dipole moment is $\sqrt{P^2 + P^2 + 2PP \cos 60^\circ} = \sqrt{3}P = \sqrt{3}qL$

20. Imagine a circle that has the dipole along its axis (i.e., the circle is perpendicular bisector of the dipole). Potential is zero on this circle.

21. Force on Q due to the dipole can be calculated easily. Same force will be applied by Q on the dipole.

22. Refer to example 32.

Potential of the conductor is $V = \frac{1}{4\pi\epsilon_0} \frac{Q}{(d+R)}$

Potential at point B is also V .

But, $V_B = V_q + V_{in}$

$\frac{1}{4\pi\epsilon_0} \frac{q}{(d+R)} = \frac{1}{4\pi\epsilon_0} \frac{q}{d} + V_{in}$

$\therefore V_{in} = \frac{-qR}{4\pi\epsilon_0(d+R)d}$

23. For writing field in the region of overlap, you can refer to the last chapter.

Potential at the centre of the left sphere is sum of the potentials due to the positive and the negative charges.

$V_1 = \frac{3}{2} \frac{1}{4\pi\epsilon_0} \frac{Q}{R} + \frac{1}{4\pi\epsilon_0} \frac{(-Q)}{d}$

Where $Q = \frac{4}{3} \pi r^3 \cdot \rho$

$\therefore V_1 = \frac{R^2 \rho}{2\epsilon_0} - \frac{R^3 \rho}{3d\epsilon_0}$

Similarly, potential at the centre of the right sphere is

$$V_2 = -\frac{R^2\rho}{2\epsilon_0} + \frac{R^3\rho}{3d\epsilon_0}$$

$$\therefore V_1 - V_2 = \frac{R^2\rho}{\epsilon_0} - \frac{2R^3\rho}{3d\epsilon_0}$$

For $d = R$

$$V_1 - V_2 = \frac{1}{3} \frac{\rho R^2}{\epsilon_0} = \frac{1}{3} \frac{\rho d^2}{\epsilon_0}$$

24. Potential is same at A and B due to the point charge Q at O . Therefore, there is no change in potential energy when q is moved from A to B . No work is done. Choice of path makes no difference.

C is a point closer to Q (than A). Therefore, PE increases when q is moved from A to C .

The electric force must have performed negative work.

Similar argument can be given for movement from C to B .

26. The charges on the inner faces of the two plates are $\left(\frac{Q_1 - Q_2}{2}\right)$ and $-\left(\frac{Q_1 - Q_2}{2}\right)$ as shown.

When both Q_1 and Q_2 are positive and $Q_1 > Q_2$, the inner face of plate 1 has positive charge. Field between the plates is directed from 1 to 2. Potential of plate 1 is higher.

When both charges are negative and $|Q_1| > |Q_2|$, the inner wall of plate 1 is negative. Field is from plate 2 to 1. Plate 2 is at higher potential.

Similarly, one can see that (c) is correct.

27. Field due to the dipole on the right is towards right at the location of the other dipole. Field is slightly higher at the location of $-q$ than that at the position of $+q$. Force on $-q$ is towards left. Hence, force is towards left.

28. Field at A is

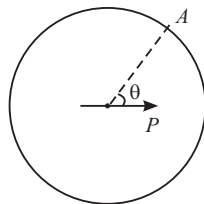
$$E = \frac{kP}{r^3} \sqrt{1 + 3\cos^2\theta}$$

$$= E_0 \sqrt{1 + 3\cos^2\theta}$$

$$\text{where } E_0 = \frac{kP}{r^3}$$

$\cos^2\theta$ can have a maximum value of 1 and its minimum value is 0.

$\therefore E$ can have a maximum value of $2E_0$ and a minimum value of E_0 .



$$\therefore E_0 \leq 10 \text{ N/C} \leq 2E_0$$

$$\Rightarrow 5 \leq E_0 \leq 10 \text{ N/C}$$

Field at all points on the sphere must lie in between E_0 and $2E_0$.

4 N/C and 32 N/C cannot be in this range.

30. Change in V_0 will lead to change in speed of the electrons. The kinetic energy acquired by the electrons in the gun depends on V_0 .

If V_0 is increased, the electrons enter the plate region with high speed. They cross the plates in smaller time. The field between the plates gets less time to deflect the electrons.

Increasing V implies increase in electric field between the plates. The electrons experience more force and deflection is more.

Field between the deflecting plates is $E = \frac{V}{d}$, where d is separation between the plates. Changing d will change E .

31. Potential at distance x from the origin is

$$V = \frac{k(-q)}{\sqrt{6+x^2}} + k \frac{2\sqrt{2}q}{\sqrt{16+x^2}}$$

$$\text{at } x = 0, V = -\frac{kq}{\sqrt{6}} + k \frac{2\sqrt{2}q}{\sqrt{16}} = kq \left[\frac{1}{\sqrt{2}} - \frac{1}{\sqrt{6}} \right]$$

Field at x is

$$E = \frac{kqx}{(6+x^2)^{3/2}} - k \frac{2\sqrt{2}qx}{(16+x^2)^{3/2}} \quad (\leftarrow)$$

at $x = 2$; $E = 0$.

Electric field is zero at $x = 0$ as well as $x = 2$.

Potential at $x = 2$ will be

$$V = k \frac{2\sqrt{2}q}{\sqrt{20}} - \frac{kq}{\sqrt{10}} = \frac{kq}{\sqrt{10}}$$

Potential at $x = 0$ is less than potential at $x = 2$ [prove this yourself]. Therefore, field is directed towards left in the region $x = 0$ to $x = 2$.

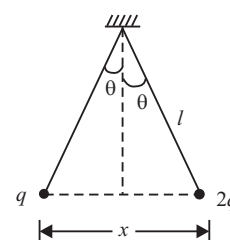
Hence, option (c) is correct.



Worksheet 3

1. (a) Both the balls exert equal and opposite force on one another. They will get deflected by same amount.
(b) Both the balls come to rest simultaneously at $\theta_2 = 60^\circ$.

$$x = 2l \sin\theta$$



Electrostatic potential energy of the system is

$$U_e = \frac{1}{4\pi\epsilon_0} \frac{q(2q)}{x} = \frac{q}{4\pi\epsilon_0 l \sin \theta}$$

As the balls move from initial position ($\theta_1 = 30^\circ$) to final position ($\theta_2 = 60^\circ$), the loss in electric potential energy is equal to gain in their gravitational potential energy.

$$\frac{q^2}{4\pi\epsilon_0 l \sin 30^\circ} - \frac{q^2}{4\pi\epsilon_0 l \sin 60^\circ} = 2mgl(\cos 30^\circ - \cos 60^\circ)$$

$$\Rightarrow q^2 \left[1 - \frac{1}{\sqrt{3}} \right] = 4\pi\epsilon_0 mgl^2 \left(\frac{\sqrt{3}}{2} - \frac{1}{2} \right)$$

$$\Rightarrow q = l\sqrt{2\sqrt{3}\pi\epsilon_0 mg}$$

2. We need not pay attention to the steps involved.

$$\begin{aligned} W_{\text{ext}} &= U_f - U_i \\ &= \left[3 \frac{kq^2}{x} \right] - \left[4 \frac{kq^2}{x} + 2 \frac{kq^2}{\sqrt{2}x} \right] \\ &= -\frac{kq^2}{a} (1 + \sqrt{2}) \end{aligned}$$

3. $E_x = -\text{slope of } V-x \text{ graph} = -\left(-\frac{20}{4}\right) = 5 \text{ Vm}^{-1}$

$$E_y = -\text{slope of } V-y \text{ graph} = -\left(\frac{16}{4}\right) = -4 \text{ Vm}^{-1}$$

$$\vec{E} = (5\hat{i} - 4\hat{j}) \text{ Vm}^{-1}$$

Force on $-q$ is

$$\vec{F} = -5q\hat{i} + 4q\hat{j}$$

$$\begin{aligned} \therefore \text{Acceleration is } \vec{a} &= \frac{\vec{F}}{m} = \frac{q}{m} (-5\hat{i} + 4\hat{j}) \\ &= (-10\hat{i} + 8\hat{j}) \text{ ms}^{-2} \end{aligned}$$

Force is independent of position as the field is uniform

4. Field near the surface of conductor is always normal to its surface. An equipotential surface is normal to the field. Therefore, it is spherical.

$$5. dV = -\vec{E} \cdot d\vec{r}$$

$$\begin{aligned} &= -[(2xy)\hat{i} + (x^2)\hat{j}] [(dx)\hat{i} + (dy)\hat{j}] \\ &= -2xydx - x^2dy \end{aligned}$$

$$\Rightarrow \int_0^{V_0} dV = - \int_{(0,0)}^{(x,y)} [(2xy)dx + x^2dy]$$

$$\text{Let } z = x^2y$$

$$\text{then } dz = 2xdy + x^2dy$$

Therefore, the above equation can be written as

$$V_0 = - \int_0^{x^2y} dz$$

$$\Rightarrow V_0 = -x^2y$$

$$\Rightarrow x^2y = -V_0$$

This is the required equation.

6. $V \propto \frac{Q}{x}$ where x is side length of the cube.

Q = charge inside the cube

$$= \rho x^3$$

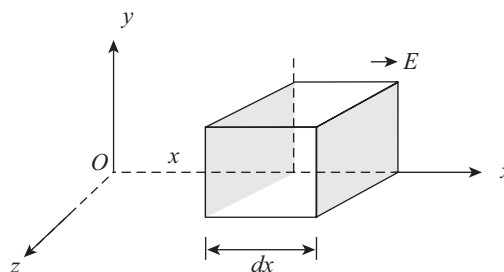
$$\therefore V \propto \frac{\rho x^3}{x} = \rho x^2$$

$$\therefore V \propto \rho x^2$$

$$\text{and } V \propto \frac{\rho}{2} (2x)^2$$

$$\therefore V = 2V_0$$

7.



Obviously, electric field is along x direction only

$$E = -\frac{dV}{dx} = 2ax$$

Consider a box shaped volume as shown. Flux of electric field through the surface of the box is

$$\begin{aligned} \phi &= -2ax \cdot A + 2a(x+dx) \cdot A \\ &= 2aA dx \end{aligned}$$

Here, A is area of the shaded surface.

If q is charge inside the box, the Gauss's law gives:

$$\phi = \frac{q}{\epsilon_0}$$

$$\Rightarrow 2aA dx = \frac{q}{\epsilon_0}$$

$$\Rightarrow q = 2aA\epsilon_0 dx$$

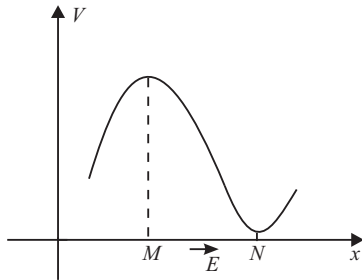
$$\therefore \rho = \frac{q}{A \cdot dx} = 2a\epsilon_0$$

8. (a) Electric field is the gradient of potential. At both M and N , $\frac{dV}{dr} = 0$. Hence, field is zero at M as well as N . Any charge will be in equilibrium at M or N .

At M , potential energy of a negative charge will be minimum. It is a position of stable equilibrium for a negative charge. If displaced slightly from M , a negative charge will be pushed back towards M .

At N , potential energy of a negative charge will be maximum. It is a position of unstable equilibrium.

- (b) Electric field between M and N will be towards N (in the direction of decreasing potential). An electron will experience force towards M . It will definitely reach M .



9. (a) Induced charge on inner wall $= -q$

Induced charge on outer wall $= q$

Potential at the centre is due to the point charge and charges on the inner and outer walls of the shell.

$$\begin{aligned} \therefore V_{\text{centre}} &= k\frac{q}{x} + k\frac{(-q)}{a} + k\frac{q}{b} \\ &= kq \left[\frac{1}{x} - \frac{1}{a} + \frac{1}{b} \right] \end{aligned}$$

- (b) The point charge (q) and induced charge ($-q$), together, produce no field at a point which is at distance $r > a$ from the centre. These two charges contribute nothing to the potential of a point outside the sphere. Potential at an outside point is only because of the charge on the outer wall. This charge is uniformly distributed spherical charge.

$$\therefore V_r = \frac{kq}{r}$$

10. (a) The potential difference depends on charge on the inner shell only. [Refer to example (35).] Therefore, there is no change in potential difference.

- (b) Before S is closed, potential of the inner shell is $V_1 = K\frac{q}{a} + k\frac{3q}{b}$

After S is closed, potential of the outer shell becomes zero. For this, charge on it must be $-q$ (refer to example 34).

Therefore, potential of inner shell will be

$$V_2 = \frac{kq}{a} + \frac{k(-q)}{b}$$

- (c) Initial and final charges on the outer shell are $3q$ and $-q$. Therefore, charge $-4q$ flows from the ground to the shell.

11. Initial charge on the smaller sphere is

$$Q_1 = 4\pi R^2 \sigma$$

Initial charge on the larger sphere is

$$Q_2 = 4\pi(2R)^2 \sigma = 16\pi R^2 \sigma$$

After connection, let the charges be q_1 and q_2 , respectively.

$$q_1 + q_2 = Q_1 + Q_2 = 20\pi R^2 \sigma \quad (i)$$

Potential of both the spheres will be same.

$$\therefore V = \frac{kq_1}{R} = \frac{kq_2}{2R}$$

$$\Rightarrow \frac{q_1}{q_2} = \frac{1}{2} \quad (ii)$$

Solving (i) and (ii), we get

$$q_1 = \frac{20}{3} \pi R^2 \sigma$$

$$\Rightarrow 4\pi R^2 \sigma_1 = \frac{20}{3} \pi R^2 \sigma$$

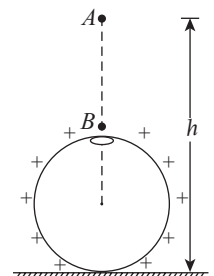
$$\Rightarrow \sigma_1 = \frac{5}{3} \sigma$$

12. After n drops have entered, the charge on the sphere will be nq . As soon as the drop touches the inner wall of the sphere, its charge spreads on the outer wall of the sphere. This charged sphere repels the falling mercury drop. A drop will just fail to enter if its speed becomes zero when it reaches the opening on the spherical surface. Using conservation of energy for $(n+1)^{\text{th}}$ drop.

$$K_A + U_A = K_B + U_B$$

$$\Rightarrow 0 + \frac{k(nq)}{(h-R)} q + mgh$$

$$= 0 + \left(\frac{k(nq)}{R} \right) \cdot q + 2mgR$$



$$\Rightarrow kq^2 n \left[\frac{1}{R} - \frac{1}{h-R} \right] = mg(h-2R)$$

$$\Rightarrow n = \frac{mgR(h-R)}{kq^2} = \frac{4\pi\epsilon_0 mgR(h-R)}{q^2}$$

13. Refer to example 50.

$$V_P = \frac{\sigma R}{\pi\epsilon_0} - \frac{\sigma R/2}{\pi\epsilon_0} = \frac{\sigma R}{2\pi\epsilon_0}$$

14. $V_1 = \frac{1}{4\pi\epsilon_0} \frac{q}{r_1}$ and $V_2 = \frac{1}{4\pi\epsilon_0} \frac{q}{r_2}$

$$\therefore V_1 - V_2 = \frac{q}{4\pi\epsilon_0} \left(\frac{1}{r_1} - \frac{1}{r_2} \right) = \frac{q}{4\pi\epsilon_0} \left(\frac{r_2 - r_1}{r_1 r_2} \right)$$

$$\Rightarrow r_2 - r_1 = \frac{4\pi\epsilon_0 (\Delta V) r_1 r_2}{q}$$

As we move outwards, the product $r_1 r_2$ increases and hence Δr increases.

15. Charge per unit length is

$$\lambda = (2\pi a \cdot 1) \sigma$$

Electric field at a distance x ($>a$) from the axis is

$$E = \frac{\lambda}{2\pi\epsilon_0 x} = \frac{2\pi a \sigma}{2\pi\epsilon_0 x}$$

$$\Rightarrow E = \left(\frac{a\sigma}{\epsilon_0} \right) \frac{1}{x}$$

$$\int_{V_0}^V dV = - \int_{x=a}^{2a} E dx$$

$$\Rightarrow V - V_0 = - \frac{a\sigma}{\epsilon_0} \int_a^{2a} \frac{dx}{x} = - \frac{a\sigma}{\epsilon_0} \ln \left(\frac{2a}{a} \right)$$

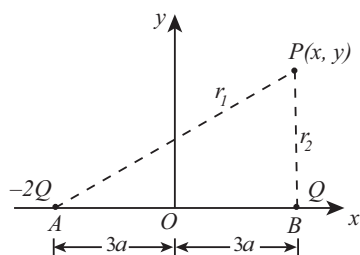
$$\Rightarrow V = V_0 - \frac{a\sigma}{\epsilon_0} \ln 2$$

16. Charge induce on the inner wall = $-q$.

Charge induce on the outer wall = q .

The system of point charge q and induced charge $-q$ has no effect for $x > R_1$.

17. (a) Let $P(x, y)$ be a point in xy plane. Potential at P is



$$V = k \frac{(-2Q)}{r_1} + k \frac{Q}{r_2}$$

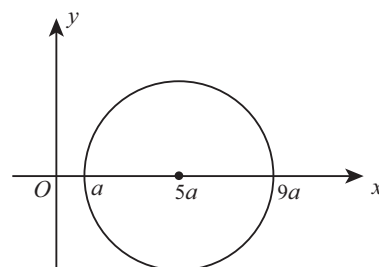
$$= kQ \left[\frac{-2}{\sqrt{(x+3a)^2 + y^2}} + \frac{1}{\sqrt{(x-3a)^2 + y^2}} \right]$$

$V = 0$ when

$$\frac{2}{\sqrt{(x+3a)^2 + y^2}} = \frac{1}{\sqrt{(x-3a)^2 + y^2}}$$

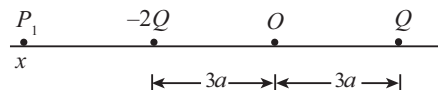
Simplification gives: $(x-5a)^2 + y^2 = (4a)^2$

This equation represents a circle of radius $4a$ having its centre at $(5a, 0)$. The circle is as shown in figure.



(b) For $x < -3a$

Distance of a point on the x axis from charge $-2Q$ is $|x - (-3a)| = |x + 3a| = -(x + 3a)$



Distance of the point from charge Q is $|x - 3a| = -(x - 3a)$

$\therefore$ Potential at point P_1 is

$$V = k \frac{-2Q}{-(x+3a)} + k \frac{Q}{-(x-3a)} = kQ \left[\frac{2}{x+3a} - \frac{1}{x-3a} \right]$$

For $-3a < x < 3a$

$$V = k \frac{-2Q}{x+3a} + k \frac{Q}{-(x-3a)} = kQ \left[\frac{1}{3a-x} - \frac{2}{3a+x} \right]$$

For $x > 3a$

$$V = k \frac{-2Q}{x+3a} + k \frac{Q}{x-3a} = kQ \left[\frac{1}{x-3a} - \frac{2}{x+3a} \right]$$

Note that $V = 0$ at $x = a$, and $x = 9a$. At a large distance from the charge system, the potential will be just like that produced due to a single $-Q$ charge.

(c) Centre of the circle is at $x = 5a$. Potential at the centre is

$$V = kQ \left[\frac{1}{x-3a} - \frac{2}{x+3a} \right]$$

$$= kQ \left[\frac{1}{5a-3a} - \frac{2}{5a+3a} \right] = k \frac{Q}{4a}$$

Potential at the circumference = 0. A positive charge released at the centre will tend to move in the direction of field (i.e., in the direction of decreasing potential). Hence, it will cross the circle.

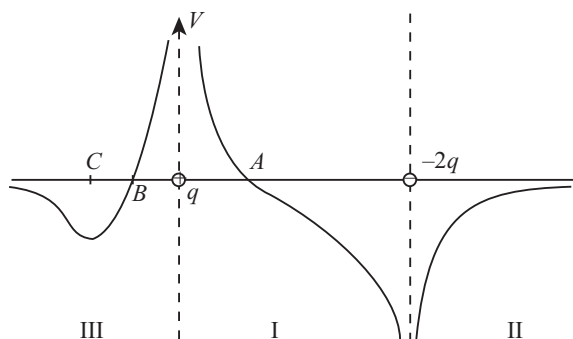
Conservation of energy gives.

$$qV_{\text{centre}} + 0 = \frac{1}{2}mu^2 + qV_{\text{circumf}}$$

$$\Rightarrow \frac{kqQ}{4a} = \frac{1}{2}mu^2 + 0$$

$$\Rightarrow u = \sqrt{\frac{qQ}{8\pi\epsilon_0 ma}}$$

18.



Between the two charges (in region I), potential is zero at point A. This point is closer to $+q$ charge. To the left of A, potential is positive and to the right of A, the potential is negative.

Very close to a point charge, the potential becomes very high.

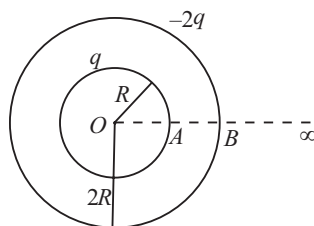
In region II, the potential is negative as a point in this region is nearer to the negative charge. At very large distance, the potential will get close to zero.

In region III, at points close to $+q$ the potential is positive. But there is a point B where potential will be zero. Beyond this point, effect of $-2q$ charge is more pronounced due to its larger magnitude. In fact, at a large distance compared to separation of the charges, the potential will be just like potential due to a $-q$ charge.

At very large distance, the potential will be zero. The potential is locally minimum at some point C.

19. Potential at O is

$$V_0 = \frac{kq}{R} + k \frac{(-2q)}{2R} = 0$$



Potential at all points from O to A is zero. For $R < r < 2R$ (i.e., between A and B)

$$V = \frac{kq}{r} + \frac{k(-2q)}{2R} = kq \left[\frac{1}{r} - \frac{1}{R} \right]$$

Since $r > R$, thus potential is negative. At B, $r = 2R$

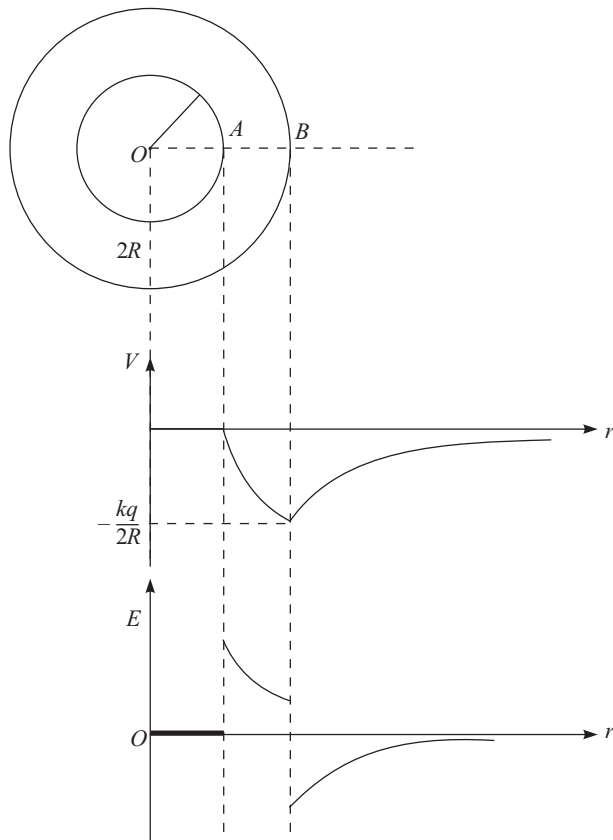
$$V = kq \left[\frac{1}{2R} - \frac{1}{R} \right] = -\frac{kq}{2R}$$

For $r \geq 2R$

$$V = \frac{k(q-2q)}{r} = -\frac{kq}{r}$$

Graph of the potential is shown below. Electric field is given by $E = -\frac{dV}{dr}$. It means negative of the slope

of $V-r$ graph is field.



$$20. \vec{r} = \overline{AB} = (4-2)\hat{i} + (-1+3)\hat{j} + (0-1)\hat{k}$$

$$= 2\hat{i} + 2\hat{j} - \hat{k}$$

Angle between $\vec{r}$ and $\vec{p}$ is given as

$$\cos\theta = \frac{\vec{r} \cdot \vec{p}}{rP} = \frac{2 \times 2 + 2 \times (-3) - 1 \times 4}{\sqrt{2^2 + 2^2 + 1^2} \sqrt{2^2 + 3^2 + 4^2}} = -\frac{2}{\sqrt{29}}$$

$\therefore$ Potential at B is

$$V = k \frac{P \cos\theta}{r^2}$$

$$= 9 \times 10^9 \times \frac{\sqrt{29} \times (-2)}{(3)^2 \times \sqrt{29}} = -2 \times 10^9 \text{ V}$$

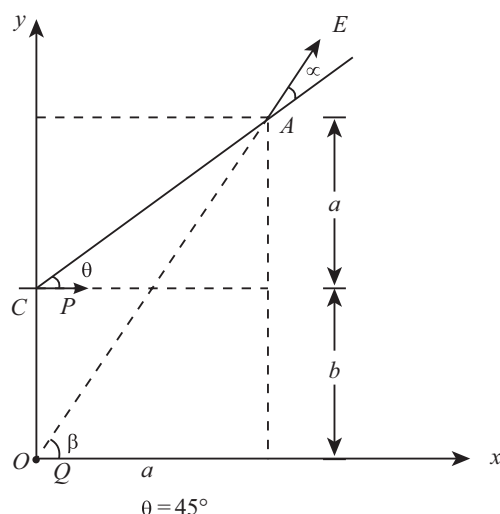
21. Look at the figure. The field at A due to the dipole and point charge Q must be equal and opposite. Field at A due to the dipole makes an angle α with the line CA, where

$$\tan\alpha = \frac{1}{2} \tan\theta$$

$$= \frac{1}{2} \tan 45^\circ$$

$$\Rightarrow \tan\alpha = \frac{1}{2} \quad (i)$$

The field due to the dipole must be along OA. A negative charge (Q) at O can create an equal field along AO at point A.



$$\beta = \theta + \alpha$$

$$\Rightarrow \tan\beta = \tan(45^\circ + \alpha)$$

$$\Rightarrow \frac{a+b}{a} = \frac{1 + \frac{1}{2}}{1 - 1 \times \frac{1}{2}}$$

$$\Rightarrow 2a = b$$

22. Using energy conservation between infinity and $r_1 = 0.2 \text{ m}$

$$e \left(k \frac{P}{r_1^2} \right) + (100 \text{ eV}) = 300 \text{ eV}$$

$$\Rightarrow k \frac{P}{(0.2)^2} = 200 \text{ V} \quad (i)$$

$$\Rightarrow P = \frac{200 \times (0.2)^2}{9 \times 10^9} = 8.9 \times 10^{-10} \text{ cm.}$$

Again using conservation of energy between infinity and r_0

$$e \left(k \frac{P}{r_0^2} \right) = 300 \text{ eV}$$

$$\Rightarrow k \frac{P}{r_0^2} = 300 \text{ V} \quad (ii)$$

From (i) and (ii),

$$\left(\frac{r_0}{0.2} \right)^2 = \frac{2}{3} \Rightarrow r_0 = 0.16 \text{ m.}$$

23. With 'S' open, the charges on inner walls of the two plates

$$\text{are } \pm \left(\frac{Q-0}{2} \right) = \pm \frac{Q}{2}.$$

Electric field between the plates is

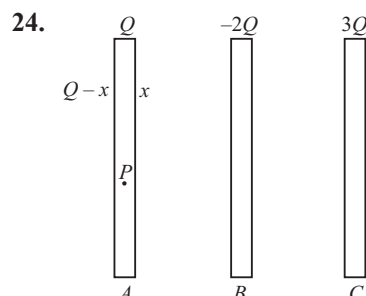
$$E = \frac{\sigma}{\epsilon_0} = \frac{Q}{2A\epsilon_0}$$

$$\text{Potential difference } V_A - V_B = E \cdot d = \frac{Qd}{2A\epsilon_0}$$

With 'S' closed, there will be no charge on outer walls of the plates. Charge $-Q$ will appear on the inner wall of B.

$$E = \frac{Q}{A\epsilon_0}$$

$$\therefore V_A - V_B = E \cdot d = 2V_0$$



Let charge on inner wall of A be x.

Field at a point P inside plate A must be zero.

$$\begin{aligned} \therefore \frac{Q-x}{2A\epsilon_0} (\rightarrow) + \frac{x}{2A\epsilon_0} (\leftarrow) + \frac{2Q}{2A\epsilon_0} (\rightarrow) \\ + \frac{3Q}{2A\epsilon_0} (\leftarrow) = 0 \\ \Rightarrow 2x = Q + 2Q - 3Q \\ \Rightarrow x = 0 \end{aligned}$$

Note that we need not know the charges separately on the two faces of B or C . Field at P due to charge on two faces of B will be same as that due to a single charge $-2Q$.

Charge on inner wall of B must be $-x$. It means there is no charge on the inner walls of A and B . There is no field between the two plates. Potential difference between them is zero.

Chapter 3 Miscellaneous Problems on Chapters 1 & 2



Match the Column

- The combined electric field due to the point charge q and the induced charge on the cavity wall is zero at all points outside the spherical cavity region. It means that field at P due to induced charges on the cavity wall is equal and opposite to the field due to point charge q .

Think of charge on inner cavity wall and the point charge q . The pair produces zero field outside the cavity. It means no work is done by the field due to this pair when a charge is moved from infinity to P . The potential is zero at P due to these charges.

The field produced by the charge on the outer wall of the conductor and Q is zero inside the conductor and the cavity.

Electric potential at C due to induced charges on the inner surface of cavity will be $-Kq/b$.

- A charge $-q_1$ is induced on the inner wall and is uniformly distributed. Charge on the outer wall is q_1 but is non uniformly distributed due to presence of q_2 . If the outside charge is moved, it will cause the charge on the outer wall to move.

If q_1 is moved away from the center, the charge on inner surface also becomes non uniform.

- The x component of electric field is given by the slope of V - x graph and y component is given by the slope of V - y graph. For example, in option (A) of column-I

$$E_x = -\frac{\partial V}{\partial x} = -1 \quad \text{and} \quad E_y = -\frac{\partial V}{\partial y} = \tan 30^\circ = \frac{1}{\sqrt{3}}$$

Now, it is easy to work out the direction of E .

- In situation A , B and C , shells I and II are not at the same potential. Hence, charge is shared between the two spheres till both acquire same potential. When charge flows in a conductor, the potential energy of system decreases and heat is produced.

In situations A and B charges shall divide in some fixed ratio, but in situation C complete charge shall be transferred to shell II for potential of shell I and II to be same.

In situation D both the shells are at same potential, hence no charge flows through connecting wire.

$$5. (A) U = \frac{1}{4\pi\epsilon_0} \frac{(-Q)^2}{2a} = \frac{Q^2}{8\pi\epsilon_0 a}$$

$$(B) U = \frac{1}{4\pi\epsilon_0} \left[\frac{(-Q) \times (-Q)}{5a/2} + \frac{(-Q)^2}{2(5a/2)} \right] = \frac{3}{20} \frac{Q^2}{\pi\epsilon_0 a}$$

$$(C) U = \frac{1}{4\pi\epsilon_0} \frac{3Q^2}{5a} = \frac{3}{20} \frac{Q^2}{\pi\epsilon_0 a}$$

$$(D) U = \frac{1}{4\pi\epsilon_0} \left[\frac{3Q^2}{5a} + \frac{(-Q)^2}{2(2a)} + \frac{(-Q) \times (-Q)}{2a} \right] \\ = \frac{27Q^2}{80\pi\epsilon_0 a}$$



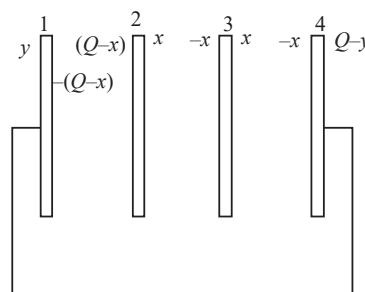
Passage Based Problems

Passage 1

Let charge on right wall and left wall of plate 2 be x and $Q-x$, respectively. Left wall of plate 3 must have $-x$ charge; its right wall will have $+x$ charge as the plate is neutral. Charge on the face of plate 4 facing plate 3 is $-x$ and that on right face of plate 1 is $-(Q-x)$.

Let charge on left face of plate 1 be y .

The total charge on plates 1 and 4 must be zero. Hence, charge on the right face of plate 4 will be $Q-y$.



The field between plates 1 and 2 is toward left and has a magnitude

$$E_1 = \frac{Q-x}{2A\epsilon_0} + \frac{Q-x}{2A\epsilon_0} = \frac{Q-x}{A\epsilon_0} \quad (\leftarrow)$$

The field between plates 2 and 3 is given as

$$E_2 = \frac{x}{A\epsilon_0} \quad (\rightarrow)$$

Similarly, the field between plates 3 and 4 is given as

$$E_3 = \frac{x}{A\epsilon_0} \quad (\rightarrow)$$

The potential difference between plates 1 and 4 must be zero

$$V_1 + E_1 d - E_2(2d) - E_3 d = V_4$$

$$\text{or, } E_1 - 2E_2 - E_3 = 0$$

$$\text{or, } Q - x - 2x - x = 0 \Rightarrow x = \frac{Q}{4} \quad [\text{Ans to 1}]$$

The potential difference between plates 1 and 2 is written as

$$V_{12} = E_1 d = \frac{3Q}{4\epsilon_0 A} \quad [\text{Ans to 3}]$$

To find y , consider a point inside plate 1. The field at this point must be zero.

$$\frac{y}{2A\epsilon_0} + \frac{3Q}{4(2A\epsilon_0)} - \frac{Q}{2A\epsilon_0} + \frac{Q}{4(2A\epsilon_0)} - \frac{Q-y}{2A\epsilon_0} = 0$$

$$\Rightarrow y + \frac{3Q}{4} - Q + \frac{Q}{4} - Q + y = 0$$

$$\Rightarrow y = \frac{Q}{2} \quad [\text{Ans to 2}]$$

Passage 2

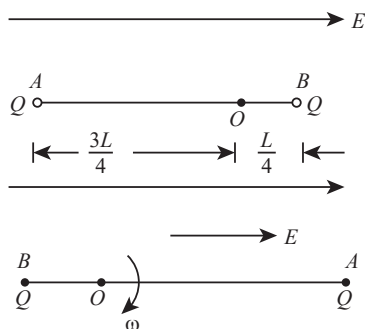
4. Net charge inside the cylinder = 0

$$\therefore \text{Flux through the cylinder} = 0$$

5. The electric field due to $-Q$ charge as well as that due to $+Q$ charge contributes positive flux at the right face.

Passage 3

6. The unstable equilibrium position is as shown in the figure. The rotation axis is passing through O . When the system is disturbed a little, the electric torque on the ball A (about O) is larger than the torque on the ball B . This causes the rod to rotate. The kinetic energy of the system will be maximum in the position shown in the second figure. After this, the larger torque on A begins to retard the rod.



The ball A moves a distance $\frac{3L}{2}$ in the direction of E and the ball B moves a distance $\frac{L}{2}$ to the left (opposite to E).

$$\text{Drop in the potential energy of ball } A = QE \left(\frac{3L}{2} \right).$$

$$\text{Gain in the potential energy of ball } B = QE \left(\frac{L}{2} \right).$$

$$\therefore \frac{1}{2} I \omega^2 + QE \frac{L}{2} = QE \frac{3L}{2}$$

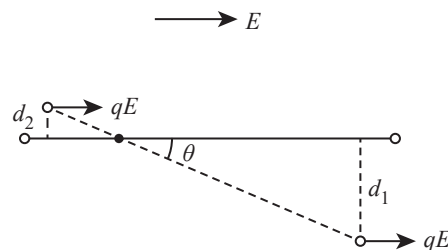
$$\text{or, } \left[m \left(\frac{3L}{4} \right)^2 + m \left(\frac{L}{4} \right)^2 \right] \omega^2 = 2QEL$$

$$\text{or, } \frac{5}{8} mL^2 \omega^2 = 2QEL$$

$$\text{or, } \omega = \sqrt{\frac{16}{5} \frac{QE}{mL}}$$

$$\therefore \text{Speed of ball } B \text{ is } v = \frac{\omega L}{4} = \sqrt{\frac{QEL}{5m}}$$

7. The final position shown in the above figure is the position of stable equilibrium. Oscillations can happen about a position of stable equilibrium only.
8. When displaced by a small angle θ from its equilibrium position, the torque on the system is given as



$$\tau = qEd_2 - qEd_1$$

$$= qE \left[\frac{L}{4} \sin \theta - \frac{3L}{4} \sin \theta \right] = -qE \frac{L}{2} \theta$$

$$[\because \sin \theta \approx \theta \text{ for small } \theta]$$

$$\Rightarrow I\alpha = -\frac{qEL}{2} \theta$$

$$\text{or, } \left[m \left(\frac{3L}{4} \right)^2 + m \left(\frac{L}{4} \right)^2 \right] \alpha = -\frac{qEL}{2} \theta$$

$$\text{or, } \alpha = -\left(\frac{4QE}{5mL} \right) \theta$$

$$\therefore T = 2\pi \sqrt{\frac{5mL}{4QE}}$$

Passage 4

9. The discs are large. The field between them will be normal to their surface and uniform. Outside the

discs, that is, in the region $x > -R$ and $x < -R$, the field will be zero.

10. The electric field is in positive x direction between the discs. The potential decreases as one moves from $x = -R$ to $x = +R$.

In the region $x < -R$ and $x > +R$, there is no field and the potential remains constant.

Passage 5

11. The field at A is $E = \frac{4\sqrt{2}\lambda R \cdot (\sqrt{3}R)}{4\pi\epsilon_0 [R^2 + 3R^2]^{\frac{3}{2}}} - \frac{\lambda}{2\pi\epsilon_0 \sqrt{3}R}$
- $$= \frac{\lambda}{\pi\epsilon_0 R} \left(\frac{3 - 2\sqrt{2}}{4\sqrt{6}} \right)$$

Acceleration of the electron will be written as

$$a = \frac{eE}{m} = \frac{\lambda e}{\pi\epsilon_0 R m} \left(\frac{3 - 2\sqrt{2}}{4\sqrt{6}} \right)$$

12. Force on the electron will be zero where the field is zero.

$$E_x = \frac{4\sqrt{2}\lambda R x}{4\pi\epsilon_0 [R^2 + x^2]^{\frac{3}{2}}} - \frac{\lambda}{2\pi\epsilon_0 x} = 0$$

Simplifying, this gives $x = R$.

13. $\int_B^A dV = - \int_R^{\sqrt{3}R} E_x dx$
- $$V_A - V_B = - \frac{\lambda}{\pi\epsilon_0} \left[\sqrt{2}R \int_R^{\sqrt{3}R} \frac{xdx}{(R^2 + x^2)^{\frac{3}{2}}} - \frac{1}{2} \int_R^{\sqrt{3}R} \frac{dx}{x} \right]$$
- $$= - \frac{\lambda}{\pi\epsilon_0} \left[\left(1 - \frac{1}{\sqrt{2}} \right) - \frac{\ln 3}{4} \right]$$

Passage 6

14. In equilibrium, $mg = k \frac{Q_1 Q_2}{h_0^2}$

The only external force on the system is gravity. Therefore, the acceleration of the COM of the system is g . After 0.1s, the velocity of the COM of the system is given as

$$v = u_{cm} + a_{cm}t$$

$$= 1 - 10(0.1) \quad \left[\because u_{cm} = \frac{0 + 2m}{2m} = 1 \frac{m}{s} \right]$$

$$= 0$$

15. The two pearls are closest when they have same velocity (say v_0). Let the minimum separation be h . When both the beads have same velocity v_0 , the COM also has velocity v_0 . Let's first find the height

gained by the COM by the time its speed changes from $\frac{u}{2}$ to v_0

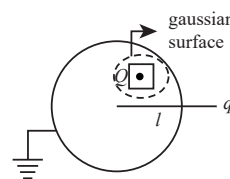
$$v_0^2 = \left(\frac{u}{2} \right)^2 - 2g \cdot \Delta h$$

$[\Delta h = \text{height gained by the COM}]$

$$\Rightarrow \Delta h = \frac{u^2}{8g} - \frac{v_0^2}{2g}$$

Passage 7

16. A closed Gaussian surface inside the conductor, as shown in the figure, will have zero flux though it. Charge inside the Gaussian surface shall be zero.



$\therefore$ Charge in the cavity wall = $-Q$

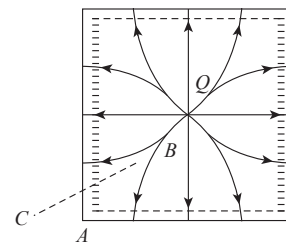
17. The sphere is grounded. Its potential must be zero. The point charge Q and $-Q$ on the cavity wall form a system which has no influence on the world outside the cavity. The potential at the centre of the sphere is due to q and charge (Q_1) on the outer wall.

$$V_{\text{centre}} = \frac{kq}{l} + k \frac{Q_1}{R} = 0$$

$$\Rightarrow Q_1 = -\frac{qR}{l}$$

The entire conductor is at zero potential.

18. The field lines inside the cavity are somewhat like that shown in the figure.



When we move from centre of the sphere (C) to A , we do not encounter any field but

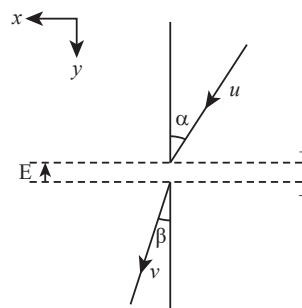
while moving from A to B we encounter a field. We move against an electric field. Potential at B that is greater than the potential at C .

Passage 8

$$u_x = u \cos \alpha, \quad u_y = u \sin \alpha.$$

The force on an electron is in y direction. Only y component of its velocity changes.

An electron gets accelerated between the two wire meshes. A potential difference V causes it to gain eV amount of kinetic energy.

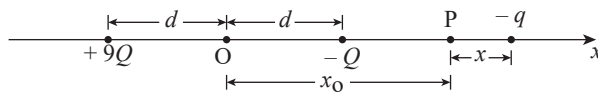


$$\begin{aligned}
 \therefore \quad \frac{1}{2}mv^2 &= \frac{1}{2}mu^2 + eV \\
 \Rightarrow \quad v^2 &= u^2 + \frac{2eV}{m} \\
 \tan \beta &= \frac{v_x}{v_y} \Rightarrow \sin \beta = \frac{v_x}{\sqrt{v_x^2 + v_y^2}} \\
 \Rightarrow \quad \sin \beta &= \frac{u \sin \alpha}{\sqrt{v^2}} \\
 \Rightarrow \quad \frac{\sin \beta}{\sin \alpha} &= \frac{u}{\sqrt{u^2 + \frac{2eV}{m}}} \\
 \Rightarrow \quad \mu &= \frac{1}{\sqrt{1 + \frac{2eV}{mu^2}}} = \frac{1}{\sqrt{1 + \frac{eV}{K}}}
 \end{aligned}
 \tag{i}$$

It can be seen that μ is independent of α and doubling the values of V as well as kinetic energy (K) does not change μ .

Passage 9

21. $x = x_0$ is equilibrium position and it is a point where the field is zero. Such a point (P) will lie to the right of charge $-Q$



$$\begin{aligned}
 E_P &= 0 \\
 \Rightarrow k \frac{2Q}{(x_0 + d)^2} &= k \frac{Q}{(x_0 - d)^2} \\
 \Rightarrow 3(x_0 - d) &= (x_0 + d) \\
 \Rightarrow x_0 &= 2d
 \end{aligned}$$

To the right of point P , the electric field is towards right, and to the left of point P , it is towards left. A negative charge kept at P will be in stable equilibrium. If it is displaced from P , it will experience a restoring force towards P . Therefore, q is negative.

22. When charge $-q$ is displaced a small distance from P , it performs oscillation. Consider it at a distance x to the right of P . Force on it is given as

$$\begin{aligned}
 F &= k \frac{Qq}{(d+x)^2} - k \frac{9Qq}{(3d+x)^2} \\
 [\text{here } q \text{ is magnitude of the negative charge}] \\
 &= \frac{kQq}{d^2} \left[\frac{1}{\left(1 + \frac{x}{d}\right)^2} - \frac{1}{\left(1 + \frac{x}{3d}\right)^2} \right]
 \end{aligned}$$

$$\begin{aligned}
 &= k \frac{Qq}{d^2} \left[\left(1 + \frac{x}{d}\right)^{-2} - \left(1 + \frac{x}{3d}\right)^{-2} \right] \\
 &= \frac{kQq}{d^2} \left[1 - \frac{2x}{d} + \frac{2x}{3d} \right] \quad [\because x \ll d] \\
 \therefore F &= -\frac{4kQq}{3d^3} x \\
 \therefore \frac{d^2x}{dt^2} &= -\left(\frac{4kQq}{3md^3}\right)x = -\left(\frac{Qq}{3\pi\epsilon_0 md^3}\right)x \\
 \therefore \text{Motion is SHM.} \\
 T &= 2\pi \sqrt{\frac{3\pi\epsilon_0 md^3}{|q|Q}}
 \end{aligned}$$

23. According to the question, the particle performs SHM with amplitude A . Its speed at the mean position is

$$\begin{aligned}
 v &= A\omega \\
 &= A \sqrt{\frac{Q|q|}{3\pi\epsilon_0 md^3}}
 \end{aligned}$$

Passage 10

24. Consider an element of angular width $d\theta$ on the ring as shown in the figure. Electric force on the element is $dF = (\lambda R d\theta)E$.

Torque of this force on the ring about its centre O is given as

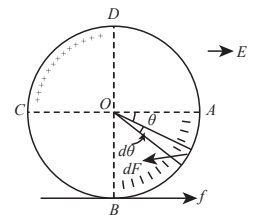
$$\begin{aligned}
 d\tau &= (dF)(R \sin \theta) = \lambda R^2 E \sin \theta \cdot d\theta \\
 \Rightarrow \quad \tau &= \int d\tau = \lambda R^2 E \int_0^{\pi} \sin \theta \cdot d\theta \\
 &= \lambda R^2 E
 \end{aligned}$$

Same torque is produced by the force acting on segment CD .

$\therefore$ Total torque is $\tau = 2\lambda R^2 E$

25. This torque is clockwise. It will impart an angular acceleration to the ring. For rolling, it must have a linear acceleration towards right. However, the resultant electric force on the ring is zero. Friction is the only force, which can provide it a linear acceleration. The friction force must be towards right. Writing equation for rotational and translational motion of the ring, we get:

$$\begin{aligned}
 \tau - f \cdot R &= I \cdot \alpha \\
 \Rightarrow 2\lambda R^2 E - f \cdot R &= mR^2 \cdot \alpha
 \end{aligned}$$



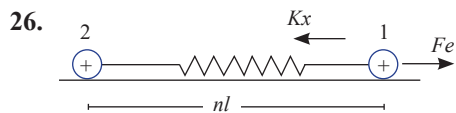
$$\Rightarrow 2\lambda RE - f = ma \quad (i)$$

$$\text{and } f = ma \quad (ii)$$

From (i) and (ii), we get

$$2\lambda RE = 2f \Rightarrow f = \lambda RE$$

Passage 11



For equilibrium of Ball 1

$$kx = \frac{1}{4\pi\epsilon_0} \frac{q^2}{(nl)^2}$$

$$\Rightarrow k(nl - l) = \frac{q^2}{4\pi\epsilon_0 n^2 l^2} \quad (i)$$

$$\Rightarrow l^3 = \frac{q^2}{4\pi\epsilon_0 n^2 (n-1)k} \quad (ii)$$

27. Frequency of oscillation when the balls are uncharged is

$$f_0 = \frac{1}{2\pi} \sqrt{\frac{k}{m}}, \text{ where } m \text{ is mass of a ball.}$$

Assume that Ball 1 is displaced by a small distance x to the right of its equilibrium position. The resultant force on the ball towards right is

$$\frac{1}{4\pi\epsilon_0} \frac{q^2}{(nl+x)^2} - k(nl+x-l)$$

$$\therefore m \frac{d^2x}{dt^2} = \frac{q^2}{4\pi\epsilon_0 (nl+x)^2} - kl(n-1) - kx$$

$$= \frac{q^2}{4\pi\epsilon_0 n^2 l^2} \left(1 + \frac{x}{nl}\right)^{-2} - kl(n-1) - kx$$

$$\therefore x \ll nl$$

$$\therefore \left(1 + \frac{x}{nl}\right)^{-2} \simeq 1 - \frac{2x}{nl}$$

$$\therefore m \frac{d^2x}{dt^2} = \frac{q^2}{4\pi\epsilon_0 n^2 l^2} \left(1 - \frac{2x}{nl}\right) - kl(n-1) - kx$$

Using (i), the above equation can be written as

$$m \frac{d^2x}{dt^2} = - \left(\frac{1}{4\pi\epsilon_0} \frac{2q^2}{n^3 l^3} + k \right) x$$

Substituting $\frac{q^2}{l^3}$ from (ii), we get

$$m \frac{d^2x}{dt^2} = - \left[\frac{2(n-1)}{n} k + k \right] x = - \left(\frac{3n-2}{n} \right) kx$$

$$\therefore \frac{d^2x}{dt^2} = - \left(\frac{3n-2}{n} \right) \frac{k}{m} x$$

$$\therefore \omega = \sqrt{\left(\frac{3n-2}{n} \right) \frac{k}{m}} \Rightarrow f = \frac{1}{2\pi} \sqrt{\left(\frac{3n-2}{n} \right) \frac{k}{m}}$$

$$\therefore \frac{f}{f_0} = \sqrt{\frac{3n-2}{n}} = \sqrt{\frac{3 \times 2 - 2}{2}} = \sqrt{2}.$$

Passage 12

28. When key K_1 is closed, the outer sphere gets grounded, and charge on it becomes $-q$, so that its potential becomes zero.

After key K_2 is closed, charge on it becomes $-q_2^1$ and its potential becomes zero.

$$k \frac{q_2^1}{R} + k \frac{(-q)}{2R} = 0$$

$$\Rightarrow q_2^1 = \frac{q}{2}.$$

Therefore, after one event involving closure and opening of the two switches, charge on the inner ball is reduced to half its original value.

29. After $(n-1)$ events, charge on the inner ball is $\frac{q}{2^{n-1}}$.

When key K_1 is closed for the n th time, charge on the outer ball becomes $-\frac{q}{2^{n-1}}$.

When K_2 is closed for the n th time, charge on the inner ball become $\frac{q}{2^n}$.

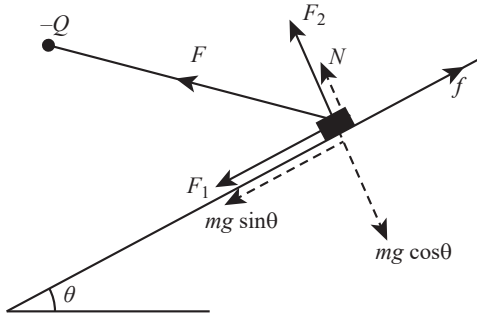
The potential of the inner ball is zero and that of the outer ball is

$$V = \frac{1}{4\pi\epsilon_0} \frac{q}{2^n (2R)} + \frac{1}{4\pi\epsilon_0} \frac{-q}{2^{n-1} (2R)}$$

$$= \frac{q}{4\pi\epsilon_0 2^n R} \left[\frac{1}{2} - \frac{1}{1} \right] = - \frac{q}{4\pi\epsilon_0 2^{n+1} R}$$

Passage 13

30.



Forces acting on the block are as shown. Work energy theorem says:

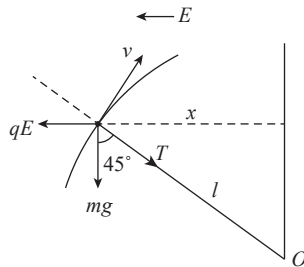
$$\begin{aligned}
 W_e + W_g + W_f &= \frac{1}{2}mv_0^2 \\
 \Rightarrow \int_A^B F_1 \cdot ds + mgh - \int_A^B \mu(mg \cot \theta - F_2) ds &= \frac{1}{2}mv_0^2 \\
 \Rightarrow \int_A^B F_1 ds + \mu \int_A^B F_2 ds &= \frac{1}{2}mv_0^2 - mgh + \mu mg \cos \theta \int_A^B ds \\
 &= \frac{1}{2}mv_0^2 - mgh + \mu mg \cos \theta \cdot \frac{h}{\sin \theta} \\
 &= \frac{1}{2}mv_0^2 - mgh + \mu mgh \cot \theta
 \end{aligned}$$

31 When the charge on the block is made negative, directions of F_1 and F_2 reverse.

$$\begin{aligned}
 \therefore \frac{1}{2}mv^2 &= - \int_A^B F_1 ds + mgh - \int_A^B \mu(mg \cos \theta + F_2) ds \\
 &= mgh - \mu mg \cos \theta \int_A^B ds - \left[\int_A^B F_1 ds + \mu \int_A^B F_2 ds \right] \\
 &= mgh - \mu mgh \cot \theta - \frac{1}{2}mv_0^2 + mgh - mgh \cot \theta \\
 \Rightarrow v &= \sqrt{4gh(1 - \mu \cot \theta) - v_0^2}
 \end{aligned}$$

Passage 14

32.



When the particle rotates through 45° , $x = \frac{l}{\sqrt{2}}$.

Electric field at the location of the particle is

$$E = \frac{\lambda}{2\pi\epsilon_0 x} = \frac{\lambda}{\sqrt{2}\pi\epsilon_0 l}$$

Writing equation for the centripetal force, we get

$$T + \frac{mg}{\sqrt{2}} - \frac{qE}{\sqrt{2}} = \frac{mv^2}{l}$$

Since the string gets slack, $T = 0$

$$\therefore \frac{mv^2}{l} = \frac{mg}{\sqrt{2}} - \frac{q\lambda}{2\pi\epsilon_0 l} \quad (i)$$

33. Work done by the electric force on q is

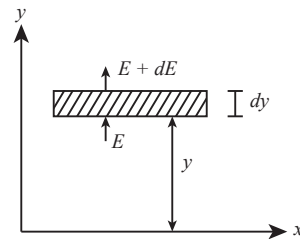
$$\begin{aligned}
 W_e &= \int_l^x qEdx = \frac{q\lambda}{2\pi\epsilon_0} \int_l^x \frac{1}{x} dx = \frac{q\lambda}{2\pi\epsilon_0} [\ln x]_l^x \\
 &= \frac{q\lambda}{2\pi\epsilon_0} \ln\left(\frac{1}{\sqrt{2}}\right) = -\frac{q\lambda \ln 2}{4\pi\epsilon_0}
 \end{aligned}$$

34. Work done by gravity + work done by electrostatic force = change in KE.

$$\begin{aligned}
 -mg \frac{l}{\sqrt{2}} - \frac{q\lambda \ln 2}{4\pi\epsilon_0} &= \frac{1}{2}mv^2 - \frac{1}{2}mu^2 \\
 \text{or, } -\frac{mg}{\sqrt{2}} - \frac{q\lambda \ln 2}{4\pi\epsilon_0 l} &= \frac{1}{2} \frac{mv^2}{l} - \frac{1}{2} \frac{mu^2}{l} \\
 \text{or, } \frac{mu^2}{l} &= \frac{mg}{\sqrt{2}} - \frac{q\lambda}{2\pi\epsilon_0 l} + \sqrt{2}mg + \frac{q\lambda \ln 2}{2\pi\epsilon_0 l} \\
 \text{or, } u^2 &= g \left(\frac{1}{\sqrt{2}} + \sqrt{2} \right) l - \frac{q\lambda}{2\pi\epsilon_0 m} (1 - \ln 2) \\
 \text{or, } u &= \sqrt{\frac{3}{\sqrt{2}} gl - \frac{q\lambda(1 - \ln 2)}{2\pi\epsilon_0 m}}
 \end{aligned}$$

Passage 15

35. Charge density charges with y co-ordinate only. Consider a box-shaped closed volume having area A and thickness dy .



Flux through the box is

$$\phi = (E + dE)A - EA = AdE$$

Applying the Gauss's law gives

$$AdE = \frac{\rho A dy}{\epsilon_0} \Rightarrow dE = \frac{ay dy}{\epsilon_0}$$

$$\Rightarrow \int_0^E dE = \frac{a}{\epsilon_0} \int_0^y y dy \Rightarrow E = \frac{ay^2}{2\epsilon_0}$$

36. There is no force on the particle along x direction. velocity component along x direction remains u . The acceleration component along y direction is

$$a_y = \frac{qE}{m} = \frac{qay^2}{2m\epsilon_0}$$

or, $v \frac{dv}{dy} = \frac{qay^2}{2m\epsilon_0}$

[here, $v = y$ component of velocity]

or, $\int_0^v v dv = \frac{qa}{2m\epsilon_0} \int_{y_0}^y y^2 dy$

or, $v = \sqrt{\frac{qa}{3m\epsilon_0} (y^3 - y_0^3)}$

37. Slope of the path = $\frac{v_y}{v_x} = \frac{v}{u}$
- $$= \sqrt{\frac{qa}{3m\epsilon_0 u^2} (y^3 - y_0^3)}$$

Passage 16

38. The potential at a point on the x -axis (for $-R < x < R$) is

$$V = \frac{Q}{4\pi\epsilon_0} \left[\frac{1}{\sqrt{R^2 + (R-x)^2}} + \frac{1}{\sqrt{R^2 + (R+x)^2}} \right] \quad (i)$$

This is minimum when $\frac{dV}{dx} = 0$.

Differentiating and simplifying gives $x = 0$.

This result can also be obtained using the symmetry of the given situation.

Putting $x = 0$ in (i) gives

$$V_0 = \frac{\sqrt{2}Q}{4\pi\epsilon_0 R}$$

39. From (i)

$$V = \frac{q}{4\pi\epsilon_0 R} \left[\left\{ R^2 + (R-x)^2 \right\}^{-\frac{1}{2}} + \left\{ R^2 + (R+x)^2 \right\}^{-\frac{1}{2}} \right]$$

or, $V = \frac{Q}{4\pi\epsilon_0 R} \left[\left\{ 1 + \left(1 - \frac{x}{R} \right)^2 \right\}^{-\frac{1}{2}} + \left\{ 1 + \left(1 + \frac{x}{R} \right)^2 \right\}^{-\frac{1}{2}} \right]$

Expanding and neglecting higher order terms, we get

$$V = \frac{q}{4\pi\epsilon_0} \left[\left\{ 1 + 1 - \frac{2x}{R} \right\}^{-\frac{1}{2}} + \left\{ 1 + 1 + \frac{2x}{R} \right\}^{-\frac{1}{2}} \right]$$

$$= \frac{Q}{4\pi\epsilon_0 R \sqrt{2}} \left[\left(1 - \frac{x}{R} \right)^{-\frac{1}{2}} + \left(1 + \frac{x}{R} \right)^{-\frac{1}{2}} \right]$$

$$= \frac{Q}{4\sqrt{2}\pi\epsilon_0 R} \left[1 + \frac{x}{2R} + \frac{3}{8} \frac{x^2}{R^2} + 1 - \frac{x}{2R} + \frac{3}{8} \frac{x^2}{R^2} \right]$$

$$\left[(1+x)^{-\frac{1}{2}} = 1 - \frac{1}{2}x + \frac{3}{8} \frac{x^2}{R^2} + \dots \right]$$

$$= \frac{Q}{4\sqrt{2}\pi\epsilon_0 R} \left[2 + \frac{3}{4} \frac{x^2}{R^2} \right] = \frac{\sqrt{2}Q}{4\pi\epsilon_0 R} + \frac{3Qx^2}{16\sqrt{2}\pi\epsilon_0 R^3}$$

$$= V_0 + ax^2$$

Where $a = \frac{3Q}{16\sqrt{2}\pi\epsilon_0 R^3}$

40. $V = V_0 + ax^2$

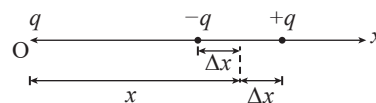
$$\Rightarrow E_x = -\frac{dV}{dx} = -2ax$$

$\therefore$ Force on the charge, when it is displaced slightly from the origin, is $F = -2aqx$

$\therefore$ Motion is SHM.

Passage 17

41. The electric potential due to q_0 at a distance x from the origin is $V = \frac{1}{4\pi\epsilon_0} \frac{q_0}{x}$



Change in potential when x changes by Δx is given as

$$\Delta V = \frac{dV}{dx} \Delta x = -\frac{q_0}{4\pi\epsilon_0 x^2} \Delta x$$

When x reduces by Δx , change in potential is given as

$$\Delta V' = -\frac{q_0}{4\pi\epsilon_0 x^2} (-\Delta x) = \frac{q_0 \Delta x}{4\pi\epsilon_0 x^2}$$

$\therefore$ PE of the dipole can be written as

$$U = q(V + \Delta V) - q(V + \Delta V')$$

$$= q(\Delta V - \Delta V')$$

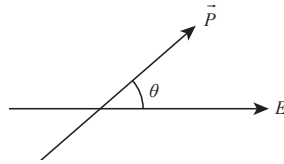
$$= -\frac{2q_0 q \Delta x}{4\pi\epsilon_0 x^2} = -\frac{q_0 p}{4\pi\epsilon_0 x^2} \quad [\because q(2\Delta x) = p]$$

At infinity, the potential energy is zero. Therefore, the PE of the dipole decreases as it moves from ∞ towards the origin. The fall in PE is equal to the gain in KE. According to the question, the KE of oscillation is negligible. We can write the translational KE to be equal to loss in PE

$$\therefore \frac{1}{2} Mv^2 = \frac{q_0 p}{4\pi\epsilon_0 x^2} \Rightarrow v = \sqrt{\frac{q_0 p}{2\pi\epsilon_0 Mx^2}}$$

42. When a dipole oscillates in an electric field, its time period of oscillation is given by

$$T = 2\pi \sqrt{\frac{I}{PE}}$$



[I = moment of inertia]

As the dipole gets nearer to the charge q_0 , the field E increases and hence the time period decreases.

Passage 18

43. The outward electrostatic pressure (i.e. force per unit area) on the shell due to its own charge is

$$P_1 = \frac{\sigma^2}{2\epsilon_0} = \frac{1}{2\epsilon_0} \left(\frac{Q}{4\pi R^2} \right)^2 = \frac{Q^2}{32\pi^2 \epsilon_0 R^4}$$

The negative charge $-q$, at the centre, produces a field on the surface of the shell given by

$$E = \frac{1}{4\pi\epsilon_0} \frac{q}{R^2}$$

This field exerts a normal force on the unit area of the shell surface given by

$$P_2 = \sigma E = \frac{Q}{4\pi R^2} \frac{1}{4\pi\epsilon_0} \frac{q}{R^2} = \frac{Qq}{16\pi^2 \epsilon_0 R^2}$$

The two hemispherical parts will remain attached if the inward pressure P_2 at every point is larger than outward pressure P_1 .

$$\Rightarrow P_2 \geq P_1 \Rightarrow \frac{Qq}{16\pi^2 \epsilon_0 R^2} \geq \frac{Q^2}{32\pi^2 \epsilon_0 R^4} \Rightarrow q \geq \frac{Q}{2}$$

44. In the critical case when $q = \frac{Q}{2}$, the force by one half shell on the other is exactly balanced by the force due to $-q$. In this case, there is no contact force between the two parts of the shell.

Charge on inner wall of the shell = $+q$

$\therefore$ Charge on the outer wall of the shell = $Q - q = q$.

$$[\because Q = 2q]$$

$\therefore$ Charge density on the outer surface = $\frac{q}{4\pi R^2}$

Chapter 4 Capacitors



Your Turn

1. If Q is doubled, the potential (V) will also double. C will not change. It depends only on geometry of the conductor.

$$\begin{aligned} 2. \quad Q &= CV = (2\mu\text{F})(1000 \text{ V}) \\ &= 2000 \mu\text{C} \\ &= 2\text{mC} \end{aligned}$$

Remember that farad $\times$ volt is coulomb.

3. Potential of the two conductors is:

$$V_1 = \frac{Q_1}{C_1} = \frac{20}{2} = 10 \text{ V}$$

$$V_2 = \frac{Q_2}{C_2} = \frac{40}{4} = 10 \text{ V}$$

Since they have equal potentials, no charge will flow between them when they are connected. No heat is dissipated.

4. If capacitance of B is C , then capacitance of A will be $2C$. After connection, the final common potential of the two spheres will be

$$V = \frac{Q_1 + Q_2}{2C + C} = \frac{-20\mu\text{C} + 20\mu\text{C}}{3C} = 0$$

Hence, final charge on both the conductors will be zero.

5. Final potential of the two spheres will be:

$$V = \frac{Q_1 + Q_2}{2C + C} = \frac{-20\mu\text{C} + 0}{3C} = -\frac{20}{3C} \mu\text{V}$$

$$\therefore q_A = (2C) V = -\frac{40}{3} \mu\text{C}$$

$$6. \quad E = \frac{V}{d} = \frac{50}{1 \times 10^{-3}} = 50,000 \text{ V/m.}$$

7. The electric field between the plates becomes $\frac{1}{5}$ times. Therefore, the potential difference also becomes $\frac{1}{5}$ times.

$$8. \quad C = \frac{\epsilon_0 A}{d}$$

$$\text{And } 2C = \frac{\epsilon_0 A}{\frac{x}{K} + (d - x)}$$

$$\Rightarrow \quad 2 \frac{\epsilon_0 A}{d} = \frac{\epsilon_0 A}{\frac{x}{3} + (d - x)}$$

$$\Rightarrow \quad \frac{2}{3}x + 2(d - x) = d$$

$$\Rightarrow \quad d = \frac{4}{3}x \Rightarrow \frac{x}{d} = \frac{3}{4}$$

9. Refer to Example 6. For the reason explained in Example 6; the capacitance of the spherical capacitor will be

$$\begin{aligned} C &= \frac{4\pi K \epsilon_0 ab}{b - a} \\ &= \frac{3 \times 0.2 \times 0.1}{9 \times 10^9 [0.2 - 0.1]} = 6.7 \times 10^{-11} \text{ F.} \end{aligned}$$

10. Electric field between the plates is independent of distance between them.

$$E = \frac{\sigma}{\epsilon_0}$$

Therefore, E will not change when distance between the plates is changed.

$\therefore$ Energy density ($u = \frac{1}{2} \epsilon_0 E^2$) will not change

11. Half the work done by the cell is lost as heat and remaining half gets stored in the capacitor.

$$12. \quad W_{\text{cell}} = 100\mu\text{J}$$

$$\Rightarrow \quad Q \cdot V = 100\mu\text{J}$$

$$\Rightarrow (5\mu\text{C}) \cdot V = 100\mu\text{J}$$

$$V = 20 \text{ V}$$

13. Charge on the capacitor after it is connected to the first cell is

$$Q_1 = CV$$

Work done by the first cell is

$$W_1 = Q_1 V = CV^2$$

Charge on the capacitor after it is connected to the second cell is

$$Q_2 = C(2V) = 2CV$$

Charge supplied by the second cell is

$$= Q_2 - Q_1 = CV$$

Work done by the second cell is

$$W_2 = (CV)(2V) = 2CV^2$$

$$\therefore \quad \frac{W_1}{W_2} = \frac{CV^2}{2CV^2} = \frac{1}{2}$$

14. Initial capacitance is $C = 10\mu\text{F}$
 Final capacitance is $2C = 20\mu\text{F}$
 Initial charge is $Q_1 = CV = (10\mu\text{F})(20\text{V}) = 200\mu\text{C}$
 Final charge is $Q_2 = 2CV = 400\mu\text{C}$

$$(a) \quad W_{\text{cell}} = \Delta QV = (Q_2 - Q_1)V \\ = (200\mu\text{C})(20\text{V}) = 4000\mu\text{J} \\ = 4\text{mJ}$$

- (b) Change in energy stored in the capacitor is

$$\Delta U = U_2 - U_1 = \frac{1}{2} (2C)V^2 - \frac{1}{2} CV^2 \\ = \frac{1}{2} CV^2 = \frac{1}{2} \times (10\mu\text{F}) \times (20\text{V})^2 \\ = 2000\mu\text{J} = 2\text{mJ}$$

$$W_{\text{extAgt}} + W_{\text{cell}} = \Delta U$$

$$\text{or, } W_{\text{extAgt}} = 2\text{mJ} - 4\text{mJ} = -2\text{mJ}$$

- (c) Electric field between the plates $E = \frac{V}{d}$
 When d is halved, E doubles.

$$\text{Energy density, } u = \frac{1}{2} \epsilon_0 E^2$$

If E doubles, u becomes 4 times.

15. Equivalent capacitance is given by

$$\frac{1}{C_0} = \frac{1}{1} + \frac{1}{2} + \frac{1}{3}$$

$$\Rightarrow C_0 = \frac{6}{11} \mu\text{F}$$

Charge on each capacitor is same as charge supplied by the cell.

$$Q = C_0 V \\ = \frac{6}{11} \mu\text{F} (66\text{V}) = 36\mu\text{C}$$

$\therefore$ Potential difference across the capacitors is

$$V_1 = \frac{Q}{C_1} = \frac{36}{1} = 36\text{V}$$

$$V_2 = \frac{Q}{C_2} = \frac{36}{2} = 18\text{V}$$

$$V_3 = \frac{Q}{C_3} = \frac{36\mu\text{C}}{2\mu\text{F}} = 12\text{V}$$

16. When two capacitors are connected in parallel, the equivalent capacitance is $C_0 = C_1 + C_2$.

When connected in series, the equivalent is less than both C_1 and C_2 .

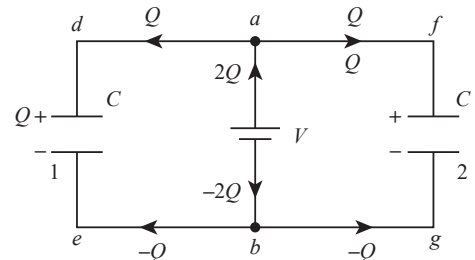
Therefore, parallel equivalent is $25\mu\text{F}$ and the series equivalent is $4\mu\text{F}$ (the smallest one).

The two capacitances are $20\mu\text{F}$ and $5\mu\text{F}$.

$$C_1 + C_2 = 25\mu\text{F} \quad \text{and} \quad \frac{C_1 C_2}{C_1 + C_2} = \frac{20 \times 5}{20 + 5} = 4\mu\text{F}$$

17. In both the diagrams, the three capacitors are in parallel.

18. The two capacitors have one of their plates connected to the point a (i.e., the positive terminal of the cell) and their other plates are connected to b (i.e., to the negative terminal of the cell). They are in parallel. Potential difference across each of them is V . We can redraw the circuit as shown. Note that the capacitor 1 can be shown anywhere along a to b . Similarly, capacitor 2 can be placed anywhere along a to b .



Charge on each of the capacitor is $Q = CV$.

- (a) Charge flow through the cell is $2Q$. It gets divided into two equal parts.

- (b) Energy stored in the two capacitors is

$$U = \frac{1}{2} CV^2 + \frac{1}{2} CV^2 = CV^2$$

- (c) Work done by the cell is $W = 2QV = 2CV^2$

$$\text{Heat generated} = W - U \\ = 2CV^2 - CV^2 = CV^2$$

19. When finding equivalent across a and b , $1\mu\text{F}$ and $4\mu\text{F}$ capacitors are in series. Their equivalent is

$$\frac{1 \times 4}{1 + 4} = \frac{4}{5} \mu\text{F}$$

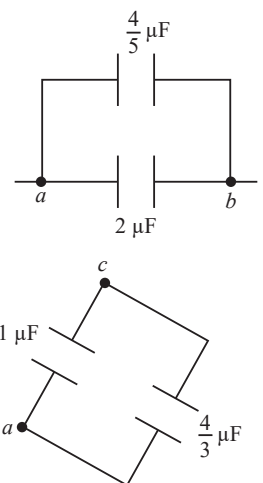
This $\frac{4}{5} \mu\text{F}$ capacitor is in parallel with $2\mu\text{F}$ capacitor.

$$\therefore C_1 = \frac{4}{5} + 2$$

$$= \frac{14}{5} \mu\text{F}$$

When finding equivalent across points a and c , $4\mu\text{F}$ and $2\mu\text{F}$ capacitors are in series. Their equivalent is

$$\frac{4 \times 2}{4 + 2} = \frac{4}{3} \mu\text{F}$$



$\frac{4}{3} \mu\text{F}$ capacitor is in parallel with $1 \mu\text{F}$ capacitor

$$\therefore C_2 = \frac{4}{3} + 1 = \frac{7}{3} \mu\text{F}$$

$$\therefore \frac{C_1}{C_2} = \frac{14}{5} \times \frac{3}{7} = \frac{6}{5}$$

20. The parallel paths are independent of one another. The $4 \mu\text{F}$ capacitor has 20V potential difference across it and the series combination of $1 \mu\text{F}$ and $2 \mu\text{F}$ also has a 20V potential difference across it. Removing the $4 \mu\text{F}$ capacitor has no effect on the rest of the circuit.

Therefore, using the result obtained in example 14 we can quickly write the potential difference across the $2 \mu\text{F}$ capacitor.

$$V_2 = \left(\frac{1}{1+2} \right) (20 \text{ V}) = \frac{20}{3} \text{ V}$$

21. Potential difference across C_1 is (refer to Example 14)

$$V_1 = \left(\frac{C_2}{C_1 + C_2} \right) V = \left(\frac{2}{1+2} \right) \times 36 = 24 \text{ V}$$

Potential difference across C_2 is

$$V_2 = \left(\frac{C_1}{C_1 + C_2} \right) V = \left(\frac{1}{1+2} \right) \times 36 = 12 \text{ V}$$

Obviously, C_1 will get punctured

22. Equivalent capacitance between points B and A is given by

$$\frac{1}{C_0} = \frac{1}{1} + \frac{1}{2} + \frac{1}{4} \quad \Rightarrow C_0 = \frac{4}{7} \mu\text{F}$$

Charge supplied by the cell is $Q = C_0 V$

$$= \left(\frac{4}{7} \mu\text{F} \right) (70 \text{ V}) = 40 \mu\text{C}$$

This is charge on each capacitor.

Potential difference across $4 \mu\text{F}$ capacitor is $\frac{40 \mu\text{C}}{4 \mu\text{F}}$
 $= 10 \text{ V}$

Plate connected to D is positively charged and the other plate (of $4 \mu\text{F}$ capacitor) is negatively charged.

$$\therefore V_D - V_A = 10$$

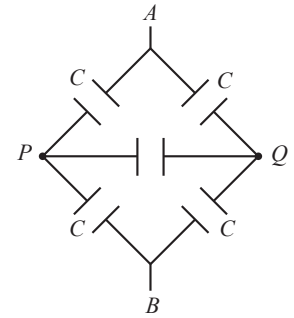
$$\Rightarrow V_D = 10 \text{ V} \quad [\because V_A = 0]$$

23. The equivalent of $4 \mu\text{F}$ and C in parallel is $(C + 4) \mu\text{F}$. The circuit is a Wheatstone bridge after we replace C and $4 \mu\text{F}$ with a single capacitor. There will be no charge on $5 \mu\text{F}$ capacitor if the Wheatstone bridge is balanced.

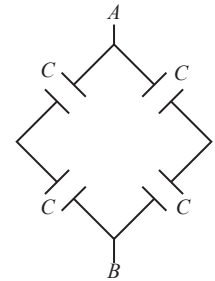
$$\frac{C_1}{C_2} = \frac{C_3}{C_4}$$

$$\text{or, } \frac{1}{C+4} = \frac{6}{48} \quad \text{or, } C = 4 \mu\text{F}$$

24. We can redraw the circuit as shown. The segment PBQ can be drawn below the line PQ . It will make no difference to the circuit. Now, it is easy to see that the circuit is a balanced Wheatstone bridge between the points A and B.



When a battery is connected across A and B, the points P and Q will be equipotential points. The capacitor between P and Q can be removed. C and C in series makes $\frac{C}{2}$. Two $\frac{C}{2}$ in parallel makes C.



$$25. C_1 = \frac{k\epsilon_0 A / 2}{d / 2} = \frac{k\epsilon_0 A}{d}$$

$$C_2 = \frac{2k\epsilon_0 A / 2}{d / 2} = 2C_1$$

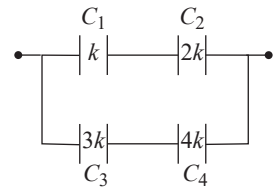
$$C_3 = 3C_1, \quad C_4 = 4C_1$$

$$C_1 \text{ and } C_2 \text{ in series give } \frac{C_1(2C_1)}{C_1 + 2C_1} = \frac{2C_1}{3}$$

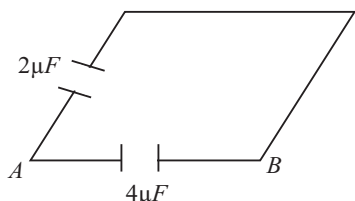
$$C_3 \text{ and } C_4 \text{ in series give } \frac{(3C_1)(4C_1)}{3C_1 + 4C_1} = \frac{12C_1}{7}$$

$$\therefore \text{Capacitance is } C = \frac{2C_1}{3} + \frac{12C_1}{7} = \frac{50C_1}{21}$$

$$= \frac{50k\epsilon_0 A}{21d}$$



26. The two $2\mu\text{F}$ capacitors and $1\mu\text{F}$ capacitors are short circuited. They can be removed. After this, we find that the two $4\mu\text{F}$ capacitors are in series.
27. The $3\mu\text{F}$ capacitor is short circuited. $1\mu\text{F}$ and $2\mu\text{F}$ capacitors are in parallel.

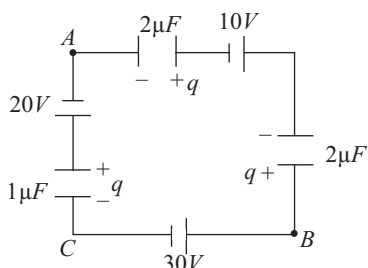


28. Both the systems are same. There are two capacitors in parallel.
29. Potential difference across both the capacitors is already same (equal to 10V). When connected, there will be no movement of charge.
30. Final common potential difference is

$$V = \frac{C_2 V_2 - C_1 V_1}{C_1 + C_2} = \frac{5 \times 100 - 1 \times 100}{5 + 1} = \frac{200}{3} \text{ V.}$$

Charge on A will be: $q_1 = C_1 V = 1 \times \frac{200}{3} = \frac{200}{3} \mu\text{C}$.

31. In a single loop circuit, all capacitors have same charge. [Example 27]



Assuming capacitors in series, the effective capacitance in the loop is given by

$$\frac{1}{C_0} = \frac{1}{2} + \frac{1}{1} + \frac{1}{2}$$

$$\text{or, } C_0 = \frac{1}{2} \mu\text{F}$$

Net emf in the loop is $\varepsilon_0 = 20 + 30 - 10 = 40\text{V}$

$$\therefore q = C_0 \varepsilon_0 = 20 \mu\text{C.}$$

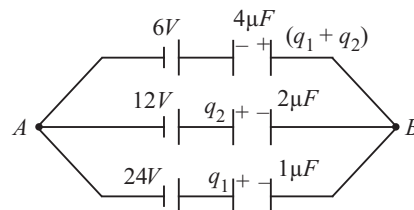
Now we move from A to B via point C .

$$V_A + 20 - \frac{20}{1} + 30 = V_B$$

$$\text{or, } V_A - V_B = -30 \text{ V}$$

32. Let charge on $1\mu\text{F}$ and $2\mu\text{F}$ capacitor be q_1 and q_2 as shown. The three plates connected to B must

have a zero total charge. Therefore, charge on the plate of $4\mu\text{F}$ capacitor connected to B is $+(q_1 + q_2)$.



We will write two loop equations and solve for q_1 and q_2 . In the upper loop having $2\mu\text{F}$ and $4\mu\text{F}$ capacitor, we start from point A and move clockwise to return at A .

$$6 + \frac{q_1 + q_2}{4} + \frac{q_2}{2} - 12 = 0$$

$$\text{or, } q_1 + 3q_2 = 24 \quad (\text{i})$$

In the lower loop, we again start at A and move clockwise

$$12 - \frac{q_2}{2} + \frac{q_1}{1} - 24 = 0$$

$$\text{or, } 2q_1 - q_2 = 24 \quad (\text{ii})$$

Solving (i) and (ii) gives

$$q_2 = \frac{24}{7} \mu\text{C} \quad \text{and} \quad q_1 = \frac{96}{7} \mu\text{C}$$

Now, moving from A to B along the middle path, we can write

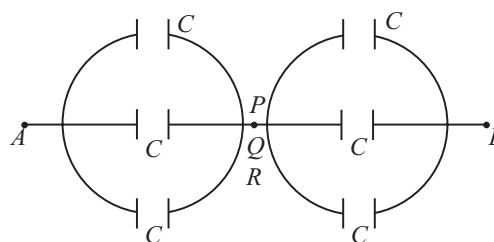
$$V_A + 12 - \frac{24}{7 \times 2} = V_B$$

$$\text{or, } V_B - V_A = \frac{72}{7} = 10.3 \text{ V}$$

33. M is the false junction that opens it and use series-parallel grouping.

34. Line AB divides the circuit into two symmetrical halves. Points P , Q and R lie on line that is perpendicular to AB . These are equipotential points. There is no potential difference across the two capacitors having capacitance $2C$. They can be removed and the points P , Q and R can be combined together.

Equivalent is $\frac{3C}{2}$.



**Worksheet 1**

1. If r is radius of the bigger drop and r is radius of each of the smaller drops, then

$$\frac{4}{3}\pi R^3 = 64\left(\frac{4}{3}\pi r^3\right)$$

or, $R = 4r$.

Capacitance of a spherical conductor $\propto$ radius.

Hence, the capacitance becomes 4 times.

2. Charge remains constant. The capacitance

$\left(C = \frac{\epsilon_0 A}{d}\right)$ decreases with increase in d . The potential difference $\left(V = \frac{Q}{C}\right)$ increases with decreasing C .

3. $U = \frac{Q^2}{2C}$. With decreasing C , the value of U will increase.

4. Charge sharing happens when potential of the two conductors are different, i.e., when

$$\frac{q_1}{C_1} \neq \frac{q_2}{C_2} \Rightarrow q_1 C_2 \neq q_2 C_1$$

Therefore, charge sharing (and hence loss in energy) will happen when $q_1 C_2 \neq q_2 C_1$

$$5. C_0 = \frac{\epsilon_0 A}{d} \quad \text{and} \quad C = \frac{\epsilon_0 A}{\frac{d}{4\epsilon_r} + \frac{3d}{4}}$$

$$\therefore C = \frac{4\epsilon_r C_0}{1 + 3\epsilon_r}$$

6. Maximum charge that $2\mu\text{F}$ capacitor can take = $2 \times 3 = 6\mu\text{C}$.

Maximum charge that $3\mu\text{F}$ capacitor can take = $3 \times 2 = 6\mu\text{C}$.

Maximum charge that $5\mu\text{F}$ capacitor can take = $5 \times 1 = 5\mu\text{C}$.

In series, all capacitors will have same charge. Maximum charge allowed will be $5\mu\text{C}$.

$$\text{Equivalent capacitance } \frac{1}{C} = \frac{1}{2} + \frac{1}{3} + \frac{1}{5}$$

$$\text{or, } C = \frac{30}{31}\mu\text{F}$$

$$\therefore V_{\max} = \frac{5\mu\text{C}}{\frac{30}{31}\mu\text{F}} = \frac{31}{6}\text{V}$$

7. $V_{\max} = (E_{\max})d = \left(19 \times 10^3 \frac{\text{V}}{\text{mm}}\right)(0.01\text{mm}) = 190\text{V}$.

8. Field due to charge on one plate = $\frac{10^5 V}{2 \text{ m}}$.

$$\text{Force on the other plate} = EQ = \frac{10^5}{2} \times 10^{-6} = 0.05 \text{ N}$$

9. $U = \frac{1}{2}CV^2 = \frac{1}{2} \times (40\mu\text{F}) \times (3000\text{V})^2 = 180\text{J}$

This energy is given to the patient in 0.002 second.

$$\therefore \text{Power given} = \frac{180}{0.002} = 90\text{kW}$$

10. The electric field between the plates $E = \frac{V}{d}$.

If d is halved, E doubles. The electric field created due to charge on any one plate is $\frac{E}{2}$ and this also doubles if d is halved.

On making the separation $\frac{d}{2}$, the capacitance doubles. Charge on the capacitor also doubles (this is the reason why E has doubled).

$$\text{Force on a plate} = \frac{E}{2} \cdot Q.$$

This force obviously becomes 4 times if the separation is halved.

11. When 2 is touched to 1, charge gets shared between them such that both of them acquire same potential

$$V = \frac{Q_2}{C_2} = \frac{Q_1}{C_1}$$

$$\text{and } Q_1 + Q_2 = Q$$

$$\therefore Q_1 = \left(\frac{C_1}{C_1 + C_2}\right)Q = \frac{Q}{3}$$

$$\text{Charge acquired by 2 is } Q_2 = \left(\frac{C_2}{C_1 + C_2}\right)Q$$

When 2 is touched to 3, the entire charge of 2 gets transferred to 3. This is the only way they can have same potential equal to zero.

After first set of operations, body 1 has $Q_1 = \frac{Q}{3}$ charge and the body 2 has no charge. Working similarly, one can see that the charge on the body 1 after second set of operations will be $\frac{Q}{3^2}$.

After N operations, charge on the first body will be $\frac{Q}{3^N}$.

12. Total charge on the capacitor, after insertion of the dielectric is $60 + 120 = 180\mu\text{C}$.

This implies that the capacitance has become 3 times. Therefore, $k = 3$.

13. Adding in series decrease the capacitance and joining in parallel increases the capacitance. Note that 8 capacitors, each of capacitance C , in series make $\frac{C}{8}$, and 8 capacitors, each of $\frac{C}{8}$, in parallel make C .

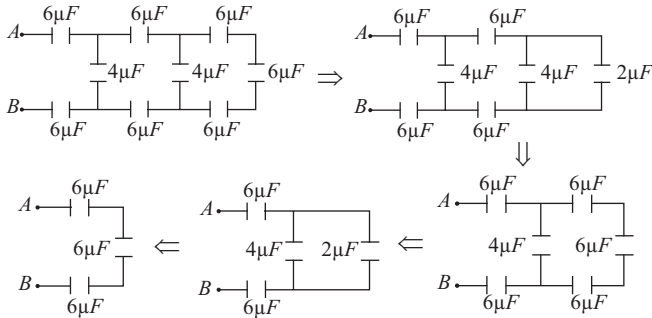
15. $C_0 = \frac{\epsilon_0 A}{d}$

$$C = \frac{\epsilon_0 A}{\frac{d}{2K} + \frac{d}{2}} \quad \text{where } k = \infty \text{ for metal}$$

$$= \frac{2\epsilon_0 A}{d}$$

$$\therefore \frac{C_0}{C} = \frac{1}{2}$$

16.



17. Capacitance of the first row = $C = 1\mu\text{F}$

$$\text{Capacitance of the second row} = \frac{C}{4}$$

$$\text{Capacitance of the third row} = \frac{C}{16}$$

And so on.

These capacitances are in parallel.

$$\therefore C_0 = C + \frac{C}{4} + \frac{C}{16} + \frac{C}{64} + \dots$$

$$= C \left[1 + \frac{1}{4} + \frac{1}{16} + \frac{1}{64} + \dots \right]$$

$$= C \frac{1}{1 - \frac{1}{4}} \quad [\text{Infinite G. P having common ratio} = \frac{1}{4}]$$

$$= \frac{4}{3} \mu\text{F} \quad [\because C = 1\mu\text{F}]$$

18. All three capacitors have same charge. Plates b, d and f have positive charge.

19. Both the sphere will be at same potential.

$$K \frac{q_1}{r} = K \frac{q_2}{2r};$$

$$\text{or, } \frac{4\pi r^2 \sigma_1}{r} = \frac{4\pi (2r)^2 \sigma_2}{2r}$$

$$\text{or, } \frac{\sigma_1}{\sigma_2} = \frac{2}{1}$$

20. Final common potential difference after the two capacitors are connected is

$$V = \frac{C_1 V_1 + C_2 V_2}{C_1 + C_2}$$

$$\text{or, } 25 = \frac{5 \times 100 + C(0)}{5 + C}$$

$$\text{or, } 5 + C = 20$$

$$\text{or, } C = 15 \text{ mF}$$

21. $E_1 = \frac{1}{2} C V^2$

When connected in series, the equivalent capacitance is $\frac{C}{2}$.

$$\therefore E = \frac{1}{2} \left(\frac{C}{2} \right) V^2 = \frac{E_1}{2}$$

22. The capacitors $2C$, C and $2C$ are in series.

Their equivalent is given by

$$\frac{1}{C_0} = \frac{1}{2C} + \frac{1}{C} + \frac{1}{2C}$$

$$\text{or, } C_0 = \frac{C}{2}$$

$\therefore$ Charge on each of these three capacitors is

$$q = C_0 V = \frac{C}{2} (60) = 30C.$$

$\therefore$ Potential difference across M and N = potential difference across the capacitance $C = \frac{q}{C} = 30V$.

Note that the presence or absence of a parallel branch (having capacitance C) has no effect on our answer.

23. Charge on $4\mu\text{F}$ capacitor is $Q = 4 \times 6 = 24\mu\text{C}$.

Equivalent of $2\mu\text{F}$ and $5\mu\text{F}$ in parallel is $7\mu\text{F}$. This $7\mu\text{F}$ in series with $3\mu\text{F}$ makes $\frac{7 \times 3}{10} = \frac{21}{10} \mu\text{F}$.

Charge on the equivalent capacitance is

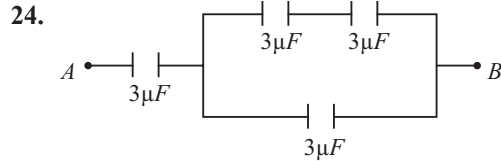
$$q = \frac{21}{10} \times 6 = \frac{63}{5} \mu\text{C}$$

This charge q gets divided on $2\mu\text{F}$ and $5\mu\text{F}$ so that potential difference across them is same. It means charges on them are in ratio of their capacitances.

$\therefore$ Charge on 5mF capacitance is $\frac{q \times 5}{5 + 2}$

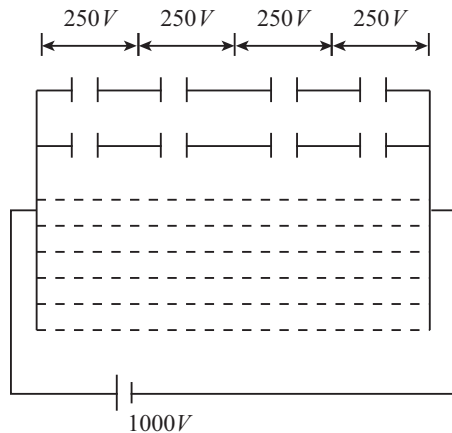
$$= \frac{63}{5} \times \frac{5}{7} = 9 \mu\text{C}$$

$$\text{Required ratio is } \frac{9\mu\text{C}}{24\mu\text{C}} = \frac{3}{8}$$



25. For peak voltage of 1000V, there must be at least 4 capacitors in series.

4 capacitors, each of capacitance $8\mu\text{F}$, in series makes $2\mu\text{F}$. If we have 8 such branches in parallel, the equivalent will become $16\mu\text{F}$.

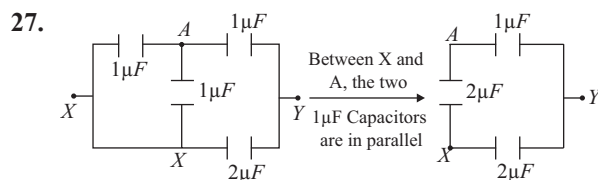


26. Equivalent of $5\mu\text{F}$ and $1\mu\text{F}$ is $6\mu\text{F}$.

The potential difference across $3\mu\text{F}$ and $6\mu\text{F}$ in series is 24V . Potential difference across $6\mu\text{F}$ is $\frac{3}{3+6} \times 24 = 8\text{V}$

$$\therefore \text{Energy in } 1\mu\text{F capacitor is } = \frac{1}{2}CV^2$$

$$= \frac{1}{2} \times 1 \times 8^2 = 32 \mu\text{J}$$



$2\mu\text{F}$ and $1\mu\text{F}$ (along XAY) are in series. Their equivalent is $\frac{2 \times 1}{3+1} = \frac{2}{3} \mu\text{F}$.

Now, $\frac{2}{3} \mu\text{F}$ and $2\mu\text{F}$ are in parallel. It makes $\frac{2}{3} + 2 = \frac{8}{3} \mu\text{F}$

28. Equivalent of $4\mu\text{F}$ and $2\mu\text{F}$ is $6\mu\text{F}$, Now $3\mu\text{F}$ and $6\mu\text{F}$ are in series and the potential difference across them is 1200V .

The potential difference across $3\mu\text{F}$ capacitor will be $\frac{3}{3+6} = 1200 = 400\text{V}$

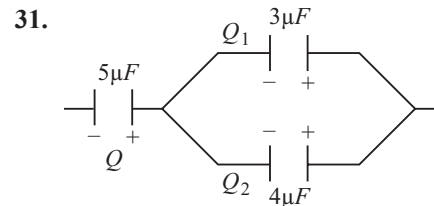
29. Charge on the first capacitor, $Q_1 = 15 \times 2 = 30\mu\text{C}$

Charge on the second capacitor, $Q_2 = 10 \times 3 = 30\mu\text{C}$
Opposite plates of the two capacitors are connected together. Hence, total charge available for distribution is $+30 - 30 = 0$.

Charge on both the capacitors will be zero, finally. Charge $30\mu\text{C}$ moves from A and neutralises the $-30\mu\text{C}$ charge present on D .

30. $3\mu\text{F}$ and $6\mu\text{F}$ in series gives $\frac{3 \times 6}{3+6} = 2\mu\text{F}$. The two $2\mu\text{F}$ in parallel gives $4\mu\text{F}$.

Total energy stored = energy stored in the equivalent capacitor $= \frac{1}{2}CV^2 = \frac{1}{2} \times 4 \times 2^2 = 8 \mu\text{J}$



$$Q = 20 \mu\text{C}$$

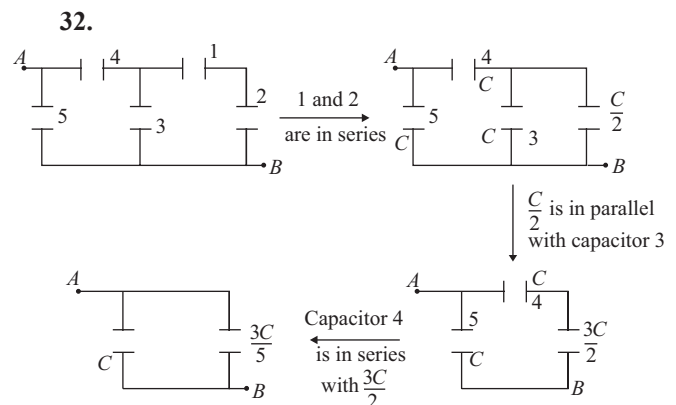
$$Q_1 + Q_2 = Q = 20 \mu\text{C} \quad (i)$$

$$\text{and } \frac{Q_1}{3} = \frac{Q_2}{4} \quad (ii)$$

Solving (i) and (ii)

$$Q_1 = \frac{3}{7} \times 20 = 8.57 \mu\text{C}$$

Obviously, charge on the right plate of $3\mu\text{F}$ capacitor is positive.

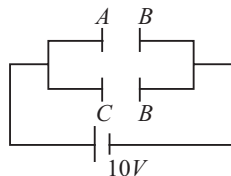


33. Maximum charge on $8\mu\text{F}$ capacitor that is allowed is $Q_1 = 8 \times 20 = 160 \mu\text{C}$.

Maximum charge on $16\mu\text{F}$ capacitor can be $Q_2 = 16 \times 80 = 1280 \mu\text{C}$. When in series, the maximum charge on both of them shall be $160 \mu\text{C}$ so that none of the capacitor punctures.

$$\begin{aligned}\therefore Q_{\max} &= 160 \mu\text{C} \\ \therefore U_{\max} &= \frac{(160)^2}{2 \times 8} + \frac{(160)^2}{2 \times 16} = 2400 \mu\text{J}\end{aligned}$$

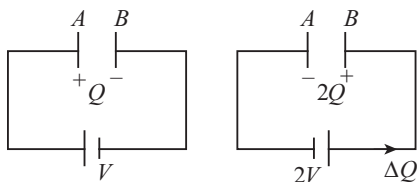
34. There are two capacitors in parallel.



$$\begin{aligned}\text{Capacitance of each capacitor is } C &= \frac{\epsilon_0 A}{d} \\ &= \frac{8.85 \times 12^{-12} \times 0.1}{0.885 \times 10^{-3}} = 10^{-9} \mu\text{F}\end{aligned}$$

$$\begin{aligned}\therefore \text{Energy stored } U &= \frac{1}{2}CV^2 + \frac{1}{2}CV^2 \\ &= 10^{-9} \times (10)^2 = 10^{-7} \text{J} = 0.1 \mu\text{J}\end{aligned}$$

35. Initial charge $Q = CV$ (Plate A is positive)



Final charge $Q' = 2CV = 2Q$ (Plate B is positive)

Charge flow through the 2V emf cell is

$$\Delta Q = 3Q = 3CV$$

$$W_{\text{cell}} = (\Delta Q)(2V) = 6CV^2$$

$$U_f = \frac{1}{2}C(2V)^2 = 2CV^2$$

$$\begin{aligned}\text{Heat} &= W_{\text{cell}} + U_i - U_f \\ &= 6CV^2 + \frac{1}{2}CV^2 - 2CV^2 = 4.5 CV^2\end{aligned}$$

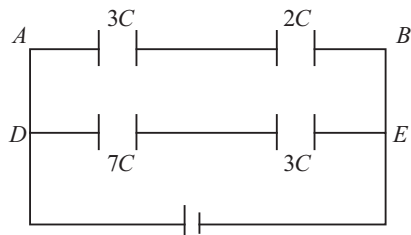
$$\frac{\text{Heat}}{U_f} = \frac{4.5 CV^2}{2 CV^2} = 2.25$$

36. First consider the branch AB.

Maximum allowed charge on $3C$ is $(3C)(10^3)$

Maximum charge allowed on $2C$ is $(2C)(2 \times 10^3)$
 $4C \times 10^3$.

For no capacitor to break down, the maximum charge on the capacitors in branch AB shall be $3C \times 10^3$.



This means that the potential difference across AB shall not exceed

$$V_1 = \frac{3C \times 10^3}{\frac{3C \times 2C}{3C + 2C}} = 2.5 \times 10^3 \text{ volt}$$

Similarly, the maximum allowed potential difference across branch DE can be calculated.

$$\text{It is } V_2 = \frac{20}{7} \times 10^3 = 2.9 \times 10^3 \text{ V}$$

Therefore, the applied potential difference shall not exceed 2.5 kV.

37. The two 1F capacitors and the $2\mu\text{F}$ capacitor are in parallel. Their equivalent is 4F .

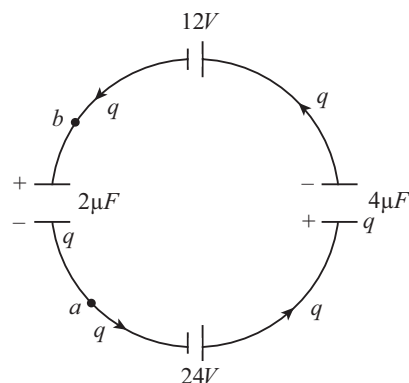
Now, 1F and 4F are in series.

Potential difference across the 4F capacitor will be

$$\frac{1}{1+4} \times 6 = \frac{6}{5} \text{ V}$$

This is potential difference across all the three capacitors in parallel.

38. Charge on the capacitors has been shown.



Using Kirchhoff's loop law

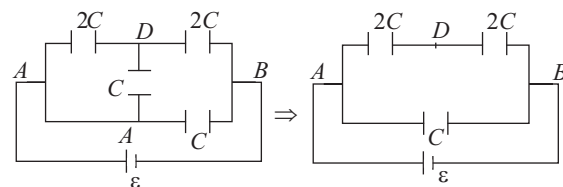
$$24 - \frac{q}{4} - 12 - \frac{q}{2} = 0$$

$$\text{or, } q = 16 \mu\text{C}$$

$$\therefore V_b - V_a = \frac{q}{2} = 8\text{V}$$

$$\text{or, } V_a - V_b = -8\text{V}.$$

- 39.

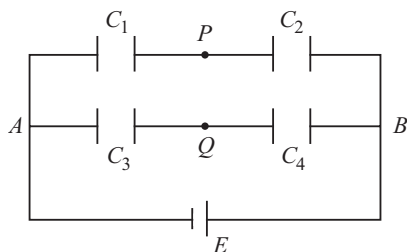


$$\text{Equivalent capacitance} = \frac{11C}{5}$$

$$\therefore \text{Charge supplied} = \frac{11C\epsilon}{5}$$

40. Potential difference between the points P and A is $V_P - V_A =$ potential difference between the plates

$$\text{of } C_1 = \frac{C_2 E}{C_1 + C_2}$$



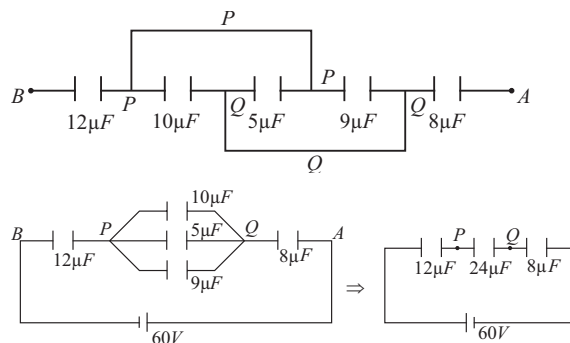
$$\text{Similarly, } V_Q - V_A = \frac{C_4 E}{C_3 + C_4}$$

$$V_P - V_Q = (V_P - V_A) - (V_Q - V_A)$$

$$= \frac{C_2 E}{C_1 + C_2} - \frac{C_4 E}{C_3 + C_4} = \frac{(C_2 C_3 - C_1 C_4) E}{(C_1 + C_2)(C_3 + C_4)}$$

41. The circuit has a balanced Wheatstone bridge along with two branches in parallel (each having two capacitance C in series).

42.



Equivalent capacitance is given by

$$\frac{1}{C_0} = \frac{1}{12} + \frac{1}{24} + \frac{1}{8};$$

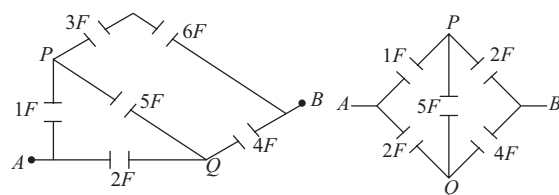
$$\text{or, } C_0 = 4\mu\text{F}$$

$$\therefore Q = C_0 V = 4 \times 60 = 240 \mu\text{C}.$$

$$\text{Potential difference } V_Q - V_P = \frac{240\mu\text{C}}{24\mu\text{F}} = 10\text{V}.$$

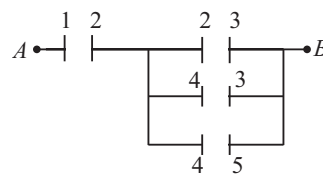
$$\therefore \text{Charge on } 5\mu\text{F is } (5\mu\text{F})(10) = 50\mu\text{C}.$$

43.



This is a balanced Wheatstone bridge.

44.

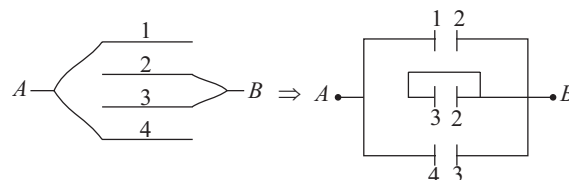


The capacitor formed between the plates 4 and 5 is out of the circuit.

$$C_0 = \frac{(2C)C}{2C + C} = \frac{2C}{3} = \frac{2\epsilon_0 A}{3d}$$

$$\therefore Q = C_0 E = \frac{2\epsilon_0 A E}{3d}$$

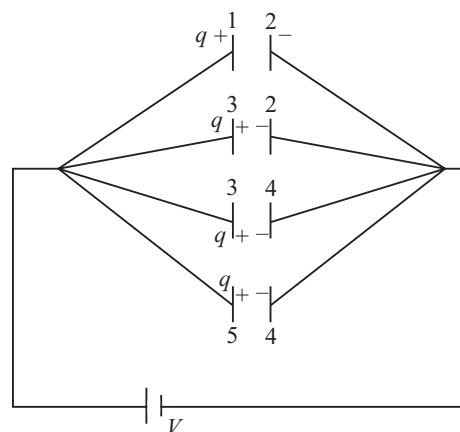
45.



Capacitor between the plates 2 and 3 is short circuited. These are only two capacitors in parallel.

$$C_0 = \frac{2\epsilon_0 A}{d}$$

46. The alternate plates are together. There are 4 capacitors in parallel.



Total charge on plate 2 is $-2q$. Charge on plate 3 is $2q$. Charge on plate 4 is $-2q$ and charge on plate 5 is $+q$.

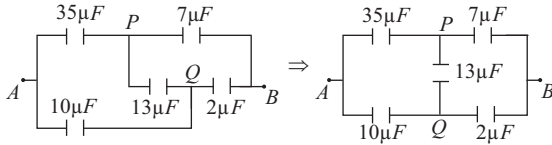
47. Spring tension $= mg$.

Force between capacitor plates $=$ spring tension $= mg$

$$\text{or, } \frac{Q^2}{2A\epsilon_0} = mg$$

$$\Rightarrow Q = \sqrt{2mgA\epsilon_0}$$

48. After combining $23\mu\text{F}$ and $12\mu\text{F}$ in parallel and two $1\mu\text{F}$ capacitors in parallel, the circuit can be redrawn as given.



This is a balanced Wheatstone bridge and $13\mu\text{F}$ capacitor can be removed.

49. Potential drops as one moves in the direction of electric field. There is no field inside the conductor and the potential does not change when one moves from $x = d$ to $x = 2d$.

The electric field is lower inside the dielectric. The rate of fall of potential is smaller between $x = 3d$ to $x = 4d$.

50. $E_0 = 200 \text{ Vcm}^{-1}$

Field is present in a region of width 3cm only after the metal bar is inserted.

$$\therefore V = E_0 (3 \text{ cm}) = 600\text{V}.$$

51. Charge on first capacitor, $Q_1 = (15\mu\text{F})(100\text{V}) = 1500\mu\text{C}$.

$$\text{Charge on the second capacitor, } Q_2 = (1\mu\text{F})(100\text{V}) = 100\mu\text{C}.$$

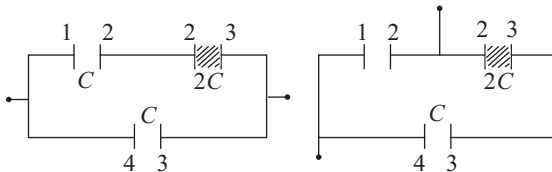
After removing the dielectric, $C_1 = 1\mu\text{F}$ and

$$C_2 = 1\mu\text{F}$$

$$\text{Final common potential difference, } V = \frac{Q_1 + Q_2}{C_1 + C_2}$$

$$= \frac{1500 + 100}{1 + 1} = 800\text{V}$$

- 52.



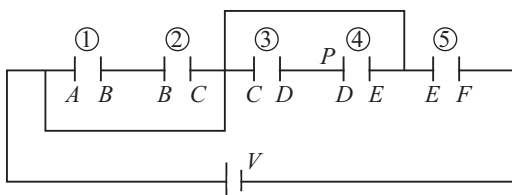
Circuit between 1 and 3

Between 2 and 4

$$C_1 = \frac{(2C)(C)}{2C + C} + C = \frac{5C}{3}$$

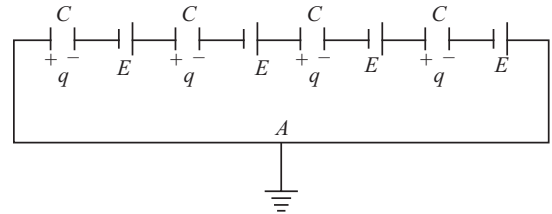
$$C_2 = \frac{(2C)(C)}{2C + C} + C = \frac{5C}{3}$$

- 53.



The series equivalent of 1 and 2 is short circuited. Similarly, the series equivalent of 3 and 4 is also shorted. The only capacitor having charge is capacitor number 5.

54. With only S_1 closed the circuit is as shown



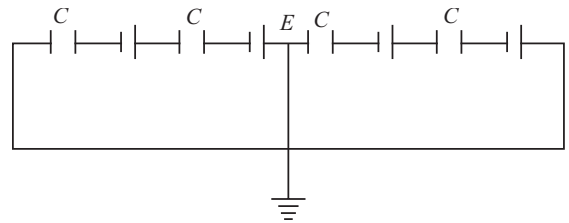
$$\text{Total emf} = 4E$$

$$\text{Effective capacitance} = \frac{C}{4}$$

$$\therefore \text{Charge on each capacitor, } Q = (4E) \left(\frac{C}{4} \right) = CE$$

[It is a single loop circuit]

After S_2 is closed, the circuit is shown.



Once again charge on each capacitor will be CE . Actually, in the first circuit (S_2 not closed),

$$V_A = V_B$$

$$V_A - \frac{q}{C} + E - \frac{q}{C} + E = V_B$$

$$\text{or, } V_A = V_B \quad [\because q = CE]$$

Joining A and B makes no difference. There is no charge flow and no heat generation.

55. If E_0 is field without dielectric, then field in the two dielectrics will be

$$E_1 = \frac{E_0}{K_1};$$

$$E_2 = \frac{E_0}{K_2}$$

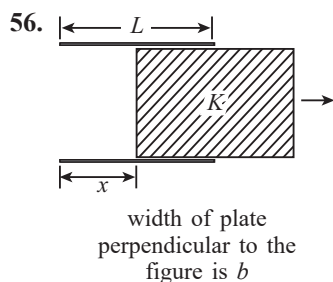
Energy stored in first dielectric is

$$U_1 = \left(\frac{1}{2} \epsilon_0 E_1^2 \right) (Ad)$$

Energy stored in the second dielectric is

$$U_2 = \left(\frac{1}{2} \epsilon_0 E_2^2 \right) (Ad)$$

$$\therefore \frac{U_1}{U_2} = \left(\frac{E_1}{E_2} \right)^2 = \frac{K_2^2}{K_1^2}$$



$$x = vt$$

$$C = \frac{\epsilon_0 b(vt)}{d} + \frac{k\epsilon_0 b(L-vt)}{d}$$

$$= K \frac{\epsilon_0 bL}{d} + \frac{\epsilon_0 bv}{d}(1-k)t$$

C decreases linearly with time. ($\because k > 1$)

57. Note that $23\mu\text{F}$ and $12\mu\text{F}$ are in parallel.

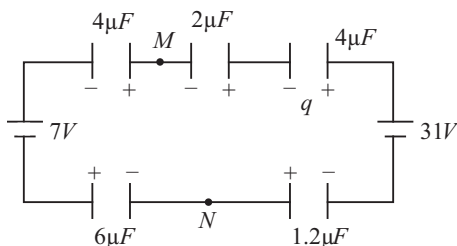
Two $1\mu\text{F}$ capacitors are also in parallel.

The circuit is a balanced Wheatstone bridge.

58. This is also a balanced Wheatstone bridge with $5\mu\text{F}$ capacitor having no charge.

59. This is a balanced Wheatstone bridge with no charge on $10\mu\text{F}$ capacitor.

60. This is a single loop circuit. All capacitors have same charge.



$$\text{Effective emf} = 31 - 7 = 24 \text{ V}$$

Effective capacitance

$$\frac{1}{C_0} = \frac{1}{4} + \frac{1}{2} + \frac{1}{4} + \frac{1}{6} + \frac{1}{1.2}$$

$$\text{or, } C_0 = \frac{1}{2} \mu\text{F}$$

$$\therefore q = (24\text{V}) \left(\frac{1}{2} \mu\text{F} \right) = 12 \mu\text{C}$$

$$V_M - \frac{q}{4} - 7 - \frac{q}{6} = V_N$$

$$\text{or, } V_M - 3 - 7 - 2 = V_N$$

$$\text{or, } V_M - V_N = 12\text{V}$$

61. Charge on capacitor A will not change. The positive and negative charges hold each other.



Worksheet 2

1. Capacitance becomes K times.

$\therefore$ Charge on the capacitor will become K times.

Force between the plates $\propto Q^2$.

$\therefore$ Force becomes K^2 times.

$$\text{Energy} = \frac{1}{2} CV^2$$

$\therefore$ Energy becomes K times.

$$\text{Electric field} = \frac{V}{d}$$

It does not change.

2. Unless all the three switches are closed, the circuit will not get completed and there will be no movement of charge.

3. Initial charge $Q_1 = CV = 10 \times 10 = 100\mu\text{C}$.

After insertion of dielectric, new capacitance = $KC = 3C$

$\therefore$ Charge on capacitor, $Q_2 = 3CV = 300\mu\text{C}$.

Increase in charge = $200\mu\text{C}$.

In a slow process, movement of charge is very slow (i.e., there is no current in the circuit). In such a process, no heat will be produced.

$$W_{\text{cell}} + W_{\text{extAgt}} = U_f - U_i$$

$$\text{or, } W_{\text{cell}} = (U_f - U_i) + (-W_{\text{extAgt}})$$

= Increase in potential energy + Work done on the external agent

Since heat = 0, the option (D) is also correct.

4. Initial charge $Q_1 = C_0 V$

With battery disconnected, the plates are brought closer. Capacitance increases to C_1 . In this process, the external agent does negative work (W_1).

$$W_1 = U_2 - U_1 = \frac{Q_1^2}{2C_1} - \frac{Q_1^2}{2C_0} = -\frac{Q_1^2}{2} \left[\frac{1}{C_0} - \frac{1}{C_1} \right]$$

When the battery is connected it supplies additional charge. Now charge on the capacitor is

$$Q_2 = C_1 V$$

In this process, heat is generated.

When the plate separation is increased, the capacitance again becomes C_0 and the charge on the capacitor becomes Q_0 . In this process, the battery gains energy equal to $(Q_2 - Q_0)V$. But the battery has already done larger amount of work

$$= \frac{Q_0^2}{2C_0} + (Q_2 - Q_1)V$$

$\therefore$ Battery is net loser of energy.

Force between the plates $\propto Q^2$.

Maximum charge on the capacitor is Q_2 when the plates are nearest.

When the plates were slowly brought closer with charge Q_0 on them, the force between them $\propto Q_0^2$
 $W_{\text{ext}} = -(\text{Force}) \times \text{displacement} \propto -Q_0^2$

When separating the plates, the charge on the plates was higher and work done by the agent is positive. Therefore, overall work done by the agent must be positive.

5. The $7\mu\text{F}$ and $3\mu\text{F}$ capacitors are in series. They have same charge. Potential difference across them is inversely proportional to their capacitance. If V_1 is potential difference across $3\mu\text{F}$ capacitor, then

$$\frac{V_1}{6} = \frac{7}{3} \Rightarrow V_1 = 14V.$$

Charge on $3\mu\text{F}$ capacitor $= (3\mu\text{F})(14V) = 42\mu\text{C}$.

The potential difference across the series combination of $7\mu\text{F}$ and $3\mu\text{F}$ is $= 14 + 6 = 20V$.

Equivalent capacitance of $3.9\mu\text{F}$, $3\mu\text{F}$ and $7\mu\text{F}$ is

$$C = \frac{7 \times 3}{7 + 3} + 3.9 = 6\mu\text{F}.$$

Note that $6\mu\text{F}$ and $12\mu\text{F}$ capacitors are in series and potential difference across the $6\mu\text{F}$ is $20V$.

$\therefore$ Potential difference across $12\mu\text{F}$ is $10V$.

$\therefore$ Emf of the cell $= 10 + 20 = 30V$.

6. Electric field between the plates is always $E = \frac{V}{d}$.

If charge density is σ_1 on the air-filled part and σ_2 on the dielectric filled part of the plates, then

$$E = \frac{\sigma_1}{\epsilon_0} = \frac{\sigma_2}{K\epsilon_0}$$

$\therefore \sigma_2 > \sigma_1$

Charge density on the air exposed part remains constant such that

$$\frac{\sigma_1}{\epsilon_0} = \frac{V}{d}$$

Charge flow $\Delta Q = (K - 1)CV$.

Energy gained by the cell $= \Delta QV = (K - 1)CV^2$

7. With switch S open, the equivalent capacitance is

$$C_1 = \frac{(2C)(C)}{2C + C} = \frac{2C}{3}$$

Charge supplied by the cell, $Q_1 = \frac{2C}{3}E$

After S is closed, a capacitor gets short circuited. The equivalent capacitance is

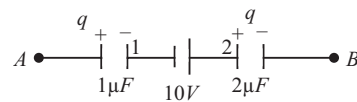
$$C_2 = 2C.$$

Total charge on the two positive plates of the capacitors in parallel is

$$Q_2 = 2CE.$$

Additional charge flow from the cell, $\Delta Q = Q_2 - Q_1$
 $= \frac{4CE}{3}$

8. It is a single loop circuit. All the capacitors will have same charge.
 9. Both the capacitors must have same charge on them.



[The charge supplied by the cell to plate 1 and 2 must be equal and opposite]

$$V_A - \frac{q}{1} + 10 - \frac{q}{2} = V_B$$

$$\text{or, } V_A - V_B + 10 = q + \frac{q}{2}$$

$$\text{or, } 5 + 10 = \frac{3q}{2}$$

$$\text{or, } q = 10\mu\text{C}$$

$$\therefore \text{Voltage across } C_1 \text{ is } V_1 = \frac{10}{1} = 10 \text{ V.}$$

$$\text{Voltage across } C_2 \text{ is } V_2 = \frac{10}{2} = 5 \text{ V.}$$

$$\frac{U_1}{U_2} = \frac{q^2}{2C_1} \times \frac{2C_2}{q^2} = \frac{C_2}{C_1} = \frac{2}{1}$$

$$10. Q = C_0V = \frac{\epsilon_0 A}{d}V$$

Charge does not change due to insertion of the dielectric. The potential difference between the plates becomes $\frac{V}{K}$ after the dielectric is inserted.

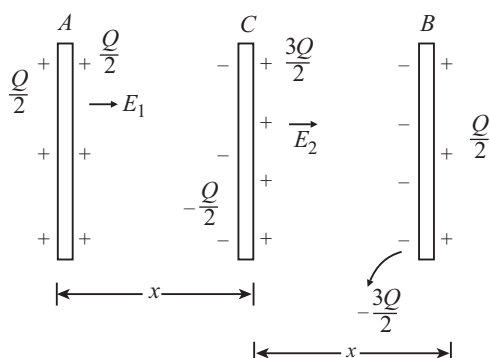
$$\therefore E = \frac{V}{dK}$$

$$W = U_f - U_i = \frac{Q^2}{2KC_0} - \frac{Q^2}{2C_0}$$

$$= \frac{(C_0V)^2}{2C_0} \left[\frac{1}{K} - 1 \right] = -\frac{1}{2}C_0V^2 \left[1 - \frac{1}{K} \right]$$

$$= -\frac{\epsilon_0 AV^2}{2d} \left(1 - \frac{1}{K} \right)$$

11. Charge distribution on the three plates will be as shown.



This will ensure that the electric field is zero inside the metal plates.

Potential difference between A and B is

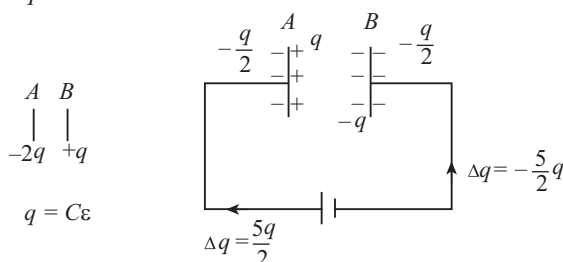
$$V_{AB} = E_1 x + E_2 x$$

$$= \frac{Q}{2A\epsilon_0} x + \frac{3Q}{2A\epsilon_0} x = \frac{2Q}{A\epsilon_0} x \quad (i)$$

Before insertion of plate C, the potential difference was same.

13. After getting connected to the cell, the potential difference between the plates will be ϵ . Charges on inner walls of A and B will be

$$\pm q = \pm C\epsilon$$



But total charge on the two plates will remain $-2C\epsilon + C\epsilon = -C\epsilon$.

Charge flow has been shown after the switch is closed.

14. C_1 and C_2 in parallel are equivalent to a capacitance $(C_1 + C_2)$.

C_3 and $(C_1 + C_2)$ are in series.

Potential difference across C_3 is $\frac{(C_1 + C_2)V}{C_1 + C_2 + C_3}$

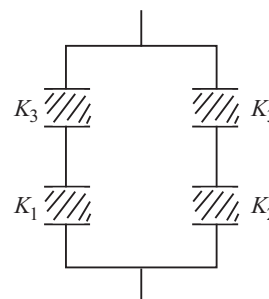
If dielectric slab is inserted in C_1 , the capacitance increases to KC_1 . Now, C_3 and $(KC_1 + C_2)$ are in series. The potential difference across capacitors in series is inversely proportional to capacitance. Therefore, potential difference across C_3 will increase.

When C_3 is shorted, the potential difference across C_1 increases to V . It means the electric field between the plates must have increased.

15. Final common potential = $\frac{CV + 2CV}{KC + 2C} = \frac{3V}{K + 2}$

No heat is produced as the process is very slow. But the stored energy decreases due to negative work done by the external agent in inserting the dielectric.

16. The equivalent shall be obtained by series and parallel combination of the 4 capacitors shown in the figure.



17. Total charge on the three plates connected to A is $= 2 \times 5 + 3 \times 20 + 5 \times 10 = 120 \mu C$.

After closing the switch, the sum of charges on the 4 plates shown inside box is still zero. This is because the total charge supplied by the cell to the system of 4 plates is zero.

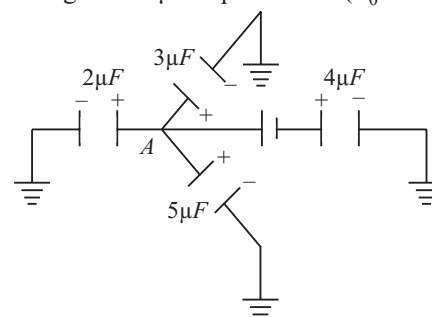
Let potential of point A be V_0 .

Charge on $2 \mu F$ capacitor $= 2V_0$

Charge on $3 \mu F$ capacitor $= 3V_0$

Charge on $5 \mu F$ capacitor $= 5V_0$

Charge on $4 \mu F$ capacitor $= 4(V_0 - 20)$



$$2V_0 + 3V_0 + 5V_0 + 4(V_0 - 20) = 120$$

$$\text{or, } V_0 = \frac{200}{14} = \frac{100}{7} V$$

$$\text{Charge on } 2 \mu F \text{ capacitor is } = 2V_0 = \frac{200}{7} \mu C$$

Sum of charge on the plates of the capacitors facing A has not changed. Therefore, sum of charges on the 4 plates connected to the ground is still $-120 \mu C$. The total charge flow to the ground is zero.



Worksheet 3

2. A charge $-q$ is induced on the cavity wall. Charge on outer surface of the ball A is q . The charge system comprising of point charge q and $-q$ charge on the cavity wall has no effect outside the cavity.

The system essentially behaves like a sphere having charge $+q$ on its surface being connected to another conductor.

- (a) Capacitance of sphere A is

$$C_1 = 4\pi\epsilon_0 R = \frac{1}{9 \times 10^9} \times 1 = \frac{1}{9} \text{ nF}. \quad (= C, \text{ say})$$

Capacitance of the other conductor is

$$C_2 = \frac{1}{18} \text{ nF} = \frac{C}{2}$$

Final common potential is

$$V = \frac{q + 2q}{C + \frac{C}{2}} = 2 \frac{q}{C}$$

$\therefore$ Charge on outer surface of A is

$$q_1 = C_1 V = 2q$$

Charge on B is $q_B = C_2 V = q$.

- (b) Initial potential of A is

$$V_1 = \frac{q}{C_1} = \frac{2nC}{\frac{1}{9} \text{ nF}} = 18 \text{ V}$$

Initial potential of B is

$$V_2 = \frac{2q}{C_2} = \frac{4nC}{\frac{1}{18} \text{ nF}} = 72 \text{ V}$$

$\therefore$ Heat = loss in electrostatic energy

$$\begin{aligned} &= \frac{C_1 C_2}{2(C_1 + C_2)} (V_1 - V_2)^2 \\ &= \frac{\frac{1}{9} \times \frac{1}{18}}{2\left(\frac{1}{9} + \frac{1}{18}\right)} (18 - 72)^2 \text{ nJ} \\ &= 54 \text{ nJ} \end{aligned}$$

3. The required capacitance is

$$C = \frac{Q}{V} = \frac{24 \times 10^{-6}}{1200} = 0.02 \mu\text{F}.$$

As the gap between the plates is made smaller, the electric field between the plates increases.

$$E = \frac{V}{d}$$

Maximum allowed E is $3 \times 10^6 \text{ V/m}$. Therefore, smallest d possible is

$$d_{\min} = \frac{1200V}{3 \times 10^6 \text{ V/m}} = 4 \times 10^{-4} \text{ m} = 0.4 \text{ mm}$$

$$\text{Now, } C = \frac{\epsilon_0 A}{d}$$

For a given value of C , smaller the value of d , smaller shall be A . Therefore, $A_{\min}$ is

$$\begin{aligned} A_{\min} &= \frac{C d_{\min}}{\epsilon_0} \\ &= \frac{0.02 \times 10^{-6} \times 0.4 \times 10^{-3}}{8.85 \times 10^{-12}} = 0.90 \text{ m}^2 \end{aligned}$$

4. The two $2\mu\text{F}$ capacitors are in parallel. They can be replaced with one single capacitor of capacitance $4\mu\text{F}$. $6\mu\text{F}$ and $12\mu\text{F}$ capacitors are in series. Their equivalent is

$$\frac{6 \times 12}{6 + 12} = 4\mu\text{F}$$

The circuit can be redrawn as shown. In this circuit, the two $4\mu\text{F}$ capacitors are in parallel and the capacitors having capacitance $8\mu\text{F}$ and $4\mu\text{F}$ are in series. Equivalent of $4\mu\text{F}$ and $4\mu\text{F}$ in parallel is $8\mu\text{F}$. Equivalent of $8\mu\text{F}$ and $4\mu\text{F}$ in series is

$$\frac{8 \times 4}{8 + 4} = \frac{8}{3} \mu\text{F}$$

The new circuit is shown. In this circuit, $1\mu\text{F}$ and $8\mu\text{F}$ are in series and their equivalent is $\frac{1 \times 8}{1 + 8} = \frac{8}{9} \mu\text{F}$.

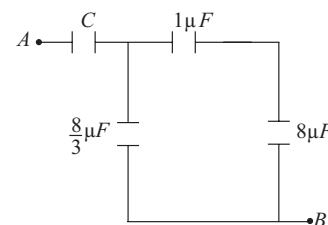
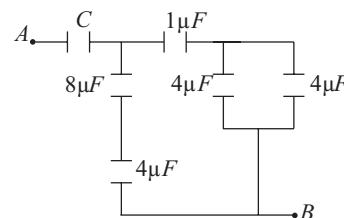
This $\frac{8}{9} \mu\text{F}$ is in parallel to $\frac{8}{3} \mu\text{F}$. Their equivalent is

$$\frac{8}{9} + \frac{8}{3} = \frac{32}{9} \mu\text{F}$$

Now C and $\frac{32}{9} \mu\text{F}$ in series result in an equivalent of $1\mu\text{F}$.

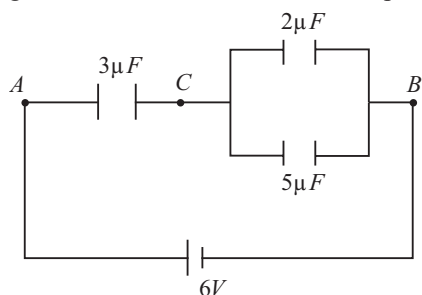
$$\therefore \frac{C \times \frac{32}{9}}{C + \frac{32}{9}} = 1$$

$$\text{or, } C = \frac{32}{23} \mu\text{F}$$



5. Note that $4\mu\text{F}$ capacitor is connected in parallel to the remaining circuit. Its presence or absence will make no difference to the remaining circuit. The potential difference between the points A and B is 6V , whether $4\mu\text{F}$ capacitor is there or not.

Equivalent of $2\mu\text{F}$ and $5\mu\text{F}$ in parallel is $7\mu\text{F}$. Now, $3\mu\text{F}$ and $7\mu\text{F}$ are in series. The 6V potential difference gets divided in inverse ratio of capacitance.



$\therefore$ Potential difference between C and D is

$$V_{CD} = \left(\frac{3}{3+7} \right) 6 = 1.8\text{V}$$

$\therefore$ Charge on $5\mu\text{F}$ capacitor is $= (5\mu\text{F}) (1.8\text{V}) = 9\mu\text{C}$.

6. (a) Take note of capacitors in parallel. The circuit can be redrawn as shown.

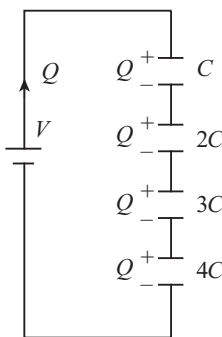
The equivalent capacitance is

$$C_0 = C + 2C + 3C + 4C = 10C$$

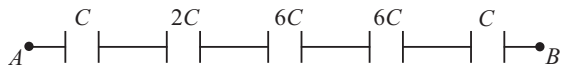
Charge supplied by cell is

$$Q = C_0 V = 10CV.$$

- (b) Obviously, charge Q gets divided into 3 equal parts among the capacitor A and the two other capacitors in parallel to it. Charge on A is $\frac{Q}{3} = \frac{10CV}{3}$.



7. The two capacitors having capacitance $3C$ are in parallel. The three capacitors having capacitance $2C$ each are in parallel. The two capacitors, having capacitance C each, are also in parallel.



$$\frac{1}{C_0} = \frac{1}{C} + \frac{1}{2C} + \frac{1}{6C} + \frac{1}{6C} + \frac{1}{C}$$

$$\text{or, } C_0 = \frac{6C}{17}$$

8. Charge on the capacitor when it is detached from the cell is $Q = C_0 V_0$.

After getting connected to the other capacitor, charge on it becomes $\frac{Q}{2}$. This means that both the capacitors have same capacitance. The final potential difference across them is $\frac{V_0}{2}$.

Heat = loss in energy of the capacitor system

$$= \frac{1}{2} C_0 V_0^2 - \left[\frac{1}{2} C_0 \left(\frac{V_0}{2} \right)^2 + \frac{1}{2} C_0 \left(\frac{V_0}{2} \right)^2 \right]$$

$$= \frac{1}{4} C_0 V_0^2$$

9. After first charging of the small capacitor, let the common potential difference be V_1 . Since the charge is conserved

$$\therefore C_0 V_0 = (C + C_0) V_1$$

$$\text{or, } V_1 = \frac{C_0 V_0}{C + C_0}$$

After first charge, the remaining charge on the large capacitor is $C_0 V_1$.

If V_2 is common potential difference after second contact, then

$$C_0 V_1 = (C + C_0) V_2$$

$$\text{or, } V_2 = \left(\frac{C}{C + C_0} \right) V_1 = \left(\frac{C_0}{C + C_0} \right)^2 V_0$$

Proceeding similarly

$$V_n = \left(\frac{C_0}{C + C_0} \right)^n V_0$$

$$\text{or, } V = \left(\frac{C_0}{C + C_0} \right)^n V_0$$

$$\text{or, } \left(\frac{C}{C_0} + 1 \right)^n = \frac{V_0}{V}$$

$$\text{or, } C = \left(\frac{V_0}{V} \right)^{\frac{1}{n}} - 1$$

10. Original capacitance is $C_0 = \frac{\epsilon_0 A}{d}$

$$= \frac{8.85 \times 10^{-12} \times 200 \times 10^{-4}}{4 \times 10^{-3}}$$

$$= 4.425 \times 10^{-11} \text{ F}$$

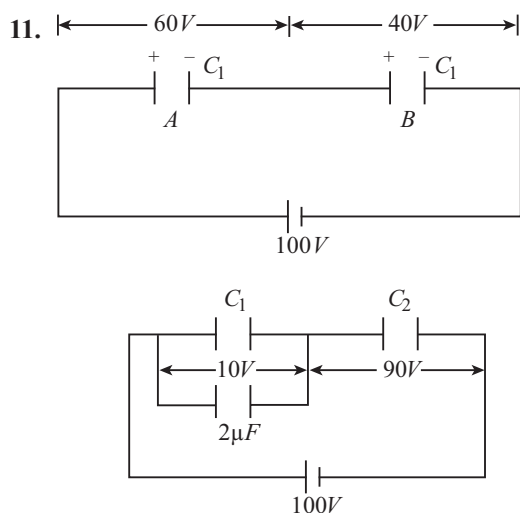
After the varnish is applied, the capacitance becomes

$$C = \frac{\epsilon_0 A}{\frac{t}{K} + (d - t)}$$

$$\begin{aligned}
 &= \frac{\epsilon_0 A / d}{\frac{t}{Kd} + 1 - \frac{t}{d}} \\
 &= \frac{C_0}{\frac{t}{Kd} + 1 - \frac{t}{d}} \\
 &= \frac{4.425 \times 10^{-11}}{\frac{0.2}{(3)(4)} + 1 - \frac{0.2}{4}} \\
 &= 4.578 \times 10^{-11} \text{ F}
 \end{aligned}$$

$$\text{Percentage change} = \frac{C - C_0}{C_0} \times 100$$

$$= \frac{4.578 - 4.425}{4.425} \times 100 = 3.46\%$$



Initially, charge is same on both the capacitors.

$$Q = C_1 V_1 = C_2 V_2$$

$$\text{or, } C_1(60) = C_2(40)$$

$$\text{or, } 3C_1 = 2C_2 \quad (\text{i})$$

After adding $2 \mu\text{F}$ capacitor, equivalent of C_1 and $2 \mu\text{F}$ is $(C_1 + 2)$.

Charge is

$$Q' = (C_1 + 2)10 = (C_2).90$$

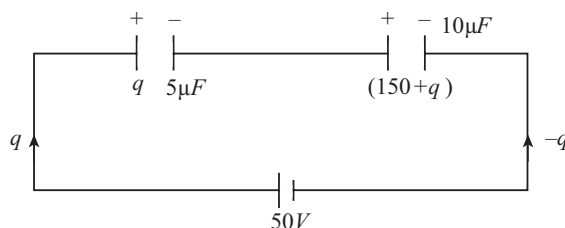
$$\text{or, } C_1 + 2 = 9C_2 \quad (\text{ii})$$

Solving (i) and (ii) gives $C_1 = 0.16 \mu\text{F}$, $C_2 = 0.24 \mu\text{F}$

12. Charge on capacitor A is $Q = (10 \mu\text{F})(15\text{V})$

$$= 150 \mu\text{C}.$$

After the switch is closed, let the cell supply a charge q . Now, the capacitor B will have charge q and A will have charge $(150 + q)$ on it.



Note that the connected plates of the two capacitors form an isolated system and the total charge on them is $150 \mu\text{C}$.

$$\frac{q}{5} + \frac{150 + q}{10} = 50$$

$$\text{or, } q = \frac{350}{3} \mu\text{C}.$$

Potential difference across the $10 \mu\text{F}$ capacitor is

$$V = \frac{150 + q}{10} = \frac{80}{3} \text{ V}$$

13. (a) Maximum electric field inside each capacitor is $E = 3 \times 10^4 \text{ Vcm}^{-1}$.

Maximum potential difference across each capacitor that can be allowed is

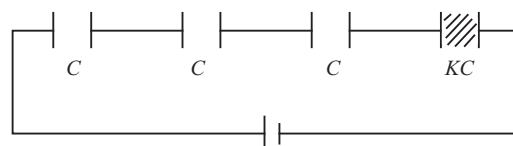
$$V = Ed = 3 \times 10^4 \times 0.7 = 2.1 \times 10^4 \text{ V}.$$

Applied potential difference = $4\text{V} = 8.4 \times 10^4 \text{ V}$

- (b) The maximum potential difference that the glass-filled capacitor can tolerate is

$$= 6.3 \times 10^4 \times 2.7$$

$$= 4.51 \times 10^4 \text{ V}.$$



All capacitors will have same charge on them. If potential difference across an air-filled capacitor is V_0 , then the potential difference across the glass filled capacitor will be $\frac{V_0}{K} = \frac{V_0}{7}$.

$$\text{capacitor will be } \frac{V_0}{K} = \frac{V_0}{7}.$$

Obviously, when $V_0 = 2.1 \times 10^4 \text{ V}$, the potential difference across the glass filled capacitor is only $0.3 \times 10^4 \text{ V}$, which is far below the voltage that it can tolerate. It means, if the applied emf is increased, the air capacitors will get punctured first. In the extreme case, potential difference across each air capacitor will be $2.1 \times 10^4 \text{ V}$ and that across the glass capacitor will be $0.3 \times 10^4 \text{ V}$. Therefore, the maximum permissible applied emf is

$$2.1 \times 10^4 + 2.1 \times 10^4 + 2.1 \times 10^4 + 0.3 \times 10^4 = 6.6 \times 10^4 \text{ V}.$$

14. Total charge available for distribution is

$$Q = 1 \times 10 + 2 \times 20 - 3 \times 30$$

$$= -40 \mu\text{C}$$

Total charge on the three plates connected to point A is $-40 \mu\text{C}$ and the total charge on all plates connected to B is $40 \mu\text{C}$.

Final common potential difference = V .

$$C_1 V + C_2 V + C_3 V = 40 \mu\text{C}.$$

$$\text{or, } V = \frac{40}{1+2+3} = \frac{20}{3} V$$

$$\therefore q_1 = C_1 V = 1 \times \frac{20}{3} = \frac{20}{3} \mu\text{C}$$

$$q_2 = C_2 V = 2 \times \frac{20}{3} = \frac{40}{3} \mu\text{C}$$

$$q_3 = C_3 V = 3 \times \frac{20}{3} = 20 \mu\text{C}$$

15. With 'S' open, the circuit has two capacitors in series connected to a cell of emf E .

Equivalent of C_1 and C_2 is

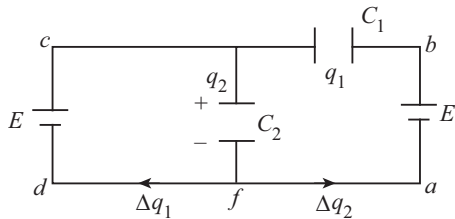
$$\frac{C_1 C_2}{C_1 + C_2}$$

Charge on each capacitor is

$$q = \frac{C_1 C_2 E}{C_1 + C_2}$$

After the switch is closed, the potential difference across C_2 is E .

$$\therefore q_2 = C_2 E.$$



A quick look in the loop abcd tells us that the net emf in the loop is zero. It means that capacitor C_1 has no charge. Another way to look at it is to assume that $V_a = V_d = 0$. Then $V_b = E$ and $V_c = E$. There is no potential difference across C_1 .

After closing 'S', the positive plate of C_1 loses its entire charge q . This charge flows into the cell. The cell must give out same charge through its negative terminal. Therefore, charge flow in wire 2 in the indicated direction is

$$\Delta q_2 = -q = -\left(\frac{C_1 C_2 E}{C_1 + C_2}\right)$$

Charge on the negative plate of C_2 changes by

$$-\left[C_2 E - \frac{C_1 C_2 E}{C_1 + C_2}\right] = -\frac{C_2^2 E}{C_1 + C_2}$$

Total charge flow at junction f must be zero.

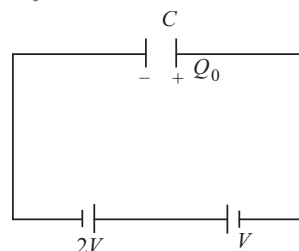
$$\therefore \Delta q_1 + \Delta q_2 - \frac{C_2^2 E}{C_1 + C_2} = 0$$

$$\text{or, } \Delta q_1 - \frac{C_1 C_2 E}{C_1 + C_2} - \frac{C_2^2 E}{C_1 + C_2} = 0$$

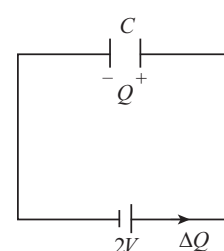
$$\text{or, } \Delta q_1 = C_2 E.$$

16. When the switch is in position 1, the effective potential difference across C is V . Charge on the capacitor is

$$Q_0 = CV.$$



Switch in position 1



Switch in position 2

After the switch is shifted to position 2, the potential difference across the capacitor is $2V$. Charge on the capacitor will be

$$Q = 2CV.$$

After the switch is shifted to position 2, a charge $\Delta Q = 2CV - CV = CV$ flows through the $2V$ cell as shown.

$$\therefore W_{\text{cell}} = \Delta Q(2V) = 2CV^2$$

Energy stored in the capacitor with switch in position 1 is

$$U_i = \frac{1}{2} CV^2$$

Energy stored with the switch in position 2 is

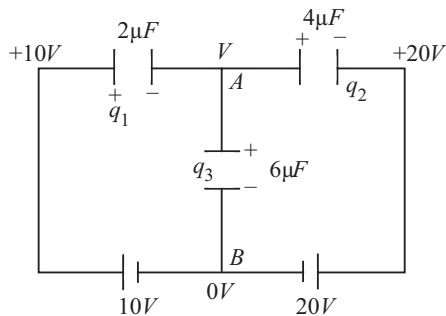
$$U_f = \frac{1}{2} C(2V)^2 = 2CV^2$$

$$\therefore \text{Heat produced} = W_{\text{cell}} + U_i - U_f$$

$$= 2CV^2 + \frac{1}{2} CV^2 - 2CV^2 = \frac{1}{2} CV^2$$

17. We will solve this question by the method illustrated in Example 30.

[We can also solve it by using loop equations].



Assume $V_B = 0$ and $V_A = V$.

Sum of charges on the three capacitor plates connected at $A = 0$

$$\text{or, } 2(V - 10) + 4(V - 20) + 6(V - 0) = 0$$

$$\text{or, } V = \frac{25}{3}V.$$

$$q_1 = 2(V - 10) = 2\left(\frac{25}{3} - 10\right) = -\frac{10}{3}\mu C$$

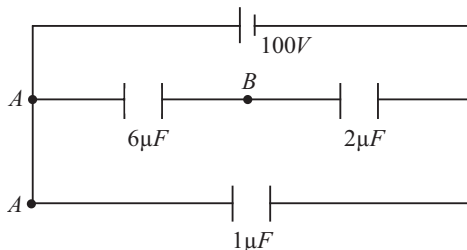
Negative sign indicates that charge on the plates of $2\mu F$ capacitors are of polarity opposite to that shown in the figure (i.e., the left plate is positive).

$$q_2 = 4(V - 20) = 4\left(\frac{25}{3} - 20\right) = -\frac{140}{3}\mu C.$$

Once again the polarity of plates is opposite

$$q_3 = 6(V) = \frac{150}{3}\mu C = 50\mu C.$$

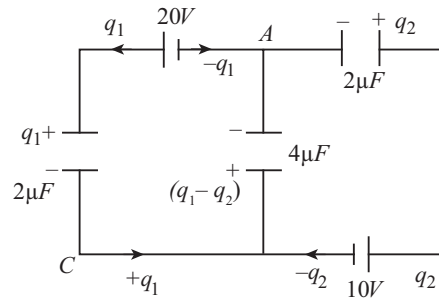
18. The two capacitors of $3\mu F$ are in parallel. Two $1\mu F$ capacitors are also in parallel. The circuit can be redrawn as shown.



$V_A - V_B =$ Potential difference across $6\mu F$ capacitor

$$= \left(\frac{2}{6+2}\right) \times 100 = 25V.$$

19. Let charge supplied by the $20V$ cell be q_1 and that by $10V$ cell be q_2 .



If x is charge on $4\mu F$ capacitor, then

$$x = (q_1 - q_2)$$

Writing loop equations in the two loops gives

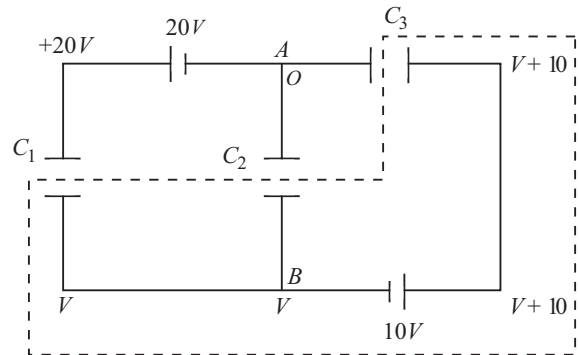
$$20 - \frac{q_1}{2} - \frac{q_1 - q_2}{4} = 0 \quad (i)$$

$$\text{and } \frac{q_1 - q_2}{4} + 10 - \frac{q_2}{2} = 0 \quad (ii)$$

Solving (i) and (ii) gives $q_1 = 35\mu C$.

Alternate Method

We will use the method described in Example 30.



Let potential of point A be 0 and potential of point B be V . Potentials of other points can be written as shown.

The system of three plates and $10V$ cell make an isolated system. Net charge on this system must be zero.

$$\therefore 2(V - 20) + 4(V - 0) + 2(V + 10 - 0) = 0$$

$$\text{or, } V = 2.5 \text{ V}$$

$$\therefore \text{Charge on } C_1 \text{ is } 2(2.5 - 20) = -35\mu C$$

Negative sign indicates that the lower plate will be negative and the upper plate will be positive.

20. When a length x of the plates are dipped in the oil, capacitance is

$$C = \frac{\epsilon_0 L(L-x)}{d} + \frac{K\epsilon_0 xL}{d}$$

$$= \frac{\epsilon_0 L^2}{d} + \frac{\epsilon_0 L(K-1)x}{d}$$

$$\text{or, } \frac{dC}{dx} = \frac{\epsilon_0 L(K-1)}{d}$$

Charge on the capacitor is $Q = CV$.

As the capacitor is lowered in the dielectric oil, its capacitance increases and hence the charge on the capacitor increases.

$$\frac{dQ}{dt} = V \frac{dC}{dt} = V \cdot \frac{dC}{dx} \cdot \frac{dx}{dt}$$

$$= V \cdot \frac{\epsilon_0 L(K-1)}{d} v$$

$$= 500 \times \frac{8.85 \times 10^{-12} \times 1(11-1)}{0.02} \cdot (0.01)$$

$$= 2.2 \times 10^{-8} \text{ Cs}^{-1}$$

21. The potential difference between A and B will be given by

$$\int_{V_A}^{V_B} dV = - \int_0^d E dx$$

$$\text{or, } V_B - V_A = - \frac{Q}{n} \int_0^d (a + bx^2) dx$$

$$\text{or, } V_A - V_B = - \frac{Q}{n} \left(ad + \frac{bd^3}{3} \right)$$

$$\therefore C = \frac{Q}{V_A - V_B} = \frac{3n}{(3ad + bd^3)}$$

22. Charge on the capacitor, $Q = KC_0 V$.

Initial energy stored in the capacitor, $U_i = \frac{1}{2} KC_0 V^2$.

Find capacitance = $\frac{C_0}{2}$

Final energy stored in the capacitor

$$U_f = \frac{Q^2}{2 \left(\frac{C_0}{2} \right)}$$

$$\text{or, } U_f = \frac{(KC_0 V)^2}{C_0} = K^2 C_0 V^2$$

$$W_{\text{ext}} = U_f - U_i = C_0 V^2 K \left(K - \frac{1}{2} \right)$$

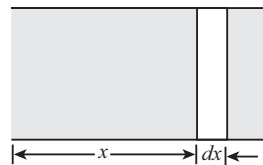
23. Consider a strip of width dx as shown. This strip forms a capacitor of plate area $L \cdot dx$. The capacitance of this small segment is

$$dC = \frac{\epsilon_0 (k_0 + ax) L dx}{d}$$

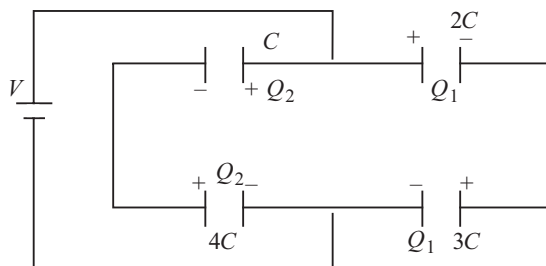
The given capacitor is made up of infinite number of such small capacitors. All these small capacitors are in parallel. Therefore, the effective capacitance is obtained by summing all these capacitances.

$$C = \int dC = \frac{\epsilon_0 L}{d} \int_0^L (K_0 + ax) dx$$

$$= \frac{\epsilon_0 L^2}{d} \left(K_0 + \frac{aL}{2} \right)$$



24. When S is open, capacitance $2C$ and $3C$ are in series. The other two capacitors are also in series.

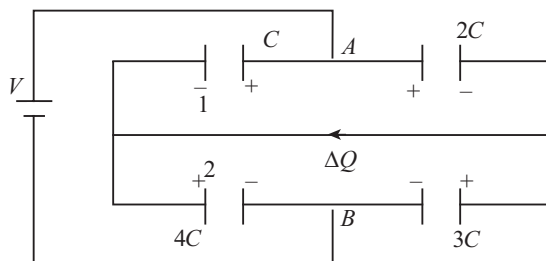


Equivalent of $2C$ and $3C$ is

$$\frac{(2C)(3C)}{2C + 3C} = \frac{6C}{5}$$

Charge on both the capacitors will be $Q_1 = \frac{6CV}{5}$.

Similarly, charge on the other two capacitors will be $Q_2 = \frac{4CV}{5}$.



After the switch S is closed, C and $2C$ will be in parallel and $3C$ and $4C$ will also be in parallel.

Equivalent of C and $2C$ is $3C$.

Equivalent of $3C$ and $4C$ is $7C$.

Series equivalent of $3C$ and $7C$ is

$$\frac{(3C)(7C)}{3C + 7C} = \frac{21C}{10}$$

$\therefore$ Charge on $3C$ and $7C$ each will be $q = \frac{21CV}{10}$.

Charge q is divided between C and $2C$ in parallel.

$\therefore$ Charge on C is $\left(\frac{C}{C+2C}\right)q = \frac{7CV}{10}$

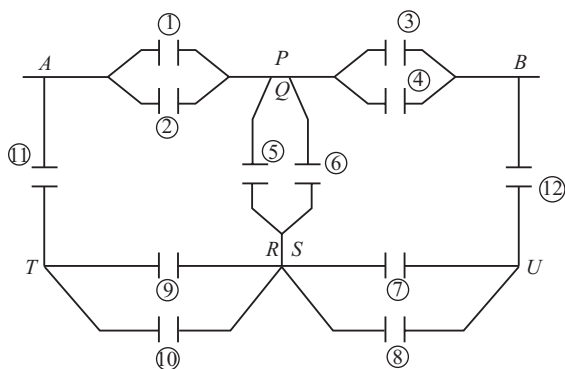
Similarly, charge on $4C$ is $\left(\frac{4C}{4C+3C}\right)q = \frac{6CV}{5}$

Total charge on plates 1 and 2 initially was zero. After the switch is closed, total charge on 1 and 2 is

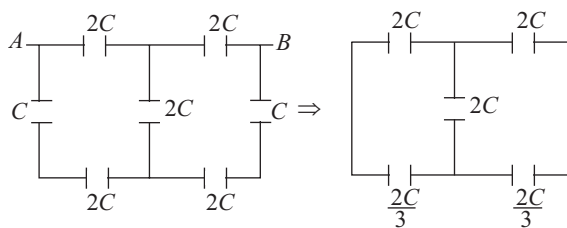
$$\frac{6CV}{5} - \frac{7CV}{10} = \frac{5CV}{10} = \frac{CV}{2}$$

Therefore, charge $\Delta q = \frac{CV}{2}$ must flow through the switch after it is closed.

25. Points P and Q are exactly symmetrical. They are at the same potential. Similarly, points R and S must be at same potential. The circuit can be redrawn as shown below.



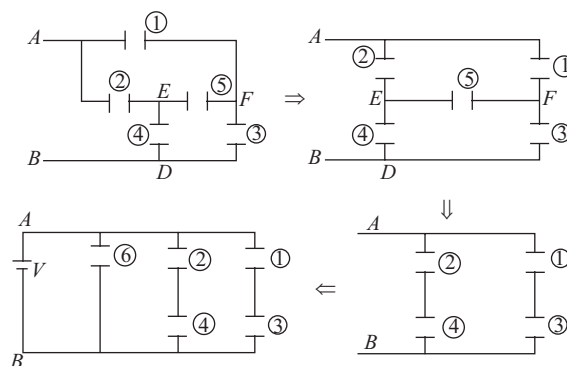
The equivalent can be calculated using series parallel grouping.



The final circuit is a balanced wheatstone bridge.

Equivalent is $C_0 = \frac{4C}{3}$

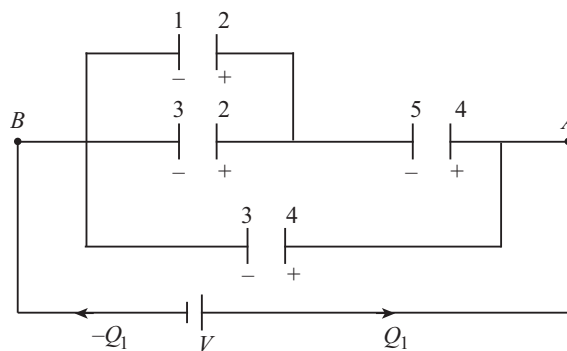
26. Capacitor 6 is directly connected across the cell. The potential difference across it is V . Hence, charge on it is CV . Capacitor 6 is connected in parallel to the remaining network. For a moment, remove capacitor 6 and simplify the circuit (as shown below) to realize that it is a balanced Wheatstone bridge. Capacitor 5 has no charge. It can be removed. In the last circuit, we have shown the complete circuit after bringing back capacitor 6.



Equivalent of capacitor 1 and 3 has an equivalent capacitance $\frac{C}{2}$. Charge on both these capacitors will be $\frac{C}{2}V$.

27. The original arrangement is same as four capacitors connected to each other as shown. Capacitance of each capacitor is

$$C = \frac{\epsilon_0 A}{d}$$

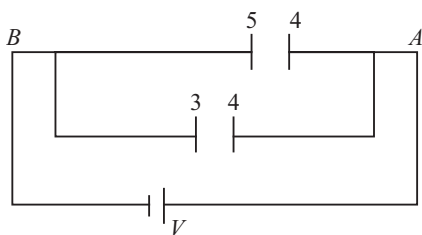


Equivalent capacitance is

$$C_1 = \frac{5C}{3} = \frac{5\epsilon_0 A}{3d}$$

Charge supplied by the cell is

$$Q_1 = C_1 V = \frac{5CV}{3}$$



After the plates 1 and 2 are connected, the capacitors between plates 1, 2 and 3, 2 get short circuited. Now, the circuit has only two capacitors in parallel.

Equivalent capacitance is $C_2 = 2C$.

Charge on each capacitor is CV .

Looking at charge on plate 4 will result in the amount of charge flow through the cell after the plates 1 and 2 are connected.

Initial charge on plate 4 = $Q_1 = \frac{5}{3}CV$.

Final charge on plate 4 = $Q_2 = 2CV$.

$\therefore$ Charge flow, $\Delta Q = 2CV - \frac{5}{3}CV = \frac{CV}{3}$

$$W_{\text{cell}} = \Delta QV = \frac{CV^2}{3}$$

Initial energy stored in the capacitor system is

$$U_i = \frac{1}{2}C_1V^2 = \frac{5}{6}CV^2$$

Final energy stored in the capacitor system is

$$U_f = \frac{1}{2}C_2V^2 = \frac{1}{2}(2C)V^2 = CV^2$$

Heat = $W_{\text{cell}} + U_i - U_f$

$$= \frac{CV^2}{3} - \frac{CV^2}{6} = \frac{CV^2}{6}$$

28. Refer to Example 8. The electric field in the region $R < r < \infty$ in the present question and that in Example 8 is same. Therefore, energy stored is also same.

29. Energy stored inside the sphere is zero, as there is no field inside the metallic sphere. Energy stored up to a radius a is basically stored in the region $R < x < a$. This energy is

$$U_1 = \frac{Q^2}{8\pi\epsilon_0} \int_R^a \frac{dx}{x^2} \quad [\text{Refer to Example 8}]$$

$$= \frac{Q^2}{8\pi\epsilon_0} \left[\frac{1}{R} - \frac{1}{a} \right]$$

Energy stored in the region $a < x < \infty$ is

$$U_2 = \frac{Q^2}{8\pi\epsilon_0} \int_a^\infty \frac{dx}{x^2} = \frac{Q^2}{8\pi\epsilon_0 a}$$

For $U_1 = U_2$, we must have

$$\frac{1}{R} - \frac{1}{a} = \frac{1}{a}$$

or, $a = 2R$.

30. Electric field at an axial distance x is

$$E = \frac{\lambda}{2\pi\epsilon_0 x}$$

Energy density at axial distance x is

$$u = \frac{1}{2}\epsilon_0 E^2 = \frac{\lambda^2}{8\pi^2\epsilon_0 x^2}$$

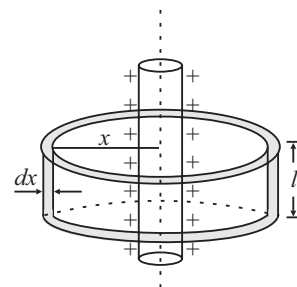
Energy stored in the cylindrical space between radii

x and $x + dx$ is

$$dU = u \cdot (2\pi x l \cdot dx) = \frac{\lambda^2 l}{4\pi\epsilon_0} \frac{dx}{x}$$

$\therefore$ Energy stored in space between $a < x < b$ is

$$U = \frac{\lambda^2 l}{4\pi\epsilon_0} \int_a^b \frac{dx}{x} = \frac{\lambda^2 l}{4\pi\epsilon_0} \ln\left(\frac{b}{a}\right)$$



31. After the two plates are moved apart by a large distance, the charge on each plate will get distributed equally on both surfaces.

$$\sigma = \frac{Q}{2A} = \frac{10 \times 10^{-6}}{2 \times 40 \times 10^{-4}} = \frac{1}{800} \mu\text{C}/\text{m}^2$$

Electric field in front of the plate is

$$E = \frac{\sigma}{\epsilon_0} = \frac{1}{800 \times 8.85 \times 10^{-12}}$$

$$= 1.4 \times 10^8 \text{ V/m.}$$

$\therefore$ Energy density

$$u = \frac{1}{2}\epsilon_0 E^2 = \frac{1}{2} \times 8.85 \times 10^{-12} \times (1.4 \times 10^8)^2$$

$$= 8.7 \times 10^4 \text{ Jm}^{-3}$$

Energy stored in a volume of 1 cm^3 is

$$= 8.7 \times 10^4 \times (1 \times 10^{-6}) = 8.7 \times 10^{-2} \text{ J}$$

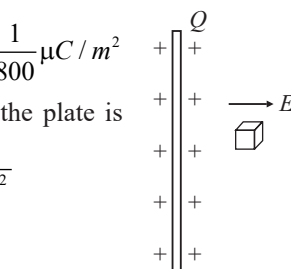
32. When brought closer, the outer faces will have equal charge

$$\frac{Q+2Q}{2} = \frac{3Q}{2}$$

The two inner faces will

$$\text{have } \pm \left(\frac{2Q-Q}{2} \right) = \pm \frac{Q}{2}$$

charge on them. [For this, please refer to Chapter 1.]



The electric field between the plates is only because of $\frac{Q}{2}$ and $\frac{-Q}{2}$ charges (effect of two $\frac{3Q}{2}$ charges cancel out). Therefore, we can say that there is a parallel plate capacitor formed between the inner faces of the two plates.

$$V = \frac{Q/2}{C} = \frac{Q}{2C}$$

- 33.** If a charge Q is given to the inner cylinder of radius a and a charge $-Q$ is given to the outer cylinder

of radius c , then charges will be induced on the inner and outer surfaces of the middle cylinder. One can prove (using the Gauss's law) that the electric field in the space between the inner and the outer cylinder will not get affected due to presence of the middle cylinder. Its presence or absence will make no difference to the capacitance between the inner and outer cylinder.

$$C = \frac{2\pi\epsilon_0 L}{\ln\left(\frac{c}{a}\right)} \quad [\text{equation (17)}]$$

Chapter 5 Miscellaneous Problems on Chapter 4



Match the Column

1. (A) Charge flow after the Capacitor is connected to the second cell is

$$\Delta Q = 2CV$$

$$W_{\text{cell}} = \Delta QV = 2CV^2$$

Change in stored energy in the capacitor is 0

$$\therefore \text{Heat} = 2CV^2$$

$$(B) \quad U_i = \frac{1}{2}CV^2$$

$$U_f = \frac{1}{2}\left(\frac{C}{2}\right)V^2$$

$$\therefore \Delta U_{\text{loss}} = \frac{1}{2}CV^2 - \frac{1}{4}CV^2 = \frac{1}{4}CV^2$$

$$(C) \quad U_i = \frac{1}{2}CV^2$$

$$U_f = \frac{1}{2}(KC)V^2 = CV^2$$

$$\therefore \Delta U = CV^2 - \frac{1}{2}CV^2 = \frac{1}{2}CV^2$$

- (D) Entire stored energy $\left(= \frac{1}{2}CV^2\right)$ is lost.

2. (A) If charge on C_1 is q , then charge on C_2 is $\frac{C_2 q}{C_3 + C_2}$

$\therefore$ Ratio of energy stored in C_1 and C_2 is

$$\frac{U_1}{U_2} = \frac{q^2}{2C_1} \times \frac{2C_2}{\left(\frac{C_2 q}{C_3 + C_2}\right)^2} = \frac{(C_3 + C_2)^2}{C_1 C_2}$$

If C_1 is increased by reducing the plate separation, the above ratio decreases.

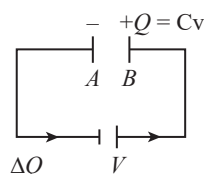
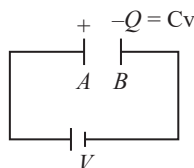
- (B) V_1 = potential difference across C_1

V_2 = potential difference across C_2 and C_3

$$C_1 V_1 = (C_2 + C_3) V_2$$

$$\text{or, } \frac{V_1}{V_2} = \frac{C_2 + C_3}{C_1}$$

$$\therefore \frac{E_1}{E_3} = \frac{V_1 / d_1}{V_2 / d_3} = \frac{d_3}{d_1} \left(\frac{C_2 + C_3}{C_1} \right)$$



If C_2 is increased (by inserting a dielectric), the above ratio increases.

- (C) After short circuiting, energy stored in C_1 is zero.

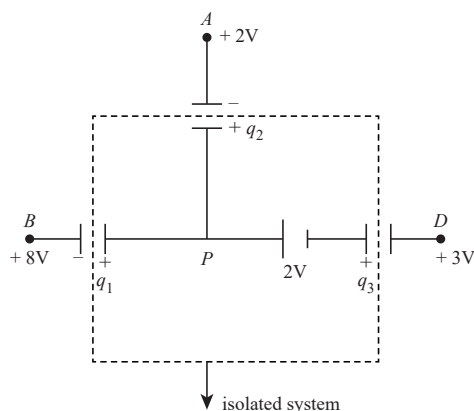
$$q_1 = \text{charge on } C_1 = q$$

$$q_2 = \text{charge on } C_2 = \frac{C_2 q}{C_3 + C_2}$$

$$\therefore \frac{q_1}{q_2} = \frac{C_3 + C_2}{C_2}$$

If C_1 changes, the above ratio will not change.

3. Let's assume charges on the three capacitors as shown.



For the isolated system shown in the figure $q_1 + q_2 + q_3 = 0$ (i)

Also,

$$V_A + \frac{q_2}{2} - \frac{q_1}{2} = V_B$$

$$\text{or, } 2 + \frac{q_2}{2} - \frac{q_1}{2} = 8$$

$$\text{or, } q_2 - q_1 = 12 \quad \text{(ii)}$$

$$\text{Similarly, } V_A + \frac{q_2}{2} - 2 - \frac{q_3}{2} = V_D$$

$$\text{or, } 2 + \frac{q_2}{2} - 2 - \frac{q_3}{2} = 3$$

$$\text{or, } q_2 - q_3 = 6 \quad \text{(iii)}$$

Solving (i), (ii) and (iii) gives

$$q_1 = -6\mu C, q_2 = 6\mu C, q_3 = 0$$

$$\text{Energy in } C_2 \text{ is } = \frac{q_2^2}{2C_2} = \frac{6^2}{2 \times 2} = 9\mu J$$

Potential at point P can be written as

$$V_p - V_D = 2$$

$$\text{or, } V_p = 3 + 2 = 5V$$

4. In experiment X , due to symmetry, both surfaces of B will have equal charge. Each surface of B has $q = 1\mu C$ charge on it. Accordingly, charge on other plates has been shown.

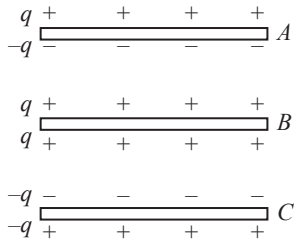
Charge on inner surface of A is $q = -1\mu C$,

$$V_{AB} = \frac{q}{C} = \frac{1\mu C}{100nF} = 10 \text{ volt}$$

Potential difference between B and C is also

$$V_{BC} = \frac{q}{C} = 10 \text{ volt}$$

In experiment, Y charge is given to plate A . The outer surfaces of A and C will have equal charges on them given by $q = \frac{2+0}{2} = 1\mu C$.

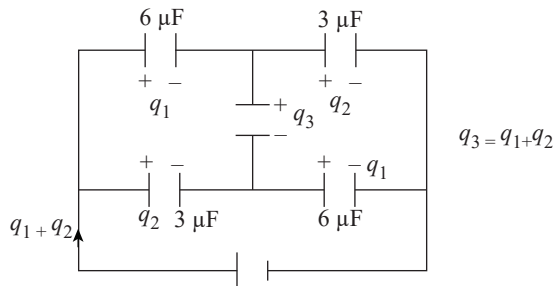


Inner face of A has $q = 1\mu C$ charge. Inner face of C has $-q$ charge. Surfaces of B have $-q$ and $+q$ charge as shown.

$$V_{AB} = \frac{q}{C} = 10 \text{ volt}$$

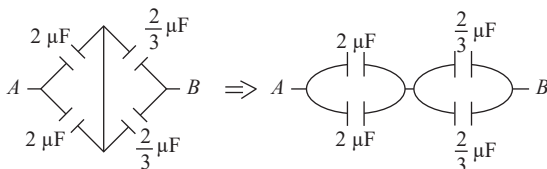
$$V_{BC} = \frac{q}{C} = 10 \text{ volt}$$

5. Taking advantage of symmetry, the charge distribution can be assumed to be as shown.



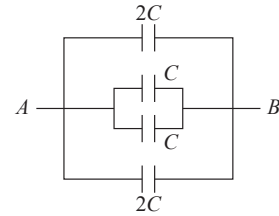
With the above diagram, one can easily match the entries in Column-I with those in Column-II

6. (A) The circuit can be redrawn as shown.



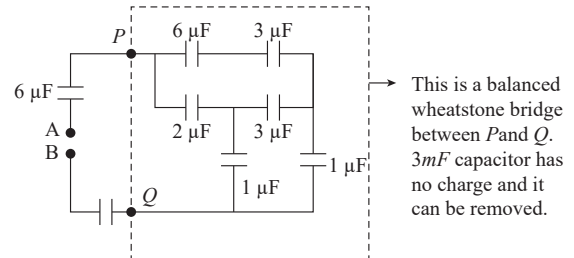
Equivalent is $1\mu F$

- (B) The equivalent circuit is as shown.



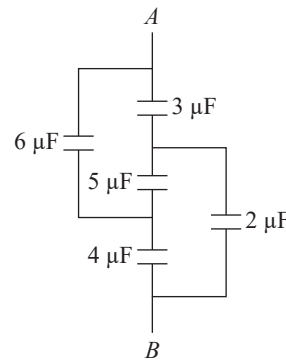
Equivalent is $6C = 6\mu F$

- (C)



The equivalent capacitance can be calculated using series-parallel grouping $C_{AB} = 1\mu F$.

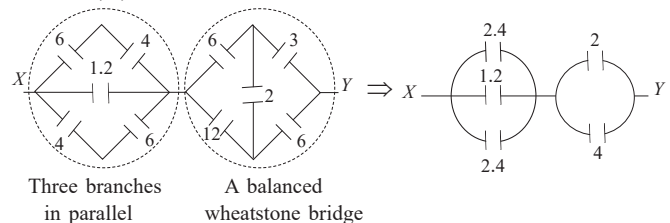
- (D)

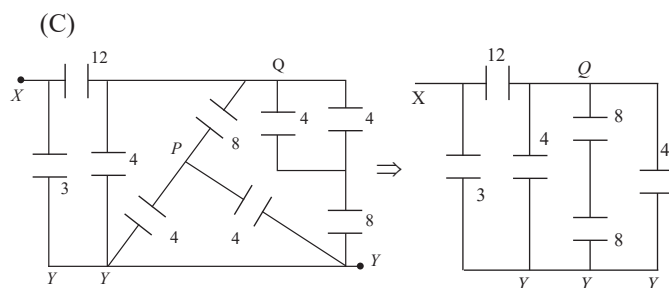
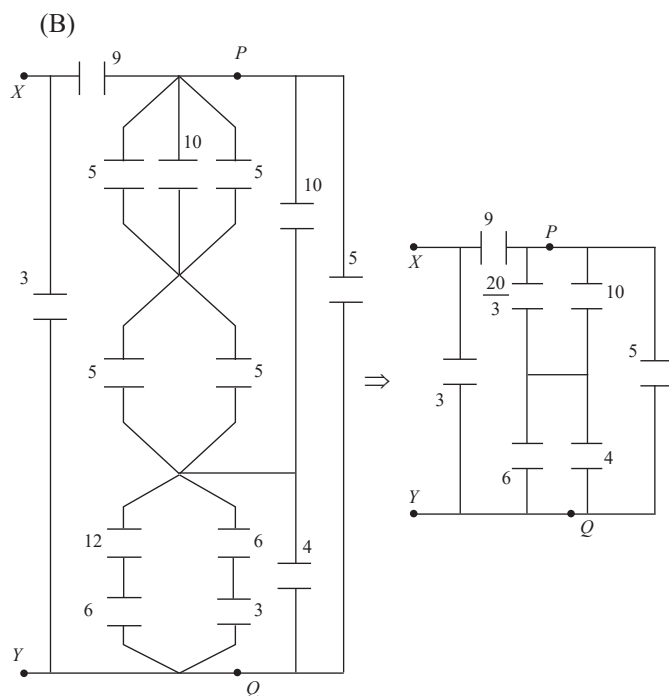


This is a balanced wheatstone bridge with $5\mu F$ capacitor having no charge.

$$C_{AB} = \frac{18}{5} \mu F$$

7. (A)





(D) $6\mu F$, $1\mu F$ and $3\mu F$ are in parallel. Two $2\mu F$ capacitors are in parallel.

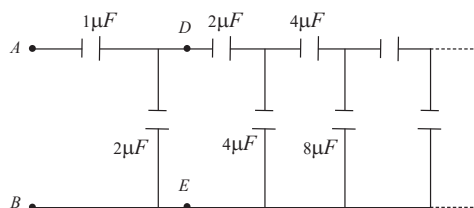


Passage Based Problems

Passage 1

- The second circuit is the exact replica of the first with every capacitance value doubled. Whenever we double every capacitance in a circuit, the effective capacitance will also get doubled. Therefore, $\frac{C_2}{C_1} = 2$.

- If we add a ladder (having two capacitance of $1\mu F$ and $2\mu F$) to the second network, then the circuit is exactly the same as the first one.



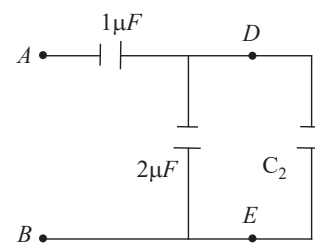
$$\therefore C_1 = \frac{1 \times (C_2 + 2)}{1 + (C_2 + 2)}$$

$$\text{or, } C_1 = \frac{2C_1 + 2}{2C_2 + 3}$$

$$\text{or, } 2C_1^2 + C_1 - 2 = 0$$

$$\text{or, } C_1 = \frac{-1 \pm \sqrt{1+16}}{4}$$

$$\text{or, } C_1 = \frac{\sqrt{17}-1}{4} \mu F$$



Passage 2

- Induced charges on the plates create an electric field (inside the plate) which is exactly equal and opposite to the external field E . The electrostatic interaction energy of the induced charges is same as energy stored in the electric field created due to the charges.

$$\therefore U = (Ad) \frac{1}{2} \epsilon_0 E^2$$

$$= \frac{1}{2} \epsilon_0 E^2 Ad$$

(i)



- When the external field is switched off, the charges on the surfaces of the metal plate disappear. Energy given by (i) is lost as heat.

Passage 3

- The final potential difference between A and B will be $V_A - V_B = V$.

Let charge q flow from B to A after S is closed; charge on A will be $(Q + q)$ and that on B will be $(2Q - q)$.

$$V_A - V_B = V$$

$$\text{or, } \frac{Q+q}{2C} - \frac{2Q-q}{C} = V$$

$$\text{or, } q = Q + \frac{2}{3} CV$$

$$\therefore Q_A = Q + q = 2Q + \frac{2CV}{3}$$

$$Q_B = 2Q - q = 2Q - \frac{2CV}{3}$$

$$6. W_{\text{cell}} = qV = QV + \frac{2}{3} CV^2$$

$$U_i = \frac{Q^2}{2(2C)} + \frac{(2Q)^2}{2C} = \frac{9Q^2}{4C}$$

$$\begin{aligned}
 U_f &= \frac{Q_A^2}{2(2C)} + \frac{Q_B^2}{2C} \\
 &= \frac{\left(2Q + \frac{2CV}{3}\right)^2}{4C} + \frac{\left(Q - \frac{2CV}{3}\right)^2}{2C} \\
 &= \frac{3Q^2}{2C} + \frac{1}{3}CV^2
 \end{aligned}$$

$$\begin{aligned}
 \therefore \text{Heat} &= W_{\text{cell}} + U_i - U_f \\
 &= QV + \frac{2}{3}CV^2 + \frac{9Q^2}{4C} - \frac{3Q^2}{2C} - \frac{1}{3}CV^2 \\
 &= QV + \frac{1}{3}CV^2 + \frac{3Q^2}{4C}
 \end{aligned}$$

Passage 4

Capacitance of A with dielectric is

$$\begin{aligned}
 C &= \frac{\left(\frac{A}{2}\right)\epsilon_0}{d} + \frac{\left(\frac{A}{2}\right)K\epsilon_0}{d} = \frac{A\epsilon_0}{2d}(1+K) \\
 &= \frac{0.04 \times 8.85 \times 10^{-12}}{2 \times 8.85 \times 10^{-4}}(1+9) = 2 \times 10^{-9} F
 \end{aligned}$$

$$\therefore U_1 = \frac{1}{2}CV^2 = \frac{1}{2}(2 \times 10^{-9})(110)^2 = 1.21 \times 10^{-5} \text{ J}$$

Work done in removing the dielectric from A is
 $W = (\text{Energy stored in } A \text{ without dielectric}) - (\text{Energy stored in } A \text{ with dielectric})$

$$= \frac{q^2}{2C_A} - 1.21 \times 10^{-5}$$

$$\text{When } q = CV = 2 \times 10^{-9} \times 110 = 2.2 \times 10^{-7} \text{ C}$$

$$C_A = \frac{\epsilon_0 A}{d} = 0.4 \times 10^{-9} F$$

$$\therefore W = \frac{(2.2 \times 10^{-7})^2}{2 \times 0.4 \times 10^{-9}} - 1.21 \times 10^{-5} = 4.84 \times 10^{-5} \text{ J}$$

Capacitance of B with dielectric is

$$C_B = \frac{K\epsilon_0 A_B}{d} = 1.8 \times 10^{-9} F$$

After A and B are connected, the final common potential difference is

$$V_0 = \frac{q}{C_A + C_B} = \frac{2.2 \times 10^{-7}}{(0.4 + 1.8) \times 10^{-9}} = 100 \text{ V}$$

$$\therefore \text{Energy stored, } U_2 = \frac{1}{2}(C_A + C_B)V_0^2 = 1.1 \times 10^{-5} \text{ J}$$

$$7. \frac{W}{U_1} = \frac{4.84}{1.21} = 4$$

$$8. U_1 - U_2 = 0.11 \times 10^{-5} \text{ J}$$

Passage 5

9. After S_1 is closed, charge on A is

$$q = C_1 V = 2 \times 180 = 360 \mu\text{C}.$$

Now, charged A is connected in parallel to the series combination of B and C (having capacitance $\frac{3 \times 6}{3+6} = 2 \mu\text{F}$).

The equivalent of B and C as well as A , both have equal capacitance of $2 \mu\text{F}$. Charge gets shared equally on them. Therefore, after closing S_2 , charge on all three capacitors will be $\frac{q}{2} = 180 \mu\text{C}$.

Now, S_2 is opened, slab in A is removed and a slab in B is inserted. New capacitance of $A = 1 \mu\text{F}$ and the new capacitance of $B = 6 \mu\text{F}$. However, charge on each capacitor is still $180 \mu\text{C}$. The situation is shown in the first figure. Total energy in the system is

$$\begin{aligned}
 U_1 &= \frac{(180)^2}{2 \times 1} + \frac{(180)^2}{2 \times 6} + \frac{(180)^2}{2 \times 6} \\
 &= 21,600 \mu\text{J}.
 \end{aligned}$$

Now, finally after S_2 is closed, let a charge q_0 flow in the circuit as shown in the second figure. The final charges on the capacitors will be as shown. Using Kirchhoff's loop law, we can write

$$\frac{180 - q_0}{1} - \frac{180 + q_0}{6} - \frac{180 + q_0}{6} = 0$$

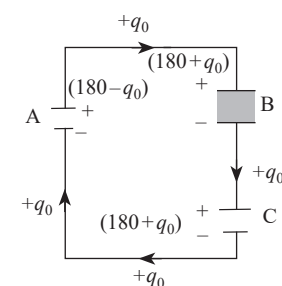
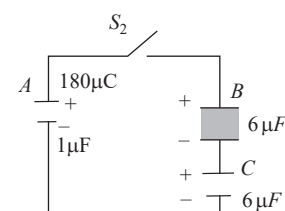
$$\text{or, } q_0 = 90 \mu\text{C}.$$

Final charge on A , B and C will be

$$q_A = 180 - q_0 = 90 \mu\text{C}$$

$$q_B = 180 + q_0 = 270 \mu\text{C}$$

$$q_C = 180 + q_0 = 270 \mu\text{C}$$



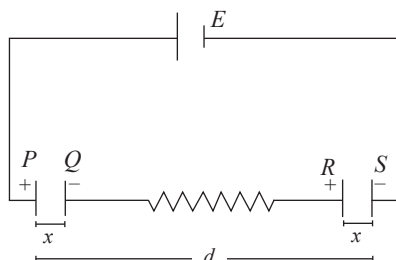
10. Heat = $U_i - U_f$

$$= 21,600 - \left[\frac{1}{2} \times \frac{90^2}{1} + \frac{1}{2} \times \frac{270^2}{6} + \frac{1}{2} \times \frac{270^2}{6} \right]$$

$$= 21,600 - 16,200 = 5400 \mu\text{J} = 5.4 \text{mJ}$$

Passage 6

11. Due to attraction between the capacitor plates, the spring stretches. Let the stretch in the spring be x . According to the question, the separation between the plates of each capacitor is also x .



$$d = x + (l + x) + x$$

$$\text{or, } x = \frac{d - l}{3}$$

$$\text{Capacitance of each capacitor is } C = \frac{\epsilon_0 A}{x} = \frac{3\epsilon_0 A}{d - l}$$

$$\therefore \text{ Charge on each capacitor is } q = \frac{C}{2} E = \frac{3\epsilon_0 A E}{2(d - l)}$$

12. Force between the capacitor plates is $F = \frac{q^2}{2A\epsilon_0}$

$$\text{or, } F = \frac{9\epsilon_0^2 A^2 E^2}{8A\epsilon_0 (d - l)^2} = \frac{9\epsilon_0 A E^2}{8(d - l)^2}$$

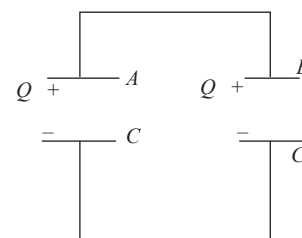
$$\text{In equilibrium } kx = F$$

$$\text{or, } k \left(\frac{d - l}{3} \right) = \frac{9\epsilon_0 A E^2}{8(d - l)^2}$$

$$\text{or, } k = \frac{27\epsilon_0 A E^2}{8(d - l)^3}$$

Passage 7

13. There are two capacitors in parallel. Initially, they have same charge Q on them.



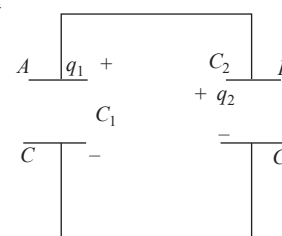
With time, the capacitance C_1 increases and C_2 decreases. For same potential difference across the two capacitors, charge on C_1 will increase and that on C_2 will decrease.

14. $C_1 = \frac{\epsilon_0 A}{d - ut}$; $C_2 = \frac{\epsilon_0 A}{d + ut}$

Total charge available for sharing between the two capacitors is $2Q$.

Potential difference across both the capacitors

$$\text{is same if } \frac{q_1}{q_2} = \frac{C_1}{C_2}$$



$$\Rightarrow q_1 = \frac{2QC_1}{C_1 + C_2} = \frac{\frac{2Q\epsilon_0 A}{d - ut}}{\frac{\epsilon_0 A}{d - ut} + \frac{\epsilon_0 A}{d + ut}}$$

$$\Rightarrow q_1 = \frac{Q(d + ut)}{d}$$

$$\therefore \frac{dq_1}{dt} = \frac{Qu}{d}$$

Chapter 6 Electric Current and Resistance



Your Turn

1. $\Delta q = i \cdot \Delta t = 2 \times 10 = 20 \text{ C.}$

$ne = \Delta q$

or, $n = \frac{\Delta q}{e} = \frac{20}{1.6 \times 10^{-19}} = 1.25 \times 10^{20}$

2. Time period of revolution, $T = \frac{2\pi r}{v}$

Frequency of revolution, $f = \frac{1}{T} = \frac{v}{2\pi r}$

$\therefore f = \frac{2.2 \times 10^6}{2 \times 3.14 \times 5.3 \times 10^{-11}} = 6.6 \times 10^{15} \text{ Hz.}$

There is only one electron, but it passes any point in the orbit 6.6×10^{15} times in 1 second. The current in total charge flow in 1 second.

$\therefore I = e \cdot f = 1.6 \times 10^{-19} \times 6.6 \times 10^{15} = 1.06 \times 10^{-3} \text{ A}$
 $= 1.06 \text{ mA.}$

3. $\Delta q = \int_0^2 i dt = \text{area under the } i-t \text{ graph.}$

$= \frac{1}{2} \times 2 \times 8 = 8 \text{ C.}$

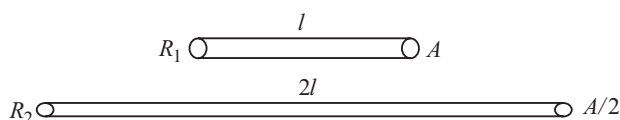
4. $\Delta q = \int_0^4 i dt = \int_0^4 (2+t) dt = \left[2t + \frac{t^2}{2} \right]_0^4 = 8 + 8 = 16 \text{ C}$

5. $v_d \propto E$ [equation (7)]
 and $i \propto v_d$ [equation (8)]

6. $R = \rho \frac{l}{A}$

Large l and small A implies higher resistance.

7. The stretched wire must have the same volume as the original wire. For this, its cross sectional area will be half of the original



$R_1 = \rho \frac{l}{A}$

$R_2 = \rho \frac{2l}{A/2} = 4 \frac{\rho l}{A} = 4R$

8. If V is volume of the wire, then

$Al = V$

or, $A = \frac{V}{l}$

Resistance, $R = \rho \frac{l}{A} = \frac{\rho l^2}{V}$

or, $R = kL^2$ [$k = \frac{\rho}{V} = \text{a constant}$]

or, $\frac{\Delta R}{R} \simeq 2 \frac{\Delta l}{l}$

or, $\frac{\Delta R}{R} \times 100 = 2 \times \left(\frac{\Delta l}{l} \times 100 \right)$
 $= 2 \times 0.2 = 0.4\%$

9. $R = \rho \frac{l}{A}$; or, $A = \frac{\rho l}{R}$

or, $A = \frac{1.7 \times 10^{-8} \times 2}{0.2} = 1.7 \times 10^{-7} \text{ m}^2.$

$\therefore \text{Volume of the wire } V = A \cdot l = 1.7 \times 10^{-7} \times 2$
 $= 3.4 \times 10^{-7} \text{ m}^3$

$\therefore \text{Mass} = V \cdot d = 3.4 \times 10^{-7} \times 9 \times 10^3$
 $= 3.1 \times 10^{-3} \text{ kg.}$
 $= 3.1 \text{ g.}$

10. $R_{\text{cu}} = R_{\text{Fe}}$

$R_{\text{ocu}} [1 + \alpha_{\text{cu}} \cdot \theta] = R_{\text{oFe}} [1 + \alpha_{\text{Fe}} \cdot \theta]$

or, $8.2[1 + 4 \times 10^{-3} \times \theta] = 7.8[1 + 5 \times 10^{-3} \times \theta]$

or, $1.05[1 + 4 \times 10^{-3} \theta] = 1 + 5 \times 10^{-3} \theta$

or, $\theta = 62.5^\circ \text{C}$

11. $R = \underset{\substack{\downarrow \\ \text{blue}}}{6} \underset{\substack{\downarrow \\ \text{white}}}{9} \times \underset{\substack{\downarrow \\ \text{black}}}{10^\circ} \pm \underset{\substack{\downarrow \\ \text{silver}}}{10\%} = (69 \pm 10\%) \Omega$

12. $j = \frac{E}{\rho}$

$\Rightarrow \frac{1}{4 \times 10^{-6}} = \frac{E}{1.7 \times 10^{-8}}$

or, $E = 4.25 \times 10^{-3} \text{ Vm}^{-1}.$

13. $R_1 = \frac{\rho l}{2A}$ and $R_2 = \frac{\rho(2l)}{A}$

$$\Rightarrow R_2 = 4R_1$$

$$P = \frac{V^2}{R}$$

For same V , the power for second wire will be $\frac{1}{4}$ of that of the first wire.

14. (a) $P = \frac{V^2}{R}$; or, $R = \frac{V^2}{P}$

$$\text{or, } R = \frac{(230)^2}{100} = 529\Omega$$

(b) $I = \frac{V}{R} = \frac{230}{529} = \frac{10}{23} \text{ A}$

(c) $P = \frac{V^2}{R} = \frac{(200)^2}{529} = 75.6 \text{ W}$

15. $I_{\max} = 5 \text{ A}$

$$\therefore P_{\max} = VI_{\max} = 100 \times 5 = 500 \text{ W.}$$

16. $P = \frac{V^2}{R}$. If length is increased 10%, R increases 10%

$$P' = \frac{V^2}{1.1R} = \frac{P}{1.1} = 0.91P$$

17. At higher values of current, a lot of heat is dissipated in the resistor and its temperature increases. This causes its resistance to increase.

18. 1Ω and 2Ω in parallel makes $\frac{1 \times 2}{1+2} = \frac{2}{3}\Omega$

$\frac{2}{3}\Omega$ in series with 3Ω has an equivalent of

$$\frac{2}{3} + 3 = \frac{11}{3}\Omega$$

$\therefore$ Current supplied by the battery is

$$I = \frac{10V}{\frac{11}{3}\Omega} = \frac{30}{11} \text{ A}$$

This current flows in 3Ω resistor.

$\therefore$ Potential difference across 3Ω resistor is

$$\frac{30}{11} \times 3 = \frac{90}{11} \text{ V}$$

Potential difference across points B and C is

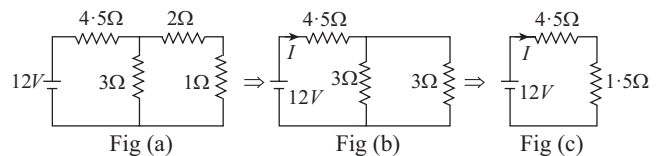
$$V = 10 - \frac{90}{11} = \frac{20}{11} \text{ V}$$

$$\therefore \text{Current in } 1\Omega \text{ resistor is } i = \frac{20/11}{1} = \frac{20}{11} \text{ A}$$

19. 1Ω and 2Ω in series makes 3Ω .

3Ω and 3Ω in parallel makes 1.5Ω .

This 1.5Ω is in series with 4.5Ω . The equivalent resistance is $1.5 + 4.5 = 6\Omega$

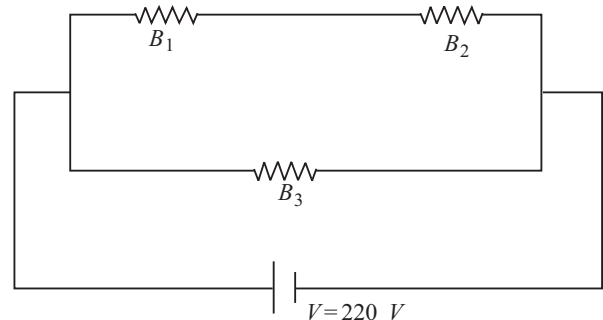


$$\text{Current } I = \frac{12V}{6\Omega} = 2 \text{ A}$$

Look at Fig (b). The current $I = 2 \text{ A}$ will divide equally in two equal parallel resistances, 3Ω each.

$\therefore$ Current in 3Ω resistance is 1 A .

20. B_3 gets its rated voltage supply of 220 V . It will give a power output of 60 W .



Both B_1 and B_2 will have less than 220 V across them. Both give power output less than their rated capacity. Since current in B_1 and B_2 is same, the bulb B_1 will be brighter as it has higher resistance ($R_{60} > R_{100}$).

21. 1Ω and 2Ω resistors are in series. Potential difference across them is proportional to the resistances. If V_1 and V_2 are potential difference across them, then

$$\frac{V_1}{V_2} = \frac{1}{2} \quad \Rightarrow \quad V_1 = \frac{1}{2} \times 12 = 6 \text{ V}$$

Emf of the cell is sum of V_1 and V_2 .

$$\therefore \varepsilon = 18\text{V}.$$

$$\text{Current in } 6\Omega \text{ resistor is } \frac{18}{6} = 3\text{A}.$$

22. The three resistors are in parallel in all the diagrams.

23. Resistance of a wire is proportional to its length. Therefore, the resistance of the smaller arc between

$$a \text{ and } b \text{ is } \frac{20\Omega}{4} = 5\Omega.$$

Resistance of the larger arc = 15Ω .

The resistances 5Ω and 15Ω are in parallel between a and b . The equivalent resistance is

$$R = \frac{5 \times 15}{5 + 15} = \frac{15}{4}\Omega$$

$$\therefore I = \frac{V}{R} = \frac{20}{15/4} = \frac{16}{3}\text{A}$$

24. Equivalent of R_2 and R_3 will be less than R_2 and R_3 . Therefore, the effective resistance connected to the battery decreases. With S open, the net resistance is $R_1 + R_2$. After S is closed the net resistance is $R_1 + R_{23}$, where R_{23} is equivalent of R_2 and R_3 and is less than R_2 .

$\therefore$ Current from the battery increases. This means that potential drop across R_1 increases after S is closed. Obviously, the potential drop across combination of R_2 and R_3 drops.

25. 3Ω , 12Ω and 12Ω are in parallel. Their equivalent is

$$\frac{1}{R} = \frac{1}{3} + \frac{1}{12} + \frac{1}{12} \Rightarrow R = 2\Omega$$

2Ω and 2Ω in series makes 4Ω .

Heat dissipated in all the resistors = Heat dissipated

$$\begin{aligned} \text{in the equivalent resistance} &= \frac{V^2}{R} \cdot t \\ &= \frac{8^2}{4} \times 30 = 480 \text{ J.} \end{aligned}$$

26. Resistance = slope of $V - I$ graph.

$$\therefore R_1 = \frac{8}{2} = 4\Omega$$

$$R_2 = \frac{4}{2} = 2\Omega$$

$$\text{and } R_3 = \frac{2}{2} = 1\Omega$$

R_2 and R_3 are in parallel. Their equivalent is

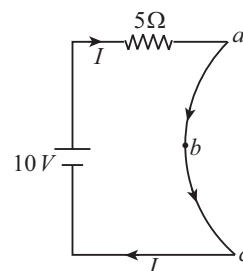
$$\frac{2 \times 1}{2 + 1} = \frac{2}{3}\Omega$$

Now, $\frac{2}{3}\Omega$ is in series with $R_1 = 4\Omega$.

$$\text{Equivalent resistance is } 4 + \frac{2}{3} = \frac{14}{3}\Omega$$

$$\therefore I = \frac{30}{14/3} = \frac{45}{7}\text{A}.$$

27. The network of 2Ω resistors have their terminals at a and c . But these two terminals are short circuited (look at the circular arc shaped wire abc). There will be no current in the network. Current will flow in the path shown in figure.



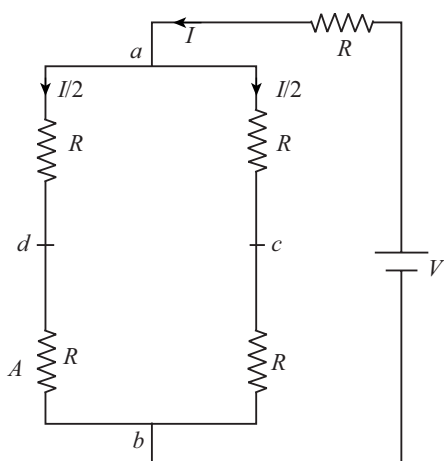
Current will not branch at a or b as there is a zero resistance path along the arc abc .

$$\therefore I = \frac{10}{5} = 2\text{A}$$

There is no current in any 2Ω resistor.

28. For maximum resistance all the resistors must be connected in series and for minimum resistance they all must be connected in parallel.

29. Between points a and b , the network is a balanced Wheatstone Bridge. The resistance between points c and d can be removed and the equivalent resistance can be calculated.



$R_{ab} = (2R) \text{ and } (2R) \text{ in parallel} = R$

$\therefore$ Effective resistance across the cell is $2R$.

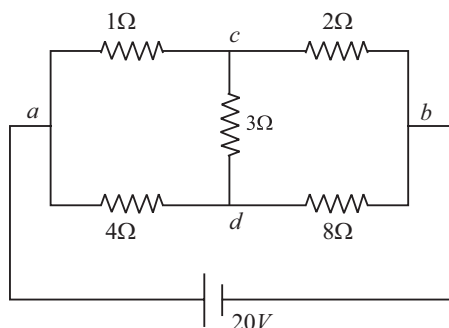
$$\therefore I = \frac{V}{2R}$$

$$\therefore V_{ab} = IR_{ab} = \frac{V}{2}$$

Current I divides equally along two parallel paths at a . The potential difference across resistor A is obviously.

$$\frac{1}{2}V_{ab} = \frac{V}{4}$$

30. The circuit can be redrawn as show. It is a balanced wheatstone bridge. There is no current in the 3Ω resistor.



3Ω resistor can be removed. Equivalent resistance is

$$R = \frac{3 \times 12}{3 + 12} = \frac{12}{5} \Omega$$

$$\therefore I = \frac{V}{R} = \frac{20}{12/5} = \frac{25}{3} \text{ A}$$

31. This is a single loop circuit.

$$E_{\text{net}} = 60 + 10 - 40 + 10 = 40\text{V}$$

$$R_{\text{net}} = 6 + 2 + 1 + 1 = 10\Omega$$

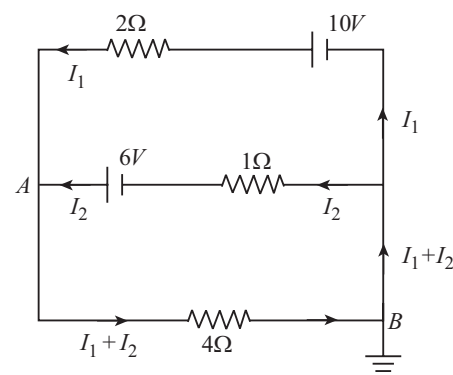
$$\therefore I = \frac{40\text{V}}{10\Omega} = 4\text{A}$$

We move from a to b in anticlockwise direction.

$$V_a - 2I - 40 - 1 \cdot I = V_b$$

$$\text{or, } V_a - V_b = 40 + 3I = 40 + 12 = 52 \text{ V.}$$

32. Let the current in the upper branch be I_1 and that in the middle branch be I_2 . The two currents join at junction A to become $I_1 + I_2$ in the third branch. We write loop equation in the upper loop. Starting from A we move clockwise.



$$2I_1 - 10 - (1)(I_2) + 6 = 0$$

$$\text{or, } 2I_1 - I_2 = 4 \quad (i)$$

Now, we write loop equation in the lower loop we again start from A and move clockwise.

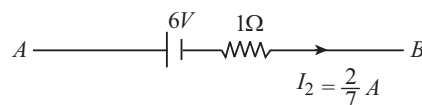
$$-6 + 1 \cdot (I_2) + 4(I_1 + I_2) = 0$$

$$\text{or, } 4I_1 + 5I_2 = 6 \quad (ii)$$

Solving (i) and (ii), we get

$$I_1 = \frac{13}{7} \text{ A and } I_2 = -\frac{2}{7} \text{ A}$$

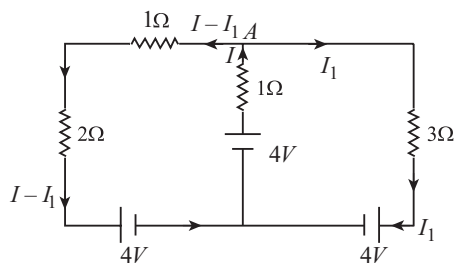
Current in middle branch is in direction opposite to that assumed.



$$V_A - 6 - 1(I_2) = V_B$$

$$\text{or, } V_A = 6 + \frac{2}{7} = \frac{44}{7} \text{ V} \quad [\because V_B = 0]$$

33. Current I branches at A . We have taken one current as I_1 and the other as $I - I_1$.



Writing loop equation in the left loop (starting from A and moving antilockwise)

$$-1(I - I_1) - 2(I - I_1) - 4 + 4 - 1(I_1) = 0$$

$$\text{or, } 3I = 4I_1 \quad (i)$$

Writing loop equation in the right loop (starting from A and moving clockwise)

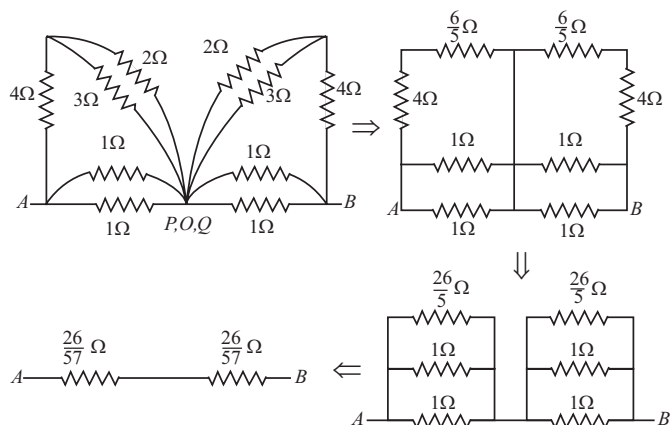
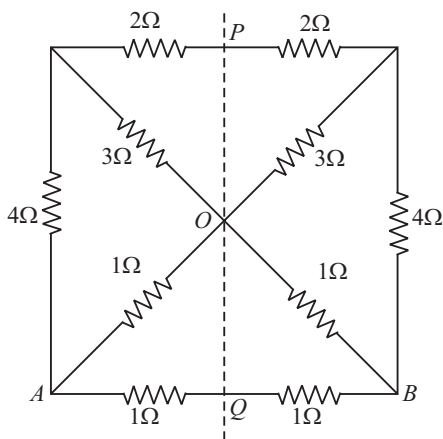
$$-3I_1 - 4 + 4 - 1(I) = 0$$

$$\text{or, } I = -3I_1 \quad (ii)$$

The equations (i) and (ii) can be true only if $I_1 = I = 0$.

Note: The net emf in each of the three possible loops is zero. In such a situation, when there is no emf in any loop of a multiloop circuit, current will be zero everywhere.

34. We have drawn the top 4Ω resistance as series combination of two 2Ω resistances. Similarly, the 2Ω resistor between A and B has been split into two 1Ω resistors. Now, the line PQ divides the entire network into two symmetrical halves. The points P , O and Q are at same potential. The circuit can be redrawn as shown below.



The equivalent resistance is $\frac{52}{57} \Omega$.

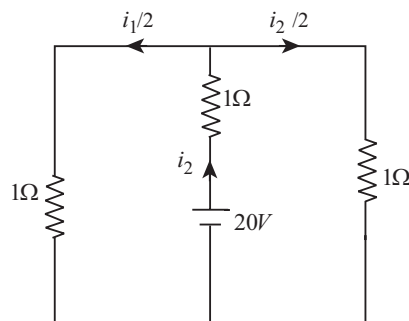
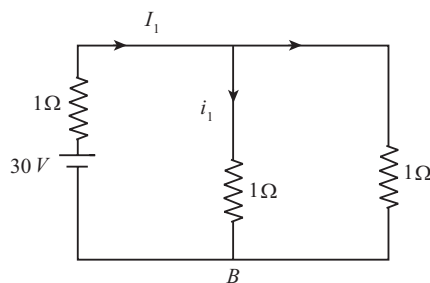
Note: You can also solve the problem by opening the junction at O .

35. With only 30V cell, the current I , is $I_1 = \frac{30}{3/2} = 20A$

$$\therefore i_1 = \frac{I_1}{2} = 10A$$

Similarly with 20V cell

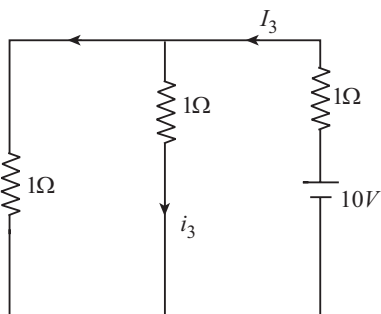
$$i_2 = \frac{20}{3/2} = \frac{40}{3} A$$



With only 10V cell

$$I_3 = \frac{10}{3/2} = \frac{20}{3} A$$

$$\therefore i_3 = \frac{10}{3} \text{ A}$$



In actual circuit the current in branch AB is algebraic sum of i , i_2 and i_3 .

$$\therefore i = 10 - \frac{40}{3} + \frac{10}{3} = 0$$

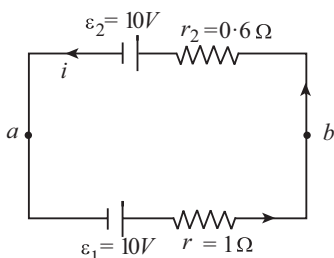
$$36. i = \frac{\varepsilon_{\text{net}}}{R_{\text{net}}} = \frac{10-2}{1+0.6}$$

$$i = 5 \text{ A}$$

$$V_a + \varepsilon_1 - ir_1 = V_b$$

$$\text{or, } V_b - V_a = \varepsilon_1 - ir_1 = 10 - 5 \times 1 = 5 \text{ V.}$$

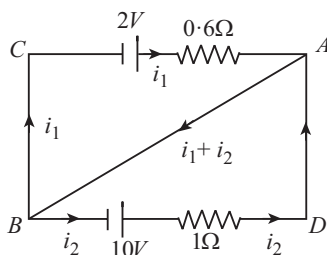
Potential difference is same across the terminals of both the cells.



37. Both the cells are short circuited. Potential difference across them must be zero.

$$V_A - V_B = 0$$

$$V_B + 2 - 0.6 i_1 = V_A$$



$$\text{or, } i_1 = \frac{2}{0.6} = \frac{10}{3} \text{ A}$$

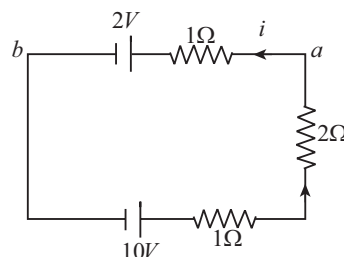
$$\text{Similarly, } V_B + 10 - (1)(i_2) = V_A$$

$$\text{or, } i_2 = 10 \text{ A}$$

From junction law (applied at point A), the current

$$\text{in path } AB \text{ is } i_1 + i_2 = \frac{40}{3} \text{ A}$$

$$38. \text{ Current is anticlockwise, } i = \frac{10-2}{1+2+1} = 2 \text{ A}$$



The 2V cell is getting recharged. Potential difference across it is given by

$$V_a - 1(i) - 2 = V_b$$

$$\text{or, } V_a - V_b = 4 \text{ V}$$

$$40. 2.5 \times 20 + 5 \times 20 + 10 \times t = 300$$

$$\text{or, } t = 15 \text{ h.}$$

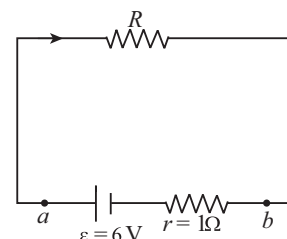
41. For maximum power in R , we must have

$$R = r = 1 \Omega$$

$$(P_R)_{\text{max}} = \frac{\varepsilon^2}{4r} = \frac{6^2}{4 \times 1} = 9 \text{ W}$$

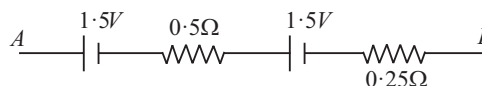
Points a and b are the terminals of the cell.

$$\text{When } R = r, V_{ab} = V_R = 3 \text{ V.}$$



42. The effective emf and resistance for group of cells in parallel is

$$\varepsilon_1 = 1.5 \text{ V and } r_1 = \frac{0.5}{2} = 0.25 \Omega.$$



This cell is in series with the third cell of emf 1.5 V and resistance 0.5 Ω.

$$\therefore \varepsilon_0 = 1.5 + 1.5 = 3.0 \text{ V}$$

$$r_0 = 0.5 + 0.25 = 0.75 \Omega.$$

43. We will group 4 cells in series. This will give emf of 6.0 V and resistance $r = 4 \Omega$.

We make the battery by connecting 10 rows, of 4 cells in series, in parallel.

$$\varepsilon_0 = 6.0 \text{ V}$$

$$r_0 = \frac{4}{10} = 0.4 \text{ } \Omega.$$

44. The polarity of the two cells is opposite. We use equation (34) with a negative sign for ε_2 .

$$\varepsilon_0 = \frac{\frac{\varepsilon_1}{r_1} - \frac{\varepsilon_2}{r_2}}{\frac{1}{r_1} + \frac{1}{r_2}} = \frac{\frac{8}{1} - \frac{4}{1}}{\frac{1}{1} + \frac{1}{1}} = 2 \text{ V}$$

$$\frac{1}{r_0} = \frac{1}{r_1} + \frac{1}{r_2} = \frac{1}{1} + \frac{1}{1}$$

or, $r_0 = \frac{1}{2}$



Worksheet 1

- Random thermal motion of the free electrons continue. When an electric field is applied, their motion gets biased in a direction opposite to $\vec{E}$.
- Free electrons are present everywhere in the wire. As soon as the electric field is set up, the electrons at every location start drifting.
- $i = (e)$ (frequency of revolution)
- Volume remains constant. Hence, the cross sectional area is halved when the length is doubled.

$$R = \rho \frac{l}{A}$$

R becomes 4 times when l doubles and A is halved.

$$5. R_0 = \rho \frac{l}{A}$$

When the length is reduced 10%, new length is $0.9l$.

$$\therefore R = \frac{\rho(0.9l)}{A} = 0.9 R_0$$

$$P_0 = \frac{V^2}{R_0}$$

$$\text{and } P = \frac{V^2}{R} = \frac{V^2}{0.9R_0} = 1.11 P_0$$

$\therefore P$ has increased by 11%.

$$6. \text{ Because } V = \frac{1}{4\pi\varepsilon_0} \frac{q}{r}$$

$$\therefore [V\varepsilon_0] = \left[\frac{q}{r} \right] = \left[\frac{AT}{L} \right]$$

$$\therefore \left[\varepsilon_0 L \frac{\Delta V}{\Delta t} \right] = [\varepsilon_0 \Delta V] \left[\frac{L}{\Delta t} \right] = \left[\frac{AT}{L} \cdot \frac{L}{T} \right] = [A]$$

$$7. j = \frac{I}{A}$$

$$\text{or, } nev_d = \frac{I}{A} \Rightarrow v_d = \frac{I}{neA}$$

$$\therefore V_d = \frac{2}{(10^{29})(1.6 \times 10^{-19})(1 \times 10^{-6})} = \frac{2}{1.6} \times 10^{-4} \frac{\text{m}}{\text{s}}$$

Time required for electrons drifting with speed v_d to travel a distance $l = 4\text{m}$ is

$$t = \frac{l}{v_d} = \frac{4 \times 1.6}{2} \times 10^4 = 3.2 \times 10^4 \text{ s.}$$

$$8. m_1 : m_2 : m_3 = 1 : 3 : 5$$

$$dA_1l_1 : dA_2l_2 : dA_3l_3 = 1 : 3 : 5 \quad [d = \text{density}]$$

$$\text{or, } A_1l_1 : A_2l_2 : A_3l_3 = 1 : 3 : 5$$

$$\text{or, } A_1(5l) : A_2(3l) : A_3(l) = 1 : 3 : 5$$

$$\text{or, } 5A_1 : 3A_2 : A_3 = 1 : 3 : 5$$

$$\text{or, } A_1 : A_2 : A_3 = 1 : 5 : 25$$

$$\therefore R_1 : R_2 : R_3 = \frac{\rho l_1}{A_1} : \frac{\rho l_2}{A_2} : \frac{\rho l_3}{A_3}$$

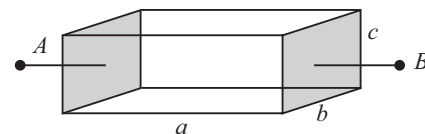
$$= \frac{l_1}{A_1} : \frac{l_2}{A_2} : \frac{l_3}{A_3}$$

$$= \frac{5}{1} : \frac{3}{5} : \frac{1}{25}$$

$$= 125 : 15 : 1$$

9. Let $a > b > c$.

$$R = \rho \frac{l}{A}$$

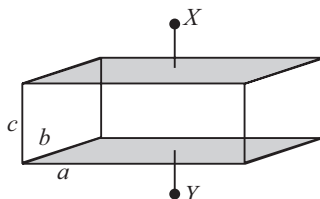


- ∴ Resistance is maximum between A and B as the length is largest and cross section is smallest.

$$R_{\max} = \rho \frac{a}{bc}$$

Resistance is smallest when c is treated as length and $A = ab$ is the cross sectional radius. Resistance is minimum between X and Y

$$R_{\min} = \rho \frac{c}{ab}$$



$$\therefore \frac{R_{\max}}{R_{\min}} = \frac{a^2}{c^2} = 4 : 1 \quad [\because a = 2c]$$

10. Current is same in both the wires. Since the cross section is same, the current density will also be same.

$$\therefore n_1 e v_1 = n_2 e v_2$$

$$\Rightarrow \frac{v_1}{v_2} = \frac{n_2}{n_1} = \frac{4}{1}$$

11. $I = (nev)(\pi r^2)$

$$I' = (ne 2v) \left(\pi \frac{r^2}{4} \right) = \frac{I}{2}$$

$$12. I = \left(nev + ne \frac{v}{4} \right) A = \frac{5}{4} nevA$$

13. For same voltage, current is lower for temperature T_2 . It means the resistance is higher for temperature T_2 .

$$\therefore T_2 > T_1$$

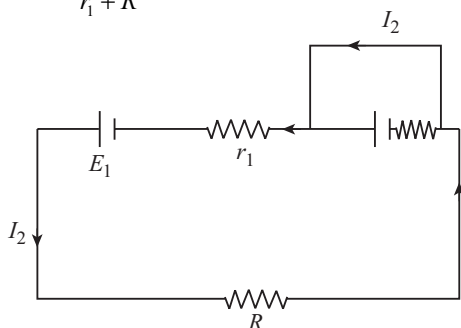
$$14. \varepsilon - Ir = V$$

$$\text{or, } \varepsilon - 0.5 \times 1 = 12.5 \quad \Rightarrow \varepsilon = 13 \text{ V}$$

$$15. I_1 = \frac{E_1 + E_2}{r_1 + r_2 + R}$$

When the battery of emf E_2 is shorted, the current is

$$I_2 = \frac{E_1}{r_1 + R}$$



We want $I_2 > I_1$

$$\text{or, } \frac{E_1}{r_1 + R} > \frac{E_1 + E_2}{r_1 + r_2 + R}$$

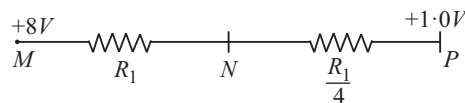
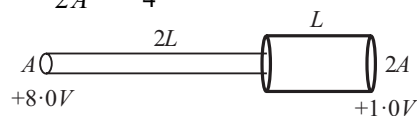
$$\Rightarrow E_1 r_1 + E_1 r_2 + E_1 R > E_1 r_1 + E_2 r_1 + E_1 R + E_2 R$$

$$\Rightarrow E_1 r_2 > E_2 (R + r)$$

$$16. \text{ Resistance of thinner wire is } R_1 = \rho \frac{2L}{A}$$

Resistance of the thicker wire is

$$R_2 = \frac{\rho L}{2A} = \frac{R_1}{4}$$



The two resistances are in series. Potential difference across them are in ratio of their resistances.

$$V_{MN} = \left(\frac{R_1}{R_1 + \frac{R_1}{4}} \right) (7) = \frac{28}{5} \text{ V}$$

$$\text{or, } V_M - V_N = 5.6$$

$$\text{or, } 8 - V_N = 5.6$$

$$\text{or, } V_N = 2.4 \text{ V}$$

$$17. R = \frac{V^2}{P}$$

$$R_F = \frac{(220)^2}{100} \text{ and } R_H = \frac{(220)^2}{1000}$$

$$\therefore R_F > R_H$$

$$18. \text{ Power of the heater } P = \frac{V^2}{R}$$

When length is cut to two-third, the resistance becomes two-third. This implies that power

dissipated becomes $\frac{3}{2}$ times.

Heat needed to boil 1 litre water is same in both the cases.

$$\therefore H = P(15 \text{ min}) = \left(\frac{3}{2} P \right) t$$

$$\Rightarrow t = 10 \text{ min.}$$

$$19. \rho \frac{l}{\pi r^2} = 5 \Omega \quad [r = 9 \text{ mm}]$$

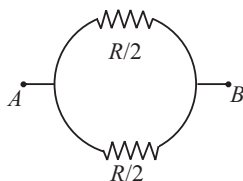
When radius is made 3mm, the new resistance is

$$\rho \frac{l}{\pi \left(\frac{r}{3}\right)^2} = 9 \times 5 = 45 \Omega.$$

6 such wires are in parallel.

$$\therefore R = \frac{45}{6} = 7.5 \Omega$$

20. Resistance of each semi circular part is $\frac{R}{2}$. The two semi circular parts are in parallel.



$$\therefore R_{eq} = \frac{R}{4}$$

21. Three 2Ω resistors are in series. Their equivalent is 6Ω . This 6Ω resistance is in parallel to a 2Ω resistance.

$$\therefore R_{eq} = \frac{6 \times 2}{6 + 2} = \frac{3}{2} \Omega$$

$$\therefore i = \frac{3V}{\frac{3}{2} \Omega} = 2A$$

The current divides in two parallel paths in the inverse ratio of the resistances.

$$\therefore i_1 = \left(\frac{2}{2+6}\right)i = \frac{1}{4} \times 2 = 0.5A$$

22. Resistances in the branch BC and CA are in series. Their equivalent is 60Ω .

This 60Ω is in parallel to 30Ω resistor in path BA .

$$\therefore R_{eq} = \frac{60 \times 30}{60 + 30} = 20\Omega.$$

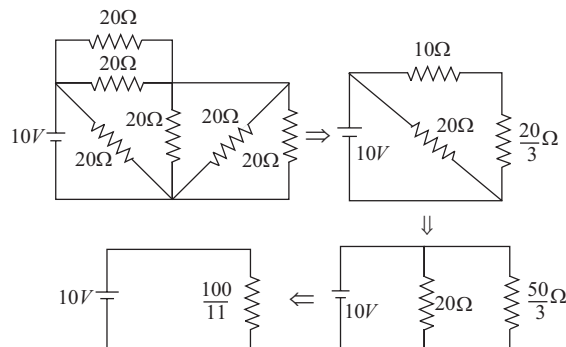
$$\therefore i = \frac{2V}{20\Omega} = 0.1A$$

$$23. R = \frac{V^2}{P}$$

$$R \propto \frac{1}{P}$$

The 2.5 W bulb has much higher resistance than the 100 W bulb. When they are connected in series the potential drop across 2.5 W bulb will be very high (much higher than the rated voltage of 110 V). It means that 2.5 W may fuse.

24.



Power dissipated in the circuit is

$$P = \frac{V^2}{R_{eq}} = \frac{10^2}{\frac{100}{11}} = 11 \text{ W}$$

$$25. H = \frac{V^2}{R}$$

R is resistance of the wire.

When the wire is cut into n equal parts, the resistance of each part will be $\frac{R}{n}$.

The equivalent resistance of n pieces connected in parallel is

$$R_{eq} = \frac{R/n}{n} = \frac{R}{n^2}$$

$$\therefore H^1 = \frac{V^2}{R_{eq}} = \frac{n^2 V^2}{R} = n^2 H$$

$$26. V = \varepsilon - ir$$

Given, $r = ki$ where k is a constant

$$\therefore V = \varepsilon - ki^2$$

The graph of V versus i is a parabola as shown in (D).

$$27. P = \frac{V^2}{R} \text{ where } P = 100 \text{ W and } V = 220 \text{ V.}$$

$$\text{Resistance of each bulb is } R = \frac{V^2}{P}.$$

$$\text{Resistance of three bulbs in parallel is } \frac{R}{3}.$$

$$\text{Total resistance is } R_{eq} = \frac{R}{3} + R = \frac{4R}{3}$$

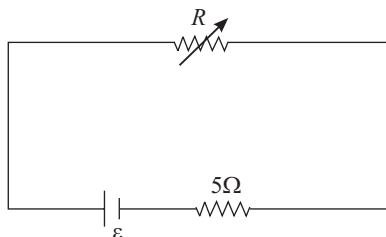
Potential difference across $\frac{R}{3}$ (i.e., three bulbs in parallel) is

$$V_1 = \left(\frac{R/3}{\frac{4R}{3}} \right) V = \frac{V}{4}$$

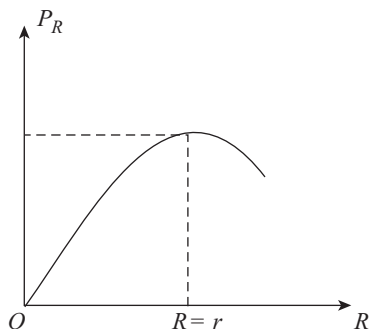
∴ Power dissipated in the last bulb is

$$P_1 = \frac{V_1^2}{R} = \frac{V^2}{16R} = \frac{100}{16} = \frac{25}{4} = 6.25 \text{ W}$$

28. When R is varied, the power consumed in it changes according to the graph shown. The power is maximum when $R = r$.



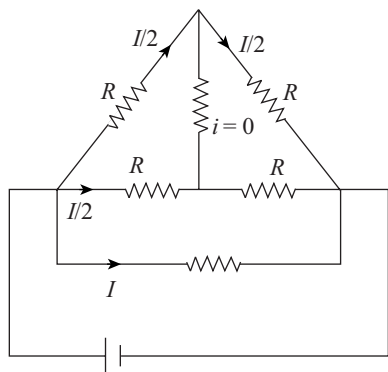
According to the question the value of R is increased from 0 to r only. Therefore the power dissipated in R will go on increasing.



29. There are 4 parallel paths having resistances 4Ω , 6Ω , 12Ω , and 4Ω .

$$\therefore \frac{1}{R_{eq}} = \frac{1}{4} + \frac{1}{6} + \frac{1}{12} + \frac{1}{4} \Rightarrow R_{eq} = \frac{4}{3} \Omega$$

30. The circuit is a balanced wheatstone bridge with another resistance R in parallel to it.



31. With S open, lamp Z is switched off. Potential difference across each of X and Y is $\frac{V}{2}$ (V = source voltage). When S is closed, the effective resistance of Y and Z combination decreases. Therefore, potential drop across X increases and that in Y decreases.

32. Let the equivalent of 100Ω and R in parallel be R_0 . For balanced bridge

$$\frac{100}{R_0} = \frac{200}{40} \Rightarrow R_0 = 20 \Omega$$

$$\text{or, } \frac{100R}{100+R} = 20 \Rightarrow 5R = 100 + R$$

$$\text{or, } R = 25 \Omega.$$

- 33.

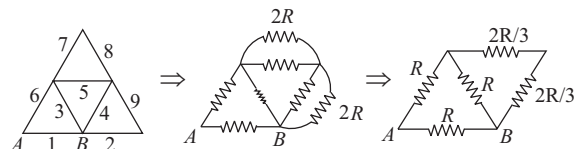
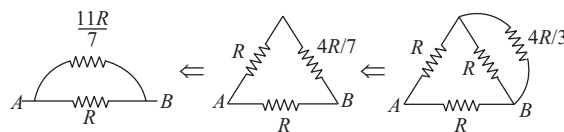


Figure – 7 and 8 are in series.
2 and 9 are in series.



$$R_{eq} = \frac{11R}{18}$$

34. The three resistors are in parallel. Their equivalent is $\frac{R}{3}$. Heat generated in the external circuit will be

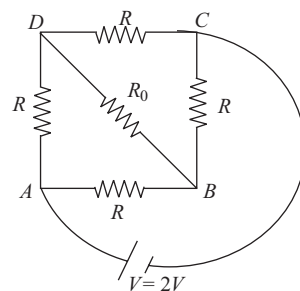
maximum when

$$\frac{R}{3} = 0.1 \Rightarrow R = 0.3 \Omega.$$

35. Resistance of each side = 1Ω . The circuit is a balanced wheatstone bridge. We can remove R_0 .

$$R_{eq} = R = 1 \Omega$$

$$\therefore P = \frac{V^2}{R_{eq}} = 4 \text{ W}$$



36. The circuit is a balanced wheatstone bridge with 7Ω resistance in series. We can remove the 10Ω resistance.

$$R_{eq} = \frac{85}{7} \Omega$$

38. Let n cells be wrongly connected.

When external cells support the battery, there are $(14 - n)$ cells supporting each other and n opposing them.

$\therefore$ Net emf is

$$\varepsilon_1 = (14 - n) \varepsilon - n\varepsilon = (14 - 2n)\varepsilon$$

$$\therefore i_1 = \frac{\varepsilon_1}{R} = \frac{(14 - 2n)\varepsilon}{14r}$$

$$\text{or, } 3 = \frac{(7 - n)\varepsilon}{7r} \quad (\text{i})$$

Similarly, when the external cells are opposing the battery, the net emf is

$$\varepsilon_2 = (12 - n) \varepsilon - (n + 2) \varepsilon = (10 - 2n) \varepsilon$$

$$\therefore i_2 = \frac{\varepsilon_2}{R} = \frac{(10 - 2n)\varepsilon}{14r}$$

$$\text{or, } 2 = \frac{(5 - n)\varepsilon}{7r} \quad (\text{ii})$$

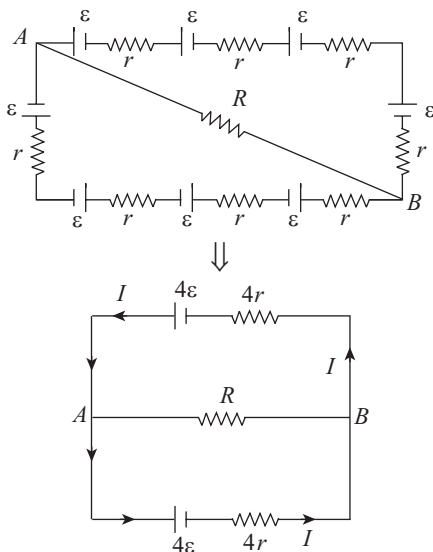
Solving (i) and (ii) given $n = 1$

$$39. I = \frac{n\varepsilon}{nr} = \frac{\varepsilon}{r} = \text{a constant}$$

$$40. I = \frac{\varepsilon}{\frac{r}{n}} = \frac{n\varepsilon}{r}$$

$\therefore I \propto n$

41. The two series grouping of the batteries have effective emf and internal resistance equal to $\varepsilon_0 = 4\varepsilon$ and $r_0 = 4r$.



These two groups are in parallel with their polarities reversed.

$\therefore$ Net emf of the battery across A and B is zero. There is no current in resistance R .

$$\text{Current in the battery loop is } I = \frac{8\varepsilon}{8r} = \frac{\varepsilon}{r}$$

$\therefore$ Potential difference across the terminals of a cell is

$$V = \varepsilon - Ir = 0$$

$$42. R = R_1 + R_2 = \rho_1 \frac{L_1}{A} + \rho_2 \frac{L_2}{A}$$

$$\Delta R = \frac{L_1}{A} \Delta \rho_1 + \frac{L_2}{A} \Delta \rho_2$$

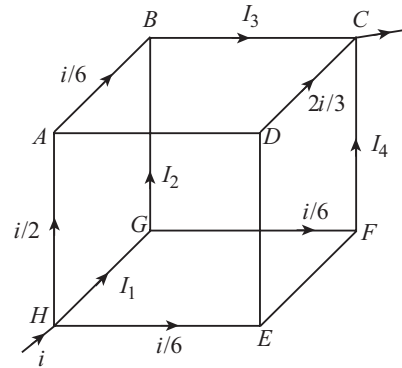
$$= \frac{L_1}{A} \rho_1 \alpha_1 \Delta \theta + \frac{L_2}{A} \rho_2 \alpha_2 \Delta \theta$$

Given $\Delta R = 0$

$$\Rightarrow \rho_1 L_1 \alpha_1 + \rho_2 L_2 \alpha_2 = 0$$

43. Consider junction at H .

$$I_1 + \frac{i}{2} + \frac{i}{6} = i$$



$$\Rightarrow I_1 = \frac{i}{3}$$

Consider junction at G

$$I_2 + \frac{i}{6} = \frac{i}{3}$$

$$\Rightarrow I_2 = \frac{i}{6}$$

At junction B :

$$I_3 = \frac{i}{6} + I_2 = \frac{i}{3}$$

At junction C :

$$I_4 + I_3 + \frac{2i}{3} = i$$

$$\text{or, } I_4 + \frac{i}{3} + \frac{2i}{3} = i \Rightarrow I_4 = 0$$

$$44. I = \frac{\alpha R_1 + \alpha R_2 + \dots + \alpha R_N}{R_1 + R_1 + \dots + R_N} = \alpha$$

∴ Potential difference across terminal of a cell is

$$V = \varepsilon - IR = \alpha R - \alpha R = 0$$

∴ Potential difference between any two points A and B is zero.

45. Let the current distribution be as shown.

Loop equation for the upper loop is

$$-10i - 3 + 6(I - i) = 0$$

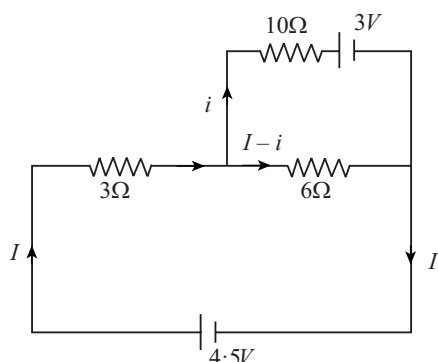
$$\text{or, } 6I - 16i = 3 \quad (i)$$

Loop equation for the lower loop is

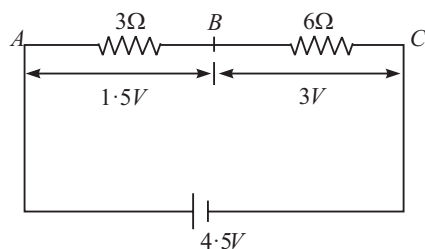
$$-3I - 6(I - i) + 4.5 = 0$$

$$\text{or, } 6i - 9I = -4.5 \quad (ii)$$

Solving (i) and (ii) gives, $i = 0$, $I = 0.5\text{A}$

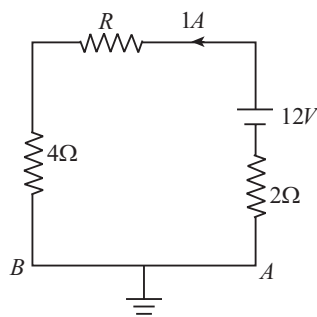


Alternate: The question can be solved by observing that the potential difference across $6\ \Omega$ resistor in the circuit shown is 3V .



If a 3V emf cell is connected across BC , it will cause no change in the circuit.

46. The circuit is not open. A and B are connected through ground.



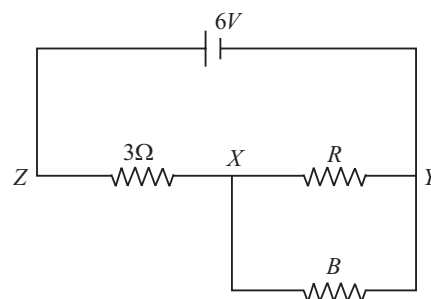
$$V_A - 2 \times 1 + 12 - R \times 1 - 4 \times 1 = V_B$$

$$V_A - V_B = 0$$

$$\therefore R = 6\ \Omega.$$

47. The bulb glows with full intensity. This means that

$$V_{XY} = 1.5\text{V}$$

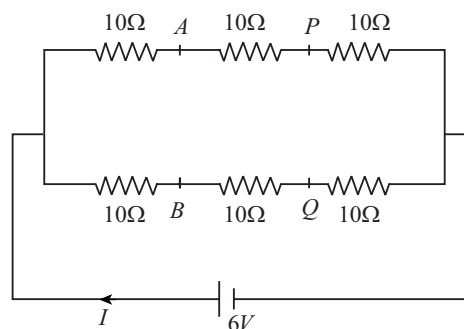


Obviously, $V_{ZX} = 4.5\text{ V}$

$$\frac{V_{XY}}{V_{ZX}} = \frac{R_{XY}}{3}$$

$$\text{or, } R_{XY} = 1\ \Omega$$

48. Consider the circuit shown here. Points A and B are at same potential. Similarly, the points P and Q are at same potential.



When we connect any resistance between A and B , there will be no current through it. Similarly, a resistance added between P and Q will have no current.

Hence, the effective circuit is as shown above.

$$R_{\text{eq}} = \frac{30}{2} = 15\ \Omega$$

$$I = \frac{6\text{V}}{15\ \Omega} = 0.4\text{ A}$$

49. When the temperature of the resistance is steady, the heat generated in it must be equal to the heat lost to the surrounding.

$$i^2 R = \lambda(\theta - \theta_0) \quad (i)$$

$$\text{Strain} = \frac{\Delta l}{l} = \frac{l\alpha(\theta - \theta_0)}{l}$$

$$= \frac{\alpha}{\lambda} i^2 R \quad [\text{ using (i)}]$$

50. $R_A = [(2r \text{ and } r \text{ in parallel}) \text{ in series with } r] \text{ in parallel with } r]$

$$= \frac{5r}{8}$$

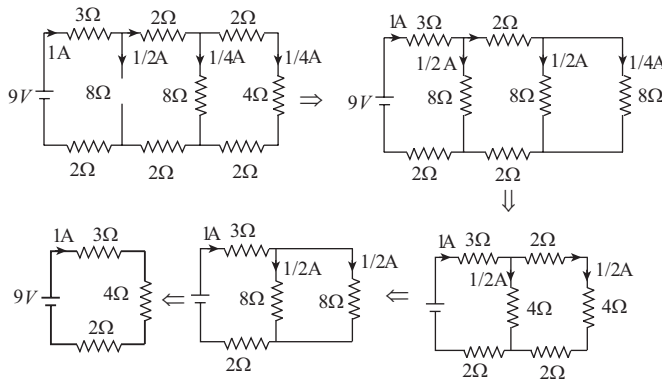
R_B = Three resistors are in parallel between the centre and b and one is in series with this combination.

$$= \frac{r}{3} + r = \frac{4r}{3}$$

In diagram (c), it is a balanced wheatstone bridge between a and b .

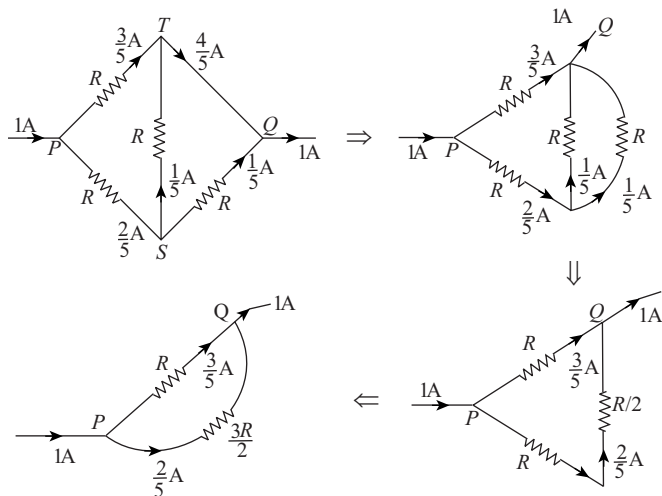
$$R_c = r$$

51.



The above diagrams show how to find equivalent resistance and the battery current. Look at the diagrams in reverse order (last first) to distribute the current in various branches.

52.



T and Q are same points. They have been merged as one single point. The circuit has been reduced and current has been distributed working from the last diagram and moving towards the first one.

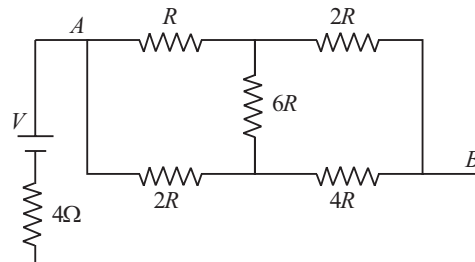
53. Resistance of the combination is $R = \frac{\rho_1 x_1}{A} + \frac{\rho_2 x_2}{A}$

Length of the combination is $x = x_1 + x_2$

$$\therefore \frac{\rho_{eq} x}{A} = \frac{\rho_1 x_1}{A} + \frac{\rho_2 x_2}{A}$$

$$\text{or, } \rho_{eq} = \frac{\rho_1 x_1 + \rho_2 x_2}{x_1 + x_2}$$

54. The circuit is as shown. There is a balanced wheatstone bridge between A and B . Resistance $6R$ can be removed. The equivalent resistance between A and B is



$$R_{AB} = 2R$$

For maximum power to be delivered, $R_{AB} = 4\Omega$
 $\Rightarrow R = 2\Omega$

55. ϵ = emf of each cell.

R = resistance of wire

$$\text{Heat dissipated in time } t, H = \frac{(3\epsilon)^2}{R} t = \frac{9\epsilon^2}{R} t$$

If m is mass of wire and S is its specific heat capacity,

$$mS\Delta T = \frac{9\epsilon^2}{R} t \quad (1)$$

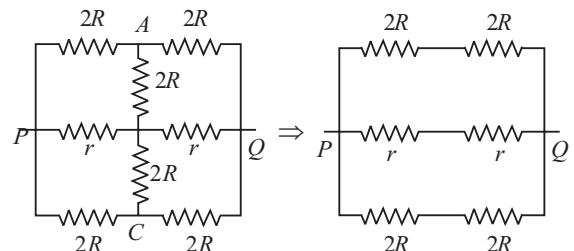
When length of wire is doubled, resistance and mass both double.

$$2mS\Delta T = \frac{(N\epsilon)^2}{2R} t \quad (2)$$

From (1) and (2)

$$N = 6$$

56. Points A , B and C are equipotential. The two resistances between A and B and that between B and C will have no current. They can be removed.



$$\frac{1}{R_0} = \frac{1}{4R} + \frac{1}{2r} + \frac{1}{4R}$$

$$\Rightarrow R_0 = \frac{2Rr}{R+r}$$



Worksheet 2

1. When there is no current, there is no directional bias in the random motion of the electrons. Over a long period of time average displacement of an electron is zero. Its average velocity is zero. Due to large population of electrons, the average of velocities of all the electrons is also zero.

2. $j = \frac{I}{A}$ and $j = nev_d$

$\therefore j$ and v_d both depend on A .

3. In first case $R_1 = \frac{30V}{10mA} = 3 \times 10^3 \Omega$

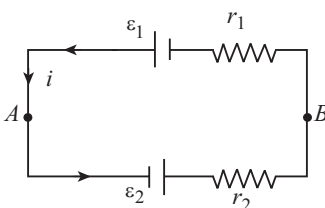
In second case $R_2 = \frac{220V}{50mA} = 4.4 \times 10^3 \Omega$

Increase in resistance happens due to rise in temperature.

4. The cell of higher emf will supply energy to the cell of lower emf. If

$\varepsilon_1 > \varepsilon_2$, the direction of current will be as shown. In this case the cell of emf ε_2 gains energy.

The rate of energy loss by one cell is equal to the rate of heat loss and the rate of energy gain by the other cell.



$$V_A - V_B = \frac{\frac{\varepsilon_1 + \varepsilon_2}{\frac{1}{r_1} + \frac{1}{r_2}}}{\frac{1}{r_1} + \frac{1}{r_2}} = \frac{\varepsilon_1 r_2 + \varepsilon_2 r_1}{r_1 + r_2}$$

5. Mobility of the ions in the electrolyte solution improves with rise in temperature. Thus internal resistance of a chemical battery falls due to rise in temperature.

The resistance of the transmission line is fixed. If I is current in it, the power loss will be $I^2 R$. Higher value of I will lead to more energy loss.

6. Current is same throughout.

$$J = \frac{I}{A} = \sigma E$$

At Q , the current density is high which means that electric field is high.

Consider a small length Δl of the conductor at P and Q . Let cross sectional area at P and Q be A_1 and A_2 , respectively. Resistance of the small length at P

$$\Delta R_1 = \frac{\rho \Delta l}{A_1}$$

$$\Delta P_1 = \text{Power loss in } \Delta R_1 = I^2 \Delta R_1 = \frac{I^2 \rho \Delta l}{A_1}$$

Similarly, power loss in small length Δl at Q is

$$\Delta P_2 = I^2 \rho \frac{\Delta l}{A_2} \Rightarrow \frac{\Delta P_1}{\Delta P_2} = \frac{A_2}{A_1}$$

Power loss per unit volume at P and Q are

$$I^2 \rho \frac{\Delta l}{A_2} \cdot \frac{1}{A_1 \Delta l} = \frac{I^2 \rho}{A_1^2} \text{ and}$$

$$I^2 \rho \frac{\Delta l}{A_2} \cdot \frac{1}{A_2 \Delta l} = \frac{I^2 \rho}{A_2^2}$$

$$\therefore \text{Ratio of power loss per unit volume is } \frac{A_2^2}{A_1^2}$$

7. $I = \frac{dQ}{dt} = a - 2bt$

$\therefore$ Current decreases linearly at a rate $-2b$.

Current becomes zero at $t_0 = \frac{a}{2b}$

Total charge flow in time t_0 is

$$Q = at_0 - bt_0^2 = \frac{a^2}{2b} - b \frac{a^2}{4b^2} = \frac{a^2}{4b^2}$$

Heat generated in R is

$$\begin{aligned} H &= \int_0^{t_0} I^2 R dt = \int_0^{t_0} (a - 2bt)^2 R dt \\ &= \frac{R}{-2b} \left[\frac{(a - 2bt)^3}{3} \right]_0^{t_0} \\ &= -\frac{R}{6b} [(a - 2bt_0)^3 - a^3] = -\frac{R}{6b} [(a - a)^3 - a^3] \\ &= \frac{a^3 R}{6b} \end{aligned}$$

8. The cross sectional radius increases linearly with distance a .

$$r = r_0 + k a \text{ where } k \text{ is a constant.}$$

$$\text{Area of cross section, } A = \pi r^2 = \pi(r_0 + ka)^2$$

$$J = \frac{I}{A} = \sigma E$$

As A increases, E must decrease.

$$E = \frac{I}{\sigma A} \quad (i)$$

Potential V must fall with distance a . consider a length Δl of the conductor at distance a from left end

$$\Delta R = \frac{\Delta l}{\sigma A}$$

$$\text{Power loss } \Delta P = I^2 \Delta R = \frac{I^2 \Delta l}{\sigma A}$$

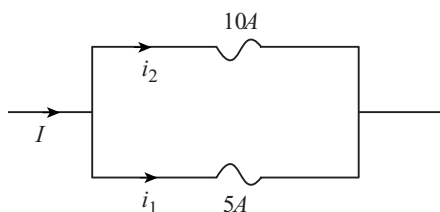
$$\begin{aligned} \text{Power loss per unit length} &= \frac{\Delta P}{\Delta l} = \frac{I^2}{\sigma A} \\ &= \frac{I^2}{\sigma \pi (r_0 + ka)^2} \end{aligned}$$

$\therefore$ (D) is correct representation

$$\text{Also } \frac{\Delta P}{\Delta l} = \frac{I^2}{\sigma A} = IE \quad [\text{from (i)}]$$

9. When connected in series, the 5A fuse will blow out when current exceeds 5A. This will break the circuit.

When connected in parallel, same current will flow through them as they have equal resistance.



When $I = 10A$, both i_1 and i_2 will be 5A. When 5A fuse blows out, the entire current ($I = 10A$) will be routed through the other fuse and it will also blow out. Hence, the combination will act as a fuse of 10A capacity.

10. When R is zero, the effective resistance of the 8Ω and resistance R ($= 0$) is zero. In this case power dissipated in 2Ω resistor is maximum.

$$P_{\max} = \frac{12^2}{2} = 72W.$$

Don't confuse this with maximum power transfer theorem. In that problem we discussed the problem of finding the external resistance when the internal resistance is fixed. Here, in this problem, the external 2Ω resistance is fixed.

11. $I_1 + I_5 = I_6$ (i)
 $I_4 + I_5 = I_2$ (ii)
 $I_4 + I_6 = I_3$ (iii)

Eliminating I_5 between (i) and (ii) gives

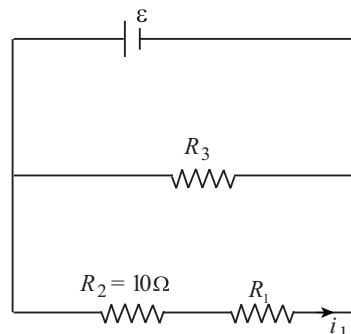
$$I_1 - I_4 = I_6 - I_2 \Rightarrow I_1 + I_2 = I_4 + I_6$$

Eliminating I_4 between (ii) and (iii) gives

$$I_5 - I_6 = I_2 - I_3$$

$$\text{or, } I_5 + I_3 = I_2 + I_6$$

12. R_2 and R_1 are in series connected in parallel to R_3 . The potential difference across R_1 is 10V and current in it is $i_1 = 500mA$.



$$\therefore R_1 = \frac{10V}{500mA} = 20\Omega$$

$$R_1 + R_2 = 30\Omega$$

$$\therefore \varepsilon = (30\Omega)(500mA) = 15V$$

$$\text{For } R_3 : R_3 = \frac{\varepsilon^2}{P} = \frac{15^2}{15} = 15\Omega$$

13. When there is a current in the gap, the arc is ignited. The potential difference across the gap is

$$V = 40 + \frac{20}{I}. \text{ The loop law gives}$$

$$40 + \frac{20}{I} + IR = V_0$$

$$\text{or, } \frac{40I + 20 + I^2 R}{I} = 100$$

$$\text{or, } RI^2 - 60I + 20 = 0$$

For I to be real

$$60^2 - 4(R)(20) \geq 0$$

$$\text{or, } \frac{3600}{80} \geq R \Rightarrow 45 \Omega \geq R.$$



Worksheet 3

1. Mass of $1m^3$ Cu = 9×10^6 g.

$$\text{Number of moles of Cu in } 1m^3 = \frac{9 \times 10^6}{63.5}$$

$$\text{Number of atoms of Cu in } 1\text{m}^3 = \frac{9 \times 10^6}{63.5} \times 6.02 \times 10^{23}$$

$$\text{Number of free electrons of Cu in } 1\text{m}^3$$

$$= \frac{9 \times 6.02}{63.5} \times 10^{29}$$

$$\text{or, } n = 8.5 \times 10^{28} \text{ m}^{-3}.$$

$$I = nev_d A \Rightarrow v_d = \frac{I}{neA}$$

$$\text{or, } v_d = \frac{10}{8.5 \times 10^{28} \times 1.6 \times 10^{-19} \times 4 \times 10^{-6}}$$

$$= 1.8 \times 10^{-4} \frac{\text{m}}{\text{s}}$$

2. When the heating element is hot (at temperature θ), its resistance is

$$R = \frac{V}{I} = \frac{230}{2.68} = 85.82 \Omega$$

$$\text{Using } R = R_0 (1 + \alpha \Delta\theta)$$

$$\text{or, } 85.85 = 75.3 (1 + 1.7 \times 10^{-4} \times \Delta\theta)$$

$$\text{or, } \Delta\theta = 821.8^\circ\text{C}$$

$$\therefore \text{Final temperature is } \theta = 27 + 821.8 \simeq 849^\circ\text{C}$$

3. Given $\frac{l_1}{l_2} = \frac{1}{2}, \frac{d_1}{d_2} = \frac{1}{3}, \frac{\rho_1}{\rho_2} = \frac{2}{1}$

$$\text{Mass } m = A_1 l_1 d_1 = A_2 l_2 d_2$$

$$\Rightarrow \frac{A_1}{A_2} = \frac{l_2 d_2}{l_1 d_1} = 2 \times 3 = 6$$

$$\frac{P_2}{P_1} = \frac{V^2/R_2}{V^2/R_1} = \frac{R_1}{R_2} = \frac{\rho_1 l_1 A_2}{A_1 \rho_2 l_2}$$

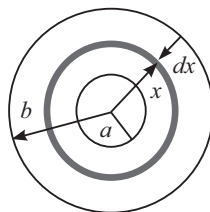
$$\text{or, } \frac{P_2}{P_1} = \left(\frac{\rho_1}{\rho_2} \right) \left(\frac{l_1}{l_2} \right) \left(\frac{A_2}{A_1} \right) = 2 \times \frac{1}{2} \times \frac{1}{6} = \frac{1}{6}$$

$$\therefore P_2 = \frac{10}{6} = \frac{5}{3} \text{ W}$$

4. (a) In this case the cross sectional area is $\pi(b^2 - a^2)$

$$\therefore R = \frac{\rho L}{\pi(b^2 - a^2)}$$

- (b) Current flows radially and the cross sectional area that it sees goes on increasing (see figure). Consider an element in shape of a thin cylinder in shape of inner radius x and outer radius $x + dx$ and length L . Resistance of the element is



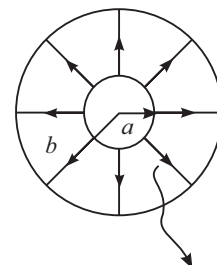
Cylindrical element has thickness dx and length L .

$$dR = \frac{\rho dx}{2\pi x \cdot L}$$

The resistances of all such elements are in series.

$$\therefore R = \int dR = \frac{\rho}{2\pi L} \int_a^b \frac{dx}{x}$$

$$= \frac{\rho}{2\pi L} \ln\left(\frac{b}{a}\right)$$



Current flows radially.

5. Ohm's law : $V = IR$

$$\text{or, } V = neV_d A \cdot \frac{\rho l}{A} = ne\rho l V_d$$

$$\therefore V_d = \frac{V}{ne\rho l}$$

$$= \frac{12 \text{ V}}{(6 \times 10^{20} \text{ cm}^{-3})(1.6 \times 10^{-19} \text{ C})(25 \Omega \cdot \text{cm})(200 \text{ cm})}$$

$$= 2.5 \times 10^{-5} \text{ cm s}^{-1}$$

Note that density of charge carriers is

$$n = 2 \times 3 \times 10^{20} = 6 \times 10^{20} \text{ cm}^{-3}$$

6. $\Delta l = l_0 \alpha \Delta\theta$

$$\text{or, } \frac{\Delta l}{l_0} \times 100 = \alpha \Delta\theta \times 100$$

$$= 1.7 \times 10^{-5} \times 1 \times 100 = 0.0017\%$$

$$\Delta A = A_0 \beta \Delta\theta = A_0 (2\alpha) \Delta\theta$$

$$\text{or, } \frac{\Delta A}{A_0} \times 100 = 2 \alpha \Delta\theta \times 100$$

$$= 2 \times 1.7 \times 10^{-5} \times 1 \times 100 = 0.0034\%$$

$$\text{Resistance of the wire is } R_0 = \rho_0 \frac{l_0}{A_0}$$

$$\therefore \frac{\Delta R}{R_0} \times 100 = \frac{\Delta \rho}{\rho_0} \times 100 + \frac{\Delta l}{l_0} \times 100 - \frac{\Delta A}{A_0} \times 100$$

$$= (\alpha R \Delta\theta \times 100) + 0.0017 - 0.0034$$

$$= 3.9 \times 10^{-3} \times 1 \times 100 + 0.0017 - 0.0034$$

$$= 0.39 - 0.0017 \simeq 0.39\%$$

Note that change in length and cross sectional area has negligible effect, for small temperature changes, on change in resistance.

7. Let the resistance of the two pieces at 0°C be R_{10} and R_{20} .

$$R_{10} + R_{20} = 30$$

(i)

At some higher temperature θ

$$R_{10} [1 + \alpha_1 \theta] + R_{20} [1 + \alpha_2 \theta] = 30$$

$$\text{or, } R_{10} \alpha_1 + R_{20} \alpha_2 = 0$$

$$\text{or, } \frac{R_{10}}{R_{20}} = -\frac{\alpha_2}{\alpha_1} = \frac{4}{0.5} = 8 \quad (\text{ii})$$

$$\text{From (i) and (ii) } R_{10} = \frac{8}{9} \times 30 = \frac{80}{3} \Omega$$

$$R_{20} = \frac{10}{3} \Omega$$

$$8. \text{ Resistance of copper } R_C = \frac{\rho_C L}{\pi r^2}$$

Refer to Q.4 (a) in this exercise. The resistance of nickel part is

$$R_N = \frac{\rho_N L}{\pi(R^2 - r^2)}$$

The two parts are in parallel.

$$\therefore \frac{1}{R} = \frac{1}{R_C} + \frac{1}{R_N}$$

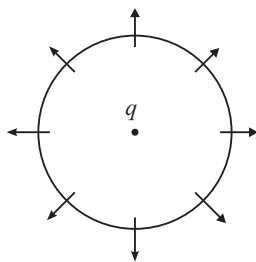
$$\text{or, } R = \frac{L}{\pi \left[\frac{r^2}{\rho_C} + \frac{R^2 - r^2}{\rho_N} \right]}$$

9. Charge inside the spherical region decreases with time. The charge leaks out uniformly from the surface of sphere of radius R . Current through the surface is

$$I = \frac{dq}{dt} = q_0 k e^{-kt}$$

$$\text{or, } j \cdot 4\pi R^2 = q_0 k e^{-kt}$$

$$\text{or, } j = \frac{q_0 k}{4\pi R^2}$$



10. Equivalent of n resistances in parallel is $R_0 = \frac{R}{n}$.

$$\text{Total resistance} = R + \frac{R}{n}$$

$$\therefore \text{ current } I = \frac{V}{R \left(1 + \frac{1}{n} \right)} = \frac{nV}{R(n+1)}$$

When there are $(n+1)$ resistances in parallel, the current changes to

$$I' = \frac{V}{R \left(1 + \frac{1}{n+1} \right)} = \frac{(n+1)V}{R(n+2)}$$

$$\text{Percentage change in current} = \frac{I' - I}{I} \times 100$$

$$\text{or, } \frac{5}{6} = \frac{\frac{(n+1)}{n+2} - \frac{n}{n+1}}{\frac{n}{n+1}} \times 100$$

$$\text{or, } \frac{5}{6} = \frac{100}{n(n+2)} \Rightarrow n^2 + 2n - 120 = 0$$

$$\Rightarrow n = 10$$

11. With light on, the current is $I = 5.0 \text{ A}$.

$\therefore$ Potential difference across terminals is

$$V = \varepsilon - Ir$$

$$\text{or, } 12 = \varepsilon - 5 \times 0.05 \Rightarrow \varepsilon = 12.25 \text{ V}$$

$$\text{Resistance of the lamp, } R = \frac{V}{I} = \frac{12}{5} = 2.4 \Omega$$

When the motor is on, current in lamp is 4.5 A .

$\therefore$ Potential difference across the lamp is

$$V_L = 4.5 \times 2.4 = 10.8 \text{ V}$$

This is also the potential difference across the battery. If current in the battery is i , then

$$10.8 = \varepsilon - ir$$

$$\text{or, } 10.8 = 12.25 - i(0.05)$$

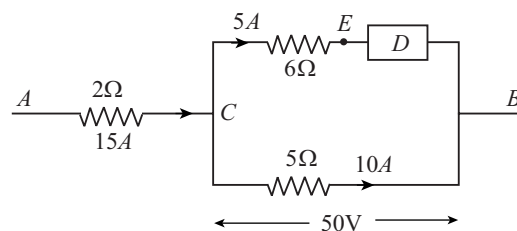
$$\text{or, } i = 29 \text{ A}$$

$\therefore$ Current in the motor is $29 - 4.5 = 24.5 \text{ A}$.

12. With switch open, current is $I = \frac{30V}{30\Omega} = 1 \text{ A}$.

The potential difference across 10Ω resistance is 10 V and that across 20Ω resistance is 20 V . It means $V_P - V_Q = 0$. When switch S is closed, two equipotential points P and Q get connected. It will make no difference to the circuit.

13. Potential drop in 5Ω resistance is $5 \times 10 = 50 \text{ V}$.
 $\therefore$ Potential drop in 2Ω resistance is 30 V .



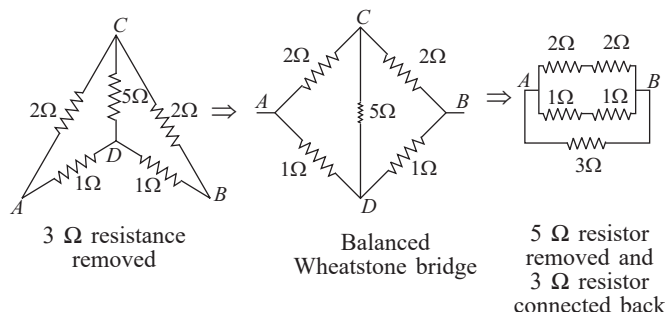
This means that the current in 2Ω resistance is 15A .
Therefore, the current in the upper branch is 5A .
Potential drop in 6Ω resistance is $6 \times 5 = 30\text{V}$.

$$\therefore V_E - V_B = 20\text{V}$$

$$\Rightarrow V_E > V_B.$$

Current in the device enters at high potential end and leaves it at low potential end. It means the electric charge loses energy in crossing the device. The device is absorbing energy at a rate given by $P = VI = 20 \times 5 = 100\text{W}$.

14. (a) Remove the 3Ω resistor and you can see a wheat stone bridge between A and B . The bridge is balanced. The 3Ω resistance is connected in parallel with this balanced bridge.

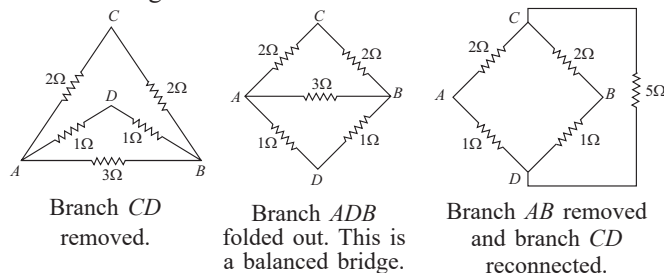


Equivalent of all resistances is given by

$$\frac{1}{R_0} = \frac{1}{4} + \frac{1}{2} + \frac{1}{3}$$

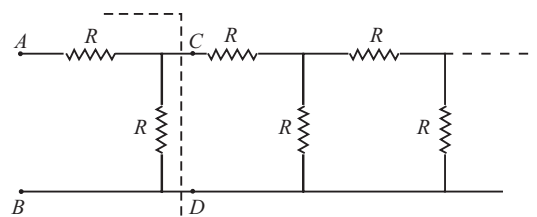
$$\text{or, } R_0 = \frac{12}{13}\Omega$$

- (b) Once again, remove the branch CD (we will connect it later in parallel). The branch ADB can be folded out. You can again see a balanced Wheatstone bridge



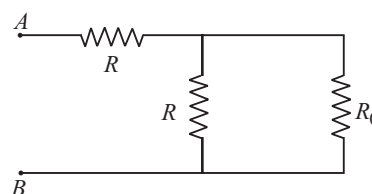
$$\frac{1}{R_0} = \frac{1}{3} + \frac{1}{3} + \frac{1}{5} \Rightarrow R_0 = \frac{15}{13}\Omega$$

15. Let the equivalent resistance between A and B be R_0 . If we remove one link, the effective resistance between the points C and D will still be R_0 .



Removing this part does not change the value of effective resistance.

We have replaced the entire network to the right of line CD with its equivalent R_0 .



Equivalent of this circuit between A and B is R_0

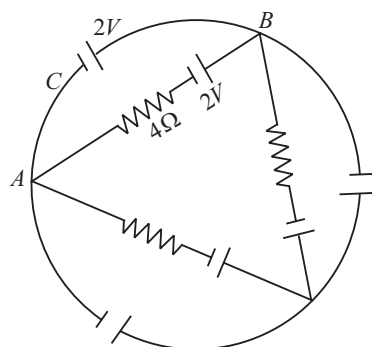
$$\text{Now, } R_{AB} = \frac{RR_0}{R + R_0} + R$$

$$\Rightarrow R_0 = \frac{RR_0}{R + R_0} + R$$

$$\text{or, } R_0^2 - RR_0 - R^2 = 0$$

$$\text{or, } R_0 = \frac{R \pm \sqrt{R^2 + 4R^2}}{2} = \frac{R}{2}(1 + \sqrt{5})$$

16. If we apply the loop law in a loop like ABC , we will write

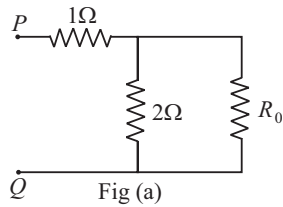


$$2 - 2 - i(4) = 0$$

$$\Rightarrow i = 0$$

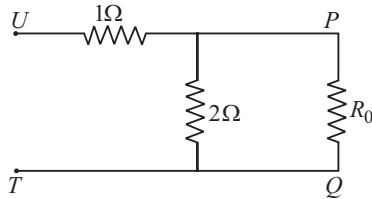
Where i = current in 4Ω resistor.

17. Consider the network shown in figure (a). The equivalent resistance between P and Q is



$$R_{PQ} = \frac{2R_0}{2 + R_0} + 1 \quad (i)$$

Now add one more link to the left of PQ . If value of R_{PQ} is R_0 , then the equivalent resistance between U and T will again be R_0 .



one link added to the left of PQ .

If we add another link to the left of UT , we again get equivalent as R_0 , and so on.

- ∴ Necessary condition for the equivalent resistance to be independent of number of links is

$$R_{PQ} = R_0$$

$$\text{or, } \frac{2R_0}{2 + R_0} + 1 = R_0$$

$$\text{or, } 2R_0^2 - R_0 - 2 = 0$$

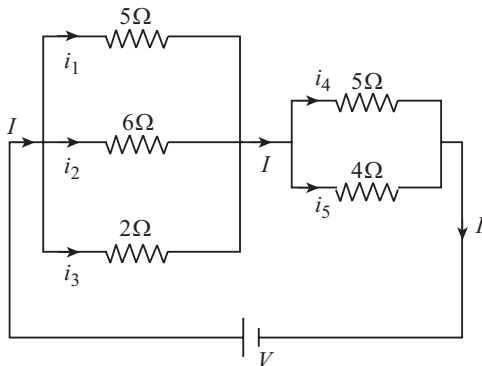
$$\text{or, } R_0 = 2 \Omega$$

18. The circuit can be redrawn as shown.

$$5i_1 = 6i_2 = 2i_3$$

$$\text{and } i_1 + i_2 + i_3 = I$$

$$\therefore i_1 = \frac{3I}{13}; i_2 = \frac{5I}{26}; i_3 = \frac{15I}{26}$$



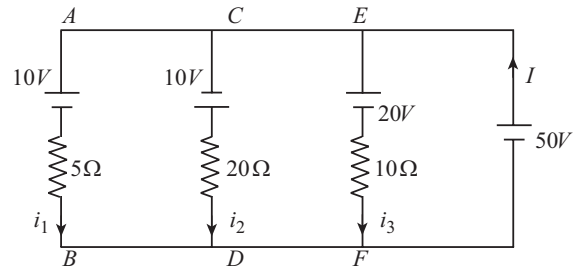
$$\text{Similarly, } 5i_4 = 4i_5$$

$$\text{and } i_4 + i_5 = I$$

$$\therefore i_4 = \frac{4I}{9}, i_5 = \frac{5I}{9}$$

One can write power dissipated in each resistor using $P = i^2 R$ and compare.

- 19.



$$V_{AB} = V_{CD} = V_{EF} = 50 \text{ V}$$

$$V_A - 10 - 5i_1 = V_B$$

$$\text{or, } V_A - V_B - 10 = 5i_1$$

$$\text{or, } 50 - 10 = 5i_1$$

$$\text{or, } i_1 = 8 \text{ A}$$

$$\text{Similarly, } V_C + 10 - 20i_2 = V_D$$

$$\text{or, } 20i_2 = 60 \text{ or, } i_2 = 3 \text{ A}$$

And writing similar equation for path EF gives

$$i_3 = 3 \text{ A}$$

$$\therefore I = i_1 + i_2 + i_3 = 14 \text{ A}$$

$$\therefore \text{Power supplied by } 50 \text{ V cell is } P_1 = (50)(I) = 700 \text{ W}$$

The 10 V cell loses energy at a rate,

$$P_2 = 10 i_2 = 30 \text{ W}$$

20. The circuit has 1Ω and 2Ω resistance in parallel.

$$\text{Their equivalent is } R = \frac{1 \times 2}{1 + 2} = \frac{2}{3} \Omega$$

$$\text{The total resistance of the circuit is } R_0 = \frac{2}{3} + \frac{2}{3}$$

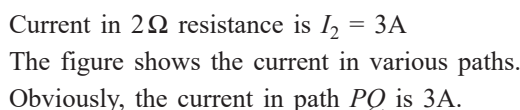
$$= \frac{4}{6} \Omega$$

$$\therefore \text{Battery current is } I = \frac{6}{4/6} = 9 \text{ A}$$

9A current divides in parallel paths in inverse ratio of resistances.

- ∴ Current in 1Ω resistance is

$$I_1 = \left(\frac{2}{1+2} \right) \times 9 = 6 \text{ A}$$



22. (a) $P = \frac{V^2}{R}$

$$V = \sqrt{PR.} = \sqrt{36 \times 100} = 60 \text{ V}$$

Diagram for Problem 1: A series circuit with a 100Ω resistor and a 50Ω resistor. The 100Ω resistor is between points A and C, and the 50Ω resistor is between points C and B. The 30V source is connected across points A and B. The voltage across the 100Ω resistor is 60V.

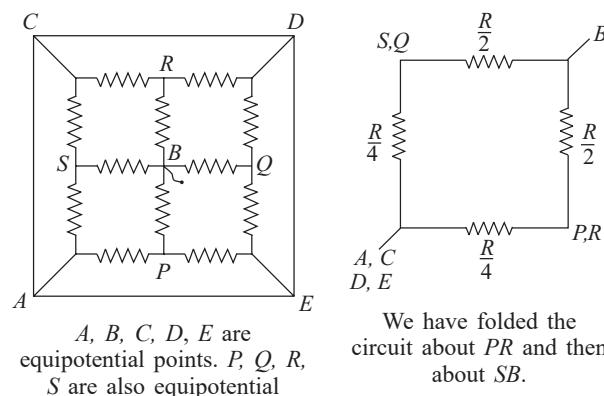
Note that above 90 V, the voltage across A and C will exceed the limiting value of 60 V.

in each of R_2 and R_3 is $\frac{V^2}{R} = \frac{30^2}{100} = 9 \text{ W}$

23. The circuit is a balanced Wheatstone bridge with a $3\ \Omega$ resistance connected in parallel to it.



25.



$$R_{eq} = \frac{3R}{8}$$

$$R_0 = 3R \quad \Rightarrow \quad R = \frac{R_0}{3}$$

The figure illustrates a circuit reduction. On the left, a network of resistors is shown with nodes A, C, E, F, D, and B. A resistor labeled '1' is between C and F, a resistor labeled '2' is between C and E, a resistor labeled '3' is between E and D, and a resistor labeled 'R' is between F and D. A diagonal resistor labeled 'R' connects nodes C and D. The equivalent circuit on the right shows a series combination of three resistors: a resistor 'R' between nodes A and C, a resistor 'R/3' between nodes C and D, and a resistor 'R' between nodes D and B.

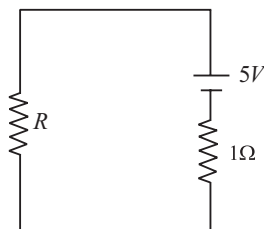
$$\therefore R_{eq} = R + \frac{R}{3} + R = \frac{7R}{3} = \frac{7R_0}{9}$$

27. We can treat two cells in parallel and regard $2\ \Omega$ resistance as their internal resistance. The emf of the equivalent cell and its internal resistance are

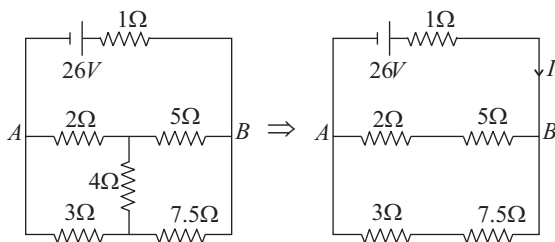
$$\varepsilon_0 = 5V$$

$$r_0 = 1\ \Omega$$

Obviously, power transfer to R is maximum when $R = 1\ \Omega$.



28. Between A and B we have a balanced wheatstone bridge. The $4\ \Omega$ resistor can be removed.



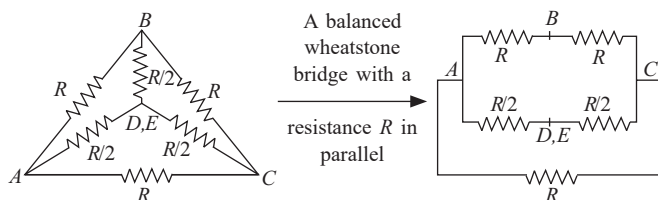
$$R_{eq} = \frac{7 \times 10.5}{7 + 10.5} + 1 = \frac{26}{5}\ \Omega$$

$$\therefore I = \frac{26V}{26/5\ \Omega} = 5A$$

$$\therefore V_A + 26 - 1 \times 5 = V_B$$

$$\text{or, } V_A - V_B = -21\ \text{Volt.}$$

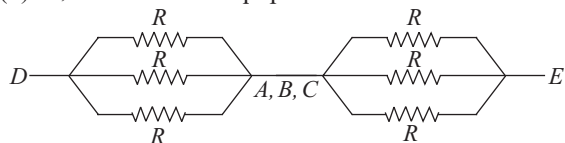
29. (a) Points D and E must be equipotential in this case.



$$\frac{1}{R_{eq}} = \frac{1}{2R} + \frac{1}{R} + \frac{1}{R}$$

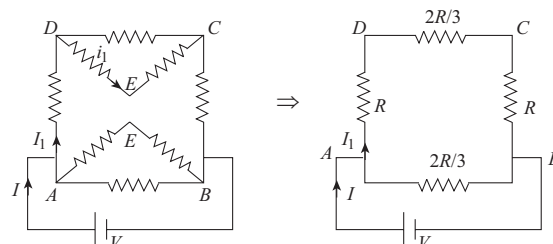
$$\text{or, } R_{eq} = \frac{2R}{5}$$

- (b) A , B and C are equipotential.



$$\therefore R_{eq} = \frac{2R}{3}$$

30. (a) When a cell is connected across AB , the current going into path AE must be same as the current coming out of path EB . Similarly, current from D to E must be same as that from E to C . This suggests that we can open the junction at E and redraw the circuit as given below.



$$R_{eq} = \frac{8R}{15}$$

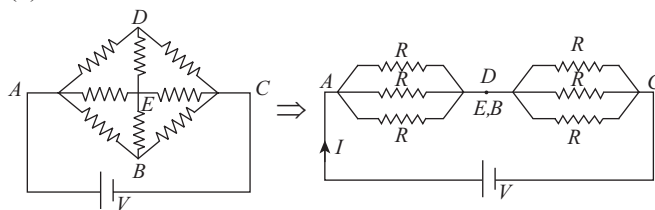
$$I = \frac{V}{R_{eq}} = \frac{15V}{8R}$$

$$I_1 = \frac{V}{8R/3} = \frac{3V}{8R}$$

I divides into two parts at D current along DEC is

$$i_1 = \left(\frac{R}{R+2R} \right) I_1 = \frac{V}{8R}$$

- (b)



Points B , E and D are equipotential.

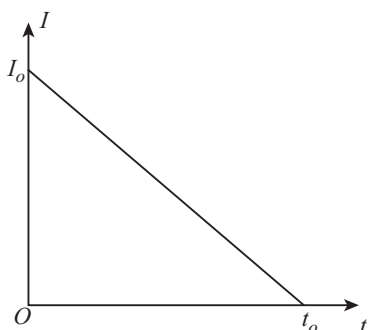
$$R_{eq} = \frac{2R}{3}$$

$$\therefore I = \frac{3V}{2R}$$

$$\therefore \text{Current in path } AE \text{ is } i = \frac{I}{3} = \frac{V}{2R}$$

31. The I - t graph is as shown.

$$I = I_0 - \left(\frac{I_0}{t_0} \right) t \quad [y = mx + c]$$



Here, I_0 is unknown. But we can find it knowing that the total charge passed in time t_0 is q .

$$q = \int_0^{t_0} I dt = \text{Area under the graph}$$

$$\text{or, } q = \frac{1}{2} I_0 \cdot t_0 \Rightarrow I_0 = \frac{2q}{t_0}$$

$$\therefore I = \frac{2q}{t_0} \left(1 - \frac{t}{t_0} \right).$$

Power dissipated in the resistance at time t is $P = I^2 R$.

Heat dissipated in a small time interval dt is $dH = P dt = I^2 R dt$.

$$\therefore \text{Total heat produced is } H = \int_0^{t_0} dH = R \int_0^{t_0} I^2 dt$$

$$\begin{aligned} \text{or, } H &= \frac{4q^2 R}{t_0^2} \int_0^{t_0} \left(1 - \frac{t}{t_0} \right)^2 dt \\ &= \frac{4}{3} \frac{q^2 R}{t_0} \end{aligned}$$

Chapter 7 Basic Electrical Measurement



Your Turn

2. We need to connect a shunt resistance in parallel to the galvanometer coil.

$$R_s = \frac{i_g R_g}{I - i_g} = \frac{10^{-3} \times 20}{10 - 10^{-3}} \simeq \frac{20 \times 10^{-3}}{10}$$

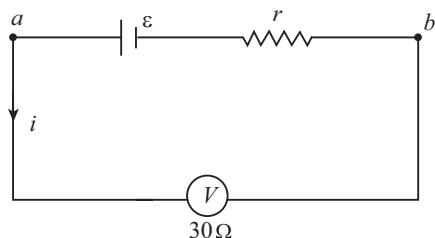
$$\therefore R_s \simeq 2\text{m}\Omega$$

Comparing to example 2, we understand that for making an ammeter of higher range, we need a smaller shunt resistance.

$$3. R = \frac{V}{i_g} - R_g = \frac{5}{5 \times 10^{-3}} - 1 = 999\Omega$$

$$4. i = \frac{1.5}{30 + 5} = 0.043\text{A}$$

$$V_{ab} = (30\Omega)(0.043\text{A}) = 1.29\text{V}$$



5. (a) Resistance of an ideal voltmeter is infinite. No current flows through it. Therefore, the potential

difference between A and C is $\frac{220}{4} = 55\text{V}$. This is the reading of the voltmeter.

- (b) Resistance of the rheostat wire between A and

$$C \text{ is } \frac{12000}{4} = 3000\Omega.$$

Effective resistance between A and C is = $\frac{3000 \times 6000}{3000 + 6000} = 2000\Omega$.

Resistance of part CB is 9000Ω .

The potential differences V_{AC} and V_{CB} will be directly proportional to the resistances R_{AC} and R_{CB} , respectively.

$$\therefore V_{AC} = \frac{2000}{2000 + 9000} \times 220 = 40\text{V}.$$

This will be the reading of the voltmeter.

$$6. (a) \frac{R_x}{R} = \frac{40}{60} \Rightarrow R_x = \frac{40}{60} \times 100 = \frac{200}{3}\Omega = 66.7\Omega$$

$$(b) \frac{R_x}{R} = \frac{40 + 4}{60 + 2} \Rightarrow R_x = \frac{44}{62} \times 100 = 71\Omega.$$

7. In the second set of measurement, the value of R_3 is almost 10 times than that in the first set of measurement. Hence, ratio $\frac{R_1}{R_2} = 10$ for the second set.

8. Use equation (12)

$$r = R \left(\frac{l_1}{l_2} - 1 \right) = 9.5 \left(\frac{76.3}{64.8} - 1 \right) = 1.68\Omega$$

9. At balance point, there is no current through the cell (in upper circuit). It makes no difference to V_{ab} even if S is closed.

$$\text{Similarly, } \frac{ER_x}{R_x + r} \propto l_2 \quad (b)$$

From (a) and (b)

$$\frac{\frac{R_x}{R_x + r}}{\frac{R}{R + r}} = \frac{l_2}{l_1}$$

$$\text{or, } \frac{R_x}{R_x + 2} = \frac{30}{50} \left(\frac{10}{10 + 2} \right)$$

$$\text{or, } R_x = 2\Omega$$

10. Potential gradient on the potentiometer wire is

$$= \frac{20\text{mV}}{100\text{cm}} = \frac{1}{5} \text{mV}(\text{cm})^{-1}.$$

$\therefore$ Potential difference across the entire length of the wire is

$$V_0 = \frac{1}{5} \times 500 = 100\text{mV}$$

$$\therefore \text{Current in the wire is } I_0 = \frac{100\text{mV}}{10\Omega} = 10\text{mA}.$$

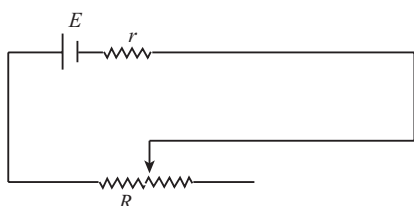
$$\text{Now, } I_0 = \frac{10\text{V}}{r + 10} \text{ or, } 10 \times 10^{-3} = \frac{10}{r + 10}$$

$$\text{or, } r = 990\Omega.$$



Worksheet 1

- Adding any resistance in parallel will increase the range because a part of the current will get diverted through the parallel path.
- The potential drop across r and R is in proportion to the resistances.



$\therefore$ When R is increased, the potential drop across R increases. When R becomes too large, the potential drop across it will be nearly E .

- The equivalent of R_2 and voltmeter resistance is

$$\frac{60 \times 120}{60 + 120} = 40 \Omega.$$

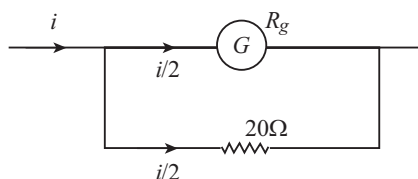
Resistance $R_1 (= 60 \Omega)$ and 40Ω are in series. Potential difference across 40Ω resistance is

$$\frac{40}{40 + 60} \times 120 = 48 \text{ V}.$$

This is what the voltmeter will read.

Note that by adding a voltmeter of small resistance, we have disturbed the circuit a lot. Without voltmeter, the potential difference across R_2 is 60 V which drops to 48 V after the voltmeter is added.

- The deflection in the galvanometer becomes half, if current through it becomes half. This will happen when $R_g = 20 \Omega$.



- Full scale deflection current is

$$i_g = \frac{0.2 \text{ V}}{20 \Omega} = 0.01 \text{ A}$$

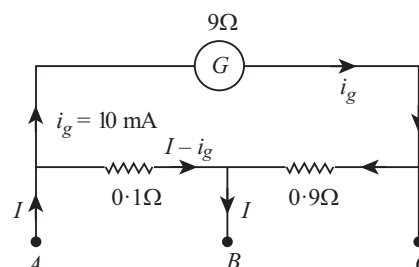
$$R_s = \frac{i_g R_g}{I - i_g} = \frac{0.01 \times 20}{10 - 0.01} = 0.02 \Omega$$

- For finding R , we must measure potential difference (V) across it and the current (I) through it and then

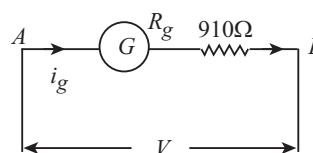
$$\text{use } R = \frac{V}{I}.$$

The diagram shown in (B) appropriately does this.

- $V_{AB} = i_g(9 + 0.9) = (I - i_g)(0.1)$
or, $10 \times 10^{-3} \times 9.9 = (I - 10 \times 10^{-3})(0.1)$
or, $I = 1 \text{ A}$



- Let V be the maximum voltage that the galvanometer can read. It means when the applied potential difference is V , the current is $i_g = 10 \text{ mA}$.

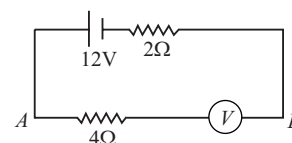


$$\Rightarrow V = i_g(90 + 910) = 10 \times 10^{-3} \times 1000 = 10 \text{ V}.$$

Each division on the scale of the voltmeter reads 0.1 V . Therefore, for a 0 to 10 V voltmeter, number

of divisions on the scale must be $\frac{10}{0.1} = 100$.

- $12 = 0.1(R + 20)$
or, $R = 100 \Omega$
- Resistance of an ideal voltmeter is infinite. Current in the circuit will be negligible. Potential drop across 4Ω and 2Ω resistances will be nearly zero. Therefore, the potential difference between A and B will be nearly 12 V .



- $\frac{R_1}{R_2} = \frac{40}{60} \Rightarrow \frac{R_1}{R_2} = \frac{2}{3}$ (i)

After a resistance of 10Ω is connected in parallel with R_2 ,

$$\frac{R_1}{\frac{10 R_2}{10 + R_2}} = \frac{50}{50} \Rightarrow R_1 = \frac{10 R_2}{10 + R_2} \quad \text{(ii)}$$

Solving (i) and (ii) gives $R_2 = 5 \Omega$, $R_1 = \frac{10}{3} \Omega$.

- If $AJ = l_1$, then

$$\frac{X}{R} = \frac{l_1}{100 - l_1} \quad \text{or,} \quad \frac{12}{18} = \frac{l_1}{100 - l_1}$$

$$\Rightarrow l_1 = 40 \text{ cm}$$

$$\text{For } R = 8\Omega$$

$$\frac{X}{R} = \frac{l_2}{100-l_2} \quad \text{or, } \frac{12}{8} = \frac{l_2}{100-l_2}$$

$$\text{or, } l_2 = 60 \text{ cm}$$

$$\therefore \text{ shift} = 20 \text{ cm}$$

14. The current in any path will not change when the switch is closed if

$$\frac{P}{R} = \frac{Q}{G},$$

where G = galvanometer resistance.

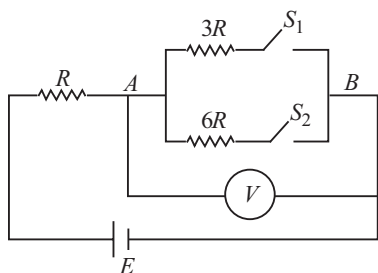
In such a case, $I_R = I_G$ and $I_P = I_Q$.

15. If the wire is uniform, the ratio of resistances of its two parts depends only on the length.

$$\frac{R_{AC}}{R_{CB}} = \frac{x}{L-x}$$

If the cross section of the wire is changed, the above ratio does not change.

17. When both S_1 and S_2 are closed, the resistance between the points A and B is least. In this case, reading of the voltmeter is least. With only S_2 closed, the resistance between A and B is maximum. In this case, reading is maximum.



18. $V_{AC} = 4 \text{ volt}$

$$\text{or, } \left(\frac{6V}{1.0\text{m}} \right) \cdot x = 4 \Rightarrow x = \frac{2}{3} \text{ m}$$

19. Current $i = \frac{11 \text{ volt}}{11\Omega} = 1\text{A}$.

Potential difference across the wire is $V_0 = iR = 1 \times 10 = 10 \text{ V}$.

$$\text{Potential gradient} = \frac{10\text{V}}{10\text{m}} = 1 \text{ V/m}.$$

20. Let V_0 = potential difference across the potentiometer wire.

$$\text{According to the problem, } E = \frac{V_0}{3}$$

$$\text{New length of wire, } L = \frac{3l}{2}.$$

Potential drop of $\frac{V_0}{3}$ will happen over a length =

$$\frac{L}{3} = \frac{l}{2}.$$

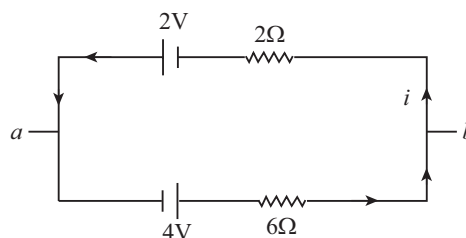
21. Obviously, $i > I_0$.

The potentiometer acts like an ideal voltmeter. It reads the true voltage across R . When a voltmeter is introduced, the overall resistance of the circuit drops and the current increases.

$$\therefore i > I_0 > I.$$

Reading of the voltmeter is less than V_0 because the combined resistance of R and voltmeter (R_v) is less than R .

22. In the external loop, the current is



$$i = \frac{6 \text{ volt}}{8\Omega} = \frac{3}{4} \text{ A}$$

$$\therefore V_a + 4 - 6i = V_b$$

$$\text{or, } V_a + 4 - 4.5 = V_b \Rightarrow V_a - V_b = 0.5 \text{ V}$$

$$\begin{aligned} \text{In potentiometer circuit, current is } I &= \frac{12 \text{ volt}}{(8 + 4 \times 4)\Omega} \\ &= \frac{1}{2} \text{ A} \end{aligned}$$

Let length AN be x .

$$R_{AN} = 4x$$

$$V_{AN} = IR_{AN} = \frac{1}{2}(4x) = 2x$$

$$\text{or, } 2x = V_a - V_b$$

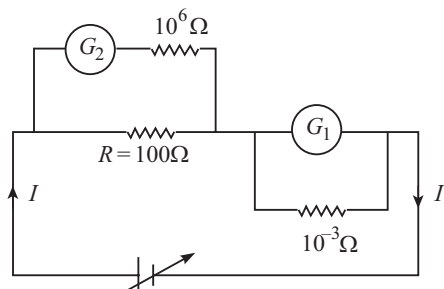
$$\text{or, } 2x = 0.5 \Rightarrow x = 0.25 \text{ m}$$



Worksheet 2

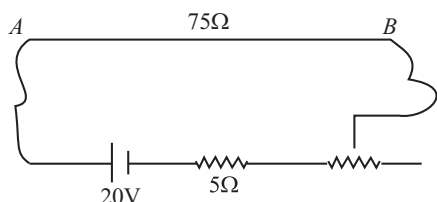
1. The battery shall be connected across A and B . The output can be taken across A and C or, C and B . As we move the slider from A towards B , the resistance R_{AC} increases and R_{CB} decreases. Therefore, the potential difference V_{AC} increases and V_{CB} decreases.

2. The arrangement has been shown. The galvanometer G_1 with $10^{-3}\Omega$ resistor in parallel acts as an ammeter. It measures current I . The galvanometer G_2 with $10^6\Omega$ resistance in series acts as a voltmeter. It measures potential difference across the resistance R .



3. For making a voltmeter of 10V range, we need to add greater resistance in series. This decreases the sensitivity and increases the range.
4. Current of 4A gets divided into R and the voltmeter. Current in R is less than 4A. Potential difference across R is 20V. This means that $R = \frac{V}{I}$ is larger than 5Ω .
6. The potential difference across the potentiometer wire is maximum when resistance of the rheostat is 120Ω .

$$(V_0)_{\max} = \left(\frac{75}{75 + 5 + 120} \right) \times 20 = 7.5\text{V}$$



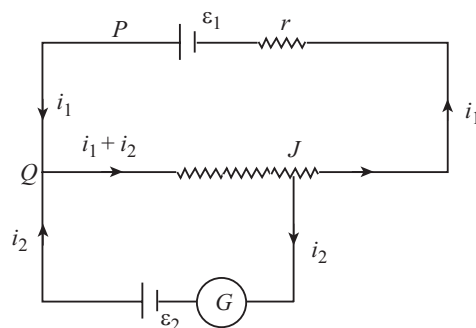
Potential difference across the potentiometer is minimum when the resistance of the rheostat is 170Ω .

$$(V_0)_{\min} = \left(\frac{75}{75 + 5 + 170} \right) \times 20 = 6\text{V}$$

The potential difference across the potentiometer can be varied between 6V and 7.5 V. We cannot measure a potential difference higher than 7.5 V using this potentiometer as no null point can be obtained.

7. Galvanometer will not show zero deflection if negative terminal of cell C is connected to the high potential end A . The other reason for no balance point is that the potential difference (V_0) across the potentiometer wire is smaller than emf of cell C . This could occur if $R \gg R_0$ or emf of $C >$ emf of D .

8. Higher the resistance of voltmeter, higher will be the current in $3\text{k}\Omega$ resistor. For case (i), current in $3\text{k}\Omega$ resistance is maximum. Potential difference V_1 is maximum in this case. In case (iii), the equivalent resistance of the two voltmeters will be less than $1\text{k}\Omega$. In this case, maximum current will get diverted into the voltmeters and current in $3\text{k}\Omega$ resistor is least. V_3 is least.
9. The situation could be as shown in the figure. This happens when ϵ_1 is too small or r is too large.



If there is a break in wire PQ , then also only i_2 will flow.

10. Potential difference across 20Ω resistor $= 20 \times 0.3 = 6\text{V}$.

$$\text{Current in } 15\Omega \text{ resistor is } = \frac{6\text{V}}{15\Omega} = 0.4\text{A}.$$

$$\text{Current in } R_1 \text{ is } = 0.8 - (0.3 + 0.4) = 0.1\text{ A}.$$

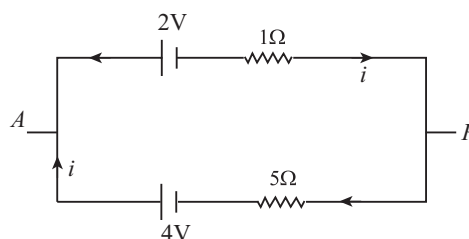
$$R_1 (0.1) = 6$$

$$\Rightarrow R_1 = 60\Omega.$$

11. When the voltmeter is ideal, it has infinite resistance. Current in the circuit is zero. There is no potential drop across the resistors shown. Therefore, potential difference across the voltmeter is 6V.

When the voltmeter is non-ideal, there will be a current in the circuit. Some potential drop will happen across the resistors shown in parallel. It means that the potential difference across the voltmeter will be less than 6V.

12. In balanced position, the potentiometer loop and the given loop of two cells are independent. Current in the loop shown is



$$i = \frac{4-2}{5+1} = \frac{2}{6} = \frac{1}{3} \text{ A}$$

$$V_A - 2 - \left(\frac{1}{3}\right)(1) = V_P \Rightarrow V_A - V_P = \frac{7}{3} \text{ V}$$

$$\text{Current in the potentiometer loop is } I = \frac{10\text{V}}{20\Omega} = \frac{1}{2} \text{ A.}$$

$$\text{Potential difference } V_{AP} = \frac{7}{3} \text{ V if}$$

$$I \left(10 \frac{\Omega}{\text{m}} \cdot x \right) = \frac{7}{3}$$

or, $x = \frac{7}{15} \text{ m}$

When both ε_1 and ε_2 are 2 V, the potential difference across A and P will be 2V under balanced condition. There will be no current in the loop of the two cells.

$$I(10 \cdot x) = 2$$

$$\text{or, } \frac{1}{2}(10 \cdot x) = 2 \Rightarrow x = 0.4 \text{ m}$$



Worksheet 3

1. Let source voltage be V .

$$\text{Current without ammeter is } I_0 = \frac{V}{R}$$

$$\text{Current with ammeter is } I = \frac{V}{R+r}$$

$$\therefore I_0 = \left(\frac{R+r}{R} \right) I = \left(\frac{100+1}{100} \right) \times 5 = 5.05 \text{ A.}$$

2. $R_g = \frac{\text{Full scale deflection voltage}}{\text{Full scale deflection current}} = \frac{75 \text{ mV}}{15 \text{ mA}}$

$$\text{or, } R_g = 5\Omega.$$

For converting it into an ammeter that can read upto $I = 0.6 \times 150 = 90 \text{ A}$, we add a shunt resistance given by

$$R_s = \frac{i_g R_g}{I - i_g} = \frac{15 \times 10^{-3} \times 5}{90 - 15 \times 10^{-3}}$$

$$\text{or, } R_s = 8.3 \times 10^{-4} \Omega.$$

For converting it into a voltmeter that can read upto 150V, we need to add a resistance in series given by

$$R = \frac{V}{I_g} - R_g = \frac{150}{15 \times 10^{-3}} - 5 = 9995 \Omega.$$

3. (a) Current $I = 0.04 \text{ A}$.

$$\therefore \text{Drop across ammeter} = 2 \times 0.04 = 0.08 \text{ V}$$

$$\text{Drop across } 3 \Omega \text{ resistance} = 3 \times 0.04 = 0.12 \text{ V}$$

$$\therefore \text{Potential drop across } 100\Omega \text{ resistor} = 3.4 - (0.08 + 0.12) = 3.2 \text{ V}$$

$$\text{Now, current through } 100\Omega \text{ resistance is } i_1 = \frac{3.2 \text{ V}}{100\Omega} = 0.032 \text{ A}$$

$$\therefore \text{Current through the voltmeter is } I_2 = 0.04 - 0.032 = 0.008 \text{ A}$$

$$\therefore \text{Resistance of the voltmeter is, } r = \frac{3.2 \text{ V}}{0.008 \text{ A}} = 400\Omega.$$

(b) For an ideal voltmeter, resistance is infinity.

$$\therefore \text{Total resistance of the circuit is } R = 3 + 2 + 100 = 105\Omega$$

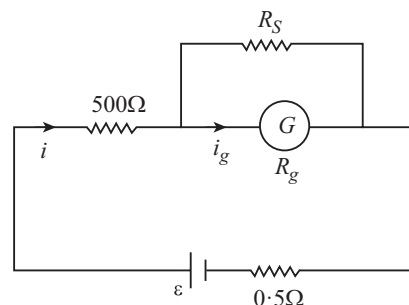
$$\therefore \text{Current, } I = \frac{3.4}{105} \text{ A}$$

Potential difference across 100Ω resistor is

$$V = 100I = \frac{340}{105} = 3.24 \text{ V.}$$

This will be the reading of the voltmeter.

4. $i = \frac{\varepsilon}{500 + 0.5 + \frac{R_g \cdot 5}{R_g + 5}}$



and

$$i_g = i \frac{R_s}{R_s + R_g}$$

$$\text{or, } 5 \times 10^{-3} = \left(\frac{\varepsilon}{500.5 + \frac{5R_g}{R_g + 5}} \right) \left(\frac{5}{5 + R_g} \right)$$

$$\text{or, } 5 \times 10^{-3} = \frac{5\varepsilon}{505.5 R_g + 2500.25} \quad (i)$$

Similarly, with $R_s = 25\Omega$,

$$20 \times 10^{-3} = \frac{25\varepsilon}{525.5 R_g + 12512.5} \quad (ii)$$

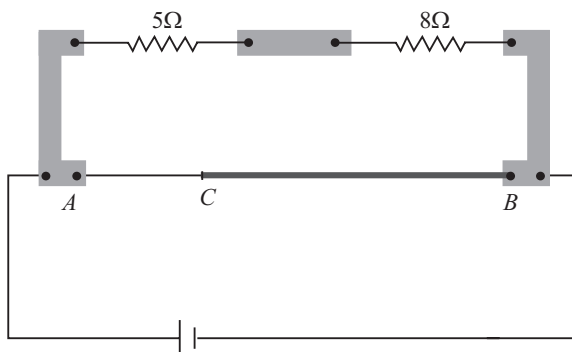
Equations (i) and (ii) can be further simplified as

$$0.51R_g + 2.5 = \varepsilon$$

$$\text{and } 0.42R_g + 10 = \varepsilon.$$

Solving these equations gives $R_g = 83.3\Omega$ and $\varepsilon \simeq 45\text{V}$.

5.



Cross-sectional area of left part = A

Cross-sectional area of right part = $4A$

$$\text{Resistance of left part } R_{AC} = \frac{\rho(30\text{ cm})}{A}$$

$$\text{Resistance of right part } R_{CB} = \frac{\rho(70\text{ cm})}{4A}$$

$$\therefore \frac{R_{AC}}{R_{CB}} = \frac{12}{7}$$

Balance will occur at a point where the ratio of resistances of the two parts of the wire is $\frac{5}{8}$. This

can happen only when the jockey is somewhere on part AC . Let the distance AJ be x .

$$R_{AJ} = \frac{\rho x}{A}$$

$$R_{JB} = R_{JC} + R_{CB} = \frac{\rho(30-x)}{A} + \frac{\rho(70)}{4A}$$

$$\text{For balance } \frac{R_{AJ}}{R_{JB}} = \frac{5}{8}$$

$$\text{or, } \frac{x}{30-x+\frac{70}{4}} = \frac{5}{8}$$

$$\text{or, } x = 18.27 \text{ cm.}$$

6. Maximum value of $R_3 = 11,110\Omega$.

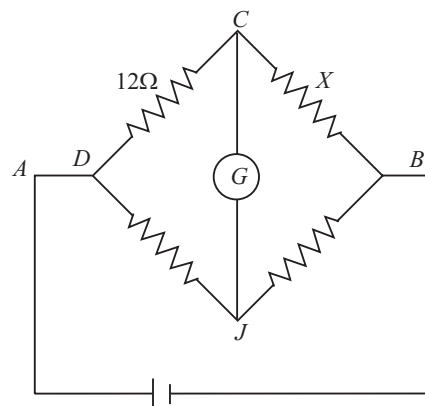
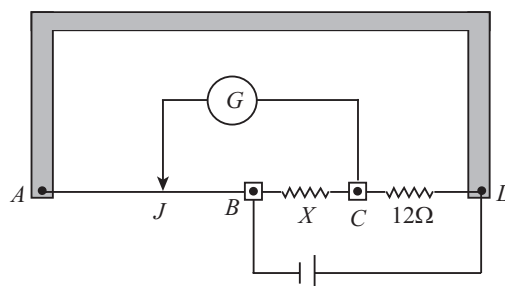
$$R_x = R_3 \left(\frac{R_2}{R_1} \right)$$

$$\text{or, } (R_x)_{\max} = (R_3)_{\max} \left(\frac{R_2}{R_1} \right)_{\max}$$

$$= 11,110 \times 100 = 1111 \text{ k}\Omega$$

$$\text{and } (R_x)_{\min} = (R_3)_{\min} \left(\frac{R_2}{R_1} \right)_{\min} = 1 \times \frac{1}{100} = 0.01\Omega$$

7. (b)



$$(c) \frac{12}{X} = \frac{R_{AJ}}{R_{BJ}} \Rightarrow \frac{12}{X} = \frac{60}{40}$$

$$\text{or, } X = 8\Omega$$

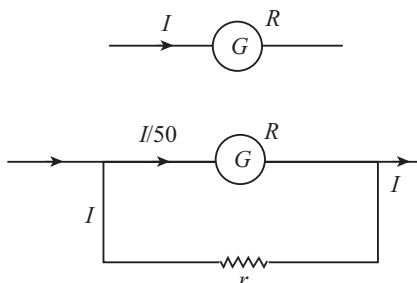
8. (a) $V_0 = (I - I_0)R_v = I_0(R_A + R)$

$$\therefore R_A + R = \frac{V_0}{I_0}; \quad \text{or, } R = \frac{V_0}{I_0} - R_A$$

(b) For $\frac{V_0}{I_0} \gg R_A$

$$R \simeq \frac{V_0}{I_0}$$

9. Sensitivity gets reduced by 50 times if the current in the galvanometer becomes $\frac{I}{50}$ when the circuit has current I . This will ensure 50 times less deflection for the same current.



$$r \cdot \frac{49I}{50} = R \cdot \frac{I}{50}$$

$$\text{or, } r = \frac{R}{49} = \frac{10,000}{49} = 204\Omega$$

10. Direct use of equation (12).

11. (a) $V_A - V_B = 6\text{V}$

or, $V_A = 6\text{V}$.

and $V_A - V_C = 4$ or, $V_C = 6 - 4 = 2\text{V}$

- (b) Potential gradient along

$$AB = \frac{6V}{100\text{cm}} = 0.06\text{V (cm)}^{-1}$$

where $AD = x$, the potential difference V_{AD} is

$$V_A - V_D = (0.06\text{V cm}^{-1})(x)$$

or, $4 = (0.06)(x)$ [$\because V_D = V_C = 2\text{V}$]

$$\text{or, } x = \frac{4}{0.06} = 66.67\text{ cm}$$

- (c) Since C and D are at the same potential, no current will flow.

(d) $V_A - V_B = 6\text{V}$ and $V_A - V_C = 7.5\text{V}$

$$\therefore V_C = -1.5\text{V}.$$

There is no point D where potential is -1.5V .

Chapter 8 Transient RC Circuits



Your Turn

1. (a) $q_0 = CV = (4\mu\text{F})(20\text{ V}) = 80\text{ }\mu\text{C}$.
 (b) $i_0 = \frac{V}{R} = \frac{20}{10 \times 10^3} = 2 \times 10^{-3}\text{ A} = 2\text{ mA}$
 (c) Time constant $\tau = RC = (10 \times 10^3)(4 \times 10^{-6}) = 0.04\text{ s}$.

We have to find current at $t = 2\tau = 0.08\text{ s}$

$$i = i_0 e^{-t/\tau} = 2e^{-2}\text{ mA} \simeq 0.28\text{ mA}$$

$$[\because e^{-1} = 0.37 \Rightarrow e^{-2} = (0.37)^2 \simeq 0.14]$$

- (d) Potential difference across R is $V_R = iR$
 $= (0.28\text{ mA})(10\text{ k}\Omega)$
 $= 2.8\text{ V}$

But $V_C + V_R = V$

$$\text{or, } V_C + 2.8 = 20$$

$$\text{or, } V_C = 17.2\text{ V}$$

2. $V_R + V_C = V$

$$\text{or, } \frac{V_C}{3} + V_C = V; \text{ or, } V_C = \frac{3V}{4}$$

$$\therefore \text{ At given time } V_R = \frac{V}{4}$$

$$\text{or, } iR = \frac{V}{4}; \quad \text{or, } i_0 e^{-t/\tau} = \frac{V}{4R}$$

$$\text{or, } \frac{V}{R} e^{-t/\tau} = \frac{V}{4R}; \quad \text{or, } e^{-t/\tau} = \frac{1}{4}$$

$$\text{or, } -\frac{t}{\tau} = \ln\left(\frac{1}{4}\right)$$

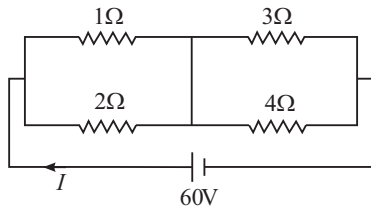
$$\text{or, } t = \tau(\ln 4) = (2\ln 2)\tau = (2\ln 2)RC$$

3. (a) At $t = 0^+$, the capacitor offers no resistance.

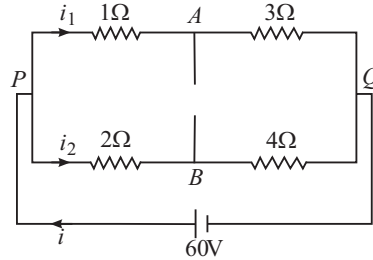
Equivalent resistance is

$$R = \frac{1 \times 2}{1+2} + \frac{3 \times 4}{3+4} = \frac{50}{21}\Omega$$

$$\therefore I = \frac{V}{R} = \frac{60+21}{50} = \frac{126}{5}\text{ A}$$



- (b) At $t = \infty$, the capacitor is like an infinite resistance.



$$R = \frac{4 \times 6}{4+6} = 2.4\Omega$$

$$\text{Current } i = \frac{60}{2.4} = 25\text{ A}$$

$$(c) V_P - V_A = \left(\frac{1}{1+3}\right) \times 60 = 15\text{ V} \quad (i)$$

$$V_P - V_B = \left(\frac{2}{2+4}\right) \times 60 = 20\text{ V} \quad (ii)$$

$$(ii) - (i) \text{ gives: } V_A - V_B = 5\text{ V}$$

This is a potential difference across the capacitor.

Charge on the capacitor is

$$q = (5\text{ V})(2\mu\text{F}) = 10\mu\text{C}$$

$$4. \text{ Equivalent resistance } R = \frac{2 \times 1}{2+1} = \frac{2}{3}\text{ k}\Omega$$

$$\text{Time constant } \tau = RC = \left(\frac{2}{3}\text{ k}\Omega\right)(3\mu\text{F})$$

$$= 2\text{ ms}$$

Current at $t = 2\text{ ms}$ is

$$i = i_0 e^{-t/\tau} = \frac{V}{R} e^{-1} = \frac{30}{\frac{2}{3} \times 10^3} e^{-1} = 45 \times 0.37\text{ mA} = 16.65\text{ mA}$$

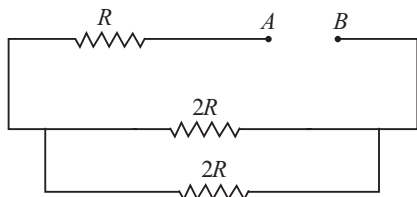
This current gets divided between two resistors in inverse ratio of resistance. Current in $2\text{ k}\Omega$ resistance

$$\text{is } \frac{1}{1+2} \times 16.65 = 5.55\text{ mA}.$$

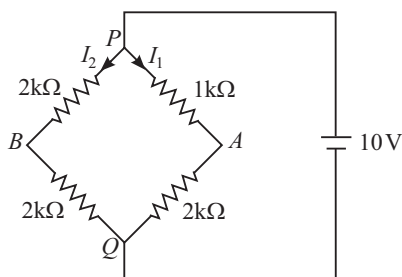
5. After shorting the cell, the equivalent resistance across A and B is

$$R_0 = \frac{(2R).(2R)}{2R+2R} + R = 2R$$

$$\therefore \tau = R_0 C = 2RC$$



6. (a) For the purpose of finding current, the effective circuit at $t = \infty$ is as shown



$$I_1 = \frac{10V}{3k\Omega} = \frac{10}{3} \text{ mA}$$

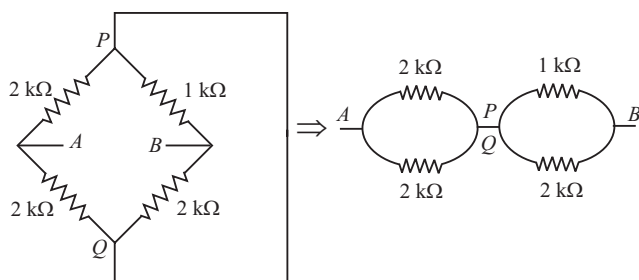
$$I_2 = \frac{10V}{4k\Omega} = \frac{5}{2} \text{ mA}$$

$$V_P - V_A = (1k\Omega) \left(\frac{10}{3} \text{ mA} \right) = \frac{10}{3} \text{ V}$$

$$V_P - V_B = (2k\Omega) \left(\frac{5}{2} \text{ mA} \right) = 5 \text{ V}$$

$$\therefore V_A - V_B = 5 - \frac{10}{3} = \frac{5}{3} \text{ V}$$

- (b) After shorting the cell, we will find equivalent resistance between A and B.



$$R_0 = \frac{2 \times 2}{2+2} + \frac{1 \times 2}{1+2} = \frac{5}{3} k\Omega$$

$$\therefore \tau = R_0 C = \left(\frac{5}{3} \times 10^3 \right) (3 \times 10^{-6}) = 5 \times 10^{-3} \text{ s}$$

Charge on the capacitor varies with time as

$$q = q_0 [1 - e^{-t/\tau}]$$

$$\text{or, } CV = CV_0 [1 - e^{-t/\tau}]$$

where V is potential difference across C .

$$V_0 = \frac{5}{3} \text{ V is final potential difference across } C.$$

$$\therefore V = V_0 [1 - e^{-t/\tau}]$$

$$\text{or, } V = \frac{5}{3} \left(1 - e^{-\frac{t}{0.005}} \right)$$

$$7. i = i_0 e^{-t/\tau} \Rightarrow \frac{i_0}{2} = i_0 e^{-t/\tau}$$

$$\text{or, } \ln\left(\frac{1}{2}\right) = -\frac{t}{\tau} \Rightarrow t = (\ln 2)\tau$$

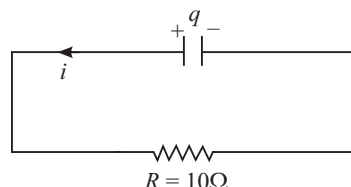
8. (a) Potential difference across the capacitor in the steady state = 1V

$$\text{Energy stored in the capacitor } U = \frac{1}{2} CV^2$$

$$= \frac{1}{2} \times (6\mu\text{F}) (1\text{V})^2 = 3\mu\text{J}.$$

This energy is dissipated as heat in the 10Ω resistor connected in parallel to the capacitor.

- (b) Rate of loss of charge = Discharging current



$$\text{Initial current } i_0 = \frac{1V}{10\Omega} = 0.1\text{A}$$

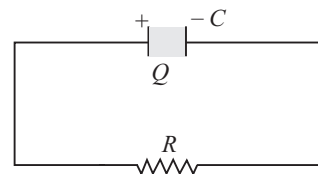
$$i = i_0 e^{-t/\tau}$$

$$\text{or, } 0.05 = 0.1 e^{-t/\tau}$$

$$\text{or, } e^{-t/\tau} = \frac{1}{2}; \quad \text{or, } \frac{t}{\tau} = \ln 2$$

$$\text{or, } t = (\ln 2)\tau = 0.69 (10 \times 6 \times 10^{-6}) \text{ s} \\ = 4.14 \times 10^{-5} \text{ s}$$

9. The effective arrangement is shown as



$$C = \frac{K\epsilon_0 A}{d}; \quad R = \frac{d}{\sigma A}$$

$$iR = \frac{Q}{C} \Rightarrow i = \frac{Q}{RC} = \frac{Q\sigma}{K\epsilon_0}$$



Worksheet 1

1. Let total resistance of the circuit be R .

$$i = i_0 e^{-t/RC}; \text{ or, } \frac{i_0}{2} = i_0 e^{-t/RC}$$

$$\text{or, } e^{-t/RC} = \frac{1}{2}; \text{ or, } \frac{t}{RC} = \ln 2$$

$$\text{or, } R = \frac{t}{(\ln 2)C} = \frac{(\ln 4) \times 10^{-6}}{(\ln 2) \left(\frac{1}{2} \times 10^{-6} \right)} = 4\Omega$$

$$\text{or, } r + 2 = 4; \text{ or, } r = 2\Omega$$

2. When charge on the capacitor is $\left(\frac{3}{4}\right)^{\text{th}}$ its maximum value, the potential difference across it is $\left(\frac{3}{4}\right)^{\text{th}}$ of

20V.

$$\therefore V_C = \frac{3}{4} \times 20 = 15 \text{ V}$$

At this moment, $V_R = 5\text{V}$

$$\text{or, } i.R = 5$$

$$\text{or, } R = \frac{5}{1} = 5\Omega$$

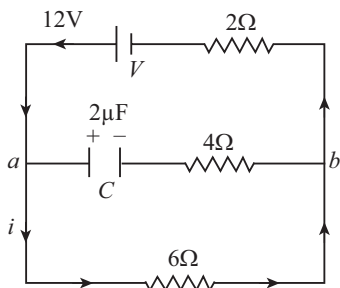
3. $V_R = nV_C$ and $V_R + V_C = V$

$$\therefore V_R = \frac{nV}{n+1}$$

$$\text{or, } iR = \frac{nV}{n+1} \Rightarrow \left(\frac{V}{R} \right) e^{-t/\tau} R = \frac{nV}{n+1}$$

$$\text{or, } \frac{t}{\tau} = \ln \left(\frac{n+1}{n} \right) \Rightarrow t = RC \ln \left(\frac{n+1}{n} \right)$$

4. In the steady state, there is no current in the branch having the capacitor. The current flows in the outer loop only.

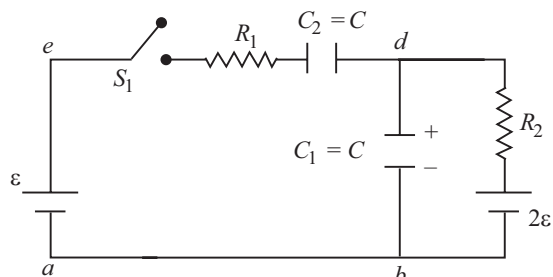


$$i = \frac{12}{6+2} = \frac{3}{2} \text{ A}$$

$$V_{ab} = (6)i = 9 \text{ V}$$

$$\therefore q = C(V_{ab}) = 2 \times 9 = 18\mu\text{F}.$$

5. When S_2 is closed, the potential difference across C_1 becomes 2ε .



Now, S_1 is closed. In the loop $abde$, the capacitor C_2 is uncharged (behaves like a zero resistance) and potential difference across C_1 is 2ε . Just after closing the switch

$$iR + \varepsilon = 2\varepsilon \Rightarrow i = \frac{\varepsilon}{R} \text{ (anticlockwise)}$$

6. The potential difference across both the capacitors is always same. If q_1 and q_2 are charges on $10\mu\text{F}$ and $5\mu\text{F}$ capacitors, respectively, then

$$q_1 = 2q_2 \text{ (always)}$$

$$\Rightarrow \frac{dq_1}{dt} = 2 \frac{dq_2}{dt} \Rightarrow i_1 = 2i_2$$

where i_1 and i_2 are currents in the two branches.

Equivalent capacitance $C = 15\mu\text{F}$

Time constant, $\tau = RC = 5 \times 15 \text{ ms} = 75 \text{ ms}$.

Current at $t = \tau$ is

$$i = i_0 e^{-1} = \left(\frac{15}{5} \right) e^{-1} = 3e^{-1}$$

Now, $i_1 + i_2 = i$ and $i_1 = 2i_2$

$$\therefore i_2 = \frac{i}{3} = e^{-1} \text{ A}$$

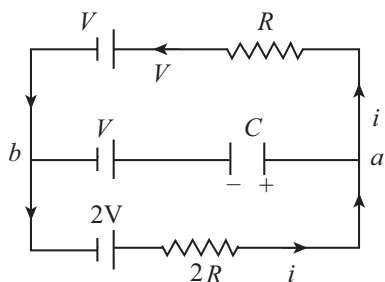
7. Immediately after closing K_2 , the capacitor offers no resistance and resistance r (in parallel to C) can be treated to be short circuited. Current is

$$I_1 = \frac{\varepsilon}{r}$$

After a long time, the capacitor acts like an infinite resistance.

$$I_2 = \frac{\varepsilon}{2r}$$

9. In the steady state, there is current in the outer loop only. Using loop law for outer loop gives
 $iR + i \cdot 2R = 2V - V$



or, $i = \frac{V}{3R}$

Moving along the upper branch, one can write

$$V_a - iR - V = V_b$$

or, $V_a - V_b = \frac{4V}{3}$

Now, moving along the middle path (from a to b) gives

$$V_a - V_C - V = V_b$$

or, $V_C = \frac{4V}{3} - V = \frac{V}{3}$

10. $I = I_0 e^{-t/\tau}$

or, $\ln I = \ln I_0 - \frac{t}{\tau}$

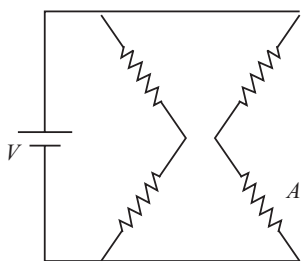
or, $\ln I = a - \frac{t}{\tau}$ [$a = \ln I_0$ (a constant)]

$\therefore$ Graph of $\ln I$ versus t is a straight line with y

intercept $= a = \ln I_0$ and slope $= -\frac{1}{RC}$.

When R is doubled, I_0 becomes half. This means that y intercept must decrease. The magnitude of the slope of the line also decreases when R is increased. Therefore, Q is the correct graph.

11. The effective circuit in the steady state is as shown in the figure. Obviously, potential difference in each of the four resistances is $\frac{V}{2}$.



12. Let us assume that $E_2 > E_1$. After a long time, when the switch is put in position 1, the capacitor loses a small amount of charge ($i_1 \Delta t$). When the switch is put into position 2, the capacitor gains same amount of charge ($i_2 \Delta t$). This way the charge on the capacitor will remain practically constant.

Let the final charge on the capacitor be q_0 . When switch is in position 1, the current is given by

$$\frac{q_0}{C} - E_1 = i_1 R_1$$

[the potential difference across the capacitor is larger than E_1]

or, $i_1 = \frac{q_0}{CR_1} - \frac{E_1}{R_1}$

When the switch is in position 2 ($E_2 > \frac{q_0}{C}$), the current is given by

$$E_2 - \frac{q_0}{C} = i_2 R_2$$

or, $i_2 = \frac{E_2}{R_2} - \frac{q_0}{CR_2}$

For charge on the capacitor to remain constant
 $i_1 = i_2$

$$\Rightarrow \frac{q_0}{CR_1} - \frac{E_1}{R_1} = \frac{E_2}{R_2} - \frac{q_0}{CR_2}$$

$$\therefore q_0 = \frac{(E_1 R_2 + E_2 R_1)C}{(R_1 + R_2)}$$

13. Maximum current will flow through the battery when the effective resistance is least. This happens at $t = 0^+$ when the capacitor offers zero resistance.

Equivalent resistance is $R_0 = \frac{R}{2}$.

$$i_0 = \frac{\varepsilon}{R/2} = \frac{2\varepsilon}{R}$$

Current through the battery will be $\frac{i_0}{2} = \frac{\varepsilon}{R}$ when the branch containing the capacitor does not conduct, i.e., when the capacitor is fully charged.

$\therefore q = C\varepsilon$



Worksheet 2

1. At the instant the switch is closed, the capacitor acts like a zero resistance and R_3 is short circuited. There is no current through R_3 .

At $t = \infty$, the equivalent resistance is

$$R_0 = \frac{2 \times 2}{2 + 2} + 1 = 2\Omega$$

$$\therefore I = \frac{10V}{2\Omega} = 5A$$

At this instant, the charge on the capacitor is maximum.

$$q_0 = (V_C)(C) = (IR_3)C = 5 \times 1 \times 1 = 5\mu\text{C}.$$

$$2. \quad i_0 = \frac{V_0}{R} \quad \Rightarrow 10 = \frac{V_0}{10}$$

$$\text{or, } V_0 = 100 \text{ V}$$

In 2 seconds, the current becomes $\left(\frac{1}{4}\right)^{\text{th}}$

$$\frac{i_0}{4} = i_0 e^{-t/\tau}, \quad \text{or, } \frac{1}{4} = e^{-\frac{2}{\tau}}$$

$$\text{or, } \ln 4 = \frac{2}{\tau}; \quad \text{or, } \tau = \frac{1}{\ln 2}$$

$$\text{or, } RC = \frac{1}{\ln 2}$$

$$\Rightarrow C = \frac{1}{10 \ln 2}$$

$$\begin{aligned} \text{Total heat produced} &= \frac{1}{2} CV_0^2 \\ &= \frac{1}{2} \times \frac{1}{10 \ln 2} \times (100)^2 \\ &= \frac{500}{\ln 2} \text{ J} \end{aligned}$$

Thermal power in the resistance

$$P = i^2 R = (i_0 e^{-t/\tau})^2 R = i_0^2 R e^{-2t/\tau}$$

$$P = P_0 e^{-t/\tau_0}$$

$$\text{where } t_0 = \frac{\tau}{2} = \frac{1}{2 \ln 2}$$

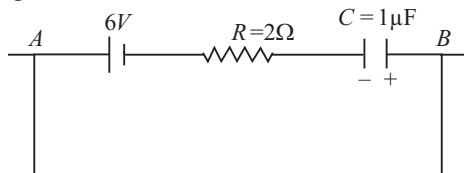
$$3. \quad \tau_1 = RC_1; \quad \tau_2 = RC_2$$

$$\therefore \tau_2 = 2\tau_1$$

Initial current is same in both the circuits as $i_0 = \frac{V_0}{R}$.

Smaller the time constant, faster is the discharge.

4. Look at the part of the circuit as shown in the figure.



$$V_A - V_B = 0$$

$$\Rightarrow 6 = \frac{q}{C} \quad [iR = 0 \text{ in the steady state}]$$

$$\Rightarrow q = 6\mu\text{C}$$

When there is no charge on the capacitor

$$V_A - V_B = 0$$

$$iR = 6 \quad [\text{current is from } B \text{ to } A]$$

$$\text{or, } i = 3\text{A}$$

$$\text{Current in the left loop is } I = \frac{3V}{3\Omega} = 1\text{A}$$

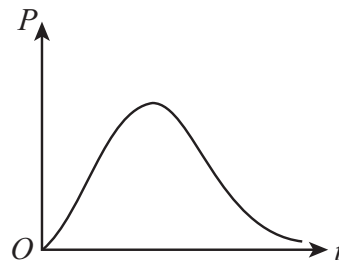
$$\therefore V_P - V_A = 1 \times 2 = 2\text{V}$$

$$5. \quad U = \frac{q^2}{2C}$$

$$\therefore \frac{dU}{dt} = \frac{q}{C} \frac{dq}{dt} = \frac{1}{C} qi$$

$$\text{or, } P = \frac{q_0 i_0}{C} (1 - e^{-t/\tau}) e^{-t/\tau}$$

The product $(1 - e^{-t/\tau}) e^{-t/\tau}$ will first increase and then decrease. To understand this, note that the product is zero at $t = 0$ as well as at $t = \infty$. But at a time $0 < t < \infty$, it is a positive number. Variation of P with t , is as shown.



The rate of heat generation in R is

$$P_R = i^2 R = i_0^2 R e^{-2t/\tau}$$

P_R goes on decreasing with time.

$$6. \quad U = \frac{q^2}{2C}$$

$$\frac{dU}{dt} = \frac{q}{C} \frac{dq}{dt}$$

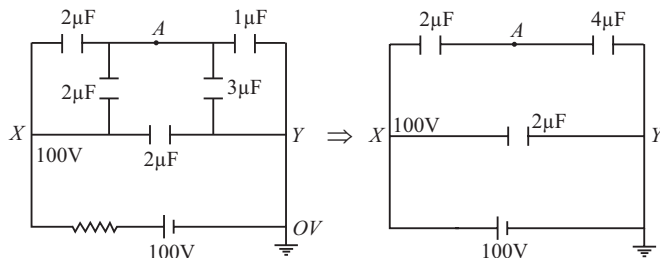
$$\text{or, } -\frac{dU}{dt} = \frac{q}{C} \left(-\frac{dq}{dt} \right) = \frac{1}{C} qi$$

This is rate of energy loss in the capacitor. It decreases as both q and i decrease with time.

Rate of energy loss of capacitor = rate of heat generation. Thus, rate of heat generation also decreases with time.


Worksheet 3

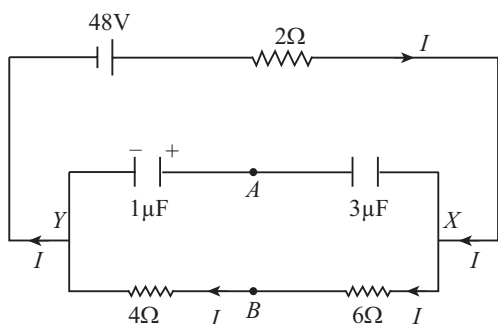
1. There will be no current in the circuit in the steady state. Potential difference across X and Y is 100V.



Obviously, $V_X - V_A = 50V$
or, $V_A = 50V$

2. There is no current in the branch having the capacitors. Current flows as shown

$$I = \frac{48}{2+6+4} = 4A$$



$$V_X - V_Y = (4A)(10\Omega) = 40V$$

$$V_X - V_B = (4A)(6\Omega) = 24V \quad (a)$$

Note that the 40 V potential difference across X and Y gets divided between the two capacitors in inverse ratio of their capacitance.

$$\therefore V_X - V_A = \left(\frac{1}{1+3}\right) \times 40 = 10V \quad (b)$$

From (a) and (b)

$$V_A - V_B = 14V$$

3. $\tau = RC = 2 \times 10 \times 10^{-6} = 20\mu s$

(a) Current at $t = 20\mu s$ is

$$i = i_0 e^{-t/\tau} = \frac{V}{R} e^{-1}$$

$$= \frac{5}{2} \times 0.37 = 0.925A$$

$$P_{\text{battery}} = iV = 0.925 \times 5 = 4.625W$$

$$(b) \text{ Rate of heat dissipation} = i^2 R = (0.925)^2 \times 2 = 1.71W$$

$$(c) U = \frac{q^2}{2C}$$

$$\begin{aligned} \text{or, } \frac{dU}{dt} &= \frac{1}{2C} 2q \left(\frac{dq}{dt} \right) = \frac{q}{C} i \\ &= \frac{q_0}{C} (1 - e^{-t/\tau}) i = V(1 - e^{-t/\tau}) i \\ &= 5(1 - e^{-1}) \times 0.925 \\ &= 5 \times 0.63 \times 0.925 = 2.91W \end{aligned}$$

$$4. V = V_0 e^{-t/\tau}$$

$$\text{or, } \frac{10}{100} = e^{-t_0/\tau} \quad (i)$$

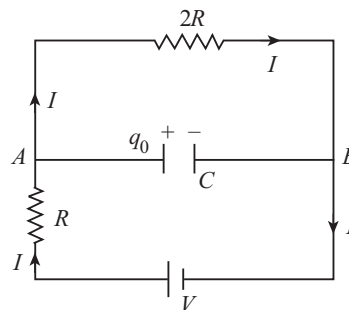
The potential difference across the capacitor at time $t = 2t_0$ is

$$\begin{aligned} V &= V_0 e^{-2t_0/\tau} = V_0 (e^{-t_0/\tau})^2 \\ &= 100 \times \left(\frac{1}{10}\right)^2 = 1V \end{aligned}$$

5. When S is closed, the potential difference across the capacitor is zero. After S is opened, charging of the capacitor begins.

$$q = q_0 e^{-t/\tau} = (CV) e^{-\frac{t}{2RC}}$$

6. Equivalent of $3R$ and $6R$ in parallel is $\frac{(3R)(6R)}{3R+6R} = 2R$. Status of the circuit, in the steady state, has been shown.



$$I = \frac{V}{3R}$$

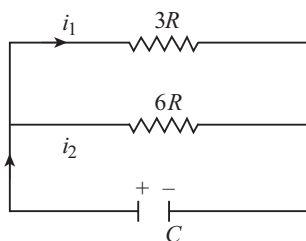
$$V_{AB} = I(2R) = \frac{2}{3}V$$

$\therefore$ Energy stored in the capacitor is

$$U = \frac{1}{2} C \left(\frac{2}{3} V \right)^2 = \frac{2}{9} CV^2$$

After S is opened, the capacitor acts as energy source that drives current in $3R$ and $6R$.

The potential difference is always equal (and decreasing) across the two resistors.



Power dissipation in a resistor is $P = \frac{V^2}{R}$. Therefore, the ratio of power dissipated in the two resistors is always in inverse proportion of the resistances.

If H_1 and H_2 represents the total heat dissipated in $3R$ and $6R$, then

$$\frac{H_1}{H_2} = \frac{6R}{3R} = \frac{2}{1} \quad (\text{i})$$

$$\text{And } H_1 + H_2 = U = \frac{2}{9} CV^2 \quad (\text{ii})$$

Solving (i) and (ii), we get

$$H_1 = \frac{2}{3} \times \frac{2}{9} CV^2 = \frac{4}{27} CV^2$$

7. The electric field $\propto q$.

$\therefore$ When the field becomes one-third, the charge on the capacitor must also have reduced to one-third of its initial value.

$$q = q_0 e^{-t/\tau}$$

$$\text{or, } \frac{q_0}{3} = q_0 e^{-t/\tau}; \quad \text{or, } t = (\ln 3)\tau$$

$$\text{or, } \frac{4.4}{1.1} = \tau; \quad \text{or, } \tau = 4 \text{ ms}$$

$$\text{or, } RC = 4 \text{ ms}; \quad \text{or, } R = \frac{4 \times 10^{-3}}{2 \times 10^{-6}} = 2000 \Omega$$

8. It is a discharging RC circuit with

$$C = \frac{K\epsilon_0 A}{d} \text{ and } R = \frac{d}{\sigma A}$$

$$\therefore \tau = RC = \frac{K\epsilon_0}{\sigma}$$

$$= \frac{5.01 \times 8.86 \times 10^{-12}}{7.4 \times 10^{-12}} = 6.0 \text{ s}$$

$$q = q_0 e^{-t/\tau}$$

$$\text{and } i = -\frac{dq}{dt} = \frac{q_0}{\tau} e^{-t/\tau}$$

At $t = 12 \text{ s}$

$$i = \frac{8.86 \times 10^{-3}}{6} e^{-12/6} = \frac{8.86}{6} (e^{-2}) \text{ mA}$$

$$= \frac{8.86}{6} (0.37)^2 \text{ mA} = 0.2 \text{ mA}$$

9. Final charge on the capacitor is $Q = CV$

$$= (10 \mu\text{F}) (40\text{V}) = 400 \mu\text{C}.$$

$$W_{\text{cell}} = QV = (400 \mu\text{C}) (40 \text{ V}) = 16000 \mu\text{J} = 16 \text{ mJ}$$

$$\text{Energy stored in the capacitor } U = \frac{1}{2} CV^2 = 8 \text{ mJ}.$$

Obviously, 8 mJ heat is dissipated during the charging process. Power dissipated in a resistance

$$= \frac{V^2}{R}, \text{ where } V \text{ is potential difference across it.}$$

The two resistors always have equal potential difference across them. Hence, power dissipated in them is inversely proportional to their resistances.

$$\frac{H_1}{H_2} = \frac{20}{40} = \frac{1}{2}, \quad (\text{i})$$

where H_1 = heat dissipated in 40Ω resistance.

H_2 = heat dissipated in 20Ω resistance.

$$H_1 + H_2 = 8 \text{ mJ} \quad (\text{ii})$$

Solving (i) and (ii) gives,

$$H_2 = \frac{2}{3} \times 8 = \frac{16}{3} \text{ mJ}.$$

10. In the steady state, there is no current through the capacitor. The remaining circuit is a balanced wheatstone bridge. The potential difference across 20Ω resistor is

$$V = \left(\frac{20}{10+20} \right) \times 100 = \frac{200}{3} \text{ V}$$

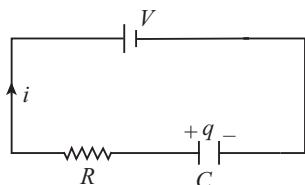
$$\therefore q = (10 \mu\text{F}) \left(\frac{200}{3} \text{ V} \right) = \frac{2000}{3} \mu\text{C} = \frac{2}{3} \text{ mC}$$

11. Capacitance with dielectric = KC .

Charge on the capacitor $Q_0 = KCV$.

After removing the slab, charge will flow out of the capacitor and finally acquire a value $Q = CV$.

Let q be the charge on the capacitor and i be the current at time t after the slab is removed



$$i = -\frac{dq}{dt}$$

$$\frac{q}{C} = Ri + V$$

$$\text{or, } R \left(-\frac{dq}{dt} \right) = \frac{q}{C} - V$$

$$\text{or, } \int_{Q_0}^q \frac{dq}{V - \frac{q}{C}} = \frac{1}{R} \int_0^t dt$$

$$\text{or, } \ln \left(\frac{V - q/C}{V - Q_0/C} \right) = -\frac{t}{RC}$$

$$\text{or, } \frac{VC - q}{VC - Q_0} = e^{-t/RC}$$

$$\text{or, } VC - q = (1 - K) VC = e^{-t/RC} \quad [\because Q_0 = KVC]$$

$$\text{or, } q = VC + VC(K - 1) e^{-t/RC}$$

$$\therefore \text{Current } i = -\frac{dq}{dt} = \frac{VC}{RC} (K - 1) e^{-t/RC}$$

$$= \frac{V}{R} (K - 1) e^{-t/RC}$$

$$\text{Heat} = \int_0^\infty i^2 R dt = \frac{V^2}{R} (K - 1)^2 \int_0^\infty e^{-\frac{2t}{RC}} dt$$

$$= \frac{1}{2} CV^2 (K - 1)^2$$

Chapter 9 Miscellaneous Problems on Chapters 6–8



Match the Column

1. (P) $\frac{FV}{s} = \frac{C}{s} = A$ $[\because CV = Q]$

(Q) $\frac{kgm}{sN} = \frac{kgms^2}{s kgm} = s$

Second is unit of time constant

(R) $\frac{V}{\Omega} = \text{ampere}$

(S) $\Omega - m$ is unit of resistivity

(T) $\Omega - m = \frac{Vm}{A} = \frac{V}{C} ms$

Hence, options (S) and (T) are same.

2. With switch S closed, one of the resistances is shorted. The potential difference across the other resistance is $V = E =$ reading of the voltmeter.

Current $I = \frac{E}{R} =$ reading of the ammeter.

Power dissipated in left resistance $P = \frac{E^2}{R}$

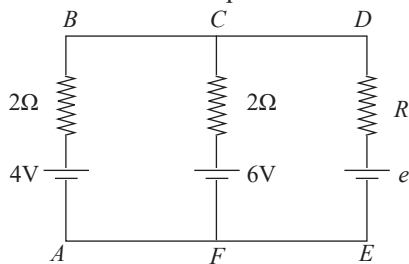
When switch S is opened, the two resistances are in series. Equivalent resistance *increases* to $2R$.

Power dissipated in one resistance *decreases* to $\frac{\left(\frac{E}{2}\right)^2}{R}$

Voltmeter reading *decreases* to $\frac{E}{2}$.

Ammeter reading *decreases* to $\frac{E}{2R}$.

3. Let us consider the resistances to be the internal resistances of the respective cells.



The equivalent emf for parallel combination of 4V cell and e volt cell is

$$E = \frac{\frac{4}{2} + \frac{e}{R}}{\frac{1}{2} + \frac{1}{R}}$$

When this equivalent emf is 6V, current in branch CF will be zero.

$$\frac{\frac{4}{2} + \frac{e}{R}}{\frac{1}{2} + \frac{1}{R}} = 6 \quad \Rightarrow \quad 2 + \frac{e}{R} = 3 + \frac{6}{R}$$

$$\Rightarrow \frac{e-6}{R} = 1 \quad \Rightarrow \quad e = 6 + R$$

$\therefore$ Current in CF will be zero if $e > 6V$

Current will be from F to C if $E < 6V$

$$\Rightarrow \frac{2 + \frac{e}{R}}{\frac{1}{2} + \frac{1}{R}} < 6 \quad \Rightarrow \quad 2 + \frac{e}{R} < 3 + \frac{6}{R}$$

$$\Rightarrow \frac{e-6}{R} < 1 \quad \Rightarrow \quad e < 6 + R$$

$\therefore$ Current can flow from F to C if $e < 6V$ and it is also possible if $e = 6V$ or $e > 6V$. For example, when $R = 10\Omega$, this is possible when $e < 16\Omega$.

Current can flow from C to F if $E > 6V$

$$\Rightarrow \frac{2 + \frac{e}{R}}{\frac{1}{2} + \frac{1}{R}} > 6 \quad \Rightarrow \quad 2 + \frac{e}{R} > 3 + \frac{6}{R}$$

$$\Rightarrow \frac{e-6}{R} > 1 \quad \Rightarrow \quad e > R + 6$$

Thus it is possible only when $e > 6V$.

To find out the direction of current in branch BA , let us find the equivalent emf of 6V cell and e volt cell.

$$E_1 = \frac{\frac{6}{4} + \frac{e}{R}}{\frac{1}{4} + \frac{1}{R}}$$

Current will flow from B to A if

$$E_1 > 4$$

$$\text{or, } \frac{\frac{3}{2} + \frac{e}{R}}{\frac{1}{4} + \frac{1}{R}} > 4 \Rightarrow \frac{3}{2} + \frac{e}{R} > 1 + \frac{4}{R}$$

$$\frac{e-4}{R} > -\frac{1}{2} \Rightarrow 2e-8 > -R \Rightarrow e > 4 - \frac{R}{2}$$

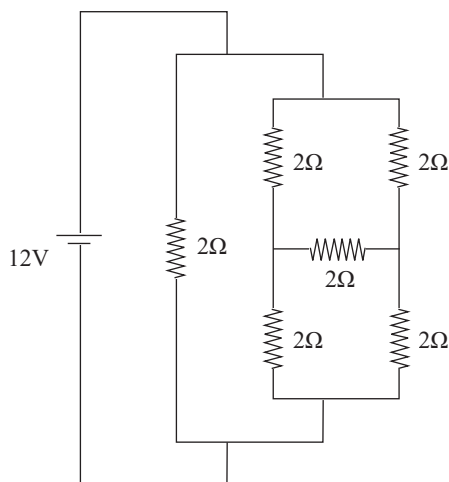
Therefore, e must be equal to or greater than 4V when R is close to zero. When $R = 8\Omega$, e must be greater than zero. It means that e can have all values above zero.

5. In A , B and C , all resistances are in parallel.

$$R_A = 1\Omega, I_A = \frac{12}{1} = 12\text{A}$$

$$R_B = \frac{2}{4} = \frac{1}{2}\Omega, I_A = \frac{12}{1/2} = 24\text{A}$$

$$R_C = \frac{2}{3}\Omega, I_C = \frac{12}{2/3} = 18\text{A}$$

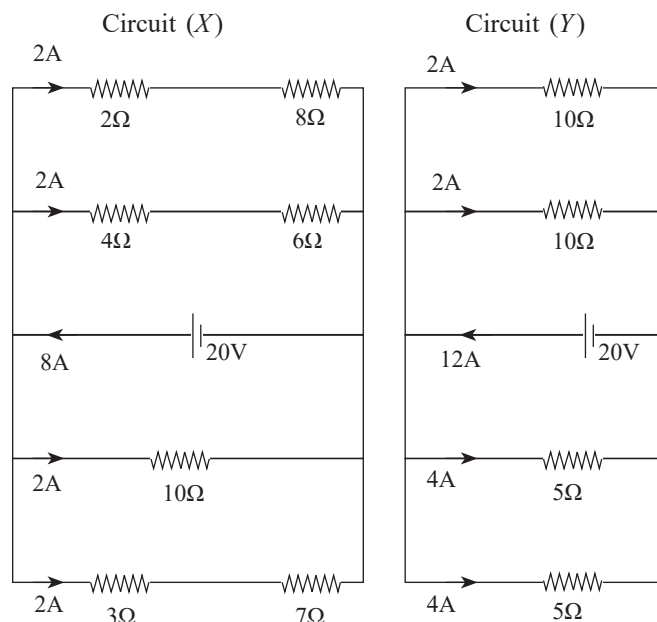


In (D), the circuit can be redrawn, as shown. There is a balanced Wheatstone bridge with another 2Ω resistance in parallel.

$$R_{eq} = 1\Omega$$

$$\therefore I_D = 12\text{A.}$$

6. In both the circuits, there are 4 parallel paths connected across the cell.



Equivalent resistance in circuit X is given

$$\text{by } \frac{1}{R_{eq}} = \frac{1}{10} + \frac{1}{10} + \frac{1}{10} + \frac{1}{10} \Rightarrow R_{eq} = 2.5\Omega$$

$$\text{Battery current } I_1 = \frac{20}{2.5} = 8\text{ A.}$$

This current is divided equally in all 4 paths.

$$\text{Power dissipated } P_1 = \frac{V^2}{R_{eq}} = \frac{20^2}{2.5} = 160\text{W}$$

In circuit Y , the equivalent resistance is given

$$\text{by } \frac{1}{R_{eq}} = \frac{1}{10} + \frac{1}{10} + \frac{1}{5} + \frac{1}{5} \Rightarrow R_{eq} = \frac{5}{3}\Omega$$

$$\text{Battery current is } I_2 = \frac{20}{R_{eq}} = \frac{20}{5/3} = 12\text{A}$$

Current in each of the 10Ω path is 2A and the current in each of the 5Ω paths is 4A.

$$\text{Power dissipated } P_2 = \frac{V^2}{R_{eq}} = \frac{20^2}{5/3} = 240\text{W}$$

7. (A) When connected in series, the cells make a battery of emf $\varepsilon_0 = 2x$

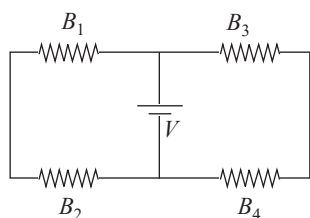
When connected in parallel, the effective emf is

$$\epsilon_0 = \frac{\frac{x}{\frac{r}{1} + \frac{r}{1}}}{\frac{r}{1} + \frac{r}{1}} = x.$$

(B) When connected in series, the rating of the combination will remain x .

When connected in parallel, the rating will become $2x$. This is because a $2x$ current will equally divide between the two fuses.

(C) With 'S' closed, bulbs B_1 and B_2 are in series connected to a V volt cell. Similarly, B_3 and B_4 are in series. Each bulb has a voltage of $\frac{V}{2}$ across it.

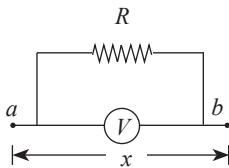


$$\text{Power in each bulb} = \frac{\left(\frac{V}{2}\right)^2}{R} = \frac{1}{4} \frac{V^2}{R} = \frac{x}{4}.$$

$\therefore$ Total power of combination = x .

With S closed, there is no potential difference between A and B . The bulbs are short circuited. No power is dissipated in them.

(D) Resistance of the voltmeter will be halved due to addition of the resistor in parallel. But the voltmeter will continue to read up to x volt only. When x volt potential difference is applied across the terminals, the full-scale deflection current will pass through the voltmeter.



8. In the first case, $Q_1 = C(2V) = 2CV$

$$U_1 = \frac{1}{2} C(2V)^2 = 2CV^2$$

Energy supplied by the cell, $E_1 = (2V)(Q_1) = 4CV^2$

Heat generated, $H_1 = E_1 - U_1 = 2CV^2$

In the second case:

Final charge on the capacitor $Q_2 = C(2V) = 2CV$

$$U_2 = \frac{1}{2} C(2V)^2 = 2CV^2$$

Energy supplied by the cell,

$$E_2 = (CV)(V) + (CV)(2V) = 3CV^2$$

[Note that the first cell supplies $q_1 = CV$ charge and the second cell supplies $q_2 = CV$ charge to make the final charge on the capacitor equal to $2CV$.]

Heat generated, $H_2 = E_2 - U_2 = CV^2$

9. $j = \sigma E$

$$I = kA = \sigma EA$$

Resistance of conductor of length l is $R = \frac{l}{\sigma A}$

$\therefore$ Power dissipated in length l is

$$P = I^2 R = \sigma^2 E^2 A^2 \frac{l}{\sigma A} = \sigma E^2 Al$$

For unit volume $Al = 1$

$$\therefore P = \sigma E^2$$

Note that σ depends on temperature.

10. With S open, two capacitors C_1 and C_2 are connected in series to a cell of emf V . Charge on each of the two capacitors is

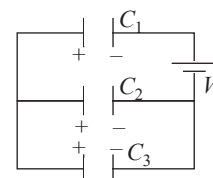
$$q_0 = \frac{CV}{2}.$$

$$\begin{aligned} \text{Energy stored in each of them is } U_{10} = U_{20} &= \frac{1}{2} C \left(\frac{V}{2} \right)^2 \\ &= \frac{1}{8} CV^2 \end{aligned}$$

After S is closed and a steady state is attained, the charges on the capacitors are

$$q_1 = \frac{2CV}{3}$$

$$q_2 = \frac{CV}{3}, \quad q_3 = \frac{CV}{3}$$



Potential differences across the three capacitors are

$$V_1 = \frac{2V}{3}, \quad V_2 = \frac{V}{3}, \quad V_3 = \frac{V}{3}$$

Energies stored in the three capacitors are

$$U_1 = \frac{1}{2} C \left(\frac{2V}{3} \right)^2 = \frac{2}{9} CV^2$$

$$U_3 = U_2 = \frac{1}{2} C \left(\frac{V}{3} \right)^2 = \frac{1}{18} CV^2$$

$$\text{Energy loss in } C_1 = U_{10} - U_1 = -\frac{7}{72} CV^2$$

$$\text{Energy loss in } C_2 = U_{20} - U_2 = \frac{5CV^2}{72}$$

$$\text{Energy gain in } C_3 = U_3 = \frac{1}{18} CV^2$$

The change in charge on the positive plate of C_1 is

$$\Delta q = \frac{2CV}{3} - \frac{CV}{2} = \frac{CV}{6}$$

$\therefore$ The cell supplied a charge $\frac{CV}{6}$ after S was closed.

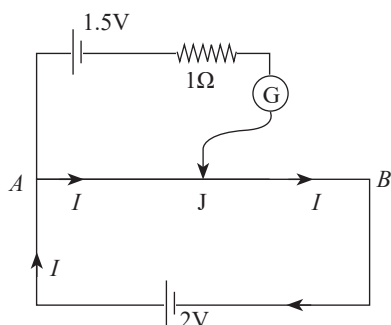
$$\text{Loss in energy of the cell} = W_{\text{cell}} = \Delta q \cdot V = \frac{CV^2}{6}$$



Passage Based Problem

Passage 1

- AB is the wire of length 10 m and resistance 30Ω . When the galvanometer shows zero deflection, the current (I) in the circuit is shown. Potential drop across wire AB is 2.0V.

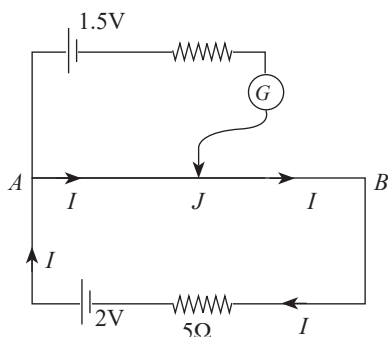


Potential gradient along AB is $\frac{2V}{10m} = 0.2V/m$.

Potential drop between A and J is 1.5V. Let the length AJ be x .

$$0.2x = 1.5 \quad \Rightarrow \quad x = 7.5 \text{ m}$$

$$2. \quad I = \frac{2V}{(30+5)\Omega} = \frac{2}{35} \text{ A}$$



Potential gradient along AB

$$= I \left(3 \frac{\Omega}{m} \right) = \frac{6}{35} \frac{V}{m}$$

Let the length AJ be x_1

$$\frac{6}{35} x_1 = 1.5 \quad \Rightarrow \quad x_1 = \frac{1.5 \times 35}{6} = 8.75 \text{ m}$$

Passage 2

- $V + V_R = 6$ (i)

When current $I = 0.4\text{A}$, the voltage drop across the rod is given by

$$I = 0.2V^{1/2}$$

$$\text{or, } 0.4 = 0.2 V^{1/2}$$

$$\Rightarrow V = 4 \text{ V}$$

$$\text{From (i), } V_R = 2\text{V}$$

$$\text{or, } IR = 2 \quad \Rightarrow \quad 0.4R = 2$$

$$\text{or, } R = 5\Omega$$

- Power dissipated in resistance $P_R = I^2 R$

Power dissipated in the rod $P_{\text{rod}} = I V_{\text{rod}}$

$$\text{or, } P_{\text{rod}} = I \left(\frac{I}{0.2} \right)^2 = \frac{I^3}{0.04}$$

$$\text{Given, } P_{\text{rod}} = 2\text{P}$$

$$\text{or, } \frac{I^3}{0.04} = 2I^2 R$$

$$\Rightarrow I = 0.08R \quad \text{(i)}$$

$$\text{Also, } P_{\text{rod}} + P_R = P_{\text{cell}}$$

$$\Rightarrow 2P_R + P_R = 6I$$

$$\Rightarrow P_R = 2I$$

$$\Rightarrow I^2 R = 2I$$

$$\Rightarrow IR = 2 \quad \text{(ii)}$$

Solving (i) and (ii), we get $R = 5\Omega$.

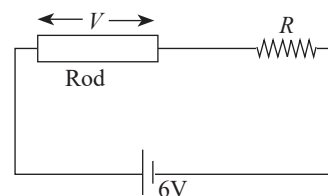
Passage 3

- When B_2 blows out, the other three bulbs will remain glowing. No other voltmeter will show zero reading.
- With B_2 glowing, the circuit is as shown in the figure.

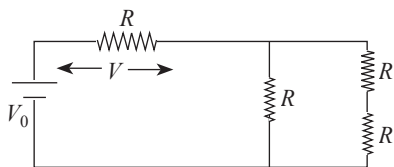
Equivalent of two parallel paths is $R_0 = \frac{2R}{3}$.

Potential drop across B_1 is

$$V = V_0 \frac{R}{R + \frac{2R}{3}} = \frac{3V_0}{5}$$



After the bulb B_2 blows out, the potential drop across B_1 becomes



$$V' = V_0 \frac{R}{R + 2R} = \frac{V_0}{3}$$

$$\therefore \frac{V}{2} < V' < V_1$$

Passage 4

7. Since there is no current in path FC , hence $V_C = V_F = 0V$.

Moving from C to E :

$$V_C - i(1) - 5 = V_E$$

[i = current in 1Ω resistor]

$$\text{or, } 0 - i - 5 = -5$$

$$\Rightarrow i = 0.$$

Moving from B to D :

$$V_B - 10 - 0 = V_D$$

$$\text{or, } V_B - V_D = 10V$$

$\therefore$ Voltmeter reads $10V$.

8. Moving from G to C gives

$$V_G + 5 - (i_1)(2) = V_C$$

[i_1 = current in 2Ω resistor above $5V$ cell]

$$\text{or, } 5 + 5 - 2i_1 = 0$$

$$\text{or, } i_1 = 5A.$$

Junction law at C tells us that the current i_1 takes the path CB . Then, this current takes the path BA . [The voltmeter has infinite resistance.]

$\therefore$ Current in A_2 is $5A$.

9. $V_C + 10 - (5)(1) = V_A$

$$\text{or, } 0 + 10 - 5 = V_A$$

$$\text{or, } V_A = 5V.$$

Passage 5

10. The rate of energy loss by cell = electrical power dissipated in the resistance = net heat and light energy lost by the conductor in steady state.

The temperature of the conductor becomes constant when the net rate of energy gain/loss becomes zero.

11. Resistance of conductor $R = \frac{\rho l}{A}$.

$$\text{Rate of heat loss} = Ks(\Delta T_1),$$

where K is a constant and s is surface area of the cylindrical surface.

In steady state:

$$\frac{V^2}{\frac{R}{2}} = K \frac{S}{2} (\Delta T_1) \quad (i)$$

After the conductor's length is halved, resistance becomes $\frac{R}{2}$ and surface area becomes $\frac{S}{2}$.

$$\therefore \frac{V^2}{\frac{R}{2}} = K \frac{S}{2} (\Delta T_2)$$

$$\text{or, } \frac{V^2}{R} = \frac{1}{4} KS (\Delta T_2) \quad (ii)$$

From (i) and (ii),

$$\Delta T_2 = 4\Delta T_1 = 24^\circ C$$

Passage 6

12. The device X obeys Ohm's law till current through it reaches $1A$. When current is $1A$, the potential difference across X is $10V$. Potential difference across 20Ω resistor is $20V$. Therefore, the source voltage is $30V$. When the current exceeds $1A$, the device X offers no resistance to change in current.

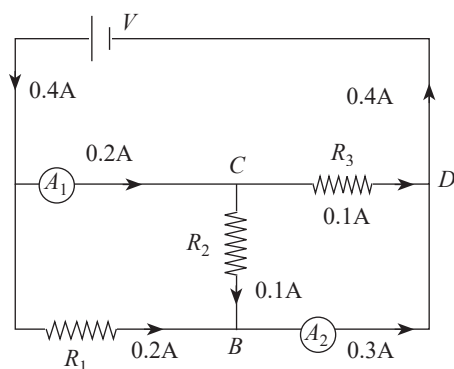
$$\frac{\Delta V}{\Delta I} = 0 \text{ for the device } X \text{ for } I > 1A.$$

The change in current, above $1A$, sees only 20Ω as effective resistance. The current rises more sharply when V is changed. Hence, graph in option (A) is correct.

13. When $V = 30V$, current is $1A$. Drop across X is $10V$ and the drop across 20Ω resistor is 20Ω . When the source voltage is increased beyond $30V$, the current increases beyond $1A$. But the voltage (V_1) across X remains constant.

Passage 7

14. The three resistors are in parallel. When $R_2 = R_3$, the current through both of them must be same. The circuit has been redrawn in a slightly different way for better understanding.



The 0.2 A current branches at C into two equal parts. Hence, current in R_2 as well as R_3 is 0.1A. Now, reading of ammeter A_2 will be 0.3A if current in R_1 is 0.2A.

Obviously, the battery current is 0.4A.

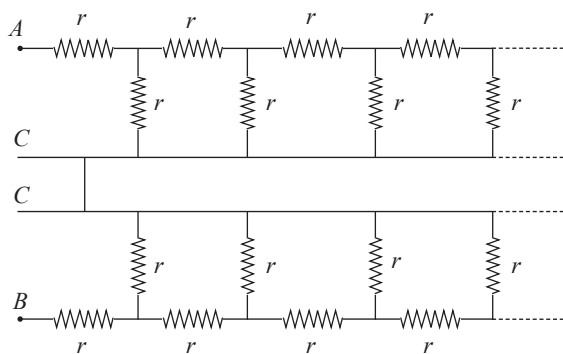
15. When $R_1 = 2\Omega$, the potential difference across it is $V = 0.2 \times 2 = 0.4\text{V}$.

Potential difference across R_2 is also 0.4 V. Current in it is 0.1 A.

$$\therefore R_2 = \frac{0.4}{0.1} = 4\Omega$$

Passage 8

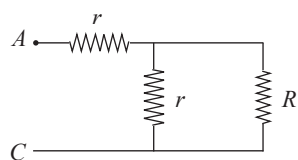
16. The given circuit can be realized by joining two identical circuits as shown.



$$V_A = 100\text{V}, V_B = 0\text{V}$$

Obviously, $V_C = 50\text{V}$

17. We will find the equivalent of upper circuit, shown in the figure between A and C. The lower circuit will have same equivalent between C and B.



Let the equivalent of upper part be R . If we remove one ladder and consider the remaining circuit, then the equivalent will be R .

$$\therefore \frac{rR}{r+R} + r = R$$

$$\text{or, } R^2 - (2r)R - r^2 = 0$$

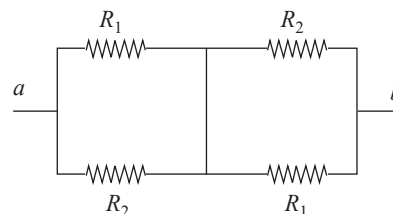
$$\text{or, } R = r \left(\frac{\sqrt{5}+1}{2} \right)$$

The equivalent between A and B will be

$$2R = (\sqrt{5}+1)r.$$

Passage 9

18. With $R_h = 0$, the network is as shown



$$\therefore R_{eq} = \frac{R_1 R_2}{R_1 + R_2} + \frac{R_1 R_2}{R_1 + R_2} = \frac{2R_1 R_2}{R_1 + R_2}$$

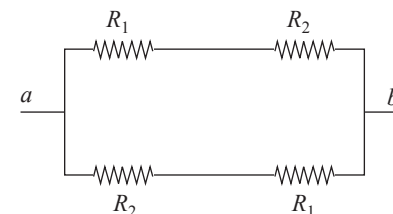
19. With $R_h = 0$;

$$IR_{eq} = V$$

$$\text{or, } 1 \left(\frac{2R_1 R_2}{R_1 + R_2} \right) = 3$$

$$\Rightarrow R_1 R_2 = 1.5(R_1 + R_2) \quad (a)$$

With $R_h = \infty$, the network is shown in the figure.



$$R_{eq} = \frac{(R_1 + R_2)}{2}$$

Current $I = 0.75\text{ A}$.

$$\therefore IR_{eq} = V$$

$$\text{or, } 0.75 \left(\frac{R_1 + R_2}{2} \right) = 3$$

$$\text{or, } R_1 + R_2 = 8$$

$$\text{From (a) } R_1 R_2 = 12$$

(b)

$$(R_1 - R_2)^2 = (R_1 + R_2)^2 - 4R_1R_2$$

$$\text{or, } (R_1 - R_2)^2 = 8^2 - 4 \times 12$$

$$\text{or, } R_1 - R_2 = \pm 4$$

$$\text{But } R_1 > R_2, \text{ hence } R_1 - R_2 = 4 \quad (\text{c})$$

Solving (b) and (c) gives

$$R_1 = 6\Omega, \quad R_2 = 2\Omega$$

Passage 10

20. In steady state, current in the circuit is $I = \frac{V}{2R}$ and the potential difference between A and B and, B and C is

$$V_{AB} = V_{BC} = \frac{V}{2}.$$

The charge on the balls maintain potential difference between them. In steady state, charge on them becomes constant and their sum will be zero. (The battery always supplies zero charge.)

21. Let charge on the ball connected to A be q_1 , that on the central ball be Q and the ball connected to C have q_2 charge on it.

$$V_A - V_B = \frac{V}{2}$$

$$\Rightarrow \frac{q_1}{4\pi\epsilon_0 a} - \frac{Q}{4\pi\epsilon_0 b} = \frac{V}{2}$$

$$\Rightarrow \frac{q_1}{a} - \frac{Q}{b} = 2\pi\epsilon_0 V \quad (\text{i})$$

$$\text{Similarly, } V_B - V_C = \frac{V}{2}$$

$$\Rightarrow \frac{Q}{4\pi\epsilon_0 b} - \frac{q_2}{4\pi\epsilon_0 a} = \frac{V}{2}$$

$$\Rightarrow \frac{Q}{b} - \frac{q_2}{a} = 2\pi\epsilon_0 V \quad (\text{ii})$$

$$\text{And } q_1 + Q + q_2 = 0 \quad (\text{iii})$$

Solving (i), (ii) and (iii) gives

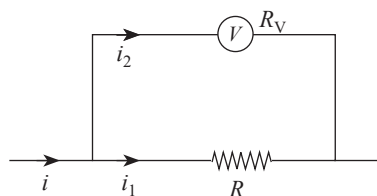
$$q_1 = 2\pi\epsilon_0 aV,$$

$$q_2 = -2\pi\epsilon_0 aV,$$

$$Q = 0$$

Passage 11

22.-23.



$$R_m = \frac{V}{i} \quad (\text{i})$$

$$V = i_1 R = i_2 R_V \quad (\text{ii})$$

$$\text{and } i_1 + i_2 = i \quad (\text{iii})$$

Hence i is ammeter reading and V is the voltmeter reading.

$$i_1 + i_2 = i$$

$$\text{or, } \frac{V}{R} + \frac{V}{R_V} = \frac{V}{R_m}$$

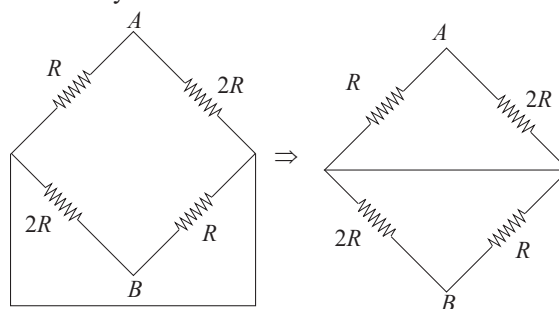
$$\text{or, } \frac{1}{R} = \frac{1}{R_m} - \frac{1}{R_V}$$

Obviously, $R > R_m$

And R does not depend on R_A .

Passage 12

24. R_0 = resistance between A and B when S is open and the battery is shorted.



$$R_0 = \frac{(2R)(R)}{2R + R} + \frac{(2R)(R)}{2R + R} = \frac{4R}{3}$$

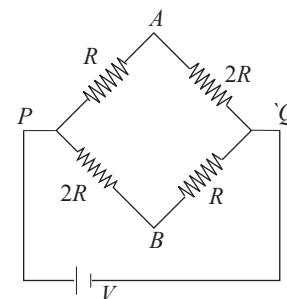
25. With switch S open,

$$V_P - V_A = \left(\frac{R}{R + 2R} \right) V$$

$$\text{or, } V_P - V_A = \frac{V}{3}$$

$$\text{and } V_P - V_B = \frac{2V}{3}$$

$$\therefore V_A - V_B = \frac{V}{3}$$

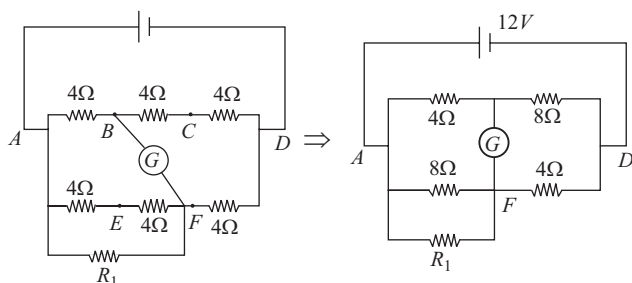


According to the question, $\epsilon = \frac{V}{3}$

$$\therefore I = \frac{\epsilon}{R + R_0} = \frac{\frac{V}{3}}{R + \frac{4R}{3}} = \frac{V}{7R}$$

Passage 13

26.

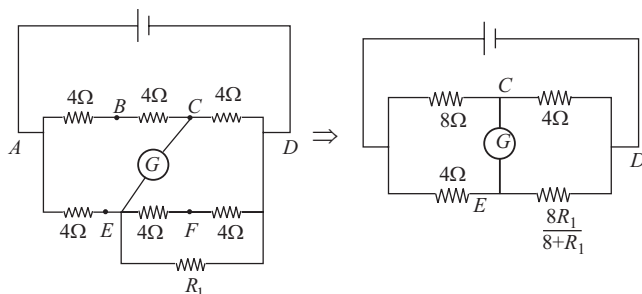


Equivalent of R_1 and 8Ω in parallel is $\frac{8R_1}{8+R_1}$. When the galvanometer shows 0 deflection, the circuit is a balanced Wheatstone bridge.

$$\frac{4}{8} = \frac{\frac{8R_1}{8+R_1}}{4} \Rightarrow 16 + 2R_1 = 8R_1$$

$$\text{or, } R_1 = \frac{8}{3}\Omega.$$

27.

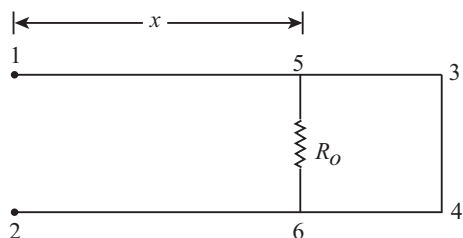


$$\frac{8R_1}{8+R_1} = \frac{8 \times \frac{8}{3}}{8 + \frac{8}{3}} = \frac{64}{32} = 2\Omega$$

The circuit is still a balanced Wheatstone bridge. Hence, galvanometer reading will be 0.

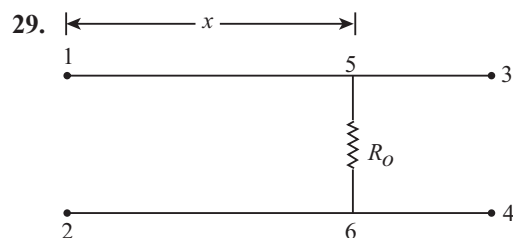
Passage 14

28. When terminals 3 and 4 are connected, the resistance of path 5-3-4-6 is in parallel with R_0 . Since R_0 is very large, the effective value of parallel combination will be nearly same as resistance of path 5-3-4-6.



$\therefore$ In this case, the resistance measured between the terminals 1 and 2 is the resistance of the cables.

$$\therefore R = 75\Omega.$$



$$\text{Resistance of segment 1 to 5 is } R_{15} = \frac{R \cdot x}{2L}$$

$\therefore$ Resistance between terminals 1 and 2 is

$$R_1 = R_{15} + R_0 + R_{26}$$

$$\text{or, } R_1 = \frac{Rx}{2L} + R_0 + \frac{Rx}{2L} = \frac{Rx}{L} + R_0 \quad (i)$$

Similarly, $R_{34} = R_{53} + R_0 + R_{64}$

$$\text{or, } R_2 = \frac{R}{2L}(L-x) + R_0 + \frac{R}{2L}(L-x)$$

$$= \frac{R(L-x)}{L} + R_0 \quad (ii)$$

Eliminating R_0 between (i) and (ii) gives

$$R_2L - R(L-x) = R_1L - R \cdot x$$

$$\text{or, } x = \frac{(R - R_2 + R_1)L}{2R} = \frac{(75 - 450 + 400) \times 1}{2 \times 75}$$

$$= \frac{1}{6} \text{ km}$$

30. Putting the value of x in (i) gives

$$R_0 = R_1 - \frac{Rx}{L} = 400 - \frac{75 \times \frac{1}{6}}{1} = 387.5\Omega$$

Passage 15

31. Let resistance of the galvanometer be R_g .

Given $R_g \ll 1000\Omega$

$$\therefore \text{ With } S_2 \text{ open, current is } i_0 = \frac{2V}{1000\Omega} = 2\text{mA}.$$

This is a full-scale deflection current. When 20Ω resistance is connected in parallel, the combined resistance of 20Ω and R_g will be smaller than R_g .

$$\therefore \text{ Again the battery current will be } i \simeq \frac{2}{1000} = 2\text{mA}.$$

But current through the galvanometer is 1 mA (as the deflection is half).

This will happen when $R_g = 20\Omega$.

The deflection in galvanometer changes by 20 div for a current of 2mA.

$\therefore$ Figure of merit is $\frac{2\text{mA}}{20\text{div}} = 0.1 \text{ mA/div}$

$$32. R = \frac{V}{i_g} - R_g = \frac{4.5}{2 \times 10^{-3}} - 20 = 2230\Omega.$$

Passage 16

$$33. i_1 = k_1 \alpha_1 \text{ and } i_2 = k_2 \alpha_2$$

When k_1 and k_2 are proportionality constants for the ammeters.

When they are in series $i_1 = i_2$

$$\Rightarrow k_1 \alpha_1 = k_2 \alpha_2$$

When they are in parallel $i_1 R = i_2 X$

$$\text{or, } k_1 \beta_1 R = k_2 \beta_2 X$$

If $\alpha_1 = 2\alpha_2$, then from (i) $2k_1 = k_2$.

Putting this in (ii) gives

$$\beta_1 R = 2\beta_2 X$$

$$\text{When } \beta_1 = 2\beta_2, R = X$$

34. Dividing (ii) by (i) gives

$$\frac{\beta_1 R}{\alpha_1} = \frac{\beta_2 X}{\alpha_2}$$

$$\text{or, } X = R \frac{\alpha_2}{\alpha_1} \frac{\beta_1}{\beta_2}$$

Passage 17

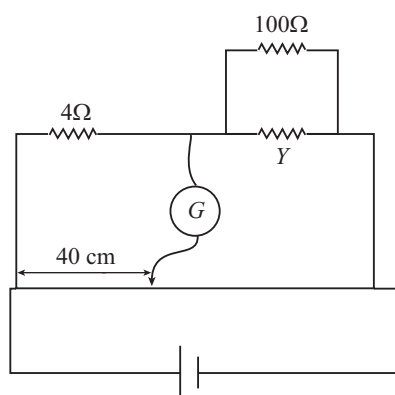
35. With rise in temperature, the resistance Y increases.

$\therefore$ Null point will move towards B .

36. Let R_0 = resistance of Y at 0°C

$$\frac{X}{R_0} = \frac{40}{60}; \text{ or, } R_0 = 4 \times \frac{6}{4} = 6\Omega$$

Let R be resistance of Y at 100°C . For balance at 100°C



$$\frac{100R}{100+R} = \frac{60}{40}$$

$$\text{or, } \frac{100R}{100+R} = 6$$

$$\text{or, } R = 6.38\Omega$$

$$\text{or, } R_0 [1 + \alpha \Delta\theta] = 6.38$$

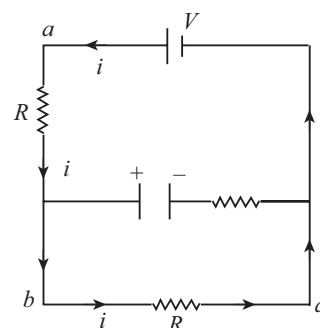
$$\text{or, } 6 [1 + \alpha \times 100] = 6.38$$

$$\text{or, } \alpha = 6.3 \times 10^{-4} \text{ K}^{-1}$$

Passage 18

37. – 38.

a and b both points are at higher potential than d . The balance point will be to the left of C in both cases.



$$V_{ab} = V_{bd} = \frac{V}{2}$$

$$V_{ad} = V$$

$$\text{And } V_{bd} = \frac{V}{2}$$

$\therefore$ Balance point will be at a distance $\frac{60\text{cm}}{2} = 30\text{cm}$ from C when terminal P is connected to b .

Passage 19

39. Let R_0 = resistance of the potentiometer wire. When S is put on, the potential difference across AB will be

$$E \left(\frac{R_0}{R_0 + r} \right)$$

$$\therefore E \left(\frac{R_0}{R_0 + r} \right) = \frac{15E}{16} \Rightarrow R_0 = 15r.$$

Resistance of 560 cm length of the wire is

$$\frac{560}{600} \times 15r = 14r.$$

The effective circuit is as shown in the figure. Note that the current through the galvanometer may not be zero. Writing loop equations gives:

$$14r(i_1 + i_2) + i_1 r = \frac{E}{2}$$

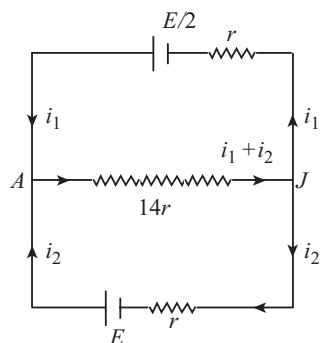
$$\text{or, } 15 i_1 + 14 i_2 = \frac{E}{2r} \quad (\text{i})$$

$$\text{And } 14r(i_1 + i_2) + i_2 r = E$$

$$\text{or, } 14 i_1 + 15 i_2 = \frac{E}{r} \quad (\text{ii})$$

Solving (i) and (ii) gives,

$$i_1 = -\frac{13E}{58r}$$



Negative sign indicates that the current is opposite to the direction shown in the figure.

40. Obviously, J_0 is the null point. In this position

$$V_{AJ} = \frac{E}{2}$$

The current in the potentiometer circuit is independent of the external circuit for this position.

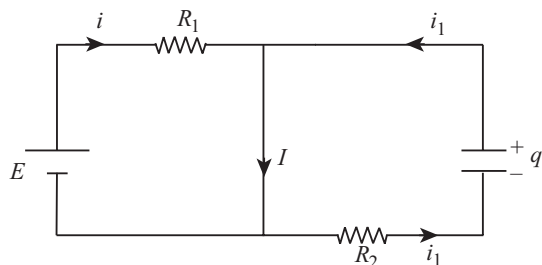
Total resistance of loop = $15r + r = 16r$.

For $V_{AJ} = \frac{E}{2}$, R_{AJ} must be $8r$.

If l is length AJ , then $\frac{l}{600} \times 15r = 8r \Rightarrow l = 320$ cm.

Passage 20

41. After the switch is closed, the potential difference across R_1 becomes E (= a constant). Therefore, the current through R_1 will be $i = \frac{E}{R_1}$ at all time after S is closed.
42. Initial potential difference across the capacitor is E and charge on it is $q_0 = EC$.



The potential difference across R_2 will always be that across the capacitor.

$$\text{Also, } i_1 = \frac{dq}{dt}$$

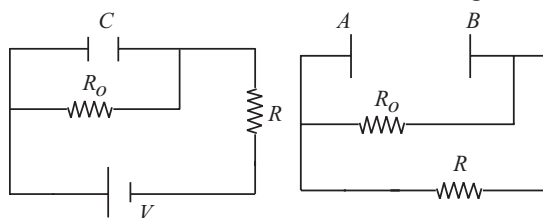
The capacitor is discharging through R_2 only.

$$\therefore i_1 = i_{10} e^{-t/\tau} = \frac{E}{R_2} e^{-\frac{t}{R_2 C}}$$

$$I = i + i_1 = \frac{E}{R_1} + \frac{E}{R_2} e^{-\frac{t}{R_2 C}}$$

Passage 21

43. The effective circuit is as shown in the figure.

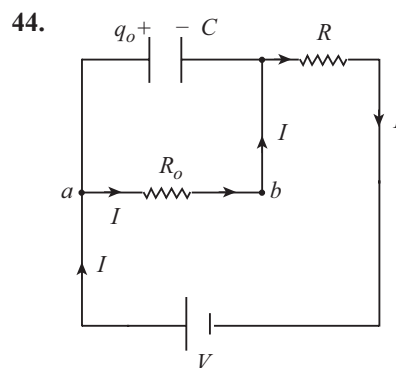


To get the effective time constant, we short circuit the cell and find the equivalent resistor across the capacitor plates A and B .

$$R_{AB} = \frac{R_0 R}{R + R_0} = \frac{R}{\frac{R}{R_0} + 1} = \frac{R}{\beta + 1}$$

$$\therefore \tau = R_{AB} \cdot C = \frac{RC}{\beta + 1}$$

The capacitor will acquire 63% charge at $t = \tau$.



After the capacitor is fully charged, the current in the circuit will be

$$I = \frac{V}{R_0 + R}$$

$$\therefore V_{ab} = R_0 I = \frac{R_0 V}{R_0 + R} = \frac{V}{1 + \beta}$$

$$\therefore \text{Final charge on the capacitor is } q_0 = CV_{ab} = \frac{VC}{1 + \beta}$$

$$\therefore q = q_0 \left[1 - e^{-\frac{t}{T}} \right] = \frac{VC}{1+\beta} \left[1 - e^{-\frac{(\beta+1)t}{RC}} \right]$$

Passage 22

45. $C_0 = \frac{\epsilon_0 A}{d_0}$

At time t , $C = \frac{\epsilon_0 A(1+t)}{d_0}$

or, $C = C_0 (1+t)$ for $0 \leq t \leq 1$

At any time t (≤ 1 s)

$$iR = \frac{q}{C}$$

or, $-\frac{dq}{dt} = \frac{q}{RC}$

or, $\frac{dq}{q} = -\frac{dt}{RC_0(1+t)}$

or, $\int_{Q_0}^q \frac{dq}{q} = -\frac{1}{RC_0} \int_0^t \frac{dt}{1+t}$

or, $\ln\left(\frac{q}{Q_0}\right) = -\frac{1}{RC_0} [\ln(1+t)]_0^t$

or, $\ln\left(\frac{q}{Q_0}\right) = -\frac{1}{RC_0} \ln(1+t)$

or, $\frac{q}{Q_0} = (1+t)^{-\frac{1}{RC_0}}$

or, $q = Q_0 (1+t)^{-\frac{1}{RC_0}}$

At $t = 1$, $q_0 = Q_0 (2)^{-\frac{1}{RC_0}}$ (i)

46. At $t = 1$, $C = 2C_0$

Charge on the capacitor at $t = 1$ is q_0 and it discharges with time constant $\tau = RC = 2RC_0$

$$\therefore q = q_0 e^{-\frac{(t-1)}{\tau}} = \frac{Q_0}{(2)^{\frac{1}{RC_0}}} e^{-\frac{(t-1)}{2RC_0}}$$

Passage 23

47. – 48.

Consider $t = 0$ when the capacitor has got discharged and the device has turned nonconducting. Now the capacitor charges according to the equation

$$V_C = V_0 \left[1 - e^{-\frac{t}{RC}} \right]$$

$$V_C = \frac{V_0}{2} \text{ when}$$

$$1 - e^{-\frac{t}{RC}} = \frac{1}{2}$$

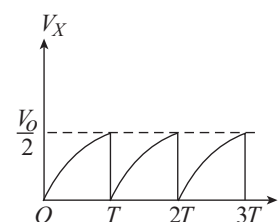
$$\Rightarrow t = (RC) \ln 2$$

Once the capacitor

acquires voltage $= \frac{V_0}{2}$, it gets discharged very fast through X .

The graphical variation of $V_C (= V_X)$ with time is as shown in the figure. The time period of this voltage wave form is

$$T = (RC) \ln 2$$

**Passage 24**

49. – 50.

50% charging of C_1 and C_2 is accomplished in same time t_0 . This implies that both the circuits have same time constants.

$$\tau_1 = \tau_2$$

$$\text{or, } RC_1 = RC_2 \Rightarrow C_1 = C_2$$

Final charge on C_2 is double that on C_1 . This implies

$$C_2 V_2 = 2C_1 V_1$$

$$\text{or, } V_2 = 2V_1$$

Now 50% charging of C_3 takes double the time for C_1 . This means

$$\tau_3 = 2\tau_1$$

$$\Rightarrow RC_3 = 2RC_1$$

$$\text{or, } C_3 = 2C_1$$

Final charge on C_3 is double that on C_1

$$\therefore V_3 C_3 = 2V_1 C_1$$

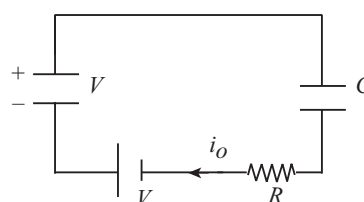
$$\text{or, } V_3 = V_1$$

Passage 25

When A is positive

The potential difference between the plates of capacitor C_1 is V_1 initially. There is no charge on C_2 and C_3 . Using Kirchhoff's loop law, we get

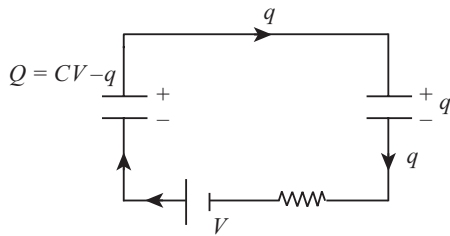
$$i_0 \cdot R = 2V \Rightarrow i_0 = \frac{2V}{R}$$



After a long time, the current in the circuit becomes zero. Let charge on the combination of C_2 and C_3 be q . Then charge on C_1 will be $CV - q$. Again using Kirchhoff's law, we get

$$\frac{CV - q}{C} - \frac{q}{C} + V = 0$$

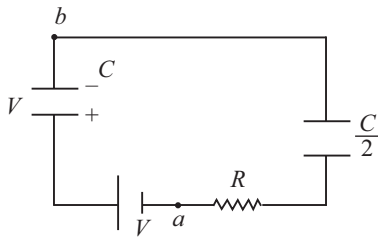
or, $q = CV$



When B is positive

The potential difference between a and b is zero. Obviously, there is no current in the circuit after the switch is closed.

There is no movement of charge in the circuit.



Passage 26

55. – 56.

When the jockey is at C , there is no potential difference between the plates. The electron moves with constant velocity along y direction given by

$$u = \frac{L}{t_0} \quad (i)$$

When the jockey is at A , plate P_1 is at a potential $\frac{V}{2}$ higher than plate P_2 . The electric field between the plates is $E = \frac{V}{2d}$ (in positive x direction)

The electron experiences a force in negative x direction and its acceleration is

$$a_x = \frac{eE}{m} = \frac{Ve}{2dm} \quad (ii)$$

At $t = \frac{t_0}{2}$, the jockey is moved to B . Now, plate P_2 is at higher potential. The acceleration of the electron is now in positive x direction given by (ii).

Since the velocity component in y direction remains unchanged, the time needed for the electron to hit the screen is still t_0 .

x component of velocity increases from zero to $-v_x$ in time $\frac{t_0}{2}$ and then it decreases to zero in further time $\frac{t_0}{2}$. Therefore, the electron hits the screen normally.

As discussed above, the velocity of electron in x direction increases in $-x$ direction and then decreases to zero. The electron always moved leftward after starting. It will hit the screen to the left of point S .

57. Initial potential difference between the plates = 0.

The jockey moves from C to A in time $\frac{t_0}{2}$. The potential difference between P_1 and P_2 increases linearly from zero to $\frac{V}{2}$ in time $\frac{t_0}{2}$.

$\therefore$ Potential difference $V(t)$ at time t ($0 \leq t \leq \frac{t_0}{2}$) is

$$V(t) = \left(\frac{\frac{V}{2}}{\frac{t_0}{2}} \right) t = \frac{V}{t_0} \cdot t$$

Electric field between the plates is $\vec{E} = \left(\frac{V}{d \cdot t_0} t \right) \hat{i}$

Acceleration of the electron in x direction is

$$a_x = -\frac{eV}{md \cdot t_0} t$$

$\therefore$ x component of velocity of the electron at time $\frac{t_0}{2}$ is

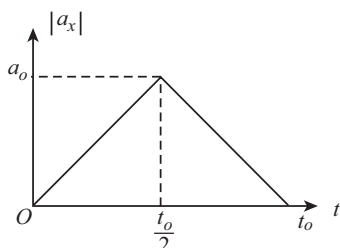
$$\begin{aligned} v_x &= \int_0^{\frac{t_0}{2}} a_x dt = -\frac{eV}{md t_0} \int_0^{\frac{t_0}{2}} t dt \\ &= -\frac{eV t_0}{8md} \end{aligned} \quad (iii)$$

Now, the jockey is moved from A to C in next interval $\frac{t_0}{2}$. The acceleration decreases linearly to zero. Obviously, the velocity v_x changes by same amount as (iii) in next interval $\frac{t_0}{2}$.

$\therefore$ Final value of v_x is $v_x = -\frac{eV t_0}{4md}$

Alternate Method

In fact, the magnitude of acceleration of the electron changes with time as shown in the graph.



$$a_0 = \frac{eV}{2md}$$

Change in velocity is equal to the area under the acceleration time graph.

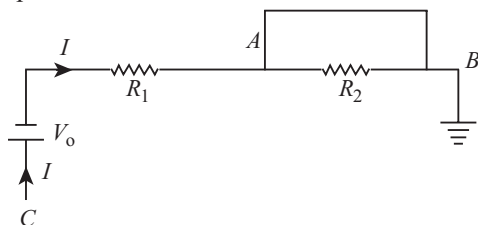
$$\therefore v_{x2} = \frac{1}{2} t_0 a_0 = \frac{eV t_0}{4md}$$

$\therefore$ Required velocity is given as

$$\vec{v} = -\left(\frac{eV t_0}{4md}\right) \hat{i} + \left(\frac{1}{t_0}\right) \hat{j}$$

Passage 27

58. The circuit given in option (A) appears to be most appropriate.



In experiment III, the resistance R_2 is short circuited. When the current I is increased, the drop in R_1 increases linearly and the potential difference V_{CB} increases (as shown in the graph).

Option (B) shows that R_1 is short circuited. If R_2 is also shorted, then the potential difference V_{CB} will become constant, which is equal to V_0 . This is not what we see in the graph of experiment III.

In option (C), $V_C < V_A$. With an increase in I , the drop in R_1 will increase and the potential difference $V_C - V_B$ will increase from $-V_0$ to become zero and then it will become positive. This is not what we see in the third graph.

Option (D) can be ruled out simply because in this case after A and B are shorted, the potential difference $V_{CB} = IR_1$. For $I = 0$, V_{CB} should be zero. This is not what we see in experiment III.

Therefore, we conclude that the circuit in option (A) is most appropriate.

59. Again consider experiment III.

When $I = 0$, $V_{CB} = 3V$

This implies that $V_0 = 3V$.

When $I = 1A$, $V_{CB} = 5V$

$$\Rightarrow IR_1 = 2$$

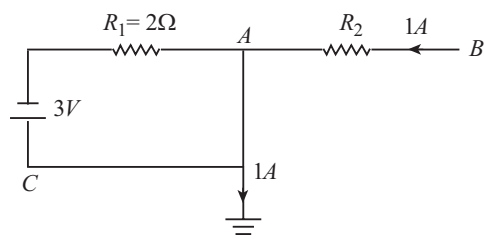
$$\Rightarrow R_1 = 2\Omega$$

60. In experiment II, when $I = 1A$, $V_{AB} = 6V$

Obviously,

$$IR_2 = 6$$

$$\text{or, } R_2 = 6\Omega$$

**Passage 28**

$$61. \frac{dQ}{dt} + \frac{1}{RC} Q = \frac{V}{R}$$

$$\text{or, } e^{\frac{t}{RC}} \frac{dQ}{dt} + \frac{1}{RC} e^{\frac{t}{RC}} \cdot Q = \frac{V}{R} e^{\frac{t}{RC}}$$

$$\text{or, } \frac{d}{dt} \left(Q \cdot e^{\frac{t}{RC}} \right) = \frac{V}{R} e^{\frac{t}{RC}}$$

$$\text{or, } \int d \left(Q \cdot e^{\frac{t}{RC}} \right) = \frac{1}{R} \int V e^{\frac{t}{RC}} dt$$

$$\text{or, } Q \cdot e^{\frac{t}{RC}} = \frac{1}{R} \int V e^{\frac{t}{RC}} dt$$

$$\text{or, } Q = \frac{e^{-\frac{t}{RC}}}{R} \int V e^{\frac{t}{RC}} dt$$

$$\therefore V_C = \frac{Q}{C} = \frac{e^{-\frac{t}{RC}}}{RC} \int V e^{\frac{t}{RC}} \cdot dt$$

$$\Rightarrow a = 1$$

62. When $t \ll RC$, then $\frac{t}{RC} \rightarrow 0$ and $e^{\frac{t}{RC}} \rightarrow 1$

$$\therefore V_C = \frac{1}{RC} \int V dt$$

Passage 29

63. If you imagine a slice of the conductor having the length L , it will be a uniform wire (having same σ everywhere). Electric field, throughout the length of the wire will be

$$E = \frac{V}{L}$$

Any other slice will also be a uniform wire (though it may have a different value of σ than the previous slice).

The electric field, everywhere inside this wire, will also be $E = \frac{V}{L}$.

Ohm's law says $j = \sigma E$.

It means, the current density will vary with θ as σ charges (though E is constant over a cross-section).

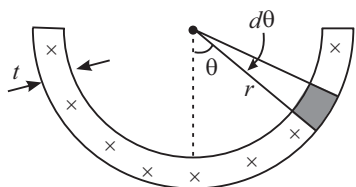
$$j = \sigma_0 E \cos \theta$$

with increasing θ , j decreases.

64. Consider a width $rd\theta$ on the semicircular section. Current through this width is

$$dI = j(dA) = j(rd\theta \cdot t)$$

$$\text{or, } dI = E\sigma_0 rt \cos \theta d\theta = \frac{V\sigma_0 rt}{L} \cos \theta \cdot d\theta$$



The current through the entire section is

$$I = \left[\frac{V\sigma_0 rt}{L} \int_0^{\frac{\pi}{2}} \cos \theta d\theta \right] \times 2 = \frac{2V\sigma_0 rt}{L}$$

$$\therefore \text{Resistance, } R = \frac{V}{I} = \frac{L}{2\sigma_0 rt}$$

Alternatively, one can work out the resistance by writing resistances of thin wire like strips of length L and obtaining their equivalent by treating them to be in parallel. Resistance of the strip shown in the above figure is

$$dR = \frac{L}{\sigma(rd\theta)t} = \frac{L}{rt\sigma_0 \cos \theta \cdot d\theta}$$

Equivalent resistance is obtained as

$$\begin{aligned} \frac{1}{R} &= \int \frac{1}{dR} = \frac{\sigma_0 rt}{L} \int_0^{\frac{\pi}{2}} \cos \theta d\theta \times 2 \\ &= \frac{2\sigma_0 rt}{L} \end{aligned}$$

$$\therefore R = \frac{L}{2\sigma_0 rt}$$

Chapter 10 Error and Measurement



Your Turn

3. (a) In a number less than 1, all leading zeros are not significant.
- (b) In a number with decimal, trailing zeros are significant.
- (c) Leading zeros should not be counted, trailing ones should be counted.

0.00 1200
↓
Significant

- (d) In a number without decimal all trailing zeros are not significant.
 - (f) In a number written in scientific notation, all digits are significant.
4. We wish to express that there are three significant digits in our measurement. We can express it in many ways. For example,
 5.00×10^2 mm, or 50.0 cm, or 0.500 m
 5. Your friend will record a reading smaller than 70.0 kg due to parallax.
 6. Reported value = Reading – zero error
 $= 12.34 - 0.01 = 12.33\text{A}.$

$$7. \bar{x} = \frac{2.0 + 2.4 + 2.2}{3} = 2.2\text{A}$$

Mean deviation is

$$\bar{\delta} = \frac{|2.0 - 2.2| + |2.4 - 2.2| + |2.2 - 2.2|}{3}$$

$$= \frac{0.4}{3} \simeq 0.1$$

∴ The measurement will be reported as $(2.2 \pm 0.1)\text{A}.$

8. Current, $I = (4.0 \pm 5\%)\text{A}$

Uncertainty, $\Delta I = 5\%$ of 4

$$= \frac{5}{100} \times 4 = 0.2\text{ A}$$

It means that most likely the current is in the range $(4.0 \pm 0.2)\text{ A}.$

Next measured value is likely to be between 3.8A and 4.2A.

9. (a) $12362 = 12400$

↓

digit to be
dropped > 5

- (b) $19.845 = 19.8$

↓

digit to be
dropped < 5

- (c) $23.750 = 23.8$ [$\because 7$ is odd]

↓

digit to be
dropped

- (d) $45.650 = 45.6$ [$\because 6$ is even]

↓

digit to be
dropped

10. Total mass = $(3.3 + 0.00315 + 0.02328)\text{ g}$
 $= 3.32643\text{ g} = 3.3\text{ kg}$

We will report the answer upto one digit after decimal.

11. $V = \frac{120.5}{4\text{ sd}} \times \frac{25.0}{3\text{ sd}} \times \frac{10}{1\text{ sd}}$
 $= 30125\text{ cm}^3$
 $= 30000\text{ cm}^3$ [Rounded off to 1 sd]

12. $T = 2 \times 3.14 \sqrt{\frac{1.00}{9.80}} = 2.00\text{ s}$

$$T = \frac{2\pi}{\sqrt{g}} \sqrt{l} = Kl^{1/2}$$

$$\therefore \frac{\Delta T}{T} \times 100 = \frac{1}{2} \frac{\Delta l}{l} \times 100$$

$$= \frac{1}{2} \times \frac{0.01}{1.00} \times 100 = 0.5\%$$

13. $T = 2\pi \sqrt{\frac{l}{g}} \Rightarrow g = 4\pi^2 \frac{l}{T^2}$

$$\therefore g = 4 \times (3.14)^2 \times \frac{1.00}{(2.00)^2} = 9.86\text{ ms}^{-2}$$

$$\frac{\Delta g}{g} = \frac{\Delta l}{l} + 2 \frac{\Delta T}{T} = \frac{0.01}{1.00} + 2 \times \frac{0.05}{2.00} = 0.06$$

$$\frac{\Delta g}{g} \times 100 = 6\%$$

$$14. K = \frac{P^2}{2m} \Rightarrow \frac{\Delta K}{K} \times 100 = 2 \frac{\Delta P}{P} \times 100$$

$$= 2 \times 0.2 = 0.4\%$$

$$15. \frac{\Delta A}{A} = 2 \frac{\Delta x}{x} + 3 \frac{\Delta y}{y} + 5 \frac{\Delta z}{z}$$

Error in z has largest effect on the error in A . Hence we should try to measure z most accurately.

$$17. \text{Zero error} = +6 \times 0.1 = +0.6 \text{ mm}$$

$$18. \text{Reading} = 43 \text{ mm} + 2(\text{LC})$$

$$= 43 + 2 \times 0.1 = 43.2 \text{ mm}$$

$$\text{Thickness} = \text{Reading} - \text{Zero error}$$

$$= 43.2 - 0.6 = 42.6 \text{ mm}$$

19. There is negative zero error in the instrument. 7th division from left is same as 3rd division from the right.

$$\text{Zero error} = -3 \times \text{LC} = -0.3 \text{ mm}$$

Reading with wire between the jaws is

$$= 4 \text{ mm} + 7 \times 0.1 = 4.7 \text{ mm}$$

Zero error is negative means that the instrument reads less than the true value.

$$\therefore \text{Diameter} = 4.7 + 0.3 = 5.0 \text{ mm}$$

$$20. \text{LC} = \frac{2 \text{ mm}}{50} = 0.04 \text{ mm}$$

21. Zero on the main scale is not visible. It means, the instrument has negative zero error. The circular scale has turned by 40 divisions [= 100 – 60] after its edge went past the zero mark on the reference line.

$$\therefore \text{zero error} = -40(\text{LC}) = -0.40 \text{ mm.}$$

$$22. \text{Reading} = 12 \text{ mm} + 28 \times 0.01 = 12.28 \text{ mm}$$

$$\text{Thickness} = 12.28 + 0.40 = 12.68 \text{ mm}$$

The instrument reads 0.40 mm less than the actual. Hence we need to add this to our reading.



Worksheet 1

$$1. V = (1.2 \times 10^{-2})^3 = 1.728 \times 10^{-6}$$

$$= 1.7 \times 10^{-6} \text{ m}^3 \text{ [Rounded to 2 significant digits]}$$

$$2. 1 \text{ VSD} = \left(\frac{n}{n+1} \right) \text{MSD} = \frac{na}{n+1}$$

$$\therefore \text{LC} = 1 \text{ MSD} - 1 \text{ VSD} = a - \frac{na}{n+1} = \frac{a}{n+1}$$

$$3. d = \frac{M}{\pi r^2 L}$$

$$\frac{\Delta d}{d} \times 100 = \left(\frac{\Delta M}{M} + 2 \frac{\Delta r}{r} + \frac{\Delta L}{L} \right) \times 100$$

$$= \left(\frac{0.003}{0.3} + 2 \times \frac{0.005}{0.5} + \frac{0.06}{6} \right) \times 100$$

$$= 1 + 2 + 1 = 4\%$$

$$4. \text{LC} = 1 \text{ MSD} - 1 \text{ VSD}$$

$$\text{or, } 0.02 \text{ cm} = 0.1 \text{ cm} - 1 \text{ VSD}$$

$$\text{or, } 1 \text{ VSD} = 0.08 \text{ cm}$$

This is possible with $n = 10$ and $m = 0.8 \text{ cm}$

$$5. \text{LC} = 1 \text{ MSD} - 1 \text{ VSD}$$

LC is smallest when 1 VSD is largest. 1 VSD will be largest when $m = 1$ (i.e., when there are $N + 1$ divisions on the vernier coinciding with N MSD)

$$8. 8.34.551 = 834.6$$

↓

digit to be dropped

There is a non-zero digit (1) to the right of 5 to be dropped. Hence preceding digit was increased by 1.

$$9. 834.501 = 834.5$$

↓

digit to be dropped is less than 5

$$10. 834.50 = 834$$

↓

digit to be dropped

even number

$$11. l = (10.0 \pm 0.1) \text{ cm}$$

$$T = \frac{(60 \pm 1) \text{ s}}{100} = (0.6 \pm 0.01) \text{ s}$$

$$g = 4\pi^2 \frac{l}{T^2}$$

$$\Rightarrow \frac{\Delta g}{g} = \frac{\Delta l}{l} + 2 \frac{\Delta T}{T} = \frac{0.1}{10} + 2 \cdot \frac{0.01}{0.6}$$

$$\text{or, } \frac{\Delta g}{g} \times 100 = 1 + 3.33 = 4.3\%$$

$$12. R = (4.22 \pm 2\%) \text{ cm}$$

$$V = \frac{4}{3} \pi R^3 = \frac{4}{3} \times 3.14 \times (4.22)^3 = 314.63$$

$$\therefore V = 315 \text{ cm}^3 \text{ [Rounded off to 3 digits]}$$

$$\frac{\Delta V}{V} \times 100 = 3 \frac{\Delta R}{R} \times 100 = 3 \times 2 = 6\%$$

13. Measurement of R needs to be most accurate as it has maximum power.

14. 4th vernier division coincides with a main scale division and the main scale reading is zero.

$$\therefore \text{Zero error} = \text{excess reading by the instrument} = +4(\text{LC}) = +0.4 \text{ mm}$$

15. The instrument has negative zero error. 6th division of vernier coincides with a main scale division. Counting from the right it is the 4th vernier division that coincides with main scale division.

$$\text{Zero error} = -4 \times \text{LC} = -0.4 \text{ mm}$$

16. $\text{LC} = \frac{1 \text{ mm}}{100} = 0.01 \text{ mm}$

Zero of main scale is not visible. The zero error is negative. The zero of the circular scale has moved ahead by 3 divisions.

$$\therefore \text{Zero error} = -3 \times \text{LC} = -0.03 \text{ mm}$$

17. Zero error is positive $= +2 \times \text{LC} = +0.02 \text{ mm}$

18. Reading = MSR + LC(CSR)
 $= 2 \text{ mm} + 0.01(44) = 2.44 \text{ mm}$

$$\begin{aligned} \text{Thickness} &= 2.44 - \text{zero error} \\ &= 2.44 - 0.02 = 2.42 \text{ mm} \end{aligned}$$



Worksheet 2

1. $\text{LC} = 1 \text{ MSD} - 1 \text{ VSD} = 1 \text{ mm} - \frac{19}{20} \text{ mm} = \frac{1}{20} \text{ mm}$

$$= 0.05 \text{ mm}$$

$$= 0.005 \text{ cm}$$

$$\begin{aligned} \text{Diameter} &= 35 \text{ mm} + 4(\text{LC}) \\ &= 35 \text{ mm} + 4(0.05) \text{ mm} \\ &= 35.20 \text{ mm} \end{aligned}$$

2. P.O. box can measure upto more number of significant digits. Accuracy of a measurement is not exactly related to precision.

3. When a resistance is connected in parallel to a galvanometer, the range of the instrument increases, the sensitivity drops.

Whether the resistance added in parallel is larger than or smaller than R , it will increase the range of the instrument.

When a resistance is added in series, it will not change the range of the instrument.

4. When the potential gradient is high, the sensitivity of the instrument drops and uncertainty increases.

We take measurement when galvanometer shows zero deflection. Simply having more divisions makes no difference.



Worksheet 3

1. (a) Mean length $\bar{x} = \frac{2.50 + 2.54 + 2.49 + 2.58 + 2.57 + 2.49}{6}$
 $= 2.528 \approx 2.53 \text{ m}$

(b) Mean deviation

$$\begin{aligned} \Delta x &= \frac{|2.50 - 2.53| + |2.54 - 2.53| + |2.49 - 2.53| + |2.58 - 2.53| \\ &\quad + |2.57 - 2.53| + |2.49 - 2.53|}{6} \end{aligned}$$

$$= \frac{0.03 + 0.01 + 0.04 + 0.05 + 0.04 + 0.04}{6}$$

$$= 0.035$$

$$(c) \text{Percentage uncertainty (error)} = \frac{\Delta x}{\bar{x}} \times 100$$

$$= \frac{0.035}{2.53} \times 100 = 1.38\%$$

2. $Y = \frac{MgL}{\pi r^2 l} = \left(\frac{4g}{\pi} \right) \frac{ML}{d^2 l} \quad \left[\because r = \frac{d}{2} \right]$

$$\therefore \frac{\Delta Y}{Y} = \frac{\Delta M}{M} + \frac{\Delta L}{L} + 2 \cdot \frac{\Delta d}{d} + \frac{\Delta l}{l}$$

$$\% \frac{\Delta Y}{Y} = \left[\frac{0.01}{3.00} + \frac{0.001}{2.890} + 2 \cdot \frac{0.001}{0.041} + \frac{0.01}{0.87} \right] \times 100$$

$$= \frac{1}{3} + \frac{0.1}{2.89} + \frac{2(0.1)}{0.041} + \frac{1}{0.87}$$

$$= 0.333 + 0.035 + 4.878 + 1.149$$

$$= 6.395 = 6.40\%$$

4. Zero error $= +7(\text{LC}) = 0.7 \text{ mm} = 0.07 \text{ cm}$

$$\text{Reading} = \text{Main scale reading} + \text{Vernier scale reading}$$

$$= 3.2 + 4 \times 0.01 \text{ cm}$$

$$= 3.24 \text{ cm}$$

$$\text{True diameter} = 3.24 - \text{Zero correction}$$

$$= 3.24 - 0.07 = 3.17 \text{ cm}$$

5. Zero error $= -0.04 \text{ cm}$ [4th division from right coincides]

$$\text{Length of bar} = \text{Reading} - \text{Zero error}$$

$$\text{or, } 4.32 \text{ cm} = \text{Reading} + 0.04 \text{ cm}$$

$$\text{or, Reading} = 4.28 \text{ cm}$$

$$\therefore \text{Main scale reading} = 4.2 \text{ cm}$$

$$\text{Vernier scale reading} = 0.08 \text{ cm}$$

$\therefore$ 8th vernier division is coinciding with a main scale division.

6. $\text{LC} = \frac{1 \text{ mm}}{50} = 0.02 \text{ mm}$

$$\text{Zero error} = -(50 - 44) (\text{LC}) = -6 \times 0.02$$

$$= -0.12 \text{ mm}$$

$$\text{Reading} = 3 \text{ mm} + 26(\text{LC})$$

$$= 3 + 26 \times 0.02 = 3.52 \text{ mm}$$

$$\text{Thickness of the plate} = 3.52 + 0.12 = 3.64 \text{ mm}$$

7. $A = x \cos \theta = 2.0 \times \cos 37^\circ = 2.0 \times 0.80 = 1.6$

$$\frac{dA}{dx} = (\cos \theta)(1) + x(-\sin \theta) \left(\frac{d\theta}{dx} \right)$$

$$\Rightarrow dA = (\cos \theta)(dx) - x(\sin \theta)d\theta$$

$$\therefore \Delta A \simeq (\cos \theta) \Delta x - (x \sin \theta) \Delta \theta$$

For calculation of error

$$\begin{aligned} \Delta A &= |\cos \theta \cdot \Delta x| + |(x \sin \theta) \Delta \theta| \\ &= 0.80 \times 0.1 + 2.0 \times 0.60 \times \frac{3.14}{180} \left[\because 1^\circ = \frac{3.14}{180} \text{ radian} \right] \\ &= 0.080 + 0.021 = 0.101 \\ \therefore A &= 1.6 \pm 0.1 \end{aligned}$$

$$8. \text{ LC} = \frac{1 \text{ mm}}{100} = 0.01 \text{ mm}$$

$$\text{Diameter, } d = 1 + 47 \times 0.01 = 1.47 \text{ mm}$$

$$\begin{aligned} \text{Surface area} &= 2\pi \left(\frac{d}{2} \right) l = \pi d l \\ &= 3.14 \times \left(\frac{1.47}{10} \text{ cm} \right) (5.6 \text{ cm}) \\ &= 2.58 \text{ cm}^2 \simeq 2.6 \text{ cm}^2 \text{ (two significant digits)} \end{aligned}$$

9. We need to calculate standard deviation (σ) as follows.

x_i	$\delta_i = x_i - \bar{x}$	δ_i^2
3.4	$3.4 - 3.4 = 0$	0
3.1	$3.1 - 3.4 = -0.3$	0.09
3.6	$3.6 - 3.4 = 0.2$	0.04
3.7	$3.7 - 3.4 = 0.3$	0.09
3.1	$3.1 - 3.4 = -0.3$	0.09
$\sum x_i = 16.9$		$\sum \delta_i^2 = 0.31$

$$\bar{x} = \frac{\sum x_i}{5} = \frac{16.9}{5} = 3.4 \text{ km}$$

$$\sigma = \sqrt{\frac{\sum \delta_i^2}{N-1}} = \sqrt{\frac{0.31}{4}} = 0.28$$

The 95% confidence lies in the range $\bar{x} \pm 2\sigma$

$\therefore$ Distance = $(3.4 \pm 0.56) \text{ km}$.

Chapter 11 Miscellaneous Problems on Chapter 10



Match the Column

1. $A = x + y \Rightarrow \Delta A = \Delta x + \Delta y$.
 $\therefore$ Uncertainty in A is higher than that in x .
 $B = x - y \Rightarrow \Delta B = \Delta x + \Delta y$
 Uncertainty in B is same as the uncertainty in A .
 Uncertainty in $\frac{A}{B}$ is
 $\Delta A + \Delta B = 2(\Delta x + \Delta y)$.
 Uncertainty in AB is also $\Delta A + \Delta B = 2(\Delta x + \Delta y)$.
 Uncertainty in $C = xy$ is $\Delta x + \Delta y$.



Passage Based Problems

Passage 1

1. $\Delta L = \sqrt{(\Delta x)^2 + (\Delta y)^2} = \sqrt{5^2 + 1^2} = 5.1 \text{ cm}$
2. $V = \frac{x}{t} = \frac{120}{20} = 6.0 \text{ m/s}$

$$\frac{\Delta V}{V} = \sqrt{\left(\frac{\Delta x}{x}\right)^2 + \left(\frac{\Delta t}{t}\right)^2} = \sqrt{\left(\frac{3}{120}\right)^2 + \left(\frac{1.2}{20}\right)^2}$$

$$= \sqrt{0.000625 + 0.0036} = 0.065$$
 In percentage, $\frac{\Delta V}{V} = 6.5\%$

Passage 2

Zero error = 0.4 mm

3. The three measurements for outer diameter are
 $D_1 = (50 + 0.5) - 0.4 = 50.1 \text{ mm}$
 $D_2 = (51 + 0) - 0.4 = 50.6 \text{ mm}$
 $D_3 = (50 + 0.8) - 0.4 = 50.4 \text{ mm}$

$$\text{Mean diameter } D = \frac{D_1 + D_2 + D_3}{3} = 50.4 \text{ mm}$$

Mean deviation,

$$\Delta D = \frac{|50.1 - 50.4| + |50.6 - 50.4| + |50.4 - 50.4|}{3}$$

$$= 0.17 \approx 0.2 \text{ mm}$$

$$\therefore D = 50.4 \text{ mm} \pm 0.2 \text{ mm}$$

4. The three values for inner diameters are

$$d_1 = 49 + 0.4 - 0.4 = 49.0 \text{ mm}$$

$$d_2 = 48 + 0.8 - 0.4 = 48.4 \text{ mm}$$

$$d_3 = 49 + 0.6 - 0.4 = 49.2 \text{ mm}$$

$$\text{Mean diameter } d = \frac{d_1 + d_2 + d_3}{3} = 48.9 \text{ mm}$$

Mean deviation,

$$\Delta d = \frac{|49.0 - 48.9| + |48.4 - 48.9| + |49.2 - 48.9|}{3}$$

$$= 0.3 \text{ mm}$$

$$\therefore d = (48.9 \pm 0.3) \text{ mm}$$

$$\text{Thickness, } t = D - d = 50.4 - 48.9 = 1.5 \text{ mm}$$

$$\Delta t = \Delta D + \Delta d = 0.2 + 0.3 = 0.5 \text{ mm}$$

$$\therefore t = (1.5 \pm 0.5) \text{ mm}$$

5. $L = 60 + 0.2 - 0.4 = 59.8 \text{ mm}$

$$L = (59.8 \pm 0.1) \text{ mm}$$

ELECTROSTATICS

AIEEE/JEE Main Questions

1. Field in front of the sheet is

$$E = \frac{\sigma}{\epsilon_0} \hat{j}$$

$$T \sin \theta = QE \text{ and}$$

$$T \cos \theta = mg$$

$$\therefore \tan \theta = \frac{QE}{mg} = \frac{Q\sigma}{\epsilon_0 mg}$$

$\therefore \sigma$ is proportional to $\tan \theta$.

2. A and B are the centres of the two rings.

Potential at A is

$$V_A = \frac{kq}{R} + \frac{k(-q)}{\sqrt{R^2 + d^2}}$$

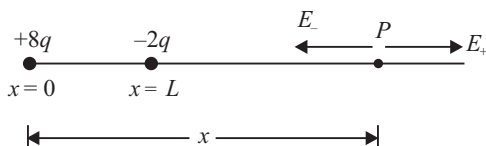
Similarly, potential at B is

$$V_B = \frac{k(-q)}{R} + \frac{kq}{\sqrt{R^2 + d^2}}$$

$$\therefore V_A - V_B = \frac{2kq}{R} - \frac{2kq}{\sqrt{R^2 + d^2}}$$

$$= \frac{q}{2\pi\epsilon_0} \left[\frac{1}{R} - \frac{1}{\sqrt{R^2 + d^2}} \right]$$

3. Let the field be zero at point P. The field due to the two charges must have equal magnitude.

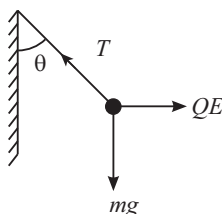


$$k \frac{8q}{x^2} = k \frac{2q}{(x-L)^2}$$

$$\text{or, } \frac{x-L}{x} = \frac{1}{2} \Rightarrow x = 2L$$

4. Field is zero inside the shell. On the surface the field is
- $\frac{kQ}{R^2}$
- . For outside points the field falls off as
- $\frac{1}{r^2}$
- .

5. Field at point P is only because of the charge inside the sphere of radius
- r_1
- . Charge (
- q
-) inside the spherical volume of radius
- r_1
- can be calculated



by adding charges on thin shells in the region $0 < x < r_1$.

$$q = \int_0^{r_1} \rho \cdot 4\pi x^2 dx = \frac{Q}{\pi R^4} 4\pi \int_0^{r_1} x^3 dx \quad \left[\because \rho = \frac{Qx}{\pi R^4} \right]$$

$$= \frac{4Q}{R^4} \cdot \frac{r_1^4}{4} = \frac{Q}{R^4} r_1^4$$

$$\therefore \text{Field at P is } E = \frac{kq}{r_1^2} = \left(\frac{Q}{4\pi\epsilon_0 R^4} \right) r_1^2$$

6. Work done by an external agent = change in potential energy of the electrons.

$$W = (V_Q - V_P)q = (-4 - 10)(-100 \times 1.6 \times 10^{-19} \text{C}) = 2.24 \times 10^{-16} \text{ J}$$

7. Refer to example 15 in Chapter 1.

$$E = \frac{\lambda}{2\pi\epsilon_0 r} = \frac{q/\pi r}{2\pi\epsilon_0 r} = \frac{q}{2\pi^2\epsilon_0 r^2}$$

8. Conceptually, the question is same as question 5. We need to find the charge enclosed inside a sphere of radius
- r
- . Only this charge will produce field at the concerned point.

$$q = \int_0^r \rho \cdot 4\pi x^2 dx = 4\pi\rho_0 \int_0^r \left(\frac{5}{4} - \frac{x}{R} \right) x^2 dx$$

$$= 4\pi\rho_0 \left[\frac{5}{4} \frac{r^3}{3} - \frac{x^4}{4R} \right]_0^r$$

$$= \pi\rho_0 r^3 \left[\frac{5}{3} - \frac{r}{R} \right]$$

$$\therefore E = \frac{q}{4\pi\epsilon_0 r^2} = \frac{\rho_0 r}{4\epsilon_0} \left(\frac{5}{3} - \frac{r}{R} \right)$$

9. Refer to example 36 in Chapter 1.

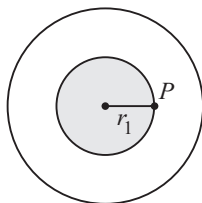
$$\epsilon_r = \frac{\rho}{\rho - \sigma} = \frac{1.6}{1.6 - 0.8} = 2$$

10. The potential inside a uniformly charged ball is given by

$$\phi = \frac{1}{4\pi\epsilon_0} \frac{Q}{R^3} \left[\frac{3}{2} R^2 - \frac{1}{2} r^2 \right]$$

$$= \frac{\rho}{3\epsilon_0} \left[\frac{3}{2} R^2 - \frac{r^2}{2} \right] \quad \left[\because Q = \rho \cdot \frac{4}{3} \pi R^3 \right]$$

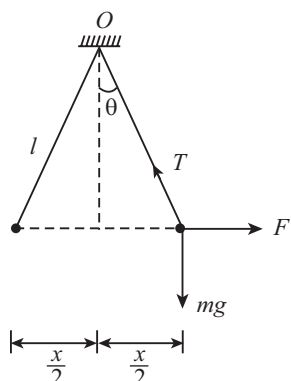
$$= \left(-\frac{\rho}{6\epsilon_0} \right) r^2 + \frac{\rho R^2}{2\epsilon_0}$$



This expression is similar to the one given in the question.

$$a = -\frac{\rho}{6\epsilon_0} \Rightarrow \rho = -6a\epsilon_0$$

12. Charge is leaking very slowly. The two balls are nearly in equilibrium all the time and their acceleration is close to zero.



$$F = T \sin \theta \text{ and } mg = T \sin \theta$$

$$\therefore \frac{F}{mg} = \tan \theta$$

$$\Rightarrow \frac{kq^2}{mg \cdot x^2} = \tan \theta$$

[q = instantaneous charge on each ball]

$$\Rightarrow \frac{kq^2}{mg \cdot x^2} \approx \frac{x}{2l}$$

$$[\because \theta \text{ is small and therefore, } \tan \theta \approx \sin \theta = \frac{x}{2l}]$$

$$\Rightarrow x^3 = \frac{2kq^2 l}{mg} \quad (i)$$

Differentiating this equation with respect to time gives

$$3x^2 \frac{dx}{dt} = \frac{4klq}{mg} \cdot \frac{dq}{dt}$$

$$\therefore \frac{dx}{dt} \propto \frac{q}{x^2} \quad \left[\frac{dq}{dt} \text{ is a constant} \right]$$

But from (i) $q \propto x^{\frac{3}{2}}$

$$\therefore \frac{dx}{dt} \propto x^{-\frac{1}{2}}$$

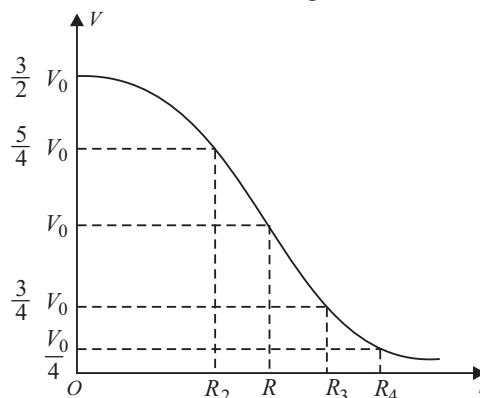
$$\Rightarrow v \propto x^{-\frac{1}{2}}$$

13. $dV = -Edx$

$$\Rightarrow \int_{V_0}^{V_A} dV = -30 \int_{x=0}^{x=2} x^2 dx$$

$$\Rightarrow V_A - V_0 = -10 \left[x^3 \right]_0^2 = -10 \times 8 = -80V$$

14. The variation of potential with distance from the centre is as shown in the figure.



Potential at the centre is $\frac{3}{2}$ times the surface potential.

Clearly, $R_1 = 0$

Also, the potential will be larger than V_0 only inside the sphere.

$\therefore$ Potential $\frac{5V_0}{4}$ will occur inside the sphere.

$\therefore R_2 < R$

R_3 and R_4 both are larger than R .

For finding R_2 , we should write.

$$V = \frac{kQ}{R^3} \left[\frac{3}{2} R^2 - \frac{1}{2} r^2 \right]$$

$$\text{or, } \frac{5V_0}{4} = V_0 \left[\frac{3}{2} - \frac{1}{2} \frac{R_2^2}{R^2} \right] \quad \left[\because V_0 = \frac{kQ}{R} \right]$$

$$\Rightarrow R_2 = \frac{R}{\sqrt{2}}$$

For finding R_3 and R_4 , we should use $V = \frac{kQ}{r}$

$$\therefore \frac{3V_0}{4} = \frac{kQ}{R_3}$$

$$\text{or, } \frac{3V_0}{4} = \frac{kQ}{R} \cdot \frac{R}{R_3}$$

$$\Rightarrow \frac{3V_0}{4} = V_0 \frac{R}{R_3}$$

$$\text{or, } R_3 = \frac{4}{3} R$$

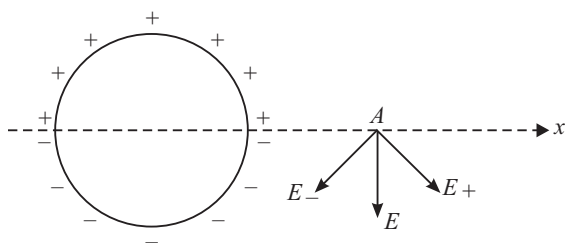
$$\text{Similarly, } \frac{V_0}{4} = \frac{kQ}{R_4} \Rightarrow \frac{V_0}{4} = \frac{kQ}{R} \frac{R}{R_4}$$

$$\text{or, } \frac{V_0}{4} = \frac{V_0 R}{R_4} \Rightarrow R_4 = 4R$$

Option (A) is correct as $2R < R_4$.

$$\text{Option (D) is also correct as } \frac{R}{\sqrt{2}} < \left(4R - \frac{4}{3}R\right)$$

15. At a point like A , the field due to the positive and negative charges will be somewhere in the indicated directions. The resultant field (E) will be perpendicular to the x axis. Only fig (b) looks appropriate.



16. Field at distance R ($a < r < b$) will depend on the charge inside the sphere of radius r .

Charge in the region $a < x < r$ is calculated as

$$q = \int_a^r \rho \cdot 4\pi x^2 dx = 4\pi A \int_a^r \frac{x^2}{x} dx$$

$$\Rightarrow q = 2\pi A[r^2 - a^2]$$

Now, total charge inside the sphere of radius r is $Q + q$

Field at distance r ($a < r < b$) is

$$E = \frac{(Q + q)}{4\pi\epsilon_0 r^2}$$

$$\Rightarrow E = \frac{Q + 2\pi A(r^2 - a^2)}{4\pi\epsilon_0 r^2} = \frac{(Q - 2\pi Aa^2)}{4\pi\epsilon_0 r^2} + \frac{A}{2\epsilon_0}$$

E will not depend on r if the first term is zero

$$Q - 2\pi Aa^2 = 0 \Rightarrow A = \frac{Q}{2\pi a^2}$$

$$17. \vec{\tau} = \vec{P} \times \vec{E}$$

$$\Rightarrow \tau \hat{k} = (P_x \hat{i} + P_y \hat{j} + P_z \hat{k}) \times (E \hat{i})$$

$$\text{or, } \tau \hat{k} = -P_y E \hat{j} + P_z E \hat{j}$$

$$\text{Obviously, } P_z = 0 \text{ and } P_y = -\frac{\tau}{E}$$

$$\text{Similarly, } -\tau \hat{k} = (P_x \hat{i} + P_y \hat{j}) \times (\sqrt{3} E_1 \hat{j})$$

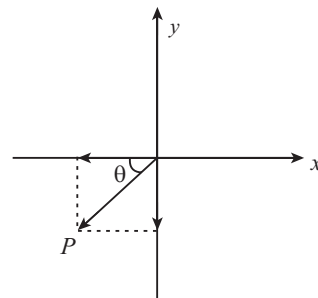
$$-\tau \hat{k} = \sqrt{3} P_x E_1 \hat{k}$$

$$\text{or } P_x = -\frac{\tau}{\sqrt{3} E_1}$$

The dipole is placed as shown.

$$\tan \theta = \frac{|P_y|}{|P_x|} = \sqrt{3}$$

$$\Rightarrow \theta = 60^\circ.$$



$$18. Q_A = \sigma(4\pi a^2); \quad Q_B = (-)\sigma(4\pi b^2); \quad Q_C = \sigma(4\pi c^2)$$

$$V_B = \frac{KQ_A}{b} + \frac{KQ_B}{b} + \frac{KQ_C}{c} \quad \left(K = \frac{1}{4\pi\epsilon_0}\right)$$

$$= K \cdot \sigma 4\pi \left[\frac{a^2}{b} - \frac{b^2}{b} + \frac{c^2}{c} \right] = \frac{\sigma(4\pi)}{4\pi\epsilon_0} \left[\frac{a^2}{b} - b + c \right]$$

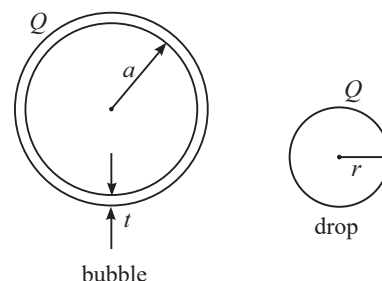
$$= \frac{\sigma}{\epsilon_0} \left[\frac{a^2 - b^2}{b} + c \right]$$

IIT JEE/ JEE Advanced Questions

$$19. E = \frac{\sigma}{2\epsilon_0} (\downarrow) + \frac{2\sigma}{2\epsilon_0} (\downarrow) + \frac{\sigma}{2\epsilon_0} (\downarrow) = \frac{2\sigma}{\epsilon_0} (\downarrow).$$

The field is independent of distance from an infinite sheet.

$$20. \text{Volume of liquid in the bubble is} \\ = 4\pi a^2 \cdot t$$



If a drop of radius r is formed, then

$$\frac{4}{3}\pi r^3 = 4\pi a^2 t$$

$$\Rightarrow r = (3a^2t)^{\frac{1}{3}}$$

Charge remains conserved. The charge on the bubble is given by

$$V = k \frac{Q}{a} \Rightarrow Q = \frac{Va}{k} \quad (i)$$

Potential of the drop is

$$V^1 = \frac{kQ}{r} = \frac{Va}{r} = \frac{Va}{(3a^2t)^{\frac{1}{3}}}$$

$$\text{or, } V^1 = V \left(\frac{a}{3t} \right)^{\frac{1}{3}}$$

21. The charge is distributed on a spherical shell of radius R_0 .

There is no field in the region $r < R_0$. Therefore, no energy is stored in this space.

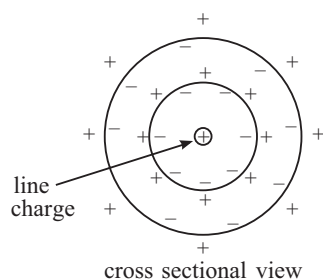
Electric field is zero for r just less than R_0 and is finite

$$\left(= \frac{kQ}{R_0} \right) \text{ for } r \text{ just larger than}$$

R_0 . Therefore, the field is discontinuous at $r = R_0$.

23. When a charge is given to the inner cylinder, it induces an equal but opposite charge on the inner wall of the outer cylinder. There is an electric field in the space between the two cylinders. Therefore, there is a potential difference.

When a charge is given to the outer cylinder, the charge will reside on its outer wall. There is no field in the space between the two cylinders. There is no potential difference between them.



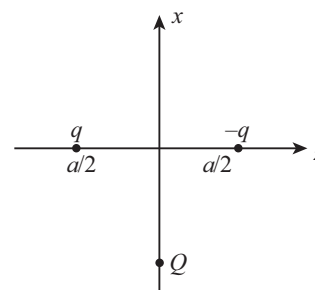
When a line charge is kept along the axis, the induced charges will appear as shown. There will be field between the two cylinders. There will be potential difference.

When both the cylinders are given same charge density, then also the outer surface of the inner cylinder and the inner wall of the outer cylinder will have equal and opposite charge.

There is a field between them. There is a potential difference.

24. Refer section 11 in Chapter 1.

25. Initial situation is as shown. The potential due to $+q$ and $-q$ at any point on x axis is zero. Potential is also zero at any point on the y axis.



$\therefore$ When Q is moved from a point on the x axis to another point on the y axis, there is no change in potential energy. No work is done.

26. The given graph means that the charge density inside the spherical nucleus remains constant till $r = a$ and thereafter decreases linearly to zero. But the charge is spherically symmetric and its total value is $Q = Ze$.

Therefore, field at $r = R$ is

$$E = k \frac{Q}{R} = k \frac{Ze}{R}$$

This is independent of a .

27. $a = 0$ implies that the charge density decreases linearly as shown in the figure. The equation of the straight line gives us the variation of ρ with r .

$$\rho = -\left(\frac{d}{R}\right)r + d$$

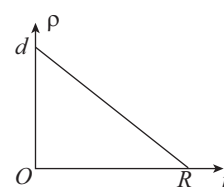
Total charge in the spherical volume can be written as sum of charges on infinite number of thin spherical shells lying between $r = 0$ and $r = R$.

$$Ze = \int_0^R \rho 4\pi r^2 dr = -\frac{4\pi d}{R} \int_0^R r^3 dr + 4\pi d \int_0^R r^2 dr$$

$$\text{or, } Ze = -\frac{4\pi d R^3}{4} + 4\pi d \frac{R^3}{3}$$

$$\text{or, } Ze = \frac{\pi d R^3}{3}$$

$$\text{or, } d = \frac{3Ze}{\pi R^3}$$



28. Electric field inside a spherical volume varies linearly with distance from the centre when the charge density in the entire volume is constant.

29. $\angle ACB = 90^\circ$

$$\therefore AC = 2R \cos 60^\circ = R$$

$$\text{And } CB = 2R \sin 60^\circ = \sqrt{3}R$$

Field due to the charges at A and B , at O , cancel out.
Field at O is that contributed by the charge at C only.

$$\vec{E} = \frac{1}{4\pi\epsilon_0} \frac{2q}{R^2} (-\hat{i}) = \frac{q}{6\pi\epsilon_0 R^2} (-\hat{i})$$

The potential energy of the system is

$$U = \frac{k\left(\frac{q}{3}\right)\left(\frac{q}{3}\right)}{2R} + \frac{k\left(\frac{q}{3}\right)\left(\frac{-2q}{3}\right)}{R} + \frac{k\left(\frac{q}{3}\right)\left(\frac{-2q}{3}\right)}{\sqrt{3}R} \neq 0$$

Force between the charges at B and C is

$$F_{BC} = \frac{k\left(\frac{q}{3}\right)\left(\frac{2q}{3}\right)}{(\sqrt{3}R)^2} = \frac{q^2}{54\pi\epsilon_0 R^2}$$

Electric potential at O is

$$V = \frac{k\frac{q}{3}}{R} + k\frac{\frac{q}{3}}{R} - k\frac{\frac{2q}{3}}{R} = 0$$

30. If q is charge inside a volume of radius $r = \frac{R}{2}$, then

the field at $r = \frac{R}{2}$ is

$$E = K \frac{q}{\left(\frac{R}{2}\right)^2}$$

Field at the surface is $E_s = k \frac{Q}{R^2}$

$$\text{Given that } E = \frac{E_s}{8}$$

$$\Rightarrow \frac{q}{\left(\frac{R}{2}\right)^2} = \frac{Q}{8R^2} \Rightarrow q = \frac{Q}{32}$$

Charge inside the sphere of volume r can be written as

$$\int_0^r \rho \cdot 4\pi x^2 dx$$

Hence, $q = \frac{Q}{32}$ implies that

$$\int_0^{\frac{R}{2}} \rho \cdot 4\pi x^2 dx = \frac{1}{32} \int_0^R \rho \cdot 4\pi x^2 dx$$

$$\text{or, } k \int_0^{\frac{R}{2}} x^a \cdot x^2 dx = \frac{k}{32} \int_0^R x^a \cdot x^2 dx$$

$$\text{or, } \left[\frac{x^{a+3}}{a+3} \right]_0^{\frac{R}{2}} = \frac{1}{32} \left[\frac{x^{a+3}}{a+3} \right]_0^R$$

$$\text{or, } \left(\frac{R}{2} \right)^{a+3} = \frac{1}{32} (R)^{a+3}$$

$$\text{or, } \left(\frac{1}{2} \right)^{a+3} = \frac{1}{32}$$

$$\text{or, } a+3 = 5 \Rightarrow a = 2.$$

31. We need to find out how much charge is there inside the cube.

Half the disc is inside the cube. That is, $3C$ charge on the disc is inside the cube.

$\frac{1}{4}$ of the rod is inside the cube. This contributes a charge $\frac{8}{4} = 2C$ inside the cube.

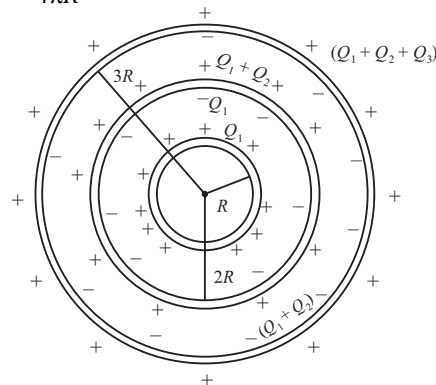
$-7C$ point charge is inside the cube, and $3C$ point charge is outside the cube.

$$\therefore \text{Total charge inside the cube} = 3C + 2C - 7C = -2C.$$

$$\therefore \text{Flux through the cube is } \frac{-2C}{\epsilon_0}.$$

32. The figure shows the three shells. Each has small thickness. The charge Q_1 resides on the outer surface of the innermost shell. Charge density on the surface is

$$\sigma_1 = \frac{Q_1}{4\pi R^2}$$



A charge $-Q_1$ is induced on the inner wall of middle shell. The outer surface of this shell will have a charge $Q_2 + Q_1$ since total charge on it is Q_2 .

A charge $-(Q_1 + Q_2)$ is induced on the inner wall of the outermost shell. Therefore, charge on its outer surface is $(Q_1 + Q_2 + Q_3)$.

The surface density of charge on the outer surface of the middle shell is

$$\sigma_2 = \frac{Q_1 + Q_2}{4\pi(2R)^2}$$

The surface charge density of the charge on the outer surface of the outermost shell is

$$\sigma_3 = \frac{Q_1 + Q_2 + Q_3}{4\pi(3R)^2}$$

According to the question, $\sigma_1 = \sigma_2 = \sigma_3$

$$\Rightarrow \frac{Q_1}{R^2} = \frac{Q_1 + Q_2}{4R^2} = \frac{Q_1 + Q_2 + Q_3}{9R^2}$$

$$\Rightarrow \frac{Q_1}{1} = \frac{Q_1 + Q_2}{4} = \frac{Q_1 + Q_2 + Q_3}{9}$$

Now, you can prove that $Q_1 : Q_2 : Q_3 = 1 : 3 : 5$

$$33. qE = mg \Rightarrow qE = \frac{4}{3}\pi r^3 \cdot d \cdot g \quad (i)$$

When there is no field, there is gravitational force and the viscous force on the falling drop. Buoyancy of air is negligible because density of air is too small. When the drop attains its terminal speed, net force on it is zero.

$$6\pi\eta r v = \frac{4}{3}\pi r^3 \cdot d \cdot g$$

$$\Rightarrow r^2 = \frac{9\eta v}{2 \cdot d \cdot g} = \frac{9 \times 1.8 \times 10^{-5} \times 2 \times 10^{-3}}{2 \times 900 \times 9.8} = \frac{9}{49} \times 10^{-10}$$

$$\Rightarrow r^2 = \frac{3}{7} \times 10^{-5} m$$

$$\text{From (i)} \quad q = \frac{4\pi r^3 d g}{3E}$$

$$= \frac{4\pi \left(\frac{3}{7} \times 10^{-5}\right)^3 \times 900 \times 9.8}{3 \times \frac{81}{7} \pi \times 10^5}$$

$$= 8.0 \times 10^{-19} \text{ C.}$$

34. This problem has not been given with intention of making the students calculate the force. All the options given in the problem have different dimensions. The option with dimension of force is obviously the correct answer.

$$[E] = \left[\frac{\sigma}{\epsilon_0} \right]; [q] = [\sigma \cdot R^2]$$

$$\therefore [F] = [E q] = \frac{\sigma^2 R^2}{\epsilon_0}$$

35. More lines are originating at Q_1 and less number of lines is terminating at Q_2 .

$$\therefore |Q_1| > |Q_2|$$

Q_1 is positive and Q_2 is negative. There will be a point to the right of Q_2 where the field will be zero.

36. Adding a constant force to a spring block system does not change its frequency of oscillations. Only the mean position gets shifted.

37. The entire potential energy of the system can be written as sum of the electrostatic PE and the surface energy of the film.

$$U = 4 \frac{kq^2}{a} + 2 \frac{kq^2}{\sqrt{2}a} + 2\gamma a^2$$

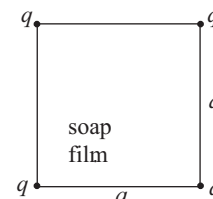
The system will be in equilibrium when $\frac{dU}{da} = 0$

$$\Rightarrow -\frac{4kq^2}{a^2} - \frac{\sqrt{2}kq^2}{a^2} + 4\gamma a = 0$$

$$\Rightarrow a^3 = \frac{kq^2}{4\gamma} (4 - \sqrt{2})$$

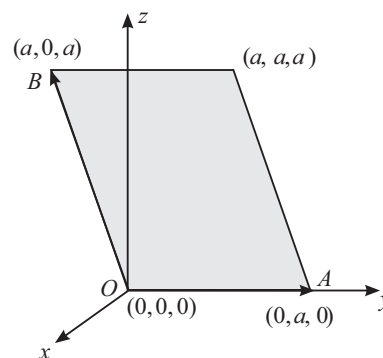
Obviously $a \propto (q)^{\frac{2}{3}}$

$$\therefore N = 3.$$



$$38. \overrightarrow{OA} = a\hat{j}$$

$$\overrightarrow{OB} = a\hat{i} + a\hat{k}$$



Area vector can be written as

$$\vec{S} = \overrightarrow{OA} \times \overrightarrow{OB} = -a^2\hat{k} + a^2\hat{i}$$

$$\therefore \phi = \vec{E} \cdot \vec{S} = (E_0\hat{i}) \cdot (-a^2\hat{k} + a^2\hat{i}) = E_0 a^2.$$

39. The charge will get shared so that both of them acquire same potential.

$$V = k \frac{Q_A}{R_A} = k \frac{Q_B}{R_B}$$

$$\Rightarrow \frac{Q_A}{Q_B} = \frac{R_A}{R_B} \quad (i)$$

$$\Rightarrow Q_A > Q_B$$

$$\text{From (i) } \frac{\sigma_A \cdot 4\pi R_A^2}{\sigma_B \cdot 4\pi R_B^2} = \frac{R_A}{R_B};$$

$$\Rightarrow \frac{\sigma_A}{\sigma_B} = \frac{R_B}{R_A}$$

$$\text{Field near the surface is } E = \frac{\sigma}{\epsilon_0}$$

$$\therefore \frac{E_A}{E_B} = \frac{\sigma_A}{\sigma_B} = \frac{R_B}{R_A}$$

$$\therefore E_A < E_B$$

40. The Gauss's law is valid only when the field follows the inverse square law.

The Gauss's law cannot be used to find field around a dipole because it is useful in symmetrical conditions only.

41. The electric field between the plates is

$$E = \frac{X}{d}$$

Apart from the horizontal electric force on the proton, there is vertical gravitational force acting on it. The resultant force F must make 45° with the vertical for the proton to move in the given direction

$$\therefore qE = mg$$

$$\Rightarrow q \frac{X}{d} = mg$$

$$\Rightarrow X = \frac{mgd}{q} = \frac{1.67 \times 10^{-27} \times 10 \times 10^{-2}}{1.6 \times 10^{-19}} = 1 \times 10^{-9} \text{V}$$

42. In option (d), the solid line graph represents variation of electric field and the graph in broken line represents the variation of the potential.

43. Linear charge density on the cylinder is

$$\lambda = \rho \text{ (volume of unit length of the cylinder)} \\ = \rho \cdot \pi R^2$$

Field at P , if there were no cavity

$$E_1 = \frac{\lambda}{2\pi\epsilon_0(2R)} = \frac{\rho \cdot \pi R^2}{2\pi\epsilon_0(2R)} = \frac{\rho R}{4\epsilon_0}$$

Field at P , contributed by the spherical charge (inside the cavity region), is

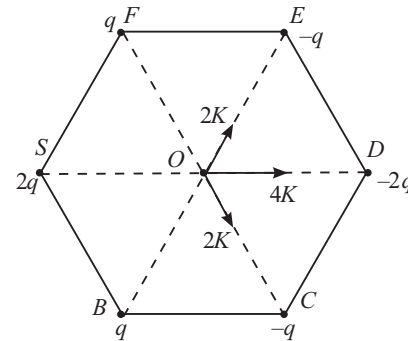
$$E_2 = \frac{1}{4\pi\epsilon_0} \frac{q}{(2R)^2} = \frac{1}{4\pi\epsilon_0} \frac{\rho \cdot \frac{4}{3}\pi \left(\frac{R}{2}\right)^3}{(2R)^2} \\ = \frac{\rho R}{96\epsilon_0}$$

$$\therefore \text{Required answer is } E_1 - E_2 = \frac{\rho R}{4\epsilon_0} \left[1 - \frac{1}{24} \right]$$

$$= \frac{23\rho R}{96\epsilon_0}$$

$$\therefore k = 6$$

44. Distance of each charge from the centre is L .



Field at O due to the charge at F is

$$k = \frac{1}{4\pi\epsilon_0} \frac{q}{L^2}$$

$-q$ charge at C produces same field at O . These two fields add to give a field $2K$ along OC .

Similarly, field is $2K$ along OE due to the charges at B and E .

Resultant of the field along OC and OE is $2K$ along OD [$\because$ these are two vectors of same magnitude at 120°].

Field due to charges $2q$ and $-2q$ is $4K$ along OD . Therefore, the resultant field is $6K$ along OD .

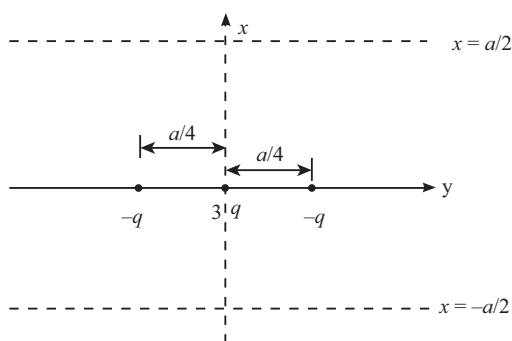
At any point on the line PR , the potential is zero. On line ST , a point is not necessarily equidistant from all the charges.

45. (a) The three charges lie on y axis as shown.

A plane at $x = \frac{a}{2}$ and the other plane at $x = -\frac{a}{2}$ are

at symmetrical locations with respect to the charges.

$\therefore$ Flux through both the planes will be same.



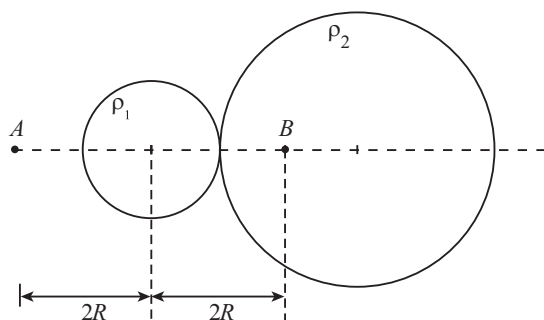
- (b) The plane $y = \frac{a}{2}$ and the plane $y = -\frac{a}{2}$ are again located symmetrically and flux through both of them must be same.

(c) $\phi = \frac{q_{net}}{\epsilon_0} = \frac{q}{\epsilon_0}$

- (d) The plane $x = \frac{a}{2}$ is a plane perpendicular to the plane of the above figure passing through $x = \frac{a}{2}$.

A plane $z = \frac{a}{2}$ is a similar plane, below the plane of the figure. Both planes must have same flux through them.

46. There can be two points A and B at a distance $2R$ from the centre of the smaller sphere. For field to be zero at A



$$\frac{1}{4\pi\epsilon_0} \frac{\rho_1 \cdot \frac{4}{3}\pi R^3}{(2R)^2} + \frac{1}{4\pi\epsilon_0} \frac{\rho_2 \cdot \frac{4}{3}\pi (2R)^3}{(5R)^2} = 0$$

$$\Rightarrow \frac{\rho_1}{\rho_2} = -\frac{32}{35}$$

For field to be zero at B :

$$\frac{1}{4\pi\epsilon_0} \frac{\rho_1 \cdot \frac{4}{3}\pi R^3}{(2R)^2} - \frac{\rho_2}{3\epsilon_0} (R) = 0$$

$$\Rightarrow \frac{\rho_1}{\rho_2} = \frac{4}{1}$$

47. $E_1 = \frac{kQ}{R^2}$; $E_2 = k \frac{2Q}{R^2}$

$$E_3 = k \frac{(4Q)}{(2R)^3} R = \frac{kQ}{2R^2}$$

$$\therefore E_2 > E_1 > E_3$$

48. Given $\frac{1}{4\pi\epsilon_0} \frac{Q}{r_0^2} = \frac{\lambda}{2\pi\epsilon_0 r_0} = \frac{\sigma}{2\epsilon_0}$ (i)

$$\Rightarrow Q = 2\pi\sigma r_0^2 \quad \text{[(a) incorrect]}$$

$$\text{And } r_0 = \frac{\lambda}{\pi\sigma} \quad \text{[(b) incorrect]}$$

$$E_1 \left(\frac{r_0}{2} \right) = \frac{1}{4\pi\epsilon_0} \frac{Q}{\left(\frac{r_0}{2} \right)^2} = \frac{Q}{\pi\epsilon_0 r_0^2}$$

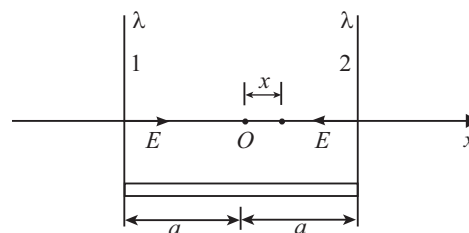
$$E_2 \left(\frac{r_0}{2} \right) = \frac{\lambda}{2\pi\epsilon_0 \frac{r_0}{2}} = \frac{\lambda}{\pi\epsilon_0 r_0} = \frac{Q}{2\pi\epsilon_0 r_0^2} \quad \text{[using (i)]}$$

$$\therefore E_1 \left(\frac{r_0}{2} \right) = 2E_2 \left(\frac{r_0}{2} \right) \quad \text{[(c) is correct]}$$

$$\text{Now, } E_3 \left(\frac{r_0}{2} \right) = \frac{\sigma}{2\epsilon_0} = E_2(r_0)$$

$\therefore$ (d) is incorrect

49. Field between the line charges is directed as shown in the figure. Field at O is zero. Charge kept at O which is in equilibrium.



The electric field's magnitude increases when we move in either direction from O . When $+q$ charge is moved to right from O it experiences force toward left. This is restoring force. When it is displaced to the left of O , it experiences a force toward right. Thus, its motion is oscillatory.

For a negative charge, the position at O is position of unstable equilibrium. When pushed to right, it will experience a rightward force due to E .

Now, consider $+q$ charge at a small distance x to the right of O .

$$m \frac{d^2 x}{dt^2} a = F_1 - F_2$$

$$= \frac{\lambda q}{2\pi\epsilon_0(a+x)} - \frac{\lambda q}{2\pi\epsilon_0(a-x)} = \frac{\lambda q}{2\pi\epsilon_0} \left[\frac{-2x}{a^2 - x^2} \right]$$

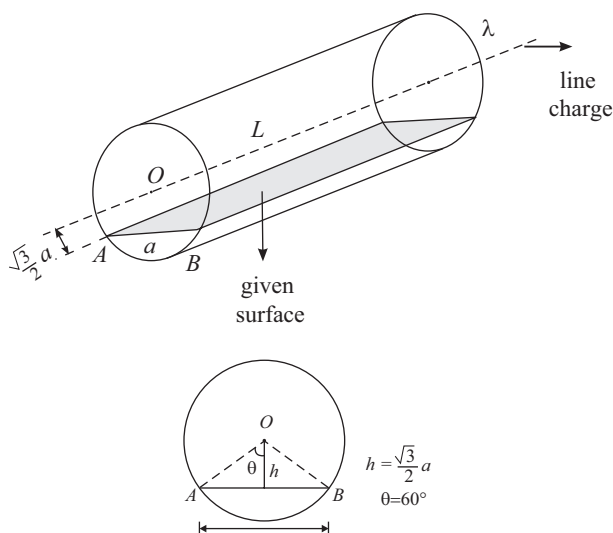
$$\Rightarrow \frac{d^2x}{dt^2} \approx -\frac{2\lambda q}{2\pi\epsilon_0 m} x \quad [\text{when } a \gg x]$$

$\therefore$ Motion is SHM if displacement is small.

50. This is a standard result that the field inside the cavity is

$$\vec{E} = \frac{\rho}{3\epsilon_0} \vec{a}$$

51. Imagine a cylinder with the line charge along its axis. Charge enclosed inside the cylinder is $q = \lambda L$.



Flux through the cylinder is $\phi_0 = \frac{\lambda L}{\epsilon_0}$

Flux through the given surface must be

$$\phi = \phi_0 \left(\frac{2\theta}{2\pi} \right) \quad [\text{see second figure}]$$

$$= \frac{\lambda L}{\epsilon_0} \left(\frac{360^\circ}{120^\circ} \right) = \frac{\lambda L}{3\epsilon_0}$$

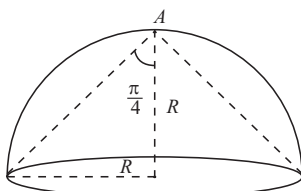
$\therefore n = 3$.

52. Let us first calculate flux through the circular base of the hemisphere.

The solid angle subtended by the circular base at the top point A is

$$\Omega = 2\pi \left(1 - \cos \frac{\pi}{4} \right)$$

$$= 2\pi \left(1 - \frac{1}{\sqrt{2}} \right)$$



$\therefore$ Flux through the base is

$$\phi = \frac{Q}{\epsilon_0} \left(\frac{\Omega}{4\pi} \right) = \frac{Q}{2\epsilon_0} \left(1 - \frac{1}{\sqrt{2}} \right)$$

The hemispherical surface and the circular base form a closed surface. Q is an outside charge. Flux due to Q through the closed surface must be zero.

$\therefore$ Flux through the hemispherical surface must be

$$\phi = -\frac{Q}{2\epsilon_0} \left(1 - \frac{1}{\sqrt{2}} \right)$$

53. $Q_{enc} = \lambda\sqrt{3}R$

$$\phi = \frac{Q_{enc}}{\epsilon_0} = \frac{\sqrt{3}\lambda R}{\epsilon_0}$$

Field due to infinitely long line charge is normal to the line charge everywhere. Therefore, the field will have no z component anywhere.

54. $E = qE_0 \sin \omega t = m \frac{dv}{dt}$

$$\int_0^v dv = \frac{qE_0}{m} \int_0^{\pi/\omega} \sin \omega t dt ;$$

$$v = \frac{qE_0}{\omega m} (-\cos \omega t) \Big|_0^{\pi/\omega}$$

$$= -\frac{qE_0}{m\omega} ((-\cos \pi) - (-\cos 0))$$

$$= \frac{2qE_0}{m\omega} = 2 \text{ m/s}$$

CAPACITOR

AIEEE/JEE Main Questions

1. Energy stored in the capacitor is used to heat the block

$$\therefore \frac{1}{2} CV^2 = ms\Delta T$$

$$\text{or, } V = \sqrt{\frac{2ms\Delta T}{C}}$$

2. There are $(n-1)$ capacitors in parallel. The equivalent capacitance is $C_0 = (n-1)C$.

3. $C_0 = \frac{\epsilon_0 A}{d} = 9 \text{ PF}$ [Air-filled capacitor]

Capacitance of the dielectric filled capacitor is

$$C = \frac{\epsilon_0 A}{\frac{d}{3(k_1)} + \frac{2d}{3(k_2)}} = \frac{\epsilon_0 A}{d} \frac{1}{\frac{1}{3 \times 3} + \frac{2}{3 \times 6}} = 40.5 \text{ PF}$$

4. If the capacitors are connected with their plates of opposite polarities together, charge on them can be zero. For this, both the capacitors must have same charge, initially.

$$(120 C_1) = (200) C_2$$

$$\Rightarrow 3C_1 = 5C_2$$

5. $E = \frac{\sigma}{k\epsilon_0}$

$$\Rightarrow 3 \times 10^4 = \frac{\sigma}{2.2 \times 8.85 \times 10^{-12}}$$

$$\Rightarrow \sigma = 3 \times 2.2 \times 8.85 \times 10^{-8} \\ = 6 \times 10^{-7} \text{ C/m}^2$$

6. The equivalent capacitance is given by

$$\frac{1}{C_0} = \frac{1}{C} + \frac{1}{3} \Rightarrow C_0 = \frac{3C}{3+C} = \frac{3}{\frac{3}{C} + 1}$$

When C increases, C_0 also increases. The charge on the equivalent increases. Charge on C is same as that on the equivalent (say Q). Charge Q gets divided between $1\mu\text{F}$ and $2\mu\text{F}$ capacitor in the ratio of capacitance.

Charge on $2\mu\text{F}$ capacitor is

$$Q_2 = \frac{2Q}{3}$$

With increase in C , Q (and hence Q_2) increases. Obviously, variation of Q (or, Q_2) with C is not linear. Therefore, options (A) and (B) cannot be true. How to decide between (C) and (D). One way is to see if the slope $\frac{dQ}{dC}$ will increase with C or decrease with it. The other way is to observe that when C is made very large (much beyond $3\mu\text{F}$), the value of equivalent capacitance C_0 approaches $3\mu\text{F}$. Charge Q

approaches $3E$ and Q_2 approaches $\frac{2}{3}(3E) = 2E$.

The graph in (D) shows that Q_2 is going to be very large for large values of C . This is not true. Hence, option (c) is most appropriate.

7. Equivalent of $3\mu\text{F}$ and $9\mu\text{F}$ is $12\mu\text{F}$.

$$\text{Equivalent of } 4\mu\text{F} \text{ in series with } 12\mu\text{F} \text{ is } C_0 = \frac{4 \times 12}{4 + 12} \\ = 3\mu\text{F}$$

Potential difference across C_0 is 8V . [Presence or absence of $2\mu\text{F}$ capacitor makes no difference].

Charge on C_0 is $Q_0 = 3 \times 8 = 24\mu\text{C}$.

Charge on $9\mu\text{F}$ capacitor is $= \left(\frac{9}{9+3}\right)Q_0 = 18\mu\text{C}$.

Total charge on $4\mu\text{F}$ and $9\mu\text{F}$ capacitor is $Q = 24 + 18 = 42\mu\text{C}$.

Field due to a point charge $Q = 42\mu\text{C}$ at a distance of 30m from it is

$$E = k \frac{Q}{r^2} = 9 \times 10^9 \times \frac{42 \times 10^{-6}}{(30)^2} = 420 \frac{N}{C}$$

8. At least four capacitors must be connected in series to withstand 1000 V . (If there are 3 or less number of capacitors in series, the potential difference across each of them will exceed 300V .)

Four $1\mu\text{F}$ capacitors in series have an equivalent capacitance of $\frac{1}{4}\mu\text{F}$. Now, we need 8 capacitors of $\frac{1}{4}\mu\text{F}$ each, connected in parallel to get an equivalent of $2\mu\text{F}$.

$\therefore$ Total number of capacitors needed $= 8 \times 4 = 32$.

9. $Q = CV$

$$= \frac{5}{3} \times 90 \times 20 = 3,000 \text{ pC} = 3nC$$

Induced charge of the dielectric wall is given by

$$Q_{in} = Q \left(1 - \frac{1}{k}\right) = 3 \left(1 - \frac{1}{5/3}\right) nC \\ = 3 \left(\frac{2}{5}\right) nC = 1.2nC$$

IIT JEE/JEE Advanced Questions

10. When the switch is in position 1, the potential difference across $2\mu\text{F}$ capacitor is V and charge on it is

$$Q_0 = 2V.$$

$$\text{Energy stored in the capacitor is } U_1 = \frac{1}{2}(2\mu\text{F})(V^2) \\ = V^2 \mu\text{J}.$$

There is no charge on $8\mu\text{F}$ capacitor.

When the switch is moved to position 2, the battery gets disconnected and the two capacitors get connected in parallel. The charge Q_0 gets divided between the two capacitors and the final common potential difference is

$$V_1 = \frac{Q_0}{2+8} = \frac{2V}{2+8} = \frac{V}{5}$$

Final energy stored in the capacitor system is

$$U_2 = \frac{1}{2}(2)\left(\frac{V}{5}\right)^2 + \frac{1}{2}(8)\left(\frac{V}{5}\right)^2 = \frac{V^2}{5}$$

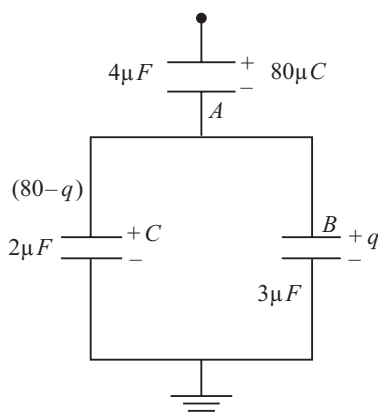
Percentage of initial energy that is lost is

$$= \frac{U_1 - U_2}{U_1} \times 100 = \frac{V^2 - \frac{V^2}{5}}{V^2} \times 100 = 80\%$$

11. Charge induced on plate A will be $-80\mu\text{C}$.

The three plates – A , B and C – form an isolated system. Total charge on them must be zero.

If charge on the plate B is q , then the charge on plate C must be $(80 - q)$. The potential difference across the $2\mu\text{F}$ and $3\mu\text{F}$ capacitors must be same.



$$\therefore \frac{q}{3} = \frac{80-q}{2} \Rightarrow q = 48\mu\text{C}.$$

12. When S_1 is closed, C_1 gets charged. Charge on it is $Q_1 = 2V_0C_1 = 2V_0C$.

Now, S_1 is opened and S_2 is closed. This connects the uncharged C_2 to the charged capacitor C_1 . Since both the capacitors are identical, they will acquire $\frac{Q_1}{2} = V_0C$ charge on each of them.

Now, S_2 is opened and S_3 is closed. Charge on C_1 will remain V_0C with its upper plate positive. But the capacitor C_2 gets connected to the cell of emf V_0 . It will acquire a charge CV_0 , with upper plate negative.

13. The potential difference between the two plates can be written as

$$V = E_1d = E_2d \Rightarrow E_1 = E_2.$$

Capacitance of the part filled with the dielectric is

$$C_1 = \frac{k\epsilon_0 \frac{A}{3}}{d} \quad (i)$$

$$\text{Total capacitance is } C = C_1 + \frac{\epsilon_0 \frac{2A}{3}}{d}$$

$$\Rightarrow C = C_1 + \frac{2C_1}{k}$$

$$\Rightarrow \frac{C}{C_1} = \frac{2+k}{k}$$

Charges on the two parts will be

$$Q_1 = C_1V \text{ and } Q_2 = C_2V = \frac{2C_1}{k}V$$

$$\therefore \frac{Q_1}{Q_2} = \frac{k}{2}$$

14. $C_1 = \frac{\epsilon_0 S}{d}$

The capacitance C_2 can be obtained as combination of 3 capacitors as shown in the figure.

$$C_A = \frac{\epsilon_1 \epsilon_0 \frac{S}{2}}{\frac{d}{2}} = \frac{\epsilon_1 \epsilon_0 S}{d}$$

$$= \frac{2\epsilon_0 S}{d}$$

$$C_B = \frac{\epsilon_2 \epsilon_0 \frac{S}{2}}{\frac{d}{2}} = \frac{\epsilon_2 \epsilon_0 S}{d} = \frac{4\epsilon_0 S}{d}$$

$$C_C = \frac{\epsilon_1 \epsilon_0 \frac{S}{2}}{\frac{d}{2}} = \frac{\epsilon_1 \epsilon_0 S}{2d} = \frac{\epsilon_0 S}{d}$$

$$\therefore C_2 = \frac{C_A C_B}{C_A + C_B} + C_C = \frac{7\epsilon_0 S}{3d}$$

$$\therefore \frac{C_2}{C_1} = \frac{7}{3}$$

15. and 16. A ball in contact with the lower plate is at potential V_0 . If C is the capacitance of the ball, charge on it will be

$$q = CV_0.$$

The ball is pushed up by the electric field between the plates.

$$E = \frac{2V_0}{d} \quad [d = \text{plate separation}]$$

$$\text{Acceleration of the ball, } a = \frac{qE}{m} = \frac{CV_0}{m} \left(\frac{2V_0}{d} \right)$$

$$\text{or, } a = \frac{2CV_0^2}{md}$$

Time needed to reach the upper plate is given by

$$d = \frac{1}{2}at^2 \quad \Rightarrow \quad t = \sqrt{\frac{2d}{a}}$$

$$\text{or, } t = \sqrt{\frac{2md^2}{2CV_0^2}} \quad \Rightarrow \quad t \propto \frac{1}{V_0}$$

When a ball touches the upper plate, its potential becomes $-V_0$. Therefore, charge on it becomes $-q = -CV_0$. It means that a charge $+2q$ is released by a ball that flows through the wire.

Now, the ball will be experiencing a downward electric force and again in time t it will be back to the lower plate. This time the ball will gain $+2q$ charge to be at potential V_0 . Again $+2q$ charge flows in the wire.

If there are n balls, a total $n(2q)$ charge flows through the wire in every t second. The average current is

$$I = \frac{2nq}{t} \propto V_0^2$$

17. With S_1 closed, C_3 gets charged. Charge on it is

$$q_0 = C_3V_0 = (1\mu\text{F})(8\text{V}) = 8\mu\text{C}.$$

Opening S_1 disconnects the cell. Closing S_2 puts C_3 in parallel with the series combination of C_1 and C_2 . Let the series equivalent of C_1 and C_2 be C_0 .

It means that capacitor C_3 ($=1\mu\text{F}$) is given a charge of $8\mu\text{C}$ and connected to an uncharged capacitance C_0 . The final common potential difference will be

$$V = \frac{8\mu\text{C}}{(1+C_0)\mu\text{F}}$$

$$\text{Charge on } C_3 \text{ will be } C_3V = \frac{8 \times 1}{1+C_0}$$

$$\text{or, } \frac{8}{1+C_0} = 5$$

$$\Rightarrow 5C_0 = 3 \quad \Rightarrow \quad C_0 = \frac{3}{5}$$

Now, C_0 is series combination of C_1 and C_2 ($=1\mu\text{F}$)

$$\therefore \frac{C_1 \times 1}{C_1 + 1} = \frac{3}{5} \quad \Rightarrow \quad C_1 = 1.5\mu\text{F}$$

$\therefore$ Dielectric constant of material inside the capacitor C_1 is 1.5.

CURRENT, ELECTRICAL MEASUREMENT AND RC CIRCUIT

AIEEE/JEE Main Questions

1. When the galvanometer reads zero, the potential difference between the points B and C is

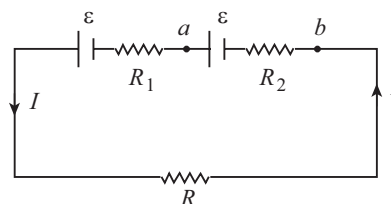
$$V_{BC} = E = 2\text{V}.$$

This is also the potential difference across the resistance X . The current supplied by E_1 is flowing through 500Ω and X and the potential drop across them are 10V and 2V , respectively.

$$\therefore \frac{I(500)}{I(X)} = \frac{10}{2} \quad \Rightarrow \quad X = 100\Omega.$$

2. Current in the circuit is

$$I = \frac{2\varepsilon}{R_1 + R_2 + R}$$



According to the question

$$V_a - V_b = 0$$

$$\Rightarrow \varepsilon = IR_2 \quad \Rightarrow \quad \varepsilon = \frac{2\varepsilon R_2}{R_1 + R_2 + R}$$

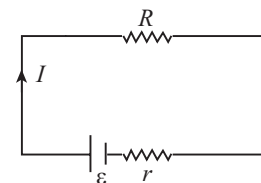
$$\Rightarrow R = R_2 - R_1$$

3. If R is a load resistance and r is the internal resistance, the current is

$$I = \frac{\varepsilon}{R + r}$$

If $r \gg R$, then

$$I \approx \frac{\varepsilon}{r} = a \text{ constant.}$$



4. Full-scale deflection current $i_g = 15 \text{ mA}$.

[$\because$ 1mA corresponds to a deflection of 10 divisions.

$\therefore$ 150 divisions correspond to a current of 15 mA].

Full-scale deflection is obtained when the applied potential difference is 75 mV .

$\therefore$ Resistance of the galvanometer is

$$R_g = \frac{75\text{mV}}{15\text{mA}} = 5\Omega$$

We have to make a voltmeter that can read upto 150V. Resistance that must be connected in series is given by

$$R = \frac{V}{i_g} - R_g = \frac{150}{15 \text{ mA}} - 5 = 9995 \Omega$$

$$5. \frac{55 \Omega}{R} = \frac{20 \text{ cm}}{80 \text{ cm}} \Rightarrow R = 220 \Omega$$

6. and 7. Consider current I entering at A and spreading uniformly over a hemispherical surface.

Current density at distance r from the point A is

$$j = \frac{I}{2\pi r^2}$$

$$\text{or, } \frac{E}{\rho} = \frac{I}{2\pi r^2}$$

$$\text{or, } E = \frac{\rho I}{2\pi r^2}$$

[This is answer to Q.7]

The potential difference between B and C can be written as

$$\int_{V_B}^{V_C} dV = - \int_a^{a+b} E dr$$

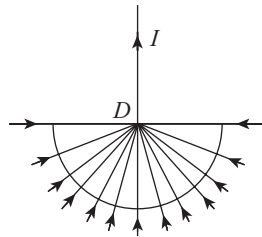
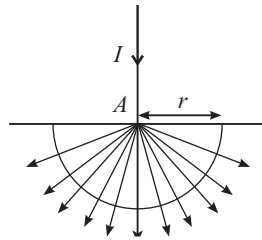
$$\text{or, } V_C - V_B = - \frac{\rho I}{2\pi} \int_a^{a+b} \frac{dr}{r^2}$$

$$\text{or, } V_C - V_B = \frac{\rho I}{2\pi} \left[\frac{1}{r} \right]_a^{a+b}$$

$$\text{or, } V_B - V_C = \frac{\rho I}{2\pi} \left[\frac{1}{a} - \frac{1}{a+b} \right] \quad (i)$$

Now, consider the current leaving the block at D . If this was the only current in the block, we can again work out the potential difference $V_B - V_C$. We get the same expression as in (i). With both the results superposed, the potential difference $V_B - V_C$ will be twice as that given in (i).

$$V_B - V_C = \frac{\rho I}{\pi} \left[\frac{1}{a} - \frac{1}{a+b} \right]$$



8. We can replace the two given batteries with a single battery of effective internal resistance and emf given by

$$\frac{1}{r_0} = \frac{1}{r_1} + \frac{1}{r_2} = \frac{1}{2} + \frac{1}{1}$$

$$\Rightarrow r_0 = \frac{2}{3} \Omega$$

$$\epsilon_0 = \frac{\frac{\epsilon_1 - \epsilon_2}{r_1 + r_2}}{\frac{1}{r_1} + \frac{1}{r_2}} = \frac{5 - 2}{\frac{1}{2} + \frac{1}{1}} = \frac{1}{3} V$$

$$\therefore I = \frac{\epsilon_0}{10 + \frac{2}{3}} = \frac{\frac{1}{3}}{\frac{32}{3}}$$

$$= \frac{1}{32} A = 0.03 A \quad [\text{From } P_2 \text{ to } P_1]$$

Alternatively, one may find the current using the Kirchhoff's laws.

9. Energy stored $\propto q^2$

$\therefore$ Energy stored becomes half means that the charge on the capacitor becomes $\frac{q_0}{\sqrt{2}}$.

$$q = q_0 e^{-t/\tau} \Rightarrow \frac{q_0}{\sqrt{2}} = q_0 e^{-\frac{t_1}{\tau}}$$

$$\text{or, } \frac{1}{\sqrt{2}} = e^{-\frac{t_1}{\tau}} \Rightarrow -\frac{1}{2} \ln 2 = -\frac{t_1}{\tau}$$

$$\text{or, } t_1 = \frac{\tau}{2} \ln 2$$

$$\text{Similarly, } \frac{q_0}{4} = q_0 e^{-\frac{t_2}{\tau}}$$

$$\text{or, } -2 \ln 2 = -\frac{t_2}{\tau}$$

$$\text{or, } t_2 = (2 \ln 2) \tau$$

$$\therefore \frac{t_1}{t_2} = \frac{1}{4}$$

10. Let the resistances be R_0 at $t = 0^\circ\text{C}$. The equivalent resistance, at 0°C , in series combination is

$$R_{s0} = 2R_0.$$

At temperature θ , the equivalent resistance will be

$$R_s = R_0 [1 + \alpha_1 \theta] + R_0 [1 + \alpha_2 \theta]$$

$$\text{or, } R_s = 2R_0 + R_0 (\alpha_1 + \alpha_2) \theta$$

$$\text{or, } R_s = 2R_0 \left[1 + \left(\frac{\alpha_1 + \alpha_2}{2} \right) \theta \right]$$

$$\text{or, } R_s = R_{s0} \left[1 + \left(\frac{\alpha_1 + \alpha_2}{2} \right) \theta \right]$$

Obviously, the effective value of α for series combination is $\frac{\alpha_1 + \alpha_2}{2}$.

When resistances are in parallel, their equivalent at 0°C is

$$R_{p0} = \frac{R_0}{2}$$

At temperature θ

$$\frac{1}{R_p} = \frac{1}{R_0(1 + \alpha_1\theta)} + \frac{1}{R_0(1 + \alpha_2\theta)}$$

$$\begin{aligned} \therefore R_p &= \frac{R_0(1 + \alpha_1\theta)(1 + \alpha_2\theta)}{2 + (\alpha_1 + \alpha_2)\theta} \\ &= \frac{R_0}{2} \left[\frac{2(1 + \alpha_1\theta)(1 + \alpha_2\theta)}{2 + (\alpha_1 + \alpha_2)\theta} \right] \end{aligned}$$

$$\text{or, } R_p = R_{p0} \left[\frac{1 + (\alpha_1 + \alpha_2)\theta + \alpha_1\alpha_2\theta^2}{1 + \left(\frac{\alpha_1 + \alpha_2}{2} \right) \theta} \right]$$

$\alpha_1\alpha_2\theta^2$ is a very small number. Let us neglect it.

$$\begin{aligned} \therefore R_p &= R_{p0} \left[1 + (\alpha_1 + \alpha_2)\theta \right] \left[1 + \left(\frac{\alpha_1 + \alpha_2}{2} \right) \theta \right]^{-1} \\ &= R_{p0} \left[1 + (\alpha_1 + \alpha_2)\theta \right] \left[1 - \left(\frac{\alpha_1 + \alpha_2}{2} \right) \theta \right] \\ &= R_{p0} \left[1 + (\alpha_1 + \alpha_2)\theta - \left(\frac{\alpha_1 + \alpha_2}{2} \right) \theta - \frac{(\alpha_1 + \alpha_2)^2}{2} \theta^2 \right] \end{aligned}$$

Again the last term inside the bracket is too small.

$$\therefore R_p = R_{p0} \left[1 + \left(\frac{\alpha_1 + \alpha_2}{2} \right) \theta \right]$$

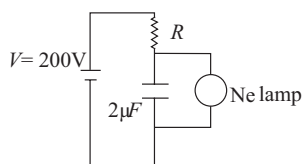
$$\therefore \alpha_{\text{eff}} = \frac{\alpha_1 + \alpha_2}{2}$$

11. Ne bulb is not conducting when it is not lit. It means there is no current in the bulb till potential difference across it becomes 120V.

The circuit is a simple RC circuit till the bulb lights. It is given that the potential difference across the capacitor becomes 120V at $t = 5\text{ s}$

$$q = q_0(1 - e^{-\frac{t}{\tau}})$$

$$\text{or, } \frac{q}{C} = \frac{q_0}{C} \left(1 - e^{-\frac{t}{\tau}} \right)$$



$$\text{or, } V = V_0(1 - e^{-\frac{t}{\tau}})$$

$$\text{or, } 120 = 200(1 - e^{-\frac{t}{\tau}})$$

$$\text{or, } -e^{-\frac{5}{\tau}} = 0.4$$

$$\Rightarrow -\frac{5}{\tau} = \ln\left(\frac{4}{10}\right)$$

$$\text{or, } \tau = \frac{5}{\ln\left(\frac{10}{4}\right)}$$

$$\text{or, } \tau = \frac{5}{\ln(2.5)} = \frac{5}{0.4 \times 2.303}$$

$$\therefore RC = \frac{5}{0.4 \times 2.303}$$

$$\text{or, } R = \frac{5}{0.4 \times 2.303 \times 2 \times 10^{-6}} = 2.7 \times 10^6 \Omega$$

12. Resistance of a bulb $R = \frac{V^2}{P}$.

$$\Rightarrow R \propto \frac{1}{P}$$

A 25 W bulb has 4 times higher resistance than 100W bulb.

When they are connected in series, the potential drop across 25W bulb will be 4 times that across the 100W bulb. The potential drop across 25W bulb will be

$$V = \frac{4}{5} \times 440 = 352\text{V}.$$

This bulb will fuse.

13. In one time constant, the voltage across the capacitor will reduce by 63% and the voltage becomes 0.37 times the original voltage.

$$0.37V_0 = 0.37 \times 25 = 9.25\text{V}$$

From the graph, it can be seen that the voltage becomes 9.25V at a time that is greater than 150s but less than 200 s.

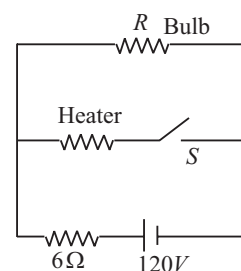
14. Resistance of 60W bulb is

$$R = \frac{(120)^2}{60} = 240\Omega$$

$$\text{Current } I = \frac{120}{6 + 240} = 0.49\text{A}$$

Voltage across the bulb is

$$V_0 = 0.49 \times 240 = 117.6\text{V}.$$



Resistance of the heater is

$$R_H = \frac{(120)^2}{240} = 60\Omega$$

Equivalent of R and R_H is $\frac{240 \times 60}{240 + 60} = 48\Omega$

Current $i = \frac{120}{48 + 6} = 2.22\text{ A}$

Drop across the bulb and the heater is

$$V = 2.22 \times 48 = 106.7\text{ V}$$

Change in voltage = $V_0 - V$
 $= 117.6 - 106.7 = 10.9\text{ V}$.

The closest answer is 10.04 V.

15. All appliances must be connected in parallel.

Current in a bulb of 40 W will be $\frac{40}{220} = \frac{2}{11}\text{ A}$

Current in a 100 W bulb is $\frac{100}{220} = \frac{5}{11}\text{ A}$

Current in 1000W heater is $\frac{1000}{220} = \frac{50}{11}\text{ A}$

Current in an 80W fan = $\frac{80}{220} = \frac{4}{11}\text{ A}$

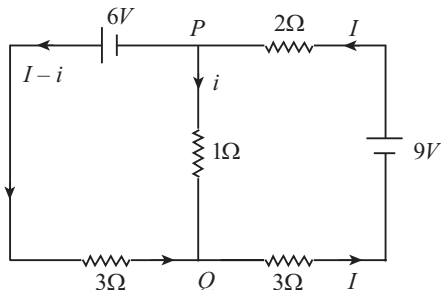
$$\therefore \text{Total current} = 15 \times \frac{2}{11} + 5 \times \frac{5}{11} + 5 \times \frac{4}{11} + \frac{50}{11}$$

$$= \frac{125}{11} = 11.4\text{ A}$$

$\therefore$ Rating of fuse needed should be larger than 11.4 A, but not too large.

Hence, the correct answer appears to be 12A.

16. Let current through 9V cell be I . It branches at P as i and $I-i$ as shown.



Using Kirchhoff's loop law in the left loop, we get

$$6 - 3(I - i) + 1(i) = 0$$

$$\Rightarrow 4i - 3I = -6 \quad \text{(i)}$$

Writing loop equation in the right loop

$$9 - 2I - 1(i) - 3I = 0$$

$$\Rightarrow 5I + i = 9 \quad \text{(ii)}$$

Solving (i) and (ii) gives $i = -\frac{3}{23} = -0.13\text{ A}$

Negative sign indicates that the current is from Q and P .

17 $I = jA = nev_d \cdot A$

or, $\frac{V}{R} = nev_d A \Rightarrow V = \frac{\rho l}{A} \cdot nev_d A$

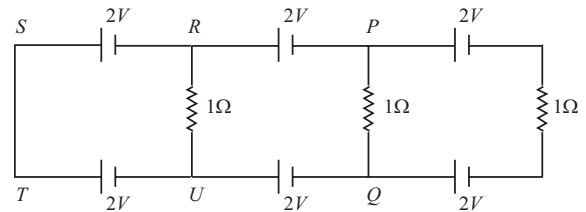
or, $\rho = \frac{V}{nev_d l} = \frac{5}{8 \times 10^{28} \times 1.6 \times 10^{-19} \times 2.5 \times 10^{-4} \times 0.1}$
 $= 1.6 \times 10^{-5}\Omega\text{m}$

18. The required shunt is

$$R_s = \frac{i_g R_g}{(I - i_g)} = \frac{1 \times 10^{-3} \times 100}{10 - 1 \times 10^{-3}} \simeq 0.01\Omega$$

19. There is no potential difference across any resistance.

For example, consider the resistance between the points P and Q . We will write potential difference across it while moving along $P R S T U Q$.



$$V_P + 2 + 2 - 2 - 2 = V_Q$$

$$\Rightarrow V_P - V_Q = 0.$$

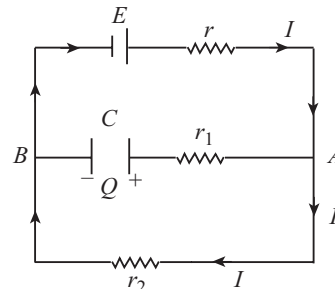
$\therefore$ There is no current in branch PQ . Similarly, there is no current in any resistor.

20. $I_g = 5\text{ mA}$; $R_g = 15\Omega$

$$R = \frac{V}{I_g} - R_g = \frac{10}{5 \times 10^{-3}} - 15 = 2000 - 15$$

$$= 1985 = 1.985 \times 10^3\Omega$$

21. In the steady state, the current flows as shown (there is no current through the capacitor branch).



$$I = \frac{E}{r + r_2}$$

$$V_A - V_B = Ir_2 = \frac{Er_2}{r + r_2}$$

This is also the potential difference across the capacitor.

$$\therefore Q = C(V_A - V_B) = \frac{CEr_2}{r + r_2}$$

22. The emf and internal resistance of the equivalent of two cells in parallel are

$$\begin{aligned} \varepsilon_0 &= \frac{\frac{\varepsilon_1 + \varepsilon_2}{\frac{1}{r_1} + \frac{1}{r_2}}}{\frac{1}{r_1} + \frac{1}{r_2}} = \frac{\frac{12 + 13}{\frac{1}{1} + \frac{1}{2}}}{\frac{1}{1} + \frac{1}{2}} \\ &= \frac{37}{3} V \end{aligned}$$

$$\frac{1}{r_0} = \frac{1}{1} + \frac{1}{2} \Rightarrow r_0 = \frac{2}{3} \Omega$$

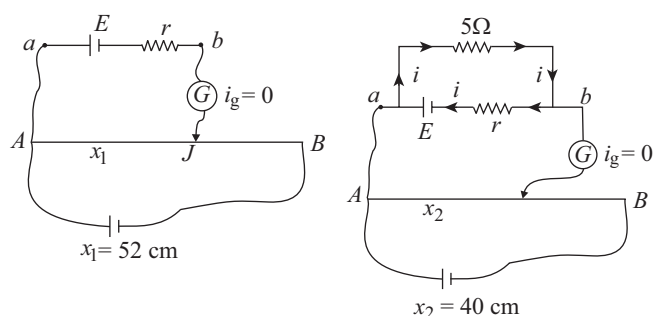
Current in resistance R is

$$I = \frac{\varepsilon_0}{R + r_0} = \frac{\frac{37}{3}}{10 + \frac{2}{3}} = \frac{37}{32} A$$

or, $I = 1.156 A$

Voltage across R is $= IR = 11.56V$

23.



$$E \propto 52 \text{ cm} \quad (i)$$

When 5Ω resistor is connected across the cell, a current i flows through the cell and the 5Ω resistor. In balanced position, there is no current through the galvanometer.

$$\therefore i = \frac{E}{r + 5}$$

$$V_{ab} = 5i = \frac{5E}{r + 5}$$

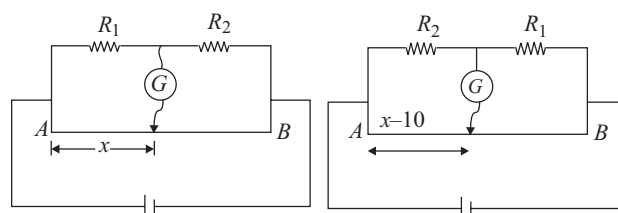
$$\therefore \frac{5E}{r + 5} \propto 40 \text{ cm} \quad (ii)$$

Taking ratio of (ii) and (i) gives

$$\frac{5}{r + 5} = \frac{40}{52}$$

$$\Rightarrow r = 1.5 \Omega$$

24. In the figure, AB is the meter bridge wire.



$$\frac{R_1}{R_2} = \frac{x}{100 - x} \quad (i)$$

$$\text{and, } \frac{R_2}{R_1} = \frac{x - 10}{100 - (x - 10)} \quad (ii)$$

From (i) and (ii)

$$\frac{100 - x}{x} = \frac{x - 10}{110 - x}$$

or, $x = 55 \text{ cm}$

$$\text{From (i) } \frac{R_1}{R_2} = \frac{55}{45} = \frac{11}{9}$$

$$\text{Also, } R_1 + R_2 = 1000$$

$$\text{or, } R_1 + \frac{9}{11} R_1 = 1000$$

$$\text{or, } R_1 = 550 \Omega$$

IIT JEE/JEE Advanced Questions

25. There cannot be a current in 2Ω resistance. There is no return path for any current in 2Ω resistance.

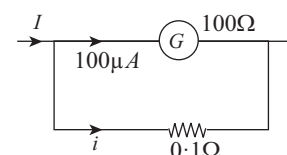
$$26. (0.1)i = (100)(100 \mu A)$$

$$\text{or, } I = 10^5 \mu A$$

$$= 100 \text{ mA}$$

$$\therefore I = 100 \text{ mA} + 100 \mu A$$

$$= 100.1 \text{ mA}$$



27. The gas container is rigid. This means no work is done by the gas. The container is thermally insulated. This means that heat supplied to the gas is zero. The entire heat generated in the resistance is used to increase the internal energy of the gas

$$\Delta U = i^2 R \Delta t = (1)^2 (100) (5 \times 60)$$

$$= 30,000 \text{ J} = 30 \text{ kJ}$$

28. $\tau = RC = 2.5 \times 10^6 \times 4 \times 10^{-6} = 10 \text{ s}$

$V_C = 3V_R$

And $V_C + V_R = 12$

or, $4V_R = 12 \Rightarrow V_R = 3V$

We know that $i = i_0 e^{-t/\tau}$

or, $iR = i_0 R e^{-t/\tau}$

or, $V = V_0 e^{-t/\tau}$

$\Rightarrow 3 = 12 e^{-t/\tau}$

or, $e^{-t/\tau} = \frac{1}{4} \Rightarrow -\frac{t}{\tau} = \ln\left(\frac{1}{4}\right)$

or, $t = \tau(2\ln 2) = 10 \times 2 \times 0.693$
 $= 13.86 \text{ s}$

29. Point B is near the centre of the meter bridge wire. When the balance point is obtained at the midpoint, the error in the value of unknown resistance X is smallest.

30. In the steady state (when the capacitor is fully charged), the current flows in the outer loop only. Current is

$$I = \frac{V}{R_1 + R_2}$$

$\therefore$ Potential difference across R_2 is

$$V_2 = IR_2 = \frac{VR_2}{R_1 + R_2}$$

This is also the potential difference across the capacitor.

$$\therefore Q_0 = CV_2 = \frac{CVR_2}{R_1 + R_2}$$

31. To get the time constant, we first find the equivalent resistance across the capacitor plates after shorting the cell.

$$R_{ab} = \frac{R_1 R_2}{R_1 + R_2}$$

$$\therefore \tau = R_{ab} C = \frac{R_1 R_2 C}{R_1 + R_2}$$

Given $Q = Q_0 [1 - e^{-\alpha t}]$

$$\alpha = \frac{1}{\tau} = \frac{R_1 + R_2}{R_1 R_2 C}$$

32. $R_{AB} = \frac{\rho \frac{l}{2}}{\pi (2r)^2} = \frac{\rho l}{8\pi r^2}$

$$R_{BC} = \frac{\rho \frac{l}{2}}{\pi r^2} = \frac{\rho l}{2\pi r^2}$$

$\therefore 4 R_{AB} = R_{BC}$

Hence, $4V_{AB} = V_{BC}$

$P_{AB} = I^2 R_{AB}$ and $P_{BC} = I^2 R_{BC}$

$\therefore P_{BC} = 4P_{AB}$

Current densities in the two sections are not equal and therefore the electric field will also not be equal.

33. In the first circuit:

$R = R_1 + R_2 = 3\Omega$, $C = C_1 + C_2 = 6\text{mF}$

$\therefore \tau_1 = RC = 18 \text{ ms}$

In the second circuit:

$$R = \frac{R_1 R_2}{R_1 + R_2} = \frac{1 \times 2}{1 + 2} = \frac{2}{3}\Omega,$$

$$C = \frac{C_1 C_2}{C_1 + C_2} = \frac{4 \times 2}{4 + 2} = \frac{4}{3}\text{mF}$$

$$\therefore \tau_2 = RC = \frac{2}{3} \times \frac{4}{3} = \frac{8}{9}\text{ms}$$

In the third circuit:

$$R = \frac{R_1 R_2}{R_1 + R_2} = \frac{1 \times 2}{1 + 2} = \frac{2}{3}\Omega, C = C_1 + C_2 = 6\text{mF}$$

$$\therefore \tau_3 = RC = \frac{2}{3} \times 6 = 4 \text{ ms}$$

34. The height of liquid level at time t is $\left(\frac{d}{3} - Vt\right)$.

Capacitance at time t is

$$C = \frac{\epsilon_0 A}{\frac{x_1}{k_1} + \frac{x_2}{k_2}} = \frac{\epsilon_0}{\frac{\frac{d}{3} - Vt}{2} + \frac{d - \left(\frac{d}{3} - Vt\right)}{1}} \quad [\because A = 1]$$

$$= \frac{6\epsilon_0}{5d + 3Vt}$$

$$\therefore \tau = RC = \frac{6\epsilon_0 R}{5d + 3Vt}$$

35. $R = \frac{\rho L}{A} = \frac{\rho L}{tL} = \frac{\rho}{t}$

$\therefore R$ is independent of L .

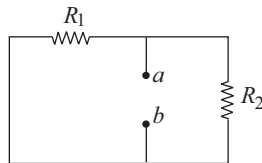
36. A voltmeter can be made by connecting a high resistance in series with a galvanometer and an ammeter can be made by connecting a small resistance in parallel to a galvanometer.

Therefore, the correct circuit is (c).

37. The bulbs must be designed to consume the rated power when their filament attains the steady state temperature. When the filaments are hot, the resistance of the filament is given by

$$R^1 = \frac{V^2}{P} \Rightarrow R' \propto \frac{1}{P}$$

$\therefore R'_{40} > R'_{60} > R'_{100}$



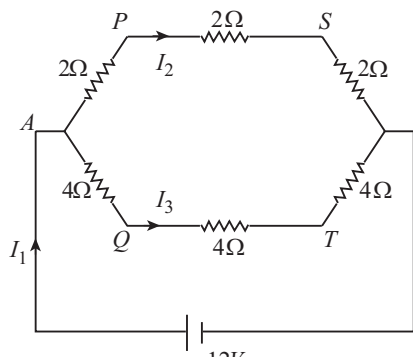
The 40W bulb will have least hot filament and the 100W bulb will have the hottest filament. It means that the temperature increase (and hence change in resistance) will be highest for the 100W bulb and smallest for the 40W bulb.

It means that the resistance of 40W bulb (R'_{40}) shall be higher than R'_{100} despite the fact that increase in resistance is more for 100W bulb. This can happen only if resistance of the 40W bulb at room temperature is much higher than the 100 W bulb.

$$\therefore R_{40} > R_{60} > R_{100} \Rightarrow \frac{1}{R_{40}} < \frac{1}{R_{60}} < \frac{1}{R_{100}}$$

$$38. \frac{x}{10} = \frac{52+1}{48+2} \Rightarrow x = \frac{530}{50} = 10.6\Omega$$

39. Points P and Q are equipotential. Similarly, the points S and T are at the same potential.



There is no current in two 1Ω resistors. Circuit has been redrawn after removing them.

$$I_2 = \frac{12}{2+2+2} = 2A$$

$$I_3 = \frac{12}{4+4+4} = 1A$$

$$\therefore I_1 = I_2 + I_3 = 3A.$$

$$V_s = V_A - (2 + 2)I_2 = V_A - 8$$

$$V_Q = V_A - 4I_3 = V_A - 4$$

$$\therefore V_s < V_Q.$$

40. Let Q be the heat needed to raise the temperature of 0.5 kg water by 40K.

$$\text{Resistance of the wire in the heater is } R = \frac{\rho L}{\pi \frac{d^2}{4}}$$

$$\text{Given } Q = \frac{V^2}{R} (4 \text{ minute}) \quad (i)$$

$$\text{The new wires have resistance } \frac{\rho L}{\pi \frac{(2d)^2}{4}} = \frac{R}{4}.$$

When the wires are connected in series, the effective resistance is $\frac{R}{2}$. Power dissipated is $\frac{V^2}{\frac{R}{2}} = \frac{2V^2}{R}$. If

t_1 , is time needed to heat the water, then

$$Q = \frac{2V^2}{R} t_1 \quad (ii)$$

From (i) and (ii), $t_1 = 2 \text{ minute}$

When the wires are in parallel, the effective resistance is $\frac{R}{8}$.

$$\text{Power dissipated} = \frac{V^2}{\frac{R}{8}}$$

If t_2 is time needed to heat the water, then

$$Q = \frac{V^2}{\frac{R}{8}} t_2$$

Using (i), $t_2 = 0.5 \text{ min.}$

41. When current in R_2 is zero, the circuit can be redrawn as shown.

When $IR_1 = V_1$, the potential difference $V_P - V_Q$ will be zero, and current in R_2 will be zero.

$$IR_1 = V_1$$

$$\Rightarrow \left(\frac{V_1 + V_2}{R_1 + R_2} \right) R_1 = V_1$$

This condition is satisfied for situations given in options (A), (B) and (D).

42. $\frac{R}{90} = \frac{x}{y}$ where $y = 100 - x$

$$\therefore R = 90 \times \frac{40}{60} = 60\Omega$$

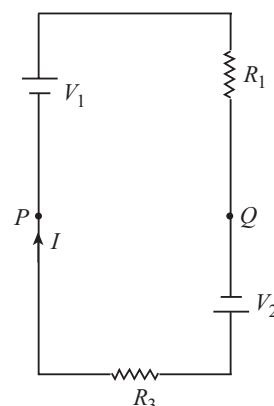
$$\text{Now, } R = 90 \left(\frac{x}{y} \right)$$

For fractional error, we can write

$$\frac{\Delta R}{R} = \frac{\Delta x}{x} + \frac{\Delta y}{y} = \frac{0.1}{40} + \frac{0.1}{60}$$

$$\therefore \Delta R = 60 \left(\frac{0.1}{40} + \frac{0.1}{60} \right) = 0.15 + 0.1 = 0.25\Omega$$

$$\therefore R = 60 \pm 0.25 \Omega.$$



43. The two bars made of Fe and Al are in parallel.

$$R_{Fe} = \frac{\rho_{Fe} l}{A_{Fe}} = \frac{1.0 \times 10^{-7} \times 50 \times 10^{-3}}{4 \times 10^{-6}} = 12.5 \times 10^{-4} \Omega$$

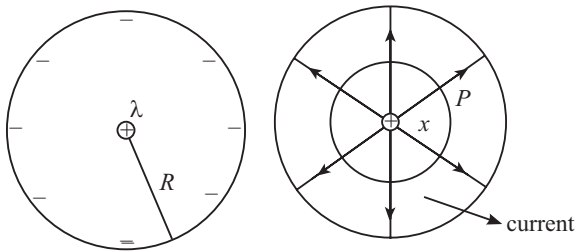
$$R_{Al} = \frac{\rho_{Al} l}{A_{Al}} = \frac{2.7 \times 10^{-8} \times 50 \times 10^{-3}}{(7^2 - 2^2) \times 10^{-6}} = 0.3 \times 10^{-4} \Omega$$

$$\therefore R = \frac{R_{Fe} R_{Al}}{R_{Fe} + R_{Al}} = \frac{12.5 \times 0.3}{12.5 + 0.3} \times 10^{-4}$$

$$= \frac{\frac{50}{4} \times \frac{3}{10}}{\frac{50}{4} + \frac{3}{10}} \times 10^{-4} = \frac{150}{512} \times 10^{-4} \Omega$$

$$= \frac{15000}{512} \mu\Omega = \frac{1875}{64} \mu\Omega$$

44. Negative charge is induced on the inner wall of the cylindrical shell. A capacitor is formed between the line charge and the shell.



Let the capacitance be C .

The material filling the space between the line charge and the shell has a resistance R . The charge leaks through this resistance. The leakage current is obviously given by

$$I = i_0 e^{-t/\tau} \quad [i = \text{current in length } l \text{ of the cylinder}]$$

Current density at a point like P will be

$$j = \frac{i}{2\pi x l}$$

$$\therefore j \propto i$$

Graphical variation of j with t is shown in fig (c).

45. As the tungsten evaporates, the filament gets thinner and thinner. Therefore, the resistance increases ($\because R \propto \frac{1}{A}$). Power consumed $P = \frac{V^2}{R}$ decreases with an increase in R .

Initially, before the filament breaks up at random locations, more heat is produced in the filament and it is hotter. According to Wien's law of blackbody radiation, a hotter body radiates more at higher frequencies (i.e., lower wavelength).

46. We will consider the voltmeter to be ideal. Its resistance is infinite and there is no current through it.

At $t = 0^+$, the uncharged capacitors offer no resistance. The effective circuit is as shown.

$$V_A = V_P \quad (i)$$

$$V_A - V_Q = V_A - V_B \quad [\because V_Q = V_B]$$

$$= 5V$$

$$\text{or, } V_P - V_Q = 5V \quad [\text{using (i)}]$$

Voltmeter has been shown to read a positive voltage when Q is at higher potential.

$\therefore$ Reading of the voltmeter = $-5V$

At $t = \infty$, the capacitors are fully charged. There cannot be a current through them. Potential drop across both the capacitors is $5V$.

$$\therefore V_A - V_P = 5V$$

$$\text{or, } V_Q - V_P = 5V$$

The voltmeter will read $+5V$.

The time constant of $40\mu F$, $25k\Omega$ combination is

$$\tau_1 = 40 \times 10^{-6} \times 25 \times 10^3 = 1 \text{ s.}$$

The time constant of $50k\Omega$, $20\mu F$ combination is

$$\tau_2 = 20 \times 10^{-6} \times 50 \times 10^3 = 1 \text{ s.}$$

Current in upper branch at time $t = \ln 2$ is

$$i_1 = \left(\frac{5}{25 \times 10^3} \right) e^{-\frac{\ln 2}{1}}$$

$$= \left(\frac{1}{5} \times \frac{1}{2} \right) \text{mA} = \frac{1}{10} \text{mA}$$

Potential drop in $25k\Omega$

$$= (25k\Omega) \left(\frac{1}{10} \text{mA} \right) = 2.5 \text{ V}$$

$\therefore$ Drop across $40\mu F$ is 2.5 V .

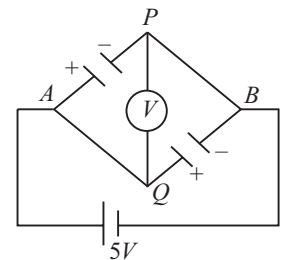
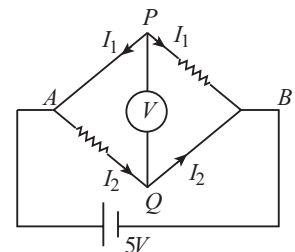
Similarly, for lower branch, current at $t = \ln 2$ is

$$i_2 = \left(\frac{5}{50 \times 10^3} \right) e^{-\ln 2} = \frac{1}{20} \text{mA}$$

$$\therefore \text{Potential drop in } 50k\Omega = (50k\Omega) \left(\frac{1}{20} \text{mA} \right) = 2.5 \text{ V}$$

$$\therefore V_P = V_Q$$

Reading of the voltmeter is zero.



Current in ammeter at 1 s is

$$i = i_1 + i_2 = \left(\frac{1}{5}e^{-1} + \frac{1}{10}e^{-1} \right) \text{mA}$$

$$= \left(\frac{1}{5} + \frac{1}{10} \right) \frac{1}{e} \text{mA}$$

$$= (i_0) \left(\frac{1}{e} \right)$$

$$\therefore \text{Initial current } i_0 = \frac{5}{25k\Omega} + \frac{5}{50k\Omega} = \left(\frac{1}{5} + \frac{1}{10} \right) \text{mA}$$

47. For maximum voltage range, the resistance should be largest. For maximum current range, resistance connected in parallel to a galvanometer must be smallest. This will happen when all resistances are connected in parallel to a galvanometer.

48. Charge on the capacitor $Q = CV_0$

$$W_{\text{batt}} = Q \cdot V_0 = CV_0^2$$

$$E_c = \frac{1}{2} CV_0^2$$

$$\text{Heat dissipated } E_D = W_{\text{batt}} - E_c = \frac{1}{2} CV_0^2$$

29. Charge supplied by the first cell, $Q_1 = C \frac{V_0}{3}$

$$\text{Work done by the first cell, } W_1 = Q_1 \frac{V_0}{3} = \frac{CV_0^2}{9}$$

Charge supplied by the second cell,

$$Q_2 = C \left(\frac{2V_0}{3} \right) - Q_1 = \frac{CV_0}{3}$$

$$\text{Work done by the second cell, } W_2 = Q_2 \left(\frac{2V_0}{3} \right)$$

$$= \left(\frac{CV_0}{3} \right) \left(\frac{2V_0}{3} \right) = \frac{2}{9} CV_0^2$$

Charge supplied by the third cell,

$$Q_3 = CV_0 - \frac{2CV_0}{3} = \frac{CV_0}{3}$$

$$\text{Work done by the third cell, } W_3 = Q_3 V_0 = \frac{CV_0^2}{3}$$

$\therefore$ Total work done by three cells

$$W = W_1 + W_2 + W_3$$

$$= \frac{CV_0^2}{9} + \frac{2}{9} CV_0^2 + \frac{CV_0^2}{3} = \frac{2}{3} CV_0^2$$

$\therefore$ Heat dissipated = $W - U$

$$= \frac{2}{3} CV_0^2 - \frac{1}{2} CV_0^2 = \frac{1}{6} CV_0^2$$

ERROR AND MEASUREMENT

AIEEE/JEE Main Questions

1. Least count = $\frac{0.5 \text{ mm}}{50} = 0.01 \text{ mm}$

$$\text{Reading} = \text{MSR} + (\text{CSR})(\text{LC})$$

$$= 3 + 35 \times 0.01 = 3.35 \text{ mm}$$

Correction for zero error:

$$\text{Diameter} = 3.35 + 0.03 = 3.38 \text{ mm}$$

2. $\text{LC} = 1\text{MSD} - 1\text{VSD}$

$$= 0.5^\circ - \frac{29}{30} \times 0.5^\circ = \left(\frac{1}{60} \right)^\circ = 1 \text{ min}$$

3. 23.023 has 5 significant digits.

0.0003 has 1 significant digits as leading zeros are not significant.

2.1×10^{-3} has 2 significant digits.

4. $\text{LC} = \frac{1 \text{ mm}}{100} = 0.01 \text{ mm}$

$$\text{Diameter} = \text{MSR} + (\text{LC})(\text{CSR}) = 0 + 0.01 \times 52$$

$$= 0.52 \text{ mm}$$

$$= 0.052 \text{ cm}$$

5. $R_{\text{eq}} = 4R$

$$\Delta R_{\text{eq}} = 4\Delta R$$

$$\text{or, } \frac{\Delta R_{\text{eq}}}{R_{\text{eq}}} = \frac{\Delta R}{R}$$

$$\Rightarrow \frac{\Delta R_{\text{eq}}}{R_{\text{eq}}} \times 100 = \frac{\Delta R}{R} \times 100 = 5\%$$

6. $R = \frac{V}{I} = V^1 I^{-1}$

$$\therefore \frac{\Delta R}{R} \times 100 = 1 \times \left(\frac{\Delta V}{V} \times 100 \right) + 1 \times \left(\frac{\Delta I}{I} \times 100 \right)$$

$$= 3\% + 3\% = 6\%$$

7. $I = e^{\frac{1000V}{T}} - 1$

$$\text{When } I = 5 \text{ mA, } e^{\frac{1000V}{T}} = 6 \text{ mA}$$

$$\text{Now, } I = e^{\frac{1000V}{T}} - 1$$

$$\text{or, } \frac{dI}{dV} = \frac{1000}{T} e^{\frac{1000V}{T}}$$

$$\text{or, } \Delta I = \frac{1000}{T} e^{\frac{1000V}{T}} \cdot \Delta V$$

$$= \frac{1000}{300} \times (6 \text{ mA}) \times 0.01 = 0.2 \text{ mA.}$$

Note that the value of temperature has been assumed to be exact at 300K. Error is propagating into the value of I due to error in measurement of V .

8. $L = (20.0 \pm 0.1)\text{cm}$

$$T = \frac{t}{100} \Rightarrow \Delta T = \frac{\Delta t}{100} = \frac{1}{100} = 0.01 \text{ s}$$

$$\therefore T = (0.90 \pm 0.01)\text{s}$$

$$\text{Now, } g = 4\pi^2 \frac{L}{T^2}$$

$$\begin{aligned} \Rightarrow \frac{\Delta g}{g} \times 100 &= \frac{\Delta L}{L} \times 100 + 2 \times \frac{\Delta T}{T} \times 100 \\ &= \frac{0.1}{20.0} \times 100 + 2 \times \frac{0.01}{0.90} \times 100 \\ &= 0.5 + 2.2 = 2.7\% \simeq 3\% \end{aligned}$$

9. $T_{\text{av}} = \frac{90+91+95+92}{4} = 92 \text{ s}$

Standard deviation is given by

$$\sigma = \sqrt{\frac{|90-92|^2 + |91-92|^2 + |95-92|^2 + |92-92|^2}{4-1}}$$

$$\therefore \sigma = \sqrt{\frac{4+1+9+0}{3}} = 2.16$$

$\therefore$ The best answer will be $(92 \pm 2)\text{s}$.

10. $\text{LC} = \frac{0.5\text{mm}}{50} = 0.01 \text{ mm}$

With the jaws closed, the zero of the circular scale overshoots by 5 divisions.

$$\therefore \text{Zero error} = -5 \times 0.01 = -0.05 \text{ mm}$$

$$\begin{aligned} \text{Thickness recorded} &= 0.5 \text{ mm} + 25 \times 0.01 \\ &= 0.75 \text{ mm} \end{aligned}$$

Applying correction for the zero error, we get thickness = $0.75 + 0.05 = 0.80 \text{ mm}$.

IIT JEE/JEE Advanced Questions

11. Side length, $l = 10 + 1(\text{LC})$

$$= 10 + 1(0.1) = 10.1 \text{ mm}$$

$$\text{Volume, } V = l^3 = (10.1)^3 \text{ mm}^3$$

$$\text{Density, } d = \frac{\text{Mass}}{\text{Volume}} = \frac{2.735\text{g}}{(10.1)^3 \text{ mm}^3}$$

$$= 0.002655 \frac{\text{g}}{\text{mm}^3}$$

$$= 2.655 \frac{\text{g}}{\text{cm}^3}$$

$$= 2.66 \frac{\text{g}}{\text{cm}^3}$$

We have reported the answer to 3 significant digits only. This is because the value of l has 3 significant digits only.

12. Zero error = $+5 (\text{LC}) = 5 \left(\frac{0.5}{50} \text{mm} \right) = 0.05 \text{ mm}$.

With balls between the jaws, the main scale reading is 2 divisions = $2 \times 0.5 = 1.0 \text{ mm}$

Circular scale reading is 25.

$$\begin{aligned} \therefore \text{Measured diameter} &= 1.0 + 25 \times 0.01 \\ &= 1.25 \text{ mm.} \end{aligned}$$

Applying zero correction, the correct diameter is = $1.25 - 0.05 = 1.20 \text{ mm}$.

13. In option (D), Δl is smallest and error in measurement of time period is also smallest. Hence, this is the correct option.

14. $Y = \frac{\text{Stress}}{\text{Strain}} = \frac{Mg}{\pi d^2 \cdot \frac{x}{L}} = \left(\frac{4LgM}{\pi} \right) \frac{1}{d^2 x}$

The quantity inside the bracket is constant

$$Y = \frac{4 \times 2 \times 9.8}{3.14} \times \frac{1}{(0.4 \times 10^{-3})^2 \times (0.8 \times 10^{-3})}$$

$$= 2.0 \times 10^{11} \frac{\text{N}}{\text{m}^2}$$

$$\frac{\Delta Y}{Y} = 2 \frac{\Delta d}{d} + \frac{\Delta x}{x}$$

$$\Delta Y = 2 \times 10^{11} \left[2 \times \frac{0.01}{0.4} + \frac{0.05}{0.8} \right]$$

$$\simeq (0.1 + 0.1) \times 10^{11} = 0.2 \times 10^{11}$$

$$\therefore Y = (2.0 \pm 0.2) \times 10^{11} \frac{\text{N}}{\text{m}^2}$$

15. $g = 4\pi^2 \frac{L}{T^2}$

$$\frac{\Delta g}{g} \times 100 = \frac{\Delta L}{L} \times 100 + 2 \frac{\Delta T}{T} \times 100$$

$$T = \frac{t}{n} \Rightarrow \Delta T = \frac{\Delta t}{n}$$

$$E_{\text{I}} = \frac{0.1}{64} \times 100 + 2 \times \frac{0.1/8}{16.0} \times 100$$

$$= 0.156 + 0.313 = 0.469\%$$

$$E_{\text{II}} = \frac{0.1}{64} \times 100 + 2 \times \frac{0.1/4}{16.0} \times 100$$

$$= 0.156 + 0.625 = 0.781\%$$

$$E_{III} = \frac{0.1}{20} \times 100 + 2 \times \frac{0.1/4}{9.0} \times 100$$

$$= 0.5 + 5.56 = 6.06 \%$$

$$\therefore E_{III} > E_{II} > E_I$$

$$16. LC = 1MSD - 1VSD = 1.0 - \frac{16}{20} = 0.2 \text{ mm}$$

$$17. \text{Diameter, } d = 2.5 + 20 \times \frac{0.5}{50} = 2.70 \text{ mm}$$

Uncertainty in value of d = least count = 0.01 mm

$$\text{density } \rho = \frac{M}{\frac{4}{3}\pi\left(\frac{d}{2}\right)^3} = KM^1d^{-3} \quad [K \text{ is a constant}]$$

$$\therefore \frac{\Delta\rho}{\rho} \times 100 = \frac{\Delta M}{M} \times 100 + 3 \frac{\Delta d}{d} \times 100$$

$$= 2\% + 3 \times \frac{0.01}{2.70} \times 100\%$$

$$= 3.1\%$$

$$18. \frac{\Delta Y}{Y} = \frac{\Delta l}{l} + 2 \frac{\Delta d}{d}$$

$$= \frac{0.005}{0.25} + 2 \cdot \frac{0.005}{0.5}$$

$$[\because \Delta l = \Delta d = LC = 0.005 \text{ mm}]$$

$$= \frac{0.005}{0.25} + \frac{0.005}{0.25}$$

$\therefore$ Contribution in error of Y is same due to errors in l and d .

$$19. LC = 1MSD - 1VSD = 0.05 \text{ cm} - \frac{2.45}{50} \text{ cm}$$

$$= 0.001 \text{ cm}$$

Note that the zero of the vernier is between 5.10 cm and 5.15 cm. This implies that the divisions on the main scale are 0.05 cm wide.

$$\therefore \text{Measured diameter} = 5.10 + 24 \times 0.001$$

$$= 5.124 \text{ cm}$$

$$20. E = A^2 e^{-\alpha t}$$

$$\text{or, } \ln E = \ln(A^2) - \alpha t$$

$$\text{or, } \frac{1}{E} \frac{dE}{dt} = \frac{2}{A} \frac{dA}{dt} - \alpha$$

$$\text{or, } \frac{\Delta E}{E} \approx \frac{2\Delta A}{A} - \alpha \Delta t$$

For maximum possible error

$$\left| \frac{\Delta E}{E} \right| = 2 \left| \frac{\Delta A}{A} \right| + |\alpha \Delta t|$$

$$\left| \frac{\Delta E}{E} \times 100 \right| = 2 \left| \frac{\Delta A}{A} \times 100 \right| + |\alpha \Delta t \times 100|$$

$$= 2 \times 1.25\% + 0.2 \frac{5 \times 1.5}{100} \times 100\%$$

$$= 4\%$$

Note that $\Delta t = 1.5\%$ of 5

21. For vernier C_1 :

$$10VSD = 9MSD = 9\text{mm}$$

$$1 \text{ VSD} = 0.9\text{mm}$$

$$\Rightarrow LC = 1MSD - 1VSD = 1\text{mm} - 0.9\text{mm} = 0.1\text{mm}$$

$$\text{Reading of } C_1 = MSR + (VSR)(LC)$$

$$= 28\text{mm} + (7)(0.1)$$

$$\text{Reading of } C_1 = 2.87\text{cm}$$

For vernier C_2 :

The vernier C_2 is abnormal. So we need to find the reading using the basic principle of a vernier calliper.

$$1 \text{ VSD} = 1.1 \text{ mm.}$$

Let the distance of zero of vernier from the 28mm mark be x . Then,

$$28\text{mm} + (1\text{mm})(8) = (28\text{mm} + x) + (1.1\text{mm})(7)$$

$$\text{solving } x = 0.3\text{mm}$$

$$\text{so reading of } C_2 = 28\text{mm} + 0.3\text{mm} = 2.83\text{mm}$$

22. Mean value of T is

$$\bar{T} = \frac{0.52 + 0.56 + 0.57 + 0.54 + 0.59}{5} = 0.56\text{s}$$

Mean error

$$\Delta T = \frac{|0.52 - 0.56| + |0.56 - 0.56| + |0.57 - 0.56| + |0.54 - 0.56| + |0.59 - 0.56|}{5}$$

$$= 0.02$$

$$\% \text{ error in } T = \frac{\Delta T}{T} \times 100 = \frac{0.02}{0.56} \times 100 = 3.57\%$$

According to the problem

$$g \propto \frac{T^2}{R-r} \Rightarrow \frac{\Delta g}{g} = 2 \frac{\Delta T}{T} + \frac{\Delta R + \Delta r}{R-r}$$

$$\therefore \frac{\Delta g}{g} \times 100 = 2 \times 3.57\% + \frac{1+1}{60-10} \times 100\% = 11\%$$

$$\text{Error in } r \text{ is } \frac{\Delta r}{r} \times 100 = \frac{1}{10} \times 100 = 10\%$$

$$23 \text{ Time} = \sqrt{\frac{2L}{g}} + \frac{L}{v}$$

$$t = \sqrt{\frac{2L}{10}} + \frac{L}{300}$$

$$\text{or, } t = \sqrt{\frac{L}{5} + \frac{L}{300}}$$

$$\text{or, } dt = \frac{1}{\sqrt{5}} \cdot \frac{1}{2L} dL + \frac{dL}{300}$$

$$\text{or, } 0.01 = \frac{dL}{2\sqrt{5}\sqrt{20}} + \frac{dL}{300}$$

$$\text{or, } 0.01 = dL \left[\frac{1}{20} + \frac{1}{300} \right]$$

$$\text{or, } 0.01 = dL \left[\frac{16}{300} \right]$$

$$\text{or, } dL = 0.01 \times \frac{300}{16}$$

$$\frac{dL}{L} \times 100\% = \frac{0.01 \times 300 \times 100\%}{16 \times 20}$$

$$= \frac{15}{16}\% \simeq 10$$

24. $r = \frac{1-a}{1+a}$

Let $1 - a = x$

and $1 + a = y$

Then $|\Delta x| = |\Delta a|$

and $|\Delta y| = |\Delta a|$

$$r = \frac{x}{y}$$

$$\Rightarrow \Delta r = r \left(\frac{\Delta x}{x} + \frac{\Delta y}{y} \right)$$

$$= \frac{x}{y} \left(\frac{\Delta a}{1-a} + \frac{\Delta a}{1+a} \right) = \left(\frac{1-a}{1+a} \right) \left(\frac{2\Delta a}{(1-a)(1+a)} \right)$$

$$= \frac{2\Delta a}{(1+a)^2}$$

25. You should try this question after studying nuclear physics.

$$26. \Delta l = \frac{Fl}{AY} = \frac{1.2 \times 10 \times 1}{\pi \left(\frac{5 \times 10^{-4}}{2} \right)^2 \times 2 \times 10^{11}} = 0.03 \text{ mm}$$

Since least count of vernier calipers is 0.1mm. So, third marking of vernier scale will coincide with a mark on the main scale.